

AI in Finance

Finance for Engineers, Scientists, and Data Scientists

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Preface

Artificial intelligence is now embedded in nearly every corner of the financial industry: credit decisions, fraud alerts, trade execution, risk limits, portfolio construction, and increasingly the generative tools analysts use to read a filing or draft a note. The people building and overseeing these systems are disproportionately trained in mathematics, statistics, physics, computer science, and engineering, not finance. They can derive a gradient, debug a training loop, and reason about a covariance matrix without much trouble. What they often lack, at least at first, is the financial vocabulary and institutional context that turns a model into something a bank, a fund, or a regulator can actually use, and use responsibly. This book is written to close that gap.

Who This Book Is For

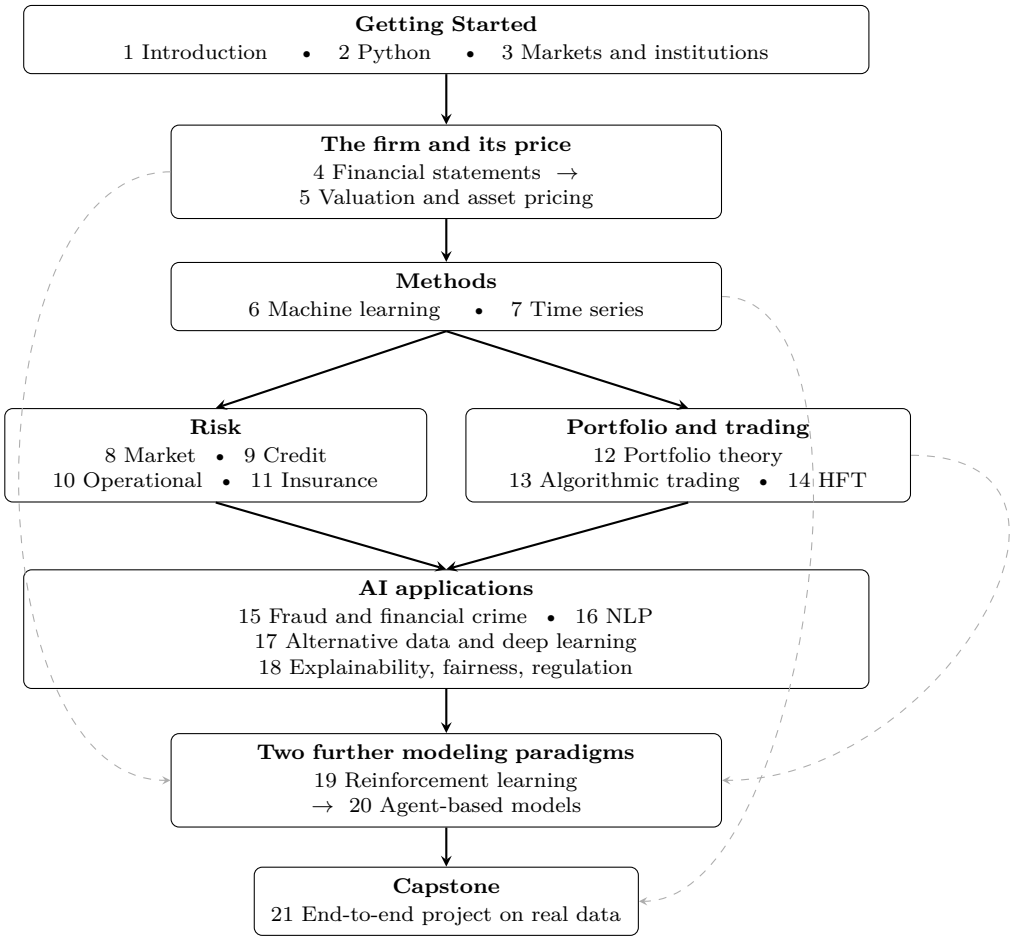
I have written this book for readers who are already comfortable with quantitative reasoning and, ideally, with basic Python, but who have limited exposure to finance and economics: graduate students and advanced undergraduates in engineering, the physical sciences, statistics, and computer science; practicing data scientists and machine learning engineers moving into a financial role; and anyone building a career on the specific combination of skills this book covers. No prior finance coursework is assumed. Chapter 3 introduces financial markets and institutions from first principles, and every subsequent chapter builds forward from there, so a reader with no finance background at all should be able to start at Chapter 1 and never feel lost.

How This Book Is Organized

The twenty-one chapters are ordered by dependency, not by alphabetical convenience or topical grouping. Chapter 2 teaches the Python tools, NumPy, pandas, scikit-learn, matplotlib, that nearly every later chapter's worked examples rely on. Chapter 3's markets and institutions precede Chapter 4's financial statements, and Chapter 4's statements feed directly into Chapter 5's valuation and asset-pricing models, which in turn recur throughout the risk, portfolio, and credit chapters that follow. The four risk chapters themselves run as a deliberate sequence: market risk (Chapter 8) models how much a position's value can move, credit risk (Chapter 9) whether a counterparty will pay at all, and operational risk (Chapter 10) the losses that remain when the institution's own processes, people, or systems fail, each defined by exactly what the one before it leaves out; insurance (Chapter 11) then closes the sequence from the other side of the table, the industry whose entire product is bearing those risks for a price, and whose actuarial machinery the operational-risk chapter has already borrowed. Machine learning fundamentals (Chapter 6) are introduced once, in depth, and then reused, not re-taught, in every later chapter that needs a regression, a classifier, or a neural network: fraud scoring, credit scoring, algorithmic trading signals, natural language processing, and the explainability methods of Chapter 18 all build on that same foundation rather than each inventing their own. Chapter 19 closes the methods sequence proper with reinforcement learning, built on the trading and portfolio machinery of Chapters 12 through 14; Chapter 20 then turns to a third modeling paradigm, agent-based models of markets and financial networks, built on that same statistical and portfolio machinery together with Chapter 19's own sequential-decision framing, immediately before Chapter 21's capstone puts every method in the book to work on one real dataset. This ordering was arrived at, in part, by actually running the companion notebooks end to end and discovering where an earlier draft asked a chapter to use a tool it had not yet been taught. The diagram below draws that structure.

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One convention holds throughout and is stated here rather than repeated in every chapter: every numerical claim in this book was computed in Python before it was written into the text, never derived by hand and checked afterward, and each is reproduced exactly in that chapter's companion notebook. Where a figure is instead quoted from published research, it is cited rather than recomputed. Readers who want to verify a number, or change an assumption and watch what happens to it, should reach for the notebook rather than re-deriving the arithmetic.



How the chapters depend on one another. Solid arrows are prerequisites. The three dashed arrows mark the places where a later group reuses an earlier one's machinery rather than rebuilding it: Chapter 5's option pricing returns in Chapters 8, 9, 12, and 19; Chapter 6's models are reused by Chapter 9 and Chapters 15 through 18 rather than retaught; and Chapter 13's execution model returns in Chapter 19. Chapter 2 underpins every companion notebook in the book and Chapter 21 draws on all of it, so neither is drawn with an arrow to each dependent chapter. An instructor short of sessions can cut most safely within a group rather than across the spine.

Beneath this chapter-by-chapter dependency, the book falls into three broader parts, marked as such in the Contents. Part I, Foundations (Chapters 1 through 7), is preparation: the finance vocabulary, institutions, financial statements, and valuation machinery a reader needs before any model can be usefully applied, together with the Python and machine learning fundamentals every later chapter draws on. Part II, Applications (Chapters 8 through 17), covers the applications a working quantitative finance professional encounters most often: market, credit, operational, and insurance risk, portfolio construction, algorithmic and high-frequency trading, fraud detection, and the natural language processing and alternative-data methods reshaping how firms extract signal from unstructured sources. Part III, Advanced Topics (Chapters 18 through 21), turns to more advanced and specialized territory: explainability, fairness, and regulation; reinforcement learning; agent-based models of markets and financial networks; and a closing capstone that applies the entire toolkit

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to one real, publicly available dataset end to end. A reader pressed for time could stop after Part II and still have a working, practitioner-level command of AI in finance; Part III is where the book pushes into territory still actively being worked out in research and in practice.

Every chapter follows the same shape: an introduction, several sections of exposition built around at least one fully worked numerical example a reader can check by hand, a discussion of the practical challenges and limitations specific to that chapter's topic, and a closing sequence of Key Takeaways, Suggested Exercises, and Further Reading pointing to the standard textbooks and original papers behind the chapter's material. Most chapters' Suggested Exercises close with a challenge question that asks the reader to redo one of the chapter's own methods on a real, freely obtainable dataset, from sources such as the UCI Machine Learning Repository, Kaggle, SEC EDGAR and the SEC's enforcement releases, or the Consumer Financial Protection Bureau's HMDA and consumer-complaint databases, or, in the handful of cases where no clean real dataset exists (a limit order book fast enough to show latency arbitrage, say), on data generated by a documented simulation instead; the point of each is to show what a real-world version of the chapter's toy example actually looks like, messy edges included.

Each chapter is sized for a single 90-minute class session, though the chapters are not all the same length. Chapters 8, 9, 12, 15, 16, and 19 carry noticeably more worked material than the rest, each running roughly a fifth longer than a typical chapter, Chapter 16 most of all, its generative-AI coverage being deliberately deeper than that of any other single topic here. An instructor who works through every worked example in those six should plan two sessions for each, or else select among their later sections. Chapter 10 is the deliberate opposite, shorter than the rest by design, and Chapter 11 is also on the short side; both leave room within a single session for exercises, discussion, or an early start on the chapter that follows. The book is designed as the backbone of a one-semester course meeting twice a week: at roughly one chapter per session, the twenty-one chapters fit into such a semester with room left over for review, examinations, and project presentations. Giving all six longer chapters a second session, about twenty-seven sessions in all, would use up most of that room, so most instructors will double only the two or three that matter most to their course. An instructor pressed for sessions can also be selective, most naturally among the more specialized chapters (10, 11, 14, 15, 17, 19, and 20), without breaking the dependency chain described above. A course meeting once a week can work from a core of about fourteen chapters, Chapters 1 through 9, then 12, 13, 16, 18, and 21, with the rest assigned as independent reading or term-project material. And the book serves a reader working through it independently at their own pace exactly as well. Three appendices follow the twenty-one chapters: a catalog of real financial datasets and APIs for readers who want to extend the book's examples with real data of their own, a glossary of finance and trading terms for readers who hit an unfamiliar term before the chapter that formally introduces it, and a career map connecting the industry's actual quantitative roles to the chapters that prepare for each, for the reader this book was written for in the first place.

Real, named case studies are woven throughout rather than confined to a single chapter: the insurer insolvencies after Hurricane Andrew in 1992 that created the catastrophe-modeling industry, the 1998 collapse of Long-Term Capital Management, the 2008 financial crisis and the Gaussian copula's role in mispricing CDO risk, the 2008 Société Générale rogue-trading loss, the 2010 Flash Crash, the 2012 Knight Capital trading malfunction, the 2012 JPMorgan "London Whale" loss, the Danske Bank Estonia money-laundering scandal, the 2019 Apple Card gender-bias controversy, and the SEC's 2022 settlement with Schwab over undisclosed robo-advisor cash allocations, among others. These are real events with real consequences, included because the lesson a formula teaches in the abstract is rarely the lesson a genuine failure teaches in practice. Chapter 21's capstone case study goes further still, working a real, publicly available consumer-credit dataset end to end, undocumented category codes, class imbalance, model comparison, calibration, explainability, and fairness auditing included, using nothing invented for the occasion.

What Makes This Book Different

Five features shape this book more than any other, and a reader deciding whether it suits them should know them in advance. First, it teaches finance from first principles to readers who are already quantitatively trained: no finance background is assumed, but comfort with quantitative reasoning and basic Python is, so the finance is explained in full while the mathematics and machine learning are allowed to move at the pace such a reader expects. Second, every number in the text is reproduced by an executed companion notebook,

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and that reproduction is checked automatically: the repository pins the library versions that produced the published outputs (`requirements.txt`) and includes a script (`tools/check_notebooks.py`) that re-executes the notebooks and reports any output that has drifted from what the book quotes. Third, each method is taught through at least one worked example small enough to check by hand, so a reader can see exactly what a model computes before trusting it at scale. Fourth, risk and regulation receive sustained depth rather than a closing chapter: four chapters on market, credit, operational, and insurance risk, and a full chapter on explainability, fairness, and the regulation of models, reflecting how much of the industry's actual work with AI is governed by them. Fifth, Appendix C maps the industry's quantitative roles to the chapters that prepare for each, so a reader building a career can see which parts of the book matter most for the job they want.

What This Book Does Not Cover

A book this broad still has to stop somewhere, and a few neighboring subjects are deliberately left to other texts. Trading strategies for cryptocurrencies and digital assets are not covered; the market-microstructure and trading chapters apply to any electronic market, and on-chain data appears in Chapter 17 as one alternative-data source among many, but crypto-specific venues, instruments, and strategies are left to specialist sources. Stochastic calculus and derivatives theory beyond the binomial model and Black-Scholes-Merton are not developed here; a reader who needs Itô calculus, exotic options, or term-structure models should turn to a standard derivatives text such as Hull's *Options, Futures, and Other Derivatives* and, beyond it, a text on stochastic calculus for finance. Quantum computing and its prospective financial applications are not discussed. Corporate finance, including capital structure, mergers and acquisitions, and deal valuation, appears only as far as Chapters 4 and 5 need it; a corporate finance text such as Brealey, Myers, Allen, and Edmans covers it properly. And accounting standards are introduced only to the depth needed to read financial statements and the expected-credit-loss rules; a reader who needs IFRS or US GAAP in detail should consult an intermediate accounting text or the standards themselves.

The Companion Notebooks

Every worked numerical example in this book has a companion Jupyter notebook that reproduces it in Python, one notebook per chapter, using ordinary, widely available libraries rather than anything exotic. Every notebook was actually executed, and its printed output, not a hand-derived approximation of it, is what the chapter text reports; where the two disagreed during drafting, the notebook's output was treated as the source of truth and the prose was corrected to match. The companion notebooks, along with the LaTeX source for this book itself, are available at:

`https://github.com/wulinsuo/AI-in-Finance`

I encourage readers to run each notebook alongside the corresponding chapter, change an input or two, and see what happens; a formula becomes far more trustworthy once you have watched it respond correctly to a change you made yourself.

Technical Notes

A few of the formulas used in this book are quoted without a full derivation, since working through every one of them line by line in the main text would bury the finance and the intuition, which is this book's actual subject, under mathematics that is not. Where a derivation is long enough to be worth doing properly but not essential to following the chapter that uses the result, I have written it up separately as a standalone technical note and placed it in the `technical notes` folder of the same GitHub repository as the companion notebooks, linked above.

These notes are entirely optional. They exist for the reader who wonders where a particular formula actually comes from and wants to see the argument worked out in full; skipping them has no effect on understanding any of the approaches covered in this book, since every chapter states the results it needs and

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shows how to use them regardless of whether the underlying derivation has been read. More notes will be added over time as new formulas in the book warrant one.

A Note on the Draft

This is a work in progress, and the watermark on every page is there to say so honestly rather than to discourage reading. I am continuing to expand, correct, and refine the material, and I welcome corrections, questions, and suggestions from readers at the address below. If you find an error, a stale cross-reference, or a passage that would benefit from a clearer example, I would genuinely like to hear about it.

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Notation

Symbols in this book follow each field's own conventions, which means a few letters carry different meanings in different chapters. The table below lists the recurring symbols and the principal uses a reader will encounter, with the chapters where each use appears; notation used only locally is defined where it appears. Typographic conventions, hats for estimates, time subscripts, units and annualization, are collected in Chapter 1's "Notation and Conventions" section.

Symbol	Principal uses
r, r_f	An interest rate or an asset's return; the risk-free rate (Chapters 3, 5, 12)
μ	Expected return or drift (Chapters 5, 8, 12)
σ, σ_t^2	Volatility of returns (Chapters 5, 8, 12); conditional variance in GARCH models (Chapter 7)
β	Regression coefficient (Chapter 6); CAPM market beta (Chapters 5, 12)
α	Significance level (Chapter 6); ARCH coefficient (Chapter 7); return in excess of a benchmark (Chapters 12, 13); learning rate (step size) of Q-learning (Chapter 19)
λ	Kyle's lambda price impact (Chapters 13, 14); Poisson intensity (Chapters 10, 11, 14); regularization strength (Chapter 6)
ρ	Correlation (Chapters 2, 8, 12); a convex risk measure (Chapter 19)
Δ, Γ	Option delta and gamma (Chapters 5, 8, 19); ΔX_t, a first difference (Chapters 1, 7)
q	Risk-neutral probability (Chapter 5)
$N(\cdot)$, $\phi(\cdot)$	Standard normal cumulative distribution function and density (Chapters 5, 8); ϕ alone, the Almgren-Chriss risk-aversion parameter (Chapters 13, 19) and an autoregressive coefficient (Chapter 7)
Z, z	Standard normal variate or z-score (Chapters 6, 8, 15); Altman Z-score (Chapter 4); Bühlmann credibility weight (Chapter 11)
ε	Regression error term (Chapters 6, 7); exploration rate in ε-greedy learning (Chapter 19)
π	A decision policy (Chapter 19); the share of informed traders in the Glosten-Milgrom model (Chapter 14)
γ	Discount factor (Chapter 19); permanent price impact in the Almgren-Chriss model (Chapter 13); a market maker's risk aversion (Chapter 14); unit-root coefficient of the Dickey-Fuller regression (Chapter 7)
η	Learning rate (step size) of gradient descent (Chapters 6, 17); temporary price-impact coefficient in the Almgren-Chriss model (Chapters 13, 19)
θ	A model's parameters (Chapters 6, 19); moving-average coefficient (Chapter 7); intensity of choice (Chapter 20)
τ	Length of one trading interval (Chapters 13, 19); average treatment effect (Chapter 6); Black-Litterman prior scaling (Chapter 12); softmax temperature (Chapter 17)

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Symbol	Principal uses
δ	A market maker's quoted spread (Chapters 13, 14); covariance shrinkage intensity (Chapter 12)
κ	Almgren-Chriss urgency parameter (Chapter 13); order-arrival decay in the Avellaneda-Stoikov model (Chapter 14)
Σ	Covariance matrix of asset returns (Chapters 2, 8, 12)
w, w_i	Portfolio weights (Chapters 2, 8, 12); weights of a linear model or neural network (Chapters 6, 17)
K, k	Option strike price (Chapters 5, 8); number of neighbors in k-nearest neighbors (Chapters 6, 9); key matrix of attention (Chapters 16, 17)
Q	Action-value function (Chapter 19); query matrix of attention (Chapters 16, 17); Ljung-Box statistic (Chapter 7); size of a metaorder (Chapter 13); Black-Litterman view returns (Chapter 12)
T	Option maturity or time horizon (Chapters 5, 8, 13); number of observations (Chapter 7)
$\mathbb{E}[\cdot]$	Expectation (written $\mathbb{E}$ throughout, never a plain E)
PD, LGD, EAD	Probability of default, loss given default, exposure at default (Chapter 9)
VaR, ES	Value at Risk and Expected Shortfall (Chapter 8)

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Abbreviations

The abbreviations below are those the text defines and uses more than in passing, collected in one alphabetical list for a reader who meets one before, or long after, the passage that spells it out. Ticker symbols, dataset and product names, and abbreviations used only within a single paragraph are left out.

Abbreviation	Meaning
ABM	agent-based model
ACF	autocorrelation function
ACH	automated clearing house
ADF	augmented Dickey-Fuller (test)
AFS	available for sale
AI	artificial intelligence
AIC	Akaike information criterion
ALM	asset-liability management
AML	anti-money laundering
AMLA	Anti-Money Laundering Authority
API	application programming interface
APO	Absolute Price Oscillator
APP	authorized push payment (fraud)
AR	autoregressive (model)
ARCH	autoregressive conditional heteroskedasticity
ARIMA	autoregressive integrated moving average
ARMA	autoregressive moving average
ARR	alternative reference rate
AUC	area under the curve
BIC	Bayesian information criterion
BISG	Bayesian Improved Surname Geocoding
BPE	byte-pair encoding
BSA	Bank Secrecy Act
CAPM	capital asset pricing model
CBDC	central bank digital currency
CCAR	Comprehensive Capital Analysis and Review
CCF	credit conversion factor
CCP	central counterparty
CDD	customer due diligence
CDO	collateralized debt obligation
CDS	credit default swap
CECL	Current Expected Credit Losses
CFPB	Consumer Financial Protection Bureau
CNN	convolutional neural network
COGS	cost of goods sold
CRO	chief risk officer
CRSP	Center for Research in Security Prices
CTR	Currency Transaction Report
CVA	credit valuation adjustment
DCF	discounted cash flow
DDM	dividend discount model

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Abbreviation	Meaning
DORA	Digital Operational Resilience Act
DQN	deep Q-network
EAD	exposure at default
EBIT	earnings before interest and taxes
EBITDA	earnings before interest, taxes, depreciation, and amortization
EBM	explainable boosting machine
ECOA	Equal Credit Opportunity Act
EDD	enhanced due diligence
EE	expected exposure
EMA	exponentially weighted moving average
EMH	efficient market hypothesis
ERC	equal risk contribution
ES	Expected Shortfall
ESG	environmental, social, and governance
ETF	exchange-traded fund
EVE	economic value of equity
EVPI	expected value of perfect information
FATF	Financial Action Task Force
FCF	free cash flow
FCFE	free cash flow to equity
FCFF	free cash flow to the firm
FDIC	Federal Deposit Insurance Corporation
FinCEN	Financial Crimes Enforcement Network
FINRA	Financial Industry Regulatory Authority
FRTB	Fundamental Review of the Trading Book
GAAP	generally accepted accounting principles
GAM	generalized additive model
GAN	generative adversarial network
GARCH	generalized autoregressive conditional heteroskedasticity
GDPR	General Data Protection Regulation
GLM	generalized linear model
GNN	graph neural network
GPD	generalized Pareto distribution
HCC	hierarchical condition categories
HFT	high-frequency trading
HHI	Herfindahl-Hirschman Index
HMDA	Home Mortgage Disclosure Act
HRP	hierarchical risk parity
HTM	held to maturity
IBNR	incurred but not reported
IFRS	International Financial Reporting Standards
IPO	initial public offering
IRB	internal ratings-based (approach)
IRRBB	interest-rate risk in the banking book
KNN	k-nearest neighbors
KRI	key risk indicator
KYC	know your customer
LCR	Liquidity Coverage Ratio
LDI	liability-driven investment
LDTI	long-duration targeted improvements
LGD	loss given default
LIME	local interpretable model-agnostic explanations
LLM	large language model
LoRA	Low-Rank Adaptation
LSTM	long short-term memory
LTCM	Long-Term Capital Management
LVaR	liquidity-adjusted VaR

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Abbreviation	Meaning
MAE	mean absolute error
MAPE	mean absolute percentage error
MARL	multi-agent reinforcement learning
MAS	Monetary Authority of Singapore
MCP	Model Context Protocol
MDP	Markov decision process
MiCA	Markets in Crypto-Assets Regulation
MoE	mixture of experts
MSE	mean squared error
NBBO	National Best Bid and Offer
NGFS	Network for Greening the Financial System
NII	net interest income
NLP	natural language processing
NOPAT	net operating profit after tax
NPV	net present value
NSFR	Net Stable Funding Ratio
OFAC	Office of Foreign Assets Control
OFI	order flow imbalance
OLS	ordinary least squares
OTC	over-the-counter (market)
P/E	price-to-earnings ratio
PACF	partial autocorrelation function
PCA	principal component analysis
PD	probability of default
PEP	politically exposed person
PFE	potential future exposure
PIN	Probability of Informed Trading
POF	proportion of failures (test)
POT	peaks over threshold
PPO	Proximal Policy Optimization
PSI	population stability index
PV	present value
RAG	retrieval-augmented generation
RLHF	reinforcement learning from human feedback
RMSE	root mean squared error
RNN	recurrent neural network
ROA	return on assets
ROC	receiver operating characteristic (curve)
ROE	return on equity
ROIC	return on invested capital
RSI	Relative Strength Index
RTGS	real-time gross settlement
RWA	risk-weighted assets
SA-CCR	standardized approach for counterparty credit risk
SAR	Suspicious Activity Report
SEC	Securities and Exchange Commission
SHAP	Shapley additive explanations
SIP	Securities Information Processor
SMOTE	synthetic minority oversampling technique
SOFR	Secured Overnight Financing Rate
SVM	support vector machine
SWES	system-wide exploratory scenario
TCA	transaction cost analysis
TF-IDF	term frequency-inverse document frequency
TWAP	time-weighted average price
UCB	upper confidence bound
VAR	vector autoregression

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Abbreviation	Meaning
VaR	Value at Risk
ViT	Vision Transformer
VIX	CBOE Volatility Index
VWAP	volume-weighted average price
WACC	weighted average cost of capital
WCDR	worst-case default rate
WOE	weight of evidence

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Part I

Foundations

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Chapter 1

Introduction to Finance and AI

1.1 Why AI in Finance

Finance has always been a quantitative discipline, but the questions it asks have outgrown the tools that were built to answer them. This part sets out what changed and how the field arrived here.

1.1.1 Why Finance Needs AI

Finance is one of the most data-rich and decision-intensive domains in the modern economy. Financial institutions collect vast amounts of information every day: prices, transactions, news, macroeconomic indicators, and firm-level reports. This information is often noisy, high-dimensional, and time-sensitive, which makes it hard to analyze with simple rules or intuition.

Financial markets are also highly competitive and dynamic, and firms, investors, and regulators must decide quickly and accurately. Methods from statistics, optimization, machine learning, and artificial intelligence have therefore become important in forecasting, risk management, trading, fraud detection, and customer service.

The central challenge in finance is not only to process data but to make decisions under uncertainty. A portfolio manager must decide how much risk to take. A bank must estimate the probability of default. A regulator must detect suspicious behavior before losses become large. In each of these cases, the decision-maker needs a model, a framework for judgment, and an understanding of the incentives and constraints of the market.

Figure 1.1 shows this process: raw data feeds a model, the model's output is checked before being trusted, and only then does it inform a decision. The book returns to that checking step when discussing why models must be interpreted rather than worshipped.

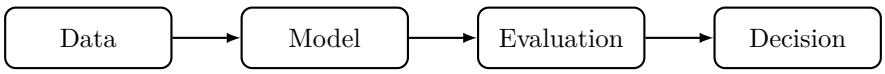


Figure 1.1: A simple pipeline from financial data to modeling, evaluation, and finally decisions

There is also an economic reason for AI's growing role. Many financial decisions are made with limited information, under time pressure, and in changing conditions, where fixed rules may miss subtle patterns in the data. Machine learning can uncover nonlinear relationships and adapt to new information, provided it is used with domain knowledge and careful validation.

Finance combines three layers of reasoning: economics, which supplies the structure of incentives; statistics, which estimates uncertain relationships from data; and implementation, which governs how a model behaves inside a real system with costs, delays, and operational constraints. A model that predicts a price movement becomes valuable only when it is linked to a trading rule, a risk limit, and the costs of execution. A credit

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model that predicts default must be read in light of borrower behavior, contract terms, and macroeconomic conditions. A strong practitioner is fluent in all three layers.

1.1.2 A Short History of Quantitative Methods and AI in Finance

Using mathematical models to make financial decisions is not new; what has changed over the past seventy years is the sophistication of the models and the scale of their data. A brief history shows why today's AI methods are the latest step in a long progression, not a sudden break.

The modern era of quantitative finance is often dated to 1952, when Harry Markowitz formalized portfolio selection as an optimization problem over expected return and variance (Markowitz 1952), work covered in depth in Chapter 12. This gave investors, for the first time, a rigorous mathematical language for the intuitive idea of diversification.

The next milestone came in 1973, when Fischer Black, Myron Scholes, and Robert Merton derived a closed-form formula for pricing options (Black and Scholes 1973; Merton 1973), the model introduced in Chapter 5. It turned derivatives trading from an informal, judgment-driven activity into something that could be systematically priced and hedged. Both developments used relatively simple, closed-form mathematics to capture an economically meaningful idea, and both remain in use today after decades of refinement.

The 1980s and 1990s saw the rise of algorithmic and electronic trading, as exchanges automated order matching and quantitative hedge funds began systematically exploiting statistical patterns in market data (Chapters 13 and 14). Around the same time, banks began applying statistical scoring models to credit decisions at scale, replacing purely judgment-based underwriting with models like those of Chapter 9.

Machine learning methods (decision trees, support vector machines, and eventually gradient boosting and random forests) entered mainstream financial use in the 2000s and 2010s as computing power and datasets grew, turning simple statistical scoring into much richer pattern recognition (Chapter 6).

Since roughly 2015, further developments have reshaped the field. Deep learning has made it practical to learn directly from large volumes of unstructured data (images, audio, and especially text), the foundation for the natural language processing methods in Chapter 16 and the alternative data methods in Chapter 17. More recently, large language models have been applied to summarizing earnings calls, extracting structured information from filings, and supporting research. Reinforcement learning has begun to be applied to sequential decisions such as trade execution, market making, and rebalancing, where an agent's actions today shape the state it faces tomorrow (Chapter 19). Agent-based modeling takes the opposite vantage point: it simulates many simple, interacting participants to see how bubbles, crashes, and contagion can emerge without any single agent intending them (Chapter 20). Table 1.1 summarizes this progression.

Period	Development
1950s	Markowitz mean-variance portfolio theory formalizes diversification mathematically
1970s	Black-Scholes-Merton option pricing enables systematic derivatives pricing and hedging
1980s–1990s	Electronic exchanges and algorithmic trading; statistical credit scoring at scale
2000s–2010s	Machine learning (trees, ensembles, support vector machines) applied to prediction and classification tasks across finance
2015–present	Deep learning for unstructured data; large language models applied to financial text; reinforcement learning for sequential decisions; agent-based models for market-wide simulation

Table 1.1: A brief timeline of quantitative methods in finance

The throughline is not that new methods replace old ones. Mean-variance optimization, Black-Scholes-Merton, and statistical scoring are all still in daily use, often alongside the newer machine learning methods. Each wave has instead expanded the range of problems that can be addressed and the data that can be used, while the economic questions (how to price risk, allocate capital, judge creditworthiness) have stayed largely

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the same. A new AI technique is generally a new way of answering an old financial question, not a reason to discard the economic reasoning behind it.

Table 1.1's AI/ML rows compress six decades of an older, independent field into two lines. AI research has its own timeline of breakthroughs and setbacks, most of it predating any financial application; Table 1.2 traces it.

Year	Development
1956	The Dartmouth Workshop, where John McCarthy and colleagues coin the term "artificial intelligence," is widely regarded as the field's founding event
1958	Frank Rosenblatt's Perceptron, the first trainable artificial neural network, generates enormous early excitement
1969–1986	Minsky and Papert's formal critique of the Perceptron's limitations helps trigger the first "AI winter", a prolonged pullback in AI funding and research interest
1986	Rumelhart, Hinton, and Williams popularize backpropagation, an efficient algorithm for training multi-layer neural networks (Chapter 6) that ends the first AI winter
1997	IBM's Deep Blue defeats reigning world chess champion Garry Kasparov, the first defeat of a sitting champion by a computer under tournament conditions
2012	AlexNet's decisive win in the ImageNet image-recognition competition demonstrates deep convolutional networks' advantage at scale, igniting the modern deep learning era (Chapter 17)
2016	DeepMind's AlphaGo defeats Go champion Lee Sedol using deep reinforcement learning (Chapter 19), a game long assumed too complex for a computer to master
2017	"Attention Is All You Need" (Vaswani et al. 2017) introduces the transformer architecture, the foundation of every large language model that follows (Chapter 16)
2020	OpenAI's GPT-3, a 175-billion-parameter language model, demonstrates that scale alone unlocks broad, few-shot task performance
2022	OpenAI's ChatGPT reaches mainstream adoption faster than any consumer software product before it, bringing generative AI into everyday use

Table 1.2: A timeline of key developments in AI and machine learning

Two patterns stand out against Table 1.2. First, progress was not steady: the field's first AI winter shows that a technique's early promise (the Perceptron) can collapse under a well-founded technical critique and take over a decade to recover, a caution worth remembering whenever a new method arrives with confident claims. Second, none of Table 1.2's milestones were built for finance: Deep Blue plays chess, AlexNet classifies photographs, AlphaGo plays Go. Each supplied a general-purpose capability (pattern recognition at scale, sequential decision-making, language understanding) that finance adopted only after it had been proven elsewhere. This is why Chapters 6, 16, 17, and 19 introduce these architectures and training algorithms as general-purpose machine learning before turning to what makes each useful in finance.

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1.2 The Financial Landscape

Before any model can be useful, the terrain it operates on has to be legible: who trades, what they trade, what data the activity leaves behind, and the handful of concepts every later chapter assumes. This part supplies that vocabulary.

1.2.1 A Brief Introduction to Financial Markets

Consider a mismatch found in every economy: a young firm has a profitable idea but no cash to build it, while a household nearby has idle savings. The household has no way to fund the idea directly, could not evaluate it if it could, and does not want to bear all the risk alone. Multiply this mismatch across millions of firms and households, with different amounts of capital, risk appetites, and time horizons, and the coordination problem becomes too large for buyers and sellers to solve by finding each other. A financial market solves this coordination problem at scale. It is a place where financial assets (stocks, bonds, currencies, commodities, and derivatives) are traded, letting individuals, businesses, and governments transfer capital, diversify risk, and price future uncertainty without every lender having to find and vet every borrower.

Financial markets do not exist only for speculation. They allocate resources across time and across economic agents: a household saving for retirement, a firm borrowing to expand, and a government issuing debt are all connected through markets with those who have capital to supply.

The value of an asset is determined by the expectations of market participants about future cash flows, risk, and liquidity. For example, the price of a stock reflects the market's assessment of a firm's future profitability, while the price of a bond reflects expectations about default risk and interest rates. Finance is therefore both empirical and forward-looking.

Some of the most important financial institutions include banks, which intermediate between savers and borrowers; asset managers, which invest funds on behalf of clients; insurance companies, which protect against specific risks; exchanges, which provide platforms for trading; and regulators, which oversee market integrity and stability. Each plays a distinct role in the financial system, and their incentives shape market behavior in subtle ways. Table 1.3 summarizes these roles.

Institution	Role
Banks	Intermediate between savers and borrowers
Asset managers	Invest funds on behalf of clients
Insurance companies	Protect against specific risks
Exchanges	Provide platforms for trading
Regulators	Oversee market integrity and stability

Table 1.3: Major financial institutions and their roles

Financial markets can be classified by what they trade. Equity markets trade ownership claims in firms. Bond markets trade debt securities. Money markets trade short-term borrowing and lending instruments, typically with maturities under one year (Chapter 3). Derivatives markets trade contracts whose value depends on other assets. Foreign exchange markets trade currencies. Each market has its own structure, participants, and incentives. Table 1.4 summarizes these market types.

Markets can also be described by the roles of their participants. Investors seek returns. Borrowers seek financing. Market makers provide liquidity by standing ready to buy and sell. Arbitrageurs exploit small pricing discrepancies. Regulators enforce rules and preserve confidence. These roles interact to create the price discovery process that makes markets informative. Table 1.5 summarizes these participant roles.

This perspective also explains why prices can move sharply on limited news. A market is not a single decision-maker but a collection of heterogeneous participants with different objectives, information, and horizons, so prices reflect a balance of competing forces rather than a simple average of known facts.

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Market	What is traded
Equity markets	Ownership claims in firms
Bond markets	Debt securities
Money markets	Short-term borrowing and lending instruments, such as Treasury bills, commercial paper, and repurchase agreements (Chapter 3)
Derivatives markets	Contracts whose value depends on other assets
Foreign exchange markets	Currencies

Table 1.4: Major types of financial markets

Participant	Objective or function
Investors	Seek returns
Borrowers	Seek financing
Market makers	Provide liquidity by standing ready to buy and sell
Arbitrageurs	Exploit small pricing discrepancies
Regulators	Enforce rules and preserve confidence

Table 1.5: Financial market participant roles

1.2.2 The Financial Data Ecosystem

Before turning to financial concepts, consider the data that feed AI applications in finance. Different data types require different tools and handling, and much of this book is organized around them. Table 1.6 summarizes the main categories.

Each data type has its own quality issues and conventions, and a recurring theme of this book is that using data well requires understanding where it comes from, not just how to load it into a model. Market data is timestamped and voluminous but expensive at high frequency and subject to microstructure noise (Chapter 2 introduces the handling of price and return data; Chapters 13 and 14 return to its highest-frequency form). Fundamental data is low-frequency and subject to reporting lag and accounting discretion (Chapter 4). Macroeconomic data is often revised after release, so a dataset built from final revised figures can overstate what was knowable in real time, a subtle look-ahead bias. Text and alternative data are the newest additions and, as Chapters 16 and 17 discuss, need far more preprocessing than the structured, tabular data of the earlier chapters.

A single financial AI system often combines several data types: a credit model may join a small business's fundamentals with its bank-account transactions, or a trading strategy may join market data with news sentiment. This is harder than it sounds. The data must be aligned in time and identity (the same firm, the same reporting period), and each source can fail or degrade independently.

1.2.3 Core Financial Concepts

A few central ideas in finance come before any AI application. They supply the vocabulary and intuition for financial problems, whether these are addressed with classical models or with machine learning.

Risk and Return

Suppose two investments are on offer: a government bond that reliably pays 3% a year, and a new company's stock that might return 30% or might lose half its value, with no way to know in advance which. Why would anyone hold the stock? Only because, on average and over time, it must offer more than the bond to compensate buyers for the chance of a large loss. This is the risk-return trade-off, a fundamental principle of finance: higher expected return usually comes with higher risk, because otherwise investors would hold the safer asset. Every investor must decide how much uncertainty to bear in exchange for potential reward, a trade-off central to portfolio choice, asset pricing, and corporate finance.

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Data type	Description and examples
Market data	Prices, quotes, trading volume, and order-book data, ranging from daily closing prices to microsecond-level tick data (Chapters 3, 13, 14)
Fundamental data	Accounting and financial statement data reported by firms, such as revenue, earnings, and balance sheet items (Chapter 4)
Macroeconomic data	Interest rates, inflation, unemployment, and other economy-wide indicators that affect asset prices broadly (Chapter 7)
Credit and transaction data	Loan repayment history, credit bureau data, and payment transaction records used in credit and fraud models (Chapters 9, 15)
Operational loss and incident data	Internal and external operational-loss event records, incident and audit narratives, and system and access logs used to model and detect operational failures (Chapter 10)
Insurance policy and claims data	Policy, exposure, claims, telematics, and catastrophe-model data used to price, reserve, and transfer insured risk (Chapter 11)
Text and news data	Earnings call transcripts, news articles, analyst reports, and social media, used for sentiment and information extraction (Chapter 16)
Alternative data	Non-traditional sources such as satellite imagery, web traffic, or shipping data that proxy for economic activity (Chapter 17)

Table 1.6: Major categories of financial data

Figure 1.2 illustrates the idea: higher expected returns are typically obtained only by accepting greater variability in outcomes.

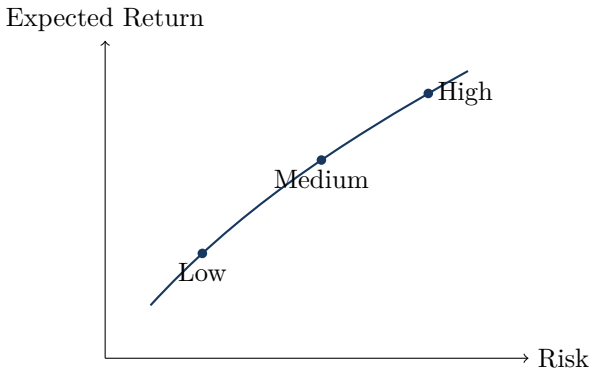


Figure 1.2: The risk-return trade-off: higher expected returns generally require accepting greater risk

Time Value of Money

Suppose someone offers to pay a debt either with $1,000 today or $1,000 in five years. Even setting aside inflation and the risk that the payer defaults, the two offers are not equivalent, and a quantitative reader's first instinct that a dollar is a dollar needs correcting: $1,000 today can be invested and grow into more than $1,000 by the time five years have passed, so accepting the delayed payment means giving up that growth

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for nothing. Money available today is worth more than the same nominal amount in the future because it can earn a return in the meantime. This principle underlies discounting, present value, interest rates, and valuation: future cash flows must be converted into present value before they can be compared fairly, a step central to bond pricing, project valuation, and investment analysis.

The future value of an amount PV invested today at rate r for n years is

$$FV = PV(1 + r)^n,$$

and Table 1.7 applies this formula to $1,000 invested at three different rates over three different horizons.

Annual rate	10 years	20 years	30 years
3%	$1,343.92	$1,806.11	$2,427.26
6%	$1,790.85	$3,207.14	$5,743.49
9%	$2,367.36	$5,604.41	$13,267.68

Table 1.7: Future value of $1,000 invested at different rates and horizons

Two patterns in Table 1.7 recur throughout the book. First, compounding is highly nonlinear in the rate: doubling the rate from 3% to 6% more than doubles the 30-year future value, and tripling it to 9% multiplies the 30-year value by more than five. Second, the horizon matters enormously: at 9%, the difference between a 10-year and a 30-year horizon is a factor of roughly 5.6, though the annual rate never changed. Present value, used constantly from Chapter 5 onward, is this same relationship read in reverse:

$$PV = \frac{FV}{(1 + r)^n},$$

discounting a future amount back to today.

Diversification

Suppose an investor holds shares in a single airline. A jump in fuel prices, a pilots' strike, or a crash investigation can each wipe out a large fraction of the stock's value, and all three risks are specific to that company and its industry. Now suppose the investor splits the same money across an airline, a software company, and a grocery chain, three businesses with largely unrelated day-to-day fortunes. A fuel-price spike still hurts the airline but does little to the other two, so news that would devastate an all-airline portfolio barely dents the diversified one. This is diversification: spreading investments across assets or risks so that no single loss can sink the portfolio. The intuition is simple (not all investments move together), but the consequence is not: diversification is one of the few strategies in finance that reduces risk without necessarily reducing expected return, which is why Chapter 12 treats it mathematically rather than as a rule of thumb. In portfolio construction, diversification reduces idiosyncratic risk, the risk specific to one company or industry, while retaining exposure to the broad economic factors that no amount of spreading across stocks can eliminate.

Market Efficiency

Suppose a quantitative analyst backtests a simple rule (buy a stock the day after it reports earnings above analyst expectations) and finds it would have produced a solid excess return over the past decade. Should the analyst expect it to keep working once real money is committed? That depends on how much of the information was already in the price before the analyst could act: if thousands of other participants saw the same earnings beat and traded within minutes, little may be left by the next morning.

The efficient market hypothesis formalizes this concern: asset prices reflect available information to a large extent. This has important consequences for forecasting, trading, and the interpretation of abnormal returns. Markets are not perfectly efficient at all times, but the framework remains a core reference point.

For quantitative readers, this motivates the distinction between signal and noise. Market efficiency does not say that no one can beat the market. It says the relevant question is often not whether a signal exists, but whether it survives competition from other informed participants, delays in data availability, transaction costs, model risk, and market frictions.

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1.3 What This Book Covers

With the landscape in place, the practical question is which financial problems machine learning improves on, and in what order they are best learned. This part maps the problems onto the chapters that treat them and fixes the notation; the preface explains the chapter order.

1.3.1 Financial Problems That AI Can Help Solve

AI is useful in finance because many important problems involve prediction, classification, optimization, and decision-making under uncertainty: forecasting asset prices or returns, predicting credit default or bankruptcy, detecting fraudulent transactions, monitoring for operational failures and misconduct, optimizing portfolios, identifying trading opportunities, and extracting information from financial text and news. None of these is a generic data problem; the underlying financial model, market structure, and regulatory context matter. A credit model must reflect the economics of default, and a trading strategy must account for transaction costs, market liquidity, and execution risk.

This is a key point for readers with a machine learning background. A model that performs well on a generic prediction task may fail in finance if it ignores institutional details: a pricing model must respect no-arbitrage conditions, and a forecasting model that ignores transaction costs, liquidity constraints, or regime changes may perform well in sample and still fail in practice. Financial data are also generated by strategic behavior: when a firm announces earnings or a central bank changes interest rates, the data record the reactions of many agents as well as economic fundamentals. Finance is a structured domain with economic logic and constraints, and it demands both predictive accuracy and economic plausibility.

Table 1.8 organizes the main problem types by machine learning task and points to where each is covered, previewing the book's structure problem-first rather than chapter-first.

Three things stand out from this taxonomy. First, several finance problems map onto more than one task type. Portfolio construction is an optimization problem but often uses regression or dimensionality reduction as an input, which is why later chapters combine methods from Chapter 6 with domain-specific structure. Second, some task types in Table 1.8 recur across many chapters (classification appears in credit, trading, and fraud) because the statistical machinery, introduced once in Chapter 6, is reused rather than re-derived for each application. Third, the table's last row is an outlier: every other row asks a model to predict, classify, or optimize for a single decision-maker, while multi-agent simulation asks what emerges when many simple decision-makers interact (Chapter 20).

1.3.2 Notation and Conventions Used in This Book

A few notational conventions are used throughout the book. Random variables and their realizations are both written in italics (for example, X or x), with the distinction clear from context. A hat denotes an estimate: $\hat{\beta}$ is an estimated coefficient and $\hat{y}$ a predicted value, while β and y denote the true, generally unobserved, parameter and outcome. Vectors and matrices are written in bold only where the distinction from scalars is not otherwise clear from context, and $\|\cdot\|$ denotes the Euclidean norm. Time subscripts are written as X_t, with $t-1$ denoting the immediately preceding period, and

$$\Delta X_t = X_t - X_{t-1}$$

denoting a first difference.

Monetary amounts are given in US dollars unless stated otherwise, and percentages are used interchangeably with decimals (5% and 0.05 mean the same thing) depending on which is clearer in context. Interest rates, returns, and growth rates are annualized unless a different frequency (monthly, daily) is explicitly stated. Every worked numerical example can be checked against the chapter's companion notebook, as the preface explains.

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Task type	Example financial problem	Covered in
Regression	Forecasting returns, volatility, an insurance claim cost, or a credit spread	Ch. 6, 7, 11
Classification	Predicting default, fraud, an operational event type, a claim's severity tier, or a trading signal's direction	Ch. 6, 9, 10, 11, 15
Time series forecasting	Forecasting interest rates, volatility, or macro indicators	Ch. 7
Distribution and tail estimation	Estimating a loss distribution and its far-tail quantiles: market VaR, operational-loss capital, insurance pricing and solvency	Ch. 8, 10, 11
Optimization	Constructing a portfolio subject to risk and constraint limits	Ch. 12
Clustering / dimensionality reduction	Grouping similar firms or compressing a large factor set into a few drivers	Ch. 6, 7, 12, 17
Sequential decision-making	Executing a trade over time to minimize impact, formalized as a Markov decision process	Ch. 13, 14, 19
Natural language processing	Extracting sentiment or information from text	Ch. 16
Anomaly detection	Flagging unusual transactions, account behavior, operational near-misses, or implausible insurance claims and coding patterns	Ch. 10, 11, 15, 17
Explainability and fairness auditing	Explaining an individual model decision and testing a model for disparate impact	Ch. 18, 21
Multi-agent simulation	Modeling how many simple, interacting participants jointly produce market-wide bubbles, crashes, or contagion	Ch. 20

Table 1.8: A taxonomy of AI problem types in finance

1.4 Reading Finance Beyond the Numbers

A financial statement is a set of numbers, but it is also a set of claims made by people with incentives. This part introduces the accounting that later chapters model, and the reasons a purely numerical reading of it will mislead.

1.4.1 Financial Statements, Cash Flows, and Valuation

Although much of modern finance is expressed through prices, rates, and stochastic models, the underlying economic reality is simpler: firms create value by generating cash flows over time. A company can be profitable on paper and still face financial distress if it does not generate enough cash to meet obligations. Financial analysis therefore begins with a careful reading of the balance sheet, the income statement, and the cash flow statement.

The balance sheet summarizes what a firm owns and what it owes at a point in time. Assets include cash, inventories, property, and intellectual property. Liabilities include debt, payables, and other obligations. Equity represents the residual claim of owners after liabilities are accounted for. The income statement shows revenues, expenses, and profit over a period of time. The cash flow statement, by contrast, explains how cash moved through the business and is often more informative than accounting earnings when one is trying to understand solvency and future financing needs.

The distinction matters because accounting profits and economic value differ. A firm may report large earnings while consuming cash through inventory accumulation, capital expenditure, or working capital needs. Investors and lenders ask not only whether the firm appears profitable, but whether it can sustain operations

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and meet future obligations. A predictive model may forecast earnings, but an analyst must also ask whether those earnings are backed by cash generation and durable competitive advantages.

Valuation rests on this logic. The value of an asset is the present value of the expected future cash flows that it will generate, discounted at an appropriate rate. In symbols, a simple present value relation is

$$V = \sum_{t=1}^{T} \frac{CF_t}{(1+r)^t}, \tag{1.1}$$

where CF_t denotes the cash flow in period t and r is the discount rate. This expression is one of the most important in finance because it unifies bond pricing, stock valuation, project appraisal, and much of corporate analysis. The challenge is that future cash flows are uncertain and the discount rate must reflect both time and risk, which is where statistics and econometrics become essential.

Valuation is rarely mechanical. A company with strong historical performance may still lose value if its market position erodes or interest rates rise sharply. A machine learning model that forecasts revenues or defaults accurately should still have its output checked against the firm's accounting structure, debt obligations, industry, and management incentives. The best systems combine statistical learning with financial reasoning and careful validation.

1.4.2 Why Finance Is Not Just a Collection of Numbers

A common misconception is that finance is the manipulation of numerical data. Finance is about incentives, contracts, and the allocation of resources under uncertainty, and the numbers in financial statements or market data are outcomes of institutional processes: decisions about borrowing, investing, pricing, regulation, and risk management. A quantitative analyst should therefore not be satisfied with a high-performing prediction model without a clear economic explanation for why it works. A pattern in stock returns may reflect risk exposure, information diffusion, transaction costs, or temporary market inefficiency. A default model may be capturing leverage, deteriorating cash flow, or sector-level stress. The signal is meaningful only when interpreted through finance.

This is also why financial modeling must be careful about causality. Correlation is not enough. A variable may be associated with future returns or defaults for reasons that do not persist once conditions change, so a sensible practitioner tests assumptions, monitors model stability, and judges whether a discovered pattern is economically plausible.

1.4.3 A Practical Perspective on the Role of AI in Finance

The purpose of AI in finance is to augment financial expertise, not to replace it. AI methods can process large and complex data, detect nonlinear patterns, and automate tasks too time-consuming for humans. Their usefulness depends on the problem formulation: a poorly defined objective yields a model that looks sophisticated but makes poor decisions. A successful practitioner translates a financial question into a statistical problem, and then the model's output back into a financial decision.

Consider the example of credit scoring. A bank may want to estimate the probability that a borrower will default over the next year. A machine learning model can use many variables, including repayment history, income, occupation, and account activity. Yet the model is most useful when it is tied to domain knowledge about loan structure, collateral, legal obligations, and the macroeconomic environment. Otherwise, it may learn patterns that reflect temporary conditions rather than durable credit risk.

This view also has organizational consequences that a purely technical treatment overlooks. Deploying an AI system in a financial institution typically requires sign-off from risk management, compliance, and sometimes a model validation team whose job is to challenge the model before it is used with real capital or customers, a process called model risk management. A model that cannot be explained to these stakeholders is often not deployable at all, whatever its statistical performance, which is part of why Chapter 18 treats explainability as a first-class technical requirement. This institutional layer can feel slow to a reader from machine learning, but a financial model's errors are rarely private. A mispriced loan, a bad trade, or a missed fraud case can affect customers, counterparties, and, in aggregate, the stability of the financial system, which is why financial AI is regulated more heavily than AI that recommends movies.

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1.4.4 Finance as a System of Incentives and Information

Section 1.4.2 made this point for a single model: a statistical pattern is meaningful only once it has an economic explanation. This section takes the idea up to the level of a whole market, where the incentives shaping how data are generated deserve naming directly. A manager may be rewarded for short-term earnings growth even when that growth is not sustainable. A lender may extend credit to a borrower who appears safe in normal conditions but becomes fragile when conditions deteriorate. A regulator may face pressure to support market confidence even when this increases moral hazard. Finance is intertwined with human behavior and institutional design.

Financial data are therefore records of strategic interaction, not passive observations. Prices aggregate the small pieces of information dispersed across many participants, which makes price data valuable, but they also encode noise, strategic behavior, and market frictions. The analyst must separate signal from noise while remembering that the data are generated by participants with incentives.

Two well-documented episodes show how badly things can go wrong when a model ignores the incentives shaping its data. Before the 2008 financial crisis, statistical models of mortgage default risk were built on data from a period of rising home prices and relatively loose underwriting. They implicitly assumed that the incentives generating that data would persist, in particular that mortgage originators bore meaningful consequences for making bad loans. The rise of mortgage securitization changed those incentives: many originators sold loans onward quickly enough to bear little of the eventual default risk, so underwriting quality deteriorated in ways the historical models never observed. The models were not statistically wrong in a narrow sense; they were estimated correctly on the data available. They failed because the economic process generating the data had shifted underneath them, the non-stationarity concern developed further in Chapter 9.

A second, smaller-scale illustration is Goodhart's Law, often summarized as "when a measure becomes a target, it ceases to be a good measure." A credit score starts as a statistical summary that correlates with creditworthiness. Once borrowers and lenders know that a numerical threshold controls loan approval or pricing, some participants will alter their behavior to cross it, whether or not their creditworthiness has changed. This is the reflexivity theme introduced in Chapter 6: unlike a model of a physical system, a financial model's targets can adapt to the model itself, one more reason financial AI requires ongoing monitoring rather than one-time validation.

1.5 Reasoning Under Uncertainty

Every financial decision is a bet made with incomplete information, which makes probability the native language of the subject. This part develops how to size such a bet, what new information is worth, and why a model's output is an argument rather than an answer.

1.5.1 Uncertainty, Probability, and Decision-Making

Finance is fundamentally about decisions under uncertainty. A lender does not know with certainty whether a borrower will repay. An investor does not know whether an asset price will rise or fall. A firm does not know whether a new project will be profitable. The question is not whether uncertainty can be eliminated, because it usually cannot, but how decision-makers should act when outcomes are uncertain and the costs of being wrong are not symmetric.

Finance runs into two different kinds of not-knowing. The economist Frank Knight drew the distinction in 1921. *Risk* describes a situation where the possible outcomes are known and their probabilities can be estimated, whether from a well-understood physical process (a fair coin) or from a long, stable history of similar events. *Uncertainty* (sometimes called Knightian uncertainty) describes a situation where even the set of possible outcomes, let alone their probabilities, cannot be fully specified in advance. A casino game is risk in this sense; a novel financial crisis, a first-of-its-kind fraud scheme, or a new technology's effect on an entire industry is uncertainty. Nearly every probabilistic model in this book (the Value at Risk models of Chapter 8, the credit-scoring models of Chapter 9, the volatility forecasts of Chapter 7) implicitly treats its subject as risk, assuming that a historical sample is enough to pin down the relevant probabilities. Before

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trusting a model's probability estimate, ask whether the situation is closer to a repeatable game with a stable, estimable distribution, or to a novel one where the model's confidence may be misplaced.

This is where probability and expected value become central. If an investor faces two possible outcomes, a gain and a loss, the expected value of the gamble may be positive even if the investor would prefer not to take the risk. This motivates the idea of risk aversion. A risk-averse decision-maker usually requires a larger expected payoff to accept a gamble with greater uncertainty, a logic that underlies portfolio choice, corporate financing, and insurance.

A simple expression for expected value is

$$\mathbb{E}[X] = \sum_i p_i x_i, \tag{1.2}$$

where p_i is the probability of state i and x_i is the associated payoff. In finance these probabilities are rarely known precisely and can change, so financial models must be updated as new information arrives.

A small numerical example makes the idea of risk aversion concrete. Suppose an investor is offered a gamble with a 50% chance of gaining \$20,000 and a 50% chance of losing \$10,000. The expected value of the gamble is

$$\mathbb{E}[X] = 0.5(20{,}000) + 0.5(-10{,}000) = \$5{,}000,$$

which is positive, so a risk-neutral decision-maker, one who cares only about expected value, should accept it. Yet many investors would decline, because the possibility of losing \$10,000 matters more to them than the expected value suggests. Economists capture this with a concave utility function $u(\cdot)$: the investor evaluates the gamble by its expected utility, $\mathbb{E}[u(X)]$, rather than its expected value. The certainty equivalent is the guaranteed amount that gives the same expected utility as the gamble. For a risk-averse investor it is always less than the \$5,000 expected value, and the gap, sometimes called the risk premium, measures how much the investor would pay to avoid uncertainty.

Making this concrete requires a specific utility function, applied to the investor's total wealth rather than the gamble's payoff alone, since a concave function is meaningful only at actual wealth levels. Suppose the investor's wealth today is $W_0 = \$50{,}000$, and take the square-root utility function $u(W) = \sqrt{W}$, a standard textbook example of a risk-averse (concave) utility function. The gamble leaves the investor with wealth $W_0 + 20{,}000 = \$70{,}000$ or $W_0 - 10{,}000 = \$40{,}000$, so

$$\mathbb{E}[u(W)] = 0.5\sqrt{70{,}000} + 0.5\sqrt{40{,}000} \approx 0.5(264.575) + 0.5(200.000) = 232.288.$$

The certainty equivalent wealth CE solves $u(CE) = \mathbb{E}[u(W)]$, that is, $\sqrt{CE} = 232.288$, giving $CE \approx \$53{,}957.51$, noticeably less than the expected wealth of $0.5(70{,}000) + 0.5(40{,}000) = \$55{,}000$. The risk premium is the difference,

$$\text{Risk premium} = \$55{,}000 - \$53{,}957.51 \approx \$1{,}042.49,$$

which has a direct interpretation: this investor would accept a guaranteed \$53,957.51 in place of the gamble, paying up to \$1,042.49 to avoid its uncertainty even though its expected wealth is higher. The same comparison of an uncertain payoff with its certainty equivalent underlies the pricing of insurance, the equity risk premium, and the risk-return trade-off introduced earlier. It reappears in Chapter 12's mean-variance framework as the reason a risk-averse portfolio manager accepts a lower expected return in exchange for lower volatility.

1.5.2 The Kelly Criterion: How Much to Bet, Not Just Whether

A positive expected value answers whether a favorable gamble is worth taking at all. A repeatable opportunity, the same favorable bet placed over and over, raises a sharper question: what *fraction* of current wealth should be wagered each time? Betting too little leaves edge on the table; betting too much, as this section shows, can make an investor with a real, positive edge poorer with certainty over time.

The question is nontrivial, rather than "bet as much as a positive edge allows," because wealth wagered repeatedly compounds rather than accumulates. A single bet's expected value is an arithmetic average across outcomes, the right tool for deciding whether to take that bet once. Wealth carried across many repetitions is a product of per-round multipliers, $(1 + f)$ after a win and $(1 - f)$ after a loss, and a quantity built from

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repeated multiplication is governed by its *geometric* mean. A bet size can raise the arithmetic average of outcomes while driving the geometric average, and with it long-run wealth, toward zero; this asymmetry, not risk aversion in the subjective sense discussed above, is what makes betting too large a fraction ruinous.

The Kelly criterion answers this for a repeatable bet with probability p of winning (paying b dollars per dollar wagered) and probability $q = 1 - p$ of losing the full amount wagered: the fraction of wealth that maximizes the long-run *geometric* growth rate of wealth is

$$f^* = p - \frac{q}{b}. \tag{1.3}$$

For an even-money bet ($b = 1$) with a 60% win probability, $f^* = 0.6 - 0.4 = 0.2$: wager 20% of current wealth on each repetition, no more and no less. The formula is quoted here rather than derived. The book's companion technical note on the Kelly criterion works the argument through in full: why f^* is a strict global maximum rather than merely a stationary point, why maximizing growth is equivalent to maximizing expected logarithmic wealth, and why the optimal fraction can rationally exceed one, recommending leverage, once a loss costs less than the entire stake.

Why "no more" matters as much as "no less" is the surprising part. The long-run growth rate of wealth from repeatedly wagering a fraction f is $g(f) = p\ln(1+f) + q\ln(1-f)$; evaluating it at three candidate fractions for this same 60%-favorable bet,

$$g(0.2) \approx 0.0201, \qquad g(0.5) \approx -0.0340, \qquad g(1.0) = -\infty,$$

shows that wagering half of wealth on every repetition, although each bet still has positive expected value, produces a *negative* long-run growth rate. An investor following this rule is mathematically certain to see wealth decay toward zero over enough repetitions, not merely higher volatility around a positive trend.

Wagering everything each time ($f = 1$) is worse still, guaranteed eventual ruin, since a single loss wipes out the entire stake. A decision-maker who evaluates only a single bet's expected value, without asking how large a fraction of wealth to commit when it repeats, can be led to a sizing decision that is ruinous. This position-sizing discipline resurfaces in Chapter 12's discussion of leverage and Chapter 13's execution-sizing decisions.

1.5.3 The Value of Information: What Is a Perfect Signal Worth?

A closely related question asks not how much to bet, but how much a decision-maker should be willing to pay for information that resolves uncertainty *before* a decision is made. Suppose a firm is deciding whether to fund a $2 million project, an idealized version of the investment decisions that Chapter 5's discounted-cash-flow and real-options material treats in detail. The project succeeds with probability 0.4, returning a net gain of $5 million, or fails with probability 0.6, losing the full $2 million invested. Without any additional information, the expected value of proceeding is

$$\mathbb{E}[\text{Proceed}] = 0.4(5) + 0.6(-2) = \$0.8 \text{ million},$$

positive, and better than the $0 from not proceeding at all, so a rational, risk-neutral firm proceeds, accepting the 60% chance of a $2 million loss along with the 40% chance of a $5 million gain.

Now suppose the firm could instead consult a perfectly accurate forecaster, at some price, who reveals with certainty which state (success or failure) will occur before the firm must commit. Knowing this, the firm would proceed only in the success state and decline in the failure state, earning

$$\mathbb{E}[\text{Proceed} \mid \text{perfect information}] = 0.4(5) + 0.6(0) = \$2.0 \text{ million},$$

since the $2 million loss is avoided entirely in the 60% of cases where failure would otherwise have occurred. The expected value of perfect information (EVPI) is the difference between these two figures,

$$\text{EVPI} = 2.0 - 0.8 = \$1.2 \text{ million},$$

the maximum a rational firm should pay for a perfectly accurate signal before deciding, since paying more would make consulting the forecaster worse than proceeding blind. No real forecast, credit model, or market signal is perfectly accurate, so EVPI is an upper bound. It answers "how much could better information possibly be worth here," a question worth asking before paying for additional data, a more sophisticated model, or a costly due-diligence process.

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1.5.4 Bayesian Updating: Learning from New Information

A related idea, which underlies how many machine learning systems in this book are best interpreted, is that beliefs about an uncertain quantity should be updated, not replaced, as new evidence arrives. Bayes' theorem formalizes this:

$$\Pr(A \mid B) = \frac{\Pr(B \mid A)\ \Pr(A)}{\Pr(B)}, \tag{1.4}$$

where $\Pr(A)$ is the prior probability of some hypothesis A before seeing evidence B, $\Pr(B \mid A)$ is the likelihood of observing that evidence if A is true, and $\Pr(A \mid B)$ is the posterior probability of A after accounting for the evidence.

An example from fraud detection (Chapter 15) shows why this matters. Suppose 1% of transactions at a bank are fraudulent, so $\Pr(\text{Fraud}) = 0.01$. A fraud detection model correctly flags 95% of actual fraud, $\Pr(\text{Flag} \mid \text{Fraud}) = 0.95$, but also flags 3% of legitimate transactions, $\Pr(\text{Flag} \mid \text{Not Fraud}) = 0.03$. Given that a transaction has been flagged, what is the probability it is actually fraudulent? Applying Bayes' theorem,

$$\begin{aligned}
\Pr(\text{Flag}) &= \Pr(\text{Flag} \mid \text{Fraud})\Pr(\text{Fraud}) + \Pr(\text{Flag} \mid \text{Not Fraud})\Pr(\text{Not Fraud}) \\
&= 0.95(0.01) + 0.03(0.99) \\
&= 0.0392, \\
\Pr(\text{Fraud} \mid \text{Flag}) &= \frac{0.95(0.01)}{0.0392} \approx 0.242.
\end{aligned}$$

Despite a model that sounds highly accurate (95% detection, only 3% false alarms), a flagged transaction is fraudulent only about 24% of the time. Fraud is so rare that even a small false-positive rate produces many more false alarms than true detections in absolute terms. The same base-rate intuition drives the discussion of imbalanced classification in Chapter 6, and the example shows quantitatively why accuracy alone is a poor way to evaluate a model applied to a rare event.

Bayesian updating also describes how a well-run financial model should be maintained: rather than fixing a model's parameters forever, or discarding the model the first time it errs, each new batch of data revises a belief incrementally, and today's posterior becomes tomorrow's prior. This is the spirit behind the ongoing model monitoring of Chapter 18 and the state-space and Kalman-filtering methods of Chapter 7, which are Bayesian updating carried out recursively over time.

The "posterior becomes the next prior" idea is easiest to see by continuing the fraud example. Suppose the flagged transaction is now compared against the customer's typical spending location, and it occurred somewhere the customer has never transacted before, a pattern found in 60% of fraud cases but only 10% of legitimate ones: $\Pr(\text{Unusual Location} \mid \text{Fraud}) = 0.6$ and $\Pr(\text{Unusual Location} \mid \text{Not Fraud}) = 0.1$. Rather than starting over, a Bayesian analysis treats the first posterior, $\Pr(\text{Fraud} \mid \text{Flag}) \approx 0.242$, as the prior for this second update:

$$\begin{aligned}
\Pr(\text{Location}) &= 0.6(0.242) + 0.1(1 - 0.242) = 0.221, \\
\Pr(\text{Fraud} \mid \text{Flag}, \text{Location}) &= \frac{0.6(0.242)}{0.221} \approx 0.657.
\end{aligned}$$

Two independent pieces of weak-to-moderate evidence, an automated flag and an unusual location, combine to raise the probability of fraud from a 1% base rate to roughly 66%, each update using the same formula with the previous posterior in place of the original prior. Figure 1.3 lays out the chain. A production fraud system (Chapter 15) accumulates evidence from several model outputs in this way before deciding whether a transaction warrants manual review, which is why several weak signals, evaluated one Bayesian update at a time, can be more reliable than any single signal.

The step that makes this chain legitimate is easy to miss. Strictly, the second update needs the likelihood of an unusual location given fraud *and* given that the transaction was already flagged, $\Pr(\text{Location} \mid \text{Fraud}, \text{Flag})$, not $\Pr(\text{Location} \mid \text{Fraud})$. Using the latter assumes that the two signals are *conditionally independent*: once the true state, fraud or not, is known, learning that the flag fired says nothing more about the location. That is rarely true. A flagging model that already leans on location-type features will tend to flag legitimate transactions made while the customer is traveling, so among flagged legitimate transactions an unusual location is more common than the 10% seen among legitimate transactions overall. If it is 30% instead, with

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the fraud likelihood unchanged at 60%, the second update gives $0.6(0.242)/[0.6(0.242) + 0.3(0.758)] \approx 0.390$, about 39% rather than 66%; and if the location signal were entirely redundant given the flag, the posterior would stay at 24%. Treating correlated signals as independent counts the same evidence twice and makes the system overconfident, which is the assumption, and the failure mode, of the Naive Bayes classifier in Chapter 6.

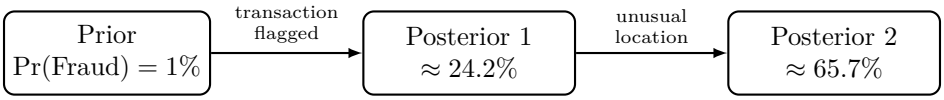


Figure 1.3: The fraud example's two-step Bayesian update chain

1.5.5 Why Models Must Be Interpreted, Not Worshipped

A common temptation in finance is to trust a model because it is sophisticated or produces attractive historical results. A model can be mathematically elegant and still economically misleading: a credit model may appear accurate in a stable period and become unstable during a recession, and a fraud model may detect obvious anomalies but miss subtle forms of misconduct that change over time. The evaluation step in Figure 1.1's pipeline is not a formality; it is where an overconfident model gets caught, or does not. Model evaluation in finance is therefore an economic and operational exercise as well as a statistical one. It asks not only whether the model fits the data but whether its assumptions make sense, its outputs are usable, and the implementation is robust.

The 1998 collapse of Long-Term Capital Management (LTCM) is a case study in this failure mode, examined further in Chapter 3. The hedge fund's partners included two Nobel laureates in economics, and its models were, by the standards of the time, extremely sophisticated. They correctly estimated how fixed-income spreads behaved under historically normal conditions, but had no mechanism for anticipating a regime in which previously uncorrelated markets, including Russian sovereign debt, moved together in a global flight to quality. At that same moment the fund's highly leveraged positions made it a forced seller into a market with too few buyers. No model in later chapters, however sophisticated, is exempt from this risk: a model is only as good as the range of conditions it was built and tested to anticipate. That is why the stress-testing and scenario-analysis techniques of Chapter 8 are a check on quantitative models generally, not a technique reserved for portfolio risk.

1.6 Limits, Challenges, and Practice

Nothing developed so far works as cleanly in production as it does on a page. This part collects the obstacles that recur throughout the book, and closes with a case in which classical finance and machine learning were combined successfully.

1.6.1 Challenges of Applying AI in Finance

Several challenges recur whenever AI methods are applied to financial problems; later chapters revisit each in depth.

Data challenges. Financial data are noisy, unevenly sampled, subject to reporting delays, and sometimes revised after the fact. Historical datasets can also suffer from survivorship bias, in which failed firms or discontinued strategies are quietly excluded, making the past look more favorable than it was.

Non-stationarity. Unlike many physical systems, financial markets change their own statistical behavior over time, in response to new regulation, new technology, changing investor composition, and the actions of other participants, including other AI systems. A model calibrated on one period can fail in the next without any error in how it was built.

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Overfitting and backtest illusions. Because financial data are limited relative to the number of strategies and features that can be tried, it is easy to find a pattern that worked in history by chance. A strategy that looks excellent in a backtest can perform poorly, or lose money, once deployed with real capital.

Interpretability and accountability. Many financial decisions, particularly credit and insurance decisions, are subject to legal requirements that they be explainable and non-discriminatory. This constrains how opaque a model can be, whatever its statistical performance (Chapter 18).

Operational and execution frictions. A model's statistical output is not a financial outcome. Transaction costs, market impact, latency, and operational failures can erode or eliminate a theoretical edge, which is why this book emphasizes economic as well as statistical evaluation.

Ethical and systemic considerations. Widely adopted AI models can create correlated behavior across many market participants, potentially amplifying booms and busts. Models used in lending, insurance, or hiring can also encode or amplify existing biases in historical data if they are not carefully audited.

These are not reasons to avoid AI in finance but to apply it carefully, with an understanding of financial institutions, incentives, and market structure alongside statistical skill, the combination this book aims to build.

1.6.2 Case Vignette: How Classical Finance and AI Combine in Practice

To make the chapter's themes concrete, consider a stylized (fictional) example that previews how the pieces of this book fit together. A mid-sized asset manager wants to build a systematic equity strategy that selects stocks likely to outperform over the coming quarter.

The manager does not start with machine learning. The process begins with the financial statement analysis of Chapter 4: screening a universe of firms on profitability, leverage, and cash generation, and discarding firms with clear signs of distress before any prediction model is involved. A purely data-driven model applied to raw price history would not make this economic judgment on its own.

Next, the manager builds a predictive model with the regression and classification tools of Chapter 6 to rank the remaining firms by expected relative performance. Its features draw on several data types from Table 1.6: fundamental ratios from the screening step, momentum and volatility measures from the time series methods of Chapter 7, and, increasingly, sentiment scores extracted from earnings-call transcripts with the natural language processing methods of Chapter 16. The model is evaluated with walk-forward validation, not a random train-test split, since a random split would let the model see information from the future relative to some of its training data.

The ranked predictions are an input to portfolio construction, not the final answer. Chapter 12's methods convert a ranking into position sizes that respect diversification, risk limits, and transaction costs, since the highest-ranked stock might be too risky, too illiquid, or too correlated with the rest to hold at a large weight. Finally, the execution methods of Chapter 13 determine how to buy and sell the positions without moving prices adversely, and the risk models of Chapter 8 monitor whether the realized portfolio behaves as expected.

At every step, a classical financial concept (distress screening, diversification, transaction costs) sits alongside a modern statistical or machine learning method, and neither is sufficient on its own. This interleaving, not the replacement of one by the other, is the organizing principle of every applied chapter that follows.

1.7 Key Takeaways

Finance is a domain in which data, uncertainty, and decision-making interact closely. Quantitative readers can progress quickly by connecting mathematical tools with financial intuition, but they must learn to interpret models in light of market structure, incentives, and institutional constraints. The most effective applications of AI in finance combine statistical skill, economic understanding, and practical judgment.

The history in Section 1.1.2 showed that today's machine learning methods extend, rather than replace, decades of classical quantitative finance, and the data ecosystem in Section 1.2.2 previewed the data types later chapters draw on, each with its own quality issues. The numerical examples (compounding in Table 1.7, the gamble and its certainty equivalent, the fraud-detection application of Bayes' theorem) each illustrate a principle that reappears in more technical form later: nonlinear compounding, risk aversion, and the danger

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of ignoring base rates and of double-counting correlated evidence. Finally, the case vignette in Section 1.6.2 is worth revisiting after finishing the book. Once Chapters 4, 6, 7, 8, 12, 13, and 16 have been covered, a reader should see how each of its steps would be implemented in practice.

1.8 Suggested Exercises

1.1 Define the difference between risk and uncertainty in the context of financial decision-making.

1.2 Explain why the time value of money matters for investment decisions.

1.3 List three types of financial markets and describe the assets traded in each.

1.4 Give one example of a financial problem where AI could be useful and one where classical statistical methods might be sufficient.

1.5 Write a short paragraph explaining why financial models should still be considered when using machine learning in finance.

1.6 Describe one financial institution and explain how it connects savers, borrowers, or investors.

1.7 Discuss why a model that performs well in a generic prediction task may still fail in a financial setting.

1.8 A gamble offers a 60% chance of gaining \$8,000 and a 40% chance of losing \$5,000. Compute the expected value, and explain, using the concept of a certainty equivalent, why a risk-averse investor might still reject it even though the expected value is positive.

1.9 Pick two of the challenges listed in Section 1.6.1 and describe, for each, a concrete financial scenario not used in the chapter in which it could cause an AI-based system to fail.

1.10 Using Table 1.7, compute the future value of \$5,000 invested for 20 years at 6%, and verify your answer is consistent with scaling the \$1,000 figure in the table.

1.11 A disease-screening test correctly identifies 90% of people who have a rare disease affecting 0.5% of the population, and has a 5% false-positive rate. Using Bayes' theorem as in Section 1.5.4, compute the probability that a person who tests positive actually has the disease, and explain why the result is counterintuitive.

1.12 Choose one entry from Table 1.8 and explain, in a short paragraph, why its financial problem could plausibly be framed as more than one of the table's task types (regression, classification, optimization, and so on).

1.13 Pick one milestone from Table 1.1 and explain, using material from later in this chapter, why the newer methods that followed it did not make it obsolete.

1.14 Continuing the sequential Bayesian updating example in Section 1.5.4, suppose a third, independent signal arrives: the transaction amount is a round number (e.g., exactly \$500), a pattern present in 40% of fraud cases but only 15% of legitimate ones. Using $\Pr(\text{Fraud} \mid \text{Flag}, \text{Location}) \approx 0.657$ as the new prior, compute the posterior probability of fraud after all three signals, and comment on how quickly it approaches certainty as independent evidence accumulates.

1.15 Using the Kelly criterion, compute the optimal betting fraction f^* for an even-money bet with a 70% win probability, and compute the long-run growth rate $g(f)$ at f^* and at $f = 0.9$, commenting on whether betting nearly everything is ever justified even when the win probability is this favorable.

1.16 A firm is deciding whether to enter a new market at a cost of \$3 million. Entry succeeds with probability 0.3, returning a net gain of \$12 million, or fails with probability 0.7, losing the full \$3 million. Compute the expected value of entering, the expected value under perfect information, and the resulting EVPI, and explain what the firm should conclude if a market-research report claiming near-perfect accuracy costs \$4 million.

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1.9 Further Reading

The following texts are useful starting points for the financial ideas introduced in this chapter:

- Brealey et al. (2025), *Principles of Corporate Finance*. A comprehensive introduction to corporate financial decision-making, including the time-value-of-money and valuation ideas this chapter introduces at a conceptual level and Chapter 5 develops formally.
- Hull (2022), *Options, Futures, and Other Derivatives*. The standard reference for derivatives pricing; useful background for the brief history of Black-Scholes-Merton option pricing sketched in this chapter's timeline and covered in full in Chapter 5.
- Bodie et al. (2024), *Investments*. Covers portfolio theory, market efficiency, and the risk-return trade-off introduced qualitatively in this chapter, previewing the quantitative treatment in Chapter 12.
- Campbell et al. (1997), *The Econometrics of Financial Markets*. A rigorous treatment of the statistical methods used to test market efficiency and analyze financial time series, going well beyond this chapter's introductory discussion.
- Goodfellow et al. (2016), *Deep Learning*. The standard reference for the deep learning methods behind the large language model and unstructured-data applications previewed in this chapter's history section and covered in Chapters 16 and 17.
- Chan (2017), *Machine Trading*. A practitioner-oriented introduction to building and testing systematic trading strategies, extending the case vignette in this chapter into the algorithmic trading material of Chapters 13 and 14.

These references cover foundational finance, asset pricing, and the use of quantitative methods in financial applications.

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Chapter 2

Python for Financial Analysis

2.1 Introduction

Python has become one of the most important tools in modern finance. It is flexible, readable, and supported by a rich ecosystem of libraries for data analysis, visualization, statistics, and machine learning. For quantitative researchers, analysts, and practitioners, Python offers a practical way to turn financial data into insight. It is also a gateway into data science more broadly, because many techniques used in finance are the same ones used in business analytics, scientific computing, and artificial intelligence.

This chapter introduces Python from a financial perspective, through common tasks: reading market data, cleaning historical prices, calculating returns, visualizing trends, and preparing data for modeling. Every concept is tied to one running numerical example, a two-asset portfolio over five trading days, so that the reader can check each computation by hand and reproduce it in code. The emphasis is on intuition and practical fluency rather than computer science theory.

The chapter builds up in layers, all on the same AssetA/AssetB price history. It starts with a one-page refresher of the language (functions, conditionals, loops, and the built-in containers) for readers who already program, then introduces the `DataFrame` and NumPy arrays that everything downstream depends on.

It then turns to pandas, the library that does most of the practical work in this book: boolean indexing, merging and grouping, resampling to a coarser frequency, and cleaning a simulated missing observation. A worked example builds and evaluates the two-asset portfolio in full, computing its mean, variance, and covariance by hand and in code, then adds a third asset to show that the logic does not change as a portfolio grows. A further section turns those daily statistics into the numbers a performance report quotes: log returns, annualized volatility, the Sharpe ratio, maximum drawdown, and returns computed from prices adjusted for splits and dividends.

The chapter closes with two practical concerns: why vectorized NumPy code can be tens of times faster than an equivalent Python loop, and how to keep a numerical workflow reproducible so that someone else, or a future self, can rerun it and get the same answer.

Sections 2.2.2 through 2.3.1 establish the vocabulary (loops, arrays, vectorized operations) that later chapters assume. Section 2.5.1's two-asset portfolio recurs by name in Chapters 8 and 12, so its covariance matrix and correlation are worth internalizing. Section 2.5.4 defines the return and performance measures later chapters quote, and Sections 2.4.4 and 2.6.1 anticipate two problems, messy data and slow unvectorized code, that resurface in nearly every applied chapter.

2.2 The Python Language

Everything later in this book is expressed in code, so the language itself has to come first. This part moves from why Python won the quantitative-finance argument to a brief refresher of its syntax and to the `DataFrame` that holds the chapter's running example.

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2.2.1 Why Python Matters in Finance

A financial analyst often works with large, messy, and changing datasets. Prices arrive continuously, statements are published irregularly, and news can affect markets within seconds. This environment needs tools that are fast, reproducible, and easy to adapt, which is where Python excels: scripts can be rerun, workflows automated, and models validated systematically.

Python also has a large scientific community. Libraries such as NumPy, pandas, Matplotlib, SciPy, and scikit-learn make numerical work efficient and lower the barrier for researchers moving from spreadsheets to a more scalable and transparent pipeline. Table 2.1 summarizes the libraries used throughout this chapter and the role each one plays.

Library	Role in financial analysis
NumPy	Fast array operations and vectorized numerical computation
pandas	Tabular and time-indexed data handling, cleaning, and aggregation
Matplotlib	Static charting of prices, returns, and diagnostics
SciPy	Statistical distributions, optimization, and hypothesis testing
statsmodels	Regression, time series models (e.g., ARIMA), and statistical inference
scikit-learn	Machine learning models, preprocessing, and model evaluation

Table 2.1: Core Python libraries for financial analysis

2.2.2 A One-Page Python Refresher

This book assumes a reader who already programs, so the language needs only a refresher, built around calculations that recur in finance. A function turns a rule into something reusable, here the future value of \$100 earning 8% a year; an `if/elif/else` chain classifies an outcome; a `for` loop repeats a step. Python's built-in containers (dictionaries for lookups, immutable tuples for fixed pairs, list comprehensions for compact filtering) hold the data before it is tidy enough for a `DataFrame`:

```
def future_value(price, rate, periods=1):
    return price * (1 + rate) ** periods

print(future_value(100.0, 0.08))             # 108.0
print(future_value(100.0, 0.08, periods=3))  # 125.97120000000001

def classify_return(r):
    if r > 0.01:
        return "strong gain"
    elif r > 0:
        return "small gain"
    elif r == 0:
        return "flat"
    else:
        return "loss"

print([classify_return(r) for r in [0.02, 0.004, 0.0, -0.015]])
# ['strong gain', 'small gain', 'flat', 'loss']

balance = 0.0                   # $200 deposited monthly at 0.4% a month
for month in range(12):
    balance = balance * (1 + 0.004) + 200.0
```

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```
print(round(balance, 2))       # 2453.51

security_info = {"ticker": "AAPL", "sector": "Technology", "price": 190.50}
quote = (100.20, 100.35)       # (bid, ask): a tuple cannot be modified
prices_list = [190.50, 145.20, 102.75]
expensive = [p for p in prices_list if p > 150]
print(security_info["sector"], round(quote[1] - quote[0], 2), expensive)
# Technology 0.15 [190.5]
```

One distinction in this block matters for the rest of the chapter. The deposit loop is sequential, since next month's balance depends on this month's, so a loop (or an equivalent closed-form formula) is the natural way to express it. Most return calculations are the opposite: the same independent operation applied to every observation. For those, the vectorized array operations of Section 2.3.1 are far faster than a loop, as Section 2.6.1 measures. Recognizing which kind of calculation is at hand is one of the most useful judgment calls a financial programmer makes.

2.2.3 Core Data Structures

An analyst tracking even a handful of securities quickly needs more than lists and dictionaries: an ordered run of daily closing prices for several assets, aligned on the same dates. For most financial analysis the key structure is therefore the pandas `DataFrame`, which organizes tabular data in rows and columns.

A `DataFrame` suits financial data because it resembles a spreadsheet: one column may contain dates, another prices, another trading volume. The chapter's running example is a small two-asset price history, built from Python lists so that it is self-contained and reproducible without any external file or internet connection:

```
import numpy as np
import pandas as pd

dates = pd.date_range("2024-01-02", periods=5, freq="B")
prices = pd.DataFrame(
    {
        "AssetA": [100.00, 102.00, 101.00, 104.00, 103.00],
        "AssetB": [50.00, 50.50, 51.26, 51.00, 52.02],
    },
    index=dates,
)
print(prices)
#             AssetA  AssetB
# 2024-01-02  100.00   50.00
# 2024-01-03  102.00   50.50
# 2024-01-04  101.00   51.26
# 2024-01-05  104.00   51.00
# 2024-01-08  103.00   52.02
```

This makes it easy to filter observations, compute statistics, and merge datasets. Here the `DataFrame` holds daily closing prices for two assets, so their performance can be compared and, later in the chapter, combined into a portfolio.

2.3 Numerical Computing with NumPy

Python's built-in containers are general-purpose and, for numerical work, slow. This part introduces the array, the data structure the rest of the book computes with, and the vectorized style of expressing conditions over it rather than looping.

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2.3.1 Working with Numerical Arrays

Computing a return one day at a time in a Python loop works for the four prices used so far, but a real price history can span thousands of trading days across hundreds of securities, and a Python-level loop over all of them is far slower than it needs to be. NumPy, the foundational package for numerical computation in Python, provides efficient arrays that support vectorized operations. Vectorization suits finance because many tasks apply the same operation to large collections of prices or returns, with no explicit loop.

Given a series of daily prices $P_0, P_1, \ldots, P_T$, the simple (arithmetic) return in period t is

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}}. \tag{2.1}$$

Applied to AssetA's five prices, this produces four daily returns:

$$R_1 = \frac{102.00 - 100.00}{100.00} = 2.00\%, \quad R_2 = \frac{101.00 - 102.00}{102.00} = -0.98\%,$$

$$R_3 = \frac{104.00 - 101.00}{101.00} = 2.97\%, \quad R_4 = \frac{103.00 - 104.00}{104.00} = -0.96\%.$$

The mean of these four returns is 0.76%, and their sample standard deviation (with $n - 1 = 3$ in the denominator) is 2.03%. In NumPy, this is computed directly on the underlying array without a Python-level loop:

```
asset_a = prices["AssetA"].to_numpy()
returns_a = (asset_a[1:] - asset_a[:-1]) / asset_a[:-1]
print(np.round(returns_a, 4))
# [ 0.02   -0.0098  0.0297 -0.0096]
print(round(returns_a.mean(), 4), round(returns_a.std(ddof=1), 4))
# 0.0076 0.0203
```

This is faster and more readable than an explicit loop, and arrays match the mathematical structure of quantitative finance, where calculations operate on vectors of observations rather than single numbers.

2.3.2 Boolean Indexing and Vectorized Conditionals

The classification function in Section 2.2.2 applied an `if/elif` chain one return at a time, the kind of loop that scales poorly as a price history grows. NumPy and pandas offer a vectorized alternative. A boolean mask, an array of `True/False` values produced by comparing an array to a condition, selects or counts observations directly:

```
up_days = returns_a > 0
print(up_days)
# [ True False  True False]
print(f"Number of up days: {up_days.sum()}")
# Number of up days: 2
print(f"Average return on up days: {returns_a[up_days].mean():.4f}")
# Average return on up days: 0.0249
```

Summing a boolean array counts the number of `True` values, since Python treats `True` as 1 and `False` as 0, and using the mask to index the array selects only the matching elements. For a conditional calculation with more than two outcomes, `np.select` generalizes the earlier `if/elif` logic into a single vectorized expression:

```
conditions = [returns_a > 0.01, returns_a > 0, returns_a == 0]
choices = ["strong gain", "small gain", "flat"]
labels = np.select(conditions, choices, default="loss")
print(labels)
# ['strong gain' 'loss' 'strong gain' 'loss']
```

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This produces the same classifications as the loop-based `classify_return` function, applied to all four returns at once. As datasets grow from a handful of observations to millions of rows of tick data, a single vectorized expression instead of a Python-level loop is the difference between a script that runs in milliseconds and one that takes minutes (Section 2.6.1).

2.4 Handling Data with pandas

Financial data arrives as tables that are misaligned, differently sampled, and incomplete. This part covers the library built for that: reshaping and joining tables, plotting what they contain, and cleaning what is wrong with them before any model sees it.

2.4.1 Data Manipulation with pandas

Real market data rarely arrives as clean as the toy price arrays used so far: a downloaded CSV may have gaps for holidays, dates out of order, or days missing for one security but not another, and NumPy knows nothing about dates. pandas is the workhorse library for time series and structured data. It loads data from CSV files, databases, or online sources; indexes by date; sorts observations; handles missing values; and computes rolling statistics, all essential because market data are often irregular and incomplete.

The pandas method `pct_change()` computes the same simple returns shown above, applied automatically to every column:

```
returns = prices.pct_change().dropna()
print(returns.round(4))
#             AssetA  AssetB
# 2024-01-03  0.0200  0.0100
# 2024-01-04 -0.0098  0.0150
# 2024-01-05  0.0297 -0.0051
# 2024-01-08 -0.0096  0.0200
```

Notice that AssetB's returns (0.0100, 0.0150, −0.0051, 0.0200) move in the opposite direction from AssetA's returns (0.0200, −0.0098, 0.0297, −0.0096) on three of the four days. This pattern reappears as a strong negative correlation later in the chapter, and it is why diversification helps so much in this example, although, as Section 2.5.1 shows, four observations pin that correlation down only loosely.

This transformation from price levels to returns underlies much of empirical finance. Returns suit many statistical models better than levels because they are approximately stationary and easier to compare across assets with very different price scales.

2.4.2 Combining and Reshaping Data: merge, groupby, and resample

Real analyses rarely involve a single tidy table; data about the same securities usually arrives from several sources and must be combined. Suppose one table holds each security's sector and price, and a second holds the number of shares held in a portfolio:

```
securities_df = pd.DataFrame([
    {"ticker": "AAPL", "sector": "Technology", "price": 190.50},
    {"ticker": "JPM", "sector": "Financials", "price": 145.20},
    {"ticker": "XOM", "sector": "Energy", "price": 102.75},
    {"ticker": "MSFT", "sector": "Technology", "price": 375.10},
])
shares = pd.DataFrame({
    "ticker": ["AAPL", "JPM", "XOM", "MSFT"],
    "shares_held": [100, 50, 200, 30],
})
```

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```
merged = securities_df.merge(shares, on="ticker")
merged["position_value"] = merged["price"] * merged["shares_held"]
print(merged)
#   ticker      sector   price  shares_held  position_value
# 0   AAPL  Technology  190.50          100         19050.0
# 1    JPM  Financials  145.20           50          7260.0
# 2    XOM      Energy  102.75          200         20550.0
# 3   MSFT  Technology  375.10           30         11253.0
```

The `merge` method aligns the two tables on the shared `ticker` column, the same logic as a SQL join, and is the standard way to combine data that arrives from different sources or systems. Once combined, `groupby` aggregates the result by a categorical column, here computing total position value by sector:

```
sector_summary = merged.groupby("sector")["position_value"].sum()
print(sector_summary)
# sector
# Energy        20550.0
# Financials     7260.0
# Technology    30303.0
# Name: position_value, dtype: float64
```

This shows at once that the portfolio's Technology exposure ($30,303, combining AAPL and MSFT) is its largest sector concentration, which would take considerably more manual bookkeeping without `groupby`.

A related operation, `resample`, aggregates a time-indexed series to a coarser frequency, for example converting daily prices to weekly prices by taking the last observation in each week:

```
extended_dates = pd.date_range("2024-01-02", periods=10, freq="B")
extended_prices = pd.Series(
    [100.00, 102.00, 101.00, 104.00, 103.00, 105.00, 107.00, 106.00, 108.00, 110.00],
    index=extended_dates, name="AssetA",
)
weekly_last = extended_prices.resample("W-FRI").last()
print(weekly_last)
# 2024-01-05    104.0
# 2024-01-12    108.0
# 2024-01-19    110.0
# Freq: W-FRI, Name: AssetA, dtype: float64

weekly_return = weekly_last.pct_change().dropna()
print(weekly_return.round(4))
# 2024-01-12    0.0385
# 2024-01-19    0.0185
```

Computing weekly returns from resampled weekly prices gives the same answer as compounding the daily simple returns within each week, $\prod_t (1 + r_t) - 1$; for the week ending January 12, both give 0.0385. It is not the same as summing the daily returns, which gives 0.0384 here and is only an approximation for simple returns. Resampling is the more direct and less error-prone method when the analysis needs only the coarser frequency.

2.4.3 Visualization

A table of a hundred daily returns is hard to read at a glance. That two assets tend to move in opposite directions, or that a return series has grown more volatile, is often invisible in a column of numbers but obvious once plotted. Charts reveal patterns in returns, volatility, trends, and correlations, and Matplotlib is the standard library for static plots.

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The chapter's `AssetA`/`AssetB` example has only four daily returns, enough to illustrate the portfolio formulas in Section 2.5.1 by hand, but too few for an informative histogram or scatter plot. To show what these charts look like, this section simulates a longer, one-year (252 trading day) two-asset price history with a similar negative correlation between the assets, used only for the three plots below:

```
rng = np.random.default_rng(6)
n_days = 252
mean_returns = [0.0006, 0.0004]
vols = [0.015, 0.010]
corr = -0.6
cov = [[vols[0]**2, corr*vols[0]*vols[1]],
       [corr*vols[0]*vols[1], vols[1]**2]]

daily_returns = rng.multivariate_normal(mean_returns, cov, size=n_days)
dates = pd.bdate_range("2024-01-02", periods=n_days)
returns_viz = pd.DataFrame(daily_returns, index=dates, columns=["AssetA", "AssetB"])
prices_viz = pd.DataFrame({
    "AssetA": 100 * (1 + returns_viz["AssetA"]).cumprod(),
    "AssetB": 50 * (1 + returns_viz["AssetB"]).cumprod(),
})
```

A line chart of the two price series and their cumulative returns shows whether the assets move together or apart. Because AssetA and AssetB start at different price levels ($100 versus $50), the left panel gives each its own y-axis with `twinx()`, so that neither series dwarfs the other on a shared scale:

```
import matplotlib.pyplot as plt

fig, axes = plt.subplots(1, 2, figsize=(10, 4))

ax_a = axes[0]
line_a, = ax_a.plot(prices_viz.index, prices_viz["AssetA"],
                    color="tab:blue", label="AssetA")
ax_a.set_ylabel("AssetA price", color="tab:blue")
ax_a.tick_params(axis="y", labelcolor="tab:blue")
ax_a.set_title("Prices")

ax_b = ax_a.twinx()
line_b, = ax_b.plot(prices_viz.index, prices_viz["AssetB"],
                    color="tab:orange", label="AssetB")
ax_b.set_ylabel("AssetB price", color="tab:orange")
ax_b.tick_params(axis="y", labelcolor="tab:orange")
ax_a.legend(handles=[line_a, line_b], loc="upper left")

(1 + returns_viz).cumprod().plot(ax=axes[1], title="Cumulative growth of $1")
plt.tight_layout()
plt.savefig("ch2_fig_prices_growth.png", dpi=150)
```

Visualization is part of the analysis, not decoration. A model may look strong in a statistical summary while a plot reveals outliers, regime shifts, or data problems that would otherwise go unnoticed. It is often the fastest way to see that two return series move in opposite directions, as AssetA and AssetB do here.

Two other chart types are used constantly in financial analysis. A histogram of returns shows the shape of the return distribution, revealing skewness or heavy tails that summary statistics alone can obscure:

```
fig, ax = plt.subplots(figsize=(6, 4))
returns_viz["AssetA"].hist(ax=ax, bins=20)
```

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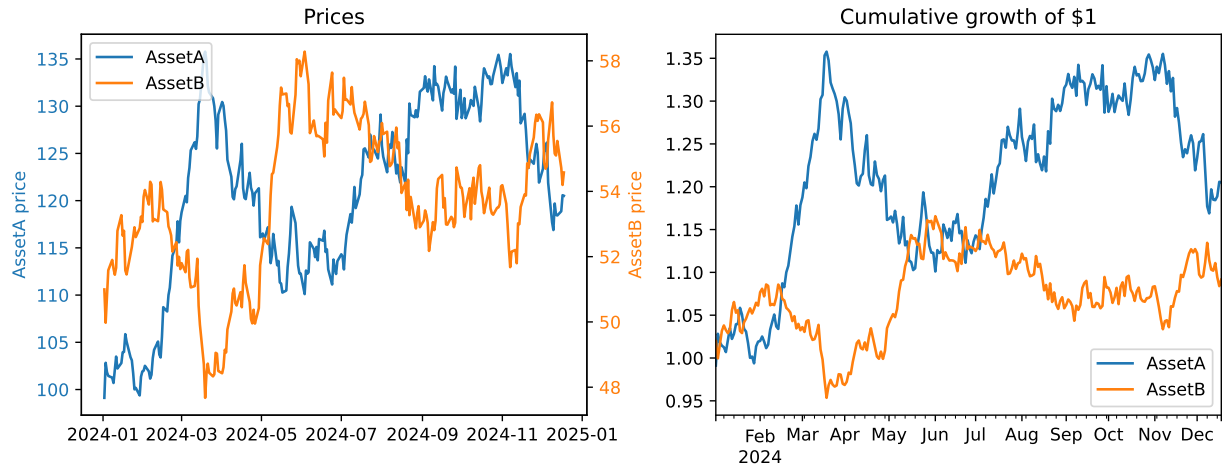


Figure 2.1: Simulated AssetA and AssetB prices, each on its own y-axis (left); cumulative growth of $1 invested in each (right), over a 252-day history

```
ax.set_title("Distribution of AssetA daily returns")
ax.set_xlabel("Return")
plt.tight_layout()
```

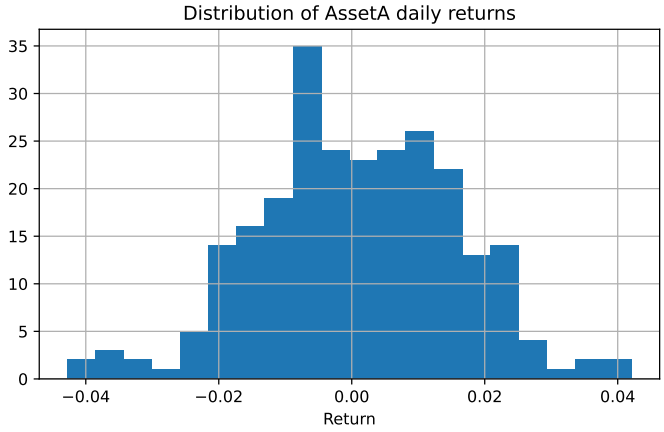


Figure 2.2: Distribution of AssetA's simulated daily returns over the 252-day history

Because the simulated returns are drawn from a normal distribution, Figure 2.2 can show neither skewness nor heavy tails, only sampling noise around a bell shape. Real daily returns typically have noticeably heavier tails than a normal distribution, the feature a histogram of real data would expose and the market-risk measures of Chapter 8 must contend with.

A scatter plot of one asset's returns against another's is the most direct way to visualize a correlation like the one computed numerically in Section 2.5.1:

```
fig, ax = plt.subplots(figsize=(5, 5))
ax.scatter(returns_viz["AssetA"], returns_viz["AssetB"], alpha=0.6)
ax.set_xlabel("AssetA return")
ax.set_ylabel("AssetB return")
```

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```
ax.set_title(f"Correlation: {returns_viz['AssetA'].corr(returns_viz['AssetB']):.3f}")
plt.tight_layout()
```

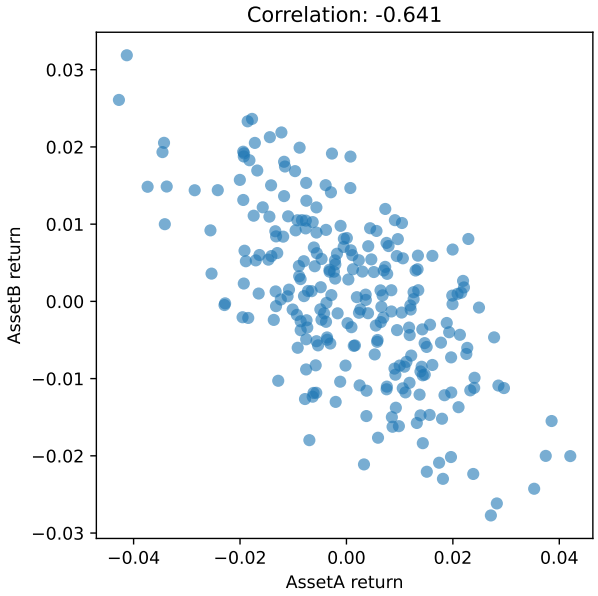


Figure 2.3: Simulated AssetA daily returns plotted against simulated AssetB daily returns

Figure 2.3 shows the cloud of 252 points sloping visibly downward, consistent with the simulated correlation of -0.641 printed in the plot title: the two assets tend to move in opposite directions. In a real analysis, the same chart is a useful diagnostic for nonlinear relationships or clusters that a single, linear correlation number cannot capture. The chapter's four-observation example (Section 2.5.1) shows the same pattern with an even stronger correlation of -0.896, but has too few points to plot informatively.

2.4.4 Data Cleaning and Preparation

Raw financial data often contains missing entries, duplicate observations, outliers, inconsistent dates, or formatting errors, so data cleaning is a major part of practical work. A well-designed pipeline may align multiple time series, fill missing observations carefully, adjust for corporate actions such as dividends and splits, and remove clearly erroneous values.

Figure 2.4 shows a simple financial data workflow, and the code below shows a typical cleaning step: forward-filling a small gap in a price series (for example, caused by a data feed outage) rather than silently dropping the observation, which would misalign the two return series.

```
raw = prices.copy()
raw.loc["2024-01-04", "AssetB"] = np.nan  # simulate a missing quote
cleaned = raw.ffill()
print(raw["AssetB"].isna().sum(), "missing value(s) before cleaning")
print(cleaned["AssetB"].isna().sum(), "missing value(s) after cleaning")
# 1 missing value(s) before cleaning
# 0 missing value(s) after cleaning
```

Forward-filling suits a short gap in an otherwise liquid series, but forward-filling an illiquid security for many consecutive days can create the false impression of zero volatility. A stock's price series must also often be adjusted for splits and dividends to produce a consistent total return series; otherwise mechanical price drops that are not economic losses distort its history, as Section 2.5.4 shows on the running example. Even a sophisticated model is only as good as the data that feed it.

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Figure 2.4: A typical financial data pipeline moves from raw data to cleaned inputs and then to modeling

2.5 Portfolio Analysis in Python

With the tools in place, this part puts them to work on the problem the rest of the book returns to repeatedly: constructing a portfolio from return data, evaluating it, and scaling the same code from two assets to many.

2.5.1 Worked Example 2.1: Building and Evaluating a Two-Asset Portfolio

This section ties the running example together into a small but complete analysis, the kind of workflow an analyst repeats constantly at larger scale. Suppose an analyst wants to evaluate an equally weighted portfolio of AssetA and AssetB.

```
weights = np.array([0.5, 0.5])
portfolio_returns = returns.dot(weights)
print(portfolio_returns.round(4))
# 2024-01-03    0.0150
# 2024-01-04    0.0026
# 2024-01-05    0.0123
# 2024-01-08    0.0052
# dtype: float64

port_mean = portfolio_returns.mean()
port_vol = portfolio_returns.std(ddof=1)
print(round(port_mean, 4), round(port_vol, 4))
# 0.0088 0.0058
```

Compare the portfolio's volatility of 0.58% to the weighted average of the two assets' individual volatilities, $0.5 \times 2.03\% + 0.5 \times 1.08\% = 1.56\%$. The portfolio is far less volatile than either asset alone, and than a naive average of their volatilities would suggest. This is the diversification effect introduced in Chapter 1, now made precise. The covariance between the two return series is

```
cov_matrix = returns.cov()
print(cov_matrix.round(6))
#          AssetA    AssetB
# AssetA  0.000414 -0.000198
# AssetB -0.000198  0.000118

corr = returns.corr().iloc[0, 1]
print(round(corr, 3))
# -0.896
```

so the two assets have a strong negative correlation of about -0.90. That estimate rests on only four pairs of returns, however, and is very imprecise: the standard Fisher-z 95% confidence interval runs from about -0.998 to $+0.467$, wide enough to include zero and even a moderately positive correlation. The example is chosen so that every number can be checked by hand, not as evidence about any real pair of assets; Chapter 12 returns to the damage this kind of estimation error does to portfolio optimization. The variance of an equally weighted two-asset portfolio is given by

$$\sigma_p^2 = w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2 w_A w_B \sigma_{AB}, \tag{2.2}$$

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and substituting $w_A = w_B = 0.5$, $\sigma_A^2 = 0.000414$, $\sigma_B^2 = 0.000118$, and $\sigma_{AB} = -0.000198$ gives

$$\sigma_p^2 = 0.25(0.000414) + 0.25(0.000118) + 2(0.25)(-0.000198) = 0.000034,$$

so $\sigma_p = \sqrt{0.000034} \approx 0.58\%$, matching the value computed directly from the portfolio return series. Agreement between the formula and the code is a useful checkpoint: whenever a hand-derived formula and a direct computation disagree, there is a bug in one of them, and reconciling the two is one of the most valuable debugging habits a quantitative analyst can build.

Running such analyses quickly, and extending them to many more assets and longer histories, is one reason Python is widely used in finance: researchers can iterate, test ideas, and communicate results reproducibly, which matters where decisions are made under time pressure.

2.5.2 Scaling Up: From Two Assets to a Small Portfolio Universe

The two-asset example was kept small so that every number could be checked by hand, but the code scales to any number of assets without modification. Adding a third asset, AssetC, requires only a new column in the original DataFrame:

```
prices["AssetC"] = [200.00, 198.00, 202.00, 199.00, 203.00]
returns3 = prices.pct_change().dropna()
print(returns3.round(4))
#             AssetA  AssetB  AssetC
# 2024-01-03  0.0200  0.0100 -0.0100
# 2024-01-04 -0.0098  0.0150  0.0202
# 2024-01-05  0.0297 -0.0051 -0.0149
# 2024-01-08 -0.0096  0.0200  0.0201
```

No line of the return, covariance, or portfolio-variance calculation changes; `pct_change()`, `.cov()`, and matrix multiplication work the same way however many columns the DataFrame has:

```
weights3 = np.array([1/3, 1/3, 1/3])
port3 = returns3.dot(weights3)
print(port3.round(4))
# 2024-01-03    0.0067
# 2024-01-04    0.0085
# 2024-01-05    0.0033
# 2024-01-08    0.0102

cov3 = returns3.cov()
port_var3 = weights3 @ cov3.values @ weights3
print(f"Equal-weighted 3-asset portfolio volatility: {np.sqrt(port_var3):.4f}")
# Equal-weighted 3-asset portfolio volatility: 0.0030
```

The expression `weights3 @ cov3.values @ weights3` is the matrix form of the portfolio variance formula, $w^\top \Sigma w$, which Section 2.6.3 restates for n assets, and it works identically whether w has 2, 3, or 200 entries. This is the payoff of writing analysis code in vectors and matrices rather than a named variable per asset: a script built for a two-asset illustration is, without rewriting, the script a researcher would run on hundreds of securities, the setting Chapter 12's optimization methods are built for.

Object-Oriented Programming: A Reusable `Portfolio` Class

The procedural style used so far, a sequence of variable assignments and function calls, works for a one-off analysis. Repeating the same mean, volatility, and variance calculations for every new portfolio, however, invites the copy-paste duplication that this chapter's reproducibility section warns against. Object-oriented programming addresses this with a `class`: a template, defined once, that bundles related data (here, a portfolio's returns and weights) with the functions that operate on it, called *methods*. Every new portfolio is then built from the same reusable blueprint rather than by copying and re-editing calculations:

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```
class Portfolio:
    def __init__(self, returns, weights):
        self.returns = returns
        self.weights = weights

    @property
    def portfolio_returns(self):
        return self.returns.dot(self.weights)

    @property
    def mean(self):
        return self.portfolio_returns.mean()

    @property
    def volatility(self):
        return self.portfolio_returns.std(ddof=1)

    def variance_from_formula(self):
        cov = self.returns.cov().values
        return self.weights @ cov @ self.weights
```

Instantiating this class on the two-asset data reproduces every number already computed by hand,

```
portfolio = Portfolio(returns, weights)
print(round(portfolio.mean, 4), round(portfolio.volatility, 4))
# 0.0088 0.0058
print(round(portfolio.variance_from_formula(), 6))
# 3.4e-05
```

but the payoff appears with the three-asset portfolio: `Portfolio(returns3, weights3)` produces the correct mean, volatility, and formula-based variance from the same class definition, with no new code. This is what "scaling up" means in an object-oriented codebase: the `Portfolio` class is a single, tested abstraction that every future portfolio, of any size, can reuse, the kind of reusable component a production research or trading system is built from.

2.5.3 Working with Time Series and Financial Data

An analyst comparing two securities that trade on different exchanges, or lining up daily prices with a monthly economic release, meets a problem that rows and columns alone do not solve: which date does each observation belong to, and what happens when one series has an entry for a day the other lacks? Many financial datasets are time series, with observations indexed by time and ordered sequentially: stock prices, interest rates, exchange rates, and macroeconomic indicators. Such data require careful handling of date indexes, alignment, and missing observations. pandas provides a time-indexed data structure that makes these operations natural, including resampling to a different frequency and computing rolling statistics:

```
rolling_vol = returns["AssetA"].rolling(window=2).std()
print(rolling_vol.round(4))
# 2024-01-03       NaN
# 2024-01-04    0.0211
# 2024-01-05    0.0279
# 2024-01-08    0.0278
# dtype: float64
```

A common workflow imports historical prices, converts them into a time series, computes returns, and then studies autocorrelation, volatility clustering, and trends. Many financial models depend on the future not being independent of the past, yet that dependence is often subtle and nonstationary, which is why Chapter 7 is needed.

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2.5.4 Returns, Risk, and Performance in pandas

Every statistic computed so far is a daily figure on simple returns from unadjusted prices. A performance report quotes something else: returns aggregated over time, risk and return stated per year, return per unit of risk, and the worst loss along the way, all computed from prices corrected for splits and dividends. This section defines each of these on the running example; later chapters use them without re-deriving them.

Simple and Log Returns

The log (continuously compounded) return over period t is

$$r_t = \ln \frac{P_t}{P_{t-1}} = \ln(1 + R_t), \tag{2.3}$$

where R_t is the simple return of Equation (2.1). The two are close for small moves, $r_t \approx R_t - R_t^2/2$, but they aggregate differently. Log returns add over time, because $\sum_t r_t = \ln(P_T/P_0)$, while simple returns compound: $\prod_t (1 + R_t) - 1 = P_T/P_0 - 1$. Simple returns, on the other hand, add across assets: a portfolio's simple return is exactly the weighted average $\sum_i w_i R_{i,t}$ of its holdings' simple returns, and no such identity holds for log returns.

```
log_returns = np.log1p(returns)          # r_t = ln(1 + R_t)
print(log_returns.round(6))
print(round(log_returns["AssetA"].sum(), 6), round(np.log(103 / 100), 6))
print(round(returns["AssetA"].sum(), 6),
      round((1 + returns["AssetA"]).prod() - 1, 6))

day1 = returns.iloc[0]                   # weights = [0.5, 0.5]
print(round(day1 @ weights, 6), round(np.log1p(day1) @ weights, 6),
      round(np.log1p(day1 @ weights), 6))
#                AssetA    AssetB
# 2024-01-03  0.019803  0.009950
# 2024-01-04 -0.009852  0.014937
# 2024-01-05  0.029270 -0.005085
# 2024-01-08 -0.009662  0.019803
# 0.029559 0.029559
# 0.030284 0.03
# 0.015 0.014876 0.014889
```

AssetA's four log returns sum to exactly $\ln(103/100) = 0.029559$, while its four simple returns sum to 0.030284, overstating the true 3.00% gain that compounding them recovers. On the first day the portfolio's simple return, 1.50%, is exactly the average of the two assets' simple returns; the average of their log returns, 0.014876, misses the portfolio's own log return, 0.014889. Hence the working rule this book follows: log returns for aggregating one series through time and for the statistical models of Chapter 7, simple returns whenever assets are combined into a portfolio.

Annualization and the Square-Root-of-Time Rule

Returns and volatilities are conventionally quoted per year, using about 252 trading days. If daily returns have mean μ_d and standard deviation σ_d, then

$$\mu_{\text{ann}} = 252\,\mu_d, \qquad \sigma_{\text{ann}} = \sqrt{252}\,\sigma_d. \tag{2.4}$$

The mean scales with time because expected returns add; the standard deviation scales with the square root of time because the variance of a sum of 252 *independent* daily returns with a common variance is $252\sigma_d^2$. Both scalings lean on assumptions. The rule is exact for log returns that are independent and identically distributed; it is wrong to the extent that daily returns are autocorrelated (positive autocorrelation makes $\sqrt{252}\,\sigma_d$ understate annual risk, negative autocorrelation overstate it) or that volatility itself clusters in calm and turbulent spells, the subject of Chapter 7's GARCH models. It is a convention for comparing numbers on a common scale, not a forecast of next year's range.

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The Sharpe Ratio

Return alone says nothing about the risk taken to earn it. The Sharpe ratio divides the average excess return over the risk-free rate r_f by the volatility of returns,

$$\mathrm{SR} = \frac{\bar{R} - r_f}{\sigma}, \tag{2.5}$$

with every quantity measured at the same frequency. A daily Sharpe ratio is annualized by multiplying by $\sqrt{252}$, which follows from Equation (2.4) and inherits its independence assumption. For orientation, the whole US stock market has earned an annualized Sharpe ratio of roughly 0.4 over the past century, and a strategy that sustains 1 after costs is considered good. Chapter 12 builds portfolio choice around this ratio, and Chapter 13 shows how easily a search over many strategies manufactures a high one by luck.

Maximum Drawdown

Volatility treats a gain and a loss of the same size alike, and it ignores the order in which they occur. An investor experiences the path: how far the account falls from its previous high before it recovers. If V_t is the value of \$1 invested at the start, the maximum drawdown is the largest such fall,

$$\mathrm{MDD} = \max_{t} \frac{\max_{s \le t} V_s - V_t}{\max_{s \le t} V_s}, \tag{2.6}$$

a path-dependent measure that Chapter 8 develops alongside Value at Risk.

Worked Example 2.2: Performance of AssetA, AssetB, and the Portfolio

The code below puts the three measures together for the two assets and the equally weighted portfolio of Section 2.5.1, with the risk-free rate set to zero for this five-day illustration, as Chapter 12 also does:

```
TRADING_DAYS = 252
series = pd.concat([returns, portfolio_returns.rename("Portfolio")], axis=1)
perf = pd.DataFrame({"mean_d": series.mean(), "vol_d": series.std(ddof=1)})
perf["mean_a"] = perf["mean_d"] * TRADING_DAYS
perf["vol_a"] = perf["vol_d"] * np.sqrt(TRADING_DAYS)
perf["SR_d"] = perf["mean_d"] / perf["vol_d"]        # risk-free rate set to 0
perf["SR_a"] = perf["SR_d"] * np.sqrt(TRADING_DAYS)
print(perf.round(4))
print(round(np.sqrt(TRADING_DAYS / len(series)), 2))  # rough std. error of SR_a
#            mean_d   vol_d  mean_a   vol_a    SR_d     SR_a
# AssetA     0.0076  0.0203  1.9079  0.3229  0.3722   5.9078
# AssetB     0.0100  0.0108  2.5186  0.1721  0.9218  14.6329
# Portfolio  0.0088  0.0058  2.2132  0.0925  1.5064  23.9138
# 7.94

def max_drawdown(r):
    wealth = np.concatenate([[1.0], (1 + r).cumprod().to_numpy()])
    running_peak = np.maximum.accumulate(wealth)
    return ((running_peak - wealth) / running_peak).max()

for name in series:
    print(f"{name:9s} maximum drawdown: {max_drawdown(series[name]):.2%}")
# AssetA    maximum drawdown: 0.98%
# AssetB    maximum drawdown: 0.51%
# Portfolio maximum drawdown: 0.00%
```

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Every entry can be checked by hand. AssetA's daily volatility of 2.03% annualizes to $2.03\% \times \sqrt{252} = 32.3\%$, a plausible figure for a single stock. Its daily mean of 0.76% annualizes to 191%, and the portfolio's daily Sharpe ratio of $0.88\%/0.58\% = 1.51$, the figure Chapter 12 quotes, annualizes to $1.51 \times \sqrt{252} = 23.9$. Those last two numbers are arithmetically correct and economically meaningless. Four observations cannot pin down a mean, and the standard error of an annualized Sharpe ratio is roughly $1/\sqrt{\text{years of data}}$, here $\sqrt{252/4} = 7.94$, so even 23.9 is barely three standard errors from zero; Chapter 13 derives this standard error and shows how badly it bites on far longer samples. The drawdowns tell the same diversification story as the volatilities: AssetA's worst fall is from \$102 to \$101, 0.98%; AssetB's is from \$51.26 to \$51.00, 0.51%; and the portfolio, which rose on all four days, never fell below a previous high at all.

Adjusted versus Raw Prices

A raw closing price series records what a share traded for, which changes for reasons that are not gains or losses to the holder. In a 2-for-1 stock split each share becomes two, and the price halves overnight; on an ex-dividend date the price drops by roughly the dividend, which the holder receives in cash. Returns computed from raw prices count both as losses. Data vendors therefore publish an adjusted price series in which every price before such an event is multiplied by a factor: the split ratio (0.5 for a 2-for-1 split), and $1 - D/P$ for a dividend D paid after a close of P. Yahoo Finance, for example, applies these split and dividend multipliers following the standards of the Center for Research in Security Prices (CRSP). The code below applies both adjustments to AssetA, first imagining that its last two prices had been reported after a 2-for-1 split, then that the \$1 fall on January 8 was the stock going ex-dividend on a \$1.00 dividend:

```
raw_a = pd.Series([100.00, 102.00, 101.00, 52.00, 51.50], index=prices.index)
split_factor = pd.Series([0.5, 0.5, 0.5, 1.0, 1.0], index=prices.index)
adj_a = raw_a * split_factor             # 2-for-1 split before the Jan 5 open
print(pd.DataFrame({"raw": raw_a.pct_change(),
                    "adjusted": adj_a.pct_change()}).dropna().round(4))

dividend = 1.00                          # AssetA goes ex-dividend on Jan 8
div_factor = 1 - dividend / 104.00       # applied to every close before then
adj_close = prices["AssetA"] * np.where(prices.index < "2024-01-08",
                                        div_factor, 1.0)
print(round(div_factor, 6), adj_close.round(4).tolist())
print(round(103.00 / 104.00 - 1, 4),
      round(adj_close.iloc[-1] / adj_close.iloc[-2] - 1, 4))
#                raw  adjusted
# 2024-01-03  0.0200    0.0200
# 2024-01-04 -0.0098   -0.0098
# 2024-01-05 -0.4851    0.0297
# 2024-01-08 -0.0096   -0.0096
# 0.990385 [99.0385, 101.0192, 100.0288, 103.0, 103.0]
# -0.0096 0.0
```

The raw series shows a 48.5% crash on January 5 that never happened to anyone's wealth; the adjusted series recovers AssetA's true 2.97% gain. In the dividend case the raw price return on January 8 is -0.96%, but the holder's total return, $(103 + 1)/104 - 1$, is zero, which the adjusted series shows. Because the factor multiplies all earlier prices by the same constant, every earlier return is unchanged. Only the level of the old prices moves, and it moves again with each new dividend, which is why an adjusted price downloaded today will not match one downloaded last year even though the returns agree. Use adjusted prices for returns and performance measures, and raw prices for anything tied to the actual traded price, such as an order's limit price or an option's strike. The `yfinance` library used in this chapter's last exercise returns split- and dividend-adjusted prices in its `Close` column by default (`auto_adjust=True`); pass `auto_adjust=False` to see the raw close alongside the adjusted one.

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2.6 Code That Holds Up

Code that produces the right number once is not the same as code that can be trusted, rerun, and handed to somebody else. This part covers performance, the errors financial data reliably provokes, the summary statistics worth computing, and the practices that make a result reproducible.

2.6.1 Why Vectorization Matters: A Performance Comparison

Section 2.3.1 claimed that vectorized array operations are faster than an equivalent Python-level loop. To demonstrate it, consider computing the cumulative growth of $1 across 100,000 simulated daily returns, once with an explicit loop and once with NumPy's vectorized `cumprod`:

```
import time

rng = np.random.default_rng(0)
simulated_returns = rng.normal(0.0005, 0.01, 100_000)

start = time.perf_counter()
total = 1.0
loop_values = []
for r in simulated_returns:
    total *= (1 + r)
    loop_values.append(total)
loop_time = time.perf_counter() - start

start = time.perf_counter()
vectorized_values = np.cumprod(1 + simulated_returns)
vector_time = time.perf_counter() - start

print(f"Loop: {loop_time:.4f}s, Vectorized: {vector_time:.4f}s, "
      f"speedup: {loop_time/vector_time:.0f}x")
# Loop: 0.0394s, Vectorized: 0.0010s, speedup: 39x
```

Both approaches produce identical final values, but the vectorized version is roughly 39 times faster on this test (the figure is machine-dependent, but a factor of 10 to 100 is typical for this kind of calculation). The Python-level loop pays interpreter overhead (argument checking, dynamic type dispatch) on each of the 100,000 iterations, while `cumprod` executes the equivalent loop once, in compiled C code, over contiguous memory. The gap widens as the dataset grows, which is why Section 2.7 treats vectorized NumPy and pandas operations as the default, reserving explicit loops for path-dependent calculations, like the sequential compounding example in Section 2.2.2, that cannot be vectorized.

2.6.2 Handling Errors and Common Pitfalls

Financial data pipelines fail constantly in small, mundane ways: a data feed returns a missing value, a ticker is misspelled, a division produces infinity because a denominator was zero. Python's `try/except` block lets a program catch such errors and respond deliberately rather than crash:

```
def safe_return(price_now, price_before):
    try:
        return (price_now - price_before) / price_before
    except ZeroDivisionError:
        return float("nan")

print(safe_return(105.0, 100.0))
# 0.05
```

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```
print(safe_return(105.0, 0.0))
# nan
```

Returning `nan` (not-a-number) rather than crashing lets downstream code, and pandas in particular, continue processing the rest of a dataset while marking the problematic observation, like the missing-value handling in Section 2.4.4. A few pitfalls recur often in financial Python code:

- **Silent `NaN` propagation.** Almost any arithmetic operation involving `NaN` produces another `NaN` rather than an error, so a single bad data point can spread through an entire calculation (a covariance matrix, for example) without warning.
- **Comparing floating-point numbers for exact equality.** Because of how computers represent decimal numbers, a calculation that should equal `0.3` may instead equal `0.30000000000000004`; use a tolerance-based comparison (`np.isclose`) rather than `==` when checking floating-point results.
- **Mutating a DataFrame while iterating over it.** Modifying rows of a DataFrame inside a loop over that same DataFrame can produce confusing, order-dependent results; it is almost always better to build a new column or DataFrame with a vectorized expression.
- **Off-by-one errors in lagged calculations.** Return, rolling-window, and lag calculations are prone to off-by-one mistakes, such as using today's price where yesterday's was intended, which can silently introduce look-ahead bias into a backtest (Section 2.7).

2.6.3 Descriptive Statistics and Portfolio Analytics

A return series that has been imported, cleaned, and plotted still leaves an obvious question: how risky does this asset look, in numbers precise enough to compare against another asset or feed into a portfolio calculation? Descriptive statistics such as averages, standard deviations, skewness, and kurtosis summarize the distribution of returns and are the starting point for risk analysis:

```
summary = returns.describe().round(4)
print(summary)
#        AssetA  AssetB
# count  4.0000  4.0000
# mean   0.0076  0.0100
# std    0.0203  0.0108
# min   -0.0098 -0.0051
# 25%   -0.0097  0.0062
# 50%    0.0052  0.0125
# 75%    0.0224  0.0163
# max    0.0297  0.0200
```

For a portfolio, one also computes correlations, covariance matrices, and portfolio volatility, as in the worked example above. These quantities are central to diversification and asset allocation, and they generalize directly to many assets: with n assets, weights $w \in \mathbb{R}^n$, and covariance matrix Σ, portfolio variance is $\sigma_p^2 = w^\top \Sigma w$, the two-asset formula in matrix form.

The same code applies unchanged whether there are 2 assets or 200. A single script can compute rolling volatilities and the annualized volatilities, Sharpe ratios, and drawdowns of Section 2.5.4 across a whole universe of securities, which matters when a researcher compares strategies, evaluates risk, or tests whether a signal survives transaction costs.

2.6.4 Reproducibility, Version Control, and Good Practice

Analysis in computational finance must be reproducible. A result that cannot be rerun by another analyst, or by the same analyst six months later, is of limited value. Python supports reproducibility through scripts, notebooks, and package management. A clear workflow includes:

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- Recording exact package versions, for example in a `requirements.txt` or `environment.yml` file, so that a colleague (or a future you) can recreate the same environment.
- Saving raw data separately from cleaned or derived data, so that a cleaning step can be re-run or audited without re-fetching from the original source.
- Documenting transformations, ideally in code and comments rather than in a separate document that can drift out of sync with the script.
- Separating analysis from presentation, so that a notebook used to explore an idea is not the same artifact used to report a result to a manager or regulator.
- Using version control (such as Git) to track changes to code over time, including which version of the code produced a given historical result.
- Fixing random seeds (as in `np.random.default_rng(0)` in Section 2.6.1) whenever a calculation involves simulation, so that two runs of the same script on the same data produce identical results.
- Writing small checks into a script or notebook, such as the put-call parity check of Chapter 5 or the accounting-identity check of Chapter 4, so that a silent bug is caught immediately rather than after it has propagated into a report or a trading decision.

2.6.5 Unit Testing: Making Checks Automatic

The "small checks" above become far more valuable once written as automatically run tests rather than one-off `print` statements an analyst must remember to inspect. A unit test is a function that exercises a piece of code and uses Python's `assert` statement to fail loudly if the result is not what is expected. Section 2.5.1's observation that the portfolio-variance formula and the direct computation agree is the kind of invariant worth encoding this way:

```
def test_portfolio_variance_matches_formula():
    weights = np.array([0.5, 0.5])
    cov = returns.cov().values
    var_A, var_B, cov_AB = cov[0, 0], cov[1, 1], cov[0, 1]
    formula_variance = (
        weights[0]**2 * var_A + weights[1]**2 * var_B
        + 2 * weights[0] * weights[1] * cov_AB
    )
    direct_variance = returns.dot(weights).var(ddof=1)
    assert abs(formula_variance - direct_variance) < 1e-10, \
        "Portfolio variance formula does not match direct computation!"
    print("Test passed: formula-based and direct portfolio variance agree.")

test_portfolio_variance_matches_formula()
# Test passed: formula-based and direct portfolio variance agree.
```

The tolerance, 10^{-10}, is loose enough to absorb ordinary floating-point rounding but tight enough to catch a real error, such as a transposed covariance term or a forgotten factor of two on the cross term. Python's own `statistics.mean` shows how quietly such bugs arise: handed a list of NumPy integers such as prices of 100, 101, and 103, it returns the truncated integer 101 rather than 101.3333, with no error or warning, a wrong answer that only a check against an independently computed value, here `np.mean`, would catch. A test like this costs a few lines and runs in a fraction of a second, but it turns a fact verified once by hand into one re-verified every time the surrounding code changes, catching a regression immediately rather than when a downstream number looks suspicious. Production financial codebases collect dozens or hundreds of such tests, run automatically before any code change is accepted with a framework such as `pytest`, but the principle is the same at any scale: encode a known-correct result as an executable check rather than a comment.

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Reproducibility is not a bureaucratic nicety; in finance it is often a legal and professional requirement. A regulator investigating a trading loss, an auditor reviewing a valuation, or a portfolio manager questioning a backtest will expect to trace a reported number back to the code and data that produced it, possibly months or years after the fact. A workflow that cannot support this scrutiny is a liability however sophisticated its models.

Good practice also means writing code that others can understand. Financial projects often involve several collaborators, long time horizons, and changing requirements, and a readable function, a clear variable name, and a documented transformation prevent errors and make the analysis easier to audit.

2.7 Challenges in Applying Python to Financial Data

Several practical challenges recur when Python is used for financial analysis.

Data quality and vendor discrepancies. Different data vendors can report slightly different prices, adjust for corporate actions differently, or use different conventions for holidays and time zones. A workflow that silently trusts a single source can propagate vendor-specific errors into every downstream result.

Look-ahead and survivorship bias. It is easy to use information that would not have been available at the time of a decision, for example by studying past performance with a company's current list of index constituents, which silently excludes firms that were delisted or went bankrupt. Both biases can make a backtest look far more profitable than any strategy could have been in real time.

Non-stationarity and changing data schemas. Return distributions, volatility regimes, and even the format of vendor data feeds change over time. Code written for one period's conventions may silently produce wrong answers, rather than an error, on a later period with a slightly different schema.

Performance at scale. A script that works on five days of data for two assets may be far too slow on years of tick-by-tick data across thousands of instruments. Vectorized NumPy and pandas operations help, but some tasks require more specialized tools, such as columnar file formats, database systems, or distributed computing frameworks.

Environment and dependency drift. A notebook that runs correctly today may fail in a year if underlying libraries change their behavior or drop older syntax, which is why pinning dependency versions and periodically re-validating old code are standard practice in production financial systems.

Latency and real-time constraints. Some applications, particularly in trading, require code to execute within milliseconds or microseconds. Python trades raw execution speed for convenience, so performance-critical components are sometimes written in compiled languages or accelerated with tools such as Numba or Cython, while Python remains the layer for research, orchestration, and analysis.

Dependency and library version conflicts. A pandas or NumPy upgrade can silently change a function's default behavior (for example, how an aggregation handles missing values), so code not re-tested against a new library version should not be assumed to behave identically, even if it runs without error.

The gap between exploratory and production code. Code written quickly in a notebook to explore an idea is rarely suitable, unmodified, for a system that runs unattended in production. Exploratory code typically lacks the error handling, logging, and automated testing a production financial system requires, and treating the two as interchangeable is a common source of operational incidents.

2.8 Key Takeaways

Python is a practical tool for financial analysis because it supports efficient computation, flexible data handling, and reproducible workflows, and its libraries take an analyst from raw market data to descriptive analysis, forecasting, and machine learning.

The worked example showed how a handful of pandas and NumPy operations on five days of prices reveal a meaningful diversification effect, and how a formula derived on paper should match a direct computation on the data. The habits introduced here (vectorized computation, careful data cleaning, reproducible workflows) scale directly to the much larger datasets of later chapters.

Four distinctions recur, largely unchanged, in every later chapter. First, the choice between a loop and a vectorized operation is not a matter of style: it depends on whether a calculation is independent across observations (vectorize it) or sequential, like the compounding balance example (a loop, or an equivalent

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closed-form formula, is required). Second, a performance number is only as meaningful as the choices behind it: simple or log returns, adjusted or raw prices, and an annualization that assumes independent returns and, on a short sample, turns a lucky week into an implausible Sharpe ratio. Third, combining data from multiple sources, the `merge` and `groupby` operations of Section 2.4.2, is a central skill, since almost no real financial analysis works from a single tidy table. Fourth, every code example in this chapter was written to be checked, by hand, as with the four AssetA and AssetB returns, or by running the companion notebook and comparing outputs line by line. Later chapters rely on this habit of verifying rather than trusting a computation once calculations become too complex to check by hand, such as the maximum-likelihood-estimated GARCH models of Chapter 7 or the gradient-boosted trees of Chapter 6.

2.9 Suggested Exercises

2.1 Write a Python function that computes the future value of an investment given a price, an annual rate, and a number of years, and use it to verify the two examples in Section 2.2.2.

2.2 Recreate the `prices` DataFrame from Section 2.2.3, add a third asset, `AssetC`, with your own five prices, and compute its returns using `pct_change()`.

2.3 Using the covariance formula for portfolio variance, compute the volatility of a portfolio with weights 0.7 on AssetA and 0.3 on AssetB, and check your answer against a direct computation on `portfolio_returns`.

2.4 Modify the data-cleaning example to simulate two consecutive missing values in AssetB, apply forward-fill, and discuss whether this is still a reasonable choice.

2.5 Plot the cumulative growth of $1 invested in AssetA, AssetB, and the equally weighted portfolio on the same chart, and describe what the chart shows about diversification that the summary statistics alone do not.

2.6 Explain why a positive correlation between two assets would change the diversification benefit computed in Section 2.5.1, and recompute the portfolio volatility assuming a correlation of $+0.90$ instead of -0.896, holding the individual volatilities fixed.

2.7 Describe one concrete situation in which look-ahead bias could enter a Python backtesting script without the analyst noticing, and propose a coding practice that would prevent it.

2.8 Rewrite the `classify_return` function from Section 2.2.2 as a single list comprehension using a conditional expression, and verify it produces the same labels for the four AssetA returns.

2.9 Using `np.select` as in Section 2.3.2, classify AssetB's four returns into "strong gain" ($> 1\%$), "small gain" ($> 0\%$), "flat," or "loss," and compare the result to AssetA's classification.

2.10 Extend the `securities_df` and `shares` example in Section 2.4.2 with one more security of your choosing, recompute `position_value`, and recompute the sector summary.

2.11 Extend the `extended_prices` series in Section 2.4.2 with five more business days of your own prices, and resample the full series to weekly frequency using both `.last()` and `.mean()`; explain when each aggregation would be the more appropriate choice.

2.12 Using the pattern in Section 2.6.1, describe (without necessarily running the code) why a loop that appends to a Python list one element at a time is typically slower than filling a pre-allocated NumPy array by index, although neither calls a vectorized function.

2.13 Using this chapter's `Portfolio` class, instantiate it with weights 0.7 on AssetA and 0.3 on AssetB, and verify that `portfolio.volatility` matches the answer you computed by hand in Exercise 2.3.

2.14 Add a second unit test, `test_weights_sum_to_one`, that asserts a portfolio's weights sum to 1.0 within a small tolerance, and explain what kind of real coding mistake this test would catch that the variance-formula test would not.

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2.15 Using Section 2.5.4, compute AssetB's four log returns and verify that they sum to ln(52.02/50.00). Then compute the equally weighted portfolio's return on the second day three ways, as the average of the two simple returns, as the average of the two log returns, and as the log of the portfolio's gross return, and explain which two of the three must agree exactly.

2.16 For the 70/30 portfolio of Exercise 2.3, compute the daily mean, the daily volatility, the annualized volatility, and the daily and annualized Sharpe ratios (risk-free rate zero). State the two assumptions behind the annualization, and use the approximation $1/\sqrt{\text{years of data}}$ to judge how much the annualized Sharpe ratio actually tells you.

2.17 Suppose AssetB paid a \$0.50 dividend and went ex-dividend on January 5, the day its price fell from \$51.26 to \$51.00. Compute that day's price return and total return, the dividend adjustment factor, and AssetB's maximum drawdown computed from raw and from adjusted prices. What does the comparison say about drawdowns measured on unadjusted data?

2.18 **Challenge (real data).** Download daily prices for two exchange-traded funds, the SPDR S&P 500 ETF (`SPY`) and the iShares 20+ Year Treasury Bond ETF (`TLT`), from January 1, 2015 through December 31, 2024 with `yfinance` (`yf.download(["SPY", "TLT"], start="2015-01-01", end="2025-01-01")`, which returns adjusted prices). (a) Compute each fund's annualized mean return, annualized volatility, and annualized Sharpe ratio, using a constant risk-free rate of 2% a year, and each fund's maximum drawdown. (b) Compute the 63-trading-day (roughly quarterly) rolling correlation between the two funds' daily returns, plot it, and describe how it changed over the decade, in particular after 2021. (c) Repeat part (a) with `auto_adjust=False` and the raw `Close` column, and explain which fund's numbers move more, and why.

2.10 Further Reading

- McKinney (2022), *Python for Data Analysis*. Written by pandas' original author, this is the definitive reference for the DataFrame, groupby, merge, and resample operations introduced in this chapter, covering many more edge cases and options than a single chapter can.
- VanderPlas (2022), *Python Data Science Handbook*. A broader tour of NumPy, pandas, and visualization that fills in the general-purpose Python and data-science background this chapter assumes, alongside its finance-specific examples.
- Hilpisch (2018), *Python for Finance*. Extends this chapter's two-asset portfolio example toward the derivatives pricing, risk management, and algorithmic trading applications covered later in this book, using the same core pandas and NumPy toolkit.
- Hilpisch (2020), *Python for Algorithmic Trading*. A practical, code-heavy treatment of building and backtesting trading systems in Python, a natural next step after this chapter's vectorization and performance material for readers headed toward Chapters 13 and 14.

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Chapter 3

Financial Markets, Institutions, and Instruments

3.1 Introduction

Financial markets are the institutions through which capital is allocated, risk is transferred, and prices are discovered. They allow households to save, firms to finance investment, governments to fund spending, and investors to diversify exposure. Anyone who wants to use data science or artificial intelligence in finance needs to understand these markets, because the objects of analysis are not abstract numbers but real contracts, incentives, and trading mechanisms.

Modern finance rests on a simple idea: people do not all have the same preferences, opportunities, or time horizons. Some want to consume today, others to save for the future, and still others will accept risk in exchange for compensation. Financial markets connect these needs. Without them, savings would be less productive, firms would find it harder to grow, and risk would be harder to manage.

This chapter introduces the structure of financial markets and the major classes of financial instruments. It begins with institutions and participants, moves to the securities traded in modern markets, and then explains how prices are formed in real trading systems, laying a foundation for the later chapters on pricing, portfolio management, risk, and trading.

Three worked examples anchor the more mechanical sections: a bid-ask spread decomposed into adverse-selection and inventory-cost components, a hand-computed walk through a small limit order book showing how an incoming order fills at one price or several, and a no-arbitrage bond-pricing example showing what happens when two ways of pricing the same cash flow disagree. Two case vignettes close the chapter: the 2010 Flash Crash, an afternoon when the mechanics of order flow and liquidity broke down in public, and the 2021 collapse of Archegos Capital Management, which reruns the chapter's margin arithmetic at a leverage of ten to one.

The institutional material (who trades, what they trade, and where) builds the vocabulary later chapters assume. Chapters 5 and 12 price and hold the instruments cataloged here; Chapters 13 and 14 return to the limit order book of Section 3.3.2 at execution and then microsecond speed; and the no-arbitrage logic of Section 3.5.3 recurs in every valuation and pricing chapter that follows.

Readers comfortable with market structure may skim the descriptive sections, but the worked examples are used again later and are worth working through by hand once.

3.2 The Market and Its Participants

A market is an institution before it is a dataset. This part sets out what financial markets are for, who shows up in them, the firms that intermediate between those participants, and the contracts they actually trade.

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3.2.1 Why Financial Markets Exist

A financial market exists because economic agents have different needs and time horizons. A household may wish to save for retirement, a firm may need funding for expansion, and a government may need resources for public projects. Markets match the supply of and demand for capital.

They do more than move money from one person to another. Markets also let people shift risk across time and across states of the world. A farmer who fears a decline in crop prices can use a futures contract to lock in a sale price. A pension fund that wants stable long-term income can purchase government bonds. A young entrepreneur who needs capital to build a company can issue equity to investors who are willing to share the upside and downside of the venture.

Financial markets also provide information. Prices reflect current supply and demand, but they also encode expectations about future cash flows, interest rates, inflation, and corporate performance. Markets are therefore systems for aggregating dispersed information into prices that guide decisions, not merely venues for exchange.

A market is more useful when participants can trade quickly and with confidence, so liquidity, transparency, and regulation matter. If a market is illiquid, prices may be unstable and trading costs may be high. If it lacks transparency, participants may face information asymmetry and make poor decisions. If it lacks rules, trust can erode and market participants may withdraw from the system.

3.2.2 Who Participates in Financial Markets

Financial market participants have different objectives and constraints. Households and retail investors save for retirement, education, or other future needs. Firms issue securities to raise funding for investment. Governments borrow to finance public spending or manage budget imbalances. Financial institutions such as banks, mutual funds, pension funds, and insurers intermediate between those who provide capital and those who need it. Chapter 1's Table 1.5 previewed these participant roles at a glance; this section develops each one in more depth.

Market makers play a particularly important role. A market maker provides liquidity by standing ready to buy and sell securities, usually quoting both a bid price and an ask price. In return for bearing inventory risk and exposure to adverse price movements, market makers earn the spread between the two prices, which is both a trading cost and compensation for providing immediacy and continuity.

Arbitrageurs and hedge funds also influence market behavior. Arbitrageurs exploit pricing discrepancies across related assets or markets. Hedge funds often pursue more complex strategies involving leverage, short selling, or macroeconomic views. Their activity can improve pricing efficiency but can also increase volatility when markets are under stress.

Finally, regulators and central banks shape the environment. Regulators enforce rules on disclosure, conduct, and market integrity. Central banks influence short-term interest rates and can intervene in crises to support liquidity or stabilize the financial system. Market outcomes therefore depend on public policy as well as private incentives.

3.2.3 Major Financial Institutions

Financial markets are organized around specialized institutions. Banks provide lending services, facilitate payments, and transform maturity and liquidity profiles. Asset managers collect capital from investors and allocate it across portfolios of securities. Insurance companies pool risk by charging premiums in exchange for compensation when specified losses occur. Exchanges provide venues where orders can be matched, and central clearinghouses reduce counterparty risk by interposing themselves between buyers and sellers. Chapter 1's Table 1.3 summarized these institutions' roles at a glance; this section develops each one further.

These institutions have different business models and incentives. A commercial bank earns money by borrowing at one rate and lending at another. An asset manager earns fees by managing assets for clients. An insurance company earns premiums and invests the proceeds while managing claims. Each therefore faces its own mix of credit, market, liquidity, and operational risk.

The financial system is fragile when institutions become highly interconnected. A bank may lend to a hedge fund, which may trade derivatives with a broker, which may depend on short-term funding from

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another bank. Such links make the system more efficient in normal conditions but transmit shocks quickly under stress, which is one reason understanding institutions is essential to understanding financial crises.

These institutions also shape the real economy. When banks tighten credit, firms may reduce investment. When insurers become more cautious, some risks become harder to insure. When exchanges improve market structure, trading costs may fall and liquidity improve. Such changes affect economic activity, corporate decisions, and household welfare.

One institutional shift of the past fifteen years deserves its own mention, because it has moved a growing share of lending outside every venue described so far. Since the 2008 crisis, tighter bank capital regulation has pushed banks out of much corporate lending. Dedicated *private credit* funds, asset managers raising long-lockup capital to lend directly to mid-sized companies, have grown into a multi-trillion-dollar market filling that space, alongside the older private equity industry that buys entire companies rather than lending to them. These private markets differ from everything else in this chapter in a way that matters greatly for this book: they generate almost none of the data its methods assume. A private loan trades rarely or never, so there is no price history. The borrower files no public statements, so there is nothing for ratio analysis beyond what the lender privately negotiates access to. And a fund holds tens of positions, not the thousands of observations a statistical credit model trains on. AI's footprint in private markets is accordingly real but narrow: mostly natural language processing applied to deal documents and covenants, and screening tools for sourcing borrowers, while underwriting and valuation remain judgment-heavy in a way public-market lending no longer is. This raises a standing question for the rest of the book: when a market's defining feature is the absence of data, how much of the toolkit still applies, and which parts fail first?

3.2.4 Financial Instruments: The Core Contracts

Financial instruments are the contracts traded in markets. Broad categories include debt securities, equity securities, derivatives, and alternative instruments, each with its own economic purpose and pricing logic. Chapter 1's Table 1.4 previewed the markets these instruments trade in; the rest of this section develops each instrument type in depth.

Debt Securities

Suppose a government wants to build a highway that will benefit taxpayers for decades but costs more than this year's tax revenue, or a corporation wants to build a factory that will not turn a profit until it is finished. Both need to raise a large sum now against a promise to repay it, with something extra, over time. Both also need that promise to be tradable, so that whoever lends today is not stuck holding an illiquid IOU for thirty years if their circumstances change. Debt securities solve this problem. A bond is a standardized, tradable loan: the investor lends money to the issuer in exchange for a stream of future payments and the repayment of principal at maturity. Bonds are issued by governments, corporations, and other entities, each with a different credit quality.

Government bonds are often viewed as relatively safe, though they still carry interest-rate and inflation risk. Corporate bonds generally offer higher yields because they are exposed to default risk. Bond pricing rests on the time value of money and discounting: a fixed-coupon bond gains value when market interest rates fall and loses value when they rise, which is why bond prices and yields move inversely.

The bond market also provides a benchmark for the broader economy. Long-term government bond yields are often read as signals of expected growth and inflation, so bond markets are tools for interpreting macroeconomic conditions as well as for financing.

Equity Securities

A bond is a poor fit for a very different kind of firm: one whose value lies almost entirely in an uncertain future, such as a startup that might be worth nothing or billions, with no stable cash flow today from which to make fixed interest payments. A bond's rigid repayment schedule could bankrupt such a firm in a lean year even if its long-run prospects remain excellent. Equity securities let investors share in the firm's fortunes rather than lend against a fixed promise. They are ownership claims: a common stock share gives the holder a claim on future profits and, in many cases, a vote on certain corporate matters. Equity investors are residual

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claimants, paid only after other obligations such as debt payments are met, so equities are generally riskier than bonds but offer greater upside.

Valuing stocks is harder than valuing bonds because future dividends and firm value are uncertain. Investors use dividend discount models, earnings-based valuation, or relative valuation. In all cases, an equity claim's value depends on expected future cash flows and the discount rate, so a stock promising strong growth may command a high price even if current profits are modest.

Equity markets also let firms raise permanent capital. Unlike debt, equity has no contractual maturity or scheduled payments. This flexibility suits firms that expect growth, but it means shareholders bear more uncertainty than lenders.

Derivatives

Neither a bond nor a stock lets a wheat farmer lock in today the price of a harvest not yet planted, or lets an airline protect itself against a jet-fuel price spike without buying and storing barrels of oil it has nowhere to put. Both need a contract whose payoff is tied to something else's future price, without either party owning the underlying asset in the meantime. Derivatives are such contracts: their value depends on the performance of another asset or variable. Common examples include forwards, futures, options, and swaps. A forward contract obligates the buyer and seller to exchange an asset at a future date for a predetermined price. A futures contract performs a similar role but is standardized and traded on an exchange. An option gives the holder the right, but not the obligation, to buy or sell an asset at a specified price before or at a certain date.

Derivatives are used for hedging, speculation, and arbitrage. A producer may use futures to lock in a selling price. A portfolio manager may use options to protect against downside risk. A trader may use derivatives to express a view on volatility or market direction. Because derivatives often embed leverage, they magnify gains and losses, which makes them tools for both risk management and risk creation.

By letting parties transfer risk precisely, derivatives expand the risk management strategies available to firms and investors. They can also create hidden exposures if participants misunderstand their payoff structure, one reason regulators and risk managers monitor them closely.

Swaps

Suppose a company has issued floating-rate debt, so its interest payments rise and fall with market rates, but its revenue is stable and it would rather know what it owes each quarter than gamble on rates. Paying off the old loan and negotiating a new fixed-rate one would be slow and expensive, and might be impossible if the lender will not renegotiate. A swap solves this without touching the underlying loan. It is an agreement between two parties to exchange cash flows according to a pre-specified formula, and the most common type is the interest rate swap.

In a plain-vanilla interest rate swap, one party pays a fixed rate on a notional principal amount while the other pays a floating rate, with only the net difference changing hands. The floating rate was historically tied to LIBOR, a benchmark discontinued after regulators found panel banks had manipulated it; transaction-based Alternative Reference Rates such as the Secured Overnight Financing Rate (SOFR) in the U.S. replaced it. The notional principal itself is never exchanged; it only determines the size of the interest payments.

Swaps are used mainly to change the interest rate exposure of an existing position without altering the underlying asset or liability. A company with floating-rate debt that prefers fixed payments can pay fixed and receive floating, converting its liability into a fixed-rate one from a cash-flow perspective. A pension fund matching long-dated fixed liabilities might do the reverse. Currency swaps, credit default swaps (which transfer credit risk, the subject of Chapter 9, rather than interest-rate risk), and total return swaps extend the same idea, exchanging one set of cash flows for another, to other kinds of risk.

Swaps trade almost entirely over-the-counter rather than on exchanges, which historically made their market opaque. Post-2008 reforms, discussed later in this chapter, pushed much of the standardized swap market toward centralized clearing to reduce the counterparty risk concentration that swaps had exposed during the financial crisis.

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Worked Example 3.1: Valuing an Existing Interest Rate Swap

A plain-vanilla interest rate swap can be valued as the difference between two bonds. The fixed leg is valued like a coupon bond, discounting each fixed payment plus a final notional repayment. The floating leg's coupon resets to the prevailing market rate, so at any reset date it is worth par (its notional value), whatever interest rates have done since the swap began. Consider a two-year swap with a \$10 million notional and a fixed rate of 5%, agreed some time ago, being valued today, immediately after a reset, when the current one-year and two-year discount rates are 4% and 4.5% respectively, lower than the 5% fixed rate locked in at initiation. The fixed leg's annual coupon is $0.05 \times 10{,}000{,}000 = \$500{,}000$, so its present value is

$$PV_{\text{fixed}} = \frac{500{,}000}{1.04} + \frac{500{,}000 + 10{,}000{,}000}{1.045^2} \approx \$10{,}095{,}933.72,$$

while the floating leg, valued at par immediately after its reset, is worth $PV_{\text{floating}} = \$10{,}000{,}000$. The swap's value to the party paying fixed and receiving floating is

$$V_{\text{fixed payer}} = PV_{\text{floating}} - PV_{\text{fixed}} = 10{,}000{,}000 - 10{,}095{,}933.72 \approx -\$95{,}933.72,$$

a loss to the fixed-rate payer and an equal gain to the floating-rate payer: market rates have fallen since the swap was struck, leaving the fixed-rate payer locked into paying 5% when the market now says 4–4.5% is fair. This is how an existing swap's mark-to-market value changes over its life without either counterparty doing anything: movement in market interest rates redistributes value between the two legs. It is the discounting logic Chapter 5 applies to bond pricing, applied to the difference of two cash flow streams.

Other Instruments

Other instruments include money market securities, preferred shares, mortgage-backed securities, and exchange-traded funds. Some are designed for short-term liquidity management, while others combine exposure to many assets in a single tradable instrument. Exchange-traded funds in particular have become widely used because they offer diversification and low-cost access to broad market exposure.

The growth of index funds and exchange-traded funds has changed the structure of investing. Many investors now buy broad baskets that track an index instead of selecting individual securities. This has lowered costs, increased participation, and changed how prices form in some markets, and it shows that modern finance is also about packaging and distributing risk efficiently.

An exchange-traded fund's price is kept close to the value of its holdings through a creation-and-redemption mechanism. Authorized institutional participants can exchange a basket of the underlying securities for new ETF shares, or vice versa, whenever the ETF's market price drifts far enough from its net asset value to make this profitable. This arbitrage, similar in spirit to the no-arbitrage logic in Section 3.5.3, keeps an ETF tracking its index closely in normal conditions, though it can break down under severe stress if the underlying securities become hard to trade.

3.2.5 Interest Rates and Central Banks

Every price in this chapter is quoted against one benchmark: the rate at which money can be borrowed overnight, which a central bank sets. The Federal Reserve announces a target range for the federal funds rate, the rate at which banks lend reserves to one another overnight. It keeps the market rate inside that range mainly by setting the interest it pays on reserves banks hold with it and the rate at which it takes cash from money funds overnight; the European Central Bank's deposit facility rate and the Bank of England's Bank Rate play the same role. No bank will lend overnight for much less than it can earn risk-free at the central bank, and no borrower with good collateral will pay much more than the central bank's lending rate. The policy rate therefore pins the whole overnight complex, including the overnight repo rate worked through later in this chapter and the SOFR benchmark, within a few basis points.

The central bank controls only the shortest rate directly. Longer rates move because the market expects the policy rate to move. Under the *expectations hypothesis*, a two-year yield is the average of the one-year rates expected over the next two years, plus a *term premium* that compensates for the risk of holding the longer bond and that here is set to zero:

$$(1 + y_2)^2 = (1 + r_1)(1 + r_2^e).$$

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Suppose the policy rate is 5% and the market expects it to stay there, so $y_2 = 5\%$ and a two-year zero-coupon bond paying \$100 is worth $100/1.05^2 = \$90.7029$. The central bank then surprises with a 50-basis-point cut, and the market comes to expect one more 25-basis-point cut next year, so the expected one-year rates become 4.50% and 4.25%. The two-year yield falls to $\sqrt{1.045 \times 1.0425} - 1 = 4.3749\%$, a move of 62.51 basis points, larger than the cut itself because it also prices next year's expected cut. The zero rises to $100/(1.045 \times 1.0425) = \$91.7926$, a gain of 1.2013% (a modified duration of 1.9048 approximates it as 1.1906%, the tool Worked Example 3.3 uses later in this chapter). A stock is a much longer-dated claim. Valued with Chapter 5's Gordon growth model, $P = D_1/(r - g)$, a stock paying $D_1 = \$4$ next year and growing at $g = 4\%$ is worth \$100 at a discount rate of 8%. If the discount rate falls by the same 50 basis points, to 7.5%, the price rises to \$114.29, a gain of 14.29%. Its sensitivity, $1/(r - g) = 25$ years, is more than ten times the two-year zero's, which is why equity markets react so strongly to central bank meetings. There is a caveat: a cut often signals that the central bank expects weaker growth, and if expected dividend growth also falls by 50 basis points, the stock's price does not move at all ($4/(0.075 - 0.035) = \$100$).

Transmission is therefore not mechanical, and the long end can even move the other way. Between September and December 2024 the Federal Reserve cut its target range by a full percentage point, yet the ten-year Treasury yield rose by roughly the same amount, as markets priced higher expected inflation, larger budget deficits, and fewer cuts ahead. With the policy rate near zero, central banks have also worked on the long end directly through *quantitative easing*, buying long-term government and mortgage bonds to compress the term premium. The effect of a policy-rate change depends on what it reveals about the future path of rates, growth, and inflation. That is why Chapter 16 parses central bank statements and Chapter 7 models the whole yield curve rather than a single rate.

3.2.6 Who Does What: The Buy Side, the Sell Side, and Rating Agencies

The participants described so far sort into two camps, a division that also organizes most careers in quantitative finance (Appendix C maps the roles). The **buy side** owns risk on behalf of investors and pays for services: asset managers and mutual funds, pension funds, insurers investing their premiums, sovereign wealth funds, endowments, family offices, and hedge funds. Their quantitative work is portfolio construction, alpha research, and risk management, and their performance is measured against a benchmark. The **sell side** sells services to the buy side and to issuers, and earns fees, spreads, and financing income rather than the return on a portfolio. Its core is the investment bank or broker-dealer, which does four things:

- **Underwriting**: helps issuers sell new stocks and bonds in the primary market, for a fee.
- **Sales and trading**: makes markets for clients in the secondary market, earning the bid-ask spread of Section 3.3.2 and bearing the inventory risk of Section 3.5.1.
- **Research**: publishes analysis of companies and markets, historically paid for through trading commissions.
- **Prime brokerage**: lends cash and securities to hedge funds, holds their collateral, and executes and finances their trades, including through derivatives such as the total return swaps of Section 3.6.4.

The two sides need each other but their incentives differ: a sell-side desk earns when clients trade and borrow, whatever the outcome for the client. Prime brokers therefore compete for a large hedge fund's business partly by offering lower margin, the competition at the center of the Archegos case at the end of this chapter.

A third group sits between issuers and investors. A **credit rating agency** publishes an opinion of an issuer's or a security's relative creditworthiness on a letter scale, from AAA through BBB and down to D (default); securities rated BBB− (Baa3 at Moody's) or higher are *investment grade*, and the rest are high yield, or junk. A rating is an ordinal ranking, not a default probability: it says AAA is safer than A, and only historical default studies, of the kind Chapter 9 uses to map ratings to probabilities, say by how much. Three agencies, S&P Global Ratings, Moody's, and Fitch, issue most ratings, and ratings are built into rules. Many investment mandates may hold only investment-grade bonds, and capital and collateral rules have long referred to ratings, so a downgrade across the investment-grade line can force selling. Since the 1970s the dominant business model has been *issuer pays*: the issuer of the security, not the investor relying on the rating, pays for it. The conflict mattered most for structured products, whose issuers could take

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a deal to whichever agency's model gave the best rating. The Financial Crisis Inquiry Commission found that from 2000 to 2007 Moody's rated nearly 45,000 mortgage-related securities triple-A, at a time when only six private-sector U.S. companies held that rating, and that 83% of the mortgage securities it rated triple-A in 2006 were ultimately downgraded (Financial Crisis Inquiry Commission 2011). The Dodd-Frank Act responded by creating an SEC Office of Credit Ratings and directing regulators to remove references to ratings from their rules. For model builders the lessons are two: a rating is a feature with a known incentive problem attached, and a label such as "AAA" can mean very different things across asset classes even when it is spelled the same.

3.3 Market Microstructure

Zooming in from the market to a single transaction reveals machinery that every later chapter's execution assumptions depend on. This part covers how orders meet, what a large one costs as it walks the book, and where the bid-ask spread comes from.

3.3.1 How Markets Are Organized

Some markets are exchanges with centralized order books; others are over-the-counter markets where participants trade directly with one another. In exchange-traded markets, buyers and sellers submit orders that are matched according to price and time priority. In over-the-counter markets, contracts are negotiated bilaterally and may be tailored to specific needs.

The distinction affects liquidity, transparency, counterparty risk, and transaction costs. Exchange-traded products are often easier to price and monitor, while over-the-counter trades offer flexibility but require more careful risk management. An exchange-listed stock can usually be traded quickly at a visible price, whereas a bespoke derivative contract may require negotiation, credit assessment, and collateral arrangements.

The study of these rules is called market microstructure. Even small changes in execution rules, fees, or order types can affect the quality of price discovery. A limit order may obtain a better price but may fail to execute. A market order executes immediately but at the cost of adverse price impact. These trade-offs are central to trading and to research in finance.

A simple measure of trading cost is the bid-ask spread:

$$\text{Spread} = P_{\text{ask}} - P_{\text{bid}}. \tag{3.1}$$

The spread is the cost of immediacy: a narrow spread means cheap trading and strong liquidity, a wide one the reverse. For quantitative researchers it is often as important as the asset's return, because it determines whether a strategy is profitable in practice.

3.3.2 Worked Example 3.2: Walking the Limit Order Book

The bid-ask spread describes the price of trading a single share. A limit order book lists several price levels on both sides, and a large order can "walk" through them, receiving a progressively worse average price. Table 3.1 shows a simplified order book for a stock.

Asks (sell orders)			Bids (buy orders)		
Level	Price	Size (shares)	Level	Price	Size (shares)
1	\$50.10	100	1	\$50.05	150
2	\$50.15	200	2	\$50.00	250
3	\$50.20	300	3	\$49.95	200

Table 3.1: A simplified limit order book

The best bid is \$50.05 and the best ask is \$50.10, so the quoted bid-ask spread is \$0.05, consistent with Equation (3.1). Figure 3.1 draws the same book as a depth chart, the standard way a trading desk visualizes

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it. Now suppose a trader submits a market buy order for 250 shares, more than the 100 shares available at the best ask. The order first consumes all 100 shares at \$50.10, then the remaining 150 shares at the next level, \$50.15:

$$\text{Average execution price} = \frac{100(50.10) + 150(50.15)}{250} = \frac{5{,}010 + 7{,}522.50}{250} = \$50.13.$$

This average price is \$0.03 above the best ask of \$50.10, roughly 6 basis points of price impact, a consequence of order size relative to the liquidity at each price level, with no new information entering the market. A trader with a large position faces a trade-off: submitting the whole order at once guarantees immediate execution at a worse average price, while breaking it into smaller pieces over time (the execution algorithms of Chapter 13) reduces price impact but exposes the trader to price movements while the order is worked.

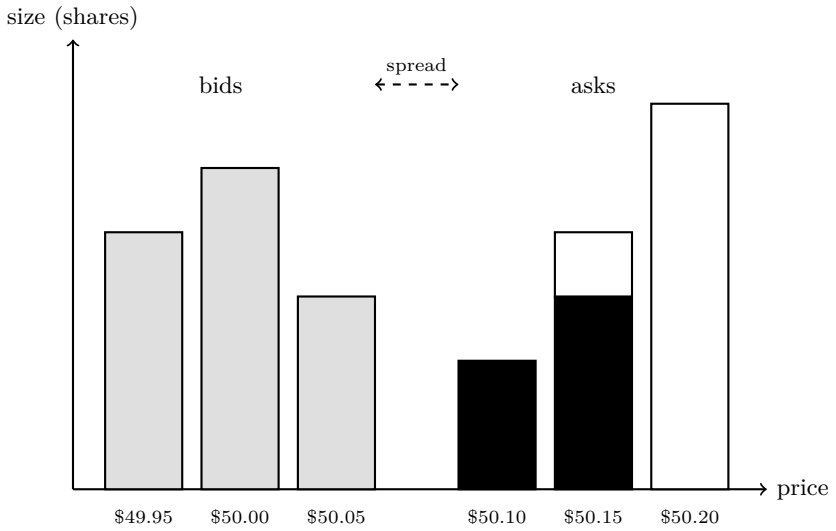


Figure 3.1: The limit order book of Table 3.1 as a depth chart. Bid sizes (light gray) sit left of the spread, ask sizes (white outline) right; the solid black portion of the two lowest ask levels marks what the 250-share market order consumes.

3.3.3 Decomposing the Bid-Ask Spread

The bid-ask spread is not pure profit for a market maker; it compensates for at least three distinct costs, and the decomposition explains why spreads widen in some conditions and narrow in others.

- **Order-processing cost**: the operational cost of handling a trade, which is largely fixed regardless of market conditions.
- **Inventory-holding cost**: compensation for the risk a market maker bears by holding a nonzero position while waiting for an offsetting order to arrive, discussed further in Section 3.5.1.
- **Adverse-selection cost**: compensation for the risk of trading with a counterparty who has better information, since an informed trader will systematically pick off stale quotes.

A trade price bounces between the bid and the ask as buy and sell orders arrive, even when the asset's underlying value has not changed. A trade at the ask followed by a trade at the bid looks, in the raw price series, like the price falling and then recovering, so a wider spread mechanically produces larger back-and-forth price changes. Roll's implied spread estimator (Roll 1984) turns this bounce into a measurement tool, recovering the effective spread from observed price changes alone, without seeing the order book,

$$\text{Spread} \approx 2\sqrt{-\text{Cov}(\Delta P_t, \Delta P_{t-1})}, \tag{3.2}$$

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because this bid-ask bounce induces negative autocorrelation in consecutive price changes, which the covariance term captures. This is a useful reminder for the time series methods of Chapter 7: not every statistical pattern in price data reflects economically meaningful predictability; some of it is a mechanical consequence of market microstructure. The book's companion technical note on Roll's estimator derives the formula. It shows that a constant half-spread applied according to the direction of each arriving trade makes observed price changes behave like a first-order moving-average process, with an autocovariance of $-s^2/4$ at lag one and zero at every longer lag, and that bounce alone can never drive the first-lag autocorrelation below $-1/2$ however wide the spread.

3.3.4 The Economics of Price Discovery

Financial prices are the output of a process in which participants with different information and objectives update their beliefs. A stock price may rise because investors revise their expectations of future cash flows, because risk premiums change, or because a large buyer enters the market. The same price change can have several interpretations, which is why finance is both quantitative and interpretive.

Because information, incentives, and trading frictions all affect prices, they do not always move smoothly. Sudden moves can reflect news, uncertainty, algorithmic trading, or liquidity shocks. In the short run, prices may deviate from fundamental values because of temporary imbalance or limited liquidity; in the long run, competition and arbitrage tend to push them toward values consistent with available information and fundamentals.

Economists and practitioners therefore distinguish fundamentals from short-run market dynamics. A fundamental analyst asks what an asset ought to be worth given expected cash flows and risk. A market microstructure analyst asks how trading rules and order flow shape the observed price path. Both perspectives are needed.

Price discovery also matters for AI applications. A model that finds patterns in observed prices without understanding the market mechanism behind them may mistake noise for signal, and a model that predicts price movements may still fail if it ignores transaction costs, market impact, and changing liquidity.

3.3.5 Why Market Structure Matters for Quantitative Finance

A portfolio manager does not merely choose assets by expected return and risk. The manager must also consider how easily those assets can be bought or sold, how much price impact trading will create, and whether the strategy survives transaction costs.

The same logic applies to credit risk models, fraud detection systems, and forecasting algorithms. A model that performs well on historical data may fail in practice if it ignores the institutional setting in which decisions are implemented. Finance is a problem of data, incentives, constraints, and execution, and the most successful applications of AI combine rigorous modeling with an understanding of the market mechanism rather than treating markets as a black box.

Suppose a machine learning model, trained only on historical price and volume data, predicts that a small-cap stock will outperform over the next week.

Before acting on this signal, a sophisticated practitioner asks institutional questions the model cannot answer. How wide is the bid-ask spread, and does the expected outperformance exceed the round-trip cost of trading, using the spread decomposition above? What share of the stock's average daily volume would the position represent, and would building it through the order-book mechanics of Section 3.3.2 move the price enough to erode the gain? Is the stock easy to borrow if the strategy requires shorting, and are there restrictions on shorting small-cap names under stress?

None of these questions is visible in the price data the model was trained on, yet each can determine whether a statistically significant signal is also economically exploitable. Market structure, not just statistical significance, decides whether a trading idea survives contact with a real market.

3.4 Market Infrastructure and Regulation

A trade is not finished when it is agreed. This part follows it through clearing and settlement, examines the businesses and rules that surround it, and connects that plumbing to where machine learning is deployed.

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3.4.1 Money Markets, Capital Markets, and the Lifecycle of a Trade

Financial markets are often divided into money markets and capital markets. Money markets are used for short-term borrowing and lending, typically for maturities of less than one year. They include Treasury bills, commercial paper, certificates of deposit, and repurchase agreements, and are crucial for liquidity management, since banks, corporations, and governments often need to finance short-term obligations or invest excess cash briefly. Chapter 1's Table 1.4 previewed money markets alongside equity, bond, derivatives, and foreign exchange markets; this section and Chapter 5 develop each of these further.

Capital markets, by contrast, are used for long-term financing. They include equity and long-term debt markets and fund business investment and public infrastructure. A firm expanding production may issue long-term bonds or equity; a government may issue sovereign debt to finance roads, schools, or defense. In both cases, the capital market separates the financing decision from the issuer's immediate cash position.

Another distinction is between primary and secondary markets. In a primary market, new securities are issued for the first time. In a secondary market, existing securities are traded among investors. The primary market raises funds for the issuer, while the secondary market sets the price of securities after issuance. A liquid secondary market reduces uncertainty for investors and so makes it easier for firms to issue securities in the first place.

The lifecycle of a trade shows why market structure matters in practice. An investor decides to buy or sell an asset, and the order is routed through a broker to an exchange or an over-the-counter dealer. In an electronic, highly liquid market the order may execute almost immediately; in a fragmented or less liquid one it may be partially filled or require price concessions. After the trade, settlement transfers ownership and exchanges cash. In many markets settlement takes a few days, during which counterparty and operational risk remain.

The process reflects the interaction of incentives, technology, and regulation. High-frequency trading firms may compete on execution speed, while institutional investors prioritize minimizing market impact. A retail investor may value simplicity and low cost, whereas a hedge fund may bear more risk for a strategic advantage. All of these objectives interact in the same marketplace.

The repurchase agreement, or repo, deserves particular attention because of its central role in the plumbing of the financial system. In a repo, one party sells a security, typically a government bond, to another while agreeing to repurchase it at a slightly higher price on a specified future date, often the next day. Economically, this is a collateralized short-term loan: the cash borrower posts the security as collateral, and the difference between the sale and repurchase price is the effective interest rate, known as the repo rate. Banks, hedge funds, and other institutions rely on the repo market to fund large securities positions daily. A disruption in repo markets, as occurred at several points during the 2008 crisis, can therefore force institutions to sell assets they would otherwise have held, transmitting stress across the financial system as described in the systemic risk discussion below.

Suppose a dealer needs to borrow \$10,000,000 overnight and posts Treasury securities as collateral in exchange for cash at an annualized overnight repo rate of 5%. Using the money market convention of a 360-day year, the interest owed for one day is

$$\$10{,}000{,}000 \times 0.05 \times \frac{1}{360} = \$1{,}388.89,$$

so the dealer repurchases the collateral the next day for \$10,001,388.89. This small daily interest, applied to an enormous notional and rolled over every business day, is why even a small rise in overnight repo rates can signal a meaningful shortage of cash for short-term collateralized lending, as in September 2019, when repo rates briefly spiked well above the Federal Reserve's target range.

Settlement conventions also matter in practice. Most equity markets settle trades on a T+1 or T+2 basis (one or two business days after the trade date), so ownership and cash do not change hands when a trade is agreed. This gap creates counterparty risk, the possibility that the other side fails to deliver cash or securities as promised, which the clearinghouses discussed below exist to mitigate.

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3.4.2 The Business of Exchanges and Market Data

It is easy to think of an exchange as neutral infrastructure, a venue where buyers and sellers meet, without asking how it makes money. Modern exchanges are for-profit businesses with several revenue streams, and these explain features of market structure that would otherwise seem arbitrary.

Listing fees are charged to companies for the right to have their shares traded on an exchange, giving exchanges an incentive to attract high-quality, high-volume issuers. Transaction fees are charged per trade, often under a "maker-taker" model in which an exchange pays a small rebate to participants who post resting limit orders (adding, or "making," liquidity) and charges a slightly larger fee to participants whose orders execute immediately against those resting orders (removing, or "taking," liquidity). This fee structure is a deliberate subsidy for market making, and it materially affects the profitability of the market makers discussed in Section 3.5.1.

Market data fees are a third, increasingly significant revenue source: exchanges sell real-time price and order-book data to trading firms, and the most detailed, lowest-latency feeds command large premiums over the free or delayed data available to retail investors. Finally, co-location fees, charged for placing a firm's servers inside or next to the exchange's data center, monetize the speed advantage discussed later in this chapter. A co-located firm receives order-flow information and executes its orders with lower latency than a remote firm, purely because data travels a shorter physical distance.

These revenue streams matter to a quantitative reader for two reasons. First, they explain why exchanges have commercial incentives to attract high-frequency trading in particular, which generates outsized transaction and data-fee revenue relative to its economic footprint; this feeds a broader regulatory debate about whether market structure balances the interests of high-frequency participants against those of long-term investors. Second, the market data used throughout this book, assumed free in every worked example so far, is a commercial product whose cost, coverage, and latency depend on how much a firm will pay, a constraint that shapes which strategies a research budget can support.

3.4.3 Clearing, Settlement, and Systemic Risk

Once a trade is executed, the market must ensure that it is honored. Clearing determines the obligations between counterparties; settlement is the final exchange of cash and securities. In exchange-traded markets, a clearinghouse often acts as the central counterparty, guaranteeing the trade if one side defaults. This reduces bilateral credit risk, but it also creates a new concentration of risk in the clearinghouse itself.

Clearing and settlement matter most under stress. If many participants must liquidate positions at once, the system may face liquidity shortages and margin calls, and a small shock can propagate through the network of institutions that depend on one another for financing and collateral. This is the core idea of systemic risk: the failure of one institution or market segment can quickly undermine the stability of the broader financial system.

The 2008 global financial crisis showed how quickly seemingly isolated problems become systemic. U.S. mortgage defaults spread through securitization structures, derivatives markets, and funding markets, eventually affecting banks, insurers, and governments worldwide. The lesson was not only that risk had been mispriced, but that the financial system had become highly interconnected and insufficiently transparent. The crisis also highlighted the importance of rules on leverage, capital, and liquidity.

A central clearinghouse manages its concentrated risk mainly through margin requirements. Each clearing member posts collateral, called initial margin, sized to cover the clearinghouse's potential loss if that member defaults under plausible adverse price moves, and posts variation margin daily (or, in volatile periods, intraday) to cover losses on open positions. This protects the clearinghouse, but a sharp price move can trigger margin calls across many participants at once, forcing coordinated selling when the market is least able to absorb it, a dynamic visible in several episodes of the 2008 crisis. Major clearinghouses have become large and interconnected enough with their member banks that regulators now treat them as systemically important institutions, subject to their own capital and risk-management standards.

Settlement is also where new market plumbing has appeared, in the form of stablecoins and tokenized securities. Both are best understood alongside the systems that already move money, so the next subsection takes them up together.

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3.4.4 Payment Infrastructure and Digital Assets

The same three questions asked of securities clearing (who owes whom, when an obligation becomes final, and who bears the risk in between) arise every time money moves. This subsection looks first at the payment rails that move money today, then at the digital assets, stablecoins and tokenized securities, that propose to move money and securities together on a new kind of ledger, and closes briefly with central bank digital currencies. The subject is plumbing and its risks. Crypto-asset trading strategies are outside this book's scope, as the preface states, and nothing here is an argument for or against holding any digital asset.

How Payments Settle

Payment systems differ along two dimensions that matter for risk: whether each payment settles individually (gross) or in batches netted against one another, and how long a payment takes to become final, meaning legally irrevocable. Table 3.2 summarizes the four main designs.

A card payment is authorized in seconds: the merchant's bank (the acquirer) sends a message through the card network to the cardholder's bank (the issuer), which checks the available balance or credit line and a fraud score before approving. The money moves later. Transactions are cleared in batches and settled between issuer and acquirer a day or two afterward, and the cardholder keeps the right to dispute a charge, a chargeback, for weeks. Card payments are therefore fast to approve but slow to become final, and that delayed finality makes card-fraud losses at least partly recoverable, the setting for much of Chapter 15. The automated clearing house (ACH) network in the United States, which carries payroll, bill payments, and direct debits, is batch-based too: banks submit files of payments, the operator nets them, and settlement occurs at fixed windows during the business day. Netting economizes on liquidity, since each bank funds only its net position, but a payment remains a promise until its batch settles, and some debits can be returned days later.

Large-value payments between banks instead settle through real-time gross settlement (RTGS) systems operated by central banks: the Fedwire Funds Service in the United States, CHAPS in the United Kingdom, and T2 in the euro area. Each payment settles individually and immediately in central bank money and is final the moment it settles. The price of that safety is liquidity: with no netting, a bank must hold or borrow enough reserves to fund every payment as it goes, which is why central banks extend intraday credit to RTGS participants.

Instant payment rails bring RTGS-style finality to everyday payments. The Clearing House's RTP network (launched in 2017) and the Federal Reserve's FedNow Service (launched in July 2023) settle individual payments within seconds, around the clock and every day of the year, as do the United Kingdom's Faster Payments and the euro area's SEPA Instant Credit Transfer scheme. Finality cuts both ways: a firm paying a supplier wants it, and so does a fraudster. In an authorized push payment (APP) scam, the victim is manipulated into sending the money personally, against a fake invoice or at the request of someone impersonating the bank. Once the payment settles on an instant rail there is no chargeback to reverse it, and the funds are typically moved onward through mule accounts within minutes. In the United Kingdom, where APP losses on Faster Payments became a policy issue, sending and receiving payment firms have since 7 October 2024 been required to reimburse most victims, up to £85,000 per claim. Chapter 15 treats APP fraud as a detection problem; the structural point here is that the faster and more final a rail, the more of the fraud screening must happen in the seconds between the payment instruction and settlement, because nothing can be clawed back afterward.

Rail	Settlement	When final	Typical use
Card networks	Batched, net	Days; disputable later	Retail purchases
ACH	Batched, net	Same day to two days	Payroll, bills, direct debits
RTGS (Fedwire, CHAPS, T2)	Individual, gross	Immediately	Large interbank payments
Instant (RTP, FedNow)	Individual, gross	Seconds, around the clock	Person-to-person, business

Table 3.2: Four payment rails compared by how they settle and when a payment becomes final

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Stablecoins: Reserves, Redemption, and Run Risk

A stablecoin is a token on a blockchain designed to hold a fixed value, almost always \$1. It does so by being redeemable on demand at par from an issuer that holds reserves of cash and short-term Treasuries against the tokens outstanding. (Algorithmic stablecoins, which tried to hold the peg without full reserves, are a different design, largely discredited by the collapse of TerraUSD in May 2022; this subsection concerns the fiat-backed reserve model that now dominates and that regulators have adopted.) A fiat-backed stablecoin is economically a money-market fund share that gives up yield in exchange for functioning as money. The issuer keeps the interest earned on the reserves, which is its business model, and the holder gets a token that moves around the clock on a blockchain and can serve as the cash leg of the tokenized-securities trades described below. The quality of its reserves is the credit question Chapter 9 would ask of any issuer.

The promise of redemption at par is also the source of fragility. A holder who doubts the reserves can redeem at \$1 today; if enough holders do, the issuer must sell reserve assets to pay them, possibly at a loss, and the loss falls on those who wait. This is the structure of a bank run (Diamond and Dybvig 1983), and stablecoins have "broken the buck" in this way. In March 2023 the issuer of USDC disclosed that \$3.3 billion of its roughly \$40 billion of reserves was held at Silicon Valley Bank, which had just failed. The token traded below 90 cents before recovering within days, once U.S. authorities guaranteed that bank's deposits.

Regulation has converged on one design, as of October 2026. In the United States, the GENIUS Act, signed into law on July 18, 2025, requires a permitted payment stablecoin issuer to hold identifiable reserves on at least a one-to-one basis, limited to cash, insured deposits, Treasury bills with a remaining maturity of 93 days or less, Treasury-backed repurchase agreements, government money-market funds, and central bank reserves; to publish the composition of its reserves monthly, examined by a registered public accounting firm; to disclose a redemption policy providing for timely redemption; and not to pay holders any interest or yield (U.S. Congress 2025). The Act takes effect on the earlier of January 18, 2027 (eighteen months after enactment) and 120 days after the federal regulators issue final implementing rules, which as of October 2026 had been proposed but not finalized. In the European Union, the Markets in Crypto-Assets Regulation (MiCA) applied its stablecoin provisions from 30 June 2024 and the rest of the regulation from 30 December 2024. It gives holders of an e-money token, the MiCA category for a stablecoin tied to a single official currency, the right to redeem at any time and at par, bars issuers from paying interest, and requires that at least 30% of the funds received be held as deposits at credit institutions, with the remainder in highly liquid, low-risk instruments (EU 2023). Full reserves, short maturities, redemption at par, and no interest: the next worked example shows why the second of these matters.

Worked Example 3.3: A Stablecoin Reserve Under a Redemption Run

An issuer has $S = \$10{,}000$ million of tokens outstanding, each redeemable at \$1, backed one for one by $C = \$1{,}500$ million of cash (15%) and $B = \$8{,}500$ million of zero-coupon Treasuries bought at a yield of $y = 4\%$. At that yield the Treasuries earn the issuer $0.04 \times 8{,}500 = \$340$ million a year, none of which is paid to holders. Two things then happen: yields rise by $\Delta y = 100$ basis points, and 30% of holders redeem, $R = \$3{,}000$ million.

Measuring the damage requires one tool from fixed income, developed fully in Chapter 5 and needed here only for the simplest bond. A zero-coupon bond paying \$1 in T years is worth $P = (1+y)^{-T}$, so $dP/dy = -\frac{T}{1+y}P$, and for a small yield change

$$\frac{\Delta P}{P} \approx -D_{\text{mod}}\,\Delta y, \qquad D_{\text{mod}} = \frac{T}{1+y}, \tag{3.3}$$

where D_{mod} is the bond's modified duration: the percentage price loss per unit rise in yield. Compare two reserve portfolios. A compliant one holds three-month bills, $T = 0.25$, inside the GENIUS Act's 93-day limit: $D_{\text{mod}} = 0.25/1.04 = 0.2404$, so the bills lose about 0.2404% (exact repricing gives 0.2390%), a mark-to-market loss of $L = 8{,}500 \times 0.002404 = \$20.43$ million. An extended one holds two-year zeros, $T = 2$: $D_{\text{mod}} = 2/1.04 = 1.9231$, a loss of about 1.9231% (exact 1.8957%), or $L = \$163.46$ million, eight times as much.

The redemption is paid first from cash and then by selling Treasuries. The issuer needs $3{,}000 - 1{,}500 = \$1{,}500$ million from sales, and at marked-down prices that means selling Treasuries that cost

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1,500/(1 − 0.002404) = \$1,503.61 million in the compliant case (a realized loss of \$3.61 million) and 1,500/(1 − 0.019231) = \$1,529.41 million in the extended case (a realized loss of \$29.41 million). The redeeming holders are paid exactly \$1 each, so the whole mark-to-market loss L, realized or not, is left to the $S - R$ tokens still outstanding. Reserves per remaining token are

$$c = \frac{S - L - R}{S - R} = 1 - \frac{L}{S - R}, \tag{3.4}$$

which gives 1 − 20.43/7,000 = \$0.9971 for the compliant issuer and 1 − 163.46/7,000 = \$0.9766 for the extended one. Table 3.3 repeats the calculation as the redeemed share grows.

Share redeemed	Three-month bills ($T = 0.25$)	Two-year zeros ($T = 2$)
0%	0.9980	0.9837
15%	0.9976	0.9808
30%	0.9971	0.9766
60%	0.9949	0.9591

Table 3.3: Reserves per remaining token, in dollars, after a 100-basis-point rise in yields, by share of tokens redeemed at par

Equation (3.4) is the arithmetic of a run. Every holder who redeems early receives a full dollar, and the same loss is spread over fewer remaining tokens, so the value left behind falls as redemptions grow: the extended issuer's remaining holders have 98.37 cents of reserves per token before anyone redeems and 95.91 cents after 60% have left. A holder who expects others to redeem should therefore redeem first, and the expectation can fulfill itself. The 93-day limit does not remove this first-mover advantage; it shrinks it by the ratio of the durations, 1.9231/0.2404 = 8, so that the compliant issuer's worst case in the table is about half a cent below par, small enough for a modest capital buffer to absorb. A larger cash buffer, such as MiCA's 30% minimum, lets the issuer meet a redemption without selling anything, which avoids the realized loss, but the economic loss on the Treasuries still held is unchanged by the decision not to sell. The same arithmetic, applied to a bank funding long-duration securities with runnable deposits, is what brought down Silicon Valley Bank in 2023. The first-order approximation slightly overstates the losses (compare 1.9231% with the exact 1.8957%), a convexity effect explained in Chapter 5.

Tokenized Securities and Money-Market Funds

Asset tokenization means issuing a conventional security (a money-market fund share, a Treasury bill, a bond) as a digital token whose ownership record is a blockchain rather than an entry at a custodian or transfer agent. The motivation is settlement economics, not speculation. A tokenized security can settle in minutes rather than days, with delivery and payment executed atomically in a single transaction so that neither leg can fail on its own, collapsing both the intermediation cost and the counterparty-risk window explained earlier. Atomic settlement requires both legs to live on the same rails, which is where a stablecoin or tokenized bank deposit supplies the cash leg. Tokenized money-market funds are the most developed case: BlackRock's USD Institutional Digital Liquidity Fund (BUIDL), launched on the Ethereum blockchain in March 2024, invests in cash, Treasury bills, and repurchase agreements and records each investor's shares as tokens that can be transferred around the clock and, in principle, posted as collateral while still earning the fund's yield.

Tokenization changes the settlement cycle, the hours during which an asset can move, and the possibility of attaching rules (who may hold the token, when it pays out) directly to the asset as code. It does not change the asset's economics: a tokenized Treasury bill has the same duration and credit risk as the bill, and the run arithmetic of Equation (3.4) applies to any token promising par redemption against assets that can fall in value. It does not by itself settle legal questions, since a holder's claim is only as good as the law's recognition of the token record and the issuer's redemption terms. And atomic settlement gives up something that batch settlement provided: because each trade must be fully funded at the moment it settles, the liquidity savings of netting disappear, the same trade-off between safety and liquidity that separates RTGS from ACH.

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For a quantitative reader the consequence is concrete. A settlement cycle collapsing from days to minutes changes collateral requirements, intraday liquidity management, and the practical meaning of counterparty exposure, all of which feed the risk models of Chapters 8 and 9. The venues where tokenized instruments trade also have formula-driven rather than dealer-driven market making, taken up in Chapter 13.

Central Bank Digital Currencies

A central bank digital currency (CBDC) is a digital form of money that is a direct liability of the central bank, held either by the public (a retail CBDC) or only by banks for settling with one another (a wholesale CBDC). Because it is central bank money, a CBDC has no reserve portfolio and none of the run arithmetic above. The concern runs the other way: in a crisis households could move deposits out of banks and into CBDC at the click of a button, which is why proposals such as the digital euro limit individual holdings. Policy has diverged, as of October 2026. China has piloted its retail e-CNY widely; the Eurosystem has been preparing a possible digital euro since November 2023, with any issuance conditional on EU legislation; and in the United States an executive order of January 23, 2025, prohibits federal agencies from establishing, issuing, or promoting a CBDC, leaving regulated stablecoins and tokenized deposits as the U.S. route to digital dollars. Central banks themselves increasingly emphasize wholesale uses, tokenized central bank reserves settling against tokenized deposits and government bonds on a shared ledger (Bank for International Settlements 2025).

For this book, the data matter as much as the designs. Instant payment rails produce the real-time transaction streams that Chapter 15's models must score before a payment becomes final, and public blockchains publish every token transfer, the on-chain data that Chapter 17 treats as an alternative data source.

3.4.5 Regulation, Disclosure, and Market Integrity

Markets need rules because they rest on trust, information, and incentives. Without regulation, participants may exploit others through fraud, insider trading, or manipulation. Disclosure requirements help investors evaluate firms and make informed decisions. Capital and liquidity rules reduce the chance that institutions will become insolvent or unable to meet obligations during stress. Market surveillance systems detect suspicious behavior and support confidence in the fairness of the trading process.

The balance between free markets and regulation is a persistent theme. Too little regulation may encourage excessive risk-taking and misconduct; too much may reduce innovation and impose unnecessary compliance costs. Effective frameworks protect integrity and stability while preserving the ability of markets to allocate capital efficiently.

For AI and data science, regulation also shapes the data environment. Financial institutions must report certain information, comply with auditing standards, and keep transparent records, which creates rich datasets for forecasting, fraud detection, and risk analysis. Regulatory constraints also limit what can be done with the data, especially where privacy, fairness, or market abuse is involved.

A few regulatory frameworks recur throughout the book and are worth knowing by name.

- **The Securities and Exchange Commission (SEC)** oversees U.S. securities markets and disclosure.
- **Regulation NMS** (National Market System) governs how orders are routed and executed across competing U.S. exchanges, directly shaping the market microstructure discussed throughout this chapter.
- **The Dodd-Frank Act**, passed in 2010 in response to the 2008 financial crisis, introduced the central clearing requirements for standardized swaps mentioned above, along with the Volcker Rule restricting proprietary trading by banks, and stress-testing requirements for large financial institutions that are directly relevant to the risk models in Chapters 8 and 9.
- **MiFID II** (the Markets in Financial Instruments Directive), in Europe, imposes extensive transparency and best-execution requirements on trading venues and investment firms.
- **Basel III** is an international framework setting minimum capital and liquidity requirements for banks, intended to make the banking system more resilient to the kind of interconnected shocks described in the systemic risk discussion above.

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None of these frameworks is a topic this book covers in depth, but recognizing their names and general purpose is useful context for the compliance-related material in Chapter 15.

3.4.6 The Link Between Market Structure and AI

A modern AI system in finance cannot be designed as if markets were simple, frictionless, and fully observable. Real markets have heterogeneous participants, delayed information, hidden inventory, and strategic behavior. An algorithm that trades on price patterns may perform well in a calm market but fail under stress, when liquidity evaporates and spreads widen. A credit model built on historical repayment data may become less reliable if underwriting standards change or the economy enters a new regime.

Financial machine learning is therefore an engineering problem embedded in an economic environment. The model is one component of a system that includes data pipelines, risk controls, execution logic, and compliance procedures, and the same prediction can have very different consequences depending on how it is used. A recommendation harmless in research may be dangerous in live trading if it creates concentration risk or triggers herding.

The most valuable AI applications in finance respect the structure of the market: they use data carefully, incorporate financial theory, and are judged by their economic and operational consequences as well as their predictive accuracy.

The connection also runs the other way: AI systems have become part of the market structure they operate within.

The high-frequency market-making firms discussed later in this chapter increasingly use machine learning to set quotes and manage inventory risk in real time. Large asset managers use natural language processing (Chapter 16) to parse earnings calls and regulatory filings faster than human analysts. Exchanges use anomaly-detection systems, related to the fraud-detection methods of Chapter 15, to flag potentially manipulative trading for regulatory review. The "other participants" whose behavior shapes liquidity, spreads, and price discovery are thus increasingly AI systems, whose collective behavior any new AI system entering the market must anticipate.

This has a subtle consequence for model validation. A backtest evaluates a strategy against data generated by the market as it existed in the past, including the mix of human and automated participants active then. If much of that liquidity and price-setting came from algorithms since retired, upgraded, or replaced, the market a new strategy faces may behave differently from its backtest, not because fundamentals changed but because the population of participants generating prices has changed. This is a subtler version of the non-stationarity concern raised in Chapter 1 and developed in Chapter 6, specific to markets with heavy algorithmic participation.

3.5 Liquidity, Leverage, and Arbitrage

Prices hold together because someone is willing to trade against a discrepancy. This part covers who provides that liquidity and on what terms, what borrowing to trade costs when it goes wrong, and the no-arbitrage logic the rest of the book prices with.

3.5.1 Liquidity, Leverage, and the Economics of Market Making

Liquidity is one of the most important but least visible features of financial markets. A liquid market allows participants to trade quickly without moving the price much. Liquidity has several dimensions:

- **Depth** is the volume that can be traded at a given price.
- **Immediacy** is the speed with which an order can be executed.
- **Resiliency** is the speed with which the market returns to a stable state after a shock.
- **Tightness** is the narrowness of the bid-ask spread.

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A liquid market is easier to use for hedging, investment, and risk management, because traders can enter and exit positions more cheaply.

Market makers do more than passively post a bid and an ask: they absorb temporary imbalances between buy and sell orders, buying when others want to sell and selling when others want to buy. This is where inventory risk bites. If the market maker buys an asset and its price falls, the position loses value; if it sells short and the price rises, losses can also accumulate quickly. The willingness of market makers to bear this risk is central to market functioning.

Market making is also shaped by information asymmetry. When some traders have better information, market makers face adverse selection. An informed trader who expects an asset to rise is more likely to buy, and one who expects it to fall is more likely to sell. The market maker, who cannot tell whether a trade conveys good or bad news, may respond by widening spreads and reducing participation, one reason trading costs often rise in uncertain periods.

Leverage is another essential concept here. A trader who borrows to increase position size magnifies both gains and losses. Margin requirements limit how large a leveraged position can become relative to the trader's equity. When prices move against the position, a margin call may force liquidation, and that forced selling can amplify price moves and add stress to the market. Leverage, useful for efficient investment and speculation, is therefore also a source of instability when risk management is weak.

3.5.2 Worked Example 3.4: Short Selling and Margin Calls

Leverage and margin are easiest to understand through a short sale. Suppose a trader believes a stock currently priced at \$50 is overvalued and borrows 100 shares to sell short, receiving proceeds of $100 \times \$50 = \$5{,}000$. Under a typical initial margin requirement of 150% of the value of the position (the U.S. Regulation T standard), the trader must maintain total account equity of at least $1.5 \times \$5{,}000 = \$7{,}500$: the \$5,000 in short-sale proceeds plus an additional \$2,500 of the trader's own cash deposited as margin.

As the stock price P moves, the trader's liability, the cost of buying back the 100 borrowed shares, is $100P$, while total account equity remains the fixed \$7,500 credit balance minus that liability:

$$\text{Equity}(P) = 7{,}500 - 100P.$$

For short positions in stocks priced at \$5 or more, FINRA Rule 4210 requires a maintenance margin of at least 30% of the current value of the short position (many brokers require more), meaning $\text{Equity}(P)/100P \geq 0.30$. Setting this ratio equal to 0.30 and solving for P gives the price at which a margin call is triggered:

$$7{,}500 - 100P = 0.30(100P) \quad \Longrightarrow \quad 7{,}500 = 130P \quad \Longrightarrow \quad P \approx \$57.69.$$

At this price (7,500/130 at full precision), equity has fallen to \$1,730.77, exactly 30% of the trader's liability, confirming the calculation. A price increase of only 15.38%, from \$50 to \$57.69, triggers a margin call on a position that started with 150% initial margin, which shows how leverage compounds a short seller's losses. The loss on the position is \$769.23, but it consumes 30.77% of the trader's own \$2,500 of capital, twice the adverse price move in percentage terms. If the trader cannot deposit more funds once the call is issued, the broker buys back some or all of the borrowed shares on the trader's behalf at whatever price is available. This is the same price-insensitive trading into a moving market that the Flash Crash vignette describes and, at the scale of a whole leveraged fund, the LTCM episode discussed later in this chapter.

Variation margin, the clearinghouse mechanism of Section 3.4.3, differs from broker margin in one important respect: rather than being called only once a position breaches a maintenance threshold, it is settled at the end of every trading day, marking every open position to its current market price.

Suppose a trader buys one stock-index futures contract with a notional value of \$50,000 and posts an initial margin of \$5,000 (10% of notional). If the index falls 2% that day, the trader's mark-to-market loss is $0.02 \times \$50{,}000 = \$1{,}000$, reducing the margin balance to $\$5{,}000 - \$1{,}000 = \$4{,}000$; the clearinghouse issues a variation margin call for \$1,000 to restore the balance to \$5,000 before trading resumes the next day. Because settlement is daily, losses on a futures position never accumulate silently as they can in the short-sale example. But a clearing member with many leveraged futures positions faces a new, potentially large cash obligation every time the market moves sharply against it, the liquidity drain described in Section 3.4.3.

Table 3.4 compares the two margin mechanics, which are easy to conflate despite their different timing and triggers.

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	Short-sale margin (Section 3.5.2)	Futures variation margin
Counterparty	Broker	Central clearinghouse
Settlement frequency	Only when a threshold is breached	Every trading day
Trigger	Equity falls below maintenance %	Any price move, however small
Typical use case	Individual equity short position	Standardized futures and swaps

Table 3.4: Two margin mechanisms compared: bilateral broker margin on a short sale versus centrally cleared futures margin

3.5.3 A Simple No-Arbitrage Example

A simple arbitrage argument shows why prices in finance are constrained by structure. Suppose an investor can buy a one-period risk-free bond that costs 100 and pays 105 one year later. The risk-free interest rate implied by this bond is

$$r = \frac{105 - 100}{100} = 0.05.$$

Now suppose there is also a claim that pays one unit of money with certainty in one year. If the claim were priced at 0.95 today, then an investor could buy the claim for 0.95 and receive 1 next year. This is a return of about 5.26%, higher than the risk-free rate available from the bond, so rational investors would buy the claim and sell the bond, pushing prices back into line. The no-arbitrage principle says that two assets with the same payoff must have the same price; otherwise investors can make a riskless profit. This idea underlies pricing in derivatives, fixed income, and valuation more broadly.

Financial prices are not arbitrary; they are linked by replication and the absence of arbitrage. A quantitative analyst should therefore not interpret a model output in isolation. A model price that violates basic consistency conditions may signal an error in the model, a missing constraint, or an overlooked transaction cost, one reason financial modeling is both statistical and economic. The same logic extends to options, swaps, and structured products, where the challenge is often to identify the replicating portfolio and the correct discounting. Such relationships make finance a systematic discipline rather than a descriptive one: even when markets look noisy, strong forces keep valuations tied to economic logic.

A second example, a forward contract, makes the same point with a replicating portfolio rather than a single comparison bond. Suppose a stock trades today at $S_0 = \$100$, pays no dividends over the next year, and the risk-free rate is $r = 5\%$. A one-year forward contract obligates its holder to buy the stock in one year at a price fixed today; the no-arbitrage forward price is

$$F_0 = S_0(1 + r) = 100(1.05) = \$105,$$

the cost of replicating the forward's payoff by borrowing \$100 today at the risk-free rate, buying the stock now, and holding it for one year (a strategy known as cash-and-carry).

If the forward instead traded in the market at $F_{\text{mkt}} = \$110$, an arbitrageur could borrow \$100, buy the stock, and simultaneously sell (go short) the forward at \$110; in one year, the arbitrageur delivers the stock against the forward for \$110, repays the loan plus interest of $100(1.05) = \$105$, and keeps a riskless profit of $\$110 - \$105 = \$5$ wherever the stock price ends up, since the forward sale fixes the delivery price in advance. Chapter 5's option and derivative pricing methods generalize this relationship. An option's value, unlike a forward's, depends on the underlying's volatility because its payoff is not a fixed, symmetric obligation, but the same replication logic underlies both: construct a portfolio of tradable assets with an identical payoff, and price the derivative to match.

A Third Example: Triangular Arbitrage in Currency Markets

No-arbitrage pricing applies just as directly when three related prices must be mutually consistent, and foreign exchange markets provide the cleanest illustration. Three exchange rates, EUR/USD, GBP/USD, and EUR/GBP, are not independent: knowing any two implies the no-arbitrage value of the third, since converting EUR to USD and then USD to GBP must produce the same result as converting EUR to GBP

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directly. Suppose the market quotes EUR/USD = 1.10 (one euro buys \$1.10) and GBP/USD = 1.25 (one pound buys \$1.25), implying a no-arbitrage EUR/GBP rate of

$$\frac{\text{EUR/USD}}{\text{GBP/USD}} = \frac{1.10}{1.25} = 0.88,$$

one euro should be worth £0.88. If the EUR/GBP market instead directly quotes 0.87, euros are priced slightly cheaper against pounds directly than the USD route implies, and a triangular arbitrage exists: starting with £1,000,000,

$$\begin{aligned}
\text{Step 1 (GBP} \to \text{EUR at 0.87):} &\quad 1{,}000{,}000/0.87 \approx €1{,}149{,}425.29,\\
\text{Step 2 (EUR} \to \text{USD at 1.10):} &\quad 1{,}149{,}425.29 \times 1.10 \approx \$1{,}264{,}367.82,\\
\text{Step 3 (USD} \to \text{GBP at 1/1.25):} &\quad 1{,}264{,}367.82 \times 0.80 \approx £1{,}011{,}494.25,
\end{aligned}$$

a riskless profit of about £11,494, or roughly 1.15%, on the round trip, realized in seconds and requiring no view on whether any of the three currencies will rise or fall. As with the bond and forward, this mispricing cannot persist. High-frequency and algorithmic traders (Chapters 13 and 14) monitor such three-way consistency continuously, and their trading pushes the rates back into alignment within moments. Triangular arbitrage in liquid major-currency pairs is therefore too small and short-lived to exploit manually; today it is mainly a textbook illustration of a real economic force rather than a strategy open to anyone without a fully automated, low-latency system.

Frictions and the Limits of Arbitrage

Financial theory often assumes frictionless markets, but real markets are full of frictions: transaction costs, taxes, delays, information asymmetry, margin requirements, and limits on borrowing. These change the economics of investment and the behavior of prices.

If trading a security is expensive, an investor may need a larger expected return to hold it. If borrowing is restricted, a trader cannot always implement an otherwise profitable strategy. If information is not shared equally, some participants act faster or better informed than others. Even small frictions can have large aggregate effects, particularly when many participants trade at once.

Practitioners therefore distinguish theoretical from realized prices. Two assets with identical cash flows should in theory trade at the same value, but may not if short selling is difficult, settlement is delayed, or the market is temporarily illiquid. Such deviations create opportunities for some participants and risk for others, a tension between theory and practice that defines much of finance.

3.6 Fragility and Market Design

The same connections that make markets efficient also transmit failure. This part covers how stress propagates, how design and speed shape that propagation, and one afternoon in 2010 when it happened in minutes.

3.6.1 Financial Crises, Contagion, and Resilience

Understanding markets also requires understanding crises. Crises are not random accidents; they often emerge when leverage is high, liquidity is fragile, and institutions are highly interconnected. A seemingly small shock can then trigger forced selling, margin calls, and sudden changes in funding conditions, and a cascade of losses can spread across firms, asset classes, and countries. This is the pattern of the 2008 crisis discussed under clearing and systemic risk: mortgage defaults, amplified through securitization, derivatives, and short-term funding markets, became systemic because institutions that seemed healthy turned out not to be resilient once funding dried up.

For AI and quantitative finance, a model can be accurate in normal conditions and still fail in a crisis if the regime changes. The data may be nonstationary, relationships between variables may weaken, and strategies may become far more costly to execute, which is why model stress testing and scenario analysis are essential.

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An earlier episode, the near-collapse of the hedge fund Long-Term Capital Management (LTCM) in 1998, teaches a related but distinct lesson: sophisticated quantitative models can fail because of leverage and crowding, not because the statistical reasoning was wrong. LTCM's strategies relied on historical relationships between related securities converging to normal levels, a reasonable premise most of the time, applied with very high leverage to turn small, reliable-looking spreads into large returns. When the 1997 Asian financial crisis and the 1998 Russian debt default triggered a broad flight to safety, those relationships moved sharply against LTCM. Because many other participants held similar positions, there was no way to exit without pushing prices further the wrong way. The near-failure was severe enough that the Federal Reserve organized a private-sector bailout to prevent broader disruption. The lesson is not that statistical models are unreliable, but that estimated relationships can break down when many participants rely on the same relationship and use leverage to size their bets on it, a crowding risk that a model built only from historical prices will rarely reveal.

3.6.2 Technology, Speed, and Market Design

Technology has transformed market design. Electronic trading, co-location, and automation have reduced costs and sped up order processing, improving access for many participants but also increasing complexity. The market now resembles a digital system in which algorithms interact at high speed and small delays can be very valuable.

Technology changes the structure of competition as well as the efficiency of trading. A participant with faster infrastructure may gain an edge without a better economic view, which is why market microstructure has become central to modern finance: price formation is shaped by trading rules, network effects, and technology as well as by fundamentals.

3.6.3 Case Vignette: The 2010 Flash Crash

On May 6, 2010, U.S. equity markets experienced one of the most dramatic intraday moves in modern history: major indices fell sharply within minutes, some stocks and exchange-traded funds briefly traded at absurd prices, pennies in some cases, and the market recovered most of its losses within the same session. The Flash Crash is a useful case study because the joint SEC-CFTC investigation identified a chain of mechanical, structural causes rather than any piece of adverse economic news.

The investigation traced the initial trigger to a large sell order in E-mini S&P 500 futures, executed by an automated algorithm designed to sell a fixed percentage of trading volume without regard to price or time. As this selling pressure pushed prices down, high-frequency trading firms, which had been providing a substantial share of market liquidity, began rapidly buying and reselling the same contracts to each other rather than absorbing the imbalance, a pattern sometimes described as "hot potato" trading.

Within minutes, many of these high-frequency firms and other liquidity providers withdrew entirely as their risk controls triggered, and depth (one of the liquidity dimensions of Section 3.5.1) evaporated on both the futures and the equity markets at once. With so little resting liquidity, ordinary sell orders walked prices down dramatically, the price-impact effect of Section 3.3.2 at a much larger and faster scale.

The Flash Crash illustrates several of this chapter's themes at once. Liquidity is not a guaranteed feature of a market but a behavior that participants can withdraw suddenly and together, especially under stress, the regime dependence of liquidity discussed in Section 3.7. Automated systems interacting with one another can produce dynamics, like the hot-potato pattern, that no individual system's designers intended or anticipated. And market fragmentation across linked venues and instruments has practical consequences: the crash began in a futures contract but spread almost instantly to equities, since closely related prices are tied together by the no-arbitrage forces of Section 3.5.3. In response, regulators introduced market-wide circuit breakers and individual stock "limit up-limit down" rules designed to pause trading automatically during extreme price moves, a structural safeguard that did not exist in May 2010.

3.6.4 Case Vignette: Archegos and Leverage Through Total Return Swaps

The Flash Crash showed liquidity vanishing in minutes. The collapse of Archegos Capital Management in March 2021 shows hidden leverage turning one stock's decline into more than $10 billion of losses at the banks

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that financed it. Archegos was the family office of Bill Hwang. According to the SEC, it grew from about $1.5 billion of capital with $10 billion of exposure in March 2020 to more than $36 billion of capital with $160 billion of exposure at its peak in March 2021, concentrated in a handful of stocks such as ViacomCBS (U.S. Securities and Exchange Commission 2022).

Most of that exposure was held through *total return swaps*, one of the swap variants mentioned earlier. In a total return swap the prime broker buys the stock as a hedge and pays the client the stock's full return, while the client pays a financing rate and posts margin. The client has the economics of owning the shares without owning them, and two consequences follow from the margin arithmetic of Section 3.5.2. First, leverage: an initial margin of m supports exposure of $1/m$ times the collateral, and a price fall of m wipes the collateral out. A cash purchase of stock is limited by the Federal Reserve's Regulation T to 50% initial margin, a leverage of 2× that survives a 50% fall. Swap margin is negotiated bilaterally, and Credit Suisse's own investigation found it held only about 10% against Archegos's swaps (Paul, Weiss, Rifkind, Wharton & Garrison LLP 2021), a leverage of 10× that survives a 10% fall. Second, invisibility: swap positions did not appear in public holdings filings, and each prime broker saw only its own book, so no one, the banks included, saw that Archegos's combined position in some stocks was a large fraction of their float.

The trigger was ordinary. On 22 March 2021 ViacomCBS closed at $100.34; after it announced a stock offering, the shares fell, Archegos could not meet its margin calls, and by the close on 26 March they stood at $48.23, a fall of 51.93%. On $10 billion of swap exposure at 10% margin that is a loss of $5,193.3 million against $1,000 million of collateral, leaving $4,193.3 million for the lenders to absorb. How it was shared depended on who sold first. Suppose, in a stylized version, that four prime brokers each hold an equal $2,500 million block, each with $250 million of collateral, and each block sold walks the price down another 12.5%, the order-book walk of Section 3.3.2 at the scale of billions. Each broker, considering only its own block, expects to sell at an average of 6.25% below the starting price, comfortably inside its 10% margin. But the four blocks hit the same order book in sequence. The first seller sells at an average 6.25% below the start and loses nothing; the second, at 18.75% below, loses $218.75 million beyond its collateral; the third, at 31.25%, loses $531.25 million; the fourth, at 43.75%, loses $843.75 million. Brokers lose $1,593.75 million in total, the fund loses $906.25 million of its $1,000 million of collateral, and the price ends 50% below where it started. This is the first-mover arithmetic of the stablecoin run in Worked Example 3.3, now among lenders: margin that looked adequate for each position alone was inadequate for all of them together.

The real losses were ordered much like that. Goldman Sachs, which sold large blocks early on Friday 26 March, was reported to expect an immaterial effect on its results, and Morgan Stanley, also an early seller, lost just under $1 billion. Credit Suisse, slower to act, lost about $5.5 billion and Nomura nearly $3 billion, and total bank losses exceeded $10 billion. Credit Suisse exited prime brokerage, and Hwang was convicted of fraud and market manipulation in July 2024 and sentenced to 18 years in prison in November 2024. For quantitative risk management the lessons are the ones LTCM taught a generation earlier: a margin model calibrated to one position's volatility ignores the liquidation impact of everyone else's identical position, and concentration that no single counterparty can see is concentration nonetheless.

3.7 Challenges in Modeling Market Structure

The institutional detail of this chapter creates specific, recurring challenges for anyone building data-driven or AI-based systems on financial markets.

Fragmented and opaque data. Much trading, especially in bonds, derivatives, and large institutional equity orders, occurs over-the-counter or in dark pools rather than on a single transparent exchange. A dataset built from public exchange feeds alone is an incomplete picture of the market, and models trained on it can miss where and how large trades occur.

Microstructure noise. At very short horizons, prices are affected by the mechanics of trading (bid-ask bounce, order-book dynamics, the strategic behavior of market makers) as well as by fundamental information. A high-frequency model that treats every price tick as a fundamental signal risks fitting this noise rather than anything economically meaningful, a problem revisited in the chapters on high-frequency trading.

Regime dependence of liquidity. Liquidity, spreads, and depth are not fixed properties of a security; they change with volatility, macroeconomic conditions, and stress, often when a trading or risk model most

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needs to work. A backtest run only over calm periods can substantially overstate a strategy's real-world performance.

Reflexivity and adversarial dynamics. Unlike a physical system, a market contains participants who adapt to observed patterns, including patterns generated by other algorithms. A signal or arbitrage relationship, once discovered and traded at scale, can weaken or disappear, so one-time model validation is not sufficient.

Modeling systemic effects. Asset-level or firm-level models rarely capture the network effects (interconnections between institutions, correlated leverage, shared counterparties) that drive systemic events such as the 2008 crisis. Modeling these system-wide dynamics remains much harder and less mature than single-asset prediction.

Realistic transaction cost estimation. A strategy's profitability depends heavily on execution costs, market impact, and available liquidity, none fully known in advance and all easy to underestimate in a backtest that assumes trades execute at the observed price with no impact.

A purely statistical view of market data that ignores institutional structure is therefore unlikely to produce robust financial models, which is why the book treats market structure as foundational before turning to prediction and optimization.

3.8 Key Takeaways

Financial markets exist to allocate capital, transfer risk, and reveal information, and their institutions and instruments shape the opportunities and constraints investors and firms face. Debt, equity, and derivatives are the building blocks of modern finance, and how they are traded has direct consequences for liquidity, pricing, and risk management.

The order-book example in Section 3.3.2 showed that, with no news and no change in fundamental value, a large order can move the average execution price simply because liquidity is finite at each price level; the Flash Crash showed the same mechanism at market-wide scale when liquidity providers withdrew together. The swaps and regulatory material of this chapter (interest rate swaps, Dodd-Frank, MiFID II, Basel III) resurfaces in the risk management chapters, since modern risk management is defined as much by regulatory capital requirements as by statistical modeling choices. Payment rails and digital assets obey the same logic as securities settlement: the faster and more final a payment, the less can be undone afterward. A stablecoin promising redemption at par is only as sound as the duration and credit quality of its reserves, as the run arithmetic of Worked Example 3.3 shows. A central bank sets only the overnight rate; longer yields and stock prices move with what that rate reveals about the expected path ahead. The Archegos case showed the first-mover arithmetic of a stablecoin run playing out among the prime brokers that financed a single, invisibly concentrated fund. Market structure is not a fixed backdrop for trading; it shapes prices, costs, and the viability of any strategy built on it.

3.9 Suggested Exercises

3.1 Explain the difference between a bond and a stock in terms of cash flow rights and risk.

3.2 Describe the role of a market maker in a financial market.

3.3 Give one example of how a derivative can be used for hedging.

3.4 Explain why liquidity matters for asset prices.

3.5 Compare exchange-traded and over-the-counter markets in terms of transparency and counterparty risk.

3.6 Discuss one reason why a trading strategy might fail even if it has good historical predictive performance.

3.7 Explain how a bid-ask spread reflects the cost of immediacy in a market.

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3.8 Choose one challenge from Section 3.7 and explain how it could cause a backtested trading strategy to look profitable historically but lose money in live trading.

3.9 Using the order book in Table 3.1, compute the average execution price and price impact (in dollars and basis points) for a market sell order of 300 shares, which will walk through the bid side of the book.

3.10 Explain, in your own words, the difference between the order-processing, inventory-holding, and adverse-selection components of the bid-ask spread, and give one market condition that would cause each component to widen.

3.11 Explain why an interest rate swap does not require the exchange of the notional principal amount, and describe a situation in which a firm would want to swap from floating-rate to fixed-rate interest payments.

3.12 Compare the 2008 financial crisis and the LTCM episode described in this chapter: what role did leverage play in each, and how did the two episodes differ in their trigger and their transmission mechanism?

3.13 Using Section 3.5.2's method, a trader shorts 200 shares at \$80 with a 150% initial margin requirement and a 30% maintenance margin. Compute the total initial account equity and the share price at which a margin call is triggered.

3.14 Using Section 3.5.3's cash-and-carry logic, a non-dividend-paying stock trades at \$40 and the risk-free rate is 4%. Compute the no-arbitrage one-year forward price, and describe the specific trades an arbitrageur would execute today and in one year if the forward instead traded at \$43.

3.15 A dealer borrows \$25 million overnight in the repo market at an annualized rate of 4.5%, using the 360-day money-market convention from this chapter's repo example. Compute the dollar interest owed for one night and the total repurchase price.

3.16 Explain how the Flash Crash illustrates the "regime dependence of liquidity" challenge from Section 3.7, using specific details from the case study.

3.17 Using this chapter's interest rate swap example, recompute the swap's value to the fixed-rate payer if the one-year and two-year discount rates were instead 5.5% and 6% (rates have risen above the 5% fixed rate rather than fallen below it), and explain in words why the sign of the swap's value to the fixed-rate payer flips relative to the original example.

3.18 Using this chapter's triangular arbitrage example, suppose the EUR/GBP rate is instead quoted at 0.89 rather than 0.87 (EUR/USD and GBP/USD unchanged). Determine whether the arbitrage cycle described in the text ($£ \to € \to \$ \to £$) is still profitable, and if not, describe which direction around the triangle would now be profitable instead.

3.19 Rework Worked Example 3.3's stablecoin run with MiCA's minimum cash share: the issuer holds $C = \$3{,}000$ million of deposits and $B = \$7{,}000$ million of zero-coupon Treasuries bought at 4%, and yields rise by 200 basis points. For three-month bills and for two-year zeros, use Equation (3.3) to compute the mark-to-market loss, then use Equation (3.4) to compute reserves per remaining token after 30% and after 60% of tokens are redeemed. How much of each redemption must be met by selling Treasuries, and how does the larger cash buffer affect the realized loss compared with the reserves per remaining token?

3.20 Using Table 3.2, explain why an authorized push payment scam is harder to reverse on an instant payment rail than an unauthorized card transaction is on a card network, and what this implies about when a fraud model must make its decision. Then name one thing that tokenizing a money-market fund changes about how its shares settle and one risk of holding the fund that tokenization leaves unchanged.

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3.21 Rework Section 3.2.5's rate-transmission example for a 25-basis-point cut that the market expects to persist, so that the expected one-year rates for the next two years are both 4.75%. Compute the new two-year yield and the percentage change in the price of the two-year zero (starting from \$90.7029), and the percentage change in the Gordon-growth stock price if its discount rate falls from 8% to 7.75% ($D_1 = \$4$, $g = 4\%$). Why does the stock move so much more than the bond?

3.22 Redo Section 3.6.4's stylized fire-sale race with a swap margin of 20% instead of 10% (four brokers, \$2,500 million blocks, 12.5% price impact per block). What leverage does the fund now have, which brokers lose money beyond their collateral, and what are total broker losses? Compare the answer with a long position bought in cash under Regulation T's 50% initial margin.

3.23 Using Section 3.2.6, classify each of the following as buy side, sell side, or neither, and state how each earns its revenue: a pension fund, an investment bank's equity derivatives desk, a credit rating agency, a quantitative hedge fund, and a prime broker. Then explain the issuer-pays conflict of interest, and why it mattered more for mortgage-backed securities before 2008 than for large corporate bonds.

3.24 **Challenge (real data).** Estimate Roll's implied spread, Equation (3.2), from daily closing prices. Download unadjusted daily closes for 3 January to 29 December 2023 for the S&P 500 ETF SPY and for the small-cap furniture maker Flexsteel Industries (FLXS) with `yfinance` (`auto_adjust=False`). (a) For each, compute the first-order autocovariance of daily price changes over the full year and, where it is negative, the Roll spread in dollars and in basis points of the average price. (b) Repeat month by month, and count the months in which the estimate is undefined because the autocovariance is positive. (c) SPY's quoted spread is about one cent. Explain what your daily-data estimates for SPY say about the estimator's reliability for liquid securities at daily frequency, and why it may be more informative for an illiquid small-cap.

3.10 Further Reading

The following books treat these topics in more depth:

- Bodie et al. (2024), *Investments*. A broad introduction to markets, instruments, and institutions that complements this chapter's overview with a fuller treatment of portfolio and asset-pricing theory.
- Hull (2022), *Options, Futures, and Other Derivatives*. The standard reference for the derivatives instruments introduced briefly in this chapter, developing their pricing and hedging in full in Chapter 5.
- Fabozzi (2007), *Fixed Income Analysis*. Goes well beyond this chapter's brief bond-market coverage into the pricing, risk, and institutional detail of fixed-income securities.
- Shleifer (2000), *Inefficient Markets*. A behavioral-finance counterpoint to the efficient-market and price-discovery discussion in this chapter, examining the frictions and limits to arbitrage that keep prices from always reflecting fundamentals.
- O'Hara (1995), *Market Microstructure Theory*. The foundational academic treatment of the order-book mechanics, bid-ask spread decomposition, and market-making economics introduced in this chapter's worked examples.
- Gorton and Zhang (2023), "Taming Wildcat Stablecoins." Argues, from the history of nineteenth-century U.S. private banknotes, that a stablecoin redeemable at par is a form of bank deposit exposed to runs, the logic behind this chapter's stablecoin worked example and the full-reserve rules that followed.
- Financial Crisis Inquiry Commission (2011), *The Financial Crisis Inquiry Report: Final Report of the National Commission on the Causes of the Financial and Economic Crisis in the United States*. The official U.S. inquiry into the 2008 crisis, including its case study of how issuer-paid ratings were assigned to mortgage securities, the background to this chapter's discussion of rating agencies.

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- Bank for International Settlements (2025), "The Next-Generation Monetary and Financial System." The Bank for International Settlements' assessment of stablecoins against the tests of sound money, and its case for tokenized central bank reserves, deposits, and government bonds on a shared ledger.

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Chapter 4

Financial Statements and Business Basics

4.1 Introduction

Financial statements are the language through which firms communicate their economic position to investors, lenders, regulators, and managers. They are structured summaries of how a business creates value, uses capital, and manages risk, not mere accounting artifacts. For an investor, analyst, trader, or data scientist, the ability to read them is indispensable: a model may forecast returns or defaults, but it is useless without the business context behind the numbers.

This chapter introduces the three core financial statements: the balance sheet, the income statement, and the cash flow statement. It explains how they fit together, why they differ, and how they support reasoning about profitability, liquidity, solvency, and value creation. The goal is not to turn the reader into an accountant but to build the intuition needed to analyze firms in a way that is financially sound and quantitatively useful.

A fictional manufacturer, Meridian Fabrication Inc., runs through the chapter as its central worked example, so that every ratio is tied to actual numbers. Its balance sheet, income statement, and cash flow statement are built up section by section, then mined for liquidity, leverage, profitability, and efficiency ratios (Section 4.4.5), an Altman's Z-Score distress assessment, and a common-size restatement that makes its cost structure comparable across firm sizes. A second company, Solstice Robotics Inc. (Section 4.5.1), is introduced for comparison, so that a ratio's meaning becomes clear by contrast: the same debt-to-equity ratio can signal prudence at one firm and fragility at another.

The ratio toolkit built here is the one that Chapter 9's credit scoring and Chapter 6's classification models later formalize statistically. Section 4.6.1 links it to Chapter 5's valuation models, since a discounted-cash-flow model is only as trustworthy as the statement forecasts feeding it. The accounting-quality cautions raised late in the chapter (reported numbers reflect choices and estimates, not just facts) apply to every later chapter that treats statement figures as clean model inputs.

4.2 Why Statements Matter, and a Running Example

Accounting is the language a firm uses to describe itself, and like any language it can be used carefully or evasively. This part explains why the statements repay close reading, and introduces the company whose numbers the rest of the chapter works through.

4.2.1 Why Financial Statements Matter

A firm is a system for converting resources into cash flows. It buys inputs, hires labor, invests in equipment, and sells products or services. In the process, it makes decisions about financing, operations, and strategy, and financial statements summarize these decisions. They let stakeholders ask essential questions: How much

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debt does the firm carry? Is it profitable? Is it generating enough cash to fund operations? What portion of the business is financed by owners versus lenders? Is growth coming from operating improvement or from financial engineering?

Finance is about the economic reality of the firm as well as price changes. A stock can look attractive because its price rises, but if the company is consuming cash and taking on excessive leverage, the apparent success may be fragile. Conversely, a firm with modest headline earnings may still be valuable if it produces durable cash flows and has strong competitive advantages. Financial statements help separate signal from noise.

The statements are analyzed together. The balance sheet shows the firm's position at a point in time, the income statement its performance over a period, and the cash flow statement the movement of cash; together they give a more complete picture than any one report. Figure 4.1 sketches how the three statements relate to one another.

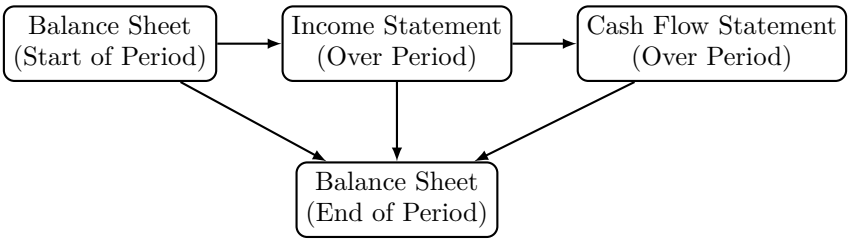


Figure 4.1: The three statements are linked: the income statement and cash flow statement explain the change in the balance sheet from the start to the end of the period

4.2.2 A Running Example: Meridian Fabrication Inc.

Meridian Fabrication Inc. is a small, fictional industrial parts manufacturer. All figures are in millions of dollars. Table 4.1 shows a simplified balance sheet at the end of two consecutive fiscal years, Table 4.2 shows the income statement for the most recent year, and Table 4.3 shows the cash flow statement for the same year. These numbers are used throughout the chapter to illustrate ratios, forecasting, and the limits of reported figures.

	Year 1	Year 2
Cash and equivalents	12.0	9.0
Accounts receivable	18.0	24.0
Inventory	22.0	30.0
Total current assets	**52.0**	**63.0**
Property, plant, and equipment (net)	60.0	66.0
Total assets	**112.0**	**129.0**
Accounts payable	14.0	17.0
Short-term debt	6.0	8.0
Total current liabilities	**20.0**	**25.0**
Long-term debt	40.0	50.0
Total liabilities	**60.0**	**75.0**
Shareholders' equity	52.0	54.0
Total liabilities and equity	**112.0**	**129.0**

Table 4.1: Meridian Fabrication Inc. – Balance Sheet ($ millions)

A first observation: Meridian's net income of $9.0 million looks respectable, but cash flow from operations is only $6.0 million, and after $14.0 million of capital expenditures the firm burned cash and had to raise $12.0

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Revenue	150.0
Cost of goods sold	96.0
Gross profit	**54.0**
Selling, general, and administrative expense	30.0
Depreciation and amortization	8.0
Operating income (EBIT)	**16.0**
Interest expense	4.0
Pretax income	**12.0**
Taxes (25%)	3.0
Net income	**9.0**

Table 4.2: Meridian Fabrication Inc. – Income Statement, Year 2 ($ millions)

Net income	9.0
+ Depreciation and amortization	8.0
− Increase in accounts receivable	6.0
− Increase in inventory	8.0
+ Increase in accounts payable	3.0
Cash flow from operations	**6.0**
Capital expenditures	14.0
Cash flow from investing	**−14.0**
Net new debt issued	12.0
Dividends paid	7.0
Cash flow from financing	**5.0**
Net change in cash	**−3.0**

Table 4.3: Meridian Fabrication Inc. – Cash Flow Statement, Year 2 ($ millions)

million of new debt to keep its cash balance from falling further. Financial statement analysis is designed to surface this kind of gap between accounting profit and cash generation, and it recurs throughout the chapter.

4.3 The Three Statements

A firm reports itself through three linked documents: what it owns and owes, what it earned, and what cash moved. This part takes each in turn and shows how they articulate with one another.

4.3.1 The Balance Sheet

The balance sheet is a snapshot of a firm's resources and claims at a specific date. It is governed by a simple accounting identity:

$$\text{Assets} = \text{Liabilities} + \text{Equity}. \tag{4.1}$$

Assets are what the firm owns or controls. They include cash, receivables, inventories, property, plant and equipment, intangible assets, and other resources expected to generate future benefit. Liabilities are obligations the firm owes to others, such as loans, accounts payable, or taxes payable. Equity is the residual interest of owners after liabilities are subtracted from assets. In Table 4.1, this identity holds in both years: $112.0 million of assets equal $60.0 million of liabilities plus $52.0 million of equity in Year 1, and $129.0 million equal $75.0 million plus $54.0 million in Year 2.

The balance sheet reveals the firm's financial structure. A firm with large cash reserves and modest debt is often more resilient than one with high leverage and weak liquidity. Significant intangible assets may

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look valuable but be hard to liquidate in distress, and high inventory creates operational risk if demand falls unexpectedly. Meridian's inventory grew from \$22.0 million to \$30.0 million, a 36% increase, and its receivables from \$18.0 million to \$24.0 million, a 33% increase, while its revenue grew only 11%, from \$135.0 million in Year 1 to the \$150.0 million of Year 2. Comparing the growth of inventory and receivables with the growth of sales is one of the simplest early-warning checks an analyst performs, and here it flags the working-capital buildup that the cash flow statement shows consuming Meridian's cash.

The balance sheet can be analyzed for liquidity (the ability to meet short-term obligations), capital structure (the mix of debt and equity financing), and asset quality (whether reported assets are likely to generate value). Analysts commonly compute current assets divided by current liabilities, debt divided by equity, and long-term debt relative to total assets as a first pass; Section 4.4.5 computes them for Meridian.

4.3.2 The Income Statement

The income statement summarizes revenues, expenses, and profit over a period such as a quarter or a year. It begins with revenue, which is the amount earned from selling goods or services. From revenue, the firm subtracts costs of goods sold and operating expenses to arrive at operating profit, often called earnings before interest and taxes (EBIT). Further deductions for interest and taxes lead to net income, which is often the headline figure quoted in the media.

For Meridian, revenue of \$150.0 million minus cost of goods sold of \$96.0 million yields gross profit of \$54.0 million, a gross margin of

$$\text{Gross margin} = \frac{54.0}{150.0} = 36.0\%.$$

After subtracting SG&A (\$30.0 million) and depreciation (\$8.0 million), operating income is \$16.0 million, an operating margin of

$$\text{Operating margin} = \frac{16.0}{150.0} = 10.7\%.$$

After \$4.0 million of interest expense and \$3.0 million of taxes (a 25% rate applied to \$12.0 million of pretax income), net income is \$9.0 million, a net margin of $9.0/150.0 = 6.0\%$.

Net income is widely reported but should not be read mechanically. Accounting profit is influenced by depreciation, amortization, one-time gains or losses, and estimates of reserves and provisions, all of which affect apparent profitability. Analysts therefore also look at operating income, adjusted earnings, and cash-based measures of performance.

Profitability is central to valuation, because investors care whether the firm can earn returns above its cost of capital. A company may report growing sales, but if margins are declining and competition is intensifying, future value may be under pressure. Profitability analysis therefore examines gross, operating, and net margins and their trend over time, not just their level in one year.

One further measure appears so often in valuation practice that it belongs alongside these margins: EBITDA, earnings before interest, taxes, depreciation, and amortization, computed by adding depreciation and amortization back to operating income. For Meridian, EBITDA $= 16.0 + 8.0 =$ \$24.0 million, an EBITDA margin of $24.0/150.0 = 16.0\%$. EBITDA is popular because it strips out depreciation and amortization, non-cash charges that vary across firms with accounting choices and asset age, making operating performance easier to compare across companies with different capital structures and depreciation schedules. It is not a substitute for free cash flow: it ignores capital expenditures and working-capital changes, which Meridian's numbers show can turn a profitable-looking income statement into a cash-consuming year. As a cross-sectional comparison tool, and as the denominator of the EV/EBITDA multiple used in Chapter 5, it is worth computing routinely alongside the three margins.

4.3.3 The Cash Flow Statement

The cash flow statement is often the most informative statement about financial health. Cash keeps a business alive, and a company can report profits and still face liquidity problems if its cash is tied up in inventory, receivables, or capital spending. The statement shows how cash is generated and used in three activities: operating, investing, and financing.

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Operating cash flow reflects cash generated from the core business. It starts from net income and adjusts for non-cash items (such as adding back depreciation) and for changes in working-capital accounts. Investing cash flow reflects purchases and sales of long-term assets, primarily capital expenditures. Financing cash flow reflects borrowing, repayments, dividends, and equity issuance. Table 4.3 shows this decomposition for Meridian: operating cash flow of \$6.0 million, investing cash flow of −\$14.0 million, and financing cash flow of \$5.0 million, which together produce a net decrease in cash of \$3.0 million, consistent with the cash balance falling from \$12.0 million to \$9.0 million on the balance sheet.

A particularly important concept is free cash flow, typically defined as

$$\text{Free cash flow} = \text{Cash flow from operations} - \text{Capital expenditures}. \tag{4.2}$$

For Meridian, free cash flow in Year 2 is $6.0 - 14.0 = -\$8.0$ million. This number tells a sharper story than net income: despite a \$9.0 million profit, Meridian generated negative \$8.0 million of free cash flow and had to borrow to cover the shortfall. Free cash flow matters for valuation because firms that generate it consistently can fund dividends, debt repayment, or reinvestment without external financing. Many valuation frameworks use free cash flows rather than accounting earnings, which non-cash charges and discretionary accounting choices can distort.

4.4 Ratio Analysis

Raw statement figures are not comparable across firms or across years; ratios are what make them so. This part builds the standard families of ratio, from short-term liquidity through leverage to the returns that reveal whether a firm creates value at all.

4.4.1 Working Capital and Liquidity

Liquidity, the ability to meet short-term obligations, is one of the clearest indicators of short-run financial health. A business that cannot pay suppliers, employees, or creditors on time may face disruption even if its long-term outlook is strong, so analysts watch current assets, current liabilities, and the cycle of converting inventory and receivables into cash.

Working capital is defined as

$$\text{Working capital} = \text{Current assets} - \text{Current liabilities}.$$

For Meridian, working capital rose from $52.0 - 20.0 = \$32.0$ million in Year 1 to $63.0 - 25.0 = \$38.0$ million in Year 2. On its own this looks like a healthy increase, but the cash flow statement shows that most of it is cash trapped in receivables and inventory rather than sitting in the bank. Too little working capital can cause distress, while too much may indicate inefficiency or, as in Meridian's case, a buildup that has not yet been converted back into cash.

Working capital is actively managed, not just measured after the fact. A treasury team can accelerate collections with early-payment discounts, negotiate longer payment terms with suppliers, or use just-in-time ordering to reduce the cash tied up in stock. Each lever maps onto one component of the cash conversion cycle computed below: shortening DSO (days sales outstanding) through faster collections, lengthening DPO (days payable outstanding) through supplier negotiation, or shortening DIO (days inventory outstanding) through leaner inventory management. Working capital financing, such as a revolving credit facility sized to bridge the gap between paying suppliers and collecting from customers, is common because this gap, as Meridian's cash flow statement shows, can consume substantial cash even for a healthy, growing business.

The cash conversion cycle is a useful way to quantify this. It is commonly expressed as

$$\text{Cash conversion cycle} = \text{DIO} + \text{DSO} - \text{DPO}, \tag{4.3}$$

where DIO measures how long inventory sits before being sold, DSO measures how long it takes to collect receivables after a sale, and DPO measures how long the firm takes to pay its own suppliers. COGS, introduced earlier as the amount subtracted from revenue to reach gross profit, is the direct cost (materials, direct labor, and manufacturing overhead) of producing what the firm sold during the period.

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DIO and DPO divide by COGS rather than revenue because inventory and accounts payable are carried on the balance sheet at cost, not at the higher price a customer pays, so a cost-basis denominator keeps numerator and denominator on the same footing. DSO divides accounts receivable, recorded at the amount billed to the customer, by revenue, the corresponding billed-price figure; swapping the denominators would mix cost-basis and price-basis quantities in one ratio. Using Year 2 year-end balances and a 365-day year (analysts often average the opening and closing balances instead, which for a firm whose working capital is growing as fast as Meridian's gives a noticeably shorter cycle),

$$\begin{aligned}
\text{DIO} &= \frac{\text{Inventory}}{\text{COGS}} \times 365 = \frac{30.0}{96.0} \times 365 \approx 114 \text{ days},\\
\text{DSO} &= \frac{\text{Accounts receivable}}{\text{Revenue}} \times 365 = \frac{24.0}{150.0} \times 365 \approx 58 \text{ days},\\
\text{DPO} &= \frac{\text{Accounts payable}}{\text{COGS}} \times 365 = \frac{17.0}{96.0} \times 365 \approx 65 \text{ days}.
\end{aligned}$$

Rounding each component to the nearest day before summing gives $114 + 58 - 65 = 107$ days, but computing the cycle from the unrounded day-counts gives $114.06 + 58.40 - 64.64 \approx 107.8$, which rounds to 108 days. The one-day gap is a rounding artifact from summing already-rounded figures, a reminder to carry full precision through a multi-step calculation and round only the final answer. Either way, Meridian's cash is tied up in the operating cycle for roughly three and a half months. A shorter cycle is generally better, since the firm recycles cash faster and relies less on external financing, and the cycle is the kind of quantity a data-driven monitoring system can track continuously across many firms or product lines.

4.4.2 Leverage and Solvency

Leverage refers to the extent to which a firm uses debt financing. Debt is often cheaper than equity and its interest may be tax-deductible, but leverage also increases financial risk: if cash flows are volatile, debt can become burdensome and force restructuring or bankruptcy.

Solvency is the ability of a firm to meet its long-term obligations. Analysts examine leverage ratios such as debt-to-equity,

$$\text{Debt-to-equity} = \frac{\text{Total debt}}{\text{Shareholders' equity}},$$

and interest coverage,

$$\text{Interest coverage} = \frac{\text{EBIT}}{\text{Interest expense}}.$$

For Meridian in Year 2, total debt is short-term debt plus long-term debt, $8.0 + 50.0 = \$58.0$ million, against equity of \$54.0 million, giving a debt-to-equity ratio of $58.0/54.0 \approx 1.07$: the firm is financed with slightly more debt than equity. Interest coverage is $16.0/4.0 = 4.0$, meaning operating income covers interest expense four times over. A coverage ratio of 4.0 is generally considered comfortable, but the trend matters: if debt keeps growing faster than operating income, coverage will fall and the cushion against a bad year will shrink.

Leverage is not inherently good or bad; it depends on the business model, cash-flow stability, and access to financing. A utility with stable revenues can carry significant debt, while a startup in a volatile industry may support little. The question is whether leverage amplifies the firm's strengths or its weaknesses, and for Meridian, needing new debt to fund a cash shortfall driven by working-capital growth is a signal worth investigating.

4.4.3 Sector-Specific Considerations in Ratio Analysis

Ratios must be interpreted in context, and a large-scale, cross-sectional model cannot apply one set of thresholds to every firm it scores. A few contrasting industries show what that context looks like.

Banks and other financial institutions are the most extreme example. A bank's balance sheet consists mostly of financial assets (loans) funded mostly by financial liabilities (deposits). A debt-to-equity ratio that would signal extreme distress for an industrial firm like Meridian, often 8 or 10 times equity, is routine for a bank, whose business model is to hold a leveraged portfolio of loans. Standard liquidity ratios such as

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the current ratio are likewise not meaningful for banks, whose balance sheets are not organized around a current/non-current distinction; regulators instead impose bank-specific capital and liquidity ratios, some introduced in the Basel III discussion of Chapter 3.

Capital-intensive industries such as utilities, airlines, and telecommunications typically carry substantial long-term debt and lower interest coverage than an asset-light firm such as Solstice Robotics, the comparison company of Section 4.5.1. Their relatively stable and predictable revenues (regulated utility rates, contracted infrastructure usage) let them service more debt safely than a firm with volatile cash flows. A debt-to-equity ratio of 1.5 might be unremarkable for a regulated utility but alarming for an unproven technology startup.

Asset-light service and technology businesses, more like Solstice than Meridian, often carry little debt, hold large cash balances, and derive most of their value from intangible assets (customer relationships, brand, intellectual property) that may barely appear on the balance sheet under conventional accounting. Market-based valuation multiples (Chapter 5) are therefore often used alongside, or instead of, book-value ratios for such firms, since the balance sheet may substantially understate the economic assets driving the business.

For any model built across many firms, ratios should be compared within a sector or peer group rather than against a universal threshold, and sector membership and capital intensity are often useful features to include directly in a predictive model.

This has a direct consequence for a distress score like Altman's Z′. A formula whose coefficients were estimated mainly on manufacturing and industrial firms will systematically misclassify even a perfectly healthy bank, whose debt-to-equity ratio would sit far above, and whose X_4 (book equity divided by total liabilities) far below, any threshold intended for an industrial company. Neither Meridian's Z′-score of 2.41 nor Solstice's 4.33, computed later in this chapter, is directly comparable to a Z′-score computed with the same coefficients for a bank or a capital-intensive utility. That is why sector-specific credit-scoring models, such as the bank-focused material in Chapter 9, are estimated on sector-specific data rather than reusing a general-purpose formula across every industry.

4.4.4 How Firms Create Value

Suppose a firm doubles its revenue and its balance sheet over five years, yet its shareholders end up no wealthier than if it had stayed the same size. Growth alone says nothing about value creation, because growth must be paid for with capital, and capital has a cost, whether lenders demand interest or shareholders demand a competitive return. A firm creates value only when it earns returns on invested capital above the cost of that capital; projects earning less than the required return destroy value however much bigger the firm becomes. This is one of the most important ideas in corporate finance, commonly expressed through economic profit,

$$\text{Economic profit} = \text{Invested capital} \times (\text{ROIC} - \text{WACC}), \tag{4.4}$$

where ROIC is the return on invested capital and WACC is the weighted average cost of capital. A simple value-creation diagram is shown in Figure 4.2.



Figure 4.2: A firm generates value when its projects earn a return above the cost of capital

This idea links financial statements to valuation. A firm that earns high returns on capital and reinvests profitably compounds value over time; one that earns low returns and uses excessive leverage may look large but create little wealth. A predictive model may forecast changes in revenue, but an analyst must still ask whether those changes translate into durable value creation.

Growth is therefore valuable when tied to profitable reinvestment and harmful when it consumes large amounts of capital with weak expected returns. Meridian grew its balance sheet from $112.0 million to $129.0 million, a 15% increase, while free cash flow turned negative. Whether this growth creates value depends on whether the additional capital will earn returns above its cost.

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The statements cannot supply a cost of capital, but they supply everything else this calculation needs. Meridian's invested capital, the sum of its interest-bearing debt and equity, is $\$58.0 + \$54.0 = \$112.0$ million (Table 4.1), and its after-tax operating profit, or net operating profit after tax (NOPAT), is EBIT $\times$ (1 − tax rate) $= 16.0 \times 0.75 = \$12.0$ million. This gives

$$\text{ROIC} = \frac{\text{NOPAT}}{\text{Invested capital}} = \frac{12.0}{112.0} \approx 10.7\%.$$

If Meridian's weighted average cost of capital, estimated using the methods of Chapter 5, were 8% (close to the 8.08% that Chapter 5 estimates from Meridian's own statements), its economic profit would be $112.0 \times (0.107 - 0.08) \approx \$3.0$ million: modestly positive, so Meridian creates a small amount of value above its cost of capital despite its cash-flow strain. The distinction is initially counterintuitive. A firm can be a (marginal) value creator by this measure while burning cash and increasing leverage, because ROIC versus WACC asks whether capital is deployed productively, while free cash flow asks whether the firm is generating or consuming cash in the short run. A machine learning model trained to predict "financial health" from statement data must be clear about which of these two questions it is answering.

4.4.5 Common Financial Ratios and What They Reveal

Financial ratios convert raw statements into interpretable measures that let analysts compare firms, track trends, and identify concerns. They fall into profitability, liquidity, efficiency, solvency, and valuation categories. Table 4.4 collects the main ratio families together with their formulas and Meridian's Year 2 values.

Ratio	Formula	Meridian Value
Current ratio	Current assets / Current liabilities	63.0/25.0 = 2.52
Quick ratio	(Current assets − Inventory) / Current liabilities	33.0/25.0 = 1.32
Gross margin	Gross profit / Revenue	54.0/150.0 = 36.0%
Operating margin	EBIT / Revenue	16.0/150.0 = 10.7%
Net margin	Net income / Revenue	9.0/150.0 = 6.0%
Debt-to-equity	Total debt / Equity	58.0/54.0 = 1.07
Interest coverage	EBIT / Interest expense	16.0/4.0 = 4.0
Asset turnover	Revenue / Total assets	150.0/129.0 = 1.16
Return on equity (ROE)	Net income / Equity	9.0/54.0 = 16.7%

Table 4.4: Summary of common financial ratios, applied to Meridian, Year 2

The current ratio of 2.52 and quick ratio of 1.32 suggest that, on paper, Meridian has ample short-term resources to cover its current liabilities. The cash flow analysis adds nuance: much of the current assets is tied up in slow-moving inventory and receivables, and the cash balance fell during the year. A ratio from a single balance sheet date can look comfortable while masking a deteriorating cash trajectory, so ratios are more informative read together with the cash flow statement and their own trend.

A particularly useful decomposition is the DuPont identity, which breaks return on equity into three economically distinct drivers:

$$\text{ROE} = \underbrace{\frac{\text{Net income}}{\text{Revenue}}}_{\text{Net margin}} \times \underbrace{\frac{\text{Revenue}}{\text{Total assets}}}_{\text{Asset turnover}} \times \underbrace{\frac{\text{Total assets}}{\text{Equity}}}_{\text{Equity multiplier}}. \tag{4.5}$$

For Meridian, Net margin = 6.0%, Asset turnover = 1.16, and the equity multiplier is 129.0/54.0 = 2.39. Multiplying these together at full precision, $0.060 \times 1.1628 \times 2.3889 \approx 16.67\%$, reproduces the directly computed ROE of 16.7%, as it must, since revenue and total assets cancel in the product; multiplying the rounded factors would open a small rounding gap, the same cascade as in the cash conversion cycle. The decomposition is diagnostic: Meridian's ROE is driven substantially by leverage (an equity multiplier above

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2) rather than by high margins or asset efficiency. Two firms can report the same ROE for very different, and not equally sustainable, reasons, and the DuPont breakdown is the standard tool for telling them apart.

The three-factor version bundles taxes, interest, and core operating profitability into a single "net margin" term, hiding whether a margin is driven by operations or by a favorable tax or financing outcome. The five-factor extended DuPont identity splits net margin further,

$$\text{ROE} = \underbrace{\frac{\text{Net income}}{\text{Pretax income}}}_{\text{Tax burden}} \times \underbrace{\frac{\text{Pretax income}}{\text{EBIT}}}_{\text{Interest burden}} \times \underbrace{\frac{\text{EBIT}}{\text{Revenue}}}_{\text{Operating margin}} \times \text{Asset turnover} \times \text{Equity multiplier},$$

isolating how much of operating profit survives interest expense and taxation. For Meridian, the tax burden is $9.0/12.0 = 0.75$ (the firm retains 75% of pretax income after taxes, consistent with the 25% tax rate used throughout this chapter), the interest burden is $12.0/16.0 = 0.75$ (the firm retains 75% of EBIT after interest expense), and the operating margin is $16.0/150.0 \approx 10.7\%$, as computed in Section 4.4.5. Multiplying all five factors,

$$0.75 \times 0.75 \times 0.107 \times 1.163 \times 2.389 \approx 16.7\%,$$

matches the directly computed ROE. The extra split is diagnostic. Meridian's two burden ratios are both 0.75, so taxes and interest erode profitability in the same proportion. A firm with a much lower interest burden (heavily levered, paying substantial interest) and a higher tax burden could show the same three-factor net margin as Meridian with a completely different cost structure, a difference only the five-factor decomposition surfaces.

Liquidity ratios such as the current ratio and quick ratio assess how easily a firm can meet short-term obligations. Efficiency ratios such as inventory turnover and receivables turnover show how quickly it converts resources into cash. Solvency ratios such as debt-to-equity and interest coverage show how vulnerable it is to distress. Valuation ratios such as price-to-earnings and price-to-book compare market expectations with fundamental performance, though they need market prices beyond the statements.

Ratios depend on context. A high leverage ratio may suit a stable utility but endanger a startup with volatile cash flows. Low inventory turnover may indicate weak sales or poor inventory management, or in some sectors a strategy of high service levels. Ratios are starting points for a conversation about the business, not final judgments.

4.5 Comparison and Common-Size Analysis

A ratio means little in isolation. This part introduces a second company and the common-size representation that makes two firms of very different scale directly comparable.

4.5.1 A Second Company for Comparison: Solstice Robotics Inc.

Ratios are most informative compared across firms or across time. Consider a second fictional company, Solstice Robotics Inc., a faster-growing, less capital-intensive firm, with the Year 2 figures in Table 4.5.

Table 4.6 places Meridian and Solstice side by side on the same ratios computed in Table 4.4.

Solstice's ROE of 26% is higher than Meridian's 16.7%, but the DuPont identity from Section 4.4.5 reveals that the two firms get there in opposite ways. Applying the same decomposition to Solstice: net margin 16.25%, asset turnover $120.0/100.0 = 1.20$, and equity multiplier $100.0/75.0 = 1.33$, so $0.1625 \times 1.20 \times 1.33 \approx 26.0\%$, matching directly.

Solstice's high ROE is driven almost entirely by superior margins (16.25% net margin versus Meridian's 6.0%), with very little help from leverage (an equity multiplier of 1.33 versus Meridian's 2.39). Meridian, by contrast, achieves a respectable but lower ROE mostly *because* of leverage. An analyst, or a machine learning model trained naively on ROE alone, would rank the two firms similarly although their financial structures, and therefore their risk profiles, are almost opposite. The DuPont decomposition exists to surface what a single summary ratio hides.

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Cash and equivalents	40.0
Accounts receivable	20.0
Inventory	10.0
Total current assets	**70.0**
Property, plant, and equipment (net)	30.0
Total assets	**100.0**
Accounts payable	15.0
Long-term debt	10.0
Total liabilities	**25.0**
Shareholders' equity	75.0
Revenue	120.0
Cost of goods sold	48.0
SG&A	40.0
Depreciation and amortization	5.0
Interest expense	1.0
Net income	19.5

Table 4.5: Solstice Robotics Inc. – Summary Financials, Year 2 ($ millions)

Ratio	Meridian	Solstice
Current ratio	2.52	4.67
Quick ratio	1.32	4.00
Gross margin	36.0%	60.0%
Operating margin	10.7%	22.5%
Net margin	6.0%	16.25%
Debt-to-equity	1.07	0.13
Interest coverage	4.0	27.0
Return on equity	16.7%	26.0%

Table 4.6: Meridian versus Solstice, Year 2

4.5.2 Common-Size Statements

A second way to make firms comparable is the common-size statement, in which every income statement line is expressed as a percentage of revenue and every balance sheet line as a percentage of total assets. Table 4.7 shows Meridian's and Solstice's income statements this way.

	Meridian	Solstice
Cost of goods sold	64.0%	40.0%
Gross profit	36.0%	60.0%
SG&A	20.0%	33.3%
Depreciation and amortization	5.3%	4.2%
Operating income (EBIT)	10.7%	22.5%
Interest expense	2.7%	0.8%
Net income	6.0%	16.25%

Table 4.7: Common-size income statements, Year 2 (percent of revenue)

Common-size statements make cost-structure differences visible in a way raw dollar figures do not: Solstice spends proportionally much less on cost of goods sold (40% of revenue versus Meridian's 64%) but more on SG&A (33.3% versus 20%), consistent with a less capital-intensive, more sales-and-marketing-driven business.

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Expressing every line relative to a common base is also the kind of feature engineering that makes statement data usable as input to the machine learning models of Chapter 6, since raw dollar figures are not comparable across firms of different sizes.

The same technique applies to the balance sheet, expressing every line as a percentage of total assets rather than of revenue. For Meridian's Year 2 balance sheet, this gives

$$\begin{aligned} \text{Cash} &= 6.98\%, & \text{Accounts payable} &= 13.18\%, \\ \text{Receivables} &= 18.60\%, & \text{Short-term debt} &= 6.20\%, \\ \text{Inventory} &= 23.26\%, & \text{Long-term debt} &= 38.76\%, \\ \text{Property, plant, and equipment} &= 51.16\%, & \text{Equity} &= 41.86\%, \end{aligned}$$

of total assets. More than half of Meridian's balance sheet, 51.16%, is tied up in fixed, illiquid property, plant, and equipment, while liabilities and equity are split 58.14% to 41.86%, a far more leveraged structure than Solstice's, whose debt-to-equity ratio is only 0.13 (Table 4.6). Common-size balance sheets reveal such structural differences at a glance, before any explicit ratio is computed.

Ratios can also be combined into a single index that predicts financial distress, rather than read one at a time as in Table 4.4. The best known is Altman's Z-score. Four of its five components are variations on ratios already computed: a liquidity measure related to the working capital of Section 4.4.1, a leverage cushion related to the debt-to-equity ratio of Section 4.4.2, and earning-power and asset-turnover measures related to the DuPont components of Section 4.4.5. The fifth measures cumulative retained profitability, and the weights are coefficients Altman estimated statistically from historical bankruptcy data rather than derived from theory. For a private (non-publicly-traded) firm like Meridian, the Z′-score variant is

$$Z' = 0.717X_1 + 0.847X_2 + 3.107X_3 + 0.420X_4 + 0.998X_5, \tag{4.6}$$

where

$$\begin{aligned} X_1 &= \text{Working capital/Total assets}, & X_2 &= \text{Retained earnings/Total assets}, \\ X_3 &= \text{EBIT/Total assets}, & X_4 &= \text{Book value of equity/Total liabilities}, \\ X_5 &= \text{Sales/Total assets}. \end{aligned}$$

Assuming, for illustration, that all of Meridian's \$54.0 million in equity represents retained earnings, the components are

$$X_1 = \frac{38.0}{129.0} = 0.295, \quad X_2 = \frac{54.0}{129.0} = 0.419, \quad X_3 = \frac{16.0}{129.0} = 0.124,$$

$$X_4 = \frac{54.0}{75.0} = 0.720, \quad X_5 = \frac{150.0}{129.0} = 1.163,$$

giving

$$Z' = 0.717(0.295) + 0.847(0.419) + 3.107(0.124) + 0.420(0.720) + 0.998(1.163) \approx 2.41.$$

Altman's original thresholds classify $Z' > 2.9$ as a "safe" zone, $1.23 < Z' < 2.9$ as a "gray" zone of ambiguous risk, and $Z' < 1.23$ as a "distress" zone with a high historical likelihood of bankruptcy within two years. Meridian's score of 2.41 falls in the gray zone, consistent with everything the chapter has found: healthy-looking ratios on the surface, a deteriorating cash position and rising leverage underneath. The score is useful because it compresses many individual signals into one number, but like any single ratio it is a starting point for investigation, not a verdict. Altman's coefficients were estimated with statistical discriminant analysis, a direct ancestor of the classification methods in Chapter 6.

The same score for Solstice Robotics (Section 4.5.1) makes the contrast concrete. Using Solstice's Year 2 figures, and again assuming all of Solstice's \$75.0 million in equity represents retained earnings, working capital is $70.0 - 15.0 = 55.0$ (current assets less accounts payable, its only current liability), so

$$X_1 = \frac{55.0}{100.0} = 0.550, \quad X_2 = \frac{75.0}{100.0} = 0.750, \quad X_3 = \frac{27.0}{100.0} = 0.270,$$

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$$X_4 = \frac{75.0}{25.0} = 3.000, \quad X_5 = \frac{120.0}{100.0} = 1.200,$$

where Solstice's EBIT of \$27.0 million matches the interest coverage figure in Table 4.6 ($27.0 \times \$1.0$ million interest expense = \$27.0 million), giving

$$Z' = 0.717(0.550) + 0.847(0.750) + 3.107(0.270) + 0.420(3.000) + 0.998(1.200) \approx 4.33.$$

Solstice's score of 4.33 sits comfortably in Altman's "safe" zone, in contrast to Meridian's 2.41, for reasons the chapter has already surfaced: Solstice carries far less leverage ($X_4 = 3.00$ versus Meridian's 0.72) and a far larger cushion of retained earnings and working capital relative to its assets. The Z-scores summarize quantitatively what the DuPont decomposition and Table 4.6 showed: firms reach their profitability through different combinations of margin, efficiency, and leverage, and a distress score is only as informative as the analyst's understanding of which component drives it.

4.6 From Statements to Forecasts

Statements describe the past; valuation needs the future. This part connects reported figures to the cash flows Chapter 5 discounts, notes how far reported numbers can be trusted, and builds a forecast from them.

4.6.1 Connecting Financial Statements to Valuation

Financial statement analysis often serves valuation. Investors value a firm by estimating the cash flows it can generate and discounting them, and the statements help identify those cash flows: revenues, margins, working capital needs, investment requirements, and financing choices all affect them.

A simple valuation logic is that a firm's value is the present value of future free cash flows,

$$V = \sum_{t=1}^{T} \frac{FCF_t}{(1+r)^t},$$

where r reflects the riskiness of those cash flows. The statements supply the raw material: revenue growth and margin trends inform the profit forecast, balance sheet data inform working capital and investment needs, and the cash flow statement shows how earnings have historically converted into cash. Projecting Meridian's free cash flow forward on its current negative trajectory would imply a far less attractive valuation than net income alone suggests, which is why practitioners insist on free cash flow rather than accounting earnings as the input to valuation.

Free cash flow, however, comes in two varieties that answer two different questions, and conflating them is a common and consequential error. Both start from CFO, the operating cash flow of Section 4.3.3. Free cash flow to the firm (FCFF) measures cash available to *all* capital providers, debt and equity holders combined, before any financing decisions,

$$\text{FCFF} = \text{CFO} + \text{Interest expense} \times (1 - \text{tax rate}) - \text{Capex}, \tag{4.7}$$

adding back after-tax interest because it is a payment *to* one class of capital provider (debtholders), not a cost of the business itself. Free cash flow to equity (FCFE) measures cash available to equity holders after debtholders have been paid and any net new borrowing received,

$$\text{FCFE} = \text{CFO} - \text{Capex} + \text{Net borrowing}. \tag{4.8}$$

Applying both to Meridian's Year 2 statements (CFO = \$6.0 million, Capex = \$14.0 million, interest expense \$4.0 million, tax rate 25%, net new debt issued \$12.0 million),

$$\text{FCFF} = 6.0 + 4.0(1 - 0.25) - 14.0 = -\$5.0 \text{ million}, \qquad \text{FCFE} = 6.0 - 14.0 + 12.0 = \$4.0 \text{ million}.$$

The two measures have opposite signs. Meridian's business, independent of how it is financed, consumed \$5.0 million more cash than it generated, but \$12.0 million of new debt more than covered that gap, leaving \$4.0

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million for equity holders. An analyst who computed only FCFE would see a healthy positive number and could miss that the core operations are cash-negative and that the positive figure exists only because of a financing decision, an increase in leverage, that carries risks this chapter has already flagged. FCFF is the natural input to an enterprise-value calculation discounted at the WACC of Chapter 5, since it is the cash available to every capital provider WACC blends together. FCFE is the input to a direct equity valuation discounted at the cost of equity alone. Pairing the wrong rate with the wrong cash flow (WACC with FCFE, or the cost of equity with FCFF) produces an error that does not announce itself, only a wrong number.

For AI applications, a machine learning system can find patterns in balance sheet or cash flow data, but those patterns must be interpreted in economic terms. A distress model should be grounded in real indicators such as falling liquidity, rising leverage, declining margins, and weak cash generation; otherwise it may capture noise rather than economic reality.

4.6.2 Accounting Quality and the Limits of Reported Numbers

Reported numbers are not always fully transparent. Accounting rules allow management some discretion, particularly in areas such as revenue recognition, depreciation, reserves, and provisioning. A firm may use aggressive accounting policies to smooth earnings or to present a stronger balance sheet than the underlying reality supports.

Careful analysts therefore ask how the statements were produced, not only what they say. They look for unusual changes in reserves, one-time charges, deferred revenue patterns, and recurring non-cash items, and compare reported figures with cash flow trends and outside information such as industry reports, competitor behavior, or macroeconomic indicators. Meridian's widening gap between net income and operating cash flow, driven by growing receivables and inventory, is a textbook signal to investigate: is the firm's revenue recognition aggressive, are customers slowing their payments, or is inventory building up because of weakening demand?

This gap between reported earnings and cash generation can itself be quantified. One simple accruals ratio measure is

$$\text{Accruals ratio} = \frac{\text{Net income} - \text{Cash flow from operations}}{\text{Total assets}}.$$

For Meridian, this is $(9.0 - 6.0)/129.0 \approx 2.3\%$: net income exceeds operating cash flow by an amount equal to 2.3% of total assets. Research on earnings quality has repeatedly found that firms with unusually high accruals ratios, whose reported earnings substantially exceed cash generation, tend on average to see earnings growth disappoint and stock returns underperform afterward. A large gap between earnings and cash warns that current profitability may be inflated by aggressive accounting, temporary working-capital effects, or both, rather than durable operating improvement. A single year's accruals ratio is weak evidence on its own. The signal becomes more informative tracked over several years and compared with the firm's own history and its peers, a sector-adjusted, longitudinal comparison that a cross-sectional machine learning model of the kind in Chapter 6 is well suited to automate.

Case Vignette: Wirecard and the Cash That Was Not There

Wirecard, a German payments processor that had joined the DAX index of Germany's largest listed companies in 2018, is the cautionary case for anyone who trusts the accruals ratio alone. On 18 June 2020 its auditor, EY, declined to sign off on the 2019 accounts because it could not confirm that €1.9 billion of cash, about a quarter of the balance sheet, sat in trustee accounts at two banks in the Philippines. Both banks said Wirecard was not their client, and the documents purporting to show the balances were forgeries. On 22 June Wirecard said the money most likely did not exist, and on 25 June it filed for insolvency. A KPMG special audit published that April had already been unable to verify the business with third-party acquiring partners that Wirecard reported as about half of its revenue and operating profit, and Financial Times reporting since early 2019 had argued that those partners and the cash attributed to them were not what they seemed (McCrum 2022). The earnings were real on paper; the cash behind them was claimed to be held by outside trustees, out of the reach of anyone who wanted to count it.

The stylized numbers below, for a fictional processor, Halcyon Payments (€ millions), show why this chapter's ratios flag such a scheme at one stage and are defeated at the next, and which simple extra check

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restores them. In Year 2, Halcyon's revenue grows 40% but receivables almost triple, so days sales outstanding, the receivables leg of the cash conversion cycle, doubles; operating cash flow falls even as profit rises.

	Year 1	Year 2	Year 3
Revenue	1,000	1,400	1,900
Net income	150	240	340
CFO (reported)	210	160	430
of which received into trustee accounts	0	0	500
Receivables	150	420	300
Cash at own banks / at trustees	400/0	450/0	380/500
Total assets	2,000	2,600	3,400
DSO (days)	54.8	109.5	57.6
CFO / EBITDA	0.84	0.41	0.80
Accruals ratio, reported	−3.00%	3.08%	−2.65%
Accruals ratio, own accounts	−3.00%	3.08%	12.06%

(EBITDA is 250, 390, and 540.) Year 2 is the pattern Meridian's analysis taught an analyst to flag: the accruals ratio turns positive at $(240 - 160)/2{,}600 = 3.08\%$ and cash conversion, CFO divided by EBITDA, falls from 0.84 to 0.41. In Year 3 the flags go quiet. The receivables are reported as collected, DSO returns to 57.6 days, cash conversion recovers to 0.80, and the accruals ratio falls to $(340 - 430)/3{,}400 = -2.65\%$, better than Year 1. But €500 million of that operating cash flow went into accounts held by third-party trustees, not into Halcyon's own banks, whose balance fell by €70 million. Recomputed on cash the firm itself controls, operating cash flow is $430 - 500 = -70$ million euros and the accruals ratio is $(340 + 70)/3{,}400 = 12.06\%$, the worst of the three years; over Years 2 and 3 together the firm reported €580 million of profit and brought €90 million of operating cash into its own accounts. The trustee balance is 56.8% of the firm's cash and 14.7% of its total assets.

Two lessons follow. A ratio is only as good as the line items under it: accruals and cash-conversion screens test whether earnings became cash, and a scheme that fabricates the cash, not just the earnings, passes them. And the defense is not a cleverer ratio but a question about existence and control: where is the cash, who holds it, and has anyone independent confirmed it directly with the bank, the step whose omission for years was at the heart of the criticism of Wirecard's audits. A machine learning model trained on reported figures inherits every assumption the statements make, including the assumption that the cash on the balance sheet exists.

Such a model may also learn patterns that are partly artifacts of accounting conventions rather than economic reality, so the analyst must understand the reporting environment and remain skeptical of simplistic interpretations.

4.6.3 Forecasting from Statements

Statement analysis is especially valuable for forecasting. Analysts project revenues, costs, margins, capital expenditures, and working capital needs, which then feed valuation models; a well-built forecast shows whether a firm is likely to create value, consume cash, or need external financing.

A simple forecasting approach ties each balance sheet and income statement line to revenue. For example, if COGS is historically 64% of revenue (as it is for Meridian: $96.0/150.0 = 64\%$), and receivables are historically 16% of revenue (24.0/150.0), then a revenue forecast for Year 3 immediately implies forecasts for COGS and receivables under a "percent of sales" model. The method is intentionally simple and can be extended with separate assumptions for margins, working capital efficiency, and capital intensity as information arrives.

Table 4.8 carries the method through, projecting Meridian's full Year 3 income statement and key working-capital lines under two assumptions: 8% revenue growth, and every other line holding its historical percent-of-revenue (or, for accounts payable, percent-of-COGS) ratio constant, with interest expense held fixed at its Year 2 dollar level since it depends on the debt outstanding rather than on sales.

Two things stand out from this forecast. First, net income is projected to grow from $9.0 million to $9.96 million, a 10.7% increase, faster than the 8% revenue growth assumption, because fixed interest expense does not grow with revenue, so a larger share of incremental operating income flows through to the bottom line, an instance of financial leverage. Second, accounts receivable and inventory are forecast to grow with

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Line	Year 2	Year 3 (forecast)	Basis
Revenue	150.0	162.0	8% growth assumption
Cost of goods sold	96.0	103.68	64.0% of revenue
Gross profit	54.0	58.32	
SG&A	30.0	32.40	20.0% of revenue
Depreciation and amortization	8.0	8.64	5.3% of revenue
Operating income (EBIT)	16.0	17.28	
Interest expense	4.0	4.0	held constant
Pretax income	12.0	13.28	
Taxes (25%)	3.0	3.32	
Net income	9.0	9.96	
Accounts receivable	24.0	25.92	16.0% of revenue
Inventory	30.0	32.40	20.0% of revenue
Accounts payable	17.0	18.36	17.7% of COGS

Table 4.8: A percent-of-sales forecast for Meridian, Year 3 ($ millions)

revenue, so, as in the Year 2 cash flow statement, this growth consumes cash even when the business performs as historically expected. A complete forecast carries the projected balance sheet into a projected cash flow statement to check whether Meridian's growth is self-funding or still needs external financing, the free-cash-flow valuation exercise of Section 4.6.1. Forecasting from statements is therefore not mechanical: the percent-of-sales method makes a specific, checkable assumption (that these ratios remain constant), and the exercise is worth as much for questioning that assumption as for computing its consequences.

That complete forecast resolves a question left open since Table 4.3. Year 3 cash flow from operations is projected net income plus depreciation, less the increase in receivables and inventory, plus the increase in payables:

$$\text{CFO}_3 = 9.96 + 8.64 - (25.92 - 24.0) - (32.40 - 30.0) + (18.36 - 17.0) = 15.64,$$

more than double Year 2's $6.0 million, because the dollar growth in net income and depreciation outpaces the dollar growth in working capital even though every underlying ratio is held constant. Holding capital expenditures at their Year 2 percentage of revenue ($14.0/150.0 = 9.33\%$) gives a Year 3 projected capex of $0.0933 \times 162.0 \approx \$15.12$ million, so

$$\text{FCF}_3 = \text{CFO}_3 - \text{Capex}_3 = 15.64 - 15.12 = \$0.52 \text{ million},$$

compared to Year 2's free cash flow of $6.0 - 14.0 = -\$8.0$ million. Under these assumptions, Meridian's free cash flow turns from sharply negative to barely positive with no change in margins or efficiency, because operating cash flow scales faster with revenue than working-capital investment once the heaviest receivables-and-inventory buildup is past. This $0.52 million is the Year 3 figure with which Chapter 5's discounted-cash-flow valuation of Meridian begins; an analyst extrapolating Year 2's deeply negative free cash flow would undervalue Meridian relative to what the same assumptions imply.

Forecasting balances structure and judgment. Some assumptions rest on historical averages, others on strategy, industry conditions, and the macroeconomic outlook. Economies of scale may be expected to improve margins, but competition can delay the improvement; planned capacity expansion can suffer delays and cost overruns. Financial forecasting is both quantitative and interpretive.

The best forecasts usually rest on simple, transparent assumptions that can be tested and updated. An overly complex model may look precise while offering little economic insight, a lesson that applies equally to financial analysis and machine learning.

4.7 Challenges in Financial Statement Analysis

Several practical challenges complicate the use of financial statements, especially as inputs to quantitative or machine learning models.

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Comparability across firms and time. Accounting standards differ across countries (for example, the International Financial Reporting Standards, or IFRS, used across much of the world, versus U.S. Generally Accepted Accounting Principles, or GAAP), and even within a single standard, firms have discretion over methods such as inventory costing (FIFO versus LIFO) or depreciation schedules. A model trained on ratios pooled across many firms may compare numbers that are not comparable unless these differences are adjusted for.

Reporting lag and low frequency. Financial statements are published quarterly or annually, weeks after the period ends. They are a slow-moving input compared with market prices or transaction data, and a model relying mainly on them reacts to changes in a firm's condition only after a considerable delay.

Restatements and revisions. Reported figures are sometimes restated after publication, because of errors, changes in accounting policy, or audit corrections. A dataset built from "as first reported" figures can differ meaningfully from one incorporating later restatements, and using the wrong vintage can introduce a subtle look-ahead bias into a backtest.

Manipulation and earnings management. Management has some latitude in how figures are presented. Detecting earnings management, for example through unusual accruals, is an active area of quantitative research, and a model that ignores this risk may be fooled by cosmetically strong statements.

Non-financial and off-balance-sheet items. Important sources of risk or value, such as operating leases (in some reporting regimes), pending litigation, environmental liabilities, or human capital, may not appear cleanly on the balance sheet, so statement-based models can miss economically material risks.

Sector heterogeneity. A ratio that signals distress in one industry may be normal in another; utilities and airlines routinely carry leverage that would alarm a software company. A cross-sectional model must account for sector context rather than apply a single threshold.

None of this means abandoning statement-based analysis. Financial statements are a valuable but imperfect and delayed signal, to be combined with market data, qualitative judgment, and, where available, higher-frequency alternative data.

4.8 Key Takeaways

Financial statements are the backbone of corporate finance. The balance sheet shows the firm's financial structure, the income statement its performance, and the cash flow statement the movement of cash; together they show whether a business is profitable, liquid, solvent, and capable of creating value.

The Meridian example showed a firm reporting solid net income and comfortable liquidity ratios while burning cash and increasing leverage, a discrepancy visible only when all three statements and their ratios are examined together. Understanding the statements comes before advanced forecasting or valuation, and before trusting any machine learning model built on accounting data. The Wirecard case marked the limit of that skepticism: ratios test whether earnings turned into cash, so a fraud that fabricates the cash passes them, and only asking who holds the cash, and whether anyone independent has confirmed it, catches it.

The comparison with Solstice Robotics reinforced a point that recurs throughout the book: a single summary statistic, whether a ratio, a Z′-score, or a model's output score, can mask very different underlying stories, which the DuPont identity and common-size statements exist to unpack.

The sector-specific discussion cautions against a single cross-firm model that ignores how differently a bank, a utility, and a technology company are financed, even though all report the same three statements.

Chapter 5's valuation methods take as input cash flows that come from the statement analysis, and the skepticism about statement quality, developed here. Chapter 9's credit models build on this chapter's leverage, liquidity, and distress-scoring ideas, with statistical estimation in place of a fixed formula like Altman's Z′-score.

4.9 Suggested Exercises

4.1 Explain the difference between assets, liabilities, and equity, and verify that the accounting identity holds for both years of Meridian's balance sheet.

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4.2 Using Table 4.1 and Table 4.2, compute Meridian's current ratio, quick ratio, and debt-to-equity ratio for Year 1, and compare them to the Year 2 values computed in the chapter.

4.3 Describe why cash flow can differ from accounting profit, using Meridian's Year 2 numbers as a concrete illustration.

4.4 Compute Meridian's free cash flow if capital expenditures had instead been \$6.0 million rather than \$14.0 million, holding all other cash flow items fixed, and discuss how this would change your assessment of the firm.

4.5 Using the DuPont identity, decompose Meridian's ROE for Year 1 (assume net income of \$7.5 million and revenue of \$135.0 million for Year 1) and compare the drivers of ROE across the two years.

4.6 Explain why a company with growing sales, like Meridian, might still face financial distress, referencing the cash conversion cycle.

4.7 Discuss two ways that accounting discretion could distort a machine learning model trained to predict corporate default using financial ratios.

4.8 Propose a percent-of-sales forecast for Meridian's Year 3 revenue, COGS, and accounts receivable, assuming 8% revenue growth and that historical ratios to revenue remain constant.

4.9 Using Table 4.5, verify the current ratio, quick ratio, and debt-to-equity ratio reported for Solstice in Table 4.6.

4.10 Explain, using the DuPont identity, why Meridian and Solstice can have different return on equity even though the gap between their net margins (6.0% versus 16.25%) is larger than the gap in their ROE (16.7% versus 26.0%).

4.11 Using Table 4.5, construct a common-size balance sheet (each line as a percent of total assets) for Solstice's Year 2 balance sheet, and compare it to Meridian's common-size balance sheet computed in this chapter, identifying the single largest structural difference between the two firms.

4.12 Compute Meridian's accruals ratio for a hypothetical Year 3 in which net income is \$11.0 million and cash flow from operations is \$4.0 million (total assets unchanged at \$129.0 million), and explain whether this year's figure is more or less concerning than Year 2's.

4.13 Using the Z′-score formula, recompute Meridian's distress score assuming Year 1 figures instead of Year 2 (Year 1 EBIT can be assumed at \$13.0 million, retained earnings equal to Year 1 equity of \$52.0 million, and other Year 1 figures from Table 4.1), and discuss whether Meridian's financial health appears to be improving or deteriorating between the two years.

4.14 Verify the Solstice Z′-score computed in Section 4.5.1 by recomputing X_1 through X_5 from Table 4.5, and explain in one sentence which single component contributes most to the gap between Solstice's score and Meridian's.

4.15 Using Table 4.8's Year 3 projections, recompute Meridian's projected free cash flow if the revenue growth assumption were 12% instead of 8% (holding every ratio in Table 4.8 fixed at its stated percentage of revenue or COGS), and explain why free cash flow does not simply scale in proportion to revenue growth.

4.16 Compute Meridian's return on invested capital and economic profit if its weighted average cost of capital were 12% instead of 8%, holding NOPAT and invested capital at their Year 2 values, and explain what this implies about whether Meridian is creating or destroying value at that higher cost of capital.

4.17 Rework the Halcyon Payments case of Section 4.6.2 for a milder Year 3, in which only €250 million of operating cash flow goes into trustee accounts, so that own-bank cash ends the year at €630 million (reported CFO of €430 million, net income €340 million, and total assets €3,400 million unchanged).

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Compute operating cash flow into the firm's own accounts, the reported and own-account accruals ratios, and the trustee share of total cash. Which of the chapter's screens would flag this year, and what single question to the auditor would settle the matter?

4.18 **Challenge (real data).** Pick any US public company and download its most recent 10-K balance sheet and income statement figures directly from the SEC EDGAR XBRL `companyfacts` API (free, no login required). (a) Compute the same ratio suite Section 4.4.5 computed for Meridian: current ratio, quick ratio, debt-to-equity, and a three-factor DuPont ROE decomposition. (b) Using this chapter's Altman Z′-score formula, classify the firm into the safe, gray, or distress zone, and comment on whether the classification matches your own intuition about the company's financial health. (c) Real XBRL filings tag the same economic concept (for example, inventory or long-term debt) under dozens of different standardized and company-specific tags across firms and years; describe at least one tag-matching or missing-data problem you encountered while pulling the figures needed for part (a), and how you resolved it.

4.10 Further Reading

- Brealey et al. (2025), *Principles of Corporate Finance*. Covers the corporate-finance logic behind cash flow, capital structure, and value creation that underlies this chapter's ratio analysis and forecasting material.
- Penman (2013), *Financial Statement Analysis and Security Valuation*. A rigorous treatment of how ratio analysis and common-size statements, introduced in this chapter with Meridian and Solstice, connect to security valuation more broadly.
- Damodaran (2012), *Investment Valuation*. Extends this chapter's percent-of-sales forecasting into the full range of valuation techniques developed formally in Chapter 5.
- Palepu et al. (n.d.), *Business Analysis and Valuation*. A case-based approach to interpreting financial statements and diagnosing accounting quality issues, complementing this chapter's Altman Z-score and distress-analysis material.
- McCrum (2022), *Money Men: A Hot Startup, a Billion-Dollar Fraud, a Fight for the Truth*. The Financial Times journalist whose reporting exposed Wirecard tells the story from the inside, a readable account of how strong reported numbers, an unquestioned audit, and a defensive regulator kept a fabricated business alive for years.

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Chapter 5

Valuation and Asset Pricing

5.1 Introduction

The preceding chapters built the pieces needed to price a financial asset: Chapter 3 described the institutions and instruments that are traded, and Chapter 4 showed how a firm's financial statements reveal the cash flows it generates. This chapter puts those pieces together and asks the central question of asset pricing: given a stream of future cash flows, or a payoff that depends on some other asset's price, what should that claim be worth today?

The chapter answers it with worked examples for the three main instrument classes introduced in Chapter 3: bonds, equities, and derivatives. Every formula comes with a computation a reader can check by hand or reproduce in the companion notebook, because a practitioner should be able to compute a price, not just describe the logic behind it.

The chapter builds in complexity. It begins with the simplest cash flow structures, a single-maturity bond and a constant-growth dividend stream, where closed-form formulas exist and every number can be verified by hand. It then introduces two ways of estimating the discount rate these formulas depend on, the term structure of interest rates and the cost of capital, before turning to the nonlinear payoffs of options, which require the replication and no-arbitrage arguments of modern derivatives pricing. The chapter can also be read as a single valuation story. Meridian Fabrication, the running example introduced in Chapter 4, first shows how a firm's financing mix determines the discount rate; it is then valued with a market-based multiple and finally from its projected cash flows, and the two values are compared. Bonds, equities, and options are not unrelated topics: the same present-value logic reappears in each.

5.2 Discounting and Fixed Income

Every valuation in this book reduces to moving cash across time and charging for the privilege. This part develops that machinery on the instruments where it is cleanest, then asks what happens to a bond's price when the rate used to discount it moves.

5.2.1 The Time Value of Money, Revisited

Chapter 1 introduced the present value relation

$$V = \sum_{t=1}^{T} \frac{CF_t}{(1+r)^t}$$

as the unifying idea behind bond pricing, stock valuation, and project appraisal. This chapter makes that idea operational. Two quantities determine the value of any claim on future cash flows: the size and timing of the cash flows themselves, and the discount rate r, which compensates the holder for the time value of money and for the risk that the cash flows might not arrive as expected. A safer stream of cash flows is discounted at a lower rate and is therefore worth more, all else equal, than a riskier stream of the same expected size.

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Why can the same relation describe such different instruments? A bond's cash flows are contractually fixed (barring default) and known in advance; a stock's cash flows are uncertain and expected to grow; an option's payoff is not just uncertain but depends nonlinearly on another asset's future price. What varies across the chapter is not the principle but the complexity of CF_t and the effort required to pin down r.

5.2.2 Bond Pricing and Yield to Maturity

Bonds come first because their cash flows are the least ambiguous: a fixed schedule of coupons and a principal repayment, specified in the bond's contract at issuance. A bond with face value F, annual coupon C, and n years to maturity is priced by discounting each promised cash flow at the market yield y:

$$P = \sum_{t=1}^{n} \frac{C}{(1+y)^t} + \frac{F}{(1+y)^n}. \tag{5.1}$$

Consider a 3-year bond with face value $F = \$1{,}000$ and an annual coupon rate of 5% (so $C = \$50$ per year), priced at a market yield of $y = 7\%$. Table 5.1 shows each discounted cash flow.

Year t	Cash flow	Discount factor $(1+y)^{-t}$	Present value
1	\$50.00	0.9346	\$46.73
2	\$50.00	0.8734	\$43.67
3	\$1,050.00	0.8163	\$857.11
		Price	**\$947.51**

Table 5.1: Pricing a 3-year, 5% annual-coupon bond at a 7% yield

The bond trades below its \$1,000 face value because the market yield (7%) exceeds the coupon rate (5%); an investor demands compensation via a lower purchase price. The yield to maturity is, conversely, the single discount rate y that makes the present value of the promised cash flows equal to the observed market price; for $n > 1$ this equation generally has no closed-form solution and is instead solved numerically (for example with a root-finding routine such as `scipy.optimize.brentq`).

This inverse relationship between price and yield is one of the most important facts in fixed income and the reason bond portfolios carry interest-rate risk. The sensitivity of price to yield is commonly summarized by duration, a weighted average of the times at which cash flows are received, weighted by the present value of each cash flow. For the bond in Table 5.1, the (Macaulay) duration is approximately 2.86 years, close to but less than the 3-year maturity, because part of the value comes from the earlier coupon payments.

Pricing Default Risk

The formula above assumes the promised cash flows arrive with certainty, a reasonable approximation for a government bond but not for a corporate bond, which may default. A simple fix is to discount the *expected* cash flow rather than the promised one, allowing for both the probability of default and the partial recovery a bondholder typically receives after a default.

Consider a simplified one-year, zero-coupon corporate bond with face value \$1,000, a risk-free rate of $r_f = 4\%$, an estimated one-year probability of default $p = 3\%$, and a recovery rate of 40% of face value if default occurs (that is, bondholders recover \$400 per \$1,000 of face value in default). The expected payoff at maturity is

$$\mathbb{E}[\text{Payoff}] = (1-p)(\$1{,}000) + p(0.40)(\$1{,}000) = 0.97(1{,}000) + 0.03(400) = \$982,$$

and discounting this expected payoff at the risk-free rate gives a price of

$$P = \frac{982}{1.04} \approx \$944.23.$$

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Equivalently, this can be expressed as a credit spread added to the risk-free rate: for small default probabilities, the required credit spread is approximately

$$p \times (1 - \text{recovery rate}) = 0.03 \times 0.60 = 1.8\%,$$

giving a corporate bond yield of roughly $4\%+1.8\% = 5.8\%$ and an approximate price of $1{,}000/1.058 \approx \$945.18$, close to but not identical to the expected-cash-flow price, since the credit-spread shortcut is itself an approximation. Both approaches price default risk the same way: by widening the discount rate, or equivalently shrinking the expected cash flow, relative to a risk-free bond with the same promised payments. The size of the adjustment depends on the probability of default and the severity of loss, the two quantities Chapter 9's credit models are built to estimate.

5.2.3 Interest Rate Risk: Using Duration to Approximate Price Changes

Duration is not just a descriptive statistic; it is the basis of a widely used first-order approximation for how a bond's price responds to a change in yield. Modified duration is defined as

$$D_{\text{mod}} = \frac{D_{\text{Mac}}}{(1+y)},$$

and the approximate percentage price change for a small yield change Δy is

$$\frac{\Delta P}{P} \approx -D_{\text{mod}}\,\Delta y. \tag{5.2}$$

For the bond in Table 5.1, whose Macaulay duration of 2.86 years is 2.8553 before rounding, $D_{\text{mod}} = 2.8553/1.07 = 2.6685$ (2.67 to two decimals; the extra digits matter for the dollar figures below). If the market yield rises from 7% to 8% ($\Delta y = +0.01$), the approximation predicts

$$\Delta P \approx -2.6685 \times 0.01 \times \$947.51 \approx -\$25.28, \qquad P_{\text{new}} \approx \$922.23.$$

Repricing the bond exactly at $y = 8\%$ using the original present-value formula gives $P_{\text{new}} = \$922.69$, so the duration approximation is off by about \$0.46, understating the price. If instead the yield falls from 7% to 5% ($\Delta y = -0.02$), the approximation gives $P_{\text{new}} \approx \$998.08$, while the exact price is $P_{\text{new}} = \$1{,}000.00$ (at $y = 5\%$, the yield equals the coupon rate, so the bond prices exactly at par). In both directions the approximation understates the true price, an effect known as convexity: the price-yield relationship curves upward, so a linear approximation always predicts too low a price, whether yields rise or fall. Duration is adequate for small yield changes; for larger moves, or where precision matters, convexity adjustments or full repricing are used.

Convexity can be estimated directly from three prices, the price today and the prices after a small yield increase and decrease, using

$$\text{Convexity} \approx \frac{P_+ + P_- - 2P_0}{P_0(\Delta y)^2},$$

and the second-order (duration-plus-convexity) approximation to the price change is

$$\frac{\Delta P}{P} \approx -D_{\text{mod}}\,\Delta y + \frac{1}{2}\,\text{Convexity}\,(\Delta y)^2.$$

Using $P_0 = \$947.5137$ (y=7%), $P_+ = \$922.6871$ (y=8%), and $P_- = \$973.2699$ (y=6%), carried to four decimals because the numerator is a small difference of large numbers,

$$\text{Convexity} \approx \frac{922.6871 + 973.2699 - 2(947.5137)}{947.5137\,(0.01)^2} = \frac{0.9296}{0.0947514} \approx 9.81.$$

Plugging in the cent-rounded prices \$922.69, \$973.27, and \$947.51 instead gives 9.92: rounding each price by less than half a cent moves the numerator from 0.9296 to 0.94, about one percent, a small-scale warning about

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numerical second differences that applies equally to the finite-difference Greeks of later chapters. Adding the convexity term to the earlier duration-only forecast for a rise to $y = 8\%$ gives

$$P_{\text{new}} \approx 947.51\left(1 - 2.6685(0.01) + \tfrac{1}{2}(9.81)(0.01)^2\right) \approx \$922.69,$$

which matches the exact repriced value to the penny, against \$922.23 from duration alone. Duration captures the slope of the price-yield relationship and convexity its curvature; using both is standard practice whenever a fixed-income portfolio's risk is measured precisely, for example in the value-at-risk calculations of Chapter 8.

Portfolio Duration

Duration extends naturally from a single bond to a portfolio of bonds: the portfolio's duration is the market-value-weighted average of the durations of its individual holdings,

$$D_{\text{portfolio}} = \sum_i w_i D_i,$$

where w_i is the fraction of portfolio market value invested in bond i, the same weighted-average structure used for portfolio return in Chapter 2 and for portfolio variance in Chapter 12. Suppose a portfolio holds 60% of its market value in the 3-year bond from Table 5.1 and 40% in a 5-year zero-coupon bond priced at a 7% yield. A zero-coupon bond has a single cash flow, so its duration equals its maturity, 5 years. Using the coupon bond's full-precision duration of 2.8553 years rather than the rounded 2.86, the portfolio duration is

$$D_{\text{portfolio}} = 0.6(2.8553) + 0.4(5.0) \approx 3.71 \text{ years}.$$

This single number lets a portfolio manager approximate the effect of a parallel rate shift on the whole portfolio with the modified-duration approximation, without repricing each bond, which matters for portfolios with hundreds of positions. Managing duration, for example by shifting the mix of short- and long-maturity bonds toward a target, is one of the most basic fixed-income risk management techniques, and it reappears in the market risk measures of Chapter 8.

5.2.4 The Term Structure of Interest Rates

Every bond example so far has used a single discount rate y for all maturities, but the market yield on a bond typically depends on its maturity, a relationship called the term structure of interest rates, or the yield curve. Table 5.2 shows a stylized, upward-sloping yield curve for government bonds of different maturities.

Maturity	1 year	2 years	5 years	10 years	30 years
Yield	4.2%	4.4%	4.8%	5.2%	5.5%

Table 5.2: A stylized government bond yield curve

An upward-sloping curve like the one in Table 5.2 is the most common shape historically and is usually attributed to two effects: investors demand a higher yield for the greater interest-rate risk of a longer-maturity bond (a term premium), and the market may expect short-term rates to rise, which raises long-term yields today. When the curve inverts, with short-term yields above long-term yields, the market is often read as expecting rate cuts, typically because it anticipates an economic slowdown. That is why inversions are watched as a potential recession indicator, the kind of macroeconomic time-series relationship studied in Chapter 7.

Once a full yield curve is available, each cash flow should in principle be discounted at the spot rate for its own maturity, not at the bond's yield to maturity, which is a complex weighted average of those spot rates. The distinction matters most for bonds with cash flows spread over a long horizon and is central to pricing and hedging any rate-sensitive instrument precisely; the single-rate approximation used in the rest of this chapter is adequate for building intuition.

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Worked Example 5.1: Extracting Forward Rates from the Spot Curve

The spot rates in Table 5.2 apply to borrowing or lending that starts today. A forward rate is the rate, implied by today's spot curve, for a loan that starts at a future date and ends at a later one, and no-arbitrage pins it down by the same replication logic as Chapter 3's no-arbitrage bond example. Investing for two years at the 2-year spot rate must pay the same as investing for one year at the 1-year spot rate and then reinvesting for a second year at the rate that can be locked in today for that year,

$$(1+y_2)^2 = (1+y_1)(1+f_{1,2}) \quad \Longrightarrow \quad f_{1,2} = \frac{(1+y_2)^2}{1+y_1} - 1. \tag{5.3}$$

Using Table 5.2's 1-year rate of 4.2% and 2-year rate of 4.4%,

$$f_{1,2} = \frac{1.044^2}{1.042} - 1 \approx 4.60\%,$$

higher than both the 1-year and 2-year spot rates, as an upward-sloping curve implies: the rate for the second year alone must exceed the 2-year average to pull that average up from the lower 1-year rate. The same logic extends to any pair of maturities; using the 2-year and 5-year spot rates, the implied 3-year forward rate spanning years 2 through 5 is

$$f_{2,5} = \left(\frac{(1+y_5)^5}{(1+y_2)^2}\right)^{1/3} - 1 \approx 5.07\%,$$

again above both the 2-year and 5-year spot rates, consistent with the curve's continued upward slope. Forward rates matter beyond the arithmetic. They are the rates a company can lock in today with a forward-starting swap or a forward-rate agreement (relatives of the swaps and forwards of Chapter 3) to hedge future borrowing costs. Under the pure expectations theory of the term structure they can also be read, subject to the term-premium caveat above, as the market's current forecast of future short-term rates.

5.3 Valuing Equity

An equity claim has no maturity and no promised payment, so its value depends entirely on what is expected of it and on the rate demanded in exchange. This part covers where that rate comes from and the two families of method built on it.

5.3.1 The Cost of Capital: CAPM and WACC

Every valuation formula in this chapter takes a discount rate r as an input, and the examples so far, like the dividend discount examples that follow, have simply been handed one. This section shows how that rate is estimated. Firms are financed by a mix of debt and equity, and each source of capital has its own cost.

The cost of debt is the most straightforward: the yield the firm currently pays on its debt (or would pay on new debt of similar risk), adjusted for the tax deductibility of interest payments, since interest expense reduces taxable income,

$$r_d^{\text{after-tax}} = r_d^{\text{pretax}}(1-\tau),$$

where τ is the firm's marginal tax rate. The cost of equity is harder to observe directly, since equity has no contractual payment, and is typically estimated using the Capital Asset Pricing Model (CAPM), covered in full in Chapter 12,

$$r_e = r_f + \beta(r_m - r_f), \tag{5.4}$$

where r_f is the risk-free rate, $r_m - r_f$ is the market risk premium (the extra return investors demand for holding the overall market rather than a risk-free asset), and β measures the stock's sensitivity to market-wide movements. The weighted average cost of capital (WACC) then blends the two financing costs in proportion to their share of the firm's total capital,

$$\text{WACC} = \frac{E}{E+D}\, r_e + \frac{D}{E+D}\, r_d^{\text{after-tax}}, \tag{5.5}$$

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where E and D are the market (or, in the absence of market prices, book) values of equity and debt.

Using Meridian Fabrication's Year 2 figures from Chapter 4,

$$\begin{aligned} E &= \$54.0 \text{ million (equity)}, & D &= \$58.0 \text{ million (total debt)}, \\ r_f &= 4\%, & \text{Market risk premium} &= 6\%, \\ \beta &= 1.2, & \text{Pretax cost of debt} &= 4.0/58.0 \approx 6.90\%, \\ \text{Tax rate} &= 25\%, \end{aligned}$$

where the pretax cost of debt is read off Meridian's own statements rather than assumed: Year 2 interest expense of \$4.0 million on the \$58.0 million of debt outstanding at year end. A rate nearly three percentage points above the risk-free rate is what one would expect for a firm whose Altman Z′-score places it in the gray zone (Chapter 4). Then

$$r_e = 0.04 + 1.2(0.06) = 11.2\%, \qquad r_d^{\text{after-tax}} = 0.0690(1 - 0.25) \approx 5.17\%,$$

$$\text{WACC} = \frac{54.0}{112.0}(11.2\%) + \frac{58.0}{112.0}(5.17\%) \approx 0.482(11.2\%) + 0.518(5.17\%) \approx 8.08\%.$$

Book equity stands in for E because Meridian's shares have no market price in this example; Section 5.5 returns to what changes when a market-based equity value is used instead. This WACC of 8.08%, close to the 8% assumed in Chapter 4's economic-profit example, is the discount rate for Meridian's free-cash-flow valuation and the one that belongs in the economic-profit formula, Invested capital × (ROIC − WACC). A firm earning a return on invested capital below its WACC of 8.08% destroys value even while reporting positive accounting profit. WACC is also sensitive to its inputs: a beta of 1.5 instead of 1.2 would raise the cost of equity to $0.04 + 1.5(0.06) = 13\%$ and the WACC to roughly 8.95%, the discount-rate uncertainty flagged as a challenge later in this chapter.

5.3.2 The Dividend Discount Model for Equities

Equity valuation follows the same discounting logic but applies it to an infinite, growing stream of dividends rather than a fixed set of coupons. The Gordon growth model prices a share as

$$P_0 = \frac{D_1}{r - g}, \tag{5.6}$$

where D_1 is the dividend expected next year, r is the required rate of return on equity, and $g < r$ is the assumed constant long-run growth rate of dividends. For example, if a firm is expected to pay a dividend of $D_1 = \$3.00$ next year, investors require a return of $r = 9\%$, and dividends are expected to grow at $g = 4\%$ per year indefinitely, the model implies a fair price of

$$P_0 = \frac{3.00}{0.09 - 0.04} = \frac{3.00}{0.05} = \$60.00.$$

The model's simplicity is also its main weakness: the price is extremely sensitive to the gap $r - g$, so small changes in the assumed growth or discount rate produce large changes in value. That sensitivity is the practical lesson: a valuation built on this model should be stress-tested over a range of plausible (r, g) pairs, not reported as a single point estimate.

A constant growth rate forever is a strong assumption, since fast-growing firms rarely sustain high growth indefinitely. The two-stage dividend discount model relaxes this by assuming a high growth rate g_1 for the first n years, followed by a lower, sustainable growth rate $g_2 < r$ thereafter. The price is the present value of the high-growth dividends plus the present value of a terminal value computed with the Gordon growth formula at the end of the high-growth period:

$$P_0 = \sum_{t=1}^{n} \frac{D_0(1+g_1)^t}{(1+r)^t} + \frac{1}{(1+r)^n}\left[\frac{D_0(1+g_1)^n(1+g_2)}{r - g_2}\right].$$

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For example, suppose a firm's most recent dividend was $D_0 = \$3.00$, it is expected to grow at $g_1 = 12\%$ for the next 3 years before settling into a stable long-run growth rate of $g_2 = 4\%$, and investors require $r = 9\%$. The high-growth dividends are

$$D_1 = \$3.36, \qquad D_2 = \$3.76, \qquad D_3 = \$4.21,$$

with a combined present value of \$9.50. The terminal value at the end of year 3 is

$$\frac{D_3(1+g_2)}{r-g_2} = 4.21(1.04)/0.05 \approx \$87.67,$$

which has a present value of \$67.70. Adding the two pieces gives

$$P_0 \approx \$9.50 + \$67.70 = \$77.20.$$

Notice that the terminal value, representing everything beyond year 3, accounts for roughly 88% of the total price; this is typical of growth-stage valuations and is why the assumed terminal growth rate g_2, not the near-term high growth rate, is usually the most consequential assumption in the model.

Preferred Stock: A Perpetuity Special Case

Preferred stock typically pays a fixed dividend indefinitely, with no growth at all, which makes the Gordon growth formula collapse to the special case $g = 0$:

$$P_0 = \frac{D}{r}.$$

This is a growing perpetuity with zero growth, sometimes called a level perpetuity. For example, a preferred share paying a fixed annual dividend of $D = \$4.50$, with investors requiring $r = 7.5\%$ given the security's risk, is worth $P_0 = 4.50/0.075 = \$60.00$. Because its dividends are fixed, preferred stock is valued more like a bond than like common stock, even though it is legally equity: for valuation, an instrument's economic behavior, here a fixed, bond-like cash flow, matters more than its legal label.

5.3.3 Relative Valuation: Multiples

Every method so far is an absolute valuation method: it estimates intrinsic value from a model of the security's cash flows. Relative valuation instead prices an asset by how similar assets are currently priced, using a valuation multiple.

The most common multiple for valuing an entire business is enterprise value to EBITDA (earnings before interest, taxes, depreciation, and amortization), since EBITDA approximates operating cash flow before financing and accounting choices and enterprise value (the value of both debt and equity claims combined) is capital-structure-neutral, unlike equity value alone. The method has three steps: compute the target firm's EBITDA, multiply it by a multiple observed for comparable, publicly traded firms, and then back out the implied equity value by subtracting net debt (total debt minus cash).

Applying this to Meridian Fabrication: EBITDA is EBIT plus depreciation and amortization, $16.0 + 8.0 =$ \$24.0 million. Suppose comparable industrial parts manufacturers currently trade at an EV/EBITDA multiple of 6.5×. The implied enterprise value is

$$EV = \text{EBITDA} \times \text{Multiple} = 24.0 \times 6.5 = \$156.0 \text{ million}.$$

Net debt is total debt minus cash, $58.0 - 9.0 = \$49.0$ million, so the implied equity value is

$$\text{Equity value} = EV - \text{Net debt} = 156.0 - 49.0 = \$107.0 \text{ million}.$$

This is nearly double Meridian's \$54.0 million book equity from Chapter 4's balance sheet, and the two need not be close: book equity is an accounting construct built from historical costs, while market-implied equity value is a forward-looking assessment of what the business is worth.

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Relative valuation is fast and grounded in observable prices, but it depends entirely on the comparable multiple. If the comparables are overvalued or undervalued, cyclically distorted, or not comparable in growth, margins, and risk, the valuation inherits the error. Absolute (discounted-cash-flow) and relative valuation are best used together: a DCF built on explicit assumptions about growth and discount rates, checked against a multiple grounded in current market pricing. A large disagreement between them is itself information, since it forces the analyst to ask whose assumptions, the DCF's or the market's, are more likely to be right.

The table below summarizes the chapter's valuation tools, including the option-pricing methods developed next, and when each is most useful.

Method	Best used for	Main inputs	Main caveat
Bond pricing	Fixed-income claims with known coupons	Coupon, maturity, face value, yield	Assumes the promised cash flows are paid as stated
Dividend discount model	Mature, dividend-paying equities	Dividend, growth, required return	Very sensitive to the gap $r - g$ and the terminal growth assumption
WACC-based DCF	Firms or projects with a mix of debt and equity	Free cash flow, growth, cost of debt, cost of equity, capital structure	Discount-rate uncertainty can dominate the final value
Multiples	Quick relative valuation	EBITDA or earnings, peer multiple, net debt	Depends on the comparables being truly similar
Binomial or Black-Scholes-Merton	Options and other nonlinear payoffs	Underlying price, strike, volatility, time, rates	Assumes simplifying conditions such as constant volatility

Table 5.3: A quick guide to the main valuation tools in this chapter

5.4 Option Pricing

An option's payoff is not linear in the underlying price, which makes discounting an expected value the wrong tool. This part builds the replication argument that replaces it, works it up to Black-Scholes-Merton, and follows the consequences into risk measurement and corporate decisions.

5.4.1 No-Arbitrage Pricing and the Binomial Option Model

Every valuation so far has followed the same recipe: find the cash flow (or expected cash flow) an instrument delivers, then discount it at a rate reflecting its risk. An option seems as if it should work the same way.

It does not, for a structural reason. A bond's coupon and a stock's expected dividend are each a single number to discount; an option's payoff depends on where the underlying price ends up, and that dependence is not a straight line. A European call option with strike K pays $\max(S_T - K, 0)$ at maturity T, where S_T is the underlying asset price at maturity; a European put pays $\max(K - S_T, 0)$.

Figure 5.1 plots both payoffs as a function of the terminal stock price S_T for a strike of $K = \$50$. The call is worthless ("out of the money") for any $S_T \leq K$ and rises one-for-one with S_T once S_T exceeds K ("in the money"); the put shows the mirror-image pattern. This kink at $S_T = K$ makes the payoff nonlinear and calls for the replication argument below instead of a discounted-cash-flow calculation.

Options are therefore priced by constructing a replicating portfolio of the underlying asset and a risk-free bond that produces the same payoff in every future state, and then invoking the no-arbitrage principle of Chapter 3: two portfolios with identical payoffs must have the same price today.

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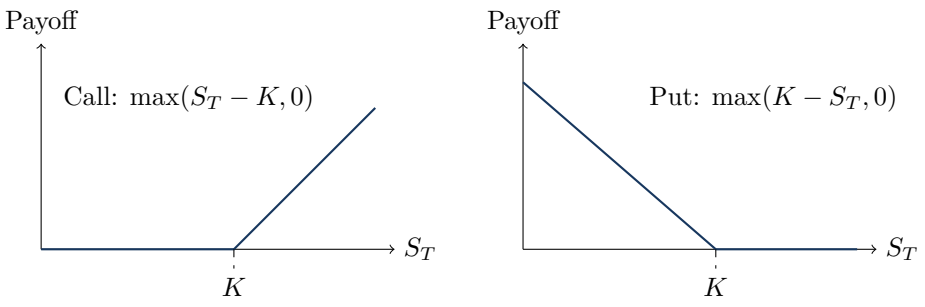


Figure 5.1: Payoff at maturity for a call option (left) and a put option (right) with strike K

The simplest version of this argument is the one-step binomial model. Suppose a stock currently trades at $S_0 = \$50$ and, over one period, either rises to $S_0 u$ or falls to $S_0 d$, with $u = 1.2$ and $d = 0.9$. A risk-free investment over the same period grows by a gross factor $R = 1 + r_f = 1.04$. Consider a call option struck at $K = \$50$. The stock will be worth $S_u = 60$ or $S_d = 45$, so the call will be worth $C_u = \max(60 - 50, 0) = \$10$ or $C_d = \max(45 - 50, 0) = \$0$. No-arbitrage pricing does not need the stock's real-world probability of rising or falling. It asks instead what probability of an up-move makes a portfolio of the stock and a risk-free bond, chosen to replicate the option's payoff in both states, price correctly today. That probability, not a forecast, values the option as a discounted expectation under the risk-neutral probability

$$q = \frac{R - d}{u - d} = \frac{1.04 - 0.9}{1.2 - 0.9} = 0.467, \tag{5.7}$$

so that

$$C_0 = \frac{1}{R}[q\, C_u + (1 - q)\, C_d] = \frac{1}{1.04}[0.467(10) + 0.533(0)] \approx \$4.49. \tag{5.8}$$

The probability q is not the investor's real-world belief about the probability of an up-move; it is the probability implied by the no-arbitrage condition, which is why this is called risk-neutral pricing. The same logic extends, by adding more steps and shrinking the time interval, to the continuous-time model presented next.

Extending to Two Steps

A single step is enough to introduce the mechanics, but a multi-step tree whose up- and down-moves are calibrated to the underlying's volatility is what converges to the Black-Scholes-Merton price, as the companion notebook shows for trees of up to 5,000 steps. Extending the same stock and option to two periods, each still with $u = 1.2$, $d = 0.9$, $R = 1.04$, produces three possible terminal prices: $S_{uu} = 50(1.2)^2 = \$72$, $S_{ud} = 50(1.2)(0.9) = \$54$, and $S_{dd} = 50(0.9)^2 = \$40.50$, with call payoffs $C_{uu} = \$22$, $C_{ud} = \$4$, and $C_{dd} = \$0$. The risk-neutral probability $q = 0.467$ is unchanged, since it depends only on u, d, and R, not on the number of steps, so the tree is solved by working backward one period at a time:

$$C_u = \frac{1}{R}[q\, C_{uu} + (1-q)\, C_{ud}] = \frac{1}{1.04}[0.467(22) + 0.533(4)] \approx \$11.92, \qquad C_d = \frac{1}{R}[q\, C_{ud} + (1-q)\, C_{dd}] \approx \$1.79,$$

$$C_0 = \frac{1}{R}[q\, C_u + (1 - q)\, C_d] \approx \$6.27.$$

Neither this two-step price of \$6.27 nor the one-step \$4.49 should be compared with the Black-Scholes-Merton price of \$6.88 computed later: the hand tree's u and d imply a lower volatility than the 30% the Black-Scholes-Merton example uses, so moving from one step to two with u and d fixed illustrates backward induction, not convergence. Convergence requires each step's moves to be calibrated to the same annualized volatility.

The $u = 1.2$ and $d = 0.9$ above were chosen for hand computation. A tree with any number of steps instead derives u and d from the annualized volatility using the Cox-Ross-Rubinstein parametrization $u = e^{\sigma\sqrt{\Delta t}}$,

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$d = 1/u$, where Δt is the length of one step, so every tree reflects the same annualized volatility σ. Table 5.4 prices the same at-the-money call ($S_0 = K = \$50$, $r = 4\%$, $\sigma = 30\%$, $T = 1$ year) with CRR trees of increasing size; the price oscillates as it converges, and the error shrinks from \$1.40 at one step to under \$0.001 by 5,000 steps, confirming that Black-Scholes-Merton is the limit of the discrete tree as the step size shrinks to zero.

Steps	Binomial price	Error vs. Black-Scholes-Merton
1	\$8.28	+\$1.40
2	\$6.21	−\$0.67
5	\$7.16	+\$0.28
20	\$6.80	−\$0.07
100	\$6.86	−\$0.01
500	\$6.874	−\$0.003
1,000	\$6.875	−\$0.001
5,000	\$6.876	−\$0.0003
∞ (Black-Scholes-Merton)	\$6.8766	—

Table 5.4: Convergence of the CRR binomial call price to the Black-Scholes-Merton price as the number of steps increases ($S_0 = K = \$50$, $r = 4\%$, $\sigma = 30\%$, $T = 1$ year)

Figure 5.2 lays out the full tree, alongside its American put counterpart introduced below, side by side: stock prices grow forward from S_0 on the left to three recombining terminal nodes on the right, while option values are computed in the opposite direction, starting from the known terminal payoffs and working backward one period at a time using q.

5.4.2 American Options and Early Exercise

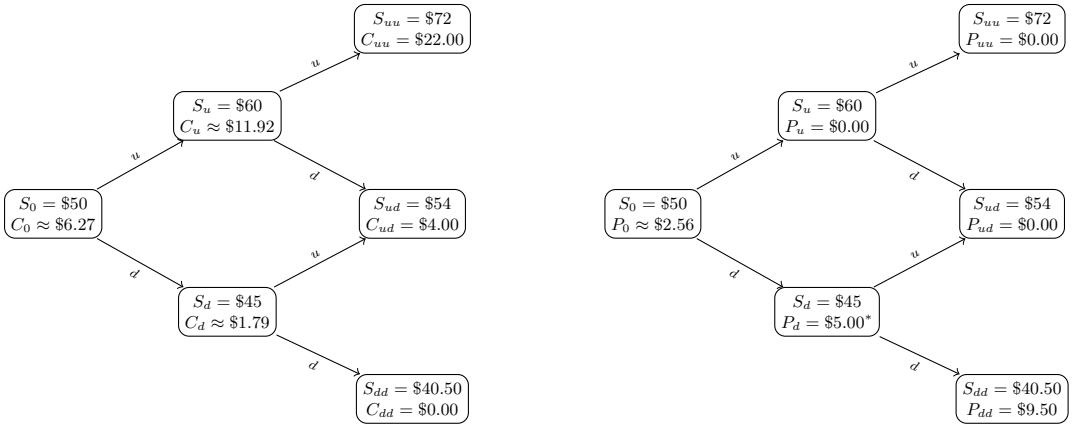
Every option considered so far has been European, exercisable only at maturity. An American option can be exercised at any time up to and including maturity, and the binomial tree extends naturally to price this feature: at each node, compare the value of exercising immediately (the intrinsic value) to the value of continuing to hold the option, and take whichever is larger.

Consider an American put with the same two-period tree and strike $K = \$50$. At maturity the payoffs are unchanged: $P_{uu} = \$0$, $P_{ud} = \$0$, $P_{dd} = \max(50 - 40.50, 0) = \$9.50$. Working backward, the European continuation value at the down node ($S_d = \$45$ after one down move) is $P_d^{\text{cont}} = [q(0) + (1-q)(9.50)]/1.04 \approx \$4.87$. But immediate exercise at that node is worth $\max(50 - 45, 0) = \$5.00$, which exceeds the continuation value, so an optimal holder would exercise early rather than wait: $P_d^{\text{American}} = \max(5.00, 4.87) = \$5.00$. At the up node, both exercise and continuation are worth \$0 (the put is out of the money after an up move), so $P_u^{\text{American}} = \$0$. Rolling back to today, the continuation value is $[q(0) + (1-q)(5.00)]/1.04 \approx \$2.56$, and since immediate exercise today is worth $\max(50 - 50, 0) = \$0$, the American put is worth $P_0^{\text{American}} \approx \$2.56$.

By contrast, the European put (never allowed to exercise early) is worth only $P_0^{\text{European}} \approx \$2.50$, computed by discounting the same terminal payoffs without ever checking intrinsic value along the way. The American put's extra \$0.07 of value (the full-precision values are \$2.5641 and \$2.4984; subtracting the cent-rounded prices would suggest \$0.06) is the early-exercise premium, and it exists because early exercise was optimal at the down node. This feature is why American puts (and American calls on dividend-paying underlyings) generally have no closed-form formula like Black-Scholes-Merton and require a tree, a finite-difference method, or a simulation that can check the exercise decision at every point in time.

In Figure 5.2 the two trees share identical stock prices, since u, d, and q do not depend on the option being priced; only the option values on the bottom line of each node differ. The asterisk at the American put's down node marks where the logic diverges: every other node is a discounted expectation of its children, but $P_d = \$5.00$ replaces a \$4.87 continuation value because immediate exercise is worth more. That single substitution is the entire early-exercise premium.

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European call American put

Figure 5.2: The two-step binomial tree for the \$50-strike European call (left) and American put (right). Both share the same stock-price tree, growing forward from S_0 under up- and down-moves $u = 1.2$ and $d = 0.9$ and recombining at the middle terminal node. Option values (bottom line of each node) are computed backward from the known terminal payoffs using the risk-neutral probability $q = 0.467$, except at the American put's down node, marked with an asterisk, where the \$5.00 intrinsic value of immediate exercise replaces the lower \$4.87 continuation value, producing the early-exercise premium discussed in the text.

5.4.3 The Black-Scholes-Merton Model

The Black-Scholes-Merton model is the continuous-time limit of the binomial approach and remains the benchmark option pricing formula in practice and theory. For a non-dividend-paying stock with current price S_0, strike K, risk-free rate r (continuously compounded), volatility σ, and time to maturity T (in years), the price of a European call is

$$C = S_0 N(d_1) - K e^{-rT} N(d_2), \qquad d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}, \tag{5.9}$$

where $N(\cdot)$ is the standard normal cumulative distribution function. Using the same at-the-money strike as the binomial example, $S_0 = K = \$50$, with $r = 4\%$, $\sigma = 30\%$, and $T = 1$ year,

$$d_1 = \frac{0 + (0.04 + 0.045)(1)}{0.30} = 0.283, \qquad d_2 = 0.283 - 0.30 = -0.017,$$

and looking up $N(0.283) = 0.6115$ and $N(-0.017) = 0.4934$ gives

$$C = 50(0.6115) - 50\, e^{-0.04}(0.4934) \approx \$6.88.$$

This is the price the CRR tree of Table 5.4, calibrated to the same $\sigma = 30\%$, approaches as its step size shrinks, not the hand-computed \$4.49 and \$6.27, whose $u = 1.2$ and $d = 0.9$ imply a lower volatility. The corresponding put price is

$$P = K e^{-rT} N(-d_2) - S_0 N(-d_1) \approx \$4.92,$$

and the two prices must satisfy put-call parity,

$$C + K e^{-rT} = P + S_0, \tag{5.10}$$

which can be checked directly: $6.88 + 50\, e^{-0.04} = 6.88 + 48.04 = \$54.92$, and $4.92 + 50 = \$54.92$, confirming consistency. Put-call parity holds regardless of any particular pricing model, since it follows from no-arbitrage alone; it is therefore a useful sanity check on any option pricing implementation, including a machine learning model trained to approximate option prices.

The quantity $N(d_1) = 0.6115$ in this example is also the option's delta, the sensitivity of the option price to a one-dollar change in the stock price, and is the first of several "Greeks" used to measure and hedge an option position's risk exposures.

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5.4.4 The Greeks: Measuring an Option's Risk Exposures

A market maker who has just sold the call option priced above is exposed to several distinct sources of risk, and each of the Greeks quantifies one of them. Four of the five Greeks below share a single building block: $\phi(\cdot)$, the standard normal probability density function,

$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2},$$

the height of the standard normal bell curve at x, distinct from $N(x)$, the cumulative area under that curve already used to price the option itself. Evaluated at the same d_1 computed above, but at full precision ($d_1 = 0.28333$, not the rounded 0.283 displayed there, since rounding before this step would cascade into every Greek that uses it),

$$\phi(d_1) = \frac{1}{\sqrt{2\pi}} e^{-0.28333^2/2} \approx 0.3832.$$

For a European call under Black-Scholes-Merton,

$$\Delta = N(d_1), \qquad \text{(sensitivity to the stock price)}$$

$$\Gamma = \frac{\phi(d_1)}{S_0\sigma\sqrt{T}}, \qquad \text{(sensitivity of delta itself to the stock price)}$$

$$\text{Vega} = S_0\phi(d_1)\sqrt{T}, \qquad \text{(sensitivity to volatility)}$$

$$\Theta = -\frac{S_0\phi(d_1)\sigma}{2\sqrt{T}} - rKe^{-rT}N(d_2), \qquad \text{(sensitivity to the passage of time)}$$

$$\rho = KTe^{-rT}N(d_2). \qquad \text{(sensitivity to the interest rate)}$$

Evaluating each at the same inputs as before ($S_0 = K = \$50$, $r = 4\%$, $\sigma = 30\%$, $T = 1$) gives $\Delta = 0.6115$ (already computed), $\Gamma = \phi(d_1)/(50 \times 0.30 \times 1) \approx 0.0255$, Vega $= 50 \times \phi(d_1) \times 1 \approx 19.16$, $\Theta \approx -3.82$ per year (about $-\$0.0105$ per calendar day), and $\rho = 23.70$ (per unit, i.e., per 100 percentage points, change in the interest rate). The shared $\phi(d_1) \approx 0.3832$ term is why gamma, vega, and theta rise and fall together. Whatever pushes d_1 toward zero, the peak of the bell curve (an at-the-money option, or higher volatility relative to the stock's distance from the strike), raises gamma, vega, and the magnitude of theta at once. Delta and rho depend only on $N(d_1)$ and $N(d_2)$ and do not share this driver.

These numbers have direct trading interpretations. Delta of 0.6115 means the option's price moves about \$0.61 for every \$1 move in the stock, which is why a market maker who sold this call would typically hedge by buying 0.6115 shares per option sold, a strategy called delta hedging. Gamma of 0.0255 measures how quickly that hedge ratio itself changes as the stock moves, so a market maker must rebalance the hedge more frequently when gamma is high. Vega of 19.16 (often quoted per percentage point of volatility, $19.16/100 \approx \$0.19$) shows the option gains or loses value from changes in volatility expectations alone, independent of the stock price; this is the channel through which the volatility clustering of Chapter 7 affects option values. Theta of $-\$3.82$ per year reflects time decay: all else fixed, the option loses value as expiration approaches, since less time remains for the stock to move in the holder's favor.

Worked Example 5.2: Delta Hedging and Its Limits

Suppose a market maker sells 1,000 of these call options and immediately delta-hedges by buying $1{,}000 \times 0.6115 = 611.5$ shares. If the stock rises by \$1, from \$50 to \$51, the hedge's stock position gains $611.5 \times \$1 = \$611.50$. Recomputing the exact Black-Scholes-Merton price at $S = \$51$ (holding every other input fixed) gives a new call price of \$7.500779, up from \$6.876632, so the short option position loses $1{,}000 \times (7.500779 - 6.876632) = \$624.15$ (the four-decimal prices \$7.5008 and \$6.8766 would give \$624.20, a rounding artifact). The hedge does not offset this loss exactly: the net result is $611.50 - 624.15 = -\$12.65$, small relative to the \$1,000-contract position but not zero, because delta is only a first-order (linear) approximation of the option's response to the stock price, the same limitation as the modified-duration approximation for bonds. Just as convexity corrected the bond-duration approximation, gamma corrects the delta approximation here: the delta-only prediction for the new option price is $C_0 + \Delta(S_1 - S_0) = 6.8766 + 0.6115(1) = \$7.4882$, while

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adding the gamma correction, $C_0 + \Delta(S_1 - S_0) + \frac{1}{2}\Gamma(S_1 - S_0)^2 = 7.4882 + 0.5(0.0255)(1)^2 \approx \$7.5009$, lands almost exactly on the true recomputed price of \$7.5008.

A market maker who rebalances only occasionally is therefore exposed to residual gamma risk between rebalances. That risk grows with the size of the stock move (the gamma term grows with its square) and is largest for at-the-money options near expiration, which have the highest gamma. That is why options market makers manage gamma as well as delta, and why one large, sudden move costs a delta-hedged book more than a series of small moves of the same total size.

5.4.5 Implied Volatility

The Black-Scholes-Merton formula takes σ as an input and produces a price as output. Market participants often run it in reverse: given an option's market price, they solve for the volatility that reproduces that price, called the implied volatility. Because the formula has no algebraic inverse, implied volatility is found numerically, with the same kind of root-finding used for yield to maturity in Section 5.2.2.

Suppose the call priced at \$6.88 under an assumed volatility of 30% is instead observed trading at \$7.50. The implied volatility $\hat{\sigma}$ solves $C_{BS}(\hat{\sigma}) = 7.50$, which a numerical solver finds to be $\hat{\sigma} \approx 33.25\%$, above the 30% assumed. The market is pricing in more uncertainty than the 30% historical-volatility assumption suggests, perhaps because of an upcoming earnings announcement, macroeconomic uncertainty, or a different view. Because implied volatility is forward-looking, reflecting current expectations rather than a historical average, it is often preferred to historical volatility for pricing related options.

If the Black-Scholes-Merton model's constant-volatility assumption held exactly, every option on the same underlying and maturity would imply the same $\hat{\sigma}$ whatever its strike: calls at $K = \$40$ and $K = \$60$ would back out the same number, because the model treats σ as a property of the stock, not of the contract. In practice, implied volatility plotted against strike traces a curve, not a flat line, in one of two common shapes. A **volatility smile** is roughly symmetric: implied volatility is lowest at the money and rises toward both deep in-the-money and deep out-of-the-money strikes, a pattern most associated with currency options, where a large move in either direction is plausible. A **volatility skew**, the shape observed for equity and equity-index options, slopes mostly downward: implied volatility is highest for low strikes (deep in-the-money calls, equivalently far out-of-the-money puts) and falls as the strike rises, so out-of-the-money puts are priced as if the stock were more volatile than at-the-money or out-of-the-money calls suggest.

Two economic explanations are offered for the equity skew, and both remain subjects of empirical research rather than settled fact. The first is the *leverage effect* (Black 1976): a falling stock price raises a levered firm's debt-to-equity ratio and therefore its equity volatility, so low-stock-price states are more volatile ones. The second is *crash risk*, sometimes called "crashophobia" (Rubinstein 1994): the pronounced equity skew now taken for granted was far less visible in listed U.S. index option prices before the October 1987 crash, and persistent demand for downside protection, out-of-the-money puts bought as portfolio insurance, has kept those puts priced at a premium (a higher implied volatility) ever since.

5.4.6 Worked Example 5.3: Computing an Implied Volatility Skew

The single implied-volatility computation above generalizes directly: solving $C_{BS}(\hat{\sigma}) =$ market price separately at each strike, holding $S_0 = \$50$, $r = 4\%$, and $T = 1$ year fixed throughout, traces out the skew itself. Table 5.5 does this for five hypothetical market prices spanning strikes of \$40 to \$60, including the \$50 at-the-money option already solved above.

The pattern in Table 5.5 is the equity skew described above, not a symmetric smile: implied volatility falls by roughly 16 percentage points, from 40.02% down to 24.00%, as the strike rises from \$40 to \$60, with the largest drop, more than six points, between the \$50 and \$55 strikes. The market is pricing more downside risk into this stock than a constant-volatility price would imply: the far-out-of-the-money \$60 call is priced as if volatility were 24%, while the deep-in-the-money \$40 call, equivalent by put-call parity to a far-out-of-the-money put, is priced as if it were 40%, sixteen points higher.

Table 5.5 fixes maturity at one year and varies only the strike, one slice through what is in general a two-dimensional object. Repeating the same exercise with $T = 0.25$ (three months) instead of one year,

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Strike (K)	Market price	d (moneyness, $S_0 - K$)	Implied volatility
\$40	\$14.23	+\$10 (in-the-money)	40.02%
\$45	\$10.57	+\$5	36.00%
\$50	\$7.50	\$0 (at-the-money)	33.25%
\$55	\$4.20	−\$5	26.92%
\$60	\$2.19	−\$10 (out-of-the-money)	24.00%

Table 5.5: Implied volatilities solved from five one-year call option market prices on the same underlying ($S_0 = \$50$, $r = 4\%$)

holding S_0, r, and the same five strikes fixed, produces a second slice:

$$K = \$40 : 47.03\%, \qquad K = \$45 : 40.04\%, \qquad K = \$50 : 33.55\%,$$
$$K = \$55 : 27.04\%, \qquad K = \$60 : 24.48\%.$$

Figure 5.3 plots both slices. Both curves slope downward, so the skew is not specific to one maturity. More informative, they are not parallel: the three-month curve sits well above the one-year curve at low strikes (47.03% versus 40.02% at $K = \$40$, a gap of nearly seven points) but the two nearly converge at high strikes (24.48% versus 24.00% at $K = \$60$, under half a point apart). The three-month skew is steeper, a well-documented pattern: near-term crash protection carries a particularly large premium, since a crash is a short, sudden event, while longer-dated options leave more time for volatility to mean-revert, flattening the skew. This joint variation across strike and maturity is what the **volatility surface** describes: a grid of implied volatilities, one for every traded combination of strike and maturity, of which Table 5.5 and its three-month counterpart are two cross-sections.

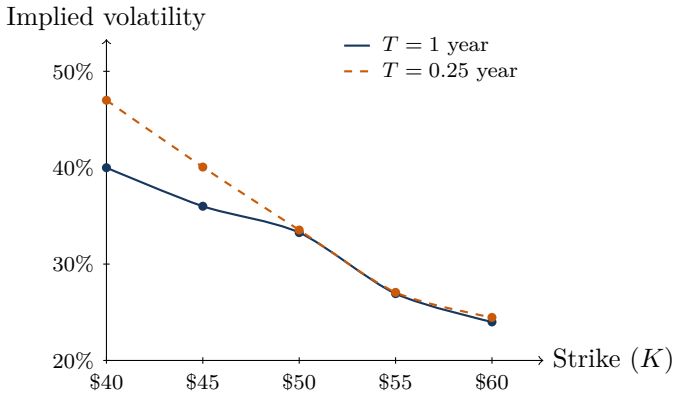


Figure 5.3: The implied volatility skew for one-year (solid) and three-month (dashed) call options on the same underlying ($S_0 = \$50$, $r = 4\%$), from Table 5.5 and its three-month counterpart

5.4.7 Why the Volatility Smile Matters for AI and Machine Learning

The skew and surface just computed are one of the most important places where finance meets machine learning. Black-Scholes-Merton assigns one volatility to every option on the same underlying, but, as Table 5.5 and Figure 5.3 showed, options with different strikes and maturities imply systematically different volatilities, in a pattern structured enough to be learned from data. Researchers and practitioners therefore use machine learning to model implied-volatility surfaces, to estimate the gap between theoretical and observed prices, and to build fast pricing or hedging tools for large option books.

The data are high-dimensional and highly structured: strike, maturity, underlying price, interest rates, and recent market regimes all matter. A model can learn which combinations predict unusually high or low

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implied volatility, but it must respect economic constraints: the fitted surface should be free of arbitrage and stable out of sample rather than memorizing recent moves. Machine learning is best viewed here as a tool for capturing patterns in the surface, not as a replacement for the no-arbitrage logic that makes option pricing coherent.

A single option's implied volatility can be distorted by events at one company, and no single option summarizes the market's volatility expectations. The CBOE Volatility Index (VIX), often called the market's "fear gauge," aggregates implied volatilities across a wide range of strikes and maturities on S&P 500 index options into one number, with a model-free methodology (following Britten-Jones and Neuberger 2000) that never inverts the Black-Scholes-Merton formula: the volatility surface condensed into one tradable index. Quoted as an annualized percentage covering the next 30 days, VIX typically trades between 12% and 20% in calm markets. Tradable through futures, options, and exchange-traded products, it recurs in this book as a market-wide risk signal: a regime filter for trading strategies in Chapter 13, and a price-based complement to the text-based sentiment measures of Chapter 16.

Figure 5.4 shows VIX's daily closing history from its 1990 inception through 9 July 2026, downloaded from Cboe's own single-file history of the index. (The companion notebook falls back to Yahoo Finance if that file cannot be reached; the two sources differ on a handful of trading days and small revisions but agree on every value quoted here.) The largest spikes are labeled: the 1998 collapse of Long-Term Capital Management amid the Russian sovereign default (Chapter 3; 45.74), the September 2001 terrorist attacks (43.74), the mid-2002 WorldCom bankruptcy and accounting-scandal selloff that closed out the dot-com bear market (45.08), the 2008 global financial crisis (80.86), the May 2010 Flash Crash (Chapter 3's own case study; 40.95), the August 2011 U.S. sovereign credit downgrade (48.00), the August 2015 selloff triggered by a surprise Chinese currency devaluation (40.74), the February 2018 "Volmageddon" collapse of short-volatility exchange-traded products (37.32), the March 2020 COVID-19 selloff (the all-time closing high of 82.69), the August 2024 yen carry-trade unwind, when a surprise Bank of Japan rate hike met a weak U.S. jobs report (65.73 intraday, 38.57 at the close), and the April 2025 tariff war (52.33, the third-highest peak of any VIX episode on record, behind only 2008 and 2020, although dozens of individual daily closes during those two crises were higher). Every spike has the same shape: VIX drifts in a narrow band, jumps to 40%-80% or more within days when a shock forces a repricing of risk, then subsides once the acute phase passes. This is the volatility clustering and mean reversion that Chapter 7's GARCH(1,1) recursion models.

Not every geopolitical shock qualifies: the June 2025 Israel-Iran conflict and the accompanying U.S. strikes on Iranian nuclear facilities left VIX only in the low twenties, a reminder that VIX prices *equity* volatility specifically, so a shock must threaten broad equity cash flows or risk appetite, not merely be geopolitically significant, to move it sharply.

The spike-and-decay pattern makes VIX a forecasting input rather than a description of the present: an unusually high VIX is informative because it is unlikely to persist. Backed out of option prices, VIX is the market's own forecast of the coming thirty days, and the standard empirical finding is that implied volatility carries information about future realized volatility beyond what past returns provide. That is why the risk models of Chapter 8 often add an implied-volatility term to their GARCH-style recursions.

Two cautions apply. VIX forecasts volatility, not direction, so it belongs in a risk or position-sizing model rather than a return forecast. And it is a biased forecast, sitting above realized volatility on average because index options embed compensation for crash risk, the same demand for downside protection that produces the skew. A model that treats VIX as unbiased will over-forecast volatility; the gap, the volatility risk premium, is a traded quantity, not a modeling error to be removed.

5.4.8 Real Options: Applying Option Pricing to Corporate Decisions

The option pricing machinery applies, with some adaptation, to corporate investment decisions, an application known as real options analysis. Many corporate decisions have an option-like structure: a firm has the right, but not the obligation, to take some action, and can choose whether and when to act as conditions evolve.

Consider a mining company deciding whether to develop a mineral deposit. A standard net-present-value calculation estimates a single expected value and approves the project only if it is positive. But the firm holds something closer to a call option: it can wait, watch commodity prices, and develop only if prices rise enough, walking away (losing what it spent on exploration and permitting) if they fall. This option to wait

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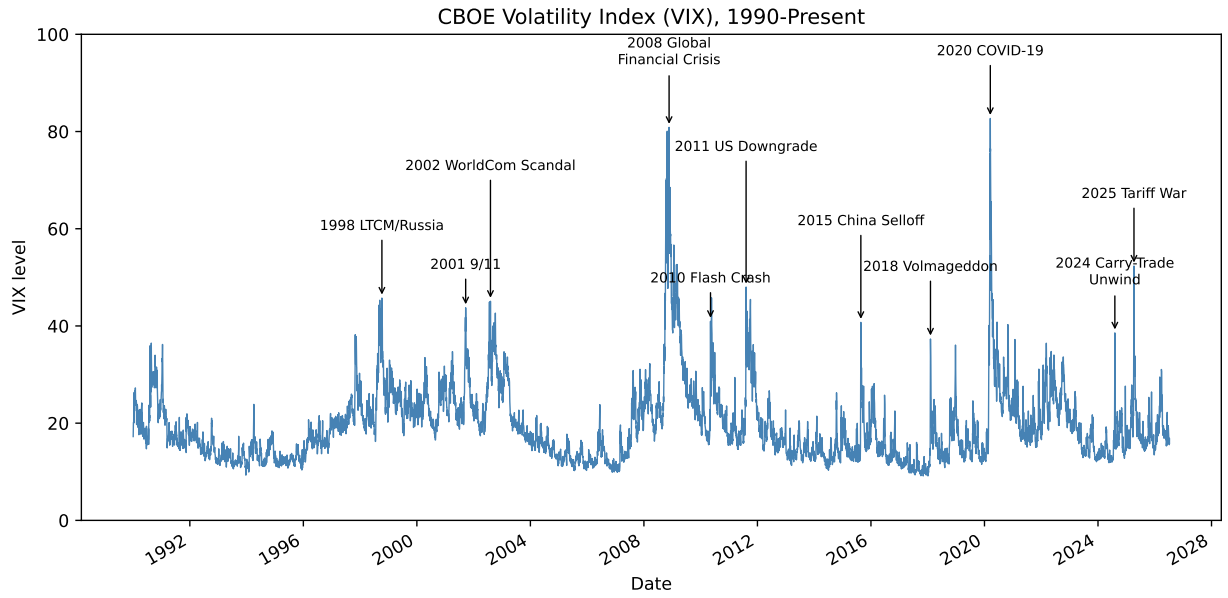


Figure 5.4: The CBOE Volatility Index (VIX), daily closes from January 1990 through 9 July 2026, from Cboe's official VIX history file. Labeled spikes correspond to the crises discussed in the text.

has value, just as an American option's flexibility (Section 5.4.2) has value beyond its European counterpart, and an analysis that ignores it can systematically undervalue the opportunity.

Other common examples include the option to expand a successful pilot project, the option to abandon a failing project and salvage its assets, and the option to switch between input sources or output products as relative prices change. In each case the underlying "asset" is the value of the cash flows the decision affects, the "strike price" is the cost of acting, and volatility plays the same role as in Black-Scholes-Merton: more uncertainty makes the option to wait, expand, or abandon more valuable, not less, since flexibility is most useful when the future is most uncertain. This runs against the naive view of uncertainty as purely a negative, and it is one of the most important lessons real options analysis offers corporate finance. The qualitative logic holds even when, as is common, the underlying does not trade and no precise Black-Scholes-Merton-style number is available.

A simple numerical illustration makes this concrete. Suppose the deposit costs $K = \$50$ million to develop and, if developed today, is worth exactly \$50 million, a marginal project with NPV $= 50.0 - 50.0 = \$0$. If the firm waits one year, the deposit's value will have moved to either \$70 million (if commodity prices rise) or \$35 million (if they fall), each with probability 0.5, and the firm develops the deposit only if doing so is still profitable at that point. The expected payoff from waiting is

$$\mathbb{E}[\text{Payoff}] = 0.5 \times \max(70.0 - 50.0,\ 0) + 0.5 \times \max(35.0 - 50.0,\ 0) = 0.5 \times 20.0 + 0.5 \times 0 = \$10.0 \text{ million},$$

which, discounted one year at a risk-free rate of 5%, has a present value of $10.0/1.05 \approx \$9.52$ million, substantially higher than the \$0 NPV of developing now. The probability 0.5 is not an arbitrary guess, and discounting at the risk-free rate is legitimate only because of it: it is the risk-neutral probability of Section 5.4.1 implied by the deposit's current value, $q = (1.05 \times 50 - 35)/(70 - 35) = 0.5$, which makes today's \$50 million equal the discounted risk-neutral expectation of the two end-of-year outcomes. A real-world probability of 0.5 would instead call for a risk-adjusted discount rate, and changing either end-of-year value while holding today's value fixed changes q as well (Exercise 5.17). Figure 5.5 shows the comparison as a one-step decision tree.

The value of waiting comes entirely from the option to walk away: the firm captures the upside if prices rise and avoids the downside if they fall. A point-estimate NPV, which implicitly commits to developing whatever prices do, cannot capture this asymmetry. A team that evaluates a marginal project by expected

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NPV alone, without asking whether waiting for information is a valuable choice, can pass over projects or approve them prematurely. Chapter 19 revisits the related American-option early-exercise decision of Section 5.4.2 with a model-free reinforcement-learning algorithm, Q-learning, which recovers the same optimal exercise policy without being told the tree's transition probabilities or payoffs.

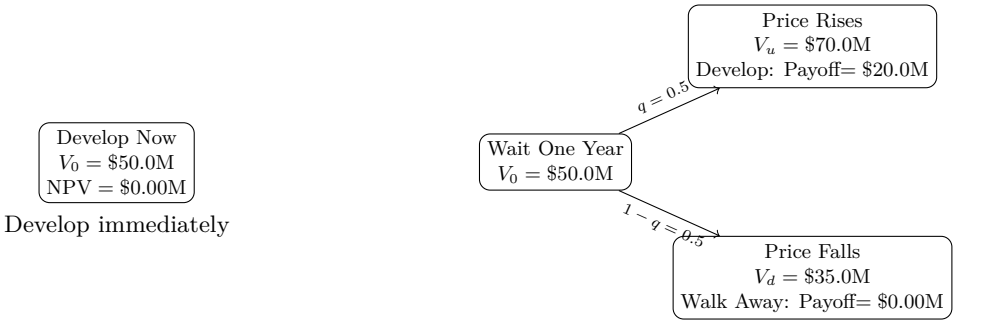


Figure 5.5: A one-step real-options decision tree for the mining deposit example: developing immediately (left) versus waiting one year to observe the commodity-price state (right)

5.5 Putting It Together: Valuing Meridian Fabrication

The chapter has now developed both major approaches to valuing a business, discounted cash flow and multiples. Applying both to Meridian Fabrication, with the WACC computed earlier, shows how differently they can answer the same question.

For the DCF approach, the valuation date is the end of Year 2, the date of Meridian's latest balance sheet, so Year 3's free cash flow arrives one period from now and Year 5's three periods from now (Figure 5.6). Chapter 4's percent-of-sales forecast supplies the first of these: free cash flow recovers from −$8.0 million in Year 2 to $0.52 million in Year 3, as working capital absorbs less cash. Beyond that one-year forecast, suppose the recovery continues as Meridian's working-capital investment matures, to $3.0 million in Year 4 and $6.0 million in Year 5, after which free cash flow grows at a stable long-run rate of $g = 3\%$ forever. Discounting at the full-precision WACC computed earlier in this chapter, $9.048/112 = 8.0786\%$ (displayed there as 8.08%),

$$PV(\text{Years 3-5 FCF}) = \frac{0.52}{1.080786} + \frac{3.0}{1.080786^2} + \frac{6.0}{1.080786^3} \approx \$7.80 \text{ million},$$

and the terminal value at the end of Year 5, using the Gordon growth formula on Year 6's free cash flow, is

$$TV_5 = \frac{6.0(1.03)}{0.080786 - 0.03} \approx \$121.69 \text{ million}, \qquad PV(TV_5) = \frac{121.69}{1.080786^3} \approx \$96.39 \text{ million},$$

discounted three periods, not five, because TV_5 sits at the end of Year 5, exactly as far from the valuation date as Year 5's own free cash flow. The DCF enterprise value is $7.80 + 96.39 \approx$ $104.19 million, and subtracting net debt of $58.0 − $9.0 = $49.0 million gives an implied equity value of approximately $55.19 million. (Discounting at the rounded 8.08% instead would give $55.16 million: the terminal value, which divides by WACC $- g$, amplifies even a rounding-level change in the discount rate.)

Compare this to the multiples-based equity value computed in Section 5.3.3: $107.0 million, nearly twice as large. Both methods are legitimate, and the gap reflects an economic disagreement, not a computational error. The DCF value is driven by Meridian's currently weak cash generation and slow projected recovery; the multiples value assumes Meridian deserves the EBITDA multiple of healthier peers, in effect assuming its cash-flow problems are temporary.

The two methods are not even fully independent. The WACC used book equity of $54.0 million as E, which the formula permits only when no market value is available; substituting the multiples-implied $107.0 million would raise the equity weight from 48.2% to 64.8% and the WACC from 8.08% to 9.08%, which would

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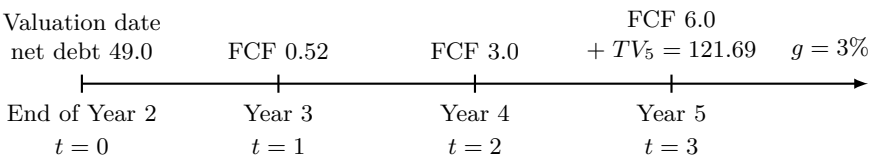


Figure 5.6: Cash-flow timeline for the Meridian DCF ($ millions). The valuation date, $t = 0$, is the end of Year 2, so Year 3's free cash flow is discounted one period, and Year 5's free cash flow and the terminal value TV_5 three periods.

lower the DCF value further and widen the gap. The choice of capital-structure weights is therefore itself a judgment about which of the two values is closer to the truth.

The DCF-implied equity value of $55.19 million is close to Meridian's $54.0 million book value of equity, while the multiples-based value is far above it. That proximity should not be over-read, since book value is a historical-cost figure and nothing forces a DCF to land near it, but it is consistent with the DCF assumptions being the more conservative starting point until there is concrete evidence, such as an improving cash conversion cycle or margin expansion, that Meridian's operating performance is converging toward that of its healthier peers.

Triangulating across methods, and explaining why they disagree, is standard practice and more useful than any single number. The next step would be to test how each estimate changes under alternative multiples, discount rates, and cash-flow scenarios; that sensitivity analysis is often more informative than the point estimate.

5.6 Challenges in Valuation and Asset Pricing

The models in this chapter are elegant and, within their assumptions, exact. Applying them to real markets introduces several recurring challenges.

Discount-rate uncertainty. Every model in this chapter takes r as given, but in practice the discount rate itself must be estimated, typically from an asset pricing model such as the CAPM or a multi-factor model (covered in Chapter 12), and estimates of the required return can vary substantially across reasonable methodologies. Since the Gordon growth model shows that price is extremely sensitive to $r - g$, uncertainty in r translates directly into wide uncertainty in fair value.

Volatility is not observed, only estimated. The Black-Scholes-Merton formula requires σ, the volatility of the underlying asset, but σ cannot be observed directly; it must be estimated from historical returns or backed out from other option prices as an "implied volatility." Different estimation windows and methods can produce meaningfully different prices for the same option, and volatility changes over time, the motivation for the GARCH-family models introduced in Chapter 7.

Model risk and the limits of closed-form pricing. Every closed-form pricing model rests on simplifying assumptions, constant volatility and continuous trading for Black-Scholes-Merton, a flat and unchanging growth rate for the Gordon growth model, no default risk in the basic bond pricing formula. Real markets violate all of these to varying degrees, and a model that is used outside the range of conditions it was built for can silently produce a confidently wrong price, a risk usually called model risk.

Credit risk complicates bond pricing. The bond pricing formula in this chapter assumes the promised cash flows are paid with certainty. A corporate bond carries default risk, so its price reflects both discounting and the probability-weighted possibility of missed payments, which requires the credit models developed in Chapter 9.

Calibration versus estimation at scale. A trading desk must reprice large portfolios of derivatives continuously as market conditions change, which requires the underlying models to be calibrated quickly and consistently across thousands of instruments. This operational requirement, not just theoretical accuracy, is a major reason simpler closed-form models such as Black-Scholes-Merton remain in wide use even though more elaborate stochastic volatility and jump-diffusion models are known to fit historical data better.

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Where machine learning fits in. Closed-form models deviate from market prices in systematic ways (the volatility smile is a symptom), so machine learning is used both to learn the deviation (the gap between model and market price) and to learn pricing functions or implied volatility surfaces end to end. These approaches capture patterns a fixed formula cannot, but they inherit the overfitting and non-stationarity concerns of Chapter 6: a pricing model trained on historical option data can fail when conditions shift away from the training regime, usually when accurate pricing matters most.

The term structure adds a further layer of estimation risk. Section 5.2.4's yield curve discussion showed that a single discount rate is itself a simplification; a full analysis requires an entire curve of spot rates, and that curve must itself be estimated (typically by "bootstrapping" it from a set of observed bond prices), which introduces additional model choices and potential errors before any single bond or derivative is even priced.

Real options are hard to value precisely. The real options perspective of Section 5.4.8 is conceptually powerful but hard to apply with a financial option's precision, since the underlying (the value of a future business opportunity) rarely has an observable price or a clean volatility estimate. Real options analysis is therefore often used qualitatively, to check whether a standard net-present-value calculation misses significant flexibility value, rather than to produce a definitive price.

Relative valuation inherits the market's own mistakes. The multiples-based approach in Section 5.3.3 depends on comparable companies being reasonably priced. If a whole sector is in a speculative bubble or an unwarranted sell-off, a multiples valuation reproduces that mispricing for the target, since it prices an asset relative to its peers, not to a first-principles estimate of intrinsic value. That is why the DCF-versus-multiples comparison of Section 5.5 is informative, especially when the two disagree.

Illiquidity complicates every valuation approach. All of the methods in this chapter implicitly assume that once a fair value is computed, an investor could transact at or near that value. For a private company, a thinly traded bond, or a large block of stock relative to average trading volume, this assumption can fail badly: the price realizable in a sale may differ meaningfully from any theoretical value, because there are too few willing counterparties, echoing the liquidity and market-impact themes of Chapter 3.

5.7 Key Takeaways

Every valuation problem in this chapter reduces to one idea: discount future cash flows, or replicate a future payoff, at a rate or under a probability measure consistent with no arbitrage. Bond pricing, the dividend discount model, and option pricing apply the same present-value logic to progressively more complex cash flows: a fixed coupon schedule, a growing perpetuity of dividends, a payoff that depends nonlinearly on another asset's price. The worked examples showed how to compute these prices and where the models break down; a practitioner who can compute a textbook price without understanding its assumptions is exposed to model risk.

The chapter also tied valuation to the statements and institutions covered earlier. The WACC example in Section 5.3.1 showed how Meridian Fabrication's balance sheet from Chapter 4 feeds into the rate used to value it. The credit-risk-adjusted bond price in Section 5.2.2 previewed the probability of default and recovery rate that Chapter 9's models estimate rather than assume. And the Greeks and implied volatility showed that even Black-Scholes-Merton is best understood as a lens for organizing risk exposures and market expectations, not a black box that outputs a single correct price. Treating elegant models as organizing frameworks, not oracles, is the mindset the rest of this book tries to instill.

The closing Meridian valuation in Section 5.5 may be the chapter's most important exercise, because it did not produce a single answer. A DCF value of $55.19 million and a multiples-based value of $107.0 million are both defensible and correctly computed, yet they disagree by nearly a factor of two.

The disagreement is itself informative: it isolates the assumption driving the difference, Meridian's projected cash-flow recovery versus the market's willingness to price it like a healthier peer, a more useful diagnostic than either number alone. It is also the situation a machine learning model trained to predict "fair value" would face silently: without an explicit decomposition like this one, a model blending both approaches could produce one confident-looking number while hiding an economically meaningful disagreement about which valuation philosophy fits this company now. Preserving that kind of interpretability is a theme of Chapter 18.

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5.8 Suggested Exercises

The exercises draw on every major tool in this chapter, and several build on the Meridian Fabrication and Solstice Robotics examples from Chapter 4, so your answers can be read alongside that earlier analysis.

5.1 A 2-year bond with face value \$1,000 and an annual coupon rate of 6% is priced at a market yield of 8%. Compute its price using the discounting formula from Section 5.2.2.

5.2 Using the Gordon growth model, find the fair price of a stock expected to pay a dividend of \$2.50 next year, assuming a required return of 10% and a long-run dividend growth rate of 6%. Then recompute the price assuming growth of 8% instead, and comment on how sensitive the price is to this two-percentage-point change.

5.3 Using the one-step binomial model with $S_0 = \$40$, $u = 1.25$, $d = 0.8$, $R = 1.05$, and strike $K = \$40$, compute the risk-neutral probability q and the fair price of a call option today.

5.4 Verify put-call parity using the Black-Scholes-Merton call and put prices computed in Section 5.4.3, and explain in your own words why put-call parity must hold regardless of which option pricing model is used.

5.5 Using the two-stage dividend discount model from Section 5.3.2, compute the fair price of a stock with $D_0 = \$2.00$, a high-growth rate of $g_1 = 15\%$ for the first 4 years, a terminal growth rate of $g_2 = 5\%$, and a required return of $r = 11\%$. What fraction of the total price comes from the terminal value?

5.6 A 3-year bond with a Macaulay duration of 2.86 years is priced at \$947.51 when $y = 7\%$. Use the modified-duration approximation to estimate its price if the yield falls to 6%, then explain, using the concept of convexity, whether you would expect the approximation to overstate or understate the exact repriced value.

5.7 Explain why the Gordon growth model is more sensitive to estimation error than the bond pricing formula in this chapter, referencing the discount-rate uncertainty challenge in Section 5.6.

5.8 Describe one concrete way a Black-Scholes-Merton-based pricing system could fail silently (produce a confident but wrong price) if market conditions moved outside the model's assumptions.

5.9 Compute Meridian's WACC using Section 5.3.1's formulas if its equity beta were 0.9 instead of 1.2, holding all other inputs (risk-free rate, market risk premium, cost of debt, tax rate, and capital structure) fixed, and explain the direction of the change.

5.10 Using the Greek formulas in Section 5.4.4, explain why gamma is largest for an option that is close to at-the-money ($S_0 \approx K$) and smaller for options that are deep in- or out-of-the-money, without necessarily recomputing the formula numerically.

5.11 A sixth call option on the same stock (still $S_0 = \$50$, $r = 4\%$, $T = 1$), with a strike of $K = \$65$, trades in the market at \$1.00. Using the same numerical root-finding approach used to build Table 5.5, solve for its implied volatility, and explain whether the result is consistent with the skew pattern already observed across the \$40–\$60 strikes in Section 5.4.6.

5.12 Using the portfolio duration formula in Section 5.2.3, compute the duration of a bond portfolio holding 30% in a bond with duration 1.5 years, 50% in a bond with duration 4.0 years, and 20% in a bond with duration 8.0 years.

5.13 Using Table 5.2, explain in your own words why a corporate bond's yield should be even higher than the government yield at the same maturity, referencing the credit-risk-adjustment concept from Section 5.2.2.

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5.14 Using the multiples method in Section 5.3.3 and Solstice Robotics' Chapter 4 figures (EBIT of \$27.0 million, depreciation and amortization of \$5.0 million, total debt of \$10.0 million, and cash of \$40.0 million), compute its implied enterprise and equity value assuming comparable technology firms trade at an EV/EBITDA multiple of 12×. Compare the implied equity value to its book equity of \$75.0 million.

5.15 Explain, using the DCF-versus-multiples comparison in Section 5.5, one reason a DCF valuation and a multiples-based valuation of the same company might disagree substantially, and describe what an analyst should do when they do.

5.16 Describe a real option (using the framework in Section 5.4.8) that a pharmaceutical company holds when deciding whether to continue funding a drug at each stage of clinical trials, and explain why higher uncertainty about the drug's eventual efficacy makes this option more, not less, valuable.

5.17 Using Section 5.4.8's option-to-wait example, recompute the value of waiting one year if the deposit's up-state value were \$90 million instead of \$70 million (down-state value, development cost, today's \$50 million value, and the risk-free rate unchanged). Do it twice: first holding the probability at 0.5, then using the risk-neutral probability q implied by the new up-state value. Show that the first calculation raises the option's present value by less than half of the \$20 million increase in the up-state value, and explain why only the second calculation is consistent with no arbitrage.

5.18 Using Section 5.5's Meridian DCF as a template, recompute the enterprise and equity value if the terminal growth rate were $g = 2\%$ instead of 3%, holding the WACC and the Year 3-5 free cash flow forecasts fixed, and explain why the terminal growth assumption has such a large effect on the total valuation.

5.19 A junior analyst argues that because the DCF and multiples valuations for Meridian disagree so much, one of the two methods must simply be "wrong" and should be discarded. Write a short paragraph, drawing on Section 5.5, explaining why this conclusion does not necessarily follow.

5.20 Using Table 5.2, compute the implied 1-year forward rate spanning years 4 to 5 using the 5-year and (assumed) 4-year spot rates of 4.8% and 4.7% respectively, and compare it to the two forward rates already computed in the text.

5.21 Using this chapter's delta-hedging example, recompute the hedge's net P&L if the stock instead falls from \$50 to \$49 (holding the number of shares hedged and all other option inputs fixed), and explain why the sign of the residual hedging error is the same as in the original up-move example even though the stock moved in the opposite direction.

5.22 **Challenge (real data).** Trace a real implied-volatility skew, as Section 5.4.6 did for hypothetical prices. Option chains in `yfinance` are available only for the current date, not historically, so record the date and time of your download and expect your numbers to differ from anyone else's. (a) For the S&P 500 ETF SPY, download the spot price, the current 13-week Treasury bill yield (ticker `^IRX`) as the risk-free rate, and the option chain (`Ticker("SPY").option_chain(expiry)`) for the first expiry at least 30 days away. (b) For strikes from about 85% to 110% of spot, take the bid-ask midpoint of the out-of-the-money option at each strike, converting each put to the equivalent call price with put-call parity, Equation (5.10), and invert the Black-Scholes-Merton formula, Equation (5.9), numerically for $\hat{\sigma}$. (c) Plot $\hat{\sigma}$ against moneyness K/S_0, and say whether you see a skew or a smile. (d) Name two simplifications in this procedure (the chapter's formula ignores SPY's dividends, and SPY options are American) and the direction in which each could bias your implied volatilities, and explain what you would do with quotes whose bid is zero, as happens outside trading hours.

5.9 Further Reading

- Hull (2022), *Options, Futures, and Other Derivatives*. The standard reference for derivatives pricing, covering the binomial model, Black-Scholes-Merton, and the Greeks in far greater depth and generality than this chapter.

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- McDonald (2012), *Derivatives Markets*. A complementary treatment of options and forwards with a strong emphasis on the no-arbitrage replication arguments introduced in this chapter.
- Damodaran (2012), *Investment Valuation*. An extensive treatment of both discounted cash flow and relative valuation, including practical guidance on estimating discount rates, growth rates, and comparable-company multiples.
- Bodie et al. (2024), *Investments*. Covers the CAPM and cost-of-capital material in this chapter within the broader context of portfolio theory and asset pricing, previewing Chapter 12.
- Fabozzi (2007), *Fixed Income Analysis*. A thorough treatment of bond pricing, duration, convexity, and the term structure of interest rates for readers who want to go well beyond this chapter's introductory treatment.
- Brealey et al. (2025), *Principles of Corporate Finance*. Covers real options and the cost-of-capital material from a corporate finance perspective, complementing the more market-oriented treatment in this chapter.
- Black (1976), "Studies of Stock Price Volatility Changes." The original leverage-effect explanation for the equity volatility skew discussed in this chapter's implied-volatility section.
- Rubinstein (1994), "Implied Binomial Trees." The "crashophobia" explanation for the post-1987 equity skew discussed in this chapter's implied-volatility section.
- Britten-Jones and Neuberger (2000), "Option Prices, Implied Price Processes, and Stochastic Volatility." The model-free methodology underlying the VIX index discussed in this chapter's volatility-surface section.

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Chapter 6

Machine Learning Fundamentals

6.1 Introduction

Machine learning has become a central part of the quantitative toolkit in finance. It provides methods for discovering patterns in data, forecasting outcomes, reducing dimensionality, and classifying observations. In principle its function is simple: use historical data to learn relationships that generalize to new situations. In practice this is hard, especially in finance, where data are noisy, relationships shift over time, and the cost of mistakes can be large.

This chapter introduces the main ideas of machine learning for readers who are mathematically trained but may be new to the subject, focusing on concepts and formulas rather than library calls. Two small worked examples run through the chapter: a regression problem relating a firm's leverage to its credit spread, and a classification problem predicting loan default. The discussion covers supervised and unsupervised learning, prediction and classification, overfitting, regularization, evaluation strategies, and the bias-variance trade-off.

By the end of the chapter, the reader will have seen how a model's parameters are found (by a closed-form solution or gradient descent), how distance-based, margin-based, probabilistic, and tree-based algorithms attack the same prediction problem from different angles, and how to choose among them and their hyperparameters without overfitting the selection itself. Every applied chapter that follows builds on this vocabulary: the credit models of Chapter 9, the portfolio methods of Chapter 12, the trading strategies of Chapters 13 and 14, the fraud detection systems of Chapter 15, the text-based methods of Chapter 16, the deep-learning extensions of Chapter 17, the explainability tools of Chapter 18, and the end-to-end case study of Chapter 21.

6.2 The Learning Problem

Two questions come before any algorithm: what a learning system does, and what kind of problem is being posed to it.

6.2.1 What Machine Learning Is

Suppose an analyst has a decade of loan-application data and wants a system that predicts default risk better than hand-written underwriting rules, and that improves as data arrives instead of being rewritten whenever economic conditions shift. Machine learning is the branch of statistics and computer science built for this: models that improve with experience, rather than a fixed decision procedure encoded by hand. A model is given data, it learns parameters from those data by minimizing some measure of error, and it is then used to make predictions or decisions on new data. The model may be used for regression, classification, clustering, dimensionality reduction, or anomaly detection.

This setup is not new. Least-squares regression, dating to Gauss and Legendre in the early 1800s, is a machine learning algorithm by this definition: it minimizes squared error over a set of parameters using data. What distinguishes the modern field is less the principle, minimizing a loss function over parameters, than

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the scale of data and computation now available. That scale favors flexible, high-capacity models (trees, ensembles, neural networks) with thousands or millions of parameters, which are harder to interpret than the small, transparent linear models of classical statistics.

In finance, machine learning can be used for several purposes. It can forecast returns or volatility, classify whether a firm is likely to default, detect fraudulent transactions, or extract sentiment from news articles. In each case the model must learn from observed patterns and generalize to future cases, a harder task than describing the past.

The challenge is that finance is not a static environment. Relationships can change because of policy shifts, market regimes, investor behavior, or macroeconomic conditions. A model that performs well in one regime may fail in another. Machine learning in finance must therefore be tested under realistic conditions, a theme that runs through this chapter and every applied chapter after it.

6.2.2 Supervised and Unsupervised Learning

Machine learning problems are often divided into supervised and unsupervised learning. In supervised learning, the outcome of interest, called the label or target, is observed in the training data. For example, one may have historical data on firm characteristics and a label indicating whether default occurred. The task is to learn a function $\hat{f}$ mapping features x to a target y, so that $\hat{f}(x) \approx y$ on new, unseen data. Common examples include linear regression, logistic regression, decision trees, and neural networks.

In unsupervised learning, no explicit target is available. The goal is to discover structure in the data. Clustering methods group similar observations together, while dimensionality reduction methods compress the data into lower-dimensional representations. In finance, unsupervised learning can help identify groups of similar assets, detect unusual trading patterns, or reduce the dimensionality of large datasets.

A third category, semi-supervised learning, uses a small amount of labeled data with a much larger amount of unlabeled data. This is common in finance, where a reliable label (confirming that a transaction was fraudulent, say, which may take a lengthy investigation) is expensive relative to the raw data. A related distinction is between discriminative models, which learn the boundary or function that predicts labels directly (linear and logistic regression and most models in this chapter), and generative models, which learn the full probability distribution of the data and can generate synthetic examples resembling the training data. Naive Bayes, introduced later in this chapter, is a simple generative model; large language models, discussed in Chapter 16, are far more elaborate examples of the same generative principle applied to text.

The choice of method depends on the question. Predicting default calls for supervised learning; finding hidden structure in a large set of securities may call for unsupervised learning. The chapter proceeds in roughly that order: supervised regression and classification first, since the applied chapters use them most, then unsupervised clustering and dimensionality reduction, and finally neural networks, which can serve either purpose depending on how they are trained.

6.3 Core Supervised Models

This part builds the standard supervised toolkit, each model worked numerically before the next is introduced.

6.3.1 Worked Example 6.1: Regression

Suppose a credit analyst wants to relate a firm's leverage, measured as debt-to-EBITDA, to the credit spread the market demands on its bonds, measured in basis points. Table 6.1 shows five observations.

Debt/EBITDA (x)	2	3	4	5	6
Credit spread, bps (y)	120	150	210	260	300

Table 6.1: Leverage and credit spread for five hypothetical issuers

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Linear regression models the target as a linear function of the feature plus noise, $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$. The ordinary least squares (OLS) estimator chooses β_0, β_1 to minimize the sum of squared residuals,

$$(\hat{\beta}_0, \hat{\beta}_1) = \arg\min_{\beta_0,\beta_1} \sum_{i=1}^{n} \left(y_i - \beta_0 - \beta_1 x_i\right)^2, \tag{6.1}$$

which has the closed-form solution

$$\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}. \tag{6.2}$$

With $\bar{x} = 4$ and $\bar{y} = 208$, the data in Table 6.1 give $\sum_i (x_i - \bar{x})(y_i - \bar{y}) = 470$ and $\sum_i (x_i - \bar{x})^2 = 10$, so

$$\hat{\beta}_1 = \frac{470}{10} = 47, \qquad \hat{\beta}_0 = 208 - 47(4) = 20.$$

The fitted model is $\widehat{\text{spread}} = 20 + 47 \times \text{leverage}$: each additional turn of debt-to-EBITDA is associated with roughly 47 additional basis points of credit spread. The fitted values are 114, 161, 208, 255, and 302 bps, close to the observed 120, 150, 210, 260, and 300. Figure 6.1 plots the five issuers with the fitted line, showing the tightness of the fit and each residual.

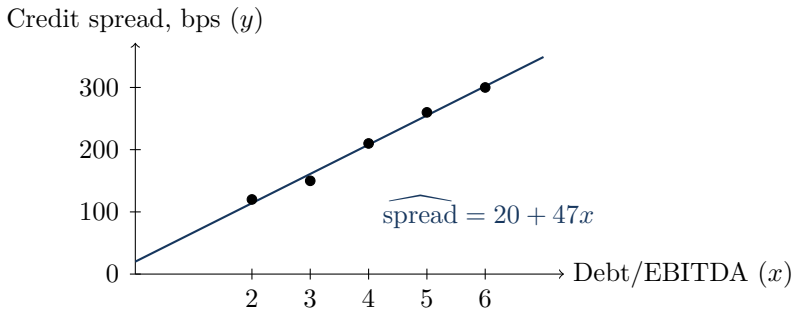


Figure 6.1: Debt-to-EBITDA leverage and credit spread for the five hypothetical issuers in Table 6.1 (black dots), with the OLS-fitted line $\widehat{\text{spread}} = 20 + 47x$ overlaid as a solid line

The residual sum of squares is

$$\text{SSE} = \sum_i (y_i - \hat{y}_i)^2 = 190,$$

and the total sum of squares around the mean is

$$\text{SST} = \sum_i (y_i - \bar{y})^2 = 22{,}280,$$

giving

$$R^2 = 1 - \frac{\text{SSE}}{\text{SST}} = 1 - \frac{190}{22{,}280} \approx 0.991,$$

so leverage explains about 99% of the variation in credit spread in this small, deliberately clean example. In sklearn, the same fit is obtained with

```
from sklearn.linear_model import LinearRegression
import numpy as np

X = np.array([2, 3, 4, 5, 6]).reshape(-1, 1)
y = np.array([120, 150, 210, 260, 300])
model = LinearRegression().fit(X, y)
print(model.intercept_, model.coef_)   # 20.0 [47.]
print(model.score(X, y))               # 0.9915...
```

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Real credit spread data would show far more scatter; the toy example makes the mechanics of fitting and evaluating a linear model transparent before real-world noise is added.

6.3.2 Standard Errors, Confidence Intervals, and p-Values for Coefficients

The point estimate $\hat{\beta}_1 = 47$ says nothing about how much confidence it deserves, and a workflow that reports only a point estimate and an R^2 leaves useful information on the table. With $n = 5$ observations and two estimated parameters $(\hat{\beta}_0, \hat{\beta}_1)$, there are $n - 2 = 3$ residual degrees of freedom, and the estimated noise variance is

$$\hat{\sigma}^2 = \text{SSE}/(n-2) = 190/3 \approx 63.33.$$

The standard error of the slope, which measures how much $\hat{\beta}_1$ would be expected to vary across hypothetical repeated samples of five issuers drawn from the same underlying relationship, is

$$SE(\hat{\beta}_1) = \sqrt{\frac{\hat{\sigma}^2}{\sum_i (x_i - \bar{x})^2}} = \sqrt{\frac{63.33}{10}} \approx 2.52.$$

Dividing the estimate by its standard error gives the t-statistic, $t = 47/2.52 \approx 18.68$, which under the null hypothesis $H_0 : \beta_1 = 0$ (leverage carries no information about credit spread) follows a t-distribution with 3 degrees of freedom. The associated p-value, the probability of a t-statistic this extreme by chance if H_0 were true, is $p \approx 0.0003$, far below the conventional 5% threshold: strong evidence, within this small and clean sample, that the slope is nonzero. A 95% confidence interval for the slope is

$$\hat{\beta}_1 \pm t^*_{0.975,3} \cdot SE(\hat{\beta}_1) = 47 \pm (3.182)(2.52) \approx (39.0,\ 55.0),$$

using the t-distribution critical value $t^*_{0.975,3} \approx 3.182$. The correct reading is that if this sampling and estimation procedure were repeated many times, about 95% of the resulting intervals would contain the true relationship between leverage and spread, not that there is a 95% chance the true β_1 falls in this specific interval. Applying the same steps to the intercept gives $SE(\hat{\beta}_0) \approx 10.68$, $t \approx 1.87$, and $p \approx 0.158$: the intercept is not statistically distinguishable from zero at the 5% level. That is unsurprising: only one of the five leverage values ($x = 2$) is anywhere near $x = 0$, so the data say little about the spread a zero-leverage issuer would command.

The contrast between slope and intercept generalizes: significance is coefficient-specific, and a model can pin down some relationships far more precisely than others. The overfitting concern of this chapter applies here too. Fitting a regression with many candidate features and reporting only those with p-values below 0.05 overstates the evidence, since testing enough coefficients against noise eventually produces a spuriously "significant" one, the multiple-testing concern revisited in Chapter 7's discussion of pairs selection for cointegration trading. A single small p-value is far more persuasive when it was the only hypothesis tested than when it is the best of dozens.

6.3.3 How Models Learn: Gradient Descent

The closed-form OLS solution above exists because linear regression's loss function has a special structure: it is quadratic in the parameters, so its minimum can be found in one step by solving a system of linear equations. Most machine learning models, including logistic regression and neural networks, lack this property, and their parameters must be found iteratively with an optimization algorithm called gradient descent. Figure 6.2 shows the idea for a single generic parameter, written θ there and standing for whichever parameters a given model has to fit.

Starting from an initial guess, the algorithm repeatedly takes a small step in the direction that most reduces the loss, the negative of the gradient (the vector of partial derivatives) of the loss with respect to each parameter, and stop once further steps no longer improve the loss. For the mean squared error loss

$$\mathcal{L}(\beta_0, \beta_1) = \frac{1}{n} \sum_i (\beta_0 + \beta_1 x_i - y_i)^2,$$

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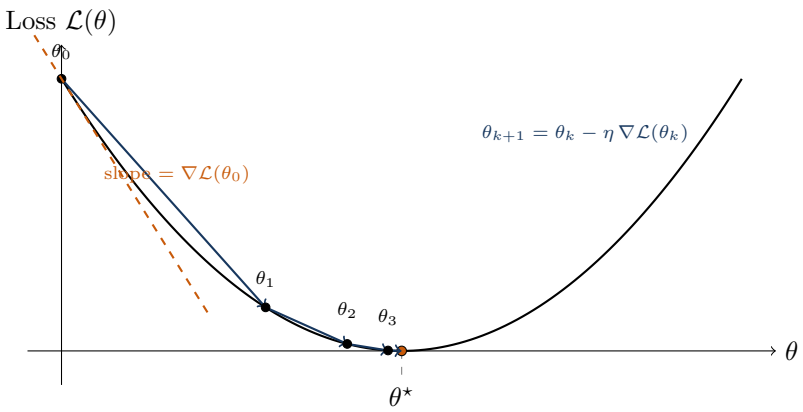


Figure 6.2: The idea of gradient descent. Starting from an initial guess θ_0, each step moves against the local slope of the loss curve, the negative gradient, illustrated here by the tangent line at θ_0, producing a sequence $\theta_0, \theta_1, \theta_2, \ldots$ that descends toward the minimizer $\theta^\star$. Steps are large where the curve is steep and shrink automatically as the curve flattens near the minimum, the pattern seen numerically in Table 6.2 below.

the gradients are

$$\frac{\partial \mathcal{L}}{\partial \beta_0} = \frac{2}{n}\sum_i (\hat{y}_i - y_i), \qquad \frac{\partial \mathcal{L}}{\partial \beta_1} = \frac{2}{n}\sum_i (\hat{y}_i - y_i)\, x_i,$$

and each iteration updates the parameters by

$$\beta_0 \leftarrow \beta_0 - \eta \frac{\partial \mathcal{L}}{\partial \beta_0}, \qquad \beta_1 \leftarrow \beta_1 - \eta \frac{\partial \mathcal{L}}{\partial \beta_1}, \tag{6.3}$$

where η is the learning rate, a small positive number controlling the step size. Applying this to the same credit-spread data as Section 6.3.1, starting from $\beta_0 = \beta_1 = 0$ with a learning rate of $\eta = 0.01$, produces the trajectory in Table 6.2.

Iteration	β_0	β_1
1	4.16	18.52
2	6.76	30.04
3	8.38	37.21
10	11.05	48.55
1,000	18.91	47.24
10,000	20.00	47.00

Table 6.2: Gradient descent converging toward the closed-form OLS solution

After 10,000 iterations, gradient descent has converged to $\beta_0 \approx 20.00$ and $\beta_1 \approx 47.00$, matching the closed-form OLS solution of Section 6.3.1 up to rounding. For a quadratic loss like this one, gradient descent is guaranteed to converge to the same global minimum, provided the learning rate is small enough to avoid overshooting. The learning rate itself involves a trade-off that recurs throughout machine learning: too small, and convergence (as seen here) can require many thousands of iterations; too large, and the updates overshoot the minimum and diverge. In practice η is one of the most important hyperparameters to tune, including for the neural networks later in this chapter, and it is rarely fixed throughout training; adaptive variants that shrink the rate over time, or use different rates for different parameters, are standard in deep

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learning software. Gradient descent is worth understanding even where a closed form exists because it scales to problems with none: logistic regression's log-loss, a neural network's loss with millions of parameters, and many other models in this book are fit with it or a close variant, arguably making it the most important algorithm in modern machine learning.

6.3.4 Worked Example 6.2: Classification

Regression predicts a continuous target; classification predicts a discrete one, such as whether a borrower defaults. Logistic regression models the probability of the positive class as

$$p(x) = \Pr(y = 1 \mid x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}, \tag{6.4}$$

so that $p(x)$ is always between 0 and 1, unlike a linear model's unbounded output. The parameters are typically chosen to minimize the log-loss (negative log-likelihood),

$$\mathcal{L}(\beta) = -\frac{1}{n} \sum_{i=1}^{n} \Big[y_i \log p(x_i) + (1 - y_i) \log \big(1 - p(x_i)\big) \Big]. \tag{6.5}$$

A note on terminology the reader will meet in industry: linear and logistic regression both belong to the generalized linear model (GLM) family, which pairs a linear combination of features with a link function and an outcome distribution suited to the target. Ordinary regression uses an identity link with a normal outcome, logistic regression a log-odds link with a binary outcome, and other widely used members a log link with Poisson-distributed counts or Gamma-distributed positive amounts. GLMs remain the regulatory workhorse of insurance pricing because their fitted coefficients read as multiplicative rating factors.

Suppose a lender fits this model to 10 past loan applicants, using a single feature, the applicant's debt-to-income ratio, and classifies an applicant as a predicted default (1) when the fitted probability $\hat{p}$ is at least 0.5 and as a predicted no default (0) otherwise; the fit itself is worked through at the end of this section. Table 6.3 shows each applicant's ratio, fitted probability, and actual and predicted outcomes.

Applicant	1	2	3	4	5	6	7	8	9	10
Debt/income (%)	48	26	44	40	33	37	22	18	31	15
Fitted $\hat{p}$	0.932	0.127	0.857	0.724	0.382	0.585	0.060	0.027	0.291	0.015
Actual default	1	0	1	0	0	1	0	0	1	0
Predicted default	1	0	1	1	0	1	0	0	0	0

Table 6.3: Debt-to-income ratio, fitted default probability, and actual versus predicted default for 10 loan applicants

Comparing the two rows produces a confusion matrix: 3 true positives (TP; applicants 1, 3, 6), 1 false negative (FN; applicant 9, an actual default the model missed), 1 false positive (FP; applicant 4, a good loan flagged as risky), and 5 true negatives (TN). Table 6.4 arranges these four counts into the standard 2×2 layout, with the model's predictions as rows and the actual outcomes as columns.

	Actual	
Predicted	Default (1)	No Default (0)
Default (1)	$TP = 3$	$FP = 1$
No Default (0)	$FN = 1$	$TN = 5$

Table 6.4: Confusion matrix for the 10 loan applicants in Table 6.3

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From these four counts, the standard classification metrics follow directly:

$$\begin{aligned}
\text{Accuracy} &= \frac{TP + TN}{n} = \frac{3 + 5}{10} = 0.80, \\
\text{Precision} &= \frac{TP}{TP + FP} = \frac{3}{4} = 0.75, \\
\text{Recall (TPR)} &= \frac{TP}{TP + FN} = \frac{3}{4} = 0.75, \\
\text{F1} &= \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} = 0.75, \\
\text{FPR} &= \frac{FP}{FP + TN} = \frac{1}{6} \approx 0.167.
\end{aligned}$$

Each metric answers a different question. Accuracy answers "what fraction of predictions were correct," which misleads when defaults are rare, since always predicting "no default" scores highly while being useless. Precision answers "of the loans flagged as risky, how many actually defaulted," which matters when false positives (turning away good customers) are costly. Recall answers "of the loans that actually defaulted, how many did the model catch," which matters when false negatives (missed defaults) are costly. A receiver operating characteristic (ROC) curve plots the true positive rate against the false positive rate as the classification threshold varies, and the area under this curve (AUC) summarizes performance across all thresholds rather than the single one used in Table 6.3.

Worked Example 6.3: Fitting the Logistic Regression and Computing the AUC

The probabilities in Table 6.3 come from choosing β_0 and β_1 to maximize the likelihood of the observed defaults under Equation (6.4), which is the same as minimizing the log-loss (6.5). There is no closed form, so the minimum is found numerically, by gradient descent or, as statistical software usually does, by Newton's method. At the minimum two first-order conditions hold, $\sum_i (y_i - \hat{p}_i) = 0$ and $\sum_i x_i (y_i - \hat{p}_i) = 0$: the fitted probabilities add up to the observed number of defaults, four, and the residuals are uncorrelated with the feature. For the ten applicants the solution is

$$\log \frac{\hat{p}}{1 - \hat{p}} = -7.2885 + 0.2063 \times \text{DTI},$$

with DTI, the debt-to-income ratio, in percentage points. The slope is read on the log-odds scale: each additional point of debt-to-income multiplies the odds of default by $e^{0.2063} = 1.229$, an increase of about 23%. The fitted probability crosses 0.5 where the log-odds are zero, at DTI $= 7.2885/0.2063 = 35.33\%$, which is why applicants 1, 3, 4, and 6 (ratios of 37% and above) are predicted to default and applicant 9, a defaulter at 31%, is missed. For applicant 6, the log-odds are $-7.2885 + 0.2063 \times 37 = 0.345$ and $\hat{p} = 1/(1 + e^{-0.345}) = 0.585$. The log-loss at the optimum is 0.401, against 0.673 for a model that ignores the feature and gives every applicant the base rate of 40%.

Two caveats belong with these numbers. First, ten observations pin the slope down poorly: its standard error, from the inverse of the Fisher information matrix, is 0.126, so $z = 1.64$ and $p = 0.10$, and the evidence that the ratio matters is suggestive rather than conclusive. Second, the fit is unpenalized, which works only because the data are not perfectly separable: applicant 4, who repaid, has a higher ratio than defaulters 6 and 9. Had every defaulter had a higher ratio than every non-defaulter, the likelihood would keep rising as $\beta_1 \to \infty$, no maximum would exist, and any reported coefficients would be artifacts of the optimizer's stopping rule or of a penalty. Library defaults matter here, as Section 6.4.1 shows for the lasso: scikit-learn's `LogisticRegression` adds a ridge penalty unless told otherwise (`C=1` by default), which shrinks this fit slightly to $-7.1781 + 0.2031 \times \text{DTI}$, and the unpenalized fit requires `C=np.inf`. This model is the workhorse of Chapters 9, 15, 18, and 21: credit scorecards, fraud scores, and the explanations built on them are logistic regressions or, increasingly, the gradient-boosted trees of Section 6.6.1.

The confusion matrix describes one threshold; the ROC curve describes all of them. Ranking the applicants by $\hat{p}$ gives the order 1, 3, 4, 6, 5, 9, 2, 7, 8, 10, in which the defaulters are 1, 3, 6, and 9. Lowering the threshold past one applicant at a time moves the curve up by 1/4 when that applicant is a defaulter and right by 1/6 when not, tracing the step curve in Figure 6.3. The area under it has an exact probabilistic

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meaning: it is the probability that a randomly chosen defaulter receives a higher score than a randomly chosen non-defaulter. With four defaulters and six non-defaulters there are $4 \times 6 = 24$ such pairs. Defaulters 1 and 3 outrank all six non-defaulters, defaulter 6 outranks five (all but applicant 4), and defaulter 9 outranks four (all but applicants 4 and 5), so

$$\text{AUC} = \frac{6 + 6 + 5 + 4}{24} = \frac{21}{24} = 0.875,$$

the same number as the area under the step curve, $\frac{1}{6} \times 0.5 + \frac{1}{6} \times 0.75 + \frac{4}{6} \times 1 = 0.875$; a tied pair, had there been one, would count one half. Credit-risk teams often report the equivalent Gini coefficient, or accuracy ratio, $2\,\text{AUC} - 1 = 0.75$, the measure Chapter 9 uses to validate its own scorecard. Because the AUC depends only on the ranking, any increasing transformation of the scores leaves it unchanged. That is both its strength, a threshold-free measure of discrimination, and its blind spot: it says nothing about whether $\hat{p} = 0.585$ really means a 58.5% chance of default, the calibration question taken up in Section 6.4.4.

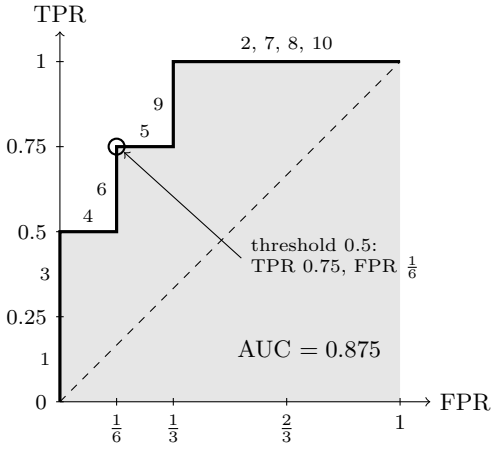


Figure 6.3: ROC curve of the fitted logistic regression for the 10 applicants of Table 6.3. Each step passes one applicant, labeled by number, in order of decreasing $\hat{p}$; the shaded area under the step curve is the AUC, the circle marks the 0.5 threshold of Table 6.4, and the dashed diagonal is the ROC curve of a random ranking.

6.3.5 K-Nearest Neighbors: A Non-Parametric Classifier

Logistic regression, like linear regression, is a parametric model: it assumes a specific functional form (a linear combination of features passed through a sigmoid) and estimates a fixed, finite set of parameters. K-nearest neighbors (KNN) is nonparametric: it assumes no functional form and classifies a new observation by a majority vote among the k most similar observations in the training data.

Consider a small loan-underwriting dataset with two features, debt-to-income ratio (%) and years of credit history, shown in Table 6.5.

To classify a new applicant with debt-to-income of 28% and 4 years of credit history, KNN computes the Euclidean distance from this query point to every training point,

$$d = \sqrt{(28 - x_1)^2 + (4 - x_2)^2},$$

giving

$$d_E = 2.24, \qquad d_A = 7.28, \qquad d_B = 8.94,$$
$$d_F = 10.44, \qquad d_C = 12.37, \qquad d_D = 14.32,$$

sorted from nearest to farthest. With $k = 3$, the three nearest neighbors are E (Default), A (Default), and B (No default), so the majority vote (2 Default, 1 No default) classifies the new applicant as a predicted default.

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Applicant	Debt/income (%)	Credit history (yrs)	Outcome
A	35	2	Default
B	20	8	No default
C	40	1	Default
D	15	10	No default
E	30	3	Default
F	18	7	No default

Table 6.5: A small loan dataset for K-nearest neighbors classification

Figure 6.4 shows the geometry: the query sits closest to a mix of Default and No-default points, and only its k nearest neighbors, not a global decision boundary, determine the prediction.

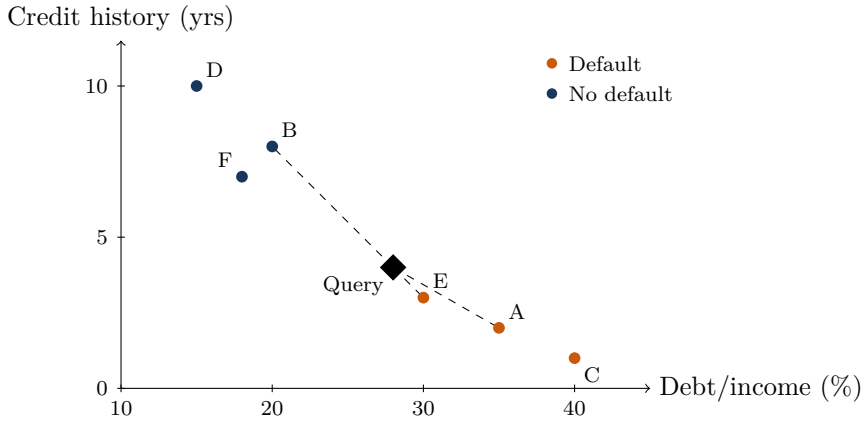


Figure 6.4: The K-nearest neighbors example of Table 6.5: six labeled training points and the query applicant (debt/income 28%, 4 years of credit history, black diamond), with dashed lines to its three nearest neighbors under $k = 3$

The choice of k controls a bias-variance trade-off like the one in Section 6.4.1. A small k (even $k = 1$) makes the prediction sensitive to individual nearby points, including noisy or mislabeled ones (high variance). A large k smooths over local structure and can underfit; when k equals the whole training set, it always predicts the majority class (high bias). KNN also degrades in high dimensions: as the number of features grows, all points become roughly equidistant, the curse of dimensionality, which is one reason KNN is used with a few carefully chosen features rather than on wide financial datasets.

6.3.6 Support Vector Machines

Suppose a credit analyst has a standardized measure of financial distress for a handful of firms and wants the dividing line between later defaulters and survivors that is least likely to misclassify a new firm near the boundary, not just any line that separates the historical data. A support vector machine (SVM) approaches classification geometrically: rather than modeling a probability (logistic regression) or voting among neighbors (KNN), it searches for the hyperplane (a line, in two dimensions) that separates the classes with the widest margin, the distance from the boundary to the nearest training point of either class. A wider margin makes the boundary less sensitive to small perturbations in the data, one reason SVMs often generalize well.

The training points on the margin, closest to the boundary, are called support vectors because they alone determine the hyperplane; every other point could be moved or removed without changing it, unlike linear regression, where every point contributes. When the classes cannot be separated perfectly, a soft-margin

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SVM lets some points fall on the wrong side of the margin, with a penalty parameter trading margin width against misclassified training points, analogous to the regularization strength λ in ridge and lasso regression.

SVMs can also be extended to nonlinear decision boundaries using the "kernel trick," which implicitly maps the features into a higher-dimensional space where a linear separator corresponds to a nonlinear boundary in the original space, without computing the higher-dimensional coordinates. In finance, SVMs have been used for bankruptcy prediction and directional price forecasting, though tree-based ensembles (Section 6.6.1) and neural networks are now more common for large tabular datasets, partly because SVMs scale poorly to very large training sets and trees need less feature preprocessing.

A simple example makes the maximum-margin idea concrete without the quadratic-programming machinery SVMs use in practice. Suppose four training points are perfectly separable along a single feature (say, a standardized measure of financial distress): two "high-risk" points at $x = 1$ and $x = 2$, and two "low-risk" points at $x = 4$ and $x = 5$. Because the classes are perfectly separated along this dimension, the maximum-margin hyperplane is the point halfway between the two closest points of opposite classes, $x = 2$ and $x = 4$, namely $x = 3$.

Writing the decision function as $wx + b$ with $w = 1$, the boundary condition $wx + b = 0$ at $x = 3$ requires $b = -3$, and checking the two nearest points confirms $w(2) + b = 2 - 3 = -1$ and $w(4) + b = 4 - 3 = 1$, the ± 1 margin boundaries the SVM optimization enforces. The points at $x = 2$ and $x = 4$ are the support vectors: they alone determine w and b, and the geometric margin, the distance from the boundary to either margin boundary, is $1/|w| = 1$, for a total margin width of 2, the full gap between the nearest opposite-class points.

The extreme points, $x = 1$ and $x = 5$, play no role in the boundary: moving the point at $x = 1$ to $x = -100$ would not change the hyperplane, whereas moving the support vector at $x = 2$ even slightly would shift it. Figure 6.5 shows the geometry.

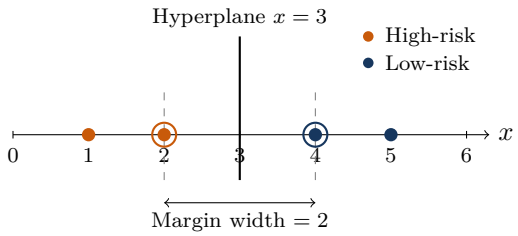


Figure 6.5: The maximum-margin hyperplane for the one-dimensional example: two high-risk points ($x = 1, 2$) and two low-risk points ($x = 4, 5$), with the support vectors circled

6.4 Generalization and Model Selection

A model that fits its training data perfectly has usually learned the noise; this part covers how that failure is detected, measured, and controlled.

6.4.1 Overfitting and Generalization

A central problem in machine learning is overfitting: a complex model fits the training data well yet fails on new data, because it has learned noise and quirks that do not recur outside the sample. Finance is especially exposed, since historical data are noisy and market regimes shift. Table 6.6 shows the pattern in a simulation. The true relationship is a gently curved quadratic, $y = 3 + 2x - 0.5x^2 + \varepsilon$, with x uniform on $[-3, 3]$ and noise ε of standard deviation 3, so no model can achieve a test MSE below the irreducible $\sigma^2 = 9$. Polynomials of increasing degree are fit by least squares to 25 training points and scored on 200 fresh test points, and the whole experiment is repeated 1,000 times on new, seeded samples; the table reports the median of each error across the repetitions. As the model grows more flexible, training error keeps falling, but test error falls only up to a point and then rises.

The minimum test error in Table 6.6 occurs at degree 2, the true degree of the relationship, not at the most flexible model, degree 9, even though degree 9 has the lowest training error in every one of the 1,000

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Polynomial degree	1	2	3	5	9
Training error (MSE)	9.62	7.68	7.33	6.61	5.14
Test error (MSE)	11.57	10.07	10.60	12.05	27.53

Table 6.6: Median training and test error over 1,000 simulated samples as polynomial degree increases

repetitions. Degree 9 also has a higher test error than degree 2 in 99.8% of them, and its errors are erratic as well as large: its mean test MSE across the repetitions is about 9,722, against a median of 27.53, pulled up by a few samples in which the fitted polynomial swings wildly between training points. This gap between training and test performance is the signature of overfitting, and the reason a model should never be selected on training performance. Figure 6.6 plots the two rows of Table 6.6 as curves.

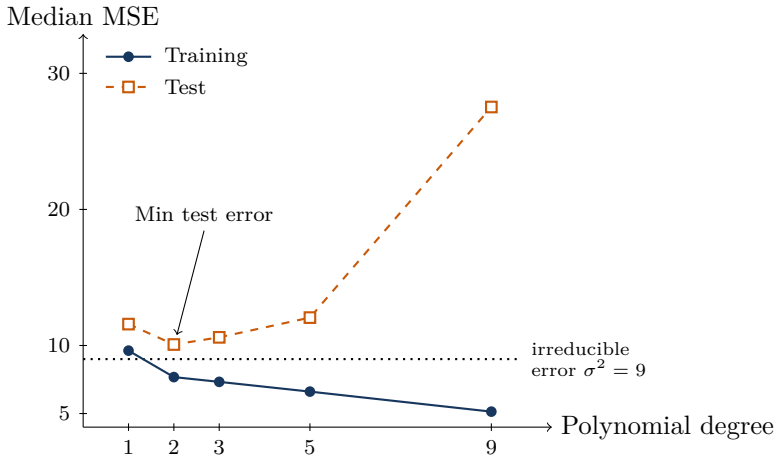


Figure 6.6: Median training and test error (MSE) from Table 6.6 as polynomial degree increases. The dotted line marks the irreducible error, and the vertical axis starts at 4.

Generalization is a model's ability to perform on unseen data, and the goal of model selection is a model that performs well out of sample, not on the training set. k-fold cross-validation formalizes this: the data are split into k roughly equal parts, and the model is trained on $k-1$ of them and evaluated on the remaining part, repeating k times so that every observation is used for validation exactly once. Figure 6.7 illustrates a single fold of 5-fold cross-validation.



Fold 3 held out for validation; Folds 1, 2, 4, 5 used for training

Figure 6.7: One iteration of 5-fold cross-validation. The procedure repeats five times, holding out a different fold each time, and the five validation scores are averaged.

For financial time series, ordinary k-fold cross-validation can mislead because it may train on the future and validate on the past, which is impossible in deployment. Walk-forward validation, in which the training window always precedes the validation window, is the standard alternative, discussed in Chapter 7.

Two subtler leaks survive even when each fold is a contiguous block of time, both from labels that span an interval. A label such as "did this loan default within 12 months" or "the stock's return over the next 20 trading days" depends on data from a window after the observation date. Training observations just

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before a validation block therefore have labels that overlap it, and those just after it may carry features computed from data inside it. *Purging* removes from the training set every observation whose label window overlaps the validation block, and an *embargo* also drops a short stretch of observations immediately after the block, long enough for serial dependence to fade (López de Prado 2018). With daily observations and 20-day forward-return labels, for instance, the 20 trading days before each validation block are purged and an embargo of similar length follows it. The cost is a somewhat smaller training set; the benefit is a validation score not inflated by the overlap, while training data from both sides of the validation block can still be used. Walk-forward validation avoids the backward leak by construction, and Chapter 13 meets the same overlap problem in backtests of trading signals.

Regularization reduces overfitting by penalizing model complexity in the optimization objective. Ridge regression adds a penalty proportional to the sum of squared coefficients,

$$\hat{\beta}^{\text{ridge}} = \arg\min_{\beta} \sum_{i=1}^{n} (y_i - x_i^\top \beta)^2 + \lambda \sum_{j=1}^{p} \beta_j^2, \tag{6.6}$$

while lasso regression uses the sum of absolute values,

$$\hat{\beta}^{\text{lasso}} = \arg\min_{\beta} \sum_{i=1}^{n} (y_i - x_i^\top \beta)^2 + \lambda \sum_{j=1}^{p} |\beta_j|. \tag{6.7}$$

In both cases, $\lambda \geq 0$ controls the strength of the penalty: $\lambda = 0$ recovers ordinary least squares, and larger λ shrinks coefficients toward zero, trading a small increase in bias for a potentially large reduction in variance. Lasso can also shrink coefficients exactly to zero, which makes it useful for feature selection when many candidate predictors are available, as is common in finance.

Worked Example 6.4: Ridge versus Lasso on a Genuine and a Spurious Feature

Sparsity is easiest to see with two features, one predictive and one pure noise, fit with both penalties at the same λ. Simulating 200 observations with $y = 2.0\,x_1 + \varepsilon$, where x_1 and x_2 are independent standard-normal features and x_2 is unrelated to y, ordinary least squares identifies both roles, $\hat{\beta}_1^{\text{OLS}} \approx 1.949$ and $\hat{\beta}_2^{\text{OLS}} \approx 0.031$, the latter attributable to sampling noise.

Before fitting, one practical trap. The library used in the companion notebook, scikit-learn, does not scale the two objectives the same way. Its `Ridge` minimizes exactly the ridge objective above, but its `Lasso` minimizes

$$\frac{1}{2n} \sum_{i=1}^{n} (y_i - x_i^\top \beta)^2 + \alpha \sum_{j=1}^{p} |\beta_j|,$$

so `Lasso(alpha=1.0)` is the lasso above with $\lambda = 2n\alpha = 400$, not $\lambda = 1$. Calling both with the default `alpha=1.0` therefore compares a lasso penalty 400 times stronger than the ridge one. Under the single objective written above, the comparison is

$$\begin{aligned} \lambda = 1: &\quad \hat{\beta}^{\text{ridge}} = (1.936,\ 0.030), &\quad \hat{\beta}^{\text{lasso}} = (1.945,\ 0.028), \\ \lambda = 400: &\quad \hat{\beta}^{\text{ridge}} = (0.543,\ -0.019), &\quad \hat{\beta}^{\text{lasso}} = (0.654,\ 0.000). \end{aligned}$$

At $\lambda = 1$ neither penalty does much, since λ is tiny next to a residual sum of squares built from 200 observations. At $\lambda = 400$ the contrast is striking: ridge shrinks both coefficients smoothly, the real one to 0.543, but leaves the spurious one nonzero, while lasso drives the spurious coefficient to *exactly* zero and keeps more of the real one (0.654). Lasso's penalty has a kink at zero that makes it optimal, past a threshold, to set a weak coefficient to zero rather than merely small, so feature selection happens as a side effect of fitting. The scaling trap is a lesson of its own: libraries scale penalties differently (by $1/n$, $1/(2n)$, or not at all), so a nominal penalty value means nothing until you have read the objective the library minimizes, and λ should be tuned by cross-validation rather than copied from a default.

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6.4.2 Bias, Variance, and the Trade-off

Suppose two analysts fit different models to the same noisy sample: one a straight line, the other a high-degree polynomial that bends to follow the training points, like the degree-9 fits in Table 6.6. Which should be trusted on new data, and why? The bias-variance trade-off answers this. For a model $\hat{f}$ trained on random training data and evaluated at a point x with true relationship

$$y = f(x) + \varepsilon,$$

where ε has mean zero and variance σ^2, the expected squared prediction error decomposes as

$$\mathbb{E}\left[(y - \hat{f}(x))^2\right] = \underbrace{\Big(f(x) - \mathbb{E}[\hat{f}(x)]\Big)^2}_{\text{Bias}^2} + \underbrace{\text{Var}\big(\hat{f}(x)\big)}_{\text{Variance}} + \underbrace{\sigma^2}_{\text{Irreducible error}}. \tag{6.8}$$

A model with high bias is too simple and systematically misses the true relationship however much data it sees, like a straight line fit to a strongly curved relationship. A model with high variance is too flexible and reacts to the noise in its particular training sample, like the degree-9 polynomial in Table 6.6. A good model balances the two, and the minimum of the test error in Table 6.6 marks that balance: the complexity at which bias and variance are jointly minimized, not the one that minimizes bias alone.

The trade-off matters especially in finance. With limited data and unstable relationships, simpler, higher-bias models can be surprisingly effective, because their lower variance makes them more robust across regimes. With large, high-quality datasets and a stable signal, more complex models may do better. Either way, models should be evaluated out of sample or walk-forward, without assuming that complexity improves results.

6.4.3 Hyperparameter Tuning and Model Selection

Every model introduced in this chapter has at least one hyperparameter, a setting chosen before training rather than learned from the data, that controls its position on the bias-variance spectrum: the ridge penalty λ, the number of neighbors k in KNN, the maximum depth of a decision tree, or the number of trees and learning rate in a gradient-boosted ensemble. Choosing them is often as consequential as choosing the model family, and it must be done without looking at the final test set, or the performance estimate no longer measures generalization.

The standard workflow, called grid search combined with cross-validation, proceeds as follows: specify a grid of candidate hyperparameter values (for example, $\lambda \in \{0.01, 0.1, 1, 10, 100\}$), and for each candidate value, run k-fold cross-validation (or, for time series data, the walk-forward validation introduced in Chapter 7) entirely within the training set, recording the average validation performance. The value with the best average cross-validated performance is selected, and only then is the model retrained on the full training set and evaluated once on the held-out test set, which the search never touched. This nesting is essential: using the test set to choose hyperparameters, even informally by trying a few values and keeping the best, reintroduces the overfitting that cross-validation was designed to prevent and inflates the reported performance.

Financial applications add two complications. First, because financial time series are non-stationary, hyperparameters selected on one period need not remain optimal later, which argues for re-running the tuning periodically. Second, the grid search is itself a form of multiple testing, like the backtest-overfitting concern of Chapter 7: a very large grid raises the chance of a setting that does well on the validation folds by luck, so practitioners often prefer a modest, economically motivated grid to an exhaustive search over hundreds of values.

6.4.4 Model Evaluation

Evaluating a model takes more than a single number, and different tasks need different metrics. For regression, common metrics include mean squared error (MSE), mean absolute error (MAE), and R^2, all illustrated in Section 6.3.1. For classification, common metrics include accuracy, precision, recall, F1, and AUC, all illustrated in Section 6.3.4. In finance one also cares about calibration (do predicted probabilities match observed frequencies), stability (does performance hold up across periods), and economic usefulness (does the output lead to a profitable or risk-reducing decision after costs).

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A default model with high accuracy may still be impractical if it is poorly calibrated, produces too many false positives, or fails to separate risk levels usefully. A trading model may perform well statistically but poorly economically once transaction costs and execution limits are counted. Evaluation in finance must therefore be tied to the decision problem, not to a generic statistical scorecard.

Calibration is easy to confuse with accuracy. Suppose a credit model assigns a default probability of 20% to a group of 50 borrowers. It is well calibrated for this group if close to 10 of the 50 (that is, $20\% \times 50$) default. If only 2 default, its probabilities are too high even if it still ranks borrowers correctly from most to least risky, a distinction that accuracy and even AUC miss, since both depend only on the ranking. Calibration matters in finance because predicted probabilities feed downstream calculations, such as the expected-loss and reserve calculations of Chapter 9, where an overstated or understated default probability becomes an overstated or understated dollar reserve.

Worked Example 6.5: Measuring and Repairing Calibration

A single number that rewards calibration is the Brier score (Brier 1950), the mean squared difference between predicted probabilities and outcomes,

$$\mathrm{BS} = \frac{1}{n}\sum_{i=1}^{n}(\hat{p}_i - y_i)^2,$$

which is zero only for perfect, certain forecasts and, unlike the AUC, penalizes a probability that is too high or too low even when the ranking is right. For the 50 borrowers above, the 20% forecast scores $\frac{1}{50}\left[10\times0.8^2+40\times0.2^2\right] = 0.160$ when 10 of them default. When only 2 default it scores $\frac{1}{50}\left[2\times0.8^2+48\times0.2^2\right] = 0.064$, while the calibrated forecast of 4% would have scored 0.038: the same borrowers and the same ranking, and a score 40% lower for getting the level right. (The two 20% scores, 0.160 and 0.064, are not comparable with each other, since they grade forecasts of different outcomes; Brier scores rank competing forecasts of the same outcomes.)

A model that ranks well but is miscalibrated need not be rebuilt; its scores can be remapped. **Platt scaling** (Platt 1999) fits a one-feature logistic regression of the outcome on the score's log-odds, $\Pr(y = 1 \mid s) = 1/\left(1+e^{-(a+b\,\mathrm{logit}(s))}\right)$ with $\mathrm{logit}(s) = \log\left(s/(1-s)\right)$, on a held-out calibration set. **Isotonic regression** (Zadrozny and Elkan 2002) instead fits the best non-decreasing step function from score to probability, which assumes no functional form but needs more data. Both preserve the order of the scores (isotonic regression up to ties), so they repair calibration without changing discrimination.

To see both at work, simulate borrowers whose true default probability is $1/(1+e^{-(-3+z)})$ for a standard-normal risk factor z, scored by a model with $s = 1/(1+e^{-(-1.8+1.6z)})$ that ranks them perfectly but overstates almost every probability, as a model trained in a period of high defaults might. On 20,000 test borrowers the default rate is 6.98% while the score averages 22.30%, and the 3,122 borrowers scored between 15% and 25% average a score of 19.62% but default at only 6.18%, the 20%-versus-4% situation above at scale. Fitting each remedy on a separate calibration set of 2,000 borrowers gives Table 6.7. Platt scaling, with fitted $a = -1.921$ and $b = 0.655$, brings the Brier score from 0.1053 to 0.0593, matching the true probabilities, because the miscalibration here has the logistic form Platt scaling assumes (the true map is $a = -1.875$, $b = 0.625$). Isotonic regression gets almost as far, to 0.0595, without assuming a form, at the price of a slightly lower AUC, since its flat steps tie borrowers the raw score had separated.

Forecast	Mean forecast	Brier score	AUC	Mean forecast, 15–25% raw band
Raw model score	22.30%	0.1053	0.7672	19.62%
Platt scaling	6.64%	0.0593	0.7672	5.49%
Isotonic regression	6.69%	0.0595	0.7641	3.89%
True probability	7.01%	0.0593	0.7672	5.97%
Observed default rate	6.98%			6.18%

Table 6.7: Recalibrating an overconfident default model on 2,000 borrowers, scored on 20,000 fresh borrowers

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Two cautions apply. The calibration set must be separate from the training data, or the remap learns the model's in-sample overconfidence instead of correcting it. And a remap fitted in one credit environment goes stale in the next, so recalibration is part of monitoring rather than a one-time fix. Niculescu-Mizil and Caruana (2005) compare the two methods across many model families; Chapter 21 applies isotonic recalibration to a real credit-card default model.

6.4.5 Quantifying Uncertainty: Split Conformal Prediction

Every model so far returns a single number: a fitted spread of $20 + 47(4.5) = 231.5$ basis points for an issuer at 4.5 turns of leverage, a default probability, a class label. A risk manager setting a limit, a trader quoting a price, or a credit officer deciding whether a case needs a second look usually needs a range as well: how far from the point forecast could the truth plausibly be? Section 6.3.2's standard errors answer a different question, how precisely the *coefficients* are pinned down. The classical prediction interval for a new observation relies on normal, constant-variance errors and a correctly specified model, and for a random forest, a gradient-boosted ensemble, or a neural network there is no such formula.

Conformal prediction (Vovk et al. 2022) puts an interval around *any* fitted model, with a coverage guarantee that needs no distributional assumptions and holds in finite samples. Its simplest and most widely used form, **split conformal prediction** (Lei et al. 2018), takes four steps:

1. Split the available data into a *proper training set* and a *calibration set* of n observations that the model never sees during fitting.
2. Fit any model $\hat{f}$ on the proper training set.
3. On the calibration set, compute a *nonconformity score* for each observation, which for regression is usually the absolute residual $s_i = |y_i - \hat{f}(x_i)|$.
4. Set $\hat{q}$ to the k-th smallest calibration score, where

$$k = \lceil (n+1)(1-\alpha) \rceil, \qquad \hat{q} = s_{(k)}, \tag{6.9}$$

   and report the interval $C(x) = \left[\hat{f}(x) - \hat{q},\ \hat{f}(x) + \hat{q}\right]$ for each new case.

Equivalently, $\hat{q}$ is the empirical quantile of the calibration scores at level $\lceil (n+1)(1-\alpha) \rceil / n$, slightly above the nominal $1-\alpha$; the $(n+1)$ in place of n is a small-sample correction whose purpose is explained below. If $k > n$, which happens when $n < 1/\alpha - 1$, the calibration set is too small to support the requested coverage and the interval is the whole real line.

The guarantee is that, provided the calibration observations and the new observation (X_{n+1}, Y_{n+1}) are **exchangeable**, meaning that their joint distribution is unchanged by reordering them (independent draws from one distribution are the leading case),

$$1 - \alpha \ \le\ \Pr\left(Y_{n+1} \in C(X_{n+1})\right) \ \le\ 1 - \alpha + \frac{1}{n+1}, \tag{6.10}$$

where the upper bound requires only that the scores have no ties. The argument is a counting one. Exchangeability makes the new observation's score s_{n+1} equally likely to occupy any of the $n+1$ rank positions among all $n+1$ scores, so the probability that it lands at or below the k-th smallest calibration score is exactly $k/(n+1)$, and choosing $k = \lceil (n+1)(1-\alpha) \rceil$ makes $k/(n+1) \ge 1 - \alpha$. Nothing in the argument depends on what $\hat{f}$ is, how well it fits, or the shape of the error distribution: a poor model produces large scores and wide intervals, but they still cover at the promised rate. The guarantee is *marginal*: the probability is over the random calibration set and new case together, so it promises 90% coverage on average, not 90% given one particular calibration set, and not 90% within every subgroup of borrowers or issuers.

Worked Example 6.6: Split Conformal Intervals for Credit Spreads

To see every number, simulate issuers from a relationship close to Section 6.3.1's fitted line, spread $= 20 + 47 \times \text{leverage} + \varepsilon$, with leverage drawn uniformly between 1 and 7 turns and ε normal with a standard

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Rank (i) and score $s_{(i)}$, bps							
(1) 1.2	(2) 4.5	(3) 4.9	(4) 9.7	(5) 11.4	(6) 14.7	(7) 15.6	(8) 16.4
(9) 17.2	(10) 20.3	(11) 20.6	(12) 21.4	(13) 23.9	(14) 25.0	(15) 27.9	(16) 28.8
(17) 29.5	(18) 29.5	(19) 34.9	(20) 37.6	(21) 37.9	(22) 38.9	(23) 44.1	(24) 44.3

Table 6.8: Sorted absolute residuals $|y_i - \hat{f}(x_i)|$ of the 24 calibration issuers, in basis points

deviation of 25 basis points. OLS on a proper training set of 40 issuers gives $\widehat{\text{spread}} = 9.91 + 50.12 \times \text{leverage}$. A separate calibration set of $n = 24$ issuers then produces the 24 absolute residuals in Table 6.8, sorted from smallest to largest.

For a 90% interval, $\alpha = 0.10$, so Equation (6.9) gives $(n+1)(1-\alpha) = 25 \times 0.9 = 22.5$, rounded up to $k = 23$, and the conformal quantile is the 23rd smallest score, $\hat{q} = 44.12$ basis points (printed as 44.1 in the table). A new issuer at 4.5 turns of leverage receives a point forecast of $9.912 + 50.122 \times 4.5 \approx 235.46$ basis points (the fitted coefficients to three decimals; the two-decimal values above would give 235.45) and the interval

$$235.46 \pm 44.12 = [191.34,\ 279.58] \text{ basis points.}$$

The width is the same for every issuer, since the score is a plain absolute residual. Because the scores are continuous, Equation (6.10) holds with the exact value $k/(n+1) = 23/25 = 0.920$, inside the guaranteed range $[0.900,\ 0.940]$.

Two checks confirm the theory. On 5,000 fresh test issuers drawn from the same process, 91.52% of the true spreads fall inside their intervals. And repeating the whole calibration step 10,000 times, each time with a new set of 24 issuers and 200 test issuers, gives an average coverage of 91.94%, essentially the predicted 92.0%. The same repetition also shows what "marginal" means in practice: across calibration draws, the coverage achieved by one particular set of 24 issuers ranges from 81.0% at the 5th percentile to 99.0% at the 95th. That spread is predicted too: given one calibration set, coverage follows a $\text{Beta}(k,\ n+1-k) = \text{Beta}(23, 2)$ distribution, whose 5th and 95th percentiles are 81.7% and 98.5%, the small remaining gap reflecting the finite 200-issuer test sets. A larger calibration set tightens this spread without changing the average.

The correction matters. Taking the plain empirical 90th percentile instead, $k = \lceil n(1-\alpha) \rceil = \lceil 21.6 \rceil = 22$, gives $\hat{q} = 38.87$ basis points and an exact coverage of only $22/25 = 0.880$, below the target: 86.66% on the 5,000 test issuers and 87.89% on average over the 10,000 repetitions. With small calibration sets, the extra $+1$ is what turns a nominal 90% interval into a real one.

Limits. Exchangeability is the whole foundation of Equation (6.10), and financial data violate it routinely. Simulating a stress regime in which the noise standard deviation doubles from 25 to 50 basis points, while keeping the same $\hat{q} = 44.12$, cuts the empirical coverage of the nominal 90% interval to 61.04%. That is the situation in which a risk manager most needs the interval to work: the regime shift, volatility clustering, and concept drift discussed in Chapter 7 and Section 6.7. Time-ordered data are not exchangeable even without a regime change, so the calibration set should come from the most recent period and be refreshed as outcomes arrive; adaptive variants that update $\hat{q}$ online as coverage errors accumulate are designed for this setting (Gibbs and Candès 2021). A second limit is that the constant width ignores heteroskedasticity: if errors are larger for highly levered issuers, a single $\hat{q}$ over-covers safe issuers and under-covers risky ones while still averaging 90%. Dividing each residual by a model of its typical size, or calibrating a quantile regression, gives intervals that widen where the model is less certain. Chapter 18's Advanced Topics returns to conformal prediction for classifiers, where the output is a set of labels and a borderline credit applicant whose set contains both "repay" and "default" is flagged for human review (see also Angelopoulos and Bates 2023).

6.4.6 Prediction versus Causation

Every model in this chapter answers a *predictive* question: given what is observed about this borrower or issuer, what outcome should be expected? Many of the decisions they feed ask a *causal* question instead: if the institution changes something, what will happen? Raising a cardholder's credit limit, offering a retention discount, calling a delinquent borrower, or cutting a loan's interest rate are interventions, and choosing among

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them requires knowing how the outcome responds to the intervention, not how the two moved together in historical data. Even the slope of 47 basis points per turn of leverage in Section 6.3.1 is an association across issuers. Highly levered firms differ from lightly levered ones in industry, asset tangibility, and cash-flow volatility, so one firm adding a turn of leverage need not see its spread rise by 47 basis points. Chapter 7's Granger causality carries the same caveat: it tests whether one series helps *predict* another, not whether it causes it.

The potential-outcomes framework makes the distinction precise. Let $D = 1$ if a customer receives an intervention (a *treatment*) and $D = 0$ otherwise, and let $Y(1)$ and $Y(0)$ be the outcomes the same customer would experience with and without it. Only one of the two is ever observed. The *average treatment effect* is $\tau = \mathbb{E}[Y(1) - Y(0)]$. Comparing the treated and untreated groups in observational data gives, when the effect is the same for everyone,

$$\mathbb{E}[Y \mid D=1] - \mathbb{E}[Y \mid D=0] = \tau + \underbrace{\mathbb{E}[Y(0) \mid D=1] - \mathbb{E}[Y(0) \mid D=0]}_{\text{selection bias}}. \tag{6.11}$$

The selection-bias term is the difference the two groups would show even if nobody were treated. It is nonzero whenever a **confounder**, a variable driving both who is treated and the outcome, differs between the groups. In consumer credit the canonical confounder is creditworthiness. Lenders grant credit-limit increases mainly to customers who look safe, and safe customers default less, so the customers who received increases would have defaulted less *even without* them.

In a linear model the size of the distortion has a simple form. Suppose the outcome is $Y = \beta_0 + \tau D + \gamma Z + \varepsilon$, where Z is the confounder, and an analyst regresses Y on D alone. The "short" regression's coefficient on D is

$$\hat{\beta}_{\text{short}} = \hat{\tau} + \hat{\gamma}\,\hat{\delta}, \tag{6.12}$$

where $\hat{\tau}$ and $\hat{\gamma}$ come from the "long" regression of Y on D and Z, and $\hat{\delta}$ from regressing Z on D, which for a binary D is the gap in the average of Z between the treated and untreated groups. This **omitted-variable bias** formula holds exactly in any sample, not just on average, so it can be checked to the last decimal.

Worked Example 6.7: A Credit-Limit Increase: Naive, Adjusted, and Randomized Estimates

Consider 20,000 cardholders with a creditworthiness score Z distributed uniformly between -1 (weakest) and $+1$ (strongest). The bank grants a limit increase with probability $\Pr(D = 1 \mid Z) = 0.5 + 0.4Z$, so its strongest customers receive one 90% of the time and its weakest 10% of the time. Each customer's probability of defaulting within twelve months is

$$\Pr(\text{default}) = 0.08 + 0.02\,D - 0.06\,Z,$$

which stays between 0.02 and 0.16. By construction, the true causal effect of a limit increase is $\tau = +0.02$: extra credit raises every customer's default probability by two percentage points.

In the simulated observational data, 50.44% of customers receive an increase. Their average creditworthiness is 0.2665, against -0.2662 for the rest, a gap of $\hat{\delta} = 0.5327$. The default rate is 8.25% among customers who received an increase and 9.80% among those who did not, so the naive comparison, identical to the short regression of default on D, says increases *lower* default by 1.55 percentage points. A predictive model would be right to use this pattern: the increase flag is a useful feature, because it carries information about creditworthiness that the bank observed when granting it. As a guide to policy, however, it points the wrong way: a lender that raised limits to reduce defaults would raise them instead.

Table 6.9 collects the estimates. Adding Z to the regression gives $\hat{\tau} = +0.0180$ and $\hat{\gamma} = -0.0629$, close to the true $+0.02$ and -0.06, and Equation (6.12) reconciles the two regressions exactly:

$$\hat{\beta}_{\text{short}} = 0.0180 + (-0.0629)(0.5327) = 0.0180 - 0.0335 = -0.0155.$$

In population terms, the gap in average creditworthiness is 0.5333, so the bias is $-0.06 \times 0.5333 = -0.032$, large enough to turn a true effect of $+0.02$ into an apparent effect of -0.012.

Adjustment works here only because the analyst observes Z exactly. With only a noisy proxy, the true score plus normal noise with a standard deviation of 0.5, the adjusted estimate is $+0.0027$: the regression

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Estimator	Estimated effect of an increase
Naive: default regressed on D alone	-0.0155
Adjusted: default regressed on D and Z	$+0.0180$
Adjusted for a noisy proxy $Z + \mathrm{N}(0, 0.5^2)$	$+0.0027$
Randomized experiment, difference in default rates	$+0.0193$
True effect τ	$+0.0200$

Table 6.9: Four estimates of the effect of a credit-limit increase on the twelve-month default rate, each from 20,000 simulated cardholders

removes some of the confounding but leaves most of the true effect hidden. In practice the bank's score is at best a proxy for the creditworthiness that drives default, and confounders it never recorded cannot be adjusted for.

A randomized experiment removes the problem by construction. Drawing 20,000 fresh customers and letting a coin flip decide who receives the increase puts 10,013 in the treated group and 9,987 in the control group, with average creditworthiness of -0.0008 and -0.0116, balanced up to chance. Their default rates are 9.94% and 8.01%, a difference of $+0.0193$. With the standard error of a difference in proportions,

$$\sqrt{\frac{0.0994(1-0.0994)}{10{,}013} + \frac{0.0801(1-0.0801)}{9{,}987}} \approx 0.0040,$$

the 95% confidence interval is $[+0.0113,\ +0.0272]$, which contains the true $+0.02$ and excludes both zero and the naive estimate.

Experiments as the gold standard. Randomization makes the treated and control groups alike, on average, in every characteristic, observed or not, so the selection-bias term in Equation (6.11) is zero and a simple difference in means estimates τ. Financial firms run such experiments routinely, as A/B tests of offers, pricing, and collection strategies, and the method has produced some of the cleanest evidence on credit. Aydin (2022) reports a field experiment at a large retail bank in Turkey: of 45,307 customers preapproved for a lender-initiated limit increase, a randomly chosen control group had the increase withheld for nine months. Treated customers spent about 11 cents per dollar of new limit in the quarter of the increase, rising to about 28 cents by the third year, a causal estimate no regression on the bank's historical limit decisions could have delivered. Experiments have costs and limits. The control group forgoes something the lender believes is good for it; consumer-protection and fair-lending rules constrain which terms can be varied and for whom; outcomes such as default take months or years to observe; and many questions, such as the effect of a recession or a central-bank rate move, cannot be randomized. Chapter 19's bandit algorithms address part of this tension: how to keep learning from experiments while limiting the cost of assigning customers to inferior options.

Observational methods. When an experiment is impossible, several methods try to recover causal effects from observational data, each replacing randomization with an assumption that must be argued rather than tested. *Difference-in-differences* compares the change in outcomes over time for a group affected by a policy against the change for an unaffected group, and requires that the two would have followed parallel trends without the policy. *Instrumental variables* exploit a variable that shifts who is treated but affects the outcome only through the treatment, such as an arbitrary change in an approval rule. *Double/debiased machine learning* (Chernozhukov et al. 2018) uses flexible models of the kind developed in this chapter to predict both the outcome and the treatment from a large set of controls, then estimates the effect from the two sets of residuals with cross-fitting, so that regularization does not bias the causal estimate; like the adjusted regression above, it still assumes every important confounder is among the controls. *Uplift modeling* predicts how much an individual customer's outcome would change if treated, so that a retention offer or a collections call goes to the customers it changes rather than those who would have responded anyway; it is most reliable when trained on experimental data. Angrist and Pischke (2009) is the standard practical introduction to these designs.

The lesson for the rest of the book is a division of labor. Predictive models, and the importance measures and Shapley values Chapter 18 uses to explain them, describe associations learned from training data. They

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are the right tool for ranking applicants, flagging transactions, or forecasting losses, and the wrong one for deciding what a new policy would do. Section 6.7's warning about feedback and reflexivity is the market-level version of the same point: once a model's output is used to intervene, the historical association it learned is no longer a reliable guide.

6.5 Working with Financial Data

Financial data breaks several assumptions the previous parts made quietly, and the features handed to a model matter more than the model.

6.5.1 Practical Considerations in Financial ML

Deploying machine learning in finance requires attention to data leakage, sample selection, and concept drift. Data leakage occurs when the model uses information that would not be available at prediction time, such as a variable recorded only after the outcome is known, producing optimistic backtest results that vanish in real time. Sample selection issues arise when the training data are not representative of the future environment, for instance training a credit model only on approved loans, which excludes information about applicants who were rejected. Concept drift occurs when the relationship between features and outcomes changes over time, for example when a recession changes which borrower characteristics matter most for default.

A compact view of the modeling lifecycle is shown in Figure 6.8.

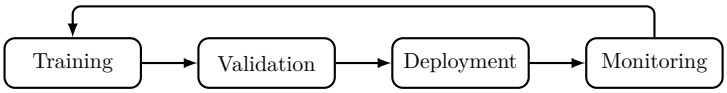


Figure 6.8: A machine learning workflow in finance should include training, validation, deployment, and continuous monitoring, with findings from monitoring feeding back into retraining

These issues often determine whether a model survives in production. Good financial modeling means not only fitting a relationship but making sure it is stable enough to stay useful, and building the monitoring to detect when it stops being useful.

6.5.2 Feature Engineering for Financial Data

Every model in this chapter takes a fixed set of numerical features, but raw financial data (prices, statement line items, transaction records) are rarely usable directly, and transforming them into informative inputs, called feature engineering, often matters as much as the choice of algorithm. Several patterns recur. Ratios, such as the debt-to-equity and margin ratios of Chapter 4, normalize dollar figures so that firms of very different sizes become comparable, as that chapter's common-size statements did.

Lagged and rolling features, such as a 30-day rolling volatility or a 12-month trailing revenue growth rate, summarize recent history into a single number a model can consume, echoing the rolling-window calculations introduced in Chapter 2. Interaction features, such as leverage multiplied by a recession indicator, let a linear model capture an effect of one variable that depends on the level of another, which a plain linear model cannot represent. Categorical features, such as industry sector, are converted to numbers by one-hot encoding (a binary column for each category), so that sector-specific behavior of the kind discussed in Chapter 4 enters the model as a signal rather than an arbitrary label. Table 6.10 and Table 6.11 make these transformations concrete for a small cross-section of three firms.

Table 6.11 shows the engineered features these raw inputs produce. The leverage ratio and year-over-year revenue growth follow directly from the raw dollar figures; the interaction term is zero for both non-recession firms whatever their leverage, but equals 1.50 for Firm 2, flagging that its elevated leverage coincides with a sector-wide downturn. One-hot encoding converts the raw Sector column into three binary indicators: Firm 1 becomes (Technology=1, Energy=0, Retail=0), Firm 2 becomes (Technology=0, Energy=1, Retail=0), and Firm 3 becomes (Technology=0, Energy=0, Retail=1).

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Firm	Sector	Debt	Equity	Rev.$_t$	Rev.$_{t-4}$	Recession	Mkt Cap
1	Tech	20	80	150	120	0	45.0
2	Energy	90	60	210	230	1	8.0
3	Retail	40	40	95	90	0	1.2

Table 6.10: Raw firm-level inputs for a small cross-section (Debt, Equity, and Revenue in $ millions; Market Cap in $ billions; Recession is 1 if the firm's primary sector was in a broad-based downturn during the period)

Firm	Leverage	Rev. growth (YoY)	Leverage×Recession	$\log_{10}$(Mkt Cap, $B)	Mkt Cap pctile
1	0.25	25.0%	0.00	1.65	100%
2	1.50	−8.7%	1.50	0.90	50%
3	1.00	5.6%	0.00	0.08	0%

Table 6.11: Engineered features derived from Table 6.10

Two further considerations are specific to finance. First, every engineered feature must respect point-in-time discipline: a feature computed from data that would not have been available at prediction time introduces data leakage, even when the computation, such as a rolling average, seems innocuous. Second, financial features are frequently skewed or heavy-tailed, and Table 6.11's market-capitalization columns make this concrete: the three firms' raw market caps span more than an order of magnitude ($1.2 billion to $45.0 billion, a ratio of 37.5×), while $\log_{10}$ of the same figures compresses this to a range of about 1.57 (0.08 to 1.65), and the cross-sectional percentile rank compresses it further still, to a bounded $[0, 100\%]$ scale regardless of the firms' raw dollar magnitudes. Such transformations make skewed or heavy-tailed features better behaved as inputs to linear and distance-based models.

Worked Example 6.8: Feature Scaling: One Applicant, Three Answers

Distance-based and penalized models need features on a common scale, as the two features of Table 6.5 show. Debt-to-income runs from 15 to 40 percentage points and credit history from 1 to 10 years, so the Euclidean distance of the K-nearest neighbors example is dominated by whichever feature has the larger numerical spread, a choice made by the units rather than the economics. Consider an applicant with a 28% debt-to-income ratio and 7 years of credit history. In the table's units, percentage points and years, the three nearest neighbors are E (distance 4.47), B (8.06), and A (8.60), two defaults and one non-default, so 3-NN predicts default. Record credit history in months instead, which carries the same information, and the three nearest neighbors become F (10.00), B (14.42), and D (38.28), all non-defaults.

Standardization removes the dependence on units by replacing each feature with its z-score,

$$z = \frac{x - \bar{x}}{s_x},$$

where $\bar{x}$ and s_x are the feature's mean and standard deviation in the training data. For the six applicants, debt-to-income has mean 26.33 and standard deviation 9.25, and credit history mean 5.17 and standard deviation 3.34 (population standard deviations, the convention of scikit-learn's `StandardScaler`), so the query applicant becomes (0.180, 0.549). In standardized units the three nearest neighbors are B (0.92), F (1.08), and E (1.22), and the prediction is no default, whether history was recorded in years or in months. Neither raw answer was wrong by the algorithm's own logic; each let the units decide how much each feature counted.

The logic reaches beyond KNN. Ridge and lasso penalize $\sum_j \beta_j^2$ or $\sum_j |\beta_j|$, and a coefficient's size depends on its feature's units, so an unscaled penalty shrinks some features more than others for no economic reason. Gradient descent converges faster when features share a scale, because the loss surface is less elongated, and SVMs and neural networks are similarly sensitive. Tree-based models, by contrast, are unaffected, since a split on $x > t$ separates the same observations however x is measured. Two rules follow. Means and standard

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deviations must be estimated on the training data only and applied unchanged to validation and test data, since computing them on the full sample is a mild form of data leakage. And heavy-tailed features are usually transformed first, with the logarithms or percentile ranks of Table 6.11, since a few extreme values inflate the standard deviation and squash everyone else's z-scores toward zero.

6.6 Beyond Linear Models

This part covers the model families that dominate practice when the relationship is not linear and the data are not tidy.

6.6.1 Tree-Based Models and Ensembles

Tree-based methods are among the most useful models in applied finance because they capture nonlinear relationships and interactions without the analyst specifying them in advance. A decision tree splits the data recursively according to feature thresholds, choosing at each step the split that most reduces impurity in the resulting groups. For classification, a common impurity measure is the Gini impurity,

$$G = 1 - \sum_{k=1}^{K} p_k^2,$$

where p_k is the proportion of observations in a node belonging to class k; $G = 0$ when a node is pure (all one class) and is largest when classes are evenly mixed. An alternative is entropy, $H = -\sum_k p_k \log_2 p_k$, which behaves similarly.

Worked Example 6.9: One Split of a Decision Tree

Growing a tree on the ten applicants of Table 6.3, with debt-to-income as the only feature, shows how a split is chosen. The root node holds four defaults and six non-defaults, so its impurity is $G = 1 - (0.4^2 + 0.6^2) = 0.48$. A candidate split sends applicants with a ratio above a threshold t to one child node and the rest to the other, and is scored by the two children's impurities weighted by their sizes. Only the midpoints between consecutive observed ratios need to be tried, nine in all. Splitting at $t = 35$, close to the logistic regression's 50% boundary of 35.33%, puts applicants 1, 3, 4, and 6 (three defaults out of four) above and the other six (one default) below:

$$G_{\text{above}} = 1 - (0.75^2 + 0.25^2) = 0.375, \quad G_{\text{below}} = 1 - \left(\left(\tfrac{1}{6}\right)^2 + \left(\tfrac{5}{6}\right)^2\right) = 0.278, \quad \tfrac{4}{10} \times 0.375 + \tfrac{6}{10} \times 0.278 = 0.317.$$

Splitting at $t = 28.5$ instead puts all four defaults and two non-defaults (applicants 4 and 5) above, and four non-defaults below:

$$G_{\text{above}} = 1 - \left(\left(\tfrac{4}{6}\right)^2 + \left(\tfrac{2}{6}\right)^2\right) = 0.444, \quad G_{\text{below}} = 0, \quad \tfrac{6}{10} \times 0.444 + \tfrac{4}{10} \times 0 = 0.267.$$

Checking all nine thresholds confirms that $t = 28.5$ has the lowest weighted impurity, a reduction of $0.48 - 0.267 = 0.213$ from the root, and it is the split scikit-learn's `DecisionTreeClassifier` chooses. The tree and the logistic regression draw their lines in different places because they optimize different things: the Gini criterion rewards the pure node of four safe applicants, while the logistic regression's 50% boundary is a by-product of fitting probabilities to every applicant. A full tree would continue by splitting each child in the same way, trying every feature at every node, until a stopping rule such as a maximum depth or a minimum node size halts it.

A single tree tends to have high variance: small changes in the training data can produce very different splits. Random forests reduce this variance by averaging many trees, each trained on a bootstrap sample of the data with a random subset of features at each split, so that the trees are decorrelated. Gradient boosting instead builds trees sequentially, each one correcting the errors of the ensemble built so far. Write $F_m(x)$ for the ensemble's prediction after m trees have been added; boosting initializes $F_0(x)$ to a simple constant

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(typically the mean of the target for regression, or the log-odds of the positive class for classification) and then grows the ensemble one shallow tree at a time,

$$F_m(x) = F_{m-1}(x) + \nu\, h_m(x), \qquad m = 1, \ldots, M, \tag{6.13}$$

where h_m is a new regression tree fit not to the target y but to the current pseudo-residuals, $y_i - F_{m-1}(x_i)$ for squared-error loss. More generally these are the negative gradient of the loss with respect to the current predictions $F_{m-1}(x_i)$, hence the name "gradient" boosting, so each new tree targets what the ensemble still gets wrong.

The learning rate $\nu \in (0, 1]$, typically well below 1 (commonly 0.01 to 0.3), shrinks each tree's contribution: a smaller ν needs more trees M to reach a given fit but tends to generalize better, the same bias-variance trade-off as the ridge and lasso penalties. Figure 6.9 shows the sequential structure. Each stage cannot be fit until the previous stage's pseudo-residuals are known, which is why random forests parallelize trivially across trees and gradient boosting does not. Rather than reducing variance by averaging, boosting reduces bias by correcting the current model's errors, and it is combined with a small learning rate and a limited number of trees to control overfitting.

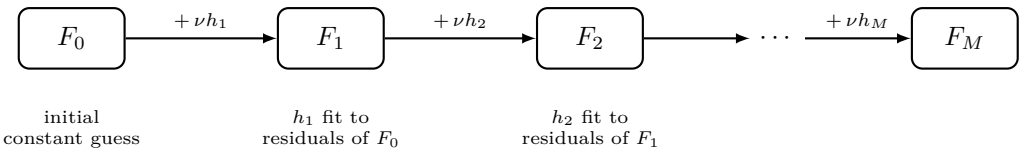


Figure 6.9: Gradient boosting as a sequential chain of shrunken trees, each fit to correct the previous stage's residuals

In finance such models are widely used for default prediction, credit scoring, and fraud detection, and they can estimate which features drive a prediction. They are not automatically more reliable than simpler models: a tree-based model can still overfit, especially on noisy data or when cross-validation ignores the time structure, so validation along the lines of Section 6.4.1, with walk-forward schemes rather than random splits, remains essential.

Table 6.12 summarizes the two ensembling strategies, which are easily confused although they solve different problems.

	Bagging (e.g., random forest)	Boosting (e.g., gradient boosting)
Trees built	In parallel, independently	Sequentially, each correcting the last
Primarily reduces	Variance	Bias
Base learner	Deep, low-bias trees	Shallow, high-bias trees
Overfitting risk	Lower, more robust to noise	Higher, requires careful tuning
Typical use case	Robust baseline with minimal tuning	Squeezing out maximum predictive accuracy

Table 6.12: Bagging (random forests) versus boosting (gradient-boosted trees)

Neither approach dominates, but the difference in overfitting risk matters in finance: a gradient-boosted model with many trees and a small learning rate can reach very low training error, and the walk-forward validation of Chapter 7, rather than a random split, is what catches overfitting before it reaches production.

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6.6.2 Naive Bayes: A Probabilistic Classifier

Chapter 1 introduced Bayes' theorem with a fraud-detection example; naive Bayes turns it into a classification algorithm. Given features $x_1, \ldots, x_p$, it estimates the probability of each class k using Bayes' theorem,

$$\Pr(y = k \mid x_1, \ldots, x_p) \propto \Pr(y = k) \prod_{j=1}^{p} \Pr(x_j \mid y = k),$$

and predicts whichever class has the highest resulting probability. The "naive" assumption is that the features are conditionally independent given the class. It is rarely true (a firm's leverage and interest coverage are clearly related) but often good enough for a fast, effective baseline, particularly for text classification such as the sentiment analysis of Chapter 16, where the features are word counts and the false independence assumption distorts less than it would for tightly coupled financial ratios. Naive Bayes trains almost instantly even on large datasets, since fitting means estimating frequencies rather than solving an optimization problem, which makes it a useful baseline before investing in a more sophisticated model.

For text, where each feature x_j counts how often a word appears, $\Pr(x_j \mid y = k)$ is naturally estimated from training-data frequencies, the fraction of class k's total word count taken by word x_j. This raw estimate has a serious flaw: a word that never appears in class k's training documents gets probability zero, and because the naive Bayes score is a *product* across features, one zero forces the whole product to zero however strongly every other word favors that class, an artifact of a small sample rather than evidence that the class is impossible. Laplace smoothing (also called add-one smoothing) fixes this by adding a small constant to every count before normalizing,

$$\Pr(x_j \mid y = k) = \frac{\text{count}(x_j, k) + 1}{\text{count}(k) + V},$$

where $\text{count}(x_j, k)$ is how many times word x_j appeared across class k's training documents, $\text{count}(k)$ is the total word count across those same documents, and V is the vocabulary size, the number of distinct words being counted. Adding one to the numerator gives every word a positive probability even if it never appeared in that class, while adding V to the denominator keeps the probabilities summing to one across the vocabulary. Chapter 16's 10-K risk-factor classifier applies this correction.

6.6.3 Clustering and Dimensionality Reduction

Not every task predicts a known outcome; sometimes the goal is to find structure in a large set of assets or transactions. Suppose a portfolio manager has a few hundred holdings and no label saying which behave similarly, and wants an algorithm that groups them from their measured characteristics rather than by eye. K-means clustering partitions observations into K groups by minimizing the within-cluster sum of squared distances to each group's center,

$$\arg\min_{\{S_k\}} \sum_{k=1}^{K} \sum_{x_i \in S_k} \|x_i - \mu_k\|^2, \qquad \mu_k = \frac{1}{|S_k|} \sum_{x_i \in S_k} x_i,$$

where S_k denotes the set of observations assigned to cluster k and μ_k is that cluster's center, the mean of its assigned points. This groups similar observations together, for example firms with similar financial profiles or trading accounts with similar behavior. Principal component analysis (PCA) is the standard dimensionality-reduction method: it finds the direction w (with $\|w\| = 1$) that maximizes the variance of the projected data, $\arg\max_w \text{Var}(Xw)$, then finds the next such direction orthogonal to the first, and so on. In finance, PCA applied to a panel of asset returns often reveals that a small number of components, frequently interpretable as a broad market factor followed by sector or style factors, explain the large majority of the total variance.

These approaches help uncover latent structure in high-dimensional data. A portfolio manager may want to know whether returns are driven by a few common factors or many idiosyncratic sources, a question taken up in Chapter 12; a fraud analyst may want clusters of unusual behavior that deserve investigation. In both cases unsupervised learning is a useful first step before more targeted modeling.

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To make K-means concrete, consider four firms described by two features, leverage and profit margin (both scaled to lie between 0 and 1):

$$F_1 = (0.20, 0.30), \qquad F_2 = (0.30, 0.25),$$
$$F_3 = (0.80, 0.05), \qquad F_4 = (0.90, 0.10).$$

Initializing two cluster centers at F_1 and F_4, the algorithm alternates between two steps: assign each point to its nearest center, then recompute each center as the mean of its assigned points. In this example, F_1 and F_2 (both low leverage, higher margin) are closest to the first center, while F_3 and F_4 (both high leverage, low margin) are closest to the second, giving updated centers $\mu_1 = (0.25, 0.275)$ and $\mu_2 = (0.85, 0.075)$. Recomputing distances with these centers changes no assignment, so the algorithm has converged after one update: the data separate into a "low-leverage, higher-margin" group and a "high-leverage, low-margin" group, purely from the geometry of the two features. Figure 6.10 shows the four firms, the initial centers, and the updated centers μ_1 and μ_2 together.

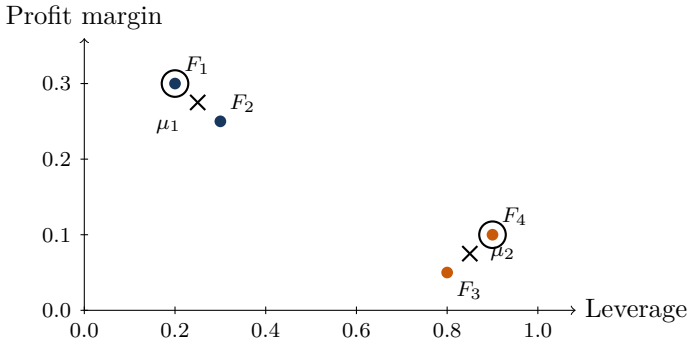


Figure 6.10: The K-means example: four firms in leverage/profit-margin space, with the initial centers (circled) and the updated centers μ_1 and μ_2

6.6.4 Neural Networks and Deep Learning in Finance

Suppose an analyst suspects that default risk depends on leverage and profitability through an interaction too intricate to write down in advance, and wants a model flexible enough to learn it from data rather than committing to a functional form as linear or logistic regression do. Neural networks are built for this kind of unspecified nonlinearity. A simple feedforward network with one hidden layer computes

$$\hat{y} = \sigma_2\big(W_2\, \sigma_1(W_1 x + b_1) + b_2\big),$$

where W_1, b_1, W_2, b_2 are learned weight matrices and bias vectors, and σ_1, σ_2 are nonlinear activation functions, such as the rectified linear unit $\mathrm{ReLU}(z) = \max(0, z)$ for hidden layers or, for a binary output layer, the sigmoid function

$$\sigma(z) = \frac{1}{1 + e^{-z}},$$

which maps any real number into the interval $(0, 1)$ so that the output can be read as a probability, and which is the same function logistic regression applied in Section 6.3.4 above. The subscripted σ_1 and σ_2 stand for whichever activation a given layer happens to use; a bare $\sigma(\cdot)$, here and in every later chapter, means this sigmoid specifically. Stacking more such layers produces a deep network. The weights are learned by minimizing a loss (mean squared error for regression, log-loss for classification) with gradient descent, the gradients computed by backpropagation, which applies the chain rule layer by layer from the output back to the input.

A small numerical example makes the forward pass (computing a prediction from an input) concrete. Suppose a network has two inputs, a hidden layer with two ReLU units, and a single sigmoid output, with

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weights

$$W_1 = \begin{pmatrix} 0.4 & -0.2 \\ 0.1 & 0.3 \end{pmatrix}, \quad b_1 = \begin{pmatrix} 0.05 \\ -0.1 \end{pmatrix}, \quad W_2 = \begin{pmatrix} 0.6 & -0.5 \end{pmatrix}, \quad b_2 = 0.1.$$

For an input $x = (1.0, 0.5)$, the hidden layer's pre-activation values are $W_1 x + b_1 = (0.35, 0.15)$, and since both are already positive, the ReLU activation leaves them unchanged: $h = (0.35, 0.15)$. The output layer's pre-activation is $W_2 h + b_2 = 0.6(0.35) - 0.5(0.15) + 0.1 = 0.235$, and applying the sigmoid function gives the final prediction $\hat{y} = 1/(1 + e^{-0.235}) \approx 0.558$.

A modern deep network uses its millions of parameters the same way, matrix multiplications and elementwise nonlinearities repeated layer after layer; its power comes not from any single computation but from learning weights across many layers that jointly capture nonlinear structure. Figure 6.11 lays out this network, with every edge and bias labeled with its weight and every neuron with the value it computes.

6.6.5 Worked Example 6.10: Backpropagation, One Layer at a Time

Section 6.3.3 introduced gradient descent for a single-layer linear regression, where the gradient of the loss with respect to each parameter could be written down directly. A multi-layer network needs the chain rule to propagate a gradient backward through each layer in turn, and the toy network above makes every step concrete. Suppose this network is being trained as a classifier and the true label for input $x = (1.0, 0.5)$ is $y = 1$ (a positive case, using the log-loss $L = -[y \ln \hat{y} + (1 - y) \ln(1 - \hat{y})]$ standard for classification). For this pairing of sigmoid output and log-loss, the gradient of the loss with respect to the output pre-activation z_2 collapses to $\hat{y} - y$:

$$\frac{\partial L}{\partial z_2} = \hat{y} - y = 0.5585 - 1 = -0.4415.$$

Backpropagation now works backward one layer at a time, applying the chain rule at each step. The output layer's weights and bias receive

$$\frac{\partial L}{\partial W_2} = \frac{\partial L}{\partial z_2} h^\top = -0.4415\,(0.35,\ 0.15) = (-0.1545,\ -0.0662), \qquad \frac{\partial L}{\partial b_2} = \frac{\partial L}{\partial z_2} = -0.4415.$$

To continue backward into the hidden layer, the gradient must first pass through W_2 to reach h, then through the ReLU activation to reach the hidden layer's pre-activations z_1:

$$\frac{\partial L}{\partial h} = \frac{\partial L}{\partial z_2} W_2 = -0.4415\,(0.6,\ -0.5) = (-0.2649,\ 0.2208), \qquad \frac{\partial L}{\partial z_1} = \frac{\partial L}{\partial h} \odot \mathrm{ReLU}'(z_1),$$

where $\mathrm{ReLU}'(z) = 1$ if $z > 0$ and 0 otherwise, and $\odot$ denotes elementwise multiplication; since both hidden pre-activations (0.35 and 0.15) are positive, $\mathrm{ReLU}'(z_1) = (1, 1)$ here and $\partial L / \partial z_1$ passes through unchanged. Finally, the first layer's weights and bias receive

$$\frac{\partial L}{\partial W_1} = \frac{\partial L}{\partial z_1} x^\top = \begin{pmatrix} -0.2649 \\ 0.2208 \end{pmatrix} (1.0,\ 0.5) = \begin{pmatrix} -0.2649 & -0.1325 \\ 0.2208 & 0.1104 \end{pmatrix}, \qquad \frac{\partial L}{\partial b_1} = \frac{\partial L}{\partial z_1} = (-0.2649,\ 0.2208).$$

All four gradients came from repeated application of the chain rule, moving backward from the loss through the output and hidden layers and reusing quantities already computed. That reuse is what makes backpropagation efficient enough for networks with millions of parameters; the alternative, perturbing each weight and re-running the forward pass, is computationally infeasible.

A gradient descent step with learning rate $\eta = 0.1$, the rule of Section 6.3.3, moves every parameter a small step opposite its gradient, $W_2 \leftarrow W_2 - \eta\, \partial L / \partial W_2$ and likewise for b_2, W_1, and b_1, nudging the prediction for this example toward $y = 1$. Repeating this update over many examples and iterations is how a deep network's weights are learned.

In finance, neural networks have been used for forecasting, option pricing, time series modeling, and text analysis. Deep learning can learn nonlinear representations from raw data, reducing the need for hand-crafted features, but it is no magic solution. Networks often need substantial data, careful tuning of architecture and learning rate, and strong regularization (such as dropout or weight decay, which is equivalent to the ridge penalty) to avoid overfitting. They are also harder to interpret than a linear model or a single tree, which matters when the model influences real, sometimes regulated, decisions. Deep learning is therefore most useful with domain knowledge and robust validation; a network that does well in the laboratory can still fail in production if the data environment shifts or the model is not monitored.

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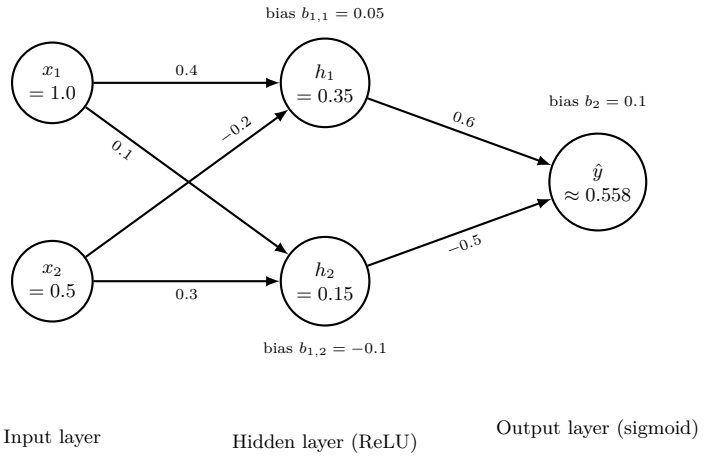


Figure 6.11: The toy feedforward network worked through above. Chapter 18 revisits it to verify LIME and integrated-gradients explanations against its exact analytical gradient.

6.7 Challenges of Machine Learning in Finance

Several challenges make financial applications of machine learning harder than many textbook demonstrations suggest.

Low signal-to-noise ratio. Financial returns are dominated by noise relative to any predictive signal, unlike many image or language tasks where the target is nearly deterministic given the input. An R^2 of a few percent can be economically meaningful for return prediction, whereas the same R^2 would be considered a failure in the clean regression example of Section 6.3.1.

Non-stationarity. The statistical relationships a model learns can change as markets evolve, as regulation changes, or as other participants adapt their own strategies. A model is trained on the past and asked to perform in a future that may not resemble it, a harder problem than the standard assumption that training and test data come from the same distribution.

Multiple testing and backtest overfitting. When researchers try many features, specifications, or trading rules and report only the best, the reported performance is biased upward, sometimes dramatically. This is the overfitting of Section 6.4.1 operating across a research process rather than a single fit, and one of the most persistent sources of disappointing live performance after a promising backtest.

Imbalanced outcomes. Events such as default, fraud, or market crashes are rare relative to normal outcomes. A classifier that always predicts "no event" can achieve high accuracy while being useless, which is why the precision, recall, and related metrics of Section 6.3.4 matter more than accuracy here. Remedies include resampling the training data (oversampling the rare class or undersampling the common one), assigning a higher misclassification cost to the rare class, and choosing the classification threshold from the relative costs of false positives and false negatives rather than defaulting to 0.5, which implicitly treats the two errors as equally costly.

Interpretability and regulatory scrutiny. Many financial applications, particularly credit decisions, are subject to regulation requiring that decisions be explainable to the affected individual and auditable by a regulator. This constrains the use of highly flexible models, such as a large ensemble or a deep neural network, or requires supplementing them with explanation methods, the subject of Chapter 18.

Feedback and reflexivity. Unlike physical systems, markets react to a successful model: if a profitable pattern becomes known and widely traded, the trading can erode or eliminate it. Past predictive power is therefore not a static property of the data; deploying the model can change it.

Hyperparameter and model-selection overfitting. Applied carelessly, Section 6.4.3's grid search is itself a source of overfitting: searching many hyperparameter combinations or model families and reporting only the best cross-validated result inflates performance like the multiple-testing problem above, one level removed.

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Computational cost at scale. Some of the most powerful techniques in this chapter, an exhaustive grid search, a large gradient-boosted ensemble, a deep neural network, are expensive to train and retrain. That matters because financial models often need refitting daily or even intraday to keep up with a non-stationary market, an engineering constraint on top of the statistical ones.

These challenges do not make machine learning inapplicable to finance; they make careful validation, economically grounded evaluation, and ongoing monitoring non-negotiable parts of any deployment.

6.7.1 Case Vignette: Zillow Offers and the Winner's Curse

Zillow Group, the U.S. real-estate listings company whose Zestimate had popularized automated home valuation, began buying homes directly in 2018. Through Zillow Offers it made cash offers, bought the homes whose owners accepted, renovated them, and resold them, so the pricing model's offer was the business. On November 2, 2021, Zillow's board decided to wind Zillow Offers down. In the third quarter the company had bought 9,680 homes and sold 3,032; it recorded a $304.4 million write-down of inventory, having bought homes "at higher prices than Zillow Group's current estimates of future selling prices," expected a further $240 million to $265 million of charges on homes it was under contract to buy, and announced a reduction of about 25% of its workforce. Its shareholder letter described the problem in forecasting terms: the business depended on forecasting home prices "accurately three to six months into the future," and "we have been unable to accurately forecast future home prices" (Zillow Group 2021).

Two mechanisms from this chapter show how a valuation model that is accurate on average can still lose money systematically. The first is the **winner's curse**, a form of the selection bias in Equation (6.11): a seller knows things about the home that the model does not and tends to accept the offers that are too high. In the companion notebook's simulation, a price model whose errors average zero, with a standard deviation of 5%, makes offers on 100,000 homes, and each seller accepts only an offer above the home's true value. Across all the homes it priced, the model's mean error is +0.16%, essentially zero, so it would pass any validation on the homes it was asked to price. On the 50.4% of homes it actually buys, it overpays by 4.13% on average, close to the small-error approximation $\sigma\sqrt{2/\pi} = 3.99\%$, and the overpayment grows in proportion to the model's error: 1.62% for a 2% error and 8.54% for a 10% error. The second mechanism is non-stationarity: a model built while prices rose steadily was asked to forecast months ahead through a turn in the market, the regime shift that in Section 6.4.5 cut the coverage of a nominal 90% interval to 61%.

Neither mechanism requires a badly built model, and the public record does not say how much each contributed at Zillow; the case shows where validation has to look. A model whose output decides which transactions happen, a pricing, lending, or underwriting model, should be evaluated on the transactions it wins, not only on the ones it scores, and its offers should be shaded for the adverse selection they attract, the same reasoning a market maker applies when quoting to better-informed traders in Chapter 14. Chapter 9's reject-inference problem is the lending version of the same selection effect.

6.8 Key Takeaways

Machine learning offers valuable tools for prediction, classification, and pattern discovery in finance, but its power must be balanced with caution. The regression and classification examples showed how ordinary least squares, logistic regression, the confusion matrix, and R^2 are computed, so that later chapters can build on precise foundations. The fitted logistic regression showed what its coefficients and probabilities mean, the AUC turned out to be a count of correctly ordered pairs, and the Brier score, Platt scaling, and isotonic regression showed that a model can rank well and still need its probabilities repaired. Overfitting, unstable relationships, and shifting regimes threaten performance in practice, and the bias-variance decomposition explains why. The most successful applications combine statistical learning with domain knowledge, careful validation, and a clear understanding of the financial decision being made.

The chapter covered a range of approaches: parametric models fit by a closed form (linear regression) or by gradient descent (logistic regression and neural networks), nonparametric models that reason from stored examples (KNN), margin-based classifiers (SVM), probabilistic classifiers built on Bayes' theorem (naive Bayes), and ensembles of trees combined in parallel to reduce variance (random forests) or sequentially to reduce bias (gradient boosting). None is universally best. The choice depends on the size and structure of the data, the importance of interpretability (Chapter 18), and the computational constraints of deployment,

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which is why Section 6.4.3's cross-validated approach to model and hyperparameter selection matters as much as the algorithms themselves.

Two further distinctions shape how these models should be used. A point forecast is incomplete without its uncertainty: split conformal prediction (Section 6.4.5) wraps any model in intervals with a finite-sample, distribution-free coverage guarantee, which holds only while tomorrow's data are exchangeable with the calibration data, as regime shifts are not. And a model that predicts well has learned associations, not causal effects: in Section 6.4.6's credit-limit example a confounder made a harmful intervention look beneficial, a distortion that only a randomized experiment, or an observational design whose assumptions hold, can undo.

6.9 Suggested Exercises

6.1 Using the data in Table 6.1, verify by hand that $\hat{\beta}_1 = 47$ and $\hat{\beta}_0 = 20$, and compute the fitted spread for a firm with debt/EBITDA of 4.5.

6.2 Add a sixth observation, debt/EBITDA of 7 and a credit spread of 500 bps, to Table 6.1, refit the regression, and discuss how much the slope and R^2 change. What does this reveal about the sensitivity of OLS to outliers?

6.3 Using Table 6.3, suppose the lender lowers its classification threshold so that applicant 5 is now also predicted to default. Recompute the confusion matrix and all five metrics, and discuss how precision and recall trade off against each other.

6.4 Explain the difference between supervised and unsupervised learning, giving one financial example of each that is not already used in the chapter.

6.5 Derive, in your own words, why a ridge penalty with a very large λ drives all coefficients toward zero, and explain what this implies about the bias and variance of the resulting model.

6.6 Describe the bias-variance trade-off using the formal decomposition in Section 6.4.2, and relate each of the three terms to a specific row of Table 6.6.

6.7 Why is walk-forward validation generally preferred over random k-fold cross-validation for financial time series?

6.8 Discuss one reason a model that performs well in-sample or in a backtest may fail out of sample or in live trading, drawing on at least two of the challenges described in Section 6.7 ("Challenges of Machine Learning in Finance").

6.9 Starting from $\beta_0 = \beta_1 = 0$, manually compute the first gradient descent update (Section 6.3.3) for the credit-spread data using a learning rate of $\eta = 0.02$ instead of 0.01, and compare the resulting (β_0, β_1) to the first iteration shown in Table 6.2.

6.10 Using Table 6.5, classify a new applicant with a debt-to-income ratio of 22% and 6 years of credit history using $k = 1$ and again using $k = 5$, and explain why the two predictions differ.

6.11 Explain, in your own words, why a wider margin in a support vector machine is generally associated with better generalization to new data, drawing an analogy to the bias-variance trade-off.

6.12 Choose one hyperparameter from any model in this chapter (for example, λ, k, or the number of boosting trees) and describe a grid of candidate values you would try, along with why searching too wide a grid could itself introduce overfitting risk, using Section 6.4.3 and Section 6.7's discussion of model-selection overfitting.

6.13 Explain why the naive independence assumption in naive Bayes is more clearly violated for the financial-ratio features used elsewhere in this chapter than for the word-count features used in text classification, and discuss whether you would still expect naive Bayes to be a reasonable baseline for a financial application.

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6.14 Using this chapter's maximum-margin example, suppose a fifth point is added at $x = 3.5$ belonging to the "low-risk" class. Compute its functional margin under the original hyperplane ($w = 1$, $b = -3$), explain why this point must become a new support vector, and recompute the resulting hyperplane and margin width.

6.15 Using this chapter's ridge-versus-lasso worked example, explain in your own words why lasso's exact zero for the spurious feature's coefficient is a direct consequence of the penalty term's shape (a kink at zero), rather than simply a coincidence of this particular dataset, and describe what would need to be true of ridge's penalty for it to ever produce an exactly zero coefficient.

6.16 Using this chapter's backpropagation worked example, recompute $\partial L/\partial z_2$, $\partial L/\partial W_2$, and $\partial L/\partial b_2$ if the true label were $y = 0$ instead of $y = 1$ (all forward-pass values unchanged), and explain in words why the sign of every downstream gradient flips as a result.

6.17 Using Table 6.8, construct the split conformal interval for the issuer at 4.5 turns of leverage at $\alpha = 0.20$ and at $\alpha = 0.05$, stating k, $\hat{q}$, and the exact coverage $k/(n+1)$ in each case. Then explain why a calibration set of 24 issuers cannot deliver a finite interval at $\alpha = 0.02$, and find the smallest calibration set that can.

6.18 The stress regime in Section 6.4.5 cut the coverage of a nominal 90% interval to 61.04%. Identify the assumption that failed, explain why recalibrating on the most recent observations helps once a new regime has persisted but cannot help on the first day of the shift, and describe how a risk manager could monitor realized coverage to detect the problem.

6.19 In Section 6.4.6's credit-limit example, suppose the bank's rule were more aggressive, $\Pr(D = 1 \mid Z) = 0.5 + 0.5Z$. Show that the population gap in average creditworthiness between treated and untreated customers becomes 2/3, and use Equation (6.12) to compute the population value of the naive estimate. Under the original rule, how large would the true effect τ have to be for the naive estimate to be exactly zero?

6.20 A card issuer wants to test whether proactive credit-limit increases raise twelve-month default. Using the standard-error formula from Section 6.4.6's randomized estimate, and assuming default rates near 8.5% in both arms, estimate how many customers per arm are needed for a standard error of 0.0025. Discuss two practical or regulatory constraints on running the experiment, and explain why a predictive default model trained on the issuer's historical data, however accurate, cannot answer the question.

6.21 Using the fitted logistic regression of the worked example on Table 6.3, $\log(\hat{p}/(1-\hat{p})) = -7.2885 + 0.2063 \times \text{DTI}$, compute the default probability of applicants with debt-to-income ratios of 30% and 42%, and find the ratio at which the fitted probability equals 20%. If the lender declines every applicant above that ratio, which of the ten applicants are declined, and what are the resulting precision and recall?

6.22 Suppose applicant 9's debt-to-income ratio in Table 6.3 were 34% instead of 31%. Without refitting the model, compute the new AUC by counting concordant pairs, and explain why no refit is needed for this calculation but would be needed to recompute the confusion matrix at the 0.5 threshold.

6.23 Using the worked example on splitting a decision tree, verify the weighted Gini impurity of the splits at $t = 42$ and $t = 20$, and explain why the split at $t = 42$, which produces a perfectly pure node, still loses to the split at $t = 28.5$. Then suggest the best second split of the node containing applicants 1 to 6 and 9 (those above 28.5%).

6.24 Standardize the two features of Table 6.5 as in the feature-scaling worked example and classify the Exercise 10 applicant (debt-to-income 22%, 6 years of credit history) with $k = 1$ and $k = 5$. Compare your answers with those of Exercise 10 in raw units, and explain which feature dominated the raw distances.

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6.25 A default model assigns 10% to 40 borrowers, of whom 3 default, and 50% to 10 borrowers, of whom 4 default. Compute its Brier score, then the Brier score after replacing each group's forecast by the group's observed default rate. Explain why a recalibration of this kind cannot raise the model's AUC.

6.26 A model predicts each stock's return over the next 20 trading days from daily features, and is evaluated with 5-fold cross-validation on contiguous blocks of 250 trading days. Explain which training observations must be purged around a validation block and why, how long an embargo you would set, and what fraction of the training data purging removes in each fold.

6.27 In the Zillow Offers vignette, the simulated overpayment on homes bought was 4.13% for a 5% model error. Using the small-error approximation $\sigma\sqrt{2/\pi}$, estimate the overpayment for a 10% error and compare it with the exact lognormal value of 8.51%. Explain why validating the model on all homes it priced would not reveal the problem, and propose a validation that would.

6.28 **Challenge (real data).** Download the UCI Bank Marketing dataset (a Portuguese bank's telemarketing campaign, roughly 45,000 client contacts, with features such as age, job, account balance, and prior campaign outcomes, and a binary label for whether the client subscribed to a term deposit). (a) Fit both a logistic regression and a k-nearest-neighbors classifier (trying at least three values of k) to predict subscription on a held-out test split, and compare their accuracy, precision, and recall. (b) Using Section 6.4.2's bias-variance framework, explain which of your k values appears to underfit and which appears to overfit, based on how training and test accuracy diverge. (c) This dataset's subscription rate is only around 11%, far from the roughly balanced toy examples used elsewhere in this chapter; discuss, referencing this chapter's overfitting and validation material, why a model that reports 89% accuracy on this dataset could still be practically useless to the bank's marketing team.

6.10 Further Reading

- Hastie et al. (2009), *The Elements of Statistical Learning*. The standard advanced reference for the statistical theory behind essentially every model in this chapter, including a much more rigorous treatment of the bias-variance decomposition and regularization.

- Murphy (2012), *Machine Learning*. A thorough, probability-first treatment that develops naive Bayes, logistic regression, and neural networks within a unified Bayesian framework, extending the Bayes' theorem introduction from Chapter 1.

- James et al. (2023), *An Introduction to Statistical Learning*. A more accessible companion to Hastie et al. (2009), with the same authors and worked examples in R and Python; a good starting point before attempting the more mathematically demanding treatment.

- Hull et al. (2025), *Machine Learning in Business*. A business- and finance-oriented introduction to many of this chapter's models, by the same Hull whose derivatives and risk-management texts are cited in Chapters 1, 3, 5, 8, and 9, with less formal derivation and more emphasis on practical business use cases than the theory-first references above.

- López de Prado (2018), *Advances in Financial Machine Learning*. Focuses specifically on the financial-data pitfalls emphasized throughout this chapter, backtest overfitting, walk-forward validation, and feature construction for financial time series, in far greater depth than a single chapter allows.

- Cristianini and Shawe-Taylor (2000), *An Introduction to Support Vector Machines and Other Kernel-based Learning Methods*. A focused treatment of the margin-maximization and kernel-trick ideas introduced only briefly in this chapter.

- Angelopoulos and Bates (2023), "Conformal Prediction: A Gentle Introduction." A short, practical introduction to conformal prediction, with code, covering the split method of Section 6.4.5, conformalized quantile regression for intervals that adapt to heteroskedasticity, and extensions to distribution shift.

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- Angrist and Pischke (2009), *Mostly Harmless Econometrics: An Empiricist's Companion*. The standard applied guide to estimating causal effects from data, covering the regression adjustment, instrumental-variables, and difference-in-differences designs pointed to in Section 6.4.6.

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Chapter 7

Time Series Analysis and Forecasting

7.1 Introduction

Almost every dataset encountered so far in this book, prices in Chapter 2, financial statement line items in Chapter 4, credit spreads in Chapter 5, has a time dimension: observations arrive in a sequence that usually carries information a model ignoring order would throw away.

This chapter makes time itself part of the model. It introduces the statistical vocabulary for describing dependence across time, stationarity, autocorrelation, and volatility clustering, and the classical models built on that vocabulary: autoregressive and moving-average models, ARIMA, GARCH, and state-space models.

Two worked examples run through the chapter: a short, hand-computable interest-rate-deviation series used to fit and forecast an autoregressive model, and the AssetA return series first introduced in Chapter 2, reused here to illustrate how GARCH captures volatility clustering. The chapter also introduces formal stationarity testing, seasonal models, and two ways of extending the analysis to several related series, cointegration and vector autoregression, and it keeps a promise made in Chapter 6: explaining why cross-validation must be modified for time series data.

The chapter follows the order in which an analyst works through a new series: establish whether it is stationary (Section 7.2.3) and how strongly it depends on its past (Section 7.2.4), then fit and forecast a model, with the forecast's uncertainty rather than a single number (Section 7.3.1). It then widens the lens to several series, cointegration (Section 7.4.1) for pairs that move together and the Kalman filter (Section 7.5.2) for a state that is only noisily observed, and closes with the walk-forward validation on which Chapters 8 and 13 depend.

7.2 Foundations of Time Series

Financial series violate the independence every earlier chapter assumed, and this part develops the vocabulary for saying how.

7.2.1 Why Time Series Analysis Matters in Finance

Financial data are overwhelmingly time series: daily prices, quarterly earnings, monthly inflation, tick-by-tick trades. Three features make them harder to work with than the typical machine learning dataset. First, observations are not independent: today's return is related to yesterday's, if only weakly, violating the independent-and-identically-distributed assumption behind many statistical procedures. Second, the variance of returns changes over time, with quiet periods following quiet ones and volatile periods following volatile ones, a phenomenon called volatility clustering. Third, the underlying process generating the data can change over time (Chapter 6's non-stationarity concern), so a model fit on one period may not describe the next.

Time series methods exist to handle these features rather than assume them away, and they underpin much of the rest of this book: the risk models in Chapter 8 need a volatility forecast, the portfolio methods

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in Chapter 12 need a covariance forecast, and the trading strategies in Chapter 13 need to know whether a signal is transient noise or a persistent pattern.

Both this chapter's models and the regression and classification models of Chapter 6 fit parameters by minimizing a loss, but a standard machine learning model assumes each training observation is drawn independently from the same distribution, the assumption a time-ordered dataset violates. A model that ignores the dependence, for example by shuffling a time series before splitting it into training and test sets, produces badly misleading performance estimates. Every technique in this chapter, from the correlogram to GARCH to walk-forward validation, exists to model, exploit, or correctly evaluate a model in the presence of this dependence.

7.2.2 Stationarity, White Noise, and Random Walks

Suppose an analyst fits a seemingly solid model relating an interest rate to its own past values, only to see it fall apart when new data arrive, not because the estimation was wrong but because the series never behaved consistently enough to model. This section makes precise the consistency a series needs for a fitted model to generalize. A time series $\{X_t\}$ is weakly stationary if its mean and variance do not change over time and the covariance between X_t and X_{t-k} depends only on the lag k, not on t itself:

$$\mathbb{E}[X_t] = \mu, \qquad \mathrm{Var}(X_t) = \sigma^2, \qquad \mathrm{Cov}(X_t, X_{t-k}) = \gamma_k \text{ for all } t. \tag{7.1}$$

Stationarity matters because most time series models, including every model in this chapter, assume that relationships estimated from the past will continue to hold; without it, there is no reason to expect a fitted model to generalize.

White noise is the simplest stationary process: a sequence ε_t with mean zero, constant variance σ^2, and zero covariance at every nonzero lag, $\mathrm{Cov}(\varepsilon_t, \varepsilon_{t-k}) = 0$ for $k \neq 0$. White noise is unpredictable by construction and serves as the building block, the "noise," in every model that follows. A stronger notion, strict stationarity, requires the entire joint distribution of any collection of observations, not just their mean and variance, to be unaffected by shifting them in time; the models in this chapter rely on weak stationarity, which is easier to verify and sufficient for the estimation methods used.

A random walk is $P_t = P_{t-1} + \varepsilon_t$, and it is not stationary: its variance grows with t (specifically $\mathrm{Var}(P_t) = t\sigma^2$ if P_0 is fixed), so its statistical behavior changes over time. Asset prices behave approximately like random walks, which is why Chapter 2 worked with returns, $R_t = (P_t - P_{t-1})/P_{t-1}$, rather than price levels: returns are much closer to stationary, since they do not inherit the growing variance of the price level. This matters: fitting a stationary-series model such as AR or ARMA to a non-stationary price level produces meaningless, unstable estimates, so the first step in almost any time series analysis is to check whether the series needs to be differenced (transformed to $\Delta X_t = X_t - X_{t-1}$) or converted to returns before modeling.

7.2.3 Testing for Stationarity: The Augmented Dickey-Fuller Test

Deciding whether a series is stationary should not rest on visual inspection alone; the standard formal test is the (augmented) Dickey-Fuller test. The test rewrites the AR(1) model

$$X_t = c + \phi X_{t-1} + \varepsilon_t$$

in first-difference form,

$$\Delta X_t = \alpha + \gamma X_{t-1} + \varepsilon_t, \qquad \gamma = \phi - 1, \tag{7.2}$$

and tests the null hypothesis $H_0 : \gamma = 0$ (equivalent to $\phi = 1$, a unit root, meaning the series is a non-stationary random walk) against the alternative $H_1 : \gamma < 0$ (meaning $\phi < 1$, so the series is stationary and mean-reverting). The "augmented" version adds lagged difference terms

$$\Delta X_{t-1}, \Delta X_{t-2}, \ldots$$

to the right-hand side to account for additional autocorrelation, but the core logic, testing whether γ is significantly negative, is unchanged.

This section and the two that follow use a single running example, Table 7.1: eight monthly observations of a short-term interest rate's deviation from its policy target, in basis points. The series is short enough that

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Month t	1	2	3	4	5	6	7	8
Deviation X_t	9.0	8.0	7.0	7.0	5.0	6.0	4.0	4.0

Table 7.1: A short interest-rate deviation series (basis points)

every calculation can be checked by hand; Section 7.2.4 computes its correlogram and Section 7.3.1 fits an AR(1) model to it.

Applying the (non-augmented) test to this series means regressing ΔX_t on X_{t-1}:

$$\hat{\gamma} = -0.290, \qquad SE(\hat{\gamma}) = 0.259, \qquad t\text{-statistic} = \frac{\hat{\gamma}}{SE(\hat{\gamma})} \approx -1.12.$$

Notice that $\hat{\gamma} = -0.290$ is consistent with the AR(1) coefficient fitted in Section 7.3.1, since

$$\hat{\phi} - 1 = 0.710 - 1 = -0.290$$

exactly. The Dickey-Fuller test does not use standard normal or t-distribution critical values, because under the null hypothesis of a unit root the test statistic has a nonstandard distribution; the relevant critical value at the 5% level (with a constant term and no trend) is approximately -2.89, not the familiar -1.96.

Since $|-1.12| < 2.89$, the test fails to reject a unit root, even though the AR(1) coefficient of 0.71 and the decaying correlogram in Figure 7.1 suggest mean reversion. The contradiction is a consequence of sample size, not a flaw in the test: with only 8 observations, the Dickey-Fuller test has little power to distinguish a stationary process from a unit root, the "limited effective sample size" challenge flagged later in this chapter. On the much longer series to which it is normally applied, its power is far greater.

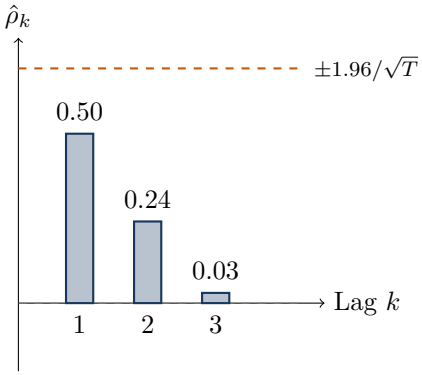


Figure 7.1: Sample correlogram for the interest-rate deviation series in Table 7.1, with the approximate 95% significance band $\pm 1.96/\sqrt{T}$ (dashed; symmetric about zero, so only its positive side is drawn)

7.2.4 Autocorrelation and the Correlogram

Suppose an analyst believes an interest-rate series is mean-reverting rather than a random walk and wants to know how far back its memory extends: does today's value mainly reflect yesterday's, or does a shock from several periods ago still leave a trace? The autocorrelation function (ACF) answers this by measuring how strongly a series is related to its own past values. At lag k, it is defined as

$$\rho_k = \frac{\gamma_k}{\gamma_0} = \frac{\mathrm{Cov}(X_t, X_{t-k})}{\mathrm{Var}(X_t)}, \tag{7.3}$$

and is estimated from a sample of T observations with mean $\bar{X}$ as

$$\hat{\rho}_k = \frac{\sum_{t=k+1}^{T}(X_t - \bar{X})(X_{t-k} - \bar{X})}{\sum_{t=1}^{T}(X_t - \bar{X})^2}.$$

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A plot of $\hat{\rho}_k$ against k is called a correlogram, one of the most common diagnostic tools in time series analysis: white noise has a correlogram indistinguishable from zero at every lag, while an autocorrelated series shows a pattern, decaying gradually for an autoregressive process and cutting off sharply for a pure moving-average process. Figure 7.1 shows the correlogram of the running example.

Because $\hat{\rho}_k$ is estimated from a finite sample, a common rule of thumb treats it as indistinguishable from zero if it falls within an approximate 95% band of $\pm 1.96/\sqrt{T}$, where T is the number of observations. For the 8-observation series this band is $\pm 1.96/\sqrt{8} \approx \pm 0.693$, larger than the sample autocorrelation $\hat{\rho}_1 = 0.497$ computed in the next section; Figure 7.1 plots the band with the correlogram.

Even the largest autocorrelation in the running example is therefore not statistically distinguishable from zero, purely because the sample is so small. This does not contradict the AR(1) pattern the correlogram shows; like the ADF test's failure to reject a unit root, it is a reminder that with 8 observations no time series diagnostic can be fully trusted, and that the qualitative pattern (geometric decay) is more informative here than any single significance test.

7.3 Forecasting Models

This part builds the classical forecasting families in order, from a single autoregressive term to the full ARIMA specification.

7.3.1 Worked Example 7.1: Fitting and Forecasting an AR(1) Model

This section works through the lifecycle of a small time series model: fitting it, checking that its correlogram behaves as theory predicts, and producing a point forecast whose uncertainty the following sections quantify. An autoregressive model of order 1, AR(1), expresses the current value as a linear function of the immediately preceding value plus noise:

$$X_t = c + \phi X_{t-1} + \varepsilon_t, \qquad \varepsilon_t \sim \text{white noise}(0, \sigma^2). \tag{7.4}$$

The process is stationary if and only if $|\phi| < 1$, in which case its theoretical autocorrelation decays geometrically, $\rho_k = \phi^k$, and its long-run (unconditional) mean is $c/(1-\phi)$.

The data are the running series introduced in Section 7.2.3, the eight monthly interest-rate deviations of Table 7.1.

The sample autocorrelations, computed directly from the formula above using the full-sample mean $\bar{X} = 6.25$, are $\hat{\rho}_1 = 0.50$, $\hat{\rho}_2 = 0.24$, and $\hat{\rho}_3 = 0.03$, plotted in Figure 7.1. The roughly geometric decay, each lag's autocorrelation a bit less than half the previous one, is the signature of an AR(1) process.

The AR(1) coefficients are estimated like any linear regression, by regressing X_t on X_{t-1} with the OLS formulas of Chapter 6:

$$\hat{\phi} = \frac{\sum_t (X_{t-1} - \bar{X}_{\text{lag}})(X_t - \bar{X}_{\text{cur}})}{\sum_t (X_{t-1} - \bar{X}_{\text{lag}})^2}, \qquad \hat{c} = \bar{X}_{\text{cur}} - \hat{\phi}\,\bar{X}_{\text{lag}},$$

where $\bar{X}_{\text{lag}}$ and $\bar{X}_{\text{cur}}$ are the means of the lagged and current values, respectively (each computed over the 7 available pairs, so they differ slightly from the full-sample mean $\bar{X} = 6.25$ used for the correlogram). Fitting this to Table 7.1 gives

$$\hat{\phi} = 0.71, \qquad \hat{c} = 1.19, \qquad R^2 = 0.60.$$

Since $|\hat{\phi}| = 0.71 < 1$, the fitted process is stationary, with an implied long-run mean of $\hat{c}/(1-\hat{\phi}) = 1.1935/0.2903 \approx 4.11$ basis points. This differs noticeably from the sample mean of 6.25: with 8 observations the AR(1) estimate carries substantial uncertainty, and the implied long-run mean is no more precise than the data behind it. The one-step-ahead forecast, given the final observation $X_8 = 4.0$, is

$$\hat{X}_9 = \hat{c} + \hat{\phi}\, X_8 = 1.19 + 0.71(4.0) \approx 4.03 \text{ basis points.}$$

Every autoregressive forecast works this way: the fitted relationship, applied to the most recent observation, produces the next point forecast.

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7.3.2 Confidence Intervals and p-Values for Model Parameters

A point estimate such as $\hat{\phi} = 0.71$ is half the story; without knowing how precisely $\hat{\phi}$ is estimated, one cannot say whether the mean reversion is a feature of the process or an artifact of a short, noisy sample.

Section 6.3.2 built the standard-error, t-statistic, p-value, and confidence-interval machinery, including how (and how not) to state what a confidence interval means. It applies unchanged, since $\hat{\phi}$ is an OLS slope of X_t on X_{t-1}. For the AR(1) regression fitted above, with $n = 7$ usable pairs and 2 estimated parameters (so $n - 2 = 5$ residual degrees of freedom),

$$SE(\hat{\phi}) = \sqrt{\frac{\hat{\sigma}^2}{\sum_t (X_{t-1} - \bar{X}_{\text{lag}})^2}} \approx 0.259, \qquad \hat{\sigma}^2 = \frac{\text{SSR}}{n-2} \approx 1.187, \qquad t = \frac{\hat{\phi}}{SE(\hat{\phi})} = \frac{0.71}{0.259} \approx 2.74,$$

where SSR is the sum of squared residuals from the fitted line. Under the null hypothesis $H_0 : \phi = 0$ (this month's deviation carries no information about next month's), this t-statistic follows a t-distribution with 5 degrees of freedom and gives $p \approx 0.041$: by the conventional 5% threshold, $\hat{\phi} = 0.71$ is barely significant, modest evidence of persistence. The corresponding 95% confidence interval,

$$\hat{\phi} \pm t^*_{0.975,\,5} \cdot SE(\hat{\phi}) = 0.71 \pm (2.571)(0.259) \approx (0.04,\ 1.38),$$

is wide, spanning nearly the whole range from very weak to very strong mean reversion and including values close to 1 (the unit-root boundary).

This is the small-sample fragility behind the Dickey-Fuller result of Section 7.2.3, which failed to reject a unit root despite $\hat{\phi} = 0.71$: a t-statistic, a p-value, and a confidence interval are three views of the same estimation uncertainty, which with 8 observations is large. Applying the identical method to the intercept gives $SE(\hat{c}) \approx 1.750$, $t \approx 0.68$, and $p \approx 0.526$, so the intercept is not statistically distinguishable from zero at conventional significance levels, a much weaker result than the slope's.

Section 6.3.2's two warnings apply unchanged: statistical significance is not economic significance, and testing many coefficients or models against the same data raises the chance that one crosses the 5% threshold by luck. The second is the multiple-testing problem that Section 7.4.1's discussion of pairs selection takes up: a p-value below 0.05 is far less persuasive when it is the best of twenty candidate pairs than when it is the only pair considered.

7.3.3 Multi-Step Forecasting and the Growth of Uncertainty

The one-step-ahead forecast above answers "what happens next period," but many financial decisions require forecasting several periods ahead. For an AR(1) model, this is done recursively: use the one-step forecast $\hat{X}_9$ as if it were the actual next observation to produce $\hat{X}_{10} = \hat{c} + \hat{\phi}\hat{X}_9$, and continue in the same way for as many steps as needed. Applying this to the fitted model ($\hat{\phi} = 0.71$, $\hat{c} = 1.19$) starting from $X_8 = 4.0$ gives the forecasts in Table 7.2.

Horizon h	1	2	3	5	10
Forecast $\hat{X}_{8+h}$	4.03	4.06	4.07	4.09	4.11

Table 7.2: Multi-step-ahead AR(1) forecasts from month 8

Two features of Table 7.2 illustrate general properties of stationary autoregressive forecasts. First, the forecasts converge toward the long-run mean of $\hat{c}/(1 - \hat{\phi}) \approx 4.11$ as the horizon grows; a stationary model's best guess about the distant future is its unconditional mean, since the information in today's deviation fades geometrically at rate $\hat{\phi}$ per period.

Second, the uncertainty around these forecasts also grows with the horizon but, unlike a random walk, does not grow without bound: the variance of the h-step forecast is proportional to $\sum_{j=0}^{h-1} \hat{\phi}^{2j}$, which converges to $1/(1 - \hat{\phi}^2) \approx 2.01$ times the one-step forecast variance as $h \to \infty$, rather than growing linearly with h as it would for a non-stationary series. This quantifies the warning in this chapter's challenges section that longer-horizon forecasts are less reliable, showing how much less reliable they are and why, for a mean-reverting series, the unreliability levels off.

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7.3.4 Diagnosing the Fit: The Ljung-Box Test and Choosing Between AR(1) and AR(2)

After fitting, the residuals should be checked for structure the model failed to capture, the diagnostic step of the Box-Jenkins methodology introduced later in this chapter. The Ljung-Box test formalizes this check: it tests the joint null hypothesis that the first h residual autocorrelations are all zero, meaning an adequately specified model should leave behind nothing but white noise, using the statistic

$$Q = n(n+2)\sum_{k=1}^{h}\frac{\hat{\rho}_k^2}{n-k}, \tag{7.5}$$

where $\hat{\rho}_k$ is the residuals' own sample autocorrelation at lag k (computed just as in Section 7.2.4, but applied to the residuals rather than the raw series) and n is the number of residuals. Under the null of a correctly specified model, Q follows a chi-squared distribution with $h-p-q$ degrees of freedom, where p and q are the numbers of AR and MA parameters estimated, subtracted because fitting them already absorbs some of the dependence.

Applying this to the AR(1) model fitted above ($\hat{\phi} = 0.71$, $\hat{c} = 1.19$), the seven residuals are

$$(0.42, 0.13, 0.84, -1.16, 1.26, -1.45, -0.03),$$

giving residual autocorrelations $\hat{\rho}_1 = -0.68$ and $\hat{\rho}_2 = 0.49$, both considerably larger in magnitude than a well-fitting model's residuals should show. With $h = 2$ lags and $p = 1$ AR parameter, $Q = 7.91$ against a chi-squared distribution with $2 - 1 = 1$ degree of freedom, giving $p \approx 0.005$, comfortably below the conventional 5% threshold. The AR(1) residuals are not white noise: some structure remains unmodeled.

Does a higher-order model capture more of it? Fitting an AR(2) model,

$$X_t = c + \phi_1 X_{t-1} + \phi_2 X_{t-2} + \varepsilon_t,$$

to the same six usable observations ($t = 3, \ldots, 8$; the sample must shrink by one further observation to accommodate the second lag, and the AR(1) model is refit on this identical six-observation sample so the two models' fit statistics are directly comparable) gives $\hat{\phi}_1 = 0.03$, $\hat{\phi}_2 = 0.88$, and a residual sum of squares of only 1.39, versus 5.60 for the AR(1) model refit on the same six observations, a dramatic improvement. Comparing the two models with the information criteria introduced in the ARIMA section below,

$$\mathrm{AIC} = n\ln(\mathrm{SSR}/n) + 2k, \qquad \mathrm{BIC} = n\ln(\mathrm{SSR}/n) + k\ln n, \tag{7.6}$$

with $n = 6$ throughout ($k = 2$ and $k = 3$ parameters for AR(1) and AR(2), respectively), gives

$$\begin{aligned} \mathrm{AIC}_{\mathrm{AR(1)}} &= 3.59, & \mathrm{AIC}_{\mathrm{AR(2)}} &= -2.76, \\ \mathrm{BIC}_{\mathrm{AR(1)}} &= 3.17, & \mathrm{BIC}_{\mathrm{AR(2)}} &= -3.39 : \end{aligned}$$

both criteria favor AR(2) by a wide margin, even after the extra parameter's complexity penalty. The conclusion deserves caution. An AR(2) model on six observations uses half the sample as estimated parameters ($k = 3$ against $n = 6$), the "limited effective sample size" and "model selection versus overfitting" concerns of this chapter's challenges section, so the much lower AIC/BIC is at least as likely to reflect a small sample fitting its own noise as a better-specified model. On a longer series the workflow, fit a candidate model, test whether its residuals are white noise, and if not compare a higher-order alternative by an information criterion, remains standard practice; only the tiny sample makes this particular conclusion (prefer AR(2)) less trustworthy than the procedure.

7.3.5 Moving Averages and Exponential Smoothing

Two much simpler forecasting tools remain widely used because of their transparency. A simple moving average forecast uses the average of the most recent m observations as the forecast for the next period,

$$\hat{X}_{t+1} = \frac{1}{m}\sum_{i=0}^{m-1} X_{t-i}.$$

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It is easy to compute and understand, but it weights all m past observations equally and drops old observations abruptly once they fall outside the window.

Exponential smoothing addresses both issues by weighting all past observations, but with exponentially decaying weight on older data:

$$\hat{X}_{t+1} = \alpha X_t + (1-\alpha)\hat{X}_t, \qquad 0 < \alpha < 1,$$

which can be expanded to show that $\hat{X}_{t+1}$ is a weighted average of every past observation, with weight $\alpha(1-\alpha)^i$ on the observation i periods back. A larger α reacts quickly to new information at the cost of more noise; a smaller α is smoother but slower. Applied to Table 7.1 with $\alpha = 0.4$ and an initial forecast $\hat{X}_1 = X_1 = 9.0$, the recursion tracks the series' decline without reacting sharply to any single observation, falling from 9.0 to a forecast for the next month of $\hat{X}_9 = 0.4(4.0) + 0.6(5.396) \approx 4.84$. The AR(1) model, by contrast, ties the forecast to the estimated mean-reversion coefficient $\hat{\phi}$ rather than only to recent history.

Because simple exponential smoothing forecasts with only a smoothed level, it always predicts a flat line, which performs poorly for a series with a clear trend, the kind of gradual decline visible in Table 7.1. Holt's linear trend method addresses this by smoothing two components separately, a level L_t and a trend T_t:

$$L_t = \alpha X_t + (1-\alpha)(L_{t-1} + T_{t-1}),$$
$$T_t = \beta(L_t - L_{t-1}) + (1-\beta)T_{t-1},$$

and forecasts h steps ahead as $\hat{X}_{t+h} = L_t + hT_t$, explicitly projecting the estimated trend forward rather than assuming it disappears. Applying this to Table 7.1 with $\alpha = 0.4$, $\beta = 0.3$, an initial level $L_1 = 9.0$, and an initial trend $T_1 = X_2 - X_1 = -1.0$, the recursion, updated once for each of X_2 through X_8, produces a final level of $L_8 \approx 3.631$ and trend of $T_8 \approx -0.706$, giving a one-step-ahead forecast of

$$\hat{X}_9 = L_8 + T_8 \approx 3.631 - 0.706 \approx 2.93.$$

This is well below both the simple exponential smoothing forecast (about 4.84) and the AR(1) forecast (4.03, Section 7.3.1), because Holt's method extrapolates the downward trend instead of assuming the series reverts toward a stable mean, as AR(1) does, or stays flat, as simple smoothing does. Which is more appropriate depends on whether the series is trending or is a mean-reverting process passing through a low patch, a judgment no formula resolves and the question the stationarity testing of Section 7.2.3 addresses: a trending (non-stationary) series suits Holt's method, a mean-reverting (stationary) one the AR(1) approach. A further extension, Holt-Winters smoothing, adds a third seasonal component to handle the kind of periodic patterns discussed in the SARIMA section below.

7.3.6 ARIMA Models

Suppose a shock, say a one-time data-reporting glitch, influences a series for exactly one more period and then vanishes, rather than decaying gradually forever as the AR(1) model implies. The AR(1) model of Section 7.3.1 belongs to a broader family that captures this kind of finite-memory dependence, among others. A moving-average model of order q, MA(q), expresses the current value as a function of recent noise terms rather than recent levels,

$$X_t = \mu + \varepsilon_t + \theta_1\varepsilon_{t-1} + \cdots + \theta_q\varepsilon_{t-q},$$

and combining autoregressive and moving-average terms gives an ARMA(p, q) model. Because many financial series (prices, but also many macroeconomic series) are non-stationary in levels but stationary after differencing, the ARIMA(p, d, q) model applies an ARMA(p, q) model to the series after differencing d times: $d = 1$ removes a linear trend, $d = 2$ removes a quadratic trend, and so on. In practice, d is rarely more than 1 or 2 for financial data, since returns (a $d = 1$ difference of log prices) are already close to stationary.

The MA(q) autocorrelation structure is the mirror image of the AR(p) one, and this contrast is what makes the ACF and PACF useful diagnostics together. For an MA(1) process $X_t = \mu + \varepsilon_t + \theta\varepsilon_{t-1}$, the theoretical autocorrelation is $\rho_1 = \theta/(1+\theta^2)$ and $\rho_k = 0$ for every lag $k > 1$; for $\theta = 0.6$, this gives $\rho_1 = 0.6/1.36 \approx 0.441$ and then an abrupt drop to zero at lag 2, in sharp contrast to the AR(1) correlogram in Figure 7.1, whose autocorrelations decay gradually across every lag rather than cutting off sharply. This is the basis of the Box-Jenkins identification rule: a correlogram (ACF) that cuts off sharply after lag q suggests an MA(q)

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model, while a partial autocorrelation function that cuts off sharply after lag p suggests an AR(p) model; a series where both decay gradually suggests a mixed ARMA(p, q) model, for which the specific orders are typically chosen by comparing information criteria across a small number of candidates rather than by reading the correlogram alone.

Choosing the orders p, d, and q is traditionally done with the Box-Jenkins methodology: examine the ACF and its partial-autocorrelation counterpart (the PACF, which measures the correlation between X_t and X_{t-k} after removing the influence of the intervening lags) to identify candidate orders, fit several candidate models, and compare them using an information criterion such as AIC or BIC, which penalize complexity so that the model with the most parameters is not chosen by default. This is the time series version of the model-selection problem of Chapter 6: more lags always fit the training data at least as well, but too many lags overfit sample noise as a high-degree polynomial does.

Many financial and economic series also exhibit seasonality, a repeating pattern at a fixed period, such as retail sales spiking every December or trading volume dipping every summer. SARIMA (Seasonal ARIMA) extends ARIMA by adding a second set of autoregressive, differencing, and moving-average terms operating at the seasonal period s, denoted SARIMA$(p, d, q)(P, D, Q)_s$: the non-seasonal part (p, d, q) captures short-run dynamics as before, and the seasonal part $(P, D, Q)_s$ captures dependence at lags $s, 2s, 3s, \ldots$. A monthly retail sales series might be modeled as SARIMA$(1, 1, 1)(1, 1, 1)_{12}$, the seasonal terms relating each month to the same month one year ($s = 12$ months) earlier on top of the month-to-month dynamics.

Seasonal differencing, $\Delta_s X_t = X_t - X_{t-s}$, is the seasonal analogue of the ordinary differencing introduced above, and it is applied when a series exhibits a seasonal unit root, an extension of the stationarity testing in Section 7.2.3 to the seasonal frequency. Most asset return series show little seasonality at the daily frequency, so this chapter's GARCH and ARIMA examples need no seasonal extension, but seasonality is common in macroeconomic series (retail sales, employment, seasonally-adjusted GDP) that feed into the fundamental analysis of Chapter 4 and the macroeconomic risk factors of Chapters 8 and 9.

7.3.7 Worked Example 7.2: The Same Diagnostics on a Real Treasury Series

Every test applied to the eight observations of Table 7.1 was inconclusive, which says more about the sample than about the tests. On a long real series the same tools do their job. The series is the month-end 10-year U.S. Treasury par yield from January 1990 to December 2023, $T = 408$ months, taken from the same Treasury file that supplies the yield-curve panel of Section 7.4.3; over the period it ranges from 0.55% to 9.04%. Table 7.3 collects the results for the yield and for its monthly change.

Statistic	Yield level	Monthly change
Dickey-Fuller t, no lags (p-value)	-2.158 (0.222)	-18.120 (< 0.0001)
Augmented Dickey-Fuller t, AIC lags (p-value)	-2.488 (0.118)	-11.281 (< 0.0001)
Autocorrelations, lags 1, 2, 3	0.986, 0.970, 0.955	0.099, -0.090, 0.033
Partial autocorrelations, lags 1, 2, 3	0.988, -0.088, 0.053	0.099, -0.101, 0.055

Table 7.3: Stationarity tests and correlograms for the month-end 10-year Treasury yield, January 1990 to December 2023 ($T = 408$; the 5% Dickey-Fuller critical value is -2.87 and the 95% autocorrelation band is ± 0.097)

Stationarity. Regressing ΔX_t on X_{t-1} exactly as in Section 7.2.3 gives a Dickey-Fuller statistic of -2.158 for the yield level, short of the 5% critical value of -2.87, and adding the three lagged differences that AIC selects moves it only to -2.488 ($p = 0.118$). The unit-root null is not rejected, just as it was not for the level factor in Section 7.4.3. This time the failure carries information, because the same test on the monthly changes rejects emphatically, at -18.12 without lags and -11.28 with the two lags AIC selects, far beyond any conventional critical value. With 400 observations rather than 8 the test has the power to tell the two cases apart, and its verdict is that the level behaves like a random walk while its changes are stationary: the diagnose-then-difference step of the workflow at the end of this chapter, carried out on real data.

Correlogram and identification. The 95% band is now $\pm 1.96/\sqrt{408} = \pm 0.097$, seven times narrower than the ± 0.693 of Section 7.2.4. The level's autocorrelations decay so slowly that they are still 0.910 at lag 6, the signature of a series at or near a unit root. Its partial autocorrelations, by contrast, cut off after the first

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lag: 0.988 at lag 1, then −0.088, 0.053, and −0.092 at lags 2 to 4, all inside the band. By the Box-Jenkins rule of the ARIMA section this identifies an AR(1). The monthly changes look much closer to white noise: their first autocorrelation, 0.099, and second partial autocorrelation, −0.101, sit just outside the band and everything else inside it, a little short-run structure that is real but too small to forecast with profitably.

The AR(1) fit. Fitting the AR(1) to the level by OLS, as in Section 7.3.1, gives $\hat{c} = 0.048$, $\hat{\phi} = 0.986$, and $SE(\hat{\phi}) = 0.0065$, with $R^2 = 0.983$. The standard error is forty times smaller than the 0.259 of the eight-point fit, and the naive 95% interval, $0.986 \pm 1.96 \times 0.0065 = (0.973,\ 0.999)$, appears to exclude a unit root. It does not, and this is the trap the Dickey-Fuller test exists to avoid: $(\hat{\phi} - 1)/SE(\hat{\phi}) = -2.158$ is the Dickey-Fuller statistic itself, and under a unit root it must be compared with −2.87, not with −1.96. Read as a stationary model, the fit implies a half-life of $\ln 0.5 / \ln 0.986 \approx 49$ months for a shock to the yield, another way of saying that 34 years of monthly data cannot distinguish very slow mean reversion from none. The high R^2 is equally uninformative, since it mostly records that this month's yield is close to last month's. The eight-point series failed every test for lack of data; the Treasury series passes or fails them for reasons the data can support.

7.4 Multivariate Methods

One series is rarely the whole story; this part models several jointly, and extracts the few factors that drive them.

7.4.1 Cointegration and Pairs Trading

Individual price series are typically non-stationary. A remarkable exception arises when two non-stationary series move together closely enough that a linear combination of them is stationary, a relationship called cointegration. Two stocks in the same industry might each wander like a random walk, yet the spread between them revert reliably to a stable relationship, because both are driven by the same industry fundamentals.

Table 7.4 shows six days of prices for two hypothetical bank stocks.

Day	1	2	3	4	5	6
Stock A	50	51	53	52	54	55
Stock B	48	49	50	50	51	52
Spread $(A - B)$	2	2	3	2	3	3

Table 7.4: Prices for two hypothetical cointegrated bank stocks

Both price series trend upward and would likely fail a stationarity test on their own. The spread, however, fluctuates narrowly around a mean of $\bar{s} = 2.5$ with a sample standard deviation of only 0.55, far more stable than either price. The standard formal test, the Engle-Granger two-step method, first regresses one price on the other to estimate the cointegrating relationship,

$$A_t = \alpha + \beta B_t + u_t,$$

which for this data gives $\hat{\alpha} = -12.5$ and $\hat{\beta} = 1.3$, and then applies the Dickey-Fuller test from Section 7.2.3 to the regression residuals $\hat{u}_t$: if the residuals reject the unit-root null (that is, they are stationary), the two series are cointegrated. One detail matters here: the test statistic must be compared with Engle-Granger critical values (tabulated by MacKinnon), not with the −2.89 used for a single observed series. Because $\hat{\beta}$ was chosen by least squares to make the residuals as small as possible, they look more stationary than an arbitrary series would, so the bar is higher: about −3.34 at the 5% level for two series in large samples, and more negative still in short ones.

The regression generalizes the simple $A_t - B_t$ spread (which assumes $\beta = 1$) to an estimated hedge ratio $\hat{\beta}$, the number of shares of B needed to offset one share of A; the simple difference is a reasonable approximation only when the two series move roughly one-for-one. Here $\hat{\beta} = 1.3$ is close enough to 1 for the raw spread in Table 7.4 to look stable, but, as the z-score below shows, not close enough to trade on.

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This relationship is the foundation of pairs trading, a strategy covered in more depth in Chapter 13. In practice the spread is standardized into a z-score, $z_t = (s_t - \bar{s})/\hat{\sigma}_s$, so that trading rules can be stated in standard deviations rather than raw units, which vary from pair to pair. Having estimated the hedge ratio, the spread to standardize is the residual $\hat{u}_t = A_t - \hat{\alpha} - \hat{\beta}B_t$, not the $\beta = 1$ difference $A_t - B_t$. For the data in Table 7.4 the residuals are $0.1, -0.2, 0.5, -0.5, 0.2, -0.1$, with mean zero (as least-squares residuals always have) and sample standard deviation 0.346, so the final residual has a z-score of $z_6 = (-0.1-0)/0.346 \approx -0.29$. (Standardizing the raw difference instead would give $(3 - 2.5)/0.548 \approx 0.91$, even the opposite sign, which shows how much the choice of hedge ratio matters.) A common rule of thumb opens a trade when $|z_t|$ exceeds roughly 2 and closes it once z_t reverts near zero, so this spread, less than a third of a standard deviation from equilibrium, would not trigger a trade.

When the spread is unusually wide relative to its historical mean and standard deviation, a trader shorts the relatively expensive stock (here, A) and buys $\hat{\beta}$ shares of the relatively cheap one (B) for every share of A, betting that the spread will revert toward its mean; when the spread is unusually narrow, the trader takes the opposite position. Because the strategy trades the spread rather than either stock's direction, it is designed to profit whether the market rises or falls, a property called market neutrality, provided the cointegrating relationship remains stable. That assumption can silently break down, a risk Chapter 13 explores for statistical arbitrage strategies.

Pairs selection is also where the multiple-testing problem flagged in Section 7.3.1 bites hardest. A trader rarely tests one pair; a more typical screen runs the Engle-Granger test on every pair in a universe of, say, 20 bank stocks, which is $\binom{20}{2} = 190$ pairs. If no pair were cointegrated, a 5% test would still flag about $190 \times 0.05 = 9.5$ of them by chance, and those false positives are the pairs whose spreads stop mean-reverting once money is on them. Two standard defenses are the Bonferroni correction, which requires each individual test to clear $0.05/190 \approx 0.00026$ so that the chance of even one false positive stays below 5%, and the less conservative false discovery rate procedure of Benjamini and Hochberg, which instead controls the expected fraction of flagged pairs that are false. Neither replaces an economic reason to expect the relationship (same industry, same funding model, same regulator) and a test on data the screen never saw.

7.4.2 Vector Autoregression: Modeling Several Series Jointly

Every model so far in this chapter describes a single series in terms of its own past. A vector autoregression (VAR) extends the AR(p) framework to several series simultaneously, allowing each variable to depend on its own past values and the past values of every other variable in the system. A VAR(1) model for two series X_t and Y_t is

$$X_t = c_1 + a_{11}X_{t-1} + a_{12}Y_{t-1} + \varepsilon_{1,t},$$
$$Y_t = c_2 + a_{21}X_{t-1} + a_{22}Y_{t-1} + \varepsilon_{2,t},$$

where the off-diagonal coefficients a_{12} and a_{21} capture cross-series dependence, for example how last month's interest rate might help predict this month's exchange rate, and vice versa. This models interconnected series such as interest rates, exchange rates, and inflation together, rather than with separate univariate ARIMA models that ignore their interdependence.

Conveniently, given the lag order, each equation of the VAR(1) system can be estimated separately by OLS, regressing X_t on X_{t-1} and Y_{t-1} and separately regressing Y_t on the same two variables. The number of lags (VAR(p) for $p > 1$) is chosen by AIC or BIC as for ARIMA orders, since a VAR with too many lags for the data overfits as readily as an over-parameterized univariate model.

VAR models are also the standard tool for testing Granger causality: variable X is said to "Granger-cause" variable Y if past values of X improve the prediction of Y beyond what past values of Y alone provide, formally tested by checking whether the coefficients on lagged X in the Y equation (here, a_{21}) are statistically significant.

Granger causality is a statement about predictive usefulness, not causality in the structural sense of Chapter 1's discussion of correlation versus causation; a variable can Granger-cause another simply because it reflects the same information sooner.

Once a VAR is fitted, an impulse response function traces how a one-time shock to one variable propagates through the system: a hypothetical one-unit shock is applied to $\varepsilon_{1,t}$ (holding $\varepsilon_{2,t} = 0$), and the model is iterated forward to see how $X_{t+1}, X_{t+2}, \ldots$ and $Y_{t+1}, Y_{t+2}, \ldots$ respond.

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This is a standard tool of macroeconomic policy analysis, tracing for example how a shock to short-term rates reaches exchange rates and eventually inflation, and the multivariate analogue of the AR(1) multi-step forecast.

A concrete example makes the mechanism tangible. Suppose a VAR(1) has already been fitted (via the equation-by-equation OLS procedure above) to changes in a short-term interest rate X_t and an exchange rate Y_t, with estimated coefficients $a_{11} = 0.6$, $a_{12} = 0.2$, $a_{21} = 0.1$, and $a_{22} = 0.7$ (intercepts omitted, since an impulse response tracks the deviation from baseline rather than the level). A one-unit shock to the interest rate at time 0, with the exchange rate unshocked ($X_0 = 1$, $Y_0 = 0$), propagates as

$$X_1 = 0.6(1) + 0.2(0) = 0.6, \qquad Y_1 = 0.1(1) + 0.7(0) = 0.1,$$

$$X_2 = 0.6(0.6) + 0.2(0.1) = 0.38, \qquad Y_2 = 0.1(0.6) + 0.7(0.1) = 0.13,$$

and continuing this recursion gives

$$X_3 = 0.254, \qquad Y_3 = 0.129, \qquad X_4 = 0.178, \qquad Y_4 = 0.116.$$

Two features stand out. First, the exchange rate, though not shocked, responds at once, rising from 0 to 0.1 through the cross-series coefficient $a_{21} = 0.1$, a spillover that separate univariate ARIMA models could never capture. Second, its response does not fade alongside the interest rate's: it rises further, from 0.1 to 0.13, before declining, because it keeps absorbing the still-elevated interest rate for a second period. This lagged peak is the signature of cross-series transmission rather than independent reversion.

Both series eventually decay toward zero, which the individual coefficients alone do not guarantee; it depends on the stability of the whole coefficient matrix

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

jointly. The multivariate generalization of the univariate AR(1) stationarity condition $|\phi| < 1$ is that every eigenvalue of A must lie strictly inside the unit circle. For this system the eigenvalues are 0.5 and 0.8, both below 1 in absolute value, so the impulse response must decay to zero rather than explode, as the values $0.6, 0.38, 0.254, 0.178$ for X and $0.1, 0.13, 0.129, 0.116$ for Y suggested.

A VAR fitted to non-stationary series (interest rate levels rather than changes, say) can easily have an eigenvalue at or above 1, and then any impulse response or multi-step forecast built from it, however precisely computed, describes an explosive process rather than an economic dynamic. This is the multivariate version of the unit-root problem the Dickey-Fuller test detects in a single series, and it is why a VAR, like an ARIMA model, is normally fit to differenced or otherwise stationary series.

7.4.3 Worked Example 7.3: PCA and Forecasting the Term Structure of Interest Rates

A VAR handles two or three series comfortably, but Chapter 5's yield curve is a whole curve, one yield for every maturity a government issues. Modeling eight or more maturities as one high-dimensional VAR is possible but wasteful, because adjacent points on the curve are very highly correlated: the 2-year and 3-year yields almost always move together.

Chapter 6 introduced principal component analysis (PCA), which finds the direction w maximizing $\mathrm{Var}(Xw)$. Applied to a panel of yield-curve maturities, it turns this redundancy into a forecasting advantage, since a few extracted factors carry nearly all of the curve's variation.

This worked example uses real data: the U.S. Treasury's daily par yield curve rates from January 2, 1990 through December 29, 2023, restricted to the eight maturities with essentially complete history over that period (3-month, 6-month, 1-year, 2-year, 3-year, 5-year, 7-year, and 10-year), giving 8,503 daily observations. Applying PCA to this panel, mean-centered but not rescaled (so that maturities with larger yield swings are not shrunk to match calmer ones), gives the explained variance ratios in Table 7.5.

The first three components explain 99.96% of the variation across 34 years of daily yield-curve moves: an eight-dimensional forecasting problem collapses, with almost no loss of information, to a three-dimensional one. This is one of the most reproduced results in fixed income, first documented by Litterman and Scheinkman

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Component	PC1	PC2	PC3	PC4	PC5
Variance explained	95.95%	3.78%	0.22%	0.03%	0.01%

Table 7.5: Explained variance ratio by principal component, 1990–2023 Treasury yield curve panel

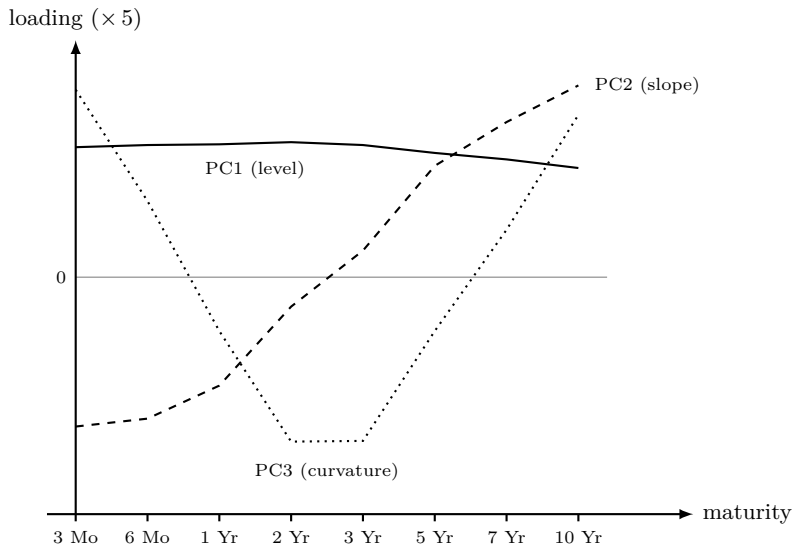


Figure 7.2: PCA loadings by maturity for the first three principal components of the 1990–2023 Treasury yield curve panel

(1991), and the components have well-known, interpretable shapes once their loadings, the weight of each maturity, are plotted against maturity.

Figure 7.2 shows this level-slope-curvature structure. PC1's loadings range only from 0.30 to 0.38 across all eight maturities, all positive and all close together: a positive PC1 shock raises every yield on the curve by roughly the same amount, a parallel shift, which is why PC1 is called the *level* factor. PC2's loadings move monotonically from -0.42 at the 3-month maturity to $+0.53$ at the 10-year maturity, so a positive PC2 shock lowers short-term yields while raising long-term ones, steepening the curve; this is the *slope* factor.

PC3's loadings are positive at both ends of the curve (0.52 at 3 months, 0.45 at 10 years) and negative in the middle (-0.46 at both 2 and 3 years), a hump-shaped pattern called the *curvature* factor, since a positive PC3 shock bends the belly of the curve down relative to both ends.

Level, slope, and curvature are themselves time series, obtained by projecting each day's curve onto the loading vectors, so they can be forecast with this chapter's univariate tools and mapped back to a curve-wide forecast through the loadings. Consider the dominant factor, PC1. Testing its level directly with the Augmented Dickey-Fuller test of Section 7.2.3 gives a test statistic of -2.295 ($p = 0.174$), failing to reject the unit-root null: like the stock prices of Section 7.4.1, the level factor is non-stationary and cannot be fed directly into an AR model. Differencing it once gives a test statistic of -16.474 ($p < 0.0001$), decisively stationary.

Fitting an AR(1) model to the differenced level factor ΔPC1_t by OLS gives

$$\Delta\text{PC1}_t = -0.0011 + 0.0411\,\Delta\text{PC1}_{t-1} + \varepsilon_t,$$

with the AR(1) coefficient statistically significant ($p = 0.0002$) thanks to the sample of 8,502 daily changes, but economically tiny: $R^2 = 0.0017$, so the previous day's move explains under one-fifth of one percent of the next. On the last trading day of the sample (December 29, 2023), the level factor stood at 3.063 after a daily change of -0.023; the fitted model forecasts the next change as $-0.0011 + 0.0411(-0.023) \approx -0.002$, an essentially negligible parallel shift. Mapped back through the PC1 loadings in Figure 7.2, this forecasts each

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of the eight maturities to move by only about -0.0007 to -0.0008 percentage points the next day, an order of magnitude smaller than a single basis point.

The result is informative, not disappointing. A significant but negligible R^2 is what an efficient, highly liquid market should produce: if daily changes in the Treasury curve's level were meaningfully predictable from their own history, large participants would trade the predictability away, the random-walk logic of Section 7.2.2 confirmed on 34 years of term-structure data. The payoff of PCA is not that it makes the level factor easy to predict; nothing makes daily interest-rate changes easy to predict. It replaces an unwieldy eight-maturity problem with a three-factor one, and whatever structure a more sophisticated model finds (a GARCH specification of Section 7.5.1 on ΔPC1_t's volatility, for instance) maps back to every maturity at once through the fixed loadings, instead of requiring eight separate models to stay mutually consistent.

7.5 Volatility and State-Space Models

Volatility is itself predictable in a way the level is not, and some of what matters is never observed directly.

7.5.1 Volatility Clustering and the GARCH Family

A defining stylized fact of financial returns is that their variance is not constant: large price moves tend to be followed by more large moves (of either sign), and calm periods tend to persist, a pattern called volatility clustering. This matters for the option pricing models of Chapter 5, which assumed a constant volatility σ; GARCH models let volatility evolve with the data instead.

The ARCH(1) model (autoregressive conditional heteroskedasticity) makes today's conditional variance a function of yesterday's squared shock,

$$\sigma_t^2 = \omega + \alpha\,\varepsilon_{t-1}^2, \qquad \omega > 0,\ \alpha \geq 0.$$

The GARCH(1,1) model, by far the most widely used volatility model in practice, adds a dependence on yesterday's own variance forecast,

$$\sigma_t^2 = \omega + \alpha\,\varepsilon_{t-1}^2 + \beta\,\sigma_{t-1}^2, \qquad \omega > 0,\ \alpha, \beta \geq 0,\ \alpha + \beta < 1. \tag{7.7}$$

The condition $\alpha + \beta < 1$ ensures the process is stationary, with a well-defined long-run variance $\omega/(1-\alpha-\beta)$ that the forecast reverts to over time; $\alpha + \beta$ close to 1, a common finding in financial data, means volatility shocks die out slowly, which is what volatility clustering looks like.

To make the recursion concrete, reuse the four AssetA returns from Chapter 2,

$$r = (0.0200, -0.0098, 0.0297, -0.0096),$$

whose sample variance was 0.000414. Take this as the seed σ_1^2, and suppose (for illustration, not estimated from only four data points) that $\omega = 0.00002$, $\alpha = 0.10$, and $\beta = 0.85$, so that $\alpha + \beta = 0.95$ and the long-run variance is $\omega/(1-0.95) = 0.0004$, a long-run volatility of 2.00%. The recursion needs shocks rather than returns: a GARCH model sits on a mean equation $r_t = \mu + \varepsilon_t$, so the shock is the return's deviation from its expected value, $\varepsilon_t = r_t - \mu$. Setting $\mu = 0$ lets the returns stand in for the shocks, the conventional treatment for daily data, where the average drift is two orders of magnitude smaller than a typical daily move and barely affects the fitted variance path. Applying the recursion with $\varepsilon_{t-1} = r_{t-1}$ on that basis gives Table 7.6.

The forecast volatility rises from 1.95% to 2.08% exactly after the large return of 2.97% at $t = 3$, and falls back toward the long-run 2.00% once smaller returns follow, the clustering GARCH is designed to capture. In practice ω, α, and β are estimated from a much longer history by maximum likelihood, but the recursion works identically once they are fitted.

The GARCH(1,1) model treats a positive and a negative return of the same magnitude as equally informative about future volatility, since the recursion depends only on r_{t-1}^2. This is a well-documented shortcoming: in equity markets especially, a large negative return is typically followed by higher volatility than an equally large positive one, an effect sometimes called the leverage effect.

Extensions such as EGARCH (exponential GARCH) and GJR-GARCH let the recursion respond to the sign of the previous shock as well as its size, at the cost of extra parameters. They are not developed here, but the symmetry built into GARCH(1,1) matters when reading its forecasts around downturns, where it is most likely to fail.

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t	Shock used, r_{t-1}	r_{t-1}^2	σ_t^2	σ_t (volatility)
1 (seed)	–	–	0.000414	2.03%
2	0.0200	0.00040	0.000412	2.03%
3	−0.0098	0.00010	0.000380	1.95%
4	0.0297	0.00088	0.000431	2.08%
5	−0.0096	0.00009	0.000396	1.99%

Table 7.6: GARCH(1,1) variance recursion using Chapter 2's AssetA returns

7.5.2 State-Space Models and the Kalman Filter

Suppose a fund re-estimates a stock's market beta from a rolling regression each quarter and finds the estimate bouncing around noticeably from one quarter to the next. Some of the bounce is estimation noise that would appear even if the true beta never changed, but some may be the true beta drifting as the business evolves, and a single quarter's estimate cannot tell the two apart.

The state-space representation models this kind of time-varying behavior by separating an unobserved ("latent") state that evolves over time from the observed data that carry noisy information about it:

$$\begin{aligned} \text{State equation:} \quad & \beta_t = \beta_{t-1} + \eta_t, \\ \text{Observation equation:} \quad & y_t = x_t^\top \beta_t + \varepsilon_t, \end{aligned}$$

where β_t is an unobserved parameter that is allowed to drift slowly over time (for example, a stock's market beta, or the level of an interest rate), and y_t is what is actually observed. The Kalman filter is the standard algorithm for estimating β_t recursively as observations arrive, updating a running estimate and its uncertainty at every step rather than re-estimating from scratch.

Each step has two parts: a prediction from the state equation alone, then an update that corrects it with the new observation, weighted by the relative reliability of prediction and observation. Concretely, given a current estimate $\hat{\beta}_{t-1}$ with variance P_{t-1}, a process (state) noise variance Q, and an observation noise variance R,

$$\begin{aligned} \text{Predict:} \quad & P_{t|t-1} = P_{t-1} + Q, \\ \text{Kalman gain:} \quad & K_t = \frac{P_{t|t-1}}{P_{t|t-1} + R}, \\ \text{Update:} \quad & \hat{\beta}_t = \hat{\beta}_{t-1} + K_t(y_t - \hat{\beta}_{t-1}), \qquad P_t = (1 - K_t)P_{t|t-1}. \end{aligned}$$

Suppose a fund's current estimate of a stock's market beta is $\hat{\beta}_0 = 1.0$ with variance $P_0 = 0.04$, the true beta is believed to drift slowly with process variance $Q = 0.001$ per period, and a single period's regression-based beta estimate is noisy with observation variance $R = 0.09$. If the next two periods' noisy beta estimates are $y_1 = 1.30$ and $y_2 = 1.25$, the filter updates as follows: at $t = 1$, $P_{1|0} = 0.041$, $K_1 = 0.041/0.131 \approx 0.313$, giving $\hat{\beta}_1 = 1.0 + 0.313(1.30 - 1.0) \approx 1.094$ and $P_1 \approx 0.028$; at $t = 2$, $K_2 \approx 0.245$, giving $\hat{\beta}_2 = 1.094 + 0.245(1.25 - 1.094) \approx 1.132$. The estimate moves toward each noisy observation without jumping all the way to it, as exponential smoothing does with a fixed weight α; the Kalman filter is an adaptive generalization of exponential smoothing whose weight K_t adjusts to the uncertainty in the current estimate relative to the noise in new observations. The gain shrinks across the two updates, from 0.313 to 0.245, as confidence in the running estimate grows, while the process noise Q reintroduces some uncertainty each step so that the filter can still adapt to real changes in beta.

Deriving the update equations is beyond the scope of this chapter, but the idea that parameters can be tracked as they evolve is directly useful in finance: a fund's exposure to a risk factor, or an asset's volatility (an alternative to GARCH known as stochastic volatility modeling), can both be framed this way.

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7.6 Validation and Accuracy

Time order makes the standard validation split invalid, so this part replaces it and asks what a forecast should be scored against.

7.6.1 Walk-Forward Validation for Time Series

Chapter 6 noted that k-fold cross-validation can mislead for time series, because a random split can train a model on data from after the validation period, information never available when a real forecast is made. Walk-forward validation keeps the training window entirely before the validation window and rolls both forward, as in Figure 7.3.

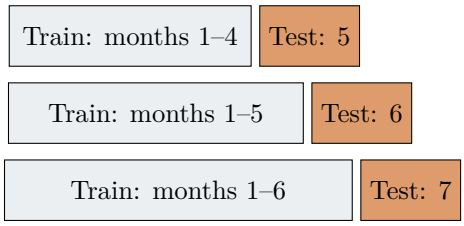


Figure 7.3: Walk-forward validation across three successive train/test splits

At each step, the model is refit (or updated) using only data available up to that point, produces a forecast for the next period, and that forecast is scored against the realized outcome once it arrives. The process rolls forward through the sample, and the reported performance is the average over these out-of-sample forecasts. It costs more than a single train-test split, since the model may be refit many times, but it is the only procedure that mimics how a forecasting model is used, one period at a time, without peeking ahead.

7.6.2 Evaluating Forecast Accuracy

The out-of-sample forecasts from walk-forward validation must be scored with a metric suited to the problem. Beyond the mean squared error used elsewhere in this book, two metrics are especially common in time series work. The root mean squared error (RMSE),

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(\hat{X}_t - X_t)^2},$$

returns the error to the original units of the series (unlike MSE, which is in squared units), making it directly interpretable, for example, as an average error of "2.1 basis points" rather than "4.4 squared basis points." The mean absolute percentage error (MAPE),

$$\text{MAPE} = \frac{100\%}{n}\sum_{t=1}^{n}\left|\frac{X_t - \hat{X}_t}{X_t}\right|,$$

expresses errors as a percentage of the actual value, which makes accuracy comparable across series of very different scales (dollar errors on a \$10 stock and a \$400 stock are not comparable, but percentage errors are). Its weakness is that it is undefined when $X_t = 0$ and unstable when X_t is near zero, a real concern for interest rates or spreads, which can pass through zero, unlike prices.

Beyond a single summary statistic, it is standard practice to compare a candidate model's walk-forward RMSE or MAPE against at least one naive benchmark, such as a forecast that simply repeats the last observed value ($\hat{X}_{t+1} = X_t$) or the simple moving average from Section 7.3.5. A sophisticated model that cannot beat such a benchmark walk-forward adds no predictive value, however compelling its in-sample fit.

A small worked comparison shows why this benchmark matters. Suppose a walk-forward evaluation produces five one-step-ahead forecasts for a price series with actual values $100, 102, 98, 105, 101$. A fitted model

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forecasts $99, 103, 97, 104, 102$, while the naive "repeat the last value" benchmark forecasts $98, 100, 102, 98, 105$ (each simply the prior period's actual value). Computing both metrics for each,

$$\text{RMSE}_{\text{model}} = 1.00, \qquad \text{RMSE}_{\text{naive}} = 4.22, \qquad \text{MAPE}_{\text{model}} = 0.99\%, \qquad \text{MAPE}_{\text{naive}} = 3.73\%,$$

the fitted model beats the naive benchmark by a wide margin, roughly a fourfold reduction in both metrics. That is encouraging, but five forecasts are far too few to show the model captures exploitable structure rather than luck. Whether one forecast's loss is significantly smaller than another's is what the Diebold-Mariano test formally assesses: it treats the period-by-period difference between the two models' losses as a series and tests whether its mean is zero, and with five such differences it has very little power. A long walk-forward history is what turns a gap like this into evidence. Had the two RMSE values been close, say 4.22 versus 4.05, the conclusion would be the opposite: the model's complexity buys almost nothing over repeating the last value, and a practitioner should ask whether it is worth the estimation risk and maintenance cost.

7.7 Challenges in Time Series Forecasting for Finance

Several challenges recur across almost every time series application in finance.

Regime changes and structural breaks. A model estimated over a calm period, including the AR(1) and GARCH parameters estimated above, can become badly miscalibrated once the underlying regime shifts, for example when a central bank changes policy or a market enters a crisis. A structural break is a sudden shift in the data-generating process itself, unlike the gradual change a stationary model allows. Formal tests detect breaks at a known or unknown date (the Chow and CUSUM tests are classical examples), but periodic re-estimation with walk-forward validation is usually a more robust defense than detecting every break in real time, since a break large enough to matter economically usually shows up as a sudden deterioration in walk-forward accuracy.

Limited effective sample size. Financial time series often look large (years of daily data) but contain far less independent information than the raw observation count suggests, because nearby observations are highly correlated. This limits how many parameters can be reliably estimated and is part of why simple, low-order models (AR(1), GARCH(1,1)) so often perform competitively against more elaborate alternatives.

Model selection versus overfitting. Adding autoregressive or moving-average terms always improves the in-sample fit. Information criteria such as AIC and BIC, with walk-forward validation, guard against a model that fits historical noise rather than a persistent pattern.

Fat tails and non-normality. Financial returns have more extreme observations than a normal distribution predicts. GARCH models are therefore often paired with a fatter-tailed error distribution (such as the Student's t), since a correctly time-varying variance does not by itself guarantee well-calibrated tail forecasts.

Multi-step forecasting compounds uncertainty. A one-step-ahead forecast, like this chapter's AR(1) example, is usually far more reliable than the same model's five- or twenty-step forecast, since uncertainty compounds with the horizon, and decisions resting on longer-horizon forecasts should treat them as considerably less certain.

Spurious cointegration and relationship breakdown. The pairs-trading strategy in Section 7.4.1 depends on a cointegrating relationship remaining stable, but with enough series to search, some pairs will appear cointegrated by chance, the multiple-testing problem again. Even an economically sound relationship can break down, for example if a merger, a change of strategy, or a regulatory shift changes one company's fundamentals relative to its peer.

Model misspecification in multivariate systems. A VAR model's conclusions, including any Granger-causality test, are sensitive to which variables are included and how many lags are used; omitting a variable that drives both series can produce a spurious Granger-causality finding, suggesting that one included variable predicts another when both are driven by the omitted one.

Classical models versus neural sequence models. Every model in this chapter, AR(1), ARIMA, GARCH, and VAR, is a classical statistical model with a small number of interpretable parameters, estimated by OLS or maximum likelihood. A recurrent neural network can in principle learn a far more flexible mapping from a series' history to its next value than any fixed linear, fixed-lag model, and Chapter 17 works through one, a hand-computed RNN forward pass. The flexibility is not free: such a model has many more parameters

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to fit from data that contain far less independent information than their observation count suggests, so it often costs more in overfitting than it returns in accuracy unless disciplined by walk-forward validation. The choice between a classical and a neural sequence model is therefore settled less by sophistication than by what a limited, noisy dataset can support without memorizing its own history.

The newest entrant in this debate sidesteps the limited-sample objection. **Time-series foundation models**, large neural sequence models pretrained once on enormous corpora of diverse time series (retail demand, energy load, traffic, weather, and much else) and then applied to a new series with little or no additional fitting, arrived in the mid-2020s under names like Chronos (Ansari et al. 2024) and TimesFM (Das et al. 2024), borrowing directly from the pretrain-then-adapt recipe Chapter 16 describes for language models. Because the parameter fitting happened on the pretraining corpus rather than the forecaster's short series, the overfitting arithmetic changes: the model brings pattern knowledge in, and on generic benchmarks it produces respectable zero-shot forecasts of unseen series. Nothing specific to finance has changed: published evaluations so far show these models competitive with, not dominant over, well-tuned classical baselines on financial series, whose low signal-to-noise ratio and regime shifts a mostly non-financial corpus does little to prepare for. Such a model is one more candidate to be judged by the same walk-forward, benchmark-against-naive discipline, not an exemption from it.

The companion notebook's optional final section applies that discipline to one such model, Chronos-Bolt-Small, a 48-million-parameter version of Chronos small enough to run on a laptop CPU, on the monthly 10-year yield of Section 7.3.7. For each month from January 2014 to December 2023 it forecasts the next month's yield zero-shot from all earlier months, the same walk-forward protocol used for the two benchmarks. Over the 120 forecasts its root mean squared error is 25.87 basis points, against 23.62 for a random walk and 23.64 for an AR(1) refit by OLS each month, and a Diebold-Mariano test rejects equal accuracy against both benchmarks ($p = 0.010$ and 0.014). Its 10%–90% forecast band covered 78.3% of the outcomes, close to the nominal 80%, so its uncertainty statements were sensible even though its point forecasts lost. One series and one decade settle nothing in general, but the outcome is the one the efficient-market reasoning of Section 7.4.3 predicts for a liquid rate: pattern knowledge imported from retail demand and electricity load does not beat a random walk on the 10-year Treasury yield.

7.8 A Practical Workflow for Building a Time Series Model

Figure 7.4 ties the chapter's tools together in the order they are typically applied.

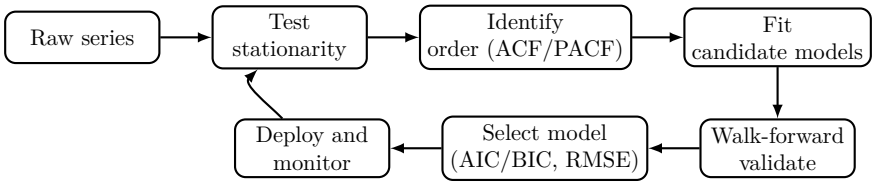


Figure 7.4: A practical time series modeling workflow

The first step is always to test stationarity (Section 7.2.3), since every model in this chapter, from AR(1) to GARCH to VAR, assumes stationarity in levels or after transformation; a non-stationary series is differenced (or converted to returns, as in Chapter 2) and re-tested.

Next, the ACF and PACF guide the choice of candidate orders, following the Box-Jenkins logic of the ARIMA section. Several candidates are fit and compared by information criteria (in-sample) and, more importantly, walk-forward validation (out-of-sample), scored by RMSE or MAPE and benchmarked against a naive forecast.

Only a model with walk-forward skill is deployed, and monitoring continues: the feedback arrow to the stationarity test reflects regime changes and structural breaks, which may require a model to be re-identified, not just re-fit, as data arrive.

The workflow applies to the single-series models (AR, ARIMA, GARCH) and the multi-series extensions (cointegration, VAR) alike; the multivariate case checks stationarity and lag orders for a whole system jointly.

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7.9 Key Takeaways

Time series methods extend the statistical toolkit of Chapter 6 to data where order matters. Stationarity makes a fitted model meaningful for the future, and checking for it, formally with the Dickey-Fuller test or informally with the correlogram, is the first step in any analysis. The AR(1) example showed coefficients estimated by the OLS logic of any regression, and the GARCH example showed volatility itself being modeled and forecast, feeding the option pricing and risk models of later chapters.

The cointegration and VAR material extended these ideas from a single series to relationships between series, previewing the pairs-trading strategies of Chapter 13 and the multivariate risk factor models of Chapter 8.

Walk-forward validation, scored by RMSE or MAPE and benchmarked against a naive forecast, follows from a principle repeated throughout this book: never evaluate a model with information it would not have had in real time, and never trust it merely for beating a strawman.

On the 8-observation running example, the Dickey-Fuller test and the ACF confidence band showed that formal significance is much harder to achieve in small samples than a chart or a plausible coefficient suggests, and that gap should make an analyst more cautious about small-sample conclusions. The same tests run on 408 months of the 10-year Treasury yield in Section 7.3.7 gave clear answers: a unit root in the level that could not be rejected, changes that were decisively stationary, and a partial autocorrelation function that pointed straight to an AR(1).

The AR(1), exponential smoothing, and Holt forecasts of the same series showed that reasonable modeling choices (mean reversion versus trend extrapolation) can give meaningfully different forecasts from identical data, and that choosing between them takes economic judgment, not just the best fit to the past.

Finally, the move from single series to relationships between series mirrors a pattern that recurs in this book: understanding one object in isolation is a prerequisite for, not a substitute for, understanding how it interacts with others, whether two cointegrated stocks here or the many assets of Chapter 12's portfolios.

7.10 Suggested Exercises

The exercises span the chapter's tools, from fitting and forecasting an AR(1) series to stationarity testing, comparing forecasting methods, and relationships between series. Several build on the worked examples, so your answers can be checked against the numbers in the text.

7.1 Explain, in your own words, why fitting an AR(1) model directly to a price series rather than a return series is likely to produce an unreliable model.

7.2 Using the data in Table 7.1, verify by hand that the sample mean is $\bar{X} = 6.25$ and that $\hat{\rho}_1 = 0.50$.

7.3 Add a ninth observation, $X_9 = 3.0$, to Table 7.1, refit the AR(1) model, and compare the new $\hat{\phi}$ and one-step-ahead forecast $\hat{X}_{10}$ to the values computed in the chapter.

7.4 Using the GARCH(1,1) recursion in Table 7.6, compute σ_6^2 if a fifth AssetA return of $r_5 = 0.035$ were observed.

7.5 Explain why $\alpha + \beta < 1$ is required for a GARCH(1,1) process to be stationary, and describe what would happen to the variance forecast if $\alpha + \beta = 1$ exactly.

7.6 Describe a concrete scenario in which k-fold cross-validation, as used in Chapter 6, would give a misleadingly optimistic evaluation of a time series forecasting model.

7.7 Discuss one reason a GARCH(1,1) model fit on five years of daily returns during a calm market might forecast volatility poorly at the onset of a financial crisis.

7.8 Using the Dickey-Fuller regression in Section 7.2.3, explain why $\hat{\gamma} = \hat{\phi} - 1$, and verify this relationship numerically using the AR(1) coefficient computed in Section 7.3.1.

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7.9 Explain why the Dickey-Fuller test failed to reject the unit-root null hypothesis in Section 7.2.3 even though the fitted AR(1) coefficient clearly indicated mean reversion, and describe what you would expect to happen to the test's conclusion if the same series had 200 observations instead of 8.

7.10 Using Table 7.4, compute the spread on a hypothetical Day 7 where Stock A trades at 56 and Stock B trades at 51, and explain, using the mean and standard deviation of the historical spread, whether a pairs trader would be more likely to open a new position or close an existing one on that day.

7.11 Explain, in your own words, the difference between Granger causality and true causal influence, and describe one financial scenario in which two series could Granger-cause each other without either one truly driving the other.

7.12 A retailer's monthly sales data shows a strong, repeating spike every December. Explain why an ordinary ARIMA model without a seasonal component would likely underperform a SARIMA model on this series, and describe what seasonal period s you would use.

7.13 Using the multi-step forecasting formulas in Section 7.3.3, compute the 4-step-ahead AR(1) forecast $\hat{X}_{12}$ from $X_8 = 4.0$, and verify it lies between the 3-step and 5-step forecasts already reported in Table 7.2.

7.14 Using Holt's linear trend method from Section 7.3.5 with $\alpha = 0.5$ and $\beta = 0.2$ (instead of $\alpha = 0.4$, $\beta = 0.3$), recompute the level and trend after processing $X_2 = 8.0$ through $X_4 = 7.0$ (three updates starting from $L_1 = 9.0$, $T_1 = -1.0$), and compare your intermediate values to those reported in the chapter.

7.15 Using the Kalman filter recursion in Section 7.5.2, compute a third update step assuming a new observation $y_3 = 1.20$ arrives, continuing from $\hat{\beta}_2 \approx 1.132$ and $P_2 \approx 0.022$ with the same $Q = 0.001$ and $R = 0.09$.

7.16 Explain, using the workflow in Figure 7.4, what step or steps you would need to repeat if a monitoring system detected that a deployed time series model's walk-forward forecast errors had grown substantially larger over the past quarter.

7.17 Using the confidence-band rule in Section 7.2.4 ($\pm 1.96/\sqrt{T}$), compute how many observations T would be needed for a sample autocorrelation of $\hat{\rho}_1 = 0.50$ to be considered statistically significant at the 95% level, and comment on how this compares to the 8 observations actually available in Table 7.1.

7.18 Using this chapter's VAR(1) impulse response example, compute X_1 and Y_1 following a one-unit shock to the exchange rate instead of the interest rate (that is, $X_0 = 0$, $Y_0 = 1$), and explain, in terms of the individual coefficients a_{12} and a_{21}, why the interest rate's one-period response to an exchange-rate shock turns out to be larger than the exchange rate's one-period response to an interest-rate shock was.

7.19 Using this chapter's RMSE/MAPE comparison, recompute both metrics for the naive benchmark if its forecasts were instead $99, 101, 99, 104, 100$ (closer to the actual values than the original naive forecasts), and discuss whether the fitted model would still show a clear advantage over this improved benchmark.

7.20 Using Section 7.3.4's Ljung-Box calculation, explain in your own words why a large Q statistic (and correspondingly small p-value) is evidence that a model's residuals still contain autocorrelation, and why this counts against the model even though the model's own R^2 or fitted coefficients might look reasonable on their own.

7.21 Section 7.3.4 found that AIC and BIC both favor the AR(2) model over the AR(1) model, yet also cautioned that this conclusion may simply reflect overfitting on only six observations. Describe a concrete piece of additional evidence (not just a lower in-sample AIC or BIC) that would make you more confident the AR(2) model is genuinely better specified, not just better fitting.

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7.22 Using the Treasury AR(1) fit of Section 7.3.7 ($\hat{c} = 0.048$, $\hat{\phi} = 0.986$), compute the implied long-run mean $\hat{c}/(1-\hat{\phi})$ and the 12-month-ahead forecast from the December 2023 yield of 3.88%, and state what fraction of the gap between the current yield and the long-run mean the forecast expects to close in a year. Explain why the Dickey-Fuller results in Table 7.3 should make you cautious about both numbers.

7.23 The pretrained forecaster in this chapter's challenges section had a higher RMSE than a random walk, and a Diebold-Mariano test rejected equal accuracy at $p = 0.010$. Explain what the test compares, why one-step-ahead forecasts allow the simple t-statistic on the loss differences without a correction for autocorrelation, and describe the evidence (series, horizons, periods) you would want before adopting such a model for a fixed-income desk.

7.24 **Challenge (real data).** Download at least five years of daily closing prices for a real, liquid stock (for example, via the `yfinance` library, as Chapter 2 and Chapter 5 already do) and compute its daily log returns. (a) Test the return series for a unit root using the augmented Dickey-Fuller test, as in Section 7.2.3, and contrast the statistical power of this test on your (likely thousands of) real observations with this chapter's own 8-observation toy example, where the test failed to reject a unit root despite a clearly mean-reverting fitted coefficient. (b) Fit a GARCH(1,1) model to the return series and plot the fitted conditional volatility alongside the raw returns, identifying at least one real historical event (an earnings surprise, a market-wide shock) that produced a visible volatility spike. (c) Compare your fitted $\alpha + \beta$ to the stationarity condition discussed in this chapter, and comment on whether real daily equity returns typically sit close to the $\alpha + \beta = 1$ boundary this chapter warns is a sign of near-nonstationary volatility.

7.11 Further Reading

- Tsay (2010), *Analysis of Financial Time Series*. Written specifically for financial applications, with extensive coverage of ARIMA, GARCH, and multivariate time series methods including cointegration and VAR models.
- Hamilton (1994), *Time Series Analysis*. The classic, mathematically rigorous graduate-level reference for the state-space and Kalman filtering material introduced only briefly in this chapter, alongside a thorough treatment of stationarity and unit-root testing.
- Engle (1982), "Autoregressive Conditional Heteroscedasticity with Estimates of the Variance of United Kingdom Inflation." The original paper introducing ARCH models, for readers interested in the primary source behind the GARCH family developed in this chapter.
- Hyndman and Athanasopoulos (2021), *Forecasting*. A highly practical, freely available online text covering exponential smoothing, ARIMA, and forecast evaluation with an emphasis on applied forecasting workflows like the one in Figure 7.4.
- Engle and Granger (1987), "Co-integration and Error Correction: Representation, Estimation, and Testing." The original paper introducing the cointegration concept and the Engle-Granger two-step testing procedure used in Section 7.4.1.
- Sims (1980), "Macroeconomics and Reality." The influential paper introducing vector autoregression as an alternative to large-scale structural macroeconomic models, foundational to the VAR and Granger-causality material in this chapter.
- Ansari et al. (2024), "Chronos: Learning the Language of Time Series." The paper introducing one of the pretrained time-series foundation models discussed in this chapter's closing challenges, applying the language-model pretraining recipe to zero-shot forecasting.
- Litterman and Scheinkman (1991), "Common Factors Affecting Bond Returns." The original paper documenting the level, slope, and curvature factor structure of the yield curve reproduced in Section 7.4.3.

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Part II

Applications

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Chapter 8

Market Risk Management

8.1 Introduction

Every model in this book so far has focused on estimating a quantity of interest: a price, a return, a probability of default, a forecast. Risk management asks a complementary question: not "what do we expect to happen," but "how bad could it get, and how often." A portfolio manager who forecasts only expected returns is flying blind, since the same expected return can come from a smooth, low-volatility strategy or from one that is usually flat but occasionally loses everything. This chapter introduces the standard toolkit for quantifying and managing that downside for market risk: Value at Risk and Expected Shortfall, maximum drawdown, volatility forecasting, backtesting, liquidity risk, and the model governance and capital adequacy concepts that let a firm and its regulators judge whether it can survive its own worst days. Chapter 9 develops the parallel toolkit for credit risk (probability of default, loss given default, credit scoring), since the two risk types call for different tools.

One worked example runs through most of the chapter: a twenty-day return history for a $10 million equity portfolio, used to compute and compare parametric, historical, Monte Carlo, and GARCH-based risk measures, backtest them, and stress-test the resulting portfolio.

The chapter builds its toolkit in the order a risk desk deploys it: a taxonomy of risk types (Section 8.2.1) to set the scope; Value at Risk (Section 8.2.2), computed three ways and extended to a full portfolio and to a volatility-adjusted version using Chapter 7's GARCH forecasts; and backtesting (Section 8.3.1) to check whether any of it worked out of sample. Later sections extend the VaR machinery in three directions: to options positions via the Greeks (the delta-normal approximation, then the sensitivity-based limits of Section 8.4), to hypothetical crises the historical sample never contained (Section 8.5.1), and, in Section 8.5.3, to a direct test of whether a more flexible machine learning model can out-forecast GARCH.

8.2 Measuring Market Risk

This part builds the standard measures in order of how much they say about the tail, each worked on the same portfolio.

8.2.1 Types of Financial Risk

Suppose a risk manager reports that the fund "lost money because of risk." The statement is true but useless: it does not say whether prices moved against a position, a counterparty failed to pay, or a position could not be sold without moving its price, three different problems calling for three different responses. Financial risk is a family of related concerns, and confusing one type for another is a common and costly mistake. Market risk is the risk of loss from movements in prices, rates, or volatilities that a firm is exposed to through the positions it holds; it is the subject of this chapter. Credit risk is the risk that a counterparty, whether a bond issuer, a loan borrower, or a derivatives counterparty, fails to meet a contractual obligation; Chapter 5 already introduced the pricing consequence of credit risk, and Chapter 9 develops the risk-measurement side of the

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same problem. Liquidity risk comes in two forms: funding liquidity risk, that a firm cannot meet its cash obligations as they come due, and market liquidity risk, that a position cannot be sold quickly without moving its price against the seller, connected to the bid-ask spread and market depth of Chapter 3. Operational risk covers losses from failed internal processes, systems, human error, or external events, including the model risk that arises when a changing environment quietly violates a model's assumptions; Chapter 9 covers a standard capital treatment for it, and Chapter 10 develops the AI methods for detecting and managing it. Underwriting risk, that claims on policies a firm has written exceed the premiums collected, is the insurer's counterpart, and Chapter 11 develops it alongside the reserving and ratemaking machinery it requires. A single adverse event often touches several categories: a large, sudden market move (market risk) can trigger defaults among leveraged counterparties (credit risk) just when many firms are forced to sell into an illiquid market (liquidity risk), the contagion dynamic behind the crises of Chapter 3.

Table 8.1 lists each risk type with its standard measurement tool and where in this book it is developed.

Risk type	Standard measurement tool	Where developed
Market risk	Value at Risk, Expected Shortfall, maximum drawdown	Sections 8.2.2–8.3.1 of this chapter
Credit risk	Probability of default, loss given default, expected loss	Chapter 9
Liquidity risk	Bid-ask spread, liquidity-adjusted VaR	Chapter 3 and later in this chapter
Operational risk	Loss distribution approach, scenario analysis, incident-text and anomaly models	Chapter 10, with the regulatory capital charge in Chapter 9
Underwriting risk	Expected loss ratio, combined ratio, chain-ladder reserves	Chapter 11
Systemic risk	Stress testing, contagion modeling	Chapter 3's financial crises discussion, this chapter's stress-testing section, and Chapter 20's network contagion models

Table 8.1: A taxonomy of financial risk types and their standard measurement tools

8.2.2 Value at Risk: Parametric and Historical Methods

Value at Risk (VaR) answers a deliberately narrow question: over a given holding period and at a given confidence level, what is the loss that will not be exceeded except with some small, specified probability? Formally, the α-level VaR of a portfolio is the smallest loss L such that

$$\Pr(\text{Loss} > L) \leq 1 - \alpha.$$

A 1-day 95% VaR of \$1 million means that, under the model's assumptions, the portfolio is expected to lose more than \$1 million on only about 5% of trading days, or roughly one day in twenty.

Table 8.2 shows twenty days of returns for a hypothetical \$10 million equity portfolio managed by a firm we will call Meridian Capital Partners.

Day	1	2	3	4	5	6	7	8	9	10
Return	0.06%	0.42%	-0.27%	-1.01%	-0.49%	-1.13%	0.13%	1.67%	-0.53%	-0.68%
Day	11	12	13	14	15	16	17	18	19	20
Return	0.65%	0.49%	0.19%	-1.06%	0.02%	0.89%	-1.55%	-0.49%	-2.22%	-1.49%

Table 8.2: Twenty days of returns for a \$10 million equity portfolio

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The sample mean return is $\bar{r} = -0.32\%$ and the sample standard deviation is $\hat{\sigma} = 0.937\%$ (with $n-1 = 19$ in the denominator, as in every standard-deviation calculation in this book). The parametric (variance-covariance) method assumes returns are normally distributed and computes VaR directly from the mean, standard deviation, and the appropriate normal quantile z_α:

$$\text{VaR}_\alpha = -(\bar{r} - z_\alpha \hat{\sigma}) \times V, \tag{8.1}$$

where V is the portfolio's dollar value. Using $z_{0.95} = 1.645$ and $z_{0.99} = 2.326$ and $V = \$10{,}000{,}000$,

$$\text{VaR}_{95\%} = -(-0.0032 - 1.645(0.00937)) \times 10{,}000{,}000 \approx \$186{,}195,$$

$$\text{VaR}_{99\%} = -(-0.0032 - 2.326(0.00937)) \times 10{,}000{,}000 \approx \$250{,}081.$$

Here, as throughout the chapter, the dollar figures are computed at full precision, with $z_{0.95} = 1.6449$, $z_{0.99} = 2.3263$, and $\hat{\sigma} = 0.9374\%$; the rounded inputs displayed reproduce them to within 0.1%. The historical method makes no distributional assumption: it sorts the observed returns and reads the percentile from the data. Sorting the twenty returns in Table 8.2 from worst to best, the second-worst return (-2.22% and -1.55% are the two worst) corresponds to the 5th percentile of a 20-observation sample, giving

$$\text{VaR}_{95\%}^{\text{historical}} = -(-0.0155) \times 10{,}000{,}000 = \$155{,}000.$$

The two estimates disagree by more than \$30,000, illustrating a persistent tension in risk measurement. The parametric method borrows strength from the whole sample and gives a smooth estimate, but it assumes normality, a poor description of returns with heavier tails, the fat-tails caveat of Chapter 7's GARCH models. The historical method assumes nothing, but with twenty observations it depends heavily on a single data point (here the -1.55% return); in practice it uses one to several years of daily data.

8.2.3 Portfolio VaR with the Variance-Covariance Matrix

The single-series parametric VaR above is a special case of a matrix formula that extends to a portfolio of several assets. Recall the two-asset covariance matrix from Chapter 2's worked AssetA/AssetB example,

$$\Sigma = \begin{pmatrix} 0.000414 & -0.000198 \\ -0.000198 & 0.000118 \end{pmatrix},$$

computed there from the two assets' daily returns, and the equal weights $w = (0.5, 0.5)$ used throughout that chapter. The portfolio's return variance is $w^\top \Sigma w$, the formula Chapter 2 verified numerically, and the parametric VaR of a portfolio worth V dollars, assuming a zero mean return (common in practice, since a one-day expected return is negligible relative to one-day volatility), is

$$\text{VaR}_\alpha \approx z_\alpha \sqrt{w^\top \Sigma w}\, V. \tag{8.2}$$

Applying this to a hypothetical \$2 million position split evenly between AssetA and AssetB, $\sqrt{w^\top \Sigma w} \approx 0.583\%$ (matching the portfolio volatility computed in Chapter 2), so

$$\text{VaR}_{95\%} \approx 1.645(0.00583)(2{,}000{,}000) \approx \$19{,}182,$$

$$\text{VaR}_{99\%} \approx 2.326(0.00583)(2{,}000{,}000) \approx \$27{,}130.$$

This matrix formulation is what scales to real trading books with hundreds or thousands of positions: a risk system maintains the covariance matrix Σ across positions and recomputes $w^\top \Sigma w$ whenever positions change, the pattern Chapter 2 introduced with `weights @ cov_matrix.values @ weights`.

Component VaR: Decomposing Portfolio Risk by Position

A portfolio VaR answers "how much risk does the whole book have," but a risk manager allocating limits across desks, or a portfolio manager deciding which position to trim, needs to know how much each position

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contributes. Component VaR answers this by decomposing total portfolio variance into per-asset contributions with the same covariance matrix,

$$\text{Component Var}_i = w_i(\Sigma w)_i, \qquad \sum_i \text{Component Var}_i = w^\top \Sigma w, \tag{8.3}$$

so that each asset's fractional share of total portfolio variance, Component $\text{Var}_i/(w^\top \Sigma w)$, scales the total portfolio VaR into a per-asset Component VaR that sums exactly back to the whole. Applying this to Section 8.2.3's two-asset position, $\Sigma w = (0.000108, -0.00004)$, so

$$\begin{aligned}\text{Component Var}_A &= 0.5(0.000108) = 0.000054,\\ \text{Component Var}_B &= 0.5(-0.00004) = -0.00002,\end{aligned}$$

summing to the portfolio variance of 0.000034 exactly as required. Scaling these fractions (1.588 and -0.588) by the total $\text{VaR}_{95\%}$ of \$19,182 gives

$$\text{Component VaR}_A \approx \$30{,}466, \qquad \text{Component VaR}_B \approx -\$11{,}284,$$

which sum back to the \$19,182 total, as the decomposition guarantees. AssetB's component is *negative*, although its standalone volatility is positive. Its strongly negative correlation with AssetA (Section 8.5.1 computes $\rho \approx -0.896$) makes it a diversifier in this portfolio: adding more AssetB, holding AssetA fixed, would *reduce* total risk, which the standalone volatilities could never reveal. That is why trading firms set and monitor risk limits with component or marginal VaR rather than standalone size or volatility: a position's risk contribution depends on how it interacts with the rest of the book.

8.2.4 Expected Shortfall: Going Beyond VaR

VaR has a well-known weakness: it answers "how much can I lose with 95% confidence" but says nothing about how bad the remaining 5% of outcomes are. Two portfolios can have identical VaR while one has a slightly worse tail and the other a catastrophically worse one. Expected Shortfall (ES), also called Conditional VaR, averages the losses in the tail beyond VaR rather than reporting its boundary:

$$\text{ES}_\alpha = \mathbb{E}[\text{Loss} \mid \text{Loss} > \text{VaR}_\alpha]. \tag{8.4}$$

For a continuous loss distribution it makes no difference whether this inequality is strict ($>$) or not ($\geq$). In a finite sample it does, since the VaR is itself one of the observed losses, so a discrete convention is needed: this book counts the VaR observation itself as part of the tail, averaging every loss at or beyond the VaR (other texts average only the $n(1-\alpha)$ losses strictly beyond it, which with twenty observations at 95% would leave just the single worst day). For the historical method applied to Table 8.2, the two worst returns (-2.22% and -1.55%) are the ones at or beyond the 95% VaR threshold, so

$$\text{ES}_{95\%}^{\text{historical}} = -\frac{(-0.0222) + (-0.0155)}{2} \times 10{,}000{,}000 = \$188{,}500.$$

The parametric Expected Shortfall under normality has a closed-form expression involving the standard normal density $\phi(\cdot)$,

$$\text{ES}_\alpha = \left(\hat{\sigma}\frac{\phi(z_\alpha)}{1-\alpha} - \bar{r}\right) \times V, \tag{8.5}$$

which for $\alpha = 95\%$ gives $\text{ES}_{95\%} \approx \$225{,}366$, larger than the parametric $\text{VaR}_{95\%}$ of \$186,195, as it must be, since ES averages over the tail rather than reporting its edge. Because ES is a coherent risk measure in a precise sense that VaR is not (it satisfies subadditivity, so diversification can never increase measured risk, which VaR can violate in pathological cases), regulators have shifted market-risk capital from VaR toward Expected Shortfall, a change taken up in this chapter's discussion of regulatory capital. Figure 8.1 shows why the two measures differ: VaR marks only the boundary of the shaded loss tail, while ES averages over the entire tail beyond it.

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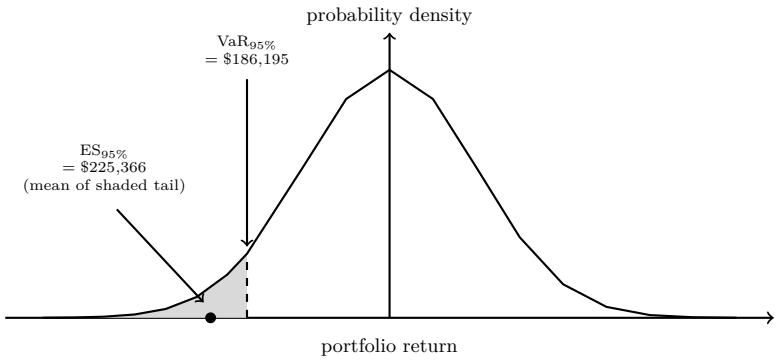


Figure 8.1: The parametric VaR and Expected Shortfall of Table 8.2's portfolio, illustrated against the assumed normal return distribution

8.2.5 Maximum Drawdown: A Path-Dependent Risk Measure

VaR and Expected Shortfall summarize the distribution of one-day losses, but neither says how much value an investor would have seen erode from its peak over an extended period. Maximum drawdown answers this. Given a portfolio value path $V_0, V_1, \ldots, V_T$, the drawdown at time t is the percentage decline from the running peak up to and including that point,

$$\text{Drawdown}_t = \frac{\max_{0 \le s \le t} V_s - V_t}{\max_{0 \le s \le t} V_s} \ge 0,$$

and the maximum drawdown is the largest drawdown along the path, $\text{MDD} = \max_{0 \le t \le T} \text{Drawdown}_t$. Unlike VaR and Expected Shortfall, which depend on the distribution of returns and not their order, maximum drawdown is path-dependent: shuffling the same twenty returns would generally change it while leaving every VaR and ES figure unchanged, because the peak and the trough depend on *when* the losses occur, not just how large they are.

Applying this to the cumulative value path implied by Table 8.2's twenty daily returns, starting from the portfolio's initial \$10,000,000, the running peak of \$10,048,025 is reached after Day 2, following two small positive returns, and is never exceeded again; the lowest point, \$9,371,210, comes at the end, Day 20, after four consecutive negative returns on Days 17 through 20. The maximum drawdown is therefore

$$\text{MDD} = \frac{10{,}048{,}025 - 9{,}371{,}210}{10{,}048{,}025} \approx 6.74\%, \qquad \text{a peak-to-trough decline of \$676,815.}$$

Figure 8.2 plots the path, its running peak, and the peak-to-trough decline. This differs materially from the cumulative twenty-day return of only -6.29%. The extra 0.45 percentage points reflect the portfolio's being briefly *up* \$48,025 before its losses began, so an investor holding from Day 2 onward saw a larger decline than a start-to-end comparison reveals. Maximum drawdown thus captures something daily VaR cannot: a portfolio whose daily losses all clear a well-calibrated 95% VaR threshold can still suffer a large, sustained drawdown if enough of them come consecutively, as in this example's final four days.

Maximum drawdown is used wherever a full return path is the object of interest; Chapter 13 uses it, alongside the Sharpe ratio and hit rate, to evaluate backtested strategies. Asset managers and hedge funds report it routinely because, unlike VaR, it needs no distributional assumption, only a value path, and it answers an investor's plain question: "how much could I have lost had I entered at the worst possible time?" Its main limitation mirrors that simplicity: a realized maximum drawdown is one observation of a path-dependent statistic, not an estimate of a distribution, so comparing two strategies' historical drawdowns says less about future worst-case risk than comparing their VaR or Expected Shortfall.

8.2.6 Monte Carlo Simulation for Risk Measurement

Monte Carlo simulation sits between the two: it assumes a distributional model, as the parametric method does, but instead of a closed-form formula it simulates many return scenarios from the model and reads

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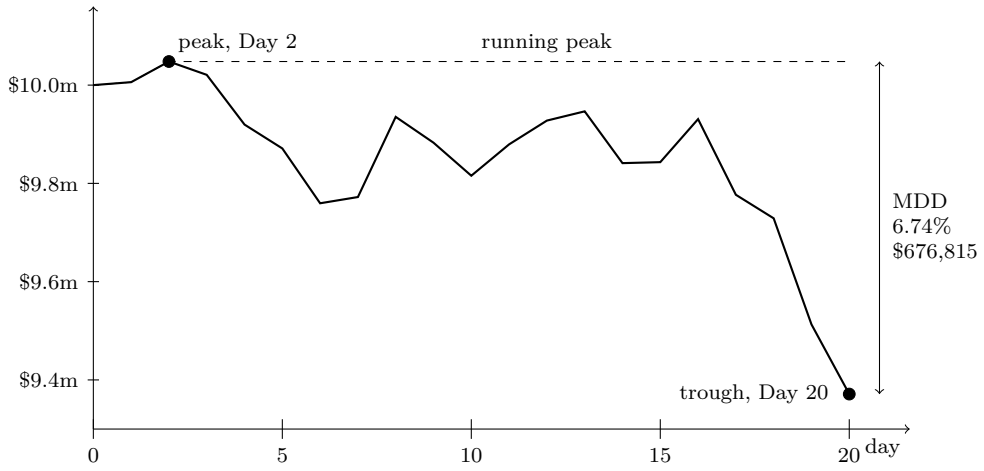


Figure 8.2: The value path of Table 8.2's portfolio, its running peak, and its maximum drawdown

VaR and ES off the simulated distribution, as the historical method reads them off the observed one. Simulating 100,000 draws from a normal distribution with the same mean and standard deviation as Table 8.2 ($\bar{r} = -0.32\%$, $\hat{\sigma} = 0.937\%$) gives $\text{VaR}_{95\%} \approx \$187{,}460$, $\text{VaR}_{99\%} \approx \$251{,}690$, and $\text{ES}_{95\%} \approx \$226{,}378$. All are close to the parametric formulas, as they should be: with enough draws, Monte Carlo VaR converges to the parametric answer whenever a closed form exists.

The method's value appears when no closed form is available, for example for a portfolio containing options (nonlinear in the underlying, a case taken up in the delta-normal discussion below) or a model with fat tails, jumps, or the time-varying volatility of Chapter 7's GARCH models. Monte Carlo remains simple there, simulating many paths under the more realistic model and reading off the empirical percentile, while a closed form either does not exist or is much harder to derive.

8.2.7 Volatility-Adjusted VaR: Connecting GARCH Forecasts to Risk Measurement

Every VaR method above uses a single volatility estimate, computed from a fixed historical window and held fixed. Because volatility clusters and evolves, Chapter 7's GARCH(1,1) model suggests an alternative: update VaR every day with the model's one-step-ahead variance forecast, an approach widely used in practice and sometimes called volatility-adjusted or "GARCH VaR."

Recall Chapter 7's GARCH(1,1) recursion applied to Chapter 2's four AssetA returns ($\omega = 0.00002$, $\alpha = 0.10$, $\beta = 0.85$): the one-step-ahead forecast volatility was 1.95% immediately before the large +2.97% return at $t = 3$, and rose to 2.08% immediately after it. Applying the same parametric VaR formula as Section 8.2.2, but with each of these two GARCH-forecast volatilities in place of a static historical standard deviation, to a hypothetical \$10 million position gives

$$\text{VaR}_{95\%}^{\text{pre-shock}} = 1.645(0.0195)(10{,}000{,}000) \approx \$320{,}775,$$
$$\text{VaR}_{95\%}^{\text{post-shock}} = 1.645(0.0208)(10{,}000{,}000) \approx \$342{,}160,$$

Full precision ($z_{0.95} = 1.64485$) gives \$320,746 and \$342,130, the values the notebook prints; the rounded z is shown for readability. an increase of about 6.7% in a single day from the updated variance forecast alone, with no change to the portfolio. A VaR based on a rolling historical standard deviation would still be averaging in the calmer pre-shock returns and would understate the risk while they remained in the window, the lag GARCH's recursive weighting avoids.

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Filtered Historical Simulation: Combining the Two Methods

The historical method and the GARCH-based method solve different problems well: the historical method captures the shape of the return distribution, its skew and fat tails, without assuming normality, while GARCH captures how far its width currently sits above or below its long-run level. Filtered historical simulation combines the two. Each historical return r_t is standardized by the GARCH volatility prevailing that day, $z_t = r_t/\sigma_t$, giving residuals roughly free of volatility clustering if the GARCH model is well specified; VaR is read off a percentile of these residuals and rescaled by *today's* GARCH forecast rather than the volatility prevailing when each return was observed.

With the two forecasts already computed, $\sigma_3 = 1.95\%$ (pre-shock) and $\sigma_4 = 2.08\%$ (post-shock), a large return observed in the calmer $t = 3$ regime, divided by that day's small σ_t, gives a large standardized residual, as it should, since the return was large *relative to its own regime*.

Chapter 2's AssetA series supplies such an observation. The +2.97% return arrived while $\sigma_3 = 1.95\%$ prevailed, the calmest forecast in the whole recursion, so its standardized residual is

$$z_3 = \frac{0.0297}{0.0195} = 1.523,$$

and rescaling that residual by the post-shock forecast $\sigma_4 = 2.08\%$ gives a hypothetical return for today's conditions of $1.523 \times 0.0208 = 3.17\%$. Plain historical simulation would have reused the 2.97% unchanged, understating the move by about twenty basis points only because the return was observed in a calmer regime than today's.

Rescaling standardized residuals by today's forecast in this way avoids both the plain historical method's problem (treating every past return as if it occurred under today's volatility) and the parametric method's (imposing normality on a shape the data may contradict). Filtered historical simulation is a practical middle ground used in production VaR systems at large trading desks, applying the same GARCH machinery across a whole return history.

8.2.8 Fat Tails: Student-*t* VaR and Extreme Value Theory

Filtered historical simulation lets the data's shape into the calculation; the more common fix replaces the normal distribution in the parametric formula with a heavier-tailed one. The standard choice is Student's t distribution with ν degrees of freedom, whose density decays like a power $|x|^{-(\nu+1)}$ rather than like the normal's $e^{-x^2/2}$, so that a five- or ten-standard-deviation day is rare but not, as under normality, effectively impossible. The smaller ν, the heavier the tails; as $\nu \to \infty$ the t becomes the normal.

The first question is what ν to use, and Table 8.2 cannot answer it. Maximizing the t likelihood over ν on the twenty returns sends the estimate off toward infinity, back to the normal, and the sample's excess kurtosis is -0.007, no fatter than a normal sample. The likelihood is also nearly flat: at $\nu = 4$ it is only 0.68 below its maximum, so the data are equally consistent with thin and with quite heavy tails. Twenty days contain almost no information about a tail that shows itself a few times a year. The tail parameter has to come from a long history or from what is known about the asset class, and estimates on long histories of daily equity returns typically fall between about 3 and 6 (McNeil et al. 2015). Take $\nu = 4$.

The scaling is where errors creep in. A standard t_ν variable has variance $\nu/(\nu-2)$, which is 2 at $\nu = 4$, so plugging its quantile into Equation (8.1) next to $\hat{\sigma}$ would double-count the spread. The t must first be rescaled to have the sample's standard deviation, with scale $s = \hat{\sigma}\sqrt{(\nu-2)/\nu} = 0.937\% \times \sqrt{1/2} = 0.663\%$. Then, with $t_{\nu,\alpha}$ the t quantile and f_ν its density,

$$\mathrm{VaR}_\alpha = -(\bar{r} - t_{\nu,\alpha}\, s) \times V, \qquad \mathrm{ES}_\alpha = \left(s\, \frac{f_\nu(t_{\nu,\alpha})}{1-\alpha}\, \frac{\nu + t_{\nu,\alpha}^2}{\nu - 1} - \bar{r} \right) \times V, \tag{8.6}$$

the t counterparts of Equations (8.1) and (8.5). Table 8.3 compares the two models at three confidence levels.

Two features matter more than the individual numbers. First, fat tails do not raise every risk figure. Holding the standard deviation fixed, the t puts more probability both very near the mean and very far from it, and less in the shoulders, so its 95% VaR is *lower* than the normal's; the two VaR curves cross at a confidence level of 97.47%, and only beyond it does the t report more risk. A desk that switches to a fat-tailed

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Confidence	Normal VaR	Student-t VaR	Normal ES	Student-t ES
95%	\$186,195	\$173,313	\$225,366	\$244,308
99%	\$250,081	\$280,373	\$281,847	\$378,056
99.9%	\$321,690	\$507,488	\$347,644	\$674,069

Table 8.3: Parametric VaR and Expected Shortfall of the \$10 million portfolio under the normal and under a Student-t with $\nu = 4$, both with the sample mean and standard deviation

model and checks only its 95% VaR can wrongly conclude that the change made it safer. Second, the gap widens rapidly in the tail: at 99.9% the t VaR is 58% above the normal's, and the t Expected Shortfall nearly double. Expected Shortfall is more sensitive to the tail model than VaR, the price of being the measure that looks into the tail.

Extreme value theory. The Student-t still imposes one shape on the whole distribution, so its tail is calibrated mostly by the ordinary days in the middle. Extreme value theory models the tail on its own. Its central result, the Pickands-Balkema-de Haan theorem, says that for a wide class of distributions the excesses of losses over a high threshold u are approximately distributed as a *generalized Pareto distribution* (GPD) with a shape ξ and a scale β, whatever the distribution looks like in the middle. A positive ξ means a power-law tail ($\xi = 1/\nu$ for a Student-t), and $\xi = 0$ an exponential, normal-like one. Fitting the GPD to the N_u losses above u out of n observations gives tail estimates

$$\mathrm{VaR}_q = u + \frac{\beta}{\xi}\left[\left(\frac{n}{N_u}(1-q)\right)^{-\xi} - 1\right], \qquad \mathrm{ES}_q = \frac{\mathrm{VaR}_q + \beta - \xi u}{1-\xi},$$

which extrapolate beyond the largest loss observed, something historical simulation can never do. This *peaks-over-threshold* method needs many exceedances, so it cannot be run on twenty days either. Simulating ten years, 2,500 days, from the $\nu = 4$ model above and setting u at the 95th percentile of losses (1.758%, with 125 exceedances) gives $\hat{\xi} = 0.013$ and $\hat{\beta} = 0.657\%$, and a 99.9% VaR of \$439,419 against a true value of \$507,488 and a normal fitted to the same sample of \$321,071. The method recovers almost two-thirds of the gap the normal leaves, but the shape estimate is far below the true $\xi = 0.25$, with a rough standard error of 0.09: even 125 exceedances pin down a tail index only loosely. That trade-off, a model built for the tail against scarce tail data, is why extreme value theory is used where the quantile is extreme and the history long, for 99.9% economic capital, for the severity distributions of Chapter 10's operational losses, and as a check on the 97.5% Expected Shortfall that now sets market-risk capital.

8.3 Validating and Extending the Measures

A risk number nobody has tested is a decoration; this part checks the measures against realized losses and extends them where the simple versions break.

8.3.1 Backtesting Value at Risk Models

A VaR model is useful only if it is calibrated, and the standard check is backtesting: comparing how often realized losses exceeded the VaR threshold (an "exception") with how often the model predicted they would. A well-calibrated 95% VaR model should produce exceptions on roughly 5% of days, no more and no fewer, since a model that is never exceeded may be too conservative, tying up capital that could be deployed productively.

Applying the parametric 95% VaR threshold from Section 8.2.2 (a daily return of -1.862%, equivalent to the \$186,195 VaR figure) to the twenty returns in Table 8.2, exactly one return, the -2.22% observation, falls below the threshold, for an observed exception rate of $1/20 = 5\%$, matching the model's target almost exactly. Applying the parametric 99% threshold (a return of -2.501%) produces zero exceptions, the natural outcome for an expected count of $0.01 \times 20 = 0.2$.

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Kupiec's proportion-of-failures (POF) test formalizes this comparison, testing the null hypothesis that the model's true exception rate equals its target p against a likelihood-ratio statistic,

$$LR_{\text{POF}} = -2\ln\left[\frac{(1-p)^{n-x}p^x}{(1-x/n)^{n-x}(x/n)^x}\right], \tag{8.7}$$

which is compared against a chi-squared distribution with one degree of freedom (critical value 3.841 at the conventional 95% test level) under the null. For the twenty-day sample above, with $p = 0.05$, $n = 20$, and $x = 1$ exception, the observed rate $x/n = 1/20 = 0.05$ exactly equals the target p, so the numerator and denominator are identical and $LR_{\text{POF}} = 0$, nowhere close to rejecting the null.

This looks like a perfect endorsement but is really a small-sample caution: one more bad day would move the rate to $2/20 = 10\%$, and even that doubled rate would not, on twenty observations, distinguish a well-calibrated model from a miscalibrated one. Kupiec's test, and the Basel traffic-light framework built on the same logic, need a much longer history, typically the one-year, 250-observation window described below, before they have real power. Like Chapter 7's Dickey-Fuller test, a test can be correctly constructed and still lack the power to detect a problem in a short sample.

Regulators take backtesting failures seriously: under the Basel framework, a bank whose internal VaR model produces too many exceptions in a rolling year can be required to hold more regulatory capital through a multiplier on the reported VaR, giving banks a financial incentive to calibrate their models rather than report an artificially low VaR.

The Basel Traffic-Light Backtesting Framework

Basel formalizes this backtesting logic into a three-zone "traffic light" system, based on the number of VaR exceptions observed over a rolling 250-trading-day window at the 99% confidence level, where $0.01 \times 250 = 2.5$ exceptions are expected on average if the model is correctly calibrated. Zero to four exceptions fall in the green zone, requiring no action beyond continued monitoring and leaving the baseline capital multiplier of 3 unchanged. Five to nine exceptions fall in the yellow zone, triggering closer scrutiny and a multiplier that rises within the zone; ten or more fall in the red zone, where the model is presumed flawed and the multiplier reaches its maximum of 4.

The framework works only over its full 250-day window: Table 8.2's twenty-day, one-exception sample cannot be classified into a zone, since the thresholds are calibrated to 250 observations, not twenty. Naively extrapolating the observed rate ($1/20 = 5\%$) to a year would suggest twelve or thirteen exceptions, well into the red zone, but that repeats the small-sample overreach just cautioned against. The traffic-light system is designed to be read only once a full 250-day window has been observed.

Backtesting Expected Shortfall

The shift from VaR to Expected Shortfall complicates backtesting. VaR backtesting reduces to counting exceptions, a binary event on which Kupiec's test and the traffic-light framework build, but ES is an average of losses *within* the tail, so a single exceedance does not say whether the ES forecast was too high, too low, or about right.

A common approach compares the *realized* loss on each exception day with the model's ES forecast for that day, averaged over the exceptions. Table 8.2's single 95% exception, the -2.22% return, realizes a loss of $0.0222 \times 10{,}000{,}000 = \$222{,}000$ against the parametric $\text{ES}_{95\%}$ forecast of \$225,366 computed earlier, a ratio of about 0.985, suggesting the forecast was, if anything, slightly conservative on this occasion.

One exception is far too little evidence to validate an ES model, but this comparison, accumulated over many exception days, is how a trading desk builds the track record needed to judge whether its ES model is calibrated, not merely whether its VaR threshold is crossed the right number of times.

8.3.2 From Daily VaR to Economic Capital: Time Scaling and Its Limits

Every VaR figure above answers a one-day question, but two related questions follow in practice: how much capital to hold against a longer regulatory horizon, and how much the institution as a whole needs over a year to survive a severe loss with high confidence. Both build on the daily VaR, through a time-scaling shortcut and a much higher confidence level respectively.

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Basel's market-risk capital framework historically scaled a bank's daily VaR up to a 10-day regulatory horizon using the square-root-of-time rule,

$$\mathrm{VaR}_T \approx \mathrm{VaR}_1\sqrt{T}, \tag{8.8}$$

which holds exactly when daily returns are independent and identically distributed, since the variance of a sum of T such returns is T times the one-day variance. Applied to the parametric 99% VaR of \$250,081 from Section 8.2.2, it gives a 10-day regulatory VaR of $250{,}081\sqrt{10} \approx \$790{,}824$ without a separate 10-day model.

The independence assumption, however, is the one Chapter 7 warned against: daily returns are rarely independent, and even modest positive autocorrelation changes how fast multi-day risk grows. If daily returns followed an AR(1) process with $\phi = 0.10$ (much weaker than the $\phi = 0.71$ Chapter 7 estimated for interest-rate deviations, but realistic for daily equity returns), the variance of a T-day cumulative return is

$$\mathrm{Var}\left(\sum_{t=1}^{T} r_t\right) = T\sigma^2 + 2\sigma^2 \sum_{k=1}^{T-1} (T-k)\phi^k,$$

which for $T = 10$ gives a 10-day volatility about 9.4% higher than the square-root-of-time rule assumes, and a 10-day 99% VaR of about \$865,413 rather than \$790,824: applied to autocorrelated returns, the shortcut understates the capital a bank should hold, in the direction that matters.

Economic capital takes the logic one step further, from a 10-day horizon to a one-year horizon at a much higher confidence level, typically 99.9% or more, consistent with the default probability of the rating (commonly single-A or double-A) a bank wants its solvency to reflect; Chapter 9 develops the analogous calculation for credit portfolios with the Vasicek model. Unlike the regulatory VaR multiplier of Section 8.3.1, a supervisory charge computed by a prescribed method, economic capital is the bank's own estimate of the capital its risk-taking needs to survive a severe, low-probability year. It serves internal capital allocation and performance measurement as much as compliance: a desk earning a given return on a smaller economic-capital charge uses the firm's scarce capital more efficiently.

8.3.3 Liquidity Risk and Liquidity-Adjusted VaR

Every VaR calculation so far has assumed a position can be liquidated at the market price, but, as Chapter 3 showed, selling costs at least half the bid-ask spread relative to the mid price, and a large order can walk through several levels of the order book at progressively worse prices. Liquidity-adjusted VaR (LVaR) adds an estimated liquidation cost to VaR,

$$\mathrm{LVaR}_\alpha = \mathrm{VaR}_\alpha + \frac{1}{2}\,(\text{spread}) \times V,$$

where the spread is a fraction of position value and the factor of one-half is the cost of crossing from the mid price to the relevant side of the quote, as in Chapter 3's round-trip cost. Applying this to the \$2 million two-asset position from Section 8.2.3, assuming an average bid-ask spread of 30 basis points, the liquidity cost adjustment is

$$\frac{1}{2}(0.0030)(2{,}000{,}000) = \$3{,}000, \qquad \mathrm{LVaR}_{95\%} = 19{,}182 + 3{,}000 = \$22{,}182,$$

an increase of roughly 16% over the standard parametric VaR of \$19,182. The adjustment matters most where it is most often ignored. In a market-wide liquidity crunch, spreads widen sharply, since they compensate market makers for inventory risk and adverse selection, both of which rise under stress. The cost of liquidating just when a firm most needs to reduce risk can therefore far exceed an adjustment calibrated on calm-market spreads, an instance of the procyclicality problem discussed later in this chapter.

Market liquidity risk, the focus above, is distinct from funding liquidity risk, the risk that a firm cannot meet its own near-term cash obligations, and Basel III addresses the latter directly with the Liquidity Coverage Ratio (LCR),

$$\mathrm{LCR} = \frac{\text{High-Quality Liquid Assets}}{\text{Total Net Cash Outflows over 30 days}} \geq 100\%,$$

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which requires a bank to hold enough liquid assets (broadly, cash and high-grade government bonds) to survive 30 days of a specified stress scenario without raising new funding. A bank with \$50 million in high-quality liquid assets and projected net cash outflows of \$40 million over a 30-day stress scenario has LCR $= 50/40 = 125\%$, comfortably above the regulatory minimum. The LCR is the funding-side complement to liquidity-adjusted VaR: LVaR asks what a position would cost to sell, the LCR whether the firm has enough liquid resources to avoid being a forced seller at all during stress. Forced selling into a stressed, illiquid market is what turned market liquidity risk into a systemic problem in the 2008 crisis discussed later in this chapter.

Basel III pairs the LCR with a second, longer-horizon ratio, the Net Stable Funding Ratio (NSFR),

$$\text{NSFR} = \frac{\text{Available Stable Funding}}{\text{Required Stable Funding}} \geq 100\%,$$

which asks the LCR's question over a one-year horizon rather than 30 days, weighted toward the structural mismatch between how long a bank's assets take to mature (loans, often years) and how quickly its liabilities could be withdrawn (deposits, often on demand). Available stable funding weights a bank's capital and liabilities by how reliably they will still be there in a year (retail deposits are weighted as more stable than short-term wholesale borrowing, for instance), while required stable funding weights its assets by how long they are locked up (a long-dated loan requires more stable funding backing it than an overnight reverse repo). Where the LCR addresses a sudden, short-term liquidity run, the NSFR addresses the slower-moving risk of a bank funding long-term assets with short-term, flighty liabilities, the maturity-transformation mismatch that turned a funding problem into a solvency problem for several institutions during the 2008 crisis once wholesale funding markets froze.

8.3.4 VaR for Nonlinear Portfolios: The Delta-Normal Approximation and Its Limits

Every VaR calculation so far has assumed a linear portfolio, whose value moves in constant proportion to the risk factor. A portfolio containing options violates this, and Chapter 5's Greeks show how.

Consider a market maker short 1,000 of the one-year, \$50-strike calls priced in Chapter 5 ($S_0 = K = \$50$, $r = 4\%$, $\sigma = 30\%$, $C_0 \approx \$6.88$, $\Delta = 0.6115$, $\Gamma = 0.0255$). The delta-normal method approximates the position's one-day loss using only delta, treating the option price as locally linear in the stock price. Converting the annual volatility to a daily figure the standard way, $\sigma_{\text{daily}} = 0.30/\sqrt{252} \approx 1.90\%$, the position's delta-implied one-day dollar standard deviation is

$$\Delta \times S_0 \times \sigma_{\text{daily}} \times 1{,}000 = 0.6115 \times 50 \times 0.0190 \times 1{,}000 \approx \$577.85,$$

giving $\text{VaR}_{95\%}^{\text{delta-normal}} \approx 1.645(577.85) \approx \$950.56$ and $\text{VaR}_{99\%}^{\text{delta-normal}} \approx 2.326(577.85) \approx \$1{,}344.08$.

Full revaluation instead reprices the same option with the same Black-Scholes-Merton formula at a stressed stock price, holding strike, rate, volatility, and time to maturity fixed (a simplification that ignores one day of time decay, isolating the delta and gamma effect this example is meant to illustrate). The worst case for a short call is the stock rising, so at the 95% stressed price $S_0(1 + z_{0.95}\sigma_{\text{daily}}) \approx \$51.55$, the same option reprices to \$7.86, a position loss of \$980.79, about 3.2% *more* than the delta-normal estimate. At the more extreme 99% stressed price of \$52.20, the option reprices to \$8.28, a loss of \$1,403.99, now about 4.5% more than delta-normal predicted.

A delta-gamma approximation closes most of this gap without requiring a full revaluation, by adding a quadratic correction term built directly from the same gamma already computed in Chapter 5,

$$\Delta P \approx \left(\Delta\,\Delta S + \tfrac{1}{2}\Gamma\,\Delta S^2\right) \times n_{\text{contracts}}. \tag{8.9}$$

With $\Delta S_{95} = S_0 z_{0.95}\sigma_{\text{daily}} \approx \$1.554$ and $\Delta S_{99} \approx \$2.198$, this gives delta-gamma losses of

$$1{,}000\left(0.6115(1.554) + \tfrac{1}{2}(0.0255)(1.554)^2\right) \approx \$981.43,$$
$$1{,}000\left(0.6115(2.198) + \tfrac{1}{2}(0.0255)(2.198)^2\right) \approx \$1{,}405.79,$$

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within a few cents of the full-revaluation figures of $980.79 and $1,403.99, compared to delta-normal's $950.56 and $1,344.08. The quadratic term costs one extra number, gamma, and recovers nearly all the accuracy of full repricing, which is why delta-gamma VaR is the standard compromise for portfolios with too many options to reprice at every stressed scenario. The growing gap is gamma's effect: delta is the option's slope at the current price, but a large move travels far enough along the price-value curve that the slope has changed by the time it is complete, and gamma measures that curvature. Delta-normal is a cheap, reasonable approximation for small moves or little gamma exposure, but it understates risk for a portfolio with meaningful gamma, and more so the larger the move, which is why the 99% gap above is proportionally wider than the 95% gap.

8.3.5 Interest-Rate Risk in the Banking Book

Every position measured so far sits in a *trading book*: held to be sold or hedged within days, marked to market every evening, its risk expressed over a one-day or ten-day horizon. Most of a commercial bank's balance sheet is different. Its loans, its deposits, and the large securities portfolio it buys with the deposits it has not lent out make up the *banking book*, which is held for the long term and accounted for largely at amortized cost. Securities a bank designates as *held to maturity* (HTM) are carried at what the bank paid for them, adjusted for amortization, whatever they would fetch today. Securities *available for sale* (AFS) are carried at fair value, but in the United States a bank outside the largest size categories may elect to keep their unrealized gains and losses out of its regulatory capital. A rise in rates that takes 18% off the market value of a bond portfolio can therefore leave both reported earnings and regulatory capital almost untouched.

The loss is real nonetheless. A bond worth $82 today is worth $82 whoever owns it; the accounting label decides only when the loss is recognized. If the bank is never forced to sell, the bond pulls back to par and the bank recovers its book value, but only by earning a below-market coupon for the remaining years: the loss arrives as thin interest margins instead of one write-down. Holding to maturity is a bet that the bank will never need the cash first, a bet about its depositors. Measuring interest-rate risk in the banking book (IRRBB) means measuring that exposure whatever the accounting says, in two complementary ways.

Economic value of equity. The first measure asks what the balance sheet is worth today, the present value of the asset cash flows minus the present value of the liability cash flows, and how much that difference moves when rates move. Applying Chapter 5's duration approximation to each side of the balance sheet gives

$$\Delta \mathrm{EVE} \approx -\Big(\sum_i D_i A_i - D_L L \Big)\, \Delta y, \tag{8.10}$$

where A_i and D_i are the value and modified duration of each class of assets, and L and D_L are the value and effective duration of the liabilities. A bank whose assets have a longer duration than its liabilities, the normal condition of an institution funding long-term loans and bonds with demand deposits, loses economic value when rates rise. The awkward term is D_L: a demand deposit has no contractual maturity, so its duration is a behavioral assumption about how long depositors will leave their money, estimated from the bank's own history.

Net interest income. The second measure asks what happens to earnings over the next year. Assets that reprice within the year, cash and floating-rate loans, start earning more at once; fixed-rate bonds do not. On the other side, banks raise deposit rates only partially and slowly, and the fraction of a market-rate move that is passed through to depositors is the *deposit beta* β. To first order,

$$\Delta \mathrm{NII} \approx \big(A_{\mathrm{reprice}} - \beta\, L_{\mathrm{dep}} \big)\, \Delta y, \tag{8.11}$$

with A_{reprice} the assets that reprice within the horizon and L_{dep} the deposits. Drechsler et al. (2017) show that low deposit betas reflect banks' market power over depositors, which is why banks treat deposits as cheap, stable, long-duration funding that hedges the duration of their assets. The two measures can point in opposite directions: a bank can be *asset sensitive*, with net interest income that rises when rates rise, while its economic value falls, because NII sees only the next twelve months and EVE sees all of them.

The supervisory standard. The Basel Committee's IRRBB standard of April 2016 (BCBS 2016a) keeps the topic in Pillar 2, the supervisory-review pillar, rather than imposing a mechanical capital charge. It prescribes six standardized rate-shock scenarios (parallel up, parallel down, steepener, flattener, short rates up, and short rates down; for the U.S. dollar the 2016 calibration sets the parallel shock at 200 basis points)

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and a *supervisory outlier test*: a bank whose largest decline in EVE across those scenarios exceeds 15% of its Tier 1 capital is flagged as an outlier, which can lead supervisors to require it to reduce the risk or hold more capital. Because the test runs on EVE, it inherits Equation (8.10)'s dependence on the deposit-duration assumption, as the worked example shows.

Worked Example 8.1: A Deposit-Funded Bank Meets a 300-Basis-Point Rise

Consider a stylized bank with \$100 billion of assets: \$8 billion of cash, \$42 billion of floating-rate loans that reset every quarter (modified duration about 0.25), and a \$50 billion securities portfolio of seven-year bonds with a 1.50% annual coupon, bought at par when seven-year yields were 1.50%. It is funded by \$91 billion of deposits and \$9 billion of equity, an ordinary leverage for a bank. Chapter 5's formulas give the bonds a Macaulay duration of 6.697 years and a modified duration of $D_{\text{mod}} = 6.697/1.015 = 6.598$. Repricing them at yields of 2.50% and 0.50% gives \$93.65 and \$106.86 per \$100 of par, so Chapter 5's three-price estimate of convexity is

$$\text{Convexity} \approx \frac{93.65 + 106.86 - 2(100)}{100\,(0.01)^2} \approx 51.27.$$

Now let yields rise 300 basis points, to 4.50%, a move of roughly the size seen in 2022. The duration approximation predicts a price change of $-6.598 \times 0.03 = -19.79\%$, and adding the convexity term gives

$$\frac{\Delta P}{P} \approx -6.598(0.03) + \tfrac{1}{2}(51.27)(0.03)^2 = -0.1979 + 0.0231 = -17.49\%,$$

while exact repricing at 4.50% gives \$82.32 per \$100, a fall of 17.68%. On \$50 billion the three estimates are losses of \$9.90 billion, \$8.74 billion, and \$8.84 billion: duration alone overstates a 300-basis-point loss by more than a billion dollars, and the convexity correction lands within \$0.1 billion of the exact figure, the same pattern Chapter 5 found for a single bond. The exact loss of \$8.84 billion is 98.2% of the bank's \$9 billion of equity. If the bonds are classified as held to maturity, none of it appears in the income statement or in regulatory capital, yet the bank's economic equity, its equity less the unrealized loss, is now $9 - 8.84 = \$0.16$ billion. Figure 8.3 sets the two balance sheets side by side.

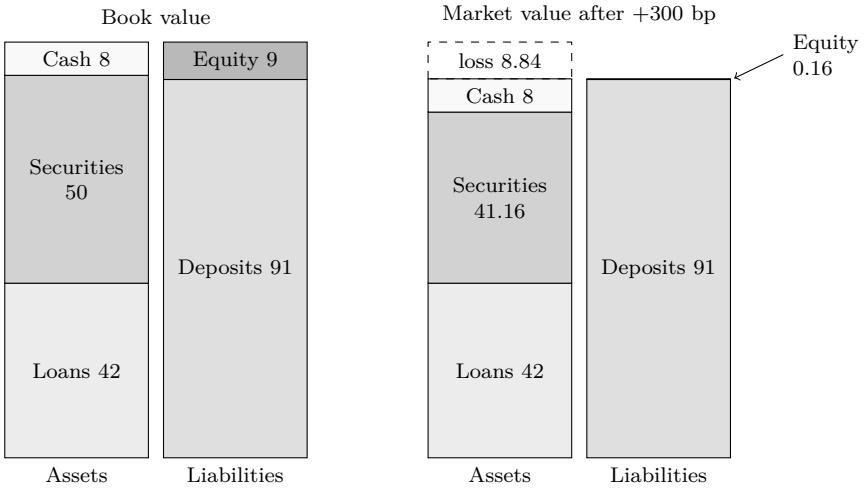


Figure 8.3: The stylized bank's balance sheet at book value and at market value after a 300-basis-point rise in rates, in \$ billions. The securities loss of \$8.84 billion leaves \$0.16 billion of economic equity, although the book-value balance sheet is unchanged.

EVE and the outlier test. Under a Basel-style parallel shock of +200 basis points, the assets' dollar duration is $50(6.598) + 42(0.25) = 340.41$, so the assets lose $340.41 \times 0.02 = \$6.81$ billion. What the deposits give back depends on their assumed duration. If the bank's deposit study says its depositors stay long enough

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to give the deposit base an effective duration of 3.5 years, their present value falls by $91(3.5)(0.02) = \$6.37$ billion, and Equation (8.10) gives $\Delta\text{EVE} = -(340.41 - 318.50)(0.02) = -\$0.44$ billion, 4.9% of Tier 1 capital and comfortably inside the 15% threshold of $1.35 billion. If the deposits are flighty instead, with an effective duration of 0.5 years, they give back only $0.91 billion and $\Delta\text{EVE} = -\$5.90$ billion, 65.5% of Tier 1 capital, a severe outlier. The balance sheet is identical in the two cases; one behavioral parameter has changed.

Net interest income. Over the next year at +300 basis points, the $50 billion of cash and floating-rate loans earns an extra $50(0.03) = \$1.50$ billion. With a deposit beta of 0.25 the deposits cost an extra $0.25(91)(0.03) = \$0.68$ billion, and Equation (8.11) gives $\Delta\text{NII} = +\$0.82$ billion: on the earnings measure this bank *gains* from rising rates, at the very moment its economic equity has all but vanished. With a beta of 0.70, the kind of pass-through a bank faces once its depositors start moving money into money-market funds and Treasury bills, the extra deposit cost is $1.91 billion and $\Delta\text{NII} = -\$0.41$ billion.

A run forces the loss into the open. Suppose a quarter of the deposits, $22.75 billion, are withdrawn. Cash covers $8 billion, so the bank must raise $14.75 billion by selling bonds at 82.32% of par, which means selling 35.8% of the portfolio, $17.92 billion of book value. The sale crystallizes 35.8% of the unrealized loss, $3.17 billion, cutting book equity to $5.83 billion with $5.67 billion of loss still unrealized. Worse, selling out of the held-to-maturity category contradicts the stated intent to hold, which under U.S. accounting rules generally forces the rest of the portfolio to be reclassified and carried at fair value, so the remaining loss stops being invisible too. The bank's solvency did not change on the day of the run: its economic equity was $0.16 billion before and after. The run removed the option to wait. And cash plus the entire securities portfolio covers only $49.16 billion of withdrawals, 54.0% of deposits; beyond that the bank must borrow against its loans or fail.

Deposits, Runs, and Liquidity

The worked example makes the depositor the decisive variable, and three features govern how a deposit base behaves. The first is the deposit beta, which is not a constant: betas are low while rates are low and the gap between deposit rates and market rates is small, and they rise as that gap widens and depositors notice it. The second is the share of *uninsured deposits*, balances above the deposit-insurance limit, which in the United States is $250,000 per depositor, per bank, per ownership category. An insured depositor has no reason to run, and an uninsured one every reason to run first: in the classic analysis of Diamond and Dybvig (1983), a bank that must liquidate long-term assets at a loss to pay early withdrawers cannot pay everyone, and the depositors who wait bear the shortfall. The third is how concentrated and connected the depositors are. A million households react to news independently and slowly; a few thousand firms advised by the same handful of investors can react together within hours. Mobile banking has removed the friction of the branch queue, and social media can carry the first signs of a run across an entire depositor network in an afternoon. Jiang et al. (2024) estimate that the rate rises of 2022 and early 2023 cut the market value of U.S. banks' assets by about 10%, roughly $2 trillion, and show that losses of that size can expose even banks with liquid assets to self-fulfilling runs by uninsured depositors.

The Liquidity Coverage Ratio of Section 8.3.3 exists to absorb these risks. Its 30-day stress scenario assigns an outflow rate to each funding category, from 5% for stable, fully insured retail deposits to 40% for unsecured deposits from nonfinancial companies, and the bank must hold enough high-quality liquid assets to cover the result (BCBS 2013). But a 30-day calibration assumes a run unfolds over weeks. The run in the vignette below took less than two days.

8.3.6 Case Vignette: Silicon Valley Bank, March 2023

Silicon Valley Bank was the worked example at full scale. Its holding company, SVB Financial Group, grew from $71 billion of assets in 2019 to $211.8 billion at the end of 2022 on a flood of deposits from technology start-ups and the venture-capital firms that funded them (Board of Governors of the Federal Reserve System 2023; SVB Financial Group 2023). Lending could not keep pace, so the bank invested the money: at the end of 2022 securities made up 55% of its assets, against 25% at comparable large banks, and 78% of those securities were classified as held to maturity. The held-to-maturity portfolio, carried at $91.32 billion, had a fair value of $76.17 billion and a weighted-average duration of 6.2 years, an unrealized loss of $15.15 billion against total stockholders' equity of $16.0 billion, or 94.7% of it, with a further $2.53 billion of unrealized

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losses in the AFS portfolio. About 94% of its $173.1 billion of deposits were uninsured, against an average of 41% at its peers. The deposit base was already turning: noninterest-bearing deposits fell from $125.9 billion to $80.8 billion during 2022, and the bank's own model now assumed a deposit beta of about 70% on its interest-bearing deposits.

The measurement failures match the worked example too. The Federal Reserve's review found that the risk appetite set by the bank's board tracked only an NII metric, not EVE; that the EVE metric had breached internal limits for years; and that in April 2022 management responded by lengthening the assumed duration of its deposits, a change that shrank the measured mismatch without taking any risk off the balance sheet (Board of Governors of the Federal Reserve System 2023). That is the deposit-duration lever of the worked example, pulled in the direction that makes the outlier disappear. The review also found that under the rules in force before 2019 the bank would have had to count its AFS losses in regulatory capital, which would have been $1.9 billion lower, and to meet the full LCR, under which it would have been about 9% short of high-quality liquid assets in December 2022.

On 8 March 2023, SVB announced that it had sold $21 billion of AFS securities at a $1.8 billion after-tax loss and planned to raise $2.25 billion of new equity. Depositors read the announcement as a sign of distress. On 9 March they withdrew about $42 billion, 24.3% of year-end deposits, in a single day, leaving the bank with a negative cash balance of about $958 million (DFPI 2023). The Federal Reserve's review describes a run "fueled by social media" and by a concentrated network of venture investors and technology firms that withdrew their deposits together "with unprecedented speed"; for comparison, Washington Mutual, the largest bank failure of 2008, lost $19 billion of deposits over 16 days. Overnight, SVB told supervisors that it expected more than $100 billion of further outflows on 10 March, which together with the first day's $42 billion would have been 82.0% of year-end deposits within two days. The bank did not have the cash or the collateral to meet them. The California Department of Financial Protection and Innovation closed it on the morning of 10 March 2023 and appointed the Federal Deposit Insurance Corporation as receiver.

On 12 March, the Treasury, the Federal Reserve, and the FDIC invoked the *systemic risk exception* of U.S. deposit-insurance law, which allowed the FDIC to protect all of SVB's depositors, insured or not, and the Federal Reserve opened a new lending facility for banks secured by government securities valued at par rather than at market (U.S. Department of the Treasury et al. 2023). Losses to the Deposit Insurance Fund were to be recovered from the banking industry through a special assessment, which the FDIC's final rule of November 2023 set at an estimated $16.3 billion for the uninsured deposits protected at SVB and at Signature Bank, which had failed two days later (FDIC 2023).

Three lessons carry beyond this bank. First, the loss existed long before the run; the accounting deferred its recognition, and the run removed the option to wait. Second, the risk number was dominated by a behavioral assumption about depositors, and governance let that assumption move when the number breached its limit, the kind of model-risk failure Section 8.6.1 is meant to catch. Third, liquidity rules calibrated to a 30-day stress were not built for a run measured in hours. Early-warning models of deposit outflows, trained on daily client-level flows and on the network of who banks with whom, are a natural use of Chapter 6's classifiers, but SVB's record suggests that a missing signal was not the binding constraint: the EVE breaches were measured, and reported to a board committee, for years.

8.4 Sensitivity-Based Risk Limits

A desk can sit inside every VaR limit it has been given and still carry the position that ends its year. The reason is structural, not a failure of calibration: VaR compresses a book into one number, and the compression hides where the risk sits. Two desks reporting the same $300,000 daily VaR can hold portfolios with nothing in common, one hedged against the coming move and the other pointed into it.

Practitioners therefore run sensitivities alongside VaR, never instead of it. A sensitivity answers a much narrower question: not "how bad could tomorrow be, all things considered," but "if this one risk factor moves by one unit, and nothing else changes, how much do we make or lose?" The answers are unglamorous, linear, and immediately actionable, which is why the trading floor lives on them.

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8.4.1 The Greeks as Limits in Their Own Right

Chapter 5 computed delta, gamma, vega, and theta for this chapter's option, and the delta-normal approximation used the first two as ingredients of VaR. On a derivatives desk each is a limit in its own right, monitored independently of VaR. Return to the market maker short 1,000 of Chapter 5's one-year \$50-strike calls ($S_0 = K = \$50$, $r = 4\%$, $\sigma = 30\%$), and read the Greeks as a risk report rather than pricing output. Table 8.4 collects them, scaled to the position and joined by rho, the one Greek Chapter 5 did not compute.

Greek	Measures exposure to	Position	Limit	Utilization
Delta	a \$1 move in the stock	−611.5	1,500	40.8%
Gamma	the change in delta per \$1 move	−25.55	50	51.1%
Vega	a one-percentage-point move in implied volatility	−\$191.62	\$250	76.6%
Theta	the passage of one day	+\$10.47	—	—
Rho	a one-percentage-point move in interest rates	−\$237.00	\$400	59.3%

Table 8.4: The full Greek profile of a short position in 1,000 one-year \$50-strike calls, with representative desk limits

A risk manager reads the table line by line. The position is short delta, so it loses money if the stock rallies. It is short gamma, meaning that delta itself moves against the position in both directions, the effect the delta-normal example isolated numerically. It is short vega, so it loses if implied volatility rises. It is long theta, collecting \$10.47 a day in time decay as long as nothing happens, the market maker's business model and the compensation for being short gamma and vega. (Chapter 5 quoted this option's theta as −\$3.82 per year; a risk report is read every morning, so desks quote theta per day, and dividing by 365 recovers the number above.) And it is short rho, though at this maturity that is the least of its concerns.

The key observation is one this chapter's VaR machinery cannot make. Every VaR figure computed for this position, delta-normal, delta-gamma, or full revaluation, shocks the *stock price* and holds implied volatility at 30%. The model is structurally blind to vega: implied volatility is not among the risk factors it perturbs, so no recalibration or backtesting will make that risk appear. Yet a five-point rise in implied volatility, from 30% to 35%, is an ordinary single-day move in a stressed market, and it would cost this position

$$1{,}000 \times \$0.1916 \times 5 = \$958,$$

against a 99% full-revaluation VaR of \$1,403.99 attributed to price moves alone. A routine volatility move, invisible to the risk report, is worth 68% of the entire price-driven 99% VaR. This is not an argument against VaR. It is an argument for reading a vega limit next to it.

8.4.2 Bucketed Sensitivities: The DV01 Ladder

The logic applies with more force to interest-rate books, where the risk factor is not a single price but a whole yield curve. The standard sensitivity there is DV01, the dollar value of a basis point: the profit or loss produced by a one-basis-point move in yields. A single aggregate DV01 for the whole book is the rates equivalent of a single VaR number, and it conceals the same thing.

Consider a desk whose positions are reported in the four standard maturity buckets of Table 8.5, with each figure giving the position's profit or loss from a one-basis-point *rise* in that bucket's yield, so that a long bond position appears as a negative number.

The net DV01 of this book is exactly zero. A report carrying only that aggregate would describe the desk as flat, and under a parallel shift it is: adding ten basis points to every yield produces gains and losses that cancel to the cent, as the middle column shows.

But the curve does not only move in parallel, and this desk is not flat. Its *gross* DV01, the sum of the absolute bucket exposures, is \$130,000 per basis point. It is long the short end and short the long end, a position that profits if the curve steepens and loses if it flattens. Applying a flattening scenario in which

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Bucket	DV01 (per bp)	Parallel +10bp	Flattening scenario
2-year	−$25,000	−$250,000	−$500,000
5-year	−$40,000	−$400,000	−$400,000
10-year	+$38,000	+$380,000	−$190,000
30-year	+$27,000	+$270,000	−$270,000
Net	$0	$0	−$1,360,000

Table 8.5: A bucketed DV01 ladder whose net exposure is exactly zero

two-year yields rise 20 basis points, five-year yields rise 10, ten-year yields fall 5, and thirty-year yields fall 10 gives

$$(-25{,}000)(20) + (-40{,}000)(10) + (38{,}000)(-5) + (27{,}000)(-10) = -\$1{,}360{,}000,$$

a loss of $1.36 million from a curve move that the net-DV01 report scores as zero exposure. Figure 8.4 shows why: under the parallel shift the bucket gains and losses offset, while under the flattening every bucket loses at once. The bucketed ladder makes this visible, which is why rate desks have limits per bucket rather than one on the total. The principle transfers to equity and credit books: aggregate where you must, but never let the aggregate be the only thing anyone looks at.

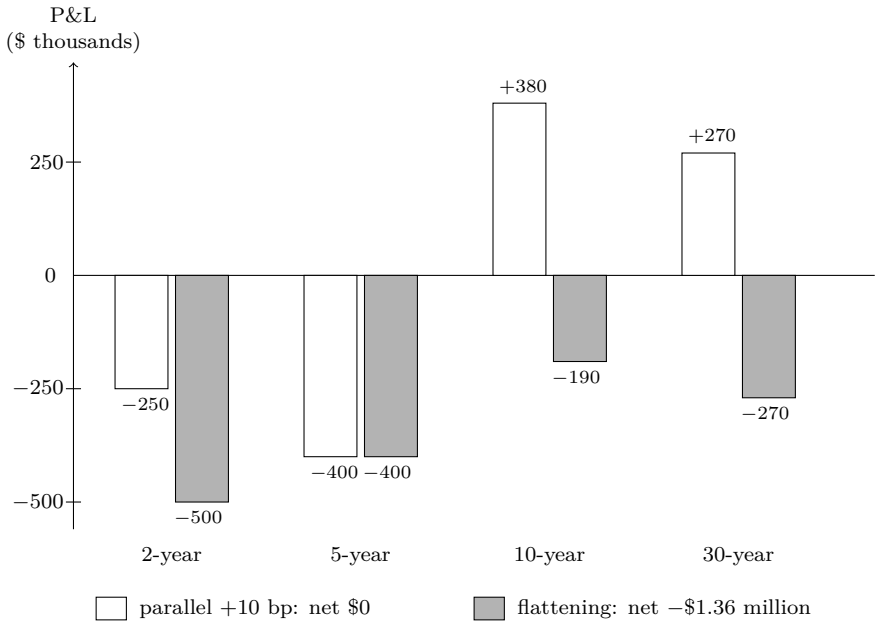


Figure 8.4: Bucket-by-bucket profit and loss of Table 8.5's ladder under a parallel +10 basis-point shift (open bars) and under the flattening scenario (shaded bars)

This is also the direction regulation has taken: the sensitivities-based method at the heart of the Fundamental Review of the Trading Book computes capital from these bucketed delta, vega, and curvature sensitivities rather than from a single portfolio-level loss distribution.

Chapter 7's principal component analysis of the yield curve explains *why* the ladder decomposes risk as it does. It found that almost all of the curve's daily variation is captured by three factors: a level factor that shifts all maturities together, a slope factor that tilts short against long, and a curvature factor. Projecting this desk's ladder onto those factors restates the whole example in one line. A net DV01 of zero is zero exposure to the *level* factor, which is why the parallel shift produces nothing; the desk's real position is a large exposure to the *slope* factor, and the flattening scenario is a slope-factor move. Bucketing by maturity

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is the practitioner's version of a factor decomposition, and a desk that has run the PCA can set limits on the three factor exposures directly, more compactly and more in line with how the curve moves than limits on four buckets treated as independent.

8.4.3 Computing Sensitivities at Scale: Adjoint Differentiation

The two preceding subsections asked for nine numbers, five Greeks and four DV01 buckets. A real trading book asks for thousands, across every instrument, risk factor, and maturity bucket, recomputed at least daily, and producing them is a computational problem with an elegant answer from machine learning.

The obvious method is to bump and reprice: value the book, add a basis point to one risk factor, value it again, and difference. It is easy and correct but does not scale, because each sensitivity costs a full revaluation. A book exposed to n risk factors costs $n+1$ valuations, and when instruments are priced by Monte Carlo, so that one valuation takes minutes, and n runs into the thousands, risk reporting becomes an overnight batch that a trader cannot interrogate during the day.

Adjoint algorithmic differentiation removes the dependence on n altogether. Its cost is bounded by a small constant multiple, typically three to five times, of a *single* valuation, no matter how many sensitivities are requested. A book that took an hour to produce one set of Greeks by bumping produces the entire set in minutes.

The mechanism is the one used to train neural networks. A pricing formula, like a network's forward pass, is a composition of elementary operations whose local derivatives are known analytically. Evaluating the formula once while recording that composition builds a computational graph. Seeding the output with a derivative of one and applying the chain rule backward through the graph then accumulates the derivative of the output with respect to every input simultaneously, in a single reverse sweep. This is reverse-mode automatic differentiation, and backpropagation is its application to a loss function instead of a price. The reason training a network with a million parameters costs only a constant multiple of a forward pass is the reason a bank can compute a thousand Greeks for the price of a few valuations: it is the same theorem.

Applying it to the Black-Scholes price of Table 8.4's option makes this concrete. One forward evaluation records a graph of roughly twenty elementary nodes; one reverse sweep then returns $\partial C/\partial S$, $\partial C/\partial \sigma$, $\partial C/\partial T$, and $\partial C/\partial r$ at once, giving a delta of 0.611539, a vega of 0.191623 per percentage point of volatility, a theta of -0.010472 per day, and a rho of 0.237003 per percentage point of rates. Each agrees with the closed form to nine decimal places, the method's other advantage: unlike a bumped finite difference, an adjoint derivative is exact up to floating-point rounding, with no step size to choose and no cancellation error to trade against truncation error.

Gamma is the instructive exception. It is a second derivative, and a single reverse sweep returns only first derivatives, so obtaining it requires differentiating the derivative, that is, applying the machinery to its own output. This is the distinction between a gradient and a Hessian in machine learning, and it carries the same cost implication: first-order sensitivities are nearly free, and second-order ones are not. Production systems handle this with nested automatic differentiation, the same construction used for Hessian-vector products when training a model with a second-order optimizer.

The technique entered finance through the work of Giles and Glasserman (2006) on Monte Carlo Greeks and is now standard on derivatives desks and in the valuation-adjustment calculations of modern derivatives infrastructure. It is not machine learning applied to finance in the usual sense, since nothing is learned from data. It is arguably more interesting: an algorithm popularized in one field answering a computational problem in the other, for reasons that lie in the structure of differentiation rather than the subject matter.

8.4.4 Case Vignette: Leveraged Gilts and the 2022 UK Pension Crisis

A UK defined-benefit pension fund owes its members payments stretching decades into the future, so the present value of its liabilities behaves like a very long bond: it rises when long-term interest rates fall. The natural hedge is to hold long-dated UK government bonds, gilts, whose prices move the same way. *Liability-driven investment* (LDI) is the strategy built on that hedge, and by 2022 more than £1 trillion was invested in it (Cunliffe 2022). Most funds wanted more interest-rate hedging than they had assets to spare, because they also held equities and other growth assets, so LDI managers supplied leverage: they bought gilts with money borrowed through repurchase agreements, or took the same exposure through swaps, against

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a cushion of the pension fund's own capital. Large schemes ran segregated mandates; smaller ones bought units in *pooled* LDI funds shared by many schemes. When gilt prices fell, the losses came out of the cushion and the banks on the other side of the repo and swap trades called for collateral in cash. The pension fund as a whole was usually no worse off, since its liabilities fell in value by more than its gilts did, but its LDI fund needed cash immediately.

On Friday 23 September 2022 the UK government announced a package of unfunded tax cuts, the "Growth Plan" or mini-budget. Long-dated gilt yields rose about 30 basis points that day, and on Monday the 30-year yield rose about 50 more, leaving it over 80 basis points above Friday morning's level. The Bank of England later told Parliament that the period saw two daily rises in the 30-year yield of more than 35 basis points, when the largest daily rise since 2000 had been 29, and that the four-day rise was more than twice the largest since 2000, set during the 2020 "dash for cash" (Cunliffe 2022). LDI managers reported that, if conditions kept worsening, they would have to sell at least £50 billion of long gilts into a market that traded about £12 billion a day. By the evening of Tuesday 27 September the 30-year yield was 67 basis points above that morning's level, and managers warned that several pooled funds would fall to negative net asset value and begin winding up the next day, with the banks that had lent to them selling the gilts held as collateral. On Wednesday 28 September the Bank announced temporary purchases of long-dated gilts, by daily auction until 14 October, and the 30-year yield fell by more than 100 basis points that day. It bought £19.3 billion in total, £12.1 billion of conventional and £7.2 billion of index-linked gilts, and sold all of them back by 12 January 2023 (Bank of England 2023).

The DV01 arithmetic of Section 8.4.2 explains the speed. Take a stylized 30-year gilt with a 3.5% annual coupon priced at par. Its modified duration is 18.39, nearly three times that of the bank's seven-year bonds in Section 8.3.5, and its convexity is 460.9. A single 35-basis-point day takes 6.16% off its price. Now put £1 billion of pension money into a pooled fund levered four times, holding £4 billion of these gilts. The fund's DV01 is £7.36 million per basis point, and one 35-basis-point day calls £247 million of collateral, 24.7% of the fund's capital, which must arrive in cash, typically by the next business day, from the trustees of many smaller schemes, few of whom could raise it that fast. A 100-basis-point rise, smaller than the four-day move of September 2022, calls £652 million, 65.2% of the capital, and a rise of 166 basis points wipes the fund out entirely; at two times leverage it would take 440 basis points. Convexity helps, since duration alone would have predicted a fall of 18.39% for a 100-basis-point rise against the exact 16.29%, but not nearly enough. In March 2023 the Bank's Financial Policy Committee concluded that LDI funds should withstand a yield rise of about 250 basis points at a minimum, on top of their buffers for ordinary daily moves (Bank of England Financial Policy Committee 2023). For this gilt a 250-basis-point rise takes 34.41% off the price, so the largest leverage that survives it is $1/0.3441 = 2.91$ times.

Three lessons connect this episode to the chapter. First, the hedge was economically sound and the risk was liquidity: the loss that mattered was the collateral call, not the scheme's funding position, so a VaR on its net exposure would have looked reassuring. Second, the danger was collective. Every leveraged fund held the same gilts and faced the same calls, so their selling raised yields and generated further calls, the fire-sale spiral that Section 8.3.3's liquidity-adjusted VaR, calibrated to one firm's spread, cannot see and Chapter 20's contagion models are built to study. Third, like Silicon Valley Bank's run, it happened in hours. A liquidity buffer sized to a 30-day horizon is irrelevant to a margin call due the next morning; the measure that matters is the yield move a fund can absorb with the cash it already holds, and that is a DV01 calculation.

8.5 Stress Testing and Machine Learning

Historical data cannot describe an event that has not happened yet, which is where scenarios and learned models both enter.

8.5.1 Stress Testing and Scenario Analysis

VaR and Expected Shortfall are backward-looking: even the parametric and Monte Carlo versions are calibrated to a historical mean and standard deviation, so they assume the future will resemble the recent past. Stress testing asks a different question: not "how bad are the worst outcomes the historical distribution suggests," but "what would happen to this portfolio under a specific, named adverse scenario," whether

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historical (such as replaying the return pattern from the 2008 financial crisis or the 2020 COVID-19 market shock) or hypothetical (such as a sudden 200-basis-point rate shock or a 30% equity market decline).

Applying a historical stress scenario, a single-day return of -7%, comparable to some of the worst individual trading days during the 2008 crisis, to the \$10 million portfolio in Table 8.2 gives a stressed loss of

$$\text{Stressed Loss} = 0.07 \times \$10{,}000{,}000 = \$700{,}000,$$

nearly three times the parametric 99% VaR of \$250,081 computed in Section 8.2.2.

The gap does not mean the VaR calculation was wrong; it is the point of stress testing. VaR describes a quantile of an assumed distribution, while a stress scenario explores an outcome a short, calm sample may never have contained. Regulators require large institutions to run such exercises (in the United States, the Federal Reserve's Comprehensive Capital Analysis and Review, or CCAR, is a prominent example) because a model calibrated on recent, calm data can understate the likelihood and severity of a crisis, the non-stationarity concern of Chapter 7.

Standard stress testing, as above, starts from a scenario and asks how much the firm would lose; reverse stress testing runs the same logic backward, starting from a specific outcome, such as the firm's capital falling below the regulatory minimum or the firm becoming insolvent entirely, and asking what combination of market moves, defaults, or funding shocks would be required to produce it. This counteracts a subtle bias in forward stress testing: designers drawing on recent history and their intuition tend to produce scenarios that resemble past crises, while reverse stress testing forces a search over the outcomes the firm cannot survive, whether or not they have ever occurred, and asks the risk committee to judge how plausible they are.

Suppose the desk running the \$10 million portfolio of Table 8.2 has a risk capital cushion of \$500,000 to absorb its trading losses before it needs capital from elsewhere in the firm. Reverse stress testing asks what single-day return would exhaust it. Since Loss $= |r| \times \$10{,}000{,}000$, solving $500{,}000 = |r| \times 10{,}000{,}000$ gives $|r| = 5\%$, a single-day move of -5% or worse. Against the -7% historical scenario, the reverse test makes explicit what the forward scenario did not: a recurrence would not merely breach the cushion but exceed it by \$700,000 − \$500,000 = \$200,000, a capital shortfall rather than a limit breach, which a risk committee needs to know before such a day arrives.

Correlation Stress Testing: Quantifying "Correlations Go to One"

The case vignette below describes correlations rising sharply in a crisis; the two-asset position of Section 8.2.3 makes the claim numerical. Chapter 2's AssetA/AssetB covariance matrix implies a calm-period correlation of

$$\rho_{\text{calm}} = \frac{-0.000198}{\sqrt{(0.000414)(0.000118)}} \approx -0.896,$$

a strongly diversifying, negative relationship, which is why the equally weighted portfolio's volatility, 0.583%, is so much lower than either asset's own volatility ($\sigma_A \approx 2.03\%$, $\sigma_B \approx 1.09\%$) individually. Stress the correlation toward the crisis-period pattern described qualitatively in the vignette, holding each asset's own volatility fixed but replacing ρ_{calm} with a stressed $\rho_{\text{stress}} = 0.5$: the stressed covariance becomes $\rho_{\text{stress}}\sigma_A\sigma_B \approx 0.0001105$, and the resulting portfolio volatility is

$$\sigma_p^{\text{stress}} = \sqrt{w^\top \Sigma_{\text{stress}} w} \approx 1.372\%,$$

more than double the calm-period figure of 0.583%, so that $\text{VaR}_{95\%}^{\text{stress}} \approx 1.645(0.01372)(2{,}000{,}000) \approx \$45{,}141$ (\$45,137 at full precision), roughly 2.35 times the calm-period parametric VaR of \$19,182 computed earlier, with neither asset's own volatility having changed at all. This is the quantitative content of the vignette's warning: a diversification benefit that looks large under calm correlations can shrink dramatically, and portfolio risk more than double, because assets that moved independently or oppositely start moving together, the mechanism that made 2008 more dangerous for diversified portfolios than their pre-crisis models suggested.

8.5.2 Machine Learning Applications in Market Risk

Every technique so far is a classical statistical model, deliberately, because a risk number feeding regulatory capital and executive decisions must be transparent and auditable. Machine learning nonetheless enters

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market risk management in several increasingly standard ways, each extending rather than replacing these tools.

Volatility forecasting is the most direct: a recurrent neural network or gradient-boosted ensemble can in principle capture nonlinear volatility patterns a fixed GARCH(1,1) cannot, feeding the volatility-adjusted VaR machinery a more flexible forecast, at the cost of the overfitting and interpretability concerns of Chapter 6.

8.5.3 A Head-to-Head Test: GARCH(1,1) versus Gradient Boosting on Asymmetric Volatility

Chapter 7 flagged a well-documented shortcoming of GARCH(1,1): its recursion depends only on ε_{t-1}^2, so it treats positive and negative shocks of the same size alike, when in practice negative shocks raise volatility more (the leverage effect). A gradient-boosted tree given the lagged return's sign as a feature can in principle learn this asymmetry, which plain GARCH(1,1) cannot represent.

The companion notebook tests this directly: it simulates 750 days of returns from a known, asymmetric GJR-GARCH-style process ($\omega = 0.000005$, $\alpha = 0.05$, $\beta = 0.85$, with an extra variance term $\gamma = 0.12$ triggered only by negative shocks), fits a plain symmetric GARCH(1,1) by maximum likelihood on the first 600 days, and separately fits a gradient-boosted regressor on the same 600 days using lagged squared *and* signed returns as features, with hyperparameters chosen by time-series-aware cross-validation rather than an arbitrary guess.

Both models forecast one step ahead over the remaining 150 days, and because the true variance path is known (it is not observable in real markets, which is why the test needs simulated data), each forecast can be scored against it.

The result surprises only if one assumes flexibility always wins: the misspecified, symmetric GARCH(1,1) achieves a forecast-variance RMSE of 2.416×10^{-5}, beating the sign-aware, cross-validated gradient-boosting model's RMSE of 3.830×10^{-5} by 36.9%, even though the gradient-boosting model had direct access to the information (the shock's sign) that GARCH(1,1) cannot use. Gradient boosting still beats a naive constant-variance benchmark (4.202×10^{-5} RMSE) by 8.8%, so it learns something, but a correctly structured recursive model, propagating variance forward as the true process does, keeps a large advantage over a generic ensemble learning from lagged features, even one given the information it needs and carefully tuned. The result is useful and sobering: it is why volatility forecasting in practice augments GARCH-family models with richer inputs (implied volatility, order flow, cross-asset signals) rather than replacing the recursion with an off-the-shelf model trained on lagged returns.

A second application addresses a computational bottleneck. Full revaluation of a large options book becomes prohibitively slow once thousands of positions must be repriced at thousands of Monte Carlo scenarios, so some desks train a neural network as a fast pricing surrogate, learned once from a pricing model such as Black-Scholes-Merton or a full Monte Carlo engine and then queried at simulation speed.

A third, newer application uses generative models, the GANs and diffusion models of Chapter 16, to synthesize plausible stress scenarios beyond anything in a firm's historical window, addressing the limitation of Section 8.5.1: a sample that never included a severe crisis cannot suggest one, but a generative model of the structure of extreme moves can propose a novel, plausible one.

In each application the discipline is unchanged: a more flexible model is validated, backtested, and governed at least as rigorously as a classical one, because its flexibility makes an undetected flaw easier to introduce and harder to catch.

8.6 Governance and Managing the Risk

Measuring a risk is not the same as owning it; this part covers who validates the numbers and what is actually done about them.

8.6.1 Model Validation and Model Risk Management

Every method in this chapter, VaR, Expected Shortfall, GARCH-based volatility forecasting, liquidity adjustment, is a model in the sense of Chapter 1: a simplified representation of reality whose outputs are only as trustworthy as its assumptions. Model risk management governs this formally, subjecting every model

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used for a material financial decision to independent scrutiny before deployment and monitoring afterward, rather than trusting it because a competent team built it.

A mature model risk management framework, closely following supervisory guidance such as the U.S. Federal Reserve and OCC's SR 11-7 (superseded in April 2026 by the interagency SR 26-2, which keeps the same pillars (Federal Reserve, OCC, and FDIC 2026)), organizes this scrutiny around three activities. Independent validation requires that a model be reviewed by staff who did not build it, checking the soundness of its assumptions (does normality make sense for this portfolio's returns), its implementation (does the code compute what the specification says), and its outcomes (does backtesting, as in Section 8.3.1, confirm its performance). Ongoing monitoring tracks performance after deployment, since a model sound at launch can degrade as markets change, and catches the degradation before it causes a material loss. Model inventory and documentation keep a current record of every model, its owner, validation status, and known limitations, since a firm that does not know what models it has, or where they are used, cannot govern the risk they pose.

In practice, a firm with hundreds of models cannot validate all of them with equal intensity, so most model risk management frameworks tier models by materiality. A model whose output could plausibly affect more than $50 million of capital or reported risk might be classified Tier 1, requiring full independent validation and annual revalidation, while a narrowly scoped model with less than $5 million of potential impact might be Tier 3, revalidated only every two to three years. This chapter's portfolio VaR model, given its role in regulatory capital, would sit in the highest tier at any bank of meaningful size.

The same governance applies to Chapter 9's credit models, and it is what separates a well-run risk function from a collection of clever models. The methods of this chapter are necessary but not sufficient, since even a correct model can mislead dangerously when applied outside the conditions under which it was validated, the failure mode of this chapter's 2008 case vignette.

8.6.2 Managing the Risk, Not Just Measuring It

Every tool so far answers "how much could we lose?" None answers the question a desk head must act on: "what are we going to do about it?" A risk report that never changes a position is an expensive newsletter; measurement earns its keep when a number crosses a threshold and someone trades differently.

A firm can do four things with a risk it has measured. It can **accept** the risk, which is a legitimate decision when the exposure is small relative to capital and is being paid for. It can **reduce** it by cutting the position. It can **hedge** it by taking an offsetting position in a related instrument, keeping the exposure it wants and shedding the exposure it does not. Or it can **transfer** it to someone else through an insurance contract or a derivative, which Chapter 11 develops from the perspective of the party accepting such risks for a living. The rest of this section works through the two middle options, which require calculation.

8.6.3 Hedging: Removing an Exposure You Do Not Want

Return once more to the short call position of Section 8.4. Its delta of -611.5 says that the position loses $611.50 for every $1 the stock rises. A market maker who wants the option's time decay without a directional view eliminates that exposure by buying 611.5 shares. The hedged book's delta is zero, and the first-order price risk is gone.

What is left is the interesting part. Delta hedging removes the linear exposure, but the position is still short 25.55 units of gamma, the rate at which the hedge goes stale as the stock moves. At the 99% one-day move of $2.198 used in the delta-normal example, the delta-hedged position still loses

$$\tfrac{1}{2} \times 25.55 \times (2.198)^2 = \$61.72,$$

compared with the $1,403.99 full-revaluation loss unhedged. The hedge removes 95.6% of the 99% loss, a large improvement but not a complete one. Two features of the residual matter more than its size. It is quadratic, so doubling the shock roughly quadruples it. And it is *one-directional*: a short-gamma position loses whether the stock rises or falls, because the sign of ΔS disappears when squared. The hedge has converted a directional bet into a bet that the stock will not move much, a different risk rather than none.

That is the general shape of hedging: a hedge exchanges a risk you do not want for a smaller, differently shaped one you are willing to hold, and the residual is what the risk report must track.

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The residual is also the opening for a machine learning treatment, which Chapter 19 applies to this position. The calculation above assumes the hedge can be rebalanced continuously and for free. It cannot: every adjustment to the share position pays a bid-ask spread, so a hedger who rebalances often enough to keep delta near zero pays away the option premium in transaction costs, while one who rebalances rarely enough to keep costs down carries the residual gamma risk just computed. There is no closed-form answer to where the optimum sits, because it depends on the cost structure, the volatility, and how much tail risk the desk is willing to tolerate. Chapter 19 solves it as a sequential decision problem, finding a rebalancing band that beats daily delta hedging on both average cost and tail risk simultaneously, and then generalizes the band to a neural hedging policy trained directly on the distribution of outcomes. The classical delta hedge computed here is the baseline that work is measured against.

8.6.4 Position Sizing: Controlling Risk Before It Exists

Hedging acts on an existing position. The cheaper control acts before the position is put on, choosing its size so that the resulting risk is the risk the firm wanted.

Volatility targeting is the standard mechanical version. Take this chapter's \$10 million portfolio, whose daily return standard deviation of 0.937% was computed in Section 8.2.2. Scaling to an annual figure in the conventional way gives

$$\sigma_{\text{annual}} = 0.937\% \times \sqrt{252} = 14.87\%.$$

Suppose the mandate this portfolio is run under specifies a 10% annualized volatility target. The portfolio is running roughly half again as much risk as authorized, and the fix requires no view on any holding: scale the whole book by the ratio of target to realized volatility,

$$\frac{10\%}{14.87\%} = 0.672,$$

holding \$6.72 million in the risky portfolio and the remaining \$3.28 million in cash. Because volatility scales linearly with position size while the composition is unchanged, this hits the target exactly, without the estimation difficulties of Chapter 12's mean-variance optimizer.

The catch makes position sizing a forecasting problem rather than arithmetic. The scaling factor is a ratio of a target to an *estimate*, so the control is only as good as the estimate, and volatility is not constant, which is what Chapter 7's GARCH models capture. Sizing on a trailing standard deviation rather than a conditional forecast that responds to yesterday's shock is the difference between de-risking after a turbulent week and de-risking during it. Section 8.5.3 compares GARCH(1,1) with a gradient-boosted tree on a series with a known asymmetric volatility response, and the winner is the model that should drive the scaling factor. Volatility targeting turns a better volatility forecast directly into a better-sized position.

A volatility-targeted portfolio rescales continually, cutting exposure as markets become turbulent and adding it back as they calm. That is intended, but it means selling into falling markets mechanically, alongside every other volatility-targeted fund doing the same arithmetic on the same day, a mechanism for the correlated forced selling that Section 8.5.1 stress-tests for.

8.6.5 Stop-Losses and Drawdown Triggers, and Why They Disappoint

The last common control is the crudest: when losses reach a threshold, cut the position, whatever any model says. Stop-losses appeal because they need no distributional assumption and cannot be gamed by a misspecified covariance matrix.

They also underdeliver, as this chapter's data shows. Section 8.2.5 computed a maximum drawdown of 6.74% for the twenty-day portfolio, reached on the final day. Suppose the desk had operated a rule liquidating the portfolio whenever its drawdown from the running peak reached 5%. Tracking the drawdown day by day, the rule would not have fired until day 19, and liquidating at that point would have avoided only the final day's −1.49% return. Out of a 6.74% peak-to-trough decline, the stop-loss rescues about 149 basis points.

The reason is generic. A drawdown trigger lags by construction: it cannot fire until the loss has been taken, so it concedes the whole distance to the threshold plus whatever happens before the fill. In a series that recovers, the same rule crystallizes a loss that would have reversed, at the point of maximum pessimism. Stop-losses are a last-resort control against catastrophic tails and operational failure, not a technique that

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improves expected outcomes. The tools that reduce risk systematically are sizing the position correctly beforehand and hedging unwanted exposures.

8.6.6 Risk Governance: The Three Lines of Defense

Every method in this chapter exists inside an organizational structure that determines whether it is used correctly, and financial institutions typically organize risk oversight with a "three lines of defense" model, illustrated in Figure 8.5.

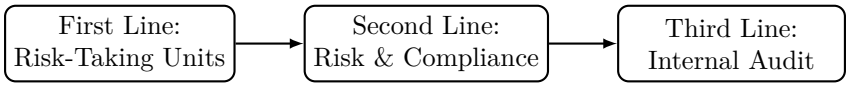


Figure 8.5: The three lines of defense in risk governance

The first line of defense is the business unit that takes the risk, a trading desk or a lending team, and is responsible for managing it day to day within approved limits. The second line is an independent risk management and compliance function, organizationally separate from the business units it oversees, that builds and maintains models like the VaR and economic capital methods introduced in this chapter and the credit-scoring methods of Chapter 9, sets risk limits, and monitors whether the first line is operating within them. The third line, internal audit, periodically and independently verifies that the first two lines function as intended, checking not just whether a number was computed correctly but whether the whole process was followed. The structure prevents the single point of failure that arises when the people earning revenue from a position also measure and report its risk, a conflict implicated in many historical risk management failures.

A model is only as trustworthy as the governance around it: a perfectly specified VaR model computed by a risk function with no real independence from the desk it monitors offers little protection. The second line is typically led by a Chief Risk Officer (CRO) with direct access to the board's risk committee, independent of the chief executive and business-line management, so that a concern can reach the board without being filtered by the managers it reflects on. This independence is not a formality. Several well-documented failures trace back to a risk function that identified a problem but lacked the standing to have it acted on before it caused a material loss, which is why governance codes and regulators treat CRO independence, board-level risk reporting, and the second line's authority to veto or unwind a position as essential.

A limit example shows how the three lines interact day to day. Suppose the second line has approved a daily 99% VaR limit of \$300,000 for the desk running the \$10 million portfolio from Table 8.2. The parametric 99% VaR of \$250,081 computed in Section 8.2.2 represents a limit utilization of $250{,}081/300{,}000 \approx 83.4\%$, well within the limit, so the first line trades normally and the second line's monitoring is routine. Now suppose realized volatility rises sharply enough, by the volatility-adjusted VaR logic above, that the next day's VaR is \$340,000, a breach of \$40,000, or roughly 13.3% over the limit. This is the event the three lines exist to catch. The breach is reported to the second line the same day, which assesses whether it reflects a temporary spike in volatility or a change in the desk's risk profile that warrants an immediate reduction. A breach that recurs or is not promptly remediated is what the third line's audit is designed to flag, even if the second line's daily response was adequate.

8.7 Case Vignette: When Value at Risk Meets a Genuine Crisis

The 2008 financial crisis is the standard cautionary tale about the limits of this chapter's methods, and the failure was not one of arithmetic. Many large banks entered 2008 with VaR models built and validated as this chapter describes: parametric or historical methods, calibrated on several years of historical returns, backtested against the kind of exception-counting procedure in Section 8.3.1, and generally passing those backtests during the calm years leading up to the crisis. The most famous symptom came early, in the August 2007 "quant quake" that preceded the main crisis. Explaining heavy losses at Goldman Sachs's own quantitative hedge funds, the firm's chief financial officer, David Viniar, told the *Financial Times*: "We were seeing things that were 25-standard deviation moves, several days in a row." A normal distribution gives a

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single 25-standard-deviation move a probability so small that it should not occur once in the history of the universe, let alone on several days in a row.

The arithmetic was not wrong; the normality assumption of Section 8.2.2 was a poor description of returns in a crisis, the fat-tails caveat of this book. A model calibrated on a multi-year window without a severe crisis will underestimate extreme moves, since it has never observed one, the finite-sample problem behind this chapter's caution about short backtesting windows. Moreover, correlations between asset classes that had looked low or stable rose sharply, a pattern summarized as "correlations go to one in a crisis," so the diversification assumed by portfolio VaR, including the variance-covariance approach of Section 8.2.3, protected far less than historical correlations suggested.

The lesson is not that VaR or Expected Shortfall are poor tools; both remain required practice at every major institution. It is that each method answers its question only as well as its assumptions hold, so stress testing, backtesting, and the judgment of an independent second line are necessary complements to any single statistical model.

The regulatory response shaped the landscape this chapter describes. The Basel Committee's Fundamental Review of the Trading Book (FRTB), published in January 2016 and revised in January 2019 (BCBS 2016b), but still only partly in force in October 2026 (Table 8.6), replaced VaR with Expected Shortfall as the basis for market-risk capital calculations. ES, unlike VaR, is sensitive to the severity of losses beyond the threshold, not only their frequency, which addresses the "25-standard-deviation event" problem: a VaR-based charge is blind to how much worse the tail is, an ES-based charge is not. FRTB also requires calibration to a period of significant historical stress rather than only recent, possibly calm data, and introduces desk-level model approval so that a bank-wide validation can no longer mask weaknesses in one desk's risk measurement. None of this makes the chapter's methods obsolete; it holds them to a more demanding, crisis-aware standard than the pre-2008 VaR framework.

Four FRTB mechanics go beyond what this chapter builds. First, the internal-model measure is Expected Shortfall at 97.5%, not 99%: under normality the two give almost the same number, since the 99% VaR multiplier is $z_{0.99} = 2.326$ while the 97.5% ES multiplier is $\phi(z_{0.975})/0.025 = 2.338$, so the change of confidence level keeps the capital level roughly where it was in calm, Gaussian conditions and charges more only where the tail is fatter than normal. Second, the single 10-day horizon of the old rules gives way to liquidity horizons of 10, 20, 40, 60, or 120 days by risk factor, so that a position in an illiquid credit spread is capitalized over the time it would actually take to exit. Third, a desk keeps internal-model approval only if it passes, alongside backtesting, a profit-and-loss attribution test showing that the risk model's own P&L tracks the front office's pricing P&L closely. Fourth, risk factors without enough real price observations to model, the non-modellable risk factors, are carved out of the ES model and charged a separate stress-scenario add-on. National implementation has lagged the standard badly, as Table 8.6 records.

8.8 Challenges in Market Risk Management

Several challenges recur across nearly every market risk management application, and Chapter 9 revisits several of them in a credit-risk setting.

Model risk in the risk model itself. Every method in this chapter, parametric VaR's normality, the GARCH recursion's parametric structure, the delta-normal method's linearity, rests on assumptions that can fail when they matter most, in a crisis, because crises are unusual events a model calibrated on typical conditions was not designed to capture. The gap between a linear approximation and full revaluation, small for a modest move, widens when the move is largest.

Correlation and tail dependence. Individual risk measures can look reasonable in isolation, but a portfolio becomes dangerous when many positions move together at the worst time, a pattern called tail dependence. The case vignette showed it: correlations that looked low and stable rose sharply in 2008, eroding the diversification the variance-covariance VaR of Section 8.2.3 assumed. Estimating tail dependence directly, rather than assuming a stable correlation matrix, is statistically difficult and highly consequential.

Procyclicality. Risk measures calibrated on recent data tend to show low risk during calm periods (encouraging more risk-taking and leverage) and high risk immediately after a shock has already occurred (forcing deleveraging precisely when the system can least afford it), an amplifying feedback loop implicated in several financial crises that no purely statistical measure resolves.

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Jurisdiction	FRTB status as of October 2026
Basel standard	Published January 2016, revised January 2019; Basel implementation date 1 January 2023
Canada, Japan	In force since November 2023 (Canada, the first jurisdiction) and March 2024 (Japan's banks)
Singapore, Hong Kong	In force since the start of 2025
European Union	Own-funds requirements postponed twice by delegated act, to 1 January 2027; a June 2026 delegated act, still under scrutiny by Parliament and Council, adds a temporary capital multiplier and operational relief for 2027–2030
United Kingdom	Standardized approach from 1 January 2027 with the rest of Basel 3.1; internal model approach from 1 January 2028
United States	2023 proposal withdrawn; re-proposed 19 March 2026 (comments closed 18 June 2026); no final rule

Table 8.6: Implementation of the FRTB market-risk standard by jurisdiction, as of October 2026 (BCBS 2020a; European Commission 2025; European Commission 2026; PRA 2026; Board of Governors of the Federal Reserve System et al. 2026)

Data quality and limited loss history. Severe crashes are fortunately rare, so the data for estimating their probability and severity are limited, the small-sample problem illustrated by this chapter's twenty-observation VaR and backtesting examples.

Regulatory and business tension. Risk models live in an organization that wants to grow revenue: a risk function that is too conservative can be overridden or under-resourced, one that is too permissive can miss a crisis, and the balance is as much a governance challenge as a statistical one.

Regulatory arbitrage. Whenever capital is calculated from a bank's own reported VaR, as under the Basel multiplier, the bank has an incentive to minimize the capital charge rather than the risk, by choosing a favorable window, confidence level, or specification that backtests well. The gap between measured and true risk can widen quietly even when every calculation is correct.

Limit-setting on a diversification-sensitive measure. Component VaR's dependence on the book's correlation structure cuts both ways. It rightly rewards diversifying positions with a smaller charge, but a desk near its limit has an incentive to seek positions that look diversifying under calm-period correlations, the positions Section 8.5.1's correlation stress test warns can start compounding losses when correlations move against the book in a crisis.

Machine learning adds a further layer of model risk. A neural network or gradient-boosted model can in principle forecast volatility or tail risk more flexibly than GARCH or a historical window, but it inherits Chapter 6's overfitting and interpretability concerns where an undetected error is unusually costly. An opaque VaR model that a regulator or validator cannot interrogate is hard to trust with a bank's capital requirement, whatever its backtested accuracy.

8.9 Key Takeaways

Market risk management complements the prediction-focused methods of earlier chapters by asking how bad outcomes can get, not just what is expected to happen. Value at Risk and Expected Shortfall summarize the downside of a return distribution by parametric, historical, or Monte Carlo methods, each trading distributional assumptions against finite-sample noise. Expected Shortfall is preferred because it accounts for the severity of losses beyond the threshold, though backtesting severity is harder than counting VaR exceptions.

Volatility need not be fixed: GARCH lets VaR update immediately after a large move, filtered historical simulation adds the historical method's distribution-free shape, and the delta-normal approximation

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understates options risk in proportion to the gamma it ignores, a gap the delta-gamma correction nearly closes.

Component VaR decomposes total risk into per-position contributions that depend on the book's correlation structure, which is why pushing correlations toward the crisis pattern can more than double measured risk with no position's own volatility changing.

No aggregate should be the only thing anyone looks at. Sensitivities sit alongside VaR because a single number discards *where* the risk is: a rate book can report a net DV01 of exactly zero while carrying a $1.36 million exposure to a curve flattening, and an options position's VaR can be computed correctly to the cent while remaining structurally blind to a routine move in implied volatility. Producing those sensitivities at scale is solved by reverse-mode automatic differentiation, the algorithm that trains neural networks.

The banking book carries the same interest-rate risk under a different accounting. A bond portfolio held to maturity is not marked to market in earnings or, for many banks, in regulatory capital, but its loss is real: a 300-basis-point rise took 17.68% off the stylized bank's seven-year bonds and left it $0.16 billion of economic equity out of $9 billion, while its net-interest-income measure showed a gain. Economic value of equity and net interest income measure the two sides of that exposure, and both rest on behavioral assumptions about depositors, their beta and effective duration, that alone can turn a supervisory outlier into a comfortable pass. Silicon Valley Bank showed what happens when concentrated, uninsured depositors test those assumptions: a loss that had been building for a year was forced into the open in two days.

Measurement is not management. Hedging exchanges an unwanted exposure for a smaller, differently shaped residual; position sizing controls risk before it is taken and is only as good as its volatility forecast; and stop-losses lag by construction and rescue far less of a drawdown than intuition suggests. None of these is self-policing: backtesting (including Basel's traffic-light zones), independent validation, and governance such as the three lines of defense are what catch a miscalibrated model before it causes a loss.

As in Chapter 5, a risk number is only as trustworthy as its assumptions, and sound practice combines several methods and an account of where they disagree rather than relying on any single measure. Chapter 9 now turns to the analogous toolkit for credit risk.

8.10 Suggested Exercises

8.1 Explain, in your own words, the difference between Value at Risk and Expected Shortfall, and describe a situation in which two portfolios could have identical VaR but very different Expected Shortfall.

8.2 Using the returns in Table 8.2, compute the parametric 90% VaR (using $z_{0.90} = 1.282$) for the $10 million portfolio, and compare it to the 95% and 99% figures computed in the chapter.

8.3 Using the historical method, compute the 90% VaR and Expected Shortfall from Table 8.2 (following the convention of Section 8.2.2, which reads the 95% VaR from the second-worst return, leaving one observation beyond it, the 90% VaR leaves two observations beyond it and so corresponds to the third-worst return out of twenty).

8.4 Using Table 8.2's cumulative value path and the definition in Section 8.2.5, compute the drawdown at Day 6 (relative to the running peak established by that point) and compare it to the chapter's maximum drawdown of 6.74%, realized only at Day 20. Explain why a portfolio can pass through a large intermediate drawdown that is later exceeded by an even larger one, rather than the path recovering.

8.5 Using Chapter 7's GARCH(1,1) recursion values at $t = 2$ ($\sigma = 2.03\%$) and $t = 5$ ($\sigma = 1.99\%$), compute the parametric 95% VaR of a $5 million position at each date, and explain why the two figures are so much closer together than the $t = 3$-versus-$t = 4$ comparison in the text.

8.6 Explain why a portfolio's 99% VaR can be smaller than its loss under a stress-test scenario, and why this is not evidence that the VaR calculation is wrong.

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8.7 A bank's VaR model produces 7 exceptions over a rolling 250-day window at the 99% confidence level. Using the Basel traffic-light framework, classify this outcome and explain what supervisory consequence it would typically trigger.

8.8 Using this chapter's delta-normal example, recompute the 95% and 99% delta-normal and full-revaluation VaR for a position that is instead *long* 1,000 calls, and explain which direction of stock-price move now represents the position's worst case.

8.9 Pick two of the challenges listed in this chapter and describe a concrete scenario, not already used in the text, in which that challenge could cause a risk management system to understate a firm's true risk.

8.10 Using Section 8.2.3's two-asset covariance matrix, recompute the portfolio's volatility and 95% VaR under a stressed correlation of $\rho = 0.8$ instead of the $\rho = 0.5$ used in the text, and compare the resulting increase to the one computed in Section 8.5.1.

8.11 A bank has \$60 million in high-quality liquid assets and projected net cash outflows of \$55 million over a 30-day stress scenario. Compute its LCR, and explain why a bank could satisfy the LCR comfortably while still having an inadequate NSFR.

8.12 A third asset, AssetC, is added to Section 8.2.3's two-asset position with a Component VaR of \$4,000. Using the fact that Component VaRs must sum to the portfolio total, explain what this implies about how AssetC is correlated with the existing AssetA/AssetB position, without needing to know AssetC's own standalone volatility.

8.13 Using this chapter's three-lines-of-defense limit example, compute the limit utilization if the approved daily VaR limit were instead \$275,000 (holding the parametric 99% VaR of \$250,081 fixed), and explain whether a subsequent rise in VaR to \$290,000 would count as a limit breach under this tighter limit.

8.14 Using this chapter's reverse-stress-testing example, compute the single-day return that would exhaust a risk capital cushion of \$800,000 instead of \$500,000 for the same \$10 million portfolio, and explain whether the -7% historical stress scenario from Section 8.5.1 would still exceed this larger cushion.

8.15 Explain, using this chapter's FRTB discussion, why a capital framework based on Expected Shortfall is less vulnerable than a VaR-based framework to a repeat of the "25-standard-deviation event" problem described in the 2008 case vignette.

8.16 Using the square-root-of-time rule from this chapter's economic-capital discussion, compute the 5-day regulatory VaR implied by the parametric 99% VaR of \$250,081, and explain why this figure is smaller than the 10-day figure computed in the chapter even though both are scaled from the same 1-day VaR.

8.17 This chapter's economic-capital discussion showed that a modest positive autocorrelation of $\phi = 0.10$ makes the square-root-of-time rule understate 10-day VaR by about 9.4%. Explain, in your own words, why a *negative* autocorrelation (mean reversion) in daily returns would instead make the square-root-of-time rule *overstate* multi-day risk, and describe a market condition in which returns might plausibly exhibit this pattern.

8.18 Using the vega in Table 8.4, compute the profit or loss on the short 1,000-call position if implied volatility rises by three percentage points while the stock price does not move at all, and express it as a percentage of the position's 95% full-revaluation VaR of \$980.79 from the delta-normal example. Explain why no amount of backtesting the VaR model would have revealed this exposure.

8.19 Apply a *steepening* scenario to the DV01 ladder of Table 8.5, in which two-year yields fall 10 basis points, five-year yields fall 5, ten-year yields rise 10, and thirty-year yields rise 15. Compute the desk's profit or loss, and explain why the sign of the result is the opposite of the flattening scenario worked in the text even though the net DV01 is zero in both cases.

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8.20 A trading book is exposed to 500 distinct risk factors. (a) How many full valuations are required to produce a complete set of first-order sensitivities by bump-and-revalue? (b) Roughly how many are required by adjoint algorithmic differentiation, and why does this figure not depend on the number 500? (c) Explain why gamma, unlike delta, is not obtained for free in the same reverse sweep, and name the machine learning computation that faces the same first-order-versus-second-order cost distinction.

8.21 Using Section 8.6.4's volatility-targeting calculation, recompute the dollar amount to hold in the risky portfolio (a) if the mandate's target volatility were 15% rather than 10%, and (b) if the daily volatility estimate were instead Chapter 7's GARCH post-shock forecast of 2.08%, holding the 10% target. Comment on the size of the position change in (b) and on what it implies about how often a volatility-targeted portfolio must trade.

8.22 Show that the residual loss on a delta-hedged short-gamma position grows quadratically, by recomputing Section 8.6.3's figure of \$61.72 for a stock move twice as large as the \$2.198 used there. Explain why the residual is a loss for *both* a rise and a fall in the stock price, and what this implies about the kind of bet a delta-hedged option seller is really making.

8.23 Using the stylized bank of Section 8.3.5: (a) estimate the loss on its securities portfolio from a 200-basis-point rise in yields, first with duration alone and then with duration and convexity, and express each as a percentage of the bank's \$9 billion of equity; (b) find the effective deposit duration at which the +200 basis point ΔEVE of Equation (8.10) exactly reaches the supervisory outlier threshold of 15% of Tier 1 capital, and discuss what your answer implies about how much of the outlier test's verdict rests on the bank's deposit study rather than on its balance sheet.

8.24 For the same bank: (a) find the deposit beta at which the +300 basis point change in net interest income from Equation (8.11) is exactly zero; (b) recompute the securities sale, the realized loss, and the remaining book equity if 40% of deposits, rather than 25%, are withdrawn; (c) explain why a board that monitors only net interest income, as Silicon Valley Bank's did, can approve a balance sheet that is close to economic insolvency.

8.25 Repeat Section 8.2.8's Student-t calculation with $\nu = 6$ instead of $\nu = 4$. (a) Compute the scale s and the 99% VaR and Expected Shortfall of the \$10 million portfolio with Equation (8.6), and compare them with Table 8.3. (b) Find the confidence level at which the $\nu = 6$ VaR crosses the normal VaR, and compare it with the 97.47% crossing for $\nu = 4$. (c) Compare the $\nu = 4$ and $\nu = 6$ VaR at 95% and at 99.9%, and explain why the choice of ν, which twenty days of data cannot settle, matters little for the first and a great deal for the second.

8.26 For the stylized 30-year gilt of Section 8.4.4: (a) find the yield rise that wipes out a pooled LDI fund levered three times; (b) compute the collateral call on a fund with £1 billion of capital levered four times when the 30-year yield rises 67 basis points, the move between the morning and the evening of 27 September 2022, both exactly and with duration alone; (c) compute the largest leverage that survives a 250-basis-point rise using duration alone, compare it with the exact 2.91 times, and explain which way the duration approximation errs and why this error is conservative for a long bond position.

8.27 **Challenge (real data).** Download at least two years of daily prices for a small portfolio of real stocks (for example, three to five components of a major index, via `yfinance`) and compute the portfolio's daily returns using fixed weights of your choosing. (a) Compute the portfolio's parametric and historical 99% one-day VaR, as in Section 8.2.2, and compare the two estimates. (b) Backtest your parametric VaR estimate over the full sample using Kupiec's test, referenced in Section 8.3.1: count the number of days on which the realized loss exceeded the VaR estimate, and determine whether the exception rate falls inside the Basel traffic-light green zone for your sample size. (c) Real portfolio returns are typically both fat-tailed and volatility-clustered, unlike the twenty-observation toy example this chapter builds on; discuss whether your backtest results suggest the parametric (normal-distribution) VaR model over- or understates risk relative to the historical method, and connect this to this chapter's own discussion of GARCH-based, volatility-adjusted VaR.

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8.11 Further Reading

- Jorion (2007), *Value at Risk*. The standard reference for VaR methodology, covering the parametric, historical, and Monte Carlo approaches introduced in this chapter in much greater depth and rigor.
- Hull (2023), *Risk Management and Financial Institutions*. A broad treatment of market, credit, and operational risk within financial institutions, extending this chapter's introductory coverage of each risk type.
- McNeil et al. (2015), *Quantitative Risk Management*. A mathematically rigorous treatment of coherent risk measures, Expected Shortfall, and the tail-dependence and correlation issues raised in this chapter's challenges section.
- BCBS (2010), *Basel III: A Global Regulatory Framework for More Resilient Banks and Banking Systems*. The primary regulatory source for the capital adequacy, backtesting, and liquidity-coverage concepts introduced in this chapter. Its market-risk successor, BCBS (2016b), *Minimum Capital Requirements for Market Risk*, is the standard produced by the Fundamental Review of the Trading Book: it replaced Value at Risk with Expected Shortfall as the regulatory measure and introduced the sensitivities-based capital charge that this chapter's treatment of sensitivity-based risk limits develops.
- Giles and Glasserman (2006), "Smoking Adjoints: Fast Monte Carlo Greeks." The paper that introduced adjoint differentiation to derivatives risk management, and the direct source for Section 8.4.3's claim that a full set of sensitivities costs a small constant multiple of a single valuation.
- Baydin et al. (2018), "Automatic Differentiation in Machine Learning: A Survey." A clear and self-contained account of forward- and reverse-mode automatic differentiation, written for the machine learning audience, making explicit the equivalence with backpropagation that Section 8.4.3 relies on.
- Crouhy et al. (2014), *The Essentials of Risk Management*. A practitioner-oriented synthesis spanning market, credit, and operational risk, model validation, and risk governance, tying together the full breadth of topics covered in this chapter and Chapter 9 in a single volume.
- BCBS (2016a), *Interest Rate Risk in the Banking Book*. The Basel standard behind Section 8.3.5's supervisory outlier test, setting out the six standardized rate-shock scenarios, the treatment of non-maturity deposits, and the disclosure requirements for EVE and NII sensitivity.
- Board of Governors of the Federal Reserve System (2023), *Review of the Federal Reserve's Supervision and Regulation of Silicon Valley Bank*. The Federal Reserve's own account of Silicon Valley Bank's failure and of the supervisory response to it, and the primary source for most of the vignette in Section 8.3.6, including the bank's EVE limit breaches and its change to the assumed duration of its deposits.
- Jiang et al. (2024), "Monetary Tightening and U.S. Bank Fragility in 2023: Mark-to-Market Losses and Uninsured Depositor Runs?" Marks the whole U.S. banking system's assets to market after the 2022 rate rises and models how unrealized losses and uninsured deposits combine into run risk, generalizing the worked example of Section 8.3.5 from one stylized bank to the entire U.S. banking system.

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Chapter 9

Credit Risk Modeling

9.1 Introduction

Chapter 8 developed the standard toolkit for market risk: Value at Risk, Expected Shortfall, and the volatility and correlation machinery behind them. This chapter turns to a related but distinct problem: credit risk, the risk that a counterparty (a bond issuer, a loan borrower, or a derivatives counterparty) fails to meet a contractual obligation. Where market risk asks how much a position's *value* could move, credit risk asks whether a counterparty will *pay at all*. That is a binary event with its own probability, its own severity, and its own regulatory and modeling apparatus: probability of default, loss given default, credit scoring, structural models, and rating systems.

The chapter proceeds from the single borrower to the whole portfolio. It begins with the three-component decomposition of expected loss and what drives each component. It then covers three distinct ways of estimating a borrower's probability of default (a statistical model fit to historical data, a structural model that reads default risk out of equity prices, and a market-implied approach that reads it out of credit derivative spreads) and extends single-borrower ratings into multi-year transition matrices. Along the way it treats climate change as a credit stress scenario and, once portfolio correlation is in place, the credit risk of derivatives, whose exposure is itself random. It closes with the portfolio, where the correlation between borrowers' defaults, not just their individual probabilities, drives extreme loss, illustrated by a worked single-factor model and by a historical episode in which underestimating that correlation contributed to a systemic financial crisis.

One worked example runs through much of the chapter: a small loan portfolio that reuses the credit-risk-adjusted bond from Chapter 5 and the loan dataset from Chapter 6.

9.2 Measuring Credit Risk

Credit risk decomposes into a probability, an exposure, and a loss given that it happens; this part builds each and scores a borrower.

9.2.1 Credit Risk Fundamentals: Expected Loss

Suppose a bank holds a bond and wants a single number summarizing how much of that exposure it should expect to lose to default, on average across many similar bonds, before knowing whether this one defaults. Credit risk is conventionally decomposed into three components whose product answers this, giving the expected loss on a credit exposure:

$$\text{Expected Loss} = \text{PD} \times \text{LGD} \times \text{EAD}, \tag{9.1}$$

where PD is the probability of default over the relevant horizon (typically one year), LGD is the loss given default, the fraction of the exposure not recovered after default, and EAD is the exposure at default, the

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dollar amount outstanding at the time of default (which can differ from the current balance for instruments like credit lines that can be drawn down further before default).

Recall the corporate bond from Chapter 5's discussion of pricing default risk: face value \$1,000, one-year default probability PD = 3%, and a recovery rate of 40% of face value, so LGD = 1 − 0.40 = 60%. Treating the full face value as the exposure at default, EAD = \$1,000, the expected loss is

$$\text{Expected Loss} = 0.03 \times 0.60 \times 1{,}000 = \$18.$$

Figure 9.1 lays out the decomposition as a pipeline; the rest of the chapter estimates each input in turn, first PD (via logistic regression and, later, the Merton model), then LGD, then EAD.

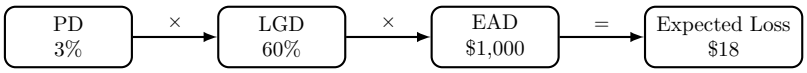


Figure 9.1: The expected-loss decomposition, applied to the Chapter 5 bond example

This matches Chapter 5's earlier calculation: the bond's expected payoff was \$982 against a promised \$1,000, and 1,000 − 982 = 18, so discounting the expected payoff and computing an expected loss are two equivalent descriptions of the same credit risk.

The two framings differ in purpose rather than substance. The pricing framing asks "what should this bond be worth today." The expected-loss framing asks "how much should a lender expect to lose, on average, and how much capital or loan-loss reserve should be held against that possibility," the accounting and regulatory question addressed later in this chapter.

EAD is the least discussed of the three components, but it deserves its own worked example, since it is not simply "the current balance" whenever a borrower has undrawn credit. Consider a committed revolving credit line with a \$500,000 total commitment, currently drawn to \$200,000, leaving \$300,000 undrawn. Many borrowers draw down a credit line further as their condition deteriorates, often because they are approaching default and want to secure cash while they can, so exposure at default must include some fraction of the undrawn commitment.

Basel's standardized approach captures this with a credit conversion factor (CCF), the assumed fraction of the undrawn commitment that converts to drawn exposure by the time of default; for an unconditionally cancellable commercial credit line, a CCF of 50% is a representative assumption, giving

$$\text{EAD} = \text{Drawn} + \text{CCF} \times \text{Undrawn} = 200{,}000 + 0.50(300{,}000) = \$350{,}000,$$

75% higher than the \$200,000 currently drawn. That is why credit lines and other commitment-based products carry more expected loss, for a given PD and LGD, than a fully drawn term loan of the same balance. Banks sophisticated enough to model EAD directly, rather than relying on a regulatory CCF table, estimate it from their own history of drawn-versus-committed balances at past defaults.

9.2.2 What Drives Loss Given Default

PD receives the most modeling attention, since it is most directly tied to the statistical and machine learning methods of Chapter 6, but LGD is just as consequential for expected loss and harder to model, since recovery outcomes depend on factors observed only after a default has occurred.

Seniority and collateral are the two most important drivers. A secured loan backed by specific collateral (real estate, equipment, inventory) typically recovers far more in default than an unsecured loan to the same borrower, since the lender has a direct claim on identifiable assets rather than competing with every other unsecured creditor. Seniority matters similarly: senior debt is repaid before subordinated debt in a bankruptcy, so a subordinated bondholder can recover substantially less than a senior bondholder of the same defaulted issuer, despite facing the same probability of default. Recovery rates are also cyclical in a way that compounds portfolio credit risk. In a recession, defaults become more frequent (PD rises) at the same time that collateral values fall and distressed-asset markets fill with sellers, so LGD rises too. PD and LGD are therefore positively correlated across the credit cycle, not independent as this chapter's simplified expected-loss and economic-capital formulas assume for tractability. A lender that estimates LGD as a

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full-cycle average, rather than conditioning it on current or forecast conditions, will understate losses during the downturns when accurate estimates matter most.

Collateral coverage, the ratio of collateral value to loan balance, is estimable from a lender's own default history, and a simple linear regression illustrates the pattern numerically. Consider four hypothetical defaulted loans:

$$\begin{aligned} &\text{Coverage} = 0.40, \quad \text{Recovery} = 25\%, \\ &\text{Coverage} = 0.60, \quad \text{Recovery} = 45\%, \\ &\text{Coverage} = 0.80, \quad \text{Recovery} = 55\%, \\ &\text{Coverage} = 1.20, \quad \text{Recovery} = 75\%. \end{aligned}$$

Fitting Recovery $= a + b\,(\text{Coverage})$ by ordinary least squares,

$$\hat{b} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = 0.60, \qquad \hat{a} = \bar{y} - \hat{b}\bar{x} = 0.05,$$

so $\widehat{\text{Recovery}} = 0.05 + 0.60\,(\text{Coverage})$, with $R^2 \approx 0.969$, a strong linear fit given only four points.

A loan with coverage exactly at 1.0 (collateral value equal to the outstanding balance) has a predicted recovery of $0.05 + 0.60(1.0) = 65\%$, implying LGD $= 35\%$. That is meaningfully lower than the 60% LGD assumed for this chapter's unsecured bond: the collateral effect made numerical. A lender with enough default history can estimate the relationship directly, applying regression to the loss side of credit risk rather than the default-probability side.

The high R^2 deserves the caution Chapter 6 raised about small samples, and the slope's standard error, computed as for that chapter's credit-spread regression, makes it concrete. With only two degrees of freedom (four observations, two estimated parameters), $\text{SE}(\hat{b}) \approx 0.0756$, giving a t-statistic of $0.60/0.0756 \approx 7.94$ and a 95% confidence interval for the slope of roughly $(0.27, 0.93)$. The slope is still significant at conventional levels ($p \approx 0.016$), but the confidence interval is enormous relative to the point estimate: any true slope from 0.27 to 0.93 is consistent with the data. Propagated through, the predicted recovery at a coverage ratio of 1.0 could plausibly range from roughly 32% to 98%, an enormous range for a single reported LGD figure. A lender relying on a regression estimated from a handful of defaults should therefore treat the fitted line as a starting point, and collect more default history or borrow estimates from a broader industry dataset before using it for capital or pricing.

9.2.3 Credit Stress Testing: Stressed PD and LGD

Credit stress testing makes this procyclicality concrete. Rather than assuming the base-case PD and LGD estimated from calm, average conditions, a stress test recomputes expected loss under a specified adverse scenario (a recession, a sector downturn, a sharp decline in collateral values) and asks how much larger it becomes.

Applying a stress scenario to the same bond from Section 9.2.1, raising the default probability from 3% to a stressed PD = 8% (about 2.7 times its base-case value) and raising loss given default to a stressed LGD = 70% (reflecting weaker collateral values in a downturn, the mechanism Section 9.2.2 described), the stressed expected loss is

$$\text{Expected Loss}^{\text{stress}} = 0.08 \times 0.70 \times 1{,}000 = \$56,$$

more than triple the base-case expected loss of \$18, although neither input moved implausibly on its own. That is why treating PD and LGD as independent, average-cycle constants can mislead capital planning: a lender reserving only against calm-period expected loss will be under-reserved in a recession, when the shortfall matters most. Regulatory exercises such as the Federal Reserve's CCAR, introduced from the market-risk side in Chapter 8, apply the same stressed-PD-and-LGD logic across a bank's entire loan book under a specified adverse macroeconomic scenario.

9.2.4 Climate Risk in Credit Portfolios

Climate change is the stress scenario that supervisors have most actively added to that list over the past decade, and it reaches a loan book through two distinct channels. *Physical risk* is the damage done by the

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climate itself: acute events such as floods, hurricanes, and wildfires, and chronic shifts such as rising sea levels and heat stress. It destroys collateral (raising LGD) and interrupts borrowers' revenue (raising PD), both in geographic clusters, so the losses arrive together rather than diversifying. *Transition risk* is the damage done by the move to a low-carbon economy: carbon prices, emissions regulation, cheaper clean technology, and shifting customer preferences erode the profitability of carbon-intensive borrowers, and they can *strand* the assets that secure their loans, oil reserves that will never be extracted and coal plants retired before the end of their economic life. Neither channel is a new kind of risk in the sense of this chapter. Both arrive as changes in PD, LGD, and EAD, which is why supervisors treat climate as a *driver* of existing credit, market, and operational risks rather than as a fourth category with its own capital charge.

What makes climate risk hard is that the statistical approach of the rest of this chapter has nothing to estimate from. The horizon runs to decades, and no historical sample contains an economy-wide carbon price at the levels policy scenarios contemplate, so no default history can say how borrowers respond. Supervisors have therefore built the discipline around scenario analysis. The Network for Greening the Financial System (NGFS), a network of central banks and supervisors, publishes reference scenarios grouped into orderly paths such as Net Zero 2050, disorderly paths such as Delayed Transition, and "hot house world" paths such as Current Policies, in which transition risk stays low and physical risk grows. Its fifth vintage, released in November 2024, estimated that a carbon price of around $300 per tonne of CO_2 would be needed by 2035 to put the world on a net-zero path by 2050 (NGFS 2024). The Bank of England's 2021 Climate Biennial Exploratory Scenario ran UK banks and insurers through three 30-year scenarios and published its results in May 2022 (Bank of England 2022). The European Central Bank's 2022 climate risk stress test covered 104 banks, and the 41 that ran its detailed bottom-up module projected about €70 billion of combined credit and market losses, a figure the ECB itself described as a substantial understatement (ECB 2022). A joint ECB and European Supervisory Authorities "Fit-for-55" scenario analysis followed in November 2024. In the United States, the Federal Reserve ran a pilot exercise with six large banks in 2023 and published a summary in May 2024 with no capital consequences (Board of Governors of the Federal Reserve System 2024), and in October 2025 the federal banking agencies withdrew their interagency climate risk-management principles for large banks altogether (Board of Governors of the Federal Reserve System et al. 2025). In Canada, OSFI's Guideline B-15 on climate risk management took effect for the largest banks from fiscal year-end 2024 (OSFI 2023). As of October 2026, then, climate scenario analysis is a firm supervisory expectation in Europe, the United Kingdom, and Canada, and a matter left to the general safety-and-soundness standards in the United States. None of these exercises sets capital directly; they are exploratory, reflecting how immature the modeling still is.

The modeling problem has a standard shape. A scenario supplies a path of macroeconomic and policy variables (a carbon price, energy prices, GDP by sector); a *transmission model* translates that path into each borrower's financial statements; and the credit models of this chapter translate the statements into PD and LGD. The judgment lives in the transmission step. A simple and transparent version measures the carbon cost a borrower would bear as a share of its earnings,

$$c = \frac{e \times p \times (1 - \pi)}{m}, \qquad \mathrm{PD}^{\mathrm{climate}} = N\big(N^{-1}(\mathrm{PD}) + \kappa\, c\big)\,, \tag{9.2}$$

where e is the borrower's emissions intensity in tonnes of CO_2 per dollar of revenue, p the carbon price, π the fraction of the cost the borrower can pass on to its customers, and m its EBITDA margin, so that c is the carbon bill as a fraction of EBITDA. The second expression shifts the borrower's default threshold by κc standard deviations, the probit scale on which the single-factor model of Section 9.4.2 moves a default threshold; κ is the threshold's sensitivity to an earnings shock, which a bank would calibrate from a Merton-style model of each sector's leverage or from its own history. Machine learning enters mainly at the data end: emissions are reported for only a minority of corporate borrowers, so they are imputed from sector, revenue, and asset data, and physical-risk exposure is mapped from satellite imagery and geocoded collateral, the methods of Chapter 17 and of the catastrophe models in Chapter 11.

9.2.5 Worked Example 9.1: A Carbon-Price Shock to a Sector Loan Book

Consider a $1 billion corporate loan book spread over four sectors, with the base-case PDs and LGDs in Table 9.1, and apply a stylized, immediate carbon price of $p = \$100$ per tonne with no pass-through ($\pi = 0$)

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and a threshold sensitivity of $\kappa = 2$, so that a carbon bill equal to half of EBITDA moves a borrower's default threshold by one full standard deviation. These are stated modeling assumptions, not estimates. For the fossil-heavy utilities, which emit 1,500 tonnes per million dollars of revenue on a 30% margin, the carbon bill is $1{,}500 \times 100/1{,}000{,}000 = 15\%$ of revenue, or $c = 0.15/0.30 = 50\%$ of EBITDA, and Equation (9.2) gives

$$\mathrm{PD}^{\mathrm{climate}} = N\big(N^{-1}(0.010) + 2 \times 0.50\big) = N(-2.326 + 1.000) = N(-1.326) = 9.24\%.$$

Sector	EAD	PD	LGD	c	$\mathrm{PD}^{\mathrm{climate}}$	EL	$\mathrm{EL}^{\mathrm{climate}}$
Oil and gas	200	2.00%	45%	30.00%	7.30%	1.800	6.571
Fossil-heavy utilities	250	1.00%	40%	50.00%	9.24%	1.000	9.236
Transportation	150	2.50%	50%	25.00%	7.21%	1.875	5.411
Commercial real estate	400	1.50%	35%	1.33%	1.60%	2.100	2.246
Total	1,000					6.775	23.464

Table 9.1: A \$100-per-tonne carbon-price shock to a four-sector loan book, with EAD and expected loss in \$ millions. Emissions intensities are 600, 1,500, 300, and 20 tonnes per \$1 million of revenue and EBITDA margins 20%, 30%, 12%, and 15%; c is the carbon bill as a share of EBITDA.

Applying EL = PD × LGD × EAD sector by sector, the book's expected loss rises from \$6.775 million to \$23.464 million, a factor of 3.46, comparable in size to Section 9.2.3's recession scenario but with an entirely different footprint. The utilities sector, the *safest* in the book on base-case PD, becomes the largest single source of stressed loss, \$9.236 million, and utilities and oil and gas together account for 77.9% of the \$16.689 million increase, while the commercial real estate book, the largest exposure, barely moves. A base-case rating says almost nothing about transition vulnerability, which is why supervisors ask for sector- and borrower-level analysis rather than a book-wide PD multiplier. Pass-through matters as much as the carbon price: if borrowers can pass half the cost on to customers, $\pi = 0.5$, stressed expected loss falls to \$12.411 million, so a single behavioral parameter that no historical data pins down roughly halves the answer.

The example is deliberately crude, and its caveats are the substance of climate credit modeling. The mapping in Equation (9.2) is static: real borrowers abate emissions, change their product mix, and invest in cleaner capacity as the carbon price rises, so a fixed intensity e overstates the long-run hit to the firms best able to adapt and understates it for those that cannot. The sensitivity κ has no data from which to estimate it in the relevant range. The emissions themselves are often imputed rather than reported, which turns a data-quality problem into a model-risk problem, and an imputed figure for a single large borrower can move the whole sector's result. Even the scenarios carry model risk: the NGFS noted in 2025 that a research paper underpinning part of its fifth vintage's physical-damage estimates had since been retracted (NGFS 2024). A climate stress result is therefore best read as conditional on a long chain of stated assumptions, and most useful for the ranking it produces (which sectors and borrowers are most exposed) rather than the headline dollar figure.

9.2.6 Credit Scoring with Logistic Regression

Expected loss requires a PD for each borrower, and Chapter 6 introduced the standard tool for producing one: logistic regression, which models the probability of a binary outcome (here, default within the horizon) as a function of borrower characteristics. Recall the small loan dataset from Chapter 6's K-nearest neighbors example, with six applicants described by debt-to-income ratio and years of credit history. Fitting a logistic regression to that same dataset,

$$\Pr(\text{Default}) = \frac{1}{1 + e^{-z}}, \qquad z = -12.152 + 0.530\,(\text{Debt/Income}) - 0.231\,(\text{Credit History}),$$

gives coefficients that make economic sense: higher leverage increases the log-odds of default (positive coefficient), while a longer credit history reduces it (negative coefficient). These coefficients come from scikit-learn's default fit, which adds an L2 (ridge) penalty with $C = 1$, and on this dataset the penalty is not

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a detail. Debt-to-income alone separates the two classes perfectly (the three defaulters sit at 30–40%, the three non-defaulters at 15–20%), and under such *complete separation* the unpenalized maximum-likelihood estimate does not exist. The likelihood keeps rising as the coefficients grow without bound, so a plain fit (statsmodels, for example) either fails with a perfect-separation error or returns arbitrarily large coefficients. The quoted values reflect the penalty as much as the data, which is why real scorecards on small or cleanly split samples use penalized or Firth-corrected logistic regression, and why this sample's perfect in-sample AUC later in the chapter is a warning sign rather than an achievement. Applying this to the same new applicant considered in Chapter 6 (debt-to-income of 28%, 4 years of credit history),

$$z = -12.152 + 0.530(28) - 0.231(4) \approx 1.761, \qquad \Pr(\text{Default}) = \frac{1}{1 + e^{-1.761}} \approx 85.3\%.$$

This is consistent with, and more precise than, the KNN classification of Chapter 6, which produced only a discrete "predicted default" label from a 2-to-1 majority vote among the three nearest neighbors. Logistic regression reports a continuous probability, the PD input the expected-loss formula requires.

Lenders often rescale a model's probability into a more interpretable credit score with a linear transformation, for example

$$\text{Score} = 850 - 550 \times \Pr(\text{Default}),$$

chosen so that a default probability near 0 produces a score near 850 (the top of a typical consumer credit-score scale) and a default probability near 1 produces a score near 300 (the bottom). For the applicant above, $\text{Score} = 850 - 550(0.853) \approx 381$, a score that would place this applicant in a high-risk tier under most lenders' underwriting policies. The scaling constants are a business choice, not a statistical one, but the logic of mapping a probability onto a scale loan officers and borrowers can interpret is universal. It connects to the calibration discussion of Chapter 6: a credit score is meaningful only if the underlying probability is accurately calibrated, not merely useful for ranking applicants.

A subtler difficulty affects every credit scoring model built this way, however well it is fit or calibrated. A lender observes loan performance only for applicants it approved, since a rejected applicant never receives a loan and never generates an outcome. A model trained only on approved applicants, like the six-applicant sample here, never sees how someone the previous policy judged too risky would have performed, a selection bias that can silently distort the fitted relationship between borrower characteristics and default risk. Reject inference is the general name for techniques that try to correct this, from simple approaches (assigning rejected applicants an assumed outcome based on their characteristics) to statistical corrections that model the approval decision alongside the default outcome. None is fully satisfying, since the counterfactual (what would have happened had a rejected applicant been approved) is unobservable. Reject inference therefore remains an imperfect compromise, and a model's performance on its training population can differ meaningfully from its performance on the full population it screens. It is a credit-specific instance of a general lesson: data generated by a decision process, here a prior lending policy's approvals and rejections, is never a neutral sample of the population that process was applied to.

9.2.7 Validating a Credit Scoring Model: The Gini Coefficient and AUC

Ranking applicants correctly from safest to riskiest is a distinct property from calibration, and it has its own standard validation metric in credit scoring: the area under the ROC curve (AUC), or equivalently the Gini coefficient, $\text{Gini} = 2\,\text{AUC} - 1$, which rescales AUC so that a purely random ranking scores 0 and a perfect ranking scores 1. Both answer the question introduced in Chapter 6: across every pair of one defaulter and one non-defaulter, how often does the model assign the defaulter a strictly higher predicted default probability?

Applying the fitted logistic regression above to all six applicants in Chapter 6's Table 6.5, sorted by predicted default probability, gives the values in Table 9.2.

Every one of the nine possible (defaulter, non-defaulter) pairs, three defaulters times three non-defaulters, is correctly ordered: each of C, A, and E's predicted probabilities exceeds each of B, F, and D's, so $\text{AUC} = 9/9 = 1.0$ and $\text{Gini} = 2(1.0) - 1 = 1.0$, a perfect ranking. This is a red flag, not a cause for celebration. The AUC is computed on the same six observations the model was fit to, with two features, so a perfect in-sample ranking is the least surprising outcome imaginable, not evidence the model would rank a new applicant accurately. A credit model's AUC and Gini coefficient are meaningful only on a held-out validation

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Applicant	Debt/income (%)	Pr(Default)	Actual outcome
C	40	0.9998	Default
A	35	0.9974	Default
E	30	0.9550	Default
B	20	0.0323	No default
F	18	0.0144	No default
D	15	0.0015	No default

Table 9.2: Predicted default probabilities for all six training applicants, sorted from highest to lowest risk

sample, following the train-test discipline of Chapter 6, and a near-perfect AUC on a tiny training sample signals overfitting rather than a property to report in a validation document.

9.2.8 Scorecards: Weight of Evidence and Points

The regression above is the right model but not the form a lender deploys. Almost every production consumer credit model is a *scorecard*: the same logistic regression, with each characteristic cut into bins, each bin replaced by a number summarizing its risk, and the result expressed as whole points. Binning lets a linear model capture a nonlinear relationship, makes the model robust to outliers and data errors, and leaves every part of it inspectable row by row.

The number replacing each bin is its *weight of evidence*, the log ratio of the share of non-defaulters to the share of defaulters falling in it,

$$\mathrm{WOE}_i = \ln\!\left(\frac{g_i/G}{b_i/B}\right), \tag{9.3}$$

with g_i, b_i the good and bad counts in bin i and G, B the portfolio totals. Table 9.3 works this for a portfolio of 1,000 loans, 900 performing and 100 defaulted.

Debt/income	Good	Bad	g_i/G	b_i/B	WOE	IV contribution
$< 15\%$	270	10	0.300	0.100	$+1.099$	0.220
15–25%	315	20	0.350	0.200	$+0.560$	0.084
25–35%	225	30	0.250	0.300	-0.182	0.009
$\geq 35\%$	90	40	0.100	0.400	-1.386	0.416
Total	900	100	1.000	1.000	—	0.729

Table 9.3: Weight of evidence and information value for a debt-to-income characteristic

The column is *monotone*, falling as leverage rises; a reversal is read as evidence that the bins are wrong or the sample too thin, and bins are merged until it disappears. The whole characteristic then summarizes into its *information value*, $\mathrm{IV} = \sum_i (g_i/G - b_i/B)\,\mathrm{WOE}_i = 0.729$, which by convention (below 0.02 useless, 0.1–0.3 medium, 0.3–0.5 strong) makes this a strong predictor; values far above 0.5 usually mean the characteristic partly restates the outcome, the credit version of the leakage problem Chapter 6 warns about.

The fitted log-odds are then rescaled so that a fixed number of points always means the same thing. Choosing PDO $= 20$ *points to double the odds* fixes the slope at $20/\ln 2 = 28.854$, and the gap between the safest and riskiest bands becomes $28.854(1.099 + 1.386) = 71.7$ points, or $71.7/20 = 3.59$ doublings, an odds ratio of $2^{3.59} = 12.0$. That checks straight off the table: the odds of being good are $270/10 = 27$ in the lowest band and $90/40 = 2.25$ in the highest, and $27/2.25 = 12.0$ exactly. The points scale adds no information; it repackages the log-odds into units a human can add up.

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9.2.9 The Scorecard Is a Generalized Additive Model

It is tempting to file the scorecard under "the classical method machine learning came to replace." That reading is wrong twice, and correcting it locates where machine learning earns its keep in credit.

Start with the bins. Choosing cut points by hand is possible but not what anyone does; the standard tool is *supervised discretization*, fitting a decision tree to the single characteristic and taking its splits as bin edges. On a simulated 20,000-loan portfolio, a depth-limited tree with a floor of 800 accounts per leaf selects cuts at 17.1%, 32.7%, 41.3%, 44.5%, and 47.3%, giving six bins whose weights of evidence run +1.054 down to −1.433, monotone with no manual merging and an information value of 0.620. The scorecard's most hand-crafted-looking step is done by a tree learner.

Now the functional form. Writing the scorecard out,

$$\ln \frac{p}{1-p} = \beta_0 + \sum_j \beta_j \,\mathrm{WOE}_j(x_j) = \beta_0 + \sum_j f_j(x_j), \tag{9.4}$$

each characteristic contributes an arbitrary piecewise-constant function of its own value, and the contributions add. That is the definition of a *generalized additive model*. A credit scorecard is a GAM fitted by hand, and has been since the 1950s, decades before the term existed.

This matters because the state of the art in interpretable machine learning, the *Explainable Boosting Machine*, fits this object automatically: it boosts shallow trees on one feature at a time, round-robin, so no interaction can form, and the accumulated trees for feature j become f_j. The companion notebook implements this in about twenty lines and fits it to the same portfolio. Comparing its learned shape function for debt-to-income against the hand-built WOE curve over a 150-point grid gives a correlation of 0.977. They are not analogous; they are two estimates of the same function.

The accuracy comparison is then informative rather than a horse race. On held-out data the scorecard scores an ROC-AUC of 0.8402, the boosted additive model 0.8296, and an unrestricted gradient-boosted model allowed to learn interactions 0.8328, against a ceiling of 0.8429 set by the true data-generating process, which does contain an interaction between leverage and file length. The scorecard lands within 0.003 of the ceiling and nothing beats it. When marginal effects are monotone, bins are well chosen, and interactions are weak, a scorecard is close to optimal and a more flexible model has little left to find. That, rather than institutional inertia, is why it remains the incumbent in regulated lending.

The corollary is the useful part, and the rest of this chapter's machine learning rests on it. ML earns its keep in credit not by fitting a better function of the same characteristics but by doing what a scorecard cannot: building features from data with no natural bins, producing a term structure rather than a single probability (as the lifetime expected-loss discussion later does), and searching a model space rather than fitting one model.

9.3 Modeling Default

This part moves from scoring an applicant to modeling default itself, from a firm's balance sheet, from ratings, and over a loan's whole life.

9.3.1 Structural Credit Models: The Merton Model

Logistic regression estimates PD statistically, from historical data on borrower characteristics and past defaults. A different approach, due to Merton (1974), estimates PD from market prices using the option pricing machinery of Chapter 5: a firm's equity can be modeled as a call option on the firm's assets, with the face value of its debt as the strike price. The analogy is exact. Shareholders have limited liability: if the firm's assets are worth more than its debt at maturity, shareholders receive the difference; if less, shareholders receive nothing and the firm defaults, with bondholders taking the (insufficient) assets. This is the payoff $\max(V_T - D, 0)$ of a European call with the firm's asset value V_T as the underlying and the debt's face value D as the strike, so the Black-Scholes-Merton formula applies, with asset value in place of a stock price and asset volatility in place of σ:

$$E_0 = V_0 N(d_1) - D e^{-rT} N(d_2), \qquad d_1 = \frac{\ln(V_0/D) + (r + \sigma_V^2/2)T}{\sigma_V \sqrt{T}}, \qquad d_2 = d_1 - \sigma_V \sqrt{T}, \tag{9.5}$$

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where E_0 is the market value of equity, V_0 is the (unobserved) market value of the firm's assets, and σ_V is asset volatility. The probability of default, the probability that assets fall below the debt's face value at maturity, is the risk-neutral probability that the option expires out of the money,

$$\Pr(\text{Default}) = N(-d_2),$$

the same $N(-d_2)$ term that appeared when Chapter 5 priced a European put. Figure 9.2 draws the payoff: nothing if assets are worth less than the debt at maturity, the excess above debt otherwise, a call option's hockey stick.

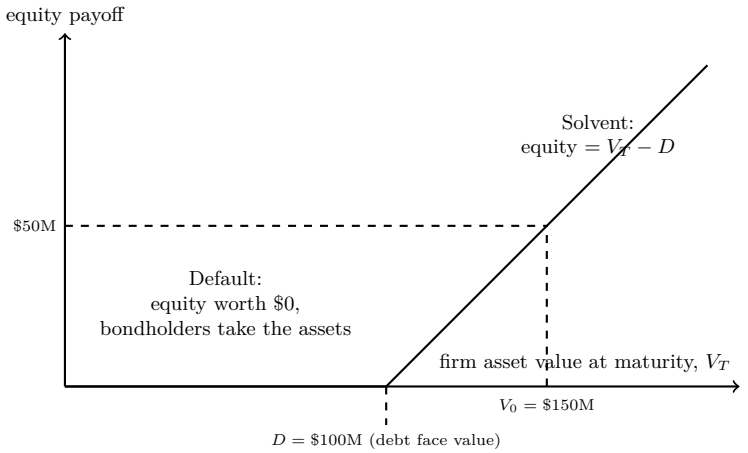


Figure 9.2: The Merton model's equity payoff at maturity, $\max(V_T - D, 0)$, for the worked example's $D = \$100$ million

Consider a firm with

$$V_0 = \$150 \text{ million (estimated asset value)}, \qquad D = \$100 \text{ million (debt due in one year)},$$
$$\sigma_V = 25\% \text{ (asset volatility)}, \qquad r = 4\% \text{ (risk-free rate)}.$$

Then

$$d_1 = \frac{\ln(150/100) + (0.04 + 0.03125)(1)}{0.25} \approx 1.907, \qquad d_2 = 1.907 - 0.25 \approx 1.657,$$

giving an implied equity value of $E_0 = 150\,N(1.907) - 100\,e^{-0.04}N(1.657) \approx \$54.37$ million and a one-year default probability of

$$\Pr(\text{Default}) = N(-1.657) \approx 4.88\%.$$

Unlike logistic regression, the Merton model requires no default history. It needs only an estimate of asset value and volatility, usually inferred from the firm's observable equity value and equity volatility by inverting the same equations. Structural models are therefore especially useful for large public firms with liquid equity and few or no historical defaults, the firms for which a statistical model would have the least data. The commercial descendant of this approach, Moody's KMV model, converts the model's "distance to default" (d_2 above) into a market-implied probability of default that updates in real time as equity prices move, without waiting for a rating agency's periodic review.

9.3.2 Inverting the Model: Recovering Asset Value and Volatility from Equity

The worked example ran the Merton model forward, from asset value and asset volatility to equity value and default probability. In practice neither V_0 nor σ_V is observable (no market quotes a firm's total asset value), so a real application runs the model in reverse, from the firm's equity price back to the unobserved asset value and volatility. This is the KMV approach, made concrete here.

The equity value equation from above, $E_0 = V_0 N(d_1) - De^{-rT}N(d_2)$, is one equation in the two unknowns V_0 and σ_V; a second equation comes from recognizing that equity, as a call option on the firm's assets, has

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its own volatility linked to the asset volatility through the option's delta, $N(d_1)$ (the same Greek introduced in Chapter 5):

$$\sigma_E E_0 = N(d_1)\,\sigma_V V_0. \tag{9.6}$$

Given a firm's observable market equity value E_0 and equity volatility σ_E (estimated from historical or option-implied stock returns, as in Chapters 5 and 7), the two equations can be solved simultaneously for V_0 and σ_V, typically with a numerical root-finder, since d_1 and d_2 depend on V_0 and σ_V nonlinearly.

As a consistency check, apply the inversion to the firm of the forward example. Its equity value was $E_0 \approx$ \$54.37 million; its implied equity volatility, from the same d_1, is $\sigma_E = N(1.907)(0.25)(150)/54.37 \approx 67.03\%$, far above the 25% asset volatility, because a leveraged equity stake is much more volatile than the assets it is a claim on. Feeding only these two observable quantities, with the known debt, rate, and maturity,

$$E_0 = \$54.37 \text{ million}, \qquad \sigma_E = 67.03\%,$$
$$D = \$100 \text{ million}, \qquad r = 4\%, \qquad T = 1 \text{ year},$$

into a numerical solver recovers $\hat{V}_0 = \$150.00$ million and $\hat{\sigma}_V = 25.00\%$ exactly, and therefore the same default probability of 4.88%.

Exact recovery is expected here, since the "observed" equity inputs were generated from the true V_0 and σ_V, but it confirms that the inversion is well posed. A real application feeds in a firm's market equity value and estimated equity volatility, with no knowledge of V_0 or σ_V, and recovers a market-implied estimate of both that updates every time the stock price moves, which makes the KMV approach useful for monitoring large public borrowers between rating reviews.

9.3.3 Market-Implied Default Probabilities: Credit Default Swap Spreads

The Merton model backs out a default probability from a firm's *equity* price. A credit default swap (CDS), introduced in Chapter 3 as a contract that transfers credit risk for a periodic premium, offers a more direct route, since a CDS spread is the market's own price for insuring against default. A standard simplification, sometimes called the credit triangle, relates the CDS spread s directly to a constant hazard rate (default intensity) h and the recovery rate R assumed on the underlying debt,

$$h \approx \frac{s}{1-R}, \tag{9.7}$$

with the cumulative default probability over a horizon T given by $\Pr(\text{Default by } T) = 1 - e^{-hT}$, treating default as a Poisson arrival with constant intensity h, the device behind the survival analysis Chapter 15 used for flagged fraud accounts. A one-year CDS trading at $s = 200$ basis points (2%) on debt with an assumed recovery of $R = 40\%$, as in this chapter's bond example, implies

$$h \approx \frac{0.02}{1-0.40} \approx 3.33\%, \qquad \Pr(\text{Default within one year}) = 1 - e^{-0.0333} \approx 3.28\%,$$

close to the PD = 3% assumed for the bond example, a useful cross-check that market-implied and assumed default probabilities broadly agree.

The constant-hazard assumption extends to any horizon. The cumulative five-year default probability implied by the same $h \approx 3.33\%$ is $1 - e^{-0.0333(5)} \approx 15.35\%$, only about 4.7 times the one-year 3.28% over a five-times-longer horizon. Survival compounds multiplicatively, so each later year's default risk applies only to borrowers still surviving, the same compounding behind the multi-year transition-matrix probabilities later in this chapter. Unlike the Merton model, the CDS-implied approach needs no volatility input, only an observed spread and an assumed recovery rate, but it depends on a reasonably liquid CDS market existing. That holds for large, actively traded corporate and sovereign names but not for the vast majority of small and mid-sized borrowers.

The chapter now has three distinct ways of arriving at a probability of default, each suited to a different population. Logistic regression (Section 9.2.6) requires a history of past borrowers and their outcomes, and it is the natural choice for retail and small-business borrowers with no traded equity or debt; its accuracy is bounded by how representative and recent that sample is. The Merton model requires no default history but

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needs liquid, exchange-traded equity, which suits it to large public companies and not to a privately held small business, the opposite population. The CDS-implied approach needs neither a default history nor an equity price, only a liquid credit derivatives market, which narrows it further to a small set of large corporate and sovereign names. A large bank therefore uses all three where each is available: logistic regression or a machine-learned variant for retail and small business, Merton-style models for large public corporates, and CDS-implied spreads as a real-time check on the most liquid names, because no single approach's data requirements are met across the whole portfolio.

When There Is No Market Price: Private Credit

The fastest-growing corner of corporate lending sits where none of the market-implied routes reach. *Private credit* is lending by non-bank funds directly to companies, typically mid-sized firms owned by private-equity sponsors, that are too large or too risky for a bank loan and too small for the bond market. The loans are negotiated bilaterally or among a small club of lenders, held to maturity, and rarely traded. The International Monetary Fund estimated the market at more than $2.1 trillion of assets and committed capital in 2023, about three-quarters of it in the United States, where its share is approaching that of syndicated loans and high-yield bonds (International Monetary Fund 2024). Its borrowers tend to be smaller and more indebted than those in the public leveraged-loan market, and the IMF reported that after the rate rises of 2022 and 2023 more than a third of them had interest costs exceeding their current earnings.

For a risk modeler the defining feature is the absence of prices. There is no CDS to invert and no traded equity for a Merton model, and the loans themselves do not trade, so a fund values them *mark-to-model*, typically once a quarter, from discounted cash flows, comparable public spreads, and the borrower's reported financials. Three consequences follow. Valuations are stale and partly subjective, and they move less than market prices would: the IMF noted that private-credit marks fell by less than leveraged-loan prices in stress periods despite lower credit quality. Returns computed from those smoothed marks understate volatility and correlation with other assets, which flatters every risk-adjusted performance number built on them. And default itself is partly a choice: lenders protected by maintenance covenants can intervene early, while competition with banks for large deals has loosened covenants, and lenders often prefer amendments, maturity extensions, or interest paid in kind to a recorded default. A PD model trained on recorded defaults then learns the lender's renegotiation policy as much as the borrower's credit quality, the selection problem of Section 9.2.6's reject inference. The tools of this chapter still apply (scoring models on borrower financials, Vasicek-style portfolio capital with an assumed correlation), but every input that elsewhere comes from a market must come from a model, and validation must start by asking what the marks themselves measure.

9.3.4 Rating Models and Transition Matrices

Suppose a bond portfolio manager holds an investment-grade bond and wants to know not just its one-year default probability but how likely it is to be downgraded before that. That is a different risk: a downgrade alone can force some funds to sell and can lower the bond's price long before any payment is missed.

Institutional credit risk (corporate bonds, sovereign debt) is often managed through a rating system, a small number of discrete risk categories (such as AAA, AA, A, BBB, down to Default) that summarize creditworthiness. Ratings are not static: a borrower can be upgraded, downgraded, or default over time, and a rating transition matrix records the probability of moving from each rating to every other rating over a fixed horizon, typically one year. Table 9.4 shows a simplified three-state transition matrix with ratings High and Medium plus an absorbing Default state (once a borrower defaults, it remains in that state, so the Default row is $[0, 0, 1]$).

From \ To	High	Medium	Default
High	0.90	0.08	0.02
Medium	0.10	0.80	0.10
Default	0.00	0.00	1.00

Table 9.4: A simplified one-year rating transition matrix

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The one-year default probability is read directly off the matrix: 2% for a High-rated borrower, 10% for a Medium-rated one. Multi-year default probabilities require treating the transition matrix as a Markov chain and computing its matrix power: the two-year transition matrix is $P^2 = P \times P$, and the entry in row i, column Default gives the cumulative two-year default probability starting from rating i. Carrying out this multiplication,

$$P^2_{\text{High}\to\text{Default}} = 0.90(0.02) + 0.08(0.10) + 0.02(1.00) = 0.046,$$

so a High-rated borrower's cumulative two-year default probability is 4.6%, more than double the one-year figure of 2%, since the borrower can now default either directly or after first migrating to the riskier Medium rating. The same calculation for a Medium-rated borrower gives a two-year default probability of 18.2%, and extending to three years ($P^3 = P^2 \times P$) gives 7.596% for a High-rated borrower. This compounding (cumulative default risk growing faster than linearly with horizon because of downgrade-then-default) is the credit analogue of the growing forecast uncertainty of Chapter 7, and it is why long-dated exposures such as a 10-year bond carry more cumulative default risk than one-year probabilities suggest.

Transition matrices also make credit risk visible before a default occurs. A bondholder whose issuer is downgraded from High to Medium has suffered a real economic loss before any missed payment, since a lower-rated bond trades at a lower price. Marking a credit portfolio to its current ratings, rather than waiting for defaults, is the logic of mark-to-market credit models such as CreditMetrics, which combine a transition matrix like Table 9.4 with bond-pricing formulas to estimate a portfolio's value distribution one year forward, including downgrades that never become defaults. This is a richer picture than the default-only economic capital of Section 9.4.1, which treats every outcome short of default as full recovery of value.

Consider a High-rated, 3-year, \$1,000 face-value bond paying a 5% annual coupon, exactly the yield High-rated bonds command, so it trades at par. One year from now it has two years of cash flows left, and its value depends on its rating then. If it remains High-rated, it is still priced at the 5% yield, giving, by Chapter 5's bond-pricing formula,

$$P_{\text{High}} = \frac{50}{1.05} + \frac{1{,}050}{1.05^2} = \$1{,}000.00,$$

unchanged, since coupon and yield coincide. If downgraded to Medium, the market demands a higher yield of, say, 7%,

$$P_{\text{Medium}} = \frac{50}{1.07} + \frac{1{,}050}{1.07^2} = \$963.84,$$

a loss of \$36.16 with no default. If the bond defaults, the bondholder recovers 40% of face value immediately, \$400.00. Weighting each outcome by Table 9.4's one-year transition probabilities for a High-rated borrower (90%, 8%, 2%) gives an expected one-year-forward value of

$$\mathbb{E}[V] = 0.90(1{,}000.00) + 0.08(963.84) + 0.02(400.00) = \$985.11,$$

with a standard deviation of \$84.16 across the three possible outcomes. The expected value sits below today's \$1,000 par purely from downgrade and default risk, separate from any change in the risk-free rate, and most of the \$84.16 dispersion comes from the rare but severe default outcome, not the likelier but modest Medium markdown: the asymmetric, fat-tailed risk a full CreditMetrics-style analysis captures across many rating states and bonds.

9.3.5 Lifetime Expected Loss: IFRS 9 and CECL

Everything so far computes expected loss over one year. Since 2018 that has not been what a bank reports: IFRS 9, the ninth International Financial Reporting Standard, and its U.S. equivalent CECL, the Current Expected Credit Loss standard, require provisioning against losses expected over an exposure's *remaining life*, which makes credit models the largest modeling estate in most banks. The change answers a crisis-era failure of incurred-loss accounting, under which a bank could not recognize a loss until impairment was objectively evident, so provisions stayed low through the boom and arrived in a rush when capital was scarcest.

IFRS 9 sorts exposures into stages: **Stage 1**, performing, at 12-month expected credit loss; **Stage 2**, having suffered a *significant increase in credit risk*, at *lifetime* loss; **Stage 3**, credit-impaired. CECL drops

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the staging and requires lifetime loss for everything from day one. Either way a one-year PD no longer suffices: what is needed is a *PD term structure*, the marginal probability of default in each future year.

The transition matrix introduced above already contains one. Take its Medium-rated borrower and Section 9.2.1's bond, EAD = \$1,000 and LGD = 60%, discounting at 5%. Raising the matrix to the nth power and reading the Medium-to-Default entry gives cumulative default probability by year n; differencing gives the marginals.

Year	Cumulative PD	Marginal PD	Undiscounted loss	Discounted loss
1	0.1000	0.1000	\$60.00	\$57.14
2	0.1820	0.0820	\$49.20	\$44.63
3	0.2502	0.0682	\$40.92	\$35.35
4	0.3078	0.0576	\$34.53	\$28.41
5	0.3570	0.0493	\$29.57	\$23.17
			Lifetime ECL	\$188.70

Table 9.5: Lifetime expected credit loss for a five-year Medium-rated exposure

Year two is checkable by hand: a Medium borrower defaults then either by staying Medium and defaulting, $0.80 \times 0.10 = 0.080$, or by being upgraded first, $0.10 \times 0.02 = 0.002$, giving 0.082 marginal and 0.182 cumulative. The marginals decline not because the borrower improves but because the survivors are a shrinking pool. In Stage 1 the bank provisions $0.10 \times 0.60 \times \$1{,}000/1.05 = \$57.14$; in Stage 2, the full \$188.70. Nothing about the borrower changes at transfer, yet $\$188.70/\$57.14 = 3.30$, and that cliff is what makes the significant-increase test one of the highest-stakes classification problems in the bank.

9.3.6 Learning the Term Structure: A Discrete-Time Hazard Model

That calculation assumes the transition matrix is constant for five years. It is not, and the standard requires a forward-looking, scenario-weighted estimate, which a fixed matrix cannot express. Here the term structure stops being arithmetic and becomes a fitted model. The standard formulation is a *discrete-time hazard*: for each loan, in each year it is still alive, model the conditional probability of defaulting during that year,

$$h(a, m) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 a + \beta_2 a^2 + \beta_3 m)}}, \tag{9.8}$$

with a the loan's age and m a macroeconomic index. The quadratic in age captures *seasoning*, the regularity that defaults are rare just after origination, peak a few years in, then decline. Fitted this way it is an ordinary logistic regression on a loan-year panel, which is why it is the workhorse: every tool from Chapter 6 applies unchanged. It is also the discrete-time counterpart of the survival analysis Chapter 15 develops for flagged accounts; the companion technical note on the Kaplan-Meier estimator derives the continuous-time version from the censored likelihood.

One subtlety decides whether it is estimable at all. If every loan originates in the same year then age and calendar year are the same variable, and no quantity of data can separate seasoning from the economy. Identification requires *staggered origination*, several cohorts observed over overlapping years, so that loans of the same age meet different economies. This is the credit instance of the age-period-cohort problem, and it constrains how the sample must be assembled rather than how the model is fitted.

Simulating six cohorts over ten calendar years, about 75,000 loan-years, recovers the data-generating parameters closely: the fitted hazard is $-3.138 + 0.600a - 0.099a^2 + 0.455m$ against a true $-3.100 + 0.550a - 0.090a^2 + 0.450m$. Evaluating it under three macroeconomic paths gives lifetime ECL of \$125.49 (benign, $m = -1.0$), \$181.79 (base, $m = 0$), and \$292.82 (adverse, $m = +1.5$); weighting these 30/50/20 gives \$187.11.

Two results deserve attention. The weighted figure lands within a few dollars of the flat matrix's \$188.70, so the two approaches agree about the central case, but only the hazard model produces the other two columns, and the \$125-to-\$293 spread is the entire content of a forward-looking provision. Second, expected loss is *convex* in the macroeconomic factor, so the average of the scenario losses exceeds the loss at the average scenario: the weighted \$187.11 exceeds the base-only \$181.79 by 2.9%. A bank modeling only its

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central forecast under-provides by that margin in calm conditions and by far more when the scenario spread widens, when it matters most. This is Jensen's inequality doing supervisory work, and it is why the standard insists on probability-weighting rather than a single best estimate.

Two extensions complete the picture. Borrowers prepay, and a loan that prepays generates no credit loss, so lifetime loss is properly a *competing risks* problem against prepayment; ignoring it overstates provisions on the exposures most likely to disappear before defaulting. And the significant-increase test is deliberately left undefined by the standard, making it a threshold-selection problem under the asymmetric costs Chapter 21 works through explicitly.

9.4 Credit Portfolios and Bank Capital

A loan book is not the sum of its loans; this part covers correlated losses, the models that price them, and the regulatory capital held against them.

9.4.1 Portfolio Credit Risk and Economic Capital

Expected loss is what a lender should expect to lose on average, already reflected in loan pricing and loss reserves. It is not by itself a risk measure, since an average says nothing about how much losses could vary in a bad year. Economic capital is the buffer held to absorb unexpected losses beyond the expected level, so that the institution remains solvent in a bad year.

Consider a simplified portfolio of 100 loans, each identical to the Chapter 5 bond example (EAD of \$1,000, PD of 3%, LGD of 60%), with defaults treated, for illustration, as independent across borrowers. The portfolio's expected loss is simply $100 \times \$18 = \$1{,}800$. Treating the number of defaults as a binomial random variable with $n = 100$ and $p = 0.03$, its standard deviation is $\sqrt{np(1-p)} \approx 1.706$ defaults, so the standard deviation of the dollar loss is

$$\hat{\sigma}_{\text{Loss}} = 1.706 \times 0.60 \times 1{,}000 \approx \$1{,}023.52.$$

Approximating the loss distribution as normal (a simplification discussed further below), the 99% Value at Risk of portfolio losses is $\text{EL} + z_{0.99}\,\hat{\sigma}_{\text{Loss}} \approx 1{,}800 + 2.326(1{,}023.52) \approx \$4{,}181$, and economic capital is defined as the unexpected loss beyond the expected level,

$$\text{Economic Capital} = \text{VaR}_{99\%}(\text{Loss}) - \text{EL} \approx 4{,}181 - 1{,}800 \approx \$2{,}381,$$

roughly 1.32 times the portfolio's expected loss. The independent-default assumption is a simplification made for tractability. In reality defaults are correlated: borrowers share the same macroeconomic conditions, and a recession raises every borrower's default probability at once. That makes the loss distribution's tail much fatter than the binomial approximation suggests, which is why regulatory capital models, such as the Basel Committee's asset-correlation framework, build in a correlation parameter. Regulatory capital, the minimum a bank must hold under frameworks such as Basel III, is similar in spirit but set with standardized formulas and correlation assumptions rather than a bank's own estimates, so that banks cannot lower measured risk by choosing favorable assumptions.

Correlation is not the only way a portfolio's risk can exceed what its expected loss suggests; concentration, how evenly exposure is spread across borrowers, matters even before correlation. The Herfindahl-Hirschman Index (HHI), $\text{HHI} = \sum_i s_i^2$ where s_i is borrower i's share of total exposure, quantifies this directly. The 100-loan portfolio above, with exposure spread evenly ($s_i = 1/100$ for every loan), has $\text{HHI} = 100 \times (0.01)^2 = 0.01$. Now compare a differently structured portfolio with the same \$100,000 total exposure and the same 3% average PD, but concentrated in only 10 borrowers of \$10,000 each: $\text{HHI} = 10 \times (0.10)^2 = 0.10$, ten times higher. Both portfolios have the same expected loss of \$1,800, since expected loss depends only on total exposure and average PD, but the concentrated one is riskier. A single default now removes 10% of the exposure rather than 1%, so one or two unlucky defaults among ten large borrowers can produce a loss far outside what the binomial approximation, built for many small exposures, suggests. Regulators and internal risk models therefore impose single-borrower and single-industry concentration limits as a complement to the correlation-adjusted capital of the Vasicek model below: a portfolio can satisfy every correlation assumption in a capital model and still be dangerously undiversified in a way that model, built around a representative borrower, never sees.

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9.4.2 Correlated Defaults: The Single-Factor (Vasicek) Model

The independent-default assumption is the simplification this chapter's case vignette shows going badly wrong at scale. The standard way to relax it, the one underlying the Basel IRB capital formulas, is the single-factor (Vasicek) model. Each borrower's default is driven by an unobserved asset value that depends on a common systematic factor Z, shared by every borrower, and an idiosyncratic factor, with asset correlation ρ controlling how much of each borrower's fortunes are tied to the shared factor. Conditional on the realized systematic factor, defaults become independent, and standard results give the α-quantile of the portfolio's default rate directly as a closed-form worst-case default rate (WCDR),

$$\mathrm{WCDR}(\alpha) = N\left(\frac{N^{-1}(\mathrm{PD}) + \sqrt{\rho}\,N^{-1}(\alpha)}{\sqrt{1-\rho}}\right). \tag{9.9}$$

The formula is quoted rather than proved here. The companion technical note on the single-factor model derives it: conditioning on the systematic factor makes defaults independent, the law of large numbers then collapses an infinitely granular portfolio's loss rate to a strictly monotone function of that factor, and the monotonicity lets the loss distribution be inverted in closed form. The note also verifies that the distribution has mean exactly PD while being strongly right-skewed: correlation moves risk into the tail without changing the average loss. Applying this to the same 100-loan portfolio (PD = 3%) with an illustrative asset correlation of $\rho = 0.20$, typical of the range Basel's own corporate risk-weight formulas use, at the 99% confidence level,

$$\mathrm{WCDR}(0.99) = N\left(\frac{N^{-1}(0.03) + \sqrt{0.20}\,N^{-1}(0.99)}{\sqrt{0.80}}\right) = N\left(\frac{-1.881 + 0.447(2.326)}{0.894}\right) \approx 17.37\%,$$

more than double the 6.97% figure the independent-default binomial approximation implied at the same 99% confidence level. Translating this into dollar terms, the correlated model's worst-case portfolio loss is $100 \times 0.1737 \times 0.60 \times 1{,}000 \approx \$10{,}422$, against the same \$1,800 expected loss as before, giving economic capital of $10{,}422 - 1{,}800 \approx \$8{,}622$, roughly 3.6 times the \$2,381 figure the independent-default assumption produced. This more-than-tripling of required capital is why regulators do not allow banks to assume independent defaults, and it is the mechanism of this chapter's case vignette: a shared systematic factor, whether a national housing downturn or a broader recession, pushes every borrower's default probability the same way at once, fattening the tail far beyond what a model of isolated coin flips suggests, even when every loan's own PD is estimated correctly.

Figure 9.3 shows the whole distribution rather than one quantile of it. With independent defaults, $\rho = 0$, an infinitely granular portfolio would lose exactly 3% every year, and the 100-loan portfolio's binomial distribution stays tightly bunched around that value: the probability that more than 10% of the loans default is 0.021%. At $\rho = 0.20$ the mean is still exactly 3%, but the distribution is pulled apart. Its mode sits near 0.25% and its median at 1.77%, so in most years the portfolio loses *less* than its expected loss, and the probability of a default rate above 10% is 5.02%, one year in twenty. A lender who judges a correlated book by the run of good years it usually produces is looking at the left half of this picture; the capital exists for the right half.

Correlation and stress compound rather than add. Applying Section 9.2.3's stressed PD = 8% to this same portfolio, still holding $\rho = 0.20$, gives

$$\begin{aligned}
\text{Stressed expected loss} &= \$4{,}800,\\
\text{Stressed WCDR}(0.99) &\approx 34.17\%,\\
\text{Worst-case loss} &= 100 \times 0.3417 \times 0.60 \times 1{,}000 \approx \$20{,}504,\\
\text{Economic capital} &= 20{,}504 - 4{,}800 \approx \$15{,}704,
\end{aligned}$$

roughly 6.6 times the base-case, independent-default economic capital of \$2,381. None of the three ingredients behind that 6.6× figure (the stressed PD, the stressed LGD embedded in expected loss, the asset correlation) is individually extreme. A bank that stresses PD and LGD while assuming independent defaults, or that accounts for correlation only under calm-period PD, captures a fraction of the combined effect, which is why regulators require the factors to be modeled jointly.

The single-factor model carries one further simplification: the closed-form WCDR assumes an infinitely granular portfolio of many small, roughly equal exposures, so that no single default materially moves the

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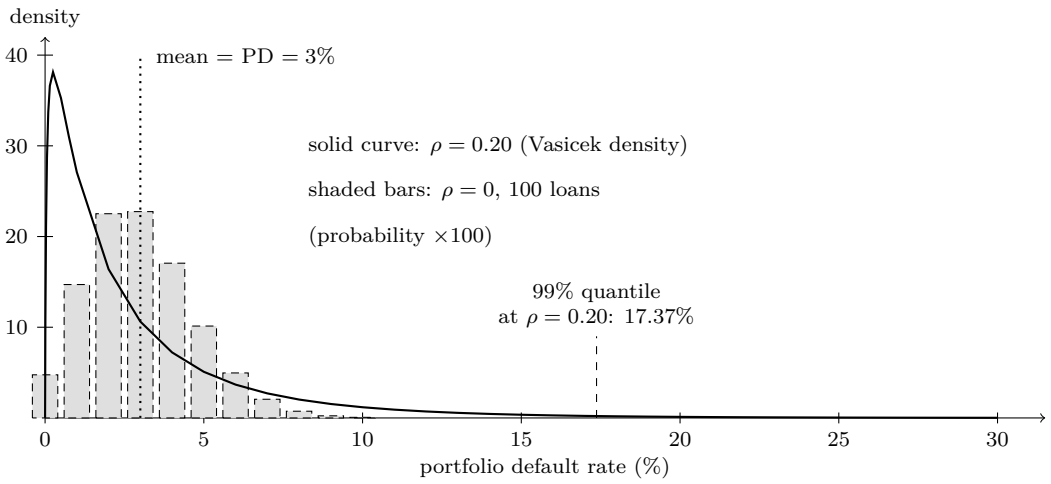


Figure 9.3: The distribution of the portfolio default rate at PD = 3%: the Vasicek density at asset correlation $\rho = 0.20$ against the 100-loan portfolio's binomial distribution under independence ($\rho = 0$), both with mean 3%

total loss. A real loan book is never quite that diversified. A handful of unusually large exposures adds name concentration risk the formula cannot see: if one of the 100 loans were ten times its size, its default would move portfolio losses far more than the granular formula assumes, even with every PD, LGD, and correlation fixed. Basel addresses this with a separate granularity adjustment, an add-on charge that grows as exposures concentrate among a few large borrowers. The closed-form result is only as trustworthy as its granularity assumption, one more instance of the chapter's theme: every formula is exact given its assumptions, and credit risk management lies largely in knowing which assumption is closest to breaking for a real portfolio.

9.4.3 The Basel Framework: A Brief History

Today's regulatory capital requirements are the product of a decades-long response to successive financial crises, summarized in Table 9.6.

The finalization's national rollout has been slower and more uneven than the Basel calendar suggests, and the dates are still moving. As of October 2026, the European Union has applied the new credit and operational-risk rules since 1 January 2025 through its Capital Requirements Regulation (CRR3), with the output floor starting at 50% and rising to 72.5% in 2030 and some transitional relief running to 2032 (European Parliament and Council of the European Union 2024b). The United Kingdom's Prudential Regulation Authority finalized its version, "Basel 3.1," in January 2026, to take effect on 1 January 2027 with the floor reaching 72.5% in 2030 (PRA 2026). In the United States, the agencies withdrew their 2023 "Basel III endgame" proposal and issued a re-proposal on 19 March 2026, whose comment period closed on 18 June 2026; no final rule had been adopted as of October 2026 (Board of Governors of the Federal Reserve System et al. 2026). Chapter 8 tracks the corresponding, and even more delayed, market-risk timetable.

Each accord reacted to its predecessor's weaknesses. Basel I's uniform risk weights did not distinguish a safer borrower from a riskier one, which Basel II addressed by permitting internal models. The 2008 crisis then revealed that those models, calibrated on the calm pre-crisis period, had let some banks hold far too little capital against risks that materialized suddenly and severely, the model-risk and procyclicality concerns of this chapter's challenges section, which Basel III's higher and higher-quality capital requirements were designed to address. Basel II's internal-ratings-based approach is built on this chapter's machinery. A bank estimates its own PD (Section 9.2.6), LGD (Section 9.2.2), and EAD for each exposure and feeds them into a regulatory risk-weight formula derived from the Vasicek model of Section 9.4.2, with the asset correlation set by the regulator rather than estimated by the bank, so that a bank cannot lower its capital by assuming a lower correlation than its portfolio exhibits.

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Accord	Key development
Basel I (1988)	Introduced a simple, standardized minimum capital ratio (8% of risk-weighted assets) applied uniformly across large internationally active banks
Basel II (2004)	Allowed large banks to use internal PD, LGD, and EAD models (much like Sections 9.2.1 and 9.2.6 of this chapter) to compute risk-weighted assets, rather than the fixed weights of Basel I
Basel III (2010–2017, phased in)	Raised capital quality and quantity requirements substantially in response to the 2008 financial crisis, added new liquidity requirements (Chapter 8's LCR and NSFR), and introduced the shift from VaR to Expected Shortfall for market-risk capital discussed in Chapter 8
Basel III finalization (2017; Basel start date 2023, national dates later)	Limited how far internal models can lower capital, through an output floor phasing in to 72.5% of standardized risk-weighted assets (by 2028 on the Basel schedule; national timetables in the text below), and replaced all of Basel II's operational-risk approaches with a single Standardized Measurement Approach

Table 9.6: A brief history of the Basel bank capital accords

9.4.4 Operational Risk Capital: The Basic Indicator Approach

Operational risk, introduced from the market-risk side in Chapter 8, is harder to quantify than market or credit risk, since no return series or default history applies uniformly across firms. A rogue trader's unauthorized position, a failed core banking system upgrade, and a mis-sold financial product are all operational risk events, but they share little statistical structure.

Basel II's simplest treatment, the Basic Indicator Approach, now of historical interest but still the clearest illustration, sidestepped this difficulty by tying capital to firm size rather than to any operational loss model,

$$K = \alpha \times \overline{\mathrm{GI}},$$

where $\overline{\mathrm{GI}}$ is the bank's average annual gross income over the past three years (a simple proxy for the overall scale of its operations) and $\alpha = 15\%$ is a fixed regulatory parameter. Under that approach, a bank with average annual gross income of \$200 million would hold

$$K = 0.15 \times 200{,}000{,}000 = \$30{,}000{,}000$$

in operational risk capital, however many operational loss events it has experienced.

Basel II also offered a finer standardized variant and an internally modeled Advanced Measurement Approach, but the 2017 Basel III finalization retired all three in favor of a single Standardized Measurement Approach, which scales capital with a size-based business indicator adjusted by the bank's own loss history; no bank may now set operational-risk capital with its own internal model, a reversal Chapter 10 examines. The size-based proxy illustrates a broader principle: when a risk is too hard to model credibly and consistently across institutions, regulators fall back on a simple, conservative, hard-to-manipulate formula rather than a model whose inputs a bank could shade in its favor, the tension between sophistication and gameability that runs through the internal-ratings-based approach to credit risk. The contrast with the rest of this chapter is instructive. PD, LGD, and asset correlation are hard to estimate, but a bank has a coherent statistical framework (logistic regression, the Merton model, the Vasicek model) for estimating each from data that plausibly exists. Operational risk events, a rogue trader, a failed software deployment, a mis-sold product, share so little common structure that no single statistical framework fits all of them well, which is why regulators settled for a size-based proxy rather than an operational-risk analogue of the PD/LGD/EAD decomposition. Chapter 10 develops both the loss-distribution machinery behind that capital charge and the AI methods (incident-text classification, anomaly detection, surveillance, early warning) a bank uses to manage it.

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9.5 Counterparty Credit Risk and Valuation Adjustments

A loan's exposure is known; a derivative's is not. This part prices the credit risk of a contract whose value to the bank, and therefore what the bank stands to lose, moves with the market.

9.5.1 Exposure Profiles: Expected and Potential Future Exposure

Everything so far has treated EAD as a number fixed, or at worst drawn down, in advance. An over-the-counter derivative breaks that assumption. An interest-rate swap is worth nothing to either side on the day it is struck, and thereafter its value drifts with interest rates, sometimes in the bank's favor and sometimes against it. If the counterparty defaults while the swap is worth $V > 0$ to the bank, the bank loses (a fraction LGD of) V, since it must replace the contract in the market at that price; if the counterparty defaults while $V < 0$, the bank still owes the full amount to the defaulted counterparty's estate and gains nothing. The exposure is therefore $\max(V, 0)$, a call-option payoff on the contract's own future value, and it is random. This is counterparty credit risk, and it is the credit risk of every derivatives desk in the book.

Two summaries of that random exposure do most of the work. The *expected exposure* at a future date t is the mean of the positive part,

$$\mathrm{EE}(t) = \mathbb{E}[\max(V(t), 0)]\,, \tag{9.10}$$

which is the input to pricing, while the *potential future exposure* $\mathrm{PFE}_\alpha(t)$ is a high quantile of the same distribution, typically $\alpha = 95\%$ or 97.5%, which is the input to credit limits, the counterparty analogue of a Value at Risk. Plotted against t, either one is an *exposure profile*, and its shape is determined by the instrument. For a swap, two forces pull in opposite directions: the uncertainty in rates grows with time, roughly as $\sqrt{t}$, while the number of remaining payments that a rate move can affect shrinks to zero at maturity. The product of the two gives the characteristic hump.

That hump can be written down in closed form under one simplifying assumption. Let rates move as an arithmetic Brownian motion with a normal volatility of σ per year (the Bachelier, or normal, model of rates), and approximate the value of a payer swap with notional N and maturity T, struck at the market rate, by the rate move times the remaining annuity, ignoring discounting within it. Then $V(t)$ is normal with mean zero and standard deviation $s(t)$, and the standard normal integral $\mathbb{E}[\max(Z, 0)] = \varphi(0) = 1/\sqrt{2\pi} \approx 0.39894$ gives

$$s(t) = N\sigma\sqrt{t}\,(T - t), \qquad \mathrm{EE}(t) = 0.39894\, s(t), \qquad \mathrm{PFE}_{95\%}(t) = 1.645\, s(t). \tag{9.11}$$

The function $\sqrt{t}\,(T - t)$ peaks at $t = T/3$, so a swap's exposure is largest about a third of the way through its life. For a five-year, \$10 million swap with $\sigma = 1\%$ (100 basis points of normal volatility per year), evaluated at the midpoint of each year, $t = 0.5, 1.5, \ldots, 4.5$, the standard deviations $s(t)$ are \$318,198, \$428,661, \$395,285, \$280,624, and \$106,066, and the expected exposures follow in Table 9.7. At $t = 1.5$, for example, $s = 10{,}000{,}000 \times 0.01 \times \sqrt{1.5} \times 3.5 = \$428{,}661$, so $\mathrm{EE} = 0.39894 \times 428{,}661 = \$171{,}011$ and $\mathrm{PFE}_{95\%} = 1.6449 \times 428{,}661 = \$705{,}084$, more than four times the expected exposure. Figure 9.4 draws both profiles. A production system, facing hundreds of trades of different types with one counterparty, cannot use a closed form, so it simulates thousands of paths of every market risk factor, revalues every trade on every path at every future date, and reads the profiles off the simulated distribution. The companion notebook does this for the one swap, with 200,000 seeded paths, and recovers the closed-form expected exposures to within about 0.6%.

9.5.2 The Credit Valuation Adjustment

A derivative's textbook price, the one Chapter 5 computes, assumes that both parties perform. The *credit valuation adjustment* (CVA) is the amount by which the value of the contract must be reduced because the counterparty might not: the market price of the counterparty credit risk. It is the expected-loss formula of Equation (9.1) applied period by period, with the expected exposure in place of a fixed EAD, a marginal default probability for each period in place of a single PD, and discounting, since the loss falls in the future. Dividing the life of the trade at dates $t_0 = 0 < t_1 < \cdots < t_n = T$,

$$\mathrm{CVA} \approx \mathrm{LGD} \sum_{i=1}^{n} \mathrm{EE}(t_i)\, \mathrm{PD}(t_{i-1}, t_i)\, \mathrm{DF}(t_i), \tag{9.12}$$

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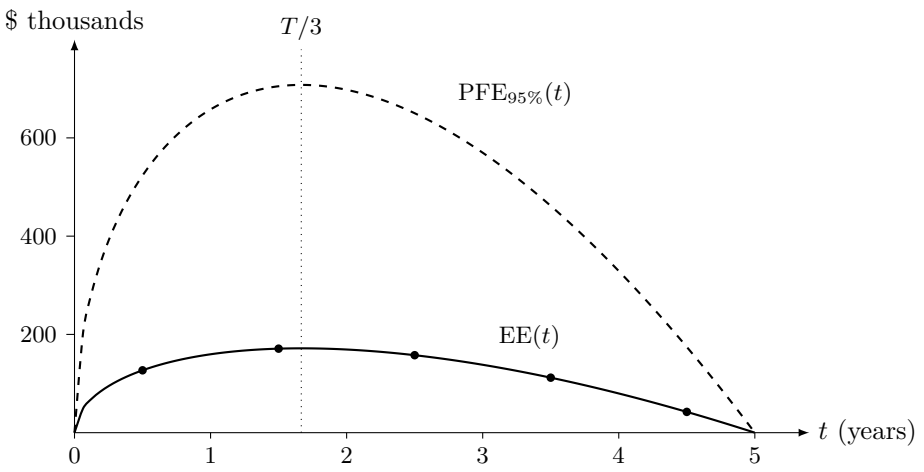


Figure 9.4: Exposure profiles of a five-year, $10 million payer swap under 1% normal rate volatility: expected exposure (solid, with the five dates used for CVA marked) and 95% potential future exposure (dashed). Both peak at a third of the swap's life.

where $\mathrm{PD}(t_{i-1}, t_i)$ is the probability that the counterparty defaults in the ith period and $\mathrm{DF}(t_i)$ is the discount factor. With a constant hazard rate h, the CDS-implied quantity of Section 9.3.3, the marginal probabilities are differences of survival probabilities, $\mathrm{PD}(t_{i-1}, t_i) = e^{-ht_{i-1}} - e^{-ht_i}$. The formula treats exposure and default as independent, which allows the expectation of their product to be written as a product of expectations; Section 9.5.5 examines what happens when they are not.

Equation (9.12) also explains why CVA is priced from *market-implied*, not historical, default probabilities. A bank that wants to hedge its CVA buys CDS protection on the counterparty, so the relevant probability is the one at which that protection trades. CVA therefore moves every day with the counterparty's credit spread even when nobody defaults, and that is where most of the losses fell in the 2008 crisis: by the Basel Committee's estimate, roughly two-thirds of banks' losses attributed to counterparty credit risk were mark-to-market CVA losses and only about one-third were actual defaults, which is why Basel III attached a separate capital charge to CVA volatility (BCBS 2011).

9.5.3 Netting, Collateral, and the Margin Period of Risk

Two contractual devices cut counterparty exposure, often sharply. The first is *netting*: under a master agreement, all the trades with one counterparty are closed out together in default and only the net amount is owed, so the exposure is $\max(\sum_j V_j, 0)$ rather than $\sum_j \max(V_j, 0)$, and trades that move in opposite directions offset one another. For two trades whose values are normal with the same standard deviation and correlation ρ, the netted expected exposure is $\sqrt{(2+2\rho)}/2$ times the sum of the two standalone expected exposures: 0.707 for independent trades, 0.500 for trades with $\rho = -0.5$, and no benefit at all only when $\rho = 1$. This is why exposure is always computed per *netting set*, never trade by trade.

The second device is *collateral*. Under a variation-margin agreement, the party that is out of the money posts cash or securities equal to the contract's value, usually daily, so a default leaves the surviving party holding collateral against its claim. Agreements often allow a *threshold* H, an amount of uncollateralized exposure each side tolerates before calling for margin, in which case the exposure becomes $\min(\max(V, 0), H)$: everything above the threshold is covered. For a normally distributed value with standard deviation s, the expected collateralized exposure has a closed form,

$$\mathbb{E}[\min(\max(V,0),H)] = s\left[\varphi(0) - \varphi(H/s)\right] + H\left[1 - N(H/s)\right], \tag{9.13}$$

which falls to zero as $H \to 0$ and recovers $s\,\varphi(0)$ as $H \to \infty$.

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Even a zero threshold does not remove the exposure entirely, because collateral arrives with a lag. Between the last margin call a defaulting counterparty met and the moment the surviving bank has closed out and replaced the trades, the *margin period of risk*, conventionally ten business days for bilateral trades, the market keeps moving and the collateral stops being topped up. The residual exposure is the value change over that window, whose standard deviation is $s(t)$ scaled by $\sqrt{\text{MPoR}/t}$ rather than the full $s(t)$, which is far smaller but never zero.

9.5.4 Worked Example 9.2: The CVA of a Five-Year Interest-Rate Swap

Price the counterparty risk on the swap of Section 9.5.1, traded with a counterparty whose one-year CDS trades at the 200 basis points of Section 9.3.3, so that $h = 0.02/(1 - 0.40) = 3.333\%$, with the chapter's LGD $= 60\%$ and the 5% discount rate of Table 9.5. Each year is one default bucket, with exposure evaluated at its midpoint. Year one's marginal default probability is $1 - e^{-0.03333} = 3.2784\%$ and its discount factor is $1.05^{-0.5} = 0.9759$, so its contribution is

$$0.60 \times 126{,}943 \times 0.032784 \times 0.9759 = \$2{,}436.83,$$

and Table 9.7 repeats the calculation for the other four years.

Year	t_i	EE(t_i)	PD(t_{i-1}, t_i)	DF(t_i)	LGD × EE × PD × DF
1	0.5	\$126,943	3.2784%	0.9759	\$2,436.83
2	1.5	\$171,011	3.1709%	0.9294	\$3,023.95
3	2.5	\$157,696	3.0670%	0.8852	\$2,568.65
4	3.5	\$111,953	2.9664%	0.8430	\$1,679.79
5	4.5	\$42,314	2.8692%	0.8029	\$584.84
				CVA	\$10,294

Table 9.7: The credit valuation adjustment of a five-year, \$10 million payer swap, uncollateralized

The CVA is \$10,294, about 10.3 basis points of notional paid up front, or 2.38 basis points a year spread over the swap's life, which is the amount a dealer adds to the rate it quotes this counterparty. The profile of the contributions follows the exposure hump, discounted and thinned by survival, with year two the largest. The Monte Carlo exposures of the companion notebook give \$10,258, and a common back-of-the-envelope shortcut, the CDS spread times the discounted expected exposures, $0.02 \times \sum_i \text{EE}(t_i)\,\text{DF}(t_i) = \$11{,}015$, lands within 7%, since it is the credit triangle of Equation (9.7) again: $s \approx h(1 - R)$.

Now suppose the two parties sign a variation-margin agreement with a threshold of $H = \$250{,}000$. At $t = 1.5$, $H/s = 250{,}000/428{,}661 = 0.583$, and Equation (9.13) gives an expected exposure of \$96,714, down from \$171,011. Repeating this at every date gives expected exposures of \$87,718, \$96,714, \$94,472, \$83,295, and \$41,986, and a CVA of \$6,763, 65.7% of the uncollateralized figure. The threshold helps least where exposure is already small: in year five the swap's value rarely exceeds \$250,000, so collateral is almost never called and the expected exposure barely changes. Removing the threshold altogether, so that only the ten-day margin period of risk remains, cuts the year-one expected exposure to \$35,905 and the CVA to \$1,743, 16.9% of the uncollateralized value. The two figures bracket the practical range: collateral removes most counterparty risk, a threshold gives part of that back, and no agreement removes the gap risk of the closeout window.

9.5.5 Wrong-Way Risk as a Correlation Stress

The independence assumption in Equation (9.12) fails in the cases that matter most. *Wrong-way risk* is a positive dependence between exposure and the counterparty's default probability: the trade is worth most to the bank in the states in which the counterparty is most likely to fail. A bank that bought CDS protection on one bank from another whose fortunes are tied to the first holds a contract worth most when its seller is weakest. A bank that receives dollars and pays local currency in a cross-currency swap with an

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emerging-market bank finds the swap most valuable after a collapse of the local currency, the same event most likely to push its counterparty into default. Its opposite, *right-way risk*, arises when exposure rises as default risk falls.

The single-factor model of Section 9.4.2 gives a clean way to quantify it, as a correlation stress. Model the counterparty's default in period i as its standardized latent variable X falling below $k_i = N^{-1}(\mathrm{PD}(t_{i-1}, t_i))$, as in Equation (9.9), and let the swap's standardized value $Y = V/s$ have correlation c with X, with $c < 0$ meaning wrong-way risk. Defaults happen in the lower tail of X, so they now come with a shifted exposure distribution. In closed form, the conditional mean of Y given default is $\mathbb{E}[Y \mid X < k] = -c\,\varphi(k)/N(k)$. For year one, $k_1 = N^{-1}(0.032784) = -1.841$ and $\varphi(k_1)/N(k_1) = 2.233$, so with $c = -0.3$ the swap's value at the moment of default is centered 0.670 standard deviations above zero rather than at zero. The expected exposure *given* default, computed in the companion notebook by integrating over the conditional distribution, is then 2.02 times the unconditional figure, and applying the corresponding multiplier in every year doubles the CVA, to \$20,952. A correlation of only -0.1 already raises it to \$13,417, and -0.5 to \$30,154.

The multiplier is large because default is a tail event, and conditioning on a tail event drags any correlated variable a long way. This is the Gaussian copula lesson of this chapter's case vignette in a different instrument: a correlation that looks modest moves the loss distribution's tail by a large factor, and it cannot be estimated with any confidence from data in which the counterparty never defaults. Regulators answer the same way they answer the asset-correlation problem, with a conservative supervisory parameter rather than a bank's own estimate: *specific* wrong-way risk, where the dependence is structural, such as a counterparty selling protection on itself or an affiliate, must be identified and treated separately, and the general multiplier described next is set conservatively partly with this effect in mind.

9.5.6 From CVA to XVA: Capital, and Where Adjoint Differentiation Pays

Regulation prices counterparty risk twice. The exposure measure for default capital is the Basel Committee's standardized approach for counterparty credit risk (SA-CCR), which sets EAD = 1.4 × (replacement cost + PFE add-on) with supervisory formulas for netting and collateral (BCBS 2014), and the separate charge for CVA volatility, rebuilt in the 2017 finalization as a basic approach and a sensitivity-based standardized approach and recalibrated in July 2020 (BCBS 2020b), is part of the Basel III finalization whose national timetables Section 9.4.3 records. Banks also now price more than credit. Alongside CVA sit a debit valuation adjustment for the bank's own default risk, a funding valuation adjustment for the cost of funding uncollateralized positions, a margin valuation adjustment for the cost of posting initial margin, and a capital valuation adjustment for the cost of the capital the trade consumes, collectively XVA (Gregory 2020). Every one of them requires the same simulation: market paths, a revaluation of every trade on every path at every date, and an aggregation by netting set.

That is where the computational tools of Chapter 8 earn most of their keep. A CVA desk hedges its CVA, and the standardized CVA capital charge is computed from CVA sensitivities, so the bank needs the derivative of a whole-portfolio, simulation-based number with respect to every curve point and volatility, often thousands of inputs. Bumping and rerunning the simulation once per input is infeasible; adjoint algorithmic differentiation delivers all of them for a small multiple of one run, which is why XVA systems were among its earliest large-scale adopters. The inner revaluations are the other bottleneck, since a path-dependent or callable trade cannot be priced in closed form at each simulated node, and the standard remedy is to replace the inner pricer with a fitted approximation, least-squares regressions on the simulated state or neural-network surrogates, the same move Chapter 11 describes for insurers' nested valuations. Huge and Savine (2020) joined the two ideas in *differential machine learning*: since adjoint differentiation produces each simulated path's pathwise derivatives almost for free, a network can be trained on both the simulated values and their derivatives, so that it learns the pricing function and its Greeks together, from far fewer simulations than training on values alone requires. It is the clearest case in this book of a machine-learning technique valuable because a financial model already supplies the gradients that other fields would have to estimate.

9.6 Machine Learning and Model Governance in Credit Risk

Every technique so far (logistic regression, the Merton model, transition matrices, the Vasicek model) is a classical statistical or structural model, chosen because a credit decision affecting a real borrower must be

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explainable to that borrower, to a regulator, and to a model validator. Machine learning nonetheless enters credit modeling in several concrete ways that extend rather than replace these tools.

Tree ensembles and gradient boosting, introduced in Chapter 6, routinely outperform logistic regression on raw discriminatory power, capturing nonlinear interactions (a debt-to-income ratio that matters far more once credit history falls below some threshold, say) that a linear logistic regression cannot represent. Chapter 21's capstone works through this comparison on a real consumer credit dataset, finding a several-point ROC-AUC improvement, at the cost of the closed-form Shapley decomposition Chapter 18 showed is available only for a model linear in the log-odds.

9.6.1 A Head-to-Head Test: Gradient Boosting versus Logistic Regression

Section 9.2.6's six-applicant sample is too small, and too cleanly separated, to show a tree-ensemble advantage; logistic regression already reaches a perfect in-sample AUC of 1.0 on it. The companion notebook instead simulates a population of 10,000 borrowers with a threshold interaction built into the true default process: debt-to-income drives default risk, but far more steeply once credit history falls below four years, a nonlinearity a logistic regression linear in both features cannot represent.

Splitting into 8,000 training and 2,000 held-out borrowers, a logistic regression fit on debt-to-income and credit history alone achieves a held-out AUC of 0.8494; a gradient-boosted tree ensemble fit on the identical two features, with no additional information, reaches a held-out AUC of 0.8634, an improvement of 1.40 percentage points, consistent in direction (if more modest in this smaller, controlled setting) with the several-point improvement of Chapter 21's real-dataset capstone. The gradient-boosting model's feature importances show why: it assigns 83% of its importance to credit history and only 17% to debt-to-income, because credit history does double duty in the true model, as a direct risk factor and as the threshold variable that determines how much debt-to-income matters, a role a single linear coefficient cannot capture.

This result sits alongside Section 9.2.9's, where a scorecard beat a gradient-boosted model, and reconciling the two is the whole argument for binning. The logistic regression beaten here is *linear in both features*, so a threshold effect is what it cannot represent. The scorecard that wins there is not: weight-of-evidence binning gives each band its own value, so the threshold is absorbed. What the tree ensemble buys here is flexibility over a straight line, not over a scorecard, and the gap closes once the straight line is replaced by the form lenders deploy. A tree's residual advantage after binning is whatever interactions remain, which in credit data are usually modest. As Section 9.2.7 emphasized, the improvement is meaningful only because it is measured on held-out data; on the 8,000 training borrowers, the ensemble's flexibility alone would inflate the gap whether or not the interaction were real.

Alternative data sources (satellite imagery, transaction and cash-flow data, web activity), developed in Chapter 17, extend credit scoring to borrowers with too little credit-bureau history for conventional scoring, a distinct capability rather than a more flexible functional form. Cash-flow underwriting makes the distinction concrete: its raw material is a stream of semi-structured merchant strings, so there are no bins to choose until a text classifier of the kind Chapter 16 develops has decided which transactions are rent, and the machine learning sits at the front of the pipeline rather than at the modeling step. Graph neural networks, introduced in Chapter 17 for a shell-company layering network, apply naturally to counterparty and supply-chain credit risk, since a firm's default risk depends partly on the health of its major customers and suppliers, a network effect no borrower-level feature table can represent. In every one of these applications the discipline of Section 9.2.7 applies: discrimination and calibration must both be validated out of sample, and added flexibility is worth its added opacity only if validation and monitoring can keep pace.

9.6.2 Monotonic Constraints: Making a Tree Ensemble Defensible

When a tree ensemble is the right choice, it still needs the property Section 9.2.8's scorecard gets for free. A boosted tree fitted to credit data will somewhere do something indefensible: a finite sample always contains regions where noise makes a few high-leverage borrowers look safer than those just below them, and the model learns the wiggle. If predicted default probability falls as debt-to-income rises, then somewhere in the book is an applicant who would be approved by borrowing *more*, and a declined applicant whose adverse action notice cannot truthfully be written.

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Monotonic constraints tell each tree that splits on a feature may move the prediction in only one direction, so the fitted function is monotone by construction rather than by inspection. Refitting the simulation above with the constraint, the unconstrained model reverses direction at 53 of 199 grid steps, and at its worst its fitted probability *falls* by more than twenty percentage points, from 62.0% to 39.6%, across a single grid step in which debt-to-income rises from 46.8% to 47.1%; the constrained model reverses nowhere. The expected accuracy cost does not appear: ROC-AUC is 0.8757 constrained against 0.8597 unconstrained, both below the 0.8812 ceiling. The constraint *improved* discrimination, because the monotone relationship is really present and forbidding reversals removes only overfitting. That is not a general law: a constraint encoding something true costs nothing and regularizes, while one encoding something false costs accuracy and deserves questioning rather than silent removal.

A monotonic constraint is, more generally, an *inductive bias*, prior knowledge about the shape of the solution built into the model class rather than learned from data, in the same family as the translation equivariance a convolutional network gets from weight sharing in Chapter 17.

9.6.3 Fairness in Lending: Searching for a Less Discriminatory Alternative

Fairness in lending is a stronger obligation than the auditing Chapter 18 develops, and one where machine learning is the tool rather than the problem. Under disparate-impact doctrine, showing that a disparity is justified by business necessity does not end the analysis: if a *less discriminatory alternative* exists, a model serving the lender's purpose comparably well with a smaller disparity, then using the original is not defensible. In U.S. consumer credit the federal footing for this doctrine narrowed in 2026, when the CFPB amended Regulation B to remove the disparate-impact "effects test" under the Equal Credit Opportunity Act, effective 21 July 2026 (CFPB 2026); disparate treatment remains prohibited, and state enforcers, the Fair Housing Act, and regimes abroad still reach disparate outcomes (Massachusetts Attorney General 2025), so the search described here remains sound practice (Chapter 18). This converts fairness from a test into a *search* over the model space, conducted by retraining under fairness regularization, transforming features that carry disparity without much signal, and varying thresholds and segmentations to trace the achievable frontier of performance against disparity. It is the third item in Section 9.2.9's corollary, and the one setting in credit where fitting many models rather than one is the compliant behavior rather than the suspicious one. The European Union now writes a comparable obligation into statute: its Artificial Intelligence Act classifies AI systems used to evaluate the creditworthiness of natural persons or to establish their credit score (other than for detecting financial fraud) as high-risk, subjecting them to risk-management, data-governance, documentation, and human-oversight requirements (EU 2024) that the Digital Omnibus has deferred to 2 December 2027 (EU 2026); Chapter 18 sets out that regulatory landscape in full.

9.6.4 Model Validation and Governance for Credit Models

Chapter 8 developed the general model risk management framework (independent validation, ongoing monitoring, model inventory and documentation), and its three-lines-of-defense structure applies unchanged to credit models: a lending or trading desk originates the risk (first line), an independent credit risk function builds and monitors the PD, LGD, and rating models (second line), and internal audit periodically verifies that the process works as designed (third line).

Credit models carry two validation concerns beyond the general framework. First, discriminatory power (Section 9.2.7's AUC and Gini coefficient) must be checked out of sample, not on the training data. Second, calibration, whether predicted probabilities match observed default rates, is distinct from discrimination, and a high AUC does not guarantee it: a model can rank borrowers correctly while being systematically over- or under-confident about the probability level, the calibration warning of Chapter 6. A validator must check both, since a model that passes discrimination but fails calibration can still produce a badly wrong dollar reserve.

A mature validation function also revisits, on a regular cycle, the chapter's other structural assumptions. Rating transition matrices, Section 9.4.1's independent-default assumption, and Section 9.4.2's asset correlation are all estimated quantities that can drift silently as conditions change: a transition matrix estimated from a benign decade understates downgrade risk once conditions deteriorate, and an asset correlation that looked reasonable in a calm period is the kind of assumption this chapter's case vignette showed failing at the worst

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moment. Validation is therefore an ongoing discipline applied to every input a model depends on, not a one-time check of its headline PD or LGD.

9.7 Case Vignette: The Gaussian Copula and the Mismodeling of CDO Credit Risk

Chapter 8's case vignette described how correlations that looked stable in calm markets rose sharply during the 2008 crisis. The credit-risk analogue concerns collateralized debt obligations (CDOs), securities that pool many loans, mortgages, or bonds and slice the cash flows into tranches of different seniority, and the correlation model Wall Street and the rating agencies used to price and rate them.

In 2000 the quantitative analyst David X. Li published an influential paper proposing the Gaussian copula as a tractable way to model default correlation across many borrowers. A single correlation parameter summarized how likely two borrowers were to default together, turning an intractable joint-default problem into one easily calibrated number, and the approach was quickly adopted to price and rate mortgage-backed CDO tranches.

The difficulty was the one this chapter's economic-capital example flags. The correlation parameter was typically calibrated on a period of calm, rising home prices, and it assumed that mortgage defaults across geographically diverse pools would remain only modestly correlated, largely independent local events. When the nationwide housing downturn arrived, defaults across the country rose together far more than assumed, the "correlations go to one in a crisis" pattern, and AAA-rated CDO tranches, built on the premise that a nationwide wave of correlated defaults was nearly impossible, suffered mass downgrades and defaults in 2007 and 2008. A widely read 2009 press account called the Gaussian copula "the formula that killed Wall Street," not because the mathematics was wrong (a copula is a legitimate way to model dependence) but because its single correlation number was calibrated on calm pre-crisis data that understates tail dependence.

The lesson is easy to oversimplify into "correlation models are untrustworthy." The Gaussian copula and Section 9.4.2's Vasicek model are close relatives: both reduce a joint-default problem to a single correlation acting through a shared systematic factor, and neither is mathematically inconsistent.

What failed was an estimation choice, made once on a sample that never contained a nationwide housing downturn and then treated as fixed rather than stress-tested the way Section 9.2.3 stress-tests PD and LGD. Stressing the correlation toward a more severe scenario, as Section 9.4.2 does for this chapter's loan portfolio, would have revealed a far larger tail loss before 2007, with the same model. The failure was organizational as much as mathematical: nothing in the copula framework prevented stress-testing the correlation, but originators, rating agencies, and investors all benefited in the short run from a favorable, low-correlation rating, so nobody was rewarded for asking the question.

As in market risk, a correlation calibrated on calm historical data is a recurring point of failure, and mathematical sophistication is no substitute for stress-testing the assumption, here default correlation, that matters most when conditions are worst.

9.8 Challenges in Credit Risk Modeling

Several challenges recur across nearly every credit risk application, extending the market-risk challenges of Chapter 8.

Correlated defaults and tail dependence. The independent-default simplification of Section 9.4.1 understates portfolio risk for pedagogical clarity; a real portfolio model must estimate default correlation, the difficulty this chapter's case vignette showed going wrong at scale. Section 9.4.2's single-factor model brings correlation back in but assumes one shared factor; a portfolio spanning several industries and geographies is better described by several correlated factors, and misspecifying how many exist, or how strongly a borrower loads on each, is statistically difficult and highly consequential for capital.

Procyclicality. PD and LGD calibrated on calm data understate risk in good times and spike once a downturn is underway, an amplifying loop, since understated capital encourages more lending when underwriting standards are already loosening. Section 9.2.3's stress test counteracts this by testing the scenario before it arrives. Section 9.4.2's combined calculation, a 6.6× increase in economic capital from

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three modest-looking assumptions, shows how severely procyclicality compounds when every input moves the same way.

Data quality and limited loss history. Credit data is plentiful in rows and thin in the dimension that constrains a model. A lender may hold records on a million loans, yet the number of complete economic cycles those loans have lived through is one, or zero, and default is a slow event: an origination cohort's true loss rate is not known until years after the underwriting decision it would judge. A decade of records can therefore amount to a single draw from the distribution a PD model is meant to describe, which is why a PD estimated from a decade that happened not to include a recession will systematically understate true credit risk. The sample is also selected rather than merely observed, since outcomes exist only for the applicants the lender chose to approve. That is Section 9.2.6's reject-inference problem as a data limitation, and no number of additional rows fixes it, because the missing borrowers are missing by construction. Section 9.2.7's perfect in-sample AUC is the same danger in miniature: with six training observations, apparent discriminatory power says almost nothing about performance on new applicants.

Regulatory and business tension. What makes this tension specific to credit is timing. A market-risk limit breach announces itself within days, whereas a loosened underwriting standard books revenue immediately and delivers its losses two or three years later, so for several reporting periods a credit function being overruled looks like one that was merely too cautious. That is why lending standards reliably loosen late in an expansion, when competition for volume is fiercest and the last downturn furthest away, and it is what the three-lines-of-defense structure of Chapter 8 is meant to counteract, by separating the unit that originates loans from the one that validates the models pricing and reserving them. A second constraint has no market-risk analogue: a consumer credit model must be explainable to the applicant it declines, under the adverse-action rules of Chapter 18, so pressure toward raw accuracy and toward transparency can pull the model's functional form in opposite directions, the trade-off monotonic constraints exist to negotiate.

Regulatory arbitrage. Whenever capital is calculated by a specific formula, whether the Basic Indicator Approach's gross-income proxy or an internal-ratings-based model, an institution has an incentive to minimize the capital charge rather than the underlying risk, a gap between measured and true risk that can widen quietly even when every calculation is technically correct. An institution allowed to choose its own asset correlation in Section 9.4.2's framework, for instance, has an obvious incentive to pick a lower ρ than its portfolio exhibits, understating capital while remaining compliant on paper.

Machine learning adds a further layer of model risk. Tree ensembles and neural networks can capture nonlinear default and loss patterns that logistic regression and the Merton model, with their fixed functional forms, cannot, but they inherit the overfitting and interpretability concerns of Chapter 6 in a setting where undetected errors are unusually costly. An opaque credit model that a regulator or validator cannot interrogate is hard to trust with a bank's capital, whatever its backtested accuracy, so Section 9.2.7's discrimination and calibration checks apply to a gradient-boosted model as much as to a hand-built logistic regression.

9.9 Key Takeaways

Credit risk decomposes into probability of default, loss given default, and exposure at default, whose product gives expected loss. Logistic regression is the standard tool for estimating PD from borrower characteristics, evaluated with the Gini coefficient and AUC for ranking power and separately for calibration. The Merton model and CDS-implied hazard rates estimate PD from market prices (equity or CDS spreads) without a default history.

LGD is not a fixed constant either: the collateral-coverage regression of Section 9.2.2 estimates recovery from a lender's own default history.

Rating transition matrices extend one-period default probabilities to multi-year cumulative estimates, and credit stress testing shows what happens to expected loss when PD and LGD move toward recession values together. That same matrix supplies the PD term structure IFRS 9 and CECL require, turning a one-year expected loss into a lifetime one: for the chapter's Medium-rated exposure the provision rises from \$57.14 to \$188.70, a 3.30× cliff crossed by reclassification alone, and a fitted discrete-time hazard model conditions that figure on the economy, where the convexity of loss in the macroeconomic factor is why the standard demands probability-weighted scenarios rather than a single central forecast.

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The chapter's most useful negative result concerns machine learning itself. A credit scorecard, weight-of-evidence bins fed to a logistic regression, is a generalized additive model fitted by hand, the object an Explainable Boosting Machine fits automatically, and on simulated data containing an interaction it lands within 0.003 AUC of the theoretical ceiling with nothing beating it. Machine learning earns its keep in credit by doing what a scorecard cannot: building features from data with no natural bins, producing a term structure rather than a probability, and searching a model space rather than fitting one model. Where a tree ensemble is the right tool, monotonic constraints buy back the scorecard's defensibility, here at no cost in accuracy.

Economic capital turns the gap between expected and unexpected loss into a capital buffer, and the Vasicek model of Section 9.4.2 shows how much larger that buffer must be once defaults are correlated. Stress and correlation combined multiplied required capital nearly sevenfold in the worked example, and the case vignette showed the same effect, neglected, contributing to a systemic mispricing of mortgage-backed credit risk.

Two extensions push the same decomposition into territory where its inputs are no longer fixed numbers. A derivative's exposure is a random variable, summarized by expected and potential future exposure profiles, and the credit valuation adjustment is expected loss applied period by period to that profile: \$10,294 on the chapter's \$10 million swap, cut to \$6,763 by a collateral threshold and to \$1,743 by full variation margin, and doubled by a wrong-way correlation of only -0.3. Climate risk, in turn, arrives through PD and LGD rather than as a new risk type, and a stylized carbon-price shock raised a sector loan book's expected loss 3.46-fold while showing that the safest borrowers on base-case PD can be the most exposed to the transition, a ranking that matters more than the headline figure given how much of the transmission rests on assumptions no history can test.

None of these methods is self-policing: independent validation of both discrimination and calibration, within a governance structure like the three lines of defense of Chapter 8, is what catches a miscalibrated model before it causes a loss. As in market risk, a risk number, whether a VaR figure or a credit score, is only as trustworthy as its assumptions, so sound practice combines several methods and accounts openly for where they disagree.

9.10 Suggested Exercises

9.1 A loan has EAD of \$500,000, PD of 5%, and LGD of 50%. Compute its expected loss, and explain how each of the three components would need to change to cut the expected loss in half.

9.2 Using the collateral-coverage regression in Section 9.2.2, compute the predicted recovery rate and implied LGD for a loan with a collateral coverage ratio of 1.50, and explain why extrapolating the fitted line this far beyond the observed data (which ranged from 0.40 to 1.20) should be treated cautiously.

9.3 Using the logistic regression coefficients in Section 9.2.6, compute the predicted probability of default and the corresponding credit score for an applicant with a debt-to-income ratio of 22% and 6 years of credit history.

9.4 Using the same bond as Section 9.2.3, recompute the stressed expected loss under a milder recession scenario with PD = 6% and LGD = 65%, and compare the resulting multiple of the base-case expected loss to the $3.11\times$ figure computed in the text.

9.5 Using Table 9.2, verify by hand that the Gini coefficient is 1.0, and explain what would happen to the computed AUC if applicant E's true outcome had instead been "No default" rather than "Default."

9.6 Using the transition matrix in Table 9.4, compute the four-year cumulative default probability for a Medium-rated borrower.

9.7 Using the Merton model from this chapter, compute the one-year default probability for a firm with asset value $V_0 = \$200$ million, debt of $D = \$120$ million due in one year, asset volatility $\sigma_V = 30\%$, and a risk-free rate of $r = 5\%$.

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9.8 A portfolio of 200 identical loans each has EAD of \$2,000, PD of 4%, and LGD of 45%, with defaults assumed independent. Compute the portfolio's expected loss, the standard deviation of portfolio losses, and its economic capital at the 99% level.

9.9 A bank's average annual gross income over the past three years is \$350 million. Using the Basic Indicator Approach, compute its operational risk capital requirement, and explain one advantage and one disadvantage of tying operational risk capital to gross income rather than to a bank's actual operational loss history.

9.10 Explain why a credit model can have a high Gini coefficient while still being poorly calibrated, and describe a concrete lending decision that would go wrong as a result even though the model ranks applicants correctly.

9.11 Explain, using the Gaussian copula case vignette, why a mathematically sound correlation model can still contribute to a systemic mispricing of risk, and describe what a stress test of the correlation assumption itself, rather than of PD or LGD individually, might have revealed before 2008.

9.12 Explain why reject inference is fundamentally an incomplete fix for the selection bias described in Section 9.2.6, and describe a concrete way a lender's approval policy could change over time such that a credit model validated on last year's approved applicants performs noticeably worse on this year's.

9.13 Using the transition matrix in Table 9.4, explain in your own words why a mark-to-market credit risk model would assign a Medium-rated bond a lower value than an identically priced High-rated bond, even in a year when neither bond actually defaults.

9.14 Pick two of the challenges listed in this chapter and describe a concrete scenario, not already used in the text, in which that challenge could cause a credit risk management system to understate a firm's true risk.

9.15 Using Section 9.4.2's single-factor formula, recompute the 99% worst-case default rate for the 100-loan portfolio under an asset correlation of $\rho = 0.12$ instead of $\rho = 0.20$, and explain in your own words why a lower asset correlation produces a lower worst-case default rate for the same average PD.

9.16 Using this chapter's mark-to-market bond example, recompute $\mathbb{E}[V]$ and its standard deviation if the Medium-rated yield were 9% instead of 7% (holding the transition probabilities, coupon, and default recovery fixed), and explain why the expected value falls by less than the price gap between the two Medium-rated scenarios might suggest at first glance.

9.17 Using this chapter's Herfindahl-Hirschman Index example, compute the HHI for a portfolio with the same \$100,000 total exposure concentrated in just 4 borrowers of \$25,000 each, and explain why this portfolio's expected loss is identical to, but its concentration risk is much higher than, the 100-loan portfolio's.

9.18 Using this chapter's EAD example, compute the exposure at default for a \$1,000,000 committed credit line currently drawn to \$400,000, assuming a CCF of 60% instead of 50%, and explain why a higher CCF assumption increases a portfolio's expected loss even if PD and LGD are unchanged.

9.19 A one-year CDS on a different issuer trades at a spread of $s = 350$ basis points, and the assumed recovery rate is $R = 35\%$. Using Section 9.3.3's credit-triangle approximation, compute the implied hazard rate and the one-year default probability, and compare the result to a Merton-model estimate for the same firm if one were available.

9.20 Combine a stressed PD = 6% with an asset correlation of $\rho = 0.15$ for the 100-loan portfolio, and compute the resulting economic capital, comparing it to both the base-case (\$2,381) and the fully stressed (\$15,704) figures computed in the text.

9.21 Explain why the CDS-implied default probability approach requires no estimate of asset volatility, unlike the Merton model, and describe the specific circumstance under which a CDS-implied estimate would not be available for a given borrower.

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9.22 Using this chapter's Merton-model inversion example, explain in your own words why the firm's implied equity volatility (67.03%) is so much higher than its assumed asset volatility (25%), and describe what would happen to that gap, wider or narrower, if the firm instead had much less debt outstanding relative to its asset value.

9.23 Using the firm in Exercise 9.7 ($V_0 = \$200$ million, $D = \$120$ million, $\sigma_V = 30\%$, $r = 5\%$), compute its Merton-model equity value E_0 and implied equity volatility σ_E, and explain what two market-observable quantities a real analyst would feed into a numerical solver, without knowing V_0 or σ_V in advance, to recover this same result in practice.

9.24 Using Table 9.3, merge the two lowest debt-to-income bands into a single band covering everything below 25%, recompute that band's weight of evidence and the characteristic's information value, and explain why merging bins can never increase information value and should therefore be done only when monotonicity or bin size demands it.

9.25 Recompute Section 9.2.8's points spread between the safest and riskiest debt-to-income bands using PDO = 25 instead of 20, and confirm that although the spread in points changes, the implied odds ratio does not. Explain what this says about which parts of a scorecard's scale are substantive and which are presentational.

9.26 Repeat Table 9.5's lifetime expected credit loss calculation for a *High*-rated borrower instead of a Medium-rated one, using the same transition matrix, LGD, exposure, and discount rate. Verify the second year's marginal probability by hand, report the resulting Stage 1 and Stage 2 provisions, and explain why the ratio between them is *larger* for the higher-quality borrower even though every provision figure is smaller.

9.27 The chapter's discrete-time hazard example weights the benign, base, and adverse scenarios 30/50/20. Recompute the probability-weighted lifetime ECL under weights of 20/50/30 and 10/50/40, and express each as an excess over the base-scenario-only figure of \$181.79. Explain why that excess grows as weight shifts toward the adverse scenario, referring to the convexity of expected loss in the macroeconomic factor.

9.28 Repeat Table 9.1's carbon-price stress assuming every borrower can pass half of its carbon cost on to customers ($\pi = 0.5$). Compute each sector's c and stressed PD by hand, confirm the book's stressed expected loss, and explain why the utilities sector's stressed PD falls by proportionally more than the transportation sector's.

9.29 Using Equation (9.2) with no pass-through, recompute the fossil-heavy utilities' stressed PD at a carbon price of \$300 per tonne, the level the NGFS associates with a net-zero path by 2035, and explain why tripling the carbon price more than triples the stressed PD. Then describe a second, LGD channel through which the same shock would raise the oil and gas sector's expected loss, and why that channel is missing from the table.

9.30 For the swap of Section 9.5.1, compute $\mathrm{EE}(t)$ and $\mathrm{PFE}_{95\%}(t)$ at $t = 1$ and $t = 4$ using Equation (9.11), and recompute the CVA of Table 9.7 for a counterparty whose CDS trades at 350 basis points with an assumed recovery of 35%, holding the exposures and discount factors fixed. By what factor does the CVA change, and why is it not simply 350/200?

9.31 Explain, using Section 9.5.5's latent-variable model, why the wrong-way multiplier on expected exposure grows as the counterparty becomes *safer* (a lower default probability, so a more negative threshold k), holding the correlation fixed. What does this imply for a bank that sells large amounts of protection to, or trades large swaps with, highly rated counterparties whose defaults would coincide with extreme market moves?

9.32 A private-credit fund reports returns from quarterly model marks that adjust only partly to news. Model the reported return as $r^*_t = w\, r_t + (1 - w)\, r^*_{t-1}$, where r_t is the true, serially independent return with standard deviation σ and $0 < w \le 1$. (a) Show that $\mathrm{Var}(r^*_t) = w\,\sigma^2/(2 - w)$ and that the reported returns have first-order autocorrelation $1 - w$. (b) For $w = 0.4$, compute the ratio of

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reported to true volatility, and the Sharpe ratio the fund would report if its true Sharpe ratio were 0.50. (c) Using the private-credit discussion in Section 9.3.3, explain why a portfolio optimizer or a Vasicek-style capital model fed with these reported returns would allocate too much to private credit, and suggest one way to undo the smoothing before using the data.

9.33 **Challenge (real data).** Download the UCI Statlog (German Credit Data) dataset (1,000 loan applicants, 20 features, a binary good/bad credit-risk label) directly from the UCI Machine Learning Repository. (a) Fit a logistic regression predicting default and report its AUC and Gini coefficient on a held-out test split, the same validation performed on the toy six-applicant model in Section 9.2.6. (b) Treating the fitted probabilities as an estimated PD and assuming a flat 60% LGD, compute the portfolio's expected loss and compare it to the realized default rate in your test split. (c) Before fitting anything, write a short data-cleaning note on at least two features whose categorical codes (for example, checking-account status) are not self-explanatory from the column names alone, the same undocumented-category-code problem Chapter 21's capstone case study encounters on a different real dataset entirely.

9.11 Further Reading

- Saunders and Allen (2010), *Credit Risk Management In and Out of the Financial Crisis*. Covers credit scoring, rating models, and portfolio credit risk with a strong emphasis on the lessons of the 2008 crisis referenced in this chapter's case vignette.

- Merton (1974), "On the Pricing of Corporate Debt: The Risk Structure of Interest Rates." The original paper establishing the structural credit model developed in Section 9.3.1.

- Gupton et al. (1997), *CreditMetrics—Technical Document*. J.P. Morgan's original technical document introducing the ratings-migration, mark-to-market portfolio credit risk framework discussed in this chapter's treatment of rating transition matrices.

- BCBS (2010), *Basel III: A Global Regulatory Framework for More Resilient Banks and Banking Systems*. The primary regulatory source for the capital adequacy concepts introduced in this chapter's discussion of economic and regulatory capital.

- Li (2000), "On Default Correlation: A Copula Function Approach." The original paper proposing the Gaussian copula model discussed in this chapter's case vignette.

- Salmon (2009), "Recipe for Disaster: The Formula That Killed Wall Street." A widely read journalistic account of the Gaussian copula's role in the 2008 financial crisis, providing helpful context for this chapter's case vignette.

- Vasicek (2002), "The Distribution of Loan Portfolio Value." The original source for the single-factor correlated-default model developed in Section 9.4.2, underlying the Basel IRB capital formulas.

- Hull (2022), *Options, Futures, and Other Derivatives*. Covers credit default swap pricing and the hazard-rate approximation used in Section 9.3.3's market-implied default probability discussion in considerably greater depth.

- Hand and Henley (1997), "Statistical Classification Methods in Consumer Credit Scoring: A Review." The standard survey of the scorecard tradition developed in Section 9.2.8, written before machine learning entered the field and therefore unusually clear about what the classical method was actually optimizing.

- Lou et al. (2012), "Intelligible Models for Classification and Regression." The paper behind the Explainable Boosting Machine, which fits by boosting the same additive form Section 9.2.9 shows a credit scorecard has always had.

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- BCBS (2015), *Guidance on Credit Risk and Accounting for Expected Credit Losses*. The supervisory expectations governing the lifetime expected-loss models discussed in this chapter, including the scenario-weighting requirement whose convexity consequence the chapter computes.
- Bellotti and Crook (2009), "Credit Scoring with Macroeconomic Variables Using Survival Analysis." Develops the discrete-time hazard approach used in this chapter on real consumer credit data, including the macroeconomic conditioning that a fixed transition matrix cannot express.
- Gregory (2020), *The xVA Challenge*. The standard practitioner text on counterparty credit risk, covering the exposure profiles, netting and collateral, wrong-way risk, and the full family of valuation adjustments introduced in Section 9.5.
- Huge and Savine (2020), "Differential Machine Learning: The Shape of Things to Come." Introduces differential machine learning, training neural-network pricing approximations on both simulated values and their adjoint-computed derivatives, the technique Section 9.5.6 connects to XVA computation.
- NGFS (2024), *NGFS Climate Scenarios for Central Banks and Supervisors: Phase V*. The reference climate scenarios used by most supervisory exercises, with the transition and physical-risk narratives behind the stress test of Section 9.2.4.
- Crouhy et al. (2014), *The Essentials of Risk Management*. A practitioner-oriented synthesis spanning market, credit, and operational risk, model validation, and risk governance, tying together the full breadth of topics covered in this chapter and Chapter 8 in a single volume.

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Chapter 10

Operational Risk Modeling

10.1 Introduction

In January 2008, Société Générale disclosed a loss of roughly 4.9 billion euros. No market view had been wrong, no borrower had defaulted, and no model had mispriced a derivative. A single trader on the bank's arbitrage desk, Jérôme Kerviel, had built enormous unauthorized directional positions and concealed them behind offsetting fictitious trades that the bank's own controls failed to catch until the positions had to be unwound into a falling market. The loss was not a market-risk loss or a credit-risk loss in the sense of the previous two chapters. It was an *operational* loss: a failure of the bank's internal processes, systems, and people.

Chapter 8 developed the machinery for market risk, how much a position's value could move, and Chapter 9 developed the machinery for credit risk, whether a counterparty will pay at all. This chapter turns to the risk that is left over once those two are accounted for, and that neither of their toolkits handles well: the risk of loss from failed internal processes, people, systems, or external events. Operational risk is at once the most intuitive of the three, everyone understands a rogue trader or a systems outage, and the hardest to model, because its events share almost none of the statistical regularity that makes market and credit risk tractable.

It would be a mistake to treat operational risk as a minor residual. For many large banks it consumes a comparable share of regulatory capital to market risk. Its worst realizations (multi-billion-dollar rogue-trading losses, mis-selling scandals that end in nine- and ten-figure settlements, systems failures that halt a payment network) have brought down or nearly brought down major institutions in a way a bad week of trading rarely does. What makes it distinctive is its character: market and credit losses arrive through prices and defaults that can be modeled with dense data, while operational losses arrive through the failure of the institution's own machinery, and the data on those failures is scarce, messy, and mostly unstructured.

That difficulty is why the chapter's central message is nuanced rather than triumphant. Artificial intelligence does not hand operational risk the kind of clean predictive model it hands credit scoring or volatility forecasting. Its contribution is asymmetric. It is powerful in the layers made of unstructured data and rare-pattern detection: reading incident reports, flagging anomalous behavior, surveilling for misconduct, warning early from leading indicators. It is limited in the layer regulators care most about, putting a credible number on the far tail. Keeping that asymmetry in view is the discipline this chapter is built around.

The chapter proceeds in three movements. It first establishes what operational risk is and why it resists a unified statistical model, then develops the classical actuarial baseline, the loss distribution approach, against which any AI method must be measured. It then works through the four layers where machine learning contributes (text, anomaly detection, conduct surveillance, and predictive early warning), each reusing methods developed earlier in the book. It closes with the ways deploying AI itself *creates* new operational risk, and with the regulatory reason AI's natural home here is risk management rather than regulatory capital.

10.2 Measuring Operational Risk

Operational risk resists a single model, and this part explains why, then builds the loss distribution anyway.

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10.2.1 What Operational Risk Is, and Why It Resists a Single Model

The Basel Committee defines operational risk as the risk of loss resulting from inadequate or failed internal processes, people, and systems, or from external events. The definition is deliberately broad, and it is conventionally organized into seven event-type categories, shown in Table 10.1, that span everything from a rogue trader to a flooded data center.

Event type	Representative event
Internal fraud	A trader conceals unauthorized positions
External fraud	A third party breaches systems and steals funds
Employment practices & workplace safety	A discrimination or wrongful-termination settlement
Clients, products & business practices	A mis-sold product or a market-manipulation fine
Damage to physical assets	A fire or natural disaster destroys a facility
Business disruption & system failures	A core banking outage halts settlement
Execution, delivery & process management	A failed trade settlement or data-entry error

Table 10.1: The seven Basel operational-risk event types, with representative losses

What makes this list awkward to model is that its rows share so little. A market-risk desk has a long, dense history of daily returns; a credit portfolio has thousands of loans with observable defaults. Operational risk has neither. Its defining events are *rare*, so any single institution observes only a handful of large losses in its entire history; *heavy-tailed*, so the few large events dominate the total and the average loss is a poor guide to the extreme one; *heterogeneous*, so a rogue-trading loss and a mis-selling fine and a data-center fire have no common driver a single model could exploit; and subject to severe *reporting and survivorship bias*, since near-misses go unrecorded, small losses are absorbed silently, and the institutions that suffer the very largest losses sometimes do not survive to contribute them to any database.

A further complication, invisible in the flat list of Table 10.1, is that the event types differ enormously in their frequency-severity profile, and in opposite directions. Execution, delivery, and process-management errors are the most *frequent* (a mistyped settlement, a duplicated payment, a reconciliation break happens daily at a large bank), but each is individually small. Clients, products, and business-practices events, such as mis-selling, market-conduct violations, and large regulatory settlements, are comparatively *rare* but individually enormous. At most large banks they dominate total operational losses by value despite being a small fraction of events by count. A model or control program calibrated to the frequent-but-small failures will therefore miss most of the actual loss, one more reason the tail, not the average, is where operational-risk attention belongs.

Chapter 9 reached the same conclusion when it introduced operational-risk capital from the credit side: a rogue trader's unauthorized position, a failed core-banking upgrade, and a mis-sold product "share so little common structure that no single statistical framework fits all of them well." No neural network will close that gap. It is the fixed constraint within which every method in this chapter, classical or AI, has to operate.

10.2.2 The Four Data Elements of Operational Risk

Because no single data source captures operational risk well, the discipline draws on four distinct elements. They explain both where the loss model of the next section gets its inputs and where artificial intelligence has the most to add.

Internal loss data is a bank's own history of realized operational losses. It is the primary input to the frequency and severity estimates of the loss distribution approach described below, but it is sparse, especially in the tail, and skewed toward the small, frequent losses that are least relevant to capital. *External loss data* is drawn from consortium databases of losses suffered by peer institutions. It supplements the tail with events a single firm has been fortunate enough not to experience, at the cost of relevance, since another firm's loss may reflect a control environment nothing like one's own. *Scenario analysis* fills the remaining gap with structured expert judgment: workshops in which experienced staff estimate the frequency and severity of

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severe hypothetical events that have not occurred but plausibly could. It is the only element that can speak to unprecedented tail risk. *Business environment and internal control factors* are the forward-looking indicators, the key risk indicators of Section 10.3.5 among them, that describe the current control environment rather than past losses.

Three of these four elements are largely text and expert judgment rather than clean numbers, which is why the natural-language and classification methods of this chapter matter so much. The incident narratives behind internal loss data, the free-text descriptions in external databases, and the qualitative reasoning of scenario workshops are all unstructured, and turning them into model inputs is the AI-heavy part of operational-risk practice. The actuarial machinery of the next section is only as good as the data these elements feed it; its most common failure in practice is not a flaw in the mathematics but a thin, biased, or miscategorized loss dataset underneath it. Figure 10.1 shows how the four elements combine to feed both the quantitative loss model and the qualitative control assessment.

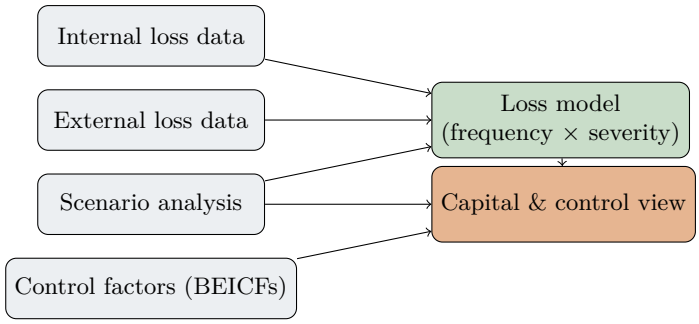


Figure 10.1: The four operational-risk data elements. Three of them are largely unstructured text and judgment, which is where AI enters.

10.2.3 The Loss Distribution Approach: Frequency Times Severity

Suppose a bank wants a single number summarizing how large its operational losses could be over the coming year, for one business line and one event type, before knowing which events will occur. The classical actuarial answer, still the workhorse of quantitative operational-risk modeling, is the loss distribution approach. It builds the annual loss out of two separately estimated pieces: how *many* loss events occur (frequency), and how *large* each one is (severity).

Frequency is typically modeled as a Poisson random variable with rate λ, the expected number of events per year. Severity is modeled as a positive, right-skewed distribution fit to the sizes of past losses; the lognormal is the standard first choice. The total annual loss S is then the sum of a random number of severity draws,

$$S = \sum_{i=1}^{N} X_i, \qquad N \sim \text{Poisson}(\lambda), \qquad X_i \sim \text{Lognormal}(\mu, \sigma), \tag{10.1}$$

a construction known as a compound Poisson distribution. Because this sum has no convenient closed form, its distribution is obtained by Monte Carlo: simulate a year by drawing N from the Poisson, then drawing N severities and adding them; repeat for hundreds of thousands of simulated years; and read the required quantiles off the resulting histogram of annual losses. This is the same Monte-Carlo-of-a-portfolio logic Chapter 8 used for market-risk Value at Risk, applied now to a compound frequency-severity sum rather than to correlated asset returns.

10.2.4 Worked Example 10.1: The Annual Loss of One Business Line

Consider a single business line with a Poisson frequency of $\lambda = 12$ loss events per year, the credibility blend of internal, external, and scenario data derived in Section 10.2.5 below, and lognormal severities with $\mu = 10.5$ and $\sigma = 1.4$ (in natural logs of dollars). Those severity parameters correspond to a median loss of about \$36,000 and, because the lognormal is right-skewed, a larger mean loss of about \$97,000: the average is pulled

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up by occasional large events. Simulating 500,000 years gives the annual-loss distribution summarized in Table 10.2.

Quantity	Annual loss
Expected annual loss	\$1,160,000
99% Value at Risk	\$4,340,000
99% Expected Shortfall	\$6,110,000
99.9% Value at Risk (regulatory capital quantile)	\$8,450,000

Table 10.2: Loss distribution approach: simulated annual operational loss for $\lambda = 12$, Lognormal$(10.5, 1.4)$ severities, 500,000 simulated years

Two features of Table 10.2 carry the entire lesson of operational-risk quantification. First, the expected annual loss, about \$1.16 million, is the number a business line would budget for as a routine cost of doing business, and it is almost irrelevant to the capital question. Second, the 99.9% quantile, the loss exceeded only once in a thousand years and the level at which regulatory operational-risk capital is conventionally set, is about \$8.45 million, roughly *seven times* the expected loss. Capital for operational risk is overwhelmingly about the tail, and a model that fits the body of the loss distribution beautifully while getting the tail slightly wrong can misstate the capital number by millions. Modeling that tail well, typically with heavy-tailed and extreme-value methods that are a specialized study in their own right, is the central technical difficulty of operational-risk capital, and it is where sparse, heavy-tailed loss data is least forgiving.

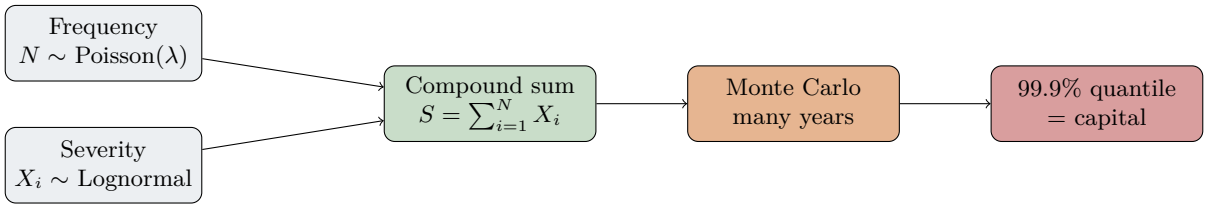


Figure 10.2: The loss distribution approach: Monte Carlo convolves frequency and severity into an annual-loss distribution, and capital is read off its far tail

10.2.5 Combining the Elements: Credibility and Bayesian Updating

Figure 10.1 draws three arrows into the loss model without saying what happens where they meet, and that junction is a real modeling decision rather than a bookkeeping step. A business line's own five-year history might imply eight or nine events a year; the consortium database of comparable peers, and the scenario workshop asked to imagine what has not yet happened, might imply twice that. Averaging the two is arbitrary, and picking whichever is preferred is worse. What is needed is a rule that leans on internal data in proportion to how much of it there is and falls back on the external and scenario view when it is thin. That is the problem credibility theory solves, which Chapter 11 develops for pricing an individual policyholder against a book-wide rate.

The operational-risk version has an especially clean form. Treat the annual event count as Poisson with an unknown rate λ, and encode the external-plus-scenario view as a Gamma prior on λ with mean λ_{prior} and a strength parameter β measured in *years of equivalent experience*, so that asserting $\beta = 3$ says the outside view is worth about three years of the bank's own history. The Gamma is conjugate to the Poisson, so after observing $\sum n_i$ events over n years the posterior mean is available in closed form, and it rearranges into the credibility blend of Chapter 11:

$$\hat{\lambda} \;=\; \frac{\alpha + \sum_i n_i}{\beta + n} \;=\; Z\,\hat{\lambda}_{\text{internal}} + (1 - Z)\,\lambda_{\text{prior}}, \qquad Z = \frac{n}{n + \beta}, \tag{10.2}$$

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with $\alpha = \beta\,\lambda_{\text{prior}}$. The credibility factor Z carries all the judgment: it rises toward one as the internal history lengthens, and the whole disagreement between the data elements is resolved by a single, auditable statement about how many years of internal experience the outside view is worth.

Work it through for the business line of Section 10.2.4. Internal loss data records 42 events over $n = 5$ years, so $\hat{\lambda}_{\text{internal}} = 8.4$. External data and the scenario workshop together point to $\lambda_{\text{prior}} = 18$, held with strength $\beta = 3$ years, so $\alpha = 54$. Then

$$Z = \frac{5}{5+3} = 0.625, \qquad \hat{\lambda} = 0.625(8.4) + 0.375(18) = 5.25 + 6.75 = 12.0,$$

which is also $(54{+}42)/(3{+}5) = 96/8 = 12.0$ read straight off the posterior, and it is the frequency Section 10.2.4 uses. That example's $\lambda = 12$ is not an assumption; it is what the four data elements produce once they are combined properly. The posterior additionally reports its own uncertainty, $\sqrt{\alpha + \sum n_i}/(\beta{+}n) = \sqrt{96}/8 \approx 1.22$ events per year, which the point estimate alone conceals.

The stakes are visible in the capital number. Running the loss-distribution Monte Carlo described above at each candidate frequency, holding the severity distribution fixed, gives a 99.9% quantile of about \$7.19 million on internal data alone, \$8.45 million on the blend, and \$10.26 million on the outside view alone. A bank that quietly used only its own loss history, the most defensible-looking choice, would understate its capital by roughly 15%, because it had been fortunate enough not to experience the events its peers had. This is the one place where a statistical method reaches inside the loss-modeling core, and it is Bayesian statistics of the kind Chapter 1 introduced rather than machine learning. What the AI layers below contribute is the raw material this blend consumes: the incident narratives, external database descriptions, and scenario write-ups all have to be classified and cleaned before any of these counts mean anything.

10.3 Where Artificial Intelligence Enters

Four distinct layers of machine learning sit on top of operational-risk data, each answering a different question.

10.3.1 Four Layers Around the Loss-Modeling Core

Nothing in the loss-distribution machinery above used machine learning, and that is deliberate: the frequency-severity core of operational-risk quantification is largely a small-data, distributional problem to which modern AI adds little. Artificial intelligence contributes elsewhere, in four layers that sit around the loss-modeling core rather than inside it, shown in Figure 10.3. Each corresponds to a different point on the timeline of an operational-risk event: the leading indicators that precede it, the near-misses and anomalies that presage it, and the incident record it leaves behind. Each reuses a family of methods developed earlier in the book.

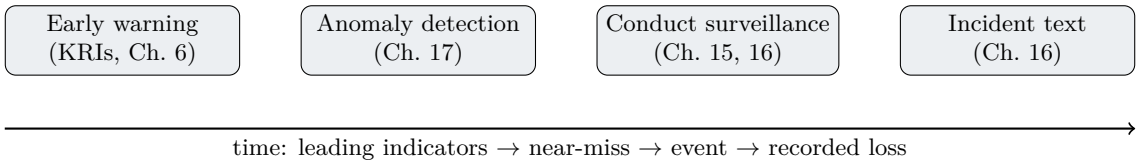


Figure 10.3: Four layers where machine learning contributes, along the timeline of a loss event

10.3.2 AI Layer 1: Text as the Primary Operational-Risk Data Source

The single most abundant operational-risk data source inside a bank is not a table of losses; it is text. Incident reports, internal-audit findings, risk-and-control self-assessment narratives, customer complaints, and help-desk tickets accumulate continuously and describe operational events in far richer detail than any structured loss database captures. But this text is unstructured. Turning it into something a loss model can use, above all sorting each incident into a Basel event type so that frequencies and severities can be estimated per category, is a text-classification task of the kind Chapter 16 developed.

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Consider a minimal illustration in the spirit of Chapter 16's Naive Bayes classifier, distinguishing two event types from short incident descriptions: internal fraud (IF) and business disruption or system failure (SF). Table 10.3 gives a tiny training set.

Event type	Incident descriptions (training)
Internal fraud (IF)	"trader hid unauthorized position"; "employee falsified approval to hide loss"; "unauthorized trade concealed by staff"
System failure (SF)	"batch system failed overnight settlement"; "software deployment caused outage"; "system outage delayed settlement"

Table 10.3: Training incidents for a two-class Naive Bayes event-type classifier

With equal class priors ($P(\text{IF}) = P(\text{SF}) = 0.5$), a vocabulary of $V = 24$ distinct words, and Laplace (add-one) smoothing, the classifier scores a new incident by summing log-probabilities word by word. For the smoothed likelihood of word w in class c,

$$P(w \mid c) = \frac{\text{count}(w, c) + 1}{\text{count}(c) + V}, \tag{10.3}$$

where $\text{count}(c)$ is the total word count in class c. Classifying the new report "unauthorized trade hidden by trader" gives log-scores of about -15.83 for internal fraud and -18.75 for system failure; exponentiating and normalizing turns these into a posterior probability of **0.949** for internal fraud. The words *unauthorized*, *trade*, and *trader* all appear in the fraud training incidents and none in the system-failure ones, so even though *hidden* was never seen in training (and contributes only through smoothing), the classifier is decisive and correct.

At production scale the same idea, moved from a hand-checkable Naive Bayes to TF-IDF representations and contextual embeddings, does three jobs at once: it routes each incident to the right Basel category so the loss distribution approach's frequency-severity model is fed clean per-category data; it clusters free-text root-cause narratives to surface recurring control weaknesses no single report would reveal; and it flags, from complaint text alone, the mis-selling and conduct issues that make up the "clients, products and business practices" event type, often the single largest source of operational losses at large banks.

The clustering job deserves emphasis, because it is where text analytics does something the structured loss database cannot. A bank's incidents are logged by whoever wrote them up, in inconsistent language, and the Basel category assigned at logging is often a coarse after-the-fact guess. Unsupervised topic modeling over the incident corpus, the same impulse as Chapter 6's clustering, groups incidents by what happened rather than by how they were filed. It routinely reveals that a scatter of separately logged "minor" incidents (a repeated reconciliation break, a recurring manual workaround) share a single underlying control failure that, uncorrected, is one bad day away from a large loss. Extracting a structured severity and root cause from each narrative, increasingly with large language models rather than keyword rules, is what turns a pile of free text into the clean per-category frequency and severity data the loss distribution approach silently assumes it was handed.

10.3.3 AI Layer 2: Anomaly Detection for Near-Misses and Control Breaks

Waiting for a recorded loss is waiting too long. Many operational failures announce themselves first as *anomalies* (a settlement process that suddenly takes an unusual path, a user account that touches systems it never normally touches, a trade booked outside its desk's usual pattern) well before they crystallize into a reportable loss. Because these near-misses are deviations from normal behavior rather than examples of a labeled class, they are a natural fit for the unsupervised anomaly-detection methods of Chapter 17, above all the autoencoder, which learns to reconstruct normal activity and flags whatever it reconstructs poorly.

The mechanism is simple: train a model on a large sample of ordinary daily control-log feature vectors, then score each new vector by its reconstruction error, the squared distance between the vector and the model's attempt to reproduce it, and flag any observation whose error exceeds a high percentile of the normal

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distribution. In a stylized five-feature illustration where normal reconstruction errors follow the expected chi-square pattern, the 99th-percentile threshold sits at about 14.9 (close to the theoretical chi-square value of 15.1), and an injected anomalous vector representing a process break produces a reconstruction error of 35.5, comfortably above the threshold and therefore flagged.

The single-vector result flatters the method, though, and a fuller test shows why. Applying the same 99th-percentile threshold (about 15.2 in this run) to a test set of 500 normal observations and 50 anomalous ones, whose feature means are shifted only moderately rather than dramatically, gives the confusion matrix in Table 10.4: the detector catches 28 of the 50 anomalies (recall 0.56) at a precision of 0.85. The precision is respectable, but the recall is the sobering number: at a threshold strict enough to keep false alarms low, the detector misses nearly half of the moderate anomalies entirely. Loosening the threshold to catch more of them would flood the investigators with false positives, the same precision-recall trade-off Chapter 15 made central to fraud detection. Anomaly detection is therefore a triage tool that narrows where humans look, not a substitute for their judgment.

	Flagged	Not flagged
Truly anomalous (50)	28 (true positive)	22 (false negative)
Truly normal (500)	5 (false positive)	495 (true negative)

Table 10.4: Anomaly detector at the 99th-percentile reconstruction-error threshold, on a test set of 500 normal and 50 anomalous observations

At production scale, on real transaction, access, and trade logs, the same logic lets an operational-risk team catch a control failure while it is still a near-miss, the point on the event timeline where intervention is cheapest. The team must still read the recall number soberly and not mistake a quiet alert queue for an absence of problems.

10.3.4 AI Layer 3: Conduct and Rogue-Trading Surveillance

The Société Générale loss that opened this chapter belongs to a long line (Barings and Nick Leeson in 1995, Société Générale and Jérôme Kerviel in 2008, UBS and Kweku Adoboli in 2011) in which a single trader's concealed positions, undetected until far too late, produced a loss large enough to threaten the institution. Each is, in retrospect, a surveillance failure. The warning signs (unusual trade cancellations, positions inconsistent with stated limits, communications hinting at concealment) existed in the bank's own data and were not connected in time.

Conduct surveillance is where two of this book's threads converge on operational risk. The trade-pattern side is anomaly detection applied to a trader's own activity, using the methods of Chapter 15's financial-crime models and Section 10.3.3's reconstruction-error approach to hunt for a book that no longer looks like its own history. The communications side is natural language processing applied to electronic communications: e-comms surveillance systems scan chat and email for the linguistic markers of collusion, concealment, or market manipulation, using the sentiment and classification machinery of Chapter 16. The 2012–2015 LIBOR and foreign-exchange manipulation cases, in which the incriminating evidence was largely trader chat, are why every major bank now runs NLP-based communications surveillance as a first-class operational-risk control.

A minimal illustration shows how the communications side works, using the same Naive Bayes machinery as Section 10.3.2. Suppose a surveillance model is trained on two classes of four short chat messages each: ordinary business talk ("please confirm the trade details," "the settlement date is tomorrow," "can you send the confirmation," "the price looks correct to me") and messages exhibiting concealment markers ("keep this off the record," "do not put this in writing," "delete this message after reading," "let us not leave a trail"). Scoring the new message "do not leave a trail on this," with a vocabulary of 36 words and equal priors, gives log-scores of about -23.86 for the suspicious class and -28.99 for the clean class, a posterior probability of **0.994** that the message warrants review. The words *do*, *not*, *leave*, and *trail* echo the concealment training examples closely enough that the classifier is nearly certain. This kind of first-pass triage lets a human analyst focus on the small fraction of communications most likely to matter, from a daily volume no team could read in full.

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The caveat parallels the fraud-detection lesson of Chapter 15. The base rate of real misconduct in trader communications is extremely low, so even a highly accurate classifier flags far more innocent messages than guilty ones. A surveillance program's usefulness therefore depends as much on ranking its alerts, and on the human review that follows, as on the raw accuracy of the classifier.

The trade-pattern side of surveillance connects operational risk to the market-microstructure material of Chapters 13 and 14. A rogue trader's book, by construction, stops looking like its own history: positions grow beyond stated limits, hedges that should offset a position are fictitious or absent, and the pattern of cancellations and amendments used to conceal the position departs from the desk's normal behavior. Each of these is detectable as an anomaly against the trader's own baseline. The most effective systems fuse the two channels: a communications flag and a trading-pattern flag arriving together are far more informative than either alone, the Bayesian-updating logic of Chapter 1 in which two independent weak signals combine into one strong one. The persistent difficulty, common to every rogue-trading case on record, is that the concealment is deliberate and adaptive. The trader knows the controls and works around them, so surveillance is an adversarial problem in which yesterday's detection rule is what tomorrow's evasion is designed to defeat.

10.3.5 AI Layer 4: Predictive Early Warning from Key Risk Indicators

The earliest point on the event timeline is before any anomaly at all, when only the *conditions* that make a loss more likely are present. Operational-risk teams track these conditions as key risk indicators: measurable quantities such as staff overtime hours, system downtime, transaction volume spikes, and employee turnover that plausibly precede loss events. The predictive question (does this business unit's current indicator profile signal elevated loss risk next quarter?) is a supervised classification problem of the kind Chapter 6 developed, and a logistic regression is the natural first model.

Consider the eight business units in Table 10.5, each described by three indicators (overtime hours in hundreds, system downtime hours, and staff turnover percentage) and labeled by whether a major loss event followed in the next quarter. A logistic regression fit to this data (after standardizing the indicators so their coefficients are comparable) gives positive standardized coefficients on all three (about 0.82 on overtime, 0.69 on downtime, and 0.77 on turnover), confirming that each is associated with higher loss probability, with overtime the strongest single signal. Those magnitudes come from scikit-learn's default fit, which adds an L2 penalty with $C = 1$, and here the penalty is doing real work. Overtime alone separates the two groups perfectly (at most 22 for the quiet units, at least 40 for the loss units), as do downtime and turnover. Under such *complete separation* the unpenalized maximum-likelihood estimate does not exist, since the likelihood keeps rising as the coefficients grow without bound. The signs are trustworthy; the exact sizes, and the ranking among three nearly equal coefficients, reflect the penalty as much as the data. Any eight-observation model is an illustration, not an estimate.

Unit	Overtime (00s hrs)	Downtime (hrs)	Turnover (%)	Loss event?
1	12	2	8	No
2	45	9	22	Yes
3	18	3	10	No
4	50	12	30	Yes
5	8	1	5	No
6	40	8	25	Yes
7	22	4	12	No
8	55	15	35	Yes

Table 10.5: Eight business units: key risk indicators and subsequent major-loss outcome

The fitted model then scores new units: a stressed unit running high on all three indicators receives a predicted loss probability of about **0.90**, while a calm unit low on all three receives about **0.07**, a spread wide enough to be useful for prioritizing where limited control-testing attention should go.

The caveat applies with particular force here. The moment a key risk indicator becomes a management target, staff have an incentive to manage the indicator rather than the underlying risk, and it ceases to

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measure what it once did: the Goodhart's Law dynamic introduced in Chapter 1. A unit told that high overtime flags it for scrutiny can stop recording overtime. Predictive early-warning models are therefore best treated as one input to human judgment, monitored for gaming, rather than as automated verdicts.

10.4 From Detection to Governance

Detecting a control breach is only useful if something changes; this part covers prevention, the risks AI itself introduces, and validation.

10.4.1 From Detection to Prevention: Process Mining

The four detection layers above catch operational failures as or after they happen; a fifth, more preventive use of data aims to remove the conditions that let them happen at all. Process mining reconstructs how a business process actually ran, step by step, from the timestamped event logs that back-office systems generate automatically, and compares that reconstructed reality against how the process was designed to run. The gaps between the two (a control step routinely skipped, an approval granted before its prerequisite, a transaction path that bypasses a four-eyes check) are the latent control weaknesses that eventually produce a loss, and they are visible in the logs long before any loss occurs.

Process mining is less glamorous than the machine-learning layers and uses simpler techniques, graph reconstruction and conformance checking rather than deep networks, but it targets the cheapest point of intervention: fixing a broken control before it is ever exploited. Where the anomaly detection of Section 10.3.3 flags individual unusual events, process mining flags systematic deviations in the process itself. It also pairs with robotic process automation, which executes routine steps deterministically and so removes the human error behind the "execution, delivery and process management" event type of Table 10.1, while introducing its own automation-failure risk.

This preventive posture also reframes where the four AI layers deliver their value. Detection reduces the *severity* of a loss by shortening the time between a failure and its discovery; prevention reduces the *frequency* of loss events by removing the conditions that cause them. In the language of the loss distribution approach, the surveillance and anomaly layers act mostly on severity, while process mining, robotic automation, and the leading-indicator layer act on frequency. A mature operational-risk program invests across both, pulling the loss distribution inward from both ends rather than only truncating its tail.

10.4.2 Entitlement Graphs and Segregation of Duties

The oldest operational-risk control of all is *segregation of duties*: no one person should be able to both create a payee and pay it, or both enter a trade and confirm it, because a single individual holding both halves can complete a fraud without needing anyone's cooperation. Every rogue-trading case in Section 10.3.4's list involved this control failing. Testing it sounds trivial, a matter of checking whether any user holds two conflicting permissions, and in a small system it is. What makes it hard at scale is that permissions are rarely granted to people directly. They are granted to roles, roles are nested inside other roles for administrative convenience, and a user inherits the transitive closure. The conflict is then a *path* through the access graph, not an edge, and a check written over directly-granted permissions cannot see it.

Table 10.6 makes the gap concrete on a deliberately small system: eight users, six granting roles, two composite roles that merely nest others (`AP_MANAGER` contains the accounts-payable clerk and supervisor roles; `OPS_ONCALL` contains treasury operations and static data), and four declared toxic permission pairs. Checking each user's directly-granted permissions flags exactly one user, `u3`, who visibly holds both halves of a conflict. Resolving the nested roles first and then applying the identical four-pair test flags four users. The three additional violations, `u4`, `u6`, and `u8`, are invisible to the direct check not because the test is wrong but because each user's conflicting permission arrives one hop away, through a composite role granted for an unrelated reason.

A ratio of four to one on eight users is not itself the finding. The point is that the missed fraction grows with the depth of role nesting. Real entitlement systems at large banks carry tens of thousands of users, thousands of roles, and nesting several levels deep, accumulated over decades of joiners, movers, and leavers whose permissions were added on each move and rarely removed. At that scale the question becomes a graph

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User	Conflicting pair held	Visible to a direct check?
`u3`	create vendor + approve payment	Yes
`u4`	create vendor + approve payment	No, via `AP_MANAGER`
`u6`	modify static data + release payment	No, via `OPS_ONCALL`
`u8`	create vendor + release payment	No, via `OPS_ONCALL`

Table 10.6: Segregation-of-duties conflicts in an eight-user system. A pairwise check on directly-granted permissions finds one; following paths through nested roles finds four.

problem, and network methods apply directly: reachability queries of the kind Chapter 15's layering-network analysis uses to trace funds through intermediaries, now tracing authority through roles, and, where the toxic combinations are not all known in advance, the graph neural networks of Chapter 17 learning which access patterns resemble those that preceded past incidents. This is the cheapest intervention point in the chapter: a conflict removed from the entitlement graph is a loss event that cannot occur through that route at all, whereas the detection layers at best notice it sooner.

10.4.3 Predictive Process Monitoring

Conformance checking compares a process against its design after the fact. The natural extension is to ask, of a case still running, what is likely to happen to it next, which turns the event log into a sequence-prediction problem of the kind Chapters 7 and 16 develop. Given the prefix of activities a payment or a trade settlement has completed so far, together with their timestamps and attributes, a model predicts the next activity, the remaining time to completion, or the probability that the case ends in a breach, an exception, or a manual write-off. Recurrent and attention-based architectures are the standard tools, since the informative signal is usually the *order* of steps and the gaps between them rather than any single step, the structure self-attention was built to exploit.

The operational-risk value is that process data becomes a leading indicator at the level of the individual case, complementing the business-unit-level indicators of Section 10.3.5: a settlement predicted to breach its deadline can be escalated while there is still time to act. The familiar caveats apply with full force. The training labels come from historical cases whose outcomes were shaped by whatever interventions the firm was already making, so the model learns the process as operated rather than as designed. And a prediction that triggers an intervention destroys the outcome it predicted, making evaluation harder than a backtest suggests. This is the same reflexivity Section 10.3.5 raised about key risk indicators, which the next section takes up as a general problem.

10.4.4 The Reflexive Turn: Artificial Intelligence as a Source of Operational Risk

Every method in the four layers above uses AI to *manage* operational risk. But deploying those same models is itself an operational activity, with its own failure modes, so artificial intelligence is increasingly a source of operational risk as well as a tool against it. This is now an explicit category in most banks' operational-risk taxonomies.

The oldest form is model risk: the risk that a model is wrong, or is used outside the conditions it was built for. This book has flagged it from Chapter 6 onward, and regulators treat it as an operational-risk discipline in its own right. A credit or trading model whose assumptions are quietly violated by a changing environment produces losses that are, in the end, operational: a failure of an internal system. The monitoring apparatus of Chapter 18 (population-stability indices, drift detection, and the explainability tools that let a human challenge a model's outputs) is best understood as an operational-risk control for models, detecting model risk before it becomes a loss.

Generative AI adds newer failure modes that did not exist when the Basel event types were written. A large language model that fabricates a plausible but false answer (hallucination), a system manipulated into ignoring its instructions through adversarial input (prompt injection), and a critical dependence on a

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single external model vendor whose service can change or fail (third-party and concentration risk) are all operational risks in the strict Basel sense: failures of systems, processes, or external parties that no traditional operational-risk framework anticipated.

A sharper version of the concern arrives once the model stops producing text and starts *taking actions*. The LLM agents with tool use that Chapter 16 describes do not draft an answer for a person to check; they call functions, and those functions can submit an order, release a payment, open a ticket, or change a configuration. That difference is not one of degree. Every control in the four detection layers assumes a window between a failure and its consequences, in which an anomaly is a near-miss rather than a loss. An agent that acts closes that window by construction: the first observable evidence is an executed transaction, not a proposal anyone declined. Two further properties compound it. An agent granted the entitlements it needs across several systems is a segregation-of-duties breach of the kind Section 10.4.2 exists to prevent: it holds both halves of a conflicting pair and, unlike a human, will never hesitate at the second half. And because it acts at machine speed and without fatigue, a single flawed instruction propagates across thousands of cases before the daily control report that would have caught it is generated. The controls that follow are conventional, the ones the operational-risk framework already knows how to write: scope each agent's permissions to the minimum its task requires, keep a human approval step on any action that moves money or commits the firm, log every tool call as an auditable event, and impose hard rate and value limits so that a failure is bounded before anyone notices it. This is the segregation, authorization, and limit discipline banks have applied to human traders since Barings, applied to a non-human actor that the existing control framework does not yet recognize as one.

A concrete example makes the mapping vivid. Suppose a bank deploys a retrieval-augmented assistant to draft responses to client queries about their products. If the model confidently states an incorrect fee or contractual term, and a client acts on it, the resulting loss is a "clients, products and business practices" operational loss. Its consequences are those of a human adviser's mis-statement, but it arrives at a scale and speed no adviser could match, since the same flawed model answers thousands of queries before anyone notices. The controls are correspondingly operational: retrieval grounding to reduce fabrication, human review of high-stakes outputs, logging for audit, and the drift monitoring of Chapter 18 to catch a model whose behavior degrades after a vendor update. Embedding AI creates less a new *kind* of risk than a new, faster *channel* for the operational risks the framework already names. As institutions embed generative methods into research, client service, and compliance workflows, mapping these failure modes onto the operational-risk framework, and building controls for them, becomes part of the operational-risk function's job.

10.4.5 Regulation: Why Artificial Intelligence Lives in Management, Not Capital

There is a striking regulatory irony at the center of this chapter. Just as machine learning was maturing into a serious tool for operational risk, the Basel Committee moved decisively *away* from letting banks use their own internal models to set operational-risk *capital*. The Advanced Measurement Approach, which had permitted internal loss-distribution models of the kind described above, was retired in the Basel III finalization in favor of the Standardized Measurement Approach. It ties capital mechanically to a size-based "business indicator" adjusted by the bank's own historical loss experience, the same philosophy as the Basic Indicator Approach Chapter 9 described, and for the same reason: the events are too heterogeneous and too sparse for regulators to trust any single institution's internal model of the tail.

The consequence is clarifying rather than discouraging. Because regulatory operational-risk capital is now formulaic, the value of this chapter's machinery, above all the four AI layers, lies in *managing* operational risk rather than in setting the regulatory number. It reduces the frequency and severity of loss events through earlier detection, better surveillance, and sharper prioritization, and it informs a bank's own internal economic-capital view even where it no longer drives the regulatory one. Artificial intelligence's home in operational risk is the risk-management function, not the capital calculation, and that is where its comparative advantage, unstructured data and rare-pattern detection, lies. Model risk itself remains squarely within supervisory scope through model-governance expectations such as the U.S. banking agencies' SR 26-2 guidance (which replaced the Federal Reserve's SR 11-7 in April 2026 (Federal Reserve, OCC, and FDIC 2026)), which is why the reflexive controls described above are not optional.

Where capital rules have retreated, operational-resilience rules have advanced, and they reach directly into the technology dependencies the reflexive turn describes. The EU's Digital Operational Resilience Act

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(DORA), applicable since 17 January 2025, sets common requirements for ICT risk management, incident reporting, resilience testing, and the management of ICT third-party risk across almost every type of EU financial firm (European Parliament and Council of the European Union 2022), and in November 2025 the European supervisory authorities designated the first nineteen "critical ICT third-party providers," among them the major cloud vendors, for direct oversight (ESAs 2025). The United Kingdom's operational-resilience rules take the outcome rather than the system as their unit: each firm identifies its important business services and sets an impact tolerance, the maximum tolerable disruption to each, and the transition period for showing it can stay within those tolerances under severe but plausible scenarios ended on 31 March 2025 (PRA 2021). Why regulators now treat a vendor as a systemic object is shown by the 19 July 2024 CrowdStrike outage, in which a faulty content update to a single security product crashed an estimated 8.5 million Windows devices, under 1% of the installed base by Microsoft's estimate, yet disrupted banks, payment systems, and exchanges, along with airlines and hospitals, around the world at once (Weston 2024). The affected firms had not failed in their own processes. The disruption came from a shared dependency, the third-party concentration risk that a firm-by-firm loss model cannot see and that DORA's oversight of critical providers is designed to address.

10.4.6 Validating and Backtesting Operational-Risk Models

Every model in this chapter must itself be validated, and operational risk makes validation unusually hard for the same reason it makes modeling hard: the events are too rare to backtest in the ordinary sense. A market-risk Value at Risk model can be backtested against roughly 250 daily outcomes a year (Chapter 8), so a 99% model should be breached about two or three times annually and a wildly miscalibrated one is caught quickly. A 99.9% operational-risk capital model, by contrast, makes a prediction about a once-in-a-thousand-year loss that no institution will ever observe often enough to test directly. Validation therefore leans on indirect checks: goodness-of-fit tests on the severity distribution, sensitivity analysis to the assumed severity and frequency distributions, benchmarking against external loss data, and structured challenge of the scenario-analysis judgments that drive the tail.

The AI layers, by contrast, are far more testable, and should be held to the same standards as any other model in this book. An incident classifier has a confusion matrix (Chapter 6); an anomaly detector has a precision-recall trade-off measured against injected or historical anomalies (Chapter 15); an early-warning model has an out-of-sample discriminatory-power measure such as the AUC of Chapter 9. Holding these models to that standard, rather than trusting them because they address an important problem, is part of the model-risk discipline of the reflexive turn: an unvalidated surveillance model that quietly stops working is not a safeguard against operational risk but a new instance of it.

"More testable" does not mean "straightforwardly testable," however. For the surveillance layer in particular, the labels needed to compute those metrics barely exist, and the ones that do are selected in a way that flatters the model. Confirmed misconduct is not merely rare; it is rare *and* observed only where some control already caught it. A recall figure computed against a bank's history of confirmed rogue-trading cases is therefore recall against the events the current surveillance system, an auditor, or a market move happened to expose. It is silent about the ones concealed successfully, which are the population the model exists to find. A model can post an excellent recall on that sample while being systematically blind to the concealment techniques that work. Chapter 9 meets the same structure in credit under the name reject inference, where a scorecard can only be validated on applicants it approved, never on the ones it turned away. The response here borrows the same instincts: evaluate against injected synthetic scenarios that encode known evasion patterns as well as against historical cases; treat the alert queue's precision, which *is* cleanly measurable from investigator dispositions, as the more trustworthy number; and read any reported recall as an upper bound rather than an estimate. In short, the detection layers can be validated well against false positives and only weakly against false negatives. That is a further reason, alongside the base-rate problem of Section 10.3.4, to treat a quiet alert queue as an ambiguous signal rather than a reassuring one.

10.5 Case Vignette: The 2012 JPMorgan "London Whale"

In 2012 JPMorgan Chase lost more than $6 billion in its Chief Investment Office through a portfolio of credit derivatives that grew so large a single trader earned the nickname "the London Whale." The loss is usually

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told as a risk-management story, and it was, but at its core it was operational. A spreadsheet-based Value at Risk model used to monitor the position contained a formula error that understated its risk; the position had grown beyond what its stated strategy could justify; and internal warnings were not escalated in time. Every element (a flawed model, a control that failed to bind, an escalation process that did not function) is an operational-risk failure in the Basel sense, even though the loss materialized through market prices.

The episode is a compact illustration of this chapter's whole argument. The model error is model risk (the reflexive turn above), of the kind Chapter 18's validation and monitoring tools exist to catch. The unnoticed drift of the position away from its mandate is a near-miss anomaly (Section 10.3.3) that trade-pattern surveillance is designed to surface. The failure to escalate is the kind of process breakdown that leading indicators (Section 10.3.5) and incident analysis (Section 10.3.2) are meant to make visible. No single AI system would have prevented the London Whale, but each layer of this chapter maps onto a point at which the failure could have been caught earlier. That is why operational-risk management is best understood as a portfolio of overlapping detection layers rather than a single model.

The aftermath reinforces the chapter's regulatory theme. JPMorgan ultimately paid roughly $920 million in coordinated regulatory settlements, an amount that dwarfs the capital any internal model would plausibly have held against this specific risk. The enforcement findings centered not on the size of the market bet but on the control and governance failures around it: risk limits that were breached and then re-calibrated rather than enforced, a valuation process that masked losses, and a model change implemented without adequate validation. That emphasis is why the supervisory response to operational risk runs through governance and model-risk discipline rather than a capital formula, and why the value of this chapter's tools lies in strengthening the controls whose failure, not the trade itself, turned a large loss into a far larger one.

10.6 Challenges in Operational Risk Modeling

The difficulties that make operational risk awkward are structural, and sound practice acknowledges them rather than modeling past them.

Data scarcity and the unquantifiable tail. The largest operational losses are the rarest, so the events that dominate the capital number are the ones about which there is least data. No amount of machine learning manufactures tail observations that do not exist, and an unprecedented event, such as a first-of-its-kind cyberattack or a novel misconduct scheme, is closer to the Knightian uncertainty of Chapter 1 than to the estimable risk any model assumes.

Reporting and survivorship bias. Loss databases record what was noticed and reported; near-misses, silently absorbed small losses, and the losses of firms that did not survive are all systematically missing, biasing any fitted frequency and severity downward in ways no purely statistical correction can fully repair.

Alert fatigue. Every detection layer in this chapter produces false positives, and the base-rate reasoning of Chapter 15 applies in full: when true events are rare, even an accurate detector generates far more false alarms than real catches, and a surveillance system that overwhelms its investigators is, in practice, no surveillance at all.

Model-on-model risk. Using AI to manage operational risk adds models whose own failures are operational risks, so the reflexive controls described above must extend to the models built to detect operational failures, a regress that is manageable but never fully closed.

Gaming and reflexivity. As the Goodhart dynamic of Section 10.3.5 shows, indicators and controls that become targets can lose their meaning, so operational-risk models require ongoing revalidation against the possibility that the behavior they measure has adapted to their existence.

The pace of new failure modes. The event types in Table 10.1 were fixed years before generative AI, cloud concentration, or large-scale ransomware became material, and technology now creates new operational-risk channels faster than the frameworks meant to catalog them. A framework that can only manage the risks it has already named is structurally behind, which is why a reflexive, continuously monitored posture matters more in operational risk than the one-time model sign-off that suffices elsewhere.

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10.7 Key Takeaways

- Operational risk is the risk of loss from failed internal processes, people, systems, or external events, distinct from the market risk of Chapter 8 and the credit risk of Chapter 9, and structurally harder to model because its events are rare, heavy-tailed, heterogeneous, and under-reported.

- The four data elements are combined, not pooled: a Gamma-Poisson credibility blend weights internal history against external and scenario data through a single factor $Z = n/(n + \beta)$, and in the worked example it turned a thin internal estimate of 8.4 events a year into the $\lambda = 12$ the loss model uses, a bank relying on internal data alone understating its capital by about 15%.

- The classical loss distribution approach builds annual loss from a Poisson frequency and a lognormal severity by Monte Carlo; in the worked example, the 99.9% capital quantile was about seven times the expected loss. Operational-risk capital is a tail problem, so modeling the severity tail with heavy-tailed and extreme-value methods is the central actuarial difficulty and the place sparse data hurts most.

- Artificial intelligence contributes in four layers around the loss-modeling core, not by replacing the actuarial loss model: incident-text classification (Chapter 16), anomaly detection for near-misses (Chapter 17), conduct and rogue-trading surveillance (Chapters 15–16), and predictive early warning from key risk indicators (Chapter 6).

- Deploying AI is itself an operational activity: model risk, drift, and generative-AI failure modes such as hallucination and prompt injection are operational risks in their own right, and the monitoring tools of Chapter 18 are their controls.

- Because Basel retired internal-model operational-risk capital in favor of a size-based formula, AI's home in operational risk is management and detection, where its strengths lie, rather than the regulatory capital calculation.

- Prevention beats detection: process mining, entitlement-graph analysis, and predictive process monitoring remove the conditions for a loss rather than shortening the time to notice one, and because permissions are inherited through nested roles, a segregation-of-duties conflict is a path through the access graph rather than an edge, invisible to a pairwise check.

- An AI system that *acts* rather than merely drafts is a different operational risk from one that generates text: it closes the detection window the four layers depend on, and holds conflicting entitlements by construction, so it needs the scoping, authorization, logging, and limit controls banks have applied to human traders since Barings.

- The AI layers must themselves be validated with the same tools as any other model (confusion matrices, precision-recall curves, out-of-sample discrimination), because an unmonitored surveillance or early-warning model that silently stops working is not a control against operational risk but a fresh source of it.

- Those validations are far stronger against false positives than false negatives: confirmed-misconduct labels exist only where a control already caught something, the same reject-inference structure Chapter 9 meets in credit, so a reported surveillance recall is an upper bound and a quiet alert queue is ambiguous rather than reassuring.

10.8 Suggested Exercises

10.1 Using the loss distribution approach with $\lambda = 8$ events per year and Lognormal($\mu = 11, \sigma = 1.2$) severities, simulate the annual loss distribution and estimate the expected loss and the 99.9% capital quantile. By what multiple does the capital exceed the expected loss, and how does that multiple change if you raise σ to 1.6?

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10.2 Explain, in terms of the frequency and severity components, why two business lines with the same expected annual operational loss can require very different amounts of capital. Which component does raising the severity σ affect, and why does it move the 99.9% quantile more than the mean?

10.3 Extend the two-class Naive Bayes classifier of Section 10.3.2 to a third event type (for example, external fraud) with three new training incidents, and classify a new report of your choosing. Show the effect of Laplace smoothing on a word that appears in only one class.

10.4 Using the key-risk-indicator logistic regression of Section 10.3.5, explain why standardizing the indicators before fitting makes the coefficients comparable across indicators, and describe one way a business unit could game each of the three indicators without reducing its actual operational risk.

10.5 Explain why an anomaly detector with a precision of 0.85 but a recall of only 0.56, as in Table 10.4, can still be operationally valuable, and describe the concrete consequence, in analyst workload and in missed events, of lowering its threshold to raise recall to 0.90.

10.6 Using Section 10.2.5's credibility blend, recompute $\hat{\lambda}$ when the scenario workshop's view is held with strength $\beta = 8$ years rather than 3, holding the internal 42 events over 5 years and $\lambda_{\text{prior}} = 18$ fixed. Report the new credibility factor Z and blended frequency, re-run the loss simulation at that frequency, and explain why a supervisor might be uncomfortable letting a bank choose β for itself.

10.7 Section 10.2.5 showed that using internal loss data alone understates the 99.9% capital quantile by about 15% for this business line. Explain why the understatement is smaller in percentage terms than the corresponding gap in *expected* loss, referring to how frequency and severity each enter the compound sum, and say what this implies for a business line whose severity σ is larger.

10.8 Add a ninth user to Table 10.6's system who holds the `TRADE_ENTRY` role plus a new composite role containing `TRADE_CONFIRM`, and confirm that the direct check misses the resulting conflict while the path-based check finds it. Then describe one realistic way an entitlement graph accumulates such conflicts without anyone making an obviously wrong decision.

10.9 This chapter's discussion of validating and backtesting operational-risk models argues that a surveillance model's reported recall should be read as an upper bound. Construct a small numerical illustration: assume 10 misconduct episodes occurred, the existing controls exposed 4 of them, and a new model flags 3 of those 4. State the recall the bank would report, the recall against all 10, and explain which of the two a model validator can actually observe.

10.10 **Challenge (real data).** Public enforcement actions are a rich, freely available source of operational-loss narratives. Download a sample of settlement or enforcement-action texts (for example, from the U.S. Securities and Exchange Commission's litigation releases or the Consumer Financial Protection Bureau's enforcement database), label each by Basel event type, and train a text classifier (TF-IDF plus logistic regression, or the embedding methods of Chapter 16) to predict the event type from the narrative. Report its confusion matrix, and discuss which event types are hardest to separate and why.

10.9 Further Reading

- Hull (2023), *Risk Management and Financial Institutions*. Includes a clear introductory treatment of operational risk and its regulatory capital, complementing the market- and credit-risk chapters cited in Chapters 8 and 9.
- Cruz et al. (2015), *Fundamental Aspects of Operational Risk and Insurance Analytics*. A thorough treatment of the loss distribution approach, frequency-severity modeling, and the extreme-value methods for the severity tail.
- McNeil et al. (2015), *Quantitative Risk Management*. A standard graduate reference spanning market, credit, and operational risk, and the extreme-value methods for heavy-tailed losses.

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- Board of Governors of the Federal Reserve System and Office of the Comptroller of the Currency (2011), *Supervisory Guidance on Model Risk Management*. The foundational supervisory statement treating model risk as an operational-risk discipline, underlying this chapter's discussions of the reflexive turn and of regulation; superseded in April 2026 by SR 26-2 (Federal Reserve, OCC, and FDIC 2026), which keeps its core pillars.
- Shevchenko (2011), *Modelling Operational Risk Using Bayesian Inference*. A book-length treatment of the problem Section 10.2.5 works in miniature, combining internal loss data, external data, and expert scenario judgment into a single frequency and severity estimate.
- van der Aalst (2016), *Process Mining*. The standard reference for the event-log reconstruction and conformance checking of Section 10.4.1, and for the predictive extensions of Section 10.4.3.

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Chapter 11

Insurance and Actuarial Modeling

11.1 Introduction

In August 1992, Hurricane Andrew crossed southern Florida and destroyed, alongside tens of thousands of homes, the insurance industry's confidence in its own loss estimates. Insurers had priced Florida windstorm coverage from a few decades of loss experience that happened to contain no direct hit on a major metropolitan area; Andrew produced insured losses several times larger than the industry's models had contemplated, and roughly a dozen insurers were rendered insolvent by a single storm. The failure was not one of underwriting discipline but of method: historical claims experience, the industry's traditional basis for pricing, is structurally silent about events larger than any it happens to contain. Out of that failure grew the modern catastrophe-modeling industry, simulation-based and data-hungry, and the lesson this chapter develops: insurance is above all a business of estimating loss distributions, and every advance in estimating them becomes an advance in how insurance is priced, reserved, and supervised.

Insurance deserves its own chapter for two reasons. First, it is enormous, by assets, premium volume, and employment of quantitatively trained people, and its analytical core, actuarial science, is the oldest data science in finance: actuaries were fitting frequency and severity distributions to loss data a century before "machine learning" existed. Second, its structure makes the AI questions crisp. An insurer prices a contract before knowing what it will cost, observes rich behavioral data about its customers, and adjudicates millions of claims a year, all inside one of the most consumer-protective regulatory perimeters in finance. Every theme of this book (predictive modeling, text and image data, fraud detection, explainability, fairness) arrives here at once.

The chapter's central claim is the asymmetry Chapter 10 found in operational risk. Machine learning does not replace the actuarial core; it feeds and sharpens it wherever the data is *granular* or *unstructured*: driver-level sensor streams instead of rating-cell averages, claim-level features instead of aggregate triangles, documents and images instead of questionnaires, and interactions no multiplicative rating plan anticipates. The classical structure (frequency times severity, credibility, the development triangle, the filed rating plan) remains the layer regulators audit and balance sheets are built on. Each section therefore gives the classical calculation first, then a hand-checkable worked example of the machine-learning layer beneath it.

Figure 11.1 maps where AI enters. The chapter first covers what makes insurance different and how an insurer makes money; then the classical core (frequency-severity pricing, the one-way-rating failure that motivated generalized linear models, credibility, chain-ladder reserving); then where machine learning changes pricing, telematics, claims, reserving, and underwriting. It widens to the balance sheet (life and health insurance, asset-liability management, catastrophe risk, solvency capital) and closes with fairness, price optimization, and proxy discrimination.

11.2 The Economics of Insurance

An insurer sells a promise and invests the proceeds, which makes its economics unlike any other financial firm's.

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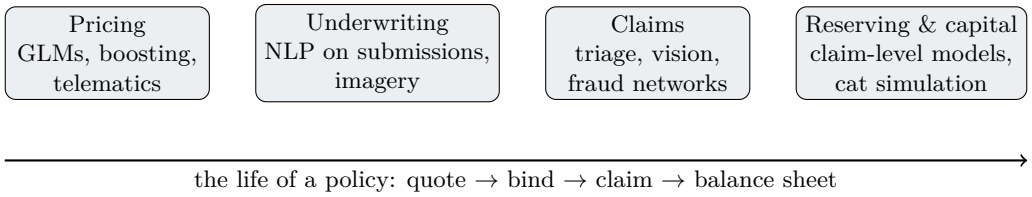


Figure 11.1: AI across the insurance value chain. Every stage feeds a filed rate or a regulated capital number, which is what makes governance structural here.

11.2.1 What Makes Insurance Different

Insurance inverts the production cycle of almost every other business: the price is collected first and the cost of goods sold, the claims, is discovered afterward, sometimes decades afterward. Everything distinctive about the industry follows from this inversion. Profitability cannot be read off current revenue; it must be *estimated*, by forecasting the ultimate cost of claims already incurred (the reserving problem of Section 11.4.1) and of contracts still in force. The product is a promise, so the industry is regulated primarily for solvency (an insurer must demonstrably be able to pay claims years away) and secondarily for fairness in pricing, since coverage is often practically or legally mandatory.

The economic engine is risk pooling, and a single calculation explains why the industry exists. Suppose each of n independent policies produces an annual loss with mean $\mu = \$720$ and standard deviation $\sigma = \$3{,}000$, a violently uncertain individual prospect, with a coefficient of variation of $3{,}000/720 \approx 417\%$. The pooled loss across $n = 10{,}000$ such policies has mean $n\mu = \$7.2$ million but standard deviation only $\sigma\sqrt{n} = \$300{,}000$, a coefficient of variation of 4.17%: aggregation has shrunk relative uncertainty by a factor of $\sqrt{n} = 100$. A loss more than 10% above expectation is a $z = 2.4$ event, roughly a 0.8% probability under a normal approximation. Individually unpredictable losses have become collectively predictable, and *that predictability, not risk-bearing appetite, is the product*: the insurer sells the law of large numbers, retail. The calculation also shows its own failure mode (Section 11.5.1): it assumed independence, and a hurricane, correlating a hundred thousand claims into one event, deletes the $\sqrt{n}$.

Two classic frictions limit the engine. Adverse selection is the tendency of higher-risk customers to seek more coverage, combated with underwriting and risk-based pricing; moral hazard is the tendency of insured parties to take more risk once covered, combated with deductibles and coverage limits. Both are information problems, which is why the industry has always been an early adopter of technology that reduces information asymmetry, from the mortality table to the telematics device.

An insurance portfolio's annual loss is a compound frequency-severity sum, the loss distribution approach of Chapter 10, since both fields face a random number of losses of random size. Historically, operational risk borrowed this machinery *from* actuarial science.

11.2.2 How an Insurer Makes Money: Float and the Combined Ratio

An insurer's operating result has two engines, illustrated in Figure 11.2. The underwriting engine is summarized by the combined ratio: losses plus expenses, divided by premiums. A combined ratio below 100% means the insurance operation is profitable in its own right. The investment engine runs on float: because premiums arrive before claims are paid, the insurer continuously holds a large pool of policyholder money (the loss reserves of Section 11.4.1 plus unearned premiums), which it invests for its own account.

A worked example shows how the engines combine. An insurer writes \$100 million of premium, incurs \$65 million of losses (a 65% loss ratio) and \$30 million of expenses (a 30% expense ratio), for a combined ratio of 95% and an underwriting profit of \$5 million. It also holds \$150 million of float invested at a 4% yield, earning \$6 million, so total pre-tax operating income is \$11 million, a 22% return on its \$50 million of supporting capital. Now rerun the year with a 103% combined ratio: underwriting loses \$3 million, yet the insurer still earns \$3 million overall, because the \$3 million underwriting loss is the *cost* of holding \$6 million worth of investable float, cheaper than any other funding the insurer could raise. This arithmetic, made famous by Berkshire Hathaway's shareholder letters, explains why insurers will write at small underwriting

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losses in competitive markets, why reserving accuracy (which determines how much float there is, and for how long) is a first-order economic question, and why asset-liability management (Section 11.4.4) belongs in the same chapter as pricing.

It also locates where machine learning can move the profit needle, because the combined ratio has only two components. Better risk selection and pricing (Sections 11.3.3–11.3.6) and earlier fraud detection (Section 11.3.7) lower the *loss* ratio; automated claims handling and NLP-driven underwriting (Sections 11.3.7 and 11.5.4) lower the *expense* ratio. On the worked book each point of combined ratio is worth $1 million of premium, pre-tax, every year. That is the business case for the industry's AI investment, and it sets this chapter's evaluation standard: an insurance model is judged by the ratio points it moves, not by its accuracy in the abstract.

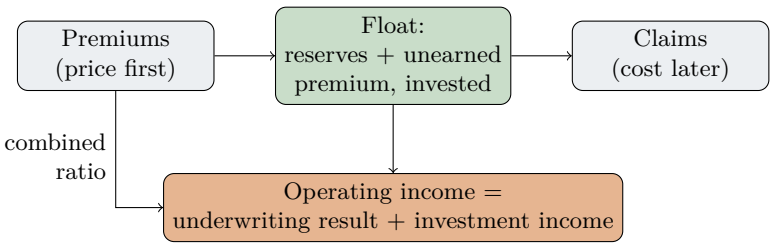


Figure 11.2: The inverted production cycle: premiums arrive before claims are paid, and the pool of policyholder money in between, the float, is invested for the insurer's account. Underwriting profitability (the combined ratio) and investment income on float are the business's two engines.

11.2.3 The Classical Pricing Baseline: Frequency Times Severity

Suppose an auto insurer must set next year's premium for a policyholder it has never observed have an accident. The classical answer prices the *expected* loss and then loads it for expenses and profit. The pure premium is

$$\text{pure premium} \;=\; \mathbb{E}[N] \times \mathbb{E}[X], \tag{11.1}$$

expected claim frequency (N) times expected claim severity (X), and the art of ratemaking is estimating how each component varies with observable characteristics of the risk.

A worked example fixes the mechanics. Suppose the book-wide base rates are a claim frequency of 0.08 claims per policy-year and a mean severity of $9,000, giving a base pure premium of $0.08 \times 9{,}000 = \$720$. Rating factors enter multiplicatively: suppose young drivers exhibit 1.8 times the base frequency, urban territories 1.25 times the base frequency, and urban claims run 1.10 times the base severity (denser traffic produces more collisions; urban repair and injury costs run higher). A young urban driver's expected frequency is then $0.08 \times 1.8 \times 1.25 = 0.18$ and expected severity $9{,}000 \times 1.10 = \$9{,}900$, for a pure premium of $0.18 \times 9{,}900 = \$1{,}782$, about 2.5 times the base risk. Loading for a 25% expense ratio and a 5% profit provision, the gross premium solves premium $\times\,(1 - 0.25 - 0.05) =$ pure premium, giving $1{,}782/0.70 \approx \$2{,}545.71$.

This arithmetic is the skeleton; the instrument that estimates the relativities is the generalized linear model of the next section. A note on vocabulary before the machine-learning comparison arrives: when the insurance industry says "traditional pricing," it means *GLM-based* pricing. The GLM is not a stepping stone to modern methods; it has been the established standard for decades, and it is the incumbent that machine learning must beat.

11.3 Pricing and Claims

This part covers how a premium is set, from the generalized linear model through credibility to telematics, and what happens when a claim arrives.

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11.3.1 GLM Ratemaking: The Traditional Approach

Where do relativities like the 1.8 and 1.25 above come from? The tempting answer, computing each factor's relativity from its own marginal experience one factor at a time, contains a trap worth working through numerically, because escaping it is what made the generalized linear model the industry's standard.

Consider a portfolio with two rating factors, age (young/experienced) and territory (urban/rural), with exposures and claim counts by cell:

Cell	Exposure	Claims	Frequency
Young, urban	200	54	0.270
Young, rural	50	9	0.180
Experienced, urban	300	45	0.150
Experienced, rural	450	45	0.100

Table 11.1: Exposures (policy-years), claim counts, and claim frequencies by rating cell

The table was built with an exactly multiplicative structure: a base frequency of 0.100, a young relativity of 1.8, and an urban relativity of 1.5 (check: $0.100 \times 1.8 \times 1.5 = 0.270$). Now compute the relativities the one-way approach recovers. Young drivers overall: $(54 + 9)/(200 + 50) = 0.252$; experienced overall: $(45 + 45)/(300 + 450) = 0.120$; one-way age relativity $0.252/0.120 = 2.10$, not 1.8. Urban overall: $(54+45)/(200+300) = 0.198$; rural: $(9 + 45)/(50 + 450) = 0.108$; one-way territory relativity 1.833, not 1.5. Both are inflated, and for the same reason: young drivers in this portfolio cluster in urban territory (200 of 250 exposures), so the age marginal absorbs part of the territory effect and vice versa, double-counting their overlap. A rating plan built from the one-way factors would charge the young urban cell $2.10 \times 1.833 = 3.85$ times base, against a true relativity of 2.70, a 43% overcharge, and would symmetrically undercharge the experienced rural cell. The mispricings then feed adverse selection as overcharged customers defect to better-priced competitors.

The remedy is simultaneous estimation, which is what the generalized linear model (GLM) does. A Poisson GLM with a log link fit to claim counts (with exposure as an offset), and a Gamma GLM fit to severities, estimate all relativities *jointly*, each factor's effect measured holding the others fixed, recovering the true 1.8 and 1.5 from Table 11.1 exactly. GLMs extended the ordinary regression of Chapter 6 to the skewed, non-negative distributions of insurance data. Spreading from British motor insurance in the 1980s and 1990s, GLM ratemaking displaced the earlier one-way and minimum-bias methods to become the industry's standard classification-ratemaking approach worldwide, the established baseline in the Casualty Actuarial Society's own rating monograph. "Traditional pricing" in today's debates thus refers to a method that was, in its time, the data-science modernization. The multiplicative-relativity output is also *filed*: in most U.S. states an insurer must submit its rating plan to a regulator, justify each factor, and use what it filed, a constraint with deep consequences for what follows.

11.3.2 Credibility: The Actuary's Bayesian Shrinkage

Between the book-wide rate and the individual risk sits a question the GLM alone does not answer: how much should one policyholder's *own* experience move their price? A commercial fleet with three years of history and one claim exhibits an observed frequency of $1/3 \approx 0.333$ against a class rate of 0.100. Charging either number is wrong, the first because three years of one fleet is noise, the second because it ignores real information. Credibility theory, actuarial science's oldest continuously used tool, prescribes the blend

$$\hat{f} \;=\; Z \times \bar{f}_{\text{own}} + (1 - Z) \times f_{\text{class}}, \qquad Z = \frac{n}{n + k}, \tag{11.2}$$

where n is the volume of the risk's own experience and k measures how noisy individual experience is relative to how much risks truly differ (formally, the ratio of expected process variance to the variance of true means across risks, the Bühlmann formulation). With $n = 3$ years and $k = 12$, the credibility weight is $Z = 3/15 = 0.20$ and the blended frequency $0.20 \times 0.333 + 0.80 \times 0.100 \approx 0.147$: the fleet's bad year moves its rate meaningfully but not catastrophically, and each additional year of clean experience will pull it back.

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The formula appears three times over in this book. It is Chapter 1's Bayesian updating, with the class rate as prior, own experience as likelihood, and Z the posterior weight on the data. It is the shrinkage of ridge regularization, pulling noisy individual estimates toward a grand mean by an amount that declines as evidence accumulates. And it is structurally the partial pooling that modern hierarchical models perform. Actuaries derived it in the 1910s–1960s because rate regulation forced them to price small risks fairly; machine learning rederived it as regularization half a century later. Credibility is this chapter's cleanest evidence that actuarial science is not a field awaiting rescue by data science but a parallel, older branch of the same subject.

11.3.3 From GLMs to Machine Learning in Pricing

The machine-learning question in pricing is therefore sharply posed. The incumbent is not a naive method but the GLM, a decades-refined statistical model with a regulatory apparatus built around it, so "machine learning versus traditional pricing" means boosted trees versus the GLM. Everything Chapter 6 established about flexible models applies, with one structural twist. Gradient-boosted trees routinely beat a GLM's predictive accuracy on large personal-lines datasets, for the familiar reason: real loss processes contain interactions (age with vehicle power, territory with annual mileage) that a multiplicative model captures only if an actuary thinks to specify them, while a tree ensemble finds them automatically. It is the lesson of Section 11.3.1's one-way trap, one level up.

11.3.4 Worked Example 11.1: The Interaction a GLM Cannot Price

Table 11.1's portfolio provides numbers with one realistic change. Suppose inexperience is more dangerous in dense traffic, so the young-urban cell's true frequency is 0.32 (64 claims on 200 exposures) rather than the multiplicative 0.27, while the other three cells keep their multiplicative values. No multiplicative model can fit these four cells: multiplicativity forces the young-urban frequency to equal base times young times urban, and the data now violate that. Refitting the Poisson GLM to the modified portfolio shows how the failure manifests, which is subtler than mispricing the interaction cell. The fitted parameters come back as a base frequency of 0.0977 with relativities of 2.05 (young) and 1.57 (urban): the interaction has *leaked into the marginals*, inflating both, because the young-urban cell dominates the exposure behind each. The fitted model then prices the young-urban cell at 0.315 against a true 0.320, only 1.6% light, but overprices young-rural at 0.201 against a true 0.180, an 11.4% overcharge on the cell least responsible for the signal, with the experienced cells absorbing smaller distortions. The mispricing lands on the *other* cells, where nobody is looking.

A depth-two decision tree, Chapter 6's simplest nonlinear model, prices all four cells exactly: its first split on age and second on territory partition the portfolio into the four cells themselves, each predicted at its own observed frequency, and a gradient-boosted ensemble reaches the same fit within a few trees. This is the GLM-versus-machine-learning pricing debate in a 2×2 table: the flexible model wins by declining to impose a structure the data contradict. On the standard academic benchmark for this comparison, a French motor liability dataset of nearly seven hundred thousand policies (this chapter's closing challenge exercise), the accuracy improvement is consistent and material, and the same result reproduces across insurers' internal books.

11.3.5 Where Machine Learning Improves on the Traditional Approach, and Where It Does Not

The interaction is one instance of a general pattern. The improvement comes through four channels, each matching a specific limitation of the traditional toolkit rather than generic "more power." First, *interaction discovery at scale*: an actuary can hand-specify the young-urban term once told about it, but not the thousands of candidate interactions among dozens of rating variables that a boosted ensemble searches automatically. Second, *nonlinear continuous effects without hand-chosen bins*: a GLM represents driver age through bands or polynomial terms chosen in advance, so the true age curve (falling steeply through the twenties, flat in mid-life, rising again past seventy) is captured only as well as those choices allow, while a tree ensemble learns its shape from data. Third, *high-cardinality categorical variables*: vehicle model and postcode take thousands of levels that a GLM must manually group (or shrink level by level with Section 11.3.2's

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credibility machinery), while tree ensembles and learned embeddings handle them natively. Fourth, and most consequentially, *features the GLM could never build*: the deep-learning methods of Chapters 16 and 17 construct features themselves, such as the braking signature in a telematics stream, the roof condition in an aerial image, or the hazard buried in a broker narrative. The largest practical gains have come less from swapping the GLM out of the filed plan than from feeding it these richer inputs. The same channels recur in reserving (Section 11.4.1), lapse modeling (Section 11.4.4), and mortality: in each case the traditional method is not wrong but *coarse*, summarizing at a level of aggregation chosen for an era of scarce data and hand computation.

The other column matters as much. Where the book of business is small, the traditional method's low variance and the credibility machinery win, Chapter 6's bias-variance logic; where the rating plan is narrow and well understood, there are few interactions left to find; and at the regulatory interface, the GLM's auditability is the point. Mature practice is therefore not replacement but a loop: fit the boosted challenger, let it reveal which interactions and curve shapes the filed GLM is missing, and *distill* those findings back into added GLM terms. The filed model then captures most of the challenger's lift while remaining a document a regulator can read.

The structural twist is that the filed-rating-plan constraint turns Chapter 18's explainability discussion into a binding production requirement. A rate filing must state what the model charges and why; an adverse underwriting decision must often be explained to the applicant under the same family of laws as adverse-action notices; and several U.S. states now require insurers to demonstrate that models built on nontraditional data do not proxy for prohibited characteristics (Section 11.5.5). The industry pattern is therefore a hybrid. Machine learning is used freely where it does not directly set the filed rate (underwriting triage, retention and demand modeling, claims analytics, challenger benchmarking), while the filed rating plan evolves more slowly, with the GLM (or a constrained, explainable boosted model) as the regulatory interface. It is this book's accuracy-versus-accountability trade-off, institutionalized more explicitly than anywhere else in finance.

11.3.6 Telematics and Usage-Based Insurance

Traditional rating variables (age, territory, vehicle) are all *proxies* for how a person drives. Telematics replaces the proxy with the behavior itself: a device or phone app records mileage, speed, time of day, and the harsh-braking and cornering events that predict collisions, and usage-based insurance prices on that record. It is the cleanest example in this book of AI shrinking an information asymmetry at the individual level: the insurer no longer infers risk from group membership but observes each driver's behavior.

In the worked example, the telematics layer is itself a GLM on behavioral features: a log-link frequency model whose fitted coefficient on harsh-braking events is $+0.163$ per event per hundred miles relative to the rating cell's average. Six months of data show the young urban driver braking harshly two fewer times per hundred miles than their cell's average, so the telematics multiplier is $e^{0.163\times(-2)} = e^{-0.326} \approx 0.722$, converting the age-based relativity of 1.8 into a personal one of $1.8 \times 0.722 \approx 1.30$. The pipeline runs raw sensor stream $\rightarrow$ engineered behavioral features (Chapter 17's feature extraction applied to accelerometer data) $\rightarrow$ a GLM coefficient $\rightarrow$ a rating relativity a regulator can read. The revised expected frequency is $0.08 \times 1.30 \times 1.25 = 0.13$, the pure premium falls to $0.13 \times 9{,}900 = \$1{,}287$, and the gross premium to $1{,}287/0.70 \approx \$1{,}838.57$, a saving of about \$707, or 27.8%, earned by demonstrated behavior rather than demographic reclassification. Run in reverse, the same calculation explains the controversy: a driver whose data reads badly faces a surcharge no traditional variable would have imposed, and continuous location and motion records raise privacy questions regulators are still working through. Enrollment is also voluntary, and drivers who expect their data to read well are the ones who opt in, so the telematics book is not a random sample of the portfolio.

11.3.7 Claims: Triage, Automation, and Fraud

Claims is where an insurer's promises are kept, and it is the most AI-saturated function in the industry. Three applications dominate. *Severity triage* predicts, from the first notice of loss, which claims will become expensive (the injury claim that will develop into litigation, the water leak that will become a mold claim), so that experienced adjusters are assigned early where they change outcomes. The target is long-tailed and

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partially censored: open claims' ultimate costs are not yet known, the censoring structure of Chapter 15's survival analysis.

A minimal triage model shows the mechanics. Suppose a logistic model scores the probability that a new auto claim ultimately exceeds $50,000, from three first-notice features: an attorney is already involved (+1.6 on the log-odds), an injury is reported (+1.3), and the claim was reported more than a week after the accident (+0.8), against an intercept of −2.2. Here σ is Chapter 6's sigmoid, $\sigma(z) = 1/(1 + e^{-z})$, not the standard deviation of the pooling calculation earlier; both meanings are standard, and a σ applied to an argument is always the sigmoid. A claim arriving with attorney and injury flags but promptly reported scores $\sigma(-2.2 + 1.6 + 1.3) = \sigma(0.7) \approx 0.668$; a promptly-adjusted no-injury claim reported late scores $\sigma(-2.2 + 0.8) = \sigma(-1.4) \approx 0.198$. The first is routed to a senior adjuster on day one, the second to straight-through processing, and the model's value is the outcome difference early senior attention makes on the claims it flags. *Straight-through processing* pays simple, well-documented claims automatically, using computer vision on damage photos to estimate repair costs and NLP on claim descriptions to verify coverage. Chapter 10's lesson applies: an automated adjudication pipeline is itself a system whose failures (overpayment at scale, systematic wrongful denial) are operational events. *Claims fraud detection* is Chapter 15's machinery transplanted: rare positives, adversarial adaptation, network structure (staged-accident rings share repair shops, clinics, and phone numbers as laundering networks share addresses), and the same precision-recall economics, since every investigated honest claim delays a legitimate payment to a customer who has just suffered a loss.

11.4 Reserving, Life, Health, and Assets

Beyond pricing, an insurer must estimate what it already owes, and manage the assets held against it.

11.4.1 Reserving: The Chain Ladder and Its Successors

An insurer's largest liability is typically not claims it knows about but claims it must *estimate*: losses incurred but not yet reported or not yet fully paid. Reserving estimates this liability, and the classical tool, the chain ladder, is worth working once by hand: it is the industry's balance-sheet backbone and an elegant use of minimal data.

Cumulative paid claims (in $000s) are arranged by accident year and development age:

Accident year	12 months	24 months	36 months
Year 1	400	600	660
Year 2	450	675	?
Year 3	500	?	?

Table 11.2: A cumulative paid-claims development triangle ($000s)

The chain ladder assumes each accident year develops proportionally the way its predecessors did. The 12-to-24-month development factor, estimated from all years observed at both ages, is $(600 + 675)/(400 + 450) = 1{,}275/850 = 1.50$; the 24-to-36-month factor is $660/600 = 1.10$. Year 2's ultimate loss is then projected as $675 \times 1.10 = 742.5$ and Year 3's as $500 \times 1.50 \times 1.10 = 825$, so the required reserve, ultimate losses minus amounts already paid, is $(742.5 - 675) + (825 - 500) = 67.5 + 325 = \$392.5$ thousand.

Machine learning enters reserving along two paths. Individual-claim reserving replaces the aggregated triangle with claim-level prediction, typically a gradient-boosted or neural regression of each open claim's remaining cost on the same first-notice features as the triage model above (injury type, attorney involvement, jurisdiction) plus its paid-to-date trajectory. It improves accuracy and, more importantly, explains *why* reserves move, information a triangle cannot contain. Neural extensions of the chain ladder itself, which let the development factors vary with claim characteristics, are an active actuarial literature (see Wüthrich and Merz in Further Reading). The chain ladder's own assumption, that development patterns are stable across accident years, breaks when claims inflation shifts or a new claims process changes settlement speed, so the drift monitoring of Chapter 18 applies to reserving models too. The stakes are high: reserves *are* the float, so

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systematic under-reserving manufactures phantom profits today at the cost of real insolvency risk later. That is why reserve opinions are among the few statistical estimates in finance that a named, professionally liable individual must sign.

The $392.5 thousand above is an undiscounted best estimate, the traditional basis for property-casualty reserves. Under IFRS 17, the international insurance-contracts standard effective for annual periods beginning on or after 1 January 2023, the same liability is measured as *fulfilment cash flows*: expected future payments, spread over the years in which they will be paid and discounted at current rates, plus an explicit *risk adjustment* for bearing uncertainty about the amount and timing of claims. On new business, a *contractual service margin* holds back the expected profit and releases it as coverage is provided, while a loss-making group of contracts recognizes its loss at once (IASB 2017). The chain ladder still supplies the cash-flow projection, but it now needs a payment pattern, a discount curve, and a view of the reserve's *distribution*, since insurers must disclose the confidence level to which their risk adjustment corresponds. Stochastic reserving and the claim-level models above can answer that question; a point estimate cannot. In the United States, the FASB's long-duration targeted improvements (LDTI), effective for most SEC-filing insurers for fiscal years beginning after 15 December 2022 and for other insurers two years later, move U.S. GAAP for life and annuity contracts in the same direction: assumptions must be reviewed at least annually, and the liability is discounted at a current upper-medium-grade (single-A) fixed-income yield updated every reporting date (FASB 2018).

11.4.2 Life Insurance, Mortality, and Longevity

Everything so far has been property-casualty insurance: short contracts, uncertain frequency and severity. In life insurance the event is certain, only its timing is not, and contracts run for decades, which moves the modeling weight to mortality curves and discounting, Chapter 5's machinery applied to human lifetimes. A one-year term policy pays $500,000 if a policyholder with annual death probability $q = 0.002$ dies during the year. Its expected present value, the net premium, is

$$q \times B \times v \;=\; 0.002 \times 500{,}000 \times \frac{1}{1.04} \;\approx\; \$961.54,$$

mortality probability times benefit times a one-year discount factor at 4%. Every life product (term, whole life, annuities) is an expected-present-value sum of such terms across future years. Level-premium products overcharge early years to prefund the later ones, creating the enormous, long-duration liabilities that drive Section 11.4.4.

The modeling frontier is the mortality curve itself. Mortality improves over time, unevenly by age and cohort. The standard stochastic model, the Lee-Carter model, decomposes the age-by-year mortality surface into age profiles and a single time-varying mortality index, a factor extraction in the spirit of Chapter 7's PCA term-structure example, and forecasts that index as a random walk with drift. The symmetric risk is longevity risk: an annuity or pension book loses when people live *longer* than projected, a slow-moving, undiversifiable trend risk transferred in practice through longevity swaps between pension funds and reinsurers. AI contributes on both fronts: mortality models enriched with cause-of-death, socioeconomic, and medical-advance information, and accelerated underwriting, which replaces the medical exam with predictive models on application data, prescription histories, and, controversially, wearable-device data. It moves a weeks-long process toward minutes for low-risk applicants while concentrating the fairness questions of Section 11.5.5 on whoever the model routes into full underwriting.

11.4.3 Health Insurance and Risk Adjustment

Health insurance runs on different economics: contracts reprice annually, so the long-tail reserving problem largely disappears, and the dominant friction is selection, since individuals know far more about their expected medical costs than any application reveals. Modern health-insurance markets manage this with one of the largest predictive-modeling systems in the world with money attached: risk adjustment. Regulators pay (or transfer between) insurers according to a model-predicted cost score for each enrollee, so that an insurer attracting sicker members is compensated rather than bankrupted.

The risk score *is* a predictive model: an additive regression of next-year cost on demographics and diagnosis indicators, whose fitted coefficients are published as score weights. In the U.S. Medicare Advantage

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program's HCC framework, a 76-year-old female enrollee might carry a demographic base of 0.45, plus 0.30 for diabetes with complications, 0.33 for heart failure, and 0.22 for COPD: a risk score of 1.30, which against a baseline payment of $10,000 brings the insurer $13,000. The incentive problem follows immediately, and it is Goodhart's law at industrial scale: the insurer controls the diagnostic coding that feeds the model, so "upcoding," documenting members as sicker than clinically warranted, converts directly into revenue. Adding one unsupported vascular-disease code at 0.13 lifts the score to 1.43, worth an extra $1,300 per member per year, or $1 billion a year across a million-member book for every tenth of a point. Both academic estimates and government audits have found aggregate coding-intensity effects of roughly this magnitude, making risk-score integrity one of the largest financial-modeling enforcement areas in the United States.

The AI story is symmetrical. On the revenue side, insurers deploy NLP on clinical charts to find *legitimately supportable* diagnoses the treating physician documented but never coded, a use that is lawful and, from the model's perspective, accuracy-improving. On the audit side, the simplest anomaly test is already decisive at scale. If an insurer codes vascular disease in 9% of 10,000 enrollees against a clinically expected regional prevalence of 6%, the excess is $z = (0.09 - 0.06)/\sqrt{0.06 \times 0.94/10{,}000} \approx 12.6$ standard errors. No plausible casemix difference explains that pattern; it flags the book for chart-level review, where Chapter 15's anomaly-detection machinery takes over. A large z proves the coding is anomalous, not that it is fraudulent, and distinguishing real from manufactured casemix is the human-review step the statistics only prioritize.

11.4.4 The Investment Side: Asset-Liability Management

An insurer's balance sheet is a leveraged bond portfolio wrapped around a promise, managed with Chapter 5's duration machinery, with the twist that both sides move with interest rates. Consider a life insurer with $110 million of assets of duration 5 against $100 million of liabilities of duration 12 (long-duration promises are the nature of the product), for a surplus of $10 million. A one-percentage-point *rise* in rates moves assets by $-110 \times 5 \times 1\% = -\$5.5$ million but liabilities by $-100 \times 12 \times 1\% = -\$12$ million, so surplus *rises* to $16.5 million; a one-point *fall* symmetrically compresses surplus to $3.5 million. The sign is the point: with liabilities longer than assets, the classic life-insurance posture, it is *falling* rates that quietly impair the industry. That is what a decade of near-zero rates did to life insurers and pension funds worldwide, and why duration-matching, immunization, and long-dated hedging are a core insurance discipline.

The machine-learning layer enters through a fact the calculation assumed away: a liability's duration is not bond mathematics, because the "bond" can walk. Policyholders hold options (to lapse a policy, surrender it for cash, or pour new money into a guarantee) and exercise them in response to rates. When market rates rise well above the credited rate, surrender becomes attractive; when guarantees are above market, policyholders cling to them. Insurers model this with dynamic lapse functions fitted to policyholder-level experience data, with the rate gap, policy age, tax penalties, and distribution channel as features, and the fitted behavior sets the liability's *effective* duration. Rerunning the worked example with behavior in it: if a one-point rate rise accelerates lapses enough to shorten effective liability duration from 12 to 9, surplus rises only to $110 - 5.5 - (100 - 9) = \$13.5$ million rather than $16.5 million; if a one-point fall slows lapses and extends duration to 14, surplus compresses to $110 + 5.5 - (100 + 14) = \$1.5$ million rather than $3.5 million. Behavior has clipped the good outcome *and* deepened the bad one, the signature of a sold option: Chapter 5's negative convexity with a human exercising it. The behavioral model, not the bond arithmetic, determines whether the hedge book is sized correctly, and the 2022–2023 rate spike gave these models their first real out-of-sample test in a generation.

Two further machine-learning deployments round out the ALM picture. Valuing blocks with embedded guarantees (variable annuities above all) requires nested simulation, with economic scenarios outside and valuation runs inside each scenario, whose cost grows multiplicatively. Insurers increasingly train neural-network surrogates (function approximators in the least-squares Monte Carlo tradition) to replace the inner valuation loop, making hedging programs computable overnight rather than over weeks; the surrogate's approximation error is itself a model risk to be validated. And the portfolio problem that remains, investing float and surplus against regulatory constraints on asset risk, is Chapter 12's mean-variance machinery under a solvency constraint. That is why insurers became early, large buyers of the private credit assets Chapter 3 described: long, illiquid, yield-bearing assets are a natural match for long, illiquid, *behaviorally modeled* liabilities.

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11.5 Capital, Catastrophe, and Fairness

The tail is where insurance is decided; this part covers catastrophe capital, solvency regulation, and the fairness questions new data raises.

11.5.1 Catastrophe Risk, Reinsurance, and Alternative Capital

Pooling fails where insurance is needed most. Catastrophe losses are not independent across policyholders (a hurricane correlates a hundred thousand claims into one event), so Section 11.2.1's $\sqrt{n}$ arithmetic collapses and the tail must be either modeled and capitalized or transferred. Transfer happens through reinsurance, insurance for insurers, typically written in layers stacked into a tower, as in Figure 11.3. For example, a reinsurer sells a $100 million layer attaching at $50 million (it pays losses between $50M and $150M from a single event). If catastrophe models put the annual probability of an event reaching the layer at 0.04 and the expected loss within the layer, given such an event, at $60 million, the layer's expected annual loss is $0.04 \times 60 = \$2.4$ million, a "rate on line" of 2.4% of the limit, before the substantial risk load such correlated, capital-intensive exposure commands. The same exposure increasingly reaches capital markets directly through catastrophe bonds, whose investors receive a spread, typically a multiple of the modeled expected loss, for bearing a tail risk largely uncorrelated with market risk: the diversification argument of Chapter 12 applied to hurricanes.

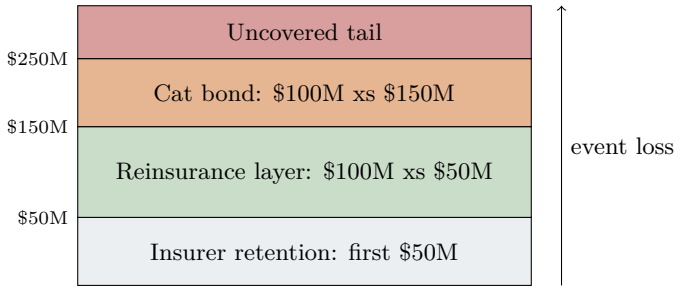


Figure 11.3: A catastrophe program as a tower: the insurer retains the first losses, traditional reinsurance covers the middle layers, and a catastrophe bond passes a higher layer to capital-market investors. Each layer's price is quoted as a rate on line: expected annual loss to the layer, plus a risk load, divided by the layer's limit.

The models behind these numbers are Hurricane Andrew's institutional legacy: simulation engines that generate tens of thousands of synthetic hurricane seasons or earthquake catalogs, propagate each event's physics over a portfolio of insured locations, and apply engineering damage functions: Monte Carlo over a compound process, Chapter 10's architecture at continental scale. AI's contribution is concentrated in the inputs. Computer vision on aerial and satellite imagery (Chapter 17's methods) now determines roof geometry, construction type, and post-event damage at the individual-building level, replacing what was once a coarse, self-reported exposure database. Climate-conditioned event generation, which draws synthetic seasons from warming scenarios rather than from the historical record alone, extends these models toward the forward-looking scenario analysis that climate stress testing requires, the physical-risk counterpart of the transition-risk stress that Section 9.2.5 applies to a bank's loan book.

That extension confronts the deepest assumption in catastrophe modeling. A simulated event catalog treats the hazard as *stationary*: the frequency and severity of hurricanes, floods, and wildfires are estimated from decades of history and assumed to describe next year, which a warming climate undermines. Warmer oceans shift the distribution of hurricane intensity and rainfall, and rising sea levels raise the base on which every storm surge builds, so a catalog calibrated on the past understates the present, and the bias compounds over the multi-decade horizons of mortgages and infrastructure. Near-term, modelers use "warm-sea-surface-temperature" or climate-conditioned event sets that reweight the historical catalog toward the conditions expected in the coming seasons. Longer-term, they condition the hazard module on climate-model output under named scenarios, the NGFS-style families supervisors use, so that a 2050 loss distribution can be

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produced alongside today's. Machine learning does most of its work in the mapping: convolutional networks trained on satellite and radar imagery delineate past flood extents and burn scars where no survey exists, and statistical downscaling translates coarse global climate projections, run on grids tens of kilometers wide, into street-level hazard estimates. The model risk is substantial and largely unvalidatable, since the out-of-sample period is the future, so such models are best used, as Chapter 10 prescribes for every structural model, for comparative scenario analysis rather than as a single forecast.

11.5.2 Parametric Insurance and Basis Risk

Parametric insurance pays a fixed amount when a measured *index* crosses a trigger (wind speed at a reference station, rainfall over a growing season, ground acceleration at a site) rather than indemnifying assessed damage. The payout requires no adjuster, so it arrives in days rather than months, and the product can be sold where claims adjustment is impractical. That is why parametric structures dominate sovereign disaster pools, index-based crop insurance across the developing world, and consumer novelties like flight-delay cover, and why they are a natural match for the satellite and sensor data of Chapter 17.

The price of simplicity is basis risk: the index and the loss are correlated, not identical. A worked example quantifies it. A parametric hurricane policy pays \$200,000 if measured wind at the reference station reaches the trigger. Suppose the modeled annual probability of the policyholder suffering material damage is 0.05; given damage, the trigger fires with probability 0.80; in the 0.95 of years without damage, it false-fires with probability 0.01. Then the policyholder faces a $0.05 \times 0.20 = 1.0\%$ annual probability of the worst outcome, real damage and no payout, while the insurer faces a $0.95 \times 0.01 = 0.95\%$ probability of paying without damage; the overall payout probability is $0.05 \times 0.80 + 0.0095 = 0.0495$, for a pure premium of $0.0495 \times 200{,}000 = \$9{,}900$. Both mismatch probabilities are design parameters. A denser sensor network or a model-based index (a "cat-in-a-grid" trigger computed from the physics models above) tightens the loss-index correlation and shrinks basis risk, which is where machine learning earns its keep in parametric design. The trade-off between trigger fidelity and trigger verifiability is Chapter 10's Goodhart discipline applied to contract design: an index must be objective and manipulation-proof because no adjuster will ever look at the actual loss.

11.5.3 Solvency: Capital Regulation as a Value at Risk

The regulatory capstone ties insurance back to Chapter 8. Modern insurance solvency regimes, Solvency II in Europe most explicitly, define required capital as a Value at Risk. The Solvency Capital Requirement is calibrated so that the insurer can absorb a 1-in-200 one-year loss: a 99.5% one-year VaR on the change in the insurer's own net asset value, aggregated across underwriting, market, credit, and operational risk modules with prescribed correlations. The calculation is market-risk machinery pointed at a balance sheet instead of a trading book. If an insurer's one-year net-asset-value change has standard deviation \$40 million, a normal approximation puts the 99.5% requirement at $2.576 \times 40 \approx \$103$ million. Everything Chapter 8 said about the normal approximation understating fat tails applies with force, since underwriting results are compound-Poisson skewed, not Gaussian. U.S. risk-based capital reaches a similar structure by formula rather than internal model, and the internal-model-versus-standard-formula tension (risk-sensitivity against comparability and gameability) is the debate Chapter 10 described operational-risk capital resolving in favor of the formula.

Machine learning enters the capital calculation through its computational bottleneck. The 99.5% one-year VaR requires the *distribution* of the insurer's net asset value a year ahead, and net asset value at year one is itself a valuation of every embedded guarantee and option on the book, under each scenario: the nested-simulation explosion of Section 11.4.4, now at whole-balance-sheet scale. Internal-model insurers therefore lean on proxy models: least-squares Monte Carlo regressions and, increasingly, neural-network approximators trained to reproduce the full valuation as a fast function of the scenario variables. A supervisor approving such an internal model is approving a machine-learning approximation of a balance sheet, so European supervisors have begun publishing guidance on ML inside internal models, and the surrogate's fitting error is examined in the far tail, the 0.5% that sets the capital number, where any regression approximates worst. Every model in this chapter (pricing, reserving, catastrophe, mortality, behavior, and the proxy models

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on top) thus feeds a regulated capital number and lives inside the model-governance perimeter of Chapters 10 and 18, with validation, documentation, and independent challenge as legal requirements.

11.5.4 Underwriting with New Data and Text

Commercial insurance underwriting is a document business: a submission arrives as a broker's narrative, loss runs from prior insurers, financial statements, and schedules of locations or vehicles, and an underwriter's first hours are spent extracting structure from paper. NLP-driven submission ingestion (extracting exposures, flagging omissions, triaging which submissions deserve an underwriter's scarce attention) has become the standard first deployment of language models in commercial carriers. Property underwriting increasingly begins from imagery rather than questionnaires, the building-level computer vision of Section 11.5.1 applied one policy at a time. Cyber underwriting, one of the industry's fastest-growing and hardest lines, prices from external scans of an applicant's security posture, an alternative-data exercise in Chapter 17's sense, applied to a peril whose correlated, catastrophe-like tail (one vulnerability, thousands of simultaneous claims) the industry is still learning to model.

11.5.5 Fairness and Proxy Discrimination

Insurance pricing discriminates by design, in the actuarial sense of charging different expected costs differently, and the law's task has always been separating legitimate risk-based distinctions from prohibited ones. AI sharpens this into Chapter 18's proxy problem: a model barred from using a protected characteristic can reconstruct it from correlated inputs, the better the richer the feature set. Insurance regulators have moved on this most concretely. Colorado's SB 21-169 requires insurers to test models and external data sources for unfairly discriminatory outcomes and to govern them accordingly. The European Union's gender ruling prohibits gender-based pricing outright even where it is actuarially predictive, a legislature deliberately overriding statistical validity with a social judgment. And the U.S. state-by-state debate over credit-based insurance scores, predictive of claims but correlated with protected characteristics, rehearsed every argument in Chapter 18's fairness section decades before machine learning raised the stakes.

Colorado's statute has since been joined by a national template. On 4 December 2023 the National Association of Insurance Commissioners adopted its Model Bulletin on the Use of Artificial Intelligence Systems by Insurers. It expects each insurer to maintain a written program governing its AI systems (covering governance and senior-management accountability, risk management and internal controls, and due diligence on third-party models and data), encourages testing for bias and unfair discrimination, and reminds insurers that decisions made with AI must still comply with existing unfair-trade-practice and unfair-discrimination laws (NAIC 2023). A bulletin binds only where a state adopts it: as of 8 October 2026 the NAIC's own tracker lists it as adopted by 27 states, the District of Columbia, and Puerto Rico, with California, Colorado, New York, and Texas regulating through their own insurance-specific rules instead (NAIC 2026). In the European Union, the Artificial Intelligence Act classifies AI systems used for risk assessment and pricing of natural persons in life and health insurance as high-risk, with the obligations Chapter 18 describes (EU 2024), now deferred by the Digital Omnibus to 2 December 2027 (EU 2026).

A second, subtler fairness frontier is price optimization: using demand models (each customer's estimated willingness to pay and propensity to shop) to set premiums *above* risk-based cost for the customers least likely to leave. The practice is standard revenue management in airlines and retail, but insurance regulators in many U.S. states have banned it. Their ground is that an insurance price must reflect expected cost and expense, not exploitable inertia, and that the customers who shop least, often older and lower-income policyholders, are those the loyalty penalty falls on hardest. The United Kingdom reached the same conclusion in 2021, prohibiting renewal prices above the equivalent new-customer price. Here the objectionable model is *perfectly accurate*: demand estimation is legitimate science, and the prohibition targets the model's use, not its errors. It is the clearest illustration in this book that model governance is about deployment decisions, not only statistical quality. Insurance supplies the financial industry's most developed case law on what "fair pricing" can mean operationally, and Chapter 18's apparatus (fairness metrics, proxy detection, counterfactual testing) supplies the tools its newest statutes effectively mandate.

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11.6 Case Vignette: Progressive's Snapshot and the Telematics Market

Usage-based insurance moved from concept to mass market when Progressive, after more than a decade of pilots dating to the late 1990s, rolled out its Snapshot program nationally in the early 2010s: a plug-in device, later a phone app, recording mileage, time of day, and hard-braking events, with discounts tied to observed behavior. Two features of the episode reward attention. First, the earliest and largest benefit was not pricing precision but *selection*. The offer of a monitoring discount attracted drivers who believed they drove well and repelled those who knew they did not, so the program improved the insurer's book before any telematics rating factor was filed: adverse selection running in reverse. Second, the program's evolution traces this chapter's regulatory arc in miniature. Early versions surcharged nothing and only discounted, because filing a telematics *surcharge* required convincing regulators of the model's fairness and accuracy, and the industry's subsequent decade of negotiating what behavioral data may be used, retained, and shared previews the governance questions every AI pricing innovation now faces.

11.7 Challenges in Insurance and Actuarial Modeling

The tail is uninsurable by experience. Hurricane Andrew's lesson recurs: the losses that matter most are those history has not yet produced, so catastrophe, cyber, and pandemic risk must be modeled from structure rather than experience, with all of the model risk Chapter 10 attaches to that move.

Feedback and selection are everywhere. Prices change who buys, monitoring changes who enrolls, and claims automation changes what is claimed; almost no insurance dataset is a random sample of the risk it is used to price, and Chapter 1's selection warnings bind with unusual force.

Long feedback loops. A mispriced auto book reveals itself in a year; a mispriced liability or life book can take a decade, so model validation must lean on the indirect methods of Chapter 10's validation discussion rather than rapid outcome feedback. The float arithmetic of Section 11.2.2 means the errors compound financially while they remain statistically invisible.

Correlated tails defeat the product's own logic. The industry sells the law of large numbers, but its gravest exposures (hurricanes, earthquakes, cyber events, pandemics) are the ones where independence fails; managing the gap between the diversifiable book and the undiversifiable tail is the industry's permanent structural problem.

Accuracy versus accountability, institutionalized. The filed-rating-plan regime makes the interpretability constraint structural: the industry's central modeling question is rarely "what is the most accurate model" but "what is the most accurate model that can be filed, explained, and audited."

Privacy as a pricing input. Telematics, wearables, and property imagery make the underwriting data itself the sensitive object, so data governance is not compliance overhead but part of product design.

11.8 Key Takeaways

- Insurance inverts the production cycle (price first, costs later) and sells the law of large numbers retail: pooling 10,000 independent risks cut relative loss uncertainty from 417% to 4.17% in the worked example, a $\sqrt{n}$ improvement that correlated catastrophe losses delete.
- An insurer earns from two engines, the combined ratio and investment income on float; in the worked example a 95% combined ratio and \$150M of float at 4% produced a 22% return on capital, and even a 103% combined ratio remained profitable because underwriting losses are the cost of cheap float.
- Classical pricing is frequency times severity with multiplicative relativities; one-way estimation double-counts correlated factors (a 43% overcharge in the worked 2×2 example), which is why the jointly estimated GLM displaced it. "Traditional pricing" today means *GLM-based* pricing, the decades-established, filed incumbent that machine learning must beat.
- Machine learning's pricing advantage is the interaction: in the worked example, a young-urban interaction that no multiplicative model can represent leaked into the GLM's fitted marginals and overcharged the

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young-rural cell by 11.4%, while a depth-two tree priced all four cells exactly. Telematics runs the same logic at driver level, a fitted coefficient of +0.163 per harsh-braking event converting sensor data into a filed relativity (two fewer events than the cell average give $1.8 \times e^{0.163\times(-2)} \approx 1.30$).

- Credibility theory, $Z = n/(n + k)$, is Bayesian shrinkage derived by actuaries decades before machine learning rederived it as regularization: the worked fleet's 0.333 observed frequency was blended to 0.147 against a 0.100 class rate.

- The chain ladder projects ultimate losses from development factors (1.50 and 1.10 in the worked triangle, a $392.5 thousand reserve); reserves are the float, so reserving error is an economic event, not an accounting one.

- Life insurance moves the weight to mortality and discounting (a $961.54 net premium in the worked term-life example), with Lee-Carter forecasting mortality as factor extraction plus a random walk. Asset-liability management is Chapter 5's duration machinery on both sides of the balance sheet, where falling rates hurt the long-liability insurer. Its ML layer is behavioral: dynamic-lapse models make liability duration a fitted model output, and in the worked example behavior clipped the favorable surplus outcome (16.5 → 13.5) while deepening the unfavorable one (3.5 → 1.5), a sold option with a human exercising it.

- Catastrophe risk is transferred through reinsurance towers and cat bonds priced from simulation models (2.4% rate on line in the worked layer), and solvency regulation caps the chapter as a 99.5% one-year VaR on the insurer's own net assets.

- Health insurance replaces long-tail reserving with a selection problem managed by risk adjustment, a regression with a budget attached, whose coding incentives (an extra 0.13 of risk score was worth $1,300 per enrollee per year in the worked example) make model integrity an adversarial, audited contest.

- Parametric insurance trades adjusters for an index, paying instantly at the cost of basis risk: in the worked example, a 1.0% annual chance of damage without payout against a $9,900 pure premium, with index design, and its manipulation-proofness, the modeling problem.

- Insurance fairness law, from credit-based insurance scores to Colorado's SB 21-169, the EU gender ruling, and the price-optimization bans, is the financial industry's most developed treatment of proxy discrimination and model *use*, and it now effectively mandates Chapter 18's technical toolkit.

11.9 Suggested Exercises

11.1 Recompute Section 11.2.1's pooling example for $n = 100$ and $n = 1{,}000{,}000$ policies. At what n does the probability of the portfolio's loss exceeding 110% of its mean fall below 0.1% (using the normal approximation), and why should you distrust that approximation in the far tail?

11.2 Recompute the worked pricing example for an experienced rural driver with frequency relativities 0.9 (age) and 0.8 (territory) and a severity relativity of 0.95. What gross premium results, and what fraction of the young urban driver's premium is it?

11.3 Using Table 11.1, verify by direct computation that the one-way relativities are 2.10 and 1.833, and then show that if the exposures had been balanced (250 in every cell, claims scaled to preserve each cell's frequency), the one-way method would have recovered the true relativities. What general condition on the exposure distribution makes one-way estimation unbiased?

11.4 In Section 11.3.4's interaction example, the GLM overcharged the young-rural cell by 11.4%. Explain the mechanism in your own words (why does an interaction concentrated in one cell distort the price of a *different* cell?), and then describe how an actuary could keep the filed model multiplicative yet still fix the mispricing (hint: an interaction term is itself another rating variable). What does your fix imply about the practical role of a boosted challenger model in a GLM-filing regime?

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11.5 A fleet has 5 years of experience with credibility constant $k = 12$. Compute its credibility weight, and the blended frequency if it observed 0.30 against a class rate of 0.10. How many years of experience would the fleet need for its own data to receive 75% weight?

11.6 Extend the chain-ladder triangle with a fourth accident year showing 12-month cumulative paid claims of 550. Compute its projected ultimate loss and the new total reserve, and state precisely which assumption you invoked about Year 4's future development.

11.7 Recompute Section 11.4.4's ALM example for an insurer that has hedged its liability duration down to 8 using swaps. What is the surplus after a one-point fall in rates, and what has the hedge cost the insurer if rates instead rise by one point?

11.8 Suppose a dynamic-lapse model of the kind Section 11.4.4 describes, fitted to policyholder experience, implies a lapse rate of $5\% + 4 \times \max(0, r - 3\%)$ per year against a 3% credited rate, where r is the market rate. Compute the lapse rate at market rates of 3%, 5%, and 6%; explain in option language (Chapter 5) why this behavior leaves the insurer short an option, the policyholder's right to surrender at book value when rates rise, which is a put on the policy's bond-like value or, equivalently, a call on rates; and describe what data a Chapter 6 classifier would need, and what drift monitoring it would require, before you would trust it to set the hedge for a one-in-a-generation rate spike.

11.9 A reinsurance layer of $50 million attaching at $150 million has a modeled 0.02 annual probability of attachment and an expected in-layer loss of $30 million given attachment. Compute its expected annual loss and rate on line, and explain why its risk load per dollar of expected loss should exceed that of Section 11.5.1's lower layer.

11.10 In the parametric example of Section 11.5.2, suppose a better index raises the trigger's hit rate given damage from 0.80 to 0.95 while leaving the false-trigger rate at 0.01. Recompute both basis-risk probabilities and the pure premium, and explain why the policyholder should be willing to pay *more* than the pure-premium increase for this redesign.

11.11 An auditor observes that an insurer's average Medicare Advantage risk score rose from 1.05 to 1.20 over five years while its members' hospitalization and mortality rates were flat. Using Section 11.4.3, lay out the innocent and the upcoding explanations for this pattern; recompute the section's prevalence z-test for an insurer coding the same condition in 7% of 2,500 enrollees against the 6% benchmark, and explain why this smaller book's identical-direction deviation is far weaker evidence; then propose one *longitudinal* test (using the flat hospitalization data) that the cross-sectional prevalence test cannot provide.

11.12 A regulator bars a rating variable that is predictive but correlates with a protected characteristic. Using Chapter 18's distinction between statistical validity and disparate impact, argue both sides: the actuarial case for the variable and the fairness case against it, and identify what empirical test would inform (though not settle) the dispute.

11.13 **Challenge (real data).** The French motor third-party liability dataset (`freMTPL2freq`, available through OpenML and the `CASdatasets` project) contains claim counts and rating features for roughly 678,000 real policies. Fit a Poisson GLM for claim frequency on driver age, vehicle age, and bonus-malus level, then fit a gradient-boosted model on the same features. Compare out-of-sample Poisson deviance, plot each model's implied age relativity, and identify one interaction the boosted model found that the GLM missed. Discuss which model you could defend in a rate filing, and what you would change about the boosted model to make it filable.

11.10 Further Reading

- Goldburd et al. (2020), *Generalized Linear Models for Insurance Rating*. The Casualty Actuarial Society's freely available practitioner monograph on GLM ratemaking, the document that defines what "traditional pricing" concretely means in Section 11.3.1.

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- Ohlsson and Johansson (2010), *Non-Life Insurance Pricing with Generalized Linear Models*. The standard compact academic treatment of the same GLM machinery.
- Wüthrich and Merz (2023), *Statistical Foundations of Actuarial Learning and its Applications*. A rigorous, freely available treatment of machine learning in actuarial work, including the GLM-versus-boosting comparison and neural reserving.
- Bühlmann and Gisler (2005), *A Course in Credibility Theory and its Applications*. The definitive treatment of Section 11.3.2's credibility machinery by its principal architect.
- Denuit et al. (2019–2020), *Effective Statistical Learning Methods for Actuaries*. GLMs, tree-based methods, and neural networks developed specifically for insurance data.
- Dickson et al. (2020), *Actuarial Mathematics for Life Contingent Risks*. The standard modern text behind Section 11.4.2's life-contingency calculations.
- Grossi and Kunreuther (2005), *Catastrophe Modeling*. The standard introduction to the simulation-based catastrophe models of Section 11.5.1 and their post-Andrew history.
- Frees (2010), *Regression Modeling with Actuarial and Financial Applications*. A bridge between this book's regression toolkit and actuarial applications, with real insurance datasets.

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Chapter 12

Portfolio Theory and Asset Allocation

12.1 Introduction

Chapter 2 introduced a two-asset portfolio to teach Python, and found, almost as an aside, that combining two negatively correlated assets produced a portfolio considerably less volatile than either asset alone. This chapter treats that observation as the central question of a whole field: given a set of assets, their expected returns, and how they move together, what is the best way to combine them?

Portfolio theory defines "best" quantitatively: the highest expected return for a given level of risk. This chapter develops the classical tools built on that definition (mean-variance optimization, the Capital Asset Pricing Model, and multi-factor models) along with the more robust methods portfolio managers use when the classical assumptions fail in real data.

Four worked examples run through the chapter. The two-asset AssetA/AssetB portfolio from Chapter 2 is extended into a full efficient frontier and a closed-form minimum-variance portfolio. The three-asset AssetA/AssetB/AssetC extension illustrates mean-variance optimization in matrix form and exposes a practical hazard the classical theory does not warn against: estimation error. A small market-model regression estimates from data the equity beta of 1.2 that Chapter 5's WACC calculation assumed. And a stylized equity/bond pair, with economically realistic volatilities and expected returns, carries the tangency, Black-Litterman, and robo-advisor examples, for which the four-day toy sample is too short and too oddly correlated.

A Brief History of Portfolio Theory

Knowing the sequence in which these ideas arrived clarifies which problem each was designed to solve. Table 12.1 summarizes the main milestones.

Year	Development
1952	Markowitz formalizes mean-variance optimization, giving diversification, already intuitive to practitioners, a precise mathematical foundation
1964	Sharpe (building on related work by Lintner and Treynor) introduces the Capital Asset Pricing Model, connecting an asset's expected return to a single market-wide risk factor
1990	Black and Litterman propose a method for blending an investor's own views with market-implied equilibrium returns, addressing a practical instability in naive mean-variance optimization
1992–1993	Fama and French introduce size and value factors alongside the market factor, launching multi-factor asset pricing models

Table 12.1: A brief history of portfolio theory

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Each development responded to a limitation of what came before. Markowitz's framework requires estimating a full covariance matrix and is highly sensitive to estimation error, which motivated both the single-factor simplification of CAPM and, decades later, Black-Litterman's more stable blending approach. CAPM's single market factor, in turn, left systematic patterns in average returns unexplained, which Fama and French's additional factors were built to capture. This chapter follows roughly the same progression.

12.2 Diversification and the Efficient Frontier

Combining assets changes the risk of the whole in a way that does not follow from the parts, and that observation is the foundation of the chapter.

12.2.1 Diversification and the Power of Combining Assets

Chapter 2 computed the covariance matrix for AssetA and AssetB directly from their return histories,

$$\Sigma = \begin{pmatrix} 0.000414 & -0.000198 \\ -0.000198 & 0.000118 \end{pmatrix},$$

implying individual volatilities of $\sigma_A = 2.03\%$ and $\sigma_B = 1.09\%$ and a correlation of $\rho_{AB} = -0.896$. For a two-asset portfolio with weight w on Asset A and $1 - w$ on Asset B, portfolio variance is

$$\sigma_p^2(w) = w^2\sigma_A^2 + (1-w)^2\sigma_B^2 + 2w(1-w)\rho_{AB}\sigma_A\sigma_B. \tag{12.1}$$

Chapter 2 evaluated this formula only at the single equal-weighted point $w = 0.5$, finding $\sigma_p = 0.58\%$, considerably below the weighted average of the two individual volatilities.

Which value of w minimizes $\sigma_p^2(w)$? Differentiating with respect to w and setting the result to zero gives a closed-form solution,

$$w_A^{\text{MV}} = \frac{\sigma_B^2 - \sigma_{AB}}{\sigma_A^2 + \sigma_B^2 - 2\sigma_{AB}},$$

which, applied to the covariance matrix above, gives $w_A^{\text{MV}} = 34.05\%$ and $w_B^{\text{MV}} = 65.95\%$. This minimum-variance portfolio has volatility $\sigma_p^{\text{MV}} = 0.322\%$, nearly half the equal-weighted portfolio's 0.583%, with no new asset and no information about expected returns, only a better combination of the same two assets.

12.2.2 The Efficient Frontier and the Global Minimum-Variance Portfolio

Sweeping w across its full range traces out every risk-return combination achievable from AssetA and AssetB, illustrated schematically in Figure 12.1. Because $\rho_{AB} < 0$, the curve bows sharply to the left of a straight line connecting the two assets, the geometric signature of diversification: some combination is less risky than either asset alone, which is possible only because the assets do not move in lockstep.

In this sample, Asset B has *both* lower risk and higher return than Asset A ($\sigma_B = 1.09\%$ versus $\sigma_A = 2.03\%$, and $\bar{r}_B = 1.00\%$ versus $\bar{r}_A = 0.76\%$), so Asset A is not itself mean-variance efficient. The figure's upper arm, from the leftmost point of the curve to Asset B, is efficient in Markowitz's sense: every portfolio on it offers the highest expected return for its level of risk. The lower arm, from that point down to Asset A, is feasible but dominated, since a portfolio directly above it offers a strictly higher return at the same risk. The leftmost point is the *global* minimum-variance portfolio. The qualifier matters: every portfolio on the frontier is a minimum-variance portfolio for its own target expected return, whereas this one minimizes variance with no return constraint at all.

The efficient frontier is a menu rather than an answer. The framework narrows the feasible region down to the solid upper segment, and the investor picks a point on it according to how much variance they are willing to bear, a choice the mathematics leaves open. The global minimum-variance portfolio is the menu's left endpoint, the choice of an investor unwilling to accept any additional variance in exchange for return, not the portfolio the analysis points toward. It nonetheless earns a practical role: it is the only point on the frontier whose weights require no expected-return estimates, only the covariance matrix, which matters given how badly the estimation-error problem of Section 12.2.3 affects estimated means.

This is the quantitative content behind the advice to diversify: every portfolio off the efficient frontier is dominated by one on it.

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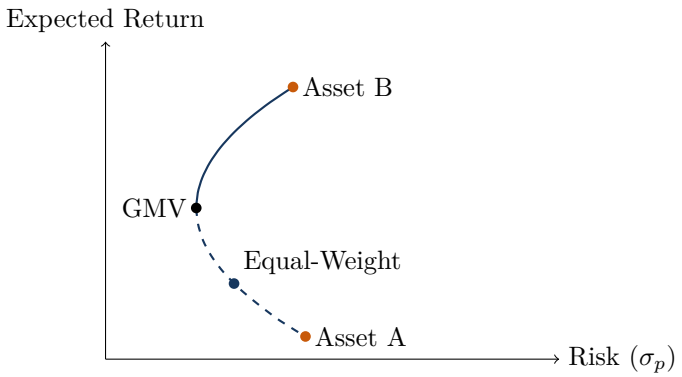


Figure 12.1: A schematic efficient frontier for the two-asset example (solid: efficient; dashed: feasible but dominated). GMV marks the global minimum-variance portfolio, the leftmost attainable point. An investor selects somewhere along the solid segment according to their risk tolerance; GMV is that menu's left endpoint, not its recommended item.

12.2.3 Mean-Variance Optimization in Matrix Form

The two-asset formulas above generalize to any number of assets using matrix notation. For n assets with covariance matrix Σ and a vector of ones $\mathbf{1}$, the (unconstrained) global minimum-variance portfolio weights are

$$w^{\text{MV}} = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}^\top \Sigma^{-1}\mathbf{1}}, \tag{12.2}$$

the direct matrix generalization of the closed-form two-asset formula in Section 12.2.1. Applying this to the three-asset covariance matrix from Chapter 2's AssetA/AssetB/AssetC extension,

$$\Sigma_3 = \begin{pmatrix} 0.000414 & -0.000198 & -0.000383 \\ -0.000198 & 0.000118 & 0.000176 \\ -0.000383 & 0.000176 & 0.000358 \end{pmatrix},$$

gives minimum-variance weights of

$$w_A \approx 44.8\%, \qquad w_B \approx 14.2\%, \qquad w_C \approx 41.0\%,$$

with a portfolio standard deviation of only about 0.050%, roughly six times lower than the equal-weighted three-asset portfolio's 0.297% computed in Chapter 2.

This result deserves suspicion rather than celebration. Chapter 2's three-asset dataset contains only four daily observations, far too few to estimate a 3×3 covariance matrix (six distinct entries) reliably. The optimizer found weights that nearly cancel the sample's realized co-movements, but it is optimizing against estimation noise as much as against real risk. This is the overfitting Chapter 6 illustrated with a high-degree polynomial: with enough free parameters relative to the data, a model (here, a portfolio optimizer) can achieve an excellent in-sample result that will not repeat out of sample. It is Markowitz optimization's central weakness. In Michaud's phrase, the optimizer is an "estimation-error maximizer": it seeks out and overweights the asset pairs whose historical correlation was most favorable, which are disproportionately likely to be pairs whose favorable correlation was a statistical accident of the sample. Sections 12.3.5 and 12.3.6 introduce two standard responses.

12.2.4 Shrinkage Estimation of the Covariance Matrix

A more direct statistical response, distinct from risk parity and Black-Litterman, is to shrink the noisy sample covariance matrix toward a simpler, lower-variance target before handing it to the optimizer. A standard

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target is a diagonal matrix built from the average sample variance, and the shrunk covariance matrix is a weighted average of this target and the raw sample estimate,

$$\Sigma^{\text{shrunk}} = \delta F + (1-\delta)\Sigma, \qquad F = \bar{\sigma}^2 I,$$

where $\delta \in [0,1]$ is the shrinkage intensity (often chosen to minimize expected estimation error using a formula due to Ledoit and Wolf, though a fixed value illustrates the idea just as well) and $\bar{\sigma}^2$ is the average of the sample variances on the diagonal of Σ. Applying $\delta = 0.5$ to the three-asset covariance matrix from Section 12.2.3, whose average variance is $\bar{\sigma}^2 \approx 0.000297$, shrinks every off-diagonal covariance term roughly in half and pulls each diagonal variance partway toward the common average. Recomputing the minimum-variance weights using this shrunk covariance matrix gives

$$w_A^{\text{shrunk}} \approx 39.2\%, \qquad w_B^{\text{shrunk}} \approx 30.5\%, \qquad w_C^{\text{shrunk}} \approx 30.3\%,$$

a far more balanced allocation than the extreme weights the raw sample covariance matrix produced. The shrunk portfolio's sample variance is necessarily higher than the unshrunk optimizer's, which is the in-sample minimum by construction. The justification for shrinkage is that this modest in-sample cost buys a large improvement in out-of-sample reliability, a bias-variance trade-off analogous to the ridge regression penalty Chapter 6 applied to an overfit polynomial.

12.3 Equilibrium Models and Factor Approaches

If everyone optimizes, prices must satisfy conditions of their own; this part develops those conditions and the factor models that relax them.

12.3.1 The Capital Asset Pricing Model and Beta Estimation

Chapter 5 introduced the Capital Asset Pricing Model, $r_e = r_f + \beta(r_m - r_f)$, to estimate Meridian Fabrication's cost of equity, using an equity beta of 1.2 that was assumed rather than derived. Beta is a regression coefficient, estimated like the OLS regressions of Chapter 6 by regressing an asset's excess returns on the market's excess returns,

$$r_{i,t} - r_f = \alpha_i + \beta_i (r_{m,t} - r_f) + \varepsilon_t. \tag{12.3}$$

Table 12.2 shows six months of hypothetical excess returns for a stock and the overall market.

Month	1	2	3	4	5	6
Market excess return	−2.0%	−1.0%	0.0%	1.0%	2.0%	3.0%
Stock excess return	−2.5%	−1.3%	0.2%	1.1%	2.6%	3.3%

Table 12.2: Six months of excess returns for a stock and the market

Applying the OLS formulas,

$$\hat{\beta} = \frac{\sum_t (r_{m,t} - \bar{r}_m)(r_{i,t} - \bar{r}_i)}{\sum_t (r_{m,t} - \bar{r}_m)^2} \approx 1.19, \qquad \hat{\alpha} \approx -0.03\%, \qquad R^2 \approx 0.992.$$

The estimated beta of 1.19 closely matches the 1.2 that Chapter 5 assumed, a reassuring if constructed confirmation, and it shows where a CAPM beta comes from in practice: not from theory alone, but from an ordinary regression. The near-zero alpha is also economically meaningful. CAPM predicts that, after accounting for market exposure, no asset should offer a systematically positive or negative excess return, so a statistically significant nonzero alpha (checked with the t-statistic and p-value machinery of Chapter 6) is read either as evidence against CAPM or as a mispricing an active manager might exploit.

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Systematic versus Idiosyncratic Risk

The $R^2 \approx 0.992$ above also has a risk-decomposition interpretation. Total return variance splits into a systematic (market-driven) component and an idiosyncratic (asset-specific) component,

$$\mathrm{Var}(r_i) = \underbrace{\beta_i^2 \,\mathrm{Var}(r_m)}_{\text{systematic}} + \underbrace{\mathrm{Var}(\varepsilon_i)}_{\text{idiosyncratic}} \ ,$$

and R^2 is the fraction of total variance that is systematic. For the stock in Table 12.2, with returns expressed as decimals, $\mathrm{Var}(r_m) = 0.000350$ and systematic variance is $\hat{\beta}^2 \,\mathrm{Var}(r_m) \approx (1.1886)^2(0.000350) \approx 0.000494$ (carrying the unrounded slope $\hat{\beta} = 1.1886$; the rounded 1.19 would give 0.000496), against total variance of about 0.000498, leaving an idiosyncratic remainder of only about 0.0000038, consistent with the regression's near-perfect fit.

CAPM's central economic claim rests on this decomposition. Idiosyncratic risk can be diversified away by holding many assets whose asset-specific shocks are independent, so a well-diversified investor should not expect compensation for bearing it. Systematic risk, by contrast, affects every asset at once through the shared market factor and cannot be diversified away. That is why CAPM's expected-return formula compensates an asset only for its beta, not its total volatility. Two assets with identical total volatility but different betas should offer different expected returns, while two assets with identical betas but very different idiosyncratic volatility should offer the same expected return, a testable and occasionally violated prediction that motivated the multi-factor extensions later in this chapter.

The Security Market Line

The Security Market Line plots CAPM's expected return, $r_f + \beta(r_m - r_f)$, as a straight line against beta, shown in Figure 12.2. Unlike the Capital Market Line of Section 12.3.2, which plots return against total risk (σ_p) and is populated only by efficient portfolios, the Security Market Line plots return against systematic risk alone, and under CAPM every correctly priced asset or portfolio, efficient or not, lies on it.

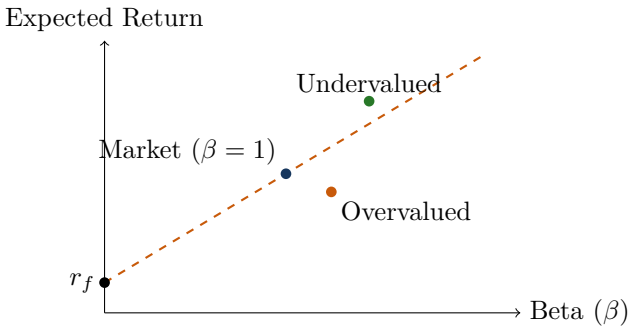


Figure 12.2: The Security Market Line plots CAPM's expected return against beta, with an undervalued asset (above the line) and an overvalued asset (below it) marked

An asset whose expected return plots above the Security Market Line is, under CAPM, undervalued: it offers more expected return than its systematic risk justifies, and the alpha from Section 12.3.1's regression is its vertical distance from the line. An asset below the line is overvalued. This is the graphical version of the active-management logic of the information ratio discussed later: an active manager searches for assets that plot persistently above the line. A manager who succeeds systematically is finding mispricing rather than taking on more systematic risk than the benchmark, a distinction the Security Market Line makes visible and a raw return comparison does not.

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12.3.2 The Capital Market Line and the Sharpe Ratio

CAPM's beta measures only one dimension of risk, co-movement with the market; the Sharpe ratio provides a complementary, portfolio-level summary of risk-adjusted performance,

$$\text{Sharpe Ratio} = \frac{\bar{r}_p - r_f}{\sigma_p}, \tag{12.4}$$

the excess return earned per unit of total risk taken. Comparing the equal-weighted and minimum-variance two-asset portfolios from Section 12.2.1, using their sample mean returns of 0.88% and 0.92% respectively and treating the daily risk-free rate as approximately zero for this short-horizon illustration,

$$\text{Sharpe}_{\text{equal}} = \frac{0.88\%}{0.583\%} \approx 1.51, \qquad \text{Sharpe}_{\text{MV}} = \frac{0.92\%}{0.322\%} \approx 2.85.$$

The minimum-variance portfolio has lower risk by construction, and here also a substantially higher Sharpe ratio, because it has a slightly higher mean return in this sample. That is not a general guarantee (the minimum-variance portfolio does not target return at all), but lower risk and competitive returns are not mutually exclusive.

A caution about units applies to every Sharpe ratio computed from the toy price series in this chapter. These are *daily* ratios, each estimated from four returns. Annualizing them by $\sqrt{252}$ would give about 24 and 45, numbers no real portfolio has ever sustained, not because these portfolios are extraordinary but because four observations cannot pin a Sharpe ratio down. For T independent returns, the standard error of an estimated Sharpe ratio is approximately $\sqrt{(1 + \text{SR}^2/2)/T}$ (Lo 2002), which here is 0.73 for the equal-weighted portfolio and 1.12 for the minimum-variance one, about as large as the estimates themselves. The calculation illustrates the formula; it is no evidence about either portfolio. Chapter 13's section on the cost of searching works out how much history a Sharpe ratio needs.

Mixing the risk-free asset with the tangency portfolio (the portfolio of risky assets with the highest Sharpe ratio) produces the Capital Market Line, illustrated in Figure 12.3: the set of risk-return combinations achievable from those two holdings.

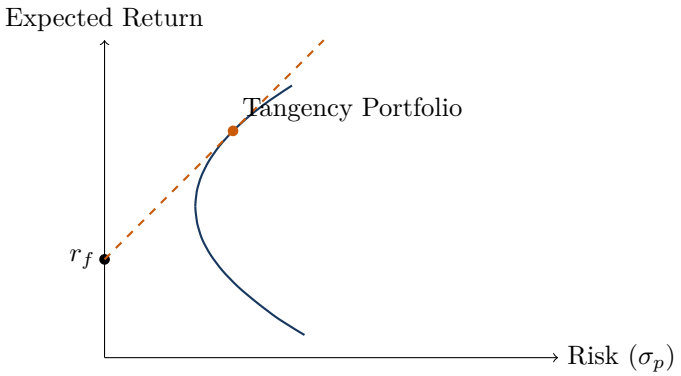


Figure 12.3: The Capital Market Line (dashed) is tangent to the efficient frontier of risky assets (solid) at the tangency portfolio

A central result of modern portfolio theory (the two-fund separation theorem) is that every investor, regardless of risk tolerance, should hold some combination of only the risk-free asset and this single tangency portfolio, differing in how much of each they hold, not in the composition of the risky portion. A more risk-averse investor holds more of the risk-free asset, while a less risk-averse investor may borrow at the risk-free rate to hold more than 100% of their wealth in the tangency portfolio, moving along the Capital Market Line without changing which risky assets they hold or in what proportion.

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12.3.3 Worked Example 12.1: Computing the Tangency Portfolio

The tangency portfolio of Figure 12.3 has a closed-form solution analogous to the minimum-variance formula, maximizing the Sharpe ratio rather than minimizing variance,

$$w^{\tan} = \frac{\Sigma^{-1}(\boldsymbol{\mu} - r_f \mathbf{1})}{\mathbf{1}^\top \Sigma^{-1}(\boldsymbol{\mu} - r_f \mathbf{1})}, \tag{12.5}$$

Applying this to AssetA and AssetB, with mean daily returns $\mu_A = 0.76\%$, $\mu_B = 1.00\%$ and, as in Section 12.3.2, a risk-free rate of approximately zero, gives

$$w_A^{\tan} = 33.76\%, \qquad w_B^{\tan} = 66.24\%,$$

strikingly close to the minimum-variance weights of 34.05% and 65.95% computed in Section 12.2.1, with a Sharpe ratio of 2.849, barely above the minimum-variance portfolio's 2.848.

This near-coincidence is specific to this dataset. With $r_f \approx 0$ and both mean returns small relative to their risk, the excess-return vector $\boldsymbol{\mu} - r_f \mathbf{1}$ that drives the tangency formula is nearly proportional to a constant vector, so $w^{\tan}$ collapses close to the minimum-variance weights (driven purely by $\Sigma^{-1}\mathbf{1}$). Applying the same formula to the equity/bond assumptions of the robo-advisor discussion later in this chapter instead ($\sigma_e = 16\%$, $\sigma_b = 5\%$, $\rho_{eb} = -0.20$, $r_e = 8\%$, $r_b = 3\%$), with a more economically meaningful risk-free rate of $r_f = 2\%$, gives a tangency portfolio of

$$w_e^{\tan} = 32.0\% \text{ equities}, \qquad w_b^{\tan} = 68.0\% \text{ bonds}, \qquad \text{Sharpe} = 0.468,$$

meaningfully different from that section's minimum-variance weights of 13.1% equities and 86.9% bonds (Sharpe ratio 0.374 at the same risk-free rate). The two portfolios coincide only when expected returns do little to distinguish the assets relative to their risk. Whenever expected returns differ meaningfully, as for equities and bonds, the two can diverge substantially, and only the tangency portfolio is the one the two-fund separation theorem recommends.

12.3.4 Multi-Factor Models: Beyond a Single Market Beta

CAPM's single market factor leaves systematic patterns in average returns unexplained: smaller companies and companies with high book-to-market ratios have historically earned higher average returns than their betas predict. Fama and French's three-factor model extends CAPM with two additional factors, SMB (Small Minus Big, the return of small-capitalization stocks minus large-capitalization stocks) and HML (High Minus Low, the return of high book-to-market "value" stocks minus low book-to-market "growth" stocks), on top of the original market factor:

$$r_{i,t} - r_f = \alpha_i + \beta_{i,\text{MKT}}(r_{m,t} - r_f) + \beta_{i,\text{SMB}}\,\text{SMB}_t + \beta_{i,\text{HML}}\,\text{HML}_t + \varepsilon_t. \tag{12.6}$$

Estimating a two-factor version (market and SMB only) requires the matrix form of OLS from Chapter 6, $\hat{\beta} = (X^\top X)^{-1} X^\top y$. Fitting this to six months of hypothetical market, SMB, and asset return data gives

$$\hat{\alpha} \approx -0.0065\%, \qquad \hat{\beta}_{\text{MKT}} \approx 1.19, \qquad \hat{\beta}_{\text{SMB}} \approx 0.35, \qquad R^2 \approx 0.999;$$

the positive SMB loading indicates the asset behaves somewhat like a small-cap stock, earning a premium beyond what its market beta predicts.

The R^2 of 0.999 should not be read as an achievement. It is an artifact of how the six months were built: the asset's returns were generated from the two factors plus noise of only 0.067% a month, about one thirty-second of the asset's own 2.17% monthly volatility, and the regression has only $6 - 3 = 3$ residual degrees of freedom with which to discover otherwise. The same construction is behind the CAPM regression's R^2 of 0.992 earlier in this chapter. A real stock's idiosyncratic volatility is typically several percent a month, so a market-and-size regression on a single stock usually explains well under half of its variance, and a fit as tight as this one on real data would itself be a reason to look for an error. The estimated alpha is equally uninformative: its standard error, 0.039%, is six times its size ($t = -0.17$).

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Each additional factor is another explanatory variable in a multiple regression, so the overfitting cautions of Chapter 6 apply: adding enough factors always improves in-sample R^2. Serious factor research therefore relies on economically motivated factors validated out of sample, not an undisciplined search over every conceivable candidate, a practice derided in the literature as data mining for a "factor zoo."

Multi-factor models have moved into everyday investment products through "smart beta" or factor-based exchange-traded funds, which build portfolios with deliberately high exposure to a factor such as value, size, momentum, or low volatility, rather than weighting holdings by market capitalization. If a factor has historically earned a premium, an investor may prefer to capture it systematically and cheaply through a rules-based fund rather than paying an active manager for discretionary stock selection. The overfitting caution applies with particular force here. A factor premium discovered by statistical search can weaken or disappear once it becomes widely known and traded, as capital chasing the same exposure competes away the mispricing it compensated for. This is Chapter 3's market-efficiency argument: an exploitable pattern tends not to remain exploitable once it is traded at scale.

12.3.5 Inverse-Volatility Weighting and Risk Parity

Section 12.2.3 showed that full mean-variance optimization is dangerously sensitive to estimation error in the covariance matrix, particularly the correlation terms, which are the hardest to estimate. The simplest risk-based allocation, inverse-volatility weighting (sometimes loosely called naive risk parity), sidesteps much of this fragility by weighting inversely to volatility alone, ignoring correlation,

$$w_i^{\mathrm{IV}} = \frac{1/\sigma_i}{\sum_j 1/\sigma_j}.$$

Applied to AssetA and AssetB,

$$w_A^{\mathrm{IV}} = 34.8\%, \qquad w_B^{\mathrm{IV}} = 65.2\%, \qquad \text{Sharpe} \approx 2.84,$$

remarkably close to the minimum-variance weights of 34.1% and 65.9% (Sharpe ratio 2.85) computed in Section 12.2.1, despite using no correlation estimate. That closeness is a feature of this dataset. For two assets, the minimum-variance weights coincide exactly with the inverse-volatility weights when the correlation is exactly -1, and AssetA and AssetB's sample correlation is -0.896, close to that limit. Holding the two volatilities fixed and raising the correlation shows how quickly the agreement breaks down: the minimum-variance weight on AssetA falls to 30.35% at $\rho = -0.5$, 22.18% at $\rho = 0$, and 2.41% at $\rho = 0.5$, while the inverse-volatility weight stays at 34.81% throughout. The case for inverse-volatility weighting is therefore robustness, not approximation: it requires estimating only n volatilities rather than $n(n-1)/2$ correlation pairs, each a source of the estimation error that made the three-asset optimization so unreliable. Risk parity proper, as practiced in industry, is usually the *equal risk contribution* (ERC) portfolio (Maillard et al. 2010), and unlike inverse-volatility weighting it does use correlations. Portfolio volatility splits exactly into one contribution per asset,

$$\sigma_p = \sum_i \mathrm{RC}_i, \qquad \mathrm{RC}_i = \frac{w_i(\Sigma w)_i}{\sigma_p}, \tag{12.7}$$

the same decomposition that underlies Chapter 8's Component VaR, and ERC chooses the weights that make every RC_i equal. Inverse-volatility weighting is the special case in which all pairwise correlations are equal. With only two assets the two methods give identical weights whatever the correlation, which is why AssetA and AssetB cannot tell them apart; the distinction first matters with three or more.

Worked Example 12.2: Equal Risk Contribution with Three Assets

Add a commodity sleeve to the equity/bond pair of Section 12.3.3: volatilities of 16%, 5%, and 20%, an equity-bond correlation of -0.20, equity-commodity 0.30, and bond-commodity 0.10. Inverse-volatility weighting gives 20%, 64%, and 16%, but the risk it produces is not balanced: by Equation (12.7), equities contribute 32.35% of portfolio risk, bonds 26.47%, and commodities 41.18%, because the commodity sleeve's positive correlation with equities adds risk that a volatility-only rule cannot see. (Equal weights are worse still: commodities then supply 58.77% of the risk and bonds 2.21%.)

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With three or more assets the ERC weights have no closed form and must be found numerically. A convenient route is that the ERC portfolio is the rescaled minimizer of the convex function $\frac{1}{2}\, y^\top \Sigma y - \frac{1}{n}\sum_i \ln y_i$ over positive y, whose first-order condition, $y_i(\Sigma y)_i = 1/n$ for every i, is the equal-contribution condition itself. Solving it gives

$$w_e^{\mathrm{ERC}} = 19.56\%, \qquad w_b^{\mathrm{ERC}} = 67.40\%, \qquad w_c^{\mathrm{ERC}} = 13.04\%,$$

for a portfolio volatility of 5.51%, against 5.90% for inverse volatility. Verifying the answer means applying the definition: at these weights, $w_i(\Sigma w)_i/\sigma_p^2$ returns 33.33% three times, and an independent solver that directly minimizes the squared differences between the contributions agrees to within 10^{-8}. ERC moved weight out of commodities and into bonds, the asset whose negative correlation with equities makes its marginal contribution to risk the smallest. For comparison, the minimum-variance portfolio of the same three assets holds 13.29%, 87.18%, and −0.46% for a volatility of 4.43%, with bonds supplying 87.18% of the risk: lower total risk, bought by concentrating it in one asset. ERC sits between the two, its usual position (Maillard et al. 2010): better balanced than inverse volatility, far less concentrated than minimum variance, and, like both, free of any expected-return estimate.

Accepting a method that is provably suboptimal under the classical assumptions in exchange for robustness when those assumptions fail is a recurring theme in practical portfolio management.

12.3.6 Black-Litterman: Blending Views with Market Equilibrium

Suppose you manage the 60/40 portfolio of Section 12.7.1 and you come to work with exactly one opinion: equities look cheap relative to bonds. This is the ordinary situation of a working investor, and the optimizer of Section 12.2.3 cannot accept it. That machinery demands a complete vector of expected returns, so before it will run you must also hand it an expected return for bonds, about which you have no opinion. Whatever number you invent, the optimizer treats it as a firmly held belief and acts on it as decisively as on the view you hold.

The second problem is that the formula has nowhere to record how sure you are. An opinion you would stake your reputation on and one you formed over coffee produce identical portfolios, though the difference between them ought to determine how far the portfolio moves from where it started.

Both problems land on the input that can least afford them. Chopra and Ziemba showed that, for a typical investor's risk tolerance, errors in estimated *means* are roughly an order of magnitude more costly, in utility lost, than errors of the same relative size in variances, which are in turn about twice as costly as errors in covariances. The expected-return vector is the most fragile input in the apparatus, and the naïve procedure requires you to fabricate most of it in order to express one real view. The same ordering is why the global minimum-variance portfolio of Section 12.2.2 is more robust than its extreme position on the frontier suggests: needing no expected returns, it is immune to the largest error source.

The Black-Litterman model answers both objections without touching the covariance matrix. It supplies a default answer to "what do I think about bonds?", so that holding no opinion becomes a position the mathematics can represent, and it lets every opinion carry an explicit statement of confidence.

The default comes from the market. Other investors' collective holdings are observable, and if they are broadly sensible they encode a set of expected returns: whatever numbers would have made that portfolio optimal to hold. Recovering them means running Section 12.2.3's machinery backward, a step called reverse optimization, which deduces the excess returns $\mathbf{\Pi}$ that make the observed market weights w_{mkt} optimal,

$$\mathbf{\Pi} = \delta\, \Sigma\, w_{\mathrm{mkt}}, \tag{12.8}$$

where δ is the market's coefficient of risk aversion. This answers the bond question with the number implied by what everyone else is already doing.

Your own opinions enter through three objects. A matrix P records which assets each view is about, so that "equities relative to bonds" can be expressed without saying anything about either alone. A vector Q gives the magnitudes claimed. And a matrix Ω states, on its diagonal, how uncertain each view is, where "I would stake my reputation on this" and "this is a hunch" become different inputs. The blend of prior and views is then

$$\mathbb{E}[R] = \left[(\tau\Sigma)^{-1} + P^\top \Omega^{-1} P\right]^{-1} \left[(\tau\Sigma)^{-1}\mathbf{\Pi} + P^\top \Omega^{-1} Q\right], \tag{12.9}$$

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the scalar τ setting how tightly the equilibrium prior is held. This is Chapter 1's Bayesian update in matrix form, with $\mathbf{\Pi}$ as the prior, Q as the evidence, and Ω^{-1} as the weight that evidence earns. A confident view (small Ω) pulls the answer toward Q; a tentative one leaves it near $\mathbf{\Pi}$.

Two consequences answer the opening problems. An investor with no views gets the market weights back unchanged, so "I have no opinion" produces a sensible portfolio rather than an arbitrary one. And an investor with a view, such as "equities will outperform bonds by 9% over the coming year," gets a portfolio tilted away from the market by an amount that depends on how firmly the view was asserted, rather than the lurch a historical-return optimizer would deliver. Black-Litterman thus combines a systematic baseline with analyst judgment that a purely statistical model like those in Chapter 6 cannot generate. The worked example that follows shows both consequences as numbers.

12.3.7 Worked Example 12.3: Blending a View into a 60/40 Portfolio

Take the equity and bond assumptions Section 12.3.3 already used ($\sigma_e = 16\%$, $\sigma_b = 5\%$, $\rho_{eb} = -0.20$, $r_f = 2\%$) and treat Section 12.7.1's classic 60/40 split as the market portfolio. That portfolio's volatility is 9.41% and, under the same 8% and 3% expected returns, its excess return is 4.00%, which pins down the risk-aversion coefficient as $\delta = 0.0400/0.0941^2 = 4.52$. Reverse optimization then implies equilibrium excess returns of

$$\Pi_e = 6.6546\%, \qquad \Pi_b = 0.0181\%.$$

Two features of this prior deserve attention. These are not the 8% and 3% assumed a moment ago, nor were they meant to be: they are the returns that would make a 60/40 investor's weights optimal, so the implied equity-minus-bond spread, $6.6546\% - 0.0181\% = 6.6365\%$, is wider than the 5.00% those assumptions embed. Bonds earn almost no equilibrium premium, not because they are unattractive but because their low volatility and negative correlation contribute almost nothing to the 60/40 portfolio's total risk, which is what a risk-based equilibrium ought to conclude.

Now suppose the investor holds the single view quoted above, that equities will beat bonds by 9%, against the 6.6365% equilibrium already implies. Setting $P = (1, -1)$, $Q = 9\%$, $\tau = 0.05$, and $\Omega = P(\tau\Sigma)P^\top$, which calibrates the view's uncertainty to the prior's own uncertainty about that same spread, the posterior expected excess returns become 7.68% for equities and -0.14% for bonds, and the resulting optimal weights are

$$w_e = 68.4\% \text{ equities}, \qquad w_b = 31.6\% \text{ bonds}.$$

The view moved the portfolio 8.4 percentage points toward equities rather than to a corner, and the size of that move tracks the confidence attached to it: halving Ω, which asserts the view more firmly, produces 71.1/28.9, while tripling it produces only 64.2/35.8. That bounded, monotone response is the point of the construction: the same 9% opinion yields a tilt the investor can dial rather than a violent swing, and setting the view aside returns exactly 60/40, the equilibrium the model began from.

12.4 Building and Running a Portfolio

A portfolio on paper becomes a portfolio in practice only after constraints, costs, currencies, and mandates are imposed on it.

12.4.1 Practical Allocation Decisions: Constraints, Rebalancing, and Transaction Costs

The portfolios developed so far ignore several frictions that matter enormously in practice. A long-only mandate prohibits negative weights (short positions), which the unconstrained formulas of Sections 12.2.1 and 12.2.3 do not rule out, and position limits or sector caps restrict concentration. Both typically require solving the optimization as a constrained quadratic program rather than with closed-form formulas.

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A concrete example shows when the long-only constraint binds. Consider two assets with $\sigma_X = 8\%$, $\sigma_Y = 20\%$, and an unusually high correlation of $\rho_{XY} = 0.9$. Applying the closed-form minimum-variance formula from Section 12.2.1 gives

$$w_X^{\text{MV}} \approx 145.5\%, \qquad w_Y^{\text{MV}} \approx -45.5\%, \qquad \sigma_p \approx 5.26\%,$$

a substantial short position in Asset Y financing a leveraged long position in Asset X, with volatility below either asset's own. A long-only mandate rules this out. Because portfolio variance decreases monotonically in w_X across the feasible range $w_X \in [0, 1]$ toward the (infeasible) unconstrained optimum, the best long-only portfolio is the corner solution $w_X = 100\%$, $w_Y = 0\%$, with Asset X's own 8% volatility. Here the constraint does not merely reduce the benefit of diversification; it eliminates it, since every feasible long-only combination is riskier than Asset X alone. The unconstrained formulas are a starting point for intuition, not a prescription: constraints can change the character of the optimal solution, not just its weights.

Even a well-constructed portfolio drifts from its target weights as assets earn different returns, so periodic rebalancing is necessary. Every rebalancing trade incurs the transaction costs of Chapter 3, at least half the bid-ask spread plus any market impact, so rebalancing that chases small deviations can destroy more value in trading costs than it preserves in risk control.

Consider a \$1,000,000 portfolio that has drifted 5 percentage points from its target weights, requiring a \$50,000 trade in each direction (selling the overweight asset, buying the underweight one) to restore them. At a one-way transaction cost of 15 basis points, each \$50,000 trade costs $\$50{,}000 \times 0.0015 = \$75$, for a total round-trip rebalancing cost of \$150, or 0.015% of the portfolio's value. That looks negligible once, but a manager who rebalances daily compounds it across hundreds of trading days per year. Practitioners therefore rebalance on a fixed calendar schedule, quarterly for example, or only when a weight drifts beyond a tolerance band, trading off precise tracking against cumulative cost drag.

Taxable investors face a further consideration: selling an appreciated position to rebalance can trigger capital gains tax, so a tax-aware strategy may tolerate more drift than risk minimization alone would recommend, in exchange for deferring the tax.

12.4.2 Solving the Constrained Problem Numerically

Once inequality constraints such as a long-only mandate or a position cap are added, the closed-form formulas of Sections 12.2.1 and 12.2.3 no longer apply, since setting a gradient to zero characterizes only an unconstrained (or purely equality-constrained) optimum. The general constrained minimum-variance problem is

$$\min_{\mathbf{w}} \ \mathbf{w}^\top \Sigma \, \mathbf{w} \quad \text{subject to} \quad \mathbf{w}^\top \mathbf{1} = 1, \quad \mathbf{w}^\top \boldsymbol{\mu} = r_0 \text{ (optional)}, \quad w_i^{\text{low}} \le w_i \le w_i^{\text{high}} \ \ \forall i,$$

where the budget constraint ($\mathbf{w}^\top \mathbf{1} = 1$) is always present, the target-return constraint is included only when tracing a specific point on the efficient frontier, and the box constraint $w_i^{\text{low}} = 0$, $w_i^{\text{high}} = 1$ recovers a long-only mandate, a tighter w_i^{high} recovers a position or sector cap, and $w_i^{\text{low}} < 0$ permits limited short selling.

This quadratic objective with linear constraints is solved with a numerical constrained optimizer, most commonly sequential quadratic programming (SLSQP), available as `scipy.optimize.minimize(..., method="SLSQP")` in Python, which iterates toward a solution rather than computing one in a single matrix step.

Applied to the AssetX/AssetY example, minimizing variance subject to the budget constraint and long-only bounds $0 \le w_i \le 1$ converges in a handful of iterations to $w_X = 100\%$, $w_Y = 0\%$, the corner solution derived by hand, a useful check that the numerical method and the closed-form reasoning agree.

One practical wrinkle is easy to miss. A generic optimizer's default convergence tolerance is tuned for objective values of ordinary magnitude, and raw daily portfolio variances, on the order of 0.0001 or smaller, are small enough that a naive call can terminate after a single iteration at the starting point, silently reporting success despite having barely moved. Rescaling the objective (for instance, working in squared-percent units) or tightening the tolerance avoids this; the companion notebook shows the failure and the fix side by side.

This machinery also revisits Section 12.2.3's three-asset minimum-variance portfolio (weights of 44.8%, 14.2%, and 41.0% in AssetA, AssetB, and AssetC, with a volatility of about 0.050%) under a position cap, a common institutional constraint the closed-form formula cannot express. Capping every weight at 40% forces

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AssetA's allocation down from its unconstrained 44.8% to the 40% cap, with the freed-up weight redistributed to the other two assets:

$$w_A^{\text{cap}} = 40.0\%, \qquad w_B^{\text{cap}} = 28.4\%, \qquad w_C^{\text{cap}} = 31.6\%, \qquad \sigma_p^{\text{cap}} \approx 0.113\%,$$

more than double the uncapped portfolio's 0.050%, the price of the diversification the cap forces.

Tracing a constrained efficient frontier under the same 40% cap adds the target-return constraint: solving for the minimum-variance portfolio achieving a 0.75% expected return, for instance, gives

$$w_A = 40.0\%, \qquad w_B = 35.1\%, \qquad w_C = 24.9\%, \qquad \sigma_p \approx 0.136\%,$$

one point on a constrained frontier that can be traced fully by sweeping the target return r_0 across a grid, as the companion notebook does.

12.4.3 Tracking Error and the Information Ratio

Many portfolio managers are evaluated relative to a benchmark, and the Sharpe ratio has a benchmark-relative analogue. The active return in any period is the portfolio's return minus the benchmark's return, $r_t^{\text{active}} = r_t^p - r_t^b$, and tracking error is the standard deviation of this active return series,

$$\text{Tracking Error} = \text{std}(r^{\text{active}}).$$

Using Chapter 2's equal-weighted AssetA/AssetB portfolio returns as the "portfolio" and AssetB's own returns as a hypothetical benchmark, the active returns across the four days are

$$r_1^{\text{active}} = 0.50\%, \qquad r_2^{\text{active}} = -1.24\%, \qquad r_3^{\text{active}} = 1.74\%, \qquad r_4^{\text{active}} = -1.48\%,$$

giving a tracking error of 1.52% and a mean active return of −0.12%. The information ratio, the active-return analogue of the Sharpe ratio, divides the two,

$$\text{Information Ratio} = \frac{\text{mean active return}}{\text{Tracking Error}} \approx \frac{-0.12\%}{1.52\%} \approx -0.08,$$

a small negative value indicating that, over this short illustrative sample, the portfolio slightly underperformed its benchmark per unit of active risk. A persistently negative information ratio suggests a manager is not adding value relative to holding the benchmark, net of fees. A positive and statistically significant one (checked with the t-statistic machinery of Chapter 6) is one standard piece of evidence that a manager's active decisions add value rather than reflect noise.

12.4.4 International Diversification and Currency Risk

Every example so far has assumed a single currency, but a portfolio that holds foreign assets adds currency risk: exchange-rate movements change the domestic-currency value of a foreign holding independently of how that asset performs in its local market. A US investor holding a European stock earns the stock's local-currency return combined with the euro's movement against the dollar, and the two can offset or reinforce each other.

International diversification is diversification applied across a wider opportunity set. An investor holding only domestic equities is implicitly betting that domestic conditions are the only risk worth diversifying, when returns across countries are far from perfectly correlated, since economies differ in industry mix, monetary policy, and business cycle. It is the mechanism of Section 12.2.1's two-asset example applied to national markets, and it is why most institutional portfolios hold meaningful international allocations despite the added currency risk.

A small numerical example shows how large the currency component can be. Suppose a US investor holds a European stock that returns 5% in euro terms over the period, while the euro simultaneously depreciates 3% against the US dollar. The unhedged US-dollar return combines both effects multiplicatively,

$$r_{\text{USD}}^{\text{unhedged}} = (1 + r_{\text{local}})(1 + r_{\text{FX}}) - 1 = (1.05)(0.97) - 1 \approx 1.85\%,$$

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more than three percentage points below the stock's local-currency performance, purely because of currency movement. A fully hedged position, using the forward contracts of Chapter 3 to lock in today's exchange rate for the future conversion, would earn approximately the full 5% (less the hedge's cost), separating the stock-picking decision from the currency decision. Whether to hedge is itself a portfolio decision. Hedged exposure keeps the diversification benefit of foreign assets without the currency bet, at the hedge's cost; unhedged exposure retains the currency bet as an additional, sometimes uncompensated, source of risk and occasionally return. Institutional investors reach different conclusions about which suits their objectives.

12.5 Machine Learning in Portfolio Construction

The classical methods in this chapter estimate their inputs (expected returns, variances, and covariances) with simple sample statistics, but each input is a prediction problem that machine learning can potentially improve. Regularized regression (ridge or LASSO) can estimate expected returns or factor exposures more stably than raw historical averages, shrinking noisy coefficients toward zero much as Section 12.2.4 shrank the covariance matrix.

12.5.1 Can Machine Learning Forecast Expected Returns?

Expected returns are the input whose estimation errors cost the most, so can the methods of Chapter 6 predict them better than a historical average? Few questions in empirical finance have been tested as thoroughly. Gu, Kelly, and Xiu ran the full comparison, putting ordinary least squares, penalized linear models, random forests, gradient-boosted trees, and neural networks on the same footing across roughly thirty thousand US stocks and ninety-four firm characteristics over six decades. Neural networks and tree ensembles won clearly, for the same reason they won in Section 12.5.2's much smaller example: they capture interactions and nonlinearities a linear specification cannot represent, and the linear benchmarks were close to useless out of sample.

The headline result is a monthly out-of-sample R^2 of roughly 0.40%, which a reader arriving from Chapter 6 will read as a catastrophic failure. It is not, and understanding why is the most transferable lesson in this section. Monthly equity volatility runs around 5% against an expected return near 0.7%, so noise dominates signal by construction, and an R^2 near half a percent is close to what is attainable. Campbell and Thompson made the argument precise: the right yardstick for a return forecast is not the fraction of variance it explains but the squared Sharpe ratio of the portfolio it funds, and by that standard a monthly R^2 of half a percent is economically substantial. Gu, Kelly, and Xiu's long-short portfolios formed on the neural-network forecasts duly earned Sharpe ratios far above what their linear competitors managed.

Three features keep the problem hard, none familiar from a standard machine learning course. The low signal-to-noise ratio is a property of the phenomenon, not the estimator. The target is adaptive and, in effect, adversarial: McLean and Pontiff found that documented anomaly returns shrink by roughly a quarter between a study's sample period and the period immediately after it, and by well over half once the result is published, so a pattern degrades because someone found it. And the data are scarcer than they look: sixty years of monthly observations amount to only a few hundred points per series, a thin diet for a method designed for millions of images.

Set beside Section 12.3.6's ordering, this yields the uncomfortable summary of the chapter's estimation problem: predictability runs opposite to importance. Variances and covariances, whose errors cost the least, are the most forecastable, which is why the volatility models of Chapter 7 work as well as they do. The level of expected returns, whose errors cost the most, is the least forecastable. In between sits cross-sectional prediction, deciding which assets will outperform which, which is materially more tractable than predicting the market's direction and is where nearly all successful applied work concentrates.

This also settles where an ML forecast should enter a portfolio process. Feeding it directly into the optimizer as $\boldsymbol{\mu}$ discards what makes it usable: the model reports not only a prediction but the accuracy of its past predictions. A forecast together with an out-of-sample error is the pair Black-Litterman asks for, a view Q and its uncertainty Ω, and routing it that way makes the tilt proportional to how much the model has earned. A predictor with a 0.40% R^2 receives a correspondingly large Ω and moves the portfolio only slightly, the correct response to a weak but real signal.

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12.5.2 A Head-to-Head Test: Ridge Regression versus OLS on Collinear Factors

Section 12.3.4's two-factor example, with six months of data and two distinct factors, is too clean to show an estimation-error problem. Real candidate factors are often correlated (momentum and a "quality" factor both partly reflect recent price strength, say), and a short history makes that collinearity far more damaging. The companion notebook extends the factor regression to twelve months and five candidate factors (market, SMB, HML, momentum, and quality), with quality *constructed* as a near-linear combination of HML and momentum (0.61 and 0.89 correlation respectively) and with known true exposures $\beta = (1.10, 0.30, 0.20, -0.10, 0.15)$, so that each estimator's error can be measured against ground truth.

Ordinary least squares, fit on all twelve months, badly overfits this short, collinear history:

$$\hat{\beta}_{\text{OLS}} = (1.078,\ -0.467,\ 0.979,\ 1.435,\ -2.322), \qquad \text{RMSE} = 1.390,$$

getting the market exposure roughly right but the SMB sign backward, wildly overstating HML and momentum, and estimating a quality exposure of -2.322 against a true value of $+0.15$. A ridge regression on the same data, with a small but nonzero penalty, recovers

$$\hat{\beta}_{\text{ridge}} = (0.957,\ 0.124,\ 0.027,\ -0.011,\ -0.075), \qquad \text{RMSE} = 0.167,$$

still imperfect, but with every coefficient pulled back to a plausible magnitude and no wild sign reversal, an 88% RMSE reduction against the five true exposures. This mirrors the covariance-shrinkage story of Section 12.2.4: the unregularized estimator is not merely less precise but misleading, fitting the collinear noise of one short sample, while a modest penalty trades a little bias for a large reduction in estimation error.

12.5.3 Beyond Return Forecasting: Hierarchical Risk Parity, End-to-End Learning, and Reinforcement Learning

Clustering methods, such as the K-means algorithm Chapter 6 applied to group firms by leverage and profit margin, can instead group assets by return co-movement before optimizing, an approach known as hierarchical risk parity. Rather than inverting a single, potentially ill-conditioned covariance matrix across all assets (the source of the fragility in Section 12.2.3), the portfolio is built recursively, allocating risk first across clusters of similar assets and then within each cluster. This tends to produce more stable, diversified allocations than full-matrix mean-variance optimization when the number of assets is large relative to the return history, a claim Section 12.5.4 tests out of sample.

More recent research trains tree ensembles and neural networks to output portfolio weights that maximize a risk-adjusted objective such as the Sharpe ratio directly, rather than first predicting returns and then optimizing. This end-to-end approach can avoid compounding prediction error with optimization error, but it inherits every overfitting and interpretability concern Chapter 6 raised, in a setting where an opaque model directly allocates capital, the model risk management concerns of Chapter 8. As elsewhere in this book, the message is not that these methods should be avoided but that they must be validated, monitored, and understood by the people responsible for the capital, not deployed because they are available.

A more experimental line of research frames allocation and rebalancing as a sequential decision problem and applies reinforcement learning: an algorithm learns an allocation policy through simulated trial and error against a reward such as risk-adjusted return net of transaction costs. This framing suits the rebalancing trade-offs discussed earlier, since an agent can learn to weigh close tracking of target weights against the cost of trading too often within a single objective. The caution applies with even greater force: an agent trained in a simulated market inherits every assumption of that simulation, and a policy that performs well there can fail unpredictably in real markets whose dynamics differ, an acute version of the non-stationarity concern raised throughout this book.

12.5.4 Worked Example 12.4: Hierarchical Risk Parity Out of Sample

Hierarchical risk parity (HRP) was introduced by López de Prado (2016), and since its justification is robustness, the test that matters is out of sample. The algorithm has three steps. First, it converts the estimated correlation matrix into a distance, $d_{ij} = \sqrt{(1 - \rho_{ij})/2}$, which is 0 for perfectly correlated assets and 1 for perfectly anti-correlated ones, and clusters the assets hierarchically on that distance, merging the two

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closest assets or groups first, then the next closest, and so on (single linkage, in the language of Chapter 6's clustering material). Second, it reorders the assets in the order the resulting tree lists them, so that similar assets sit next to one another and the reordered covariance matrix is close to block-diagonal. Third, it allocates by *recursive bisection*: split the ordered list in half, give each half a variance V computed with inverse-variance weights inside it, divide the capital between the halves in the proportions $\alpha = 1 - V_{\text{left}}/(V_{\text{left}} + V_{\text{right}})$ and $1 - \alpha$, and repeat inside each half until single assets remain. No matrix is ever inverted, which is the point.

The test universe is simulated, so that the true covariance matrix is known and every portfolio's out-of-sample risk can be measured exactly: twenty assets in four clusters of five, labeled A to D, with annual volatilities between 10% and 35%, correlations of 0.4 to 0.7 within a cluster, and 0.05 to 0.30 across clusters. On one 60-day estimation window, the clustering recovers the four groups exactly and lists cluster B, then A, then C, then D. The first bisection therefore separates A and B from C and D, with half-variances of 4.72×10^{-5} and 4.21×10^{-5} (daily units), which sends $\alpha = 47.1\%$ of the capital to the A and B half; the next two splits give 62.4% of that half to B and only 18.8% of the other half to C, the most volatile cluster. The final HRP weights run from 0.84% to 12.88%, all long. The sample minimum-variance portfolio built from the same 60 days puts 37.00% into a single asset and holds seven short positions totaling −62.55%. Measured against the true covariance matrix, its volatility is 9.34%, against 8.35% for HRP.

One window is an anecdote, so the companion notebook repeats the exercise a thousand times, each time estimating on 60 days and then holding the portfolio for the next 250. Table 12.3 reports the results.

Portfolio	Mean realized vol.	Std. dev.	95th pct. of true vol.
Equal weight	10.64%	0.46%	10.66%
Inverse variance	8.35%	0.39%	8.65%
Sample minimum variance	8.21%	0.64%	9.18%
Hierarchical risk parity	8.10%	0.42%	8.52%
Minimum variance, true covariance known		6.74%	

Table 12.3: Out-of-sample annualized volatility of portfolios estimated from 60 days of data and held for 250 days, over 1,000 simulated trials with twenty assets in four clusters. The standard deviation is that of realized volatility across trials; the last column is computed from the true covariance matrix.

HRP has the lowest average out-of-sample volatility, but only narrowly, 8.10% against 8.21% for the sample minimum-variance portfolio, and it beats that portfolio in only 54.8% of the individual trials. Its advantage is in the tail. The minimum-variance portfolio's risk varies across trials about half again as much as HRP's (0.64% against 0.42%), and in its worst twentieth of trials its true volatility exceeds 9.18%, against 8.52% for HRP: an optimizer that inverts a noisy matrix is sometimes excellent and sometimes badly wrong, while HRP is consistently decent. All four methods fall well short of the 6.74% available to an investor who knew the true covariance matrix, a direct measure of what estimating twenty volatilities and 190 correlations from 60 days costs.

The ranking is not fixed, either; it depends on how much data the optimizer is given. Averaged over 500 trials per window length, the sample minimum-variance portfolio's true volatility is 9.54% with 40 days of data, 8.23% with 60, 7.37% with 120, and 7.01% with 250, while HRP's barely moves, from 8.19% to 7.95%. HRP wins decisively when the number of assets is large relative to the history, ties at around 60 days, and loses once the history is long enough for the inverse to be trusted, because its bisection ignores the correlations across the halves it splits, information a well-estimated covariance matrix would use. That is the fair summary of HRP, and of every robust method in this chapter: it does not beat optimization, it beats optimization *on too little data*. In a real universe of hundreds of assets re-estimated every month, too little data is the ordinary situation.

12.6 Robo-Advisors: Automating Asset Allocation

A robo-advisor packages nearly every tool of this chapter (a risk-based target allocation, periodic rebalancing, tax awareness) into a single automated, low-cost retail product delivered through a web or mobile interface. Betterment and Wealthfront, both founded in 2008, pioneered the category; incumbents including Vanguard

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and Schwab entered a few years later, typically pairing the same automated core with an optional human advisor for a higher fee, a hybrid model this section returns to below.

The appeal is that the classical machinery, once encoded correctly, does not require a highly paid professional to execute, so automating it can extend a disciplined version of institutional-grade asset allocation to investors with far smaller balances than traditional advice has served.

12.6.1 From a Risk Questionnaire to a Target Allocation

A new client typically completes a short questionnaire covering age, time horizon, income stability, and stated comfort with market declines, which the platform converts into a target stock/bond split. A common industry heuristic sets the equity share at 110 minus the investor's age, so a 35-year-old investor receives a suggested allocation of 75% equities and 25% bonds.

This chapter's machinery shows what the heuristic is and is not doing. Treating "equities" and "bonds" as two broad asset classes with illustrative volatilities $\sigma_e = 16\%$, $\sigma_b = 5\%$, and correlation $\rho_{eb} = -0.20$, Section 12.2.1's closed-form minimum-variance formula gives

$$\begin{aligned} \text{Minimum-variance:} &\quad 13.1\% \text{ equities},\ 86.9\% \text{ bonds},\ \sigma_p = 4.43\%, \\ \text{Questionnaire-based 75/25:} &\quad 75\% \text{ equities},\ 25\% \text{ bonds},\ \sigma_p = 11.81\%, \end{aligned}$$

the questionnaire split roughly two and a half times riskier than the minimum-variance point.

This is not a mistake in either number; it reflects two different objectives. Using illustrative expected returns of 8% for equities and 3% for bonds, and the same 2% risk-free rate as the tangency and Black-Litterman examples,

$$\begin{aligned} \text{Minimum-variance:} &\quad \mathbb{E}[r_p] = 3.65\%,\ \text{Sharpe} = 0.374, \\ \text{Heuristic 75/25:} &\quad \mathbb{E}[r_p] = 6.75\%,\ \text{Sharpe} = 0.402, \end{aligned}$$

so the heuristic improves on the minimum-variance point's risk-adjusted return. The minimum-variance portfolio ignores expected returns and holds far less equity than the 32.0%/68.0% tangency portfolio, whose Sharpe ratio of 0.468 is the highest these two assets can deliver. The 75/25 split overshoots the tangency weights in the other direction, with more than twice the tangency equity share, giving up some Sharpe ratio for substantially higher expected growth. A young investor with a long horizon and years of contributions ahead is pointed to a point beyond the tangency portfolio on the efficient frontier of Figure 12.1, a trade such an investor is usually willing to make.

This is a practical departure from the two-fund separation theorem, under which every investor holds the same tangency portfolio, using leverage if necessary to hold more than 100% of it. A robo-advisor does not offer a young client a leveraged position in a bond-heavy tangency portfolio; it shifts the mix of risky assets toward equity, because most retail investors neither can nor will borrow at the risk-free rate to lever up a conservative portfolio. As a client approaches a stated goal, typically retirement, the target allocation is walked back toward bonds along a predetermined glide path, the same lifecycle logic behind a conventional target-date fund.

12.6.2 Rebalancing and Tax-Loss Harvesting at Scale

The \$1,000,000 portfolio, 5-percentage-point drift, 15-basis-point cost example of Section 12.4.1 is the calculation a robo-advisor's rebalancing engine performs continuously across every client account, trading only once a client's drift tolerance band is breached rather than on a calendar schedule.

Tax-loss harvesting, a hallmark of the taxable (non-retirement) accounts these platforms offer, extends the automation to the client's tax bill. When a position falls in value, the platform sells it to realize the loss and immediately buys a similar, but not "substantially identical," security, avoiding the wash-sale rule that would disallow the loss, while keeping market exposure essentially unchanged.

Suppose a client holds a \$50,000 equity-fund position that has fallen 10%, a \$5,000 realized loss. Under U.S. tax rules, up to \$3,000 of a net capital loss can offset ordinary income in the year it is realized, and any remainder offsets capital gains. At a 32% marginal income-tax rate and a 15% long-term capital-gains rate, this single harvesting event saves $\$3{,}000 \times 32\% = \$960$ against ordinary income and $\$2{,}000 \times 15\% = \$300$

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against capital gains, a first-year tax saving of $1,260 on this one position. It is generated automatically and, across hundreds of thousands of accounts, far more consistently than a human advisor reviewing holdings once or twice a year could match. Some platforms extend this to direct indexing, holding an index's individual stocks rather than a single fund, which multiplies the tax lots available for harvesting at any time.

12.6.3 Fee Structure and Long-Horizon Impact

A typical robo-advisor charges roughly 0.25% of assets annually, against roughly 1% for a traditional human advisor. Compounded over a long horizon, the gap is far larger than it appears. Starting from $100,000 and assuming a 7% gross annual return sustained for 30 years,

$$\begin{aligned} \text{0.25\% fee (robo-advisor):} &\quad \$709{,}637, \\ \text{1\% fee (traditional advisor):} &\quad \$574{,}349, \\ \text{0\% fee (hypothetical benchmark):} &\quad \$761{,}226. \end{aligned}$$

The lower-cost option preserves $135,288 more than the traditional-fee alternative over three decades, purely from a 75-basis-point difference in annual cost: fee drag through compounding, quantified with Chapter 5's time-value-of-money tools.

None of this makes a robo-advisor strictly superior to a human one. The risk questionnaire is a blunter instrument than a skilled advisor's judgment about a client's full financial picture; tax situations, estate planning, and major life events do not fit a set of multiple-choice questions. Self-reported risk tolerance is also unstable, shifting with recent market performance and question framing, so some platforms now refine a client's risk profile from ongoing behavior rather than treating the onboarding survey as permanent. This is where machine learning does distinct work in a robo-advisor. A classifier trained on a client's observed behavior, such as whether they reduced contributions or tried to sell equities in a past downturn, can produce a revealed-preference risk profile that predicts future behavior better than the questionnaire alone. An optimization layer decides, lot by lot and continuously, which tax lots to harvest and which substitute to buy, a large-scale constrained optimization no human reviewing accounts manually could replicate at that frequency. It is also why the industry has moved toward hybrid models, such as Vanguard's Personal Advisor Services and Betterment's premium tier, which layer a human advisor on the automated core for the situations the algorithm handles least well.

12.6.4 Case Vignette: Schwab Intelligent Portfolios and the Cost of Undisclosed Cash Drag

In June 2022, the U.S. Securities and Exchange Commission announced that Charles Schwab & Co. and an affiliated adviser would pay $187 million, $52 million in disgorgement and prejudgment interest and a $135 million civil penalty, to settle charges concerning its robo-advisor, Schwab Intelligent Portfolios. From March 2015 through November 2018, the SEC found, Schwab's algorithm allocated a materially large share of client portfolios to cash, averaging about 12.5% across accounts and ranging from roughly 6% for the most aggressive risk profiles to nearly 24.9% for the most conservative ones, without disclosing that Schwab's own internal analysis showed this allocation would very likely underperform a portfolio holding less cash under most market conditions. Schwab profited directly, earning an estimated $46 million from the spread between what it paid depositors on that cash and what it earned investing it, a consideration its marketing materials did not adequately disclose as shaping how the portfolios were built.

The episode illustrates a point Chapter 8's model-risk governance and Chapter 18's explainability material both made: an algorithmic allocation model is still a model, and a cash allocation is still an asset allocation decision, one this chapter's machinery would need to justify on risk-return grounds, not accept because the operator has a balance-sheet incentive to hold more of it. A robo-advisor's principal selling point is that automation removes the conflicts of interest and biases a commission-driven human advisor might introduce. That fails if the automation is built, silently, around the operator's own incentives. The governance discipline Chapters 8 and 15 developed for other financial models (independent validation, clear disclosure, and ongoing monitoring against the model's stated objective) therefore applies to a robo-advisor's allocation engine with no less force than to a bank's credit-scoring or trading model.

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12.7 Institutional Allocation: Illiquidity and ESG Mandates

Large institutions face allocation questions a retail platform does not: horizons long enough to hold illiquid assets, and mandates that impose non-financial constraints on the portfolio.

12.7.1 Case Vignette: From the 60/40 Portfolio to the Endowment Model

The history of asset allocation illustrates several of this chapter's themes. For decades, a simple 60% equities / 40% bonds allocation was the default institutional and individual portfolio, an informal application of the diversification logic of Section 12.2.1: equities and bonds have historically been imperfectly correlated (bonds have often, though not always, done relatively well in equity downturns), so a fixed mix captured much of diversification's benefit without any matrix algebra.

Beginning in the 1990s, several large university endowments, most prominently Yale's under David Swensen, moved toward what became known as the endowment model: substantial allocations to less liquid alternative assets (private equity, venture capital, hedge funds, real assets) and less in public equities and bonds. The logic applies this chapter's diversification and factor ideas: alternative assets have return drivers only partly explained by the market factor that drives public equities, so adding them can move a portfolio's position on the efficient frontier of Figure 12.1 favorably, not merely diversify within existing risk factors.

The benefit comes with a cost a mean-variance framework can understate: illiquidity. An endowment with a long horizon and predictable, modest spending can hold assets that cannot be sold for years, and be compensated with an illiquidity premium, a higher expected return for giving up the ability to trade. A pension fund or individual with nearer-term or less predictable liquidity needs faces a different trade-off and cannot copy the endowment model's allocations without its liquidity profile. That mismatch has caused real difficulty for institutions that adopted illiquid strategies without attention to their own funding needs.

12.7.2 ESG Constraints and the Cost of a Tilt

Many institutional investors now layer environmental, social, and governance (ESG) considerations onto asset allocation, either by excluding sectors or companies (negative screening) or by tilting weights toward higher-scoring companies within a sector (positive tilting), sometimes formalized as an additional factor alongside market, size, and value.

From a mean-variance perspective, any constraint or tilt not motivated by risk and expected return, including an ESG constraint, can only leave a portfolio on or below the unconstrained efficient frontier of Figure 12.1, never above it.

Whether ESG investing is nonetheless value-enhancing is disputed in the empirical literature and turns on questions this chapter's framework cannot settle. One is whether ESG characteristics proxy for otherwise-unpriced risks, in which case ESG exposure is a disguised risk factor capturable within the CAPM and multi-factor machinery. Another is whether investors will accept a lower expected return for values-alignment, a non-financial objective the mean-variance framework was never built to represent.

12.7.3 Worked Example 12.5: The Cost of an ESG Tilt

Section 12.4.2's numerical machinery makes the "on or below the efficient frontier" claim precise. Extend the three-asset AssetA/AssetB/AssetC covariance matrix with an illustrative ESG score for each asset on the 0-to-100 composite scale rating providers use: AssetA (a carbon-intensive legacy business) scores 40, AssetB (a clean-technology firm) scores 85, and AssetC (a diversified industrial) scores 60.

The unconstrained long-only minimum-variance portfolio, unchanged from Section 12.2.3, weights these 44.8%, 14.2%, and 41.0%, giving a portfolio ESG score of only $0.448(40) + 0.142(85) + 0.410(60) \approx 54.6$, dragged down by AssetA's large weight despite its low score.

Adding a minimum-portfolio-ESG-score constraint to the identical SLSQP optimizer, $\mathbf{w}^\top \mathbf{esg} \geq \mathrm{ESG}_{\min}$, alongside the same budget and long-only constraints, reallocates weight away from AssetA and toward AssetB

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as the floor rises:

$$\begin{aligned}
\text{ESG} \geq 65: &\quad w = (36.0\%,\ 48.8\%,\ 15.2\%), &\quad \sigma_p \approx 0.216\%,\\
\text{ESG} \geq 70: &\quad w = (31.8\%,\ 65.4\%,\ 2.8\%), &\quad \sigma_p \approx 0.315\%,\\
\text{ESG} \geq 75: &\quad w = (22.2\%,\ 77.8\%,\ 0.0\%), &\quad \sigma_p \approx 0.484\%,
\end{aligned}$$

volatility rising from the unconstrained 0.050% to more than four times that at the mildest floor tested, and nearly ten times at the strictest. Every additional point of required ESG score is bought with additional variance, since the optimizer is no longer free to choose the covariance-minimizing weights.

These ESG scores were assigned by hand to make the trade-off visible; a harder practical problem lies upstream, in choosing which score to use at all. Berg et al. (2022) compare ESG ratings for the same companies across six major providers and find an average pairwise correlation of only 38% to 71%, far lower than the near-perfect agreement credit rating agencies typically show for the same bond. The divergence traces mostly to disagreement about which metrics to measure and how to weight them, not to noisy measurement of an agreed concept.

A "high ESG" tilt built on one provider's scores can therefore differ meaningfully, in composition and in the volatility cost computed above, from one built on another's for the same universe: a model-risk concern inside the governance discipline of Chapters 8 and 18, applied to a data input rather than a fitted model.

12.8 Challenges in Portfolio Theory and Asset Allocation

Several challenges recur across nearly every practical application of this chapter's methods.

Estimation error in expected returns. Expected returns are far harder to estimate than volatilities or correlations, since return estimates are much noisier relative to their magnitude over any realistic sample. An optimizer fed noisy expected returns and a noisy covariance matrix compounds both errors, the problem Black-Litterman (Section 12.3.6) was designed to address.

Non-stationarity of correlations. As Chapter 8's discussion of the 2008 financial crisis noted, correlations can rise sharply in a crisis, exactly when diversification is needed most, so a portfolio built on calm-period estimates can protect far less in a crisis than the mathematics, calibrated on calm data, suggests.

Factor model instability. Factors that explain returns well in one period need not do so in the next, and multi-factor models, estimated on history, assume relationships that can and do fail as market structure and investor behavior evolve.

Crowding. When many investors independently adopt similar factor tilts or risk models, through smart-beta products, shared research, or convergent proprietary models, their positions can become highly correlated without anyone realizing it. A shock to one crowded strategy can then force mutually reinforcing selling across many ostensibly independent portfolios.

Liquidity and capacity constraints. Optimal weights may be unachievable for a large portfolio, since trading a large position in a less liquid asset moves its price, the market-impact concern of Chapter 3. A strategy that looks optimal on paper can underperform once realistic trading costs and capacity limits are included.

Behavioral and institutional deviations from the theory. Investors and institutions deviate from the optimal portfolio for reasons the classical theory does not capture: career risk (a manager who deviates from the benchmark and is wrong suffers far more than one who hugs it and is wrong), regulatory capital treatment of specific asset classes, and biases such as overconfidence or loss aversion.

Time-varying risk and return. Every formula in this chapter treats volatilities, correlations, and expected returns as constants estimated once from a sample. Chapter 7 showed that volatility clusters and evolves over time, and correlations and factor exposures likely shift too, so a framework that re-estimates its inputs infrequently can work from stale, sometimes badly wrong, assumptions.

Survivorship bias in factor research. Factor premiums are often estimated from databases that include only firms that survived to be recorded, quietly excluding bankruptcies and delistings. A strong historical premium may look much less attractive once those return-dragging failures are accounted for.

Multiple testing across the factor literature. Research has collectively tested an enormous number of candidate factors against the same limited history, so the multiple-testing concern of Chapters 6 and 7

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applies to the published literature as a whole. Some factors will appear significant by chance alone, and clearing a significance threshold in one study does not make a factor exploitable.

Model risk in portfolio construction itself. Every optimizer, shrinkage estimator, and factor model in this chapter is a model requiring the governance discipline of Chapter 8 (independent validation, ongoing monitoring, documentation of known limitations), just like a bank's VaR or credit-scoring model. A process that treats its optimizer as a black box, rather than as one input checked against simpler benchmarks and economic intuition, is exposed to undetected model risk.

Liquidity mismatches. Any allocation to illiquid assets must be sized against the investor's own liquidity needs, not only the expected diversification benefit. An institution that mismatches the two can be forced to sell its most liquid holdings at the worst time because its illiquid ones cannot be sold, a dynamic related to the funding-liquidity risk of Chapter 8.

Benchmark and objective misspecification. The tracking error and information ratio of Section 12.4.3 are only as meaningful as the benchmark. A manager evaluated against a benchmark that does not reflect the mandate can appear to add or destroy value as an artifact of the comparison, a problem with no statistical fix, since the right benchmark is a judgment about what the portfolio is trying to achieve.

None of these challenges argues against the classical or robust methods; each is a reason to apply them with statistical rigor and institutional judgment, treating an optimizer's output as an informed starting point rather than a final answer.

12.9 Key Takeaways

Portfolio theory formalizes an intuition Chapter 2 first illustrated numerically: combining imperfectly correlated assets can reduce risk without sacrificing return, and mean-variance optimization finds the best combination, given accurate inputs. The efficient frontier traces every such combination; the Capital Market Line and two-fund separation theorem show that mixing the risk-free asset with a single tangency portfolio dominates every other combination; CAPM ties expected return to an estimable market beta (which Section 12.3.1 showed matches Chapter 5's WACC assumption) and splits total risk into systematic and idiosyncratic parts; and multi-factor models extend this logic to further systematic patterns.

The central practical warning is that full mean-variance optimization is acutely sensitive to estimation error, especially with limited data, which is why covariance shrinkage, risk parity, hierarchical clustering, and Black-Litterman exist as more robust, if theoretically suboptimal, alternatives. The sharpest form of that fragility is an inversion: the inputs rank in opposite order on how much their errors cost and how well they can be forecast. Covariances are the most predictable and least damaging to get wrong, expected returns the least predictable and by an order of magnitude the most damaging, which is why so much of the practical machinery avoids estimating expected returns at all.

Real portfolio management adds constraints that can qualitatively change optimal weights, rebalancing disciplined by transaction costs, tracking error and information ratios for benchmark-relative evaluation, and currency considerations, all of which the classical mathematics sets aside but a real allocation decision cannot ignore.

Robo-advisors automate this toolkit for retail investors: a risk questionnaire maps to a point on the efficient frontier, not necessarily the minimum-variance point; rebalancing and tax-loss harvesting run continuously at scale; and machine learning learns a client's revealed risk tolerance from behavior rather than a one-time questionnaire. The Schwab Intelligent Portfolios case showed that automating an allocation decision does not exempt it from the governance and disclosure discipline required of every other financial model.

12.10 Suggested Exercises

12.1 Using the two-asset formula in Section 12.2.1, verify by hand that $w_A^{\text{MV}} = 34.05\%$ minimizes portfolio variance, and compute the resulting portfolio's expected return.

12.2 Two assets have $\sigma_X = 15\%$, $\sigma_Y = 25\%$, and $\rho_{XY} = 0.20$. Compute the minimum-variance portfolio weights and the resulting portfolio volatility.

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12.3 Explain, using Figure 12.1, why a rational investor would never hold a portfolio on the dashed (inefficient) segment of the frontier, even if they were extremely risk-averse.

12.4 Using the data in Table 12.2, verify by hand that the estimated beta is approximately 1.19, and compute the CAPM-implied cost of equity assuming $r_f = 3\%$ and a market risk premium of 7%.

12.5 A portfolio has an average daily return of 0.05% and a daily standard deviation of 1.1%. Assuming a daily risk-free rate of approximately zero, compute its Sharpe ratio, and compare it to the Sharpe ratios computed in Section 12.2.1.

12.6 Three assets have volatilities of 10%, 20%, and 30%. Compute their inverse-volatility weights.

12.7 Explain, in your own words, why Michaud describes mean-variance optimization as an "estimation-error maximizer," referencing the three-asset example in Section 12.2.3.

12.8 Using Section 12.2.4's shrinkage formula and the three-asset covariance matrix of Section 12.2.3, recompute the minimum-variance weights at a shrinkage intensity of $\delta = 0.25$ rather than the $\delta = 0.5$ used in the text, and state whether they fall between the unshrunk weights and the $\delta = 0.5$ weights. Then explain, without computing anything further, what the weights must become as $\delta \to 1$, and why.

12.9 Describe a specific scenario in which an investor using Black-Litterman with no strong views would end up holding a different portfolio than one using naïve historical-mean mean-variance optimization, and explain why.

12.10 Using Section 12.3.7's equity/bond setup, recompute the Black-Litterman weights for an investor who believes equities will beat bonds by 7% rather than 9%, holding τ and Ω unchanged. Compare the resulting tilt away from 60/40 with the text's, and explain why a view only slightly above the equilibrium spread of 6.6365% moves the portfolio so little.

12.11 A portfolio manager rebalances a two-asset portfolio back to its 60/40 target weights every time the actual weight drifts by more than 5 percentage points. Explain, referencing Chapter 3's discussion of transaction costs, why this rule might outperform both a policy of rebalancing daily and a policy of never rebalancing at all.

12.12 Using Section 12.4.2's method, recompute the three-asset minimum-variance portfolio under a 50% (rather than 40%) position cap, and state whether the cap still binds on AssetA and, if so, by how much the resulting volatility differs from both the uncapped and the 40%-capped cases.

12.13 Explain, in your own words, why a numerical optimizer can terminate after a single iteration and report "success" on a badly scaled objective function, referencing Section 12.4.2's naive-versus-rescaled comparison, and describe one way to detect this failure mode besides comparing against a known closed-form answer.

12.14 Pick two of the challenges listed in this chapter and describe a concrete scenario, not already used in the text, in which that challenge could cause a mean-variance-optimized portfolio to perform poorly out of sample.

12.15 A portfolio earns returns of 1.5%, −0.5%, 2.0%, and 0.5% over four periods, while its benchmark earns 1.0%, 0.5%, 1.0%, and 0.5% over the same periods. Compute the tracking error and information ratio, and interpret the sign of the information ratio.

12.16 A US investor holds a Japanese bond that returns 2% in yen terms while the yen appreciates 4% against the US dollar over the same period. Compute the unhedged US-dollar return, and explain why it differs from the bond's local-currency return.

12.17 Explain, referencing the hierarchical risk parity idea in this chapter, why clustering assets before optimizing might produce more stable portfolio weights than applying full mean-variance optimization to all assets simultaneously.

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12.18 Using the Security Market Line in Figure 12.2, explain what it means for an asset to plot above the line, and describe how this relates to the nonzero alpha estimated in Section 12.3.1.

12.19 Explain why a factor's historically estimated return premium might weaken after it becomes the basis of a widely traded smart-beta product, referencing the market-efficiency discussion in Chapter 3.

12.20 Using the robo-advisor discussion's method, compute the 110-minus-age heuristic equity weight for a 50-year-old investor, and, using the same illustrative $\sigma_e = 16\%$, $\sigma_b = 5\%$, $\rho_{eb} = -0.20$, $r_e = 8\%$, $r_b = 3\%$ assumptions with $r_f = 2\%$, compute that portfolio's volatility, expected return, and Sharpe ratio.

12.21 A client holds a \$30,000 position that has fallen 8%. Using the robo-advisor discussion's tax-loss-harvesting method and a 24% marginal income-tax rate and 15% long-term capital-gains rate, compute the first-year tax savings from harvesting this loss.

12.22 Using the robo-advisor discussion's fee-drag example, recompute the 30-year future value of a \$100,000 initial investment at a 6% (rather than 7%) gross annual return for both the 0.25% and 1% fee cases, and state whether the dollar gap between them grows or shrinks relative to the text's example.

12.23 Explain, referencing Section 12.3.2's two-fund separation theorem, why a robo-advisor's practice of shifting a client's equity/bond mix based on risk tolerance is a departure from what that theorem, taken literally, would recommend.

12.24 Two assets have $\sigma_X = 12\%$, $\sigma_Y = 18\%$, $\rho_{XY} = 0.10$, $\mu_X = 9\%$, $\mu_Y = 6\%$, and $r_f = 1\%$. Using Section 12.3.3's formula, compute the tangency portfolio weights, and explain whether you would expect them to be closer to or further from the minimum-variance weights than the AssetA/AssetB example in the text, before actually computing the minimum-variance weights to check.

12.25 Explain, in your own words, why the tangency portfolio and the minimum-variance portfolio coincided so closely for AssetA and AssetB but not for the equity/bond example in Section 12.3.3, referencing the role r_f and the spread in expected returns each play in the tangency formula.

12.26 Using the three-asset equity/bond/commodity example of Section 12.3.5, verify with Equation (12.7) that the weights 19.56%, 67.40%, and 13.04% give each asset one third of the portfolio's risk. Then compute the risk shares of the equal-weighted portfolio (1/3 in each asset) and explain which asset dominates its risk, and why.

12.27 Section 12.5.4 found that hierarchical risk parity beats the sample minimum-variance portfolio with 40 days of data but loses to it with 250. Explain why the ranking reverses, and say which method you would expect to do better for a 500-stock universe re-estimated each month from one year (252 days) of daily returns, and why.

12.28 A strategy has a true daily Sharpe ratio of 0.10. Using the standard-error formula quoted in Section 12.3.2, how many daily observations are needed for the standard error of its estimated Sharpe ratio to fall to 0.05, half the quantity being estimated? Express the answer in years of 252 trading days, and compare it with the four observations behind this chapter's toy Sharpe ratios.

12.29 **Challenge (real data).** Download at least five years of monthly returns for five to ten real stocks (via `yfinance`) together with the Fama-French three-factor data for the same period (freely available from the Ken French Data Library). (a) Estimate each stock's CAPM beta by regressing its excess returns on the market factor, as in Section 12.3.1, and compare the range of estimated betas to the single illustrative beta of 1.19 used in this chapter. (b) Construct the minimum-variance and tangency portfolios from your assets' sample covariance matrix, as in Sections 12.2.1 and 12.3.3, and comment on whether the resulting weights look reasonable or exhibit the extreme, concentrated positions Michaud's "estimation-error maximizer" critique (Section 12.2.3) warns about. (c) Regress each stock's excess return on all three Fama-French factors (market, size, value) rather than the market alone, and discuss whether the market-only CAPM alpha from part (a) shrinks once size and value exposures are accounted for.

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12.11 Further Reading

- Markowitz (1952), "Portfolio Selection." The original paper introducing mean-variance optimization, foundational to nearly every method in this chapter.
- Sharpe (1964), "Capital Asset Prices: A Theory of Market Equilibrium under Conditions of Risk." The original derivation of the Capital Asset Pricing Model used throughout Section 12.3.1.
- Fama and French (1993), "Common Risk Factors in the Returns on Stocks and Bonds." The original paper introducing the SMB and HML factors used in this chapter's multi-factor model example.
- Black and Litterman (1992), "Global Portfolio Optimization." The original treatment of the Bayesian view-blending approach summarized in Section 12.3.6.
- Grinold and Kahn (2000), *Active Portfolio Management*. A practitioner-oriented treatment connecting mean-variance theory to real active-management decisions, including risk parity and other robust allocation methods introduced in this chapter.
- Bodie et al. (2024), *Investments*. A comprehensive textbook treatment of portfolio theory, CAPM, and factor models, extending this chapter's introductory coverage of each topic.
- Ang (2014), *Asset Management*. A modern, practitioner-oriented synthesis of factor models, smart beta, and the practical allocation considerations, including illiquidity and rebalancing, discussed later in this chapter.
- Michaud and Michaud (2008), *Efficient Asset Management*. The original source for the "estimation-error maximizer" critique of mean-variance optimization discussed in Section 12.2.3, along with resampling-based methods for producing more robust portfolios.
- Gu et al. (2020), "Empirical Asset Pricing via Machine Learning." The comprehensive machine learning bake-off behind Section 12.5.1, including the neural-network and tree results and the long-short portfolios built on them.
- Campbell and Thompson (2008), "Predicting Excess Stock Returns Out of Sample: Can Anything Beat the Historical Average?" The argument that a return forecast should be judged against the Sharpe ratio it produces rather than the variance it explains, which is what makes Section 12.5.1's tiny R^2 figures economically meaningful.
- McLean and Pontiff (2016), "Does Academic Research Destroy Stock Return Predictability?" The source of Section 12.5.1's post-publication decay estimates, and the clearest evidence that a return-forecasting target adapts to being modeled.
- Chopra and Ziemba (1993), "The Effect of Errors in Means, Variances, and Covariances on Optimal Portfolio Choice." The source of Section 12.3.6's ordering of the three input-estimation errors by how much utility each destroys.
- Ledoit and Wolf (2004), "Honey, I Shrunk the Sample Covariance Matrix." The standard practitioner reference for Section 12.2.4's shrinkage estimator, including the formula for choosing the shrinkage intensity δ optimally rather than fixing it by hand as the text's example does.
- D'Acunto et al. (2019), "The Promises and Pitfalls of Robo-Advising." An empirical study of robo-advised accounts documenting the gap between clients' stated and revealed risk tolerance discussed in this chapter's treatment of robo-advisors.
- SEC (2022), *Schwab Subsidiaries Misled Robo-Adviser Clients about Absence of Hidden Fees*. The regulatory announcement underlying this chapter's Schwab Intelligent Portfolios case vignette.
- Berg et al. (2022), "Aggregate Confusion: The Divergence of ESG Ratings." The empirical study of cross-provider ESG rating disagreement underlying this chapter's ESG-tilt worked example.

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- Maillard et al. (2010), "The Properties of Equally Weighted Risk Contribution Portfolios." The standard analysis of the equal-risk-contribution portfolio of Section 12.3.5, including why its volatility lies between those of the minimum-variance and equal-weighted portfolios.
- López de Prado (2016), "Building Diversified Portfolios that Outperform Out of Sample." The original description of hierarchical risk parity and of the Monte Carlo comparison that Section 12.5.4 reproduces in simplified form.
- Lo (2002), "The Statistics of Sharpe Ratios." The sampling theory of the Sharpe ratio behind Section 12.3.2's caution about the four-day toy ratios, including why ratios cannot in general be annualized by a square-root rule.

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Chapter 13

Algorithmic Trading and Market Microstructure

13.1 Introduction

Every strategy considered so far in this book, from the DuPont-decomposed valuation of Chapter 4 to the mean-variance portfolios of Chapter 12, has treated buying and selling an asset as instantaneous and free, so that the financial or statistical idea could be presented without execution detail. Chapter 3 showed this is not right: every trade costs at least half the bid-ask spread, and a large enough order walks through several levels of the order book at a progressively worse price. This chapter treats execution as a subject in its own right: how a trading signal is generated, how an order is worked in the market, how a strategy's historical performance should be evaluated, and how market microstructure shapes each of those decisions.

Several worked examples run through the chapter, each grounded in earlier material. A moving-average crossover strategy, backtested on a short synthetic price series, shows how a signal becomes a sequence of positions and then a profit-and-loss series. The limit-order-book walk from Chapter 3 is extended into a comparison of aggressive, single-shot execution with a patient, multi-slice alternative, motivating the TWAP and VWAP benchmarks, and into a regression estimate of Kyle's lambda that gives price impact an economic interpretation.

The market-making intuition of Chapter 3 is developed into a worked Avellaneda-Stoikov inventory example, showing how and why a market maker's quotes should skew as its position grows.

The pairs-trading setup of Chapter 7 becomes a backtested trade, entered and exited with the z-score rule that chapter only described, and is then generalized into a multi-stock statistical arbitrage example built on Chapter 12's factor regressions.

Chapters 6 and 7 supplied the tools to generate a trading signal; Chapter 12 supplied the logic for combining assets and signals into a position. This chapter supplies the missing link: everything that happens after a model or optimizer has decided what to hold, and before that decision becomes a completed trade at a real, imperfect price. Treating execution as instantaneous and free is the simplifying assumption every earlier chapter made for clarity; this chapter shows what it costs when it is relaxed.

13.2 Trading Signals and a First Strategy

Every algorithmic strategy begins with a signal, a rule that turns observable data into a view about where a price is going. This part classifies the signals the industry trades and works one of the simplest through, from raw prices to a realized return.

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13.2.1 A Taxonomy of Trading Strategies and Signals

Suppose two traders notice the same pattern in a stock's recent price history but reach opposite conclusions: one buys, betting a recent rise will continue; the other sells short, betting the rise has gone too far and must reverse. Both can be running a coherent strategy; they disagree about the economic source of the pattern.

Strategies are usually organized by that source. Momentum strategies bet that a recent price trend will continue, on the theory that information diffuses gradually through a market rather than being reflected in prices instantly. Mean-reversion strategies bet the opposite: that a price, or a relationship between two prices as in this chapter's pairs trade, has moved too far from a fundamental or statistical anchor and will revert, an application of Chapter 7's stationarity concepts. Market-making strategies (Section 13.4) take no directional view, profiting from repeatedly capturing the bid-ask spread while managing the resulting inventory risk. Statistical arbitrage generalizes mean reversion to many assets, often using Chapter 12's factor models to find a stable relationship.

Each strategy type produces the same output: a signal, a number (or a direction) indicating whether and how much to buy or sell at a given time. Converting that signal into a position, and the position into executed trades, is the concern of most of this chapter, a step with its own economics, costs, and risks, separate from whether the signal is any good.

13.2.2 Worked Example 13.1: A Moving-Average Crossover Strategy

A moving-average crossover strategy, one of the oldest and most transparent momentum strategies, holds a long position whenever a short-window moving average is above a longer-window one, and is flat otherwise, on the theory that the crossing signals a strengthening upward trend. Table 13.1 applies a 3-day short average and a 6-day long average to twelve days of hypothetical prices.

Day	Price	MA(3)	MA(6)	Signal	Strategy Return
1	100	–	–	0	–
2	101	–	–	0	0.00%
3	103	101.33	–	0	0.00%
4	106	103.33	–	0	0.00%
5	110	106.33	–	0	0.00%
6	115	110.33	105.83	1	0.00%
7	119	114.67	109.00	1	3.48%
8	118	117.33	111.83	1	−0.84%
9	114	117.00	113.67	1	−3.39%
10	109	113.67	114.17	0	−4.39%
11	104	109.00	113.17	0	0.00%
12	100	104.33	110.67	0	0.00%

Table 13.1: A 3-day/6-day moving-average crossover strategy

The signal turns on (long) at day 6, when the 3-day average first exceeds the 6-day average, and turns off (flat) at day 10, when it first falls back below. Crucially, the strategy return in each row applies the *previous* day's signal to the *current* day's price return. A signal computed from today's close cannot be traded until the next available price, the point-in-time discipline Chapter 6 emphasized against data leakage; using today's signal against today's return would silently look ahead.

Compounding the strategy returns gives a cumulative return of −5.22% over the twelve days, while buying and holding from day 1 to day 12 (starting and ending at 100) returns 0%. In this sample the crossover strategy underperforms buy-and-hold: it captures part of the days-6-through-8 uptrend but stays long one day too long into the days-9-and-10 reversal, the whipsaw risk that afflicts every momentum strategy at a trend reversal. The example is deliberately not a success story: a backtest evaluation (Section 13.6.1) must be prepared to find that a plausible strategy does not work on the available data.

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13.3 Execution and Market Impact

A signal is only half a strategy. Turning a desired position into an actual one means sending orders into a market that moves in response, so this part develops the mechanics of execution, the cost of demanding liquidity too quickly, and the models that balance that cost against the risk of trading too slowly.

13.3.1 Order Types and the Mechanics of Execution

Chapter 3 introduced the market order (executed immediately at the best available price, walking through the book if necessary) and the limit order (executed only at a specified price or better, potentially never filling at all). Converting a signal like the one in Section 13.2.2 into a position requires choosing among these order types, a direct trade-off: a market order guarantees execution but accepts whatever price impact results, while a limit order controls the price but risks never filling if the market moves away before the order reaches the front of the queue. A stop order becomes a market order once a trigger price is reached, commonly used to implement a predetermined exit rule (a stop-loss) without continuous monitoring.

An execution system must also decide how to route an order, since modern equity markets consist of many competing exchanges and alternative venues rather than one centralized book, a fragmentation Chapter 3 touched on in its discussion of Regulation NMS. A smart order router evaluates liquidity and price across every venue and splits or directs an order to capture the best aggregate execution, a harder problem than the single-book walk but built on the same economics of price and available size at each level.

A large order also raises a tension the order types above do not resolve: posting the full order as a visible limit order reveals the trader's intentions to other participants, who can respond, for example by front-running the anticipated remaining demand. Iceberg orders address this by displaying only a small portion of the order at a time, revealing more as the visible portion fills. Dark pools go further: private venues that match buy and sell orders without displaying any pre-trade price or size, reporting only completed trades after the fact.

Both mechanisms trade away pre-trade transparency, one of the market-quality benefits Chapter 3 attributed to well-functioning exchanges, for reduced information leakage and, in principle, lower market impact for large orders. It is a trade-off, not a free improvement: when a large fraction of volume trades in non-displayed venues, the lit order book becomes a less complete picture of supply and demand, a concern regulators have studied as dark-pool volume has grown. In 2025, for the first time, more than half of U.S. equity volume executed off-exchange, most of it not in dark pools but through wholesale market makers that pay retail brokers for their order flow and internalize it. Chapter 14's section on U.S. equity market structure as of 2026 dates that figure and the rule changes that followed it, among them a half-penny tick, lower access-fee caps, new round-lot sizes, and expanded execution-quality reports under Rule 605, all of which change the inputs a smart order router works with.

Figure 13.1 places these decisions in sequence. A signal is generated, an order type is chosen, a venue is selected, and the resulting execution is measured, with that measurement feeding back into how the next signal is traded. The rest of this chapter works through that loop one stage at a time; every stage introduces costs the signal never sees.

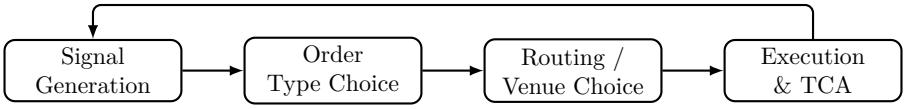


Figure 13.1: The path from a trading signal to a completed, evaluated trade: each stage introduces its own decisions and costs, and transaction cost analysis at the end feeds back into how future signals and execution choices are made

13.3.2 Market Impact and the Cost of Trading Fast

Chapter 3's limit-order-book example showed that a 250-share market order, larger than the 100 shares available at the best ask of $50.10, walked through a second price level and executed at an average price of $50.13, a price impact of about 6 basis points relative to the best ask. This cost arises entirely from the

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order's size relative to available liquidity; it is not a fee but the consequence of consuming the supply at each successive price level.

The alternative is to break the order into smaller pieces spaced over time, letting the book replenish between slices. Suppose the 250-share order is split into five 50-share slices, each small enough to fill within the 100 shares at the best ask, and suppose the best ask stays at \$50.10 between slices (a simplification; in reality the book may not fully replenish, and the price may drift for reasons unrelated to this trader).

This patient approach achieves an average price of \$50.10, eliminating the price impact, but at a cost the single-shot execution does not bear: while five slices are worked, the price can move for unrelated reasons, a risk called timing risk or execution risk. The aggressive strategy pays a known impact cost immediately; the patient one pays an expected impact cost of zero but accepts an uncertain timing-risk cost. Neither is universally superior; the choice depends on order size relative to liquidity, the asset's volatility, and urgency, the trade-off formalized in Section 13.3.5.

13.3.3 The Kyle Model: A Formal Model of Price Impact

The order-book walk demonstrates price impact mechanically but does not explain *why* a market maker sets prices that respond to order flow. The classic model of Kyle (1985) supplies the explanation: a market maker cannot distinguish an informed trader (who knows something about the asset's true value) from a liquidity trader (who trades for unrelated reasons, such as needing cash), and must set a single price schedule that breaks even on average against this mixture. Its rational response is to move the price in proportion to the net signed order flow it observes,

$$\Delta P = \lambda \times (\text{Net Order Flow}) + \varepsilon, \tag{13.1}$$

where net order flow is buy volume minus sell volume over some interval and λ (Kyle's lambda) measures the price impact per unit of net order flow, estimated by regressing observed price changes on net order flow.

Table 13.2 shows six intervals of net order flow (in thousands of shares) and the resulting price change (in cents).

Interval	1	2	3	4	5	6
Net order flow (000s)	5	−3	8	−6	2	4
Price change (cents)	10	−7	15	−11	3	9

Table 13.2: Net order flow and price change for estimating Kyle's lambda

Applying the same OLS formula used since Chapter 6,

$$\hat{\lambda} = \frac{\sum_t (\text{Flow}_t - \overline{\text{Flow}})(\Delta P_t - \overline{\Delta P})}{\sum_t (\text{Flow}_t - \overline{\text{Flow}})^2} \approx 1.95 \text{ cents per thousand shares}, \qquad R^2 \approx 0.99.$$

A larger λ means prices move more sharply for a given net buying or selling pressure, which Kyle's model interprets as informed trading being a larger fraction of order flow, or liquidity (the order-book depth available to absorb flow) being thinner. This gives price impact an economic interpretation: it is compensation the market maker requires for the risk of unknowingly trading against better-informed counterparties, the adverse-selection component of the spread introduced in Chapter 3, now a single estimable slope.

One caution, because everything built on that slope inherits it: $\hat{\lambda}$ is estimated as a constant, and it is not one. Impact per unit of order flow varies with the time of day, widest around the open and narrowest in the quiet afternoon; with the volatility regime; and with order size, since a larger order walks into thinner depth. A production pre-trade cost model therefore fits a conditional impact surface, regressing realized impact on order size, participation rate, quoted spread, volatility, and time of day, Chapter 6's supervised-learning setup pointed at execution. Kyle's constant λ makes the economics legible; the fitted surface is what a desk budgets against.

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13.3.4 TWAP and VWAP: Basic Execution Algorithms

The multi-slice approach in Section 13.3.2 is a simple version of the time-weighted average price (TWAP) algorithm: split an order into equal slices spread evenly across a horizon, without reference to how volume is distributed over it. Table 13.3 shows five hypothetical intervals with market prices and total trading volume.

Interval	1	2	3	4	5
Price	\$50.00	\$50.05	\$50.10	\$50.08	\$50.12
Market volume	10,000	12,000	8,000	15,000	9,000

Table 13.3: Prices and market volume over five execution intervals

A trader executing a 1,000-share order via TWAP trades 200 shares in each interval regardless of volume, achieving an average execution price equal to the simple average of the five prices,

$$\text{TWAP execution price} = \frac{50.00 + 50.05 + 50.10 + 50.08 + 50.12}{5} = \$50.07.$$

The volume-weighted average price (VWAP) benchmark, by contrast, weights each interval's price by that interval's share of total market volume,

$$\text{VWAP} = \frac{\sum_t P_t V_t}{\sum_t V_t} = \frac{50.00(10{,}000) + \cdots + 50.12(9{,}000)}{10{,}000 + 12{,}000 + 8{,}000 + 15{,}000 + 9{,}000} \approx \$50.068,$$

close to the TWAP execution price, a difference of about 0.4 basis points. The gap turns on which intervals carried the volume: the two most expensive intervals, the third at \$50.10 and the fifth at \$50.12, drew the least volume, 8,000 and 9,000 shares, so volume weighting puts the least weight where the price was worst for a buyer. Interval 4 carries the largest volume, 15,000 shares, but at \$50.08 it sits a cent *above* the TWAP average and nudges the weighted price the other way. A VWAP-following algorithm would have traded proportionally less in the two expensive intervals than TWAP did, capturing a slightly better average price. In practice, a VWAP algorithm trades according to a historical intraday volume profile, since the day's volume is not known in advance. The theory is that trading when the market is naturally most liquid minimizes the trader's footprint and price impact.

That contains a forecasting problem, and it is the hard part of the algorithm. Trading to VWAP requires knowing how the day's volume will be distributed before the day is over, and a historical average profile is the crudest forecast of it. Volume departs from that profile for partly predictable reasons: an index rebalance, an earnings release, an options expiry, or a heavy opening auction all bend the curve in ways a static profile cannot anticipate. Predicting the intraday volume curve is a supervised regression problem in Chapter 7's sense, with the day's realized volume so far as the most informative feature, and it demands the same walk-forward evaluation. An execution desk treats this forecast, not the weighting formula, as where a VWAP algorithm is won or lost.

13.3.5 Optimal Execution: Balancing Impact and Timing Risk

Section 13.3.2's comparison of aggressive and patient execution is a simplified version of the optimal execution problem formalized by Almgren and Chriss. Given an order of fixed size that must be completed by a fixed deadline, how should it be split over time to minimize expected market impact cost plus the variance introduced by timing risk? Impact cost is typically modeled as increasing in the trading rate, since trading faster consumes more of the liquidity at each price level, as in Chapter 3's order-book walk. Timing risk increases with the time the order remains unexecuted, since a longer horizon exposes the remaining quantity to more adverse price movement, the variance-grows-with-horizon logic of Chapter 7's multi-step forecasts.

The framework's output is not a single number but an efficient frontier analogous to Chapter 12's risk-return frontier. For each level of acceptable timing risk there is an optimal trajectory that minimizes expected cost, typically front-loaded, trading more aggressively early and tapering off. A more risk-averse trader, who weights the variance of the outcome more heavily, will choose a faster, more front-loaded execution than a less risk-averse trader facing the same order and market.

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This formalizes the trade-off Section 13.3.2 described qualitatively: neither extreme is generally optimal, and the answer depends on impact sensitivity and price volatility, which must be estimated from data, with all the estimation-error caveats Chapter 12 raised.

13.3.6 Worked Example 13.2: The Almgren-Chriss Optimal Trajectory

Almgren and Chriss formalize the trade-off above by minimizing a risk-adjusted total cost, expected temporary-impact cost plus ϕ times its variance from timing risk, where $\phi \geq 0$ is the trader's risk-aversion parameter (written ϕ here rather than the λ of Almgren and Chriss's own paper, to keep it distinct from Kyle's lambda above),

$$\min_{x_1,\ldots,x_{N-1}} \underbrace{\eta \sum_{j=1}^{N} \frac{(x_{j-1}-x_j)^2}{\tau}}_{\text{expected impact cost}} + \phi \underbrace{\sigma^2 \tau \sum_{j=1}^{N} x_j^2}_{\text{timing-risk variance}}, \tag{13.2}$$

where x_j is the number of shares still unexecuted after interval j (so $x_0 = X$, the full order, and $x_N = 0$, fully executed), τ is the length of each interval, η is the temporary-impact cost coefficient, and σ is the volatility of the share *price*, in dollars per share per unit of time, so that $\sigma^2\tau$ is the variance of the price change over one interval. The units matter: σ multiplies share counts in the variance term, so it must be a price volatility, not a percentage return volatility. The closed-form solution to this minimization is

$$x_j = X \frac{\sinh(\kappa(T - t_j))}{\sinh(\kappa T)}, \qquad \kappa = \sqrt{\frac{\phi\sigma^2}{\eta}}, \tag{13.3}$$

with $T = N\tau$ the total horizon; κ governs how front-loaded the trajectory is, growing with risk aversion ϕ and volatility σ and shrinking with impact cost η. The trajectory is stated without proof; the companion technical note on the Almgren-Chriss model derives it from the discrete first-order condition, a linear second-order difference equation, and verifies the two limiting cases: uniform, TWAP-style trading as risk aversion vanishes, and immediate liquidation as it grows without bound.

Applying this to Section 13.3.2's 250-share order split across the same $N = 5$ intervals, with each interval taken as the unit of time ($\tau = 1$) and illustrative parameters $\sigma = \$0.02$ per share per interval, $\eta = 0.01$, and a modest risk aversion of $\phi = 0.5$ (so $\kappa \approx 0.1414$), the optimal trajectory leaves

$$\begin{aligned} \text{Unexecuted shares:} &\quad x = (250,\ 194.2,\ 142.4,\ 93.4,\ 46.2,\ 0), \\ \text{Traded per interval:} &\quad (55.8,\ 51.9,\ 49.0,\ 47.1,\ 46.2), \end{aligned}$$

mildly front-loaded relative to the perfectly uniform 50-share-per-interval TWAP schedule of Section 13.3.2. Raising risk aversion tenfold to $\phi = 5$ (so $\kappa \approx 0.4472$) sharpens this front-loading considerably,

$$\text{Traded per interval:} \quad (92.8,\ 60.9,\ 41.3,\ 30.1,\ 25.0),$$

nearly double the first-interval quantity of the low-risk-aversion trajectory: a more risk-averse trader accepts a higher expected impact cost up front to shrink the variance from the part of the order still exposed to price moves. Figure 13.2 plots both trajectories against the uniform TWAP baseline, showing how the front-loading grows with ϕ.

Scoring the optimal trajectory, the uniform TWAP schedule, and the single-shot execution of Section 13.3.2 on the same risk-adjusted objective at $\phi = 0.5$ makes the trade-off concrete:

$$\begin{aligned} \text{Optimal trajectory:} &\quad \text{cost} = 139.4, \\ \text{Uniform TWAP:} &\quad \text{cost} = 140.0, \\ \text{Fully aggressive single-shot:} &\quad \text{cost} = 625.0, \end{aligned}$$

the TWAP schedule barely worse since $\phi = 0.5$ is only mildly risk-averse, and the single-shot execution over four times worse, since it eliminates timing risk but pays a quadratically larger impact cost for trading the whole order in one interval. Both extremes are feasible points, but neither is optimal once risk aversion

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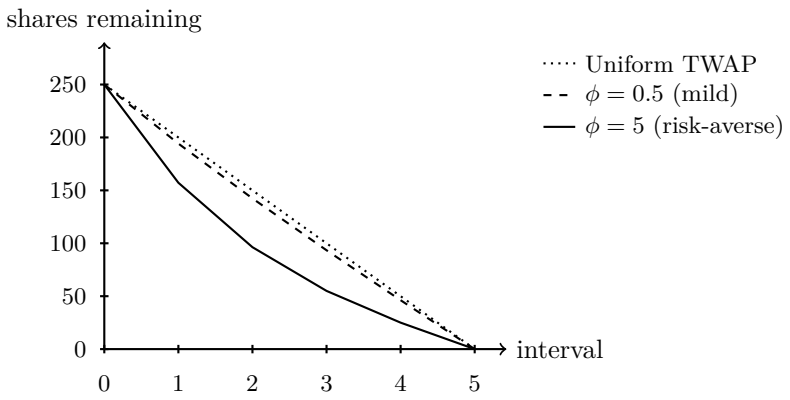


Figure 13.2: Almgren-Chriss optimal unexecuted-share trajectories at two risk-aversion levels, against the uniform TWAP baseline

is weighed against expected cost, and the front-loaded shape of the optimum is not a generic property of "spreading an order out" but the output of this closed-form solution.

The costs above are in dollars, with ϕ measured per dollar, and the units are a common source of error. On the \$50 stock of the order-book example, $\sigma = \$0.02$ per share per interval is a return volatility of 0.04% per interval; with five intervals to the trading day it scales by the square-root-of-time rule to 0.0894% a day and 1.42% a year, a deliberately calm stock chosen to keep the numbers readable. Going the other way, a typical 25% annual volatility on the same stock converts to $0.25 \times \$50/\sqrt{252 \times 5} = \$0.3521$ per share per interval. The mistake the units guard against is entering a percentage directly: reading $\sigma = 0.02$ as a 2% return volatility on a \$50 stock means $\sigma = \$1.00$, which makes $\kappa = 7.07$, fifty times larger, and an optimal schedule that sells 249.8 of the 250 shares in the first interval.

The objective in Equation (13.2) also contains only the *temporary* impact of each slice, the premium paid to consume liquidity, which leaves the price for later slices untouched. Almgren and Chriss also model a *permanent* impact: every share sold lowers the price for all later shares by γ, because the market reads the order flow as information. With linear permanent impact its expected cost is $\gamma X^2/2$ minus a small term, $\frac{\gamma}{2}\sum_j (x_{j-1} - x_j)^2$, that depends on the schedule in the same way as the temporary cost and so lowers the effective temporary coefficient from η/τ to $\eta/\tau - \gamma/2$. At an illustrative $\gamma = \$0.002$ per share per share sold, the permanent cost is $\$62.50 - \$12.56 = \$49.94$ for the optimal trajectory and $\$62.50 - \$12.50 = \$50.00$ for TWAP: a substantial cost, but almost the same for every gradual schedule, which is why the optimal trajectory is usually stated with temporary impact alone. Permanent impact matters a great deal for *how much* to trade, as the next section shows, and hardly at all for *how to spread* a given order.

The closed form's assumptions are stronger than its elegance suggests. The trajectory is optimal only if impact is linear in the trading rate, volatility is constant across the horizon, and the trader has no view on where the price is heading. All three are false in ways any execution desk would recognize, and relaxing any one destroys the closed form. What survives is the problem: choosing a sequence of trade sizes to minimize expected cost plus a risk penalty is a sequential decision problem, solvable by learning a policy from simulated episodes instead of solving a difference equation. Chapter 19 does this, recovering the trajectory above for the same parameters with tabular Q-learning, then extending it to frictions the closed form cannot express.

13.3.7 The Square-Root Law and Strategy Capacity

Kyle's model and the Almgren-Chriss objective both treat impact as linear in the amount traded, a reasonable local approximation for a single small slice. For a whole *metaorder*, the full parent order a fund works over hours or days, the best-documented empirical regularity in market microstructure says otherwise. Across stocks, markets, and decades, the average price impact of a metaorder of Q shares grows with the square root of its size,

$$\frac{\Delta P}{P} \approx Y\, \sigma_{\text{daily}} \sqrt{\frac{Q}{V}}, \tag{13.4}$$

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where V is the stock's average daily volume, σ_{daily} its daily return volatility, and Y a constant of order one (Tóth et al. 2011). Studies of large institutional order sets find the same concave shape, with fitted exponents close to one half; Almgren et al. (2005), on Citigroup's U.S. equity orders, preferred a slightly steeper 3/5. Two features make the law useful. Volatility and daily volume, both easy to measure, carry most of the cross-sectional variation in impact. And concavity changes the arithmetic of size: with $Y = 1$ and 2% daily volatility, an order for 1% of daily volume costs about $0.02 \times \sqrt{0.01} = 20$ basis points, and one for 4% costs 40. Quadrupling the order doubles the cost per share, so the total dollar cost rises eightfold, as $Q^{3/2}$.

The square-root law describes the *peak* impact at the end of a metaorder, which is largely temporary: once buying stops, a substantial part of the price move reverts over the following hours, and the part that remains is the permanent, information-driven component of the previous section (Bacry et al. 2015). The distinction is economically sharp. A fund pays the peak on every share it trades but keeps only the permanent part as a mark-to-market gain on the position it built.

The law's most important application is the question every strategy eventually faces: how much capital can it run? Consider a strategy that rebalances monthly into 50 stocks, each trading \$50 million a day, turning over its whole book at each rebalance. Every trade carries a gross expected alpha of 50 basis points of the value traded and pays a 3 basis point half-spread plus square-root impact with $Y = 1$ and 2% daily volatility. With assets under management A, each stock receives a trade of $A/50$, so the net alpha per trade is

$$\alpha_{\text{net}}(A) = 50 - 3 - 200\sqrt{\frac{A/50}{\$50 \text{ million}}} \quad \text{basis points.}$$

At \$10 million the trades are 0.4% of daily volume and net alpha is 34.35 basis points; at \$100 million they are 4% of volume and only 7.00 basis points survive; net alpha reaches zero at \$138.1 million, and at \$200 million each rebalance loses money. The capacity that matters to the manager is not that break-even point but the size that maximizes total dollar profit, $A\,\alpha_{\text{net}}(A)$. Because impact grows as $\sqrt{A}$, profit $A(a - k\sqrt{A})$ is maximized where $k\sqrt{A} = 2a/3$, which leaves exactly one third of the pre-impact edge a as net alpha. Here that is \$61.4 million, with 15.67 basis points of net alpha per trade and about \$96,000 of profit per rebalance, \$1.15 million a year. Growing the fund to \$100 million would cut its profit per rebalance to \$70,000, though still well short of break-even. Gross alpha that looks robust in a backtest at small size can support only a modest fund, and the square-root law puts a number on it, which a linear-impact model, with its constant cost per share, cannot do.

13.4 Inventory Models and Market Making

Chapter 3 introduced market making: a market maker profits from the bid-ask spread by standing ready to buy and sell, but bears inventory risk from holding a nonzero position while waiting for an offsetting order. The Avellaneda-Stoikov model formalizes this by having the market maker adjust both quotes based on current inventory, not just on fair value. A market maker with a large long position, from selling less than buying, should skew both quotes downward: a lower bid makes it less attractive for others to sell to it, and a lower ask makes it more attractive to buy from it. That works the inventory back toward zero rather than waiting for random order flow to do so.

The model formalizes this skew through a reservation price, the price at which the market maker would be indifferent to a marginal trade given its inventory, which sits below the mid price when it is long and above it when short,

$$r = s - q\,\gamma\,\sigma^2\,(T - t), \tag{13.5}$$

where s is the current mid price, q is the current inventory (positive if long, negative if short), γ is a risk-aversion parameter, σ^2 is the asset's return variance, and $T - t$ is the time remaining in the trading horizon. The optimal bid and ask are centered on this reservation price, not on the mid price, and separated by an optimal spread

$$\delta = \gamma\sigma^2(T - t) + \frac{2}{\gamma}\ln\left(1 + \frac{\gamma}{\kappa}\right), \tag{13.6}$$

where κ governs how sensitively order arrival rates respond to how competitively the market maker quotes. Consider a market maker with mid price $s = \$100$, long $q = 5$ units, $\gamma = 0.1$, $\sigma = 2$, one unit of time

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remaining ($T - t = 1$), and $\kappa = 1.5$. The reservation price is $r = 100 - 5(0.1)(2^2)(1) = \$98.00$, two dollars below the mid price, reflecting the desire to reduce the long position. Both formulas are quoted rather than derived; the companion technical note on the Avellaneda-Stoikov model derives the reservation price from exponential (CARA) utility applied to a normally distributed terminal mark-to-market value, and the per-side markup from the Poisson arrival intensity, which is why the spread decomposes into an inventory-risk term that vanishes as the horizon closes and an order-arrival term that does not. The optimal spread works out to $\delta \approx \$1.69$, giving

$$\begin{aligned} \text{Inventory-aware quotes:} &\quad \text{bid} = r - \delta/2 \approx \$97.15,\ \text{ask} = r + \delta/2 \approx \$98.85, \\ \text{Naive quotes (ignoring inventory):} &\quad \text{bid} = \$99.15,\ \text{ask} = \$100.85, \end{aligned}$$

the inventory-aware quotes both \$2.00 lower, the reservation-price adjustment carried through to each side. The ask becomes more attractive to a buyer (encouraging trades that reduce the long position) and the bid less attractive to a seller (discouraging trades that would increase it), while the \$1.69 spread, and so the expected compensation per round trip, is unchanged.

This skewing connects to the spread decomposition of Chapter 3: the inventory-holding component of the spread is the compensation a market maker requires for the risk that its inventory moves against it before an offsetting trade arrives, and Avellaneda-Stoikov derives the optimal quote adjustment as a function of inventory, remaining horizon, and volatility. The skew grows with γ, σ^2, and the position q: a more risk-averse market maker, a more volatile asset, or a larger position all call for a more aggressive skew, since inventory risk is more urgent to offload when it is larger or more dangerous to hold. A market maker who quotes a fixed spread around fair value regardless of position is exposed to accumulating, unhedged directional risk, an instance of the risk management discipline of Chapter 8.

The caveat of Section 13.3.5 applies here too, for the same reason. Avellaneda-Stoikov is solvable in closed form because order arrivals are Poisson with fixed intensity, volatility is constant, and the only state that matters is inventory and time remaining. A market maker in a real book faces arrivals that cluster, counterparties whose order flow carries information the model's symmetric arrivals cannot represent, and a choice of venue on top of the choice of quote. Learned quoting policies exist to handle these, which is why Chapter 19 rebuilds this model as a reinforcement learning problem. The closed form is still the right thing to meet first: it explains *why* quotes should skew toward inventory and by how much, a question no learned policy will answer for you.

13.5 Arbitrage and Relative-Value Strategies

Momentum and market making both take a view on a single instrument. A third family instead trades the relationship between instruments, holding offsetting positions so that the strategy profits from relative mispricing rather than from market direction.

13.5.1 Worked Example 13.3: A Pairs-Trading Backtest

Chapter 7 introduced pairs trading with six days of prices for two hypothetical bank stocks, finding a spread mean of $\bar{s} = 2.5$, a standard deviation of $\hat{\sigma}_s = 0.548$, and a final z-score of 0.91, too small to trigger a trade under the common rule of entering when $|z| > 2$. Extending the series four more days, with new prices for Stock A of \$58, \$56.50, \$54.80, and \$54.50, and for Stock B of \$54, \$53, \$52, and \$52, produces the z-scores in Table 13.4, computed with the mean and standard deviation from the original six days (in-sample parameters applied out of sample, the walk-forward discipline).

Day	7	8	9	10
Spread ($A - B$)	4.00	3.50	2.80	2.50
z-score	2.74	1.83	0.55	0.00

Table 13.4: Out-of-sample spread and z-score for the extended pairs-trading example

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Day 7's z-score of 2.74 exceeds the entry threshold of 2, triggering a trade: short 1 share of Stock A and long 1 share of Stock B (the $\beta \approx 1$ simple-spread convention Chapter 7 used), betting that the wide spread will revert toward its mean. By day 10 the spread has reverted to 2.50, the in-sample mean, triggering an exit.

The trade's profit is the decline in the spread over the holding period,

$$\text{P\&L} = \text{Spread}_{\text{entry}} - \text{Spread}_{\text{exit}} = 4.00 - 2.50 = \$1.50 \text{ per share-pair},$$

since shorting the spread profits in proportion to how much it narrows. Relative to the capital required (the entry prices of the two legs, $58 + 54 = \$112$), this is a return of about 1.34% over the three-day holding period. The trade illustrates the complete pairs-trading lifecycle: estimating the relationship in-sample, monitoring it out-of-sample, entering when the spread deviates unusually far from its mean, and exiting once it reverts. The strategy depends on the cointegrating relationship remaining stable, and one successful trade in a four-day extension is not evidence that it will continue to hold.

13.5.2 Statistical Arbitrage Beyond Pairs

The pairs trade of Section 13.5.1 generalizes to more than two assets using Chapter 12's factor models rather than a two-asset regression. Instead of one hedge ratio β between two stocks, a statistical arbitrage strategy regresses each stock in a universe on common factors (the market, size, and value factors of the multi-factor model, or a larger set extracted by principal component analysis) and trades the regression residual, the return no common factor explains. A stock whose residual has drifted unusually far from its historical mean is a candidate to trade against a basket of peers, the same mean-reversion idea diversified across many residual bets.

Consider three stocks, W, X, and Y, in the same sector, each regressed on the six months of market excess returns used for the CAPM beta regression in Chapter 12. Using the first five months to estimate each stock's beta and residual volatility, Table 13.5 shows each stock's estimated beta and its month-6, out-of-sample residual z-score.

Stock	Estimated β	Month-6 residual	z-score
W	0.98	+1.26%	+13.3
X	1.21	−0.15%	−1.8
Y	0.81	+0.09%	+1.1

Table 13.5: Estimated betas and out-of-sample residual z-scores for three sector peers

Stocks X and Y behave as their betas predict, with residuals within ordinary in-sample noise. Stock W returned far more than its beta of 0.98 and the month's 3% market move would explain, a residual z-score of over 13. A deviation that large suggests either stock-specific news (in which case the move may be permanent and should not be traded against) or a temporary dislocation (in which case a statistical arbitrage desk would short Stock W against a market-neutral basket of its sector peers, betting the residual reverts).

This is the multi-asset generalization of the pairs trade in Section 13.5.1: rather than trading one stock against one other, the position is effectively long the sector and short this stock's idiosyncratic move, diversifiable across many residual signals computed the same way across a universe of stocks.

The generalization changes the risk profile. A single pairs trade is exposed to the risk that its cointegrating relationship breaks down; a book trading hundreds of residual relationships diversifies away much of that single-relationship risk, in Chapter 12's portfolio-level sense, provided the bets are not all driven by a common, unmodeled factor that would make them break down together.

That caveat is not hypothetical. Such a correlated breakdown was central to the August 2007 "quant quake", when several large, otherwise unrelated quantitative equity funds suffered simultaneous, severe losses over a few days. It was widely attributed to many funds holding similar statistical arbitrage positions and being forced to unwind them at once, the crowding-driven, self-reinforcing price move that Chapter 12's discussion of factor crowding described for long-term allocation rather than short-term trading.

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13.6 Evaluating a Strategy

A strategy that looks profitable on paper has cleared a much lower bar than one that is tradable. This part develops the metrics that summarize a backtest, the traps that make a backtest lie, and the post-trade accounting that reveals what the strategy cost to run.

13.6.1 Performance Metrics

Evaluating a backtested strategy, whether the moving-average crossover of Section 13.2.2 or the pairs trade of Section 13.5.1, requires more than a cumulative return. The Sharpe ratio of Chapter 12 applies directly to a strategy's return series, dividing mean return by return volatility; for the moving-average strategy's eleven return periods (mostly zeros, since it is flat for seven of them), the mean return is -0.47% against a sample standard deviation of 2.03%, a Sharpe ratio of about -0.23.

Maximum drawdown, introduced in Chapter 8, measures the largest peak-to-trough decline in cumulative value over the backtest, a risk that a return or Sharpe-ratio summary can miss. The moving-average strategy's cumulative value peaks at a 3.48% gain (after day 7) and then falls 8.40% from that peak by day 10, to a cumulative loss of 5.22% ($109/119 - 1 = -8.40\%$, since the strategy is long from the \$119 close to the \$109 close), so its maximum drawdown is 8.40%. That is arguably a more decision-relevant number than the final -5.22%, since it shows how much value an investor would have seen erode from the strategy's peak, not just where it ended.

The hit rate, the fraction of active (non-flat) days that were profitable, is 25% for the moving-average strategy, one profitable day out of four. It is a useful diagnostic because a strategy can have a positive Sharpe ratio with a hit rate below 50% if its winners are sufficiently larger than its losers, or vice versa. Like accuracy in Chapter 6's classification metrics, hit rate alone can mislead when separated from the size of the gains and losses.

13.6.2 Common Backtesting Pitfalls

A credible backtest must also avoid data traps that make a strategy look far better than it would have performed in real time. Survivorship bias, introduced in Chapter 1, is especially dangerous here: testing only on stocks still trading today silently excludes every company that went bankrupt or was delisted, the population a strategy most needs to be tested against, since avoiding disasters is at least as valuable as picking winners.

Look-ahead bias occurs whenever a backtest uses information that was not available when the decision was made, for example finalized, restated financial statements instead of the preliminary figures public at the time, or, in this chapter's moving-average example, using today's signal against today's return rather than lagging it by one period as Section 13.2.2 did.

Both biases share the root cause of Chapter 6's data leakage: information from outside the available information set silently improves backtested performance in a way no real-time process could replicate. Both require the same discipline: skeptical attention to what data would have been known, and in what form, at each point the backtest simulates a decision.

13.6.3 Worked Example 13.4: The Cost of Searching

Both pitfalls above are properties of the data. The third, and in practice the most damaging, is a property of the researcher. The 3-day/6-day crossover of Section 13.2.2 scored a Sharpe ratio of -0.23, and almost nobody stops there: the next step is a 5-day/20-day pair, then a 10-day/50-day pair, then a stop-loss rule, reporting whichever combination came out best. That is not a backtest of one strategy. It is a search across many, and the figure it reports is the maximum of many noisy estimates rather than any single strategy's edge.

How damaging that is turns on a prior question: how far from zero can a worthless strategy's Sharpe ratio wander on luck alone? Any estimate from a finite sample differs from what it estimates, and the typical size of that gap is its *standard error*, the price of watching a random process for a finite time. For a strategy

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with no edge over T periods, the measured Sharpe ratio is centered on zero with a standard error of $1/\sqrt{T}$ per period, which on annualizing becomes

$$\text{standard error of the annualized Sharpe ratio} \approx \frac{1}{\sqrt{\text{years of data}}}. \tag{13.7}$$

The observation count cancels, leaving only calendar span: re-running a five-year backtest on five-minute bars multiplies the observations by about seventy-eight and buys no extra ability to tell an edge from luck, because it shrinks the per-period Sharpe ratio by the same factor. Only a longer history helps. The companion technical note on the sampling distribution of the Sharpe ratio derives both results from the fact that a measured Sharpe ratio is the ordinary one-sample t-statistic divided by $\sqrt{T}$.

Five years of daily data therefore gives a standard error of 0.447. Testing one strategy against it is Chapter 6's hypothesis test, and at the conventional one-sided 5% level the figure must clear 1.645 standard errors, the distance beyond which only 5% of a bell curve lies, or $1.645 \times 0.447 = 0.74$; one worthless strategy in twenty does. Search N configurations, though, and the reported number is the largest of N draws, which lands further out the more draws are taken, without ceiling. Table 13.6 gives that expected best in standard errors, the same figure as a Sharpe ratio, and the *honest* hurdle the winner must beat for the search *as a whole* to be significant. The naive 0.74 controls the chance that one pre-specified strategy looks good by luck; the honest hurdle controls the chance that *any* of the N does.

Configurations N	Expected best, in standard errors	Expected best Sharpe	Honest 5% hurdle
1	0.00	0.00	0.74
10	1.54	0.69	1.15
100	2.51	1.12	1.47
1,000	3.24	1.45	1.74

Table 13.6: Best-of-N in-sample Sharpe ratios when every strategy searched is worthless, five years of daily data

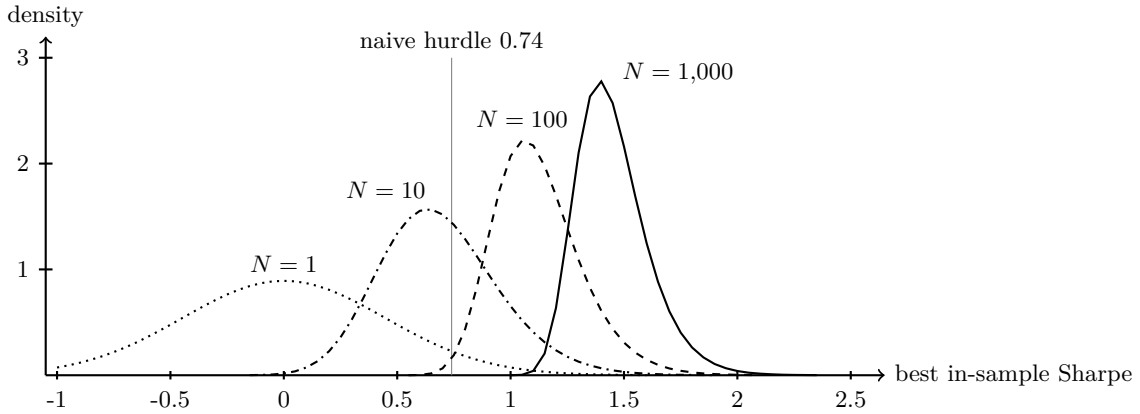


Figure 13.3: Density of the best annualized in-sample Sharpe ratio among N worthless strategies, five years of daily data (standard error 0.447). The vertical line is the naive one-strategy hurdle of 0.74; the curves are dotted ($N = 1$), dash-dotted ($N = 10$), dashed ($N = 100$), and solid ($N = 1{,}000$).

The table reports averages; Figure 13.3 shows the whole distribution behind them. Each added order of magnitude in N moves the density of the winning Sharpe ratio to the right and makes it narrower: the most likely best-of-ten figure is 0.64, the most likely best-of-a-thousand 1.39. The naive hurdle of 0.74 is cleared by luck alone with probability 5% when one strategy is tested, 40.1% when ten are, and 99.4% when a hundred are, so for a search of that size clearing it says almost nothing.

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Take $N = 100$: five short windows, five long windows, and four exit rules, a search most researchers would call modest. The expected best in-sample annualized Sharpe ratio is 1.12 though every one of the hundred is worthless by construction, and a Monte Carlo of ten thousand such searches averages 1.118 against the analytic 1.121. That clears the naive 0.74 hurdle comfortably and would usually be presented as a strong result, whereas the honest hurdle for the best of a hundred is 1.47, almost twice. Out of sample the question is settled: the winner earns a mean Sharpe ratio of +0.004, loses money outright 49.6% of the time, and the in-sample to out-of-sample rank correlation across the hundred is +0.001. The winner of a search is not better than what it beat. It is luckier, and luck does not persist into the next sample.

Three responses follow, and none of them is to stop searching. Report N, since a backtest presented without the number of configurations tried cannot be evaluated. Compare the reported figure against the best-of-N hurdle rather than zero, as the deflated Sharpe ratio of Bailey and López de Prado does, discounting a figure by how much a search of that size would inflate it and adjusting for the non-normality of real returns. And recognize that neighboring grid points are highly correlated (a 49-day and a 50-day moving average produce nearly identical positions), so the number of *independent* trials sits well below the raw count: correlating every pair at 0.5 lowers the mean best Sharpe ratio to 0.80, what about eighteen independent trials would give rather than a hundred.

What none of it can manufacture is data: for a strategy whose *true* annualized Sharpe ratio is 0.5, a respectable edge, separating it from zero in a single pre-specified test takes about eleven years of daily returns, and clearing a hundred-configuration search about forty-three. Little strategy research runs on anything near that history, the strongest argument for pre-specifying a few economically motivated candidates, as Section 13.2.1 framed them, rather than letting a grid search decide what the hypothesis should have been.

13.6.4 Implementation Shortfall and Transaction Cost Analysis

The gap between a strategy's theoretical performance, computed from the prices a signal was based on, and its realized performance, computed from the prices at which trades were executed, is called implementation shortfall, Execution Price − Decision Price. For the TWAP execution in Section 13.3.4, if the decision to buy was made at \$50.00 (the decision price) but the order was executed via TWAP at \$50.07, the implementation shortfall is \$50.07 − \$50.00 = \$0.07 per share, 14 basis points, or \$70 on the 1,000-share order. This number bundles every friction in this chapter: the market impact of Section 13.3.2, the timing risk of Section 13.3.5, and any price drift between decision and execution unrelated to the trade. It is the standard metric of transaction cost analysis (TCA) for judging whether a desk's execution adds or subtracts value relative to the decisions it implements. A strategy profitable on decision prices can prove unprofitable once realistic shortfall is subtracted, which is why serious backtesting should model transaction costs explicitly rather than assume execution at the prices a signal was computed from.

13.7 Machine Learning and Event-Driven Signals

Every signal so far, the moving-average crossover and the pairs-trading z-score, is a simple, interpretable rule with a handful of parameters. The classification and regression methods of Chapter 6 can generate richer ones: a gradient-boosted tree or neural network trained on many engineered features (lagged returns, rolling volatilities, volume features) can capture nonlinear patterns a fixed-form rule cannot represent. This part develops the features such a model is given and the discipline that keeps a flexible model honest, then turns to signals that arrive as events rather than continuous series.

13.7.1 A Head-to-Head Test: Gradient Boosting versus Logistic Regression on a Regime Interaction

The twelve-day price series of the feature-engineering catalog below is far too short to fit any model, so testing the claim above requires a longer history. The companion notebook builds 5,000 days of returns with volatility clustering, using the GARCH-style recursion of Chapter 7, and layers a regime interaction on top: three-day momentum continues in calm, low-volatility stretches but reverses in turbulent ones. A logistic regression's additive coefficients cannot represent this, since one coefficient on momentum cannot be

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positive in one regime and negative in another. A logistic regression and a gradient-boosted classifier are trained on the same three features from Table 13.7, on the first 3,500 simulated days, and evaluated on the final 1,496, following the walk-forward discipline, since time-ordered data cannot be split randomly without leaking future information.

Logistic regression reaches a held-out AUC of 0.5130, barely above a coin flip's 0.50, because its additive structure cannot capture a momentum effect that flips sign with the regime. Gradient boosting reaches 0.5328, a modest improvement that holds up across random seeds, with feature importances of 0.423 (momentum), 0.388 (volatility), and 0.189 (RSI) showing it draws on momentum and volatility jointly. Both AUCs are deliberately modest: financial returns have a low signal-to-noise ratio, as Chapter 6 cautioned, so an AUC in the low 0.50s is a realistic stand-in for a real signal's small edge. Gradient boosting captures an interaction a linear-in-its-features model structurally cannot, even though the edge, like most real ones, is small.

13.7.2 A Feature-Engineering Catalog for Trading Signals

Chapter 6 developed the general feature patterns (ratios, lagged and rolling windows, interaction terms, categorical encodings) used across this book's applications. Trading signals draw on a narrower catalog built almost entirely from price, volume, and order-book history, in four families. **Price-based** features summarize recent direction and magnitude: momentum, the trailing return over a fixed lookback window, and realized volatility, the rolling standard deviation of returns over the same window, with the moving averages of Section 13.2.2 a third. **Technical indicators** repackage price history into a bounded, more interpretable scale: the Relative Strength Index (RSI), the Absolute Price Oscillator (APO), the difference between a fast and a slow exponential moving average, and Bollinger Bands, which wrap a moving average in a pair of volatility-scaled upper and lower bands, all three worked through numerically below.

Volume-based features, raw volume, dollar volume (price times volume), and volume relative to its own recent average, capture how much conviction accompanies a price move, which no price-based feature can see. **Microstructure-based** features draw on Chapter 3 and this chapter: the bid-ask spread and the order-flow imbalance behind Kyle's model in Section 13.3.3 change on the timescale of individual trades rather than daily closes and carry information a price-based feature never sees.

13.7.3 Momentum, Realized Volatility, and RSI

Table 13.7 computes three of these features from the twelve-day price series of Section 13.2.2: a 3-day momentum ($\text{Mom3}_t = P_t/P_{t-3} - 1$), a 3-day realized volatility (the rolling standard deviation of the three most recent daily returns), and a 3-period RSI, $\text{RSI} = 100 - 100/(1 + RS)$ where RS is the average of the window's positive price changes (gains) divided by the average magnitude of its negative changes (losses).

Day	Price	Mom3 (%)	Vol3 (%)	RSI3
4	106	6.00	0.781	100.00
5	110	8.91	0.732	100.00
6	115	11.65	0.667	100.00
7	119	12.26	0.450	100.00
8	118	7.27	2.328	90.00
9	114	−0.87	2.835	44.44
10	109	−8.40	1.493	0.00
11	104	−11.86	0.523	0.00
12	100	−12.28	0.313	0.00

Table 13.7: Three engineered trading-signal features, computed from Section 13.2.2's twelve-day price series

The three columns tell complementary stories about the same twelve days. Momentum tracks the trend, rising through day 7 and falling steadily thereafter, mirroring the price series.

Realized volatility does something momentum cannot. It stays low throughout the smooth days-4-through-7 uptrend (0.45% to 0.78%) and then spikes, to 2.3% and 2.8%, at the days 8 and 9 reversal that Section 13.2.2

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identified as the crossover strategy's whipsaw. That is an instance of the volatility clustering of Chapter 7's GARCH material: volatility rises around regime changes, not uniformly through time, which is why realized volatility is a useful predictive feature and not merely a risk-management input.

RSI shows a limitation alongside its usefulness. Because every change in the days-4-through-7 window is positive, RSI saturates at 100 for four consecutive days, the conventional "overbought" signal, but cannot distinguish a mild uptrend from an explosive one once every recent change has the same sign; it falls as quickly to a saturated 0 ("oversold") once every recent change turns negative. A model trained only on RSI could not tell days 4 through 7 apart, despite their different momentum. Practitioners therefore feed a model both a bounded indicator like RSI and the underlying continuous features, such as momentum and realized volatility, letting the model learn how much weight each deserves rather than discarding what a saturated indicator throws away.

13.7.4 Two Further Technical Indicators

Two further indicators are common enough to work through alongside RSI, both computed from an exponentially weighted moving average (EMA) rather than the simple moving average of Section 13.2.2: an EMA weights recent prices more heavily, updating as $\mathrm{EMA}_t = \lambda P_t + (1-\lambda)\mathrm{EMA}_{t-1}$ with smoothing factor $\lambda = 2/(N+1)$ for an N-period window, seeded at the first available price.

The **Absolute Price Oscillator (APO)** is the difference between a fast and a slow EMA, $\mathrm{APO}_t = \mathrm{EMA}_t^{\mathrm{fast}} - \mathrm{EMA}_t^{\mathrm{slow}}$. Industry practice typically uses a 10-day fast and 40-day slow window (or, in the related MACD indicator, a 12-day and 26-day pair smoothed by a signal line), but Table 13.8, which also carries the Bollinger Bands introduced next, uses a 3-day and 6-day pair, scaled down to the twelve-day series as Section 13.2.2 scaled its moving-average windows.

Day	Price	EMA3	EMA6	APO	SMA5	Upper	Lower
1	100	100.000	100.000	0.000	100.000	100.000	100.000
2	101	100.500	100.286	0.214	100.500	101.500	99.500
3	103	101.750	101.061	0.689	101.333	103.828	98.839
4	106	103.875	102.472	1.403	102.500	107.083	97.917
5	110	106.938	104.623	2.314	104.000	111.266	96.734
6	115	110.969	107.588	3.381	107.000	117.040	96.960
7	119	114.984	110.849	4.136	110.600	122.234	98.966
8	118	116.492	112.892	3.600	113.600	123.447	103.753
9	114	115.246	113.208	2.038	115.200	121.575	108.825
10	109	112.123	112.006	0.117	115.000	122.043	107.957
11	104	108.062	109.719	−1.657	112.800	124.071	101.529
12	100	104.031	106.942	−2.911	109.000	122.023	95.977

Table 13.8: Two further technical indicators on the same twelve-day price series: the Absolute Price Oscillator (3-day and 6-day EMA) and Bollinger Bands (5-day SMA, $k = 2$)

The APO column tells the same trend story as momentum through an EMA's smoother lens: it is positive and climbing throughout the days-1-through-7 uptrend, peaking at 4.136 at day 7's local high, then falls and turns negative by day 11 once the fast EMA crosses below the slow one, a lagging but interpretable confirmation of the reversal the crossover strategy was built to detect.

Bollinger Bands wrap a simple moving average in bands a fixed number of standard deviations above and below it: $\mathrm{Middle}_t = \mathrm{SMA}_t$, $\mathrm{Upper}_t = \mathrm{SMA}_t + k \cdot \sigma_t$, and $\mathrm{Lower}_t = \mathrm{SMA}_t - k \cdot \sigma_t$, where σ_t is the rolling standard deviation of price (not return) over the same window as the moving average and k is conventionally set to 2. Industry practice typically uses a 20-day window; the table instead uses a 5-day window, for the same reason it shortened the EMA windows above.

The band width (upper minus lower) is itself a volatility indicator, telling the same story as Table 13.7's Vol3 column: the bands are tight while days 1 through 5 climb smoothly, then widen sharply from day 6 and stay wide through day 12, tracking the days-8-and-9 spike Vol3 flagged and the GARCH-style clustering of

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Chapter 7, expressed geometrically around the price chart. Both indicators transform the rolling mean and dispersion already summarized by momentum and realized volatility, so neither supplies information a model could not in principle extract from those two. Their value is the packaging: a chart a human trader can read at a glance, which is why both remain popular in practitioner tools even though a machine-learning pipeline can work from the underlying features.

13.7.5 Putting the Features to Work

Computing indicators is only the first step. The four topics below cover what surrounds them in a working strategy: a market-wide regime filter, two data disciplines, the move from features to a portfolio, and out-of-sample validation.

A Market-Wide Feature: The VIX

Every indicator so far is computed from a single asset's price history and looks backward. The CBOE Volatility Index (VIX), introduced in Chapter 5 as the market-implied volatility aggregated across S&P 500 index options, is a different kind of feature: a market-wide, forward-looking measure of expected volatility over the next 30 days.

Strategies routinely use the VIX level, or its rate of change, as a regime filter on top of asset-specific signals. A momentum strategy's whipsaw risk, of which Section 13.2.2's days-8-and-9 reversal is a small-scale illustration, tends to be worse when VIX is elevated and the market unsettled. A strategy might therefore reduce position sizes, widen its stop-losses, or stand aside once VIX crosses a threshold, rather than treating every day as statistically identical. Because VIX is tradable, through futures, options, and exchange-traded products, some strategies also trade VIX-linked instruments as a hedge against the market volatility that erodes asset-specific strategies' returns.

Two Disciplines: Point-in-Time Data and Stationarity

Two disciplines from earlier chapters apply to every feature in Table 13.7. First, the point-in-time rule Section 13.2.2 enforced applies to every engineered feature: a momentum, volatility, or RSI value computed from prices through day t can only trade day $t+1$'s return, never day t's. A backtest that lines up a feature with the same day's return it was partly computed from commits the data leakage Chapter 6 warned against, only harder to spot because the feature looks like an input rather than an answer.

Second, Chapter 7's stationarity material explains why every feature in the table is a *transformation* of price rather than the raw price level. A price can trend arbitrarily far from its starting value over a dataset (the non-stationarity of the unit-root discussion), so a model trained on raw price levels from one period would have to extrapolate to price ranges it never saw. Momentum, RSI, and volume ratios are normalized to a roughly stationary, comparable scale regardless of price level or period, which is why they, rather than price, are handed to a model.

From Features to Portfolio Construction

A quantitative equity manager typically does not predict an absolute return but trains a model to produce a cross-sectional ranking of a universe of stocks from most to least attractive over the next period, since a ranking is often easier to learn reliably and more useful for portfolio construction than a point forecast of each return. The ranking feeds into the mean-variance and risk-parity machinery of Chapter 12: the optimizer takes the model's scores as its expected-return input, combining a machine-learning forecast with classical risk-management discipline, the layered, classical-plus-modern pipeline emphasized throughout this book.

Beyond price and volume history, increasingly important feature sources are order-book microstructure data, such as the order-flow imbalance (the net difference between resting buy and sell volume at the top of the book) behind Kyle's model in Section 13.3.3, and the alternative data of Chapter 17, such as satellite imagery, web traffic, or transaction data, which can supply predictive information independent of the price and volume history every other participant observes.

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Overfitting and Walk-Forward Validation

The overfitting concern of Section 13.6.3 applies with particular force here, for an additional reason: a flexible model does not merely search a grid of parameters but an enormous implicit space of functional forms, so its effective N is larger than any explicit configuration count and harder to deflate for. Walk-forward validation, introduced in Chapter 7 because a random train-test split leaks future information in time-ordered data, is essential for any machine-learning trading signal. Even a walk-forward-validated signal must still be evaluated net of implementation shortfall and market impact before its statistical performance counts as a tradable economic edge.

Is This Signal Causal?

A signal that survives walk-forward validation has earned the right to be called predictive, not causal; Chapter 6's section on prediction versus causation (Section 6.4.6) explains the difference. For pure trading it often does not matter: a feature that reliably precedes returns is worth trading whether it moves prices or merely shares a cause with whatever does. It matters the moment the strategy's own behavior, or the market's, changes the relationship. A momentum signal may work because slow-moving capital chases recent winners; if a regime filter, a change in index rules, or other funds trading the same feature alter that flow, the correlation the model learned can vanish without any warning visible in its historical fit. The questions Chapter 6 asks of a credit-limit decision apply here too. What mechanism would make the feature move prices? Is there a confounder, such as market-wide volatility, that drives both the feature and the return? Would the relationship survive an intervention, such as the strategy itself trading at scale? A signal backed by a plausible mechanism, and still present after the obvious confounders are controlled for, is more likely to persist out of sample than one discovered only because it fit.

These methods apply across a wide range of horizons, from a multi-day pairs trade to an execution algorithm working an order over an afternoon. At the shortest end, high-frequency trading pushes this chapter's ideas (order types, market impact, inventory management) down to milliseconds or microseconds, where the physical latency of transmitting information and orders becomes a first-order concern. With its own technology, market-making techniques, and regulatory debates, high-frequency trading receives its own treatment in Chapter 14.

13.7.6 News and Event-Driven Trading

A distinct signal category trades on the arrival of new information: an earnings announcement, a central bank rate decision, a merger announcement, or a news headline. Because the content of such events is largely unpredictable (a firm's earnings-release date is known in advance, but not what it will say), event-driven strategies focus less on predicting when a signal will arrive and more on reacting to it faster and more accurately than other participants.

This is where the natural language processing methods of Chapter 16 meet this chapter's execution machinery. Extracting a sentiment score or structured summary from an earnings call transcript or news wire is only half of an event-driven strategy. The other half is converting that signal into a position quickly enough to matter (a slowly generated signal may arrive after the price has already moved) and executing the trade while managing the market impact and implementation shortfall of Sections 13.3.2 and 13.3.5. Those costs are often most severe right after a major news event, when liquidity is thin and many participants are trading on the same information at once.

13.8 Market Structure and Safeguards

Automated trading interacts with other automated trading, sometimes in ways nobody designed. This part covers the safeguards the market imposes on that interaction, and one episode in which a firm's own missing safeguards destroyed it in forty-five minutes.

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13.8.1 Circuit Breakers and Trading Halts

Chapter 3 described the 2010 Flash Crash, in which the Dow Jones Industrial Average fell and then recovered nearly 1,000 points within minutes, driven in significant part by algorithmic systems interacting in ways no participant had designed or anticipated. One regulatory response to that episode, and to the risk that automated trading can amplify a price move far beyond what fundamentals justify, is the circuit breaker: an automatic, exchange-wide (or single-stock) trading halt triggered once price moves beyond a threshold within a time window. It gives participants, human and algorithmic, a pause to assess whether a move reflects new information or a mechanical, self-reinforcing feedback loop like the Flash Crash.

Circuit breakers interact with the execution and market-making concepts of this chapter. A market maker running the Avellaneda-Stoikov framework of Section 13.4 must decide how to handle a position when trading is suddenly halted, since the model assumes it can trade continuously, an assumption a halt violates by design. Similarly, a TWAP or VWAP schedule spread across a day needs explicit logic for a halt partway through, either pausing and resuming or reassessing the remaining plan, since proceeding as though nothing happened could execute the remaining slices at prices that no longer reflect the conditions the schedule was designed for.

13.8.2 Case Vignette: The Knight Capital Trading Glitch

On August 1, 2012, Knight Capital Group, then one of the largest market makers in US equities, deployed new trading software to its production servers. A deployment error left obsolete code active on one of eight servers; combined with the new software, it caused the firm's systems to send a rapid, unintended stream of erroneous orders into the market. Within approximately 45 minutes, before the problem was identified and the system halted, Knight had accumulated large, unintended positions across dozens of stocks. Unwinding them cost about $440 million, more than its capital cushion could absorb; its share price fell by roughly three-quarters within two days, and the loss forced an emergency capital infusion and ultimately a rescue merger with a competitor.

The episode illustrates the technology and latency risk flagged in this chapter's challenges section, and the failure was not a flawed trading strategy. Knight's algorithms were not making a bad prediction or a poorly calibrated risk-reward trade-off; a deployment error caused the firm's execution infrastructure, the machinery this chapter has treated as neutral plumbing, to behave in a way no human intended or approved.

The speed that makes algorithmic and high-frequency trading profitable is what turned a deployment bug into a $440 million loss in well under an hour: a human-paced operation making the same error would likely have intervened long before losses reached that scale. The lesson generalizes to the model risk management framework of Chapter 8: an algorithmic trading system requires the same independent testing, staged deployment, real-time monitoring, and kill-switch capability as a bank's VaR or credit model, because the speed that makes automation valuable makes an undetected error catastrophic.

13.9 Challenges in Algorithmic Trading and Market Microstructure

Several challenges recur across nearly every algorithmic trading application.

Backtest overfitting. With enough data and candidate strategies, some rule will appear profitable by chance, the multiple-testing concern raised in Chapters 6, 7, and 12. Section 13.6.3 put a number on it: searching a hundred worthless configurations yields an expected best in-sample Sharpe ratio of 1.12, and the winner has no out-of-sample edge. A strategy found by searching many parameter combinations must be validated out of sample, and deflated for the size of the search, far more skeptically than one derived from a pre-specified economic hypothesis.

Market impact is itself uncertain and strategy-dependent. The clean numbers in Section 13.3.2 come from a simplified, static order book; real impact depends on volatility, on whether others can detect and react to a large order being worked, and on liquidity conditions that vary over time, making it harder to estimate than a strategy's other inputs.

Latency and technology risk. Execution and market-making strategies depend on fast, reliable technology, and a software bug, a connectivity failure, or delayed market data can turn a sound strategy

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into a source of real, sometimes rapid, losses: the operational risk of Chapter 10 in a domain where a failure compounds within seconds rather than days.

Adverse selection from faster participants. A posted limit order risks being filled by a faster or better-informed participant when that is disadvantageous, the adverse-selection component of the bid-ask spread introduced in Chapter 3; the more of this risk a strategy bears, the wider the effective spread it must earn to break even.

Regulatory and market-structure change. Algorithmic and high-frequency trading operate within an evolving regulatory framework (Regulation NMS and its analogues), and a strategy whose profitability depends on a specific market-structure feature, such as an exchange's fee structure or order-matching rule, can see that edge disappear or reverse if the rules change.

Crowding in statistical arbitrage. The 2007 quant quake of Section 13.5.2 is a documented instance of a general hazard. When many firms independently discover and trade similar statistical relationships, their positions become correlated in a way none can observe from its own book. A shock or forced unwind at one large fund can then trigger simultaneous, self-reinforcing losses across ostensibly unrelated strategies, the crowding risk Chapter 12 raised for long-horizon factor investing, unfolding over hours rather than months.

Exchange fee structures and rebates. Many exchanges charge a fee to remove liquidity (executing against a resting order) and pay a rebate to add it (posting a resting limit order that another participant later executes against). This maker-taker pricing model creates incentives independent of the trading signal. Profitability can depend materially on whether a strategy posts passive limit orders or crosses the spread, and two strategies with identical signals can have different net profitability purely because of how each interacts with an exchange's fee schedule. A realistic backtest, following the transaction-cost discipline of Section 13.6.1, should account for fees and rebates explicitly, since a few basis points of fee difference, compounded across thousands of trades, can separate a strategy that is profitable after costs from one that only appears to be. The fee schedule is one more component of implementation shortfall, no less real for being easy to overlook.

Capacity constraints on strategy scale. A strategy backtested at a small notional size can look highly attractive on a risk-adjusted basis. But the market impact costs of Section 13.3.2 grow faster than linearly with order size, as $Q^{3/2}$ in total dollars under the square-root law of Section 13.3.7. The same strategy at ten or a hundred times the capital can see its net returns deteriorate sharply, or turn negative, well before it exhausts its statistical edge; that section's example supports at most $61.4 million profitably. Capacity is an economic limit on how much capital any trading idea, however sound, can absorb.

13.10 Key Takeaways

Converting a trading signal into a position, and a position into a profit-and-loss series, means confronting the microstructure realities of Chapter 3: the bid-ask spread, the limited depth at each price level, and the price impact of consuming that depth too quickly. The Kyle model interprets price impact as compensation for trading against better-informed counterparties; TWAP and VWAP are standard execution benchmarks; and the Almgren-Chriss framework formalizes the trade-off between market impact and timing risk.

Market making becomes an explicit inventory-management problem, and pairs trading becomes a complete, backtestable strategy once entry and exit rules are applied out of sample, a logic that extends to statistical arbitrage across many assets using the factor models of Chapter 12. The execution and market-making models are closed-form answers bought with strong assumptions, and Chapter 19 revisits each as a sequential decision problem solved by learning rather than algebra.

Evaluating a strategy requires more than a return figure: the Sharpe ratio, maximum drawdown, and hit rate each reveal a different dimension of performance, and each must be computed net of realistic transaction costs and implementation shortfall, since the gap between theoretical and realized performance often separates a strategy that looks good on paper from one worth trading.

The Knight Capital vignette and the discussion of circuit breakers make a point that recurs throughout this book: the technology and market structure around a trading decision are not neutral plumbing but sources of risk in their own right, deserving the governance discipline Chapter 8 developed for risk models.

A statistically sound backtest is necessary but never sufficient. Fees, capacity limits, crowding, and the risk of a software or operational failure all sit between a promising historical result and a profitable, sustainable trading operation. The machine learning signals of Chapter 6 and the natural-language and alternative-data

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sources of Chapters 16 and 17 must clear the same hurdles before a sophisticated model becomes a trade that makes money net of these frictions.

The next chapter narrows the focus to the millisecond-scale end of this spectrum, where every idea introduced here still applies but is compressed into a timescale where speed itself becomes a primary competitive advantage.

13.11 Suggested Exercises

13.1 Using Table 13.1, verify by hand that the 3-day and 6-day moving averages at day 8 are 117.33 and 111.83, and explain why the strategy signal remains long (1) at that point.

13.2 Recompute the moving-average crossover strategy's cumulative return using a 2-day short average and a 4-day long average on the same twelve-day price series, and compare the resulting signal timing to the 3-day/6-day version in the text.

13.3 A 500-share market order walks through an order book with 200 shares available at \$30.00 and additional shares available at \$30.05. Compute the average execution price and the price impact in basis points relative to the best price.

13.4 Using Table 13.3, compute the TWAP and VWAP execution prices for a 2,000-share order, and explain why they differ.

13.5 Explain, in your own words, why an inventory-skewing market maker (Section 13.4) adjusts both its bid and ask when it accumulates a large position, rather than adjusting only the side of the quote nearest the current inventory.

13.6 Using the extended pairs-trading data in Table 13.4, compute the trade's profit and loss if the position were instead closed at day 9 rather than day 10, and compare it to the day-10 exit computed in the text.

13.7 A strategy has a Sharpe ratio of 1.2 and a maximum drawdown of 18%, while a second strategy has a Sharpe ratio of 0.8 and a maximum drawdown of 5%. Discuss which strategy an investor with a low tolerance for large losses might prefer, and why Sharpe ratio alone would not reveal this consideration.

13.8 A portfolio manager's decision price is \$100.00 and the resulting order is executed at an average price of \$100.12 on a 5,000-share order. Compute the implementation shortfall in basis points and in total dollars.

13.9 Using Table 13.7's twelve-day price series, try to compute the 3-day momentum, 3-day realized volatility, and 3-period RSI at day 3, and explain why none of them is yet defined there under the chapter's definitions, while all three first become available at day 4.

13.10 Explain, referencing Table 13.7's RSI column, why an indicator that saturates at 0 or 100 can lose information a model might otherwise use, and describe why feeding a model both RSI and the underlying momentum feature side by side addresses this limitation.

13.11 Using Table 13.8, verify by hand that day 8's APO is 3.600, and explain why APO does not turn negative until day 11, four days after day 7's price high, rather than exactly at it.

13.12 Using Table 13.8, explain why the band width at day 7 (Upper minus Lower) is so much larger than at day 2, referencing this section's discussion of Table 13.7's Vol3 column.

13.13 Explain why walk-forward validation, introduced in Chapter 7, is essential when evaluating a machine-learning-based trading signal, and describe a specific way that a random train-test split could produce a misleadingly optimistic backtest.

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13.14 Pick two of the challenges listed in this chapter and describe a concrete scenario, not already used in the text, in which that challenge could cause an algorithmic trading strategy to lose money despite a statistically sound backtest.

13.15 Using Table 13.2, verify by hand that Kyle's lambda is approximately 1.95 cents per thousand shares, and interpret what a much larger estimated lambda (say, 10 cents per thousand shares) would suggest about that market compared to the one in the text.

13.16 Explain the difference between an iceberg order and trading in a dark pool, and describe one advantage and one disadvantage each has relative to posting a single, fully visible limit order.

13.17 Describe how a market maker using the inventory-skewing logic of Section 13.4 should adjust its behavior if trading in the underlying stock is suddenly halted by a circuit breaker, and explain what risk this halt introduces that continuous trading does not.

13.18 Using the Avellaneda-Stoikov formulas in Section 13.4, recompute the reservation price, optimal spread, bid, and ask for a market maker with the same parameters as the worked example except that it is short $q = -8$ units instead of long 5, and explain why the resulting skew now points in the opposite direction.

13.19 Explain why the Knight Capital case vignette is better understood as a technology and governance failure than as a bad trading strategy, and describe one specific model risk management practice from Chapter 8 that, if followed, might plausibly have prevented or limited the loss.

13.20 A statistical arbitrage fund holds 200 similar mean-reversion positions across a sector. Explain, referencing the 2007 quant quake, why this diversification might provide less protection than the number of positions alone would suggest.

13.21 Using Table 13.5, explain why Stock W's large residual z-score alone does not tell a trader whether to put on the suggested trade, and describe what additional information (beyond the regression output) an analyst would want before shorting Stock W against its sector peers.

13.22 A backtest of a value strategy is run only on stocks currently listed on a major exchange, using their full, publicly available price history. Identify the specific bias this introduces, referencing this chapter's discussion of backtesting pitfalls, and describe one concrete step a researcher could take to correct it.

13.23 Using the Almgren-Chriss worked example, compute κ and the resulting optimal trajectory x_j for the same 250-share order and $N = 5$ intervals if risk aversion were $\phi = 2$ instead of 0.5 or 5, and state whether the resulting schedule is more or less front-loaded than both trajectories computed in the text.

13.24 Explain, in your own words, why the fully aggressive single-shot execution in the Almgren-Chriss worked example has zero timing-risk variance, and why this alone does not make it the risk-adjusted optimal strategy even for a highly risk-averse trader.

13.25 **VWAP under a mistaken volume forecast.** Section 13.3.4 argues that a VWAP algorithm is only as good as its forecast of the day's volume curve. Using Table 13.3's prices and realized volumes, suppose an algorithm works a 1,000-share buy order according to a forecast volume profile of 15%, 20%, 25%, 15%, and 25% across the five intervals. (a) Compute the realized-volume VWAP benchmark and the average price the algorithm actually achieves. (b) Report the shortfall against that benchmark in cents per share, in basis points, and in dollars on the full order. (c) Identify which intervals' forecast errors caused the shortfall and which one partially offset it, and explain why an error in an interval whose price sits close to the VWAP costs almost nothing.

13.26 Using the square-root law and the capacity example of Section 13.3.7, suppose the strategy's gross alpha per trade is 30 rather than 50 basis points, with everything else unchanged. (a) Compute the impact cost of a trade equal to 2% of a stock's daily volume. (b) Find the assets under management

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at which net alpha falls to zero, and the size that maximizes dollar profit. (c) Explain why a 40% cut in gross alpha shrinks the profit-maximizing fund size by much more than 40%.

13.27 A stock trades at \$20 with an annual return volatility of 30%, and an execution desk splits each trading day into ten intervals. Convert this volatility into the units the Almgren-Chriss objective of Section 13.3.5 requires, and explain what would go wrong if $\sigma = 0.30$ were entered into the formula directly.

13.28 A researcher finds that stocks with a high order-flow imbalance at the close tend to rise the next morning, and the signal survives walk-forward validation. Following this chapter's discussion of whether a signal is causal, name one mechanism that would make the relationship causal, one confounder that could produce it without any causal link, and one change in market conditions under which you would expect it to stop working.

13.29 **Challenge (real data).** Download at least two years of daily prices for a real, liquid stock (via `yfinance`) and implement this chapter's moving-average crossover strategy (Table 13.1) using a 20-day short window and a 50-day long window. (a) Backtest the strategy's cumulative return against a simple buy-and-hold benchmark over the full sample, and compute both strategies' Sharpe ratios. (b) Using walk-forward validation (Chapter 7) rather than a single in-sample backtest, split your sample into several sequential training/testing windows and report whether the strategy's out-of-sample Sharpe ratio holds up as well as its in-sample Sharpe ratio. (c) Compute the feature set in Table 13.7 (momentum, realized volatility, RSI) for your real price series, and discuss whether the volatility clustering the twelve-day example showed around its crossover reversal appears at your strategy's entry and exit points.

13.12 Further Reading

- Almgren and Chriss (2001), "Optimal Execution of Portfolio Transactions." The original formalization of the market-impact-versus-timing-risk trade-off developed in Section 13.3.5.
- Almgren et al. (2005), "Direct Estimation of Equity Market Impact." An estimation of permanent and temporary impact from a large set of institutional U.S. equity orders, the empirical counterpart of Section 13.3.7's square-root law.
- Tóth et al. (2011), "Anomalous Price Impact and the Critical Nature of Liquidity in Financial Markets." A theory of why metaorder impact grows with the square root of size, together with fresh empirical support for the law used in Section 13.3.7.
- Appel (2005), *Technical Analysis*. Written by the creator of the MACD indicator, the closely related, signal-line-smoothed cousin of the Absolute Price Oscillator used in Section 13.7.4's feature-engineering worked example.
- Avellaneda and Stoikov (2008), "High-Frequency Trading in a Limit Order Book." The original inventory-based market-making model summarized in Section 13.4.
- Bailey and López de Prado (2014), "The Deflated Sharpe Ratio: Correcting for Selection Bias, Backtest Overfitting, and Non-Normality." The source of the deflation adjustment recommended in Section 13.6.3, and the standard reference on backtest overfitting.
- Bollinger (2001), *Bollinger on Bollinger Bands*. Written by the creator of the Bollinger Bands indicator used in Section 13.7.4's feature-engineering worked example.
- Chan (2013), *Algorithmic Trading*. A practitioner-oriented treatment of strategy development and backtesting, including the pairs-trading and mean-reversion strategies extended in this chapter.
- Harris (2003), *Trading and Exchanges*. A comprehensive treatment of order types, market structure, and execution that extends this chapter's introductory coverage and connects directly back to Chapter 3.

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- Kissell (2013), *The Science of Algorithmic Trading and Portfolio Management*. Covers transaction cost analysis and implementation shortfall, introduced in this chapter, in considerably greater quantitative depth.
- Kyle (1985), "Continuous Auctions and Insider Trading." The original paper formalizing the price-impact model summarized in Section 13.3.3.
- Wilder (1978), *New Concepts in Technical Trading Systems*. The original source introducing the Relative Strength Index used in Section 13.7.3's feature-engineering worked example.
- Guéant et al. (2013), "Dealing with the Inventory Risk: A Solution to the Market Making Problem." A rigorous, closed-form treatment extending the Avellaneda-Stoikov inventory model of Section 13.4 to more realistic order-arrival assumptions.
- Patterson (2012), *Dark Pools*. A journalistic but well-researched account of the rise of dark pools, algorithmic trading, and the market-structure debates touched on in this chapter's discussion of order routing and venue choice.

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Chapter 14

High-Frequency Trading

14.1 Introduction

Chapter 13 developed execution algorithms, market impact, and inventory-based market making at the pace of orders worked over minutes or hours. This chapter pushes those ideas down to milliseconds and microseconds, where the time it takes information and orders to travel between computers matters as much as the economic logic of what those orders say. High-frequency trading (HFT) is not a separate branch of finance with its own theory. It is the same market making, statistical arbitrage, and order-routing machinery at a timescale where speed itself is a competitive advantage, and where the operational risks of the Knight Capital vignette become the central concern.

Several worked examples run through the chapter. A latency-arbitrage calculation puts a dollar figure on what a faster participant can extract from a slower one during the window a speed advantage creates. A Glosten-Milgrom sequential-trade model and its probability-of-informed-trading (PIN) statistic extend Chapter 1's Bayes'-theorem introduction to a market maker updating its beliefs trade by trade. A market-making profitability example extends Chapter 13's Avellaneda-Stoikov inventory model to a full day of quote refreshes, showing how thin per-trade spread capture adds up to a meaningful daily profit, and how adverse selection erodes it. Further examples cover ETF-basket and cross-market statistical arbitrage, a co-location break-even calculation, and an order-flow-imbalance regression that extends Chapter 13's Kyle's-lambda framework to a predictive setting.

The race to zero latency (Section 14.2.2) explains why microseconds are worth paying for. The latency-arbitrage and Glosten-Milgrom examples (Sections 14.3.1 and 14.3.2) quantify what that speed is worth, and the inventory and daily P&L sections (Sections 14.4.1 and 14.4.2) turn a single quote refresh into a full day's economics. Later sections turn to market plumbing and regulation: queue position and price-time priority (Section 14.5.5), the order-flow-imbalance regression a desk would use to anticipate the next tick (Section 14.7.1), and the risk controls, flash-crash case study, and regulatory debate that ask whether speed makes markets better or merely faster.

14.2 The High-Frequency Landscape

High-frequency trading is less a single strategy than a family of them sharing one constraint: they must act before anyone else can. This part establishes what the label covers, and why the pursuit of speed became an arms race with a physical floor.

14.2.1 What High-Frequency Trading Is, and Why It Matters

Suppose two firms are both trying to trade the same stock a fraction of a second after the same piece of market-moving news breaks. One firm's edge is not a better forecast of where the stock is headed next quarter; it has no such forecast. Its servers sit physically closer to the exchange's, and its software reacts in microseconds rather than milliseconds. High-frequency trading describes a cluster of practices built around winning this

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kind of race: extremely short holding periods (often seconds or less), high order-to-trade ratios (many quotes posted and canceled for every one that executes), and reliance on speed, low-latency infrastructure, and often co-location at exchange data centers, rather than superior information about fundamental value. HFT firms are not trying to predict next quarter's price, or even next minute's. They are trying to react to information (an order-book update, a trade print, a price change on a correlated instrument) faster than anyone else.

This distinction reframes what HFT contributes to and extracts from a market. A large share of HFT activity is market making in the sense of Chapters 3 and 13: posting both a bid and an ask, earning the spread, and managing the inventory risk (the Avellaneda-Stoikov problem), at a much higher frequency and with much shorter holding periods. A second category is latency arbitrage: detecting a price change on one venue or instrument and trading on a slower-to-update one before its quotes catch up. A third, more controversial category depends on other participants' reaction times or order-handling logic, sometimes shading into manipulative practices such as spoofing (posting orders with no intention of executing them, to influence others' behavior), which is illegal in most jurisdictions and discussed in this chapter's regulatory section.

Table 14.1 organizes the main strategy categories together with the economic mechanism each exploits and where this chapter develops it further.

Strategy type	Economic mechanism	Developed in
Market making	Earn the spread; manage inventory risk with continuously updated quotes	Sections 14.4.1, 14.3.2
Latency arbitrage	Trade against stale quotes before they update, exploiting a pure speed advantage	Section 14.3.1
Statistical arbitrage	Trade a fast, model-implied relationship between related instruments before it reverts	Chapter 13's Section on statistical arbitrage, extended to microsecond horizons
Manipulative practices	Create a false impression of supply or demand to move prices, then trade on the resulting move	This chapter's regulatory case vignette

Table 14.1: A taxonomy of high-frequency trading strategies

Every category in Table 14.1 except the last shares a common structure with the book's classical models: a well-specified economic mechanism (spread capture, speed-based adverse selection, or statistical mean reversion) that can be formalized and measured. The last category differs in kind, not degree. It is not an aggressive trading strategy but conduct that securities and commodities law define as manipulative, however it is implemented.

14.2.2 The Race to Zero: Latency, Co-location, and the Physics of Speed

At millisecond and microsecond timescales, the speed of light becomes a binding physical constraint on trading, not a metaphor. Two consequences follow. First, HFT firms pay substantial fees for co-location, placing their servers inside or adjacent to an exchange's data center, since every meter of cable adds a delay that matters at this timescale. Second, firms connecting two distant trading hubs have raced to use whatever medium transmits data fastest, since a firm whose data arrives a few milliseconds sooner can act on a price-moving event before a slower competitor has even received it.

A calculation shows why this is worth real money. Signals in fiber-optic cable travel at only about 68% of the speed of light in vacuum, because the glass core has a refractive index of about 1.47, which slows light to roughly $c/1.47$; a microwave signal transmitted through open air, by contrast, travels at close to the vacuum speed of light, about 99.97% of c. Over a corridor of roughly 1,200 kilometers (comparable to the distance

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between major financial hubs such as New York and Chicago), the round-trip transmission times are

$$t_{\text{fiber}} = \frac{2 \times 1{,}200 \text{ km}}{0.68c} \approx 11.77 \text{ ms}, \qquad t_{\text{microwave}} = \frac{2 \times 1{,}200 \text{ km}}{0.9997c} \approx 8.01 \text{ ms},$$

a difference of roughly 3.8 milliseconds. This is why firms have invested heavily in microwave and even millimeter-wave relay networks between trading hubs despite their weather sensitivity and lower bandwidth: a persistent few-millisecond advantage lets whoever owns or leases the fastest link act on new information before competitors, an advantage the latency-arbitrage example below prices in dollars.

14.3 Latency Arbitrage and Informed Trading

The simplest way to profit from speed is to trade against a price somebody else has not yet updated. This part works that mechanism through numerically, then asks what a market maker facing possibly-informed counterparties should do about it.

14.3.1 Worked Example 14.1: Latency Arbitrage and Stale Quotes

Consider a stock quoted at a bid of \$99.95 and an ask of \$100.05 by two market makers, Firm S (slow, with a data-to-order latency of 500 microseconds) and Firm F (fast, with a latency of 50 microseconds). Suppose a piece of public news causes the stock's fundamental value to jump instantly to \$100.10. Firm F's systems detect this and cancel or update its stale quotes within 50 microseconds; Firm S takes 500 microseconds to do the same, leaving its old ask of \$100.05 executable for an extra $500 - 50 = 450$ microseconds after the price has already moved.

During that window, Firm F can buy against Firm S's stale \$100.05 ask, immediately holding shares worth \$100.10: a risk-free profit of \$0.05 per share, owed purely to the 450-microsecond latency gap and not to any view about the stock. Figure 14.1 lays out the sequence. If Firm F can execute 5,000 shares against Firm S's stale quote before it updates, and this kind of event happens roughly 20 times in a trading day across F's traded universe,

$$\text{Profit per event} = 5{,}000 \times \$0.05 = \$250, \qquad \text{Daily profit} = 20 \times \$250 = \$5{,}000.$$

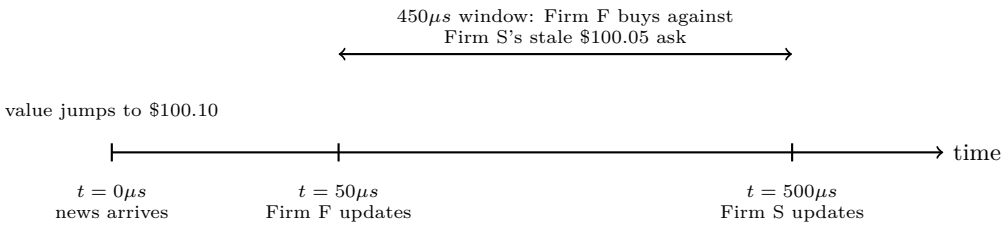


Figure 14.1: The sequence of events behind the latency-arbitrage worked example (not drawn to scale)

For Firm S this is not a mysterious loss but the quantifiable cost of adverse selection from Chapter 13's discussion of Kyle's model. Its quotes are being picked off by counterparties with better information, here faster rather than more informed in the traditional sense. The \$0.05 per-share loss is the kind of cost the inventory-holding and adverse-selection components of the bid-ask spread (Chapter 3) exist to compensate for. A slower market maker that ignores this risk will find its realized spread capture persistently below its quoted spread, because a predictable fraction of its fills are the ones a faster competitor chose to make happen.

14.3.2 A More Advanced Model: Sequential Trade and Bayesian Learning

Chapter 13's Kyle model treats informed and uninformed order flow as arriving in a single batch and derives a single price-impact coefficient, λ, from it. The Glosten-Milgrom model is better suited to how a high-frequency

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market maker operates: trades arrive one at a time, and the market maker updates its beliefs about the asset's true value after each one, using the Bayesian updating rule of Chapter 1's fraud-detection example.

Suppose an asset's true value V is equally likely, before any trading, to be $V_H = \$51$ or $V_L = \$49$, so the unconditional expected value is \$50. A fraction $\pi = 10\%$ of order flow comes from informed traders, who know V exactly and buy when $V = V_H$ and sell when $V = V_L$. The remaining 90% comes from uninformed traders, who buy or sell with equal probability regardless of V.

A competitive market maker sets its ask equal to the asset's expected value *conditional on the next order being a buy*, and its bid equal to the expected value conditional on a sell, since each quote must break even on average against a mix of informed and uninformed counterparties. Using Bayes' theorem,

$$\Pr(V_H \mid \text{Buy}) = \frac{\Pr(\text{Buy} \mid V_H)\ \Pr(V_H)}{\Pr(\text{Buy})} = \frac{[\pi(1) + (1-\pi)(0.5)](0.5)}{0.5} = 0.55, \tag{14.1}$$

so observing a buy order raises the market maker's belief that $V = V_H$ from the 50% prior to a 55% posterior, and the resulting break-even quotes are

$$\text{Ask} = 0.55(51) + 0.45(49) = \$50.10, \qquad \text{Bid} = 0.45(51) + 0.55(49) = \$49.90,$$

a \$0.20 spread. This is not an arbitrary markup. In this two-value setup it equals $\pi(V_H - V_L) = 0.10 \times \$2.00 = \$0.20$: the entire spread compensates for adverse selection risk, and a higher fraction of informed traders π widens it, the same conclusion Kyle's λ reaches by a different route.

The model's power is in what happens next: today's posterior becomes tomorrow's prior, the recursive logic behind Chapter 7's Kalman filter. If a second order also arrives as a buy, the market maker updates again, starting from the 55% prior rather than 50%,

$$\Pr(V_H \mid \text{Buy}, \text{Buy}) = \frac{[\pi(1) + (1-\pi)(0.5)](0.55)}{0.505} \approx 0.599,$$

where the denominator $0.55(0.55) + 0.45(0.45) = 0.505$ is the probability of a second buy under the updated prior, no longer the exact one-half of the symmetric first round. The revised posterior pushes the next ask up to approximately $0.599(51) + 0.401(49) \approx \$50.20$ and the next bid up to exactly \$50.00.

Two consecutive buy orders, with no other information, have moved the whole quote range up by about ten cents. A market maker who cannot tell informed from uninformed flow should rationally react this way: each additional buy makes it more likely that informed buyers who know $V = V_H$ dominate the flow, and the quotes drift toward V_H. This is the mechanism behind the observation that a market maker (or an algorithm) "leans into" a run of one-directional trades, and it is why high-frequency market-making systems continuously re-estimate fair value from recent order flow rather than quoting a static price all day.

The Probability of Informed Trading (PIN)

Glosten-Milgrom updates beliefs trade by trade; a related model by Easley, Kiefer, O'Hara, and Paperman condenses the same idea into a single daily statistic, the probability of informed trading. On any given day, an information event occurs with probability α, in which case informed traders arrive at rate μ; uninformed buyers and sellers arrive continuously at rate ε each, information event or not. Averaging over these arrival rates gives

$$\text{PIN} = \frac{\alpha\mu}{\alpha\mu + 2\varepsilon}. \tag{14.2}$$

For a stock with $\alpha = 30\%$, $\mu = 150$ informed trades per day on event days, and $\varepsilon = 100$ uninformed buy (and separately, sell) arrivals per day,

$$\text{PIN} = \frac{0.30(150)}{0.30(150) + 2(100)} = \frac{45}{245} \approx 18.4\%,$$

meaning roughly 18% of this stock's trades are estimated to come from informed traders: a single summary of the adverse-selection risk that Section 14.3.1 and the Glosten-Milgrom example quantified trade by trade. A high-frequency market maker uses PIN differently. Rather than moving quotes after every trade, PIN (re-estimated periodically, since it needs enough trades to estimate reliably) sets how aggressively to widen quotes and how much inventory risk to tolerate in a name, a slower-moving risk parameter layered on top of the trade-by-trade adjustments.

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14.3.3 Statistical Arbitrage at High Frequency: ETF-Basket Arbitrage

Chapter 13 introduced statistical arbitrage as trading a model-implied relationship between related instruments, illustrated with a multi-day pairs trade. At high frequency, the most common version exploits a more mechanical relationship. An exchange-traded fund (ETF) is by construction a claim on a specified basket of securities, and creation and redemption lets authorized participants exchange ETF shares for the basket (or vice versa) at the fund's net asset value, which tethers the ETF's price to the real-time fair value of its holdings. This is the no-arbitrage logic Chapter 3 introduced for a risk-free bond and a certain-payoff claim.

Suppose an ETF trades at \$150.05 while the real-time, weighted fair value of its underlying basket of stocks is \$150.00, a \$0.05 premium (0.033%). An HFT firm can sell the (relatively expensive) ETF and simultaneously buy the (relatively cheap) underlying basket, or a close proxy such as index futures, capturing the premium as the two prices converge, through creation-redemption or ordinary trading pressure. On a \$10 million position, this captures

$$\$10{,}000{,}000 \times 0.033\% \approx \$3{,}333,$$

and if smaller versions of this same mispricing recur roughly 100 times in a trading day, at an average position of \$200,000 each,

$$\text{Daily profit} \approx 100 \times (\$200{,}000 \times 0.033\%) \approx \$6{,}667.$$

As in the no-arbitrage bond example, two claims on the same cash flows must trade at the same price. The strategy needs HFT-level speed because the mispricing is small and short-lived. A participant slower than the fastest competitors will find the gap closed by the time its order arrives: the stale-quote dynamic of Section 14.3.1, applied to a relationship between two instruments.

14.4 High-Frequency Market Making

Chapter 13's inventory model assumed a dealer with time to think. This part pushes the same machinery down to horizons where quotes are revised thousands of times a day, and works a full day's profit and loss.

14.4.1 Market Making at High Frequency: Extending the Inventory Model

Chapter 13's Avellaneda-Stoikov example computed a market maker's optimal quote skew using a time horizon $T - t$ of a full unit (one day, in that illustration). High-frequency market making applies the same formula,

$$r = s - q\,\gamma\,\sigma^2\,(T - t), \tag{14.3}$$

but with $T - t$ a tiny fraction of that horizon, since an HFT market maker refreshes its fair value and quotes continuously rather than once per day. Holding inventory q, risk aversion γ, and volatility σ fixed at Chapter 13's values ($q = 5$, $\gamma = 0.1$, $\sigma = 2$), shrinking the horizon from $T - t = 1$ (one full day) to $T - t = 0.001$ (roughly one-thousandth of a day, on the order of a minute) shrinks the reservation-price skew proportionally,

$$\text{skew}(T{-}t = 1) = 5(0.1)(2^2)(1) = 2.00, \qquad \text{skew}(T{-}t = 0.001) = 5(0.1)(2^2)(0.001) = 0.002,$$

a full 1,000-fold reduction. This is the mathematical reason high-frequency market making holds little inventory for little time. Because the inventory-risk penalty scales linearly with the remaining horizon, a market maker who refreshes quotes (and ideally flattens inventory) thousands of times per day can run with much higher risk aversion or larger momentary inventory than one refreshing once per day: each exposure is held for a far shorter time. It is also why the technology investments of Section 14.2.2 pay. A firm that updates quotes and manages inventory a few hundred microseconds faster operates with a shorter $T - t$, hence a smaller optimal skew for the same risk, so it can quote a tighter effective spread while bearing the same risk per unit of time.

Does the *total* optimal spread, not just the inventory skew, shrink by the same factor? The two are easy to conflate. Recall Chapter 13's optimal spread formula,

$$\delta = \gamma\sigma^2(T - t) + \left(\frac{2}{\gamma}\right)\ln\left(1 + \frac{\gamma}{\kappa}\right), \tag{14.4}$$

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has two different terms: an inventory-risk term that scales with $T - t$ like the skew above, and an order-arrival-elasticity term governed by κ that does not depend on $T - t$ at all. Holding $\kappa = 1.5$ fixed alongside the same $\gamma = 0.1$ and $\sigma = 2$, the total spread at $T - t = 1$ is

$$\delta = 0.1(4)(1) + \left(\frac{2}{0.1}\right)\ln\left(1 + \frac{0.1}{1.5}\right) \approx 0.400 + 1.291 = 1.691,$$

matching Chapter 13's \$1.69 figure, while at $T - t = 0.001$ it is $\delta \approx 0.0004 + 1.291 = 1.291$, a reduction of only about 23.6%, not the 99.9% reduction the skew alone showed. The elasticity term, 1.291, is a floor the total spread cannot shrink below however short the inventory-holding horizon. It compensates for a different risk, how sensitively order arrivals respond to quote competitiveness, which has nothing to do with how long inventory is held.

Refreshing quotes thousands of times per day all but eliminates the inventory-driven component of the spread, but not the spread itself. A competitive HFT quote must still compensate for adverse selection from faster or better-informed counterparties, the risk that Section 14.3.2's Glosten-Milgrom model and Section 14.3.1's stale-quote example quantify from two angles.

14.4.2 Worked Example 14.2: Daily Profit and Loss from High-Frequency Market Making

Extending Section 14.4.1's logic to a trading day, suppose a high-frequency market maker in a liquid stock completes 50,000 fills over the course of one day, each fill one side of a round trip (a buy soon offset by a sell, or vice versa), so 25,000 round trips in all, at an average size of 100 shares. A round trip earns the full spread of \$0.01 per share, i.e. a half-spread of \$0.005 per share on each side, so

$$\text{Gross spread capture} = 50{,}000 \times 100 \times \$0.005 = \$25{,}000.$$

Not every fill is profitable, however: suppose 5% of these fills are with informed ("toxic") counterparties, the other side of Section 14.3.1's latency arbitrage, and that these trades lose an average of \$0.02 per share rather than earning the half-spread,

$$\text{Adverse selection loss} = (50{,}000 \times 0.05) \times 100 \times \$0.02 = \$5{,}000,$$

leaving a net daily profit of \$25,000 − \$5,000 = \$20,000, with adverse selection consuming 20% of gross spread capture. This is the day-to-day version of the trade-off Chapter 13's Kyle model formalized: realized profitability depends not just on the quoted spread and volume, but on the fraction of volume from faster or better-informed counterparties. That is why market makers invest in the technology of Section 14.2.2. Lower latency does not just create offensive latency-arbitrage opportunities; it reduces how often one's own quotes are the stale ones picked off.

14.5 The Economics of Speed

Speed costs money, and how much it is worth has a numerical answer. This part prices the infrastructure, the message traffic, and the queue position that high-frequency strategies depend on, alongside the case in which the pursuit crossed into manipulation.

14.5.1 Cross-Market Latency Arbitrage: Futures Leading the Cash Market

Section 14.3.1 illustrated latency arbitrage within a single stock's quotes, and the ETF-basket example extended it to an ETF and its basket. A third, equally common form exploits a well-documented fact: broad-market index futures, such as the E-mini S&P 500 contract, typically incorporate new macroeconomic information a few milliseconds before the cash market (index-member stocks and index ETFs such as SPY). Futures are highly liquid, traded in one venue, and heavily used by macro-sensitive participants reacting to news in real time.

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Suppose index futures jump by 2.00 index points on a surprise economic release, from 5,000.00 to 5,002.00. Since an index ETF such as SPY is constructed to track roughly one-tenth of the underlying index level, this implies a fair-value move in the ETF of approximately $2.00/10 = \$0.20$. If SPY's quotes take, say, 300 microseconds longer to reflect this than futures, a firm monitoring futures can buy SPY at its stale price and capture the difference once SPY's quotes catch up.

On a 1,000-share trade,

$$\text{Profit per event} = 1{,}000 \times \$0.20 = \$200, \qquad \text{Daily profit at 30 events/day} = 30 \times \$200 = \$6{,}000.$$

This is one of the most important reasons HFT firms watch futures order books as a leading indicator for equity and ETF trading. It also explains why equity index ETFs' quoted spreads often widen in the seconds after major scheduled economic releases. Slower cash-market makers, aware that faster participants watching futures may already know the "correct" new price, protect themselves by quoting wider until their systems catch up: the adverse-selection-driven widening that Chapter 3 and the Glosten-Milgrom model predict after any event that raises the perceived share of informed flow.

14.5.2 The Economics of Co-location: A Break-Even Calculation

Section 14.2.2 called co-location a necessary investment without quantifying when it pays for itself. Suppose an exchange charges \$25,000 per month for a co-located server rack and \$10,000 per month for a direct, low-latency market data feed (rather than the slower, free consolidated feed most retail-facing brokers use), a total fixed cost of \$35,000 per month. Using the \$250-per-event profit figure from Section 14.3.1, the firm needs only

$$\frac{\$35{,}000}{\$250} = 140 \text{ profitable latency events per month, or about 6.7 per trading day,}$$

to break even on the infrastructure cost.

If the firm instead captures latency-arbitrage opportunities at the rate assumed earlier, 20 events per day across a 21-trading-day month, its monthly profit from this one mechanism is $20 \times 21 \times \$250 = \$105{,}000$, three times the infrastructure cost. This is why co-location and direct feeds, despite their recurring cost, are essential infrastructure rather than a discretionary expense for latency-sensitive strategies: the break-even bar is low relative to what a persistent speed advantage can capture, and exchanges have a commercial incentive to keep offering and upgrading these paid advantages. The \$10,000-per-month direct feed does double duty. It also makes the SIP-versus-direct-feed arbitrage of Section 14.6.1 possible, since a firm on the slower consolidated feed cannot see the stale-quote window that mechanism exploits. Adding that mechanism's \$3,000-per-day profit (over 21 trading days, \$63,000 per month) to the \$105,000 from latency arbitrage brings the monthly return on the same \$35,000 investment to \$168,000, nearly five times its cost. The broader pattern: one fixed infrastructure investment typically unlocks several smaller sources of latency-based profit at once, which is part of why the large fixed costs of competing at this level are easier to justify than they first appear.

14.5.3 Case Vignette: Prosecuting Spoofing

Not every controversial high-frequency practice is merely aggressive; some are illegal outright, and the legal line is worth locating precisely. Spoofing means placing a large, visible order with no intention of letting it execute, to create a false impression of supply or demand that other participants (increasingly, other algorithms) react to, then canceling it moments later. It was made explicitly illegal in U.S. futures markets under the Dodd-Frank Act of 2010. The first criminal conviction under that provision came in 2015, against Michael Coscia, a trader who had used the technique in commodity futures in 2011: placing large orders on one side of the book to move the price, executing a smaller real order on the other side at the resulting favorable price, and canceling the large orders before they could be filled.

The case clarifies the boundary this chapter has drawn throughout. Legitimate high-frequency market making (Sections 14.4.1 and 14.3.2) posts genuine quotes the firm is willing to have executed, even if many are canceled and replaced as fair-value estimates update. Spoofing posts orders *because* the firm does not want them executed, turning the order book's role as a public signal of supply and demand against the

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participants who rely on it. High cancellation rates and order-to-trade ratios are not by themselves evidence of spoofing, since legitimate market making produces them too. What distinguishes spoofing legally is intent, evidenced by patterns such as consistently canceling large orders moments before they would be filled, which is why regulators and exchanges monitor cancellation patterns as closely as executed trades.

14.5.4 Order-to-Trade Ratios and the Economics of Message Traffic

An HFT market maker generates far more messages (new orders, cancellations, modifications) than executed trades, since a firm updating its quotes as fair-value estimates change cancels and reposts far more often than it transacts. Suppose a firm sends 2,000,000 order messages in a trading day and executes 40,000 trades,

$$\text{Order-to-trade ratio} = \frac{2{,}000{,}000}{40{,}000} = 50 : 1.$$

Many exchanges impose a fee once this ratio exceeds a threshold, to discourage message traffic that consumes shared infrastructure without contributing proportionally to executed volume.

At an illustrative allowed ratio of 20:1 and a fee of \$0.01 per message above that threshold,

$$\text{Allowed messages} = 40{,}000 \times 20 = 800{,}000, \qquad \text{Excess messages} = 2{,}000{,}000 - 800{,}000 = 1{,}200{,}000,$$

$$\text{Total fee} = 1{,}200{,}000 \times \$0.01 = \$12{,}000.$$

This fee is a market-based response to a real externality. Each message imposes a small processing cost on the exchange's shared matching engine whether or not it results in a trade. A firm that posts and cancels very many orders relative to its executed volume therefore imposes costs a per-trade commission does not capture, whether its motive is legitimate quote management or influencing others' perception of supply and demand.

14.5.5 Queue Position and the Economics of Price-Time Priority

Suppose two traders each place a limit order to buy at the same price at nearly the same moment, yet one fills within seconds while the other waits nearly a minute. Nothing differs between the orders except where each landed in a queue. Most equity and futures exchanges allocate fills using strict price-time priority: better prices execute first, and at the same price, the earliest order fills first. This rule has a large, underappreciated effect on high-frequency strategy design: an order's expected time to execution depends not only on how much volume trades at its price, but on how many shares are queued ahead of it, which a trader can read from the order book's depth data.

Suppose a trader submits a limit buy order at the best bid, where 400 shares from other orders are already queued ahead of it, and incoming marketable sell orders trade through that price level at an average rate of 600 shares per minute (10 shares per second). Treating this rate as roughly constant over a short interval, the expected time until the queue ahead of the trader's order is fully executed is

$$\text{Expected wait} = \frac{400 \text{ shares}}{10 \text{ shares/second}} = 40 \text{ seconds},$$

and if the trader's own order is for 100 shares, the expected time until it is completely filled is $(400+100)/10 = 50$ seconds. This is why two economically identical limit orders can have very different expected execution times, and why sophisticated participants track queue position, not just price, as a first-class input.

It also reveals a cost of canceling and replacing that the order-to-trade fee did not capture: a canceled order that is resubmitted, even at the same price, loses its time priority and joins the back of a new queue. If only 250 shares are queued ahead at resubmission, the expected wait falls to $250/10 = 25$ seconds. That sounds like an improvement, but only because the trader gave up a position it had already waited to earn; holding the original order, partway through the original 40-second queue, would typically have meant a shorter wait still. HFT market-making systems are therefore engineered to modify order size in place, or to reprice only when the price improvement outweighs the lost queue position. Matching-engine details, such as whether an order keeps time priority after a partial cancellation or a price-preserving modification, are strategic details HFT firms study as closely as the message-fee schedules of Section 14.5.4.

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14.6 Market Structure and Market Quality

Whether high-frequency trading improves markets depends on how the market is wired, and on what improvement is measured against. This part covers the data-feed architecture that creates the fastest participants' advantage, and the metrics by which the result is judged.

14.6.1 Direct Data Feeds, the Consolidated Tape, and Reg NMS

Section 14.2.2 explained why firms pay for co-location and fast transmission, but not why a data feed's speed matters independently of an exchange's matching latency. In U.S. equity markets, every exchange publishes its own direct data feed and also contributes its quotes to a consolidated feed, the Securities Information Processor (SIP). The SIP aggregates quotes from all exchanges into the National Best Bid and Offer (NBBO) that Regulation NMS (National Market System) requires brokers to respect when routing and executing orders.

Because the SIP must collect, synchronize, and republish quotes from every exchange, it is structurally slower than any exchange's direct feed, historically by low single-digit milliseconds, the same order of magnitude as the fiber-versus-microwave gap of Section 14.2.2. A firm subscribing to every exchange's direct feed therefore sees the true NBBO before the consolidated quote does, and can react to a price change before a SIP-only participant even sees it. This is the stale-quote mechanism of Section 14.3.1 again, arising from data infrastructure and regulatory design rather than physical distance.

Reg NMS's order protection rule, which prohibits trading through a protected quote displayed on another exchange, was meant to guarantee execution quality across a market fragmented over dozens of venues. But it also created a durable incentive for direct-feed investment. A broker or venue that must check the NBBO to comply needs it as quickly as possible, and a firm that has it fractionally sooner can trade profitably on the gap. This illustrates a recurring theme of this book: a well-intentioned regulation, here protecting investors from trade-throughs, can create incentives its designers did not intend, here a latency arms race in market data.

A numerical illustration, structured like Section 14.3.1's stale-quote example but at the millisecond scale of the SIP-versus-direct-feed gap, shows the mechanism. Suppose a stock's true NBBO moves the moment a large order executes on one exchange, visible instantly on that exchange's direct feed, but the SIP-based NBBO, and every participant relying on it, does not reflect the new price for 3 milliseconds.

A firm on direct feeds can trade against counterparties still quoting or routing off the stale NBBO throughout that window; if it captures \$0.02 per share on 10,000 shares each time this occurs, and the pattern recurs roughly 15 times per trading day across its traded universe,

$$\text{Profit per event} = 10{,}000 \times \$0.02 = \$200, \qquad \text{Daily profit} = 15 \times \$200 = \$3{,}000,$$

a smaller daily figure than Section 14.3.1's microsecond-scale example, since a window based on a known, structural feed delay is more heavily competed away than a novel news event. But it recurs far more predictably, because the SIP-versus-direct-feed gap exists every trading day, not only when news arrives. That regularity is why direct feeds from every exchange are close to a baseline requirement for any serious HFT operation rather than one latency-reduction technique among several: the SIP's structural delay is a known, exploitable constant that any firm without direct feeds pays every day, whether or not it ever meets a dramatic latency-arbitrage event.

14.6.2 Measuring Market Quality: Effective Spreads and Price Discovery

The quoted bid-ask spread of Chapter 3 is not always what a trader pays: an order can receive price improvement inside the spread, or a large order can walk through several price levels. The effective spread corrects for this by comparing the execution price to the midpoint prevailing when the order arrived,

$$\text{Effective Spread} = 2 \times |P_{\text{execution}} - P_{\text{midpoint}}|. \tag{14.5}$$

If the midpoint at order arrival is $(\$50.07 + \$50.08)/2 = \$50.075$ and a buy order executes at the full quoted ask of \$50.08,

$$\text{Effective spread} = 2 \times |50.08 - 50.075| = \$0.01,$$

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identical to the quoted spread: no price improvement.

If instead the order executes at \$50.078, perhaps because a competing market maker's hidden or "odd-lot" liquidity intervened,

$$\text{Effective spread} = 2 \times |50.078 - 50.075| = \$0.006,$$

narrower than the \$0.01 quoted spread. Academic studies of market quality frequently find that effective spreads have narrowed considerably as HFT market making has grown. Proponents cite this as evidence that HFT liquidity provision benefits ordinary investors through lower trading costs. Critics respond that narrower average spreads coexist with periodic liquidity withdrawal and adverse-selection costs, so an average can mask what happens in the moments of stress when execution quality matters most.

14.6.3 Worked Example 14.3: Roll's Implied Spread from Bid-Ask Bounce

Both measures above need quote data. Chapter 3 introduced a related idea in passing: bid-ask bounce, prices alternating between bid and ask as buy and sell orders arrive, induces negative autocorrelation in price changes even when fundamental value has not moved. Roll's model (Roll 1984) turns this artifact into a measurement tool: if bounce alone drives the negative autocorrelation, the spread can be recovered from transaction prices, with no quote data at all.

The model assumes each trade price is the efficient price m_t plus or minus half the spread, $P_t = m_t + c\,I_t$, where $c = s/2$ is half the spread and $I_t \in \{+1, -1\}$ is the trade-direction indicator (buy or sell), drawn independently and with equal probability from trade to trade, and independent of the efficient price's random-walk innovations. Under these assumptions the first-order autocovariance of price changes reduces to

$$\operatorname{Cov}(\Delta P_t,\, \Delta P_{t-1}) = -c^2 = -\frac{s^2}{4}, \tag{14.6}$$

so whenever the sample covariance is negative, as bounce predicts, inverting the relationship recovers the spread,

$$\hat{s} = 2\sqrt{-\widehat{\operatorname{Cov}}(\Delta P_t,\, \Delta P_{t-1})}. \tag{14.7}$$

Suppose a stock's true, constant bid-ask spread is \$0.02 (bid \$50.00, ask \$50.02), and 20 consecutive trades arrive, each equally likely to hit the bid or the ask. Price changes bounce between $-\$0.02$, \$0, and $+\$0.02$ depending on whether consecutive trades land on the same or opposite sides. The sample covariance between each price change and the previous one, across the 18 consecutive pairs from 19 price changes, is $\widehat{\operatorname{Cov}}(\Delta P_t, \Delta P_{t-1}) \approx -0.0000889$, and therefore

$$\hat{s} = 2\sqrt{0.0000889} \approx \$0.0189,$$

within about 6% of the true \$0.02 spread, recovered from trade prices without observing a single quote. The gap is not a computational error. Roll's estimator is unbiased only in expectation over infinitely many i.i.d. trades; a finite sample lands above or below the true value from sampling noise, and twenty trades is a realistic size for a quiet stock over a short window. A larger sample shrinks the gap, as it shrinks the standard error of any estimate.

Roll's model has a well-known limitation visible in its formula: the estimator is undefined whenever the sample covariance is non-negative, which happens with some regularity in short or quiet samples. Information arrival, momentum, and other sources of serial correlation compete with, and can overwhelm, the weak bounce signal the model relies on. Practitioners therefore prefer the effective-spread and quote-based measures above, which use richer data and do not depend on this occasionally violated assumption, and reserve Roll's estimator for settings where quote data is not available.

14.6.4 U.S. Equity Market Structure as of 2026

The feed, spread, and queue mechanics of this chapter operate inside a market structure that U.S. regulators have kept reshaping since the Tick Size Pilot (discussed in the regulatory-debate section) ended in 2018. Five developments matter most, as of October 2026, for anyone building or evaluating an execution or high-frequency system in U.S. equities; Table 14.2 dates each one.

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Retail order flow trades off-exchange. In 2025, for the first time, more than half of U.S. equity volume executed away from the exchanges: trades reported to FINRA's Trade Reporting Facilities made up 50.6% of consolidated volume, and 81.3% of that off-exchange volume was executed by principal dealers rather than in dark pools (Cboe 2026). Much of the principal-dealer share is retail. Retail brokers route their customers' marketable orders to a handful of wholesale market makers, which pay for the flow (payment for order flow) and fill it from their own inventory (internalization) at prices slightly better than the NBBO. This is the adverse-selection logic of Section 14.3.1 in reverse: retail orders are on average uninformed, so a wholesaler can fill them inside the spread and still earn more than on a lit exchange, where its fills are disproportionately the ones a faster or better-informed trader chose. The SEC's 2022 proposals to force retail orders through order-by-order auctions before internalization (the Order Competition Rule) and to adopt its own best-execution standard were withdrawn in June 2025 (SEC 2025b), leaving disclosure as the current U.S. response.

Execution-quality disclosure: amended Rule 605. The amendments adopted in March 2024 (SEC 2024a) extend Rule 605's monthly execution-quality reports to every broker-dealer with 100,000 or more customer accounts, add a human-readable summary report, size orders by notional value and by fractional, odd-lot, or round-lot status rather than share count, and require statistics such as effective over quoted spread and realized spreads at horizons from 50 milliseconds to 5 minutes. The 50-millisecond realized spread nearly measures the speed-driven adverse selection this chapter quantifies. After a postponement from December 2025, compliance began on August 1, 2026 (SEC 2025a), so the first reports in the new format, covering August 2026, appeared in the autumn of 2026.

Tick size, access fees, and round lots. In September 2024 the SEC amended Regulation NMS (SEC 2024b) to introduce a $0.005 minimum pricing increment for stocks priced at $1.00 or more whose time-weighted average quoted spread is $0.015 or less, to cut the cap on exchange access fees from $0.003 to $0.001 per share for those stocks, and to accelerate the round-lot and odd-lot definitions of the 2020 market data infrastructure rule. For the most liquid names this is the opposite of the Tick Size Pilot: a one-cent tick binds there, so long queues form at the best price and the race for queue position of Section 14.5.5 becomes the competition; a half-penny tick shortens those queues and lets price compete where time priority did. The round-lot definition took effect on November 3, 2025 (100 shares for stocks up to $250, then 40, 10, and 1 share in tiers above $250, $1,000, and $10,000), so quotes in high-priced stocks now count toward the NBBO at smaller sizes. The tick and fee-cap changes were stayed by the SEC pending litigation; after the D.C. Circuit denied the exchanges' petitions in October 2025, the SEC delayed compliance first to November 2026 and then, by an order of June 2026, to the first business day of November 2027 (SEC 2026). As of October 2026 the half-penny tick is adopted but not yet in force.

Settlement and speed bumps. U.S. equities moved from T+2 to T+1 settlement on May 28, 2024 (SEC 2023); the European Union will follow on 11 October 2027 under Regulation (EU) 2025/2075 (EU 2025), and the United Kingdom has set the same date, with its implementing regulations published in draft. A shorter cycle cuts counterparty exposure and margin, not trading latency. Market design aimed at latency itself is older: in June 2016 the SEC approved IEX as a national exchange (SEC 2016) with a 350-microsecond delay on incoming orders, created by coiled fiber, which lets IEX reprice its pegged orders before a fast trader's order can reach them, removing on one venue the stale-quote window of Section 14.3.1. Budish, Cramton, and Shim's frequent batch auctions (Budish et al. 2015) go further, replacing continuous trading with auctions held many times a second, so that speed differences smaller than the batch interval earn nothing; they remain a proposal rather than the operating model of U.S. stock exchanges.

Consolidated data. The SEC's 2020 market data infrastructure rule (SEC 2020) was meant to enrich the consolidated feed (odd-lot quotes, depth of book) and to replace the exclusive SIPs with competing consolidators. Its content changes are arriving, the round-lot definition in 2025 and each exchange's best odd-lot quotes on the SIPs from 2026, but competing consolidators have not replaced the exclusive SIPs, so the SIP-versus-direct-feed gap of Section 14.6.1 persists. The European Union, which had no consolidated tape at all, selected Ediphy (fairCT) as its bond tape provider in July 2025 and authorized EuroCTP to run the tape for shares and ETFs on 27 July 2026, with a transition period to 30 September 2026 (ESMA 2026).

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Measure	Adopted	In effect (status, October 2026)
T+1 settlement, United States	Feb. 2023	28 May 2024
Amended Rule 605 disclosure	Mar. 2024	1 Aug. 2026 (postponed from Dec. 2025)
Round-lot definition (Reg NMS)	Dec. 2020; accelerated Sept. 2024	3 Nov. 2025
Half-penny tick, lower access-fee caps	Sept. 2024	Delayed to first business day of Nov. 2027
Order Competition Rule; Regulation Best Execution	Proposed Dec. 2022	Withdrawn June 2025
T+1 settlement, EU	Oct. 2025	11 Oct. 2027
T+1 settlement, UK	Draft regulations, 2026	11 Oct. 2027 (planned)
EU consolidated tape, shares and ETFs	EuroCTP authorized 27 July 2026	Transition period to 30 Sept. 2026

Table 14.2: Market-structure and settlement changes since the Tick Size Pilot, as of October 2026

14.7 Prediction and the Decision Pipeline

Everything so far has reacted to observable state. This part turns to forecasting the next price move from order flow, and to the trade-off between a model's accuracy and the microseconds it costs to evaluate.

14.7.1 Predictive Modeling: Order Flow Imbalance

Not every HFT model is a formal microstructure model like Kyle's or Glosten-Milgrom; many production signals are simple regressions, in the spirit of Chapter 6, fit to order-book data. The most widely used is order flow imbalance (OFI): the change in resting buy-side depth minus the change in resting sell-side depth at the top of the book between two snapshots. A book growing relatively bid-heavy should predict a price increase over the next few ticks, and vice versa. Table 14.3 shows six snapshots with their OFI and the subsequent price change.

Snapshot	1	2	3	4	5	6
OFI (contracts)	50	−30	80	−60	20	40
Next price change (cents)	2	−1	3	−2	1	2

Table 14.3: Order flow imbalance and the subsequent price change

Fitting the OLS regression of Chapter 6,

$$\Delta P = \alpha + \beta\,\mathrm{OFI} + \varepsilon, \tag{14.8}$$

gives

$$\hat{\alpha} \approx 0.22 \text{ cents}, \qquad \hat{\beta} \approx 0.0369 \text{ cents per contract of imbalance}, \qquad R^2 \approx 0.99,$$

an unusually tight fit for illustration; real order-book data produces a much noisier, though typically still significant, relationship. For a new snapshot with OFI $= 65$, the model predicts a price move of $0.22 + 0.0369(65) \approx 2.62$ cents over the next interval. With the standard-error procedure of Chapter 6 and only $6 - 2 = 4$ degrees of freedom, $\mathrm{SE}(\hat{\beta}) \approx 0.0016$, giving $t \approx 23.0$ and a 95% confidence interval for $\hat{\beta}$ of roughly $(0.032, 0.041)$. The coefficient is estimated unusually precisely only because this dataset was constructed to fit almost perfectly. A production OFI model, fit to thousands of noisier snapshots, should show a wider confidence interval and a lower, though often still significant, R^2.

This is the simplest member of the machine learning family, and it illustrates the latency-versus-complexity trade-off developed next. A single-feature linear regression can be evaluated in a few machine instructions. A richer model on dozens of order-book and trade-history features, such as a gradient-boosted tree or neural network, may predict more accurately but takes measurably longer to compute. That cost matters enormously at HFT timescales and much less at the daily or weekly horizons of most of Chapter 6's examples.

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14.7.2 The Decision Pipeline, and Machine Learning's Latency Cost

Every high-frequency trading decision, whether market making, latency arbitrage, order-flow-imbalance prediction, or the sequential-trade updating of Section 14.3.2, passes through the same pipeline (Figure 14.2): market data arrives, a fair-value or signal model consumes it, a risk check validates the resulting order, and the order goes to the exchange. Every stage adds latency, so at HFT timescales the choice of model is a latency decision, not merely an accuracy decision.

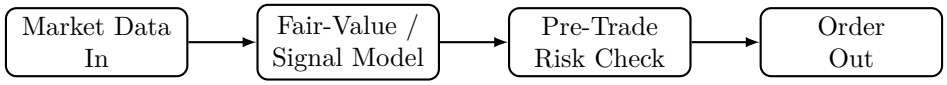


Figure 14.2: The high-frequency trading decision pipeline

Chapter 6 evaluated models from linear regression to deep neural networks mainly on predictive accuracy. At HFT timescales, inference latency, how long a model takes to produce a prediction from new data, becomes a competing objective of comparable importance. A threshold rule or small logistic regression typically predicts in low single-digit microseconds, a shallow tree or small gradient-boosted ensemble in perhaps ten to a few tens of microseconds, and a moderate deep network in the high tens of microseconds to low milliseconds, depending on architecture and hardware. These are order-of-magnitude figures, not benchmarks; Section 14.7.3 measures both sides of the trade-off on a simulated order book. If a firm's decision budget, the time between receiving data and needing to act before a competitor, is only tens of microseconds, a more accurate but slower model can be strictly worse than a less accurate faster one, since the correct decision made after a competitor has traded captures nothing. This is why the simplest models, not the most sophisticated, dominate the most latency-sensitive layers of high-frequency trading. Complex models, including deep learning, are reserved for slower decisions, such as which symbols to commit market-making capital to today, where microseconds do not matter but accuracy does.

Concept Drift and Model Retraining at High Frequency

Inference speed is only half of the latency-sensitive machine learning problem; the other half is how quickly what the model has learned goes stale. Chapter 7's non-stationarity, a series' statistical properties changing over time, appears here as concept drift. The relationship between order flow imbalance and subsequent price changes in Section 14.7.1 is not a physical law but an empirical regularity that depends on the mix of market participants, prevailing volatility, and even the stock's tick size regime, all of which change far faster than the daily or monthly retraining cycles of slower applications.

A high-frequency model like the OFI regression is therefore retrained far more often than one built on daily data, sometimes several times a day, on a recent rolling window rather than the full trading history, so that $\hat{\alpha}$ and $\hat{\beta}$ reflect current conditions rather than a stale regime. This creates its own trade-off. A short window adapts quickly to regime change but gives noisier estimates from fewer observations, the bias-variance trade-off of Chapter 6 in a different setting, while a longer window gives stabler estimates that may lag a real shift in market structure. Firms handle this with automated monitoring of a live model's out-of-sample accuracy, triggering an automatic retrain, or at worst shutting the signal down through the kill switch described next, the moment accuracy falls below a preset threshold. At HFT timescales no human can notice a model's slow decay and intervene before meaningful losses accumulate.

14.7.3 Worked Example 14.4: Accuracy Against Latency on a Simulated Order Book

The latency figures quoted above are orders of magnitude; this example measures both sides of the trade-off on the same data. The companion notebook simulates a five-level limit order book event by event, 200,000 events with a fixed random seed. Each event is a market order (20% of events), a new limit order (40%, more likely near the top of the book), or a cancellation (40%, more likely in deeper queues). Market-order signs are persistent, flipping with probability 0.15 per order, the way a large order worked in slices produces a run of trades on one side. When the best ask or best bid queue is emptied, the mid-price ticks up or down and

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the book shifts by one level, which happens 15,130 times. The features are the order flow imbalance at each of the five levels over the previous 20 events, the multi-level generalization of Section 14.7.1's top-of-book OFI (Cont et al. 2014; Xu et al. 2020), and the label is whether the next mid-price change is up or down. Sampling every ten events gives 19,998 labeled snapshots, of which the first 70% train both models and the last 6,000 test them, in time order.

A logistic regression on the five features reaches a held-out accuracy of 58.78%. A gradient-boosted classifier with 100 trees of depth three, 1,452 nodes in all, reaches 57.68%. Five further seeds give differences between -0.25 and $+0.05$ percentage points, and a slower-learning booster (300 trees of depth two) reaches 58.02%, so the result is not one unlucky draw: on these features the flexible model has nothing to find. The reason is visible in the fitted models. Almost all of the information sits in the top level, which receives the largest logistic coefficient (0.0354 per share, against at most 0.0064 for the deeper levels) and 69% of the boosted model's feature importance, and the probability of an up-move rises smoothly with it. That resembles the linear OFI-price relation Cont et al. (2014) found in real order books, and on a near-linear relation boosting spends its flexibility on noise.

Latency is measured by scoring one event at a time, the way a trading loop sees the data. Written as lean code, the logistic model is one five-term dot product and a comparison; the boosted model must walk 100 trees. In the notebook's Python implementation the logistic model takes under a microsecond per event and the boosted one roughly 25 to 30, a ratio of 35 to 50 across runs. These timings depend on the machine and are excluded from the notebook's reproducibility check; compiled production code would be faster for both models, but the ratio tracks the operation count and survives. One more lesson: calling the library's general-purpose `predict()` on a single row costs between about 70 and 320 microseconds, depending on the model and the run, more than the lean version of *either* model, so the software layer around a model can cost more than the model itself.

Why can the simpler model win? Suppose each signal is worth one tick when right and costs one tick when wrong, so the expected edge per signal is $2a - 1$ for accuracy a, but only if the order arrives before the opportunity disappears. If an opportunity's remaining life is exponentially distributed with a mean of 50 microseconds, a model that is Δ microseconds slower keeps only a fraction $e^{-\Delta/50}$ of its edge. The logistic model's edge is $2(0.5878) - 1 = 0.1757$ ticks per signal. The boosted model's is 0.1537 even at equal speed, falls to 0.1390 if it is 5 microseconds slower, and to 0.1030 at 20 microseconds. Turned around, a model 20 microseconds slower must be 63.10% accurate just to break even, 4.32 points better than the logistic model, and even a 5-microsecond gap demands 0.92 points. Gains of that size are rare at high frequency, where the signal-to-noise ratio is low and near-linear structure dominates. Two further advantages point the same way: a six-parameter model can be refit several times a day on a short rolling window, and each decision can be read off its coefficients when a risk manager asks why it traded. The flexible model belongs at slower layers, choosing which symbols to quote or how wide to set limits, where a few microseconds cost nothing and accuracy is the only objective.

14.8 Risk Controls and Systemic Episodes

A system that can trade thousands of times a second can also lose money thousands of times a second. This part covers the controls that stop it, and the afternoon on which the interaction of many such systems briefly emptied the market.

14.8.1 Risk Controls: Kill Switches and Pre-Trade Risk Checks

Chapter 13's Knight Capital vignette showed how a firm can lose an enormous amount of money in well under an hour when automated infrastructure malfunctions: the speed that makes automated strategies profitable also makes an undetected error catastrophic before a human can intervene. HFT firms and exchanges have built several layers of technical risk control in response. Pre-trade risk checks validate every outgoing order against limits (maximum order size, maximum position, a fat-finger price collar around the current price) before it reaches the exchange, rejecting any violation whatever the upstream trading logic intended. A kill switch, ideally independent of the trading logic, can immediately halt all order flow from a firm or strategy, either automatically (on a risk-limit breach, such as an unusually large position or an abnormal message rate) or manually by a supervisor.

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A price-collar example shows how a pre-trade check intervenes. Suppose a stock's last traded price is \$50.00, and a firm's pre-trade risk check enforces a fat-finger collar of $\pm 2\%$ around that reference price, rejecting any outgoing order priced outside $[\$49.00, \$51.00]$. If a software defect causes the trading algorithm to submit a buy order at \$45.00, perhaps from a decimal-place error in an upstream price feed, the pre-trade check rejects it before it reaches the exchange, whatever flawed logic generated the price. Without the check, an order at \$45.00 against a \$50.00 market would either execute at a wildly favorable price for a counterparty (an avoidable loss) or fail to fill while consuming exchange bandwidth and risking a message-traffic problem of its own. The collar turns the failure into a rejected message logged for review, the kind of independent, code-level safeguard the Knight Capital incident lacked.

Knight Capital was not a failure of trading strategy but of deployment and risk-control discipline: obsolete code was left active on production servers, and no automated control caught the runaway order flow before it had built hundreds of millions of dollars in unintended positions. This connects to the model risk management framework of Chapter 8. An algorithmic trading system, like a VaR or credit-scoring model, requires independent testing before deployment, staged rollout, and continuous automated monitoring with the authority to intervene, because the speed that makes high-frequency trading viable makes a human-in-the-loop safeguard, by itself, too slow to prevent a catastrophic loss.

14.8.2 The 2010 Flash Crash Revisited: The Hot-Potato Effect

Chapter 3 introduced the May 2010 Flash Crash, in which the Dow Jones Industrial Average fell and then recovered nearly 1,000 points within minutes. The subsequent regulatory investigation identified a specific mechanism, sometimes called the "hot-potato effect," through which high-frequency trading amplified the initial shock. As a large sell order began moving prices down, several high-frequency firms, each managing inventory risk as Section 14.4.1's model prescribes, rapidly bought and resold the same shares among themselves, passing inventory from firm to firm within fractions of a second rather than absorbing it, with no single firm's net position changing much. This passing inflated trading volume without providing the risk-absorbing liquidity the volume figures suggested. As more firms hit the same inventory and risk limits at once, many withdrew from market making within moments of each other, removing liquidity at the moment it was most needed. The episode is a systemic version of a risk Chapter 8 discussed for an individual VaR model. An individually sensible rule (reduce inventory and widen quotes when risk rises) can destabilize the market when many participants with similar models and conditions follow it simultaneously. That is the crowding dynamic Chapter 12 raised for factor investing and Chapter 13 for statistical arbitrage, compressed into seconds rather than months. Circuit breakers, introduced in Chapter 13, exist to interrupt this feedback loop by forcing a pause before a self-reinforcing price move can fully unwind.

14.9 The Regulatory Debate: Does HFT Help or Hurt Markets?

High-frequency trading remains contested among regulators, academics, and market participants, and both sides deserve a fair statement. Proponents point to the narrower effective spreads of the market-quality discussion, faster price discovery (information reaches prices sooner when fast participants constantly monitor and react to it), and the increase in continuously available liquidity relative to the slower, more manual market making HFT has largely displaced. Critics make three arguments. The Flash Crash's hot-potato effect shows that HFT liquidity can be illusory when it is needed most. The adverse-selection costs quantified in Sections 14.3.1 and 14.4.2 are borne by slower participants, including ordinary investors' pension and mutual funds. And manipulative practices such as spoofing, prosecuted in numerous cases, create a false impression of supply or demand that other participants, often algorithms, react to.

Both sides can be right about different aspects of the same activity. HFT market making of the kind formalized here plausibly tightens effective spreads and improves liquidity most of the time, while HFT's aggregate behavior in stress events, and a minority's use of manipulative strategies, plausibly create periodic costs and risks that average market-quality statistics understate. This is why regulatory responses have been targeted rather than blanket: circuit breakers address flash-crash risk, order-to-trade fees address message traffic, and spoofing prosecutions address manipulative intent, rather than one rule treating "high-frequency trading" as a single undifferentiated activity.

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Approaches also differ across jurisdictions. The European Union's MiFID II is more structural than the largely case-by-case U.S. enforcement model. It requires algorithmic-trading firms to register as such, to test their algorithms before deployment in a way regulators can audit (an externally verifiable version of internal model risk management), and to maintain minimum resting times for certain order types to discourage very high cancellation rates.

The SEC's Tick Size Pilot was designed to test one of this chapter's structural claims directly. Would a wider tick size, by making price-time priority (Section 14.5.5) a coarser advantage, reduce HFT-driven quote competition and thicken displayed liquidity, or would it merely widen effective spreads? Running from 2016 to 2018, the pilot widened the minimum price increment from the standard \$0.01 to \$0.05 for roughly 1,200 small-capitalization stocks split across several test groups with different additional rules (some also had a "trade-at" rule forcing certain orders to route to the quoting exchange rather than be internalized). Quoted and effective spreads widened materially for pilot stocks relative to a control group, as a wider tick mechanically predicts, but displayed depth and overall market quality showed no clear improvement large enough to offset that, a result closer to critics' prediction than to proponents' hoped-for thicker, more stable liquidity. The SEC did not make the widening permanent; in 2024 it moved the opposite way for the most liquid stocks, adopting a half-penny tick that, as of October 2026, is not yet in force (Section 14.6.4). Even a carefully designed regulatory experiment can produce a result too ambiguous, or too specific to the stocks and rules tested, to settle a question as broad as "does high-frequency trading, on net, help or hurt ordinary investors."

14.10 Challenges in High-Frequency Trading

Several challenges recur across nearly every high-frequency trading application, and many connect to themes developed elsewhere in this book.

The arms race has diminishing, and unevenly distributed, returns. Every dollar spent shaving microseconds off latency, through co-location, microwave links, or specialized hardware, raises the cost of competing at the top of the speed distribution without necessarily adding liquidity or price discovery for the market as a whole. Critics call this a socially wasteful arms race. Participants answer that the fastest firm captures a disproportionate share of the latency-arbitrage profits of Section 14.3.1, making the investment individually rational even if its social value is debatable.

Technology and operational risk dominate strategy risk. As the Knight Capital vignette and the kill-switch discussion emphasized, the dominant risk in high-frequency trading is often not wrong economic logic but an implementation that fails in an unanticipated way faster than humans can intervene, a risk that demands engineering and governance discipline as much as modeling skill.

Backtesting at microsecond resolution is extraordinarily data- and infrastructure-intensive. The walk-forward validation of Chapter 7 still applies, but a microsecond-sensitive strategy needs tick-by-tick, timestamp-accurate data and a simulator that faithfully reproduces queue position and latency, a much higher bar than the daily or minute-level data of earlier chapters' slower strategies.

Regulatory and market-structure change can eliminate an edge overnight. A strategy that depends on a specific fee schedule, order-to-trade rule, or co-location arrangement (the market-structure dependence flagged in Chapter 13) is especially exposed in high-frequency trading, where regulators and exchanges adjust such rules frequently, often in response to the practices this chapter describes.

Crowding and correlated risk management compound during stress. The hot-potato mechanism of the 2010 Flash Crash is a standing risk whenever many participants run similar inventory logic; a firm's risk model can be well-calibrated and still contribute to a destabilizing outcome in a shock, the systemic risk Chapter 8 raised for correlated VaR models across institutions.

High fixed costs create barriers to entry and market concentration. Co-location, direct feeds, and specialized hardware are largely fixed costs that do not scale down for a smaller firm, so high-frequency market making and latency arbitrage are dominated by a small number of well-capitalized specialists rather than open to anyone with a good idea. This raises its own policy question, distinct from liquidity and manipulation: whether a market structure that rewards infrastructure spending this heavily produces the most efficient possible price discovery, or merely the best attainable given an arms race that smaller, possibly equally skilled participants cannot afford to enter.

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A predictive signal's own success can erode it. An order-flow-imbalance model, or any short-horizon signal, gets priced into the market by every sophisticated participant who arrives at a similar signal. This is concept drift with a twist: the drift is caused not by an exogenous change in conditions but endogenously, by the signal being traded profitably at scale. It is a faster, more severe version of the crowded-factor concern Chapter 12 raised for slower cross-sectional signals. At HFT timescales a novel predictive relationship can be discovered, exploited, and arbitraged away within weeks or months, which is why continuous signal research is an operating cost of this business, not a one-time model-building exercise.

14.11 Key Takeaways

High-frequency trading applies the market-making, execution, and inventory-management ideas of Chapter 13 at a timescale where physical latency is a first-order economic variable and operational risk is the central concern. Latency arbitrage turns a pure speed advantage (co-location, a faster transmission medium, or a faster data feed under Reg NMS) into profits from slower participants' stale quotes, within a single stock, between an ETF and its basket, or between index futures and the cash market. The Avellaneda-Stoikov model shows why a much shorter horizon lets a market maker run with a much smaller inventory-risk penalty.

Under price-time priority, queue position, not just price, is a first-class variable in HFT strategy design. Order-to-trade fees and effective spreads are two tools regulators and researchers use to judge whether high-frequency activity contributes liquidity or mainly consumes shared infrastructure.

The 2010 Flash Crash's hot-potato effect shows how individually rational, Chapter-13-style inventory management can destabilize the market when many participants follow similar rules at once.

As with every method in this book, the response is not to avoid the technology but to govern it: independent testing, staged deployment, automated kill switches, and monitoring for concept drift are the speed-adjusted analogue of the model risk management discipline of Chapter 8, not optional extras.

14.12 Suggested Exercises

14.1 Using Section 14.3.1, recompute the daily latency-arbitrage profit if Firm F can execute against 800 shares per event and such events occur 35 times per day, holding the \$0.05 price jump fixed.

14.2 Explain, in your own words, why a persistent millisecond-scale latency advantage translates into a repeatable dollar profit rather than a one-time gain, referencing the stale-quote mechanism in Section 14.3.1.

14.3 Using the reservation-price formula in Section 14.4.1, compute the skew for $q = 8$, $\gamma = 0.15$, $\sigma = 1.5$, and $T - t = 0.01$, and compare it to the chapter's example.

14.4 A high-frequency market maker completes 80,000 fills (40,000 round trips) per day at an average size of 150 shares, capturing a half-spread of \$0.004 per share on each fill (a full spread of \$0.008 per round trip), with 4% of fills toxic at an average loss of \$0.03 per share. Compute gross spread capture, adverse selection loss, and net daily P&L.

14.5 A firm sends 3,500,000 order messages and executes 50,000 trades in a day. Compute its order-to-trade ratio, and its fee at an allowed ratio of 25:1 and a per-excess-message fee of \$0.008.

14.6 An order executes at \$25.02 when the prevailing midpoint was \$25.015. Compute the effective spread, and explain whether this trade received price improvement relative to a \$0.02 quoted spread.

14.7 Explain, using the hot-potato mechanism in this chapter, why high trading volume during a market stress event does not necessarily indicate that genuine, risk-absorbing liquidity was available.

14.8 Describe two specific technical risk controls a high-frequency trading firm should have in place to prevent a Knight-Capital-style incident, and explain what each one would have needed to do differently on August 1, 2012 to prevent the loss described in Chapter 13.

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14.9 Discuss, in a short paragraph, why the same empirical finding (narrower average effective spreads) can be cited by both proponents and critics of high-frequency trading as evidence for their position, referencing this chapter's regulatory debate section.

14.10 Pick two of the challenges listed in this chapter and describe a concrete scenario, not already used in the text, in which that challenge could cause a high-frequency trading strategy to lose money or draw regulatory scrutiny despite a sound underlying economic rationale.

14.11 Using Section 14.5.5, compute the expected wait time for a 200-share limit order joining a queue of 900 shares at the best bid, given an average execution rate of 15 shares per second, and explain why canceling and resubmitting that order at the same price could increase its expected time to fill.

14.12 Explain, using Section 14.7.1's concept-drift discussion, why a predictive model like the order-flow-imbalance regression needs to be retrained far more frequently than the credit-scoring model in Chapter 6, and describe one practical way a firm could detect that retraining is needed without human intervention.

14.13 Index futures jump by 3.50 points and the corresponding ETF tracks one-tenth of the index level. Using the cross-market latency arbitrage logic in this chapter, compute the implied ETF fair-value move, the profit on a 2,000-share trade, and the daily profit at 25 such events per day.

14.14 Using this chapter's SIP-versus-direct-feed example, recompute the daily profit if the per-share capture were \$0.015 instead of \$0.02 but the number of daily events rose to 22 (reflecting a more liquid, more frequently mispriced stock), and compare the result to the original \$3,000 figure.

14.15 Using this chapter's order-flow-imbalance regression, compute the width of the 95% confidence interval for $\hat{\beta}$ (the upper bound minus the lower bound), and explain why this width would be expected to grow substantially if the same regression were refit on a much noisier, more realistic sample of order-book snapshots.

14.16 Using this chapter's fat-finger price-collar example, determine whether an outgoing order at \$48.50 would be accepted or rejected under the same \$50.00 reference price and $\pm 2\%$ collar, and recompute the acceptable price range if the collar were widened to $\pm 3\%$.

14.17 Using this chapter's co-location break-even calculation, recompute the combined monthly profit from latency arbitrage and the SIP-versus-direct-feed mechanism if the SIP-based arbitrage instead occurred only 10 times per day rather than 15, and state the resulting ratio of total monthly profit to the \$35,000 monthly infrastructure cost.

14.18 Using Section 14.4.1's spread-floor calculation, recompute the total optimal spread δ and the percentage reduction from its $T - t = 1$ value if κ were instead 3.0 (a market where order arrivals respond less sensitively to quote competitiveness), holding $\gamma = 0.1$ and $\sigma = 2$ fixed, and explain whether the spread floor rises or falls relative to the $\kappa = 1.5$ case in the text.

14.19 Explain, in your own words, why the inventory-risk term of the optimal spread shrinks to nearly zero as $T - t$ shrinks while the order-arrival-elasticity term does not, and describe what kind of market-making cost the elasticity term is actually compensating the market maker for.

14.20 Using Section 14.6.3's Roll's-model example, explain what it would mean, in terms of the underlying assumptions of the model, if the sample covariance of price changes for a different stock's trade sequence came out positive rather than negative, and why the formula $\hat{s} = 2\sqrt{-\widehat{\mathrm{Cov}}(\Delta P_t, \Delta P_{t-1})}$ would fail in that case.

14.21 Using the edge calculation of Section 14.7.3 (one tick won or lost per signal, opportunity life exponential with a mean of 50 microseconds, logistic edge 0.1757 ticks), suppose a richer model reaches 60.5% accuracy but is 10 microseconds slower than the logistic model. (a) Compute its edge per signal and say which model you would deploy. (b) Find the accuracy at which the two models break even at this latency gap. (c) Recompute part (a) for a market in which opportunities last 200 microseconds on average, and explain what the change says about where flexible models belong.

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14.22 Section 14.7.3 found that a library's general-purpose single-row prediction call cost more than the lean version of either model. Explain why a general-purpose call carries this overhead, why it matters more at high frequency than for the daily models of Chapter 6, and what a firm would do about it before deploying a model in the latency-critical layer of Figure 14.2.

14.23 **Challenge (synthetic data).** Real, order-by-order limit-order-book data at the time resolution this chapter's examples assume (LOBSTER, Databento, WRDS NYSE TAQ) is paid or license-restricted, so generate a synthetic session instead: simulate 30 minutes of trading with Poisson order arrivals, limit orders arriving at $\lambda_L = 20$ per second spread uniformly across 10 price levels on each side, and market orders arriving at $\lambda_M = 2$ per second. (a) Reconstruct the resulting limit order book at 1-second snapshots and compute the realized bid-ask spread and quoted depth over the session, comparing your synthetic session's average spread to the 1-2 cent NBBO spread this chapter assumes for a liquid large-cap stock. (b) Inject a single large order arriving 50 milliseconds ahead of the rest of the market reacting to a simulated price-moving news event, and measure the profit it could extract using this chapter's co-location break-even calculation. (c) Discuss which of your simulation's design choices, arrival rates, order-size distribution, the absence of any strategic order cancellation, are least realistic relative to how this chapter describes real high-frequency venues behaving, and what real dataset, even a paid one, would be needed to validate or replace each assumption.

14.13 Further Reading

- Aldridge (2013), *High-Frequency Trading*. A comprehensive practitioner-oriented treatment of the strategies, technology, and risk management introduced in this chapter.
- Lewis (2014), *Flash Boys*. A widely read journalistic account of the latency arms race and co-location practices discussed in Section 14.2.2, useful context despite its explicitly critical perspective on the practices it describes.
- Budish et al. (2015), "The High-Frequency Trading Arms Race: Frequent Batch Auctions as a Market Design Response." An influential academic analysis of the latency-arbitrage mechanism formalized in Section 14.3.1 and a proposed market-design alternative to continuous trading.
- U.S. Securities and Exchange Commission and Commodity Futures Trading Commission (2010), *Findings Regarding the Market Events of May 6, 2010*. The official joint regulatory report documenting the hot-potato effect and other mechanisms behind the Flash Crash discussed in this chapter.
- Menkveld (2013), "High-Frequency Trading and the New Market Makers." An empirical study of high-frequency market making and its effect on market quality, directly relevant to this chapter's discussion of effective spreads.
- Cont et al. (2014), "The Price Impact of Order Book Events." The empirical study establishing the near-linear relation between order flow imbalance and short-horizon price changes that underlies Sections 14.7.1 and 14.7.3.
- Xu et al. (2020), "Multi-Level Order-Flow Imbalance in a Limit Order Book." The extension of order flow imbalance to several levels of the book, the feature set used in Section 14.7.3.
- Roll (1984), "A Simple Implicit Measure of the Effective Bid-Ask Spread in an Efficient Market." The original derivation of the bid-ask-bounce spread estimator worked through in Section 14.6.3.
- SEC (2005), *Regulation NMS*. The primary regulatory source for the order protection rule and consolidated-tape (SIP) requirements discussed in this chapter's treatment of direct data feeds.
- O'Hara (2015), "High Frequency Market Microstructure." A survey of how high-frequency trading has reshaped classical market microstructure theory, including the Glosten-Milgrom and PIN models developed earlier in this chapter.

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Chapter 15

Fraud Detection, Compliance, and Financial Crime

15.1 Introduction

Chapter 1 previewed this chapter with a single number: a fraud detection model that correctly flags 95% of actual fraud and mistakenly flags only 3% of legitimate transactions still produces a flagged transaction that is fraudulent only about 24% of the time, because fraud is rare enough that even a small false-positive rate generates many more false alarms than true detections. That number is the central operating constraint of every fraud detection and anti-money-laundering (AML) system, and this chapter is built around managing it. Chapter 6 developed classification models and the confusion matrix using loan default as the running example; Chapter 9 applied logistic regression, scoring, and rating models to credit risk. This chapter applies the same machinery to fraud, laundered money, and other financial crime. There the base rate is typically far lower than a loan portfolio's default rate, and the consequences of a false positive or a missed detection differ.

Four worked examples run through the chapter. Chapter 1's Bayes' theorem calculation is scaled up to a day's transactions, turning a posterior probability into an alert queue a compliance team must work through. A small fraud-scoring logistic regression shows how a few transaction features combine into a fraud probability, and where such a model still misses a case. A Kaplan-Meier survival analysis, new to this book, estimates how long flagged accounts stay active before fraud is confirmed, introducing censored observations. Finally, a structuring example shows how AML rules are built around a legal reporting threshold, and how criminals evade it.

15.2 Detecting Fraud

Fraud is a rare-event classification problem, and this part builds a detector while confronting how rare the positives are.

15.2.1 The Landscape of Financial Crime and Where AI Fits

Suppose a bank builds a fraud-scoring model that catches nearly every stolen-card transaction and considers its detection obligations satisfied, only to face a regulatory fine for missing a money-laundering scheme the model was never designed to catch. "Fraud detection" understates the breadth of what financial institutions are legally required to monitor for.

Table 15.1 organizes the main categories of financial crime this book touches on, the detection approach each typically relies on, and where it is developed.

Every row in Table 15.1 shares the same statistical structure: a rare, adversarial event must be distinguished from an overwhelming volume of legitimate activity, and the adversary adapts once a detection method becomes known. A credit-scoring model's borrowers, Chapter 9's subject, are not trying to defeat the model; a money launderer designing a transaction pattern to stay under a reporting threshold is. Financial crime

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Crime type	Typical detection approach	Developed in
Card and payment fraud	Real-time transaction scoring, anomaly detection	Sections 15.2.2, 15.2.3
Account takeover	Behavioral biometrics, device and login anomaly detection	Section 15.4.1
Synthetic identity fraud	Cultivation-pattern and shared-attribute linkage; unsupervised detection where labels are absent	Section 15.3
Authorized push payment (scam) fraud	Deviation from the customer's own payment history; mule detection on the receiving side	Section 15.3
Money laundering	Rule-based transaction monitoring, network analysis, structuring detection	Section 15.5.1
Market manipulation (spoofing, layering)	Order-book pattern surveillance	Chapter 14's spoofing case vignette
Insider trading	Anomaly detection linking trading activity to material news events	Chapter 1's Bayesian updating, extended here

Table 15.1: A taxonomy of financial crime and detection approaches

detection therefore combines the statistical tools of Chapters 6 and 9 with rules, network analysis, and a regulatory reporting apparatus with no counterpart in credit scoring.

A second difference between the rows decides more about which methods are available than the crime type does: what *labels* each problem produces as a by-product of its operation. Card fraud sits at one extreme. A cardholder disputes a charge within days, the dispute is ground truth rather than an opinion, and millions arrive automatically, so a supervised model can be retrained continuously against real outcomes. Money laundering sits at the other. The only label a bank ordinarily obtains is that one of its own analysts filed a Suspicious Activity Report, which records a suspicion rather than a confirmed crime, and law enforcement almost never reports back what came of it. A model fitted to those filings learns its institution's alerting behavior rather than criminality, and the true prevalence of laundering, the base rate every calculation in the next section depends on, is unmeasurable.

This asymmetry is why the methods in this chapter are distributed as they are. Supervised learning dominates where outcomes are observed; unsupervised detection, rules, and network structure carry the weight where they are not; and machine learning enters anti-money laundering largely through alert triage, ranking work the rules have already produced, because analyst dispositions are the one label that problem generates in quantity. Ask of each method below what it is trained on, and whether that is an outcome or an opinion.

15.2.2 Anomaly Detection in Transactions

Suppose a bank wants to stop a stolen credit card from being drained before the cardholder notices it is missing, deciding in real time whether each new charge looks like the cardholder or a thief. The simplest systems built for this decision are rule-based: a transaction is flagged if it exceeds a fixed dollar threshold, occurs in a country the cardholder has never transacted in, or arrives within seconds of another transaction from the same account in a distant location (a velocity check).

Rule-based systems are transparent and easy to audit, which matters for the regulatory reasons discussed later, but they are brittle: a threshold set to catch fraud will also catch a legitimate customer's unusual but innocent purchase. A publicly known threshold (such as the $10,000 reporting trigger of Section 15.5.1) can be engineered around.

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Statistical anomaly detection asks instead whether a transaction is unusual relative to the customer's own history, computing something like a z-score for amount, location, or timing against that customer's rolling mean and standard deviation, or using Chapter 6's unsupervised clustering to flag transactions that resemble none of the customer's usual behavioral clusters. Both approaches produce the same output as Chapter 6's classifiers, a binary flag or a risk score thresholded into one, and so face the same trade-off Chapter 1's Bayes' theorem example quantified.

Worked Example 15.1: The Bayes' Theorem Result at Full Transaction Volume

Chapter 1's example assumed a bank where 1% of transactions are fraudulent, a model correctly flags 95% of actual fraud, and incorrectly flags 3% of legitimate transactions, and found that a flagged transaction is fraudulent only about 24.2% of the time. That posterior's practical significance is easy to underestimate until it is scaled up to the volume a real institution processes.

Suppose the bank handles 500,000 transactions in a day. At a 1% fraud rate, this implies 5,000 fraudulent and 495,000 legitimate transactions, and applying the model's 95% true-positive rate and 3% false-positive rate gives

$$TP = 5{,}000 \times 0.95 = 4{,}750, \qquad FN = 5{,}000 \times 0.05 = 250,$$
$$FP = 495{,}000 \times 0.03 = 14{,}850, \qquad TN = 495{,}000 \times 0.97 = 480{,}150.$$

The model flags $4{,}750 + 14{,}850 = 19{,}600$ transactions a day, of which only 4,750 are fraud, giving

$$\text{Precision} = \frac{4{,}750}{19{,}600} \approx 0.2423, \qquad \text{Recall} = \frac{4{,}750}{5{,}000} = 0.95,$$

matching, up to rounding, Chapter 1's 24.2% posterior probability, since precision and $\Pr(\text{Fraud} \mid \text{Flag})$ are the same quantity computed two ways. The F_1 score, $2 \times 0.2423 \times 0.95/(0.2423 + 0.95) \approx 0.386$, is a useful single-number summary for comparing models, but it is the raw count, 19,600 flagged transactions in one day, that a compliance team must staff and review; Section 15.6.1 returns to this cost.

This is why fraud detection systems almost never rely on a single model or threshold. A bank that lowered its false-positive rate to improve precision would, by the same arithmetic, also lower recall and let more fraud through, unless it also improved the model's discriminative power. That is the motivation for combining distinct signals (transaction-level scoring, behavioral risk tiering, and survival modeling) rather than relying on any one alone.

A Second Application: Insider Trading Surveillance

The same Bayesian structure applies beyond transaction fraud. Market surveillance teams monitor for insider trading (trading ahead of material non-public information) largely by flagging abnormal trading volume or price movement in the days before a public announcement (an earnings release, a merger, a regulatory decision). This is the anomaly-detection logic of Chapter 14's order-flow-imbalance model, applied to a much lower-frequency, event-driven setting.

Suppose that, across all announcements a surveillance team reviews in a year, insider trading precedes about 2% of them, a volume-anomaly detector flags 75% of these cases, and it also flags 8% of announcements with no insider trading, a higher false-positive rate than Section 15.2.2's transaction-fraud example because pre-announcement volume is noisier than a single card transaction. Applying Bayes' theorem as in Chapter 1,

$$\Pr(\text{Flag}) = 0.75(0.02) + 0.08(0.98) = 0.0934, \qquad \Pr(\text{Insider} \mid \text{Flag}) = \frac{0.75(0.02)}{0.0934} \approx 0.1606,$$

so a flagged announcement reflects insider trading only about 16.1% of the time, even lower than Section 15.2.2's 24.2%. The interesting question is *why*: this example's true-positive rate (75%) is worse than the transaction-fraud example's (95%), which alone would push precision down only modestly, but its false-positive rate (8%) is more than double the other's (3%), and that difference dominates.

At a low base rate, precision is far more sensitive to the false-positive rate than to the true-positive rate, since the false-positive rate is multiplied by the (much larger) population of non-events; design effort is therefore almost always better spent tightening false-positive rates than chasing marginal gains in true-positive rates, a lesson that applies equally to the transaction-fraud and AML alert systems in this chapter.

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15.2.3 Worked Example 15.2: Fraud Scoring with Logistic Regression

Section 15.2.2's example treated the detection model as a black box with a fixed true- and false-positive rate; this section builds one explicitly, using the logistic regression Chapter 6 introduced and Chapter 9 applied to credit scoring. Table 15.2 records ten transactions, each with three features: the transaction amount expressed as a z-score relative to the cardholder's typical spending, the distance from the cardholder's home address in hundreds of kilometers, and a late-night indicator (1 if the transaction occurred between midnight and 5 a.m. local time), together with whether the transaction was fraudulent.

Transaction	Amount z-score	Distance (100s km)	Late night	Actual fraud	Predicted
1	3.1	8.0	1	1	1
2	0.2	0.1	0	0	0
3	2.8	6.5	1	1	1
4	1.9	6.0	0	0	0
5	0.4	0.2	0	0	0
6	3.4	9.2	1	1	1
7	0.1	0.4	0	0	0
8	0.6	0.1	1	0	0
9	1.2	0.3	1	1	0
10	0.3	0.2	0	0	0

Table 15.2: Ten transactions used to fit a fraud-scoring logistic regression

Fitting a logistic regression

$$\Pr(\text{Fraud}) = \sigma(\beta_0 + \beta_1 z + \beta_2 d + \beta_3 n) \tag{15.1}$$

using the sigmoid function introduced in Chapter 6, to these ten observations gives coefficients $\hat{\beta}_0 \approx -2.57$, $\hat{\beta}_1 \approx 0.83$ (amount z-score), $\hat{\beta}_2 \approx 0.15$ (distance), and $\hat{\beta}_3 \approx 0.82$ (late night), all with the intuitive sign: an unusually large purchase, far from home, late at night, each raises the predicted fraud probability.

These coefficients carry one caveat. The ten transactions are linearly separable (a linear support vector machine classifies all ten correctly), and on separable data plain maximum likelihood has no finite solution, because scaling up any separating coefficient vector always raises the likelihood. The fit above therefore uses scikit-learn's default L2 (ridge) penalty with $C = 1$, which shrinks the coefficients toward zero as Chapter 6's ridge regression shrinks its slopes, acting like a prior that keeps a ten-point model from claiming certainty it has not earned. Weakening the penalty shows what it is holding back: at $C = 10^6$ the coefficients run to $(27.08, -4.15, 12.70)$ with an intercept of -36.10, every transaction is classified correctly, and the distance coefficient even turns negative. The penalized fit is the more sensible model, but transaction 9's miss below is partly a product of the penalty, not of the data alone.

At a 0.5 threshold, the model flags transactions 1, 3, and 6 (true positives) and clears 2, 4, 5, 7, 8, and 10 (true negatives), but assigns transaction 9 a predicted probability of only about 33%, missing a fraud case (a false negative) despite the late-night flag, because its amount and distance were unremarkable. There are no false positives in this small sample, giving

$$\text{Precision} = \frac{3}{3+0} = 1.00, \qquad \text{Recall} = \frac{3}{3+1} = 0.75,$$

$$F_1 = \frac{2(1.00)(0.75)}{1.00 + 0.75} \approx 0.857,$$

a recall figure that, by design, mirrors Chapter 6's loan-default confusion matrix: three true positives and one false negative out of four positive cases.

Transaction 9 illustrates a structural limitation: a fraud that looks unremarkable in isolation will not be caught by a model that looks at one transaction at a time, however well it is fit. That is why Section 15.4.1 layers an account-level behavioral risk score on top of transaction scoring, and why Section 15.4.2 tracks how account risk accumulates over time.

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15.2.4 The Threshold Choice: A Precision-Recall Trade-off

The 0.5 threshold is a modeling choice, not a mathematical requirement, and moving it trades precision against recall as Chapter 6 described. Table 15.3 recomputes the same ten-transaction confusion matrix at two alternative thresholds.

Threshold	Precision	Recall	F_1
0.30	0.800	1.000	0.889
0.50	1.000	0.750	0.857
0.85	1.000	0.500	0.667

Table 15.3: Precision-recall trade-off at three classification thresholds

Lowering the threshold to 0.30 also flags transaction 4 (predicted probability 0.470, legitimate), a false positive, but catches transaction 9 as well, raising recall to 100% while precision falls from 1.00 to 0.80. Raising the threshold to 0.85 does the opposite: transaction 3 (predicted probability 0.822) falls below the bar and becomes a second false negative, so recall drops to 50% while precision stays perfect. Neither threshold is objectively correct: a bank sets its threshold by weighing the cost of a missed fraud against the cost of another false-positive investigation, not by any property of the model.

The choice becomes determinate once a cost is attached to each kind of error. Suppose reviewing a flagged transaction costs \$50 of investigator time and a missed fraud costs \$200 on average, so that operating at threshold τ costs

$$C(\tau) = \$50 \times (\text{transactions flagged}) + \$200 \times (\text{frauds missed}). \tag{15.2}$$

On the ten fitted probabilities this gives \$350 at the default $\tau = 0.5$ (three reviews and one missed fraud) and \$500 at $\tau = 0.85$, against a minimum of \$250 at $\tau = 0.25$, where five transactions are flagged and none is missed. The default threshold is 29% more expensive than the best, and the optimum is no accident of this sample: flagging pays whenever the expected loss avoided exceeds the review cost, $p \times \$200 > \$50$, so the cost-minimizing threshold is the ratio $\$50/\$200 = 0.25$. A threshold of 0.5 is correct only when a missed fraud costs exactly twice a review. Section 15.6.1 runs the calculation at the scale of a real alert queue.

15.2.5 Class Imbalance: Reweighting and Resampling

The examples so far used ordinary logistic regression to keep the statistical issue, extreme class imbalance, transparent. Production systems add more expressive models and dedicated imbalance techniques, which solve different parts of the problem and are often combined.

Cost-Sensitive Learning: Class Weighting

The most direct fix reweights the training loss so that a missed fraud case (a false negative) counts for more than a false positive. Refitting Section 15.2.3's logistic regression on the ten transactions of Table 15.2, with each class weighted inversely to its frequency,

$$\text{Weight (legitimate)} = 10/(2 \times 6) \approx 0.83, \qquad \text{Weight (fraud)} = 10/(2 \times 4) = 1.25,$$

shifts the fitted coefficients to

$$\hat{\beta}_0 \approx -2.25, \qquad \hat{\beta}_1 \approx 0.86, \qquad \hat{\beta}_2 \approx 0.13, \qquad \hat{\beta}_3 \approx 0.85,$$

and raises transaction 9's predicted probability from 33% to about 41.7%, still short of 0.5. At the unchanged 0.5 threshold, class weighting alone makes this small sample's confusion matrix worse: transaction 4 crosses into a false positive, giving precision 0.75, recall unchanged at 0.75, and $F_1 \approx 0.75$, below the original model's 0.857. Class weighting reshapes the decision boundary, but it does not guarantee that a missed case crosses the existing threshold, and on a sample this small its effect is noisy.

Lowering the threshold to 0.40 alongside the reweighted coefficients (compare Table 15.3) does catch transaction 9 (now at 41.7%) while still flagging transaction 4, giving precision 0.80, recall 1.00, and $F_1 \approx 0.889$,

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the same operating point the unweighted model reached at a threshold of 0.30. For a linear model, class weighting and threshold selection move the decision boundary in closely related ways, so a team tuning one should revisit the other.

Resampling: Undersampling and SMOTE

A second family of techniques changes the training data rather than the loss function. Random undersampling discards a random subset of majority-class (legitimate) observations until the training set is more balanced: cheap, but wasteful, since it throws away information about legitimate behavior that helps the model avoid false positives. The Synthetic Minority Oversampling Technique (SMOTE) instead generates synthetic fraud examples: for each real fraud case, it finds its nearest neighbors among other fraud cases (Chapter 6's k-nearest-neighbors distance calculation, used here to construct training points rather than to classify) and creates synthetic points along the line segments connecting them. This adds minority-class examples without discarding legitimate data, but a synthetic point interpolated between two fraud cases is not a real transaction. If the two cases do not represent a coherent fraud pattern, SMOTE can teach a model a feature combination that corresponds to no real fraud behavior, a subtler version of the overfitting risk Chapter 6 introduced.

Applying SMOTE to Table 15.2's four fraud cases (transactions 1, 3, 6, and 9), asking for two synthetic points with $k = 3$ neighbors, produces:

$$\begin{aligned}
&\text{Synthetic point 1:} \quad (1.836,\ 2.763,\ 1), \quad 39.7\% \text{ from transaction 9's } (1.2, 0.3, 1) \\
&\qquad\qquad\qquad\qquad\ \text{toward transaction 3's } (2.8, 6.5, 1), \\
&\text{Synthetic point 2:} \quad (3.263,\ 8.654,\ 1), \quad 54.5\% \text{ from transaction 1's } (3.1, 8.0, 1) \\
&\qquad\qquad\qquad\qquad\ \text{toward transaction 6's } (3.4, 9.2, 1),
\end{aligned}$$

verified by solving for each point's interpolation fraction λ along the amount-z-score axis and confirming it reproduces the distance coordinate exactly. Neither point is an observed transaction: both fall between similar fraud cases here, but SMOTE cannot know whether that holds for every pair it selects on a larger, messier dataset.

15.2.6 More Expressive Models: Ensembles and Unsupervised Anomaly Detection

Logistic regression's score is additive in its features, which is part of why Section 15.2.3's coefficients are easy to interpret, but real fraud often depends on *combinations* of features that an additive model can only approximate. Chapter 6's tree-based ensembles, Random Forest and gradient boosting, split the feature space recursively and can represent a rule like "flag this transaction only if the amount is unusual *and* it occurs at an unusual location *and* at an unusual time, but not if only one of these is true," a nonlinear interaction a single logistic coefficient vector cannot express. This flexibility does not solve class imbalance (a gradient-boosted model trained on the same skewed data shows the same base-rate problem unless combined with class weighting or resampling), and it costs interpretability, which Section 15.6.2 returns to as a regulatory constraint.

Since ten transactions are too few to show an ensemble advantage, the companion notebook simulates 5,000 transactions with a three-way interaction built into the fraud-generating process: amount, distance, and lateness each contribute modestly on their own, but fraud probability jumps only when all three are unusual at once. It then evaluates a logistic regression and a gradient-boosted ensemble, with matched class weighting and no information about the interaction structure, on a held-out test set using precision-recall AUC, the metric argued for below.

Logistic regression, unable to represent the three-way interaction, reaches a PR-AUC of 0.2847; gradient boosting, which can split on combinations of features, reaches 0.3795, a 33.3% relative improvement, with feature importances of 0.408 (amount), 0.490 (distance), and 0.102 (lateness) confirming it uses all three features jointly. This is a real improvement, not the in-sample perfection Chapter 9's Gini/AUC validation example warned against, because it is measured on data the ensemble never saw.

A fundamentally different approach abandons supervised labels. Isolation Forest and autoencoder-based anomaly detection learn what *normal* behavior looks like from the (overwhelmingly legitimate) data itself,

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without being told which cases were confirmed fraud, and flag observations that are isolated or poorly reconstructed relative to that norm. This matters because confirmed-fraud labels can take weeks to arrive (Section 15.4.2), and adversarial adaptation means a labeled training set always describes yesterday's fraud. An unsupervised detector needs no label to flag a new pattern, but its flag only says "this looks unusual," not "this is fraud," so it typically has even lower precision than the supervised models and is best used as one more signal alongside them, not a replacement.

Fitting an Isolation Forest to the same simulated training transactions, without the fraud labels, and scoring the held-out transactions confirms this works in practice: the 34 held-out fraud cases receive a mean anomaly score of 0.5525 versus 0.4658 for legitimate transactions, and the unsupervised score alone achieves a ROC-AUC of 0.7428 against labels it never saw, modest but real discriminatory power obtained without a single confirmed fraud label.

Evaluating Models Under Imbalance: Precision-Recall AUC, Not ROC-AUC

The receiver operating characteristic (ROC) curve plots the true-positive rate against the false-positive rate across every threshold, and its area (ROC-AUC) is a common summary of a classifier's ranking ability. Under severe class imbalance, ROC-AUC can look deceptively good: because the false-positive rate is a share of an enormous legitimate population, even a model that ranks fraud only modestly well can show a high ROC-AUC, the base-rate distortion of Section 15.2.2. The precision-recall (PR) curve and its area (PR-AUC) avoid this blind spot, since precision is computed against the *predicted* positives rather than the huge negative population, which makes PR-AUC the more informative summary for rare events.

Figure 15.1 draws both curves for the two simulated models on the same 1,000 held-out transactions, 34 of them fraudulent, and the two summaries disagree. ROC-AUC prefers logistic regression, 0.7895 against 0.7316 for gradient boosting; PR-AUC reverses the order, 0.3795 against 0.2847. The metrics look in different places. Among the 20 transactions each model ranks highest, gradient boosting finds 14 frauds and logistic regression 10, a precision of 0.70 against 0.50; that is the top of an alert queue, where a bank operates, and it is where the precision-recall curve puts its weight. Logistic regression earns its ROC advantage far down the ranking: at a 40% false-positive rate it finds 26 of the 34 frauds against gradient boosting's 22, but only by flagging 386 legitimate transactions, a precision near 6% that no fraud team would accept. ROC-AUC weights every false-positive rate equally, so it rewards the ordering of transactions no one will review, while the operating region, the first percent or two of false positives, is a thin sliver at the plot's left edge. The right-hand panel also shows the baseline the ROC plot hides: a no-skill classifier sits on the diagonal in ROC space, a respectable-looking half of the area, but at a precision of 0.034, the base rate, in precision-recall space.

15.3 Two Fraud Types That Defeat Transaction-Level Scoring

The preceding part assumed a fraudulent transaction is, in principle, distinguishable from a legitimate one given enough features and a good enough model. Two of the fastest-growing categories of fraud violate that assumption, each in a different way, and both show where the machinery developed above stops being sufficient.

15.3.1 Synthetic Identity Fraud: The Case With No Labels

A synthetic identity is not a stolen one. The fraudster assembles a person who does not exist, typically pairing a real but unused identity number (often a child's, a recent immigrant's, or a deceased person's) with a fabricated name, date of birth, and address. Because no real person owns the composite, no one ever notices it, disputes a charge, or files a report.

That fact changes the detection problem. The account is then *cultivated*: it takes a small credit line, pays every bill on time, and is rewarded with progressively higher limits, as a creditworthy new customer would be. Suppose four tradelines are grown from a \$500 to a \$10,000 limit over eighteen months, with about \$120 a month paid across all of them,

$$\text{Paid in} = 18 \times \$120 = \$2{,}160, \qquad \text{Exposure at bust-out} = 4 \times \$10{,}000 = \$40{,}000,$$

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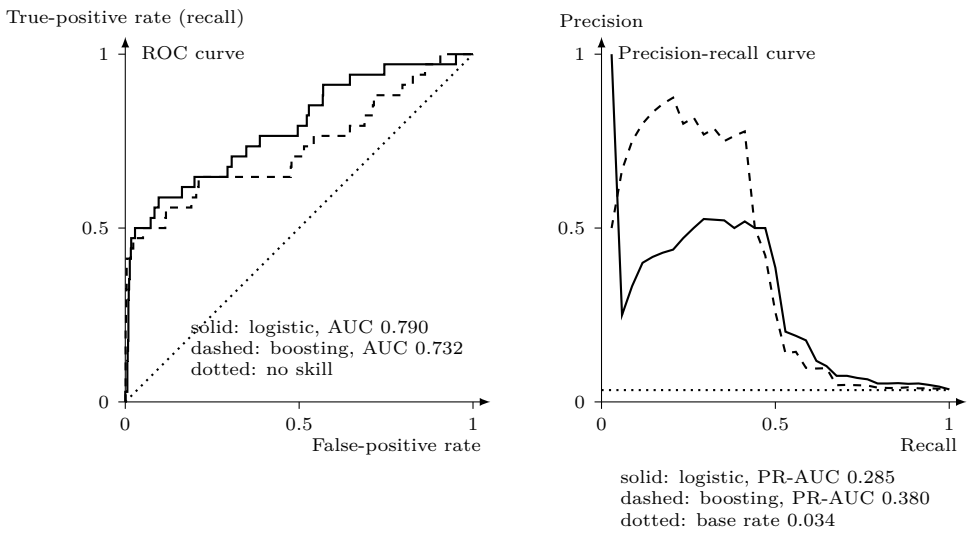


Figure 15.1: ROC and precision-recall curves for the logistic regression and gradient-boosting fraud models on 1,000 held-out transactions with 34 frauds. ROC-AUC ranks logistic regression first; PR-AUC, which weights the top of the alert queue, ranks gradient boosting first.

so the "bust-out," drawing every line to its limit and disappearing, nets roughly $37,840 on an outlay of $2,160, a return of about eighteen times. The behavioral risk score of Section 15.4.1 would have said throughout: velocity normal, geography stable, activity consistent with the profile declared at onboarding. Every input this chapter has built points the wrong way, because during cultivation the account is a model customer. The score spikes only at the bust-out, when the money is gone.

The deeper problem is the labels. A bust-out is usually written off as credit loss, since there is no victim and no obvious moment of theft. If 5% of a lender's 100,000 annual charge-offs are synthetic identities, then 5,000 fraud cases a year sit in the training data labeled "default," and a supervised fraud model never learns them. This is the structural blindness of Chapter 9's reject inference, in a more damaging form: the measured fraud base rate is understated, so every precision figure computed against those labels is optimistic by an unknown margin.

It is also the strongest argument in this chapter for the unsupervised methods of Section 15.2.6. An Isolation Forest or autoencoder needs no confirmed fraud label, and the cultivation pattern (several thin-file accounts sharing an address, device fingerprint, or phone number, all maturing on similar trajectories) is the kind of structure an anomaly detector or the network analysis of Section 15.5.1 can surface without one.

15.3.2 Authorized Push Payment Fraud: The Case Where Every Feature Is Legitimate

In an authorized push payment scam the customer is deceived, by a fake invoice, an impersonated bank official, or a romance or investment pretext, into sending the money themselves. The credentials are correct, the device is the customer's usual one, the authentication succeeds, and the instruction is genuine. This is transaction 9 of Table 15.2 carried to its limit: a payment indistinguishable from a legitimate one on every feature a transaction-level model observes, because it *is* a legitimate instruction from the account holder.

This was long treated as a customer-education problem, since a bank that correctly executed an authorized instruction had done nothing wrong. Mandatory reimbursement regimes, most prominently the United Kingdom's, which since 7 October 2024 has required the sending and receiving firms to split reimbursement of qualifying scam losses equally, moved the loss onto the bank's books and made it an expected-cost calculation.

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The economics then become favorable quickly. Suppose a bank processes 500,000 faster payments a month, 0.02% of which are scams, at an average loss of $4,200, and that it bears half of the reimbursement,

$$\text{Scams} = 0.0002 \times 500{,}000 = 100, \qquad \text{Bank's exposure} = 100 \times \$4{,}200 \times 0.5 = \$210{,}000 \text{ per month.}$$

A model operating at 60% recall and 5% precision, poor by this chapter's earlier standards, catches 60 scams while raising $60/0.05 = 1{,}200$ alerts. At 15 minutes per intervention (a call to the customer before the payment is released) and a fully loaded analyst cost of $45 an hour (the rule of thumb Section 15.6.1 also uses),

$$\text{Cost} = \frac{1{,}200 \times 15}{60} \times \$45 = \$13{,}500, \qquad \text{Loss prevented} = 60 \times \$4{,}200 \times 0.5 = \$126{,}000,$$

a net benefit of $112,500 a month, or $1.35 million a year. The break-even recall is 6.43%: the model pays for itself once it stops six or seven of the hundred monthly scams, a far more forgiving operating point than anything in Section 15.6.1, because a missed case now costs the bank $2,100 rather than nothing.

Because the instruction itself carries no signal, detection has to look elsewhere: at deviation from the customer's own payment history (a first-time payee, an unusual amount, an account that has never sent a faster payment), at context such as a payment initiated during or right after a long inbound call, and above all at the *receiving* side, where mule accounts show the short-dwell, high-pass-through pattern of Section 15.5.1's layering example. The victim's transaction looks normal; the beneficiary's does not.

Worked Example 15.3: Scoring a Real-Time Payment Under a Latency Budget

The calculation above treated the model as a recall and a precision. On an instant-payment rail it must return a decision while the customer waits, and the institutions around it have changed quickly. In the first eighteen months of the United Kingdom's regime, from 7 October 2024 to 31 March 2026, the Payment Systems Regulator reports that 88% of the money lost to APP scams, £316 million, was returned to victims; reimbursement is capped at £85,000 per claim, and the sending firm may charge an excess of up to £100 (PSR 2026). Firms received a matching tool: since 30 October 2024 a UK payment firm may delay an outbound payment by up to four business days when it has reasonable grounds to suspect APP fraud (FCA 2024), which makes a *hold* (a payment paused while the customer is called) a lawful alternative to releasing or refusing it. The problem is spreading with the rails. The Federal Reserve's FedNow service went live in July 2023 and raised its per-payment limit to $10 million in November 2025 (Federal Reserve Financial Services 2025), and the European Union's Instant Payments Regulation requires euro-area payment firms to check, from October 2025, that the payee's name matches the account before a transfer is sent (European Parliament and Council of the European Union 2024d). In the United States, by contrast, Regulation E's consumer protections are built around *unauthorized* transfers, so a payment a deceived customer authorized generally falls outside them, and who bears the loss is set by network rules and each bank's policy, not a federal reimbursement mandate.

Consider one payment, scored at the moment the customer presses "send." Because the evidence arrives as separate signals, the natural first model is the Naive Bayes form of Chapter 1's sequential Bayesian updating: if the signals are independent given the payment's true class, the posterior log-odds is the prior log-odds plus one log-likelihood ratio per signal. Suppose that a new payee appears in 85% of scams but 10% of legitimate payments, a phone call in progress during the session in 60% against 3%, a payee account opened within the last 30 days in 70% against 2%, and a payee account that forwards more than 90% of what it receives within 24 hours in 75% against 3%, and that the amount's z-score against the customer's history is distributed $N(1.5, 1)$ for scams and $N(0, 1)$ for legitimate payments, so that its likelihood ratio is $e^{1.5z - 1.125}$. The payment goes to a new payee, its amount is ordinary ($z = 0.5$), and no call is in progress. From the sender's side alone, starting from the 0.02% scam rate,

$$\begin{aligned} \text{log-odds} &= \ln \tfrac{0.0002}{0.9998} + \ln \tfrac{0.85}{0.10} + (0.75 - 1.125) + \ln \tfrac{0.40}{0.97} \\ &= -8.517 + 2.140 - 0.375 - 0.886 = -7.638, \qquad \Pr(\text{scam}) = 0.048\%, \end{aligned}$$

about 2.4 times the prior, but far too low to act on. The payee's account, however, was opened nine days ago and forwards nearly everything it receives, adding $\ln 35 = 3.555$ and $\ln 25 = 3.219$:

$$\text{log-odds} = -7.638 + 3.555 + 3.219 = -0.864, \qquad \Pr(\text{scam}) = 29.66\%.$$

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The two halves of the evidence disagree by a factor of more than 600: everything suspicious about this payment is on the receiving side.

Whether to hold follows from the same cost comparison as Equation (15.2). A hold costs a 15-minute call at \$45 an hour plus an assumed \$5 of customer friction, \$16.25 in all, and the call breaks the scam in an assumed 70% of cases, so with the bank bearing half of a \$4,200 loss a hold pays when $p \times 0.70 \times \$2{,}100 > \$16.25$, that is, above

$$p^* = \frac{\$16.25}{\$1{,}470} = 0.01105.$$

The sender-side score implies an expected saving of $0.00048 \times \$1{,}470 = \$0.71$, so the payment is released; the full score implies \$436.00, so it is held. The threshold of about 1.1% is the reason a model with 5% precision paid for itself above: under mandatory reimbursement a bank should rationally hold payments it believes are almost certainly genuine.

The payee-side signals are also the slow ones. They live at the receiving firm or a shared intelligence service, and fetching them costs a network round trip inside the decision budget. Suppose 220 milliseconds of the budget remain once the sender-side model has run, and the lookup's latency is lognormal with a median of 80 ms and a log standard deviation of 0.6. The lookup then misses the budget with probability

$$\Pr\left(Z > \frac{\ln(220/80)}{0.6} = 1.686\right) = 0.0459,$$

which rises to 0.1240 with 160 ms left and 0.2496 with 120 ms, and falls to 0.0138 with 300 ms. Table 15.4 shows what the system should do when the answer is late. The companion notebook simulates 5,000,000 payments containing 1,000 scams, scores each with the model above, and rescales to the earlier example's 500,000 payments and 100 scams a month, counting \$16.25 per hold, \$2,100 per scam released, and 30% of \$2,100 for each scam held but not stopped.

Policy	Holds	Recall	Precision	Monthly cost (\$)
Release every payment	0	0.000	–	210,000
Sender-side score only	1,089.6	0.519	0.0476	151,413
Full score, lookup always on time	659.8	0.856	0.1297	94,890
Full score, on timeout use sender score	675.6	0.832	0.1231	98,674
Full score, on timeout hold	23,557.5	0.860	0.0037	466,389

Table 15.4: Monthly outcomes of five holding policies for 500,000 instant payments containing 100 scams, at the hold threshold $p^* = 0.01105$ and a 4.59% chance that the payee lookup misses the latency budget

Three lessons follow. First, the receiving-side signals improve recall and precision at once, from 51.9% to 85.6% and from 4.8% to 13.0%, while *reducing* holds from about 1,090 to 660 a month, and they cut the monthly cost by \$56,523; on the sender side alone the scam and the genuine payment look alike. Second, when the lookup is late the right fallback is the next-best score, which costs only \$3,784 a month against a system that is never late. Third, the instinct to fail safe by holding whatever cannot be scored in time is ruinous: 4.59% of 500,000 payments is 22,950 extra holds, and the policy costs more than twice as much as releasing everything. The latency budget is therefore a modeling parameter, not an engineering detail: it decides how often the model's best evidence is available. Two cautions apply to the numbers. The likelihood ratios are assumptions, and a young account and a high pass-through rate are both symptoms of a mule, so treating them as independent overstates the evidence; the 29.66% is too high, and a fitted logistic regression, which learns that correlation, would temper it. And the 70% stop rate is the least certain parameter, since a customer coached by a scammer will often insist that the payment is genuine.

15.3.3 Generative AI as an Accelerant: Deepfakes and Forged Identities

Generative AI has handed both preceding fraud types a cheap accelerant. Image generators can produce a convincing face for a nonexistent person, or alter a real identity document, and pair it with stolen or

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fabricated data: the synthetic identity of Section 15.3.1, now with a photograph and a document that pass the remote identity verification know-your-customer (KYC) rules require at account opening. Voice cloning and real-time video impersonation give an authorized push payment scam what it used to lack, a familiar face and voice. In the best-documented case, reported by Hong Kong police in February 2024, a finance employee in the Hong Kong office of a multinational, later confirmed as the British engineering firm Arup, joined a video conference in which the chief financial officer and several colleagues were all deepfakes, and made 15 transfers totaling HK$200 million, about US$25 million, to accounts the fraudsters controlled (Chen and Magramo 2024; Magramo 2024). Every payment was authorized by a real employee on the firm's own systems: Section 15.3.2's pattern at corporate scale.

Regulators have taken note. On 13 November 2024 the U.S. Financial Crimes Enforcement Network (FinCEN) issued alert FIN-2024-Alert004, reporting a rise since 2023 in suspicious activity reports describing suspected deepfake media, above all falsified identity documents used to circumvent verification, and asking institutions to tag such reports with the key term "FIN-2024-DEEPFAKEFRAUD" (FinCEN 2024a). Its red flags read like a feature list for a detection model: a photo inconsistent with the customer's other identifying information, identity documents inconsistent with each other, a third-party webcam plugin or a sudden request to change communication method during a live verification check, a refusal to use multi-factor authentication, and device or location data that do not match the documents.

The industry response has three layers, and only the first inspects the media itself. *Liveness detection* prompts the applicant to respond in a live audio or video session and looks for signs of synthesis or injected video; it is an arms race, and FinCEN's alert notes that criminals may answer live prompts with tools that generate synthetic audio and video, although the responses may still reveal inconsistencies. *Out-of-band verification*, confirming an instruction through an independently known channel (a call back to a number already on file, not one supplied in the request), defeats impersonation however good the fake; it would have stopped the Hong Kong transfers. *Behavioral and device signals*, the account-level features of Section 15.4.1 and the receiving-side mule detection above, observe what the fraudster does rather than looks like, and so are the layer least eroded by better generators. The lesson generalizes: once the artifact a control inspects can be forged at near-zero cost, detection has to move to signals that remain expensive to fake.

15.4 Risk Tiering and Survival Analysis

Beyond a single score, this part ranks customers by behavior and models time-to-detection.

15.4.1 Behavioral Risk Frameworks and Customer Risk Tiering

Chapter 9's credit-scoring model mapped a predicted default probability into letter-grade ratings; in fraud and AML, institutions apply an analogous mapping, customer risk tiering, built from behavioral features accumulated over the whole account relationship rather than a single transaction.

Typical inputs include the account's transaction velocity (how this month's frequency and size of transactions compare to its own trailing average, an account-wide analogue of Section 15.2.3's amount z-score), geographic diversity of counterparties, newly added payees, and, for anti-money laundering (Section 15.5.1), whether the occupation and expected volume the customer declared at onboarding under Know-Your-Customer (KYC) requirements are consistent with observed activity.

Table 15.5 shows an illustrative scheme built from a composite behavioral score on a 0-100 scale, combining such features with weights calibrated from the institution's own fraud and AML case data, like Chapter 9's rating buckets.

Composite score	Risk tier	Typical monitoring response
0–29	Low	Standard, periodic account review
30–59	Medium	Increased transaction monitoring sensitivity
60–84	High	Manual review of new activity; tighter velocity limits
85–100	Enhanced due diligence	Senior compliance review; may restrict or close account

Table 15.5: An illustrative customer risk-tiering scheme

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A concrete composite score makes Table 15.5 usable. Suppose an institution combines three behavioral sub-scores, each normalized to a 0-100 scale: a transaction-velocity sub-score (how far this month's activity deviates from the account's trailing average), a geographic-diversity sub-score (how unusual the mix of counterparty locations is), and a KYC-consistency sub-score (how far observed activity deviates from what the customer declared at onboarding), weighted 40%, 35%, and 25% based on the institution's backtested case outcomes.

For an account with a velocity sub-score of 70 (volume well above its recent average), a geographic-diversity sub-score of 50 (a moderate increase in counterparty locations), and a KYC-consistency sub-score of 90 (activity far exceeding what was declared), the composite score is

$$0.40(70) + 0.35(50) + 0.25(90) = 28.0 + 17.5 + 22.5 = 68.0,$$

placing the account in the High tier of Table 15.5, triggering manual review and tighter velocity limits, even though no single sub-score (velocity at 70, geographic diversity at 50) would necessarily have triggered that response alone.

This is the point of a composite score: the KYC-consistency mismatch, the largest contributor at 22.5 of the 68.0 total, combines with two moderate signals to cross a review threshold that no single signal reached, the accumulating, multi-signal evidence of Chapter 1's sequential Bayesian updating example.

The weights are written here as though a committee chose them, and in many institutions one did. But calibrating them from historical case data is supervised learning: given a labeled history of which accounts eventually warranted a report, the weights are fitted coefficients from the logistic regression of Section 15.2.3, and the tiers discretize its fitted score, as Chapter 9's letter grades discretize a predicted default probability.

Tiering changes what the system optimizes for: rather than asking whether *this* transaction looks fraudulent, it asks whether *this account* has become risky enough to warrant tighter monitoring of everything it does. That is the right framework for a case like transaction 9 of Table 15.2, unremarkable alone but possibly the first sign of a compromised account whose score has begun drifting upward.

Regulators (Section 15.5.1) generally require such ongoing, risk-based due diligence rather than a one-time onboarding check, because a customer's risk, like credit risk in Chapter 9's rating-transition framework, evolves.

15.4.2 Survival Models: Time to Detection and Account Risk Over Time

Every classification model in this book, including Section 15.2.3's logistic regression, answers a yes-or-no question at a single point in time. A new question, and one institutions care about operationally, is how long a flagged account survives before fraud is confirmed and the account is closed. The answer helps a compliance team prioritize alerts and a fraud team estimate how much exposure a compromised account accumulates before detection.

Survival analysis, developed in biostatistics to study time until death or disease recurrence, answers this kind of question for data in which the outcome, here fraud confirmation, has not yet happened for every observation. The key complication, which ordinary regression cannot handle, is censoring: some accounts are closed for unrelated reasons (voluntary closure, a non-fraud risk exit) or are still open when the data are pulled, so their time to fraud confirmation is known only to be *at least* the observed time.

Dropping censored observations, or treating them as if fraud were confirmed at the moment of censoring, both bias the estimate; the Kaplan-Meier estimator instead uses every observation up until the point it is censored. The estimator computes the survival function, the probability an account remains unconfirmed (open, or closed for reasons unrelated to fraud) beyond time t, as a product of conditional survival probabilities at each observed event time,

$$\hat{S}(t) = \prod_{t_i \le t} \left(1 - \frac{d_i}{n_i}\right), \tag{15.3}$$

where n_i is the number of accounts still under observation (not yet closed or censored) just before time t_i, and d_i is the number of confirmed-fraud closures at time t_i. This book's companion technical note on the Kaplan-Meier estimator shows that this product-limit formula is not an ad hoc device but the exact nonparametric maximum-likelihood estimator of the survival function once right censoring is written into the likelihood, which is also why the curve steps down only at event times, never at censoring times.

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15.4.3 Worked Example 15.4: Kaplan-Meier Survival of Flagged Accounts

Suppose a bank's fraud team tracks 12 accounts flagged by Section 15.4.1's risk-tiering system, recording the days until a confirmed fraud closure (an event) or until an unrelated closure or the end of the observation window (censored, marked +): 5, 8, 8^+, 12, 15, 15^+, 20, 24^+, 30, 30^+, 35, 40^+. Table 15.6 works through the Kaplan-Meier calculation at each day on which a confirmed fraud closure occurred.

Day t_i	At risk n_i	Fraud closures d_i	Factor $(1 - d_i/n_i)$	$\hat{S}(t_i)$
5	12	1	0.9167	0.9167
8	11	1	0.9091	0.8333
12	9	1	0.8889	0.7407
15	8	1	0.8750	0.6481
20	6	1	0.8333	0.5401
30	4	1	0.7500	0.4051
35	2	1	0.5000	0.2025

Table 15.6: Kaplan-Meier survival calculation for 12 flagged accounts

Two features of Table 15.6 stand out. First, the number at risk, n_i, drops not only at a fraud closure but also whenever a censored observation passes, which is why n_i falls from 12 to 9 between day 5 and day 12 although only two closures occurred in between: the account censored on day 8 also leaves the risk set, without contributing a d_i term.

This is how censored observations contribute information: they stay in the risk set for as long as they are observed, without being counted as either a confirmed event or a survivor beyond their last observed day.

Second, the estimated probability that a flagged account remains unconfirmed beyond 35 days is only about 20.3%. Roughly 80% of accounts eventually confirmed as fraud are therefore confirmed within a little over a month of being flagged. A compliance team can use that number to set service-level targets for completing an investigation, and to recognize that an account still unresolved well beyond that horizon is atypical, an idea developed next.

Figure 15.2 draws the estimate as a step function, flat between events and dropping only on the seven days a fraud closure occurred; the five censored accounts appear as tick marks where they leave the risk set without moving the curve (the three censored on event days 8, 15, and 30 leave just after that day's drop). The steps grow as the curve moves right, from 1/12 of the remaining probability at day 5 to half of it at day 35, because each event removes one account from a shrinking risk set; the curve first falls to one half at day 30, the estimated median time to confirmation. The dashed band is the 95% interval derived in the next subsection.

15.4.4 How Precise Is $\hat{S}(t)$? Greenwood's Formula

A survival estimate like $\hat{S}(35) \approx 0.2025$ is half the story without a sense of how precisely 12 accounts pin it down.

Greenwood's formula gives the standard error of the Kaplan-Meier estimator at any time t,

$$\widehat{\mathrm{Var}}[\hat{S}(t)] = \hat{S}(t)^2 \sum_{t_i \le t} \frac{d_i}{n_i(n_i - d_i)}, \tag{15.4}$$

built from the at-risk counts n_i and event counts d_i already in Table 15.6. Applying this at $t = 20$, where $\hat{S}(20) \approx 0.5401$, the summed term $\sum d_i/(n_i(n_i - d_i))$ across the five event times up to and including day 20 is approximately 0.08175, giving $\widehat{\mathrm{Var}}[\hat{S}(20)] \approx 0.5401^2(0.08175) \approx 0.02385$ and a standard error of SE ≈ 0.154, so a normal-approximation 95% confidence interval is $0.540 \pm 1.96(0.154) \approx (0.237, 0.843)$, wide enough to span nearly the entire plausible range.

At $t = 35$, the final event time, $\hat{S}(35) \approx 0.2025$ with SE ≈ 0.165, giving a 95% interval of approximately (0.000, 0.526) once the lower bound is truncated at zero (a known limitation of the normal approximation

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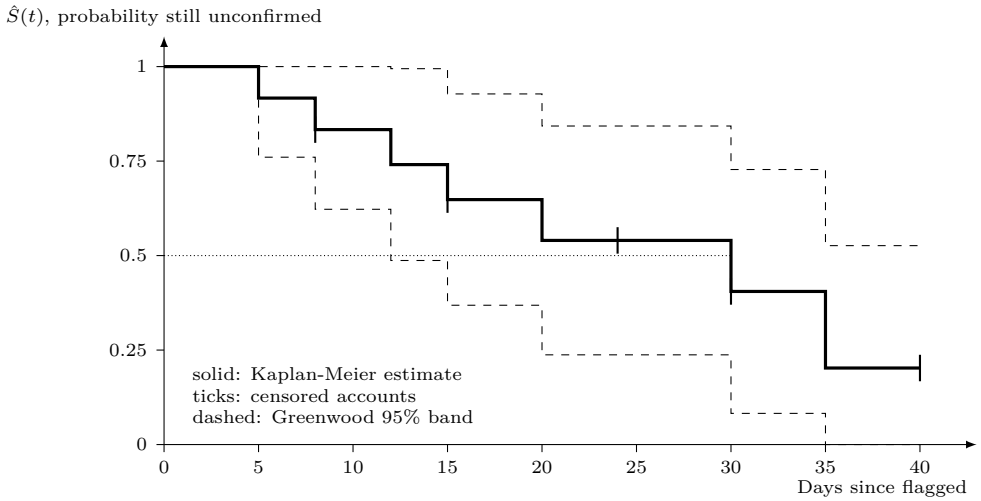


Figure 15.2: Kaplan-Meier survival curve for the 12 flagged accounts of Table 15.6, with censoring marks and the Greenwood 95% band

applied to a probability): the point estimate of roughly 20% is consistent with a 35-day survival probability anywhere from essentially zero to over 50%.

The dashed band of Figure 15.2 applies the same calculation at every event day, widening as the risk set shrinks. This is the small-sample caution of Chapter 7's AR(1) confidence interval and Chapter 9's collateral-coverage regression: with 12 accounts and 7 confirmed closures, the curve's later points carry large uncertainty however precise their four decimal places look.

The 80%-confirmed-within-35-days figure is a planning number from a small pilot sample, not a population parameter, and its interval will narrow as outcomes accumulate.

This is where the survival curve earns its prioritization role, since what orders an alert queue is a *conditional* probability read off $\hat{S}(t)$. A freshly flagged account is confirmed within its first fifteen days with probability $1 - \hat{S}(15) = 0.352$, whereas an account still unresolved at day 20 is confirmed by day 35 with probability $1 - \hat{S}(35)/\hat{S}(20) = 1 - 0.2025/0.5401 = 0.625$, nearly twice as likely to convert over the coming fortnight as the fresh flag is over its first. An investigator should therefore open the aging case before the newest one, the opposite of the instinct to work a queue newest-first, and this follows from Table 15.6 without any additional data.

Kaplan-Meier is nonparametric and uses no covariates, so it can say *when* flagged accounts are confirmed but not *which* ones. The learning extensions supply the missing half. Cox proportional hazards multiplies a baseline hazard by a function of an account's features, and survival forests and gradient-boosted survival models drop the proportionality assumption, so the same censored data yield a per-account predicted time to confirmation, which is what an alert-ranking system consumes. Chapter 9's discrete-time hazard formulation is the credit-risk counterpart of the same idea.

15.5 Money Laundering and Sanctions

This part turns to laundering typologies, sanctions screening, and detection shared across institutions.

15.5.1 Anti-Money Laundering: Structuring, Monitoring, and Regulation

Money laundering, disguising the origin of criminal funds so that they appear to come from a legitimate source, differs from transaction fraud: the funds are often not stolen from the institution, and the concealed crime frequently occurred entirely outside the financial system (drug trafficking, corruption, sanctions evasion) before the money reached a bank.

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In the United States, the Bank Secrecy Act requires banks to file a Currency Transaction Report (CTR) for any cash transaction exceeding $10,000, and a Suspicious Activity Report (SAR) for any transaction pattern, regardless of size, that appears designed to evade reporting requirements or otherwise suggests criminal activity; equivalent frameworks exist in most jurisdictions under the international standards set by the Financial Action Task Force (FATF). The European Union is replacing its patchwork of nationally transposed directives with a 2024 legislative package, published in the Official Journal on 19 June 2024: a directly applicable AML Regulation that applies from 10 July 2027, a sixth AML Directive, and a new EU Anti-Money Laundering Authority (AMLA) seated in Frankfurt, which began operating in July 2025 and, as of October 2026, is expected to select around 40 high-risk obliged entities during 2027 for direct supervision from 2028 (European Parliament and Council of the European Union 2024c; European Parliament and Council of the European Union 2024a; AMLA 2025).

Worked Example 15.5: Detecting Structuring

Structuring (sometimes called "smurfing") is the practice of breaking a large cash transaction into several smaller ones, each below the CTR reporting threshold, to avoid triggering a report. Suppose a customer makes five cash deposits of $9,500 each over three days, each individually $500 below the $10,000 CTR threshold,

$$\text{Total deposited} = 5 \times \$9{,}500 = \$47{,}500,$$

an amount that would trigger a CTR as a single deposit but, spread across five deposits each under the threshold, triggers none. This is why AML monitoring cannot rely on Section 15.2.2's single-transaction rules: "flag any transaction over $10,000" is by construction blind to a pattern engineered to stay under it.

Structuring detection instead applies a rule across a rolling window, for example flagging any customer with three or more cash deposits within seven days each within $1,000 of the threshold, which the five $9,500 deposits would trigger even though no single deposit would. This is the account-level monitoring philosophy of Section 15.4.1: the signal is a pattern across transactions and time, not any single transaction.

15.5.2 Placement, Layering, and Integration: A Network Example

Structuring is the classic technique for the first of the three stages of laundering: *placement*, introducing cash into the financial system, the deposit pattern above. *Layering* follows: moving the placed funds through a chain of transactions and accounts, often across institutions and jurisdictions, to obscure their criminal origin.

Integration is the final stage, reintroducing the layered funds into the legitimate economy, for example through a real estate purchase or an investment in a business, after which they are practically indistinguishable from legitimately earned money.

Layering is where the network-based detection of Table 15.1 becomes necessary, because no single account's activity, viewed in isolation, looks suspicious. Suppose $500,000 in placed funds moves through four shell entities, Alpha, Beta, Gamma, and Delta, each receiving the funds and forwarding 98% of what it received onward within 24 hours (retaining 2%, nominally a service or consulting fee), a repeating pattern that Table 15.7 traces hop by hop.

Entity	Received	Forwarded (98%)	Retained (2%)
Alpha	$500,000.00	$490,000.00	$10,000.00
Beta	$490,000.00	$480,200.00	$9,800.00
Gamma	$480,200.00	$470,596.00	$9,604.00
Delta	$470,596.00	$461,184.08	$9,411.92

Table 15.7: A four-hop layering chain: funds received, forwarded, and retained at each shell entity

After four hops, $461,184.08, over 92% of the original placed amount, reaches its destination, with only $38,815.92 lost to retained "fees," having passed through four legally separate entities in under a week. Figure 15.3 draws this chain as the network it is, rather than as four individually unremarkable rows in a table.

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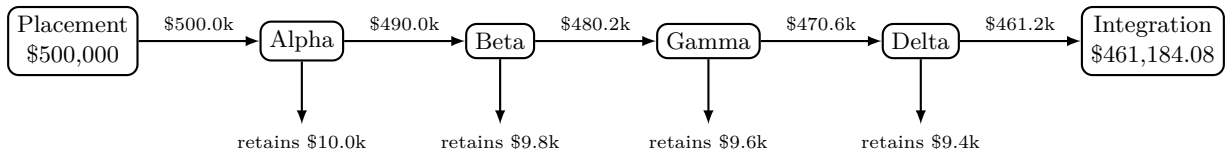


Figure 15.3: The four-hop layering chain of Table 15.7 as a network

No individual hop in Table 15.7 looks wrong; each is a plausible business payment of a plausible size. What is anomalous is the pattern viewed as a network, and each of its three features is a number rather than an impression: a pass-through ratio of exactly 0.98 at every hop, where a real operating business retains and redeploys most of what it receives; a dwell time under 24 hours, against the weeks a real payables cycle takes; and, at every interior node, exactly one party paying in and one party paid out, an in-degree and out-degree of one in the vocabulary of graphs, where a real supplier has many customers and many suppliers. Each has innocent explanations on its own. All three at once, repeated four hops in a row, do not.

That conjunction is a reason to look, not a conclusion. A factoring company, an escrow agent, or a freight forwarder shows high pass-through and short dwell as ordinary business, and a sole-source subcontractor does have one customer and one supplier; the network's shape alone cannot supply a reason to believe the money is criminal. Three further tests do that work. The first is *commercial substance*: a real intermediary has employees, premises, and counterparties outside this chain, whereas Alpha has a bank account, an incorporation date shortly before the flows began, and a registered address shared with dozens of unrelated entities. The second is *common control*: shared directors or beneficial owners, a shared address, telephone number, or device, or four accounts opened at one branch within days of each other, which turns four legally separate companies into one hand moving its own money and has no innocent reading. The third is the fee. Two percent of principal, identical at every hop, is a cut rather than a price, because a real service is charged for work performed and does not scale with the sum passing through. Only with such corroboration does an alert amount to the suspicion of criminal origin a Suspicious Activity Report requires; the graph by itself reports, as Section 15.2.6 said of any unsupervised flag, that something is unusual, not that it is a crime.

This is why anti-money-laundering systems increasingly supplement Section 15.4.1's per-account scoring with graph-based analysis of the relationships between accounts, since a layering scheme is constructed so that every individual node looks unremarkable.

As a learning problem, this is node and subgraph classification on the transaction graph. Community detection isolates tightly connected clusters that ordinary commerce does not produce, and graph neural networks, which Chapter 17 develops on this layering network, learn each account's representation from its neighbors' features as well as its own, scoring an account on the company it keeps. The supervision is weak, since confirmed laundering is rare and arrives late, so in practice these models rank leads for investigators rather than deciding anything.

Figure 15.4 places structuring and layering detection alongside the chapter's statistical and behavioral tools, showing how a monitoring system combines several signals before a case reaches a human investigator.



Figure 15.4: The fraud and AML monitoring pipeline

15.5.3 Ongoing Due Diligence and Adverse-Media Screening

AML regulation also imposes an ongoing due-diligence obligation that goes beyond fraud detection's largely reactive posture. Customer Due Diligence (CDD) at onboarding establishes a customer's expected activity profile (Section 15.4.1's KYC inputs); Enhanced Due Diligence (EDD) applies heightened scrutiny to higher-risk customers such as politically exposed persons or those in high-risk jurisdictions; and ongoing monitoring

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continuously compares behavior against that profile, the time-varying risk view Section 15.4.2's survival framework formalizes.

Two of these obligations are increasingly discharged with the language models of Chapter 16. Adverse-media screening, checking whether a customer appears in news reporting tied to fraud, corruption, or sanctions evasion, is a document-classification problem of the kind that chapter's Naive Bayes and embedding methods address, and it inherits this chapter's precision problem in acute form, since most name matches in news text are the wrong person. Drafting the suspicious-activity report itself, a structured account of what was observed and why it appears suspicious, is now commonly assisted by a large language model working from the alert's records. Both uses are governed rather than autonomous: a filed SAR is a legal statement by the institution, so the model proposes, an investigator signs, and the model is subject to the validation regime of Section 15.6.2.

15.5.4 Sanctions Screening: A Matching Problem, Not a Lookup

A bank must not move money for a party on a government sanctions list, and unlike almost every other obligation in this chapter, this one admits no risk appetite: in the United States liability under the Office of Foreign Assets Control (OFAC) regime is strict, so a bank that processes a prohibited payment has violated the law whether or not it knew, intended, or profited.

It is tempting to conclude that sanctions screening is a database lookup rather than a modeling problem. The legal *consequence* of a match is binary, but the *match itself* is not, and conflating the two is the most common way to misunderstand where the difficulty lies. A watchlist records a name in one romanization; a payment message carries whatever spelling, word order, punctuation, and truncation the originating bank's system produced. Nothing guarantees the two strings agree even when they name the same person, and nothing prevents two different people from having names that differ by a single letter.

The standard tool for comparing two strings that should match but do not is the edit (Levenshtein) distance $d(a, b)$, the minimum number of single-character insertions, deletions, or substitutions turning one string into the other, normalized into a similarity score in $[0, 1]$,

$$\mathrm{sim}(a, b) = 1 - \frac{d(a, b)}{\max(|a|, |b|)}. \tag{15.5}$$

Two short surnames show the mechanics. Turning `SMITH` into `SMYTH` requires one substitution, so $d = 1$ and $\mathrm{sim} = 1 - 1/5 = 0.800$; turning `SMITH` into `SCHMIDT` requires four edits against a longer target, giving $\mathrm{sim} = 1 - 4/7 \approx 0.429$. A screening engine flags any pair scoring above some threshold and sends it to an analyst.

Table 15.8 applies this to a watchlist entry, `Mohammed Al-Rashid`, against seven incoming counterparty names, four of which are the same individual and three of which are not. Scores are computed twice: after simple normalization (uppercase, punctuation stripped), and after also sorting the name's tokens alphabetically, a standard step to defeat surname-first ordering.

Counterparty name	Same person?	Normalized	Token-sorted
Mohammad Al-Rashid	yes	0.9412	0.9412
Mohamed Alrashid	yes	0.9412	0.9412
Muhammad Al-Rasheed	yes	0.7778	0.7778
Al-Rashid, Mohammed	yes	0.2941	1.0000
Mohammed Al-Rashad	no	0.9412	0.9412
Ahmed Al-Rashid	no	0.7647	0.2941
Mohammed Al-Karim	no	0.7647	0.7647

Table 15.8: Name-similarity scores against the watchlist entry `Mohammed Al-Rashid`

Token sorting earns its place: `Al-Rashid, Mohammed` is the same name written surname-first, and plain edit distance scores it 0.2941, below every non-match in the table, because reversing two words costs almost

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as many edits as rewriting the string. Sorting the tokens raises it to 1.0000. The same step pushes `Ahmed Al-Rashid`, a different person, down from 0.7647 to 0.2941, improving separation in both directions.

The result that matters is the one no preprocessing fixes. `Mohammed Al-Rashad`, a different individual whose family name differs from the target's by a single vowel, scores 0.9412, tying *exactly* with two of the four true matches, while the weakest true match, `Muhammad Al-Rasheed`, scores only 0.7778. The highest-scoring non-match outranks a true match, and no threshold separates the two classes: set the bar at 0.90 and the engine clears a sanctioned party while flagging an innocent one; set it at 0.75 and it catches everything by flagging almost everyone.

This is not a defect a better string distance would repair. A name alone does not identify a person, and two of these strings are too close for any function of the characters to tell apart. Production systems therefore treat the string score as a first-stage filter and resolve the remainder with secondary identifiers (date of birth, nationality, passport or national identity number, address), which is why the Know-Your-Customer records collected at onboarding (Section 15.4.1) double as screening inputs.

Here strict liability bites. Section 15.2.3's threshold choice was an economic trade-off; here one missed true match is a legal violation however favorable the arithmetic, so a bank cannot buy precision with a little less recall. It sets the threshold low enough to make a miss nearly impossible and absorbs the resulting false positives, which is what makes screening expensive.

Suppose a mid-sized bank screens 1,200,000 payment messages a month against consolidated watchlists, that 15 of them involve a sanctioned party, and that a threshold set for near-certain recall produces a 0.5% false-positive rate against the remaining traffic. Then

$$\text{Alerts} = 0.99(15) + 0.005(1{,}199{,}985) \approx 6{,}015, \qquad \text{Precision} = \frac{14.85}{6{,}015} \approx 0.00247,$$

so roughly 0.25% of screening alerts are true matches, against the 3.0% alert-to-SAR conversion Section 15.6.1 computes for transaction monitoring: a precision twelve times worse. At 10 minutes per screening alert (half an AML alert's time, since most are cleared on sight) and the same \$45 hourly cost used above,

$$\text{Analyst hours} = \frac{6{,}015 \times 10}{60} \approx 1{,}002, \qquad \text{Monthly cost} \approx 1{,}002 \times \$45 \approx \$45{,}100,$$

or roughly \$541,000 a year. This is a second alert queue, three-quarters the size of the 8,000-alert transaction-monitoring queue analyzed in Section 15.6.1, running at one-twelfth its precision, at every institution that moves money across borders. Supervised learning enters here because labels, scarce almost everywhere else in this chapter, are abundant in screening: every alert is adjudicated by an analyst, so a bank accumulates hundreds of thousands of match and no-match decisions a year. A classifier trained on those dispositions scores a candidate pair on far more than string distance: how common the name is in the relevant population, whether date of birth, nationality, and document number agree, whether the two spellings belong to the same transliteration family, and which kind of list entry was matched. Because it *ranks* rather than thresholds, it can hold recall fixed while pushing most false positives below the line at which the queue is worked.

If a reader takes one operational fact from this chapter, it should be this: in most banks, name screening, not transaction monitoring, is the largest single source of compliance alerts, and reducing its false-positive rate without touching recall is among the highest-value applications of machine learning in financial crime compliance.

15.5.5 Privacy-Preserving Collaboration: Federated Learning Across Institutions

The layering networks above expose a weakness of single-institution detection: a laundering scheme spreads its transactions across many banks so that no single bank sees enough of the pattern. The obvious remedy, pooling transaction data, is largely prohibited: bank-secrecy and data-protection law restricts sharing customer data, and no bank wants its records in a competitor's hands. In federated learning, institutions jointly train one shared model *without any of them transmitting its underlying data.* Each bank fits the current shared model to its own transactions and sends back only the parameter estimates, and a coordinator combines them, most simply by federated averaging (a sample-size-weighted mean), into the next version of the model.

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A hand-checkable round shows the mechanics. Suppose three banks jointly train the fraud-scoring logistic regression of Section 15.2.3 on a shared structuring-intensity feature, holding 2,000, 5,000, and 3,000 labeled transactions respectively. Their local coefficient estimates come back as 0.90, 1.30, and 1.10, the smallest bank's being the noisiest. Federated averaging weights each estimate by its sample share,

$$\hat{\beta}_{\text{fed}} = \frac{2{,}000 \times 0.90 + 5{,}000 \times 1.30 + 3{,}000 \times 1.10}{10{,}000} = \frac{11{,}600}{10{,}000} = 1.16,$$

against an unweighted mean of 1.10; the weighting lets the estimate lean toward the banks with the most evidence. The payoff is clearest from the smallest bank's seat: scoring a transaction with structuring intensity $x = 2$ (with intercept -3), its own local model flags it with probability $\sigma(-3 + 0.90 \times 2) \approx 0.2315$, while the federated model assigns $\sigma(-3 + 1.16 \times 2) \approx 0.3363$, a sharper detector obtained without a single customer record leaving any bank.

Model updates are not automatically harmless: an estimate computed from a small dataset can leak information about the records behind it, so production systems add differential privacy, calibrated random noise added to each update before transmission so that no observer can reliably infer whether any single transaction was in the data. Aggregation makes this affordable: perturbing the three local estimates by illustrative noise draws of $+0.04$, -0.02, and $+0.01$ changes the federated coefficient only from 1.16 to 1.161, because independent noise largely cancels in a weighted average while masking each contribution. The privacy-utility trade-off does bind (more noise, more privacy, a worse model), and federated systems raise new operational questions (a malicious participant could poison the shared model through its updates, the adversarial dynamic of Chapter 18's robustness discussion moved into the training loop). But the core bargain, most of the pooled-data detection gain with none of the pooled-data legal exposure, is why central banks have begun testing it. The BIS Innovation Hub's Project Aurora, concluded in May 2023, combined privacy-enhancing technologies, machine learning, and network analysis on a synthetic dataset of domestic and cross-border payments with embedded laundering patterns, and found this collaborative analysis more effective than today's siloed, rules-based monitoring: it detected up to three times more laundering in complex schemes and cut false positives by up to 80% (BIS Innovation Hub 2023). Those are proof-of-concept results on simulated data, not production evidence; as of October 2026 a second phase is still under way.

15.6 Economics and Governance

A detection model's value is decided by the cost of its false positives and by who is accountable for it. This part is deliberately the shortest in the chapter: detection methods earn their length because they can be taught with worked examples, whereas the failures that produce enforcement actions are organizational, and are better shown once, by the case vignette that follows.

15.6.1 Alert Economics and the Cost of Low Precision

Section 15.2.2 showed that one bank's fraud model can generate 19,600 flagged transactions in a day; AML monitoring systems, which layer structuring rules, behavioral risk scores, and sanctions screening on top of transaction scoring, typically generate a comparably large alert volume relative to the number that warrant a filed report.

Suppose a bank's combined monitoring system generates 8,000 alerts in a typical month, of which 240 are ultimately investigated and filed as SARs,

$$\text{Alert-to-SAR conversion rate} = \frac{240}{8{,}000} = 3.0\%,$$

well below Section 15.2.2's 24.2% fraud-flag precision, since each additional rule or model adds its own false-positive rate. If each alert takes an average of 20 minutes to review, including the 97% ultimately cleared as false positives,

$$\text{Analyst hours per month} = \frac{8{,}000 \times 20 \text{ minutes}}{60} \approx 2{,}667 \text{ hours},$$

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roughly 16 to 17 full-time analysts working a standard month just to clear this one system's queue. At a fully loaded analyst cost of $45 per hour (a common rule of thumb including salary, benefits, training, and overhead), this is a monthly labor cost of

$$2{,}666.67 \times \$45 \approx \$120{,}000,$$

or roughly $1.44 million annually to staff alert review for one system at one institution, before the cost of data infrastructure, the models, or the SAR-filing process. This dollar figure makes precision a direct cost-reduction lever rather than a statistical nicety.

If a model improvement doubled precision from 3.0% to 6.0% while holding recall (and therefore the 240 SARs filed) fixed, the same 240 cases would require only $240/0.06 = 4{,}000$ alerts, half of the original 8,000, cutting analyst hours to $4{,}000 \times 20/60 \approx 1{,}333$ and monthly labor cost to $1{,}333 \times \$45 \approx \$60{,}000$, half of the original $120,000, since cost scales linearly with alert count once recall is held constant.

This is why a bank sets its compliance technology budget and its model-development priorities jointly: a data-science team's precision improvement translates dollar for dollar into the compliance department's staffing budget, as few other model metrics do. It is also why the layered approach of this chapter, transaction scoring (Section 15.2.3), behavioral risk tiering (Section 15.4.1), and survival-informed prioritization (Section 15.4.2), matters as much for cost as for detection accuracy. A system that better ranks alerts by true risk, even at unchanged recall, lets a fixed staff spend its hours on the alerts most likely to be real: Chapter 6's precision-recall trade-off expressed in analyst-hours and dollars.

15.6.2 Governance: Model Risk Management and the Three Lines of Defense

Chapters 8 and 9 introduced model risk management (independent validation, ongoing performance monitoring, and clear accountability for a model's use) for market and credit risk models; Chapter 14 revisited it for algorithmic trading, building on Chapter 13's Knight Capital vignette. Fraud and AML models carry an additional obligation: a poorly governed AML program exposes the institution to direct regulatory and criminal liability, not merely financial loss, since failing to detect and report money laundering is in most jurisdictions a legal violation whether or not the institution profited.

Accountability follows the three lines of defense framework of Chapter 8, unchanged: business units own the first line, an independent compliance and risk function the second, internal audit the third. What differs here is what each line is accountable *for*. The second line builds and validates this chapter's scoring and monitoring models rather than VaR models, and the third line's testing is itself subject to regulatory examination, so a weak audit function is not merely an internal control gap but a finding a supervisor can act on.

This structure answers the failure mode of the case vignette below: a monitoring system can exist on paper, and even generate large alert volumes, while failing in practice if the first line lacks the incentive or staff to act, the second line's challenge is weak, or the third line's audits are not independent of the business they review.

Model Monitoring: Precision Drift and Adversarial Adaptation

Chapter 14 introduced concept drift, a model's fitted relationship growing stale as conditions change, as a reason high-frequency trading models need frequent retraining. Fraud and AML models face a distinct version: because financial crime is adversarial (Section 15.2.1), fraudsters who learn, directly or by trial and error, which patterns a bank's model flags will shift away from them, degrading the model's precision even if the population of legitimate customers has not changed. Table 15.9 illustrates a stylized six-month trajectory for the fraud-scoring model of Section 15.2.3, holding its classification threshold fixed.

If the second line has set a precision floor of 20%, below which the alert burden of Section 15.6.1 is judged too costly relative to the fraud caught, Table 15.9 crosses it between month 3 and month 4. That should trigger a pre-defined response (refitting on recent data, adding features that capture the behavioral shift, or tightening the threshold) rather than letting the model run unchanged because no single month's decline looks alarming. This is the ongoing monitoring Chapter 8's model risk management framework requires, and the second line should define the floor in advance and act once it is breached, not leave the decision to first-line staff focused on today's alert queue.

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Month	1	2	3	4	5	6
Precision	24.2%	22.5%	20.8%	19.6%	18.3%	17.1%

Table 15.9: An illustrative six-month precision drift at a fixed classification threshold. The trajectory is constructed to show the shape, not measured from a real portfolio.

15.7 Case Vignette: The Danske Bank Estonia Case

Between roughly 2007 and 2015, the Estonian branch of Danske Bank, Denmark's largest bank, processed an estimated 200 billion euros in payments for non-resident customers, many connected to Russia and other former Soviet states, including large volumes of transactions with no plausible legitimate economic purpose. The branch had inherited this portfolio through an earlier acquisition and, rather than winding it down as a high-risk legacy business, kept serving and even expanding it, drawn by the fee income of a small branch's outsized transaction volume. A whistleblower first raised concerns in 2013; an external investigation the bank later commissioned found extensive, longstanding failures in customer due diligence, ongoing monitoring, and internal escalation of red flags, the first- and second-line failures the three-lines-of-defense framework of Section 15.6.2 is designed to prevent. Estonian authorities ultimately required the branch's closure in 2019, and in 2022 Danske Bank pleaded guilty in the United States to fraud charges connected to the case, agreeing to forfeit and pay roughly $2 billion.

The case is instructive because it was not primarily a failure of statistical modeling. Rule-based transaction monitoring of the kind Section 15.5.1 describes was reportedly in place. But it generated so many alerts relative to the staff available to review them (the problem of Section 15.6.1) that a large fraction went uninvestigated or were cleared without adequate scrutiny. It is also a case study in correspondent banking risk: much of the flagged activity moved through other banks acting as intermediaries for often hard-to-identify beneficial owners, which makes a single bank's customer due diligence, however well executed, an incomplete safeguard. The case's lasting legacy has been closer supervisory focus, across the European Union and beyond, on the governance question of Section 15.6.2: not whether a monitoring system technically exists, but whether an institution can demonstrate, to an independent third line and ultimately a regulator, that alerts are investigated and reported at a standard consistent with the risk its customer base presents.

A Current Parallel: TD Bank (2024)

The same failure recurred, at larger scale, at TD Bank, the U.S. arm of Canada's Toronto-Dominion Bank. On 10 October 2024 TD Bank, N.A. became the largest bank in U.S. history to plead guilty to Bank Secrecy Act program failures, and the first to plead guilty to conspiracy to commit money laundering. The Justice Department resolution carried about $1.89 billion in criminal fines and forfeiture; coordinated actions by FinCEN (a record $1.3 billion civil penalty), the Office of the Comptroller of the Currency, and the Federal Reserve brought the total to roughly $3 billion, with independent monitors and an OCC cap holding the bank's total U.S. assets at their 30 September 2024 level (DOJ 2024; FinCEN 2024b; OCC 2024). The failures were transaction-monitoring failures in the literal sense: from January 2018 to April 2024 the bank left 92% of its transaction volume, about $18.3 trillion, unmonitored; its monitoring scenarios went essentially unchanged from 2014 through late 2022; and three laundering networks moved more than $670 million through its accounts. Prosecutors traced this to a "flat cost paradigm" that held compliance spending flat while the bank's profits and risk grew. That is Section 15.6.1's alert economics resolved in the worst way: rather than buying precision to make the queue affordable, the bank kept it affordable by not generating alerts, and the second and third lines of Section 15.6.2 did not stop it.

15.8 Challenges in Fraud Detection, Compliance, and Financial Crime

Several challenges recur across nearly every application in this chapter.

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Extreme class imbalance makes accuracy a misleading metric. As Section 15.2.2 showed, even an accurate-sounding model produces many false positives when the event is rare, which is why this chapter, like Chapter 6, insists on precision, recall, and alert economics rather than raw accuracy.

The adversary adapts, which most of this book's other models do not have to contend with. Borrowers (Chapter 9) are not trying to defeat a credit-scoring model; a launderer engaged in structuring (Section 15.5.1) is. A fraud or AML model can therefore decay not just through the concept drift of Chapter 14 but through targeted evasion once a threshold becomes known or guessable.

Data across institutions and jurisdictions is fragmented. As the case vignette's correspondent-banking structure shows, any single institution observes only its own slice of a transaction chain that may span several banks and countries. That is why cross-institutional information sharing, within the limits of privacy and banking secrecy law, materially improves detection, and why the federated learning of Section 15.5.5 is worth its complexity.

False positives carry a real customer and reputational cost, not just an operational one. Every wrongly frozen account or declined transaction is a legitimate customer temporarily denied their own funds, a cost absent from any precision or recall statistic but material to customer relationships and, in aggregate, to regulators judging whether a program's false-positive rate is proportionate.

Explainability requirements constrain model choice. As Chapter 18 develops, a suspicious activity report or account closure frequently must be explained to a regulator or, in some jurisdictions, the customer, which limits how much a program can rely on the least interpretable models of Chapter 6, despite their potential accuracy advantage.

Governance failures, not modeling failures, are the more common root cause of large-scale enforcement actions. The Danske Bank Estonia case, like the Knight Capital vignette in Chapter 13, shows that the binding constraint is frequently not the detection model's sophistication but whether the organization around it (staffing, escalation, independent challenge, and audit) functions as the three-lines-of-defense framework of Section 15.6.2 assumes on paper.

15.9 Key Takeaways

Fraud detection and anti-money-laundering compliance apply the classification tools of Chapters 6 and 9 to a domain defined by extreme class imbalance and an adversary that adapts to known detection rules. Beyond a single logistic regression, production systems draw on the toolkit of Sections 15.2.5 and 15.2.6: class weighting and resampling such as SMOTE address the imbalance, tree-based ensembles capture nonlinear interactions an additive model misses, unsupervised anomaly detection catches patterns no confirmed label yet describes, and precision-recall AUC, not ROC-AUC, is the metric suited to a rare-event problem.

At a low base rate, even an apparently accurate model generates a large false-positive queue, and precision is far more sensitive to the false-positive rate than to the true-positive rate, so alert economics, not just accuracy, must guide how a program is designed and staffed.

A single-transaction model cannot catch a fraud that looks unremarkable in isolation, which motivates account-level risk tiering and survival analysis (the Kaplan-Meier estimator), which uses both confirmed events and censored observations to describe how account risk resolves over time.

Two growing fraud types defeat transaction-level scoring outright. A synthetic identity has no victim, so its losses are booked as credit loss and never become fraud labels, the strongest case for unsupervised detection and cross-account linkage. An authorized push payment scam is authorized by the customer, so detection must move to the customer's own baseline and the beneficiary account; mandatory reimbursement made it an expected-cost calculation, and on an instant-payment rail, when receiving-side evidence arrives late, falling back to the sender-side score is cheap while holding by default is ruinous.

Sanctions screening is a matching problem, not a lookup: a non-match can tie exactly with true matches on name similarity, so secondary identifiers must finish the job, and because liability is strict, recall cannot be traded for precision. Name screening is therefore typically a bank's largest and least precise alert queue.

Anti-money-laundering regulation adds a reporting and due-diligence apparatus (CTRs, SARs, KYC, sanctions screening) built around legal thresholds that launderers evade through placement (structuring), layering (shell-entity chains), and integration. The Danske Bank Estonia case shows that the binding

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constraint on a compliance program is at least as often organizational as statistical: governance discipline separates a monitoring system that works on paper from one that works in practice.

15.10 Suggested Exercises

15.1 A different fraud model achieves a 90% true-positive rate and a 1% false-positive rate at the same 1% fraud prevalence Chapter 1 assumed. Using Bayes' theorem, compute the resulting precision, and explain why a lower false-positive rate improved precision more than the higher true-positive rate in Chapter 1's original model.

15.2 Using the insider trading surveillance example in Section 15.2.2, recompute Pr(Insider | Flag) if the false-positive rate is reduced from 8% to 4%, holding the 2% base rate and 75% true-positive rate fixed, and compare the size of this improvement to what a similarly sized improvement in the true-positive rate alone would produce.

15.3 Using Table 15.2, suppose an eleventh transaction has an amount z-score of 2.2, a distance of 4.0 (hundreds of km), and occurs during the day (late-night flag 0). Using the fitted coefficients in Section 15.2.3, compute the transaction's predicted fraud probability and classification at a 0.5 threshold.

15.4 Using transaction 9 in Table 15.2, explain why a transaction-level model can have perfect precision and still leave a bank exposed to fraud, and how Section 15.4.1's behavioral risk tiering could catch the missed case.

15.5 Using Section 15.2.5's class-weighting example, explain why raising transaction 9's predicted probability from 33% to 41.7% did not by itself fix the false negative at the 0.5 threshold, and why class weighting plus a lower threshold reached the same operating point as lowering the unweighted model's threshold to 0.30.

15.6 Explain in a short paragraph why ROC-AUC can overstate a fraud model's practical usefulness under extreme class imbalance, and why precision-recall AUC does not share this weakness.

15.7 Using Table 15.6's method, compute the Kaplan-Meier survival estimate for a new sample of 8 flagged accounts with observed times (days, + denotes censored) of 4, 6^+, 10, 14, 14^+, 22, 28^+, 33, showing your calculation at each confirmed-fraud event time.

15.8 A customer makes four cash deposits of $9,200 each within five days. Compute the total deposited, and explain, using Section 15.5.1, why no individual deposit triggers a Currency Transaction Report although the total would if deposited at once.

15.9 Using Table 15.7's method, compute a five-hop layering chain starting from $300,000, with each entity retaining 3% and forwarding the rest, and report the amount reaching the final destination and the total retained across all five hops.

15.10 A bank's monitoring system generates 12,000 alerts per month, of which 180 become filed SARs, and each alert takes an average of 25 minutes to review. Compute the alert-to-SAR conversion rate and the total analyst-hours required per month, and compare both figures to Section 15.6.1's worked example.

15.11 Using Section 15.6.2's three-lines-of-defense framework, identify which line (first, second, or third) each of the following failures most directly represents, and why: (a) a branch employee ignores an internal escalation policy for a suspicious customer; (b) an independent audit team reports directly to the same regional manager whose unit it is auditing; (c) the compliance function never independently tests whether the transaction-monitoring model's alert thresholds are still appropriate.

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15.12 Pick two of the challenges listed in this chapter and, for each, describe a concrete scenario, not used in the text, in which it could cause a fraud or AML program to fail despite a well-designed statistical model, and explain briefly why more computing power or a more flexible model would not by itself resolve either scenario.

15.13 Explain, using the Danske Bank Estonia case vignette, why correspondent banking relationships make financial crime detection harder for any single institution, and describe one mechanism, such as cross-institution information sharing or a shared utility for sanctions and beneficial-ownership screening, that could mitigate this limitation.

15.14 Compare the class-imbalance problem in this chapter's fraud examples to the loan-default example in Chapter 6: using its confusion matrix (a 75% true-positive rate and, among the six actual non-defaulters, one false positive), compute what the loan portfolio's default rate would need to be, at the same true-positive and false-positive rates, for precision to fall to this chapter's 24.2%.

15.15 Using Greenwood's formula, compute $\widehat{\mathrm{Var}}[\hat{S}(12)]$ and the resulting 95% confidence interval for $\hat{S}(12) \approx 0.7407$, compare its width to the interval computed at $t = 20$ in the text, and explain why the earlier interval is narrower even though both come from the same 12 accounts.

15.16 Using Section 15.5.4's similarity definition, compute sim for the pairs `MOHAMMED ALRASHID`/`MUHAMMED ALRASHID` and `MOHAMMED ALRASHID`/`MOHAMMED ALRASHEED`, and explain why the second pair, which involves a different family name, nonetheless scores higher than `Muhammad Al-Rasheed` does in Table 15.8.

15.17 Using Section 15.5.4's screening-volume example, recompute the monthly alert count, precision, analyst hours, and dollar cost if a better matching engine cut the false-positive rate from 0.5% to 0.2% while holding recall at 99%, and express the annual saving relative to the $541,000 in the text. Then compute what the same 60% reduction in false positives would save in Section 15.6.1's transaction-monitoring queue, and explain why a bank might still prioritize the screening engine even though the transaction-monitoring saving is the larger.

15.18 Explain why a bank cannot manage its sanctions-screening alert volume as Section 15.2.3 manages fraud-scoring alerts, by raising the threshold, and describe what a screening program can change instead to reduce false positives without reducing recall.

15.19 Using Section 15.3's cultivation arithmetic, recompute the bust-out for a synthetic identity that grows *six* tradelines to a $7,500 limit each over 24 months while paying $150 a month across all of them. Report the amount paid in, the exposure at bust-out, the net loss, and the return on the fraudster's outlay, and explain why the return multiple is lower than the text's despite the larger loss.

15.20 Suppose the reimbursement rules of Section 15.3 were tightened so the sending bank bore 100% of authorized push payment losses rather than half. Holding the 500,000 monthly payments, 0.02% scam rate, $4,200 average loss, 60% recall, 5% precision, 15-minute intervention and $45 hourly cost fixed, recompute the loss prevented, the net monthly benefit, and the break-even recall, and comment on what the change implies about how much detection effort a bank would rationally fund under each regime.

15.21 Using Figure 15.1 and the operating points quoted with it, compute the precision of each model at a false-positive rate of 1%, where logistic regression reaches a true-positive rate of 0.294 and gradient boosting 0.412 on the 34 frauds and 966 legitimate transactions. Compare the result with the two models' ROC-AUC ranking, and explain which ranking a fraud team that can review about 25 alerts per 1,000 transactions should trust.

15.22 Using Section 15.3.2's real-time scoring example, recompute the sender-side log-odds and scam probability for the same payment if a phone call *is* in progress during the session, and decide whether the payment should be held before the payee lookup returns. Then recompute the hold threshold p^* if a call stops only 50% of scams, and state whether the decision changes. Finally, using Table 15.4,

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explain why holding every payment whose payee lookup times out is worse than releasing every payment.

15.23 **Challenge (real data).** Download the Kaggle "Credit Card Fraud Detection" dataset (Université Libre de Bruxelles; roughly 285,000 European cardholder transactions over two days, with 492 confirmed frauds and features already anonymized via PCA). Its 0.172% fraud rate is far more extreme than Table 15.2's toy example. (a) Fit a logistic regression with and without class weighting, as in Section 15.2.5, and report precision, recall, F_1, and precision-recall AUC for both, since accuracy is uninformative at this imbalance. (b) Using Section 15.6.1's cost-based framework, with a \$50 review cost per flagged transaction and a \$200 average loss per missed fraud, find the threshold that minimizes total expected cost, and compare its alert volume to a naive 0.5 threshold's. (c) Since the features are anonymized PCA components, this dataset cannot support the domain-driven feature engineering (for example, distance from home) of this chapter's toy example; discuss what information a real bank's fraud team has that this privacy-scrubbed dataset withholds.

15.11 Further Reading

- Bolton and Hand (2002), "Statistical Fraud Detection: A Review." A foundational survey of statistical approaches to fraud detection, including the class-imbalance and base-rate issues central to Section 15.2.2.
- Chawla et al. (2002), "SMOTE: Synthetic Minority Over-sampling Technique." The original paper introducing the resampling technique discussed in Section 15.2.5.
- Klein and Moeschberger (2003), *Survival Analysis*. A standard, thorough reference on the Kaplan-Meier estimator and censoring introduced in Section 15.4.2.
- FATF (2012), *International Standards on Combating Money Laundering and the Financing of Terrorism & Proliferation*. The primary international regulatory framework underlying the KYC, CDD, and reporting obligations discussed in Section 15.5.1.
- OFAC (2019), *A Framework for OFAC Compliance Commitments*. The regulator's own statement of what a sanctions compliance program must include, and the primary source for the strict-liability standard underlying Section 15.5.4.
- FinCEN (n.d.), *Bank Secrecy Act Regulations and Guidance*. The primary U.S. regulatory source for the CTR and SAR reporting requirements discussed in this chapter.
- Bruun & Hjejle (2018), *Report on the Non-Resident Portfolio at Danske Bank's Estonian Branch*. The external investigation commissioned by Danske Bank itself, the primary source for the factual account in this chapter's case vignette.
- IIA (2020), *The IIA's Three Lines Model*. The current authoritative statement of the governance framework developed in Section 15.6.2.

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Chapter 16

Natural Language Processing in Finance

16.1 Introduction

Every earlier chapter has worked with numbers that arrive already structured: prices, returns, ratios, transaction amounts. Yet much of the information that moves markets starts as an earnings-call transcript, a paragraph in a 10-K's risk factors, a news wire, or a social-media post. It must be converted into features before any of the tools of Chapters 6 through 15 apply. This chapter develops that conversion and the techniques built on it, collectively natural language processing (NLP).

Several earlier threads converge here. Chapter 1 noted that deep learning learns directly from unstructured text, and that large language models (LLMs) extend it to summarizing earnings calls and extracting information from filings. Chapter 3 observed that large asset managers use NLP to parse filings faster than human analysts. Chapter 6 introduced Naive Bayes as a generative classifier for word counts, of which LLMs are far more elaborate examples. Chapter 13 flagged that a sentiment score is only half of an event-driven strategy; executing the trade is the other half. This chapter makes those threads concrete.

Nineteen worked examples run through the chapter. The first build representations: bag-of-words and TF-IDF vectors for a few earnings-call sentences, and a sentiment score from a subset of the Loughran-McDonald dictionary, the lexicon used in finance research and practice. Those feed an event-driven trading signal, evaluated as Chapter 13 evaluates any strategy, and a Naive Bayes classifier that sorts 10-K risk-factor sentences into categories. The chapter then turns to modern architectures: word-embedding geometry, including a vector-arithmetic analogy, and a hand-computed scaled dot-product self-attention calculation, the core of the transformer, extended to show how a causal mask changes a token's representation across encoder-only, decoder-only, and encoder-decoder architectures. Then come a retrieval-augmented generation example grounding an answer in a retrieved passage, with precision@k and faithfulness metrics; a forward diffusion step for synthetic stress scenarios; a Gunning Fog readability comparison; a low-rank adaptation (LoRA) parameter count; a three-step LLM agent that computes and cross-checks a financial ratio; how per-step reliability compounds over a multi-step agent task; and an API-versus-self-hosting cost comparison showing when quantization's memory savings become deployment savings.

Because generative AI (LLMs, retrieval, prompting, and synthetic data from GANs and diffusion models) is developed here in more depth than any earlier topic, this chapter runs longer than the rest of the book.

16.2 Text as Data

Most financial information arrives as prose, and this part turns that prose into numbers a model can use.

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16.2.1 Text as Data in Finance

Financial text differs from the numerical data used so far in three ways that matter for modeling. First, it is unstructured: a sentence has no predefined fields the way a balance sheet line does, so the first step is always to impose structure. Second, it is domain-specific: sentiment analysis trained on product reviews or social media misreads financial language. "Liability" is neutral accounting terminology, and "losses were smaller than expected" is good news phrased with two words, losses and smaller, that a generic model would likely score as negative. Third, financial text is produced under legal and strategic incentives: management chooses what to disclose, how to phrase it, and how much to hedge, so the text is a strategic object, not a neutral report.

Table 16.1 catalogs the main sources of financial text and where each is developed.

Source	Typical use	Developed in
Earnings call transcripts	Management tone and forward guidance, extracted as a sentiment score	Sections 16.2.3, 16.3.1
Regulatory filings (10-K, 10-Q)	Risk-factor classification, disclosure readability	Sections 16.3.2, 16.2.4
News wires and press releases	Event detection, sentiment scoring for event-driven strategies	Section 16.3.1
Social media and analyst reports	Aggregate sentiment signals, higher noise, higher volume	Section 16.4.4

Table 16.1: Sources of financial text and where they are used in this chapter

Figure 16.1 sketches the pipeline from raw text to a model-ready representation to an application. Every technique in this chapter fills in the middle step.

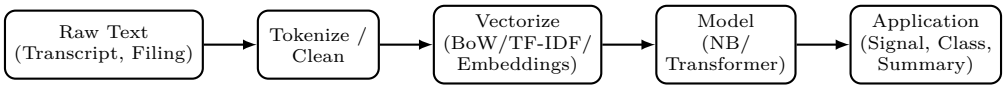


Figure 16.1: The text-to-application pipeline this chapter develops

The tokenize step decides what the units of text are. Classical methods split text into words at spaces and punctuation, so a word unseen in training is unknown. Transformer models instead use subword tokenization, most often byte-pair encoding (BPE) (Sennrich et al. 2016): start from single characters (or bytes) and repeatedly merge the most frequent adjacent pair into a new entry until the vocabulary reaches a target size, typically tens of thousands to a few hundred thousand entries. Common words become one token, while a rare one, a new ticker or a drug name, splits into familiar pieces, so no input is out of vocabulary. Numbers are where this matters in finance. A tokenizer chops a figure such as 1234567 into chunks set by merge frequencies, not place value, so the digit groups need not line up with thousands and millions. Singh and Strouse (2024) show the cost: merely forcing digits into groups of three from the right, as thousands separators do, raised GPT-4's accuracy on an addition task from 84.4% to 98.9%. The practical rule, applied in Section 16.6.3, is to treat a language model as a reader of numbers, not a calculator: let it extract figures, and hand the arithmetic to code.

16.2.2 From Text to Numbers: Bag-of-Words and TF-IDF

Suppose an analyst wants to feed thousands of earnings-call transcripts into one of Chapter 6's classifiers, each of which expects a fixed-length numeric vector, not a paragraph. The simplest conversion is the bag-of-words representation: fix a vocabulary, and represent each document by the count of each term it contains, discarding word order. This is drastic ("growth despite headwinds" and "headwinds despite growth" produce identical

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vectors), but it is a surprisingly effective baseline, and it is the "word counts" representation Chapter 6 anticipated for Naive Bayes.

A raw count, however, overweights common words that appear in almost every document and discriminate little. Term frequency-inverse document frequency (TF-IDF) corrects for this. For term t in document d, drawn from a collection of N documents,

$$\text{tf-idf}(t, d) = \text{tf}(t, d) \times \ln\left(\frac{N}{\text{df}(t)}\right), \tag{16.1}$$

where $\text{tf}(t, d)$ is the raw count of t in d and $\text{df}(t)$ is the number of documents containing t at least once. A term that appears in every document has $\text{df}(t) = N$ and therefore an idf of $\ln(1) = 0$, its tf-idf weight vanishes regardless of how often it appears, while a term confined to a single document receives the largest possible idf weight.

Worked Example 16.1: Bag-of-Words and TF-IDF for Four Earnings-Call Sentences

Consider four short earnings-call sentences, treated as four documents ($N = 4$):

D_1: "revenue growth exceeded expectations this quarter"
D_2: "management is cautious about revenue growth given the headwinds"
D_3: "management remains confident despite the headwinds this quarter"
D_4: "expectations were not met and revenue declined"

Restricting attention to six vocabulary terms gives the term-frequency matrix in Table 16.2, alongside each term's document frequency, idf, and resulting tf-idf weight in D_1.

Term	D_1	D_2	D_3	D_4	df	idf $= \ln(4/\text{df})$	tf-idf in D_1
revenue	1	1	0	1	3	0.2877	0.2877
growth	1	1	0	0	2	0.6931	0.6931
headwinds	0	1	1	0	2	0.6931	0.0000
expectations	1	0	0	1	2	0.6931	0.6931
quarter	1	0	1	0	2	0.6931	0.6931
management	0	1	1	0	2	0.6931	0.0000

Table 16.2: Term frequency, document frequency, idf, and TF-IDF weight in D_1

The first four columns of Table 16.2 are the bag-of-words representation. Over the vocabulary (revenue, growth, headwinds, expectations, quarter, management) they give $D_1 = (1, 1, 0, 1, 1, 0)$ and $D_2 = (1, 1, 1, 0, 0, 1)$, the vectors a classifier would receive without weighting. They make the loss of word order concrete: any rearrangement of D_1's words produces the same vector, and D_1 and D_4 share two terms although one reports growth and the other a decline.

"Revenue" appears in three of the four documents, so it earns a modest idf of $\ln(4/3) \approx 0.2877$; every other term appears in two, giving a larger idf of $\ln(4/2) = \ln 2 \approx 0.6931$. "Headwinds" and "management" do not appear in D_1, so their tf-idf weight in D_1 is zero whatever their idf, while "growth," "expectations," and "quarter" each occur once in D_1 and contribute their full 0.6931. A downstream classifier, such as the Naive Bayes model of Section 16.3.2, would receive this in place of raw counts.

16.2.3 Financial Sentiment Analysis and the Loughran-McDonald Dictionary

Suppose a trader wants to know, within seconds of an earnings call ending, whether management sounded upbeat or defensive, without waiting for an analyst's summary. The most direct way to score sentiment is a lexicon-based approach: maintain a list of positive and negative financial words, count how many of each appear in a document, and combine the counts into a tone score,

$$\text{tone} = \frac{\#\text{positive words} - \#\text{negative words}}{\#\text{total words}}. \tag{16.2}$$

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Generic lexicons, built from product reviews or general text, perform poorly in finance for the reason Section 16.2.1 raised: "tax," "cost," or "liability" read as negative in everyday English but are neutral in a filing. The Loughran-McDonald dictionary, built from 10-K filings, classifies words into finance-specific categories (positive, negative, uncertainty, litigious, and others) by how they function in financial disclosure.

Worked Example 16.2: Scoring Two Earnings-Call Transcripts

Using a small subset of real Loughran-McDonald positive words ({strong, growth, exceeded, confident, improved}) and negative words ({cautious, headwinds, decline, declined, weak, concern, concerns}), consider two short transcripts:

> Transcript A: "revenue growth was strong this quarter and results exceeded expectations across every segment we remain confident heading into next year"
> Transcript B: "management is cautious about headwinds and results declined modestly we have some concerns about demand next quarter"

Transcript A contains 20 words, of which 4 (growth, strong, exceeded, confident) are positive and none are negative, giving a tone score of $(4-0)/20 = 0.20$. Transcript B contains 17 words, of which none are positive and 4 (cautious, headwinds, declined, concerns) are negative, giving a tone score of $(0-4)/17 \approx -0.2353$. Scores like these feed the event-driven trading signal developed next.

Worked Example 16.3: Scoring Central-Bank Communication

Company disclosures are not the only text markets parse by machine. The Federal Open Market Committee's post-meeting statement is arguably the most market-moving recurring document in finance, and it is scored with the machinery above, with one twist: the axis is not positive versus negative but *hawkish* versus *dovish*, language signaling tighter versus easier future monetary policy. The sign convention depends on the asset: a "strong" labor market is good news about the economy but hawkish news for bond prices. A general-purpose dictionary, and even Loughran-McDonald, therefore misreads central-bank language as generic lexicons misread 10-Ks, and practitioners maintain purpose-built hawkish and dovish word lists instead.

Using the hawkish list {strong, elevated, increases, tightening, restrictive} and the dovish list {moderated, eased, easing, patient, softening}, consider two consecutive stylized statements:

> January: "inflation remains elevated and the committee anticipates that ongoing increases in the target range will be appropriate the labor market remains strong"
> March: "inflation has moderated in recent months and the committee will be patient as it assesses incoming data the labor market has shown signs of softening"

The January statement contains 22 words, of which 3 (elevated, increases, strong) are hawkish and none are dovish, giving a net-hawkishness score of $(3-0)/22 \approx +0.1364$. The March statement contains 25 words, of which none are hawkish and 3 (moderated, patient, softening) are dovish, scoring $(0-3)/25 = -0.1200$. What a rates trader cares about, though, is neither level but the *change*: a drift of $-0.1200 - 0.1364 \approx -0.2564$, a decisive swing from hawkish to dovish. Every participant has already read the previous statement, so the information in a new one lies almost entirely in what changed. That is why coverage of an FOMC release is dominated by redline comparisons against the prior statement, and why the tone *difference* is the natural trading signal: the event-driven logic of the next section applied to the policy calendar. Research versions, scoring decades of central-bank communication and measuring the market reaction to its unanticipated component, are now a standard tool in empirical monetary economics (see Further Reading).

16.2.4 Disclosure Analysis: Readability and the Gunning Fog Index

Regulators require public disclosures to be understandable to ordinary investors (the U.S. Securities and Exchange Commission's "Plain English" rule is one such requirement), and readability metrics measure whether a passage meets that bar. The Gunning Fog index estimates the years of formal education a reader needs to understand a passage on a first reading,

$$\text{Fog} = 0.4 \left[\frac{\text{words}}{\text{sentences}} + 100 \times \frac{\text{complex words}}{\text{words}} \right], \tag{16.3}$$

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where a complex word is (approximately) one with three or more syllables.

Worked Example 16.4: Two Disclosure Paragraphs Compared

Consider two short passages. Paragraph A: "The firm's sales grew this year. Profits also rose. We expect this trend to grow next year." This has 17 words across 3 sentences and zero complex words (every word has one or two syllables), giving Fog $= 0.4\,[(17/3) + 100(0/17)] = 2.27$.

Paragraph B: "The Company's operations are subject to numerous regulatory, competitive, and macroeconomic uncertainties. These uncertainties could materially and adversely affect anticipated profitability and operational continuity." This has 24 words across 2 sentences, of which 14 are complex (numerous, regulatory, competitive, macroeconomic, uncertainties, materially, adversely, anticipated, profitability, operational, continuity, and others), giving Fog $= 0.4\,[(24/2) + 100(14/24)] = 28.13$.

A Fog index above roughly 17 corresponds to a post-graduate reading level; Paragraph B's 28.13 is far beyond that. The toy paragraphs are built to contrast sharply, but the direction matches a well-documented finding: 10-K risk-factor sections and other mandatory disclosures routinely score in this elevated range, dense, technical, and long-sentence.

Readability metrics give compliance teams an automatable way to flag disclosures that may miss the plain-English standard, and give researchers a candidate predictive signal (some published research finds that harder-to-read disclosures are associated with worse subsequent firm performance, consistent with strategic obfuscation).

The Fog index is also useful during drafting, not only after the fact. Rewriting Paragraph B in plainer language, "Our business faces many risks from rules, competition, and the economy. These risks could hurt our profits and how well we operate," gives 22 words across 2 sentences with only 3 complex words (competition, economy, operate), so

$$\text{Fog} = 0.4\,[(22/2) + 100(3/22)] \approx 9.85,$$

close to Paragraph A's score and nowhere near Paragraph B's original 28.13, while covering the same categories of risk and the same warning about profitability. This is the use a Plain English rule intends to encourage: the legal and business meaning did not change, only sentence structure and word choice, yet the Fog score fell by nearly two-thirds. A readability check built into the drafting workflow can therefore flag an overly complex passage before it is filed, not after a regulator or researcher measures it.

16.3 Classical Text Models

Before transformers, text was classified and traded on with methods that remain fast, auditable, and often sufficient.

16.3.1 Worked Example 16.5: An Event-Driven Sentiment Trading Signal

A sentiment score is only half of an event-driven strategy: the signal must still be evaluated like any trading strategy, and eventually converted into an executed position, which is Chapter 13's concern. This section stops at generating and evaluating the signal.

Suppose eight earnings-call events produce the tone scores and next-day stock returns in Table 16.3. A simple trading rule takes a long position when tone is positive and a short position when tone is negative, so the strategy's return on each event is sign(tone) $\times$ return.

The strategy is correct (its position's sign matches the return's) in 6 of 8 events, a 75% hit rate. Events 4 and 7 are misses: in event 4, mildly negative tone preceded a small gain, and in event 7, mildly positive tone a small loss. In both the tone score was close to zero and only weakly informative: a sentiment signal is noisy, not a deterministic predictor. Comparing the two strategies (Sharpe-like ratio: mean divided by standard deviation, with no annualization applied here),

$$\begin{aligned} \text{Signal-driven strategy:} &\quad \text{mean} = 0.775\%,\ \text{std} = 0.8481\%,\ \text{Sharpe-like} = 0.9138, \\ \text{Buy-and-hold:} &\quad \text{mean} = 0.075\%,\ \text{std} = 1.1829\%,\ \text{Sharpe-like} = 0.0634. \end{aligned}$$

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Event	Tone score	Next-day return (%)	Signal	Strategy return (%)	Correct?
1	+0.15	+1.2	Long	+1.2	Yes
2	−0.10	−0.8	Short	+0.8	Yes
3	+0.08	+0.3	Long	+0.3	Yes
4	−0.05	+0.4	Short	−0.4	No
5	+0.20	+2.1	Long	+2.1	Yes
6	−0.18	−1.5	Short	+1.5	Yes
7	+0.02	−0.2	Long	−0.2	No
8	−0.12	−0.9	Short	+0.9	Yes

Table 16.3: Eight earnings-call events: tone score, next-day return, signal, and strategy return

The signal adds value on this small sample, with a higher and more consistent return, even though it is wrong a quarter of the time. Turning it into an executed position, at the right size and without excessive market impact, is the execution problem of Chapter 13.

The tone score is a text-based sentiment measure, worth placing beside a price-based one: the CBOE Volatility Index (VIX) from Chapter 5, the market's "fear gauge," which aggregates option prices into a forward-looking measure of expected market-wide volatility without reading any text. The nickname needs care: VIX is mostly a forecast of how much the index will move, and only partly a reading of how investors feel, because an option's price compensates both for expected volatility and for the discomfort of bearing it. Only that second, risk-compensation component behaves like a sentiment score; the headline index blends the two. The measures are complementary. A tone score captures sentiment specific to one company's call, which VIX cannot see, while VIX moves on macroeconomic or geopolitical news that names no company and that a firm-specific model was never trained to detect. A more sophisticated strategy conditions on both, using VIX as a market-wide risk-regime signal, as Chapter 13 describes, beside the firm-specific tone score: a positive tone is a different proposition in a calm market than in one gripped by elevated uncertainty.

Worked Example 16.6: An Event Study: Abnormal Returns Around One Call

Table 16.3 judged the signal against raw next-day returns, which mix the stock's reaction to its own news with its ordinary co-movement with the market. On a day the market rises 1%, a stock with a beta above one rises too, whatever management said. The event study (Campbell et al. 1997) separates the two, and the evidence that text moves prices, including the research behind the Loughran-McDonald dictionary, is almost all stated in its terms. It has three steps. First, fit the market model

$$R_t = \alpha + \beta R_{m,t} + \varepsilon_t$$

over an estimation window that ends well before the event, so the news cannot contaminate the fit; here 120 trading days, days −130 to −11. Second, over a short event window, compute each day's abnormal return $AR_t = R_t - (\hat{\alpha} + \hat{\beta} R_{m,t})$, the part of the return the market does not explain, and add them into the cumulative abnormal return CAR. Third, test it: if the event carried no news, each AR_t is noise with the residual standard deviation $\hat{\sigma}$ of the fit, so over an L-day window

$$t = \frac{CAR}{\hat{\sigma}\sqrt{L}}, \tag{16.4}$$

which ignores the small additional error from estimating $\hat{\alpha}$ and $\hat{\beta}$ on 120 days.

Apply this to event 5 of Table 16.3, a tone score of +0.20 followed by a +2.1% return. The companion notebook simulates the estimation window from a market model with $\alpha = 0.02\%$, $\beta = 1.30$, and $\sigma = 1.40\%$ a day, so the fit can be checked against the truth: it returns $\hat{\alpha} = 0.0260\%$, $\hat{\beta} = 1.2949$, $\hat{\sigma} = 1.4215\%$, and $R^2 = 0.5022$. The call is held after the close on day 0, so the reaction falls on day +1, and the event window is days −1 to +1, with stock returns of −0.1%, 0.3%, and 2.1% and market returns of −0.2%, 0.4%, and

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0.9% (expected returns computed from the unrounded estimates):

$$\begin{aligned}
\text{Day } -1: &\quad AR = -0.10 - (0.0260 + 1.2949 \times (-0.20)) = -0.10 - (-0.2330) = 0.1330\%, \\
\text{Day } 0: &\quad AR = 0.30 - (0.0260 + 1.2949 \times 0.40) = 0.30 - 0.5439 = -0.2439\%, \\
\text{Day } +1: &\quad AR = 2.10 - (0.0260 + 1.2949 \times 0.90) = 2.10 - 1.1914 = 0.9086\%,
\end{aligned}$$

so $CAR(-1,+1) = 0.7977\%$, with standard error $1.4215\sqrt{3} = 2.4621\%$, $t = 0.324$, and a two-sided p-value of 0.7465.

Two lessons carry back to the trading signal. First, the market explains 56.7% of the 2.1% that Table 16.3 credited to tone: the stock's expected return on day +1, given the market's 0.9% rise and its beta of about 1.3, was already 1.19%, and the abnormal part is 0.91%. A signal scored on raw returns gets credit for every market rally that follows a positive call, so the hit rate and Sharpe-like ratio above should be recomputed on abnormal returns before anyone trades on them. Second, one event proves nothing. With a daily idiosyncratic volatility of 1.42%, even day +1's abnormal return alone has $t = 0.639$, and a three-day CAR of 0.80% is indistinguishable from noise. Averaging over N comparable events shrinks the standard error by $\sqrt{N}$: the same average CAR would give $t = 1.02$ over 10 events and $t = 2.05$ over 40. That is why published evidence on text and returns pools thousands of filings or calls, and why eight events cannot establish that a sentiment signal works.

16.3.2 Naive Bayes and Document Classification

Suppose a compliance team must sort thousands of 10-K risk-factor sentences into "Credit Risk" or "Market Risk," a task too large to do by hand each filing season and too nuanced for a keyword rule. Chapter 6 introduced Naive Bayes as a generative classifier that estimates

$$\Pr(y = k \mid x_1, \ldots, x_p) \propto \Pr(y = k) \prod_{j=1}^{p} \Pr(x_j \mid y = k) \tag{16.5}$$

and noted its suitability for text, where features are word counts and the conditional-independence assumption, while not literally true, distorts less than it might for correlated financial ratios. This section applies it using only the counts of six vocabulary words (default, borrower, counterparty, volatility, rate, currency).

Table 16.4 shows the six training sentences, three per class, written in the style of a 10-K risk-factors disclosure.

Sentence	Label
"A default by a major borrower or counterparty could trigger a chain of default across our loan portfolio."	Credit Risk
"Our borrower base includes several highly leveraged parties, and a default by any significant borrower could impair our credit portfolio."	Credit Risk
"We are exposed to counterparty risk if a significant counterparty were to default on its obligations under our agreements."	Credit Risk
"Significant volatility in interest rate markets, combined with broader market volatility, could adversely affect the value of our trading positions."	Market Risk
"Changes in the benchmark rate or the market rate environment, combined with volatility in trading conditions, could affect our net interest income."	Market Risk
"Fluctuations in the currency exchange market and adverse currency movements, together with changes in the base lending rate, could reduce our reported earnings."	Market Risk

Table 16.4: Training sentences for the 10-K risk-factor Naive Bayes classifier

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Counting runs over the six vocabulary terms only, so the class total below is a vocabulary-token count, not a sentence length. Credit Risk contributes `default` four times, `borrower` three, and `counterparty` three, with the three market terms absent, giving $N_{\text{Credit}} = 10$; Market Risk contributes `rate` four times, `volatility` three, and `currency` twice, with the three credit terms absent, giving $N_{\text{Market}} = 9$. Chapter 6's Section 6.6.2 Laplace (add-one) smoothing correction then divides by $N_c + V$ with vocabulary size $V = 6$, that is by 16 for Credit Risk and 15 for Market Risk, giving the estimated word probabilities

$$\begin{aligned}
\Pr(\text{default} \mid \text{Credit}) &= 0.3125,\\
\Pr(\text{borrower} \mid \text{Credit}) = \Pr(\text{counterparty} \mid \text{Credit}) &= 0.25,\\
\Pr(\text{volatility} \mid \text{Credit}) = \Pr(\text{rate} \mid \text{Credit}) = \Pr(\text{currency} \mid \text{Credit}) &= 0.0625,\\
\Pr(\text{rate} \mid \text{Market}) &= 0.3333,\\
\Pr(\text{volatility} \mid \text{Market}) &= 0.2667,\\
\Pr(\text{currency} \mid \text{Market}) &= 0.20,\\
\Pr(\text{default} \mid \text{Market}) = \Pr(\text{borrower} \mid \text{Market}) = \Pr(\text{counterparty} \mid \text{Market}) &= 0.0667,
\end{aligned}$$

reflecting each class's training vocabulary.

Worked Example 16.7: Classifying an Ambiguous Sentence by Hand

Consider the held-out test sentence "A default by a key counterparty during a period of heightened market volatility could be exacerbated by continued volatility in broader financial markets," with word counts (default: 1, counterparty: 1, volatility: 2, all others: 0), an ambiguous case mentioning credit-risk terms and, twice, a market-risk term. With equal class priors $\Pr(\text{Credit}) = \Pr(\text{Market}) = 0.5$, the (unnormalized) class scores are

$$\begin{aligned}
\text{score}(\text{Credit}) &= 0.5 \times 0.3125^1 \times 0.25^1 \times 0.0625^2 = 0.00015259,\\
\text{score}(\text{Market}) &= 0.5 \times 0.0667^1 \times 0.0667^1 \times 0.2667^2 = 0.00015802.
\end{aligned}$$

Normalizing gives $\Pr(\text{Credit} \mid \text{doc}) \approx 0.4912$ and $\Pr(\text{Market} \mid \text{doc}) \approx 0.5088$: the model predicts Market Risk, narrowly and incorrectly, since the true label is Credit Risk. In such near-toss-ups the independence assumption matters most: the two "volatility" mentions tip a mixed-signal sentence to the wrong side by less than two percentage points of posterior probability.

Applying the same model to a test set of six sentences (three of each true class) produces the confusion matrix in Table 16.5.

	Predicted Credit	Predicted Market
Actual Credit	2 (TP)	1 (FN)
Actual Market	0 (FP)	3 (TN)

Table 16.5: Confusion matrix for the 10-K risk-factor Naive Bayes classifier (positive class: Credit Risk)

Precision is $2/2 = 1.00$, recall is $2/3 \approx 0.6667$, and $F_1 \approx 0.80$: every sentence labeled Credit Risk was Credit Risk, but the model missed one, the ambiguous sentence worked through above.

16.3.3 Word Embeddings: From Sparse Counts to Dense Vectors

Bag-of-words and TF-IDF vectors are sparse and high-dimensional: with a vocabulary of tens of thousands of words, most entries in any document's vector are zero, and nothing records that "stock" and "equity" are related while "stock" and "bond" are not: every word is a separate dimension. An embedding is a fixed-length, continuous-valued vector standing in for a discrete object, here a word, chosen so that geometric closeness reflects closeness in meaning. Word embedding methods (Word2Vec and GloVe are the classical examples)

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learn a dense, low-dimensional vector for each word from co-occurrence patterns in a large corpus, so that words used in similar contexts get similar vectors. Cosine similarity,

$$\cos(u, v) = \frac{u \cdot v}{\|u\| \, \|v\|} \tag{16.6}$$

is the standard way to measure how similar two embedding vectors are.

Worked Example 16.8: Cosine Similarity of Toy Embeddings

To illustrate the geometry without training a model, assign six financial terms small hand-chosen two-dimensional vectors:

$$\begin{aligned} \text{stock} &= (0.90, 0.20), & \text{equity} &= (0.85, 0.25), \\ \text{bond} &= (0.10, 0.90), & \text{debt} &= (0.15, 0.85), \\ \text{bullish} &= (0.80, -0.30), & \text{bearish} &= (-0.75, -0.35). \end{aligned}$$

Computing cosine similarity between related pairs gives $\cos(\text{stock}, \text{equity}) \approx 0.9977$ and $\cos(\text{bond}, \text{debt}) \approx 0.9980$, both close to 1: the vectors point in nearly the same direction. Unrelated or opposed pairs differ: $\cos(\text{stock}, \text{bond}) \approx 0.3234$ is only weakly similar, and $\cos(\text{bullish}, \text{bearish}) \approx -0.7000$ is strongly negative, capturing near-opposites. In an embedding trained on financial text, the same structure, related concepts clustering and opposites pointing apart, emerges from co-occurrence alone, without the model being told what any word means.

A more striking property is that vector *arithmetic* can capture relationships between words, not just similarity, and this toy example was built so that the classic "analogy" relation holds exactly. Computing stock − equity + bond,

$$(0.90, 0.20) - (0.85, 0.25) + (0.10, 0.90) = (0.15, 0.85) = \text{debt},$$

reproduces the debt vector exactly, because the near-synonym pairs stock/equity and bond/debt were placed at the identical offset, $(0.05, -0.05)$. In a trained embedding space, the same arithmetic famously (if imperfectly) captures "king − man + woman ≈ queen." In a financial space, an analogous relation might hold between a country and its currency, letting an analyst query by relationship rather than single-word similarity. Real embeddings only approximate this, and the approximation is weaker for rare or abstract financial terms, but the toy example isolates the geometric idea before the noise of a high-dimensional trained model obscures it.

16.4 Transformers and Large Language Models

Attention changed what a language model can represent, and this part develops it from the mechanism up to retrieval and prompting.

16.4.1 Attention and the Transformer Architecture

Embeddings give each word a fixed vector regardless of context, but the same word can mean different things in different sentences ("the Fed will likely raise rates" versus "the flood raised water levels"). Transformer models address this with self-attention. Each token's embedding x is linearly projected, using three weight matrices W^Q, W^K, W^V learned in training and shared across every token, into a query vector $Q = xW^Q$, a key vector $K = xW^K$, and a value vector $V = xW^V$: the same embedding viewed through three learned lenses. Intuitively, a token's query asks "what am I looking for," its key advertises "what I offer to a token looking for this," and its value is the information it contributes once attention focuses on it: a search query, an index entry, and a payload.

A token's contextual representation is then a weighted combination of every token's value vector, with weights determined by how well each token's key matches the query,

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V, \qquad \text{softmax}(z)_i = \frac{e^{z_i}}{\sum_j e^{z_j}}, \tag{16.7}$$

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where d_k is the key dimension, included to keep dot products from growing too large as d_k increases, and softmax rescales raw scores into positive weights summing to one, so the combination of value vectors is an interpretable average rather than an arbitrary weighted sum.

This mechanism, applied across many layers and attention "heads," is the core building block of BERT, GPT, and every other modern transformer language model. Figure 16.2 diagrams the three-projection, one-combination structure every attention layer repeats.

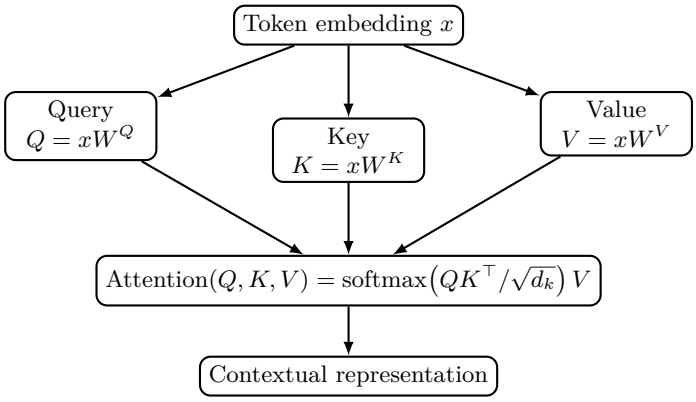


Figure 16.2: The self-attention computation as a diagram

Worked Example 16.9: Self-Attention for a Three-Token Sentence

Consider the toy sentence "stocks rallied today," with $d_k = 2$. Rather than training projection matrices W^Q, W^K, W^V, this example hand-chooses the resulting keys and values, standing in for what a trained model would produce: $K_{\text{stocks}} = (1, 0)$, $K_{\text{rallied}} = (0, 1)$, $K_{\text{today}} = (1, 1)$, and $V_{\text{stocks}} = (2, 0)$, $V_{\text{rallied}} = (0, 2)$, $V_{\text{today}} = (1, 1)$. "stocks" and "today" share the same first key component, so a query matching on that component attends to both equally, the effect this example illustrates.

The representation of "stocks" uses its query $Q_{\text{stocks}} = (1, 0)$. The raw scores $Q \cdot K_i/\sqrt{2}$ are 0.7071 against "stocks," 0 against "rallied," and 0.7071 against "today," since "rallied"s key is orthogonal to the query. Softmax turns these into attention weights of 0.4011, 0.1978, and 0.4011, which sum to 1, and the weighted sum of value vectors is

$$0.4011\,(2, 0) + 0.1978\,(0, 2) + 0.4011\,(1, 1) = (1.2033,\ 0.7967).$$

Figure 16.3 draws these weights, with edge thickness proportional to each token's contribution. "Today," a different word, contributes exactly as much to the new representation of "stocks" as "stocks" itself, because both keys align with the query, while "rallied" contributes far less, its value $(0, 2)$ pulled in only weakly.

This mechanism lets a transformer represent "raise" differently next to "rates" than next to "water levels": a trained W^K would align "rates" and "raise" in a financial context and "water" and "raise" in an environmental one, as this example's keys were aligned by construction.

Multi-Head Attention and Positional Encoding

A single attention calculation forces every token to distribute its focus by one notion of relevance, but relevance is not one-dimensional: local grammatical structure (which word modifies which noun) and long-range dependency (which earlier passage a later reference points back to) are different relationships, and one attention pattern cannot represent both well. Two refinements give the mechanism used in practice. First, a transformer layer computes several attention calculations in parallel, called attention heads, each with its own learned Q, K, V projections, and concatenates their outputs. One head might attend mainly to nearby tokens (local syntax), another to the subject of a distant verb (long-range dependencies, such as linking a hedge described early in a paragraph to a number reported much later), and the model combines these perspectives. Second, the calculation above treats the input as an unordered set: nothing in $Q \cdot K$

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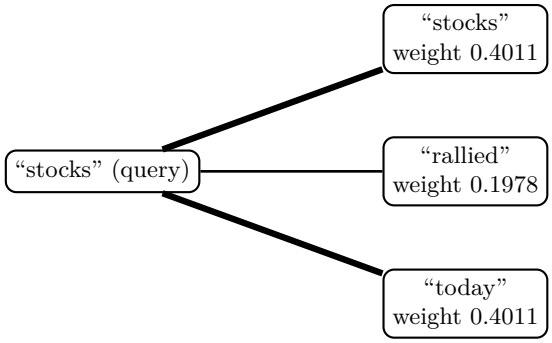


Figure 16.3: Self-attention weights for the query token "stocks" over the three-token sentence "stocks rallied today," with edge thickness proportional to attention weight

depends on position, so "the bank raised rates" and "rates raised the bank" would receive identical weights given identical token embeddings. Positional encoding (PE) fixes this by adding a position-dependent vector to each embedding before attention, most commonly

$$PE_{(pos,2i)} = \sin\left(\frac{pos}{10000^{2i/d}}\right)$$

and

$$PE_{(pos,2i+1)} = \cos\left(\frac{pos}{10000^{2i/d}}\right)$$

for position pos and embedding dimension index i, so that word order survives into a mechanism that would otherwise discard it.

16.4.2 Encoder-Only, Decoder-Only, and Encoder-Decoder Architectures

Self-attention is the building block of every transformer, but different financial NLP tasks arrange it differently, and the three resulting architecture families are not interchangeable. An encoder-only model (BERT and its financial variant FinBERT are the standard examples) lets every token attend bidirectionally to every other token, the unmasked computation above, and is trained with a masked-language-modeling objective (predicting a hidden word from its context in both directions). This makes encoder-only models well suited to classification and embedding tasks, such as the sentiment scoring of Section 16.2.3 and the document classification of Section 16.3.2, where the full passage is available at once and the output is a label or vector, not new text. A decoder-only model (the GPT family and the LLMs of the rest of this chapter) restricts each token to attending to itself and earlier tokens, a causal mask, since it is trained to predict each next token from what precedes it, the one-token-at-a-time process of Section 16.4.4. An encoder-decoder model (T5-style architectures) uses both: a bidirectional encoder reads the input, and a causal decoder generates the output while cross-attending to the encoder's output, a natural fit for input-to-output tasks such as translating a filing or condensing an earnings call into a summary.

Worked Example 16.10: How a Causal Mask Changes a Token's Representation

Return to "stocks rallied today" and compute the representation of "rallied" with its query $Q_{\text{rallied}} = (0, 1)$, first with full bidirectional attention (as an encoder-only model would) and then with a causal mask hiding the later token "today" (as a decoder-only model would). Bidirectionally, the raw scores $Q_{\text{rallied}} \cdot K_i/\sqrt{2}$ are 0 against "stocks," 0.7071 against "rallied," and 0.7071 against "today," giving softmax weights of 0.1978, 0.4011, and 0.4011 and an output of

$$0.1978\,(2, 0) + 0.4011\,(0, 2) + 0.4011\,(1, 1) = (0.7967,\ 1.2033).$$

Under a causal mask, "today" is removed, leaving the scores against "stocks" (0) and "rallied" (0.7071); renormalizing these two via softmax gives weights of 0.3302 and 0.6698, and an output of

$$0.3302\,(2, 0) + 0.6698\,(0, 2) = (0.6605,\ 1.3395).$$

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The same word in the same sentence gets two representations, $(0.7967, 1.2033)$ versus $(0.6605, 1.3395)$, purely from architecture: an encoder-only model's "rallied" may reflect that "today" follows it, while a decoder-only model's, used to predict what comes *after* "rallied," cannot peek at "today" without trivializing its training objective.

So swapping a decoder-only LLM into a task an encoder-only model was designed for, or vice versa, is not a matter of using "a big enough model": the two produce different token representations by construction, not by training data.

16.4.3 How These Models Are Trained, and Why They Eventually Need Retraining

No one hand-writes the rule that tells a model which distant word a pronoun refers to, or which number a sentence's context makes plausible; the model must learn it, and the only training signal available at that scale is the text itself, with no human labels. None of the weight matrices in Figure 16.2, W^Q, W^K, W^V, or the many more in a transformer's feedforward and output layers, is hand-specified; each is learned by ordinary gradient descent (Chapter 6's Section 6.3.3), applied to a self-supervised objective computed from raw, unlabeled text at enormous scale.

An encoder-only model is pretrained by masking a random fraction of tokens and predicting each from its bidirectional context; a decoder-only model is pretrained on the single objective of predicting each next token from everything before it, the causal-masked computation just carried out by hand, repeated across a very large corpus. No human labels the output token by token; the text supplies the signal, which is why these models can be pretrained on far more data than any supervised model in this book.

Pretraining is not, however, a one-time event. Financial language drifts: new instruments, regulations, tickers, and company names appear, established terms acquire new meanings (a "pandemic bond" meant nothing before 2020), and the pretraining vocabulary can gradually stop matching what a live system encounters, a text analogue of the concept drift of Chapter 14's Section 14.7.2 and Chapter 18's population-stability index. Three responses are common, roughly in increasing order of cost: prompting or retrieval (Section 16.4.5) supplies current information to a frozen model; parameter-efficient fine-tuning updates a small number of added parameters on a modest domain dataset; and continued pretraining resumes the self-supervised objective on fresh text, updating the weights more thoroughly at higher cost. The choice depends on whether the problem is a knowledge gap (a missing recent fact, best fixed by retrieval) or a shift in how language is used (better fixed by continued pretraining or fine-tuning), a diagnosis that requires the ongoing monitoring Chapter 18 established for every model.

16.4.4 Large Language Models in Finance: Capabilities, Generative Modeling, and Limits

Suppose an analyst is handed a hundred-page merger filing an hour before a decision, or a portfolio manager wants a plain-English answer to a question buried in it. None of this chapter's classification or scoring tools, built to output a single label or score, can do this. Chapter 6 noted that LLMs are far more elaborate examples of the generative principle behind Naive Bayes: both learn a probability distribution, Naive Bayes over word counts given a class, an LLM over the next token given all preceding tokens, and both can be used generatively. An LLM is, mechanically, a very large stack of the self-attention layers of Section 16.4.1, trained on enormous volumes of text to predict the next token, then adapted (through instruction tuning and related techniques) to follow natural-language prompts.

In finance, this supports tasks well beyond classification and scoring: summarizing a long earnings call, extracting a table of figures from an unstructured filing, drafting a research note, or answering a question about a specific document. These tasks come with a different failure mode: hallucination, fluent, confident text that is factually wrong, such as a number invented rather than retrieved, a misattributed filing detail, or a summary that omits a covenant violation buried in a footnote.

A classifier fails with a wrong label and a computable error rate; an LLM fails with a wrong but plausible answer, harder to detect automatically and dangerous where one invented figure in a research note can be consequential. The next four sections develop the leading mitigation for hallucination (retrieval-augmented

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generation), how to instruct a generative model (prompting), a use of generative modeling that has nothing to do with text (synthetic data), and how an institution governs all of this.

Beyond Text: Code Generation and Multimodal Applications

Two further capabilities go beyond text-in, text-out tasks. Code generation applies next-token prediction to programming languages. An analyst can describe a calculation in plain English, for example "load this CSV of daily returns and compute the annualized Sharpe ratio and maximum drawdown," and receive working Python code, extending the pandas and NumPy analysis this book has built since Chapter 2. The code still needs scrutiny before it runs against real portfolio data: a subtly wrong annualization factor or an off-by-one window is a bug, not a hallucination in the text sense, but no less consequential.

Multimodal models extend the architecture to inputs beyond text, reading a scanned PDF page, a chart in an investor presentation, or a table image directly, without a separate, brittle optical-character-recognition step. This matters because much financial disclosure (annual report exhibits, scanned older filings, chart-heavy investor decks) is not available as clean machine-readable text.

Reasoning Models and Long Context Windows

Two developments since 2024 have changed how these models are used. Reasoning models are trained, largely by reinforcement learning on problems whose answers can be checked automatically, such as mathematics and code, to write out a long chain of thought before answering (DeepSeek-AI 2025): the chain-of-thought prompting of Section 16.4.7, learned rather than requested. Accuracy then rises with test-time compute, the number of reasoning tokens spent per question, which turns cost and latency into a dial. Reconciling two differently defined metrics across a pair of filings justifies a long think; labeling a headline's sentiment does not. Reasoning tokens are usually billed as output tokens, the expensive side of Section 16.6.8's cost formula. Reasoning lowers error rates without removing them, and a displayed trace is not guaranteed to be the computation that produced the answer, so a reviewer should treat it as evidence, not proof.

Context windows, the number of tokens a model can read at once, have grown from a few thousand to hundreds of thousands and, as of October 2026, a million or more for several models, enough to hold an entire 10-K. This changes what retrieval is for, not whether it is needed. A question about one filing can now often be answered by reading the whole filing, which removes the risk of retrieving the wrong passage. But a firm's corpus runs to millions of documents; every token in the prompt is paid for on every query; and models use long contexts unevenly: Liu et al. (2024) found accuracy highest when the relevant passage sits near the start or end of the context and markedly lower when it sits in the middle. Retrieval also keeps citations precise and lets access control be enforced document by document (Section 16.5.2).

Reading Loan Documents in Private Credit

One extraction task shows why reading, not writing, is the hard part. Private credit, loans made directly to mostly mid-sized companies by funds and other nonbank lenders rather than syndicated by banks, had grown to about $2.1 trillion of assets and committed capital worldwide by 2023, roughly three-quarters of it in the United States, and the IMF noted signs of weaker underwriting and looser covenants as these lenders competed with banks for large deals (International Monetary Fund 2024). The loans rarely trade and are documented in bespoke agreements of a hundred pages or more, so a lender monitoring several hundred borrowers must answer the same questions from each agreement and quarterly compliance certificate: what is the maximum leverage, how is it measured, and how close is the borrower? These are the summarization and extraction tasks above, and the danger lies in the definitions. Suppose a borrower has $250 million of net debt, reported EBITDA of $40 million, and a maintenance covenant capping total net leverage at 5.50 times "Consolidated EBITDA," which the agreement defines to include add-backs, here $5 million of projected cost synergies and $3 million of restructuring charges. With every add-back, leverage is $250/48 = 5.208$ times, a pass with 5.3% of headroom, since the covenant needs only $250/5.50 = \$45.45$ million. If the definition caps add-backs at 15% of EBITDA before them, the measure is $40 + 6 = 46$ and leverage 5.435 times, still a pass but with 1.2% of headroom. On reported EBITDA alone it is $250/40 = 6.250$ times, a breach. A model that summarizes the position as "leverage of about 5.2 times against a 5.5 times covenant" has answered from the wrong clause, and has done so fluently. The safe design is the one this chapter keeps arriving at:

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the model extracts the definition, the cap, and each add-back as structured fields, each tied to the clause it came from (the retrieval and citation of the next subsection); a calculator computes the ratio (the tool use of Section 16.6.3); and an analyst reviews every borrower near its limit.

16.4.5 Retrieval-Augmented Generation

Retrieval-augmented generation (RAG) is the most widely deployed mitigation for hallucination in finance. Rather than relying on facts an LLM absorbed in training, possibly outdated, generic, or never seen, a RAG system first retrieves the most relevant passages from a trusted, current collection (a bank's own filings, policy manuals, or licensed research), then conditions the answer on them.

This turns open-ended generation into something closer to open-book comprehension: the model answers from the retrieved text, and its output can be traced to a source passage, which is what an auditor or compliance reviewer needs. Retrieval applies Section 16.3.3's embedding geometry: embed the query and every candidate document in the same space, rank documents by cosine similarity to the query, and pass the top passages to the model.

Figure 16.4 lays out the pipeline in two phases. Offline, the collection is split into chunks, tagged with metadata, embedded, and indexed. Online, each query is matched against the index, candidates are re-ranked, and the best few passages go into the prompt with the question; the answer cites those passages and is checked claim by claim before anyone sees it. The worked example below walks through retrieval on a toy index, the discussion after it covers chunking, hybrid retrieval, and re-ranking, and Section 16.4.6 develops the two final checks.

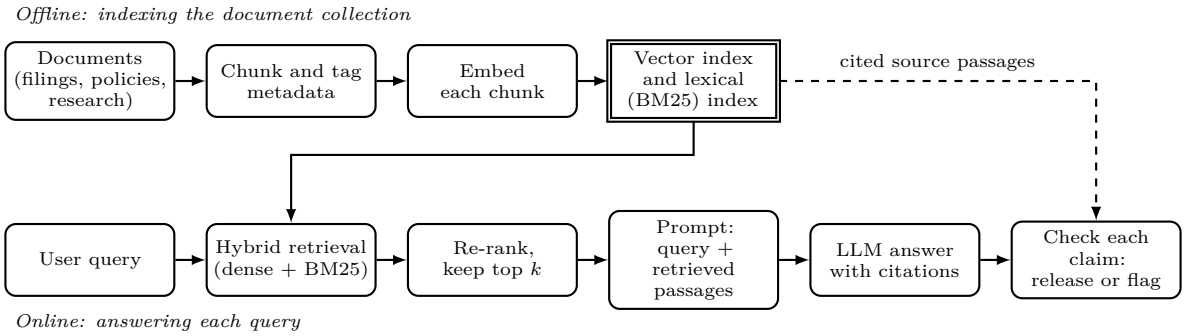


Figure 16.4: A retrieval-augmented generation pipeline. Documents are chunked, embedded, and indexed offline; each query is answered from retrieved and re-ranked passages, and the answer is checked against its cited sources before release.

Worked Example 16.11: Retrieving the Right Passage from a Toy Knowledge Base

Consider a knowledge base of four short passages about two companies, embedded in a hand-illustrative three-dimensional scheme whose dimensions loosely represent revenue, leverage, and other-company topics (Table 16.6).

A user asks, "What was Company X's revenue in Q3?", embedded as $(0.95, 0.05, 0.00)$.

Computing cosine similarity between the query and each document gives

$$\begin{aligned}\cos(\text{query}, \text{Doc1}) &\approx 0.9983,\\ \cos(\text{query}, \text{Doc4}) &\approx 0.8008,\\ \cos(\text{query}, \text{Doc2}) &\approx 0.1625,\\ \cos(\text{query}, \text{Doc3}) &= 0.0000.\end{aligned}$$

Doc1 is the top match by a wide margin, and only Doc1 (or, at a higher retrieval count, Doc1 and Doc4) would be passed to the model, which answers from Doc1's text: "Company X's Q3 revenue was $420 million,

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Document	Passage	Embedding
Doc1	"Company X reported Q3 revenue of $420 million, up 8% year over year."	$(0.90, 0.10, 0.00)$
Doc2	"Company X's debt-to-equity ratio increased to 1.8 in Q3."	$(0.10, 0.90, 0.00)$
Doc3	"Company Y announced a new product launch in Q4."	$(0.00, 0.00, 0.90)$
Doc4	"Company X's Q3 net margin improved to 12.5% from 10.1%."	$(0.60, 0.50, 0.00)$

Table 16.6: Toy knowledge base for the retrieval-augmented generation worked example

up 8% year over year," grounded in a verifiable source rather than invented from training data. Doc2 and Doc3, though they mention Company X or a business topic, rank far below Doc1 and Doc4, since neither answers a revenue question, the discrimination retrieval needs to make.

Chunking and Hybrid Retrieval

A production system differs from this toy in two ways. First, documents must be split into chunks before embedding, and chunk size is a trade-off: too small, and a figure is separated from the sentence saying which quarter it belongs to; too large, and one embedding averages over several topics and matches none well. Common practice is chunks of a few hundred tokens with some overlap, cut along the document's structure (sections together, tables whole) and tagged with metadata such as company, filing date, and section, so a query can be filtered to the right filing before similarity is computed. Second, dense embeddings encode what a passage is about and are weak at exact matches on the identifiers finance depends on: a ticker, a CUSIP, a clause number, "Item 1A." Hybrid retrieval therefore runs a lexical ranker, usually BM25 (Robertson and Zaragoza 2009), a refined descendant of Section 16.2.2's TF-IDF, beside the dense retriever and merges the two ranked lists (for example by reciprocal rank fusion). The top few dozen candidates then go to a re-ranker, a model that reads the query and each passage together and scores relevance more accurately than either retriever, before the best handful reach the generator.

The companion notebook's optional "at scale" section repeats this retrieval with a real sentence-embedding model over a thirty-passage knowledge base, side by side with a TF-IDF ranking, and scores Section 16.2.3's two transcripts with a transformer classifier fine-tuned on financial news. It needs the extra packages listed in the repository's `requirements-dl.txt` and downloads the pretrained model weights on its first run.

16.4.6 Evaluating a RAG System: Precision@k and a Faithfulness Score

Retrieving *a* document is not retrieving the *right* one, and a production RAG system needs to check retrieval and generation separately, not only spot-check final answers. Precision@k measures retrieval alone: what fraction of the top k retrieved documents is relevant? For the revenue question, only Doc1 answers it; Doc4 (net margin) and Doc2 (leverage) concern Company X but do not answer *this* question. Ranking by the cosine similarities already computed (Doc1, then Doc4, then Doc2, then Doc3),

$$\text{Precision@1} = \frac{1}{1} = 100\%, \qquad \text{Precision@2} = \frac{1}{2} = 50\%, \qquad \text{Precision@3} = \frac{1}{3} \approx 33.3\%,$$

since only one document in each of the top 1, 2, and 3 is relevant. This is a quantitative reason to keep k small and the similarity threshold strict: retrieving more documents mechanically dilutes precision, and every non-relevant passage passed to the model is another chance for it to draw on irrelevant context.

A second check addresses generation: given the right passage, did the answer stick to it? A faithfulness score compares the answer's embedding to the source passage's, using the same cosine geometry as retrieval. The grounded answer, "Company X's Q3 revenue was $420 million, up 8% year over year," embeds almost

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identically to Doc1, at $(0.89, 0.12, 0.00)$, giving

$$\text{Faithfulness} = \cos\big((0.89, 0.12, 0.00),\ \text{Doc1}\big) \approx 0.9997,$$

close to a perfect match, since the answer restates Doc1. A hallucinated answer that invents a leverage claim absent from Doc1 embeds instead at $(0.30, 0.85, 0.00)$, closer to Doc2's topic than Doc1's:

$$\text{Faithfulness} = \cos\big((0.30, 0.85, 0.00),\ \text{Doc1}\big) \approx 0.4349,$$

far below the grounded answer. A production system can flag any answer scoring below, say, 0.85 against its source for rejection or human review. That makes the score a useful *topical-drift* check: it catches an answer that has wandered to a different subject.

It does not catch the error that matters most in finance, a wrong number on the right topic. Suppose the model answers "Company X's Q3 revenue was $240 million, down 8% year over year," transposing two digits and reversing the direction. The answer is about the same subject as Doc1, so in this topic-based embedding it sits where the grounded answer sits and scores the same 0.9997, passing the 0.85 threshold. Real embedding models behave the same way, because they encode what a sentence is about far more strongly than its digits. Even bag-of-words vectors built from the sentences (Section 16.2.2) give the wrong answer a cosine of 0.7333 against Doc1, against 0.8667 for the grounded one: lower, but nowhere near the leverage answer's 0.4349, and a threshold strict enough to reject it would also reject correct answers phrased differently.

Faithfulness therefore needs a second, *claim-level* check. Extract every verifiable claim from the answer (numbers, entities, dates, directions of change) and require each to appear in, or be computable from, the source. Here the check is mechanical: the grounded answer's three claims (420, 8, "up") are all in Doc1, while the wrong answer's (240, 8, "down") pass on only one of three, and it is rejected. Production systems go beyond exact matching by asking a natural-language-inference (entailment) model whether the source entails each claim, or a second LLM to judge each claim ("LLM-as-judge"), and frameworks such as RAGAS package claim-level faithfulness scores with retrieval metrics like precision@k. Together, the two checks give the validation practice Section 16.5.2 calls for, "sample many outputs and check them against source documents for factual grounding," a computable form.

Finance-specific benchmarks test the whole pipeline. FinanceBench (Islam et al. 2023) poses 10,231 questions about public companies' filings, each with an answer and its supporting evidence passage (150 released openly); in its original 2023 evaluation, GPT-4-Turbo with retrieval answered 81% of the sampled questions incorrectly or refused, a sobering baseline. Published scores age quickly and can leak into later models' training data, so an institution should also keep a private question set from its own documents. When an LLM grades answers at that scale, the judge's biases need controlling. Zheng et al. (2023) document three: position bias (favoring the answer shown first), verbosity bias (favoring longer answers), and self-enhancement bias (favoring its own answers). The remedies are to swap the order and count only verdicts that survive the swap, to grade against the claim list or a reference answer rather than holistically, to use a judge from a different model family, and to calibrate the judge against a human-labeled sample, the validation against ground truth Chapter 18 requires of any model.

16.4.7 Prompting Strategies for Financial Tasks

How a task is phrased to a generative model, its prompt, materially affects output quality, and three strategies recur in finance. Zero-shot prompting states the task: "Classify the sentiment of the following earnings-call excerpt as positive, negative, or neutral: *[excerpt]*." This works for common, well-defined tasks but can be inconsistent where domain conventions the model has not reliably learned apply, the finance-vocabulary issue of Section 16.2.1.

Few-shot prompting puts a few worked examples in the prompt to show the labeling convention wanted: two or three (excerpt, label) pairs, perhaps from Section 16.2.3's transcripts, followed by a new excerpt to classify. This resembles Chapter 6's supervised learning, except the "training" happens within a single prompt, with no weight updates, and the "training set" can be two or three examples.

Chain-of-thought prompting asks the model to reason step by step before answering, for example: "First identify the reported revenue figure and the prior-year figure, then compute the percentage change, then state whether this represents an acceleration or deceleration relative to the previous quarter's growth rate, showing

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each step." For multi-step numerical tasks, such as computing a ratio from figures buried in a paragraph or reconciling two differently defined metrics, this measurably reduces errors compared with asking for the number directly, because it puts the intermediate steps in the output where they can be checked, much as this book shows every intermediate step of a calculation, a practice established in Chapter 7.

16.5 Generative AI in Practice

Generation raises problems consumption does not, from synthetic data to governance to domain adaptation.

16.5.1 Generative Models for Synthetic Financial Data

A different application of generative modeling has nothing to do with text: generating synthetic numerical data (stress-test scenarios, order-book sequences, extra examples of a rare event class) that resembles real data closely enough to train or test other models. Two architectures dominate.

A generative adversarial network (GAN) pits two neural networks against each other: a generator G maps random noise z, drawn from a simple fixed distribution p_z such as a standard normal, to synthetic samples $G(z)$, and a discriminator D tries to distinguish real samples x, drawn from the data distribution p_{data} being imitated, from generated ones. They are trained jointly via

$$\min_G \max_D \; \mathbb{E}_{x\sim p_{\text{data}}}[\log D(x)] + \mathbb{E}_{z\sim p_z}\left[\log\left(1 - D(G(z))\right)\right]. \tag{16.8}$$

Each term rewards D for half its job: the first for assigning high probability to real samples, the second for assigning low probability to fakes. Only the second involves G, so the generator's objective is to drive $D(G(z))$ up while the discriminator pushes it down. As training proceeds, each improves against the other, and at convergence the discriminator can no longer reliably tell generated samples from real ones. In finance, GANs have been used to augment a fraud model's minority class, extending Chapter 15's resampling techniques such as SMOTE with synthetic rather than interpolated examples, and to generate limit-order-book sequences for backtesting execution algorithms without exposing client order flow.

The companion notebook evaluates this minimax game on Chapter 15's fraud-scoring data: a fixed, illustrative discriminator $D(x) = \sigma(2.0x - 5.0)$ assigns transaction 1's fraud amount, standardized to 3.1, a probability of being real of $D(x_{\text{real}}) = 0.7685$. (Chapter 15 names this standardized feature with the conventional z of a standard score; here z is the generator's noise input, so the feature is called by its role.) An early-training generator, still producing samples near a typical legitimate transaction's scale, might output $G(z) = 1.0$, which the discriminator flags as almost certainly fake, $D(G(z)) = 0.0474$, a gap of 0.7211 from $D(x_{\text{real}})$.

A later generator producing a fraud-scale sample, $G(z) = 2.9$, gets $D(G(z)) = 0.6900$, closing the gap to 0.0786: the generator's loss, $-\log D(G(z))$, falls from 3.0486 to 0.3711, a numerical measure of how much harder it has become to catch.

GANs are also notoriously hard to train. Because generator and discriminator are optimized against each other rather than toward one shared objective, training can oscillate instead of converging, or fall into mode collapse: the generator finds a few samples that reliably fool the discriminator and stops exploring the rest of the distribution, producing fraud examples or stress scenarios that are realistic but too uniform to augment a thin dataset. This instability, which not every generative architecture shares, is one reason diffusion models, described next, have become the more common choice for new synthetic-data applications despite being costlier to sample from.

Diffusion models take a different approach, motivated by physical diffusion: a forward process gradually adds noise to real data over many steps until it is indistinguishable from pure noise, $x_t = \sqrt{\alpha_t}\, x_0 + \sqrt{1-\alpha_t}\, \varepsilon$, where x_0 is the original data point, ε is standard-normal noise, and α_t decreases from near 1 (step 0, almost no noise) toward 0 (fully noised) as t increases. A neural network is trained to reverse this, removing noise one step at a time, so that running the learned reverse process from pure noise produces a new sample resembling the original distribution.

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Worked Example 16.12: One Forward Diffusion Step

Consider a toy stress-test scenario variable, a multiplier applied to a baseline loss, with baseline (undiffused) value $x_0 = 1.00$ (no stress) and a noise draw $\varepsilon = -0.50$. At an early diffusion step with $\alpha_t = 0.80$,

$$x_t = \sqrt{0.80}\,(1.00) + \sqrt{0.20}\,(-0.50) = 0.6708.$$

At a later, noisier step with $\alpha_t = 0.55$,

$$x_t = \sqrt{0.55}\,(1.00) + \sqrt{0.45}\,(-0.50) = 0.4062.$$

As α_t falls from 0.80 to 0.55, the scenario value moves from 0.6708 down to 0.4062, drifting from the baseline of 1.00 toward the noise draw of -0.50; continuing to $\alpha_t \to 0$ would carry x_t to ε itself, pure noise with no trace of the original scenario. The two coefficients are square roots chosen so that $\left(\sqrt{\alpha_t}\right)^2 + \left(\sqrt{1-\alpha_t}\right)^2 = 1$: they hold the *variance* of x_t constant as noise replaces signal, and they are not a weighted average of x_0 and ε, since they sum to more than one. A same-signed noise draw can therefore push x_t temporarily outside the range spanned by x_0 and ε before it settles toward ε. A trained diffusion model learns to run this in reverse, starting from noise and reconstructing a plausible, never-observed stress scenario that Chapter 8's stress testing could use to probe losses beyond anything in the historical sample.

16.5.2 Governance and Risk Management for Generative AI

Chapter 15 developed a three-lines-of-defense governance framework for fraud and compliance models; generative AI needs the same discipline, applied to risks deterministic models did not raise. Three deserve attention.

First, non-determinism: the same prompt submitted twice can produce two outputs, which breaks validation approaches that assume a fixed input gives a reproducible output, and pushes validation toward sampling many outputs. Second, prompt injection: adversarial text in a retrieved document, an uploaded file, or ordinary user input can try to hijack the system, for example an instruction hidden in a retrieved filing telling the model to ignore its task, an attack surface with no counterpart in earlier chapters' classifiers. Third, data leakage: a RAG system with access to sensitive internal documents can, if not carefully scoped, surface information to a user who should not see it, an inadvertent access-control failure.

These risks change how the third line's independent validation, introduced in Chapter 15, operates. Rather than checking predictions against a fixed labeled test set, it must sample many outputs, check them against source documents for factual grounding (the check Section 16.4.5's retrieval supports), and test for prompt injection and data leakage, before deployment and on an ongoing basis as the model, document collection, or user population changes. Regulators have generally applied existing model risk management expectations rather than writing new rules, although the U.S. interagency guidance that replaced SR 11-7 in April 2026, SR 26-2, explicitly leaves generative and agentic AI outside its scope pending separate treatment (Federal Reserve, OCC, and FDIC 2026). In practice a generative AI deployment is still expected to meet the same expectations, documented validation, ongoing monitoring, and a clear escalation path, though the tools for meeting them differ.

16.6 Advanced Topics: Fine-Tuning, Agents, and Domain Adaptation

The base LLM of Section 16.4.4, trained only to predict the next token, is not yet the instruction-following assistant of Section 16.4.5 and this chapter's case vignette. Four further ideas close that gap.

16.6.1 Fine-Tuning and Reinforcement Learning from Human Feedback

A model pretrained only to predict the next token is an unreliable assistant: asked a question, it may as easily continue writing in the same style as answer it, since nothing in next-token prediction teaches it to follow an instruction or to prefer a helpful, accurate response over a fluent but unhelpful one. After pretraining, a model is typically trained in two further stages: supervised fine-tuning on curated (prompt, ideal response)

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pairs, teaching it to follow instructions, and reinforcement learning from human feedback (RLHF), which uses human preference judgments between pairs of candidate outputs to train a reward model, then adjusts the LLM to produce outputs that reward model scores highly. The reward model itself is typically trained as a preference classifier: given two candidate outputs with reward scores r_A and r_B, the Bradley-Terry model estimates the probability that a human would prefer output A as

$$\Pr(A \succ B) = \text{sigmoid}(r_A - r_B) = \frac{1}{1 + e^{-(r_A - r_B)}}. \tag{16.9}$$

Worked Example 16.13: Reward-Model Preference Probability

Suppose a reward model scores a source-accurate earnings-call summary at $r_A = 2.1$ and a fluent but subtly inaccurate one at $r_B = 0.8$. The preference probability is $\text{sigmoid}(2.1 - 0.8) = \text{sigmoid}(1.3) \approx 0.7858$: the reward model favors the accurate summary about 78.6% of the time, and RLHF nudges the LLM's outputs toward what the reward model scores highly, in aggregate across many such comparisons.

16.6.2 Parameter-Efficient Fine-Tuning: Low-Rank Adaptation (LoRA)

RLHF and supervised fine-tuning describe *what* a model is trained to do after pretraining; a practical question is *how*. Updating every weight, full fine-tuning, requires storing a second full copy of the parameters (the update) beside the original, and repeating this for every task or client is prohibitively expensive at the scale of a modern LLM.

Low-Rank Adaptation (LoRA) is the most widely used solution: instead of learning a dense update ΔW for a $d \times d$ weight matrix, it constrains the update to a low-rank decomposition, $\Delta W = BA$, where B is $d \times r$, A is $r \times d$, and the rank r is far smaller than d. Only B and A are trained; W is frozen, and the adapted forward pass uses $W + BA$ in place of W.

Worked Example 16.14: Parameter Reduction from Low-Rank Adaptation

Consider a single $d = 4{,}096$ weight matrix, a realistic dimension for one projection inside a modern LLM. Full fine-tuning of it requires learning all $d^2 = 4{,}096^2 = 16{,}777{,}216$ entries of ΔW. Using LoRA with rank $r = 8$, the trainable parameters are instead only those of B ($d \times r$) and A ($r \times d$),

$$2 \times d \times r = 2 \times 4{,}096 \times 8 = 65{,}536,$$

a reduction factor of $16{,}777{,}216/65{,}536 = 256\times$ for this matrix, with a comparable reduction for every matrix LoRA is attached to.

This is why LoRA has become the default way to adapt a large model to a bank's internal document style or an asset class's terminology. It avoids both the storage cost of a duplicate model per use case and the compute cost of updating billions of parameters. The frozen base model, possibly quantized as described below, can be shared across many clients or tasks, with only a small task-specific B and A pair, tens of thousands of parameters rather than billions, swapped in per use case.

16.6.3 LLM Agents and Tool Use

Computing and cross-checking a company's P/E ratio requires looking up its price, pulling its latest earnings from a filing, dividing, and comparing against a sector benchmark, a sequence no single LLM call, which produces text in one pass, can carry out. An agent extends a single LLM call into a multi-step loop: the model can call an external tool (a retrieval system, a calculator, a database query, or a trading API), observe the result, and choose a next action, repeating until it produces a final answer. This composes Section 16.4.5's retrieval with ordinary computation.

In practice the loop runs on function calling, also called tool calling. The developer describes each tool by a name, a plain-language description, and a JSON schema for its arguments; when the model wants a tool, it emits a structured call, in effect `calculator(57.87/3.15)`, and the application checks the call against the schema, runs it, and returns the result as a new message. The same machinery provides structured outputs:

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constraining a model to produce JSON conforming to a schema, which turns "extract the figures from this filing" into typed fields a downstream system can validate rather than prose it must parse. The Model Context Protocol (MCP), an open standard introduced by Anthropic in November 2024 and transferred in December 2025 to the Agentic AI Foundation, a Linux Foundation project (Anthropic 2024; Linux Foundation 2025), standardizes how tools and data sources are offered to models, so that one connector to a market-data service or a document store can serve any compliant application.

Worked Example 16.15: A Three-Step Agent Computing and Cross-Checking a P/E Ratio

Suppose an agent is asked, "What is Company X's price-to-earnings ratio using its Q3 diluted EPS and quarter-end share price?" The agent first calls a retrieval tool (Section 16.4.5's mechanism), retrieving two facts: diluted EPS of \$3.15 and a quarter-end closing price of \$57.87. It then calls a calculator tool rather than doing the arithmetic itself, computing $\mathrm{P/E} = 57.87/3.15 \approx 18.37$. This removes a whole category of arithmetic-hallucination risk: a calculator's division is exact by construction, while an LLM can make an arithmetic error despite retrieving correct inputs.

A more robust agent takes a third step, calling an independent tool to cross-check the result before presenting it. Suppose it queries a market-data vendor's reported P/E for Company X and retrieves 19.10. Compared with its own value,

$$\text{Discrepancy} = \frac{19.10 - 18.37}{18.37} \approx 3.97\%,$$

exceeds a pre-set tolerance of, say, 2%, so rather than presenting either number as final, the agent flags the disagreement, reporting both figures with a note that the gap may reflect a definitional or timing difference (the vendor may use trailing rather than diluted EPS, or a different price snapshot) worth investigating before either is used in a client-facing report.

The cross-check costs one tool call but turns a possibly wrong number into either a validated one or a flagged discrepancy, the independent-verification discipline Chapter 8's model-validation framework requires of a classical risk model.

An agent that can act also inherits Section 16.5.2's risks in sharper form, because a successful prompt injection now triggers tool calls rather than bad text. Indirect prompt injection, instructions planted in a web page, an email, or a retrieved filing that the agent reads as data, was demonstrated against deployed applications by Greshake et al. (2023), and prompt injection heads the OWASP Foundation's 2025 list of risks for LLM applications, with "excessive agency" also on it (OWASP Foundation 2024). No model-level defense is yet reliable, so the controls are architectural: least-privilege permissions (a research agent gets read-only data tools, and no trading or payment tool); every tool result and retrieved document treated as untrusted data, never as instructions; human approval before any irreversible or external action, such as sending an email, placing an order, or moving money; hard limits on spending and on the number of calls; and a complete log of every tool call for audit, the agent-level counterpart of Chapter 15's three lines of defense.

16.6.4 Agentic AI in Finance: Planning, Memory, Evaluation, and Failure

The P/E agent took three steps, each checkable by eye. The systems banks began piloting in 2025 and 2026 take tens of steps across several internal systems (refreshing a client's know-your-customer file, reconciling a failed trade, assembling a credit memo), and at that length the questions change. How does the agent decide what to do next? What may it touch? What does it remember between steps and tasks? How is it evaluated, when the output is a sequence of actions? And how does it fail? This subsection takes each in turn; its worked example shows why the last dominates: reliability that looks excellent per step becomes mediocre per task.

The reason-act-observe loop. Most agents run some version of the ReAct pattern of Yao et al. (2023), shown in Figure 16.5. At each step the model writes a short reasoning trace (what it knows, what it still needs), emits one action, a structured tool call, and receives the tool's output as an observation appended to its context. The loop ends when the model gives a final answer, escalates to a human, or exhausts a step or spending budget. Two planning styles sit on top of this loop. In an interleaved style the agent decides

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one step at a time, which adapts to surprises but can wander; in a plan-then-execute style it first writes the whole plan ("retrieve the client's ownership records, screen each beneficial owner against the sanctions list, compare against the file on record, draft the exceptions"), executes it, and replans only when a step fails, which a reviewer can read in advance and check afterward. Regulated workflows tend to prefer the second for that reason.

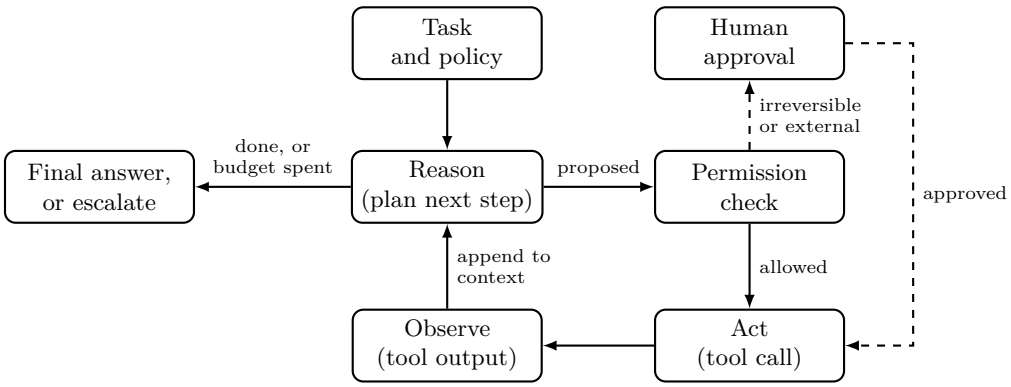


Figure 16.5: The reason-act-observe loop of an LLM agent, with a permission check between the model's proposed action and its execution, and human approval for irreversible or external actions

Permissions and sandboxing. The permission check in Figure 16.5 is where least privilege becomes engineering. Tools are sorted by consequence: read-only tools (a filing search, a market-data lookup) run freely; write tools whose effects can be undone (drafting a document, opening a ticket) run with logging; and tools whose effects cannot be undone or leave the firm (messaging a client, releasing a payment, submitting an order) run only after a named human approves that specific call, with its arguments, not the task in general. Code the agent writes runs in a sandbox, an isolated container with no network access or credentials, so a wrong or injected script damages nothing outside it. And the agent acts under its own service identity, never a borrowed human login, so every action is attributable and its entitlements reviewable like anyone else's; BNY, for example, gives each of its more than one hundred agent "digital employees" its own login credentials (Axios 2025).

Memory. An agent has two kinds. Short-term memory is the context window, the running transcript of thoughts, actions, and observations, which is finite and degrades before it is full: models use information at the start and end of a long context more reliably than in the middle (Liu et al. 2024), so long tasks summarize old steps. Long-term memory is an external store, typically a vector database searched with Section 16.4.5's retrieval, holding past interactions, client preferences, or lessons from earlier tasks. That is where finance-specific risks collect. A planted instruction written into memory persists into every later task that retrieves it, turning a one-time prompt injection into a standing one; a stored fact goes stale (a client's address, a counterparty's rating) and is retrieved with the same confidence as a fresh one; and memory shared across clients or desks can carry material nonpublic information across the information barriers a bank must maintain. Memory therefore needs the same retention schedule, access control, and periodic review as any other client record.

Evaluating agents. An agent is evaluated on a suite of realistic tasks, with two kinds of measurement. The first is outcome: did the task end in the correct state? The τ-bench benchmark (Yao et al. 2024), for instance, compares the database at the end of each simulated customer-service conversation with the annotated goal state, rather than trusting the agent's report, and asks whether the agent succeeds on all of k repeated attempts, its pass^k reliability; when it was published in June 2024, even the strongest function-calling agents solved fewer than half the tasks, and their pass^8 in the retail domain was below 25%. The second is trajectory review, reading the steps themselves, because a correct end state reached by a wrong path (a forbidden data source consulted, an approval skipped, a calculation done by the model instead of the calculator) is a failure outcome scoring cannot see. In practice a firm reports task success rate, the rate and

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appropriateness of escalations, cost and steps per task, and the findings of humans reviewing a sample of trajectories, the agent-level version of Chapter 6's held-out evaluation.

Worked Example 16.16: How Per-Step Reliability Compounds

Suppose a task requires k steps, each completed correctly with probability p, and, as a first approximation, that errors at different steps are independent. The task succeeds only if every step does:

$$\Pr(\text{task succeeds}) = p^k. \tag{16.10}$$

Table 16.7 evaluates this for two per-step reliabilities. An agent that is right 95% of the time at each step completes a ten-step task correctly only $0.95^{10} = 59.87\%$ of the time, and a twenty-step task 35.85% of the time. Even at 99% per step, one task in five fails at twenty steps. Inverting the equation shows how demanding a task-level target is: a 95% task success rate requires $p = 0.95^{1/k}$ per step, which is 0.98979 at five steps, 0.99488 at ten, and 0.99744 at twenty.

Per-step reliability p	$k = 5$	$k = 10$	$k = 20$
0.95	0.7738	0.5987	0.3585
0.99	0.9510	0.9044	0.8179

Table 16.7: Probability that a k-step agent task succeeds, p^k, for independent steps

Now add a verification step at the end of each task, an independent check of the kind the P/E agent ran, which catches a fraction c of the tasks that went wrong and also, occasionally, flags a task that went right, at a false-alarm rate f. Every flag goes to a human reviewer. Over N tasks the expected review load R and the expected number of silent failures S, wrong outputs that pass the verifier unflagged, are

$$R = N\left[c\,(1 - p^k) + f\,p^k\right], \qquad S = N\,(1 - c)\,(1 - p^k). \tag{16.11}$$

Take $N = 1{,}000$ ten-step tasks, a verifier that catches $c = 80\%$ of failures, and a false-alarm rate of $f = 2\%$. At $p = 0.95$, 401.3 of the tasks contain an error; the verifier flags $0.80 \times 401.3 = 321.0$ of them and $0.02 \times 598.7 = 12.0$ good tasks besides, so humans review 333.0 tasks, a third of the volume, and $0.20 \times 401.3 = 80.3$ wrong outputs still go through unflagged. At $p = 0.99$, only 95.6 tasks fail; humans review $76.5 + 18.1 = 94.6$ tasks, under a tenth of the volume, and 19.1 errors go through silently. Without a verifier, all the failures, 401.3 or 95.6, would be silent.

Three lessons follow. First, per-step reliability is the economic lever: moving from 95% to 99% per step cuts the review load by a factor of about 3.5 and silent failures by about 4.2, which decides whether an agent saves analyst time or merely moves it into review. Second, a verifier converts silent failures into review work but never removes them; the residual S is what the firm accepts, and c must be estimated on tasks with known errors, not assumed. Third, as the agent improves, more of what the verifier flags is false alarm: the flags' precision, $321.0/333.0 = 96.4\%$ at $p = 0.95$, falls to $76.5/94.6 = 80.9\%$ at $p = 0.99$, because the same 2% false-alarm rate applies to many more good tasks, the base-rate effect of Chapter 1's Bayes' theorem example. Independence is the assumption to doubt. Errors from one shared cause, a misread instruction or a stale fact in memory, cluster in the same tasks, which leaves more tasks clean than p^k predicts but makes those errors systematic; and a verifier built on the agent's own model likely shares its blind spots, so its true c on those errors is below its average.

Failure modes. The worked example is the arithmetic of the most basic failure, small errors compounding over many steps. The others are behavioral. An agent can loop, retrying a failing call or re-planning without progress until a budget stops it, which is why step and spending limits are design requirements. It can call a tool with a hallucinated argument, a plausible but wrong account number or ticker, which a schema check catches only if the value is malformed. It can pursue a proxy for the goal, closing reconciliation breaks by booking offsetting entries rather than finding their causes, the reward hacking Chapter 19 warns about. It can report success on a task it only partly completed, which is why evaluation checks the end state, not the agent's account. It can escalate too often to be useful or too rarely to be safe. And it remains exposed to

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prompt injection, now with memory to make it persistent. Industry forecasters expected many projects to fail on these grounds: in June 2025 Gartner predicted that over 40% of agentic AI projects would be canceled by the end of 2027, citing escalating costs, unclear business value, and inadequate risk controls (Gartner 2025).

What banks are doing. As of October 2026, the public record shows banks moving from pilots to bounded deployments with three common features: the work is internal, middle- and back-office rather than client-facing; the agent drafts or prepares rather than decides; and a named human remains accountable. Citi began piloting agentic capabilities in its in-house Stylus Workspaces platform for 5,000 employees in September 2025, letting one prompt run a multi-step task such as researching a client and compiling a profile from public and internal sources (Torres 2025). BNY described more than one hundred agent "digital employees" working alongside staff by October 2025, with tasks such as validating payment instructions (Axios 2025). In February 2026 Goldman Sachs said it had spent six months working with embedded Anthropic engineers to build agents for trade and transaction accounting and for client vetting and onboarding, the know-your-customer work Chapter 15 described, and that the effort was still at an early stage (CNBC 2026). In August 2026 DBS extended an agentic corporate-credit tool from a 150-person pilot to about 1,500 bankers: specialized agents handle more than 70 tasks to produce a review-ready first draft of a credit memo from annual reports, industry research, and internal records, while relationship managers and credit risk managers remain responsible for the final memo; the bank's stated aim is to cut by at least 30% the time spent on credit memos and related work, which it says can take up to 40% of a relationship manager's time (DBS 2026). These are the firms' own descriptions and targets, not independently measured results. Each applies this subsection's arithmetic: a credit memo assembled from dozens of retrieval and drafting steps is a long-k task, and the human sign-off is the verifier. How supervisors expect such systems to be governed is the subject of Chapter 18.

16.6.5 Mixture-of-Experts: Scaling Without Proportional Compute

Making a language model more capable usually means making it larger, but a larger dense feedforward network makes every token more expensive, whether or not its context needed the full capacity. A mixture-of-experts (MoE) architecture replaces the large feedforward network in each transformer layer with several smaller "expert" networks and a router that sends each token to a handful of them, so capacity grows without a proportional increase in compute per token.

Worked Example 16.17: Sparse Activation Fraction

Consider a MoE layer with 8 experts of 6 billion parameters each (48 billion total expert parameters) that routes each token to its top 2 experts. Each token activates $2 \times 6 = 12$ billion of the 48 billion expert parameters, an active fraction of $12/48 = 0.25$, or 25%: the capacity of a 48-billion-parameter layer at the per-token cost of a 12-billion-parameter one.

16.6.6 Domain-Adapted Financial Language Models: BloombergGPT

Section 16.2.3's Loughran-McDonald dictionary made a lexicon-based version of a broader point: tools built on general-purpose text underperform on financial language. BloombergGPT, announced by Bloomberg in 2023, applies the principle to a full-scale LLM: a 50-billion-parameter model trained from scratch on a mixed corpus of over 700 billion tokens, roughly 363 billion of Bloomberg's own financial data (news, filings, and other proprietary text spanning decades) and roughly 345 billion of general-purpose text, with 569 billion tokens sampled for training. It outperformed similarly sized general models on financial benchmarks (sentiment analysis, financial named-entity recognition, and financial question answering) while remaining competitive on general benchmarks, confirming at LLM scale the domain-specificity argument of Section 16.2.1.

The later evidence is more mixed. Within months, Li et al. (2023) found that general-purpose GPT-4, with no finance-specific pretraining, could outperform BloombergGPT on financial text tasks, although fine-tuned smaller models still beat it at structured extraction such as named-entity and relation recognition. Practice has broadly followed: as of October 2026, a general frontier model with retrieval over the institution's own documents is a common starting point, and domain-specific training tends to earn its cost for narrow, high-volume tasks, for data that cannot leave the firm, or for small models that must run in-house. The

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domain-specificity argument survives; what has changed is where domain knowledge is most cheaply supplied, at retrieval and fine-tuning time rather than by pretraining from scratch.

16.6.7 Quantization: Deploying a Large Model at Lower Cost

An institution that wants to run a model like BloombergGPT on its own infrastructure rather than send every query to an external API, a real consideration given Section 16.5.2's data-leakage concern, faces a direct memory and compute cost. Quantization reduces it by storing each parameter with fewer bits than were used in training, trading a little accuracy for a large reduction in memory.

Worked Example 16.18: Memory Footprint Under Quantization

A 50-billion-parameter model held at 16-bit precision (2 bytes per parameter) requires $50 \times 10^9 \times 2 = 100 \times 10^9$ bytes, 100 gigabytes, just to store its weights, before activations or a long conversation's key-value cache. Quantizing to 8-bit precision (1 byte per parameter) halves this to 50 gigabytes, and 4-bit precision (0.5 bytes per parameter) brings it to 25 gigabytes, a fourfold reduction from 16-bit. This trade-off, some accuracy lost for a model that fits on cheaper hardware and answers faster, often decides whether an institution can run a domain-adapted model in-house at all.

16.6.8 Worked Example 16.19: API Cost versus Self-Hosting at Scale

Quantization's payoff is easiest to see by pricing the alternative, framed as a break-even volume because API prices move far faster than this book's print cycle. The two cost structures differ in kind. An API bills per token, so its cost is purely variable, and output tokens cost more than input tokens, typically four to six times as much, since generating a token takes more compute than reading one. Self-hosting is mostly a fixed cost: suppose the 4-bit-quantized 50-billion-parameter model runs on one dedicated cloud GPU node reserved around the clock at $20 an hour, or $480 a day ($14,400 a month), whether it is busy or idle, with capacity for 4,000 queries an hour, 96,000 a day.

Let a query carry n_{in} input tokens (the prompt plus any retrieved context) and n_{out} output tokens, with API prices p_{in} and p_{out} per million tokens. Self-hosting wins once daily query volume Q exceeds

$$Q^* = \frac{\$480}{(n_{\text{in}}\, p_{\text{in}} + n_{\text{out}}\, p_{\text{out}})/10^6}.$$

Take an institution's financial-research assistant handling 40,000 queries a day at 800 input and 200 output tokens each. The prices are the parameter that dates. At the 2023 list price of the original GPT-4, $30 per million input tokens and $60 per million output tokens, a query cost $0.036, the API bill came to $1,440 a day, three times the self-hosted node, and the break-even was only 13,333 queries a day. As of October 2026, published list prices for mainstream models span roughly $0.05 to $10 per million input tokens and $0.50 to $50 per million output tokens, from the smallest models to the frontier ones, with premium reasoning tiers priced higher still. At an illustrative frontier-tier price of $5 in and $25 out, a query costs $0.009, the same workload costs $360 a day through the API, cheaper than the $480 node, and self-hosting wins only above 53,333 queries a day. At an illustrative mid-tier price of $2 in and $10 out, the break-even rises to 133,333 queries a day, more than the node can serve, and at a small model's $0.10 in and $0.50 out, about $7 a day buys the whole workload. Input and output asymmetry matters as much as the headline rate: tilt the same 1,000 tokens toward long generated answers, 200 in and 800 out, and the frontier-tier cost per query rises to $0.021, cutting the break-even to 22,857 queries a day.

What survives price changes is the structure, not any single number. Falling per-token prices have pushed the break-even volume up roughly fourfold or more since 2023, so a pure cost case for self-hosting now needs very high volume, an output-heavy workload, or a premium model with a high API price. The memory-footprint calculation still makes self-hosting feasible at all: a model that fits on one GPU only at 4-bit precision keeps the fixed cost at one node rather than a multi-GPU cluster. And cost is rarely the whole decision: Section 16.5.2's data-leakage concern can favor self-hosting even below the break-even, because every query stays inside the institution's infrastructure.

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16.7 Case Vignette: AI @ Morgan Stanley

In September 2023, Morgan Stanley announced that AI @ Morgan Stanley Assistant, built together with OpenAI on GPT-4, was fully live for its wealth management financial advisors and support staff. In this chapter's terms the assistant is a retrieval-augmented generation system: it gives advisors natural-language access to the firm's "intellectual capital," roughly 100,000 research reports and internal documents, retrieving the passages relevant to a question and generating an answer grounded in them, the mechanism Section 16.4.5 worked through on four toy documents. Morgan Stanley later reported over 98% adoption among advisor teams, and that the share of research content advisors could effectively find and use rose from roughly 20% to roughly 80%, a direct measure of the retrieval problem at large scale.

The firm then applied the same models to a structurally different task: AI @ Morgan Stanley Debrief, which uses OpenAI's Whisper speech-to-text model with GPT-4 to turn a client meeting recording, made only with the client's consent, into a structured summary, client notes filed automatically into the customer relationship management system, and a draft follow-up email an advisor reviews and sends. This is the summarization capability of Section 16.4.4, deployed on sensitive input (a recorded client conversation) rather than a public filing, which is why consent, access control, and human review before anything reaches a client are first-class design requirements, the discipline Section 16.5.2 argued a deployment needs from the outset.

A third tool, AskResearchGPT, extended the same retrieval-augmented pattern into the firm's institutional research business: a RAG system validated for one internal use tends to generalize to others faster than building anew. The case is instructive because its two headline results, near-total adoption and a fourfold increase in effective document accessibility, came not from a more powerful model than other firms could license, but from the unglamorous work this chapter has emphasized: a well-curated retrieval corpus, retrieval grounded in the cosine-similarity geometry of Section 16.3.3, and a governance structure (consent for Debrief's recordings, human review of drafted outputs) that made deployment at scale defensible.

16.8 Challenges in Natural Language Processing in Finance

Several challenges recur across the techniques this chapter developed, and many connect to themes from earlier chapters.

Domain-specific vocabulary defeats generic tools. Lexicons and language models trained on general text misread financial language (Section 16.2.1), which is why finance-specific resources like the Loughran-McDonald dictionary exist and why even a large general model benefits from finance-specific fine-tuning or careful prompting.

A weak signal is still noisy, not deterministic. Section 16.3.1's signal beat buy-and-hold on mean return and Sharpe-like ratio while still being wrong on two of eight events: an NLP-derived signal is a probabilistic input to a strategy, not a certainty, as Chapter 6 emphasized about model evaluation.

Hallucination is a different failure mode from misclassification. A classifier's error rate is computable from a confusion matrix (Section 16.3.2); an LLM's fluent, wrong answers (Section 16.4.4) have no equally clean error rate, since detecting a hallucination often requires the expertise the tool was meant to save. Retrieval-augmented generation (Section 16.4.5) mitigates this without eliminating it: a model can still misread or misquote a correctly retrieved passage.

Generative AI introduces risks classical models never raised. Non-deterministic outputs, prompt injection, and data leakage (Section 16.5.2) have no counterpart in the deterministic models of Chapters 6 through 15, and require validation by statistical sampling of outputs rather than checking a single fixed prediction.

Explainability and audit requirements constrain deployment. As Chapter 18 develops, a trading decision or disclosure classification influenced by an NLP model may need to be explained after the fact, which is harder for the attention-based architectures of Section 16.4.1 than for the inspectable probabilities of Section 16.3.2's Naive Bayes classifier.

Data licensing and evaluation are harder than they look. High-quality financial text, especially real-time news and licensed feeds, is expensive and contractually restricted, and evaluating an NLP system is harder than evaluating a numerical model, since two humans often disagree about the "correct" sentiment or summary of a passage, so even the ground truth is somewhat subjective.

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16.9 Key Takeaways

Financial text must be converted into numerical features before this book's earlier tools apply: bag-of-words and TF-IDF (Section 16.2.2) give sparse count vectors, the Loughran-McDonald lexicon (Section 16.2.3) gives a finance-specific sentiment score, and word embeddings (Section 16.3.3) give dense vectors that capture semantic similarity geometrically.

A lexicon-based sentiment score is a useful but noisy trading signal: Section 16.3.1's 75% hit rate still beat buy-and-hold on average return and consistency, and generating the signal is only half the task Chapter 13 develops. It should be judged by an event study, on abnormal rather than raw returns and pooled over many events, since the market explained more than half of one event's reaction and a single event's cumulative abnormal return was indistinguishable from noise. Naive Bayes applies directly to document classification (Section 16.3.2), including ambiguous cases where its independence assumption shows most.

Self-attention (Section 16.4.1) lets transformer models, including the LLMs of Section 16.4.4, build context-dependent representations of text, and their generative capability supports summarization and extraction well beyond classification, at the cost of a new failure mode, hallucination, that a computable error rate cannot fully capture. Retrieval-augmented generation (Section 16.4.5) is the leading mitigation, grounding answers in verifiable source passages, while prompting strategy (Section 16.4.7), zero-shot, few-shot, or chain-of-thought, shapes how reliably a model performs a task without changing the model.

Generative modeling extends beyond text: GANs and diffusion models (Section 16.5.1) generate synthetic numerical data, minority-class fraud examples or stress scenarios beyond the historical sample, by fooling an adversarial discriminator or reversing a noising process. All of this requires governance (Section 16.5.2) that extends Chapter 15's three lines of defense to risks, non-determinism, prompt injection, data leakage, that deterministic models never presented.

In Section 16.6, RLHF's reward modeling turns human preferences into a trainable signal, and agents compose retrieval with computation through function calling (delegating arithmetic to a calculator tool), with least-privilege permissions and human approval as safety controls. Over a long task, reliability compounds as p^k (Section 16.6.4), so an agent right 95% of the time per step completes a ten-step task only about 60% of the time, and per-step reliability, not headline capability, sets how much human review an agent saves or creates. Mixture-of-experts architectures decouple model capacity from per-token compute, and BloombergGPT carries the domain-specificity argument to LLM scale, although general frontier models with retrieval have since often matched it. Finally, readability metrics like the Gunning Fog index (Section 16.2.4) turn a regulatory concern, whether disclosure is understandable, into an automatable measurement.

16.10 Suggested Exercises

16.1 Using Table 16.2's method, compute the idf and tf-idf weight (in D_2) for a seventh term, "confident," which appears once in D_3 only, and explain why its idf is the largest of any term considered so far.

16.2 Using Section 16.2.3's method and word lists, compute the tone score for the sentence "results were weak and management remains cautious about the coming quarter despite improved margins."

16.3 Using Table 16.3's method, add a ninth event with tone score $+0.11$ and next-day return -0.6%, and recompute the strategy's hit rate, mean return, and Sharpe-like ratio.

16.4 Explain, using events 4 and 7 in Table 16.3, why a tone score close to zero is a weaker signal than one far from zero, and propose a modification to the trading rule that would account for this (for example, only trading when |tone| exceeds some threshold).

16.5 Using Section 16.3.2's fitted word probabilities, classify a new sentence with word counts (default: 0, borrower: 1, counterparty: 1, volatility: 0, rate: 1, currency: 0), showing the unnormalized class scores and the resulting posterior probabilities.

16.6 Using Table 16.5, explain in words why this classifier's precision is perfect (1.00) while its recall is not, and describe a business setting in which that trade-off (never a false alarm, but occasionally missing a true case) would be preferable to the reverse.

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16.7 Using Section 16.3.3's toy embeddings, add a seventh term, "dividend" $= (0.70, 0.40)$, compute its cosine similarity to "stock" and to "bond," and explain which existing term it should be geometrically closest to and why.

16.8 Using Section 16.3.3's toy embeddings, compute bearish − bullish + stock, and explain what analogy this attempts to capture, and why, unlike the stock − equity + bond example, the result does not land exactly on any of the chapter's six labeled terms.

16.9 Using Section 16.4.1's worked example, recompute the self-attention output for the token "rallied" (using query $Q_{\text{rallied}} = (0, 1)$ instead of Q_{stocks}), and explain which token's value vector now receives the most weight.

16.10 Using Section 16.4.2's causal-versus-bidirectional worked example, compute "today"'s own contextual representation (using query $Q_{\text{today}} = (1, 1)$) under a causal mask, noting that a token at the last position of a sequence can attend to every earlier token, and explain why, for the very last token in a sentence, a causal mask produces the same output a fully bidirectional model would.

16.11 Explain, in a short paragraph, why an LLM's hallucination failure mode is harder to catch automatically than a classification model's error rate, referencing Section 16.4.4.

16.12 Using Section 16.2.4's method, compute the Gunning Fog index for a passage of your own construction with 30 words, 2 sentences, and 6 complex words, and compare it to both worked paragraphs' scores.

16.13 Compare this chapter's Naive Bayes document classifier (Section 16.3.2) to Chapter 6's original loan-default Naive Bayes discussion: explain, in your own words, why the conditional-independence assumption is more defensible for word-count features in a document than it would be for a firm's financial ratios.

16.14 Using Section 16.4.5's worked example, add a fifth knowledge-base document, Doc5, "Company X's Q3 operating expenses rose 5% to $310 million," with embedding $(0.75, 0.30, 0.00)$, compute its cosine similarity to the same query, and state where it ranks among the five documents.

16.15 Using Section 16.4.6's precision@k example, recompute Precision@4 if Doc5 from the previous exercise (also not relevant to a revenue question) is retrieved fourth, and explain in words why adding a fifth, similarly irrelevant document could never increase precision at any k, only decrease it or leave it unchanged.

16.16 Using Section 16.4.6's faithfulness-score example, an answer embeds at $(0.80, 0.25, 0.00)$. Compute its faithfulness score against Doc1, and state whether it would pass a review threshold of 0.90.

16.17 Write a zero-shot prompt, a two-example few-shot prompt, and a chain-of-thought prompt, following Section 16.4.7's patterns, for the task of classifying whether a sentence from a 10-K risk-factors section describes credit risk or market risk (the task Section 16.3.2 solved with Naive Bayes instead).

16.18 Using Section 16.5.1's worked example, compute x_t for a third diffusion step with $\alpha_t = 0.30$, holding $x_0 = 1.00$ and $\varepsilon = -0.50$ fixed, and confirm that x_t has moved even closer to ε than the two steps already computed.

16.19 Explain, in a short paragraph, why validating a generative AI system requires sampling many outputs rather than checking a single prediction against a labeled test set, referencing Section 16.5.2 and contrasting it with how Chapter 6 evaluates a classification model.

16.20 Pick two of the challenges listed in this chapter and describe a concrete scenario, not already used in the text, in which that challenge could cause an NLP-based financial application to fail despite technically correct code.

16.21 Using the case vignette, explain why AI @ Morgan Stanley Debrief's requirement of client consent before recording is best understood as a governance control (Section 16.5.2) rather than a legal formality, and describe how human review before sending a draft follow-up further limits the hallucination risk of Section 16.4.4.

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16.22 Using Section 16.6's Bradley-Terry worked example, recompute the preference probability if the inaccurate summary's reward score r_B rises from 0.8 to 1.7 (holding $r_A = 2.1$ fixed), and explain in words why the preference probability moves toward 0.5 rather than away from it.

16.23 Using Section 16.6's agent example, explain why the agent calling a calculator tool to compute 57.87/3.15 is preferable to the LLM performing that division itself, and describe a different two-step financial task (not P/E) that would similarly benefit from separating retrieval from computation.

16.24 Using Section 16.6's mixture-of-experts example, recompute the active parameter fraction for a layer with 16 experts of 4 billion parameters each, routing each token to its top 4 experts, and compare the resulting total capacity and active-compute cost to the chapter's original 8-expert example.

16.25 Using Section 16.6's quantization worked example, compute the memory footprint of a 175-billion-parameter model at FP16, INT8, and INT4 precision, and state the reduction factor from FP16 to INT4.

16.26 Using Section 16.6's LoRA worked example, recompute the number of trainable parameters and the reduction factor relative to full fine-tuning if the rank r is increased from 8 to 32 (holding $d = 4{,}096$ fixed), and explain in words the trade-off a practitioner faces when choosing a larger rank.

16.27 Using Section 16.6.4's worked example and equations (16.10) and (16.11), (a) compute the task success probability for a fifteen-step task at $p = 0.98$ per step; (b) for $N = 1{,}000$ ten-step tasks at $p = 0.99$, recompute the review load R, the silent failures S, and the precision of the verifier's flags if the verifier is improved to $c = 0.90$ while its false-alarm rate falls to $f = 0.01$; and (c) state which of the two changes, raising c or lowering f, matters more for the review load at this level of p, and why.

16.28 A bank deploys an agent that refreshes know-your-customer files in about twelve steps (retrieve the file, pull registry records, screen each beneficial owner against sanctions lists, compare, draft exceptions). Explain why scoring the agent on its own report that each file was "completed" would overstate its success rate, propose an outcome check on the end state and one trajectory-review rule that would catch a correct file reached by an unacceptable path, and say which tools in the task you would place behind human approval.

16.29 Using Section 16.3.1's event study and Equation (16.4), recompute the day +1 abnormal return, $CAR(-1, +1)$, and its t-statistic if the market had risen 1.5% on day +1 instead of 0.9%, holding every other return and the fitted $\hat{\alpha} = 0.0260\%$, $\hat{\beta} = 1.2949$, and $\hat{\sigma} = 1.4215\%$ fixed. Explain what the result implies about crediting event 5's +2.1% to the call's tone, and how many events with the text's original average CAR of 0.7977% would be needed to reach $t = 2$.

16.30 A private-credit borrower has $300 million of net debt, reported EBITDA of $52 million, add-backs of $9 million (projected synergies) and $4 million (restructuring), and a covenant capping total net leverage at 5.25 times Consolidated EBITDA, with add-backs capped at 20% of EBITDA before add-backs. Using Section 16.4.4's private-credit example, (a) compute leverage on reported EBITDA, with all add-backs, and with the capped add-backs, and state which pass; (b) find the reported EBITDA at which the capped test is breached and the percentage fall from $52 million that this represents; and (c) list the fields an LLM should extract from the credit agreement, with what evidence, before a calculator performs (a) and (b).

16.31 **Challenge (real data).** Download the risk-factors (Item 1A) section of a real company's 10-K from SEC EDGAR's full-text search (free, no login required), and the Loughran-McDonald word list (free from the University of Notre Dame's Software Repository for Accounting and Finance). (a) Apply the tone-scoring method of Section 16.2.3 to the text, reporting the tone score and the positive and negative words matched. (b) Compute the Gunning Fog index (Section 16.2.4) for the same passage, and compare it to both worked-example scores. (c) Real 10-K risk-factor sections are written by legal and investor-relations teams with every incentive to sound cautious whatever the firm's condition; discuss whether a negative tone score on such boilerplate should be read the same way as one on the earnings-call language of this chapter's worked examples.

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16.11 Further Reading

- Loughran and McDonald (2011), "When Is a Liability Not a Liability? Textual Analysis, Dictionaries, and 10-Ks." The original paper introducing the finance-specific sentiment dictionary used in Section 16.2.3.
- Hansen and McMahon (2016), "Shocking Language: Understanding the Macroeconomic Effects of Central Bank Communication." A research-grade version of Section 16.2.3's central-bank worked example, using text methods to separate what the Federal Reserve says from what markets expected it to say.
- Jurafsky and Martin (n.d.), *Speech and Language Processing*. A comprehensive, freely available reference covering bag-of-words, TF-IDF, and embedding methods (Sections 16.2.2 and 16.3.3) in far greater depth.
- Vaswani et al. (2017), "Attention Is All You Need." The original transformer paper introducing the self-attention mechanism worked through by hand in Section 16.4.1.
- Mikolov et al. (2013), *Efficient Estimation of Word Representations in Vector Space*. The original Word2Vec paper underlying the embedding geometry illustrated in Section 16.3.3.
- Li (2010), "The Information Content of Forward-Looking Statements in Corporate Filings—A Naïve Bayesian Machine Learning Approach." An application of the Naive Bayes text classification approach used in Section 16.3.2 to real corporate filings.
- Lewis et al. (2020), "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks." The original paper introducing the retrieval-augmented generation approach worked through in Section 16.4.5.
- Wei et al. (2022), "Chain-of-Thought Prompting Elicits Reasoning in Large Language Models." The original paper on the chain-of-thought prompting strategy discussed in Section 16.4.7.
- Goodfellow et al. (2020), "Generative Adversarial Networks." The original paper introducing the generator-discriminator framework discussed in Section 16.5.1.
- Ho et al. (2020), "Denoising Diffusion Probabilistic Models." The paper underlying the forward and reverse diffusion process worked through by hand in Section 16.5.1.
- Board of Governors of the Federal Reserve System and Office of the Comptroller of the Currency (2011), *Supervisory Guidance on Model Risk Management*. The foundational U.S. model risk management guidance, superseded in April 2026 by SR 26-2 (Federal Reserve, OCC, and FDIC 2026), which keeps its pillars but leaves the generative AI governance questions raised in Section 16.5.2 outside its scope.
- Morgan Stanley (n.d.), *"Key Milestone in Innovation Journey with OpenAI" and "Launch of AI @ Morgan Stanley Debrief"*. The primary source for the retrieval-augmented generation and meeting-summarization deployment described in this chapter's case vignette.
- Ouyang et al. (2022), "Training Language Models to Follow Instructions with Human Feedback." The paper introducing the RLHF pipeline and reward modeling worked through by hand in Section 16.6.
- Yao et al. (2023), "ReAct: Synergizing Reasoning and Acting in Language Models." A foundational paper on the agent tool-use loop illustrated in Section 16.6's worked example.
- Yao et al. (2024), *τ-bench: A Benchmark for Tool-Agent-User Interaction in Real-World Domains*. The agent benchmark discussed in Section 16.6.4, which scores an agent on the end state of each task and introduces the pass^k measure of reliability across repeated attempts.
- Shazeer et al. (2017), "Outrageously Large Neural Networks: The Sparsely-Gated Mixture-of-Experts Layer." The original paper introducing the sparse routing architecture discussed in Section 16.6.

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- Wu et al. (2023), *BloombergGPT: A Large Language Model for Finance*. The paper describing the domain-adapted financial language model discussed in Section 16.6.
- Islam et al. (2023), *FinanceBench: A New Benchmark for Financial Question Answering*. The benchmark of filing-based financial questions discussed in Section 16.4.6, and a practical template for building an institution's own evaluation set.
- SEC (1998), *A Plain English Handbook*. The SEC's own guidance underlying the readability and disclosure-analysis discussion in Section 16.2.4.

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Chapter 17

Alternative Data and Deep Learning in Finance

17.1 Introduction

Chapter 1's table of financial data set aside one row, alternative data, for this chapter: non-traditional sources such as satellite imagery, web traffic, or shipping data that proxy for economic activity. Chapter 3 noted that large asset managers already parse filings and calls faster than human analysts, and Chapter 4 that alternative data can supplement the delayed, discretionary numbers in financial statements. This chapter turns to data sources that did not exist, in usable form, a generation ago, and to the deep learning architectures built to extract signal from them. It extends Chapter 6's feedforward network and Chapter 16's transformer architecture to two more data shapes, grid-structured images and ordered sequences, and works through the full pipeline from raw data to a usable signal, treating the biases specific to this data as a first-class topic.

Thirteen worked examples run through the chapter. The first five follow a signal from data to trade: a vendor subscription's cost against the trading edge it buys; a regression of a retailer's revenue growth on a satellite-derived car-count proxy; a multi-source earnings-surprise signal, evaluated as Chapter 13 evaluates any strategy; the same signal rebuilt to respect real data-delivery lag; and a peer-relative normalization of the satellite proxy. Three hand-computed networks follow: a two-dimensional convolution over a satellite image patch, a recurrent neural network forward pass over web-traffic readings, and an autoencoder that flags a data-quality anomaly. A Loughran-McDonald tone score built from a company's own risk-factor disclosures shows the appeal and the limits of an in-house ESG-style signal. The last four go deeper: a transfer-learning fine-tuning step, a graph neural network update over a shell-company network, a cross-attention fusion of three data modalities, and a contrastive-pretraining calculation that learns without labels.

The chapter follows the arc of a real alternative-data project. It begins with what alternative data is and how a fund acquires and evaluates it (Sections 17.2.1 and 17.2.2), then turns to the analytics: a single-source regression, a multi-source signal, the point-in-time discipline that keeps that signal honest, and the seasonal and peer-relative adjustments that make raw readings usable (Sections 17.2.3 through 17.3.2).

It then covers the deep learning architectures that extract features from raw images and sequences (Section 17.4), the data-quality and governance problems specific to this data (Section 17.3.3), an in-house ESG-style text signal as a partial, auditable response to those problems (Section 17.5.1), and the more advanced architectures where the field is moving: transfer learning, graph neural networks, cross-attention fusion, and contrastive pretraining (Section 17.5.2). A case vignette and a closing set of challenges tie these techniques to the economic and regulatory stakes of using data that, in most cases, did not exist in usable form before this decade.

17.2 The Alternative-Data Landscape

This part establishes what alternative data is, who sells it, and works two signals end to end.

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17.2.1 What Counts as Alternative Data

Suppose a hedge fund wants to know how a retailer's holiday-quarter sales are trending three months before the retailer reports earnings. The information is valuable if the fund can act on it early and worthless once everyone else has it. Traditional market, fundamental, and macroeconomic data (Chapter 1's Table 1.6) cannot answer this: market data reflects only what the market already knows, and fundamental data about this quarter will not exist until the retailer files it.

Alternative data exists to fill this gap. In industry usage, it means any data source used to inform an investment or credit decision that falls outside the traditional trio of market, fundamental, and macroeconomic data. The term is a moving target: yesterday's alternative data becomes today's mainstream input once enough participants adopt it, so the boundary is set by current practice rather than by any technical property. Table 17.1 catalogs the categories most widely used.

Category	Examples	Typical financial use
Satellite and aerial imagery	Parking-lot car counts, crop health indices, oil-storage tank levels, construction progress	Retail sales nowcasting, commodity supply estimation, real-estate tracking
Web and app data	Website traffic, app downloads and reviews, job postings, e-commerce pricing	Revenue nowcasting, hiring-trend and competitive signals
Transaction and card-panel data	Aggregated, anonymized consumer card-spending panels	Same-store sales nowcasting ahead of earnings
Geolocation and foot traffic	Mobile-device location pings at store or venue locations	Foot-traffic proxies for retail, dining, and travel
Shipping and logistics	Vessel-tracking (AIS) data, port congestion, rail and truck telemetry	Commodity flow and trade-volume forecasting
Social and search data	Social-media posts, review platforms, search-trend indices	Consumer-sentiment and brand-momentum signals
On-chain (blockchain) data	Public transaction ledgers, wallet flows, exchange inflows and outflows, stablecoin issuance	Crypto asset-flow signals, exchange-risk monitoring, AML and fraud forensics

Table 17.1: Major categories of alternative data

Two properties distinguish alternative data from the data earlier chapters used. First, it usually arrives already processed by a vendor (a satellite imagery provider sells car counts, not raw pixels), so the modeler inherits whatever assumptions and errors are baked into that proprietary step, typically without seeing it. Second, it comes at a frequency and granularity (individual stores, individual weeks) that quarterly, firm-level fundamental data does not offer. That is why it is valuable: it lets an analyst estimate a firm's performance before the firm reports it.

The table's last category inverts both properties. **On-chain data**, the complete public transaction ledger a blockchain such as Bitcoin's or Ethereum's maintains by construction, records every transfer, wallet balance, and smart-contract interaction, free of charge and with no vendor between the analyst and the raw record. Funds trading crypto assets (Chapter 13's tokenization discussion introduces the instruments) read signals from it: net flows into and out of exchange-controlled wallets as a proxy for imminent selling or accumulation, and stablecoin issuance as a proxy for capital entering the ecosystem. Risk and compliance teams watch the same ledger to monitor exchange solvency and trace stolen or laundered funds. What the analyst inherits instead of a vendor's opaque preprocessing is pseudonymity: every party is an unlabeled address, so the commercial value lies in entity labeling and address clustering, the graph techniques Chapter 15 applies to shell-company layering chains. On-chain data is thus the rare category where the raw data is free and complete and the interpretive layer is the scarce, priced asset, the reverse of a satellite feed, whose pixels are expensive and whose car counts are the deliverable.

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17.2.2 The Alternative-Data Vendor Ecosystem

Unlike the market and fundamental data of Chapters 2 through 5, which arrive through established exchange feeds and regulatory filings, alternative data is bought in a young and heterogeneous vendor market, and how a fund evaluates and integrates a vendor matters almost as much as the data itself. Procurement typically runs through three stages.

First, the fund identifies a hypothesis (does satellite-derived parking-lot data predict Cascade's revenue?) and requests a trial dataset covering a historical period the vendor already has on file. Second, it backtests the trial data against already-reported outcomes, the exercise Section 17.2.3 works by hand, before paying for a live subscription. A vendor unwilling to provide a meaningful historical trial, or whose trial performance does not hold up when checked, is a warning sign. Third, if the trial convinces, the fund negotiates a license that specifies not just price but exclusivity (who else subscribes, and how many), delivery latency (Section 17.3.3 explains why this matters), and historical backfill (how far back the data goes, which bounds the fund's own model validation).

This procurement cycle is itself a source of adverse selection. A vendor has every incentive to showcase the period and the firms where its data performed best, since that is what sells subscriptions, so strong trial performance is not evidence that the data will perform as well later or on other firms. This is Chapter 6's multiple-testing concern about a researcher's model search, applied to a vendor's marketing.

A subscription decision comes down to whether the expected trading edge, scaled to the capital deployed on it, exceeds the data's licensing cost. Suppose a fund would deploy a \$50 million position whenever Section 17.2.4's signal fires long, which happened in five of the eight quarters of Table 17.3 below, and that the signal's incremental edge over an unconditional baseline is the 0.5425 percentage points computed there (the 0.88% average return on signaled quarters minus the 0.3375% average across all quarters). With four earnings releases per year and a 5/8 signal frequency, the fund expects about $4 \times 5/8 = 2.5$ signaled events a year, for an expected incremental profit of $\$50\text{M} \times 0.54\% \times 2.5 \approx \$678{,}000$. Against Section 17.6's subscription-cost range, a satellite feed at \$250,000 per year plus a web and card-panel bundle at \$150,000, data costs total \$400,000, for a net expected benefit of about \$278,000 a year. The edge is modest, and Section 17.3.3 warns that it will shrink as more funds license the same feeds. Even the gross figure flatters the signal: the 0.5425-point edge is measured on the same eight quarters the regression was fitted to, so \$678,000 is an in-sample upper bound on what the signal would earn on quarters it has not seen.

The same arithmetic drives a build-versus-buy decision many larger funds eventually face. Once trading volume across many similar signals is large enough, an in-house satellite-analytics or web-scraping team can be cheaper than paying a vendor's margin on the same imagery or logs, at the cost of taking on the computer-vision and data-engineering work Section 17.4 describes. Smaller funds typically lack the scale to justify this fixed cost and remain vendor customers, one more way in which alternative data's advantages concentrate among larger participants.

17.2.3 Worked Example 17.1: Satellite Imagery and Retailer Revenue

Suppose a hypothetical home-improvement retailer, Cascade Home & Garden, is followed by a data vendor that counts cars in a sample of its store parking lots from satellite imagery, as Section 17.6 describes being done for real retailers since 2010. Table 17.2 lists eight quarters of year-over-year car-count growth alongside Cascade's subsequently reported year-over-year revenue growth.

Fitting the simple regression Revenue growth $= \alpha + \beta\,$(Car-count growth) by ordinary least squares, following Chapter 6's Section 6.3.1 method, gives $\bar{x} = 1.7125$, $\bar{y} = 1.6625$, $S_{xy} = \sum(x_i - \bar{x})(y_i - \bar{y}) = 43.92375$, $S_{xx} = \sum(x_i - \bar{x})^2 = 67.28875$, so $\hat{\beta} = S_{xy}/S_{xx} = 0.6528$ and $\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x} = 0.5446$.

The fitted line, $\widehat{\text{Revenue growth}} = 0.5446 + 0.6528\,(\text{Car-count growth})$, achieves $R^2 = 0.870$: car-count growth alone explains 87% of the quarter-to-quarter variation in Cascade's revenue growth in this sample. That is strong for a single proxy, though 13% of the variation remains unexplained.

Suppose satellite data for a ninth quarter shows 5.0% car-count growth three weeks before Cascade's earnings release. The regression predicts $\widehat{\text{Revenue growth}} = 0.5446 + 0.6528(5.0) = 3.809\%$, a tradeable estimate of a number the market will not see for three more weeks.

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Quarter	Car-count growth (%)	Revenue growth (%)
Q1	2.1	1.5
Q2	−1.5	−0.3
Q3	4.8	4.5
Q4	0.6	1.0
Q5	6.2	4.0
Q6	−3.0	−1.0
Q7	3.5	3.8
Q8	1.0	−0.2

Table 17.2: Satellite car-count growth and Cascade's reported revenue growth, by quarter

17.2.4 Combining Alternative Data Sources: A Multi-Source Earnings Signal

A real alternative-data desk rarely uses a single source. It combines several into one model, to average out each source's idiosyncratic noise and to capture aspects of the business a single proxy misses. Figure 17.1 sketches the resulting pipeline.

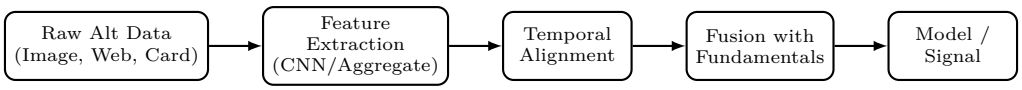


Figure 17.1: The alternative-data pipeline this chapter develops

Extending Section 17.2.3's example, suppose Cascade is also tracked by a web-traffic vendor (year-over-year growth in visits to Cascade's e-commerce site) and a card-panel vendor (year-over-year growth in aggregated card spending at Cascade's stores). Table 17.3 lists all three alternative-data growth rates alongside the earnings surprise (actual revenue growth minus the analyst consensus available at the time) and Cascade's stock return on the trading day after the announcement.

Quarter	Car (%)	Web (%)	Card (%)	Surprise (%)	Fitted (%)	Signal	Return (%)
Q1	2.1	3.0	1.5	0.5	0.48	Long	1.1
Q2	−1.5	−0.5	−1.0	−1.3	−1.08	Flat	−0.6
Q3	4.8	5.5	3.0	2.5	1.95	Long	2.0
Q4	0.6	1.5	0.8	0.0	−0.44	Flat	0.3
Q5	6.2	4.0	4.5	2.0	2.46	Long	−0.4
Q6	−3.0	−2.0	−1.8	−1.5	−1.92	Flat	−1.4
Q7	3.5	2.5	2.0	2.3	1.66	Long	1.5
Q8	1.0	2.0	0.5	−1.2	0.18	Long	0.2

Table 17.3: Multi-source alternative data, earnings surprise, and next-day return, by quarter

Regressing surprise on the three alternative-data growth rates by ordinary least squares, using the normal equations

$$\hat{\beta} = (X^\top X)^{-1} X^\top y, \tag{17.1}$$

with X's columns an intercept, car-count growth, web-traffic growth, and card growth, gives

$$\hat{\beta} = (-0.291,\ 1.092,\ -0.113,\ -0.792), \qquad R^2 = 0.837.$$

The negative coefficient on card growth is not a mistake. With car-count and web-traffic growth already in the regression, card growth's explanatory power is largely redundant, and its small residual coefficient partly offsets collinear noise the other two share. This is common when several correlated proxies for the

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same quantity enter one regression, and a reminder that individual coefficients in a multi-source model are not clean, isolated effects the way Chapter 6's Section 6.3.1 single-variable coefficient is.

Column "Fitted" reports each quarter's fitted surprise, and column "Signal" converts a positive fitted surprise into a long position taken ahead of the release (flat otherwise), the logic Chapter 16's Section 16.3.1 applied to a sentiment score. Five of the eight quarters signal long; four of those (Q1, Q3, Q7, Q8) are followed by a positive next-day return and one (Q5) by a negative return despite a strongly positive fitted surprise, an 80% hit rate.

The signaled quarters average a 0.88% next-day return versus 0.34% across all eight quarters. The edge is real within this sample, though in-sample, since it is measured on the quarters the regression was fitted to, and it is probabilistic, as Chapter 6 and Chapter 16's Section 16.3.1 both emphasize: Q5's miss shows that even a well-fitted multi-source model cannot rule out a price reaction driven by something the data could not capture, such as guidance issued alongside the release. A ninth quarter with car, web, and card growth of 5.0%, 4.5%, and 3.5% yields a fitted surprise of

$$-0.291 + 1.092(5.0) - 0.113(4.5) - 0.792(3.5) = 1.888\%,$$

again a tradeable estimate available before the official release.

17.3 Working with Alternative Data

Alternative data breaks alignment, scale, and quality assumptions that price data never did.

17.3.1 Point-in-Time Alignment and the Cost of Look-Ahead Bias

Section 17.2.4's regression silently assumed that every growth rate in Table 17.3 was in hand before that quarter's earnings release. Card-panel data shows why that assumption fails even when delivery is fast. Commercial card panels typically arrive within days or weeks of the period they cover, but the history a vendor sells today is not the history a subscriber saw in real time. Late transactions are added after the first release, the panel's composition shifts as card issuers join or leave, and vendors routinely revise and re-weight past figures, backfilling the entire history with today's panel and methodology. A backtest that treats today's revised series as if each figure had been known in that form at the time commits look-ahead bias: it uses information it could not have had.

To see how much this matters, refit the regression on quarters Q2 through Q8 (dropping Q1, which has no prior quarter to lag from) two ways. The naive version uses each quarter's contemporaneous card growth, as before:

$$\hat{\beta}_{\text{naive}} = (-0.290,\ 1.102,\ -0.119,\ -0.802), \qquad R^2 = 0.837,$$

with four quarters signaling long (Q3, Q5, Q7, Q8) and a 75% hit rate. The ideal fix is each quarter's first-release vintage, the figure as originally delivered, but a backtest often lacks those vintages. A common conservative substitute, used here, is the card growth reported one quarter earlier, a figure certainly in hand before the release:

$$\hat{\beta}_{\text{realistic}} = (0.251,\ 0.424,\ -0.100,\ -0.286), \qquad R^2 = 0.873,$$

but Q8 no longer signals long once the lag is respected, leaving three quarters (Q3, Q5, Q7) with a 66.7% hit rate and a higher average return (1.03% versus 0.83%) on that smaller, more selective set of trades.

The lesson is not that the lagged version is worse (its R^2 and its average signaled return are both higher) but that it is a *different* model making *different* trading decisions on the same quarters. A backtest of the naive version reports performance for trades the fund could never have placed, since the card figures they depend on were not yet in hand in that form, whether that backtest happens to look better or worse. This is why Section 17.3.3 treats delivery timing and revisions as data-quality issues rather than modeling nuances. The direction of the bias is not predictable in general, but ignoring delivery timing and data vintages always invalidates a backtest as a measure of tradeable performance. Point-in-time discipline, using for every historical date only the data vintage available on that date, is the standard defense.

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17.3.2 Feature Engineering for Alternative Data

Raw alternative-data readings, like the market and fundamental data of Chapters 2 through 5, are rarely used as delivered. Section 17.3.1 addressed when a reading becomes usable; this section addresses what must be done to it first.

Seasonal adjustment. Web traffic, foot traffic, and card volumes carry strong calendar effects (day-of-week patterns, holiday spikes, back-to-school and year-end seasons) unrelated to a firm's underlying performance. Comparing a reading with the same period a year earlier, as year-over-year growth rates such as Section 17.2.3's already do, removes most of this, provided the calendar aligns. Thanksgiving week, for instance, falls on a different date each year, so the alignment should be by trading day or holiday-relative position rather than by calendar date.

Peer-relative normalization. A retailer's car-count or web-traffic growth reflects both firm-specific performance and sector-wide conditions (gasoline prices, consumer spending, weather). Subtracting a sector-average benchmark isolates the firm-specific component a stock-picking strategy cares about.

Suppose sector-average car-count growth for Cascade's peers over the eight quarters of Table 17.2 is $(1.0, -0.5, 2.0, 0.3, 2.5, -1.0, 1.5, 0.5)$; subtracting it from Cascade's own gives a peer-adjusted series $(1.1, -1.0, 2.8, 0.3, 3.7, -2.0, 2.0, 0.5)$. Refitting the regression on this series gives $\hat{\beta} = 1.064$, $\hat{\alpha} = 0.678$, and $R^2 = 0.867$, essentially unchanged from the raw series' $R^2 = 0.870$. Sector-wide movements were small relative to Cascade's own variation in this sample, so the adjustment barely changes the fit. It matters far more when the whole sector moves together, in a consumer-spending downturn or a fuel-price spike that depresses driving-based foot traffic, where an unadjusted signal would mistake a sector-wide move for firm-specific news.

17.3.3 Data Quality, Bias, and Integration Challenges

Alternative data's advantages (frequency, granularity, and an informational edge before official reporting) come with quality problems that have no close analogue in the market and fundamental data earlier chapters used.

Coverage bias. A satellite vendor can cover only stores it has usable imagery for, typically larger, freestanding locations with visible parking lots, so its sample underrepresents smaller-format or mall-based stores, and a model trained on the covered subset may not generalize to the retailer as a whole.

Vendor black-box processing. As Section 17.2.1 noted, a vendor typically sells an already-processed feature, so an analyst inherits whatever computer-vision or aggregation errors are baked into its pipeline without being able to inspect them. An undisclosed methodology change between periods can look, in the data, exactly like a change in the underlying business.

Delivery lag and look-ahead bias. A backtest that assumes alternative data was available on the date it describes, rather than the often later date it reached subscribers, uses information that was not tradeable in real time. Section 17.2.4's multi-source signal is useful only if all three sources reach the desk before the earnings release, not merely before the quarter ends, and only in the vintage that arrived then, not as later revised.

Licensing cost and alpha decay. High-quality alternative data is expensive and often licensed exclusively or semi-exclusively. Section 17.6 documents a 4-5% informational edge around earnings from satellite car counts; as more funds license the same feed, that edge is competed away, the erosion of a widely traded pattern Chapter 6 described. An alternative-data edge has a shelf life.

Privacy and legal exposure. Card-panel and geolocation data are aggregated and anonymized before sale, but reconstructing individual behavior from supposedly anonymized data has repeatedly proven possible in other domains, so a vendor's privacy practices, and a buyer's own legal exposure, are due-diligence questions that market and fundamental data never raised.

The line between alternative data and material non-public information is not always bright. Card-panel and web-traffic vendors aggregate across many cardholders or visitors so that no single company's non-public information is sold, but the boundary is a legal judgment, not a technical threshold. A narrowly scoped feed, transaction data from only a handful of a firm's largest customers, say, can approach material non-public information even though it never touches the firm's systems and no insider leaks anything. This is

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a new compliance risk that Chapter 15's insider-trading discussion, framed around information leaking from inside a firm, did not anticipate.

Is this signal causal? Car counts do not cause revenue, and Section 17.2.3's regression never claimed they did: both are effects of customers visiting stores. A trading signal needs only prediction, the job Chapter 6's Section 6.4.6 assigns to predictive models. The causal question still matters, for two reasons. The first is durability. A proxy that predicts because it measures the business (cars in the lot are shoppers) should keep working as long as the business model does. One that predicts because a third variable drove both it and the target during the sample, a run of fine spring weather lifting both foot traffic and garden-center sales, say, is exposed to the confounding Section 6.4.6 describes, and fails the first time that variable moves on its own. Asking what mechanism links a signal to cash flows is the cheapest robustness test available, and a signal with no plausible mechanism, found by screening many series, is more likely the multiple-testing artifact of Section 17.2.2. The second reason is intervention. A retailer that reads the same car counts to decide where to open stores or how much to spend on promotions is asking a causal question the regression cannot answer, for the same reason that Chapter 6's credit-limit flag predicted lower default while causing higher default. As with Chapter 7's Granger causality, a signal that predicts carries information; it says nothing about how the information arises.

17.4 Neural Networks for Structured and Unstructured Data

Section 17.2.3 treated the vendor's car count as a finished number, but the vendor had to produce it from raw satellite pixels, and the web-traffic and card figures likewise start from raw logs or transaction records. Chapter 6's feedforward network, in which every input feeds every hidden unit with its own weight, ignores two kinds of structure alternative data routinely has: the two-dimensional grid of an image and the ordering of a time-stamped log. Specialized architectures exploit each, and a third needs no labeled target at all.

17.4.1 Convolutional Neural Networks for Grid-Structured Data

Suppose a vendor wants a model to count cars in a million-pixel satellite tile rather than paying an analyst to do it by eye. A feedforward network could treat each pixel as an input feature, but it would need a separate weight connecting each pixel to each hidden unit, and it would have to learn separately that a cluster of bright pixels in the top-left corner and the identical cluster in the bottom-right both mean a car, since nothing in its structure says that a pattern's meaning does not depend on where it sits.

A convolutional neural network (CNN) exploits this redundancy. A useful pattern (a car's outline, an edge, a texture) can appear anywhere in the frame, so instead of a separate weight for every pixel-hidden-unit pair, a CNN learns a small filter (or kernel) and slides it across the image, reusing the same weights at every position. This weight sharing makes CNNs practical: a 1000×1000-pixel tile has a million pixels, far too many for per-pixel weights to be learned from any realistic training set, but a 2×2 filter has four weights regardless of image size.

Consider a stylized 4×4 satellite image patch of a single parking lot, with pixel intensities between 0 (dark asphalt) and 1 (a car's bright roof), containing two clusters of parked cars in opposite corners:

$$I = \begin{pmatrix} 0.1 & 0.1 & 0.8 & 0.9 \\ 0.1 & 0.1 & 0.9 & 0.8 \\ 0.7 & 0.8 & 0.1 & 0.1 \\ 0.8 & 0.9 & 0.1 & 0.1 \end{pmatrix}.$$

Convolving this image with a 2×2 averaging filter

$$K = \begin{pmatrix} 0.25 & 0.25 \\ 0.25 & 0.25 \end{pmatrix}$$

at stride 1 (sliding the filter one pixel at a time, computing the elementwise product with each 2×2 patch and summing) produces a 3×3 feature map. The top-left entry, for instance, averages the patch

$$\begin{pmatrix} 0.1 & 0.1 \\ 0.1 & 0.1 \end{pmatrix}$$

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to 0.1; the entry one column to the right averages

$$\begin{pmatrix} 0.1 & 0.8 \\ 0.1 & 0.9 \end{pmatrix}$$

to $(0.1 + 0.8 + 0.1 + 0.9)/4 = 0.475$. Carrying this out for all nine positions gives

$$\text{Feature map} = \begin{pmatrix} 0.100 & 0.475 & 0.850 \\ 0.425 & 0.475 & 0.475 \\ 0.800 & 0.475 & 0.100 \end{pmatrix}.$$

The two corner entries where a car cluster sits, 0.850 (top right) and 0.800 (bottom left), stand out above the two empty corners (0.100 each) and the mixed middle entries: the filter highlights regions of concentrated brightness.

A trained CNN learns many such filters at once (edge, corner, and texture detectors) and stacks several convolutional layers, so that later layers respond to increasingly complex combinations of earlier features. It typically applies a ReLU activation and a pooling step (taking, say, the maximum of each 2×2 block) after each convolution, shrinking the map while keeping its strongest responses, before feeding the result into feedforward layers of the kind Chapter 6 described to produce a single number, an estimated car count.

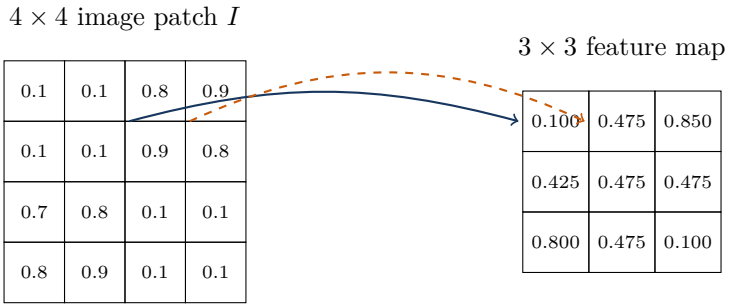


Figure 17.2: Sliding a 2×2 averaging filter across the 4×4 satellite patch I at stride 1

Stacking convolutional layers naively runs into a training problem once a network gets deep: gradients computed by backpropagation (Chapter 6) must flow back through every layer, and in a deep plain CNN they can shrink toward zero before reaching the earliest layers, leaving those layers essentially untrained. This is the vanishing-gradient problem.

The residual network (ResNet), the architecture that made very deep CNNs practical, adds a shortcut connection carrying each block's input to its output. The block then learns only a residual adjustment to what passes through unchanged, and gradients flow through the shortcut even when the block's own transformation contributes little. This is the instinct behind Chapter 12's shrinkage methods (do not force a model to learn from nothing what it could inherit and adjust), applied to network depth rather than portfolio weights.

17.4.2 Recurrent Neural Networks for Sequential Data

Suppose an analyst wants to judge whether Cascade's web traffic is accelerating into its next earnings release from the last several weeks of readings, since a single week cannot distinguish a steady climb from a spike about to reverse. A feedforward network could take the weeks as separate features, but it would treat them as an unordered bag, with nothing telling it that three weeks ago came before last week.

A recurrent neural network (RNN) addresses this. It processes a sequence one step at a time, maintaining a hidden state h_t that is passed forward and updated at every step,

$$h_t = \tanh(w_x x_t + w_h h_{t-1} + b), \tag{17.2}$$

so that h_t is, in principle, a function of every input seen so far, $x_1, \ldots, x_t$. This gives the network a memory that Chapter 7's autoregressive and GARCH recursions share in spirit (each defines a current value as a function of a prior state plus new information) and that Chapter 6's feedforward network lacks.

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Suppose Cascade's e-commerce site shows three successive weekly readings of normalized web-traffic growth in the final three weeks before its next earnings release, $x_1 = 0.5$, $x_2 = -0.2$, $x_3 = 0.9$, and a single-hidden-unit RNN with weights $w_x = 0.6$, $w_h = 0.4$, $b = 0.1$, and initial hidden state $h_0 = 0$. Step by step:

$$h_1 = \tanh(0.6 \times 0.5 + 0.4 \times 0 + 0.1) = \tanh(0.400) = 0.3799,$$
$$h_2 = \tanh(0.6 \times (-0.2) + 0.4 \times 0.3799 + 0.1) = \tanh(0.13196) = 0.1312,$$
$$h_3 = \tanh(0.6 \times 0.9 + 0.4 \times 0.1312 + 0.1) = \tanh(0.69248) = 0.5996.$$

Notice that h_2 is close to zero not because x_2 was extreme (it was mildly negative) but because it also carries a damped memory of h_1: the hidden state blends past and present rather than reacting to the latest input alone.

Passing the final hidden state through an output layer, $\hat{y} = \sigma(w_y h_3 + b_y)$, where $\sigma(z) = 1/(1 + e^{-z})$ is Chapter 6's sigmoid, with $w_y = 1.2$ and $b_y = -0.3$, gives $\hat{y} = \sigma(1.2 \times 0.5996 - 0.3) = \sigma(0.4195) = 0.6034$, which might be read as a 60.3% estimated probability that Cascade's upcoming quarter shows accelerating revenue growth. Figure 17.3 unrolls this computation across its three time steps, showing how each hidden state depends on the current input and the previous hidden state.

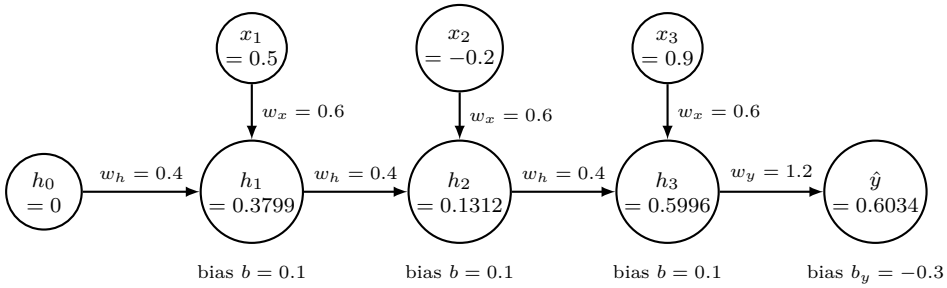


Figure 17.3: The single-hidden-unit RNN unrolled across its three time steps

In practice, a plain RNN's memory decays quickly over long sequences, since each step multiplies an early input's running influence by another factor less than one. The long short-term memory (LSTM) architecture addresses this by maintaining a separate cell state c_t alongside the hidden state, updated as

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, \tag{17.3}$$

where

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f)$$

is a forget gate controlling how much of the previous cell state to retain,

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i)$$

is an input gate controlling how much of a new candidate value

$$\tilde{c}_t = \tanh(W_c[h_{t-1}, x_t] + b_c)$$

to admit, and $\odot$ denotes elementwise multiplication; a third, output gate

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o)$$

then controls how much of c_t is exposed as the hidden state

$$h_t = o_t \odot \tanh(c_t). \tag{17.4}$$

The three gates use σ, whose output lies in $(0, 1)$ and reads as a fraction retained or admitted; the candidate uses tanh, whose output lies in $(-1, 1)$ and carries a signed value to write into the cell state. The worked

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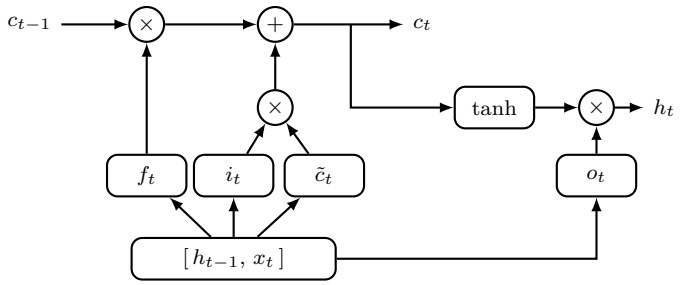


Figure 17.4: The LSTM cell. The cell state crosses the top of the diagram scaled once and added to once, never passed through a squashing function, which is why information survives along it; the three gates below all read the same concatenated $[h_{t-1}, x_t]$.

example below turns on that difference: from identical arguments it computes $i_1 = \sigma(0.25) = 0.5622$ but $\tilde{c}_1 = \tanh(0.25) = 0.2449$. Figure 17.4 puts these six equations in one picture, with the cell state running along the top and the gates feeding it from below.

Because the cell state's update is additive rather than fully multiplicative, a forget gate near 1 lets information pass across many time steps largely undamped, fixing the memory decay the single-unit example illustrates. The cost is four times as many weights as the plain recurrent unit, one set each for the forget, input, and output gates and the candidate.

A worked example makes this concrete. Apply a single-unit LSTM to the plain RNN's three-week input sequence ($x_1 = 0.5$, $x_2 = -0.2$, $x_3 = 0.9$, $h_0 = 0$, $c_0 = 0$), with illustrative gate weights $w_{f,h} = 0.5$, $w_{f,x} = 0.3$, $b_f = 2.0$ (a large bias, chosen so the forget gate stays close to 1 and the memory effect is visible) and $w_{i,h} = w_{i,x} = w_{c,h} = w_{c,x} = w_{o,h} = w_{o,x} = 0.5$ with all other biases at 0. The first step gives

$$f_1 = \sigma(2.15) = 0.8957, \qquad i_1 = \sigma(0.25) = 0.5622, \qquad \tilde{c}_1 = \tanh(0.25) = 0.2449,$$
$$c_1 = f_1 c_0 + i_1 \tilde{c}_1 = 0.1377, \qquad o_1 = \sigma(0.25) = 0.5622, \qquad h_1 = o_1 \tanh(c_1) = 0.0769.$$

Continuing this recursion through the second and third steps gives

$$f_2 = 0.8785, \qquad c_2 = 0.0912, \qquad h_2 = 0.0441,$$
$$f_3 = 0.9082, \qquad c_3 = 0.3537, \qquad h_3 = 0.2092;$$

passing h_3 through the same output layer as the plain RNN example ($w_y = 1.2$, $b_y = -0.3$) gives $\hat{y} = \sigma(1.2 \times 0.2092 - 0.3) = 0.4878$.

The forget gates $f_2 = 0.8785$ and $f_3 = 0.9082$ are the mechanism to isolate. Their product, $0.8785 \times 0.9082 \approx 0.7979$, is the fraction of c_1 that survives two steps later through the forget gate's multiplicative decay, against $w_h^2 = 0.4^2 = 0.16$ for the plain RNN's two-step decay of its hidden state's dependence on h_1.

Roughly 80% of c_1's information survives to c_3 in the LSTM, against roughly 16% in the plain RNN. Information still decays in an LSTM, since the forget gate is strictly less than 1, but a trained forget gate can sit close enough to 1 that memory persists over many more steps than a plain RNN's fixed recurrent weight allows. Figure 17.5 draws the first step as a cell diagram, with all four gates computed from the same inputs before they combine into the new cell and hidden states.

17.4.3 Autoencoders for Unsupervised Anomaly Detection

Suppose a data-quality analyst wants to flag quarters whose alternative-data readings look unusual, without defining "unusual" in advance and without a single labeled example of a bad reading. The CNN and RNN examples cannot address this unsupervised problem, since both are trained toward a known target.

An autoencoder is trained without any label. It learns an encoder that compresses an input to a small bottleneck representation and a decoder that reconstructs the input from it, with reconstruction error as the only training signal. Trained on typical data, it reconstructs anomalous inputs poorly, since its bottleneck

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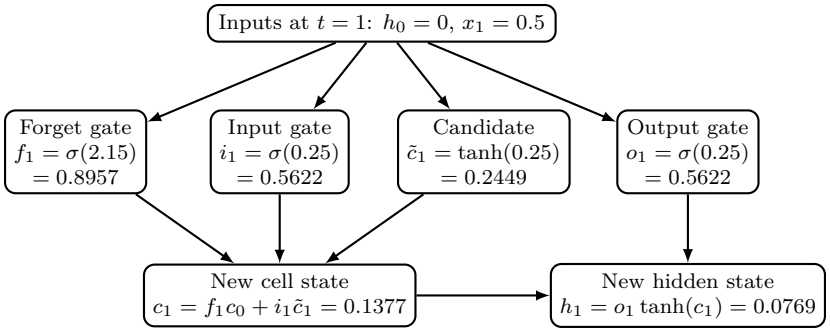


Figure 17.5: The LSTM cell's first time step, computed from the worked example in the text

was never shaped to represent what made them unusual. Reconstruction error is thus a natural anomaly score, extending Chapter 15's classification-based anomaly detection to a setting with no labeled anomalies.

Consider a simple linear autoencoder, encoder weights $w = (0.5, 0.3, 0.2)$ applied to Table 17.3's three alternative-data growth rates $x = (\text{car}, \text{web}, \text{card})$ for each quarter, producing a single bottleneck code $z = w \cdot x$, and a tied-weight decoder that reconstructs $\hat{x} = z\,w$. For Q4, $x = (0.6, 1.5, 0.8)$, so $z = 0.5(0.6) + 0.3(1.5) + 0.2(0.8) = 0.91$, and $\hat{x} = 0.91(0.5, 0.3, 0.2) = (0.455, 0.273, 0.182)$, giving a reconstruction error $\sum(x_i - \hat{x}_i)^2 = (0.6 - 0.455)^2 + (1.5 - 0.273)^2 + (0.8 - 0.182)^2 = 1.908$.

Carrying this out for all eight quarters and dividing each quarter's error by its total energy $\sum x_i^2$ (so that quarters with larger growth rates are not flagged just for being larger) gives relative errors from 39.4% (Q7) to 58.7% (Q4). Q4 is the quarter the autoencoder reconstructs worst, not because its numbers are largest (Q5's raw error of 30.9 is far larger) but because Q4's web-traffic growth (1.5%) is large relative to its car-count and card growth (0.6% and 0.8%), a pattern this bottleneck direction does not represent well.

One caution: the weights $w = (0.5, 0.3, 0.2)$ are illustrative, chosen for easy arithmetic, not trained. A trained tied-weight linear autoencoder learns a unit-length weight vector (the data's first principal direction), and only a unit-length w makes $\hat{x} = z\,w$ a true projection. This w has length 0.6164, so every reconstruction above is shrunk toward zero and every error inflated. Rescaling w to unit length, $(0.8111, 0.4867, 0.3244)$, cuts the relative errors to between 1.6% (Q7) and 32.9% (Q4); the levels fall sharply, but Q4 is still the worst-reconstructed quarter, which is the point of the example.

This is the quarter a data-quality analyst should investigate first (is the web-traffic reading an error, a one-time promotional spike, or an early signal the other two sources have not yet picked up?) before trusting it in a downstream model.

Every network in this section runs on hand-picked weights and a single forward pass, so that each number can be checked by hand. The companion notebook's optional "at scale" section trains the real thing in PyTorch: an LSTM forecaster validated walk-forward against simpler baselines and an autoencoder that rediscovers the unit-length direction above from 3,000 simulated rows; it needs the extra packages listed in the repository's `requirements-dl.txt`.

17.4.4 Transformers as a Unifying Architecture

Chapter 16 developed the transformer and its self-attention mechanism (Section 16.4.1) for text, where each token attends to every other token. The mechanism generalizes: a vision transformer (ViT) divides an image into fixed-size patches, treats each patch as a token, and applies self-attention across patches instead of convolution across pixels. A satellite image classifier can then weigh a distant part of the image against the current patch directly, which a CNN's local filters do only through many stacked layers.

The companion notebook splits the CNN example's 4×4 image into four non-overlapping 2×2 patches, treats each as a token, and computes scaled dot-product self-attention across them. Querying from patch 2 (the top-right car cluster, mean brightness 0.85) against keys built from each patch's brightness, patch 2 attends to itself with weight 0.6174 and to patch 3 (the bottom-left car cluster, mean brightness 0.80, the

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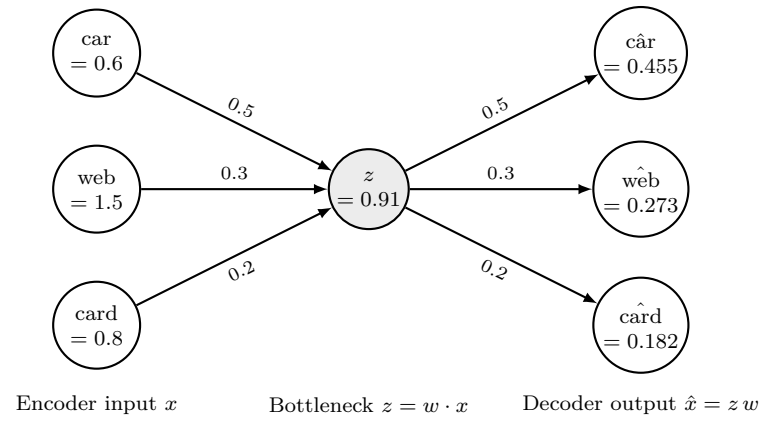


Figure 17.6: The linear autoencoder applied to Q4's three alternative-data growth rates

farthest patch in the image) with weight 0.3817, while the two empty patches 1 and 4 (mean brightness 0.10 each) receive a combined weight of only 0.001.

Patch 2's most relevant neighbor is thus patch 3, a content-similar patch in the opposite corner, not either empty patch nearer to it, and self-attention finds that relevance in one step rather than through several stacked convolutional layers. In current practice CNNs remain common for smaller or highly structured image tasks (including much car-counting and object-detection work), while transformer-based architectures increasingly dominate large-scale image classification and the web-traffic and card-transaction sequence modeling that once used RNNs. A single architectural family, self-attention, now spans text (Chapter 16), images, and sequences.

Training discipline for deep learning in finance. Every network in this section has a handful of weights; a production LSTM or CNN has tens of thousands to millions, fitted to short, noisy, nonstationary financial data. Four habits keep such a model honest. *Dropout* sets a random fraction of hidden units (commonly 10–50%) to zero at each training step, so that no unit can lean on a particular partner and the trained network behaves like an average of many thinner ones; it is switched off at prediction time. *Weight decay* adds Chapter 6's ridge penalty to the loss, shrinking every weight toward zero. *Early stopping* records the loss on a held-out validation block after each pass through the training data and keeps the weights from the epoch where it was lowest, since the training loss keeps falling long after the network has started to fit noise. *Walk-forward splits* make every validation and test block strictly later than the training data, as Chapter 7 requires of any forecast. Random folds let the network learn from the future of the periods it is scored on, and labels that overlap in time call for a gap between training and test blocks as well (López de Prado 2018).

The companion notebook's optional LSTM cell applies the last two habits to 1,500 simulated observations whose dynamics switch between momentum after a rise and reversal after a fall, a nonlinearity no linear autoregression can represent. Averaged over three walk-forward test blocks, the committed run's root-mean-squared errors are 1.128 for a naive last-value forecast, 1.074 for a linear AR(2), 1.066 for gradient-boosted trees, and 1.050 for the LSTM, against a noise floor of 1.037 that even the true model cannot beat. The LSTM wins by 2.3% over the AR(2) on a problem built so that a linear model must fail. A rerun on another machine gives similar rather than identical digits, since a trained network's weights depend on floating-point details that seeding does not fix.

Even that narrow win is the exception for the data shape most finance problems arrive in, a table of features per firm, borrower, or transaction. Comparing deep networks with tree ensembles on 45 tabular datasets of around 10,000 rows each, after extensive tuning of both, Grinsztajn et al. (2022) found that random forests and gradient-boosted trees remained stronger. They traced the gap to three properties of tabular data: many uninformative features, which trees ignore and networks must learn to ignore; columns whose individual meaning matters, which a network's indifference to rotations of the feature space throws away; and targets with sharp, irregular jumps, which a tree fits with one split and a network smooths over.

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Chapter 21's real-data credit bake-off, where the two tree ensembles lead, is the typical finance case. The exception is return prediction at very large scale: in the Gu, Kelly, and Xiu comparison Chapter 12 discusses, on tens of thousands of stocks over six decades, shallow neural networks edged out the tree ensembles (Gu et al. 2020). The working rule: start with gradient boosting on tabular data, and reach for a deep network when the input is an image, a sequence, or text, or when the dataset is large enough to pay for one.

17.5 Applications and Advanced Methods

This part builds a signal from disclosure text, then surveys the deep-learning methods the data invites.

17.5.1 Building an ESG Signal Directly from Disclosure Text

Chapter 12 showed that six major ESG rating providers agree only weakly (average pairwise correlations of 38% to 71%), mostly because they measure different facts about a company and weight them differently, not because they measure an agreed concept noisily. Section 17.2.2's build-versus-buy logic applies here as it does to satellite imagery: rather than buying a vendor's ESG score and inheriting its undisclosed methodology, a fund can build a narrow, transparent environmental-tone signal from a company's public disclosures with the text-analysis tools of Chapter 16.

Suppose Meridian Fabrication, the manufacturer whose ratios Chapter 4 analyzed, files a 10-K whose risk-factors section runs to several pages. Chapter 16's Naive Bayes classifier (Section 16.3.2) can be extended with a third class, Environmental Risk, trained on a few sentences using vocabulary (emissions, environmental, climate, remediation) that the original Credit Risk and Market Risk training sentences never mention. The retrained classifier flags four sentences in Meridian's filing as Environmental Risk with high confidence, each keying on at least one environment-specific term:

> S1: "Management remains confident that planned facility upgrades will reduce our emissions, but continued regulatory headwinds could still increase compliance costs."
> S2: "We remain cautious about the pace of emissions reduction given persistent supply chain headwinds affecting our environmental capital projects."
> S3: "Emissions declined for the third consecutive year, reflecting strong progress on our environmental targets despite a challenging regulatory backdrop."
> S4: "Physical risks from climate change, including potential flooding at our coastal facility, remain a weak point in our disaster preparedness planning and a continuing concern for the board."

Applying Chapter 16's Loughran-McDonald dictionary tone score (Section 16.2.3) to the four sentences, with the same positive word set {strong, growth, exceeded, confident, improved} and negative word set {cautious, headwinds, decline, declined, weak, concern, concerns} Chapter 16 used to score its two earnings-call transcripts, gives

$$\begin{aligned}
&\text{S1: } (1-1)/20 = 0.0,\\
&\text{S2: } (0-2)/19 \approx -0.1053,\\
&\text{S3: } (1-1)/19 = 0.0,\\
&\text{S4: } (0-2)/28 \approx -0.0714,
\end{aligned}$$

an average environmental tone score of -0.0442 across the four sentences, a mildly cautious reading of Meridian's own environmental disclosure.

This exercise surfaces a new data-quality problem specific to building ESG signals from generic financial-sentiment tools. The Loughran-McDonald lexicon was built to score revenue and earnings language, where "declined" is reliably bad news, but in sentence S3 the same word describes good environmental news, emissions falling for a third consecutive year, and the lexicon cannot tell the difference. S3's score of 0.0 understates how positive the sentence is. Fixing this needs a lexicon built for environmental disclosure, one that distinguishes

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a decline in emissions from a decline in revenue, and no standardized public dictionary of that kind yet exists the way Loughran-McDonald's does for general financial tone.

Comparing this result with Chapter 12's worked example makes the deeper point. That example scored three fictional assets on a common, if arbitrary, 0-to-100 composite scale and varied only the *weight* providers place on the same E, S, and G sub-scores, one of three sources of divergence Berg, Kölbel, and Rigobon identify. This section's -0.0442 tone score is not on that scale and not comparable to it: it measures something narrower (the tone of one filing's risk-factors section) by a different method (word counting rather than a proprietary composite), the *scope* and *measurement* divergence the same paper finds behind most of the disagreement. An in-house signal like this one does not resolve the disagreement; it adds a fourth, differently scoped measurement to a field that already has too many inconsistent ones.

What it buys instead is auditability: every step from raw sentence to final number can be inspected, recomputed, and defended to a model validator, as Chapter 8's model-risk-management framework requires.

17.5.2 Advanced Topics: Transfer Learning, Graphs, and Fusion

Four further techniques extend Section 17.4's toolkit in directions that matter increasingly in practice: reusing an already-trained model instead of training from scratch, exploiting network structure between entities, combining several data modalities into one representation, and pretraining without labels.

Transfer Learning: Fine-Tuning a Pretrained Vision Backbone

Section 17.3.3 noted that a new alternative-data feed typically has a short history, too little to train a large CNN's millions of weights from scratch. Transfer learning sidesteps this: a CNN already trained on a large general image dataset, which has learned generic filters for edges, textures, and shapes, is reused as a frozen feature extractor, and only a small final layer mapping its features to the task (car count, cloud cover, construction status) is trained on the limited task data. Because the frozen backbone receives no updates, this training is Chapter 6's gradient descent applied to far fewer parameters than a full network has.

Suppose a pretrained backbone, frozen, produces a two-dimensional feature vector $f = (0.7, 0.3)$ for a new satellite image patch, and the task is to fine-tune a final logistic layer, currently $w = (0.4, 0.4)$, $b = -0.1$, to predict whether the patch contains a car cluster (label $y = 1$ for this patch). The current prediction is

$$\hat{y} = \sigma(w \cdot f + b) = \sigma(0.4 \times 0.7 + 0.4 \times 0.3 - 0.1) = \sigma(0.30) = 0.5744.$$

Following Chapter 6's gradient descent update with learning rate $\eta = 0.5$, the gradient of the log-loss with respect to w is

$$(\hat{y} - y)f = (0.5744 - 1)(0.7, 0.3) = (-0.298, -0.128),$$

and with respect to b is $\hat{y} - y = -0.426$, giving updated weights

$$w \leftarrow (0.4, 0.4) - 0.5(-0.298, -0.128) = (0.549, 0.464)$$

and

$$b \leftarrow -0.1 - 0.5(-0.426) = 0.113.$$

Recomputing the prediction with these updated weights gives

$$\hat{y} = \sigma(0.549 \times 0.7 + 0.464 \times 0.3 + 0.113) = \sigma(0.6365) = 0.6540,$$

closer to the true label of 1 after a single gradient step on a handful of parameters. This is why a fund with only a few hundred labeled satellite patches can still fine-tune a usable classifier, rather than needing the millions of labeled images a CNN trained from scratch would require.

Graph Neural Networks for Ownership and Counterparty Networks

Chapter 15 modeled a shell-company layering network as a static sequence of hops. A graph neural network (GNN) treats the network as a computational structure: each node's representation is updated by aggregating its neighbors', so risk propagates through the network instead of being assessed for each entity in isolation.

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This is the structure alternative data increasingly captures (common addresses, shared directors, correlated transaction timing) and a purely tabular model discards. A single GNN layer updates node v's representation as

$$h_v = \sigma(w_1 x_v + w_2 \cdot \text{mean}_{u \in N(v)}(x_u) + b), \tag{17.5}$$

where $N(v)$ is the set of v's neighbors in the graph.

Consider a four-node layering chain, $A \to B \to C \to D$, extending Chapter 15's shell-company example, with initial suspicion scores $x_A = 0.9$ (a flagged placement account), $x_B = 0.2$, $x_C = 0.1$, and $x_D = 0.05$ (an apparently unremarkable integration account), and neighbor sets

$$N(A) = \{B\},\ N(B) = \{A, C\},\ N(C) = \{B, D\},\ N(D) = \{C\}.$$

With weights $w_1 = 0.6$, $w_2 = 0.8$, $b = -0.05$, one round of mean-aggregation gives

$$h_B = \sigma\big(0.6 \times 0.2 + 0.8 \times \tfrac{0.9+0.1}{2} - 0.05\big) = \sigma(0.47) = 0.6154$$

and

$$h_C = \sigma\big(0.6 \times 0.1 + 0.8 \times \tfrac{0.2+0.05}{2} - 0.05\big) = \sigma(0.11) = 0.5275,$$

compared to $h_A = \sigma(0.65) = 0.6570$ and $h_D = \sigma(0.06) = 0.5150$. Node B's updated score, 0.615, is far above its raw feature of 0.2 because it is one hop from the flagged account A, while node C, two hops away, reaches only 0.528, barely above D's 0.515: after one round, A's signal has not yet arrived there, and a second round is needed to carry it that far. A model that scored each account independently, as Chapter 15's original logistic regression did, would miss this network-propagated risk.

Multi-Modal Fusion via Cross-Attention

A single financial decision often draws on several modalities at once: an image feature from a CNN, a sentiment embedding from Chapter 16's text pipeline, and a tabular summary like Section 17.2.4's regression inputs. Combining them well matters as much as extracting each. Cross-attention, a direct extension of Chapter 16's self-attention (Section 16.4.1), treats one modality's representation as the query and the others as key-value pairs to attend over:

$$\text{Attention}(Q, K, V) = \text{softmax}(QK^\top / \sqrt{d_k})V,$$

as before, but fusing modalities instead of word tokens.

Suppose a tabular query vector $Q = (0.6, 0.8)$ (summarizing Cascade's alternative-data growth direction) attends over an image modality with key $K_{\text{img}} = (0.9, 0.1)$ and value $V_{\text{img}} = (0.7, 0.2)$, and a text-sentiment modality with key $K_{\text{txt}} = (0.2, 0.9)$ and value $V_{\text{txt}} = (-0.3, 0.6)$ (a mildly cautious sentiment reading). With $d_k = 2$, the scaled dot-product scores are

$$Q \cdot K_{\text{img}} / \sqrt{2} = 0.62 / 1.4142 = 0.4384$$

and

$$Q \cdot K_{\text{txt}} / \sqrt{2} = 0.84 / 1.4142 = 0.5940;$$

applying softmax gives attention weights 0.4612 on the image modality and 0.5388 on the text modality. The fused representation is

$$0.4612(0.7, 0.2) + 0.5388(-0.3, 0.6) = (0.161, 0.416),$$

a single vector that leans slightly toward the cautious text signal, and that a downstream model (a trading rule like Section 17.2.4's, a credit decision, a fraud flag) can consume in place of either modality, without the analyst hand-specifying how much weight each deserves.

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Self-Supervised Contrastive Pretraining

Section 17.5.2's transfer-learning example assumed a pretrained backbone was available. Contrastive pretraining is one common way to train such a backbone without any human labels, which matters for satellite imagery, where labeled examples (this patch contains three cars) are far scarcer than raw imagery. The idea is to take two randomly augmented views of the same image, a small crop and rotation, say, and train the encoder to pull their embeddings together (a positive pair) and push a different image's embedding away (a negative pair), using only same-versus-different as the training signal.

Suppose an encoder produces embedding $a = (0.6, 0.8)$ for one crop of a satellite patch, $p = (0.55, 0.83)$ for a second, differently augmented crop of the *same* patch, and $n = (0.1, 0.9)$ for a crop of a different patch. Using Chapter 16's cosine similarity (Section 16.3.3), $\cos(a, p) = 0.9983$ and $\cos(a, n) = 0.8614$: two views of the same scene are closer than a different scene, as intended.

A contrastive loss converts these similarities into a probability that the positive pair is correctly identified among the candidates,

$$\Pr(\text{positive}) = \frac{e^{\cos(a,p)/\tau}}{e^{\cos(a,p)/\tau} + e^{\cos(a,n)/\tau}}, \tag{17.6}$$

a softmax over similarities analogous to the cross-attention softmax above, with a temperature τ controlling how sharply the model is pushed to separate positive from negative. With $\tau = 0.1$, $\Pr(\text{positive}) = e^{9.983}/(e^{9.983} + e^{8.614}) = 0.797$, well above chance (50%) before any training. Training drives this probability toward 1 by adjusting the encoder so that same-image crops end up closer together and different-image crops farther apart.

Trained this way on a large unlabeled archive of satellite imagery, the encoder becomes the frozen backbone that the transfer-learning example fine-tuned with a handful of labeled examples, closing the loop between the chapter's two ways of coping with scarce labels.

17.6 Case Vignette: Satellite Imagery and the Rise of Alternative Data

The parking-lot example this chapter has used throughout is not hypothetical in its origins. In 2009, brothers Tom and Alex Diamond conceived the idea of counting cars in retailer parking lots from satellite imagery to anticipate revenue, and founded RS Metrics to sell the data, starting with McDonald's locations and later covering Lowe's, Home Depot, and dozens of other retailers. The approach reached wider attention in 2010, when UBS analyst Neil Currie bought parking-lot counts for 100 Walmart stores and published them ahead of Walmart's earnings release, an early public demonstration that satellite data could anticipate a reported number. A later academic study using RS Metrics imagery from 2011 to 2017, covering 44 major U.S. retailers including Walmart, Target, Costco, and Whole Foods, confirmed that year-over-year change in car counts predicts quarterly sales, and put the resulting informational edge at 4% to 5% in the three trading days around an earnings announcement, a substantial return for so short a window and a large-scale counterpart to Section 17.2.3's worked example.

A separate study of how that coverage spread across retailers found effects beyond rewarding the subscribing funds: retailers that gained satellite coverage subsequently saw more informed short selling, less-informed individual buying, and lower stock liquidity around their earnings reports. Unequal access to alternative data can widen the information gap between sophisticated and ordinary investors without making prices more accurate any faster, the access-asymmetry concern Section 17.3.3 raises about licensing cost and exclusivity.

Satellite data extends well beyond retail. Orbital Insight built a comparable business in commodities: by measuring the shadows cast by the floating roofs of many crude-oil storage tanks (a floating roof sits on the stored oil, so its height reveals how full the tank is), the firm estimates inventory across more than 27,000 tanks worldwide. During the demand shock of the early 2020s pandemic, this work showed global storage running at roughly 55% of an estimated six-billion-barrel capacity, a data point no government agency could have produced as quickly. Satellite-based financial data services are estimated to generate over $500 million in annual revenue, within an alternative-data market on which Wall Street is estimated to spend roughly $3 billion a year, and a single institutional satellite subscription can run from $50,000 to $500,000 per year, so this edge is available mainly to well-capitalized participants able to pay for exclusive or near-exclusive access.

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17.7 Challenges in Alternative Data and Deep Learning in Finance

Several challenges recur across this chapter's techniques and connect directly to themes from earlier chapters.

A short history limits statistical confidence. Sections 17.2.3 and 17.2.4 fit regressions on only eight quarters, because that is all the history a new alternative-data feed typically has. Chapter 6's warnings about overfitting with few observations apply with particular force, since a vendor's dataset is often too new to have lived through a full economic cycle.

Non-stationarity compounds with vendor changes. Financial relationships shift as markets evolve (Chapter 6), and alternative data adds a second source of instability: a vendor can silently change its imagery source, computer-vision model, or store coverage, so a model's apparent degradation may reflect a change at the vendor rather than in the financial relationship.

Deep architectures trade interpretability for representational power. A CNN's learned filters and an RNN's hidden state are far harder to interpret than Section 17.2.3's single regression coefficient, a tension Chapter 18 develops, and one that matters more when the input is as opaque to manual review as a stack of satellite pixels.

Integration is usually the hard part, not modeling. Aligning several sources in time and by firm identity, and confirming that each was available before the date being modeled, typically takes more effort than fitting the model, the reverse of the market-data-only settings of Chapters 12 and 13.

Cost and exclusivity concentrate the advantage. Section 17.6's subscription costs mean that alternative data's edge accrues mostly to well-resourced funds able to license several exclusive feeds, an access asymmetry with no counterpart in the public market and fundamental data most of this book uses.

Model risk governance now has to reach the vendor, not just the model. Chapter 15's three-lines-of-defense framework validates a single model's inputs, outputs, and ongoing performance. An alternative-data pipeline adds a layer it must now cover, the vendor relationship, since a model's validation is only as good as the vendor's coverage, delivery timing, and methodology upstream of it, none of which the fund controls or, usually, can fully observe.

17.8 Key Takeaways

Alternative data (satellite imagery, web traffic, card-transaction panels) fills the row Chapter 1's data table reserved for it: non-traditional sources that proxy for economic activity before it is officially reported. A single proxy, satellite car counts, explained 87% of a retailer's quarterly revenue variation in Section 17.2.3's example, and combining several sources produced a useful but imperfect earnings-surprise signal (an 80% hit rate, not 100%) in Chapter 13's evaluation framework. Raw readings typically need seasonal adjustment and peer-relative normalization first (Section 17.3.2).

Extracting signals from raw data, rather than from a vendor's processed number, needs architectures beyond the feedforward network: convolutional networks (Section 17.4) exploit an image's grid through weight-shared filters, recurrent networks exploit a sequence's order through a hidden state, and transformer-based self-attention increasingly unifies both roles. None of this removes alternative data's quality problems (Section 17.3.3): coverage bias, vendor black-box processing, look-ahead bias from delivery lag, and cost-driven alpha decay, which a modeler must treat as seriously as the modeling itself.

Respecting delivery lag changes which trades a model recommends, not merely how good its backtest looks (Section 17.3.1), and an autoencoder can turn the same toolkit on the data itself to flag a quarter worth double-checking. Rather than buying a vendor's ESG score, whose disagreement with other providers Chapter 12 documented at 38% to 71% pairwise correlation, a fund can build a narrow, auditable environmental-tone signal from a company's own disclosures (Section 17.5.1), at the cost of adding one more differently scoped measurement to the field.

Section 17.5.2 went deeper: transfer learning fine-tunes a CNN's frozen features on a small sample, contrastive pretraining trains that backbone with no labels at all, graph neural networks let risk propagate across related entities that Chapter 15's layering chain could only describe, and cross-attention fuses image, text, and tabular modalities into one representation without hand-specified weights.

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17.9 Suggested Exercises

17.1 Using Section 17.2.3's method, refit the regression after adding a ninth quarter with car-count growth of -2.0% and revenue growth of -1.5%, and report the new slope, intercept, and R^2.

17.2 Explain, using Table 17.2, why an R^2 of 0.870 from a single alternative-data proxy is considered strong in this context, contrasting it with Chapter 6's discussion of low signal-to-noise ratios in financial prediction generally.

17.3 Using Section 17.2.4's fitted coefficients, compute the predicted surprise for a tenth quarter with car, web, and card growth of -2.0%, -1.0%, and -1.5% respectively, and state whether the resulting signal is long or flat.

17.4 Explain, using Q5 in Table 17.3, why a strongly positive fitted surprise (2.46%) can still be followed by a negative next-day return (-0.4%), and describe one piece of earnings-day information the alternative-data model in this chapter cannot capture.

17.5 Using Section 17.4's convolution example, recompute the 3×3 feature map using a 2×2 filter of all 0.5s instead of all 0.25s, and explain how the result relates to the original feature map.

17.6 Using the same convolution example, apply 2×2 max pooling (taking the maximum of each non-overlapping 2×2 block) to the original 3×3 feature map's top-left 2×2 block, and state the resulting pooled value.

17.7 Using Section 17.4's RNN example, recompute h_1, h_2, and h_3 if w_h is increased from 0.4 to 0.8 (holding w_x, b, and the inputs fixed), and explain in words how a larger w_h changes how much the hidden state is influenced by earlier time steps versus the current input.

17.8 Explain why a convolutional neural network's weight sharing makes it far more practical than a plain feedforward network for a 1000×1000-pixel satellite image, referencing the parameter-count argument in Section 17.4.

17.9 Using this chapter's case vignette, explain why the 2010 UBS Walmart parking-lot report is a natural precedent for Section 17.2.3's worked example, and describe why the same informational edge tends to shrink as more funds license the same data source.

17.10 Pick two of the data-quality issues in Section 17.3.3 and describe a concrete scenario, not already used in the text, in which that issue could cause an alternative-data-based trading or credit model to fail despite technically correct code.

17.11 Compare this chapter's satellite car-count regression (Section 17.2.3) to Chapter 5's binomial and Black-Scholes-Merton option-pricing models in terms of what each treats as the source of uncertainty, and explain why alternative-data models are typically evaluated by out-of-sample predictive accuracy rather than by no-arbitrage argument.

17.12 Using Section 17.3.1's method, explain why the realistic, lagged-card model's higher R^2 (0.873 versus 0.837) does not mean it is simply the better trading model, referencing the difference in which quarters each version actually signals.

17.13 Using Section 17.3.2's method, recompute the peer-adjusted regression's slope and R^2 if the sector-average car-count growth for Q5 is revised from 2.5% to 5.0% (a sector-wide surge that quarter), holding all other quarters fixed, and explain why the peer-adjusted model's R^2 falls even though Cascade's own reported revenue growth has not changed.

17.14 A vendor offers a trial dataset covering only the three quarters (of Table 17.3) in which its data predicted the following quarter's earnings surprise correctly. Using Section 17.2.2's discussion, explain what is wrong with evaluating the vendor's data quality using only this trial.

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17.15 Using Section 17.2.2's cost-benefit method, recompute the expected net annual benefit if the fund can only deploy a $20 million position per signaled event (instead of $50 million), holding the signal frequency, edge, and data costs fixed, and state whether the subscription is still worth its $400,000 annual cost.

17.16 Using Section 17.4's autoencoder example, compute the reconstruction error and relative error for Q2 (car $= -1.5$, web $= -0.5$, card $= -1.0$) using the same encoder/decoder weights $w = (0.5, 0.3, 0.2)$, and state whether Q2 would be flagged ahead of Q7 (relative error 39.4%) for data-quality review.

17.17 Using Section 17.5.2's transfer-learning example, perform a second gradient-descent step starting from the updated weights $w = (0.549, 0.464)$, $b = 0.113$, and report the new prediction, confirming that it continues to move toward the true label of 1.

17.18 Using Section 17.5.2's graph neural network example, compute h_A and h_D's updated scores using the same weights, and explain, using the network's chain structure $A \to B \to C \to D$, why h_D's score barely moves from its raw feature while h_B's moves substantially.

17.19 Using Section 17.5.2's cross-attention example, recompute the fused output if the text modality's key becomes $K_{\text{txt}} = (0.8, 0.2)$ (a much less distinctive key vector, more similar to K_{img}), and explain in words why the two attention weights move closer to 50/50.

17.20 Explain why a graph neural network's mean-aggregation step (Section 17.5.2) is a more direct computational analogue of "guilt by association" than Chapter 15's original, per-account fraud-scoring logistic regression.

17.21 Using Section 17.5.2's contrastive-pretraining example, recompute Pr(positive) if the temperature τ is raised from 0.1 to 1.0 (holding the two cosine similarities fixed), and explain in words why a higher temperature makes the model's implied preference for the true positive pair less extreme.

17.22 Using Section 17.4's LSTM worked example, recompute the forget gate f_1 and the resulting fraction of c_1 surviving to c_3 (that is, $f_2 \times f_3$) if the forget-gate bias b_f were 0.5 instead of 2.0, holding every other weight fixed, and explain why a smaller bias makes the LSTM behave more like the plain RNN example earlier in this section.

17.23 Explain why the plain RNN's memory-decay problem is a consequence of its hidden state being fully overwritten by a multiplicative recursion at every step, while the LSTM's cell state update is additive, and describe what would happen to the LSTM's own memory retention if its forget gate were trained to a value close to 0 instead of close to 1.

17.24 Using Section 17.5.1's method, recompute the tone score for a fifth sentence, "The board remains confident that emissions will decline further even as regulatory headwinds intensify," using the same positive and negative word lists, and explain why this sentence's mixed vocabulary makes its tone score a poor summary of whether the sentence is good or bad news.

17.25 Explain why Section 17.5.1's in-house tone score and Chapter 12's vendor-style ESG scores are not directly comparable even though both are commonly described as "an ESG measurement," and describe which one of Berg, Kölbel, and Rigobon's three sources of rating divergence, scope, measurement, or weight, this incomparability illustrates.

17.26 The training-discipline passage at the end of Section 17.4.4 reports mean walk-forward test RMSEs of 1.128 (naive), 1.074 (AR(2)), 1.066 (gradient boosting), and 1.050 (LSTM) against a noise floor of 1.037. (a) Compute each model's excess RMSE over the noise floor, and the percentage of the naive forecast's excess that each of the other three models removes. (b) Explain why the LSTM's 2.3% improvement over the AR(2) and its share of the removable error tell different stories, and which of the two a model-risk reviewer should ask to see. (c) Explain why reshuffling the 1,500 observations into five random folds, instead of walk-forward blocks, would be expected to flatter the two flexible models, the LSTM and gradient boosting, more than the naive forecast and the AR(2), given that each input is a window of the ten preceding values.

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17.27 A vendor's foot-traffic series for a chain of garden centers has predicted the chain's quarterly revenue growth well over five years. Using the "Is this signal causal?" paragraph of Section 17.3.3, describe one mechanism through which the series measures revenue directly and one confounder that could produce the same association without it, and propose a check on the data that would help tell the two apart. Then explain why the chain's own management, deciding whether a promotion that raises foot traffic will raise revenue, needs a different kind of evidence from a fund trading the stock.

17.28 **Challenge (real data).** Download at least two years of weekly Google Trends search-interest data (via `pytrends` or manually from trends.google.com, no API key required) for a public retailer's brand name, and quarterly reported revenue for the same company from the SEC EDGAR XBRL `companyfacts` API (as in Chapter 4's challenge exercise). (a) Following Section 17.2.3's method, regress the company's quarter-over-quarter revenue growth on the average quarter-over-quarter growth in search interest, and report the fitted slope, intercept, and R^2. (b) Using Section 17.3.1's distinction between a realistic, lagged signal and a look-ahead-biased one, identify what reporting lag your search-interest data is actually available at relative to the revenue figure it is meant to predict, and explain whether your regression in (a) would still be usable in real time. (c) Following Section 17.2.2's cost-benefit framework, discuss one alternative-data quality issue from Section 17.3.3 other than look-ahead bias that is specific to Google Trends data, such as index normalization or query ambiguity, and explain how it could produce a spuriously strong or weak regression fit.

17.10 Further Reading

- Counts (2019), *How Hedge Funds Use Satellite Images to Beat Wall Street, and Main Street*. A research summary of the RS Metrics parking-lot car-count studies underlying Section 17.2.3 and this chapter's case vignette.
- Katona et al. (2025), "On the Capital Market Consequences of Big Data: Evidence from Outer Space." The academic study of RS Metrics satellite imagery (2011-2017, 44 U.S. retailers) quantifying the earnings-window informational edge discussed in Section 17.6; it also documents the short-selling, liquidity, and information-asymmetry effects of satellite coverage discussed in this chapter's expanded case vignette.
- Feng and Fay (2022), "An Empirical Investigation of Forward-Looking Retailer Performance Using Parking Lot Traffic Data Derived from Satellite Imagery." A complementary study of satellite-derived parking-lot data's power to predict retailer sales, underlying Section 17.2.3's worked example.
- LeCun et al. (2015), "Deep Learning." A survey covering convolutional and recurrent neural network architectures underlying Section 17.4.
- Goodfellow et al. (2016), *Deep Learning*. A comprehensive textbook treatment of convolution, recurrence, and the attention mechanisms Chapter 16 and Section 17.4 both draw on.
- Grinsztajn et al. (2022), "Why Do Tree-Based Models Still Outperform Deep Learning on Typical Tabular Data?" The benchmark of tree ensembles against deep networks on 45 tabular datasets behind the training-discipline passage in Section 17.4.4, with an analysis of why trees keep winning on that kind of data.
- Hochreiter and Schmidhuber (1997), "Long Short-Term Memory." The original paper introducing the LSTM architecture mentioned as an extension of Section 17.4's plain recurrent network.
- Dosovitskiy et al. (2021), "An Image Is Worth 16x16 Words: Transformers for Image Recognition at Scale." The original vision transformer paper underlying Section 17.4's discussion of transformers as a unifying architecture.
- Kolanovic and Krishnamachari (2017), *Big Data and AI Strategies: Machine Learning and Alternative Data Approach to Investing*. An industry survey of alternative data sources and their use in systematic investing, covering the categories in Table 17.1.

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- Berg et al. (2022), "Aggregate Confusion: The Divergence of ESG Ratings." The empirical study of cross-provider ESG rating disagreement discussed in Chapter 12 and revisited in Section 17.5.1 from the standpoint of building an in-house text-derived signal rather than buying a vendor's score.
- Hinton and Salakhutdinov (2006), "Reducing the Dimensionality of Data with Neural Networks." A foundational paper on autoencoders, underlying Section 17.4's reconstruction-error anomaly-detection example.
- Pan and Yang (2010), "A Survey on Transfer Learning." A comprehensive survey of the pretrained-backbone fine-tuning approach worked through in Section 17.5.2.
- Kipf and Welling (2017), "Semi-Supervised Classification with Graph Convolutional Networks." A foundational paper on the graph neural network mean-aggregation update used in Section 17.5.2's shell-company network example.
- Chen et al. (2020), "A Simple Framework for Contrastive Learning of Visual Representations." The SimCLR paper underlying Section 17.5.2's contrastive-pretraining worked example.
- He et al. (2016), "Deep Residual Learning for Image Recognition." The original ResNet paper underlying Section 17.4's discussion of shortcut connections and the vanishing-gradient problem in deep CNNs.

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Part III

Advanced Topics

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Chapter 18

Explainability, Fairness, and Regulation

18.1 Introduction

Every applied chapter so far has flagged the same unresolved obligation and deferred it here. Chapter 1 warned that a model too opaque to explain to a risk committee is often not deployable, whatever its statistical performance. Chapter 5 showed that collapsing two defensible valuations into one confident-looking number can hide an economically meaningful disagreement. Chapter 6 noted that regulation limits how opaque a credit or insurance model can be, and that interpretability is itself a criterion for choosing among algorithms.

Chapter 15 observed that a suspicious activity report or an account closure often must be explained to a regulator or a customer, which limits how far a fraud or anti-money-laundering program can lean on its least interpretable models; Chapter 16 made the same point about decisions driven by an attention-based language model, and Chapter 17 about the filters and hidden states of convolutional and recurrent networks. This chapter pays off these deferred obligations, not as an afterthought bolted onto a finished model, but as a technical and regulatory discipline with its own methods, worked examples, and open problems.

The chapter is organized around three questions a financial institution must answer before, during, and after deploying a model: *why* a model produced a given output (interpretable and explainable AI); whether it treats people and groups *fairly* under one or more competing legal and ethical definitions; and whether it keeps working as intended once deployed, monitored, and eventually attacked or gamed (robustness, governance, and responsible AI).

Twelve worked examples run throughout, each reusing a model or dataset from an earlier chapter, because explainability and fairness are properties of a model's behavior on real cases. Permutation feature importance and partial dependence work on Chapter 15's fraud-scoring logistic regression; an exact Shapley-value decomposition and a counterfactual explanation address Chapter 9's declined loan applicant; a locally interpretable surrogate is built around Chapter 6's toy feedforward network. A two-group fairness audit, a calibration-by-group check, and adverse-action reason codes rest on the same credit-scoring model; an adversarial "structuring" perturbation extends Chapter 15's money-laundering vocabulary to model evasion; and a population stability index measures score drift, complementing Chapter 15's precision-drift monitoring. Going deeper, a cost-weighted algorithmic-recourse comparison and an integrated-gradients attribution revisit Chapter 6's network to show where a naive local-gradient shortcut fails and a proper path integral does not.

A case vignette on the 2019 Apple Card gender-bias controversy and its regulatory aftermath ties these methods to a real explainability and fairness dispute, and a second, on Upstart Network's engagement with the CFPB, shows the same concerns handled proactively. A final governance subsection turns from models that score to AI agents that act, and sets out the controls such systems need where frameworks written for scoring models only partly reach them.

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18.2 Why Explanation Matters

An unexplained model is not merely opaque; in lending and insurance it is often illegal.

18.2.1 Why Financial AI Needs a Theory of Explanation

Suppose a bank's model validation team asks a data scientist why the credit model just denied a particular applicant, and the honest answer is "because that is what several hundred decision trees, averaged together, produced." The answer is true, but it satisfies no one: not the applicant, who is legally entitled under fair-lending rules to specific reasons for the denial; not the regulator, who must verify that the model is not proxying for a protected characteristic; and not the risk committee, which cannot sign off on a decision it does not understand.

Two related but distinct words recur throughout this chapter, and casual usage often treats them as interchangeable. **Interpretability** is a property of a model itself: a model is interpretable to the degree that a human can see, from its structure alone and without any additional tool, how its inputs map to its outputs. A linear regression's coefficients, Chapter 6's credit-spread regression among them, are interpretable in this sense: each coefficient answers "how much does the prediction change per unit change in this feature."

Explainability, by contrast, is a property of a method applied *after* a model is built: an explanation method takes any model, interpretable or not, and produces a human-comprehensible account of a specific prediction or of the model's general behavior. A deep neural network's millions of weights admit no direct reading, but the network can still be explained, approximately, by the post-hoc methods this chapter develops. Every explanation method here falls into one of four cells formed by two distinctions, illustrated in Figure 18.1: whether it explains the model's behavior in general (*global*) or a single prediction (*local*), and whether it is built into the model's structure (*intrinsic*) or applied afterward to a fitted model (*post-hoc*).

	Intrinsic	**Post-hoc**
Global	Linear/logistic regression coefficients (Ch. 6, 8); a single decision tree	Permutation importance; partial dependence (Section 18.3.1)
Local	One applicant's own linear score decomposition (additive by construction)	Shapley values; LIME (Sections 18.3.2, 18.3.5)

Figure 18.1: A taxonomy of explanation methods along two axes: global versus local, and intrinsic versus post-hoc

The trade-off this taxonomy makes visible is the central tension of the chapter. Chapter 6 observed that the most flexible models, large ensembles, deep neural networks, and transformers (Chapter 16) among them, are typically the least interpretable, while the most interpretable ones, a linear regression or a shallow decision tree, fit a narrower class of relationships. Post-hoc explanation methods do not eliminate this trade-off; they manage it, producing an approximate, human-readable account of a black-box model without changing the model. The approximation is useful (a bank can keep the gradient-boosted model that best predicts default and still produce an adverse-action notice for a declined applicant), but it is an approximation, and Section 18.3.5's worked example shows both how well it can work near the point it is built around and how it degrades away from it.

18.2.2 Intrinsically Interpretable Models Revisited

Before the post-hoc methods, consider choosing a model that needs no post-hoc explanation at all, the left column of Figure 18.1. Chapter 6 introduced its two workhorses: linear and logistic regression, whose

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coefficients are the explanation, and the single decision tree, whose if-then splits read directly as a chain of rules ("if debt-to-income exceeds 30% and credit history is under two years, predict default"). Chapter 9's credit-scoring model and Chapter 15's fraud-scoring model, both logistic regressions, were built this way on purpose, and Section 18.3.2's exact, closed-form Shapley decomposition depends on it. A bank that accepts a small accuracy cost for this direct interpretability avoids the entire post-hoc apparatus.

Two refinements extend this intrinsic column well beyond a single linear model or tree. A **generalized additive model (GAM)** replaces logistic regression's linear combination $\beta_0 + \sum_j \beta_j x_j$ with a sum of flexible, individually plotted functions, $\beta_0 + \sum_j f_j(x_j)$. Each feature's effect can then be nonlinear (a U-shaped effect of age on default risk, say) while the model stays additive, so each f_j can still be inspected on its own, like a linear coefficient. A **credit scorecard**, the industry-standard format behind Chapter 9's 850-to-300 score transformation, goes the other way. It bins each feature into a few ranges (debt-to-income under 20%, 20-30%, over 30%, for instance), assigns each bin an integer number of points from a table, and sums the points across features. A loan officer, auditor, or regulator can verify the result by hand with addition and table lookups. Both formats give up some predictive power, the trade-off of Section 18.2.1, to stay entirely in the intrinsic column and need none of the machinery of Sections 18.3.1 through 18.3.5. That is why scorecards remain the dominant format in consumer credit underwriting, even though tree ensembles and neural networks routinely beat them on held-out accuracy.

18.3 Explanation Methods

This part works the three standard families numerically on models built earlier in the book.

18.3.1 Global Explanations: Permutation Importance and Partial Dependence

Suppose a risk manager wants a one-sentence answer to which of the fraud model's three features it relies on, before deciding whether to keep buying all three data feeds. A global explanation answers this by describing the model's behavior across its input space, or at least across a representative dataset, rather than for one case.

Permutation feature importance asks an empirical question: how much does the model's performance degrade if one feature's values are shuffled across the dataset, breaking that feature's relationship with the outcome while leaving every other feature and the true labels untouched? A large drop marks a feature the model relies on; no change marks one it could apparently do without, at least on this data.

Recall Chapter 15's fraud-scoring logistic regression, fit to ten transactions with three features (amount z-score, distance from home in hundreds of kilometers, and a late-night indicator) and coefficients $\hat{\beta}_0 \approx -2.57$, $\hat{\beta}_1 \approx 0.83$, $\hat{\beta}_2 \approx 0.15$, $\hat{\beta}_3 \approx 0.82$, achieving 90% accuracy and an F_1 of 0.857 at its 0.5 threshold.

Permuting a feature means reassigning its ten values to the ten transactions in a different order; the simplest way by hand is to reverse the column, so that transaction 1 receives transaction 10's value, transaction 2 receives transaction 9's, and so on, while every other feature and every fraud label stays put. Table 18.1 reports the resulting accuracy and F_1 with each feature permuted in turn.

Feature permuted	Accuracy	Precision	F_1
(baseline, no permutation)	0.900	1.000	0.857
Amount z-score	0.300	0.000	undefined
Distance from home	0.900	1.000	0.857
Late-night indicator	0.900	1.000	0.857

Table 18.1: Permutation feature importance for the fraud-scoring model

Reversing the amount z-score column is catastrophic: accuracy collapses from 90% to 30%, and the three transactions flagged after permutation (5, 8, and 10) are all false positives. None of the four frauds is caught, so precision and recall both fall to zero and F_1 is undefined (conventionally reported as zero). Reversing the distance column or the late-night column, by contrast, changes none of the ten predictions, so accuracy and F_1 match the unpermuted baseline.

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The conclusion is not that distance and late-night timing are irrelevant to fraud (Chapter 15's coefficients show that both raise predicted fraud probability), but that in this ten-transaction sample the amount z-score so dominates the linear combination that no rearrangement of the other two features pushes any transaction across the 0.5 threshold. Permutation importance always carries this caveat: it measures a feature's contribution to performance *on the data at hand*, and a feature that looks unimportant here could prove decisive on a larger sample.

A complementary global tool, **partial dependence**, asks a different question: holding every other feature at its empirical values, how does the model's average prediction change as one feature is swept across a range? Formally, the partial dependence of the fraud model on the amount z-score z is $\mathrm{PD}(z) = \frac{1}{n}\sum_{i=1}^{n} \sigma\big(\hat\beta_0 + \hat\beta_1 z + \hat\beta_2 d_i + \hat\beta_3 n_i\big)$, averaging over the ten transactions' actual distance and late-night values while only z varies. Table 18.2 evaluates this at six values of z.

Amount z-score	-1	0	1	2	3	4
Average predicted probability	0.095	0.185	0.320	0.489	0.663	0.807

Table 18.2: Partial dependence of predicted fraud probability on amount z-score

The curve crosses the 0.5 threshold between $z = 2$ and $z = 3$, at approximately $z \approx 2.07$: an otherwise typical transaction (average distance and late-night status across the sample) is predicted fraudulent once its amount is roughly two standard deviations above the cardholder's usual spending. Partial dependence and permutation importance answer different questions and can disagree. A feature can have a smooth, informative partial-dependence curve and still show zero permutation importance on a specific sample, as distance did above, because partial dependence describes the model's functional form while permutation importance describes how much one dataset's performance metric depends on that feature.

18.3.2 Local Explanations: Shapley Values for a Single Decision

A global explanation tells a modeler or regulator how a model behaves in general; it does not answer the question a declined applicant asks, which is why *my* application was denied. A **Shapley value**, borrowed from cooperative game theory (Shapley 1953) and popularized in machine learning as SHAP (SHapley Additive exPlanations), answers it by dividing credit for one prediction's deviation from a baseline among the features that produced it. For a model linear in its inputs, as Chapter 9's credit-scoring logistic regression is in the log-odds, the Shapley value has a clean closed form. The contribution of feature j is

$$\phi_j = \hat\beta_j (x_j - \bar x_j), \tag{18.1}$$

the feature's coefficient times its deviation from a reference baseline. These contributions sum *exactly* to the difference between the individual's predicted log-odds and the baseline's, a property called efficiency that any valid additive attribution must satisfy.

Recall Chapter 9's credit-scoring model,

$$z = -12.152 + 0.530\,(\text{Debt/Income}) - 0.231\,(\text{Credit History})$$

(coefficients shown rounded; the fitted model itself carries full floating-point precision), applied to an applicant with a 28% debt-to-income ratio and 4 years of credit history. A natural baseline is the mean applicant in Chapter 6's six-applicant reference population (debt-to-income $\bar x_1 = 26.33\%$, credit history $\bar x_2 = 5.17$ years). Comparing the two,

$$\begin{aligned} \text{Applicant:}\quad & z \approx 1.761,\ \Pr(\text{Default}) \approx 85.3\%,\ \text{score} \approx 381, \\ \text{Baseline:}\quad & z_{\text{baseline}} \approx 0.608,\ \Pr(\text{Default})_{\text{baseline}} \approx 64.7\%,\ \text{score} \approx 494, \end{aligned}$$

the applicant's score well below most lenders' approval cutoffs, and the baseline itself subprime but far less severe.

The Shapley decomposition of the gap between them is

$$\underbrace{1.761}_{z_{\text{applicant}}} = \underbrace{0.608}_{z_{\text{baseline}}} + \underbrace{0.530\,(28 - 26.33)}_{\text{debt-to-income contribution} = 0.883} + \underbrace{(-0.231)\,(4 - 5.17)}_{\text{credit-history contribution} = 0.270},$$

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and $0.608 + 0.883 + 0.270 = 1.761$ up to rounding, verifying efficiency. Both features push predicted risk above the baseline: the applicant's debt-to-income is above the reference average (more leverage) and the credit history is shorter (less seasoning). But debt-to-income's contribution (0.883) is more than three times credit history's (0.270), so debt-to-income is the dominant driver of this applicant's elevated risk, not one factor among equals. Figure 18.2 draws the decomposition as a waterfall chart, the standard presentation of a SHAP explanation.

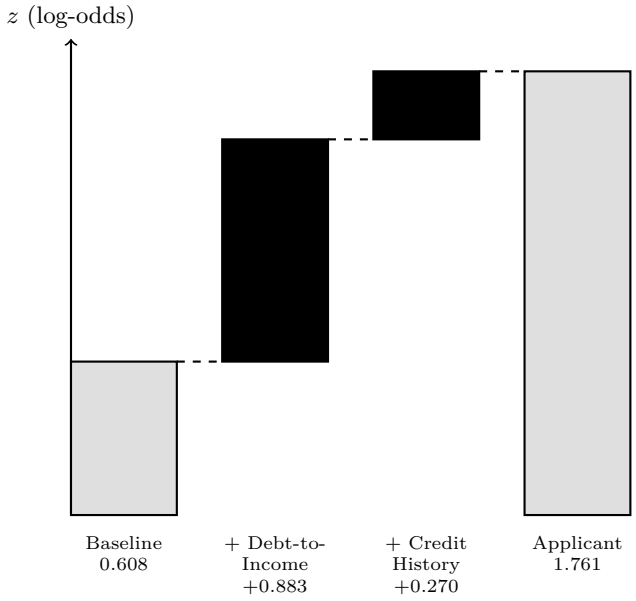


Figure 18.2: The Shapley decomposition as a waterfall chart

18.3.3 From Shapley Values to Adverse Action Reason Codes

This decomposition is, in substance, what U.S. law already requires a lender to produce. The Equal Credit Opportunity Act and its implementing Regulation B require a lender who denies credit to give the applicant the specific, principal reasons for the denial, an *adverse action notice*, rather than a bare rejection. Ranking the Shapley contributions by magnitude produces this: "high debt-to-income ratio relative to similar applicants" as the principal reason (contribution 0.883), and "limited length of credit history" as a secondary reason (contribution 0.270). It works cleanly here because the model is linear in the log-odds. For a nonlinear model such as a gradient-boosted ensemble, the same additive decomposition can still be computed with the general Shapley formula, which averages a feature's marginal contribution over every ordering of the other features and is far more expensive; that is why fast approximations, and simpler surrogates like Section 18.3.5's, remain an active research area.

The Consumer Financial Protection Bureau addressed machine-learning models in two circulars. Circular 2022-03 (CFPB 2022) stated that the adverse-action requirement applies in full to credit decisions based on complex algorithms: a creditor cannot defend a vague notice on the ground that its model is too opaque, so a model whose reasons cannot be extracted is one the creditor should not use. Circular 2023-03 (CFPB 2023) closed the obvious workaround: Regulation B's sample checklist of reasons ("insufficient income," "length of employment") does not suffice when the reason the model relied on, a behavioral or alternative-data feature, is not on the list, and the creditor must state it specifically, even in words of its own. Both circulars were among the 67 guidance documents the Bureau withdrew on 12 May 2025 (CFPB 2025), but a withdrawn interpretation is not a repealed rule: the duty to state the specific, principal reasons sits in the statute and in Regulation B's adverse-action provision, and as of October 2026 neither has changed. The circulars remain the clearest statement of what that duty demands of a black-box model, and so of why the Shapley-to-reason-code mapping above is needed.

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18.3.4 Counterfactual Explanations: How Much Would Need to Change?

A related, often more actionable, explanation answers a different question: not why was I denied, but what is the smallest change that would have gotten me approved? Suppose this lender approves any applicant with a credit score of at least 620, equivalent to

$$\Pr(\text{Default}) \leq \frac{850 - 620}{550} \approx 41.8\%,$$

or $z \leq -0.330$. Holding the applicant's 4 years of credit history fixed and solving

$$-12.152 + 0.530\,(\text{DTI}) - 0.231(4) = -0.330$$

for debt-to-income gives DTI $\approx 24.05\%$, about 3.95 percentage points below the applicant's actual 28%. This *counterfactual explanation* (reduce your debt-to-income ratio by roughly four percentage points, holding everything else fixed, and this model would approve you) is often more useful to a declined applicant than a coefficient or a Shapley value, because it states a concrete target rather than a ranked list of factors. It shares the caveat of Section 18.3.5's local surrogate: it describes what would happen if only debt-to-income changed and nothing else about the applicant shifted with it, a convenient assumption that need not hold in reality.

18.3.5 Local Surrogate Models: LIME for a Genuinely Nonlinear Model

Shapley values have an exact closed form for Chapter 9's credit-scoring model only because it is linear in the log-odds; most flexible models in this book, Chapter 6's ensembles and Chapter 17's CNNs and RNNs, are not. **LIME** (Local Interpretable Model-agnostic Explanations) treats the model as a black box instead: it samples points near the input to be explained, records the black box's predictions there, and fits a simple interpretable model, ordinarily a weighted linear regression, to those input-output pairs. The local model's coefficients are reported as the explanation, valid by construction only in the neighborhood sampled.

Recall Chapter 6's small feedforward network, two inputs, a hidden layer of two ReLU units, and a sigmoid output, with weights

$$W_1 = \begin{pmatrix} 0.4 & -0.2 \\ 0.1 & 0.3 \end{pmatrix},\ b_1 = \begin{pmatrix} 0.05 \\ -0.1 \end{pmatrix},\ W_2 = \begin{pmatrix} 0.6 & -0.5 \end{pmatrix},\ b_2 = 0.1,$$

which at $x_0 = (1.0, 0.5)$ produces hidden activations $h_0 = (0.35, 0.15)$ and a prediction $\hat{y}_0 \approx 0.558$. Both ReLU units are active at x_0 (their pre-activations are positive), so in a small enough neighborhood the network is exactly linear, and the example doubles as a check on LIME: the fitted surrogate should recover the true local behavior almost exactly. Table 18.3 lists eight perturbed points around x_0, each within ± 0.1 of x_0 in each coordinate, with the network's exact prediction at each.

Perturbed input (x_1, x_2)	Hidden activations	Prediction $\hat{y}$
$(1.1, 0.5)$	$(0.39, 0.16)$	0.5632
$(0.9, 0.5)$	$(0.31, 0.14)$	0.5538
$(1.0, 0.6)$	$(0.33, 0.18)$	0.5518
$(1.0, 0.4)$	$(0.37, 0.12)$	0.5651
$(1.1, 0.6)$	$(0.37, 0.19)$	0.5565
$(0.9, 0.4)$	$(0.33, 0.11)$	0.5605
$(1.1, 0.4)$	$(0.41, 0.13)$	0.5698
$(0.9, 0.6)$	$(0.29, 0.17)$	0.5471

Table 18.3: Perturbed samples around $x_0 = (1.0, 0.5)$ and the network's exact prediction

Fitting an ordinary least-squares surrogate

$$\hat{y} \approx a + b_1 x_1 + b_2 x_2 \tag{18.2}$$

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to these eight samples gives $a \approx 0.545$, $b_1 \approx 0.0468$, $b_2 \approx -0.0666$. The analytical local gradient, obtained by differentiating the network directly (valid because both ReLU units stay active throughout this neighborhood), is $\frac{\partial \hat{y}}{\partial x} = \hat{y}_0(1 - \hat{y}_0)\, W_2 W_1 \approx (0.0469, -0.0666)$, matching the LIME slope to three decimal places. The surrogate evaluated at x_0 gives $\hat{y}_{\text{surrogate}}(x_0) \approx 0.558$, matching the network's output 0.558: the local fidelity LIME is designed to deliver. The agreement follows from the neighborhood being small enough that no ReLU unit crosses its activation boundary.

Perturbing further away breaks this: at $x = (1.0, -0.1)$, the second hidden unit's pre-activation is exactly zero and the ReLU clips it, giving a true prediction of $\hat{y} \approx 0.594$, while the surrogate extrapolated to the same point predicts approximately 0.598, a discrepancy that widens further out. LIME's neighborhood radius is therefore a modeling choice with real consequences. Too narrow, and the explanation is accurate but says nothing about the model's broader behavior. Too wide, and it averages over a region where the black box is no longer linear, understating the kinked behavior that motivated using a deep model in the first place.

18.4 Regulation, Fairness, and Robustness

Explanation is a legal requirement before it is a technical one, and fairness is measured, not asserted.

18.4.1 International Regulatory Approaches to Explainability

Section 18.3.3 described the U.S. approach to mandated explanation, the adverse-action notice, which is principles-based: it specifies the outcome (specific, principal reasons for a denial) without prescribing a technical method, so Section 18.3.2's Shapley decomposition and Section 18.2.2's summed scorecard points are equally acceptable. The European Union's approach is broader in scope and, in places, more prescriptive. Article 22 of the General Data Protection Regulation (GDPR) restricts decisions based solely on automated processing that produce legal or similarly significant effects, and the Court of Justice of the EU held in 2023 that a credit bureau's score can itself be such a decision when lenders rely on it decisively (CJEU 2023). How much individualized explanation Articles 13–15 then demand, a global description of the model's logic or an account of the specific decision, remains contested among legal scholars, a rare case in this book where the technical machinery of Sections 18.3.2 and 18.3.5 is more settled than the legal question of how much of it is required. The EU's Artificial Intelligence Act (EU 2024) moves from a data-protection framing to a product-safety one: it lists credit scoring and life and health insurance pricing as high-risk systems that must be documented, overseen, and tested *before* deployment, rather than explained one decision at a time afterward.

Table 18.4 sets side by side the regimes a credit or insurance-pricing model is most likely to meet, as of October 2026. Much of it is supervisory guidance rather than statute, and it moves quickly: five of its rows changed during 2026 alone.

The table's most consequential pattern is divergence. Between 2023 and 2026 the EU, the UK, Canada, and Singapore all moved toward more explicit expectations for AI models in credit and insurance, while the U.S. federal government moved the other way on fairness. Executive Order 14281 of April 2025 (Executive Order 14281 2025) directs federal agencies to deprioritize disparate-impact enforcement, and the April 2026 amendment of Regulation B (CFPB 2026) withdrew federal reliance on disparate impact under the Equal Credit Opportunity Act while leaving the adverse-action explanation duty untouched; the government-wide AI-safety agenda of Executive Order 14110 (2023) had already been revoked on 20 January 2025. That does not make fairness testing optional for a U.S. lender. The Fair Housing Act, under which the Supreme Court recognized disparate-impact claims in 2015, state fair-lending and consumer-protection laws (Section 18.4.2), and the regimes abroad still reach outcomes, and a model built for one market is often deployed in several. A firm operating in both the EU and the U.S. therefore holds two compliance postures on the same model: explaining individual outcomes after the fact on one side, and certifying the system's design, documentation, and human oversight before it scores a real applicant on the other. The same toolkit, Sections 18.3.1 through 18.3.5, serves both. The U.S. banking agencies' replacement of SR 11-7 by SR 26-2 (Federal Reserve, OCC, and FDIC 2026) in April 2026 changes less for this chapter than its headline suggests: the new guidance keeps conceptual soundness, ongoing monitoring, and outcomes analysis, makes the intensity of validation explicitly risk-based, and leaves generative and agentic AI out of scope pending separate treatment, so the scoring and fraud models of this chapter remain inside it.

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Jurisdiction	Instrument	What it asks of a credit or pricing model	Status (October 2026)
EU	AI Act (Reg. 2024/1689), general provisions	Prohibited practices (e.g., social scoring); AI literacy; separate obligations for general-purpose AI models	In force 1 Aug 2024; prohibitions from 2 Feb 2025; GPAI obligations from 2 Aug 2025
	AI Act, Annex III, points 5(b)–(c)	Credit scoring of natural persons (fraud detection excluded) and life and health insurance risk assessment and pricing are high-risk: risk management, data governance, technical documentation, logging, human oversight, accuracy and robustness; deployers assess fundamental-rights impact	Originally 2 Aug 2026; the Digital Omnibus (proposed 19 Nov 2025; Reg. 2026/1744, in force 27 Jul 2026) defers it to 2 Dec 2027
US federal	ECOA and Regulation B	Specific principal reasons for every adverse action; no discrimination on a prohibited basis	2026 amendment, effective 21 Jul 2026, removes the disparate-impact "effects test"; disparate treatment still prohibited
	CFPB Circulars 2022-03 and 2023-03	Neither a complex algorithm nor the sample-form checklist excuses vague adverse-action reasons	Issued 2022 and 2023; withdrawn 12 May 2025; the underlying duty is unchanged
	SR 11-7 (2011), now SR 26-2 / OCC Bulletin 2026-13	Model risk management: conceptual soundness, ongoing monitoring, outcomes analysis, independent validation, scaled to risk	SR 26-2 superseded SR 11-7 on 17 Apr 2026; generative and agentic AI out of scope
US states	Colorado SB 24-205, replaced by SB 26-189	Original: duty of care against algorithmic discrimination in consequential decisions (incl. lending, insurance), impact assessments. Replacement: notice, plain-language explanation within 30 days of an adverse decision, data correction, human review	SB 24-205 never took effect (delayed to 30 Jun 2026, then repealed); SB 26-189 signed 14 May 2026, main duties from 1 Jan 2027
	NYDFS Insurance Circular Letter No. 7	Insurers using AI or external consumer data in underwriting and pricing must show it is not unfairly discriminatory, test for disparate impact, govern and document it, and oversee vendors	Issued 11 Jul 2024
UK	PRA SS1/23	Five model risk management principles for banks, explicitly covering AI and machine-learning models	Effective 17 May 2024
	FCA Consumer Duty	Good outcomes for retail customers: fair price and value, avoiding foreseeable harm; reaches algorithmic pricing and decisioning	Open products 31 Jul 2023; closed products 31 Jul 2024
Canada	OSFI Guideline E-23	Enterprise-wide model risk management over the model lifecycle, explicitly including AI and machine-learning models, at federally regulated banks and insurers	Final 11 Sep 2025; effective 1 May 2027
Singapore	MAS FEAT principles, Veritas, and Guidelines on AI Risk Management	FEAT (2018): fairness, ethics, accountability, transparency, with Veritas assessment methods; Guidelines: AI inventory, risk-materiality assessment, lifecycle controls, board oversight	FEAT voluntary; Guidelines issued 7 Oct 2026, effective 7 Oct 2027 (some sections by 7 Oct 2028)
China	PIPL, Art. 24	Transparent, fair automated decisions with no unreasonable differential treatment in prices or terms; right to an explanation and to refuse solely automated decisions with significant effects	Effective 1 Nov 2021
	Algorithmic Recommendation Provisions	Algorithm filing; option to switch off personalization; no unreasonable price discrimination from user profiles	Effective 1 Mar 2022
	Generative AI Interim Measures	Lawful training data, content labeling, security assessment and filing for public-facing generative AI services	Effective 15 Aug 2023

Table 18.4: AI and model-risk regimes relevant to credit and insurance-pricing models, as of October 2026. Compiled from the instruments themselves (EU 2024; EU 2026; CFPB 2026; CFPB 2022; CFPB 2023; CFPB 2025; Federal Reserve, OCC, and FDIC 2026; Executive Order 14281 2025; Colorado General Assembly 2026; NYDFS 2024; PRA 2023; FCA 2022; OSFI 2025; MAS 2026; NPC Standing Committee 2021).

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One further development closes the loop: regulators now run models themselves. Supervisory technology, SupTech, is AI deployed by the supervisor rather than the supervised: securities regulators mine trading and filing data for the manipulation patterns Chapter 14 described, financial-intelligence units apply network analytics to the suspicious-activity reports Chapter 15's institutions file, and central banks apply text methods to their supervisory correspondence to triage which firms need attention. For a model builder, the consequence is symmetry: a bank's AI system is increasingly examined *by* an AI system, and the governance artifacts this chapter argues for (documentation, monitoring records, explanation apparatus) are what that examination consumes. It also brings this chapter's problems to the other side of the table: a supervisor's model can be biased, opaque, or wrong at system-wide scale, which is why supervisory bodies have begun publishing governance standards for their own tools that look strikingly like the SR 11-7 discipline they impose on the firms they oversee.

18.4.2 Fairness in Financial AI

Explainability answers what drove a specific decision; fairness asks a logically separate question, whether a model's decisions, however well explained, treat different groups equitably. Regulators have historically operationalized fairness through two statistical definitions that are not entirely compatible, and theoretical work has shown that the incompatibility is, in general, a mathematical impossibility rather than a matter of insufficiently clever engineering.

Disparate impact and the four-fifths rule. A lending, hiring, or underwriting practice exhibits disparate impact if it produces substantially different outcome rates across groups, even when no feature encodes group membership and no discriminatory intent can be shown. U.S. regulatory guidance operationalizes "substantially different" with the *four-fifths rule*: if the group with the lower favorable-outcome rate receives that outcome less than 80% as often as the group with the higher rate, the *adverse impact ratio*,

$$\text{AIR} = \frac{\min(\text{rate}_A, \text{rate}_B)}{\max(\text{rate}_A, \text{rate}_B)}, \tag{18.3}$$

falls below 0.80, disparate impact is presumed and the practice invites regulatory scrutiny regardless of intent. The rule comes from employment law: it was written into the 1978 Uniform Guidelines on Employee Selection Procedures (EEOC 1978) as a rule of thumb for hiring tests, and fair-lending analysts borrowed it as a screening heuristic, a flag for where to look harder rather than a legal threshold for credit. Its federal footing in lending narrowed in 2026, when the CFPB amended Regulation B to state that the Equal Credit Opportunity Act does not support disparate-impact liability (Section 18.4.1), but the Fair Housing Act, state law, and lenders' own model-risk programs keep the ratio in routine use.

Equalized odds and error-rate parity. A second family of definitions asks whether the model is equally *accurate* for each group: whether its false-negative rate (missing an applicant who will default, in this chapter's running example) and false-positive rate are similar across groups. A model can satisfy the four-fifths rule while its false-negative rate is far higher for one group, or the reverse. A theoretical literature (Chouldechova 2017; Kleinberg et al. 2017), developed for criminal-justice risk scores but directly applicable here, proves something stronger: unless the two groups have identical underlying default rates, no model can simultaneously satisfy calibration, equal false-negative rates, and equal false-positive rates across groups. A regulator or model-risk function must therefore choose explicitly which fairness definition a model is held to, rather than assume that satisfying one satisfies the others.

Table 18.5 makes this tension concrete with a small fairness audit of Chapter 9's credit-scoring model applied to two hypothetical four-applicant groups, Alpha and Beta, at the same 620-score approval threshold used in Section 18.3.4 ($\Pr(\text{Default}) \leq 41.8\%$).

Group Alpha's approval rate is $3/4 = 75\%$; Group Beta's is $1/4 = 25\%$, giving an adverse impact ratio of $0.25/0.75 = 0.333$, far below the 0.80 threshold: on this evidence alone, a clear four-fifths-rule violation calling for further fair-lending investigation. The error rates tell a different story. Of Group Alpha's two future defaulters (Alpha-2 and Alpha-3), the model approves Alpha-2, whose moderate debt-to-income and credit history keep her predicted probability just under the threshold, a false-negative rate of 50%. Of Group Beta's two future defaulters (Beta-2 and Beta-4), the model denies both, a false-negative rate of 0%. The group approved far more often (Alpha) is the group whose risk the model is worse at catching, while the group facing the harsher approval rate (Beta) is, on this small sample, screened perfectly. A regulator focused

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Applicant	Debt/income	Credit history	Pr(Default)	Decision	Will actually default
Alpha-1	15%	9 yrs	0.2%	Approve	No
Alpha-2	22%	6 yrs	13.2%	Approve	**Yes**
Alpha-3	30%	3 yrs	95.5%	Deny	Yes
Alpha-4	18%	8 yrs	1.1%	Approve	No
Beta-1	25%	4 yrs	54.3%	Deny	No
Beta-2	33%	2 yrs	99.2%	Deny	Yes
Beta-3	20%	5 yrs	6.2%	Approve	No
Beta-4	38%	1 yr	>99.9%	Deny	Yes

Table 18.5: A fairness audit of the credit-scoring model on two synthetic four-applicant groups

only on approval-rate parity would flag the model for disadvantaging Beta; one focused only on error-rate parity would note that Alpha's approved population carries more missed risk. Reconciling the two, rather than optimizing either in isolation, is the judgment call that the impossibility result says no purely statistical fix can avoid.

The impossibility result names a third statistic, calibration, worth checking directly. Both groups share an actual default rate, $2/4 = 50\%$ for Alpha and $2/4 = 50\%$ for Beta, the special case the theorem carves out as the one setting where calibration, equal false-negative rates, and equal false-positive rates could in principle coexist.

Yet a calibration-in-the-large check, comparing each group's average *predicted* default probability with its actual default rate, shows the model is not equally calibrated across them:

$$\begin{aligned} \text{Group Alpha predictions:} &\quad 0.19\%,\ 13.26\%,\ 95.50\%,\ 1.14\% \quad (\text{mean } 27.5\%,\ \text{actual } 50\%), \\ \text{Group Beta predictions:} &\quad 54.3\%,\ 99.2\%,\ 6.2\%,\ 99.96\% \quad (\text{mean } 64.9\%,\ \text{actual } 50\%), \end{aligned}$$

the model under-predicts risk for Alpha and over-predicts it for Beta, although both groups' realized outcomes are identical. This does not violate the impossibility theorem, which concerns idealized population distributions rather than any one small sample, but it shows how the tension appears in practice: equal base rates do not force equal calibration, equal error rates, or equal approval rates in a finite population, and an audit that checked only the adverse impact ratio, only the error-rate gap, or only calibration would each tell an incomplete part of the same story.

Individual Fairness: A Third Paradigm

Both definitions above are *group* fairness notions: they compare an aggregate statistic, an approval rate or an error rate, across predefined groups. A third paradigm, **individual fairness** (Dwork et al. 2012), asks whether two individuals who are similar with respect to the task are treated similarly, regardless of group. Formally, this needs a task-specific similarity metric, a distance between applicants that captures what "similar with respect to creditworthiness" means, and a model is individually fair if it maps similar inputs to similar outputs under that metric.

Table 18.5's Alpha-2 and Beta-3 show why this lens matters: their predicted default probabilities, 13.2% and 6.2%, are broadly comparable despite their different groups, and both are approved (though a stricter similarity threshold could flag even this gap as worth investigating). The practical difficulty is the one this parenthetical hints at. Choosing the similarity metric requires the same value judgment that group definitions push onto the choice of protected groups and outcome statistics: which features are legitimately relevant to creditworthiness. Individual fairness therefore relocates the value judgment of Section 18.4.2's impossibility result rather than resolving it, from "which group statistic to equalize" to "which similarity metric to adopt."

Fairness Testing in Practice: Proxies and Less Discriminatory Alternatives

Table 18.5 took group membership as known. In most U.S. consumer lending it is not: Regulation B generally forbids a lender to ask an applicant's race or ethnicity, except for the mortgage data collected under the

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Home Mortgage Disclosure Act, so a fair-lending analyst testing an auto-loan or credit-card model must first estimate it. The standard tool is **Bayesian Improved Surname Geocoding (BISG)** (Elliott et al. 2009), which the CFPB adopted and published for its own fair-lending analysis (CFPB 2014). It is Bayes' rule from Chapter 1: the Census Bureau's surname list supplies a prior probability of each race and ethnicity given the applicant's surname, and the racial composition of the applicant's census block group updates it,

$$\Pr(\text{race} \mid \text{surname}, \text{location}) \propto \Pr(\text{race} \mid \text{surname}) \times \Pr(\text{location} \mid \text{race}),$$

under the assumption that surname and location are independent given race. The output is a probability for each group, not a label, and good practice carries those probabilities through the audit, weighting each applicant's outcome by them, rather than rounding each applicant into one group. Proxy error is not random, since it depends on where people live and how distinctive their surnames are, so a disparity measured through BISG can be biased in either direction (Chen et al. 2019); an audit should report it as an estimate with that caveat.

Finding a disparity starts the analysis rather than ending it. U.S. disparate-impact doctrine runs in three steps: a disparity is shown, the lender shows a legitimate business need for the practice, and the question becomes whether a **less discriminatory alternative** would serve that need about as well. For a machine-learning model the third step is a search problem. Many models fit the same data almost equally well, the phenomenon called model multiplicity (Black et al. 2022), so the analyst retrains over different feature sets, hyperparameters, constraints, and thresholds and looks among the near-equally accurate candidates for the one with the highest adverse impact ratio (Gillis et al. 2024). The CFPB no longer presses that search under the Equal Credit Opportunity Act after 2026, but state enforcers still treat untested disparities as a violation: in July 2025 the Massachusetts Attorney General settled with the student lender Earnest Operations for $2.5 million over algorithmic underwriting alleged to produce disparate outcomes for Black, Hispanic, and non-citizen applicants, and over inaccurate adverse-action notices (Massachusetts Attorney General 2025). The settlement required the kind of model governance and fairness testing this section describes.

18.4.3 Case Vignette: The Apple Card Gender-Bias Controversy

In November 2019, software entrepreneur David Heinemeier Hansson posted that the Apple Card, issued by Goldman Sachs, had given him a credit limit twenty times his wife's, although the couple filed joint tax returns and his wife had a longer credit history and a comparable or better credit score. Apple co-founder Steve Wozniak reported a similar disparity with his own spouse. The claims spread rapidly and prompted the New York State Department of Financial Services to open a formal inquiry into whether Goldman Sachs's credit-limit algorithm violated fair-lending law.

The regulator's findings, published in 2021, illustrate this chapter's distinctions clearly. The investigation found no evidence that the algorithm used sex or gender as an input, and no evidence of intentional discrimination under the Equal Credit Opportunity Act, the claim the controversy had focused on.

It did find that Goldman Sachs's underwriting process lacked the testing and monitoring Section 18.4.2 formalizes. The bank could not demonstrate after the fact that its model did not produce disparate outcomes correlated with sex, because it had not tested for this in the way a four-fifths-rule audit like Table 18.5's would. And its customer service representatives, asked to explain a specific credit-limit decision, could offer no model-derived reason of the kind Section 18.3.3's reason codes produce, only that "the algorithm decided." The regulator recommended not a fine but stronger fairness testing and clearer consumer-facing explanations of algorithmic credit decisions, the two capabilities this chapter builds by hand. The episode corrects a common misreading of fair-lending law. Leaving a protected characteristic out of a model's inputs, sometimes called "fairness through unawareness," is neither necessary nor sufficient to avoid disparate impact. Facially neutral inputs correlated with a protected characteristic, such as a joint account structure reflecting gendered patterns in whose name assets are held, can reproduce the disparity a cruder model would produce by using the characteristic directly.

Postscript (October 2024). The Apple Card came back before a regulator for a different reason. On 23 October 2024 the CFPB ordered Goldman Sachs to pay a $45 million civil penalty and at least $19.8 million in redress to consumers, and Apple a $25 million penalty (CFPB 2024). The orders did not concern credit limits or sex. Apple had failed to send tens of thousands of customers' transaction disputes to Goldman, Goldman had not investigated the disputes it did receive as federal law requires, and both had misled customers about

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interest-free monthly installment plans for Apple devices; Goldman was also barred from launching a new credit card without a credible plan to comply with the law. (The Bureau terminated Apple's order on 22 September 2025, after Apple had paid its penalty.) In 2021 the gap was a model nobody had tested for disparate outcomes or could explain to a customer; in 2024 it was a dispute system launched four days after Goldman's board had been warned it was not fully ready. Neither was a defect in the scoring model's inputs; both were failures of the governance around it, the subject of Section 18.5.1.

18.4.4 Adversarial Robustness in Financial Models

A model's outputs can be manipulated deliberately, not just misunderstood, and Chapter 15's money-laundering vocabulary already has a name for this: *structuring*, splitting a transaction to fall just under a reporting or detection threshold. The idea applies directly to evading a fraud-scoring model rather than a currency-reporting requirement.

Recall Chapter 15's transaction 1, with an amount z-score of 3.1, a distance of 800 kilometers from the cardholder's home, and a late-night flag, correctly flagged as fraud with predicted probability 88.3%. Holding distance and late-night status fixed and solving $-2.57 + 0.83\,z + 0.15(8.0) + 0.82(1) = 0$ for z gives $z \approx 0.663$. A fraudster who could disguise the amount so that it appears only 0.663 standard deviations above the cardholder's typical spending, for instance by splitting it into several smaller purchases at the same distant, late-night location, would push the prediction just under the 0.5 threshold and evade detection. The required change is large, a reduction of 2.44 in the amount z-score, because the transaction's distance and timing already carry so much of the model's suspicion. This is a more reassuring finding than the classic adversarial examples in image classification, where an imperceptibly small pixel-level perturbation (Goodfellow et al. 2015) can flip a deep network's prediction. A low-dimensional model, linear in the log-odds, with several independently suspicious features is much harder to game through any single feature than a high-dimensional deep network. It remains vulnerable, though, to an adversary who manipulates several features jointly (a smaller amount, a nearby location, business hours), an attack this single-feature analysis does not rule out.

This is why Chapter 17's point about vendor black-box processing and Chapter 15's point about adversarial adaptation both matter here. Any model whose scoring logic can be inferred by the population it scores will see its inputs shift in response, whether the inference comes from a fraud ring or from market participants learning which alternative-data patterns a fund's model rewards. That is the concept-drift dynamic Chapter 14 introduced for high-frequency trading models and Chapter 15 quantified for fraud-model precision.

Adversarial robustness concerns an adversary crafting an input to fool the model. It differs from the stress testing of Chapters 8 and 9, which subjects a model or portfolio to an extreme but plausible scenario such as a severe recession or a market crash. Both ask how a model behaves outside the data it was trained on, but stress testing probes a shift in the whole population (unemployment rising across the board, say), while adversarial robustness probes a targeted manipulation of one case, and a model can be robust to one and vulnerable to the other. A fraud model retrained often enough to absorb an economy-wide shift in spending can still be gamed one transaction at a time, as shown above. And a credit model that behaves sensibly in a simulated recession offers no guarantee against the manipulation risk Section 18.3.4 raised: an applicant who learns the threshold and adjusts a self-reported input to cross it, without any change in creditworthiness.

18.5 Governance and Supervisory Engagement

A model in production drifts, and this part covers what is monitored and what is done about it.

18.5.1 Model Governance and Monitoring in the AI Era

Chapters 8 and 9 introduced model risk management, independent validation and ongoing monitoring for market and credit risk models, and Chapter 15 formalized the three-lines-of-defense structure (business unit, independent compliance and risk function, internal audit) for fraud and anti-money-laundering programs. U.S. bank regulators codify this discipline for essentially every model a bank relies on in SR 11-7 (Board of Governors of the Federal Reserve System and Office of the Comptroller of the Currency 2011) (superseded in April 2026 by SR 26-2, which keeps all three pillars; Section 18.4.1). It names conceptual soundness, ongoing monitoring, and outcomes analysis as the three pillars of sound model risk management, matching the three

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capabilities this chapter builds: understanding why a model produces an output, tracking whether it keeps working, and checking its outcomes for fairness.

Chapter 15's precision-drift table tracked a fraud model's precision falling from 24.2% to 17.1% over six months as fraudsters adapted, but precision is a *lagging* indicator: computing it requires investigators to confirm which flagged transactions were fraudulent, which takes time.

The **population stability index (PSI)** offers a complementary, *leading* indicator that needs no labels, only the model's score outputs: it compares the current period's score distribution with a baseline established when the model was validated. Formally,

$$\mathrm{PSI} = \sum_i (c_i - b_i) \ln(c_i / b_i), \tag{18.4}$$

summed over score bins i with baseline proportion b_i and current proportion c_i. Table 18.6 illustrates this for the fraud-scoring model, comparing its validation-time score distribution across four bins against a later snapshot.

Score bin	$[0, 0.25)$	$[0.25, 0.50)$	$[0.50, 0.75)$	$[0.75, 1.00]$	Total PSI
Baseline proportion	0.50	0.25	0.15	0.10	
Current proportion	0.34	0.29	0.22	0.15	
Bin contribution	0.062	0.006	0.027	0.020	0.115

Table 18.6: Population stability index for the fraud-scoring model's score distribution

A total PSI of 0.115 falls into the conventional "moderate shift, worth investigating" band of credit-risk monitoring (below 0.10 is treated as no material change; above 0.25 as a major shift typically requiring retraining or revalidation), well before precision has moved enough to be distinguishable from noise. A smaller share of transactions now scores in the lowest bin and a larger share in every higher bin: the score population has shifted upward, a sign that the transactions the model sees have changed (a new customer segment, a different merchant mix, an emerging fraud pattern), and PSI reports it before enough investigated cases accumulate to move precision. The adaptive evasion of Section 18.4.4 would look different. Structuring pushes only the fraudulent transactions' scores *down*, and because fraud is a small share of transactions, that shift can hide inside an aggregate distribution whose PSI barely moves, one reason to monitor scores by segment as well as in aggregate. This is the early warning SR 11-7's ongoing-monitoring pillar is meant to produce, and why a mature governance program tracks both a leading, label-free statistic like PSI and a lagging, outcome-based one like precision.

The National Institute of Standards and Technology's AI Risk Management Framework (NIST AI RMF 1.0; NIST 2023) organizes this chapter's concerns (explainability, fairness, robustness, and governance) into a voluntary framework built around four functions: *govern* (accountability and policy: Section 18.5.1 and Chapter 15's three lines of defense), *map* (a model's context and likely impact, including on demographic groups: Section 18.4.2), *measure* (the methods of Sections 18.3.1 through 18.3.5 and 18.4.4), and *manage* (acting on what measurement reveals, from adverse-action notices to model retirement). The framework is deliberately not finance-specific and not a checklist to complete once. It is best read as an organizing structure for a discipline that, as the Apple Card vignette (Section 18.4.3) showed, tends to fail in the gaps between the four functions: an algorithm measured for accuracy but never mapped for demographic impact, or governed on paper with no one able to produce an explanation when a regulator asked for one.

18.5.2 Governing Systems That Act

Every model this chapter has governed so far produces a number (a score, a probability, a flag) that a person or a separate process then acts on, and every tool it has built, from reason codes to the population stability index, governs that number. The LLM agents of Chapter 16's section on agentic AI break this arrangement: they call tools, and the tools change records, send messages, and move money. Chapter 10 made the central point: an AI that acts closes the detection window every downstream control depends on, so the first evidence of a failure is an executed action rather than a declined proposal. This subsection asks what that changes for governance.

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Delegated authority. Banks govern people through delegated authority: what each person may approve, sign, or trade, up to what amount, is written down, granted by someone entitled to grant it, and reviewed. An agent needs the same instrument, a mandate that states its tasks, its tools, the data and counterparties it may touch, and its value thresholds, and names the human who owns it. Two rules follow. An agent's authority can never exceed that of the person delegating to it, and an agent serving several people must not accumulate the union of their entitlements, which would recreate the segregation-of-duties breach Chapter 10 warned an agent holds by construction. FINRA's 2026 oversight report, published in December 2025 and the first to address AI agents directly, lists this among their risks: agents that act beyond the scope and authority the user intended (FINRA 2025).

Accountability for actions rather than predictions. Validating a scoring model asks whether its outputs are accurate, stable, and fair. Validating an agent must also ask, for each action, whether it was authorized, whether it was correct, and who answers for it. The firm does: an agent is not a legal person, and neither the model vendor nor the agent absorbs the firm's obligations to its clients or supervisors. In practice this turns outcomes analysis, SR 11-7's third pillar, into the trajectory review Chapter 16 described: reading samples of what the agent did, step by step, not only what it concluded.

Action logs and audit trails. An action that cannot be reconstructed cannot be reviewed, contested, or corrected. An agent's log must be richer than a model's prediction log: the instruction it received, the context it retrieved, each tool call with its arguments and result, each human approval with the approver's identity, and the versions of model, prompt, and tool set in force. This is the agent counterpart of the EU AI Act's requirement that high-risk systems automatically record events over their lifetime (Article 12) (EU 2024), and of FINRA's expectation that firms track agent actions and decisions (FINRA 2025). The log is also the raw material for the supervision by machine described in Section 18.4.1.

Limits and kill switches. Because an agent acts at machine speed, its failures must be bounded before anyone notices them: caps on the value of any single action, on actions per hour, and on total spending, and a kill switch that halts the agent and leaves its work in a known state. Electronic trading learned this expensively, as Chapter 14's account of Knight Capital showed, and the EU AI Act's human-oversight article requires that an overseer be able to interrupt a high-risk system through a "stop" button or similar procedure that brings it to a halt in a safe state (Article 14(4)(e)) (EU 2024).

Human approval for irreversible actions. Least privilege decides what an agent may do alone; approval governs the rest. Approval should bind to one specific action and its arguments (this payment, this amount, this beneficiary) rather than to the agent's task in general, and it should be required wherever an action cannot be undone or leaves the firm. Singapore's Model AI Governance Framework for Agentic AI, launched in January 2026, makes such checkpoints part of keeping humans meaningfully accountable (IMDA 2026). Approval is a control only if the approver actually decides. Chapter 16's worked example showed that a ten-step agent at 95% per-step reliability sends about a third of its tasks to review; a reviewer facing that volume who approves nearly everything is a rubber stamp, so approval and override rates are themselves monitored, as this section monitors a model's score distribution.

Change management. An agent is a composition (a model, its prompts, tools, memory, and permissions), and a change to any one is a change to the system. Vendors revise and retire hosted models on their own schedule, and a new version can change behavior on tasks the old one handled reliably, so a governed agent pins its versions, reruns its evaluation suite before promoting any change, and records the result. Adding a tool deserves particular care, because it changes what the agent can do, not just how well: it is a change in authority and should pass through the same approval as a change to a human's entitlements. The Upstart vignette that follows shows the same tension in a classical model: a supervisory assurance covers the system as reviewed, and a fast-changing system sits uneasily inside it.

Third parties. Most agents rest on a foundation model from an outside vendor and reach data through connectors, increasingly built on the Model Context Protocol Chapter 16 described, that may themselves come from third parties. Each is a third-party relationship in the sense of the U.S. banking agencies' 2023 interagency guidance, with its life cycle of due diligence, contract, ongoing monitoring, and exit (FRB, FDIC, and OCC 2023): the contract should give the bank notice of model changes, limits on the vendor's use of its data, and rights to the information it needs to validate. A connector is also an attack surface, since everything it returns enters the agent's context, so vetting connectors defends against the prompt injection Chapter 16 described and is more than a procurement step.

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How existing frameworks apply. As of October 2026, no framework a bank answers to was written for agents, and the existing ones reach them unevenly. In the United States, SR 26-2 (Federal Reserve, OCC, and FDIC 2026), which replaced SR 11-7 in April 2026, expressly leaves generative and agentic AI outside its scope pending separate treatment. That leaves no dedicated U.S. model-risk guidance for agents, but it does not leave agents ungoverned: a classical model an agent calls, such as a credit score or a fraud model, remains inside SR 26-2; the agent itself falls under operational-risk, third-party, and conduct expectations; and many banks extend their model-risk programs to agents voluntarily. The EU AI Act (EU 2024) has no agent category. It classifies by use, so an agent that scores the creditworthiness of natural persons is a high-risk system with the logging and human-oversight obligations above, applicable from December 2027 under the Digital Omnibus (Table 18.4), while the general-purpose model underneath carries its provider's own obligations, in force since August 2025. The NIST AI Risk Management Framework (NIST 2023), with its Generative AI Profile of July 2024 (NIST 2024), supplies the govern-map-measure-manage structure, and in February 2026 NIST's Center for AI Standards and Innovation launched an AI Agent Standards Initiative on agent security, identity, and interoperability (NIST 2026). ISO/IEC 42001, published in December 2023 (ISO/IEC 2023), is a certifiable management-system standard that covers agents as it covers any AI system, by requiring an organization to inventory, assess, and control them, though it prescribes no agent-specific controls. The most agent-specific texts are supervisory and voluntary: FINRA's 2026 report and Singapore's framework, both cited above. Table 18.7 collects the controls this subsection argues for, with the framework provision each most directly answers.

Control	What it requires	Framework hook (as of October 2026)
Mandate and owner	Written scope of tasks, tools, data, counterparties, and value thresholds; a named accountable human	FINRA 2026 (scope and authority); IMDA (human accountability)
Agent identity and least privilege	Own service credentials; entitlements reviewed like a person's; no accumulated conflicting entitlements	Segregation of duties (Chapter 10); IMDA (bound risks upfront)
Approval of irreversible actions	Approval bound to one action and its arguments; approval and override rates monitored	EU AI Act Art. 14; IMDA checkpoints
Action log	Instruction, retrieved context, every tool call and result, approvals, versions in force	EU AI Act Art. 12; FINRA 2026 (track actions)
Limits and kill switch	Caps on value per action, actions per hour, and spending; halt to a safe state	EU AI Act Art. 14(4)(e)
Pre-deployment evaluation	Task suite scored on end state; trajectory review; prompt-injection tests	NIST AI RMF (measure); SR 26-2 by analogy
Change control	Pinned model, prompt, and tool versions; evaluation rerun before promotion; new tools approved as new authority	ISO/IEC 42001; SR 26-2 by analogy
Vendor and connector oversight	Change notification, data-use limits, audit rights; connectors vetted	2023 interagency third-party guidance
Ongoing monitoring	Task success, escalation, override, and cost rates; drift in inputs	SR 26-2 by analogy; NIST AI RMF (manage)

Table 18.7: A control checklist for AI agents in a regulated financial institution. "By analogy" marks guidance that excludes agentic AI but whose discipline banks commonly extend to it.

None of these controls is new. Each is the discipline a bank already applies to a human employee, a trading algorithm, or an outsourced service, written down for an actor that combines features of all three. What is new is the order: for a scoring model, monitoring can follow deployment, because a human still stands between the score and its consequence; for an agent, the mandate, limits, log, and kill switch must exist before its first action, because afterward there may be nothing left to intercept.

18.5.3 Case Vignette: Regulatory Sandboxes and the Upstart No-Action Letter

Section 18.4.3 showed a regulator responding to a public controversy after deployment; the CFPB's engagement with Upstart Network shows the same concerns, whether an AI underwriting model produces disparate impact and can be adequately explained, addressed in advance. In 2017 the Consumer Financial Protection Bureau granted Upstart a no-action letter, a formal assurance that the agency would not bring

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an enforcement action against specific practices. Upstart is an online lender that uses machine learning for credit underwriting, including alternative data such as educational attainment. The letter was conditioned on Upstart regularly reporting detailed fair-lending test results to the CFPB, including disparate-impact statistics like Section 18.4.2's adverse impact ratio, broken out by race, ethnicity, and sex.

The CFPB issued a renewed letter in November 2020, which also required Upstart to notify the agency of significant model changes before making them. That condition ended the arrangement. In April 2022 Upstart told the CFPB it intended to add a significant number of new variables to its underwriting model; the Bureau asked for more time to evaluate the changes; and Upstart, citing the need to keep its models accurate during a period of rapid economic change, asked for the letter to be terminated, which the CFPB did by order in June 2022. The ending is itself a lesson in monitoring: a supervisory assurance covers the model as reviewed, and a model that must be retrained quickly sits uneasily inside it. The contrast with the Apple Card case is instructive. Upstart's reporting reportedly showed its model approving a meaningfully higher share of applicants, at similar or lower average interest rates, than a traditional credit-score-only model would have, evidence that a more flexible model can *expand* credit access rather than merely risk disparate impact. That evidence was credible, and available to the regulator at all, only because Upstart had committed in advance to the ongoing fairness testing and reporting that Sections 18.4.2 and 18.5.1 develop. A regulatory sandbox or no-action letter does not waive the underlying obligations (disparate-impact testing, adverse-action explanation, ongoing monitoring); it is a supervised space in which a firm demonstrates that a novel model meets them before, rather than only after, a wide public rollout.

18.6 Advanced Topics

Four further topics extend the chapter's toolkit: making a counterfactual explanation actionable rather than merely valid; a causal rather than statistical definition of fairness; extending explanation methods to the deep architectures of Chapters 16 and 17; and attaching a finite-sample uncertainty guarantee to individual predictions.

18.6.1 Algorithmic Recourse: Making Counterfactuals Actionable

Section 18.3.4's counterfactual, reduce debt-to-income by about four percentage points, is only one way to cross the approval threshold. Holding debt-to-income at the applicant's actual 28% and solving for credit history instead gives approximately 13.0 years, an increase of about 9.0 years from the actual 4. Both solve the same equation, $z \leq -0.330$, but they are not remotely equally achievable: an applicant can pay down debt over months, while nobody can shorten the wait for nine more years of credit history.

Algorithmic recourse (Ustun et al. 2019) formalizes this by attaching a cost to changing each feature and recommending the counterfactual with the lowest total cost rather than the nearest one in raw feature space. With debt-to-income costing 1 unit per percentage point (roughly, the difficulty of paying down that much debt) and credit history 5 units per year (it can only be waited out), the debt-to-income path costs 3.95 and the credit-history path $9.04 \times 5 = 45.19$, quantifying what the raw numbers already suggest: debt-to-income is the only path a lender should recommend. The lesson generalizes. A production counterfactual system should distinguish mutable features an individual can act on, such as debt-to-income, savings, or a co-signer, from effectively immutable ones: age, the passage of time, and any feature correlated with a protected characteristic. Recommending recourse through an immutable feature is unhelpful, and it can raise the fair-lending concerns of Section 18.4.2.

Recourse is a causal question. A counterfactual explanation is a statement about the model: move debt-to-income to 24.05% and this scorer approves. Recourse is advice to act in the world, and an action is an intervention in the sense of Chapter 6's Section 6.4.6. Actions rarely move one feature alone. An applicant who pays down debt out of savings lowers debt-to-income but also the savings balance a richer model would score; one who takes a second job raises income but may cut short their tenure with their current employer. The nearest counterfactual treats features as independent dials, while real actions move several at once through causal links, so a recommendation computed as if one feature moved can fail to deliver approval once the knock-on changes are scored. Recourse is therefore best framed as the cheapest intervention on a causal model of the applicant's features rather than the nearest point across the decision boundary (Karimi et al. 2021). A second gap is subtler: crossing the threshold need not change the applicant's true default

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risk. Chapter 6's credit-limit example showed a feature, the limit-increase flag, that predicts lower default while causally raising it; recourse through a feature of that kind, such as opening a new account purely to lengthen recorded credit history, earns approval without earning creditworthiness. The loss falls on the lender and, repeated across many applicants, becomes the drift Section 18.5.1 monitors. Recommending actions that move a real cause of repayment, such as paying down debt from income, keeps approval and actual risk aligned.

18.6.2 Counterfactual Fairness: A Causal Perspective

Section 18.4.2's disparate-impact and equalized-odds definitions are purely statistical: they compare outcome or error rates across groups with no model of *why* the rates differ. **Counterfactual fairness** (Kusner et al. 2017) asks a causal question instead: would this individual's decision have been different had their protected attribute (sex, race) been different, holding fixed everything about them that is not causally downstream of that attribute?

This reframing picks up where the Apple Card vignette (Section 18.4.3) left off. A joint account's credit history can be causally downstream of a couple's gender-influenced decision about whose name assets are held under, so a model using joint-account history is not necessarily counterfactually fair even though it never reads sex or gender. Holding the actual, gender-influenced account history fixed while imagining a different gender is the counterfactual comparison this definition requires, and the correlation-only reasoning of Section 18.4.2 cannot express it. Operationalizing it requires a causal graph specifying which features are downstream of the protected attribute, a much harder modeling commitment than any statistical metric, which is why counterfactual fairness is more common as a lens for diagnosing cases like Apple Card's after the fact than as a routine pre-launch test.

The definition is Chapter 6's Section 6.4.6 applied to a protected attribute. Treat the attribute as the "treatment" in the potential-outcomes framework, and counterfactual fairness asks that this applicant's decision be the same under both potential values of it; the causal graph plays the role that knowing the confounder Z played in that section's credit-limit example. Two lessons carry over. First, dropping the protected attribute is the analogue of the short regression: its influence still flows through correlated features, joint-account history or a ZIP code, as creditworthiness flowed into the limit-increase flag, and the omitted-variable formula describes how a variable left out of a regression leaves its effect behind in the variables kept in. Second, that section's noisy-proxy estimate, which recovered only $+0.0027$ of a true $+0.02$ effect, warns that adjusting for an imperfectly measured variable removes only part of a bias: a fairness repair that conditions on a coarse proxy for the legitimate driver of risk can leave much of the protected attribute's influence in place while appearing to have removed it. One thing does not carry over: there is no randomized experiment to fall back on, since a protected attribute cannot be assigned at random. Counterfactual fairness therefore rests on an assumed causal graph, which must be argued for and documented because it cannot be tested.

18.6.3 Explaining Deep Models: Saliency, Integrated Gradients, and the Faithfulness Problem

Section 18.3.5's LIME surrogate treats the network as a black box; when the model's own gradients are available, as they are for any network trained by backpropagation, a more direct family of methods computes an attribution straight from those gradients rather than from a sampled local regression. The simplest version, a *saliency map*, is the gradient of the output with respect to each input, the analytical local gradient Section 18.3.5 already computed, $(0.0469, -0.0666)$ at x_0.

Multiplying this gradient by the full distance to a baseline, $x_0 - x' = (1.0, 0.5)$ from $x' = (0, 0)$, gives a naive attribution of $(0.047, -0.033)$, summing to 0.014, but the network's output changes by $0.558 - 0.532 = 0.026$ between these two points, nearly double the naive estimate. The discrepancy is not rounding. The straight-line path from x' to x_0 crosses a kink: the second hidden unit's ReLU is inactive at x' (pre-activation -0.1) and switches on at $t = 0.4$ of the way along, so a gradient evaluated only at the endpoint x_0 cannot see the nonlinearity the path passes through earlier.

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Integrated gradients (Sundararajan et al. 2017) fixes this by integrating the gradient along the entire path rather than evaluating it once,

$$\mathrm{IG}_i(x) = (x_i - x'_i) \int_0^1 \frac{\partial F}{\partial x_i}\big(x' + t(x - x')\big)\, dt. \tag{18.5}$$

Because this network's pre-activations are piecewise linear in t along the straight path, the integral has an exact closed form on each of the two segments (before and after the ReLU switches on at $t = 0.4$), using the fact that $\sigma'(a + bt)$ integrates to $\sigma(a + bt)/b$. Carrying this out gives $\mathrm{IG}_1 = 0.0520$ and $\mathrm{IG}_2 = -0.0260$, summing to 0.0260, which matches the true output change up to rounding: the *completeness* axiom integrated gradients is designed to satisfy and the endpoint-gradient shortcut failed. This matters for the deep architectures of Chapters 16 and 17: CNNs and attention-based transformers admit the same gradient-based treatment, and integrated gradients (or a close variant) is the standard way to produce a saliency map for an image classifier or a token-level attribution for a language model.

A subtler warning concerns attention weights, since Chapter 16 introduced self-attention and it is tempting to read a high attention weight as "the model relied on this token." Empirical work (Jain and Wallace 2019) has shown that attention weights often fail a basic test of any explanation: alternative attention distributions can be found that leave a transformer's output essentially unchanged, so the observed weights are not uniquely responsible for the prediction in the way an explanation implies.

The same faithfulness concern extends to a large language model's own natural-language, chain-of-thought explanation of its reasoning. Research on LLM-generated rationales (Turpin et al. 2023) has found cases where a model's stated reasoning does not match the factor that changed its answer: the explanation is fluent and plausible but not faithful, in this chapter's technical sense, to the computation that produced the output. Gradient-based methods like integrated gradients, computed from the model's own mathematics rather than from a separately generated explanation, therefore remain the more trustworthy default for a high-stakes financial deployment of Chapter 16 or Chapter 17's architectures, however fluent a model's self-reported explanation may read.

18.6.4 Conformal Prediction: Honest Uncertainty for Individual Decisions

Every explanation method in this chapter says why the model predicted what it did; none says how *sure* the model is entitled to be. Chapter 6's Section 6.4.5 introduced split conformal prediction for a regression: hold out a calibration set, take the $\lceil (n+1)(1-\alpha) \rceil$-th smallest of its nonconformity scores, and wrap any model's point forecast in an interval whose coverage holds in finite samples, provided the data are exchangeable. Two things change when the model is a credit classifier deciding one applicant at a time.

First, the output becomes a set of labels. With one minus the probability the model assigns to the true label as the nonconformity score, and $\hat{q}$ the same calibration quantile, an applicant with predicted default probability $\hat{p}$ has a prediction set containing "repay" if $\hat{p} \le \hat{q}$ and "default" if $1 - \hat{p} \le \hat{q}$. A singleton set is a decision the model can stand behind at the chosen coverage; the set {repay, default} marks a case it cannot resolve, a natural trigger for the human review that Section 18.5.1's governance framework provides for.

Second, the guarantee Chapter 6 proved is marginal, an average over all applicants, and an average can hide a group the model serves badly. The companion notebook simulates an applicant pool in which 20% have thin credit files, whose default risk depends mostly on information the lender does not observe, fits a logistic regression on a score and a thin-file indicator, and calibrates on 2,000 applicants, 376 of them thin-file, at $\alpha = 0.10$. Pooled calibration gives $\hat{q} = 0.7025$. Averaged over 2,000 fresh calibration sets, its sets cover 90.07% of all applicants, as promised, but 91.75% of thick-file applicants and only 83.43% of thin-file ones: the guarantee holds on average while failing the group the model knows least about. *Group-conditional* (or Mondrian) calibration computes a separate quantile within each group, here 0.6903 from the 1,624 thick-file applicants and 0.7795 from the 376 thin-file ones, restoring 90.17% and 90.11% coverage. The repair has a visible price: the share of thin-file applicants whose set is {repay, default}, and who are therefore referred to a human, rises from 40.0% to 60.9%, while the thick-file share falls from 21.5% to 16.6%. That is the honest answer: the model cannot decide most thin-file cases at 90% confidence, and pooled calibration had hidden this behind overconfident singletons.

Three cautions apply. Group-conditional calibration needs the group at decision time, and calibrating separately by a protected attribute would use that attribute in the decision, raising the disparate-treatment

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concerns of Section 18.4.2; in practice a lender calibrates on legitimate operating segments, such as file thickness, product, or channel, and *audits* coverage by protected group in the spirit of that section's tests. Each group also needs enough calibration observations, since a quantile estimated from 376 cases varies more from one calibration set to the next than one estimated from 1,624. And the guarantee lasts only as long as the calibration data stay relevant: under the drift that Section 18.5.1's population stability index monitors, yesterday's quantile silently loses today's coverage, as Chapter 6's stress-regime example showed for a regression interval.

18.7 Challenges in Explainability, Fairness, and Regulation

Several challenges recur across this chapter's methods and connect directly to themes from earlier chapters.

Explanation methods are themselves approximations, not ground truth. Section 18.3.5's worked example showed a local surrogate matching the true model almost exactly, and showed how that agreement degrades once a perturbation crosses a kink in the underlying function. A post-hoc explanation is only as trustworthy as the neighborhood or reference baseline it was built around, a modeling choice with no universally correct answer.

Multiple fairness definitions cannot generally be satisfied simultaneously. Section 18.4.2's worked example, and the impossibility results underlying it, mean that a bank's choice of fairness criterion is an explicit policy decision with real distributional consequences, not a detail a sufficiently sophisticated model can sidestep.

Proxy discrimination survives the removal of protected characteristics. Section 18.4.3's case vignette showed that omitting a protected characteristic from a model's inputs does not guarantee outcomes free of disparate impact along that characteristic, since facially neutral features can correlate with it closely enough to reproduce the pattern.

Explainability and model capability can be in tension. Chapters 6, 16, and 17 each observed that some of the most powerful models (deep ensembles, attention-based transformers, convolutional and recurrent networks) are among the least interpretable. Post-hoc methods manage that tension rather than eliminate it, which is why some regulated uses, consumer credit decisions in particular, still favor interpretable models over marginally more accurate black boxes.

Monitoring must anticipate adaptive, not just random, drift. Sections 18.4.4 and 18.5.1 show that in an adversarial financial setting the population being scored can learn from and react to the model itself, so drift detection is an ongoing arms race rather than a one-time validation exercise.

Governance frameworks require organizational capability, not just documentation. Section 18.4.3 showed that a bank can have a governance framework on paper and still fail the underlying obligation, producing no usable explanation when a customer or regulator asked for one, which is why SR 11-7 and the NIST AI RMF both emphasize ongoing practice over a one-time compliance checklist.

18.8 Key Takeaways

Interpretability is a property of a model's structure; explainability is a property of a method applied to a model. The distinction matters because most of this book's powerful models, Chapter 6's ensembles, Chapter 16's transformers, and Chapter 17's CNNs and RNNs, are not interpretable and depend on the post-hoc methods of Section 18.2.1's taxonomy. Global methods (Section 18.3.1) describe a model's behavior in general, and Chapter 15's fraud model showed that a feature can matter in principle yet show zero measured importance on a specific sample. Local methods answer a single case's question. An exact Shapley decomposition (Section 18.3.2) of a model linear in the log-odds, like Chapter 9's credit scorer, yields adverse-action reason codes and a counterfactual explanation almost for free. LIME (Section 18.3.5) extends the idea to nonlinear models such as Chapter 6's toy network, with validity guaranteed only in a local neighborhood whose size is itself a modeling choice.

Fairness (Section 18.4.2) is not a single statistical target: the four-fifths rule and error-rate parity can point in different directions on the same model and data, as the two-group audit showed, and theory proves the tension cannot generally be resolved away. The 2019 Apple Card controversy (Section 18.4.3) showed

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a widely used consumer credit product failing the fairness-testing and explanation obligations this chapter formalizes without ever using a protected characteristic as an input.

Section 18.4.4 extended Chapter 15's structuring vocabulary from evading a reporting threshold to evading a model, and Section 18.5.1 showed that a leading indicator like the population stability index can flag a shift in the scored population before enough labeled outcomes accumulate for a lagging indicator like precision. Together they tie explainability, fairness, robustness, and governance into the single discipline that SR 11-7 and the NIST AI Risk Management Framework formalize. AI agents that act rather than score (Section 18.5.2) extend it from outputs to actions: a mandate and a named owner, least privilege, approval bound to each irreversible action, a complete action log, hard limits with a kill switch, and change control over model, prompts, and tools, all in place before the first action, at a time when SR 26-2 leaves agentic AI out of scope and the EU AI Act reaches agents only through their use.

Section 18.6 pushed each thread deeper: cost-weighted recourse showed that not every valid counterfactual path is achievable; integrated gradients showed where a naive gradient shortcut understates a deep model's attribution, and why attention weights and chain-of-thought rationales cannot always be trusted as faithful explanations of Chapter 16's models; and the CFPB's Upstart no-action letter (Section 18.5.3) showed the same obligations met proactively, through structured regulatory engagement, rather than after a public controversy like Apple Card's.

18.9 Suggested Exercises

18.1 Using Table 18.1's method, permute the late-night indicator using a different fixed rearrangement (a cyclic shift by one position rather than a full reversal) and recompute the confusion matrix. Does the conclusion that late-night timing shows zero permutation importance on this sample still hold?

18.2 Explain why a feature can have a strong, monotonic partial-dependence curve (Table 18.2) while simultaneously showing zero permutation importance (Table 18.1) on the same dataset.

18.3 Using Section 18.3.2's method, compute the exact Shapley decomposition for a second applicant with a 32% debt-to-income ratio and 7 years of credit history, using the same reference baseline, and verify the efficiency property (that the two contributions plus the baseline log-odds sum exactly to this applicant's own log-odds).

18.4 Rank the two contributions from Exercise 18.3 and state which reason code would appear first on this applicant's adverse action notice.

18.5 Using Section 18.3.4's method, compute the counterfactual credit history (holding debt-to-income fixed at 28%) that would have gotten Chapter 9's original applicant approved at the 620-score threshold, and compare the required change to the required debt-to-income change computed in the text.

18.6 Using Table 18.3's network, compute the network's exact prediction at $x = (1.05, 0.55)$ and compare it to the LIME surrogate's prediction at the same point, reporting the discrepancy.

18.7 Explain why the LIME surrogate fitted in Section 18.3.5 recovers the analytical local gradient almost exactly, referencing which structural feature of the network (the sign of each hidden unit's pre-activation) makes this possible.

18.8 Using Section 18.4.2's formula, compute the adverse impact ratio if Group Beta's approval rate in Table 18.5 were instead $2/4 = 50\%$ (holding Group Alpha's rate fixed), and state whether the four-fifths rule would still be violated.

18.9 Using Table 18.5, compute each group's false-positive rate (the fraction of applicants who will not actually default but are nonetheless denied), and discuss whether this statistic points in the same direction as the false-negative-rate comparison in the text.

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18.10 Explain, using the impossibility result cited in Section 18.4.2, why a lender cannot generally fix Table 18.5's adverse impact ratio and its false-negative-rate gap with a single adjustment to the model's classification threshold alone.

18.11 Using Section 18.4.3's case vignette, explain why removing sex or gender as a model input does not, by itself, guarantee that a credit-limit algorithm's outcomes are free of gender-correlated disparate impact.

18.12 Using Section 18.4.4's method, compute the amount z-score reduction needed to flip Chapter 15's transaction 6 (amount z-score 3.4, distance 9.2, late-night flag 1) from fraud to not-fraud, holding distance and late-night status fixed, and compare the required reduction to transaction 1's.

18.13 Explain why the fraud-scoring model in Section 18.4.4 is harder to game with a single-feature perturbation than a deep image classifier is with a pixel-level perturbation, referencing the number of independently suspicious features each transaction carries.

18.14 Using Table 18.6's formula, compute the total PSI if the current-period distribution instead matched the baseline distribution exactly, and explain what this result confirms about the PSI formula's behavior when there is no drift at all.

18.15 Using the same PSI formula, compute the total PSI for a more extreme drift scenario in which the current distribution is $(0.20, 0.25, 0.30, 0.25)$ (holding the baseline fixed), and state whether this would fall in the "no material change," "moderate shift," or "major shift" band described in the text.

18.16 Explain why the population stability index is described as a leading indicator and Chapter 15's precision-drift table as a lagging indicator, and describe one practical consequence of a model governance program monitoring only the lagging one.

18.17 Using Section 18.5.1's description of SR 11-7 and the NIST AI RMF, map each of this chapter's twelve worked examples to the framework function (govern, map, measure, manage) it most directly supports, and justify any example you assign to more than one function.

18.18 Compare Section 18.3.2's exact, closed-form Shapley decomposition to Section 18.3.5's sampled, approximate LIME surrogate, and explain the specific structural property of the credit-scoring model (as opposed to the toy neural network) that makes the closed form available in the first place.

18.19 Using Section 18.6's method, compute the algorithmic-recourse cost of a third counterfactual path that changes both debt-to-income and credit history by equal fractions of their individually required changes (that is, half of the debt-to-income path and half of the credit-history path combined), and compare its total weighted cost to the two single-feature paths in the text.

18.20 Using Section 18.6's integrated-gradients derivation, recompute IG_1 and IG_2 using a baseline of $x' = (0.5, 0.25)$ instead of $(0, 0)$, and verify that the two attributions still sum to $\hat{y}(x_0) - \hat{y}(x')$ at this new baseline, illustrating that such an attribution is defined only relative to a chosen reference point.

18.21 Using Section 18.4.2's calibration-in-the-large check, recompute each group's average predicted default probability if Alpha-3's predicted probability were instead 60% rather than 95.5% (holding every other applicant's prediction fixed), and state whether Group Alpha's calibration gap (average predicted probability versus its 50% actual default rate) widens or narrows as a result.

18.22 Explain why the fact that Group Alpha and Group Beta share an identical 50% actual default rate in Table 18.5 does not, by itself, guarantee that a model calibrated on the combined population will also be equally calibrated within each group separately.

18.23 A bank plans an agent that reconciles failed trades: it reads the trade and settlement records, identifies the cause of each break, drafts a correcting entry, and books entries below $10,000 itself. Using Section 18.5.2 and Table 18.7, write the agent's mandate (tasks, tools, data, value thresholds, owner), classify each of its tools as read-only, reversible, or irreversible, and specify which actions require human approval and what the action log must record for an auditor to reconstruct a booked entry a year later.

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18.24 The vendor that supplies the foundation model under the agent in the previous exercise announces that the model version the bank uses will be retired in ninety days. Using Section 18.5.2's discussion of change management and third parties, describe the steps the bank should take before switching to the new version, and explain why SR 26-2's exclusion of agentic AI does not relieve it of doing so.

18.25 Using Section 18.6.4's rule, find the range of predicted default probabilities $\hat{p}$ for which the prediction set is {repay}, {default}, or {repay, default} under the pooled quantile $\hat{q} = 0.7025$, and again under the thin-file quantile $\hat{q} = 0.7795$. Explain why the wider thin-file quantile raises the share of thin-file applicants referred to human review, and why no applicant's set can be empty when $\hat{q} \geq 0.5$.

18.26 Chapter 6's Section 6.4.5 showed that, given one calibration set, coverage follows a Beta(k, $n+1-k$) distribution. (a) Using Section 18.6.4's group sizes, n = 1,624 thick-file and n = 376 thin-file calibration applicants at $\alpha = 0.10$, compute k for each group and the mean and standard deviation of the corresponding Beta distribution (a Beta(a, b) variable has mean $a/(a+b)$ and variance $ab/[(a+b)^2(a+b+1)]$). (b) Explain what the difference in standard deviations means for a validator who checks thin-file coverage on a single calibration set. (c) Explain why a lender should not instead calibrate separately by race or sex, and what it should do to check coverage across those groups.

18.27 **Challenge (real data).** Download a recent year of HMDA (Home Mortgage Disclosure Act) loan application register data for a single large lender (freely available from the Consumer Financial Protection Bureau's HMDA data browser, no login required). (a) Using Section 18.4.2's adverse impact ratio formula, compute the approval-rate ratio between two applicant groups of your choosing (for example, by race or ethnicity, both reported in the data), and state whether the four-fifths rule is violated, as it was in Table 18.5's toy example. (b) Fit a simple logistic regression predicting loan approval from the legitimate underwriting variables HMDA reports (income, loan amount, debt-to-income category), and recompute the adverse impact ratio using the model's predictions instead of the raw historical approval decisions, discussing whether the disparity narrows, widens, or stays about the same. (c) HMDA data reports the reason lenders gave for each denial; compare the stated denial reasons for the two groups you chose, and discuss whether this real, lender-reported information changes your interpretation of any disparity found in part (a), relative to the toy example in the text, where no such reason codes were available to explain the underlying gap.

18.10 Further Reading

- Lundberg and Lee (2017), "A Unified Approach to Interpreting Model Predictions." The original SHAP paper unifying Shapley-value-based feature attribution, underlying Section 18.3.2.
- Ribeiro et al. (2016), ""Why Should I Trust You?": Explaining the Predictions of Any Classifier." The original LIME paper underlying Section 18.3.5.
- Shapley (1953), "A Value for n-Person Games." The original cooperative-game-theory paper defining the Shapley value that Section 18.3.2 applies to a single credit decision.
- Molnar (2022), *Interpretable Machine Learning*. A comprehensive, freely available treatment of permutation importance, partial dependence, Shapley values, and LIME, covering the full taxonomy in Figure 18.1.
- Doshi-Velez and Kim (2017), *Towards a Rigorous Science of Interpretable Machine Learning*. A foundational discussion of what interpretability means and how to evaluate it rigorously, underlying the distinction drawn in Section 18.2.1.
- Barocas et al. (2023), *Fairness and Machine Learning*. A comprehensive treatment of statistical fairness definitions, including disparate impact and equalized odds, underlying Section 18.4.2.
- Chouldechova (2017), "Fair Prediction with Disparate Impact: A Study of Bias in Recidivism Prediction Instruments." One of the two papers establishing the impossibility of simultaneously satisfying calibration and error-rate parity across groups, cited in Section 18.4.2.

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- Kleinberg et al. (2017), "Inherent Trade-Offs in the Fair Determination of Risk Scores." The companion impossibility result to Chouldechova (2017), cited in Section 18.4.2.
- Dwork et al. (2012), "Fairness Through Awareness." The original individual-fairness definition discussed as a third fairness paradigm in Section 18.4.2.
- Board of Governors of the Federal Reserve System and Office of the Comptroller of the Currency (2011), *Supervisory Guidance on Model Risk Management*. The foundational U.S. bank regulatory guidance on model risk management underlying Section 18.5.1.
- Federal Reserve, OCC, and FDIC (2026), *Revised Guidance on Model Risk Management*. The April 2026 interagency guidance that superseded SR 11-7, discussed in Section 18.4.1.
- EU (2024), *Regulation (EU) 2024/1689 Laying Down Harmonised Rules on Artificial Intelligence (Artificial Intelligence Act)*. The EU regulation whose high-risk obligations for credit scoring and insurance pricing are summarized in Table 18.4.
- NIST (2023), *Artificial Intelligence Risk Management Framework (AI RMF 1.0)*. The govern/map/measure/manage framework discussed at the close of Section 18.5.1.
- FINRA (2025), *2026 FINRA Annual Regulatory Oversight Report*. The first U.S. supervisory report to treat AI agents directly, listing the autonomy, scope-and-authority, and auditability risks discussed in Section 18.5.2.
- IMDA (2026), *Model AI Governance Framework for Agentic AI*. Singapore's voluntary governance framework for agentic AI, organized around bounding risks, human accountability, technical controls, and end-user responsibility, discussed in Section 18.5.2.
- Goodfellow et al. (2015), "Explaining and Harnessing Adversarial Examples." The foundational paper on adversarial perturbations in deep learning, contrasted with the fraud-model perturbation in Section 18.4.4.
- NYDFS (2021), *Report on Apple Card Investigation*. The regulatory report underlying the case vignette in Section 18.4.3.
- Ustun et al. (2019), "Actionable Recourse in Linear Classification." The foundational algorithmic-recourse paper underlying the cost-weighted counterfactual comparison in Section 18.6.
- Karimi et al. (2021), "Algorithmic Recourse: From Counterfactual Explanations to Interventions." The argument that recourse should be computed as a minimal intervention on a causal model of the applicant's features rather than as the nearest counterfactual, behind the causal discussion in Section 18.6.1.
- Kusner et al. (2017), "Counterfactual Fairness." The original causal counterfactual-fairness definition discussed in Section 18.6.
- Sundararajan et al. (2017), "Axiomatic Attribution for Deep Networks." The original integrated-gradients paper underlying Section 18.6's worked example on Chapter 6's network.
- Jain and Wallace (2019), "Attention is not Explanation." The empirical case against reading Chapter 16's attention weights directly as explanations, discussed in Section 18.6.
- Turpin et al. (2023), "Language Models Don't Always Say What They Think: Unfaithful Explanations in Chain-of-Thought Prompting." The empirical study of unfaithful LLM-generated rationales discussed in Section 18.6.
- Angelopoulos and Bates (2023), "Conformal Prediction: A Gentle Introduction." A practical introduction to conformal prediction, including prediction sets for classifiers and group-conditional coverage, extending Chapter 6's split-conformal method to Section 18.6.4's credit decisions.
- CFPB (2017), *CFPB Announces First No-Action Letter to Upstart Network*. The regulatory announcement underlying the case vignette in Section 18.5.3.

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Chapter 19

Reinforcement Learning in Finance

19.1 Introduction

Every model built so far in this book, however sophisticated, has answered a single question once. Chapter 6's classifiers predict a label from a feature vector; Chapter 9's credit models predict a default probability; Chapter 13's Almgren-Chriss trajectory prescribes an entire trading schedule, but it does so by solving one optimization problem in advance and then executing the resulting plan. None of these approaches change their answer in response to how the world responds to the decisions already made.

A portfolio manager, a market maker, and a trading desk do not operate this way: they observe an outcome, update their beliefs, and choose their next action accordingly, over and over, for as long as the position stays open. Reinforcement learning is the branch of machine learning built for this setting, sequential decision-making under uncertainty, where an agent's own actions influence the future states it will face, and where the right action today depends on the consequences it sets up for tomorrow, not only on today's immediate reward.

This chapter formalizes that setting, the Markov decision process, and develops the two families of algorithms built to solve it: value-based methods, culminating in Q-learning and its neural-network extension, and policy-based methods, culminating in the policy-gradient algorithms that power most modern applications. It builds on machinery this book has already developed: Chapter 5's binomial-tree early-exercise decision, Chapter 12's portfolio rebalancing problem, and Chapter 13's optimal-execution and market-making models are all special cases of the general sequential-decision problem, simple enough that a closed-form or backward-induction solution exists. Reinforcement learning is what a practitioner reaches for when the problem is still sequential but no longer simple enough for a closed form, a common situation once market impact, transaction costs, and a client's actual risk preferences no longer fit the clean functional forms those chapters assumed.

The chapter's central worked example makes this connection concrete. It reframes Chapter 5's American put early-exercise decision as a three-state Markov decision process, solves it with model-free Q-learning instead of backward induction, and shows that the learned answer converges to the value the tree already established: a small, verifiable proof that reinforcement learning recovers a known answer, before the chapter asks the reader to trust it on problems with no closed form.

The chapter proceeds in three stages. Sections 19.2.1 through 19.2.6 build the Markov decision process framework, the Bellman equation, and the vocabulary (model-based versus model-free, on-policy versus off-policy) needed to place every algorithm that follows. Sections 19.3.1 through 19.3.3 develop the algorithms, from the verifiable tabular Q-learning example to deep Q-networks and policy-gradient methods for problems too large for a table. The remaining sections turn to application and assessment: optimal execution, market making, and portfolio management (Sections 19.4.1 through 19.4.6), an industry case study (Section 19.5.1), and the evaluation methodology and risks (Sections 19.5.2 and 19.6) that separate a system that merely impresses in a backtest from one that is safe to deploy with real capital.

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19.2 The Reinforcement Learning Framework

This part sets up the sequential decision problem and the value functions that make it tractable.

19.2.1 Markov Decision Processes: States, Actions, Rewards, and Policies

Consider a market maker deciding, moment by moment, how to skew its bid and ask quotes as its inventory evolves, or a trader deciding how many shares of a large order to release this minute versus holding back for a better price later. In both cases today's decision changes tomorrow's starting position, so the decisions cannot be solved one at a time, as most models in Chapters 12 and 13 solve them. A reinforcement learning problem is formalized as a Markov decision process, a tuple $(\mathcal{S}, \mathcal{A}, P, R, \gamma)$, to capture this kind of interlocking, path-dependent problem.

At each discrete time step t, an **agent** observes a **state** $s_t \in \mathcal{S}$, a summary of everything about the environment relevant to the decision (a market maker's current inventory and the prevailing bid-ask spread; a trading agent's remaining order size and elapsed time; a portfolio manager's current holdings and estimated risk factors). The agent selects an **action** $a_t \in \mathcal{A}$ (post a quote at a given skew; sell a given number of shares this period; rebalance toward a given target weight) according to a **policy** $\pi(a \mid s)$, the probability of taking action a in state s, which maps each state to a distribution over the available actions with $\sum_a \pi(a \mid s) = 1$. A policy that puts all its probability on one action per state is deterministic: the agent always does the same thing from the same state.

The environment then transitions to a new state s_{t+1} according to a transition function $P(s_{t+1} \mid s_t, a_t)$ and returns a scalar **reward** r_{t+1} (realized profit and loss net of costs; a small quadratic risk penalty; an exercise payoff), governed by a reward function $R(s_t, a_t, s_{t+1})$. Figure 19.1 depicts this loop; it repeats until the episode ends, either at a fixed horizon (the market closes, the order is fully executed) or when the agent reaches an absorbing state (a position is liquidated, an option is exercised).

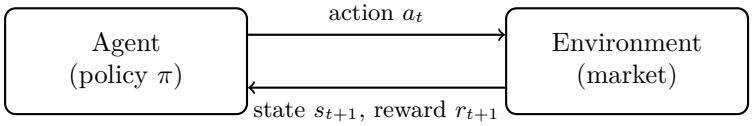


Figure 19.1: The reinforcement learning loop

The word "Markov" is a precise mathematical assumption: $P(s_{t+1} \mid s_t, a_t)$ and $R(s_t, a_t, s_{t+1})$ depend only on the current state and action, not on the history that led there. Every stochastic process this book has used to model prices makes the same assumption, from the binomial tree's independent up-and-down moves in Chapter 5 to the random-walk and GARCH innovations of Chapter 7; it is a familiar assumption under a new name, now applied to an agent's decisions as well as to prices. Whether a proposed state representation is Markovian is itself a design choice: a market maker's state that includes only the current mid-price, and not recent order flow, discards information a real market maker would use, which is why Chapter 14's order-flow-imbalance regression and Chapter 13's Kyle's-lambda framework are natural sources of state features for a trading agent, not only standalone predictive signals.

The agent's goal is to choose a policy that maximizes the expected **return**, the discounted sum of future rewards,

$$G_t = r_{t+1} + \gamma r_{t+2} + \gamma^2 r_{t+3} + \cdots = \sum_{k=0}^{\infty} \gamma^k r_{t+k+1}, \tag{19.1}$$

where the **discount factor** $\gamma \in [0, 1]$ plays the role of the discount factor $1/(1+r)$ in Chapter 5's valuation formulas: a reward received later is worth less today, both because of the time value of money and because a distant reward is a noisier, harder-to-credit consequence of the current action. A myopic agent with $\gamma = 0$ maximizes only the immediate reward, the one-shot optimization earlier chapters' models perform; the closer γ is to 1, the more the current choice is shaped by consequences many steps away, the structure a sequential problem like optimal execution or ongoing portfolio management requires and a one-shot model cannot represent.

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19.2.2 From One-Shot Optimization to Sequential Decisions

Why the models of Chapters 12 and 13 did not need this machinery clarifies what reinforcement learning adds. The Almgren-Chriss optimal-execution model (Chapter 13's Section 13.3.5) assumes a known functional form for market impact, quadratic in the trade rate, and a known risk-aversion parameter ϕ; given those, the optimal trajectory follows in closed form in one step, with nothing to learn from experience, because the model already specifies how the environment responds to every action. Chapter 13's Avellaneda-Stoikov inventory model (Section 13.4) is more dynamic, recomputing the market maker's quotes at every instant as inventory and time-to-close evolve, but its update rule is a closed-form formula derived from an assumed order-arrival process, not a policy learned by trial and error. Chapter 12's Section 12.4.1 allocation discussion goes furthest toward the sequential nature of real portfolio management, noting that rebalancing recurs and that transaction costs make each decision depend on the current position, but it treats each rebalancing as a separate, myopic optimization. Each model is right within its assumptions and remains the correct tool whenever they hold: a well-estimated impact function and a well-specified risk-aversion parameter make a closed-form solution more accurate and far cheaper to compute than a learned one.

Reinforcement learning becomes the better tool when that closed form is unavailable: when the true market-impact function fits no simple parametric form, when transaction costs interact with the portfolio's history in ways too complex to solve analytically, or when the reward the agent cares about, a client's realized utility net of taxes, fees, and tolerance for drawdowns, does not reduce to a quadratic cost. In each case the problem is still a sequential decision problem as formalized in Section 19.2.1; what changes is that no one can write down P and R in closed form, which rules out the direct optimization of Chapters 12 and 13 and motivates learning a policy from simulated or historical experience.

19.2.3 The Simplest Case: Multi-Armed Bandits

A trading desk sending orders to market faces a choice most descriptions of execution skip: which venue to send each order to. A large-cap U.S. equity trades on a dozen or more venues, and the same order routed to two of them comes back at different prices, because each venue has its own resting liquidity, fee schedule, and population of counterparties. A desk that knew each venue's average price improvement would route everything to the best one. No desk knows this, and the only way to find out what a venue delivers is to send it orders, so the desk pays, in worse fills, for every order spent learning rather than earning.

This tension separates reinforcement learning from Chapters 6 and 7. A supervised model is handed a fixed dataset and never influences what it sees next; here the agent's own choices determine which data it observes, so an action that looks bad early may never be tried again and its true value never discovered. The problem in its purest form is the **multi-armed bandit**, named for a row of slot machines with unknown payout rates: at each step the agent picks one of K actions, called **arms**, and receives a noisy reward drawn from that arm's fixed but unknown distribution.

The bandit is the right place to start because of what it leaves out. There is one state, which never changes; the reward depends only on the action just taken; and no action moves the agent anywhere. Section 19.2.5's Bellman equation, with its discounting, bootstrapping, and credit assignment across a trajectory, is unnecessary, because $Q(a)$ is the mean reward of arm a, estimable by averaging. The one difficulty left is the one this chapter meets everywhere: the **exploration-exploitation trade-off**, choosing between the arm that currently looks best and one whose value is still uncertain.

19.2.4 Worked Example 19.1: Routing Orders Across Three Venues

A desk routes each arriving order to one of three venues. Measured in basis points of price improvement against the arrival mid-price, venue A delivers a mean of 1.2, venue B a mean of 2.0, and venue C a mean of 0.5, each with a standard deviation of 2.0 basis points. Those means are stated here so the example can be checked; the desk does not know them, and must infer them from the fills it receives. Since the standard deviation of a single fill is 2.5 times the 0.8 basis-point gap between the best venue and the second best, one order tells the desk almost nothing.

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The desk keeps an estimate $Q(a)$ of each venue's mean and a count $N(a)$ of orders sent there, updating after each fill by the running-average rule

$$Q(a) \leftarrow Q(a) + \frac{1}{N(a)}\big[r - Q(a)\big], \tag{19.2}$$

which is the sample mean of the rewards seen from a, written so it can be updated one fill at a time without storing them. After sending one order to each venue to get started, it follows an ε-greedy policy with $\varepsilon = 0.2$: with probability 0.2 route to a uniformly random venue, otherwise route to whichever venue currently has the highest Q. Table 19.1 traces the first twelve orders.

Order	Rule	Venue	Fill	$Q(A)$	$Q(B)$	$Q(C)$
1	warm-up	A	1.45	1.450	—	—
2	warm-up	B	1.74	1.450	1.740	—
3	warm-up	C	1.78	1.450	1.740	1.780
4	explore	A	1.92	1.685	1.740	1.780
5	exploit	C	2.39	1.685	1.740	2.085
6	exploit	C	−2.03	1.685	1.740	0.713
7	exploit	B	2.08	1.685	1.910	0.713
8	exploit	B	1.56	1.685	1.793	0.713
9	exploit	B	0.54	1.685	1.480	0.713
10	exploit	A	0.57	1.313	1.480	0.713
11	exploit	B	4.09	1.313	2.002	0.713
12	explore	C	3.23	1.313	2.002	1.342

Table 19.1: The first twelve orders of the venue-routing bandit. Fills and estimates are in basis points of price improvement; a dash marks a venue not yet tried. Every estimate follows from the running-average rule applied to the fills above it, so each row can be checked by hand.

The opening rounds show the failure mode that makes exploration necessary. The single warm-up order to venue C returned 1.78 basis points, a good fill drawn by chance from the *worst* of the three venues, and that one lucky draw made C the greedy choice. Orders 5 and 6 went there. What rescued the desk was not a clever exploration rule but repetition: order 6 came back at -2.03, dragging $Q(C)$ from 2.085 down to 0.713 and demoting C below both rivals. Exploiting a wrong estimate long enough tends to correct it, because every exploit is also a sample. By order 12 the desk has $Q(B)$ highest and is routing to the best venue, though $Q(A) = 1.313$ and $Q(C) = 1.342$ remain badly estimated from three and four orders respectively.

Run the same policy for 5,000 orders and the estimates sharpen to $Q(A) = 1.25$, $Q(B) = 1.99$, and $Q(C) = 0.40$, against true means of 1.2, 2.0, and 0.5. The desk routes 86.2% of its orders to B and realizes an average of 1.826 basis points per order.

That average is visibly short of the 2.0 basis points a desk that knew the answer would earn. The shortfall, 868 basis points in total across the 5,000 orders, is the **regret**: what learning cost, measured against an oracle that never had to learn. Most of it comes not from the early mistakes, which are cheap and brief, but from the ε-greedy rule itself, which keeps routing one order in five at random long after the answer is settled. That permanent tax is

$$\varepsilon \times \big(2.0 - \tfrac{1}{3}(1.2 + 2.0 + 0.5)\big) = 0.2 \times 0.7667 = 0.1533 \text{ basis points per order},$$

which is why the asymptotic average is $0.8 \times 2.0 + 0.2 \times 1.2333 = 1.8467$ rather than 2.0, and why the realized 1.826 sits just below it. The remedy is to explore hard early and taper off: decaying ε over the same 5,000 orders lifts the share routed to B from 86.2% to 95.2% and the average fill from 1.826 to 1.943 basis points, while keeping the early exploration that found B. Every later worked example in this chapter uses a decayed exploration schedule for this reason.

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Exploring Where It Pays: Upper Confidence Bounds and Thompson Sampling

Decaying ε fixes *how often* the desk explores but not *where*: an exploratory order is still as likely to go to venue C, which the data condemned long ago, as to venue A, which might yet be competitive. Two better rules direct exploration toward the venues whose value is still uncertain. The **upper confidence bound** (UCB) rule of Auer et al. (2002) routes each order to the venue with the largest optimistic estimate,

$$a_t = \arg\max_a \left[Q(a) + \sigma \sqrt{\frac{2 \ln t}{N(a)}} \right], \tag{19.3}$$

where $\sigma = 2.0$ basis points is the fill noise and t the order number. The bonus is large for a venue sampled rarely and shrinks as $N(a)$ grows, so a venue is revisited only for as long as its plausible upside could still beat the leader; the slowly growing $\ln t$ guarantees that no venue is abandoned forever on the strength of a few unlucky fills. **Thompson sampling** takes a Bayesian route to the same end (Russo et al. 2018). Under a flat prior and the known noise σ, the posterior for venue a's mean is normal with mean $Q(a)$ and standard deviation $\sigma/\sqrt{N(a)}$; before each order the desk draws one plausible mean from each venue's posterior and routes to the largest draw. A venue therefore receives each order with exactly the probability that it is the best venue given the fills so far, so exploration fades on its own as the posteriors tighten.

Run on the same three venues, with the same warm-up and the same seed as the ε-greedy desk, the two rules route 96.7% (UCB) and 98.5% (Thompson sampling) of 5,000 orders to venue B, against 86.2% for fixed and 95.2% for decayed ε, and earn average fills of 1.961 and 1.971 basis points against 1.826 and 1.943. Realized regret falls from 868 basis points to 286 (decayed ε), 195 (UCB) and 144 (Thompson sampling). A single run mixes the policy's choices with the luck of the fills, so Figure 19.2 averages the *expected* regret, the sum over orders of the gap between venue B's mean and the chosen venue's mean, across 200 seeds. The fixed-ε curve is a straight line: over the last 1,000 orders it adds 0.152 basis points per order, the permanent 0.1533 tax computed above. Decayed ε bends but is still adding 0.028 per order at the end, while UCB and Thompson sampling add only 0.005 and 0.003, finishing at 103 and 74 basis points of cumulative regret against 794 for fixed ε. Their curves flatten in the way theory predicts: Lai and Robbins (1985) showed that no rule can keep regret growing more slowly than the logarithm of the number of rounds, and UCB and Thompson sampling both achieve that logarithmic rate (Lattimore and Szepesvári 2020).

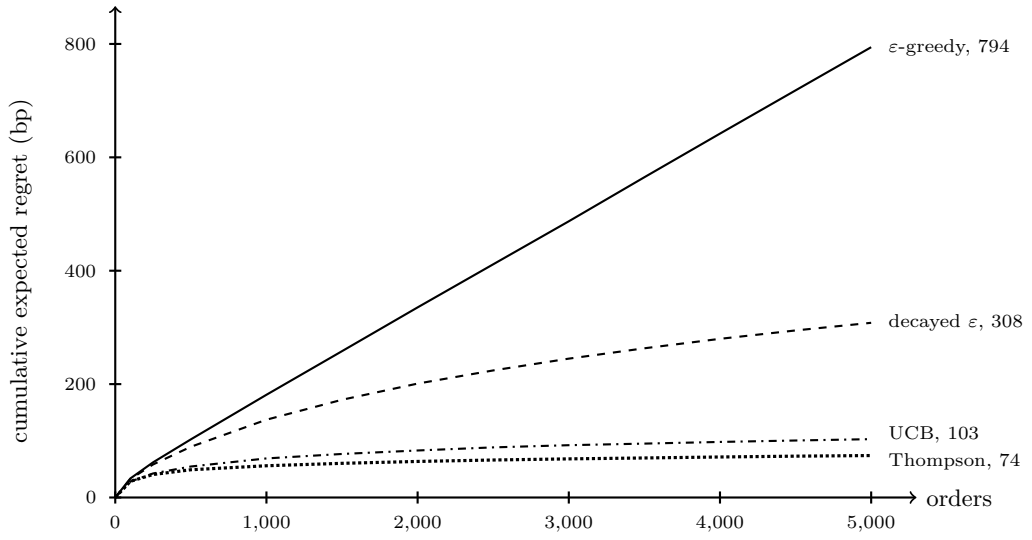


Figure 19.2: Cumulative expected regret of four venue-routing rules, averaged over 200 seeds of 5,000 orders each. Fixed ε (solid) grows linearly; decayed ε (dashed) bends; UCB (dash-dot) and Thompson sampling (dotted) flatten toward logarithmic growth.

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When the Best Venue Depends on the Order: Contextual Bandits

The three-venue desk assumed that one venue is best for every order. In practice the answer depends on what the desk observes when the order arrives: a small order may do best on a venue with tight quotes but little depth, a large one on a venue whose hidden liquidity absorbs size, and the ranking can shift with the spread, the time of day, or recent volatility. A **contextual bandit** lets the reward depend on a feature vector x_t observed before acting and learns a value $Q(a, x)$ for each action as a function of context, typically a regression per arm such as $Q(a, x) = \theta_a^\top x$. The exploration ideas carry over: LinUCB, introduced by Li et al. (2010) for news-article recommendation, adds to each arm's regression prediction a bonus proportional to that prediction's standard error, which is equation (19.3) with the regression's uncertainty in place of $\sigma/\sqrt{N(a)}$. A contextual bandit is still not a full reinforcement learning problem, since the context arrives from outside and today's routing decision does not change tomorrow's context. That makes it attractive for routing, pricing, and recommendation decisions that recur thousands of times a day, and it is also its limit: once an action changes the next state, as when a large order's impact moves the price the rest of the order will face, the problem needs the machinery of the next subsection.

Two threads run forward from here. Add a next state, so that today's action changes tomorrow's situation, and averaging is no longer enough: the agent must reason about consequences that arrive later, which Section 19.2.5's Bellman equation supplies and Section 19.3.1's Q-learning example puts to work. Keep the single state but shrink the gap between arms relative to the noise, and even this simplest problem becomes hard: Section 19.4.4 returns to a bandit whose best and second-best arms differ by so little against so much noise that 200,000 rounds still leave the agent on the wrong one.

19.2.5 Value Functions and the Bellman Equation

Suppose two agents follow the same policy but start from different states, one with a small, comfortable inventory and one deep in an overextended position: the policy alone cannot say how much trouble each is in, only what each plans to do next. Two functions answer this by summarizing how good a state, or state-action pair, is under a policy π: the **state-value function** $V^\pi(s) = \mathbb{E}_\pi[G_t \mid s_t = s]$, the expected return starting from state s and following π thereafter, and the **action-value function** $Q^\pi(s, a) = \mathbb{E}_\pi[G_t \mid s_t = s, a_t = a]$, the expected return from taking action a in state s and following π afterward.

Because the return decomposes into an immediate reward plus a discounted continuation, both functions satisfy a recursive relationship, the **Bellman equation**, named for Richard Bellman's original work on dynamic programming:

$$V^\pi(s) = \sum_a \pi(a \mid s) \sum_{s'} P(s' \mid s, a) \Big[R(s, a, s') + \gamma V^\pi(s') \Big]. \tag{19.4}$$

The **Bellman optimality equation** specializes this to the best possible policy, replacing the expectation over actions with a maximization:

$$Q^*(s, a) = \sum_{s'} P(s' \mid s, a) \Big[R(s, a, s') + \gamma \max_{a'} Q^*(s', a') \Big]. \tag{19.5}$$

A reader who worked through Chapter 5's two-step binomial tree (Section 5.4.2) has already solved a Bellman optimality equation by hand under another name. At each node, backward induction compares the value of exercising immediately with the discounted expected value of continuing, and takes the larger: the $\max_{a'}$ above, applied to the actions "exercise" and "hold." Section 19.3.1 makes the correspondence exact by reformulating that tree as a Markov decision process and re-deriving the same option value. Dynamic programming methods, value iteration and policy iteration, solve the Bellman equation directly whenever P and R are fully known, as Chapter 5's tree does. Reinforcement learning's model-free methods, developed next, estimate Q^* from sampled transitions and rewards alone, which makes them applicable where, as in most real trading and portfolio problems, no one can specify P and R in closed form.

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19.2.6 A Taxonomy of Reinforcement Learning Methods

Two further distinctions organize the many reinforcement learning algorithms in use, and fixing them now makes it easier to place each method this chapter introduces. Figure 19.3 previews where each fits; π_θ there denotes a policy with tunable parameters θ, the form Section 19.3.3 develops.

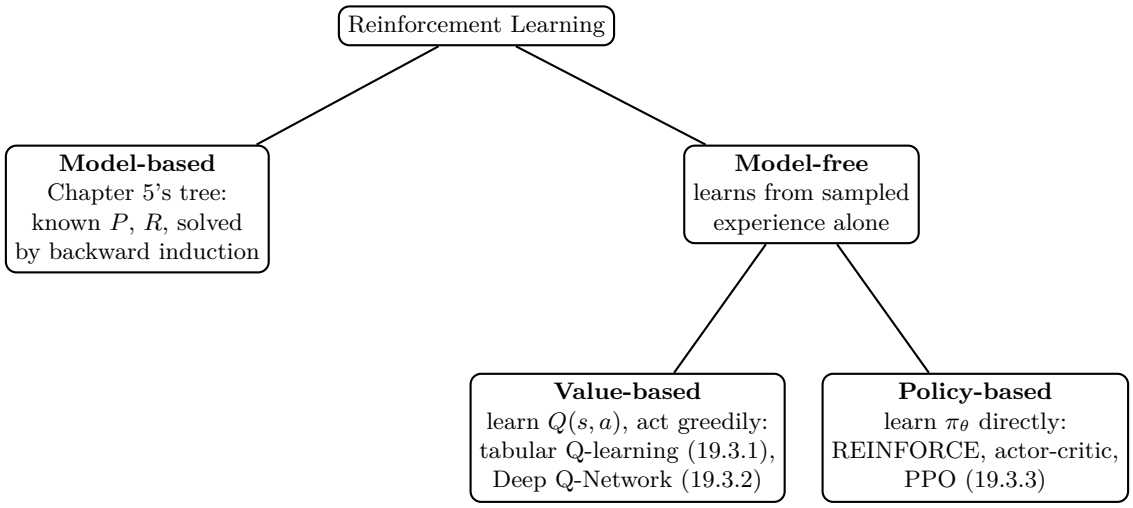


Figure 19.3: A map of the reinforcement learning methods this chapter develops

Model-based methods have access to, or explicitly learn, the transition function P and reward function R, and use them to plan. Chapter 5's binomial tree is model-based: q, u, d, and every terminal payoff are known in advance, so backward induction computes the exact optimal value with no simulated experience. **Model-free methods** never estimate P or R; they learn a value function or a policy from sampled transitions and rewards, treating the environment as a black box that can be queried (by simulating an episode) but not inspected. Every method in the rest of this chapter, tabular Q-learning, deep Q-networks, and the policy-gradient family, is model-free, which is why each remains usable in the settings Section 19.2.2 described, where no closed-form P and R exist for a real limit order book or a real client's multi-period utility.

A third, hybrid category, sometimes called **learned-model-based methods**, trains an approximate model of P and R from data (rather than assuming it, as the tree does) and plans against it, gaining some of model-based planning's sample efficiency without requiring P and R in closed form. This approach underlies some of reinforcement learning's most prominent successes outside finance, including game-playing systems that combine a learned model with tree search, but it introduces the sim-to-real gap Section 19.5.2 discusses, since a plan is only as good as the learned model behind it.

Off-policy methods learn about a policy different from the one actually generating their training data. Q-learning's update rule is a clear example: its bootstrapped target uses $\max_{a'} Q(s', a')$, the value of the best available action, regardless of which action the ε-greedy behavior policy, one that takes a random action with probability ε and the currently best-valued action otherwise, actually selected next. This lets a single stream of exploratory, partly random experience be reused to learn about the greedy-optimal policy while the agent is still exploring, which is why Section 19.3.1's worked example converges to the correct answer despite spending a meaningful fraction of its 100,000 episodes taking random actions. Off-policy methods have a clear data-efficiency advantage, since one batch of experience can be reused across many rounds of learning, which is why DQN's replay buffer (Section 19.3.2) works at all.

On-policy methods instead evaluate and improve the policy being followed. The classic on-policy analogue of Q-learning, SARSA (State-Action-Reward-State-Action), replaces $\max_{a'} Q(s', a')$ with $Q(s', a')$ for the action a' the policy selects next, so its learned values reflect the still-exploring policy rather than the greedy target. Policy-gradient methods (Section 19.3.3) are typically on-policy too, since their gradient estimator is valid only for returns experienced under the current policy π_θ. On-policy methods are generally

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simpler to reason about and more stable to train, since they never correct for a mismatch between the policy that generated the data and the one being evaluated.

19.3 Methods

From a lookup table to a neural policy, this part builds the algorithms in order of the problems that forced each one.

19.3.1 Tabular Q-Learning: Recovering Chapter 5's Early-Exercise Decision

This subsection works tabular Q-learning on a problem whose answer is known: Chapter 5's American put early-exercise decision, solved first by backward induction and then learned from simulated experience alone.

Setting Up the Markov Decision Process

Chapter 5's American put example used a two-step binomial tree with $S_0 = K = \$50$, up and down factors $u = 1.2$ and $d = 0.9$, and a gross risk-free return $R = 1.04$ per period, giving a risk-neutral up-probability

$$q = \frac{R - d}{u - d} = \frac{0.14}{0.30} \approx 0.4667$$

and a per-step discount factor $\gamma = 1/R \approx 0.9615$.

That tree has three decision points: today, at $S_0 = \$50$; after one up move, at $S_u = \$60$; and after one down move, at $S_d = \$45$. At each of these three states the holder chooses between two actions: **exercise**, receive the intrinsic value $\max(K - S, 0)$ immediately and end the episode, or **hold**, receive nothing now and let the tree evolve one more step.

The two terminal nodes reachable from the up state, $S_{uu} = \$72$ and $S_{ud} = \$54$, both pay $\max(50 - S, 0) = \$0$; of the two reachable from the down state, S_{ud} again pays \$0, while $S_{dd} = \$40.50$ pays $\max(50 - 40.50, 0) = \$9.50$.

Together these define a Markov decision process with states $\mathcal{S} = \{S_0, S_u, S_d\}$ and actions $\mathcal{A} = \{\text{exercise}, \text{hold}\}$. Transition probabilities are q and $1 - q$, rewards come from the intrinsic-value and terminal-payoff structure above, and $\gamma = 1/R$ discounts every step as Chapter 5's tree did.

Figure 19.4 redraws Chapter 5's tree in this chapter's language, labeling each decision node with its two available actions rather than with a single option value.

Chapter 5 solved this problem by backward induction, working from the known terminal payoffs back to today. Restated in this chapter's notation, that solution is

$$\begin{aligned}
Q^*(S_u, \text{exercise}) &= \$0.00, & Q^*(S_u, \text{hold}) &= \gamma\big[q(0) + (1-q)(0)\big] = \$0.00,\\
Q^*(S_d, \text{exercise}) &= \$5.00, & Q^*(S_d, \text{hold}) &= \gamma\big[q(0) + (1-q)(9.50)\big] \approx \$4.8718,\\
Q^*(S_0, \text{exercise}) &= \$0.00, & Q^*(S_0, \text{hold}) &= \gamma\big[q\,V^*(S_u) + (1-q)\,V^*(S_d)\big] \approx \$2.5641,
\end{aligned}$$

where $V^*(S_u) = \max(0.00, 0.00) = \$0.00$ and $V^*(S_d) = \max(5.00, 4.8718) = \$5.00$, since exercising is optimal at the down node. The optimal policy is therefore to hold at S_0 and S_u and exercise at S_d, giving an American put value of $V^*(S_0) \approx \$2.5641$, the value Chapter 5 reports.

The Q-Learning Update Rule

Backward induction requires knowing q, γ, and every terminal payoff in advance, the closed-form knowledge Section 19.2.2 argued is unavailable in most real trading and portfolio problems. **Q-learning** instead estimates Q^* from simulated experience alone, never using q in its update rule.

Starting from an arbitrary initial table $Q(s, a)$ (here, all zeros), the agent repeatedly simulates an episode: at each state it chooses an action using an ε-greedy policy (with probability ε, act randomly to keep exploring;

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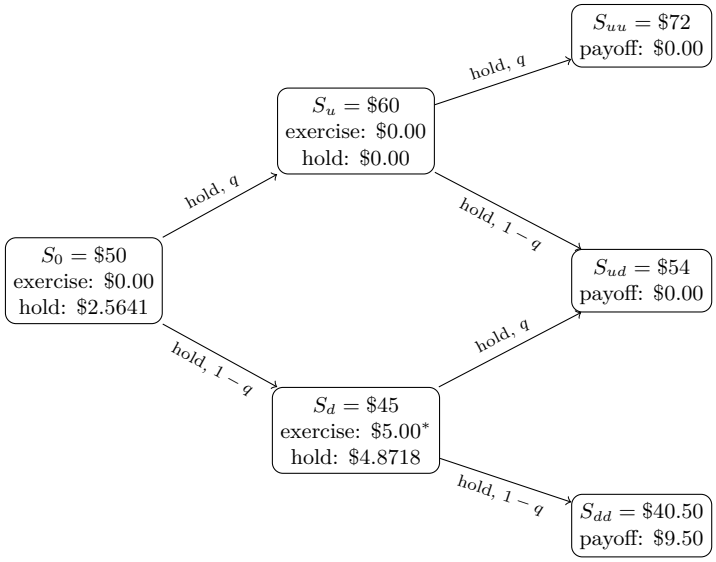


Figure 19.4: The early-exercise decision of Chapter 5's American put, redrawn as a three-state Markov decision process, with the optimal action at each node marked by an asterisk

otherwise, act greedily with respect to the current Q-table), observes the resulting reward r and next state s', and updates

$$Q(s,a) \leftarrow Q(s,a) + \alpha\Big[r + \gamma \max_{a'} Q(s',a') - Q(s,a)\Big], \tag{19.6}$$

where α is a learning rate. This update rule, **temporal-difference learning**, nudges $Q(s,a)$ toward a bootstrapped target, the observed reward plus the discounted value of the best action available in the next state, without requiring a model of the transition probabilities.

The agent learns q's effect on the optimal decision by observing many simulated up and down moves and seeing which actions paid off, learning a relationship from data as every other machine learning method in this book does.

The First Two Learning Steps

Starting from a Q-table initialized entirely to zero, with a learning rate of $\alpha = 0.1$, suppose the first simulated episode reaches S_d, chooses **hold** (all actions still look equally worthless, so the choice is effectively random), and the simulated down-move occurs, landing on the terminal payoff of \$9.50 at S_{dd}. Since this transition ends the episode, the bootstrapped target's continuation term is zero, and the update is

$$Q(S_d, \text{hold}) \leftarrow 0 + 0.1\big[9.50 + \gamma(0) - 0\big] = 0.1(9.50) = \$0.9500,$$

the first nonzero entry the table ever acquires. Now suppose a later episode reaches S_0, chooses **hold**, and the simulated move is down, transitioning to S_d with an immediate reward of \$0 (holding earns nothing until a terminal payoff is reached). The bootstrapped target now uses the current, already partially learned table, $\max_{a'} Q(S_d, a') = \max(0, 0.9500) = 0.9500$, giving

$$Q(S_0, \text{hold}) \leftarrow 0 + 0.1\big[0 + \gamma(0.9500) - 0\big] \approx 0.1(0.9615)(0.9500) \approx \$0.0913.$$

Two updates in, the table already shows the shape of the true solution, $Q(S_d, \text{hold})$ positive and $Q(S_0, \text{hold})$ smaller but nonzero, though both values are still far from their closed-form targets of \$4.8718 and \$2.5641.

The 100,000-episode simulation below repeats this mechanism: each terminal payoff nudges the value of the state that led to it, and that nudge propagates one more step backward the next time an earlier state is visited, diffusing the terminal reward's information back to S_0 without ever consulting q.

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The companion notebook simulates this process for 100,000 episodes, using a learning rate that decays with each state-action pair's visit count and an exploration rate ε that decays from 0.30 to 0.02 over training, standard choices that ensure enough early exploration to discover the exercise decision at S_d while converging to a stable, greedy policy. Table 19.2 compares the learned Q-table to the closed-form values derived above.

State-action	Closed-form Q^*	Learned Q (Q-learning)	Relative gap
$(S_0, \text{exercise})$	\$0.0000	\$0.0000	—
(S_0, hold)	\$2.5641	\$2.5328	1.22%
$(S_u, \text{exercise})$	\$0.0000	\$0.0000	—
(S_u, hold)	\$0.0000	\$0.0000	—
$(S_d, \text{exercise})$	\$5.0000	\$5.0000	0.00%
(S_d, hold)	\$4.8718	\$4.9234	1.06%

Table 19.2: Closed-form Bellman-optimal Q-values (from backward induction) versus tabular Q-learning's learned estimates after 100,000 simulated episodes

Comparing the Learned and Closed-Form Solutions

The learned policy matches the closed-form optimal policy exactly: hold at S_0 (\$2.5328 > \$0.0000), exercise at S_d (\$5.0000 > \$4.9234), and, at S_u, both actions are learned as worth \$0.0000, an exact tie that matches Chapter 5's own observation that "both exercise and continuation are worth \$0" at that node, so which action the tie-break happens to favor is economically inconsequential.

Every "exercise" value converges exactly, with no residual error, since exercising ends the episode immediately and its reward is observed without any noise; every "hold" value converges to within roughly 1% of the true value but not exactly, since a "hold" estimate is a bootstrapped average built from a finite number of noisy sampled continuations rather than an exact expectation computed from a known q.

This gap is not a bug to be engineered away; it is the fundamental price of model-free learning, and the most important lesson of this example: when the transition probabilities and payoffs are known in closed form, as they are in Chapter 5's tree, backward induction is strictly better, exact, instantaneous, and needs no simulated data, and reinforcement learning should not be used. Reinforcement learning earns its keep only in the more realistic setting, developed throughout the rest of this chapter, where no such closed form exists and simulated or historical experience is the only source of information available.

19.3.2 From Tables to Function Approximation: Deep Q-Networks

Tabular Q-learning stores one number for every state-action pair, which works only because Section 19.3.1's example has three states and two actions. A realistic trading state (inventory, elapsed time, recent order-flow imbalance from Chapter 14, and a handful of price-based features) takes values in a continuous or very large discrete space, so an explicit table can be neither stored nor visited often enough to fill in.

The standard fix, introduced by Mnih et al. (2015) and known as the **Deep Q-Network** (DQN), replaces the table with a neural network $Q(s, a; \theta)$, the feedforward architecture Chapter 6's Section 6.6.4 introduced, trained to approximate the same Bellman-optimal Q^* function. The network takes a state (or a fixed-size feature vector describing one) as input and outputs one value per possible action; training minimizes the squared temporal-difference error,

$$\mathcal{L}(\theta) = \mathbb{E}\Big[\big(r + \gamma \max_{a'} Q(s', a'; \theta^-) - Q(s, a; \theta)\big)^2\Big], \tag{19.7}$$

using the gradient-descent machinery Chapter 6's Section 6.3.3 developed for supervised learning. Here θ^- is a periodically updated copy of the network's own parameters (a "target network") that keeps the bootstrapped target from chasing a constantly moving estimate. It is one of two stabilization tricks that made deep Q-learning reliable enough to train at all; the other is storing past transitions in a large "replay buffer" and resampling them at random rather than training only on the most recent experience.

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The update rule is otherwise identical to tabular Q-learning: the network generalizes across similar states instead of memorizing each one, so the agent can act sensibly in a state it has never visited, provided it resembles ones it has seen, the same generalization argument that justifies every other neural network in this book.

Deep Q-networks are also much more finicky to train than tabular Q-learning, which matters given this book's repeated warnings (Chapter 12's badly scaled optimizer example among them) that a plausible-looking numerical method can fail to converge without any error message.

A DQN's training loss can fall steadily while the learned policy remains poor, since the loss measures how well the network fits its own, still-inaccurate bootstrapped targets, not how good the policy is. The only reliable check is to evaluate the policy's performance directly: the discipline Section 19.3.1's example applied by comparing learned Q-values with an independently known closed-form answer, and that Section 19.5.2 generalizes into an evaluation methodology for problems where no such answer exists.

19.3.3 Policy Gradient Methods and Continuous Actions

Q-learning and its deep variant learn a value function and derive a policy from it by acting greedily. This works when the action set is small and discrete (exercise or hold, buy or sell) but breaks down when the natural action is continuous, such as how many shares to trade this period or what target weight to give an asset, since $\max_{a'} Q(s', a')$ has no simple closed form once a' ranges over a continuum.

Policy gradient methods sidestep this by parameterizing the policy directly, $\pi_\theta(a \mid s)$, typically as a neural network outputting the parameters of a continuous distribution (a mean and standard deviation, for instance), and adjusting θ to increase the expected return, using the policy gradient theorem,

$$\nabla_\theta J(\theta) = \mathbb{E}_{\pi_\theta}\Big[\nabla_\theta \log \pi_\theta(a \mid s)\, G_t\Big], \tag{19.8}$$

which says, informally, that actions followed by a higher-than-expected return should be made more likely, and actions followed by a lower-than-expected return should be made less likely, with the size of the nudge proportional to how surprising the return was.

Actor-critic methods reduce the noise in this estimate by learning a value function (the "critic") alongside the policy (the "actor") and using it to judge whether a given return was better or worse than expected from that state, rather than judging it against the raw, highly variable return alone.

Proximal Policy Optimization (PPO), introduced by Schulman et al. (2017), is the most widely used modern policy-gradient algorithm. It constrains each policy update to stay close to the previous policy, preventing the large, destabilizing updates that plagued earlier policy-gradient methods, and its combination of reasonable sample efficiency and training stability has made it the default starting point for continuous-action problems across robotics, game-playing, and, increasingly, the trading and portfolio applications developed next.

PPO is also the reinforcement-learning step inside the large language models of Chapter 16. In reinforcement learning from human feedback, the policy π_θ is the language model itself, an action is the response it generates to a prompt, and the reward is the score a learned reward model assigns to that response, less a penalty for drifting too far from the model's starting point; Ouyang et al. (2022) optimized this objective with PPO. The penalty plays the same role as PPO's own constraint on each update: both keep a policy from exploiting the quirks of an imperfect reward, the reward-hacking risk Section 19.6 raises for trading agents.

Table 19.3 summarizes the four families of methods, the setting each suits, and the finance application each most naturally serves in the sections that follow.

19.3.4 Learning from Expert Demonstrations: Imitation and Inverse Reinforcement Learning

Every method so far learns by trial and error: an agent that starts knowing nothing must discover which actions are good through the ε-greedy exploration of Section 19.3.1 or its neural-network analogues.

When a record of a skilled decision-maker's past choices exists, an experienced trader's execution decisions, say, there is a faster route: **behavioral cloning** treats the expert's historical state-action pairs as an ordinary supervised training set and fits a policy $\pi_\theta(a \mid s)$ to predict the expert's action from the state with Chapter 6's

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Method	Learns	Best suited to	Natural finance application
Tabular Q-learning	Action-value function $Q(s, a)$	Small, discrete state and action spaces	Section 19.3.1's early-exercise decision; small discretized inventory or execution problems
Deep Q-Network (DQN)	Neural-network approximation of $Q(s, a; \theta)$	Large or continuous state spaces, discrete actions	Execution and market-making agents (Section 19.4.1) with rich order-book state features
Policy gradient (REINFORCE, actor-critic)	A parameterized policy $\pi_\theta(a \mid s)$ directly	Continuous action spaces	Trade-size or quote-skew decisions that vary continuously rather than over a small discrete menu
Proximal Policy Optimization (PPO)	A parameterized policy, with a stabilized update rule	Continuous actions, need for reliable, stable training	Portfolio-rebalancing agents (Section 19.4.3) and production execution systems such as Section 19.5.1's LOXM

Table 19.3: A summary of the reinforcement learning methods introduced in this chapter

classification machinery. It sidesteps the exploration problem by never requiring the agent to try an action the expert did not.

Behavioral cloning has a well-known weakness: it is only as good as the expert it imitates, and it can fail badly in states the expert rarely visited. A classifier trained on the expert's typical behavior has no guidance for recovering once the agent's own small errors push it into unfamiliar states, and the errors compound over a long trajectory in a way a single supervised prediction's never do.

Inverse reinforcement learning takes a different route: rather than imitating the expert's actions, it infers the reward function the expert appears to optimize, then solves the resulting Markov decision process with this chapter's methods. The resulting policy can, in principle, generalize to states the expert never visited. The cost is a harder, less well-posed estimation problem, since many reward functions can rationalize the same observed behavior.

Both techniques bear on the applications that follow. Section 19.5.1's LOXM, reported to be trained on "a large history of past trades together with simulated trading scenarios," plausibly combines demonstration-based learning, bootstrapping an initial policy from historical human and algorithmic execution decisions, with further trial-and-error refinement in simulation.

This hybrid is common because it attacks reinforcement learning's sample inefficiency: starting from an expert-informed policy leaves the trial-and-error stage a far smaller gap to close, which justifies the added complexity in a high-stakes application.

19.4 Financial Applications

This part points the machinery at execution, allocation, and hedging, recovering earlier chapters' answers where they exist.

19.4.1 Reinforcement Learning for Optimal Execution and Market Making

Optimal execution was among the earliest documented applications of reinforcement learning in finance. Nevmyvaka et al. (2006) applied tabular Q-learning, structurally like Section 19.3.1's example but with a far larger state space built from order-book features, to historical limit order book data. The learned execution policy reduced trading costs relative to the TWAP and VWAP benchmarks of Chapter 13's Section 13.3.4, without being given a closed-form market-impact model to optimize against.

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The state in a modern version of this problem typically includes remaining inventory, time remaining in the execution window, the current bid-ask spread, and recent order-flow imbalance; the action is how many shares to submit, and at what limit price, this period; and the reward is the negative implementation shortfall (Chapter 13's Section 13.6.4 transaction-cost-analysis measure) realized on that period's fill.

Unlike Almgren-Chriss, which commits in advance to a schedule regardless of how the market behaves, a learned execution policy can condition its trade size on the order book it observes, trading faster when liquidity is unusually deep and slowing down when the book thins, an adaptive behavior no closed-form schedule replicates.

19.4.2 Worked Example 19.2: Q-Learning Recovers the Almgren-Chriss Optimal Trajectory

Section 19.3.1 verified Q-learning against a known answer on a three-state problem. The same check is worth doing on a problem closer to trading practice: Chapter 13's Almgren-Chriss worked example, the 250-share order split across $N = 5$ intervals with impact coefficient $\eta = 0.01$, per-interval price volatility $\sigma = \$0.02$ per share, and risk aversion $\phi = 0.5$, whose closed-form optimal trajectory left unexecuted shares $(250,\ 194.2,\ 142.4,\ 93.4,\ 46.2,\ 0)$ at a total risk-adjusted cost of 139.4.

Reformulated as a Markov decision process, the state at interval j is the inventory x_{j-1} still unexecuted, discretized to the nearest 5 shares (a grid of 51 possible inventory levels), and the action is how many of those shares, again in multiples of 5, to sell this interval.

The reward for selling a_j shares and leaving $x_j = x_{j-1} - a_j$ unexecuted is the negative of Chapter 13's per-interval cost,

$$r_j = -\left[\eta\,\frac{a_j^2}{\tau} + \phi\sigma^2\tau\,x_j^2\right],$$

with $\tau = 1$ matching Chapter 13's normalization, and the last interval ($j = 5$) forces full liquidation, $x_5 = 0$, as the closed-form problem requires. Unlike Section 19.3.1's tree, this environment is **deterministic**: an action leaves no uncertainty about the resulting state or reward, since Almgren-Chriss prices timing risk analytically through the variance term $\phi\sigma^2\tau x_j^2$ rather than by exposing the agent to a randomly sampled price path.

Reinforcement learning is still the right tool here, for a different reason: not to cope with stochastic transitions, but to search roughly 5,300 discretized state-action possibilities, far too many to enumerate by hand, without being given the closed-form $\sinh(\cdot)$ trajectory Chapter 13 derived analytically.

Exact backward-induction dynamic programming over the same grid, computable directly since only 51 inventory levels and 4 real decision points are involved, gives an optimal discretized trajectory of $(250,\ 195,\ 145,\ 95,\ 45,\ 0)$ shares at a total cost of 139.52, close to but not identical to the continuous closed-form values, as expected from rounding a continuous optimum onto a 5-share grid.

Tabular Q-learning, trained model-free for 600,000 episodes with an ε-greedy behavior policy decaying from $\varepsilon = 0.5$ to 0.05 and a visit-count-decayed learning rate, **converges to this same trajectory and this same cost**, 139.52, not merely close to it.

This differs from Section 19.3.1's example, where the learned "hold" values settled within roughly one percent of their closed-form targets: because this environment is deterministic, every reward Q-learning observes is noise-free, so once a state-action pair has been explored enough to identify its best continuation, the learned value converges to it exactly.

Figure 19.5 shows the backward-induction computation. Dynamic programming does not work through the five intervals in the order a trader experiences them; it starts from the one stage with no real decision left, the forced liquidation at x_4, and solves backward, using each solved stage's value function to solve the one before it, until $V_1(x_0)$ gives the answer for the full problem.

Only after the backward pass is complete can the optimal trajectory be traced forward, choosing at each state the action that attains its computed value: the bottom row of the figure.

Table 19.4 places the Q-learning trajectory alongside the uniform TWAP schedule and the fully aggressive single-shot execution Chapter 13 also scored: the efficient-frontier ordering Chapter 13 established analytically emerges from an agent never given the cost formula.

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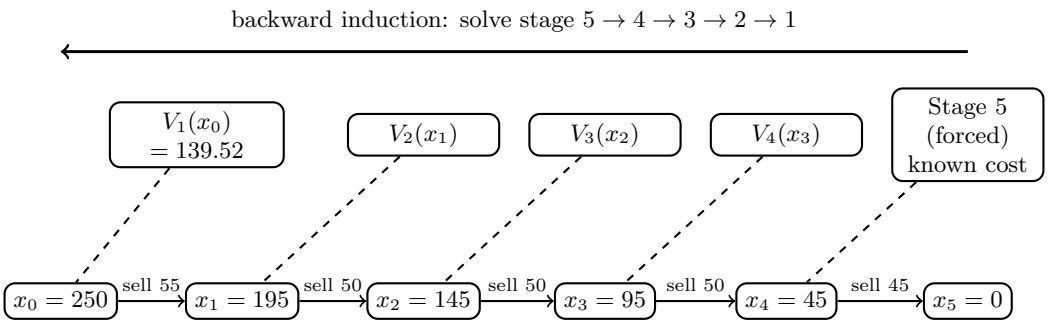


Figure 19.5: Backward-induction dynamic programming for the execution problem

Strategy	Unexecuted-share trajectory	Total cost
Fully aggressive (single shot)	$(250, 0, 0, 0, 0, 0)$	625.00
Uniform TWAP	$(250, 200, 150, 100, 50, 0)$	140.00
Q-learning (learned, discretized)	$(250, 195, 145, 95, 45, 0)$	139.52

Table 19.4: Total risk-adjusted execution cost under three strategies for the same 250-share, $N = 5$-interval order ($\eta = 0.01$, $\sigma = \$0.02$ per share per interval, $\phi = 0.5$)

Figure 19.6 plots the three trajectories. The fully aggressive strategy liquidates instantly, the uniform TWAP schedule is a straight line, and the Q-learning trajectory sits just below TWAP from the first interval on, with fewer shares left unexecuted, trading a little faster than uniform while inventory is largest and tapering to match TWAP by the final interval: the mild front-loading Chapter 13's continuous closed-form solution also produced.

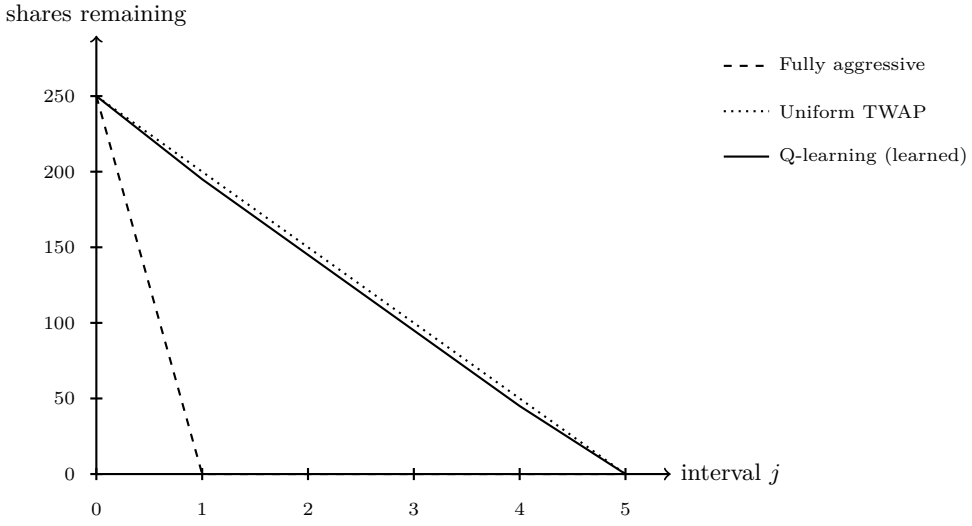


Figure 19.6: Unexecuted-share trajectories under the three strategies of Table 19.4

The tabular agent needed a 5-share grid to keep its Q-table finite. The companion notebook's optional "at scale" section drops the grid: a neural-network policy trained with PPO (Section 19.3.3) from the `stable-baselines3` library chooses a continuous fraction of the remaining inventory to sell, and is scored

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against the exact closed-form trajectory and TWAP across three training seeds; it needs the extra packages listed in the repository's `requirements-dl.txt`.

Extending the Framework to Market Making

Market making fits the same framework. Chapter 13's Avellaneda-Stoikov model derives a closed-form optimal quote skew from an assumed order-arrival intensity function and a known risk-aversion parameter.

A reinforcement-learning market maker instead learns a quoting policy from simulated or historical order flow. The state again includes inventory, time-to-close, and order-book depth; the action is the bid and ask quotes (or, in a simpler discrete version, a few candidate skews); and the reward is realized spread capture net of inventory risk and adverse selection, the concern Chapter 14's Glosten-Milgrom model formalizes. The appeal matches the execution case: a learned policy can adapt to order-arrival patterns that deviate from Avellaneda-Stoikov's assumed functional form, at the cost of needing much representative training data and giving up the closed-form model's transparency about why a quote was chosen, a trade-off the chapter's closing sections return to.

Designing States, Actions, and Rewards in Practice

Neither problem above specifies its Markov decision process uniquely, and the design choices matter enormously to whether the agent learns anything useful. A state that omits recent order-flow imbalance discards the signal Chapter 14's order-flow-imbalance regression showed carries information about near-term prices, leaving the agent weaker than a well-informed human trader or a simple rule that conditions on that signal. A state with too many redundant or noisy features slows learning, forcing the agent to discover which features matter: the feature-engineering problem of Chapter 6's Section 6.5.2, made harder because the agent must learn it from its own trial and error rather than from a labeled training set.

Reward design carries the same tension in sharper form, because, as Section 19.6 shows, an imperfect reward produces not a slightly worse agent but one that is often confidently wrong about what it is optimizing. A common compromise is to shape the reward with a densely observed intermediate signal, a small penalty proportional to inventory held at each step, say, alongside the sparse terminal profit and loss. This speeds learning, since the agent gets feedback every step rather than only at the end of a long execution episode. But if the shaping term is not scaled carefully against the true objective, the agent can over-optimize the proxy, holding needlessly little inventory, at the expense of the goal it was meant to support. There is no universal recipe; the balance is validated as any other model choice in this book is, by comparing the policy's out-of-sample performance with the closed-form or rule-based benchmark it should improve on, as Section 19.5.2 recommends.

19.4.3 Reinforcement Learning for Portfolio Management

Chapter 12's mean-variance framework solves a single-period allocation problem: given today's estimated expected returns and covariance matrix, find the weights that optimize a risk-return trade-off today. Section 12.4.1 acknowledges that real portfolios are rebalanced repeatedly and that transaction costs make each rebalancing depend on the current position, but stops at flagging this rather than solving the multi-period problem.

Framed as a Markov decision process, the state is the portfolio's current holdings together with estimated risk-factor exposures (Chapter 12's Section 12.3.4 multi-factor model is a natural source of these features); the action is a new target weight vector, or equivalently the trades needed to reach it. The reward combines the period's realized return with an explicit penalty for transaction costs and, if desired, for realized volatility or drawdown, letting the agent learn a rebalancing policy that trades off today's allocation quality against the cost of getting there, rather than treating each rebalancing as a fresh, myopic optimization as Chapter 12 does.

This framing is more general than Chapter 12's static optimization, and the generality is its cost. A reinforcement-learning portfolio agent needs either a large amount of representative historical data or a realistic market simulator to train in, and a poorly specified reward (one that ignores drawdown, or rewards raw return without risk adjustment) will produce a policy that optimizes the wrong objective with no warning, a version of the "reward hacking" problem Section 19.6 discusses.

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In practice, most institutional applications of reinforcement learning to portfolio management remain closer to research than production: they test whether an adaptive rebalancing policy can beat Chapter 12's static mean-variance benchmark net of realistic transaction costs, rather than replacing it. Reinforcement learning has so far shown clearer, more reproducible gains in the execution and market-making applications of Section 19.4.1, where the reward (realized trading cost) is comparatively unambiguous, than in full portfolio management, where the right reward is an open question tied to a client's often unstated risk preferences.

Multi-period portfolio choice has a classical closed-form precedent that clarifies what a learned policy would need to rediscover. Chapter 1's Section 1.5.2 introduced the **Kelly criterion** for a single repeated bet; maximizing the expected *logarithm* of long-run wealth rather than expected wealth extends to repeated rebalancing under a power-utility objective, giving an allocation that avoids both the ruin risk of over-betting and the excessive caution of under-betting. This is a sequential decision problem with a closed-form answer under simplifying assumptions (constant, known return and variance), the kind of further step Section 19.2.2 noted Chapter 12 could have taken. It also supplies a sanity check: on a simplified simulated market where the Kelly fraction is known exactly, a correctly implemented reinforcement-learning portfolio agent should converge toward it, a check with the same purpose as Section 19.3.1's comparison against Chapter 5's closed-form option value, before the agent is trusted on a real market with no closed form to check against.

19.4.4 Worked Example 19.3: Why Model-Free Search Struggles on the Kelly Bet

Running that sanity check, rather than asserting it would work, is instructive because it does not go as cleanly as Sections 19.3.1 and 19.4.2. Chapter 1's repeated bet, an even-money wager ($b = 1$) won with probability $p = 0.6$, has closed-form optimal fraction $f^* = p - q/b = 0.20$ and long-run growth rate $g(f^*) \approx 0.0201$. As a Markov decision process this is a single, repeated state (the optimal fraction does not depend on current wealth, so wealth need not be tracked), with action f on a grid of 5% increments, $f \in \{0.00, 0.05, \ldots, 0.95\}$, and an immediate, noisy reward $r = \ln(1 + f)$ with probability p or $r = \ln(1 - f)$ with probability q, drawn independently every period, with no next state and so no bootstrapping. It is a multi-armed bandit, structurally identical to Section 19.2.4's venue-routing problem with twenty arms in place of three, but with no hope of an easy answer, for a reason worth stating before the numbers.

The difficulty is the ratio of the reward's noise to what there is to learn: the gap between the optimal action's growth rate and its nearest neighbors on the grid, $g(0.20) - g(0.15) \approx 0.0013$ and $g(0.20) - g(0.25) \approx 0.0013$, is tiny, while a single bet's outcome swings between $\ln(1.20) \approx 0.182$ and $\ln(0.80) \approx -0.223$, a standard deviation roughly 150 times larger than the gap Q-learning must resolve.

Trained for 200,000 simulated bets with the same ε-greedy, visit-count-decayed setup used throughout this chapter, tabular Q-learning settles on $f = 0.15$, not $f^* = 0.20$, leaving $g(0.20) - g(0.15) \approx 0.0013$ of achievable growth rate on the table, about 6.4% of the optimal rate: a real, if modest, miss, unlike the exact or near-exact recovery of Sections 19.3.1 and 19.4.2. Figure 19.7 shows why: the growth-rate curve is nearly flat across the 0.10-to-0.30 range, so the true optimum and Q-learning's answer sit almost on top of each other.

The contrast is what a **model-based** approach, Section 19.2.6's other category, does with the same 200,000 observations. Rather than averaging opaque rewards over a large discretized action grid, it exploits the problem's compact underlying model: a single unknown parameter, the win probability p, plugged into Chapter 1's closed form.

Counting wins and losses across the same 200,000 simulated bets gives an observed win frequency $\hat{p} = 0.6003$, and substituting into $\hat{f}^* = \hat{p} - (1 - \hat{p})/b$ gives $\hat{f}^* \approx 0.2006$, within 0.0006 of the true optimum, an order of magnitude closer than Q-learning's answer from the same data. This is Section 19.2.6's model-based-versus-model-free distinction made numerical, and it sharpens Section 19.2.2's argument: reinforcement learning is the right tool once no compact model is available, but when one is, as here, where a single Bernoulli parameter stands for the whole problem, exploiting that structure can be far more sample-efficient than a black-box search over a discretized action grid.

Not every problem that can be written as a Markov decision process is best solved as one; recognizing when a problem's structure should be modeled directly rather than learned model-free is part of the practitioner's job, not something reinforcement learning relieves them of.

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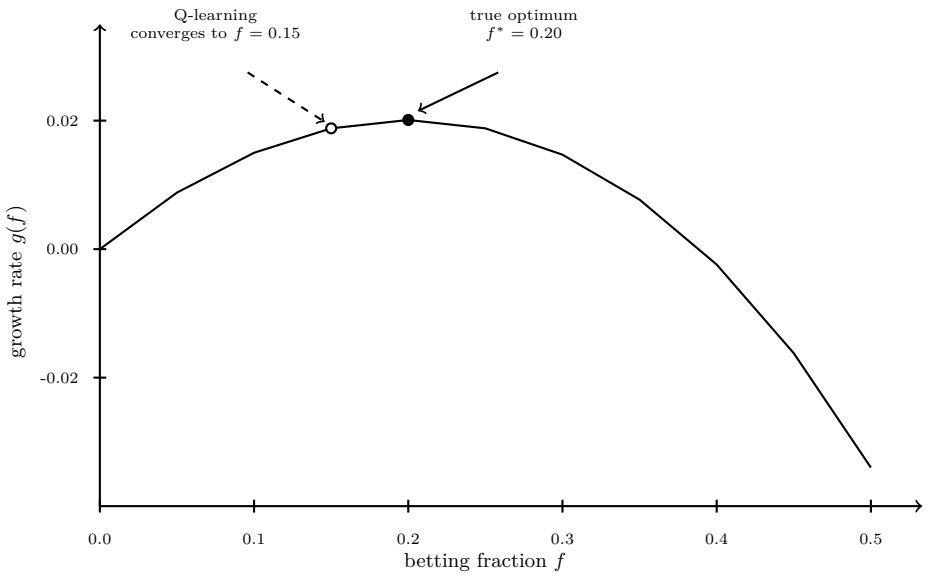


Figure 19.7: The Kelly growth-rate curve $g(f) = p\ln(1+f) + q\ln(1-f)$ for $p = 0.6$, $b = 1$

19.4.5 Deep Hedging: Learning to Hedge Under Frictions

Chapter 5's replication argument prices an option by continuously holding Δ shares of the underlying, and Chapter 8's delta-normal example measures an option book's risk through the same delta. Both assume something no trading desk enjoys: that the hedge can be rebalanced continuously and for free. A real desk pays a bid-ask spread and fees on every rebalancing trade, which creates a dilemma the frictionless theory defines away. Rebalance after every tick and transaction costs devour the option premium; rebalance rarely and the position runs unhedged in between, exposed to the gap risk Chapter 8 quantified. Deciding *when* and *by how much* to adjust the hedge as the underlying moves and expiry approaches is a sequential decision problem under uncertainty, this chapter's subject.

Cast in Section 19.2.1's terms, the state is what the hedger observes, time to maturity τ, the underlying price S_t, and the position h_t currently held; the action is the new position h_{t+1} to hold until the next decision point; and the objective is defined on the terminal outcome: let W_T be the hedger's final wealth after the option is settled and the hedge unwound, premium, trading gains, and all transaction costs included. **Deep hedging** chooses the policy to minimize a convex risk measure of the shortfall,

$$\min_{\pi} \ \rho(-W_T^{\pi}), \qquad \text{for example } \rho = \mathrm{ES}_{95\%}, \tag{19.9}$$

where the risk measure is the Expected Shortfall Chapter 8 introduced, now serving not to measure a fixed position's risk but as the training objective a policy is optimized against.

A small worked example makes the trade-off concrete with numbers this book has already computed. Take Chapter 8's market maker, short the one-year \$50-strike call priced in Chapter 5 ($S_0 = K = \$50$, $r = 4\%$, $\sigma = 30\%$, $C_0 \approx \$6.88$, $\Delta_0 = 0.6115$), and have them hedge it over the option's life on 10,000 simulated price paths (geometric Brownian motion with real-world drift $\mu = 8\%$, 252 daily steps, seed 7 in the companion notebook), paying a proportional transaction cost of 50 basis points of traded value on every trade, the friction an illiquid underlying imposes. The policy class is a simple one-parameter family: rebalance to the current Black-Scholes-Merton delta only when the held position has drifted more than a *band* b away from it, so $b = 0$ is the textbook daily delta hedge and larger b trades less often. The first trade is hand-checkable: the hedger buys $\Delta_0 = 0.6115$ shares at \$50, paying $0.005 \times 0.6115 \times \$50 \approx \$0.15$ in costs. Table 19.5 reports, across the 10,000 paths, the mean terminal profit and loss, its standard deviation, the mean total transaction cost, and the 95% Expected Shortfall of the terminal loss.

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Policy	Mean P&L	Std. dev.	Mean cost	$\text{ES}_{95\%}$
No hedge	−\$1.32	\$12.54	\$0.00	\$39.30
Daily delta ($b = 0$)	−\$1.60	\$0.59	\$1.56	\$3.01
Band $b = 0.02$	−\$1.37	\$0.61	\$1.34	\$2.82
Band $b = 0.06$	−\$0.95	\$0.70	\$0.93	\$2.56
Band $b = 0.20$	−\$0.60	\$1.92	\$0.53	\$5.76

Table 19.5: Hedging a short one-year \$50-strike call under 50-basis-point proportional transaction costs, across 10,000 simulated paths. The band policy with $b = 0.06$, found by searching the band grid for the lowest $\text{ES}_{95\%}$, beats the textbook daily delta hedge on mean cost and tail risk simultaneously.

The table repays a careful read. Leaving the position unhedged is catastrophic: a \$12.54 standard deviation and a \$39.30 Expected Shortfall against a \$6.88 premium collected. The textbook daily delta hedge cuts that risk twenty-fold, at a price the frictionless theory never mentions: \$1.56 of expected transaction costs, nearly a quarter of the premium. The band policy at $b = 0.06$, selected by searching the grid for the lowest $\text{ES}_{95\%}$, cuts mean costs by 40% *and* carries less tail risk than hedging every day, because each skipped rebalance saves a cost that would otherwise be paid with certainty, while the extra hedging error it accepts is small and diversifies across the path. Push the band too wide and unhedged risk floods back in, as the $b = 0.20$ row shows. The friction level, not taste, dictates the answer: rerun with a 5-basis-point cost and the ES-optimal band shrinks to 0.01, essentially the textbook hedge, so Black-Scholes-Merton delta hedging is not wrong but is the frictionless limit of a more general policy that the data-driven search recovers.

One feature of this simulation needs a warning before the method goes further. Its paths grow at the real-world drift $\mu = 8\%$, not at the risk-free rate, because a hedger who wants to know how much it could lose needs the distribution of outcomes it will face: risk is measured under the **real-world measure**, often written P. Prices are not. Chapter 5's Black-Scholes-Merton value is an expectation under the risk-neutral measure Q, in which the underlying drifts at r, and the two answers differ by far more than rounding. Discounting the call's expected payoff along the simulator's own drift gives $e^{-rT}\mathbb{E}^P[(S_T - K)^+] = \$8.18$, against the \$6.88 arbitrage-free price, an overstatement of \$1.30, or 18.9%, because the faster-growing stock finishes in the money more often. An agent rewarded with discounted payoffs on P-paths, and then asked what the option is worth, will learn that wrong number, and nothing in its training signal will tell it so. Reinforcement learning used for pricing must therefore either simulate under Q, calibrated to traded prices, or, as deep hedging does, price through the hedge itself: Buehler et al. (2019) define the price as the cash amount that makes the hedger indifferent, under its risk measure, between writing the option and not, an indifference price that coincides with the risk-neutral price only in the frictionless, complete-market limit. Under frictions the gap between them is real, and it is the price of risk that cannot be hedged away.

Deep hedging proper, introduced by Buehler et al. (2019), replaces the one-parameter band with a neural network policy $\pi_\theta(\tau, S_t, h_t)$ trained by stochastic gradient descent over simulated paths to minimize the same kind of convex risk measure, the step from a hand-designed policy family to a learned one that Section 19.3.2 made for value functions. The gain is generality, not magic. The network is never shown a delta, a gamma, or any other Greek, yet in the frictionless limit it rediscovers the Black-Scholes-Merton hedge, and away from that limit it can absorb what no closed-form hedge handles gracefully: several hedging instruments at once, transaction costs of realistic shape, discrete trading times, position limits, and payoffs with no replicating formula. Reports of production use at major derivatives desks followed within a few years of the paper, making hedging, alongside Section 19.5.1's optimal execution, one of the two applications in this chapter where learned policies have reached practice. The caveats developed earlier bind here with full force: every path the policy learns from comes from a simulator, so the learned hedge is only as good as the simulator's realism (Section 19.5.2's sim-to-real gap), and a neural hedging policy inherits the governance and explainability burden Chapter 18 attaches to any model whose decisions cannot be read off a formula.

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19.4.6 Multi-Agent Dynamics and the Limits of a Single-Agent Framework

Every Markov decision process in Section 19.2.1 assumes a single agent facing a fixed, if stochastic, environment, an assumption real markets violate: the environment a trading agent faces is largely made up of other agents, some human and increasingly many algorithmic, each adapting to the same price and order-flow signals. Chapter 14's hot-potato effect during the 2010 Flash Crash illustrates the dynamic: individually reasonable inventory-management rules, followed by many participants at once, produced a collectively destabilizing outcome that no single rule, viewed on its own, would have predicted. An agent trained against a static historical simulator implicitly assumes other participants' behavior stays fixed, an assumption that weakens as the agent's own trading grows, since a strategy profitable at small scale can behave very differently once it moves the order flow it was trained to react to.

Multi-agent reinforcement learning extends the framework to several learning agents adapting to one another, and is an active research area for the market-making and execution problems of Section 19.4.1, since real limit order books are populated by competing, adapting participants. As of this writing, multi-agent methods are much harder to train reliably than single-agent ones: the convergence guarantees of tabular Q-learning in Section 19.3.1's simple setting generally fail when the environment changes because other agents are learning too. Most production trading systems, including Section 19.5.1's LOXM, are best understood as single-agent policies trained against a historical or simulated environment, with the gap between that assumption and the real market showing up as the sim-to-real and non-stationarity concerns of Section 19.6.

A more tractable middle ground treats other participants' aggregate behavior as part of the state. A single-agent method from Sections 19.3.1 through 19.3.3 can then respond to features such as recent order-flow imbalance or realized volatility that summarize what others are collectively doing, without modeling each of them. Most of this chapter's real applications, execution and market making among them, rely on this compromise.

19.5 From Simulation to Deployment

A policy trained in simulation has proved nothing yet, and this part covers what would.

19.5.1 Case Vignette: JPMorgan's LOXM

In 2017, JPMorgan disclosed LOXM, an equity trade execution system trained using reinforcement learning on a large history of past trades together with simulated trading scenarios, initially deployed on the bank's European cash equities desk.

Structurally, LOXM addresses the problem of Section 19.4.1. At each point during an order's execution it decides how much of the remaining order to place, at what price, and how long to wait before reassessing, balancing the risk of trading too aggressively (unnecessary market impact) against trading too passively (missing the current opportunity and executing later at a worse price). That is the impact-versus-timing-risk trade-off Almgren-Chriss solves in closed form in Chapter 13, learned here from data rather than derived from an assumed cost function. The system was reportedly trained on millions of historical and simulated trading scenarios so that it could adapt to prevailing market conditions rather than follow one fixed schedule, the advantage Section 19.4.1 attributes to a learned execution policy.

LOXM illustrates both sides of this chapter's central tension in one deployment. It shows that reinforcement learning for execution is more than an academic exercise: a major bank judged a learned policy reliable enough for live client order flow, the practical validation closed-form models have had for decades and any machine-learning alternative must earn. But a deep reinforcement-learning execution policy is far harder to explain to a regulator or internal model validator than an Almgren-Chriss schedule, whose logic reduces to a handful of interpretable parameters, a governance burden that Section 19.6 and Chapter 18's explainability toolkit address and that any institution deploying such a system must take on explicitly.

LOXM was an early example, not an isolated one. Since 2017, industry reporting has described similar reinforcement-learning execution research and deployments at several other major banks and trading firms, consistent with the chapter's claim that optimal execution, alongside Section 19.4.5's hedging application, has the clearest record of reaching production, because its reward, realized execution cost against a well-defined

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benchmark, is comparatively unambiguous to specify, the property Section 19.4.3 found still missing from full portfolio management.

19.5.2 Simulation, Backtesting, and Off-Policy Evaluation

Every agent in this chapter learns by trial and error, which raises an operational question none of this book's supervised models faced: where does the trial and error happen? Training on live markets is rarely acceptable, since a partially trained policy will take some bad actions while exploring, and a bad trading action costs real money in a way a still-training classifier's bad prediction does not. The standard alternative is to train in a **simulator**, a model of the market built from historical data that can be reset and replayed without financial consequence; Chapter 13's backtesting framework (Section 13.6) is a close cousin. It shares that framework's core limitation: a simulator built from history reflects how the market behaved, including in response to other participants' historical order flow, not how it would respond to a new agent's own actions at scale. This **sim-to-real gap** is at least as serious for a reinforcement-learning trading agent as backtest overfitting is for the supervised models Chapter 13 warned about.

A subtler problem is **off-policy evaluation**: given historical data generated by one policy (a human trader's decisions, say, or an older execution algorithm), how well would a newly trained policy have performed, without running it? Historical data reflects only the actions the deployed policy took, not the outcomes of actions it did not take: the missing-counterfactual problem Chapter 18's algorithmic-recourse discussion raised for credit decisions, now in a sequential setting where one wrong counterfactual estimate can compound over many steps. Techniques such as importance sampling reweight historical outcomes to estimate what a new policy would have achieved, but their reliability degrades quickly as the new policy diverges from the one that generated the data. A novel learned policy is therefore rolled out cautiously: first in simulation, then on a small fraction of live order flow, with continuous monitoring for the performance drift that Chapter 15's precision-drift and Chapter 18's population-stability-index discussions developed, and never at full scale on the strength of a backtest or an off-policy estimate alone.

This staged approach, simulator, then a small live allocation, then a growing one, is sometimes called **curriculum deployment**. It mirrors the graduated rollout that Chapter 8's three-lines-of-defense model and Chapter 18's staged-deployment recommendation argue for any high-stakes model, reinforcement-learning policy or static classifier. What is new is that the policy may keep updating during the rollout, a live, still-learning system rather than a frozen model. Monitoring must therefore detect not only a fixed policy drifting from its backtested expectation but an adapting policy drifting toward behavior nobody has validated, a strictly harder problem than those Chapters 15 and 18 posed.

19.5.3 Evaluating Trained Policies: Metrics and Variance Across Training Runs

Chapter 13's Section 13.6.1 developed a standard toolkit, Sharpe ratio, maximum drawdown, hit rate, for evaluating a strategy's backtested performance, and every metric applies unchanged to a reinforcement-learning policy. What reinforcement learning adds is a source of variability a closed-form model like Almgren-Chriss lacks. Every method in this chapter is trained by a stochastic process (random initialization, ε-greedy exploration, and, for the neural-network methods of Sections 19.3.2 and 19.3.3, stochastic gradient descent), so two training runs of the same algorithm on the same data, differing only in their random seed, can converge to meaningfully different policies with meaningfully different realized performance.

Section 19.3.1's worked example shows this in miniature. It was trained with a fixed seed (seed = 7) so that its reported numbers, \$2.5328 and \$4.9234 among them, would be reproducible for a reader rerunning the companion notebook. A different seed would give slightly different learned values, still close to the closed-form \$2.5641 and \$4.8718 but not identical to those in Table 19.2. In a three-state problem with a known answer this is a curiosity; in a large trading application with no closed form it is a serious concern, since a single run's backtested performance could be a favorable draw rather than a property of the algorithm the institution intends to rely on.

The standard remedy is to train several independent runs from different seeds and report performance as a distribution rather than a single number. The practice, standard in machine learning, has not been needed elsewhere in this book, whose supervised and closed-form models are deterministic given their data.

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It extends the caution of Chapter 6's confidence intervals and Chapter 7's out-of-sample RMSE comparisons to the uncertainty introduced by the training process itself.

19.5.4 Practical Considerations: What Deployment Actually Requires

The methods above can leave the impression that adopting reinforcement learning is mainly an algorithmic choice, DQN or PPO, Q-learning or actor-critic, when in practice the harder and more expensive decisions are organizational and infrastructural.

A trustworthy simulator is usually the binding constraint, not the algorithm. Every method here needs a large volume of training experience, and Section 19.5.2 explained why it rarely comes from live markets. Building a simulator accurate enough to train a useful policy, with realistic order-book dynamics, price impact, and plausible reactions from other participants, is a substantial engineering project, often larger than the learning algorithm itself, and few institutions can justify it unless the decision is important and recurring enough. That is why optimal execution at a large bank, a high-volume, high-value recurring decision, adopted early, while more bespoke, lower-frequency problems largely have not.

The required skills must complement, not replace, those of this book's earlier chapters. A team building a reinforcement-learning trading system needs the market-microstructure fluency of Chapters 13 and 14, the statistical and machine-learning foundations of Chapter 6, and, on top of both, expertise in the distinct engineering discipline of reinforcement learning: stable training, careful hyperparameter selection, and the evaluation methodology of Section 19.5.2. That combination is still rare, and most institutions build it gradually. This is one more reason closed-form models like Almgren-Chriss and Avellaneda-Stoikov remain the right default for most applications: they need the microstructure expertise this book develops, but not the specialized reinforcement-learning engineering on top of it.

The relevant cost-benefit comparison is against the closed-form alternative, not against doing nothing. A closed-form model, once specified, costs almost nothing to run and is immediately auditable by the model-risk-management process of Chapter 8; a reinforcement-learning policy costs far more to build, validate, and monitor, and the extra cost is justified only when the closed-form model's simplifying assumptions demonstrably cost real money, through the market-impact and reward-specification gaps Section 19.2.2 identified. Consistent with this book's compute-first approach, build and validate the closed-form baseline first, know what it costs in realized performance, and only then invest in a learned alternative, with a pre-specified bar for how much improvement would justify the added complexity and governance burden.

Compute cost is a recurring line item, not a one-time expense. Training a deep Q-network or a PPO policy over the hundreds of thousands or millions of simulated episodes a realistic problem requires consumes far more computing resources than fitting any supervised model elsewhere in this book, and the cost recurs every time the policy is retrained, on a schedule to keep pace with the non-stationarity Section 19.6 describes or in response to detected drift in live performance. This retraining cost, together with the monitoring Section 19.5.2 requires, belongs in the same total-cost-of-ownership comparison as the simulator investment above.

19.6 Challenges in Reinforcement Learning for Finance

Reinforcement learning inherits every model-risk concern raised elsewhere in this book and adds several that are specific to its sequential, trial-and-error structure.

Reward specification is the entire problem in disguise. An agent optimizes the reward it is given, not the objective its designer had in mind, and the gap between them is usually invisible until the agent finds and exploits it, a failure mode called **reward hacking**. An execution agent rewarded only for beating a benchmark price, with no penalty for the market impact it leaves for the next order, can learn to trade in ways that look good on that metric while making the market worse for the desk's future orders; a portfolio agent rewarded for raw return with no risk adjustment can learn to take tail risk a real client would never accept. Specifying a reward that captures an institution's objective, including the parts that are hard to quantify, is consistently harder than designing the learning algorithm.

Non-stationarity compounds ordinary concept drift. Chapters 15 and 18 warned that a supervised model's accuracy can degrade as the world changes. A reinforcement-learning trading agent faces the same

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risk with a twist: other market participants adapt to *its* behavior, particularly once a learned strategy is large enough to influence prices. The environment then changes partly *because of* the agent's own past actions, an adversarial dynamic no supervised classifier in this book has had to face.

Reinforcement learning is far more sample-hungry than the supervised methods elsewhere in this book. Section 19.3.1's worked example needed 100,000 simulated episodes to estimate six numbers to within roughly one percent, with only three states and two actions. Chapter 6's supervised classifiers reach useful accuracy from training sets orders of magnitude smaller relative to their problem's apparent complexity, because a supervised label gives direct feedback on every example, while a reinforcement-learning reward, especially a delayed terminal one, says little per episode about which of many earlier actions deserves credit, the credit-assignment difficulty discussed below. This inefficiency is why Section 19.5.4's simulator investment is not optional infrastructure but the prerequisite for any of this chapter's methods to work outside small, hand-verifiable settings.

Exploration is expensive and risky in a way it is not in a game. The ε-greedy exploration of Section 19.3.1's worked example costs nothing beyond simulated time; on a live trading desk the same exploration means deliberately taking actions believed to be suboptimal, in real markets, with real capital, to learn whether they are better than believed. This risk profile differs fundamentally from a research agent exploring a video game, and it is the main reason simulation-based and off-policy training (Section 19.5.2) are close to mandatory for any serious financial application.

Explainability and governance do not disappear because a policy performs well. A deep reinforcement-learning policy inherits the opacity concerns Chapter 18 raised for deep neural networks, compounded because its behavior is defined over a trajectory of decisions rather than a single prediction, so a SHAP-style, single-decision explanation is only a partial account. A model validator or regulator evaluating a system like Section 19.5.1's LOXM needs an account of why the policy behaves as it does across a representative range of market conditions, not just a performance summary: the model-risk-management discipline of Chapter 8, extended to explainability in Chapter 18, applied to a policy that acts rather than predicts.

Credit assignment gets harder the longer the horizon. A supervised classifier gets immediate feedback on every prediction; Chapter 6's confusion matrix scores a decision the moment it is made. A reinforcement-learning agent often learns whether an early action was good only many steps later, when the reward arrives: the **credit assignment problem**. Section 19.3.1's worked example largely sidesteps it, since its longest episode is two decisions deep, but over a trading day's execution decisions or a multi-quarter rebalancing episode it can be hard to tell which earlier action caused a later gain or loss, which slows learning and demands far more data than a supervised problem of similar apparent complexity.

Simulators are themselves models, with their own model risk. The sim-to-real gap of Section 19.5.2 is more than a data-availability inconvenience. A simulator built from historical data encodes its own assumptions about prices and order flow, and an agent trained only against a flawed simulator learns a policy optimal for that simulator's biases rather than the real market, the same point as Chapter 8's warning that a risk model is only as trustworthy as its assumptions. Validating the simulator against out-of-sample market behavior, not just the agent trained inside it, is part of the same model-risk discipline.

The regulatory and legal framework for autonomous trading policies is still catching up. Chapter 8's model-risk guidance and Chapter 18's explainability requirements were written largely for static, single-prediction models. A policy that adapts over time, possibly learning after deployment, raises questions those frameworks do not yet answer: how often must a live, adapting policy be re-approved, and what counts as an adequate explanation of a policy rather than a prediction? Existing supervisory guidance addresses neither with the specificity it applies to a fixed logistic regression or gradient-boosted tree, so institutions deploying reinforcement learning in a regulated trading or advisory context should expect to extend, and in places invent, their own governance answers.

19.7 Key Takeaways

Reinforcement learning formalizes sequential decision-making as a Markov decision process, with states, actions, rewards, and a discount factor, and its objective, maximizing expected discounted return, generalizes the one-shot optimizations this book has relied on since Chapter 5.

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The Bellman equation expresses a state's or action's value as immediate reward plus discounted continuation value, and Chapter 5's binomial-tree backward induction already solves a Bellman optimality equation by hand. The chapter's central worked example recovered that tree's known value, \$2.5641, with model-free tabular Q-learning; the learned and closed-form solutions agree to within roughly one percent after 100,000 simulated episodes.

Deep Q-networks and policy-gradient methods, including actor-critic algorithms and PPO, extend the value-based and policy-based logic to large or continuous state and action spaces where a table is impossible to maintain.

Optimal execution and market making, Chapter 13's closed-form models, become reinforcement-learning problems once market impact and order arrival are too complex for a clean parametric form, a generalization validated in research (Nevmyvaka et al. 2006) and in industry practice (JPMorgan's LOXM). Portfolio management admits the same framing but remains harder, chiefly because its reward is harder to specify than an execution cost.

All these applications inherit reinforcement learning's distinctive risks: reward misspecification, non-stationarity driven by other agents' adaptation, the cost and danger of live exploration, severe sample inefficiency relative to this book's supervised methods, and an explainability burden that compounds Chapter 18's concerns. All argue for simulation-based training, staged deployment, and continuous monitoring rather than treating a trained policy as a finished product.

Markets are not single-agent environments: other participants adapt to a learned policy as it adapts to them, which is why most production systems remain single-agent approximations of a multi-agent problem. The practical bottleneck to adopting these methods is usually a trustworthy simulator and the engineering skill to train against it reliably, not the choice of algorithm.

19.8 Suggested Exercises

19.1 Using Section 19.3.1's three-state Markov decision process, verify by hand that $Q^*(S_d, \text{hold}) = \gamma[q(0) + (1-q)(9.50)] \approx \$4.8718$, and explain why immediate exercise is optimal at that state despite continuation having a positive expected value.

19.2 Suppose the down-move factor in Section 19.3.1's tree were $d = 0.8$ instead of 0.9, with $u = 1.2$ and $R = 1.04$ unchanged. Recompute q, the terminal payoff at the down-down node, and the closed-form value $V^*(S_0)$, and state whether the optimal policy at S_0 changes.

19.3 Explain why a discount factor of $\gamma = 1$ would be inappropriate for an infinite-horizon trading agent that earns a small positive expected reward every period, and describe what specifically goes wrong with the return G_t in that case.

19.4 Using this chapter's Q-learning update rule, compute by hand the new value of $Q(S_0, \text{hold})$ after a single simulated transition from S_0 to S_d with reward 0, given a current table of $Q(S_0, \text{hold}) = 2.00$, $Q(S_d, \text{exercise}) = 5.00$, $Q(S_d, \text{hold}) = 4.50$, a learning rate $\alpha = 0.1$, and $\gamma \approx 0.9615$.

19.5 Compare Almgren-Chriss (Chapter 13, Section 13.3.5) and a reinforcement-learning execution agent (Section 19.4.1) along three dimensions: the data required to use each, the interpretability of the resulting trading schedule, and each approach's ability to adapt mid-execution to an order book that behaves differently than assumed.

19.6 Describe a reward function for a portfolio-rebalancing agent (Section 19.4.3) that would encourage the reward hacking Section 19.6 warns about, and propose a specific, concrete change to that reward function that would discourage it.

19.7 Explain why off-policy evaluation (Section 19.5.2) becomes less reliable the more a newly trained policy's behavior diverges from the policy that generated the available historical data, and describe one practical consequence this has for how a bank should stage the rollout of a new trading agent.

19.8 A market-making desk considers replacing its Avellaneda-Stoikov quoting model (Chapter 13, Section 13.4) with a learned reinforcement-learning policy. List three pieces of evidence a model validator

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should require before approving this replacement, drawing on both this chapter's discussion and Chapter 8's model-risk-management framework.

19.9 Explain why maximizing expected log-wealth (the Kelly criterion, Section 19.4.3) produces a different, generally more conservative allocation than maximizing expected wealth directly, and describe why this makes the Kelly fraction a useful sanity check for a reinforcement-learning portfolio agent trained on a simplified simulated market.

19.10 Using Section 19.4.6's discussion, explain why a reinforcement-learning execution agent trained against a fixed historical simulator might perform worse than expected once deployed at large enough scale to influence the very order flow it was trained on, and describe one practical step a firm could take to detect this problem before it causes significant losses.

19.11 Explain why behavioral cloning (Section 19.3.4) can fail badly in a state the expert trader rarely visited, even though it performs well on states similar to the expert's typical behavior, and describe how inverse reinforcement learning's approach, inferring a reward function rather than directly imitating actions, is intended to address this specific weakness.

19.12 Section 19.2.4's desk has sent four orders to venue A, giving $Q(A) = 1.60$, and six to venue B, giving $Q(B) = 1.85$. (a) A fifth order to A fills at 3.00 basis points; apply the running-average rule to compute the updated $Q(A)$, and state which venue the greedy policy now selects. (b) A single fill has a standard deviation of 2.0 basis points. Compute the standard error of each of the two estimates and explain why the switch in part (a) is far weaker evidence than it looks. (c) A fourth venue D, whose true mean of 2.4 basis points beats every existing venue, is now connected, and the desk keeps $\varepsilon = 0.2$ with exploration spread uniformly over all four. Compute the new permanent shortfall $\varepsilon(\mu^* - \bar{\mu})$, compare it to the three-venue figure, and explain why adding a venue better than all the others nonetheless raises the per-order cost of a fixed exploration rate.

19.13 After 200 orders, Section 19.2.4's desk has sent $N = (12, 180, 8)$ orders to venues A, B, C, with estimates $Q = (1.40, 1.95, 0.60)$ basis points; the fill noise is $\sigma = 2.0$. (a) Using equation (19.3) with $t = 201$, compute each venue's upper confidence bound and state where UCB routes order 201. (b) Write down each venue's Thompson-sampling posterior (flat prior, known σ), and compute the probability that venue A's draw exceeds venue B's, and likewise for C. (c) UCB and Thompson sampling disagree about order 201 in a characteristic way. Describe the disagreement and explain why both rules still route most of the next thousand orders to B.

19.14 Section 19.4.5's simulator draws price paths under the real-world drift μ. (a) Using the companion notebook's closed form for $e^{-rT}\mathbb{E}[(S_T - K)^+]$ under drift μ, compute this quantity for the \$50-strike call at $\mu = 4\%$, 8% and 12%, and identify which value is the arbitrage-free price. (b) Explain why the hedging-risk figures in Table 19.5 are correctly computed under the real-world drift while a price is not. (c) A colleague proposes training a reinforcement-learning agent to quote option prices by rewarding it with discounted payoffs on historical price paths. State what number it will converge to and propose one fix.

19.15 **Challenge (real data).** Download at least three years of daily prices for a real, liquid stock (via `yfinance`). (a) Discretize the state space into a small number of buckets based on the sign and magnitude of the trailing five-day return, in the spirit of Section 19.3.1's three-state example, and train a tabular Q-learning agent with actions {buy, hold, sell} and a reward equal to the next day's realized return (long) or its negative (short), using this chapter's Q-learning update rule. (b) Compare the trained agent's cumulative return over the full sample to a simple buy-and-hold benchmark, then re-run training on only the first half of the sample and evaluate the resulting policy on the second half, and report whether the agent's apparent advantage survives this split. (c) Using Section 19.5.2's discussion of off-policy evaluation, explain why backtesting a single historical price path is an especially weak form of policy evaluation for a trading agent, and describe one practical safeguard, drawing on Section 19.4.6 or Chapter 8's model-risk-management framework, a firm should require before deploying such an agent with real capital.

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19.9 Further Reading

- Sutton and Barto (2018), *Reinforcement Learning*. The standard textbook treatment of the Markov decision process framework, the Bellman equation, and value-based and policy-based methods developed throughout this chapter.
- Lattimore and Szepesvári (2020), *Bandit Algorithms*. A modern, rigorous treatment of the bandit problem, including the regret bounds behind Section 19.2.4's comparison of ε-greedy, UCB, and Thompson sampling, and contextual bandits.
- Mnih et al. (2015), "Human-Level Control through Deep Reinforcement Learning." The original Deep Q-Network paper underlying Section 19.3.2's function-approximation extension of tabular Q-learning.
- Schulman et al. (2017), *Proximal Policy Optimization Algorithms*. The algorithm underlying the modern policy-gradient methods discussed in Section 19.3.3.
- Nevmyvaka et al. (2006), "Reinforcement Learning for Optimized Trade Execution." An early and influential application of tabular Q-learning to real limit-order-book execution data, underlying Section 19.4.1's discussion.
- Almgren and Chriss (2001), "Optimal Execution of Portfolio Transactions." The closed-form optimal-execution model this chapter's Section 19.2.2 contrasts with a learned, model-free alternative.
- Avellaneda and Stoikov (2008), "High-Frequency Trading in a Limit Order Book." The closed-form inventory-based market-making model Section 19.4.1 generalizes into a learned quoting policy.
- Buehler et al. (2019), "Deep Hedging." The paper that introduced Section 19.4.5's risk-measure-based, neural-network approach to hedging under transaction costs and other frictions.
- Abbeel and Ng (2004), "Apprenticeship Learning via Inverse Reinforcement Learning." The foundational paper on inverse reinforcement learning discussed in Section 19.3.4.

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Chapter 20

Agent-Based Models in Finance

20.1 Introduction

Every model in this book so far has taken one of two forms. Chapter 6's machine learning models estimate a function from a fixed dataset, implicitly assuming that the data-generating process does not change while the model learns. Chapter 12's portfolio theory and Chapter 19's reinforcement-learning agents each reason about a single decision-maker, or a small number of them, optimizing against an environment that is either fixed or, at most, adapts slowly.

Neither approach is built to answer a question that matters a great deal in practice: what happens when many participants, each following a simple, individually reasonable rule, interact repeatedly in the same market? Agent-based models (ABMs) answer that question. Rather than solving for an equilibrium in closed form or estimating a statistical relationship from historical data, an ABM specifies a population of heterogeneous agents, each governed by a simple behavioral rule, lets them interact in a simulated market, and studies whatever aggregate pattern emerges: a stable price, a bubble, a crash, a cascade of defaults.

This chapter's central claim is that emergence is the point, not a side effect. Chapter 14's account of the 2010 Flash Crash described a version of this: no single participant's inventory-management rule was unreasonable in isolation, yet many such rules operating simultaneously produced a collectively destabilizing hot-potato effect that no single participant intended or could have predicted from their own rule alone. An ABM can reproduce that kind of story directly, agent by agent, rather than only describing it after the fact.

Several worked examples run through this chapter, each reproduced in the companion notebook under the same compute-first discipline as the rest of the book. A zero-intelligence double-auction example shows that a market mechanism can produce an efficient outcome even when every participant in it behaves close to randomly.

A fundamentalist-chartist example shows how the same market can either converge smoothly toward fair value or spiral into a self-reinforcing bubble, depending only on the mix of trading strategies, with no change to the market mechanism, and a 500-period extension shows fat tails and volatility clustering emerging from two simple, fixed behavioral rules switching in and out of dominance. A financial-contagion example, built on the clearing-vector logic behind real central-bank stress-testing models, shows a shock to one institution cascading into the default of a second that was never exposed to the original shock, and a second network topology, compared against the first under the same shock, shows that the same total loss can be concentrated into a few outright defaults or spread thinly across many weakened survivors, depending only on the network's shape.

A final pair of examples previews the AI-driven alternative to fixed-rule agents, developed in Section 20.3.5. The first is a seller who learns which of two ask prices is more profitable by trial and error, using Chapter 19's Q-learning, and shows the limits of trial-and-error learning when an unlucky early experience locks in the wrong lesson. The second is a fundamentalist-chartist market in which the population mix is switched not by a threshold the modeler imposes but by agents comparing each strategy's recent profit through a discrete-choice rule, so that the bubble-triggering threshold derived earlier is crossed endogenously rather than by assumption.

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This places agent-based models among the three modeling paradigms the book has now covered. Chapter 6's supervised and unsupervised models answer "given this data, what function best describes it," and are validated by predictions on data the model never saw. Chapter 19's reinforcement-learning agents answer "given this environment, what sequence of actions maximizes reward," and are validated by realized performance, in a backtest or in production.

Agent-based models answer a third question: "given these individual rules, what pattern emerges when many such agents interact," and, as Section 20.4.1 develops, they are validated far more weakly than the other two, by whether a simulated pattern resembles an observed one rather than by a held-out prediction or a realized return. None of the three is more fundamentally correct; they answer different questions, and recognizing which question a financial problem poses (prediction, sequential decision, or emergence) is one of this book's recurring practical lessons. Table 20.1 lays the three side by side.

	Supervised/unsupervised ML (Ch. 6)	Reinforcement learning (Ch. 19)	Agent-based models (this chapter)
Core question	What function best fits this data?	What action sequence maximizes reward?	What pattern emerges from many interacting rules?
Input	A fixed, historical dataset	An environment to interact with	A set of behavioral rules and a market mechanism
Validation	Held-out prediction accuracy	Realized backtest or live performance	Resemblance to observed stylized facts (weakest of the three)
Typical output	A trained predictive model	A trained policy	A simulated history and its emergent statistics

Table 20.1: Three modeling paradigms used in this book

A reader who has followed this book's emphasis on machines that learn might object that every rule above, zero-intelligence, fundamentalist, chartist, is fixed by the modeler in advance, which sits oddly in a book about AI in finance. The fixed-rule design is a deliberate methodological choice, not a limitation of agent-based modeling, and it keeps Table 20.1's third column honest: separating the question of what pattern emerges from a given set of rules from the question of how an agent's rule might itself be learned keeps each worked example small enough to hand-check, the discipline Chapters 1 through 19 apply everywhere else.

That second question, learning the rule itself, is Chapter 19's question, and Section 20.3.5 connects the two. It argues that agent-based models and multi-agent reinforcement learning (MARL) are two responses to the limitation Chapter 19 identifies in its own single-agent framing, that training a policy against a fixed historical simulator assumes every other participant's behavior stays fixed, and it makes the connection concrete through the learning-seller example previewed above, in which one agent in an otherwise fixed-rule market learns its own quoting rule with Chapter 19's Q-learning. Every fixed-rule agent elsewhere in the chapter is a transparent, tractable special case of that more general world, in which agents learn rather than follow a rule handed to them and which is much harder to train and validate, not a rejection of it.

Each application follows the same escalating structure: state the mechanism and a small, hand-checkable illustration, then show what happens when the mechanism is scaled up or repeated. Sections 20.2.2 and 20.2.4 build market-level price dynamics from individual trading rules; Section 20.3.1 scales the latter to hundreds of periods to show statistical patterns emerging; Sections 20.3.2 and 20.4.2 apply the same logic to a network of banks. Section 20.3.5 compares the fixed-rule agents with Chapter 19's learned alternative, and Section 20.4.4 distills the common workflow into a reusable template. The chapter closes with the practical limitations, summarized in Section 20.4.1 and the Challenges section, that keep agent-based models a specialized complement to the book's other methods rather than a replacement for them.

20.2 Why Agent-Based Models

Replacing the representative agent with many heterogeneous ones changes what a model can produce.

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20.2.1 From Representative Agents to Heterogeneous, Interacting Ones

Much of classical finance, and much of this book, reasons in terms of a representative agent: Chapter 12's CAPM derives an equilibrium as though a single, rational, mean-variance investor stood in for the whole market; Chapter 7's GARCH models fit a single statistical process to an entire return series without asking which market participants' behavior produced that process. These simplifications are enormously useful, and this book relies on them throughout, but by construction they cannot answer certain questions.

A representative-agent model cannot easily explain why a market that is efficient on an average day can suffer a Flash Crash on a particular afternoon, because the average behavior built into the model has already smoothed away the heterogeneity that made the crash possible. Agent-based modeling starts at the other end: instead of solving for what a single rational agent would do, it specifies what a population of different, typically boundedly rational, agents do, and asks what pattern emerges from their interaction.

This bottom-up approach is well suited to the phenomena that are hardest to explain with a representative agent: the fat tails and volatility clustering Chapter 7's GARCH models describe statistically but do not explain mechanistically; asset-price bubbles and crashes that unfold over time rather than instantaneously; and the kind of systemic, contagious risk Chapter 8's stress-testing framework treats largely at the level of a single institution's portfolio rather than as a property of an interconnected system.

Agent-based modeling did not originate in finance. Its most famous early instance is the segregation model of Schelling (1971), built with a checkerboard and coins of two colors: each agent mildly prefers to have at least some same-type neighbors, far short of wanting a segregated neighborhood, and moves to a nearby empty square only if that preference is unmet. Simulating this individually reasonable rule across many agents produces, almost without exception, a starkly segregated checkerboard, an outcome none of the agents asked for. The financial applications in this chapter apply the same logic to markets: individually reasonable rules, simulated across many interacting agents, produce an aggregate outcome (efficiency, a bubble, a cascade of defaults) that is a property of the interaction rather than of any agent's intentions.

The idea reached finance through the Santa Fe Institute's Artificial Stock Market, built in the early 1990s by Arthur, Holland, LeBaron, Palmer, and Tayler and described in Arthur et al. (1997), widely regarded as the first serious agent-based model of a financial market. Instead of the two fixed strategy types of Section 20.2.4's model, its agents carried evolving collections of predictive rules, more like a population of competing forecasting models, and poorly performing rules died out while successful ones proliferated, an early instance of the performance-based selection Brock and Hommes later formalized more tractably. Its central finding, that the market could switch between a calm, efficient regime and a volatile, trend-chasing regime depending on how quickly agents adapted their rules, foreshadows the mechanism behind Section 20.3.1's worked example, and the project is credited with establishing agent-based computational finance as a research area separate from the social-science tradition of Schelling's model.

Agent-based models now play a recurring role in the finance industry, which this chapter's later sections examine. Central banks and other macroprudential regulators run network-based models with behavioral feedbacks, such as the Bank of England's RAMSI in the case vignette below, to stress-test a banking system's interconnected exposures rather than one institution at a time, now a standard part of the macroprudential toolkit. Exchanges and microstructure researchers use zero-intelligence and heterogeneous-agent markets, the subject of Sections 20.2.2 and 20.2.4, to learn how much of observed market efficiency, or its breakdown in an event like the Flash Crash, comes from the trading mechanism rather than from participants' sophistication. Quantitative trading desks build synthetic, agent-populated limit-order-book markets to test how a new execution algorithm of the kind Chapter 13 develops is likely to move prices and interact with others before risking capital on it, a cheaper and safer proving ground than live trading. And a growing body of research combines agent-based markets with the multi-agent reinforcement learning of Section 20.3.5, training an execution or market-making policy inside a simulated population of agents rather than against static historical data alone.

Figure 20.1 sketches the simulation loop every example in this chapter follows: a population of agents, each with its own rule, observes the market state; each agent acts (submits an order, updates a belief, decides whether to pay a debt); a market mechanism aggregates the actions into a new state; and the loop repeats. Nothing is estimated from historical data the way Chapter 6's models are; everything is specified as a rule and simulated forward.

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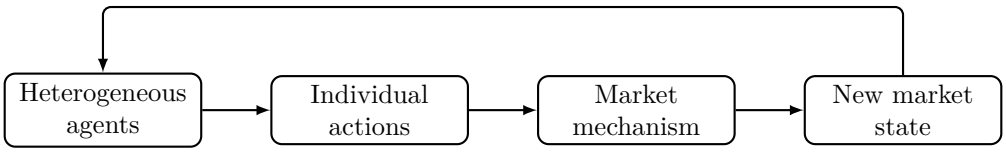


Figure 20.1: The basic agent-based simulation loop

20.2.2 Zero-Intelligence Traders and the Double Auction

The most striking early result in agent-based market modeling, due to Gode and Sunder (1993), is that a double auction, the continuous matching of bids against asks behind the limit order book of Chapter 3, can produce a highly efficient outcome even when no trader in it is intelligent at all: a "zero-intelligence" trader submits a bid or ask drawn at random from a budget-feasible range, with no strategic reasoning about price or other traders. Consider two buyers with private valuations $V_{B1} = \$52$ and $V_{B2} = \$48$, and two sellers with private costs $C_{S1} = \$45$ and $C_{S2} = \$50$, for one unit of the same asset each. The efficient allocation, the one a fully rational market would produce, is easy to find: buyer 1 and seller 1 should trade, since $V_{B1} > C_{S1}$ generates a surplus of $52 - 45 = \$7$, while buyer 2 and seller 2 should not, since $V_{B2} < C_{S2}$ would destroy value. Total efficient surplus is therefore \$7.

Now suppose each buyer i submits a random bid drawn from $[0, V_{Bi}]$ and each seller j a random ask drawn from $[C_{Sj}, 100]$, ranges that keep each trader budget-feasible (a buyer never bids above its valuation, a seller never asks below its cost), with \$100 an arbitrary price ceiling. Nobody knows anyone else's valuation. Suppose that in one simulated round the draws are $b_1 = \$50$ (buyer 1), $b_2 = \$30$ (buyer 2), $a_1 = \$47$ (seller 1), and $a_2 = \$65$ (seller 2). The standard double-auction clearing rule sorts bids from highest to lowest and asks from lowest to highest, then matches and trades every pair whose bid meets or exceeds the ask:

$$\begin{aligned} \text{Bids (descending):} &\quad 50,\ 30, \\ \text{Asks (ascending):} &\quad 47,\ 65. \end{aligned}$$

The top bid, \$50, meets the top ask, \$47 ($50 \geq 47$), so buyer 1 and seller 1 trade; the second bid, \$30, does not meet the second ask, \$65 ($30 < 65$), so buyer 2 and seller 2 do not. By matching random bids against random asks, the clearing mechanism has selected the efficient pair, buyer 1 with seller 1, and rejected the inefficient one. Whatever transaction price rule is used, for instance the midpoint of the matched bid and ask, $(50 + 47)/2 = \$48.5$, the realized surplus, computed from true valuations rather than random bids, is $V_{B1} - C_{S1} = 52 - 45 = \$7$, the full efficient surplus, captured although neither trader bid or asked anything close to its true valuation. The lesson, confirmed systematically in Gode and Sunder's experiments across many traders and rounds, is that a market's efficiency can come largely from the structure of the trading mechanism rather than from the sophistication of its traders, a counterintuitive result relative to the fully rational representative agent of Chapter 12's equilibrium models. This single round shows that the mechanism *can* select the efficient pair from random bids, not that it *always* will. As the companion notebook's supplementary experiments with this two-buyer, two-seller market confirm, a single random draw from a wide range is a noisy way to reach that outcome; Gode and Sunder's robust result relies on many traders and many trading opportunities per period, not on one exchange like the one worked above.

20.2.3 Building a Simple ABM in Python

Every agent-based model in this chapter reduces to two ingredients Chapter 2 already teaches how to code: a data structure for each agent's state, and a function that updates the market given every agent's action. The zero-intelligence double auction above takes only a few lines:

```
import random

def zero_intelligence_round(buyers, sellers, seed=None):
    rng = random.Random(seed)
    bids = [(rng.uniform(0, v), v) for v in buyers]      # random bid, true value
```

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```
    asks = [(rng.uniform(c, 100), c) for c in sellers]    # random ask, true cost
    bids.sort(key=lambda x: -x[0])
    asks.sort(key=lambda x: x[0])
    trades = []
    for (bid, val), (ask, cost) in zip(bids, asks):
        if bid >= ask:
            trades.append({'price': (bid + ask) / 2, 'surplus': val - cost})
    return trades

trades = zero_intelligence_round(buyers=[52.0, 48.0], sellers=[45.0, 50.0], seed=2)
print(trades)   # [{'price': 48.91, 'surplus': 7.0}]
```

Every other example in the chapter follows the same pattern: Section 20.2.4's price dynamics are a one-line update rule called in a loop; Section 20.3.2's clearing vector is a short function computing each bank's payment given what it receives; and Section 20.3.1's 500-period simulation is the same fundamentalist-chartist loop with an `if` statement selecting each period's regime. None of this needs tools beyond Chapter 2; what distinguishes agent-based modeling is not the code but the habit of specifying individual rules and letting the population-level pattern emerge from running them, rather than solving for it in closed form. The brevity matters: it is what makes a fixed-rule agent fully interpretable, since each of these short functions is a rule a reader can audit line by line, as a Chapter 19 policy's learned weights are not, a contrast Section 20.3.5 develops once both approaches are on the table.

This is also why the companion notebook, not the chapter text, is the place to explore an ABM's behavior. Each worked example reports one set of parameter values and, where relevant, one random seed. Changing any of them (the switching threshold, the population shares, the interbank network's shape, the buyer valuations the learning seller faces) produces a different simulated history, and the only way to build intuition for how sensitive a result is to an assumption is to make the change and rerun, the hands-on exploration this book's preface encourages for every companion notebook.

20.2.4 Heterogeneous Agents, Bubbles, and Crashes: Fundamentalists versus Chartists

Zero-intelligence traders show that a market mechanism can be efficient despite unsophisticated participants; they do not explain bubbles and crashes, since random bidding has no memory and no tendency to extrapolate a trend. The next step in the agent-based tradition, associated with Brock and Hommes and with Lux and Marchesi, introduces two purposeful agent types whose population shares can shift over time. In plain terms, a fundamentalist behaves like a value investor: the further the price sits below what the asset is worth, the more eagerly the fundamentalist buys, and the further above, the more eagerly it sells, a demand rule that always leans toward pulling the price back to fundamental value. A chartist behaves like a momentum trader: the faster the price has just risen, the more eagerly the chartist buys, betting that the trend continues, with no opinion about what the asset is worth. Formally, fundamentalists believe the price will revert toward a known fundamental value p_f and submit demand proportional to the mispricing,

$$d_t^f = \alpha(p_f - p_t), \tag{20.1}$$

while chartists (trend followers) extrapolate the most recent price change forward, submitting demand proportional to recent momentum,

$$d_t^c = \beta(p_t - p_{t-1}). \tag{20.2}$$

A simple market-impact rule aggregates the two populations' demand, weighted by their respective population shares n_f and $n_c = 1 - n_f$, into a single aggregate demand,

$$D_t = n_f\, d_t^f + n_c\, d_t^c,$$

which a linear price-impact equation converts into a price update,

$$p_{t+1} = p_t + \lambda D_t, \tag{20.3}$$

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where λ controls how strongly aggregate demand moves the price. The "Aggregate demand" column of Table 20.2 below reports D_t at each step. The fundamental value, the two behavioral rules, and the price-impact equation are held fixed across the two scenarios below; only the population mix (n_f, n_c) differs.

Worked Example 20.1: Convergence versus a Self-Reinforcing Bubble

Let $p_f = 100$, $\alpha = 0.3$, $\beta = 1.0$, $\lambda = 1.5$, and start both scenarios from the same two initial prices, $p_{-1} = \$93$ and $p_0 = \$95$ (a modest existing uptrend). In Scenario A, fundamentalists are the large majority, $n_f = 0.8$, $n_c = 0.2$. Table 20.2 works the first three steps by hand.

t	p_t	d_t^f	d_t^c	Aggregate demand D_t	p_{t+1}
0	95.000	1.500	2.000	1.600	97.400
1	97.400	0.780	2.400	1.104	99.056
2	99.056	0.283	1.656	0.558	99.893

Table 20.2: Scenario A: fundamentalist-dominated market ($n_f = 0.8$, $n_c = 0.2$), converging toward $p_f = 100$

The price rises from \$95.00 toward the fundamental value of \$100 in shrinking steps, \$2.40, then \$1.66, then \$0.84: the damped convergence a stabilizing fundamentalist majority should produce, since the fundamentalist demand pulling the price toward p_f weakens as the price approaches it.

Scenario B changes only the population mix, to a market with no fundamentalists at all, $n_f = 0$, $n_c = 1$. With $n_f = 0$ the price update collapses to $p_{t+1} = p_t + \lambda\beta(p_t - p_{t-1})$, and since $\lambda\beta = 1.5 \times 1.0 = 1.5 > 1$, each period's price change is 1.5 times the previous one, an explosive recursion with no anchor to p_f:

$$p_1 = 95 + 1.5(95 - 93) = 95 + 3.00 = 98.00,$$
$$p_2 = 98 + 1.5(98 - 95) = 98 + 4.50 = 102.50,$$
$$p_3 = 102.5 + 1.5(102.5 - 98) = 102.5 + 6.75 = 109.25.$$

The price moves further from the fundamental value of \$100 every period, by \$3.00, then \$4.50, then \$6.75, a self-reinforcing bubble driven by momentum-chasing, with each move larger than the last. The market mechanism did not change between Scenario A and Scenario B; only the composition of the trading population did. This kind of regime-dependent behavior, stable in one population mix and explosive in another, is what a single representative agent cannot produce, since a representative agent is always the same agent.

20.2.5 Why the Market Tips: A General Stability Condition

Scenario A converged and Scenario B exploded, but they also differed in n_c, which raises the question of how much chartist weight tips the market from one regime to the other. The answer is the kind of population-level fact this chapter keeps returning to. It is written into neither behavioral rule, since neither demand equation mentions a critical population share, yet their combination produces a sharp, calculable threshold that a representative-agent model could not even pose, having no population mix to vary (Section 20.2.1). Intuitively, a chartist's demand today depends on how much the price just rose, but that demand also pushes the price up tomorrow, feeding a fresh, possibly larger, round of chartist demand the day after; whether the loop fizzles out or feeds on itself depends on its strength relative to the fundamentalist demand pulling the other way, a balance the threshold below makes precise. Substituting the two demand equations into the price-update rule turns the model into a linear recursion in the two most recent prices alone,

$$p_{t+1} = \lambda n_f \alpha p_f + (1 - \lambda n_f \alpha + \lambda n_c \beta)\, p_t - \lambda n_c \beta\, p_{t-1}, \tag{20.4}$$

and the standard stability analysis for a recursion of this form (worked out in full, and confirmed against a parameter sweep, in the companion notebook) shows that, for every parameter combination used in this section, the market is stable if and only if

$$\lambda n_c \beta < 1. \tag{20.5}$$

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Scenario A's coefficient was $\lambda n_c \beta = 1.5 \times 0.2 \times 1.0 = 0.30 < 1$, comfortably stable; Scenario B's, with no fundamentalists diluting the chartist share at all, was $\lambda n_c \beta = 1.5 \times 1.0 \times 1.0 = 1.50 > 1$, explosive. Solving $\lambda n_c \beta = 1$ for the population share gives the exact critical threshold,

$$n_c^* = \frac{1}{\lambda \beta} = \frac{1}{1.5 \times 1.0} \approx 0.667,$$

so this market tips from convergent to explosive once chartists exceed roughly two-thirds of the trading population, holding λ and β at the values used throughout this section. The companion notebook confirms the threshold: at $n_c = 0.60$, just below n_c^*, the price oscillates around p_f with shrinking amplitude, settling close to \$100 within about 60 periods, while at $n_c = 0.70$, just above n_c^*, it oscillates with growing amplitude, reaching 87.7 after only 15 periods and still widening. Near the threshold, convergence is oscillatory rather than the smooth, monotonic approach of Scenario A's $n_c = 0.2$, because the recursion's roots are a complex-conjugate pair whenever n_c is this large: the market converges not just more slowly but differently, overshooting p_f and swinging back before settling. This kind of sharp regime change, a market that behaves completely differently once one parameter crosses a critical value, motivates the endogenous switching mechanism behind Section 20.3.1's simulation: real markets do not sit at a fixed n_c, and a population that drifts back and forth across a threshold like n_c^* alternates between the qualitatively different regimes worked out above.

20.3 Emergence and Systemic Risk

The point of these models is what they produce without being told to, including contagion nobody designed.

20.3.1 Emergent Stylized Facts: Fat Tails and Volatility Clustering from Simple Rules

Chapter 7's GARCH(1,1) model captures volatility clustering by imposing an autoregressive structure on the variance of returns: large moves tend to follow large moves because the model's parameters say so. Heterogeneous agent models like Section 20.2.4's reproduce the same stylized fact, along with the excess kurtosis (fat tails) Chapter 7 discusses when introducing GARCH, without specifying any variance process. Suppose the population mix (n_f, n_c) evolves over time, shifting toward chartists whenever chartist strategies have recently been more profitable, the performance-based strategy switching Brock and Hommes formalize. The market then alternates endogenously between calm, fundamentalist-dominated periods like Scenario A and volatile, chartist-dominated episodes like Scenario B, and clustered volatility and heavy-tailed returns emerge from the switching itself rather than from an assumption built into the model. The distinction matters in practice: a GARCH model is a superb tool for forecasting volatility once fitted, Chapter 7's use case, but it does not explain why volatility clusters, while an agent-based model offers one candidate mechanism, at the cost of being much harder to fit confidently to a historical dataset, a tension Section 20.4.1 returns to.

Worked Example 20.2: Volatility Clustering from Endogenous Regime Switching

The companion notebook makes this concrete by extending the fundamentalist-chartist model to 500 simulated periods, with two bounded (non-explosive) regimes rather than Scenario B's deliberately unbounded pure-chartist case: a calm regime ($n_f = 0.85$, $n_c = 0.15$, normally distributed price shocks with standard deviation 0.15) and a volatile regime ($n_f = 0.25$, $n_c = 0.75$, shock standard deviation 0.50), each added to the price update of equation (20.3). The simulation keeps $\alpha = 0.3$ and $p_f = 100$ but uses $\beta = 1.1$ and $\lambda = 1.0$, not the $\beta = 1.0$ and $\lambda = 1.5$ of Section 20.2.4, and the difference matters: by condition (20.5), Section 20.2.4's values would give the volatile regime $\lambda n_c \beta = 1.5 \times 0.75 \times 1.0 = 1.125 > 1$, an explosive market, whereas the simulation's values give $1.0 \times 0.75 \times 1.1 = 0.825$ in the volatile regime and 0.165 in the calm one, both safely below 1. The simulation switches into the volatile regime whenever the most recent price move exceeds a fixed threshold, a simple proxy for a situation in which a chartist strategy would just have outperformed a fundamentalist one, and reverts to the calm regime otherwise. Both regimes are individually stable: the price never diverges the way Scenario B's pure-chartist market did, ranging from \$93.93 to \$107.77 over the full 500 periods. But the two regimes' realized volatility differs sharply. The simulation spends 213 periods in the calm regime, with a return variance of 0.147, and 287 periods in the volatile regime, with a return variance of

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0.903, a variance ratio of 6.14× between two regimes driven by the same shock process and differing only in which population of traders is active. The full 500-period return series has a Pearson kurtosis of 4.67, well above the Gaussian benchmark of 3.0 (an excess kurtosis of 1.67), confirming fat tails, and the autocorrelation of squared returns, the standard volatility-clustering diagnostic Chapter 7 applies to real return series, is 0.658 at a lag of one period and still 0.216 at a lag of five, both far from the zero that i.i.d. returns would produce. None of these statistics (the variance ratio, the excess kurtosis, the squared-return autocorrelation) was a modeling target; all three emerged from a simulation whose only design choice was which population of fixed behavioral rules was active each period.

20.3.2 Agent-Based Models of Systemic Risk and Financial Contagion

The examples so far model a single market's price. A separate application, and for regulators arguably a more important one, models a financial system of interconnected institutions, extending Chapter 8's stress-testing framework from a single portfolio to a network. The idea is intuitive: if a bank cannot pay all it owes, every bank counting on that payment is now short of cash, and may in turn be unable to pay everyone counting on *it*, so one institution's shortfall can pass through a network of obligations to institutions that never dealt with the original problem. Eisenberg and Noe (2001) formalize this chain reaction as a clearing vector: given a network of interbank obligations and each bank's external (non-interbank) assets, compute how much each bank actually pays, recognizing that a bank not paid in full by others may be unable to pay its own obligations in full. Writing $\bar{p}_i$ for bank i's total interbank obligation, e_i for its external assets, and π_{ji} for the fraction of bank j's obligations owed to bank i, the clearing vector p solves

$$p_i = \min\left(\bar{p}_i,\ e_i + \sum_j \pi_{ji}\, p_j\right) \quad \text{for every bank } i, \tag{20.6}$$

so each bank pays its obligations in full if its resources allow, and otherwise pays everything it has.

20.3.3 Worked Example 20.3: A Shock That Cascades Through the Network

Consider three banks connected in a simple chain: Bank A owes Bank B \$50, and Bank B owes Bank C \$40; Bank C has no obligations to anyone. Each bank also holds external assets: \$60 for A, \$10 for B, and \$5 for C. Figure 20.2 shows this network.

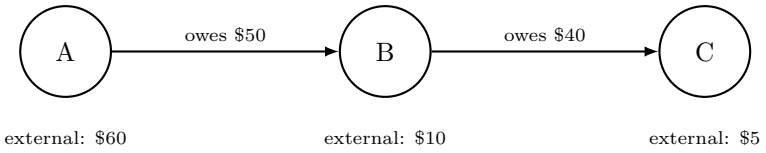


Figure 20.2: A three-bank interbank obligation network

In the baseline, with no shock, every bank pays in full. A's resources are its \$60 of external assets (no one owes A anything), covering its \$50 obligation to B and leaving A with net worth $60 - 50 = \$10$. B's resources are its \$10 of external assets plus the \$50 from A, \$60 in total, covering its \$40 obligation to C and leaving B with net worth $60 - 40 = \$20$. C's resources are its \$5 of external assets plus the \$40 from B, a net worth of \$45. Total system-wide net worth is $10 + 20 + 45 = \$75$.

Now suppose an external shock, a bad loan or a trading loss unrelated to the interbank network, destroys \$45 of Bank A's external assets, leaving A with \$15. Recomputing the clearing vector: A's resources are \$15 against its \$50 obligation, so A pays only $\min(15, 50) = \$15$ to B, a default with a 30% recovery rate, leaving A with net worth \$0. B's resources are its \$10 of external assets plus only \$15 from A (not the \$50 it was owed), \$25 in total, against its \$40 obligation to C; B can pay only $\min(25, 40) = \$25$, so *B also defaults*, with a 62.5% recovery rate, although B suffered no external shock: its default is entirely a consequence of A's. C's resources are its \$5 plus only \$25 from B (not the \$40 it was owed), for a final net worth of \$30, down from \$45, a real loss even though C owes nothing and never defaults.

Table 20.3 summarizes the cascade. Total system-wide net worth falls from \$75 to $0 + 0 + 30 = \$30$, a loss of \$45, equal to the original shock: the network neither creates nor destroys value, it redistributes one

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institution's loss across every institution downstream of it, in a pattern invisible to a stress test of Bank A's portfolio alone.

Bank	External (shocked)	Owed	Received	Paid	Net worth (baseline)	Net worth (shocked)
A	15	50	0	15	10	0
B	10	40	15	25	20	0
C	5	0	25	0	45	30

Table 20.3: The default cascade following a $45 shock to Bank A's external assets (all figures in $)

20.3.4 Case Vignette: The Bank of England's RAMSI Model

The contagion mechanism worked by hand in Section 20.3.3 is a simplified version of machinery central banks actually use. The Bank of England's RAMSI (the Risk Assessment Model for Systemic Institutions) is a large-scale, top-down stress-testing model of the UK banking sector: top-down in that the same model and assumptions apply to every major bank, so results can be compared across banks and added up to a view of the whole system. Work on it began before the 2008 financial crisis, whose Gaussian-copula CDO mispricing Chapter 9 examines; a prototype was presented at central-bank conferences in 2007, and the model was used in the IMF's 2011 assessment of the UK financial sector. Each bank enters with its own balance sheet and income statement, projected forward under macroeconomic scenarios and linked to the others through interbank exposures, the network-level cascade of Section 20.3.3's three-bank example at a far larger scale.

RAMSI is not a full agent-based model in this chapter's sense. Its banks do not optimize, and its developers describe them as acting largely passively: each follows simple behavioral rules, such as a target capital ratio that governs how retained earnings are reinvested. What carries it beyond Section 20.3.3's clearing vector is a set of feedback effects on top of those rules. A bank whose fundamentals are projected to weaken pays more for funding and, past certain thresholds, is shut out of funding markets altogether, and banks perceived to resemble it face a higher chance of being shut out too. A bank in trouble may also sell assets, pushing down their prices and causing mark-to-market losses at every other bank holding them, a channel absent from the pure balance-sheet cascade, where no bank's response could feed back into the market prices other banks' balance sheets depend on. The distinction matters in practice: a pure Eisenberg-Noe network captures what regulators call first-round, direct contagion (a defaulted counterparty's unpaid obligations), while fire-sale and funding feedbacks capture second-round, indirect contagion (one bank's distress damaging other, still-solvent banks through prices and funding markets), the individually reasonable but collectively destabilizing dynamic of Section 20.2.1's Schelling and Flash Crash discussions. In the Bank's prototype these feedbacks turn jointly normal shocks into a bimodal distribution of system-wide outcomes, with a small second peak of contagious defaults in the tail, which neither the three-bank chain nor the hub-and-spoke example, both purely first-round mechanisms, can reproduce.

Models of this kind stress-test the UK banking system as a network rather than one institution at a time, extending Chapter 8's single-portfolio framework to the contagious, second-round loss an institution-by-institution stress test misses: in the worked example, a stress test of Bank B alone, which received no direct shock, would have reported B as healthy right up until it wasn't. Other central banks and the IMF have developed comparable network-based systemic-risk tools, some with agent-based behavioral components, and system-wide stress testing is now a standard, if specialized, part of the macroprudential toolkit alongside the single-institution VaR and stress-testing methods of Chapter 8.

20.3.5 Agent-Based Models versus Multi-Agent Reinforcement Learning: Fixed Rules versus Learned Behavior

Chapter 19's discussion of multi-agent dynamics observes that an agent trained against a static historical simulator implicitly assumes every other participant's behavior stays fixed, an assumption the Flash Crash's hot-potato effect shows can fail badly. Agent-based models and multi-agent reinforcement learning (MARL) are two responses to that observation rather than competitors. An ABM specifies each agent's rule directly (fundamentalist, chartist, zero-intelligence) and asks what emerges from many fixed rules interacting; a

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MARL system lets each agent's behavior be learned, typically with Chapter 19's Q-learning or policy-gradient machinery, adapting to the others' evolving behavior. The ABM gives up realism about how traders learn in exchange for transparency and tractability, since every rule in this chapter is a single, interpretable equation; MARL gives up that transparency for agents that behave more like real, adaptive participants, at the cost of the training instability and weak convergence guarantees Chapter 19 flags as MARL's central open problem. A growing body of research combines the two, building ABMs in which some or all agents follow reinforcement-learning policies and using the simulated market as the kind of training environment Chapter 19's execution and market-making sections rely on, an active research direction rather than a settled methodology. Table 20.4 summarizes the practical trade-offs.

	Agent-based model	Multi-agent reinforcement learning
Agent behavior	Fixed, hand-specified rule	Learned via Chapter 19's Q-learning or policy-gradient methods
Interpretability	High: every rule is a single, readable equation	Low: behavior is implicit in a trained policy's weights
Adaptivity	None, unless the modeler adds explicit switching (Section 20.2.4)	Agents adapt continuously to each other's evolving behavior
Convergence guarantees	Not applicable; the model is simulated, not trained	Weak once multiple agents learn simultaneously (Chapter 19)
Computational cost	Low to moderate	High, and grows sharply with agent count
Typical role in this book	Exploring what mechanisms are capable of producing an observed pattern (Section 20.4.1)	Training a single production policy against a realistic simulated environment

Table 20.4: Agent-based models (fixed rules) versus multi-agent reinforcement learning (learned behavior)

20.3.6 Worked Example 20.4: A Learning Seller in the Double Auction

To make the ABM-versus-MARL distinction concrete, reconsider Section 20.2.2's double auction, but replace one zero-intelligence seller with a seller who learns, in the trial-and-error spirit of Chapter 19's bandit examples. Each round, a new buyer arrives with a valuation drawn from $\{45, 50, 55, 60, 65\}$ with equal probability, facing a seller with cost \$40 who chooses between two fixed ask levels: Arm H (ask \$58) or Arm Lo (ask \$47). A trade occurs whenever the ask is at or below the buyer's valuation, earning the seller ask minus cost; a rejected ask earns nothing. The true expected profit of each arm, unknown to the seller but computable from the valuation distribution, is

$$\mathbb{E}[\text{profit}_H] = \Pr(v \geq 58) \times (58 - 40) = \tfrac{2}{5} \times 18 = 7.2,$$
$$\mathbb{E}[\text{profit}_{Lo}] = \Pr(v \geq 47) \times (47 - 40) = \tfrac{4}{5} \times 7 = 5.6,$$

so Arm H, despite trading only half as often, is the better arm: its larger profit per trade more than compensates for its lower fill rate, a trade-off that neither arm's trade frequency reveals on its own.

The learning seller explores both arms twice (rounds 1 through 4, alternating H, Lo, H, Lo) and then plays greedily, choosing whichever arm has the higher running-average realized profit. Table 20.5 traces one representative 12-round history. After the four exploration rounds, Arm H's running average stands at \$9.00 against Arm Lo's \$3.50, so the seller plays Arm H for the rest of the sequence, the true optimal choice, and its running average, swinging between \$6.00 and \$10.80 along the way, happens to finish the twelfth round at the true value of \$7.20.

This is the ABM-versus-MARL distinction of Table 20.4 within a single example: every other seller in Section 20.2.2 follows a fixed rule (draw a random ask), while this one adapts its behavior to realized experience, the shift from a specified rule to a learned policy that separates the table's two columns. A single

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Round	Buyer value	Arm played	Profit	Avg. profit, H	Avg. profit, Lo
1	45	H	0	0.00	–
2	45	Lo	0	0.00	0.00
3	60	H	18	9.00	0.00
4	55	Lo	7	9.00	3.50
5	55	H	0	6.00	3.50
6	60	H	18	9.00	3.50
7	60	H	18	10.80	3.50
8	45	H	0	9.00	3.50
9	55	H	0	7.71	3.50
10	45	H	0	6.75	3.50
11	55	H	0	6.00	3.50
12	65	H	18	7.20	3.50

Table 20.5: One 12-round history of the learning seller in the double auction

12-round history shows only so much. In this valuation sequence, each arm's first exploration draw met a \$45 buyer and earned nothing, and the deciding second draws went H's way: a \$60 buyer accepted the high ask, while the \$55 buyer facing the low ask earned the seller only \$7. That locked the seller onto the correct arm, but on two observations per arm, a margin far too thin to trust. A less fortunate exploration phase could as easily have left a misleadingly low early estimate for the better arm, favoring the worse one for a while: the explore-exploit tension that makes reinforcement learning harder in practice than this clean run suggests.

20.3.7 Discrete-Choice Switching Replaces the Fixed Threshold

The learning seller replaces one agent's fixed rule with something learned. The same question applies to Section 20.3.1's regime-switching market, where the population mix (n_f, n_c) evolved only according to a threshold rule handed down by the modeler (switch to the volatile regime whenever the last price move exceeds a fixed cutoff), not according to anything a fundamentalist or chartist experienced. Brock and Hommes's own mechanism, named in Sections 20.2.1 and 20.3.1 and derived in full in this book's companion technical note on the Brock-Hommes model, replaces that external threshold with the comparison a real trader would make: each period, each strategy's population share updates by a discrete-choice (logit) rule driven by that strategy's recently realized profit, so the switching Section 20.3.1 imposed from outside emerges from many agents each comparing two numbers and choosing, probabilistically, whichever strategy has recently paid off better.

Reusing Section 20.2.4's market, $p_f = 100$, $\alpha = 0.3$, $\beta = 1.0$, $\lambda = 1.5$, $p_{-1} = 93$, $p_0 = 95$, define each strategy's realized profit at t as the price change just observed times the demand that strategy would have submitted one period earlier,

$$\pi_{f,t} = (p_t - p_{t-1})\, d^f_{t-1}, \qquad \pi_{c,t} = (p_t - p_{t-1})\, d^c_{t-1},$$

and update the fundamentalist population share by the binary logit rule

$$n_{f,t} = \frac{e^{\theta \pi_{f,t}}}{e^{\theta \pi_{f,t}} + e^{\theta \pi_{c,t}}}, \qquad n_{c,t} = 1 - n_{f,t}, \tag{20.7}$$

where $\theta \geq 0$ is the intensity of choice (written θ rather than the technical note's β, to avoid clashing with this chapter's chartist coefficient β): $\theta = 0$ freezes the population at its starting split regardless of performance, while a large θ sends nearly the whole population to whichever strategy did even slightly better last period.

Starting from an uninformed $n_{f,0} = n_{c,0} = 0.5$, no fitness history yet exists to compare, and $\theta = 0.5$, Table 20.6 works the first three periods by hand.

The first period's fitness gap is already informative: with $\Delta p_1 = 2.625$, the chartist demand submitted the period before, $d^c_0 = 2.0$, earned a realized profit of $\pi_{c,1} = 2.625 \times 2.0 = 5.250$, against the fundamentalist's $\pi_{f,1} = 2.625 \times 1.5 = 3.938$; chartists were not just more numerous but more profitable, since their demand

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t	p_t	$n_{f,t}$	$n_{c,t}$	d_t^f	d_t^c	D_t	p_{t+1}
0	95.000	0.500	0.500	1.500	2.000	1.750	97.625
1	97.625	0.342	0.658	0.713	2.625	1.972	100.583
2	100.583	0.056	0.944	−0.175	2.958	2.783	104.757

Table 20.6: Discrete-choice switching ($\theta = 0.5$) replacing Section 20.3.1's fixed threshold. Each period's population shares are computed from the fitness realized over the immediately preceding period, then used to set that period's aggregate demand.

was better aligned with the up-move that materialized. The logit rule converts this into $n_{c,1} = 0.658$, already at the edge of the critical threshold $n_c^* \approx 0.667$ Section 20.2.5 derived for the same λ and β. One more period of chartist outperformance, $\pi_{c,2} = 7.764$ against $\pi_{f,2} = 2.107$, pushes the population past the threshold to $n_{c,2} = 0.944$, and the price increments confirm that the market has crossed into the unstable regime: $\Delta p_1 = 2.625$, $\Delta p_2 = 2.958$, $\Delta p_3 = 4.174$, growing rather than shrinking, Scenario B's explosive signature, reached here with no threshold rule in the model, only agents comparing two realized numbers each period and updating a choice probability.

This result fits neither column of Table 20.4 unchanged. It is not Chapter 19's Q-learning: no agent maintains a state-action value table or bootstraps a return estimate as the learning seller does. It is a population-level analogue of the same idea, an evolutionary or replicator dynamic in which strategies that recently paid off are used by more of the population next period. That matters for interpretability: a logit population share is a single number a reader can audit by hand, as this worked example did, unlike a trained neural-network policy's weights, so discrete-choice switching sits nearer the transparent, fixed-rule end of the spectrum than the learning seller does, even though both replace a rule the modeler would otherwise impose with something the agents determine. The companion technical note develops the mechanism with a fully specified CARA-utility demand system instead of this chapter's linear demand rules and derives, in closed form, the equilibrium population share and its stability condition as a function of the intensity of choice, an analogue of this section's n_c^* expressed as a threshold on θ rather than on the population share, since in that richer model the share is an endogenous function of θ rather than a free parameter. Brock and Hommes's result goes further: beyond the critical intensity of choice, the market does not merely become unstable, its dynamics undergo a period-doubling bifurcation into deterministic chaos, price behavior statistically indistinguishable from noise although generated by a fully deterministic, rational system, the technical note's "rational route to randomness." Readers who want the rigorous derivation should turn to the technical note.

20.3.8 When the Agents Are Algorithms: Collusion, Herding, and Supervisory Simulation

The learning seller faced buyers who never learned back. Put two learning algorithms in the same market and each becomes part of the other's environment, the multi-agent setting of Table 20.4, and the outcome can surprise their designers. Calvano et al. (2020) let Q-learning pricing agents, the algorithm of Chapter 19, compete repeatedly in a standard oligopoly model of price competition. The agents consistently learned to charge prices above the competitive level without communicating with one another, and they sustained those prices the way a cartel would: a price cut by one agent was met by a finite phase of punishment, followed by a gradual return to the high price. The result held under cost and demand asymmetries, different numbers of competitors, and several forms of uncertainty. Nothing in the agents' code mentions cooperation; it emerges from the interaction, which is this chapter's central claim in a form a competition authority must take seriously. Competition law has traditionally looked for evidence of an agreement, and two pricing algorithms that never exchange a message leave none. The finding comes from simulation, and how often it occurs in real markets is still debated, but it is a direct warning for any institution that lets learning algorithms set quotes, loan rates, or deposit rates in a concentrated market.

A second concern runs in the opposite direction: not algorithms that learn to coordinate, but algorithms that are already alike. The Financial Stability Board's November 2024 report on artificial intelligence warned that "the widespread use of common AI models and data sources could lead to increased correlations in trading,

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lending, and pricing" and so amplify market stress (Financial Stability Board 2024), and the Bank of England's Financial Policy Committee pointed in April 2025 to the risk of "firms taking increasingly correlated positions and acting in a similar way during a stress, thereby amplifying shocks," made worse by reliance on a few vendor-provided models (Bank of England, Financial Policy Committee 2025). This is **herding** produced by a **model monoculture**, and this chapter has already measured why it matters. The fundamentalist-chartist market stays stable only while the trend-following share stays below $n_c^* = 1/(\lambda\beta) \approx 0.667$; if many firms adopt the same model and that model reacts to a recent price move the same way, the effective n_c is set by the model's adoption rather than by a diversity of independent judgments, and adoption alone can carry it across the threshold. A population's composition is exactly what an agent-based model varies and a representative-agent model cannot, which is why simulating many AI-driven traders is a natural way to study the question.

Central banks have extended the same approach well beyond RAMSI. Bank of England staff built an agent-based model of the UK housing market in which first-time buyers, owner-occupiers, buy-to-let investors, and renters are each simulated individually and calibrated to household survey and housing market micro-data (Baptista et al. 2016); in it a larger buy-to-let sector amplifies house-price cycles, and a portfolio limit on high loan-to-income mortgages attenuates them, a policy question about population composition that only a heterogeneous-agent model can pose. The Bank's **system-wide exploratory scenario** (SWES), launched in June 2023 and reported on 29 November 2024, went further and used real institutions as the agents (Bank of England 2024). Around 50 firms, banks and non-banks among them, were given a hypothetical scenario, a sudden, sharp, and severe shock to global financial markets triggered by geopolitical tensions, and over several rounds reported how they would respond to it and to one another's responses. The first of the exercise's six conclusions was that firms' collective actions amplify the initial shock. Where Section 20.4.1 warns that an agent-based model's behavioral rules are assumptions the modeler must defend, SWES replaced the assumed rules with each participant's own account of its behavior, at the price of an exercise that cannot be rerun at will the way a simulation can.

20.4 Building and Validating a Model

An agent-based model that matches nothing is a simulation, not a model; this part covers the difference.

20.4.1 Calibrating and Validating Agent-Based Models

Chapter 6 validates a model by holding out data the model never saw and checking its predictions against it, a well-defined procedure because the model makes a specific, checkable prediction for each held-out observation. Agent-based models are far harder to validate in this sense: an ABM produces an entire simulated history, and the object of interest is usually a qualitative or statistical property of that history (does it produce fat tails, a plausible contagion cascade?), not a point forecast. The standard approach, the method of simulated moments, chooses the model's free parameters (here, α, β, λ, and the rule governing n_f and n_c) so that the simulated data's statistical moments, its volatility, kurtosis, and autocorrelation structure, match the same moments of real market data as closely as possible, rather than maximizing a likelihood as Chapter 6's models typically do.

This procedure deserves the skepticism Chapter 6 applies to overfitting: an ABM with enough free behavioral parameters can often be tuned to match almost any set of stylized facts, which shows that the model is flexible, not that its mechanism (fundamentalists and chartists switching by relative profitability, say) is the true explanation for those facts. A model that reproduces fat tails is not thereby confirmed; a different ABM, or different parameters in the same one, might reproduce them through an entirely different mechanism. This is a substantive limitation, and the main reason this book uses agent-based models to explore which mechanisms can produce an observed pattern, alongside institutional case studies like the Flash Crash and the RAMSI vignette, rather than as forecasting tools validated the way Chapter 6's classifiers and Chapter 7's time-series models are.

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Concretely: Matching a Target Moment

Section 20.3.1's regime-switching simulation happened to produce a Pearson kurtosis of 4.67 from a switching threshold of 0.25; nothing forced that value. Method-of-simulated-moments calibration treats the switching threshold (and the model's other free parameters) as a dial to search over, rerunning the simulation at several threshold values and recording the resulting kurtosis, as Table 20.7 does.

Switching threshold	Pearson kurtosis
0.15	3.87
0.20	3.61
0.25	4.67
0.30	7.42
0.35	8.60

Table 20.7: Simulated kurtosis as a function of the switching threshold (500 periods, otherwise identical to Section 20.3.1)

The direction is not the one intuition suggests: across the sweep, kurtosis *rises* as the threshold rises, that is, as the volatile regime becomes *harder* to trigger. The rise is not perfectly monotone (kurtosis dips from 3.87 to 3.61 between the two lowest thresholds), and the dip is itself instructive: each row is a single 500-period run, and kurtosis from one short simulated history is noisy enough to reverse a small step, which is why practical method-of-simulated-moments work averages each moment over many seeds before comparing it with data. A low threshold puts the market into the volatile regime often, raising overall variance but blending the two regimes together; a high threshold makes the volatile regime rare, so when it does trigger it produces a few sharp outliers against a calm baseline, the mixture of a common calm regime and a rare extreme one that drives kurtosis up. Suppose an analyst wanted the simulation to match an asset's historical kurtosis of 6.0: Table 20.7 brackets the target between the thresholds 0.25 and 0.30, which land at 4.67 and 7.42, nearly equidistant on either side, so a finer search between them is the next step. Finding a threshold that approximately matches 6.0 would establish only that this mechanism, among many other threshold values, population-share assumptions, or ABM structures, is *capable* of producing that kurtosis, not that real traders switch strategies by anything like this rule, and, as the reversed direction shows, not that the calibrated parameter's economic interpretation ("how readily the market panics") behaves as casual intuition suggests. That gap between matching a moment and confirming a mechanism is Section 20.4.1's central caution, here in computed numbers.

20.4.2 Advanced Topics: Network Topology and the Limits of a Simple Chain

Section 20.3.3's three-bank example used the simplest possible network, a chain, in which every bank has one creditor and at most one debtor. Real interbank networks look nothing like a chain: they are typically core-periphery structures, a few large, heavily interconnected banks at the core surrounded by many smaller banks connected mainly to the core, and the network's shape matters for how a shock propagates, not just its size. Acemoglu, Ozdaglar, and Tahbaz-Salehi's widely cited "robust-yet-fragile" result formalizes a counterintuitive finding: for small shocks, a more densely connected network is more stable, since each bank's exposure to any one counterparty's distress is diluted across many relationships, the diversification logic of Chapter 12's portfolio theory. Past a certain shock size, the same dense interconnection becomes a contagion channel that transmits a large shock further and faster than a sparser network would, because a large loss at one interconnected core bank has many more counterparties to infect.

20.4.3 Worked Example 20.5: Chain versus Hub-and-Spoke

Replace Section 20.3.3's chain with a hub-and-spoke network hit by the same \$45 shock, so the two topologies can be compared. Figure 20.3 sets them side by side. A hub bank H owes \$30 to each of three peripheral banks P_1, P_2, P_3 (\$90 in total), H holds \$100 of external assets, and each P_i holds \$5 of external assets and

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owes nothing. In the baseline, H pays all three in full, leaving H with net worth $100 - 90 = \$10$ and each P_i with $5 + 30 = \$35$, for a total system net worth of $10 + 3(35) = \$115$.

Now apply the same \$45 shock used against Bank A in Section 20.3.3, this time to H's external assets, reducing H's resources to \$55 against its \$90 obligation. Eisenberg-Noe clearing requires H to pay each equally senior creditor the same pro-rata share: H can cover only $55/90 = 61.1\%$ of what it owes, so each P_i receives \$18.33 instead of \$30. H's net worth falls to $55 - 55 = \$0$, a default, as Bank A's did in the chain. But now every peripheral bank absorbs a slice of the loss at once rather than in sequence: each P_i ends with net worth $5 + 18.33 = \$23.33$, down from \$35, a 33.3% loss for each of the three. Table 20.8 summarizes both networks. Total system-wide loss is \$45 in both cases, equal to the shock regardless of topology, a general property of this clearing mechanism without bankruptcy costs. Topology does not change how much value a shock destroys; it changes how the loss is distributed. The chain concentrates outright default on two institutions in sequence (A, then B), with C absorbing a modest, once-removed spillover. The hub-and-spoke network concentrates default on a single institution, the hub, while spreading a survivable loss across every peripheral bank at once. No peripheral defaults, since none had an obligation to miss, but all three are meaningfully weaker. Neither topology is simply "worse." Which one a regulator should worry about more depends on whether the greater concern is a few outright failures cascading through a chain of counterparties or one well-connected institution's distress passing a loss to everyone depending on it, the topology-dependent judgment RAMSI-style network models are built to inform.

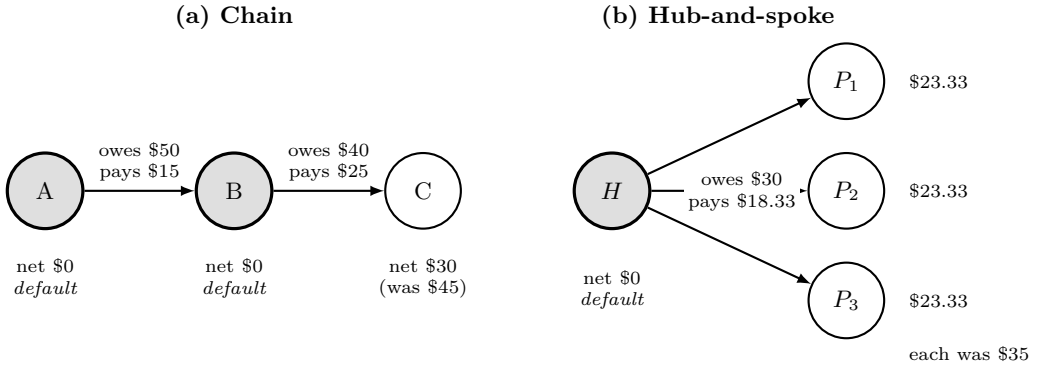


Figure 20.3: The same \$45 shock under two topologies. Shaded circles default outright; each arrow gives what is owed and what is actually paid after Eisenberg-Noe clearing. The total loss is \$45 either way, but the chain destroys two institutions one after another while the hub-and-spoke destroys one and weakens three at once.

Network	Institutions defaulting outright	Total system loss	Loss distribution
Chain (Section 20.3.3)	2 of 3 (A, then B)	\$45	Concentrated, sequential
Hub-and-spoke (above)	1 of 4 (H only)	\$45	Diffuse, simultaneous

Table 20.8: The same \$45 shock, chain versus hub-and-spoke

A full treatment of the robust-yet-fragile transition would simulate the clearing vector across many more network structures and shock sizes than these two, the large-scale exercise real central-bank network models like RAMSI are built to run.

20.4.4 A Practical Workflow for Building an Agent-Based Model

Every example in this chapter, across three different applications (market efficiency, price dynamics, and financial contagion), followed the same workflow. Stated as a template, it is a recipe for building a new ABM and a concrete account of what it means to ask Table 20.1's question, "what pattern emerges from many interacting rules," rather than fitting a function to data or training a policy against an environment.

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1. **Specify the agents and their behavioral rules.** Decide what types of agents populate the model (zero-intelligence traders, fundamentalists and chartists, banks in a network) and write down each type's rule as an explicit function of the current market state, as Sections 20.2.2 through 20.3.2 did with a single equation or a short block of code per agent type.

2. **Specify the market mechanism.** Decide how individual agent actions aggregate into a new market state: a double-auction clearing rule, a price-impact equation, an Eisenberg-Noe clearing vector. This is the second box in Figure 20.1's simulation loop, and is usually the piece of the model most directly grounded in institutional fact rather than behavioral assumption.

3. **Choose free parameters and initial conditions.** Every worked example above had a handful of free parameters, α, β, λ, the population shares (n_f, n_c), a switching threshold, that are not derived from anything but set by the modeler. Report these explicitly and be prepared to defend each, since Section 20.4.1's calibration critique applies to every one of them.

4. **Simulate forward and collect statistics.** Run the loop for enough periods that the statistic of interest, an efficiency percentage, a variance ratio, a kurtosis, a default count, stabilizes, and report it alongside the raw simulated series wherever possible, as this chapter's companion notebook does for all four worked examples.

5. **Calibrate against real data if a real comparison is available.** Where a target real-world moment exists, Section 20.4.1's method of simulated moments searches over the free parameters from step 3 until simulated and real moments match as closely as possible.

6. **Interpret the result as a candidate mechanism, not a confirmed explanation.** A model that reproduces a target pattern has shown that its mechanism is *capable* of producing that pattern; Section 20.4.1's central caution, the chapter's most important methodological point, is that this is weaker evidence than a validated prediction from Chapter 6's train-test framework, and should be presented and used accordingly.

20.5 Challenges in Agent-Based Modeling for Finance

Several challenges explain why ABMs remain a specialized tool rather than a default choice in this book. Computational cost is real: simulating a large, heterogeneous population over many periods, especially when agents are themselves learned with the reinforcement-learning methods of Section 20.3.5, can be far more expensive than fitting a closed-form or statistical model. Every worked example here used two to four agents because a hand-checkable illustration must; a research-grade financial ABM typically simulates thousands to millions, and since every agent is updated every period, cost grows at least linearly, and often faster, in both agent count and horizon, a scaling concern this book's other methods (a fitted regression or a trained network) mostly do not share. General-purpose ABM frameworks such as Mesa (Python) and NetLogo exist to manage this complexity, but building and validating a large financial ABM remains a much larger engineering effort than fitting one of Chapter 6's models to a comparable dataset. Calibration and validation (Section 20.4.1) remain unresolved: the method of simulated moments can match target statistics without establishing that the mechanism is correct, and there is no counterpart to Chapter 6's train-test split for an entire emergent history. Reproducibility across studies is weaker than for most of this book's methods, since two groups' ABMs of nominally the same phenomenon often differ in dozens of rarely specified implementation details (agent count, update order, tie-breaking rules) that can materially change the results. Finally, regulatory and industry acceptance lags the academic literature. RAMSI-style tools have earned a place in the macroprudential toolkit because central banks can defend their assumptions for a narrow, systemic-risk purpose, but agent-based models remain far from routine in the validation and governance framework built around Chapter 15's three lines of defense and Chapter 18's explainability toolkit. A rule-based simulation of thousands of interacting agents is harder to explain to a validator or regulator than a fitted regression or classifier.

None of these challenges is a reason to avoid agent-based models; they are reasons to be precise about what a given ABM has shown, the discipline Section 20.4.4's final step asks for. A model that reproduces the Flash Crash's hot-potato dynamics, the fat tails of Section 20.3.1's simulation, or RAMSI's contagion cascades

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is a useful demonstration that a specific, named mechanism is *sufficient* to produce an observed pattern, valuable evidence in its own right and often the only tractable way to reason about emergent phenomena. It is not, on its own, evidence that the mechanism is *necessary*, the one true explanation among the many candidates that might produce the same pattern, and every application in this chapter should be read with that distinction in mind.

20.6 Key Takeaways

- Agent-based models simulate a population of heterogeneous, boundedly rational agents interacting repeatedly, and treat the aggregate pattern that emerges, efficiency, a bubble, a cascade, as the object of study, in contrast to this book's representative-agent equilibrium models and data-fitted statistical models.

- Market efficiency can emerge from a trading mechanism itself rather than from participant sophistication: zero-intelligence traders bidding at random within budget constraints, cleared through an ordinary double auction, recovered the fully efficient trade in this chapter's worked example.

- The same market mechanism can produce either smooth convergence to fundamental value or a self-reinforcing, explosive bubble, depending only on the population mix of fundamentalist and chartist traders, holding every other parameter fixed.

- Heterogeneous-agent models can reproduce fat tails and volatility clustering as an emergent consequence of strategy switching, rather than by directly imposing an autoregressive variance structure the way Chapter 7's GARCH models do: this chapter's 500-period simulation produced a 6.14× variance ratio between regimes, a Pearson kurtosis of 4.67 against a Gaussian benchmark of 3.0, and a squared-return autocorrelation of 0.658 at lag one, from two fixed behavioral rules switching in and out of dominance.

- A clearing-vector approach to interbank obligations, the same network-contagion logic at the core of the Bank of England's top-down RAMSI stress-testing model, can show one institution's shock cascading into a second institution's default even when the second institution received no direct shock, a systemic risk a single-institution stress test cannot see.

- Agent-based models are far harder to calibrate and validate than this book's other methods, since matching a target set of statistical moments demonstrates flexibility, not that the model's particular mechanism is the true explanation for the observed pattern.

- Network topology shapes contagion independently of shock size: the "robust-yet-fragile" result shows dense interconnection dampens small shocks through diversification but can transmit large shocks further and faster than a sparser network would; the same $45 shock produced two outright defaults in a chain network but only one, spread more diffusely, in a hub-and-spoke network of the same total size.

- The fundamentalist-chartist market's stability is governed by a single exact threshold, $n_c^* = 1/(\lambda\beta)$: below it the price converges, above it a bubble grows without bound, and near it convergence becomes oscillatory rather than smooth, a sharp regime change from one parameter crossing a critical value.

- Agent-based models answer a third kind of question this book's other two paradigms do not: not "what function best fits this data" (Chapter 6) or "what action sequence maximizes reward in this environment" (Chapter 19), but "what pattern emerges when many simple, interacting rules are simulated together."

- Even a calibrated parameter's economic interpretation can run against intuition: raising the fundamentalist-chartist model's switching threshold, making the volatile regime *harder* to trigger rather than easier, raised simulated kurtosis from 3.87 to 8.60 across a parameter sweep, the opposite of what naive intuition suggests, which is why a calibrated ABM's mechanism should never be trusted just because its output moments look right.

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- A fixed-rule ABM and a learning agent can share the same market: replacing one zero-intelligence seller with a simple trial-and-error learner that tracks realized profit converged on the true better strategy in this chapter's worked example, but a single unlucky exploration phase can just as easily lock a purely greedy learner onto the wrong one permanently, with no further exploration to self-correct it.
- A market's switching threshold can be crossed by assumption or by learning: Section 20.3.1's regime switch was a rule the modeler imposed from outside, while Section 20.3.7's discrete-choice mechanism let the population cross the same $n_c^* \approx 0.667$ threshold on its own, because agents kept reallocating toward whichever strategy had just earned more, reaching $n_{c,2} = 0.944$ and an accelerating price after only two periods of chartist outperformance.
- When the agents are themselves learning algorithms, emergence cuts both ways: Q-learning pricing agents can learn to collude without communicating, and many firms relying on the same AI models can herd, the population-composition risk that the FSB and the Bank of England now flag and that agent-based models, including the Bank's housing-market model and its system-wide exploratory scenario, are built to study.

20.7 Suggested Exercises

20.1 Using the zero-intelligence double-auction setup of Section 20.2.2, suppose buyer 2's random bid is $b_2 = \$49$ instead of \$30, with every other draw unchanged. Recompute the clearing outcome and explain, in terms of total realized surplus, whether the market mechanism still selects the efficient allocation.

20.2 Section 20.2.5 derives the stability threshold $n_c^* = 1/(\lambda\beta)$ for $\lambda = 1.5$, $\beta = 1.0$. Recompute n_c^* for $\beta = 2.0$ instead, holding $\lambda = 1.5$ fixed, and use the companion notebook to confirm your answer by simulating the market just above and just below the new threshold.

20.3 Extend the three-bank contagion network of Section 20.3.3 by adding a fourth bank, D, to whom Bank C owes \$20, with D holding \$15 of external assets and no other obligations. Recompute the clearing vector under the same \$45 shock to Bank A's external assets, and report whether the cascade reaches D.

20.4 Section 20.4.3's hub-and-spoke example used three peripheral banks. Redo the calculation with six peripheral banks instead, keeping the hub's total obligation at \$90 (so each peripheral is now owed \$15) and its external assets at \$100, under the same \$45 shock. Report each peripheral's percentage loss and compare it to the three-peripheral case's 33.3%, and explain the direction of the difference.

20.5 Section 20.4.1 argues that matching a target set of statistical moments does not confirm an ABM's underlying mechanism. Propose one additional form of evidence, beyond matching moments, that would make you more confident a particular agent-based mechanism is the right explanation for an observed market phenomenon.

20.6 Using the companion notebook's regime-switching simulation, lower the switching threshold from 0.25 to 0.15 and re-run the 500-period simulation. Report the new variance ratio and kurtosis, and explain, in terms of how often the market now enters the volatile regime, why the direction of the change makes sense.

20.7 Section 20.3.6's learning seller happened to settle on the better arm after exploring. Recompute its running averages after exploration if rounds 1 through 4 instead see buyer values $50, 50, 50, 50$ (using the companion notebook to check your work), report which arm the seller would then play for the rest of the sequence, and explain why this outcome is a failure of the explore-exploit process rather than merely bad luck that later self-corrects.

20.8 Using Section 20.3.7's discrete-choice market, recompute $n_{f,1}$, $n_{c,1}$, $n_{f,2}$, and $n_{c,2}$ with a lower intensity of choice, $\theta = 0.2$ instead of 0.5, holding every other parameter and the same $p_{-1} = 93$, $p_0 = 95$ fixed, and report the first period at which $n_{c,t}$ exceeds the critical threshold $n_c^* \approx 0.667$

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derived in Section 20.2.5, comparing it to the $\theta = 0.5$ case worked in the text. Does halving the intensity of choice postpone the crossing, change its severity, or both?

20.9 Section 20.3.8 links model monoculture to the stability threshold of Section 20.2.5. Suppose a fraction m of a market's participants adopt the same AI trading model, which behaves exactly like a chartist with $\beta = 1.0$, while the remaining participants keep Scenario A's split of 80% fundamentalists and 20% chartists, with $\lambda = 1.5$. (a) Write the market's chartist share n_c as a function of m and find the smallest adoption rate m at which the market becomes unstable. (b) Repeat with the remaining participants split 90% and 10%, and explain what the comparison says about whether diversity among the non-adopters protects the market. (c) Explain why the Bank of England's system-wide exploratory scenario asked participating firms to report their own responses rather than assigning them behavioral rules, and name one limitation of that design relative to a simulation.

20.10 **Challenge (real data).** Download the BIS locational banking statistics (the cross-border claims table, freely available at stats.bis.org with no registration required) for five or six reporting countries. (a) Treating each country's aggregate cross-border claims on the others as a simplified interbank exposure network, construct a liabilities matrix analogous to Section 20.3.3's three-bank example (using each country's claims on another as that other country's liability), and compute the Eisenberg-Noe clearing vector following an assumed shock that wipes out 10% of one country's external, non-interbank assets. (b) Compare the resulting loss cascade to this chapter's toy three-bank result, and discuss whether the real network's exposures look more chain-like or more hub-and-spoke, referencing Section 20.4.3's topology comparison. (c) Explain one specific way in which country-aggregated BIS data understates the fragility of the true institution-level network (for example, netting across many banks within a country, or omitting domestic interbank exposures entirely), and describe, referencing this chapter's RAMSI case vignette, what institution-level supervisory exposure data a central bank could use to close that gap.

20.8 Further Reading

- Schelling (1971), "Dynamic Models of Segregation." The original checkerboard segregation model discussed in Section 20.2.1, the earliest and best-known agent-based model outside finance.
- Arthur et al. (1997), "Asset Pricing Under Endogenous Expectations in an Artificial Stock Market." The Santa Fe Institute Artificial Stock Market discussed in Section 20.2.1, widely regarded as the founding model of agent-based computational finance.
- Gode and Sunder (1993), "Allocative Efficiency of Markets with Zero-Intelligence Traders: Market as a Partial Substitute for Individual Rationality." The original zero-intelligence double-auction result worked through by hand in Section 20.2.2.
- Brock and Hommes (1997), "A Rational Route to Randomness." The performance-based strategy-switching framework underlying Section 20.2.4's fundamentalist-chartist model.
- Lux and Marchesi (1999), "Scaling and Criticality in a Stochastic Multi-Agent Model of a Financial Market." An influential heterogeneous-agent model reproducing fat tails and volatility clustering, discussed in Section 20.3.1.
- Eisenberg and Noe (2001), "Systemic Risk in Financial Systems." The clearing-vector framework for financial contagion worked through by hand in Section 20.3.3.
- Acemoglu et al. (2015), "Systemic Risk and Stability in Financial Networks." The "robust-yet-fragile" network-topology result discussed in Section 20.4.2.
- Zhang et al. (2021), "Multi-Agent Reinforcement Learning: A Selective Overview of Theories and Algorithms." A survey of the multi-agent reinforcement learning methods contrasted with fixed-rule agent-based models in Section 20.3.5, including the convergence difficulties that arise once multiple agents learn simultaneously.

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- Burrows et al. (2012), *RAMSI: A Top-Down Stress-Testing Model*. The real-world central-bank stress-testing model, top-down and network-based, discussed in the case vignette above.
- Calvano et al. (2020), "Artificial Intelligence, Algorithmic Pricing, and Collusion." The simulation study, discussed in Section 20.3.8, in which Q-learning pricing agents learn to sustain supracompetitive prices without communicating.
- Bank of England (2024), *The Bank of England's System-Wide Exploratory Scenario Exercise Final Report*. The Bank of England's report on its system-wide exploratory scenario, in which around 50 real firms played the agents of a market-stress simulation.
- Axtell and Farmer (2025), "Agent-Based Modeling in Economics and Finance: Past, Present, and Future." A comprehensive, up-to-date survey of the field from two of its leading figures, and the best single starting point for a reader who wants to go beyond this chapter's introductory treatment.
- LeBaron (2006), "Agent-Based Computational Finance." An earlier survey of the agent-based finance literature specifically, including the calibration and validation challenges raised in Section 20.4.1.
- Farmer and Foley (2009), "The Economy Needs Agent-Based Modelling." An argument, from two leading practitioners, for agent-based approaches as a complement to equilibrium modeling in macroeconomics and finance.

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Chapter 21

Case Studies and Capstone Projects

21.1 Introduction

Every dataset used so far in this book, Cascade Home & Garden's satellite-derived car counts, Meridian Fabrication's financial statements, the ten-transaction fraud-scoring sample, was built for one purpose: to make a method's mechanics checkable by hand, free of the mess real data carries. That was the right choice for teaching each method in isolation, but it leaves a gap this closing chapter fills. A real financial dataset, of the kind a data scientist downloads from a public repository such as the UCI Machine Learning Repository or Kaggle, arrives with undocumented category codes, an imbalanced target, correlated and redundant features, and no guarantee that the model that wins on one metric also wins on every other metric that matters.

This chapter works one such dataset end to end, using only tools this book has already built: feature engineering and a model bake-off spanning linear and tree-based methods, from Chapter 6; Chapter 18's SHAP and permutation-importance methods, now run with the actual `shap` library rather than by hand; a cost-sensitive threshold choice extending Chapter 15's alert economics; and a fairness audit extending Chapter 18's synthetic two-group example to real data. One topic is new, calibration, motivated by Chapter 9's caveat that a credit score is meaningful only if the probability underneath it is accurate. The chapter closes with governance considerations and further capstone project ideas for the reader.

Two more advanced analyses go beyond a first pass through the pipeline: a systematic hyperparameter search measuring how much tuning gains over a single reasonable configuration, and an intersectional fairness cut that splits the test population two ways at once and uncovers a clear four-fifths-rule violation that the chapter's single-attribute audit, run first, missed.

The dataset is the *Default of Credit Card Clients* dataset (Yeh and Lien 2009): 30,000 Taiwanese credit card accounts with 23 raw features and a binary label, default on payment the following month, hosted on the UCI Machine Learning Repository and widely distributed on Kaggle under the same name. Unlike every other dataset in this book, it was not built to illustrate anything, and working through it surfaces problems no fictional dataset would: undocumented category codes that must be cleaned before any model is fit, a real 22.1% default rate rather than a convenient round number, and, as the fairness audits show, a model that passes a single-attribute fairness check comfortably while failing an intersectional one, although no protected attribute is among its inputs.

The chapter follows the order a data scientist follows on a new dataset. It begins by inspecting the raw data closely enough to find its undocumented quirks (Section 21.2.1) and engineering features from raw monthly statements rather than from a clean feature table (Section 21.2.2). It then compares several model families on the same held-out data (Section 21.3.1) and checks whether the winner's probabilities can be trusted at face value (Section 21.3.2). From there it explains that model's behavior with real software (Section 21.3.4), converts its output into a lending decision under an explicit cost assumption (Section 21.3.5), and audits that decision for fairness, first along one dimension (Section 21.3.6) and then along two at once (Section 21.4.4). A discussion of what deploying the pipeline would require (Section 21.4.1) is followed by a reusable template distilled from it (Section 21.4.5) and by further capstone projects to apply the template to (Section 21.4.6). Those end the chapter, and the book, on the note Chapter 1 opened with: these tools are only as trustworthy as the discipline applied in using them.

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21.2 The Data

The capstone begins where every real project does: with a dataset that is not what its documentation claims.

21.2.1 The Capstone Dataset and Its Imperfections

The raw dataset records, for each of 30,000 accounts, a credit limit (`LIMIT_BAL`), demographic fields (`SEX`, `EDUCATION`, `MARRIAGE`, `AGE`), six months of repayment status codes (`PAY_1` through `PAY_6`, where -1 indicates payment made duly and a positive integer indicates the number of months a payment is overdue), six months of billed amounts (`BILL_AMT1`–`BILL_AMT6`), six months of amounts actually paid (`PAY_AMT1`–`PAY_AMT6`), and the target, default on the following month's payment.

Loading the data surfaces the first imperfection the fictional datasets never had: the data dictionary documents `EDUCATION` as taking values 1 (graduate school), 2 (university), 3 (high school), or 4 (other), yet the column also contains 14 rows coded 0, 280 rows coded 5, and 51 rows coded 6, none of which the documentation defines. `MARRIAGE` shows the same pattern: documented as 1 (married), 2 (single), or 3 (other), it also contains 54 rows coded 0.

Neither anomaly is disclosed in the documentation, and a modeler who does not notice them will silently feed a model category codes with no defined meaning. Following standard practice for this well-studied dataset, both sets of undocumented codes are folded into their "other" categories (0, 5, and 6 into `EDUCATION`'s category 4; 0 into `MARRIAGE`'s category 3) before any modeling, a cleaning step with no equivalent elsewhere in this book because every other dataset was built clean.

The repayment-status columns carry two further quirks of the same kind. First, the raw file has no `PAY_1` at all: the most recent month's status column is named `PAY_0`, while the billed and paid amounts run from `BILL_AMT1` and `PAY_AMT1`, so the companion notebook renames `PAY_0` to `PAY_1` to line the three six-month series up. Second, the dictionary defines the status codes as -1 (paid duly) and 1 through 9 (months overdue) only, yet the columns are dominated by two undocumented codes, -2 and 0, which together make up 66.9% of all monthly status entries and appear at least once for 25,939 of the 30,000 accounts. They are commonly read as "no balance to pay" and "revolving credit, minimum payment made," but nothing in the official documentation says so. This chapter leaves them as they are; since both rank below 1, neither counts as a delinquency in the `max_delinquency` feature defined below, which is a modeling decision a reader should notice rather than inherit silently.

Table 21.1 summarizes four representative raw fields before any cleaning or feature engineering; it is worth inspecting closely, the habit Chapter 6 recommended before fitting any model.

Field	Mean	Std. dev.	Min	Median	Max
`LIMIT_BAL` (NT$)	167,484	129,748	10,000	140,000	1,000,000
`AGE` (years)	35.5	9.2	21	34	79
`BILL_AMT1` (NT$)	51,223	73,636	−165,580	22,382	964,511
`PAY_AMT1` (NT$)	5,664	16,563	0	2,100	873,552

Table 21.1: Summary statistics for four raw fields (before cleaning)

`BILL_AMT1`'s negative minimum is another undocumented quirk: a negative billed amount means the account was in credit, having overpaid the previous month, a legitimate state for a revolving credit account but one that a modeler unfamiliar with the data might mistake for a data-entry error and drop or clip without justification. `PAY_AMT1`'s heavy right skew, a median of just NT$2,100 against a mean of NT$5,664 and a maximum of NT$873,552, is the kind of distribution Section 21.2.2's ratio features are designed to normalize before they reach a linear model, though the tree-based models of Section 21.3.1 are much less sensitive to skew than logistic regression, since a tree splits on a feature's rank order rather than its raw scale.

Two more properties of the raw data matter before any model is fit. The first is the class balance. A 22.1% default rate is much higher than a typical consumer credit-card portfolio, which more commonly runs in the low single digits: a published research dataset's class balance reflects how and when it was sampled, not the balance another institution's portfolio will show. This data covers a six-month window in 2005, shortly before a Taiwanese consumer-credit crisis, so a model built on it should not be assumed to transfer

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its absolute predicted probabilities to a different portfolio without recalibration, even if its ranking of riskier against safer accounts transfers reasonably well.

The second property is redundancy. Several raw features are highly correlated by construction: the six `BILL_AMT` columns track one account's revolving balance from month to month, so they move together. That matters for Section 21.3.4's explanations, because an importance score assigned to any one of several correlated features is partly arbitrary when the model can substitute one for another with little effect on its predictions. Chapter 6 raised this issue as multicollinearity in linear regression, and it carries over, in a different technical form, to tree-based models and their explanations.

21.2.2 From Raw Fields to Model-Ready Features

Chapter 6 introduced feature engineering in the abstract; this section performs it on raw fields for the first time. Five engineered features summarize six months of repayment and billing history in a form a model can use more directly than six separate monthly columns. `max_delinquency` is the worst (maximum) repayment-status code across the six months, capturing whether an account was ever seriously overdue even if its latest month looks clean. `avg_utilization` is the six-month average of billed amount divided by credit limit, a standard credit-risk signal Chapter 9 introduced but never computed from raw statements. `total_bill` and `total_paid` are the six-month sums of billed and paid amounts, and `pay_to_bill_ratio` is their ratio, capturing how much of what an account owed it paid back over the window; it is clipped at 5 to bound the influence of a few accounts with a large payment relative to a near-zero bill. These five, together with 20 of the 23 raw fields, give 25 features. The three raw fields left out are `SEX`, `MARRIAGE`, and `AGE`: taking U.S. fair-lending law as the reference point, as Chapter 18 did, the Equal Credit Opportunity Act and its Regulation B forbid basing a credit decision on sex or marital status and allow age only in narrow ways (for instance, as a variable in a demonstrably and statistically sound scoring system that never treats applicants aged 62 or older less favorably), so the cleanest choice is to keep all three out of the model and hold `SEX` and `AGE` aside as attributes for the fairness audits of Sections 21.3.6 and 21.4.4.

The 30,000 accounts are split 80/20 into a training set of 24,000 and a held-out test set of 6,000, stratified by the default label so both sets keep the 22.1% default rate, with a fixed random seed so that the split, and every result in this chapter, is reproducible from the companion notebook. Every model comparison, calibration check, explanation, threshold choice, and fairness audit below is computed on this held-out test set, never on data a model was fit to. Chapter 6 introduced that discipline as essential to detecting overfitting, and it matters more once a real dataset's quirks make it easy to believe a model has learned something it has only memorized.

A real dataset also makes another discipline easy to violate: every engineered feature, `max_delinquency` included, is computed from information available when each account's repayment history was recorded, not from anything that depends on knowing whether the account defaulted. Chapter 17's point-in-time discipline applies here in a subtler form: a feature engineered carelessly from a dataset that also contains the outcome, for instance a rolling average that accidentally includes a future month's billing data, can leak the label into the features and produce a model that looks excellent on a test set drawn from the same leaking dataset yet fails once deployed on new applicants for whom the leak is unavailable. This dataset's structure makes such leakage easy to avoid, since every raw field precedes the single target month, but the habit of checking for it explicitly, rather than assuming a clean-looking dataset has none, is one to carry into Section 21.4.6's further projects, several of which (Lending Club in particular) are much easier to leak by accident.

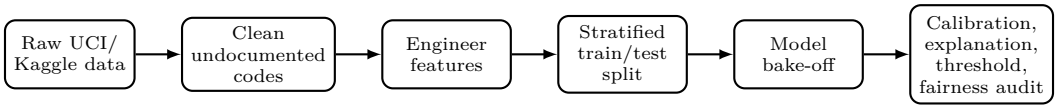


Figure 21.1: The end-to-end pipeline this chapter works through by hand

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21.3 Modeling and Evaluation

This part runs a full bake-off, then asks the questions that decide whether any of it is usable.

21.3.1 A Model Bake-Off: Logistic Regression, Random Forests, and Gradient Boosting

Four models are fit on the training set and evaluated, once, on the untouched test set: a plain logistic regression (Chapter 6's workhorse, standardized features, no adjustment for class imbalance), a class-weighted logistic regression (Chapter 15's first remedy for imbalanced classification, applied here directly rather than to a stylized example), a random forest (300 trees, Chapter 6's bagging-based ensemble), and gradient-boosted trees via XGBoost (300 boosting rounds, Chapter 6's sequential, bias-reducing ensemble family, named but not fit by hand in earlier chapters).

Table 21.2 reports seven metrics for each.

Model	Accuracy	Precision	Recall	F_1	ROC-AUC	PR-AUC	Brier
Logistic (plain)	0.807	0.683	0.238	0.353	0.720	0.494	0.146
Logistic (balanced)	0.731	0.426	0.615	0.503	0.721	0.485	0.205
Random Forest	0.816	0.660	0.350	0.458	0.775	0.554	0.136
XGBoost	0.818	0.665	0.356	0.464	0.778	0.557	0.135

Table 21.2: Held-out test set performance for four models on the credit-default dataset

The plain logistic regression's accuracy (80.7%) looks respectable in isolation, the trap Chapter 15 warned against for a rare-event problem: with a 22.1% default rate, a model far too conservative about flagging defaults can still post high accuracy by predicting the majority class, and this model's recall of only 23.8% shows it misses more than three-quarters of the accounts that default at the standard 0.5 threshold. Class weighting flips the trade-off as Chapter 15 described: recall rises to 61.5% while precision falls from 68.3% to 42.6%, a much better F_1 (0.503 versus 0.353) but a distinctly worse Brier score (0.205 versus 0.146). That pairing is the book's first concrete evidence that class weighting is not a free improvement: reweighting the training loss to counter imbalance also pulls the model's output away from a well-calibrated probability, a cost Chapter 15 never quantified with a metric like the Brier score, since its worked example used a balanced confusion-matrix illustration rather than a held-out calibration check.

The two tree ensembles dominate both discrimination metrics, ROC-AUC and PR-AUC, over either logistic regression. The comparison also gives Chapter 6's bagging-versus-boosting distinction its first empirical instance: XGBoost's sequential, error-correcting boosting edges out the random forest's parallel, variance-reducing bagging on every metric in the table, though by a narrower margin (ROC-AUC 0.778 versus 0.775) than the gap either ensemble opens over logistic regression. That pattern matches a finding well documented in applied machine learning: gradient boosting and random forests are far closer competitors with each other on tabular data than either is with a linear model, though boosting often comes out slightly ahead.

The comparison also corrects a common oversimplification, that a logistic regression is always the "safe, interpretable" choice and a tree ensemble the "powerful, opaque" one. Both ensembles beat both logistic regressions on every ranking-based metric by roughly five to six points of ROC-AUC while remaining explainable in Chapter 18's sense, with the tooling Section 21.3.4 applies below, though not in the coefficient-reading sense that made Chapter 9's linear credit-scoring model interpretable by construction.

A lender choosing among these four models is therefore not trading accuracy for interpretability along a single dial; it is trading a small amount of post-hoc explanation overhead (Section 21.3.4's SHAP computation) for a measurable gain in discrimination and, as Section 21.3.2 shows next, calibration.

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21.3.2 Is the Model's Probability Actually a Probability? Calibration in Practice

Chapter 9 noted that a credit score is meaningful only to the extent that the default probability underneath it is well calibrated, not merely useful for ranking applicants from safest to riskiest; this section makes that check concrete.

Calibration asks whether, among all accounts a model assigns a predicted default probability of, say, 30%, roughly 30% actually default. Table 21.3 bins the test set into ten deciles of predicted probability for the plain logistic regression and for XGBoost, and compares each decile's average predicted probability to its actual observed default rate.

	Logistic (plain)		XGBoost	
Decile (low to high)	Mean predicted	Mean actual	Mean predicted	Mean actual
1	0.040	0.135	0.036	0.045
5	0.167	0.118	0.131	0.152
6	0.185	0.120	0.159	0.183
10	0.577	0.670	0.725	0.697

Table 21.3: Calibration by predicted-probability decile: plain logistic regression versus XGBoost

XGBoost's predicted and actual rates stay within three percentage points of each other in every decile shown (and, in the companion notebook's full table, in every decile not shown), consistent with its lower Brier score in Table 21.2. The plain logistic regression shows a visible gap in its lowest decile, an average predicted probability of 4.0% against an actual default rate of 13.5%, more than three times higher: the safest-looking accounts by this model's own ranking are much riskier than its probabilities suggest, a miscalibration a lender relying on the raw probability, not just the ranking, would want caught before deployment.

Figure 21.2 draws all ten deciles of both models as a **reliability diagram**, together with the isotonic recalibration of the next subsection. Unlike every other figure in this book except those of Chapters 2 and 5, it and the two that follow are the companion notebook's own matplotlib output rather than redrawn diagrams, because on real data the scatter of the points is itself the evidence. A perfectly calibrated model lies on the diagonal. XGBoost's points hug it from the safest decile to the riskiest. The plain logistic regression's zigzag across it at the low end, where its four lowest deciles predict between 4.0% and 14.2% while the observed default rates jump between 8.7% and 16.8%, sag below it through the middle deciles, and finish well above it in the top decile, 57.7% predicted against 67.0% observed.

This is why Section 21.3.5's cost-sensitive threshold and Section 21.3.6's fairness audit both use XGBoost's probabilities rather than the plain logistic regression's: both depend on the predicted probability meaning what it claims, not merely on the model's ability to rank risky accounts above safe ones.

21.3.3 Fixing Miscalibration: Isotonic Recalibration

The plain logistic regression's calibration gap raises the next question: can it be corrected without discarding the model? **Isotonic recalibration** fits a separate, nonparametric, monotonically increasing function that remaps a model's raw probabilities onto better-calibrated ones. The remapping is learned from the data the original model was fit on, but evaluated out-of-fold, Section 21.2.2's train/test discipline applied one level deeper, to the recalibration step itself. Fitting the remapping directly on the test set, as a careless analyst might, would leak the very data the recalibration is meant to be validated against.

Applied to the plain logistic regression, scikit-learn's `CalibratedClassifierCV` with isotonic regression, five-fold cross-validated entirely within the training set, improves the test-set Brier score from 0.1460 to 0.1419, a modest but real reduction, while leaving ROC-AUC essentially unchanged (0.720 before, 0.724 after). That is what an isotonic remapping should do: because the transformation is monotonic, it cannot change how the model ranks any two applicants, only how well its stated probability matches reality at a given rank. The improvement is concentrated where Table 21.3 found the problem.

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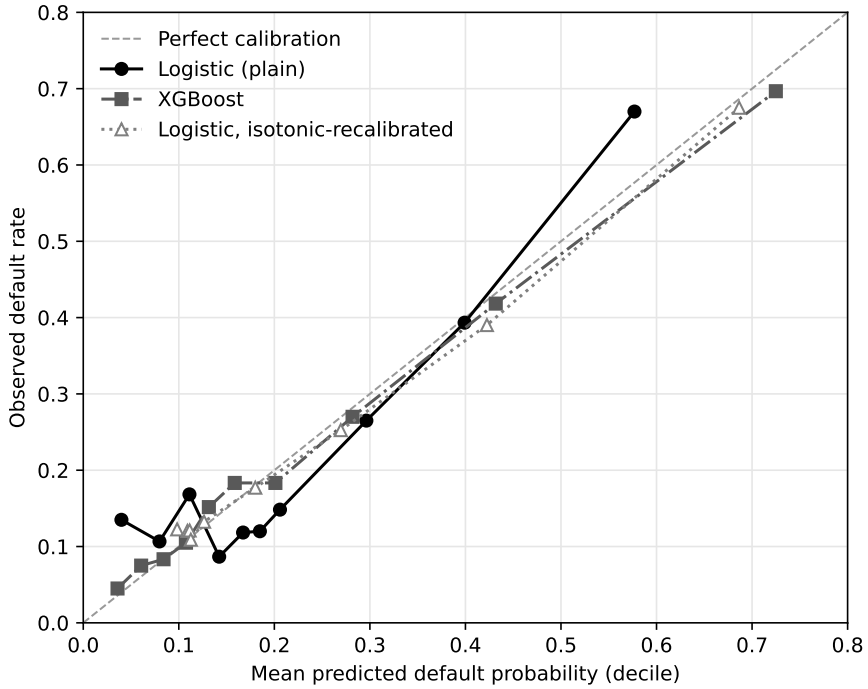


Figure 21.2: Reliability diagram on the 6,000-account test set: mean predicted default probability against observed default rate, by decile of each model's own predictions. The recalibrated logistic regression has nine bins because many of its scores tie.

Recomputing the reliability table on the recalibrated probabilities for the lowest-probability bin,

$$\begin{aligned} \text{Recalibrated:} &\quad \text{predicted} = 9.8\%,\ \text{actual} = 12.3\%,\ \text{ratio} \approx 1.2, \\ \text{Original (uncalibrated):} &\quad \text{predicted} = 4.0\%,\ \text{actual} = 13.5\%,\ \text{ratio} > 3, \end{aligned}$$

The lesson is general: a model with good discriminative power but poor calibration, the plain logistic regression's situation here, need not be replaced to become a trustworthy probability estimate, only recalibrated, a much cheaper fix than retraining a different model family, and one that leaves all the original coefficients, and so the model's adverse-action interpretability, intact.

Recalibration does not fix everything: XGBoost's Brier score (0.135) remains lower than the recalibrated logistic regression's (0.142), since XGBoost's advantage was never only a calibration artifact. Its higher ROC-AUC in Table 21.2 reflects superior discriminative power that no monotonic remapping can extract from a linear model's more limited functional form.

Calibration matters beyond this chapter's threshold and fairness exercises. A bank reporting expected credit losses under IFRS 9 or the analogous U.S. CECL framework multiplies a probability of default directly into a provisioning calculation, Chapter 9's EL = PD × LGD × EAD decomposition. A systematically miscalibrated probability, not merely a poorly ranked one, therefore becomes an over- or understated loss provision in the bank's financial statements, the kind of number Chapter 4 taught how to read and Chapter 9 how to model, and one that depends on a property, calibration, that neither accuracy nor ROC-AUC guarantees.

Section 21.3.4 returns to this formula with real provisioning numbers, once this chapter's own low-risk and high-risk applicants have been introduced.

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21.3.4 Explaining a Real, Nonlinear Model: SHAP and Permutation Importance

Chapter 18 computed an exact Shapley decomposition by hand for Chapter 9's credit-scoring model only because that model was linear in the log-odds; XGBoost is not. The `shap` library's `TreeExplainer` algorithm computes exact Shapley values efficiently for tree ensembles by exploiting the tree structure directly, rather than relying on the linear shortcut or on sampled perturbations as Chapter 18's LIME example did.

Table 21.4 compares the resulting global feature ranking, mean absolute SHAP value across 1,000 held-out accounts, against Chapter 18's other global method, permutation importance (here, the drop in ROC-AUC when a feature is shuffled), computed on the full test set.

Rank	Permutation importance (feature)	Mean |SHAP| (feature)
1	`PAY_1` (0.0337)	`max_delinquency` (0.444)
2	`max_delinquency` (0.0318)	`PAY_1` (0.339)
3	`avg_utilization` (0.0118)	`avg_utilization` (0.167)
4	`BILL_AMT1` (0.0094)	`BILL_AMT1` (0.130)
5	`LIMIT_BAL` (0.0057)	`LIMIT_BAL` (0.118)

Table 21.4: Top features by two global importance methods, XGBoost model

The two methods agree on which five features matter (the most recent repayment status, the worst repayment status over six months, utilization, the most recent bill, and credit limit) and order ranks three through five identically, but disagree on the top two: permutation importance ranks the most recent month's repayment status (`PAY_1`) above the worst status across six months, while SHAP ranks them the other way around. Neither ranking is wrong; they answer different questions, as Chapter 18 cautioned: permutation importance describes a feature's contribution to a specific performance metric on specific data, while SHAP describes each feature's average marginal contribution to individual predictions. A modeler who reports only one of the two risks a false precision about which single feature "matters most."

A single applicant's explanation shows at the individual level what Section 21.3.1's aggregate table cannot. The first test-set applicant is assigned a predicted default probability of 13.07% by XGBoost.

SHAP's additive decomposition operates on the model's raw log-odds output rather than on the probability, as Chapter 18's exact Shapley decomposition did for the linear credit-scoring model. This applicant's base value, the average log-odds prediction across the training population, is -1.272, and summing the SHAP contributions of every feature gives a total log-odds of -1.895. The sigmoid converts that back to 13.07%, matching the model's output: the completeness property Chapter 18 checked by hand for a linear model, now confirmed for a nonlinear ensemble. The applicant's largest contributions are

$$\begin{aligned} \texttt{max_delinquency} = 0 \text{ (no serious delinquency)}: &\quad \text{SHAP} = -0.309, \\ \texttt{avg_utilization} = 12.1\% \text{ (low)}: &\quad \text{SHAP} = -0.215, \\ \text{Credit limit} = \text{NT\$50,000 (modest)}: &\quad \text{SHAP} = +0.175, \\ \text{Most-recent repayment status} = -1 \text{ (paid duly)}: &\quad \text{SHAP} = -0.151, \end{aligned}$$

the modest credit limit contributing positively to predicted risk since, in this dataset, lower limits are associated with higher-risk accounts.

Figure 21.3 draws the whole decomposition as a **waterfall**: starting from the base value of -1.272, each bar moves the running log-odds by one feature's SHAP value, hatched bars raising predicted risk and plain bars lowering it. The seven largest contributions are shown individually; the remaining 18 features together contribute -0.247, and the bars end at -1.895, the model's actual output, which is the completeness property made visible.

Chapter 18's adverse-action reason codes followed automatically from a linear model's coefficients; for a deployed ensemble, producing the same kind of individualized explanation requires this extra step, running `TreeExplainer` against the fitted model, before a bank can issue a model-derived reason to an applicant under the Equal Credit Opportunity Act.

A contrasting high-risk applicant from the same test sample makes the method's value clearer. This applicant is assigned a predicted default probability of 92.2%, and the same decomposition gives a base value

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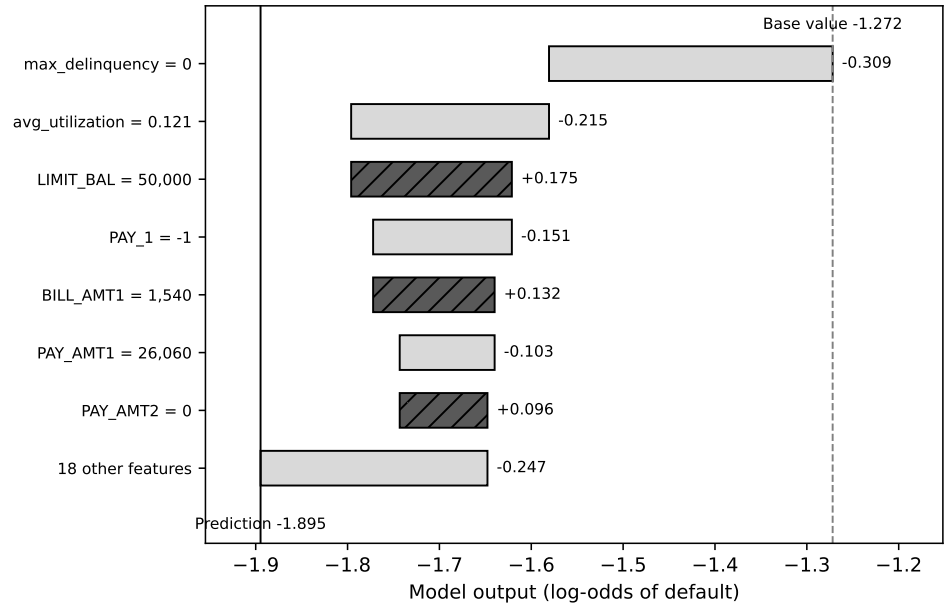


Figure 21.3: SHAP waterfall for the low-risk applicant (predicted default probability 13.07%), in log-odds. Hatched bars increase predicted risk; plain bars decrease it. The dashed line marks the base value and the solid line the model's output.

of −1.272 and a summed log-odds of 2.470 (whose sigmoid is, again exactly, 92.2%). Here the dominant contributions run the other way:

$$\begin{aligned} \text{Most-recent repayment status} = 3 \text{ (three months overdue)}: &\quad \text{SHAP} = +1.009, \\ \text{Worst-ever repayment status} = 7 \text{ (severely delinquent)}: &\quad \text{SHAP} = +0.921, \\ \text{Credit limit} = \text{NT\$10,000 (the dataset's minimum)}: &\quad \text{SHAP} = +0.303, \end{aligned}$$

with `pay_to_bill_ratio` = 0 (repaid essentially nothing over six months) adding a further +0.151, and the most-recent-repayment contribution the single largest seen in either applicant's decomposition.

Side by side, the two explanations read almost as mirror images: no delinquency history and full repayment push one applicant's risk sharply down, while recent, severe delinquency and near-zero repayment push the other's sharply up. That individualized contrast is what an adverse-action notice or a credit-committee review needs, and what an aggregate importance ranking like Table 21.4 cannot provide.

Worked Example 21.1: From Predicted Probability to a Provisioning Number

Section 21.3.2 promised to return to Chapter 9's EL = PD × LGD × EAD decomposition with real numbers; the two applicants supply them. Each applicant's credit limit stands in for exposure at default, a standard simplification for a revolving line assumed fully drawn at default, and, with no collateral information for an unsecured card balance, the example reuses Chapter 9's illustrative LGD = 60%.

All amounts in this dataset are in New Taiwan dollars (NT$), not U.S. dollars; at the 2005 exchange rate of roughly NT$32 to the U.S. dollar, the low-risk applicant's NT$50,000 limit is worth only about US$1,550. For the low-risk applicant, PD = 13.07077% (the 13.07% reported earlier, carried to more decimals to reproduce the figure below exactly by hand) and EAD = NT$50,000 (this applicant's `LIMIT_BAL`), giving

$$\text{EL}_{\text{low-risk}} = 0.1307077 \times 0.60 \times \text{NT\$50,000} = \text{NT\$3,921.23},$$

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while the high-risk applicant, with PD = 92.19913% (similarly carried to more decimals) and a much smaller EAD = NT$10,000 (a lower credit limit, itself consistent with this applicant's weaker profile), gives

$$\text{EL}_{\text{high-risk}} = 0.9219913 \times 0.60 \times \text{NT\$10,000} = \text{NT\$5,531.95}.$$

The high-risk applicant's expected loss is only about 1.41× the low-risk applicant's, a far smaller multiple than the 7.05× gap in predicted default probability (92.20%/13.07%), because the high-risk applicant's credit limit, one-fifth of the other's, largely offsets the higher probability: an interaction among the three inputs to expected loss that looking at PD alone would miss. A bank provisioning against its whole card portfolio sums a version of this calculation across every account, so a systematic miscalibration of the kind Section 21.3.2 found in the plain logistic regression, understating risk in the lowest-probability decile, would understate provisions for the bank's safest-looking accounts, the segment a supervisor examining provisioning adequacy would be least likely to scrutinize.

`TreeExplainer` is also fast: it computes exact Shapley values for a tree ensemble in time roughly proportional to the number of trees times the square of tree depth, rather than the exponential cost of the general, model-agnostic Shapley formula Chapter 18 mentioned, which is why it can run here against a several-hundred-tree ensemble rather than only against the small linear toy models of Chapter 18's hand computations. This matters operationally: a bank issuing thousands of adverse-action notices a month needs an explanation method fast enough for that volume, and a sampling-based method like LIME, useful for a single illustrative explanation, does not scale as cleanly as an algorithm built for the model family being explained.

21.3.5 Choosing an Operating Threshold: A Real Cost-Sensitive Analysis

Every result so far used the default 0.5 threshold. Chapter 15's alert-economics section argued that the right threshold is a business decision balancing the cost of a missed default against the cost of denying a good customer, not a property of the model, and this section quantifies that trade-off. Assigning an illustrative cost of $500 to a missed default (a false negative, an account that defaults but is approved) and $50 to wrongly denying a good customer (a false positive, a tenth of the missed-default cost, since a denied good customer typically costs foregone interest rather than lost principal), the expected total cost across the test set at threshold t is $\text{Cost}(t) = 500 \cdot FN(t) + 50 \cdot FP(t)$.

Table 21.5 sweeps this across a grid of thresholds using XGBoost's predicted probabilities.

Threshold	False negatives	False positives	Total cost
0.05	29	4,071	$218,050
0.10 (cost-minimizing)	129	2,905	$209,750
0.20	450	1,218	$285,900
0.35	667	510	$359,000
0.50 (naive default)	854	238	$438,900
0.60	918	179	$467,950

Table 21.5: Cost-sensitive threshold sweep, XGBoost model, held-out test set

The cost-minimizing threshold on this grid is $t = 0.10$, far below the naive 0.5 default, at a total expected cost of $209,750, versus $438,900 at the naive threshold, more than double. That value could have been predicted without a sweep. Denying an account with predicted default probability p is cheaper than approving it whenever $p \times \$500 > (1 - p) \times \$50$, so for a calibrated model the cost-minimizing threshold is

$$t^* = \frac{c_{FP}}{c_{FP} + c_{FN}} = \frac{50}{50 + 500} = 0.091, \tag{21.1}$$

the same expected-cost comparison Chapter 15 used to derive its fraud-review threshold, adapted to a decision in which a correctly denied defaulter costs nothing. Only the ratio of the two costs matters, not their level or currency. The grid's empirical minimum at 0.10 sits right next to this analytic value (at exactly 0.091

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the test-set cost is \$214,100, the small difference being sampling noise in 6,000 accounts), a further check on Section 21.3.2's finding that XGBoost's probabilities are well calibrated.

Figure 21.4 plots the cost at every threshold on the 0.01-spaced grid behind Table 21.5. Its shape carries a practical message the table only hints at: the curve is nearly flat around its minimum, from \$218,050 at $t = 0.05$ to \$209,750 at $t = 0.10$, and then climbs steeply and almost linearly toward \$438,900 at the naive threshold. Setting the threshold somewhat too low is cheap here; setting it far too high is expensive.

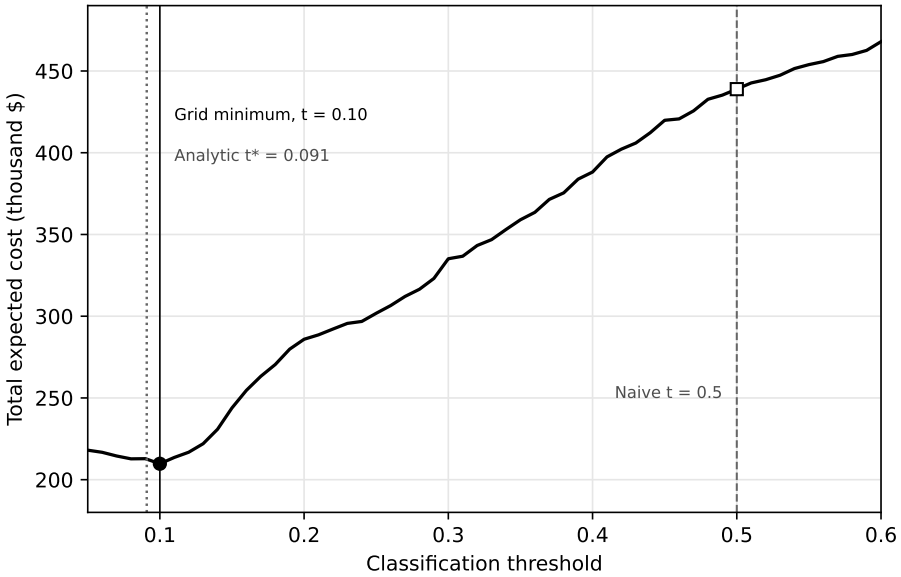


Figure 21.4: Total expected cost on the test set, $500\,FN(t) + 50\,FP(t)$, against the classification threshold t, XGBoost model. The solid vertical line marks the grid minimum at $t = 0.10$, the dotted line the analytic threshold $t^* = 0.091$, and the dashed line the naive $t = 0.5$.

The gap between \$209,750 and \$438,900 is the lesson: choosing a threshold by convention rather than by the institution's cost structure leaves real money on the table, much more here than the gap between any two models in Table 21.2, which is why Chapter 15 treated threshold choice as at least as consequential as model choice. The optimal threshold depends entirely on the assumed 10-to-1 cost ratio; a lender who judged missed defaults and denied good customers comparably costly would land much closer to 0.5. This is a business judgment, not a number the chapter's methods can supply.

The resulting operating point deserves a plain statement: at $t = 0.10$ the model approves only 31.6% of the test accounts, against 88.2% at the naive 0.5 threshold, and turns away 62.2% of the accounts that would not have defaulted. No real card lender operates anywhere near so strict a cutoff. The toy cost ratio drives it: charging a missed default only ten times a wrongly denied good customer, against a 22.1% base default rate, pushes the threshold below the probability most accounts are assigned. A real lender's cost of denying a good customer includes the lifetime value of the relationship, and the cost of a missed default is proportional to the exposure rather than flat, so a realistic cost model would place the threshold, and the approval rate, very differently.

The threshold sweep is a cost-weighted walk along the receiver operating characteristic curve Chapter 6 introduced. Each threshold in Table 21.5 is one point on that curve, trading fewer false negatives for more false positives as the threshold falls. What this section adds is not a new statistical concept but a conversion of that trade-off into dollars, so the operating point is chosen by comparing two cost figures rather than by picking a point on a curve that looks like a reasonable compromise.

A lender would also stress-test this choice, as Chapter 9 stress-tested a credit portfolio: if a recession doubled the cost of a missed default relative to a denied applicant, the cost-minimizing threshold would fall

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further, flagging more accounts as high-risk, the scenario analysis a model-risk function should run before committing to a single operating threshold in production.

21.3.6 A Real Fairness Audit

Chapter 18's fairness audit used a synthetic eight-applicant dataset built to make a point cleanly; this section runs the same audit, the four-fifths rule and an error-rate comparison, on the real test set at the cost-minimizing threshold of Section 21.3.5, split by the dataset's `SEX` field. Table 21.6 reports the results.

Group	n	Approval rate	False-negative rate	False-positive rate
Male	2,402	30.9%	8.9%	62.5%
Female	3,598	32.1%	10.3%	62.0%

Table 21.6: A fairness audit by sex, XGBoost model at the cost-minimizing threshold ($t = 0.10$)

The adverse impact ratio, computed from the unrounded approval rates as $30.85\%/32.13\% = 0.960$, clears the 80% four-fifths threshold of Chapter 18's fairness section by a wide margin, and some gap is expected, since male accounts in the test set default more often (23.4% against 21.3%). The error rates show the tension Chapter 18's synthetic example illustrated, now on real data: the group with the higher approval rate (female accounts) also has the higher false-negative rate, missing 10.3% of accounts that go on to default, against 8.9% for male accounts, so the group favored by the headline approval rate is the group whose default risk this threshold captures less completely. Neither comparison is "the" fairness result; both are properties of the same model and threshold, and, as Chapter 18's impossibility result predicts, no adjustment to this one threshold can bring both gaps to zero unless the groups' underlying default rates coincide.

Leaving the protected attributes out of the model mattered: the same XGBoost configuration refit with `SEX`, `MARRIAGE`, and `AGE` as inputs, which the companion notebook runs for contrast, approves 28.6% of male and 33.9% of female accounts at the same threshold, an adverse impact ratio of only 0.843, while scoring essentially no better (test ROC-AUC 0.779 against 0.778). But removing the attributes did not remove the difference. The model never sees `SEX`, yet its decisions still differ measurably by sex, the pattern of Chapter 18's Apple Card vignette: "fairness through unawareness" narrows a disparity only as far as the remaining features fail to encode the omitted attribute. What the audit does not establish is why the residual difference exists: whether it reflects the groups' different default rates or another feature correlated with sex closely enough to reproduce the pattern. Income proxies and credit-limit history are the leading suspects in this dataset, though testing that rigorously would require the counterfactual-fairness framing of Chapter 18's advanced topics, not merely this correlational check. Section 21.4.4 then shows that this single-attribute finding, reassuring as its 0.960 ratio looks, does not remotely tell the whole story once a second characteristic is added.

21.4 From Model to Project

A model is not a project; this part covers deployment, and the template for doing all of it again.

21.4.1 From Model to Deployment: Governance Considerations

Chapter 18's governance section organized model governance around SR 11-7's three pillars, conceptual soundness, ongoing monitoring, and outcomes analysis, which SR 26-2, the guidance that superseded it in April 2026, keeps intact (Federal Reserve, OCC, and FDIC 2026), and this pipeline maps onto all three.

Conceptual soundness is Section 21.3.1's model bake-off and Section 21.3.2's calibration check, documented well enough that an independent validator, Chapter 15's second line of defense, could reproduce every number in Table 21.2 from the companion notebook without re-deriving the modeling choices. Outcomes analysis is Section 21.3.6's fairness audit, which this chapter's numbers show passing on one cut (0.960 by sex) and failing on another (Section 21.4.4's sex-by-age cut), so a one-time check at deployment is not enough; ongoing

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monitoring, with Chapter 18's population stability index on this feature set and a recurring rerun of both audits, is what would catch a shift in the applicant pool moving either ratio before an examiner does.

None of this replaces human judgment: Section 21.3.5's cost assumptions, the $500-to-$50 ratio driving the threshold, are a business decision no algorithm here can make for the institution, the kind of accountable, documented judgment that Chapter 15's three-lines-of-defense framework and the NIST AI RMF's "govern" function exist to make explicit.

A deployment checklist follows from the chapter's sections. Before the model scores a real applicant, an institution would need Section 21.2.1's cleaning logic and Section 21.2.2's feature engineering encoded in a reproducible data pipeline rather than a one-off notebook cell. It would need Section 21.3.1's model selection and Section 21.4.3's hyperparameter search documented well enough for a validator to rerun them and obtain the same numbers. It would need Section 21.3.2's calibration check and Section 21.3.5's cost assumptions re-approved on a recurring basis, since calibration and an institution's true cost ratio both drift as the applicant population and business environment change. And it would need the fairness audits of Sections 21.3.6 and 21.4.4, the intersectional cut included, rerun on every new cohort of applicants, because Section 21.4.4 showed that a single-attribute check passing today is no guarantee that a deeper cut will not fail tomorrow.

21.4.2 Advanced Topics

Two further analyses go beyond a first pass: a systematic search over XGBoost's hyperparameters, to see how much Section 21.3.1's single configuration left on the table, and a fairness audit that splits the test set two ways at once, revealing a disparity Section 21.3.6's single-attribute audit could not see.

21.4.3 Hyperparameter Tuning: Diminishing Returns in Practice

Section 21.3.1's XGBoost model used a single, manually chosen configuration (300 trees, maximum depth 4, learning rate 0.05). A randomized search over 30 candidate configurations varied tree depth (3 to 6), learning rate (0.01 to 0.1), the number of boosting rounds (100 to 500), the row and column subsampling fractions, and the minimum child weight controlling how conservatively a tree can split. Every candidate was evaluated by five-fold cross-validation on the training set alone, never touching the test set: Section 21.2.2's train/test discipline applied one level deeper, since a hyperparameter search must also be validated on data it never saw.

The best configuration found, depth 4, learning rate 0.1, 100 rounds, no row subsampling, 60% column subsampling, and a minimum child weight of 5, achieved a cross-validated ROC-AUC of 0.786, and, refit on the full training set and evaluated once on the held-out test set, a test ROC-AUC of 0.781, a PR-AUC of 0.554, and a Brier score of 0.135. Compared against Section 21.3.1's untuned XGBoost model (test ROC-AUC 0.778, PR-AUC 0.557, Brier 0.135), the result of a reasonably thorough search is not even a uniform improvement: ROC-AUC rises by three thousandths, PR-AUC falls by three thousandths, and the Brier score does not move at the third decimal, all changes far smaller than the gap either tree ensemble opened up over logistic regression in Table 21.2.

This is a useful, if unexciting, finding. Once a model family is well chosen and its configuration set to sensible, moderate values, as Section 21.3.1's untuned model already was, further search on a dataset of this size and feature set faces diminishing returns. The gap between the cross-validated score (0.786) and the held-out test score (0.781) is a small reminder that a search's selected score is mildly optimistic, having been chosen among 30 candidates because it performed best on that particular split.

21.4.4 Intersectional Fairness: What Aggregation Can Hide

Section 21.3.6's audit split the test set only by sex. Splitting it a second way at the same time, by sex and by an age cutoff at 30 years, reveals a pattern the single-attribute audit could not, shown in Table 21.7.

The most disadvantaged subgroup by approval rate, young men under 30, is approved only 23.9% of the time, against 33.5% for the most favored subgroup, men 30 and over. That is an adverse impact ratio, from the unrounded rates, of 23.92%/33.53% = 0.713, a clear four-fifths-rule violation, and it stands in sharp contrast to Section 21.3.6's sex-only audit, which found a ratio of 0.960, comfortably on the right side of the same threshold. The two findings are not in tension; they answer different questions. Averaging young and

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Sex	Age group	n	Approval rate	False-negative rate
Male	Under 30	669	23.9%	4.7%
Male	30 and over	1,733	33.5%	10.5%
Female	Under 30	1,243	30.0%	10.1%
Female	30 and over	2,355	33.3%	10.4%

Table 21.7: An intersectional fairness cut: sex crossed with age group

older men together in the sex-only audit hides that young men, on their own, face a starkly lower approval rate than any other subgroup, a disparity the single-attribute view cannot see because it never asks. Nor is this simply an age effect that a separate age audit would have caught at the same strength: among women the under-30 group is approved at 30.0% against 33.3%, a ratio of about 0.90, so the shortfall is concentrated where the two characteristics meet. It also arises in a model that, as Section 21.2.2 specified, never sees `AGE` at all, a second and sharper instance of Section 21.3.6's point that leaving an attribute out does not keep the features that correlate with it, shorter credit histories and lower assigned limits among the youngest borrowers, from carrying it back in; the most obvious such feature here is `LIMIT_BAL`, since the youngest borrowers, and young men above all, hold markedly lower credit limits (a median of NT$80,000 for men under 30 in the test set, against NT$150,000 for men 30 and over). This is the failure mode intersectional or subgroup analysis exists to catch, and that fair-lending examiners and model validators test for: a model can pass every single-attribute audit an institution runs and still produce a stark disparity for a subgroup defined by two or more characteristics, and the only way to find it is to look for it, since Table 21.6's aggregate numbers give no hint that it exists.

A thorough fairness audit therefore cannot stop at a single-attribute cut, however clean its result; it must cross every combination of characteristics an institution has reason to worry about, a much larger testing burden than a single audit table suggests, and, as this dataset shows, not merely a theoretical one. The 2026 change in U.S. federal law does not remove that burden. Since 21 July 2026 Regulation B no longer carries a disparate-impact "effects test" under the Equal Credit Opportunity Act (CFPB 2026), so a ratio below 0.80 is not by itself a federal fair-lending violation; but the four-fifths rule remains the standard screening heuristic, state enforcers and regimes abroad still reach disparate outcomes (Chapter 18's table of regulatory regimes), and model-risk governance expects such a disparity to be found and explained either way.

21.4.5 A Template for Completing a Capstone Project

Section 21.4.6 lists several real datasets a reader could build a capstone project on. This section is the template for doing so: a stage-by-stage checklist distilled from this chapter's pipeline (Figure 21.1), so that a reader starting from any dataset on that list, or one of their own, has a concrete standard to work toward rather than an open-ended instruction to "build a model." Each stage in Table 21.8 names the tool this chapter or an earlier one developed for it, along with a deliverable, a specific artifact, table, or number, that the stage must produce before the project moves on. A project that cannot produce every deliverable is not finished, whatever its headline accuracy.

Stage 9: Deploy and Monitor

The first eight stages end with a model that is correct in a notebook. Stage 9 asks whether it stays correct once it runs unattended, and it is the stage a capstone project is most tempted to describe rather than demonstrate. Section 21.4.1's checklist said what an institution would need before the model scores a real applicant; Stage 9 turns that list into four checks, each one an earlier stage's discipline restated as a property of a running system (Sculley et al. 2015).

Feature parity: training-serving skew. Stage 2's leakage rule, use only information available at decision time, is enforced in a notebook by the analyst's care and in production by the plumbing, and the two can quietly disagree. A data warehouse that already holds the decision day computes a "three-day average balance" over days $t-2$ to t; a live service, which cannot see day t yet, computes it over $t-3$ to $t-1$. For a customer whose daily balances are $1,000, $1,000, $1,000 and then $200 on the decision day, the

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Stage	What this stage requires	Deliverable to produce and report
1. Problem framing and dataset selection	State the business decision the model will inform in one sentence (approve or deny a loan, flag or clear a transaction, and so on); confirm the chosen dataset has a genuine, non-leaking target label and enough rows to support a genuinely held-out test set.	A one-paragraph problem statement naming the target variable, its base rate, and the decision the model's output will feed into.
2. Data cleaning and feature engineering	Inspect every column's summary statistics and value counts before fitting anything, exactly Section 21.2.1's method; document any undocumented or anomalous codes found and how each was resolved; engineer at least one derived feature not present in the raw columns, and explicitly check it (and every other feature) for information that would not genuinely have been available at the moment a real prediction would be made, Section 21.2.2's leakage discipline.	A short data-quality memo listing every anomaly found and the cleaning decision made, plus the engineered feature list with a one-sentence leakage justification for each.
3. Train/test discipline and model bake-off	A stratified train/test split with a fixed, reported random seed; at least one intrinsically interpretable model (linear or logistic regression) and at least one tree ensemble, evaluated once, on the held-out test set alone, on accuracy, precision, recall, F_1, ROC-AUC, PR-AUC, and Brier score, Section 21.3.1's method.	A results table in the same format as Table 21.2, plus both sets' target base rates confirming the stratification held.
4. Calibration check	A reliability table (deciles of predicted probability against actual outcome rate) for at least the best-discriminating model, Section 21.3.2's method.	A calibration table in the format of Table 21.3, plus one sentence stating whether the model's raw probability, not merely its ranking, can be trusted at face value.
5. Explainability	A global importance ranking (permutation importance or mean absolute SHAP value) and at least one individual, local explanation for a specific prediction, Section 21.3.4's method.	A top-five global feature ranking and one specific case's local explanation, with its completeness check (SHAP values plus base value reproducing the model's actual output) shown explicitly.
6. Threshold and cost analysis	An explicit, stated assumption for the dollar cost of a false negative and a false positive, and the resulting cost-minimizing threshold found by sweeping a grid of candidate thresholds, Section 21.3.5's method.	The two assumed costs, the resulting threshold, and its total expected cost compared explicitly against the naive 0.5 threshold's cost.
7. Fairness audit	At least one single-attribute fairness cut (approval rate, the four-fifths adverse impact ratio, and a false-negative/false-positive rate comparison) and, wherever the data contains a second relevant characteristic, one intersectional cut crossing two attributes at once, Sections 21.3.6 and 21.4.4's method.	An approval-rate table and adverse impact ratio for at least one single-attribute cut, and, if attempted, one intersectional cut, with an explicit statement of whether either violates the four-fifths rule.
8. Governance write-up	A short memo addressing the three SR 11-7 pillars (now carried by SR 26-2) for this specific model: what conceptual soundness was checked, what ongoing monitoring (a population stability index or a re-audit cadence) is proposed, and what outcomes analysis would be needed before this model could be trusted with a real decision, Section 21.4.1's method.	A half-page governance memo naming a concrete monitoring metric, its planned check frequency, and who (which of Chapter 15's three lines of defense) would be responsible for acting on it.
9. Deploy and monitor	Show that the model computed in production is the model that was validated: one shared definition of every feature, versioned artifacts and lineage that let any past decision be rebuilt, a staged release with a rollback trigger fixed in advance, and monitoring whose alarms start a governed retraining rather than an automatic one.	A deployment plan naming the feature-parity check, the artifact and lineage record, the release stages and rollback trigger, and the monitored metrics with the values that trigger review and retraining.

Table 21.8: A nine-stage template for a capstone project, distilled from this chapter's own pipeline

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two definitions give $733.33 and $1,000.00, and a one-feature scorecard PD $= 1/(1 + e^{-(1.2-0.0025x)})$ turns them into 34.7% and 21.4%. Against a 30% cutoff the model was validated to decline this applicant and production approves them, understating the probability of default by 13.3 points, 38% of the correct figure, with no error raised anywhere. The deliverable is a parity test: score a sample of recent applicants through the training code and through the serving path, and require identical feature values. A feature store that holds one definition of each feature, computed once and served to both, is the standard structural answer.

Lineage: the rebuild test. Stage 5's explanation must be regenerable months later, since an adverse-action notice explains the decision the model made then, not the one it would make now. That requires versioning the whole artifact, preprocessing, fitted parameters, feature definitions, and Stage 6's threshold, together with the identity of its training data and the library versions, so that any past score can be rebuilt exactly. The deliverable is a rebuild of one past decision from stored records alone.

Release: shadow, challenger, rollback. A validated model is a hypothesis about production. Run it first in shadow, scoring live applications while its outputs affect nothing, which is also the cheapest way to catch skew; then let it decide a defined slice of applications as a challenger to the incumbent; then widen the slice in stages. The rollback trigger, a named metric crossing a named value, is fixed before the release, because a trigger chosen while a model is misbehaving tends to be chosen to justify keeping it, and the incumbent stays runnable throughout.

Monitoring and retraining triggers. Monitor the inputs and score distribution with Chapter 18's population stability index, using its conventional bands, below 0.10 no material change, 0.10 to 0.25 worth investigating, above 0.25 requiring revalidation, and monitor calibration on realized defaults as they mature, since calibration usually decays first, together with Stage 7's approval-rate ratios on each new cohort. An alarm should start a retraining run whose output re-enters the release sequence as a challenger, with Stages 4 through 7 repeated before promotion; it should not trigger an automatic refit. Automatic retraining turns a governed model change into an unreviewed one, and in lending it learns from outcomes observed only for applicants the incumbent approved, a feedback loop that can entrench the incumbent's errors rather than correct them.

A project report built on this template falls into five sections, usable directly as a report or presentation outline. The **executive summary** states Stage 1's problem statement and the headline result in one paragraph a non-technical reader could act on. **Data and methodology** covers Stages 2 and 3, written so that another analyst could reproduce the cleaning, feature engineering, and model choices exactly. **Results** presents Stages 3 and 4's tables together, so a reader can judge discrimination and calibration side by side rather than seeing only the most favorable number. **Explainability and fairness** covers Stages 5 through 7, which belong together: an explanation of what drives the model's decisions is also evidence for or against a fairness concern, as Section 21.3.6's discussion of `LIMIT_BAL` and Section 21.4.4's intersectional finding showed. **Limitations and recommendations** carries Stage 8's governance memo and Stage 9's deployment plan, with an explicit, named list of what the project did not check.

That last list is the one most often skipped, and it matters most. Section 21.4.4 showed that an unchecked intersectional cut can hide a violation the report's headline numbers give no hint of. A capstone report that does not name its blind spots is not more trustworthy for omitting them; it is less honest about a limitation every model in this chapter shares.

Two failure modes are the most common ways a project satisfies this template's letter while missing its point. The first is treating each stage as an isolated checkbox rather than letting later stages inform earlier ones: Section 21.3.2's finding that the plain logistic regression's safest decile was miscalibrated should change which model Stages 6 and 7 are run on, not be reported and then ignored in favor of the model with the best Stage 3 accuracy. The second is stopping the fairness audit (Stage 7) at the first cut that passes, the mistake this chapter would have made had Section 21.4.4 never been run; a single-attribute cut that passes cleanly is a reason to look harder for an intersectional disparity, not to stop looking.

21.4.6 Capstone Project Ideas

The pipeline built in this chapter generalizes to other public financial datasets: clean, engineer features, split, bake off models, calibrate, explain, choose a threshold, audit for fairness. Each of the following is a capstone project on a real, freely available dataset, several of them also distributed on Kaggle under the names given.

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- **Credit card fraud detection.** The Kaggle/Worldline/ULB "Credit Card Fraud Detection" dataset (284,807 European card transactions, 492 confirmed frauds, features anonymized via PCA) is far more imbalanced than this chapter's 22% default rate, a severe 0.17% positive rate that makes Chapter 15's precision-recall framing, and this chapter's calibration and cost-sensitive threshold methods, much more consequential than here; a reader completing this project should expect the naive 0.5 threshold to be almost useless, and should find that Section 21.3.5's analytic rule $c_{FP}/(c_{FP} + c_{FN})$ still locates the cost-minimizing threshold only if the model's probabilities are calibrated, which is far harder to achieve and to check with 492 positives than with this chapter's 6,636.

- **Consumer lending at greater scale.** The Lending Club loan dataset (millions of originated peer-to-peer loans with real interest rates, loan grades, and repayment outcomes) extends this chapter's pipeline to a dataset large enough that model training time itself becomes a design constraint, and includes enough temporal history to test a model's stability across a full economic cycle, the "non-stationarity" concern Chapter 6 and Chapter 17 both raised only in the abstract.

- **An algorithmic trading backtest on real prices.** Historical daily prices for the S&P 500 constituents (freely available from sources such as Yahoo Finance or Stooq) let a reader rebuild Chapter 13's pairs-trading or Chapter 12's mean-variance framework on real, not fictional, return series, confronting transaction costs, survivorship bias (delisted constituents silently missing from many free datasets), and the gap between an in-sample backtest and true out-of-sample performance.

- **Financial sentiment at scale.** The Financial PhraseBank dataset (thousands of analyst-labeled sentences from financial news, by agreement level) or a Twitter financial-sentiment dataset let a reader rebuild Chapter 16's Loughran-McDonald scoring and Naive Bayes classifier on labeled data, and compare a hand-built lexicon with a fine-tuned transformer, which Chapter 16 only described.

- **A second, independent fraud dataset for cross-dataset generalization.** The IEEE-CIS Fraud Detection dataset (a Kaggle competition dataset with several hundred anonymized features spanning transaction, identity, and device information) is large and high-dimensional enough to make Section 21.3.4's SHAP-based explanation necessary rather than illustrative, since no analyst could manually inspect several hundred features' relationships to a prediction the way this chapter's 25-feature model still just barely permits.

- **A robo-advisor: automated, risk-based portfolio construction.** Using free historical price data for a diversified set of low-cost exchange-traded funds (the kind of instrument Chapter 3 introduced), a reader can rebuild the core of a robo-advisor, following Chapter 12's own worked treatment directly: a short risk-tolerance questionnaire mapped to a target asset allocation, the mean-variance optimizer (or its risk-parity variant) computing the actual portfolio weights on real rather than illustrative return data, periodic rebalancing back to those target weights as prices drift, and, for a realistic version, tax-loss harvesting logic and a fee-drag comparison against a comparable actively managed alternative, extending Chapter 12's stylized numbers to a real portfolio and, ideally, a second real case besides Schwab's to compare against. This project differs from every other one in this list in an instructive way: it is not a classification or regression problem at all, but an optimization and decision-rule problem, so Chapter 18's SHAP and permutation-importance tools do not directly apply, and the natural "explainability" question becomes instead whether the resulting portfolio weights are traceable back to the questionnaire's stated risk tolerance and Chapter 12's stated optimization objective, a different, allocation-based notion of interpretability worth contrasting explicitly with this chapter's classification-based one.

- **Derivative pricing and delta-hedging on real options data.** Historical options-chain data for a liquid underlying such as the S&P 500 ETF (available from sources including Yahoo Finance, CBOE historical data files, and options-focused Kaggle datasets) lets a reader confront Chapter 5's Black-Scholes-Merton and binomial models with actual market prices rather than an assumed volatility input: back out each option's implied volatility from its traded price, plot the resulting volatility smile or skew across strikes (a pattern Chapter 5's constant-volatility assumption cannot produce at all, making its own limitation directly visible), and then implement and backtest a discrete-time delta-hedging

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strategy, rebalancing a hedge position at fixed intervals and tracking the resulting hedging error against the theoretical, continuous-rebalancing ideal Chapter 5 derives. This project is a natural pair with Chapter 14's market-microstructure material, since a real hedging desk's rebalancing frequency is itself constrained by the transaction costs Chapter 14 modeled, so a reader can extend the backtest to show how hedging error and transaction cost trade off against each other as the rebalancing interval shrinks.

- **Portfolio risk management through a real historical crisis.** Real daily return data for a multi-asset portfolio (a mix of equity, bond, and commodity ETFs, again freely available from sources such as Yahoo Finance or Stooq) lets a reader rebuild Chapter 8's Value-at-Risk and Expected Shortfall machinery, parametric, historical-simulation, and Monte Carlo, on real returns rather than a small illustrative covariance matrix, and then backtest the resulting VaR estimates using Chapter 8's own exception-counting method across a window that spans a real historical stress period, the 2008 financial crisis or the 2020 pandemic crash, both of which are included in most free multi-year daily price histories. The point of choosing a window with a real crisis in it deliberately is to test the assumption every one of Chapter 8's VaR methods quietly makes, that recent historical volatility and correlation are a reasonable guide to near-term risk, against a period where that assumption visibly failed, correlations across asset classes that looked low in calm markets rose sharply during the crisis itself, a phenomenon a reader can verify directly by comparing the portfolio's rolling covariance matrix estimated inside versus outside the crisis window, before feeding either estimate into Chapter 12's mean-variance or risk-parity optimizer and comparing the resulting portfolios' actual, realized risk during the crisis itself.
- **A smaller-scale replication for a first project.** The Statlog German Credit dataset (1,000 applicants, 20 features), also mirrored on Kaggle as "German Credit Risk," is small enough to rerun this entire chapter's pipeline, cleaning, feature engineering, bake-off, calibration, SHAP, threshold choice, fairness audit, in an afternoon, making it a natural first capstone for a reader who wants to practice the full workflow once, end to end, before tackling one of the larger datasets above.

21.5 Discussion Questions

1. Table 21.2 shows the class-weighted logistic regression achieving the best F_1 score among the two logistic models but the worst Brier score of all four models. If you were advising a lender that needed to report a single default probability to a regulator for every applicant, which of the four models would you recommend, and would your answer change if the lender only needed to rank applicants from safest to riskiest rather than report a specific probability to anyone?

2. Section 21.2.1 cleaned two undocumented category codes by folding them into an existing "other" category. Discuss an alternative cleaning strategy (for instance, treating each undocumented code as missing data and imputing it, or as its own separate category) and what evidence in the data would help you decide which approach is more defensible.

3. Section 21.3.4 showed permutation importance and SHAP disagreeing on whether `PAY_1` or `max_delinquency` is more important. If a regulator asked your institution which single feature drives most of your credit model's decisions, how would you respond given this disagreement, and does the question itself presuppose something that may not be true of the model?

4. Section 21.3.5's cost-minimizing threshold depends entirely on the assumed \$500-to-\$50 cost ratio. Propose a concrete way an institution could estimate these two costs from its own historical data, and discuss whether the same cost ratio should apply uniformly to every applicant or could reasonably vary by loan size.

5. Section 21.3.6's adverse impact ratio (0.960) passes the four-fifths rule comfortably, yet the same model refit with `SEX`, `MARRIAGE`, and `AGE` as inputs scored 0.843, and Section 21.4.4's sex-by-age cut fails (0.713). Discuss what additional analysis, beyond recomputing these same ratios next quarter, an internal model risk function should perform to decide whether the passing result reflects a stable, defensible model, and how much of the remaining disparity it should attribute to proxies for the omitted attributes rather than to genuine differences in default risk.

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6. Choose one of Section 21.4.6's capstone project ideas and walk through Table 21.8's nine stages for it explicitly, one sentence per stage naming the specific dataset column, model choice, or metric that stage would use, and flag any stage whose deliverable that project's dataset cannot actually support (for instance, a dataset with no protected-characteristic field at all, which would leave Stage 7 unable to produce a real fairness cut).

7. Section 21.4.6's robo-advisor and portfolio-risk-management projects are not classification problems and have no natural confusion matrix, ROC curve, or SHAP value. For either one, propose a concrete way to check whether the resulting system is behaving as intended, given that this chapter's usual evaluation toolkit does not directly apply, and identify which of Table 21.8's nine stages would need to be reinterpreted rather than dropped entirely.

8. Section 21.4.4's intersectional cut checked sex crossed with age. Propose a third characteristic from this dataset worth crossing with sex in a real fair-lending examination, justify your choice, and discuss what evidence would tell you whether it is worth checking before you actually run the numbers.

9. Section 21.4.5 names two failure modes: treating each template stage as an isolated checkbox, and stopping a fairness audit at the first cut that happens to pass. Describe a plausible way a well-intentioned project team could fall into each failure mode without realizing it, and propose one concrete process change (not a new statistical method) that would catch each one before a project is reported as finished.

21.6 Challenges in End-to-End Financial AI Projects

Several challenges recur across this chapter's pipeline and connect directly to themes from earlier chapters.

Real data is never clean by default. Section 21.2.1's undocumented category codes have no counterpart in any earlier chapter's fictional dataset, and a project that skips this step inherits whatever silent errors those codes introduce into every downstream model.

The best model by one metric is not always the best by every metric. Table 21.2's class-weighted logistic regression has the best F_1 and the worst Brier score of the four models, so "best model" is well defined only once the deployment use case fixes which metric matters.

Different explanation methods can disagree, and both can be right. Section 21.3.4's permutation-importance-versus-SHAP disagreement on the top two features is not a bug in either method but a consequence of the two answering different questions, the caution Chapter 18 raised and this chapter confirms empirically.

A threshold choice is a business decision with real financial consequences, not a modeling detail. Section 21.3.5's more than twofold cost difference between the naive and cost-optimal thresholds dwarfs the gap between any two models in the bake-off, a quantified instance of Chapter 15's alert-economics argument.

Removing a protected attribute is not the same as removing its effect. Leaving `SEX`, `MARRIAGE`, and `AGE` out of the model, as fair-lending law requires, raised Section 21.3.6's sex-only adverse impact ratio from 0.843 to 0.960 at almost no cost in accuracy, yet the model still treats the sexes measurably differently and, never having seen `AGE`, still approves young men at well under four-fifths of the rate of older men, which is why Section 21.4.1 treated ongoing outcome monitoring, not the choice of inputs alone, as the real governance obligation.

A single-attribute fairness audit can pass while an intersectional one fails. Section 21.4.4's sex-and-age cut found a clear four-fifths-rule violation (0.713) that Section 21.3.6's sex-only audit (0.960, passing) gave no hint of, so an institution that stops testing after the first clean-looking result may never discover a disparity that exists for a subgroup it never checked.

Not every financial AI project is a classification problem, and explainability means something different for each. Section 21.4.6's robo-advisor project has no confusion matrix, ROC curve, or natural SHAP value, since it is an optimization problem rather than a supervised-learning one, so "explaining" its output means tracing portfolio weights back to a stated risk tolerance and optimization objective, a different task from those Section 21.3.4's methods were built for, and a reminder that this book's explainability toolkit, powerful as it is for classification and regression, is not a universal solvent.

An end-to-end pipeline is only as trustworthy as its reproducibility. Every number in this chapter came from a fixed random seed and a documented sequence of cleaning, feature engineering, and modeling

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choices, recorded in the companion notebook so that Section 21.4.1's independent-validation obligation can actually be met.

21.7 Key Takeaways

The chapter's dataset, the UCI/Kaggle Default of Credit Card Clients data, was the book's first not built to illustrate anything, and it surfaced problems the fictional datasets never could: undocumented category and status codes and a misnamed column to clean (Section 21.2.1), a class imbalance whose remedy (class weighting) traded a better F_1 score for a worse Brier score (Section 21.3.1), and a calibration gap in the plain logistic regression's safest-looking decile that a rank-based evaluation would never have surfaced (Section 21.3.2).

Running Chapter 18's explanation toolkit with real software, `shap`'s `TreeExplainer` and scikit-learn's permutation importance, on a nonlinear gradient-boosted model confirmed the completeness property Chapter 18 verified by hand and exposed a disagreement between two legitimate importance rankings (Section 21.3.4).

A cost-sensitive threshold cut expected cost by more than half relative to the naive 0.5 default (Section 21.3.5), though at an operating point far stricter than any real lender would accept, an artifact of the toy cost ratio that the analytic threshold $c_{FP}/(c_{FP} + c_{FN})$ makes transparent. The model, built without `SEX`, `MARRIAGE`, or `AGE`, passed a sex-only fairness audit comfortably but still showed a residual sex difference and the approval-rate-versus-error-rate tension Chapter 18 demonstrated with synthetic data. Leaving out a protected attribute is not fairness, and a single deployment-time check is not governance; ongoing monitoring is (Section 21.4.1).

Section 21.4.2 went deeper in two directions. A systematic hyperparameter search found no consistent gain over Section 21.3.1's single XGBoost configuration, a reminder that model-family choice and feature engineering matter far more than fine-tuning once a sensible configuration is in place. An intersectional cut by sex and age then uncovered a stark, four-fifths-rule-violating disparity for young male applicants that Section 21.3.6's single-attribute audit, run first and reported as passing, gave no hint of: the book's sharpest illustration of why an audit's scope, not just its arithmetic, determines what it can find.

The breadth this one pipeline drew on is the point of a capstone chapter. Chapter 2's pandas and NumPy fundamentals ran every line of the code. Chapter 6's train/test discipline, feature engineering, bagging and boosting, and overfitting caution structured Sections 21.2.2 and 21.3.1. Chapter 9's calibration caveat became Section 21.3.2's central finding. Chapter 15's class-imbalance remedies and cost-sensitive threshold framing became the empirical results of Sections 21.3.1 and 21.3.5. And Chapter 18's explainability and fairness toolkit, moved from hand computation on a linear model to real software on an ensemble, became Sections 21.3.4, 21.3.6, and 21.4.4.

A single real dataset, messier than any this book constructed, was enough to put nearly every tool the book developed to simultaneous use. None of these methods was meant to be practiced in isolation, and a real financial AI project rarely allows using only one at a time.

Section 21.4.5 distilled the pipeline into a nine-stage template: problem framing, data cleaning and feature engineering, model bake-off, calibration, explainability, threshold and cost analysis, fairness audit, a governance write-up, and a deployment and monitoring plan that keeps the validated model the one actually running. Each stage names its required deliverable, so that a reader knows what "finished" means.

The chapter, and the book, close with Section 21.4.6's further project ideas, each a real, freely available dataset, spanning fraud detection, consumer lending, algorithmic trading, financial sentiment, robo-advisory, derivatives hedging, and credit scoring at a different scale. They are for a reader now ready to run the same disciplined pipeline independently on data this book did not hand-pick: clean, engineer, split, compare, calibrate, explain, price the threshold, and audit for fairness.

21.8 Suggested Exercises

21.1 Using Section 21.2.1's method, tabulate the default rate separately for accounts with an undocumented `EDUCATION` code (0, 5, or 6) before cleaning, and compare it to the default rate for documented codes

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1 through 4, discussing whether folding the undocumented codes into category 4 seems reasonable given what you find.

21.2 Using Table 21.2, compute each model's precision-recall trade-off directly (precision divided by recall) and rank the four models by this ratio, then explain what a very high or very low ratio implies about a model's operating point.

21.3 Using Section 21.3.2's method, compute the calibration gap (mean predicted probability minus mean actual default rate) for every decile of the random forest's predictions, and state whether the random forest's calibration more closely resembles the plain logistic regression's or XGBoost's in Table 21.3.

21.4 Section 21.2.2 kept `AGE` out of the model. Refit the XGBoost model with `AGE` added back as a 26th feature, use Section 21.3.4's method to compute its mean absolute SHAP value and its rank among the 26 features, and rerun Table 21.7's audit. Discuss, in light of Chapter 18's discussion of proxy features correlated with protected characteristics, whether leaving `AGE` out actually removed age-related information from the model's decisions.

21.5 Using Section 21.3.5's method, recompute the cost-minimizing threshold if the cost of a missed default is revised from \$500 to \$1,000 (holding the \$50 false-positive cost fixed), and explain in words the direction you expect the optimal threshold to move before recomputing it.

21.6 Using Section 21.3.6's method, recompute the adverse impact ratio and the false-negative-rate gap at the naive 0.5 threshold instead of the cost-minimizing 0.10 threshold, and discuss whether the choice of threshold changes which group appears more favored by each of the two statistics.

21.7 Using Section 21.4.6's description of the Kaggle credit card fraud dataset, explain why a 0.17% positive rate makes accuracy an even less useful headline metric than it was for this chapter's 22.1% default rate, referencing Chapter 15's original caution about accuracy under class imbalance.

21.8 Using Table 21.7's method, compute the adverse impact ratio for a third cut, sex crossed with `EDUCATION` (collapsed to graduate school versus all other categories), and state whether this cut reveals a four-fifths-rule violation as stark as the sex-and-age cut in the text.

21.9 Section 21.4.3 found no consistent improvement from hyperparameter tuning. Propose one plausible reason a much larger or higher-dimensional dataset (such as the IEEE-CIS fraud dataset in Section 21.4.6) might show a larger gain from the same kind of search, referencing the bias-variance discussion in Chapter 6.

21.10 Using Section 21.3.3's isotonic recalibration result, explain why the recalibrated logistic regression's ROC-AUC barely changed (0.720 to 0.724) even though its Brier score improved measurably, referencing the monotonicity property isotonic regression is constrained to satisfy.

21.11 Explain why fitting an isotonic recalibration function directly on the test set, rather than via cross-validation on the training set alone, would invalidate the resulting calibration check, referencing Section 21.2.2's train/test discipline.

21.12 Using Stage 9's training-serving skew example in Section 21.4.5, with the same balances of \$1,000 on days $t-3$ through $t-1$ and the same scorecard and 30% cutoff: (a) recompute both probabilities of default if the decision-day balance is \$600 rather than \$200, and state whether the two code paths now reach the same decision; (b) find the decision-day balance below which the trained model declines the customer while the serving path still approves; (c) explain why a monitoring dashboard that tracks only the average served PD would not reveal this skew, and propose the test that would.

21.9 Further Reading

- Yeh and Lien (2009), "The Comparisons of Data Mining Techniques for the Predictive Accuracy of Probability of Default of Credit Card Clients." The original paper introducing the dataset this chapter's case study is built on.

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- UCI Machine Learning Repository (n.d.), *Default of Credit Card Clients Data Set*. The public hosting page for the dataset used throughout this chapter, including its full data dictionary.
- Chen and Guestrin (2016), "XGBoost: A Scalable Tree Boosting System." The original paper describing the gradient-boosting implementation used in Section 21.3.1.
- Lundberg and Lee (2017), "A Unified Approach to Interpreting Model Predictions." The SHAP paper underlying the `TreeExplainer` algorithm used in Section 21.3.4, previously cited in Chapter 18 for its linear-model application.
- Niculescu-Mizil and Caruana (2005), "Predicting Good Probabilities with Supervised Learning." A foundational study of which model families tend to produce well-calibrated probabilities, directly relevant to Section 21.3.2's findings.
- Provost and Fawcett (2013), *Data Science for Business*. A practitioner-oriented treatment of the cost-sensitive threshold reasoning developed in Section 21.3.5.
- Dua and Graff (2019), *UCI Machine Learning Repository*. The broader repository this chapter's dataset is drawn from, and a starting point for several of Section 21.4.6's further project ideas.
- Sculley et al. (2015), "Hidden Technical Debt in Machine Learning Systems." The classic account of why machine-learning systems accumulate maintenance debt in data and configuration rather than code, the concerns behind Stage 9 of Section 21.4.5's template.
- Buolamwini and Gebru (2018), "Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification." The landmark empirical demonstration that a system can pass single-attribute fairness checks while still showing a stark disparity for an intersectional subgroup, the same pattern Section 21.4.4 found directly in a lending context.

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Appendix A

Datasets and APIs for Financial AI

A.1 Introduction

Every chapter in this book is built around a worked numerical example, and several of the later chapters, most notably Chapter 21's UCI credit-card default case study, are built around a real, publicly available dataset rather than a fictional one. This appendix collects, in one place, the data sources a reader is most likely to reach for when reproducing this book's examples with real data or extending them into an independent project: what each source covers, how it is typically licensed or priced, and the specific API or library used to pull it into Python.

This appendix is deliberately organized around *access mechanism* as much as content, since the practical difference between a dataset a reader can query in five minutes with a free API key and one that requires an institutional subscription is often the difference between a project that gets started and one that does not. Where this book's own companion notebooks already reach outside the book's fictional running examples for real data, chiefly Chapter 5's CBOE Volatility Index (VIX) history (Section A.2, downloaded live via the `yfinance` library) and Chapter 21's UCI "Default of Credit Card Clients" dataset (Section A.6), this appendix cross-references the relevant chapter directly.

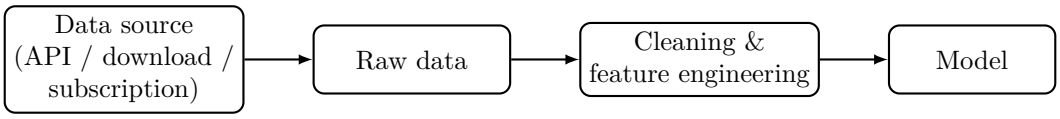


Figure A.1: Where the datasets and APIs in this appendix sit in a typical workflow: they supply the raw data, but the cleaning and feature-engineering steps introduced in Chapter 2's data-cleaning section and used throughout this book still have to be applied before the data reaches a model

A.2 Market Price and Trading Data

Table A.1 summarizes the most commonly used sources for equity, index, foreign exchange, and derivatives price data. Yahoo Finance, accessed through the `yfinance` Python library, is the source this book itself uses: Chapter 5's implied-volatility section downloads the actual daily history of the VIX this way, with no API key required. It is an unofficial wrapper around Yahoo Finance's public website rather than a documented, versioned API, which is precisely why it is free; the trade-off, discussed further in Section A.13, is that it can change behavior or break without notice whenever Yahoo changes its own site.

Where CRSP and Compustat are US-centric, Refinitiv Datastream (a product of the London Stock Exchange Group since its 2021 acquisition of Refinitiv) is the closest institutional equivalent for international and cross-country work: decades of equity, index, bond, and FX price history across essentially every major market, not just the US, together with the Worldscope database of global company fundamentals bundled

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into the same platform. Like WRDS, it is licensed rather than freely available, typically through a university or employer subscription, and is accessed either through its legacy Excel add-in or through the Datastream Web Service (DSWS), a REST API with an official `DatastreamPy` Python client.

Source	Access	API / library	Notes
Yahoo Finance	Free	`yfinance` (PyPI)	Equities, ETFs, indices, FX, crypto, options chains; used in Chapter 5 of this book (the VIX example), and a natural real-data upgrade for Chapter 2's synthetic AssetA/AssetB series; unofficial, can break without notice
Alpha Vantage	Free tier + paid	REST API, key required (`alphavantage.co`)	Equities, FX, crypto, and technical indicators; free tier is rate-limited to a handful of calls per minute
Nasdaq Data Link (formerly Quandl)	Free + paid tiers	REST API (`nasdaqdatalink.com`)	Broad marketplace of market, commodity, and economic datasets from many contributors
Polygon.io	Free limited tier + paid	REST and WebSocket API	Real-time and historical US equities, options, FX, and crypto, including tick-level data
Tiingo	Free tier + paid	REST and WebSocket API	End-of-day prices, fundamentals, and news
CBOE DataShop	Paid	Web portal, bulk download	Official source for VIX methodology data and listed options data
WRDS: CRSP, TAQ	Institutional subscription (typically via a university)	Web query tool, plus SAS/Python/R client	CRSP is the academic gold standard for US equity returns; TAQ provides tick-by-tick trade and quote data used in Chapter 14's microstructure examples
Refinitiv Datastream (LSEG)	Institutional subscription (typically via a university or employer)	Excel add-in, or the Datastream Web Service (DSWS) REST API via `DatastreamPy`	The standard source for international/cross-country equity, index, bond, and FX history, plus Worldscope global fundamentals in the same platform; fills the gap CRSP and Compustat leave outside the US

Table A.1: Market price and trading data sources

A.3 Options and Derivatives Data

Options data deserves its own treatment because it is structurally different from equity price data: a single underlying can have hundreds of simultaneously listed contracts across many strikes and expirations, and

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the quantity most useful for the work in Chapter 5 (pricing, the Greeks, implied volatility) and Chapter 8 (delta-normal and delta-gamma VaR for an options position) is not the raw quoted price but the implied volatility and Greeks computed from it, which not every source provides pre-computed.

OptionMetrics, accessed through WRDS as the IvyDB US database, is the main dataset academic options research is built on, occupying essentially the same role for options that CRSP occupies for equities (Section A.2): daily, exchange-sourced, precomputed implied volatilities and Greeks for every listed US equity and index option going back to 1996, standardized enough that a result computed on it is directly comparable across studies. It is not free, licensed only through an institutional WRDS subscription, which is precisely why Table A.2 also lists several lighter-weight alternatives, from a free but shallow current-chain snapshot to paid practitioner-oriented vendors, for a reader without WRDS access.

A.4 Fundamental and Financial Statement Data

The U.S. Securities and Exchange Commission's EDGAR system is the single most important free source of fundamental data for U.S. public companies, and it is directly relevant to Chapter 4's treatment of the income statement, balance sheet, and cash flow statement. Beyond the human-readable filings, the SEC exposes two machine-readable APIs at `data.sec.gov`: a full-text search API for locating specific language across filings (directly usable for Chapter 16's 10-K risk-factor classification example), and an XBRL "company facts" API that returns every structured financial figure, revenue, COGS, CFO, and so on, that a company has ever reported, tagged and time-stamped, for a given ticker or CIK (Central Index Key). Because these figures come directly from a company's own regulatory filings, they require no scraping or vendor licensing, only a descriptive `User-Agent` header identifying the requester, which the SEC requires of all automated EDGAR access.

A.5 Macroeconomic and Interest Rate Data

FRED, the Federal Reserve Bank of St. Louis's Federal Reserve Economic Data service, is the standard free source for the interest-rate and macroeconomic series this book uses conceptually in Chapter 5's term-structure discussion and Chapter 8's stress-testing scenarios. A free API key registered at `fred.stlouisfed.org` grants access to its REST API, and the community-maintained `fredapi` Python package wraps it in a single function call that returns a time-indexed `pandas` series, matching the data structures this book's notebooks already use throughout.

A.6 Credit, Lending, Insurance, and Consumer Finance Data

This is the category with the deepest overlap with this book's own worked examples. The UCI "Default of Credit Card Clients" dataset is already Chapter 21's own running case study, downloaded programmatically by that chapter's notebook rather than committed to this repository. The other datasets in Table A.5 are natural extensions for the capstone project ideas Chapter 21 closes with.

A.7 Fraud, AML, and Regulatory Compliance Data

Genuinely labeled fraud data is scarce, since financial institutions rarely release real transaction-level fraud labels, which is why the two datasets in Table A.6 recur so often in both industry benchmarks and this book's own Chapter 15 and Chapter 21 capstone list. Operational-risk loss data proper is even scarcer: the industry's event-level loss databases (the ORX consortium foremost among them) are member-only, which is why Chapter 10's real-data challenge falls back on the public enforcement and complaint text sources in the same table, regulatory documents rich in the operational-loss and misconduct narratives that a bank's internal incident data contains.

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Source	Access	API / library	Notes
Yahoo Finance	Free	`yfinance`'s `Ticker.option_chain()`	Current listed option chains (bid, ask, volume, open interest) for a given expiration; no deep history, and implied volatility is Yahoo's own estimate rather than a modeled surface
CBOE DataShop	Paid	Web portal, bulk download	Official historical end-of-day and intraday options quotes directly from the exchange that lists most US index and equity options
Polygon.io Options API	Free limited tier + paid	REST and WebSocket API	Historical and real-time options chains with server-computed Greeks and implied volatility
WRDS: OptionMetrics (IvyDB US)	Institutional subscription	Web query tool, plus SAS/Python/R client	The academic standard for US options research: daily implied volatility surfaces and Greeks, precomputed, going back to 1996; directly usable for Chapters 5 and 8
ORATS	Paid	REST API	Historical implied volatility surfaces, Greeks, and options analytics aimed at practitioners rather than academics
Deribit	Free, public endpoints	REST and WebSocket API, no key required for public data	The largest crypto options venue; full order book and historical settlement data for BTC and ETH options, useful for practicing the same Black-Scholes-Merton and Greeks machinery from Chapter 5 on an asset class with far higher realized volatility
CME DataMine	Paid	Bulk download / API	Historical futures and futures-options data (rates, commodities, equity index futures) directly from the exchange

Table A.2: Options and derivatives data sources

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Source	Access	API / library	Notes
SEC EDGAR	Free	REST API (`data.sec.gov`); full-text search and XBRL company-facts endpoints	Every US public company's own filings; ties directly to Chapters 4 and 16
Financial Modeling Prep	Free tier + paid	REST API	Pre-parsed income statements, balance sheets, cash flow statements, and ratios
WRDS: Compustat	Institutional subscription	Web query tool, plus SAS/Python/R client	Standardized, restated fundamentals used throughout academic finance research
Macrotrends, StockAnalysis.com	Free (web only)	No official API	Convenient for spot-checking a single company's numbers by hand; not suitable for programmatic bulk use

Table A.3: Fundamental and financial statement data sources

Source	Access	API / library	Notes
FRED	Free, key required	REST API (`fred.stlouisfed.org`); `fredapi` package	Treasury yields, GDP, inflation, unemployment; directly usable for Chapters 5 and 8
World Bank Open Data	Free	REST API (`worldbank.org`)	Global macro and development indicators across countries
IMF Data	Free	REST API	Global macro and financial-stability indicators
U.S. Bureau of Labor Statistics	Free, key optional	REST API (`bls.gov`)	US employment and price-inflation series

Table A.4: Macroeconomic and interest rate data sources

A.8 Text, News, and Sentiment Data

Chapter 16's treatment of sentiment scoring rests on the Loughran-McDonald Master Dictionary, a word list purpose-built for financial text (ordinary sentiment dictionaries misclassify routine financial vocabulary, such as "liability" or "tax," as negative), freely downloadable from its maintainer's University of Notre Dame page. Table A.7 lists this and the other text sources most relevant to that chapter.

A.9 Alternative Data

Chapter 17's case vignette on satellite-derived retail and commodity signals (RS Metrics, UBS, and Orbital Insight's oil-storage monitoring) is representative of this entire category: the underlying vendor data is almost always enterprise-priced and licensed rather than available through a simple API key, since it is expensive to produce and its main customers are institutional investors, not students. Table A.8 separates the handful of

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Source	Access	API / library	Notes
UCI Default of Credit Card Clients	Free	Direct download via `urllib`	Chapter 21's own dataset; Taiwanese credit-card default data
UCI German Credit Data	Free	Direct download	Small, classic credit-scoring dataset, conceptually close to Chapter 6 and 9's examples
Lending Club loan data	Free (via Kaggle mirror)	Kaggle API	Loan-level peer-to-peer lending data; a Chapter 21 capstone idea
HMDA (Home Mortgage Disclosure Act) data	Free	REST API via the Consumer Financial Protection Bureau's data browser (`consumerfinance.gov`)	Mortgage application-level data widely used for fair-lending analysis; directly usable for Chapter 18's fairness audits
French Motor Third-Party Liability (`freMTPL2freq`)	Free	OpenML (`fetch_openml`) or the `CASdatasets` R package	The standard actuarial ratemaking benchmark: roughly 678,000 real motor policies with claim counts and rating features; Chapter 11's challenge exercise

Table A.5: Credit, lending, and consumer finance data sources

alternative-data sources that are genuinely free or low-cost from the enterprise vendors that are not, so a reader does not spend time chasing a free tier that does not exist.

A.10 High-Frequency and Market Microstructure Data

Reconstructing an actual limit order book, the exercise Chapter 3 introduces by hand and Chapters 13 and 14 build trading strategies on top of, requires message-level data that most retail-facing APIs simply do not provide. LOBSTER, an academic data service built on top of NASDAQ's own historical ITCH data, is the most accessible source of genuine, reconstructed limit-order-book data: it offers a free academic tier (typically a handful of sample days for a limited set of tickers) alongside paid access to its full history.

A.11 Factor, Index, and Portfolio Data

No source is more central to Chapter 12's factor-model examples than the Ken French Data Library, maintained by Dartmouth's Tuck School of Business, which has published the Fama-French factor returns (market, size, value, and their later extensions) and a large set of industry-sorted portfolios as free, direct-download CSV and ZIP files for over two decades. It has no formal REST API, but its file layout and URLs have been stable for so long that the `pandas-datareader` package includes a dedicated reader for it.

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Source	Access	API / library	Notes
IEEE-CIS Fraud Detection (Kaggle)	Free (Kaggle competition dataset)	Kaggle API	Real, anonymized e-commerce transaction fraud data; a Chapter 15/21 capstone idea
PaySim	Free	Kaggle API	Synthetic mobile-money transaction data modeled on a real provider's logs; well suited to practicing the structuring and layering detection introduced in Chapter 15
OFAC Sanctions Lists	Free	Bulk data files and a REST search API via the US Treasury	Sanctioned-party screening lists used in AML compliance workflows
CFPB Consumer Complaint Database	Free	REST API and CSV export via the Consumer Financial Protection Bureau (`consumerfinance.gov`)	Millions of consumer complaints about financial products, many with free-text narratives; the proxy Chapter 10's challenge solution uses for classifying operational-risk narratives by category
SEC litigation releases and administrative proceedings	Free	Bulk index files and full-text search via `sec.gov`	Enforcement-action texts rich in operational-loss and misconduct narratives; the other real-text source Chapter 10's challenge exercise suggests

Table A.6: Fraud, AML, and regulatory compliance data sources

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Source	Access	API / library	Notes
SEC EDGAR full-text search	Free	REST API (`data.sec.gov`)	Search across filing text; directly usable for Chapter 16's 10-K risk-factor classification example
Loughran-McDonald Master Dictionary	Free	Direct CSV/text download	The sentiment word list underlying Chapter 16's scoring examples
Financial PhraseBank	Free	Hugging Face `datasets` library, or direct download	Sentence-level, analyst-labeled financial news sentiment
GDELT Project	Free	REST API and BigQuery access	Global news event and tone database, updated continuously
Reddit API	Free, registered app required, rate-limited	`praw` (Python Reddit API Wrapper)	Social sentiment data, e.g., retail-investor forums
X (formerly Twitter) API	Paid tiers	REST and streaming API	Social sentiment; access has moved to paid tiers since 2023, unlike the largely free access assumed by older tutorials

Table A.7: Text, news, and sentiment data sources

Source	Access	API / library	Notes
Google Trends	Free, unofficial	`pytrends` (unofficial wrapper)	Search-interest series, often used as an attention or sentiment proxy
Orbital Insight, RS Metrics	Enterprise, paid	Vendor portal/API	Satellite-derived alternative data referenced in Chapter 17's case vignette; not accessible without a commercial contract
Similarweb, Sensor Tower	Paid	Vendor API	Web-traffic and app-usage panel data
MSCI ESG, Sustainalytics, Refinitiv ESG	Paid	Vendor API	ESG scores and underlying metrics

Table A.8: Alternative data sources

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Source	Access	API / library	Notes
LOBSTER	Free academic tier + paid	Web portal, bulk download (`lobsterdata.com`)	Reconstructed limit order book data for NASDAQ-listed stocks; directly usable for Chapters 3, 13, and 14
Databento	Pay-as-you-go	REST API, historical and live feeds	Modern, lower-cost alternative to legacy tick-data vendors for US equities, futures, and options
WRDS: NYSE TAQ	Institutional subscription	Web query tool, plus SAS/Python/R client	Tick-by-tick trade and quote data; the traditional academic source for Chapter 14-style microstructure research

Table A.9: High-frequency and market microstructure data sources

Source	Access	API / library	Notes
Ken French Data Library	Free	Direct CSV/ZIP download; also wrapped by `pandas-datareader`	Fama-French factor returns and industry portfolios; directly usable for Chapter 12
iShares, SPDR ETF holdings	Free	Daily CSV download from the issuer's own website	Actual ETF basket composition; usable for Chapter 14's ETF-basket arbitrage example
S&P Dow Jones Indices	Free (delayed) + paid (full)	Vendor portal	Index membership, methodology documents, and historical levels

Table A.10: Factor, index, and portfolio data sources

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A.12 Putting It Together: A Chapter-by-Chapter Dataset Map

Table A.11 condenses the sections above into a single reference: for each chapter of this book, which of the sources above is the most natural place to look for real data to extend that chapter's fictional examples. Chapter 21's own closing list of capstone project ideas already names several of these sources directly; this table extends that same mapping to every other chapter.

A.13 Practical Challenges in Sourcing Financial Data

Pulling real data into a project is rarely as simple as calling an API and moving on; several recurring problems are worth flagging explicitly.

Point-in-time and look-ahead bias. Most free APIs return a security's or company's data as it is known *today*, not as it was actually known on the date being studied. Fundamentals get restated after the fact, index membership changes, and a company's own name or ticker can change; naively joining "current" fundamentals to historical prices silently leaks future information into a backtest, the look-ahead-bias problem Chapter 17 raises directly.

Survivorship bias. A free API that only lists currently-traded tickers implicitly excludes every company that has since been delisted, gone bankrupt, or been acquired, which biases any historical universe built from it toward survivors. CRSP's paid, point-in-time data is the academic standard specifically because it avoids this.

Licensing and redistribution restrictions. Most vendor terms of service prohibit redistributing raw data even for non-commercial or educational use, which is why this book's own real-dataset chapters (Chapter 21, and this appendix's own Chapter 5 VIX example) download data programmatically at run time rather than committing it to the repository. The same discipline is worth following in any derivative project.

API and vendor instability. Unofficial APIs such as `yfinance` work by interacting with a public website rather than a documented, contractually stable interface, so they can change behavior or stop working without notice whenever the underlying site changes. Even official free tiers are periodically discontinued or replaced outright, as has happened repeatedly across the market-data-vendor landscape. Every specific vendor and endpoint named in this appendix should therefore be treated as illustrative rather than permanent: verify current availability and terms directly with the provider before building anything on top of one.

Rate limits and cost tiers. Free tiers are typically rate-limited (a fixed number of calls per minute or per day), which matters when a project needs to pull data for hundreds of tickers rather than one; budgeting for a paid tier, or batching and caching requests locally, is often unavoidable at scale.

A.14 Further Reading

- SEC (n.d.), *EDGAR Application Programming Interfaces*.
- Federal Reserve Bank of St. Louis (n.d.), *FRED API Documentation*.
- French (n.d.), *Data Library*.
- LOBSTER (n.d.), *Limit Order Book Data Documentation*.
- Kaggle (n.d.), *Datasets and Public API Documentation*.
- Dua and Graff (2019), *UCI Machine Learning Repository*.
- Loughran and McDonald (2011), "When Is a Liability Not a Liability? Textual Analysis, Dictionaries, and 10-Ks." The paper underlying the Loughran-McDonald Master Dictionary.

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Chapter	Most relevant data sources
1. Introduction to Finance and AI	(conceptual; no dataset needed)
2. Python for Financial Analysis	Yahoo Finance / `yfinance`, for real practice alongside the chapter's synthetic AssetA/AssetB series
3. Financial Markets, Institutions, and Instruments	LOBSTER (limit order book reconstruction), Yahoo Finance
4. Financial Statements and Business Basics	SEC EDGAR (filings and XBRL company facts)
5. Valuation and Asset Pricing	FRED (yield curve), Yahoo Finance / `yfinance` (already used for the VIX in Section A.2), OptionMetrics or Deribit for real options chains (Section A.3)
6. Machine Learning Fundamentals	UCI German Credit Data, or any dataset above used generically for classification practice
7. Time Series Analysis and Forecasting	FRED (macro series), Yahoo Finance (price series)
8. Market Risk Management	Yahoo Finance and FRED, for a real-portfolio VaR backtest; OptionMetrics or Polygon.io for a real delta-normal/delta-gamma VaR options example (Section A.3)
9. Credit Risk Modeling	UCI Default of Credit Card Clients, Lending Club loan data, HMDA data
10. Operational Risk Modeling	CFPB Consumer Complaint Database, SEC litigation releases and administrative proceedings
11. Insurance and Actuarial Modeling	French Motor Third-Party Liability (`freMTPL2freq`, via OpenML or `CASdatasets`)
12. Portfolio Theory and Asset Allocation	Ken French Data Library, Yahoo Finance
13. Algorithmic Trading and Market Microstructure	LOBSTER, Yahoo Finance intraday data (where available)
14. High-Frequency Trading	LOBSTER, Databento, WRDS NYSE TAQ
15. Fraud Detection, Compliance, and Financial Crime	IEEE-CIS Fraud Detection, PaySim, OFAC sanctions lists
16. Natural Language Processing in Finance	SEC EDGAR full-text search, Loughran-McDonald dictionary, Financial PhraseBank, GDELT, Reddit API
17. Alternative Data and Deep Learning in Finance	Google Trends, satellite/alt-data vendors (enterprise), Similarweb
18. Explainability, Fairness, and Regulation	HMDA data, UCI Default of Credit Card Clients (as in Chapter 21's own capstone fairness audit)
19. Reinforcement Learning in Finance	Simulated market environments by design; LOBSTER or Yahoo Finance intraday data for a real extension of the chapter's execution and market-making agents
20. Agent-Based Models in Finance	Simulated agent populations by design; LOBSTER order-book data for a real zero-intelligence-trader comparison, or BIS locational banking statistics for a real interbank exposure network
21. Case Studies and Capstone Projects	The full list above; see Chapter 21's own closing section for specific project pairings

Table A.11: A chapter-by-chapter map from this book's topics to the datasets and APIs most relevant to them

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Appendix B

Glossary of Finance Terms

This glossary is a quick-reference point, not a substitute for the chapter that actually develops a term. Each entry gives a one- or two-sentence working definition and points to the chapter (or chapters) where the concept is developed in full, usually with a worked numerical example; use the glossary to get unstuck on an unfamiliar term encountered before its formal introduction, then follow the pointer for the real treatment. Entries are alphabetized for lookup, since a reader hitting an unfamiliar term mid-chapter needs to find it quickly, not read the glossary front to back. This glossary deliberately does not redefine standard statistics and machine learning vocabulary (regression, neural network, gradient descent, and the like) that this book's intended reader already knows; it covers finance- and trading-specific jargon, regulatory terms, and a handful of quantitative-finance terms of art (e.g., "backtest," "look-ahead bias") that have a specific meaning in this context beyond their everyday use.

Adverse selection The risk that a market maker's counterparty knows something the market maker does not, so that trading against better-informed counterparties is systematically unprofitable; one of the three components Chapter 3 decomposes the bid-ask spread into.

Alpha The portion of an investment's return that is not explained by its exposure to broad market risk (beta); loosely, "skill" or "edge" rather than just being exposed to a rising market. Chapter 12 defines alpha formally as the intercept in a factor-model regression.

Alternative data Non-traditional data sources, such as satellite imagery, web traffic, or shipping data, used as an early proxy for economic activity that has not yet appeared in official statistics or financial statements. See Chapter 17.

AML (Anti-Money Laundering) The set of laws, regulations, and institutional controls designed to detect and prevent the disguising of criminal proceeds as legitimate funds. See Chapter 15.

Backtest Testing a trading strategy or model against historical data to estimate how it would have performed, a central tool of Chapter 13 that is also a common source of overly optimistic, overfit results if not done carefully.

Backtest overfitting The distortion that arises when a strategy is selected as the best of many configurations tested on the same history, so that its reported performance reflects the maximum of many noisy estimates rather than a genuine edge. Chapter 13 shows that searching a hundred worthless strategies over five years of daily data yields an expected best Sharpe ratio of about 1.12. The deflated Sharpe ratio corrects for it.

Balance sheet A financial statement showing what a company owns (assets), what it owes (liabilities), and the residual claim of its owners (equity) at a single point in time. See Chapter 4.

Basel Accords A series of international banking regulatory agreements (Basel I, II, III) setting minimum capital requirements and risk-management standards for banks, spanning market, credit, and operational risk. See Chapters 8, 9, and 10.

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Basis point (bp) One hundredth of one percentage point ($0.01\% = 0.0001$); the standard unit for quoting small differences in interest rates, spreads, and fees. See Chapter 3.

Beta A measure of how sensitive an asset's returns are to the overall market's returns; a beta of 1.5 means the asset tends to move 1.5% for every 1% move in the market. First used informally in Chapter 5, then estimated directly from data in Chapter 12.

Bid-ask spread The gap between the highest price a buyer is currently willing to pay (the bid) and the lowest price a seller is currently willing to accept (the ask); a direct, observable cost of trading and a proxy for an asset's liquidity. See Chapter 3.

Black-Scholes-Merton model A closed-form formula for pricing European call and put options as a function of the underlying's price, the strike price, time to maturity, volatility, and the risk-free rate, derived independently by Black and Scholes and by Merton in 1973; the model whose Greeks (delta, gamma, vega, theta) are the standard language for describing an option position's risk. See Chapter 5.

Bond A debt security in which the issuer (a company or government) promises to make periodic coupon payments and repay the bond's face value at maturity, in exchange for the money the bondholder lent upfront. See Chapter 5.

BSA (Bank Secrecy Act) The primary U.S. law requiring banks to maintain records and file reports (CTRs and SARs) that assist in detecting money laundering. See Chapter 15.

Call option A contract giving its holder the right, but not the obligation, to buy the underlying asset at a fixed strike price on or before a specified date; the holder profits if the underlying's price rises above the strike. See Chapter 5.

CAPM (Capital Asset Pricing Model) A model relating an asset's expected return to its beta and the market's expected excess return; the foundational single-factor model that later multi-factor models extend. See Chapter 12.

Cash flow statement A financial statement reconciling a company's cash balance from one period to the next, broken into operating, investing, and financing activities. See Chapter 4.

Catastrophe bond A bond whose investors lose principal if a specified catastrophe (a hurricane or earthquake meeting defined criteria) occurs, the proceeds going to the issuing insurer or reinsurer; a way of passing modeled tail risk directly to capital markets, paying a spread largely uncorrelated with ordinary market risk. See Chapter 11.

CDD / EDD (Customer / Enhanced Due Diligence) CDD is the baseline process of verifying a customer's identity and establishing their expected activity profile at onboarding; EDD applies heightened scrutiny to higher-risk customers, such as politically exposed persons. See Chapter 15.

CDS (Credit Default Swap) A derivative contract in which the protection buyer pays a periodic premium in exchange for a payout if a specified borrower defaults; effectively insurance against default. See Chapter 9.

CECL (Current Expected Credit Losses) The U.S. accounting standard requiring a lender to provision, from the day a loan is originated, against the credit losses expected over its entire remaining life rather than only the next twelve months. Its international counterpart is IFRS 9. See Chapter 9.

Chain ladder The classical actuarial method for estimating loss reserves: arrange cumulative claims by accident year and development age, estimate average development factors from completed history, and project each open year to its ultimate cost. See Chapter 11.

Circuit breaker An exchange rule that automatically halts trading in a security (or an entire market) after a sufficiently large price move, intended to give participants time to reassess rather than trade on a possible error or panic. See Chapter 14.

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Clearinghouse An intermediary that stands between the two counterparties to a trade after execution, guaranteeing settlement and reducing counterparty risk. See Chapter 3.

COGS (Cost of Goods Sold) The direct costs attributable to producing the goods or services a company sells, subtracted from revenue to compute gross profit. See Chapter 4.

Collateral Assets a borrower pledges to a lender to secure a loan, which the lender may seize if the borrower defaults; the basis for how loss given default depends on how well an exposure is secured. See Chapters 3 and 9.

Co-location Renting rack space physically inside (or adjacent to) an exchange's own data center specifically to minimize the network latency of sending and receiving orders. See Chapter 14.

Cointegration A long-run statistical relationship between two or more non-stationary time series whose particular linear combination is itself stationary; the basis for pairs-trading strategies. See Chapters 7 and 13.

Combined ratio An insurer's losses plus expenses divided by its premiums; below 100% the underwriting operation is profitable in its own right, and even slightly above 100% an insurer may profit overall from investment income on float. See Chapter 11.

Counterparty The other party to a financial transaction or contract; counterparty risk is the risk that this other party fails to honor its obligations. See Chapters 3 and 9.

Coupon The periodic interest payment a bond issuer promises to pay its bondholders, usually expressed as an annual percentage of the bond's face value. See Chapter 5.

Credibility theory The actuarial formula for blending an individual risk's own claims experience with its class average, weighting by $Z = n/(n + k)$; Bayesian shrinkage derived by actuaries decades before machine learning rederived it as regularization. See Chapter 11.

Credit rating A letter grade (e.g., AAA, BBB, CCC) assigned by a rating agency summarizing an issuer's or bond's creditworthiness. See Chapter 9.

Credit spread The extra yield a risky bond pays over a comparable risk-free bond, compensating the lender for default risk; wider spreads mean the market perceives more credit risk. See Chapters 6 and 9.

CTR (Currency Transaction Report) A report a U.S. financial institution must file for any cash transaction exceeding $10,000, regardless of whether the transaction looks suspicious. See Chapter 15.

Dark pool A private trading venue that does not publicly display its order book before a trade executes, used mainly by large institutional traders seeking to limit the market impact of a large order. See Chapter 3.

Default A borrower's failure to make a required payment on a debt obligation as promised; the event whose probability (PD), severity (LGD), and exposure (EAD) together determine expected credit loss. See Chapter 9.

Derivative A financial contract whose value is derived from the price of some other, underlying asset (a stock, a bond, a rate, a commodity), rather than having independent value of its own; options, futures, and CDS are all derivatives covered in this book. See Chapters 3 and 5.

Disparate impact A legal theory under which a facially neutral policy or model is nonetheless unlawful if it produces a substantially worse outcome for a protected group, regardless of intent; the four-fifths rule is the most common numerical screen for it. In U.S. consumer credit, a 2026 amendment of Regulation B removed the theory's "effects test" under ECOA, effective 21 July 2026, though the Fair Housing Act and some state laws still apply it. See Chapter 18.

Diversification Combining assets whose returns are not perfectly correlated so that a portfolio's overall risk is less than the weighted average of its individual assets' risks. See Chapters 2 and 12.

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Dividend A cash (or stock) payment a company distributes to its shareholders, typically out of earnings. See Chapter 3.

Dividend Discount Model (DDM) A valuation method that prices a stock as the present value of all its expected future dividend payments. See Chapter 5.

Drawdown The decline in a portfolio's value from its most recent peak to a subsequent trough, usually expressed as a percentage; maximum drawdown is a common risk metric because it captures the pain of a losing streak in a way volatility alone does not. See Chapter 13.

Duration A bond's approximate percentage price sensitivity to a small change in interest rates; a duration of 5 means the bond's price falls by roughly 5% for every 1 percentage point rise in yields. See Chapter 5.

EAD (Exposure at Default) The estimated amount a lender is exposed to at the moment a borrower defaults, one of the three components (with PD and LGD) of expected loss. See Chapter 9.

EBITDA Earnings Before Interest, Taxes, Depreciation, and Amortization; a rough proxy for a company's operating cash-generating ability that strips out financing decisions, tax jurisdiction effects, and non-cash accounting charges. See Chapter 4.

ECOA (Equal Credit Opportunity Act) The U.S. law prohibiting credit discrimination on the basis of race, sex, and other protected characteristics, and requiring lenders to provide specific reasons when denying credit. Since 21 July 2026 its implementing Regulation B no longer recognizes disparate-impact liability; disparate treatment remains prohibited. See Chapter 18.

Efficient frontier The set of portfolios that offer the highest expected return for each level of risk (or, equivalently, the lowest risk for each level of expected return). See Chapter 12.

ESG (Environmental, Social, and Governance) A set of non-financial criteria used to evaluate a company's or portfolio's exposure to environmental, social, and governance-related risks and practices. See Chapter 12.

Exchange A regulated, centralized venue where buyers and sellers meet to trade standardized securities under a common set of rules. See Chapter 3.

Expected Loss The average loss a lender expects on a credit exposure, computed as the product of probability of default, loss given default, and exposure at default. See Chapter 9.

Expected Shortfall (ES) The average loss in the worst scenarios beyond the VaR threshold; unlike VaR, it answers "how bad, on average, is a bad day" rather than just "how bad is a day that is bad enough." Also called Conditional VaR (CVaR). See Chapter 8.

Face value (par value) The amount a bond's issuer promises to repay the bondholder at maturity, used as the basis for computing coupon payments. See Chapter 5.

Factor model A model explaining an asset's returns as a linear combination of exposures to a small number of common risk factors (market, size, value, momentum, and so on), extending CAPM's single-factor idea. See Chapter 12.

FATF (Financial Action Task Force) An intergovernmental body that sets international standards for combating money laundering and terrorist financing. See Chapter 15.

Fat-finger error A large, unintended order caused by human data-entry error (an extra zero, a misplaced decimal) rather than a deliberate trading decision; exchanges use price collars specifically to limit the damage such an error can cause. See Chapter 14.

Four-fifths rule A screening threshold from the EEOC's 1978 Uniform Guidelines on employee selection, borrowed by fair-lending analysts as a heuristic rather than a legal test for credit: if a protected group's selection or approval rate is less than 80% of the most-favored group's rate, the practice is flagged as having a possible disparate impact. See Chapter 18.

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Free Cash Flow (FCF) The cash a company generates after covering the capital expenditures needed to sustain its operations; the actual cash available to pay down debt, pay dividends, or reinvest. See Chapters 4 and 5.

FRTB (Fundamental Review of the Trading Book) A Basel Committee framework overhauling how banks must measure and hold capital against market risk, developed in response to weaknesses the 2008 financial crisis exposed in earlier VaR-based approaches. See Chapter 8.

Futures contract A standardized, exchange-traded agreement to buy or sell an asset at a predetermined price on a specified future date, marked to market daily. See Chapter 3.

GAAP (Generally Accepted Accounting Principles) The accounting standards used for financial reporting in the United States, the counterpart to the International Financial Reporting Standards used across much of the rest of the world. Differences between the two are one reason ratios pooled across firms in different jurisdictions may not be comparable. See Chapter 4.

GARCH Generalized Autoregressive Conditional Heteroskedasticity; a time-series model in which today's volatility depends on past volatility and past shocks, capturing the empirical fact that volatile periods tend to cluster together. See Chapter 7.

Greeks The sensitivities of an option's price to various underlying factors: delta (price), gamma (delta's own sensitivity to price), vega (volatility), and theta (time). See Chapter 5.

Gross margin Gross profit (revenue minus cost of goods sold) divided by revenue, expressed as a percentage. See Chapter 4.

Hedge A position taken specifically to offset the risk of an existing position, reducing (rather than eliminating) exposure to an adverse price move. See Chapters 5 and 8.

IFRS 9 The international accounting standard governing provisions for credit losses, which sorts every exposure into one of three stages and requires 12-month expected credit loss for performing exposures but *lifetime* expected loss once credit risk has increased significantly since origination. See Chapter 9.

Implied volatility The volatility level that, when plugged into an option-pricing model, reproduces the option's actual observed market price; a market-implied forecast of future volatility rather than a measure of past volatility. See Chapter 5.

Integration The final stage of money laundering, in which layered funds are reintroduced into the legitimate economy (e.g., through a real estate purchase) and become largely indistinguishable from legitimately earned money. See Chapter 15.

Inventory risk The risk a market maker bears that its current net position (long or short) moves against it before an offsetting trade arrives; one of the three components of the bid-ask spread and the central concern of inventory-based market-making models. See Chapters 3 and 13.

IPO (Initial Public Offering) A company's first sale of shares to the public, converting it from privately to publicly held. See Chapter 3.

Key risk indicator (KRI) A measurable quantity, such as staff overtime hours, system downtime, or employee turnover, tracked because it plausibly signals elevated operational-loss risk before any loss occurs; the input to predictive early-warning models of operational risk. See Chapter 10.

KYC (Know Your Customer) The due-diligence process of verifying a customer's identity and understanding their expected financial activity, the foundation of a bank's anti-money-laundering program. See Chapter 15.

Kyle's lambda A measure of price impact: the price change per unit of net order flow, estimated by regressing price changes on signed trading volume. See Chapters 13 and 14.

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Latency arbitrage A strategy that profits from being faster than other participants at reacting to new information or price changes, rather than from superior forecasting. See Chapter 14.

Layering The second stage of money laundering, in which placed funds are moved through a chain of transactions and accounts, often across institutions and jurisdictions, specifically to obscure their criminal origin. See Chapter 15.

Leverage The use of borrowed money to increase the potential return (and risk) of an investment or a company's balance sheet. See Chapters 4 and 8.

LGD (Loss Given Default) The fraction of an exposure a lender expects to lose if a borrower defaults, after accounting for any recovery (e.g., from collateral). See Chapter 9.

Limit order An order to buy or sell at a specified price or better, which rests in the order book until it either executes against an incoming order or is canceled. See Chapter 3.

Liquidity How easily an asset can be bought or sold close to its current price without moving that price; a liquid market has narrow spreads and deep order books. See Chapter 3.

Long position Owning an asset outright, profiting if its price rises; the ordinary way of holding an investment, as opposed to a short position. See Chapter 3.

Look-ahead bias A backtesting error in which a strategy or model is (often inadvertently) given access to information that would not actually have been available at the time of the simulated decision, producing an unrealistically good result. See Chapter 17.

Loss distribution approach The classical actuarial method for quantifying operational risk: build the distribution of a year's total operational losses from a separately estimated frequency distribution (how many loss events occur) and severity distribution (how large each one is), typically by Monte Carlo simulation, then read capital off a far-tail quantile. See Chapter 10.

Margin Collateral a trader must post to open or maintain a leveraged position, protecting the counterparty (or exchange) against the trader's potential losses. See Chapter 3.

Margin call A broker's demand for additional collateral when a leveraged position's losses erode the trader's existing margin below a required maintenance level. See Chapter 3.

Market impact The extent to which placing a trade itself moves the price against the trader, distinct from the price movement that would have happened anyway. See Chapter 13.

Market maker A firm or algorithm that continuously quotes both a bid and an ask, profiting from the spread while bearing inventory risk. See Chapters 3 and 13.

Market order An order to buy or sell immediately at the best currently available price, trading certainty of execution for uncertainty of price. See Chapter 3.

Maturity The date on which a bond's principal is repaid, or an option or futures contract expires; time to maturity is a key input to bond pricing, duration, and option valuation. See Chapter 5.

Mean reversion The tendency of a price or spread to drift back toward its historical average after departing from it; the statistical basis for pairs trading and related strategies. See Chapters 7 and 13.

Merton model A structural credit-risk model treating a firm's equity as a call option on its assets, with default occurring if asset value falls below the face value of debt at maturity. See Chapter 9.

Model risk The risk that a model is wrong, misused, or misunderstood, and that a decision relying on it is therefore wrong even though the model appeared to function correctly; regulators treat it as an operational-risk discipline in its own right. See Chapters 8 and 10.

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NBBO (National Best Bid and Offer) The highest bid and the lowest offer available across all US equity exchanges, consolidated and published by the Securities Information Processor; Regulation NMS requires brokers to respect it when routing and executing orders. See Chapter 14.

Net Present Value (NPV) The present value of an investment's expected future cash flows minus its upfront cost; a positive NPV indicates the investment is expected to create value. See Chapter 5.

No-arbitrage The principle that two portfolios with identical future payoffs must have the same price today, since otherwise a riskless profit could be locked in; the foundation of most derivatives-pricing theory. See Chapters 3 and 5.

Notional (value) The total value a derivative contract controls or references, as distinct from the (usually much smaller) amount of capital actually posted to enter the position. See Chapter 5.

Operational risk The risk of loss resulting from inadequate or failed internal processes, people, and systems, or from external events, a rogue trader, a systems outage, a mis-sold product, as distinct from the market risk of price moves and the credit risk of counterparty default. See Chapter 10.

Option A derivative contract giving its holder the right, but not the obligation, to buy or sell an underlying asset at a fixed strike price by a specified date, in exchange for an upfront premium; see *call option* and *put option* for the two basic types. See Chapter 5.

Order book The list of all currently resting limit orders for a given security, organized by price level, showing the depth of buying and selling interest at each price. See Chapter 3.

Order flow The stream of incoming buy and sell orders a market or market maker observes; signed order flow (buys minus sells) is the input to price-impact measures such as Kyle's lambda. See Chapters 13 and 14.

OTC (Over-the-Counter) Trading conducted directly between two counterparties rather than on a centralized, regulated exchange. See Chapter 3.

Pairs trading A strategy that takes offsetting long and short positions in two historically related securities, betting that a temporary divergence between them will revert to its historical relationship. See Chapter 13.

PD (Probability of Default) The estimated likelihood that a borrower will fail to meet its debt obligations over a given time horizon. See Chapter 9.

PIN (Probability of Informed Trading) A market microstructure measure of the fraction of order flow estimated to come from traders with private information, as opposed to uninformed liquidity traders. See Chapter 14.

Placement The first stage of money laundering: introducing illicit cash into the financial system, for example through structured deposits. See Chapter 15.

Portfolio A collection of financial assets (stocks, bonds, cash, derivatives, or any combination) held together by an investor; this book's running two- and three-asset portfolios are the simplest possible example. See Chapters 2 and 12.

Present Value (PV) The value today of a future cash flow (or stream of cash flows), computed by discounting it back at an appropriate rate. See Chapter 5.

Prime broker A bank or securities firm providing a bundle of services (financing, securities lending, clearing) to institutional clients such as hedge funds. See Chapter 3.

Primary market The market in which securities are first sold by their issuer to investors (e.g., an IPO), as opposed to a secondary market where existing securities change hands between investors. See Chapter 3.

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Pure premium The expected loss cost of an insurance policy, expected claim frequency times expected claim severity, before loading for expenses and profit; the risk-based core of an insurance price. See Chapter 11.

Put option A contract giving its holder the right, but not the obligation, to sell the underlying asset at a fixed strike price on or before a specified date; the holder profits if the underlying's price falls below the strike. See Chapter 5.

Recovery rate The fraction of a defaulted exposure a lender ultimately recovers, typically through collateral or bankruptcy proceedings; loss given default is simply one minus the recovery rate. See Chapter 9.

Reinsurance Insurance purchased by insurers: contracts (often written in layers above an attachment point) that transfer part of an insurer's losses, especially correlated catastrophe losses that defeat ordinary pooling, to a reinsurer. See Chapter 11.

Repo (Repurchase Agreement) A short-term, collateralized loan structured as a sale of a security with an agreement to repurchase it later at a slightly higher price, the effective interest being the difference. See Chapter 3.

Retained earnings The cumulative portion of a company's net income that has been kept (not paid out as dividends) since the company's founding. See Chapter 4.

Risk-free rate The theoretical return on an investment with zero risk of default, typically proxied in practice by a short-term government bond yield; the baseline every risk premium (equity risk premium, credit spread) is measured against. See Chapters 5 and 12.

ROA (Return on Assets) Net income divided by total assets, measuring how efficiently a company generates profit from everything it owns. See Chapter 4.

ROE (Return on Equity) Net income divided by shareholders' equity, measuring the return generated on shareholders' invested capital. See Chapter 4.

ROIC (Return on Invested Capital) After-tax operating profit divided by the total capital (debt plus equity) invested in the business, often compared against WACC to judge whether a company is creating or destroying economic value. See Chapter 4.

Robo-advisor An automated investment-advisory service that builds and manages a portfolio for a client based on a short risk questionnaire, typically using mean-variance or similar optimization rather than a human advisor's judgment. See Chapter 12.

Rogue trading Unauthorized trading by an employee who conceals positions or losses from their institution's own controls, historically the source of several multi-billion-dollar operational losses (Barings in 1995, Société Générale in 2008, UBS in 2011) and the canonical example of internal fraud among the Basel operational-risk event types. See Chapter 10.

SAR (Suspicious Activity Report) A report a financial institution must file for any transaction pattern, regardless of size, that appears designed to evade reporting requirements or otherwise suggests criminal activity. See Chapter 15.

Sanctions screening Checking counterparties against government-maintained lists of prohibited individuals, entities, and countries; a binary legal-compliance check rather than a statistical risk model. See Chapter 15.

Scenario analysis Structured expert judgment estimating the frequency and severity of severe hypothetical events that have not occurred but plausibly could; in operational risk, the one data element able to speak to genuinely unprecedented tail events, and in market risk, a companion to statistical stress testing. See Chapters 8 and 10.

Secondary market The market in which investors trade securities among themselves after their original issuance; most day-to-day trading (e.g., on a stock exchange) happens here. See Chapter 3.

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Sharpe ratio A portfolio's average excess return (over the risk-free rate) divided by the standard deviation of that return; the standard measure of return earned per unit of risk taken. See Chapters 12 and 13.

Shell company An entity with no significant independent operations or assets, often used to obscure the true ownership or purpose of funds flowing through it. See Chapter 15.

Short position (short selling) Selling a borrowed asset now with the obligation to return an equivalent asset later, a bet that the price will fall. See Chapter 3.

SIP (Securities Information Processor) The consolidated feed that aggregates quotes from every US equity exchange into the single published NBBO; because it must collect and synchronize those quotes first, it is structurally slower than any individual exchange's own direct feed, which is what makes latency arbitrage against it possible. See Chapter 14.

Slippage The difference between the price a trader expected when deciding to trade and the price actually achieved once the order executes. See Chapter 13.

SR 11-7 and SR 26-2 U.S. supervisory guidance on model risk management. SR 11-7 (Federal Reserve and OCC, 2011) built it on conceptual soundness, ongoing monitoring, and outcomes analysis, backed by independent validation; on 17 April 2026 it was superseded by SR 26-2 (OCC Bulletin 2026-13), issued jointly with the FDIC, which keeps those pillars, scales validation to a model's risk, and leaves generative and agentic AI outside its scope. See Chapters 8 and 18.

Stationarity A time series property in which the statistical characteristics (mean, variance, autocorrelation) do not change over time; many forecasting models require stationarity, or a transformation (such as differencing) to achieve it. See Chapter 7.

Stock (equity) A security representing fractional ownership of a company, entitling the holder to a proportional share of its residual value and (usually) a vote in corporate matters; also called equity, as opposed to debt. See Chapter 3.

Stress testing Evaluating how a portfolio, model, or institution would perform under an extreme but plausible adverse scenario, rather than relying solely on statistical measures estimated from typical historical data. See Chapter 8.

Strike price The fixed price at which an option holder may buy (call) or sell (put) the underlying asset. See Chapter 5.

Structuring (smurfing) Deliberately breaking a large cash transaction into several smaller ones, each below a regulatory reporting threshold, specifically to avoid triggering a report. See Chapter 15.

Survivorship bias The distortion introduced by analyzing only entities (companies, funds, strategies) that still exist today, silently excluding the failures that were removed along the way. See Chapters 1 and 13.

Systematic risk (and idiosyncratic risk) Systematic risk is the portion of an asset's risk driven by broad market-wide factors and cannot be diversified away; idiosyncratic risk is asset-specific and shrinks toward zero as a portfolio is diversified. This distinction is what CAPM's beta measures and diversification exploits. See Chapter 12.

10-K filing A U.S. public company's annual report filed with the SEC, containing audited financial statements alongside extensive qualitative disclosure, including a risk-factors section; a primary real-world source of the text financial NLP methods are applied to. See Chapter 16.

Telematics Driving-behavior data (mileage, speed, time of day, harsh braking) collected by a device or phone app and used to price usage-based auto insurance on observed behavior rather than demographic proxies. See Chapter 11.

Terminal value The estimated value of all cash flows beyond an explicit forecast horizon, typically the largest single component of a discounted-cash-flow valuation. See Chapter 5.

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Three lines of defense A governance framework dividing risk-management responsibility across business units (first line), independent risk and compliance functions (second line), and internal audit (third line). See Chapters 8 and 15.

Tick size The minimum allowed price increment a security can move by on an exchange. See Chapter 14.

Transition risk The financial impact on a company or portfolio of the policy, technology, and consumer-preference shifts associated with a transition to a lower-carbon economy, as distinct from physical climate risk. See Chapter 8.

TWAP / VWAP Time-Weighted and Volume-Weighted Average Price; two common execution benchmarks (and the algorithms designed to track them) that split a large order into smaller pieces over time, either evenly (TWAP) or in proportion to expected trading volume (VWAP). See Chapter 13.

Underlying (asset) The asset a derivative contract's value is based on, e.g., the stock an option gives the right to buy or sell. See Chapters 3 and 5.

Underwriting The process of evaluating and pricing the risk of extending credit or issuing insurance, historically performed by a human underwriter and increasingly informed by statistical scoring models. See Chapters 3, 9, and 11.

Unit root A property of a non-stationary time series in which shocks have a permanent rather than temporary effect; tested for directly (e.g., with the Augmented Dickey-Fuller test) before fitting many time-series models. See Chapter 7.

VaR (Value at Risk) A single dollar (or percentage) figure summarizing the loss a portfolio is not expected to exceed over a given time horizon at a given confidence level. See Chapter 8.

Volatility A statistical measure (typically the standard deviation of returns) of how much an asset's price fluctuates; higher volatility means larger, more frequent price swings. See Chapters 2 and 5.

Volatility clustering The empirical tendency for periods of high volatility to be followed by more high volatility, and calm periods to be followed by more calm, rather than volatility being independent from one period to the next. See Chapter 7.

WACC (Weighted Average Cost of Capital) A company's blended cost of financing, combining the cost of debt and the cost of equity in proportion to how much of each the company uses; the standard discount rate for valuing a firm's overall cash flows. See Chapter 4.

Walk-forward validation A time-series-appropriate alternative to a random train-test split, in which a model is repeatedly trained on data only up to a point in time and tested on the period immediately following, mimicking how the model would actually have been used in real time; essential for avoiding look-ahead bias in trading-signal backtests. See Chapters 7 and 13.

Working capital Current assets minus current liabilities; a measure of a company's short-term liquidity and ability to fund its day-to-day operations. See Chapter 4.

Yield The return an investor earns on a bond if held to maturity, accounting for both coupon payments and any difference between the purchase price and face value. See Chapter 5.

Yield curve A plot of yields against maturity for otherwise-comparable bonds (typically government bonds), whose shape (upward-sloping, flat, inverted) is widely watched as a signal about market expectations for future interest rates and growth. See Chapters 5 and 7.

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Appendix C

A Career Map: Roles, Chapters, and Preparation

This book's stated audience is a reader with quantitative training who wants a career in the financial industry, and the chapters have been building the technical vocabulary that industry runs on. What the chapters do not do, because no single chapter is the right place for it, is describe the industry's job landscape: which roles exist, what each one does all day, and which parts of this book matter most for each. This appendix closes that gap. It is a map, not a syllabus; job titles vary considerably across firms (the same work may be called "quantitative researcher" at a hedge fund, "strats" at an investment bank, and "data scientist" at a fintech), so the descriptions below are organized around what the work is rather than what any particular firm calls it.

One framing helps before the table. Broadly, quantitative roles in finance divide into those that *make* risk (research and trading roles, judged on returns), those that *measure and police* risk (risk management, model validation, and compliance roles, judged on rigor and independence), and those that *build the machinery* both sides run on (engineering and data roles, judged on reliability and scale). All three groups hire people with this book's intended background, the mix of statistics, programming, and financial context, but they reward different temperaments, and the interview processes differ accordingly: the first group leans hardest on probability, brainteasers, and market intuition; the second on statistical rigor, regulatory awareness, and the ability to explain a model's limits; the third on software engineering and system design, with the finance tested more lightly.

C.1 Reading the Map

A few patterns in Table C.1 deserve comment. Chapters 2 and 6, Python and machine learning fundamentals, appear implicitly in every row; they are the price of admission everywhere and are omitted from rows only where a role's distinctive preparation lies elsewhere. Chapter 18, explainability and fairness, appears in every role that touches a regulated decision about a person, which is the pattern the chapter itself predicts: the closer a model sits to a consumer outcome, the more its explanation apparatus matters relative to its accuracy. And Chapter 21's capstone appears wherever the answer to "what should I build to demonstrate competence?" is a complete, end-to-end project rather than familiarity with a single method; a worked capstone, cleaned data, calibration checks, and fairness audit included, is the most credible artifact a candidate without industry experience can bring to an interview, because its nine-stage template mirrors how the work is structured inside a firm.

The risk-and-validation column of the industry deserves a special word, because candidates with strong ML backgrounds systematically underrate it. Model validation and risk roles are often easier to enter than front-office research, teach the firm's full model landscape faster than any other seat, are hired against a regulatory requirement rather than a profit forecast (making them notably more stable across market cycles), and increasingly involve hard technical work: challenging a deep-learning surveillance model requires understanding it at least as well as its builder did. A reader whose favorite chapters were 8 through 11 and 18 should take that as information about where they would thrive.

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Role	What the job actually is	Core chapters
Quantitative researcher (buy side)	Finding and testing predictive signals in market, fundamental, text, and alternative data; building and validating backtests	2, 5–7, 12, 13, 16, 17
Execution and e-trading quant	Designing the algorithms that execute orders and make markets: impact models, order routing, market-making logic	3, 13, 14, 19
Market risk quant	Measuring how much the firm can lose to market moves: VaR and ES models, stress testing, backtesting, limits	5, 7, 8
Credit risk modeler	Estimating who will default and what it costs: PD/LGD/EAD models, scorecards, portfolio credit models, capital	4, 6, 9, 18, 21
Actuary and insurance data scientist	Pricing, reserving, claims, and catastrophe modeling for insurers: GLMs and their machine-learning challengers inside a filed-rating regime	5, 6, 10, 11, 18
Model validator (second line)	Independently challenging every model above: testing assumptions, benchmarking, documentation, SR 11-7/SR 26-2 discipline	6–11, 18
Operational-risk and surveillance analyst	Quantifying and detecting operational failures: loss models, anomaly detection, conduct and communications surveillance	6, 10, 15, 16
Fraud and financial-crime data scientist	Catching fraud, money laundering, and market abuse in transaction and network data, at extreme class imbalance	6, 14, 15, 17, 18
NLP and alternative-data researcher	Turning filings, calls, news, satellite and web data into investable signals with point-in-time discipline	13, 16, 17
Fintech and lending ML engineer	Building production credit, robo-advisory, and payments models end to end, with fairness and explainability constraints	2, 6, 9, 12, 18, 21
AI governance and compliance specialist	Setting and enforcing the rules AI systems must satisfy: model inventories, monitoring, fairness audits, regulatory response	10, 15, 16, 18

Table C.1: A map from common quantitative-finance roles to the chapters that prepare for them

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C.1.1 Roles That Have Taken Shape by 2026

The ten rows above describe seats that have existed for a decade or more. Table C.2 adds seven that have grown, in the last few years, from a side responsibility into a job title of their own, each mapped to the chapters that prepare for it in the same way.

Each of the seven follows from a development this book has traced. The validator and governance rows exist because the models changed faster than the control functions around them: a validator trained on regressions must now challenge a gradient-boosted credit model, a language model's free-text output, or a policy learned by trial and error, and Chapter 20's discussion of algorithmic collusion and herding shows why an institution running many AI agents needs someone watching what they do together, not only one at a time. The LLM engineer is the production counterpart of Chapter 16's generative-AI sections, where most of the work turns out to be evaluation and control rather than the model itself. Payments fraud became its own specialty as instant-payment systems spread, since a payment that settles in seconds cannot be recalled once a scam is noticed, so the decision has to be right before authorization. The treasury and ALM quant rose in standing after the 2023 failure of Silicon Valley Bank, which Chapter 8 examines as an interest-rate-risk and deposit-run event rather than a credit one. And climate-risk and central-bank research roles have pulled in the scenario analysis, stress testing, and agent-based methods of Chapters 8, 11, and 20. What the seven share is the pattern of the original table: each pairs a specific financial problem with the machine-learning tools now applied to it, and each rewards a candidate who can explain where those tools fail.

C.2 Preparing, Concretely

Three uses of this book map onto how candidates are evaluated. First, the worked examples are interview questions: computing a two-asset portfolio variance by hand, walking through a limit-order-book fill, explaining why the 99.9% operational-loss quantile dwarfs its mean, deriving a Shapley decomposition for a small linear model, and reasoning through Bayes' theorem at a realistic base rate are all standard fare, and every one of them is a worked example somewhere in Chapters 1 through 21. Second, the challenge exercises are portfolio projects: each one pairs a method from a chapter with a real, freely downloadable dataset, which is the scope of take-home assignments many firms now use in place of a second interview round. Third, the Further Reading lists are the canon: Hull for derivatives and risk, Bodie-Kane-Marcus for investments, Chan and Hilpisch for systematic trading, and the original papers behind XGBoost, SHAP, and the transformer are the references interviewers themselves learned from, and recognizing them signals fluency the way no self-description can.

The map's final message is the book's own: every role in Table C.1 is being reshaped by the same forces this book has traced chapter by chapter, and the durable advantage is not any single tool but the combination the intended reader has been building all along, enough statistics to be skeptical, enough programming to be self-sufficient, and enough financial context to know what the numbers mean and what happens when they are wrong.

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Role	What the job actually is	Core chapters
AI model validator	Independently challenging machine-learning, language, and learned-policy models rather than classical ones: out-of-time and stress testing, calibration and fairness checks, evaluation of generative output, and the limits of explanation tools	6, 8, 16, 18, 19, 21
AI governance and agent supervisor	Overseeing AI agents that act rather than advise: what each agent may do, human approval points, audit trails, kill switches, and monitoring for collective behavior across agents	10, 16, 18, 19, 20, 21
LLM engineer (financial services)	Building retrieval-augmented assistants, document extraction, and agent tooling on large language models, with evaluation, guardrails, and data controls a regulator will accept	2, 6, 16, 17, 18
Payments-fraud data scientist	Scoring card and instant-payment transactions in the milliseconds before authorization, including scams in which the customer authorizes the payment, using network and behavioral features	3, 6, 15, 17, 18, 21
Climate-risk analyst	Translating physical and transition risk into credit, market, and insurance exposures: scenario analysis, carbon-price shocks, catastrophe models	9, 8, 11, 12, 17
Treasury and ALM quant	Managing a bank's or insurer's balance sheet as a whole: interest-rate risk in the banking book, deposit behavior, liquidity and funding, hedging	3, 5, 7, 8, 11
Central-bank research economist or data scientist	Forecasting, financial-stability monitoring, system-wide stress tests, and text analysis of policy communication, increasingly with agent-based and machine-learning tools	3, 7, 8, 16, 20

Table C.2: Seven roles that have taken shape as distinct jobs by 2026, and the chapters that prepare for them

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Bibliography

Every chapter's own "Further Reading" section lists sources with a short note on how each connects to that chapter's material. This bibliography consolidates every source cited in the book into a single alphabetized list; several sources are cited in more than one chapter and appear here only once. Full author names, publishers, editions, and (for articles) volume, issue and page numbers were verified rather than guessed; a handful of entries for which no single edition, page range or exact date could be confirmed are given in the fullest form that could be verified, without an invented detail.

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